



# Stage 04 — Operational Target Investigation

S04\_operational\_target\_investigation.ipynb

## Configuración del Entorno

### Importación de librerías

```
In [2]: import sys
import re
#Instalación de librería pandas_market_calendars
#!{sys.executable} -m pip install -q pandas_market_calendars
#print("Librería instalada: pandas_market_calendars")

from functools import reduce
# Utilidades generales
from datetime import datetime, timedelta
import os
import glob
import requests
import warnings
warnings.filterwarnings('ignore')

# Manejo y procesamiento de datos
#import ta
import pandas as pd
import numpy as np
from tabulate import tabulate
import matplotlib.pyplot as plt
# Calendario de mercados
#import pandas_market_calendars as mcal

#from ta.momentum import StochasticOscillator, ROCIndicator
#from ta.volatility import BollingerBands, AverageTrueRange

#from scipy.stats import spearmanr
from __future__ import annotations # permite type hints modernos (Python < 3.11
import json # para guardar el summary como JSON
import os # para leer variables de entorno
from pathlib import Path # manejo robusto de rutas
import pandas as pd
```

### Rutas de archivos

```
In [3]: from pathlib import Path

# Buscar la raíz del proyecto
```

```
PROJECT_ROOT = Path.cwd()

while PROJECT_ROOT.name != "neural_profit_local":
    PROJECT_ROOT = PROJECT_ROOT.parent

print("Project root:", PROJECT_ROOT)
```

Project root: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local

## Carga de dataset mnq\_intraday.parquet

```
In [4]: MNQ_INTRADAY_PATH = PROJECT_ROOT / "data" / "02_mnq_intraday"
MNQ_INTRADAY_PATH.mkdir(parents=True, exist_ok=True)
MNQ_INTRADAY_PARQUET = MNQ_INTRADAY_PATH / "mnq_intraday.parquet"
```

```
In [5]: def load_mnq_parquet(dataset_path):
os.path.exists(dataset_path)
print("Archivo encontrado en disco. Cargando dataset local...")
mnq_parquet = pd.read_parquet(dataset_path)
return mnq_parquet

mnq_intraday = load_mnq_parquet(MNQ_INTRADAY_PARQUET)
```

Archivo encontrado en disco. Cargando dataset local...

## Función para ver información de dataset

```
In [6]: from __future__ import annotations

from dataclasses import dataclass
from typing import Any, Dict, Optional, Tuple

import pandas as pd

def mnq_dataset_info(
    df: pd.DataFrame,
    *,
    name: str = "mnq_raw",
    tz_assume_if_naive: Optional[str] = None, # ej: "UTC" o "America/New_York"
    day_def: str = "calendar", # "calendar" (fecha calendario) o "trading" (día)
) -> Dict[str, Any]:
    """
    Resume un dataset OHLCV con DatetimeIndex (ideal para mnq_raw).

    - Si el índice es tz-naive:
        - Si tz_assume_if_naive != None, lo localiza a esa tz.
        - Si no, reporta "tz-naive" (no se puede afirmar horario UTC).
    - Devuelve dict con métricas principales (y lo imprime bonito si se desea).
    """
    if not isinstance(df.index, pd.DatetimeIndex):
        raise TypeError(f"{name}: se requiere DatetimeIndex, recibido: {type(df.index)}")

    idx = df.index

    # --- timezone / UTC info ---
    tzinfo = idx.tz
    if tzinfo is None:
```

```

        tz_status = "tz-naive (sin zona horaria)"
    if tz_assume_if_naive:
        idx = idx.tz_localize(tz_assume_if_naive)
        tzinfo = idx.tz
        tz_status = f"localizado como {tzinfo}"
    else:
        tz_status = f"{tzinfo}"

    # --- rango temporal ---
    ts_min = idx.min()
    ts_max = idx.max()

    first_day = ts_min.date()
    last_day = ts_max.date()

    # --- días ---
    if day_def == "calendar":
        total_days = (pd.Timestamp(last_day) - pd.Timestamp(first_day)).days + 1
    elif day_def == "trading":
        total_days = idx.normalize().nunique()
    else:
        raise ValueError("day_def debe ser 'calendar' o 'trading'")

    # --- columnas ---
    columns = list(df.columns)

    # --- checks útiles ---
    n_rows = len(df)
    n_cols = df.shape[1]
    n_missing = int(df.isna().sum().sum())
    missing_by_col = df.isna().sum().to_dict()
    dup_index = int(idx.duplicated().sum())
    is_monotonic = bool(idx.is_monotonic_increasing)

    # Frecuencia estimada (puede fallar si hay huecos grandes)
    freq = pd.infer_freq(idx[: min(50000, len(idx))]) # muestra grande pero aco

    # Cobertura por día (min/max de hora del día, en tz del índice)
    # (Útil para ver si es 24/7 o horario de sesión)
    tod = pd.Series(idx.time)
    # Convertimos time a minutos del día para resumen robusto
    tod_minutes = pd.Series([t.hour * 60 + t.minute for t in tod])
    typical_minute_min = int(tod_minutes.min())
    typical_minute_max = int(tod_minutes.max())

    # Rango promedio de filas por día (sólo días con datos)
    rows_per_day = df.groupby(idx.normalize()).size()
    rows_per_day_stats = {
        "days_with_data": int(rows_per_day.shape[0]),
        "rows_per_day_min": int(rows_per_day.min()),
        "rows_per_day_p50": float(rows_per_day.median()),
        "rows_per_day_max": int(rows_per_day.max()),
    }

    # Si hay tz, también mostramos rango en UTC
    if tzinfo is not None:
        ts_min_utc = ts_min.tz_convert("UTC")
        ts_max_utc = ts_max.tz_convert("UTC")
        utc_range = (str(ts_min_utc), str(ts_max_utc))
        utc_note = "El índice está tz-aware; el horario UTC es inequívoco."

```

```

else:
    utc_range = None
    utc_note = "El índice es tz-naive; no se puede asegurar si está en UTC s

info: Dict[str, Any] = {
    "name": name,
    "shape": (n_rows, n_cols),
    "columns": columns,
    "index_type": type(df.index).__name__,
    "index_tz": tz_status,
    "utc_note": utc_note,
    "datetime_min": str(ts_min),
    "datetime_max": str(ts_max),
    "first_day": str(first_day),
    "last_day": str(last_day),
    "total_days": int(total_days),
    "day_definition": day_def,
    "utc_range_if_applicable": utc_range,
    "is_index_monotonic_increasing": is_monotonic,
    "duplicated_timestamps_in_index": dup_index,
    "inferred_freq_sample": freq,
    "missing_total_cells": n_missing,
    "missing_by_col": missing_by_col,
    "rows_per_day_stats": rows_per_day_stats,
    "time_of_day_minutes_range": {
        "min_minute_of_day": typical_minute_min,
        "max_minute_of_day": typical_minute_max,
    },
}
}
return info

def print_mnq_dataset_info(info: Dict[str, Any]) -> None:
    """Imprime el dict de mnq_dataset_info de forma ordenada."""
    print(f"Dataset: {info['name']}")
    print(f"Shape: {info['shape']}")
    print(f"Columns: {info['columns']}")
    print(f"Index: {info['index_type']} | TZ: {info['index_tz']}")
    print(f"Datetime min/max: {info['datetime_min']} -> {info['datetime_max']}")
    print(f"First/Last day: {info['first_day']} -> {info['last_day']}")
    print(f"Total days ({info['day_definition']}): {info['total_days']}")
    #print(f"Inferred freq (sample): {info['inferred_freq_sample']}")
    #print(f"Index monotonic increasing: {info['is_index_monotonic_increasing']}")
    #print(f"Duplicated timestamps in index: {info['duplicated_timestamps_in_ind")
    #print(f"Missing total cells: {info['missing_total_cells']}")
    #print(f"Missing by col: {info['missing_by_col']}")
    #print(f"Rows/day stats: {info['rows_per_day_stats']}")
    print(f"Time-of-day range (minutes): {info['time_of_day_minutes_range']}")
    print(f"UTC note: {info['utc_note']}")
    if info["utc_range_if_applicable"] is not None:
        print(f"UTC range: {info['utc_range_if_applicable'][0]} -> {info['utc_

```

```

In [7]: info = mnq_dataset_info(mnq_intraday, name="mnq_intraday", tz_assume_if_naive=No
print_mnq_dataset_info(info)

```

Dataset: mnq\_intraday  
 Shape: (1024062, 9)  
 Columns: ['date', 'minute\_of\_day', 'regime\_id', 'open', 'high', 'low', 'close', 'volume', 'contract']  
 Index: DatetimeIndex | TZ: America/New\_York  
 Datetime min/max: 2020-01-02 04:30:00-05:00 -> 2026-04-17 16:00:00-04:00  
 First/Last day: 2020-01-02 -> 2026-04-17  
 Total days (trading): 1482  
 Time-of-day range (minutes): {'min\_minute\_of\_day': 270, 'max\_minute\_of\_day': 960}  
 UTC note: El índice está tz-aware; el horario UTC es inequívoco.  
 UTC range: 2020-01-02 09:30:00+00:00 -> 2026-04-17 20:00:00+00:00

## Validación temporal de dataset mnq\_intraday

Valida lo esencial para series temporales antes de construir targets: orden cronológico, duplicados, consistencia por día, monotonía de minute\_of\_day, gaps temporales y saltos sospechosos. Esto es importante porque en trading hay que validar cuidadosamente los timestamps para evitar look-ahead bias, y porque en series temporales el orden secuencial es parte central del problema.

```

In [8]: import pandas as pd
import numpy as np

# =====
# Validación temporal de mnq_intraday
# Ejecutar inmediatamente después de:
# mnq_intraday = load_mnq_parquet()
# =====

def validate_mnq_intraday(df: pd.DataFrame, verbose: bool = True) -> dict:
    """
    Valida consistencia temporal básica para un dataset intradía.

    Chequeos:
    1) Índice datetime válido
    2) Orden cronológico global
    3) Duplicados de timestamp
    4) Consistencia de columna `date`
    5) Monotonía de `minute_of_day` dentro de cada día
    6) Duplicados de `minute_of_day` dentro de cada día
    7) Saltos temporales negativos o nulos
    8) Gaps intradía distintos de 1 minuto
    """

    result = {
        "ok": True,
        "checks": {},
        "summary": {},
        "artifacts": {}
    }

    df = df.copy()

    # -----
    # 0) Verificaciones básicas de estructura
    # -----
    required_cols = ["date", "minute_of_day", "open", "high", "low", "close", "v
  
```

```

missing_cols = [c for c in required_cols if c not in df.columns]
if missing_cols:
    raise ValueError(f"Faltan columnas requeridas: {missing_cols}")

if not isinstance(df.index, pd.DatetimeIndex):
    raise TypeError("El índice del DataFrame debe ser un pd.DatetimeIndex")

if df.empty:
    raise ValueError("El DataFrame está vacío")

# -----
# 1) Orden global del índice
# -----
is_monotonic = df.index.is_monotonic_increasing
has_unique_index = df.index.is_unique
duplicated_index = df.index[df.index.duplicated()].unique()

result["checks"]["index_is_monotonic_increasing"] = bool(is_monotonic)
result["checks"]["index_is_unique"] = bool(has_unique_index)
result["summary"]["n_duplicated_timestamps"] = int(len(duplicated_index))
result["artifacts"]["duplicated_timestamps"] = duplicated_index

# Si no está ordenado, mostramos evidencia pero no reordenamos silenciosamen
if not is_monotonic:
    diffs_ns = pd.Series(df.index.view("i8")).diff()
    bad_order_pos = np.where(diffs_ns <= 0)[0]
    result["artifacts"]["bad_global_order_positions"] = bad_order_pos[:20]
    result["ok"] = False

if not has_unique_index:
    result["ok"] = False

# -----
# 2) Consistencia entre index.date y columna `date`
# -----
# Normalizamos ambos a fecha sin hora
index_dates = pd.Index(df.index.tz_localize(None).date if df.index.tz is not
col_dates = pd.to_datetime(df["date"]).dt.date

date_match = (index_dates == col_dates).all()
result["checks"]["date_column_matches_index_date"] = bool(date_match)

if not date_match:
    mismatch_mask = index_dates != col_dates
    mismatches = df.loc[mismatch_mask, ["date", "minute_of_day", "close"]].h
    result["artifacts"]["date_mismatches_head"] = mismatches
    result["summary"]["n_date_mismatches"] = int(mismatch_mask.sum())
    result["ok"] = False
else:
    result["summary"]["n_date_mismatches"] = 0

# -----
# 3) Diferencias temporales globales
# -----
# Trabajamos en segundos
diffs_sec = pd.Series(df.index).diff().dt.total_seconds()

n_non_positive_diffs = int((diffs_sec.iloc[1:] <= 0).sum())
result["checks"]["all_global_time_diffs_positive"] = (n_non_positive_diffs == 0)
result["summary"]["n_non_positive_global_diffs"] = n_non_positive_diffs

```

```

if n_non_positive_diffs > 0:
    bad_diff_rows = df.iloc[np.where((diffs_sec <= 0).fillna(False))[0][:20]]
    result["artifacts"]["non_positive_global_diffs_head"] = bad_diff_rows
    result["ok"] = False

# -----
# 4) Validación por día
# -----

daily_stats = []
bad_minute_order_days = []
duplicate_minute_days = []
intraday_gap_rows = []

grouped = df.groupby("date", sort=False)

for day, g in grouped:
    g = g.copy()

    # 4.1 orden del índice dentro del día
    idx_mono = g.index.is_monotonic_increasing

    # 4.2 minute_of_day creciente dentro del día
    mod_diff = g["minute_of_day"].diff()
    minute_order_ok = bool((mod_diff.iloc[1:] > 0).all())

    # 4.3 duplicados de minute_of_day dentro del día
    dup_mod = g["minute_of_day"].duplicated().sum()
    has_dup_mod = dup_mod > 0

    # 4.4 gaps intradía del índice
    idx_diff_sec = pd.Series(g.index).diff().dt.total_seconds()
    gap_mask = (~idx_diff_sec.isna()) & (idx_diff_sec != 60)

    n_intraday_gaps = int(gap_mask.sum())

    if not minute_order_ok:
        bad_minute_order_days.append(day)

    if has_dup_mod:
        duplicate_minute_days.append(day)

    if n_intraday_gaps > 0:
        gap_info = g.loc[gap_mask, ["date", "minute_of_day", "open", "high",
                                     "gap_seconds"]] = idx_diff_sec[gap_mask].values
        intraday_gap_rows.append(gap_info)

    daily_stats.append({
        "date": day,
        "n_rows": len(g),
        "index_monotonic": bool(idx_mono),
        "minute_of_day_monotonic": minute_order_ok,
        "n_duplicate_minute_of_day": int(dup_mod),
        "n_intraday_gaps_not_60s": n_intraday_gaps,
        "minute_min": int(g["minute_of_day"].min()),
        "minute_max": int(g["minute_of_day"].max()),
    })

    if not idx_mono or not minute_order_ok or has_dup_mod:
        result["ok"] = False

```

```

daily_stats_df = pd.DataFrame(daily_stats)

result["artifacts"]["daily_stats"] = daily_stats_df
result["summary"]["n_days"] = int(daily_stats_df.shape[0])
result["summary"]["days_with_bad_minute_order"] = int(len(bad_minute_order_d
result["summary"]["days_with_duplicate_minute_of_day"] = int(len(duplicate_m
result["summary"]["days_with_intraday_gaps_not_60s"] = int((daily_stats_df["

result["checks"]["all_days_have_monotonic_minute_of_day"] = (len(bad_minute_
result["checks"]["no_duplicate_minute_of_day_within_day"] = (len(duplicate_m
result["checks"]["all_intraday_steps_are_60s_within_day"] = bool(
    (daily_stats_df["n_intraday_gaps_not_60s"] == 0).all()
)

result["artifacts"]["bad_minute_order_days"] = bad_minute_order_days
result["artifacts"]["duplicate_minute_days"] = duplicate_minute_days

if intraday_gap_rows:
    result["artifacts"]["intraday_gaps_head"] = pd.concat(intraday_gap_rows,
else:
    result["artifacts"]["intraday_gaps_head"] = pd.DataFrame()

# -----
# 5) Resumen global
# -----
result["summary"]["n_rows"] = int(len(df))
result["summary"]["start"] = df.index.min()
result["summary"]["end"] = df.index.max()

# -----
# 6) Reporte por pantalla
# -----
if verbose:
    print("=" * 70)
    print("VALIDACIÓN TEMPORAL DE mnq_intraday")
    print("=" * 70)
    print(f"Rows                : {result['summary']['n_rows']}")
    print(f"Days                  : {result['summary']['n_days']}")
    print(f"Start                 : {result['summary']['start']}")
    print(f"End                   : {result['summary']['end']}")
    print("-" * 70)
    print(f"Index monotonic       : {result['checks']['index_is_monotonic']}")
    print(f"Index unique          : {result['checks']['index_is_unique']}")
    print(f>Date == index.date    : {result['checks']['date_column_matche']}")
    print(f"Global diffs > 0      : {result['checks']['all_global_time_di']}")
    print(f"minute_of_day monotonic : {result['checks']['all_days_have_mono']}")
    print(f"No dup minute_of_day  : {result['checks']['no_duplicate_minut']}")
    print(f>Intraday steps = 60s   : {result['checks']['all_intraday_steps']}")
    print("-" * 70)
    print(f"Duplicated timestamps : {result['summary']['n_duplicated_time']}")
    print(f>Date mismatches       : {result['summary']['n_date_mismatches']}")
    print(f"Non-positive global diffs: {result['summary']['n_non_positive_gl']}")
    print(f"Bad minute order days  : {result['summary']['days_with_bad_min']}")
    print(f"Dup minute_of_day days : {result['summary']['days_with_duplica']}")
    print(f"Days with !=60s gaps   : {result['summary']['days_with_intrada']}")
    print("-" * 70)
    print(f"DATASET OK            : {result['ok']}")
    print("=" * 70)

```



```

    if result["summary"]["n_duplicated_timestamps"] > 0:
        print("\nDuplicated timestamps (head):")
        print(pd.Index(result["artifacts"]["duplicated_timestamps"][:10]))

    if result["summary"]["n_date_mismatches"] > 0:
        print("\nDate mismatches (head):")
        print(result["artifacts"]["date_mismatches_head"])

    if result["summary"]["days_with_bad_minute_order"] > 0:
        print("\nDays with bad minute_of_day order (head):")
        print(result["artifacts"]["bad_minute_order_days"][:10])

    if result["summary"]["days_with_duplicate_minute_of_day"] > 0:
        print("\nDays with duplicate minute_of_day (head):")
        print(result["artifacts"]["duplicate_minute_days"][:10])

    if not result["artifacts"]["intraday_gaps_head"].empty:
        print("\nIntraday gaps != 60 seconds (head):")
        print(result["artifacts"]["intraday_gaps_head"])

    return result

# =====
# Ejecución inmediata
# =====
validation = validate_mnq_intraday(mnq_intraday, verbose=True)

# Si quiere abortar automáticamente cuando haya problemas críticos:
critical_checks = [
    "index_is_monotonic_increasing",
    "index_is_unique",
    "date_column_matches_index_date",
    "all_global_time_diffs_positive",
    "all_days_have_monotonic_minute_of_day",
    "no_duplicate_minute_of_day_within_day",
]

failed_critical = [k for k in critical_checks if not validation["checks"].get(k,

if failed_critical:
    raise ValueError(
        "Validación temporal fallida. Checks críticos con error: "
        + ", ".join(failed_critical)
    )

```

```

=====
VALIDACIÓN TEMPORAL DE mnq_intraday
=====
Rows                : 1024062
Days                : 1482
Start               : 2020-01-02 04:30:00-05:00
End                 : 2026-04-17 16:00:00-04:00
-----
Index monotonic     : True
Index unique        : True
Date == index.date  : True
Global diffs > 0    : True
minute_of_day monotonic : True
No dup minute_of_day : True
Intraday steps = 60s : True
-----
Duplicated timestamps : 0
Date mismatches       : 0
Non-positive global diffs: 0
Bad minute order days : 0
Dup minute_of_day days : 0
Days with !=60s gaps  : 0
-----
DATASET OK          : True
=====

```

El dataset `mnq_intraday` se encuentra correctamente estructurado para la construcción de targets temporales.

En particular, se verificó que:

- el índice `datetime` está ordenado cronológicamente en forma ascendente,
- no existen timestamps duplicados,
- la columna `date` coincide con la fecha derivada del índice,
- no hay saltos temporales negativos ni diferencias no positivas,
- dentro de cada jornada, `minute_of_day` es estrictamente creciente,
- no existen duplicados de `minute_of_day` dentro de un mismo día,
- y todos los pasos intradía son consistentes con una frecuencia de 1 minuto.

Por lo tanto, el dataset queda validado como temporalmente consistente y apto para construir targets forward del tipo `delta_h` y `ret_h`, siempre que dichos horizontes se calculen respetando los límites de cada jornada y evitando cruces entre días.

# 1. Introducción

El objetivo de este capítulo es investigar targets operativos candidatos para futuros modelos predictivos aplicados al contrato intradiario MNQ.

A diferencia de etapas posteriores, en este capítulo todavía no se evaluará la relación entre variables predictivas y targets. Es decir, no se analizarán features, indicadores técnicos, correlaciones, Information Coefficient ni modelos de Machine Learning.

El foco de este stage será exclusivamente el target: cómo se construye, qué representa, qué distribución tiene, qué tan balanceado es, qué tan estable resulta a lo largo del tiempo y qué tan razonable parece desde una perspectiva operativa.

Esta separación es importante porque, antes de preguntarnos qué variables pueden predecir algo, primero necesitamos definir con claridad qué queremos predecir.

En este capítulo se investigarán tres familias de targets:

DIR = Dirección significativa con umbral  
BAR = Resultado por barreras TP/SL  
OPC = Target operativo combinado

**Este stage toma como insumo los thresholds operativos calibrados previamente en el Stage 03, los cuales serán utilizados para transformar excursiones futuras del precio en targets discretos evaluables.**

## 1.1. DIR: Dirección significativa con umbral

El target **DIR** tiene como objetivo clasificar la dirección relevante del movimiento futuro del precio dentro de un horizonte operativo **H**.

La pregunta que busca responder es:

**¿El precio se movió lo suficiente hacia arriba, hacia abajo o no presentó un movimiento relevante dentro del horizonte evaluado?**

Los valores posibles del target son:

| Valor | Interpretación                                                        |
|-------|-----------------------------------------------------------------------|
| 1     | Movimiento alcista significativo. Posible escenario LONG.             |
| -1    | Movimiento bajista significativo. Posible escenario SHORT.            |
| 0     | Movimiento no significativo. No existe una dirección operativa clara. |

La clave de **DIR** es que no intenta predecir simplemente si el precio sube o baja en términos absolutos. Su objetivo es identificar si el movimiento futuro supera un umbral mínimo de relevancia operativa.

De esta forma, se filtran movimientos pequeños que podrían corresponder a ruido de mercado y se concentra el análisis en desplazamientos que, al menos desde el punto de vista del target, tienen mayor sentido operativo.

Es importante remarcar que **DIR** no representa todavía una estrategia de trading ni una orden de ejecución. Es una etiqueta candidata que permitirá estudiar si existen movimientos futuros suficientemente relevantes y direccionales para ser modelados posteriormente.

## 1.2. BAR: Resultado por barreras TP/SL

El target **BAR** tiene como objetivo evaluar el resultado operativo de una operación hipotética dentro de un horizonte **H**, considerando la secuencia interna del precio.

La pregunta que busca responder es:

**Si se entra en una operación, ¿qué ocurre primero dentro del horizonte evaluado: Take Profit, Stop Loss o ninguno de los dos eventos?**

Los valores posibles del target son:

| Valor | Interpretación                                                                        |
|-------|---------------------------------------------------------------------------------------|
| 1     | Se alcanza primero el Take Profit. Trade exitoso.                                     |
| -1    | Se alcanza primero el Stop Loss. Trade fallido.                                       |
| 0     | No se alcanza ni TP ni SL dentro del horizonte. Trade sin resolución operativa clara. |

A diferencia de un target basado únicamente en el precio final del horizonte, **BAR** analiza la trayectoria interna del precio. Esto permite determinar qué evento ocurre primero dentro de la ventana futura: el objetivo de ganancia, la barrera de pérdida o ninguno de los dos.

Por este motivo, **BAR** se aproxima más a una lógica real de ejecución. No solo interesa hacia dónde se movió el precio, sino si ese movimiento habría permitido alcanzar un objetivo operativo antes de tocar una barrera de riesgo.

Es importante remarcar que **BAR** requiere definir previamente la dirección de la operación hipotética. Para una operación LONG, el Take Profit se ubica por encima del precio de entrada y el Stop Loss por debajo. Para una operación SHORT, ocurre lo contrario. Por lo tanto, este target depende de una regla previa que determine el lado de la operación evaluada.

### 1.3. OPC: Target operativo combinado

El target **OPC** representa un target operativo combinado. Su objetivo es integrar dos dimensiones del problema:

DIR = dirección operativa del movimiento futuro

BAR = resultado operativo asociado a esa dirección

A diferencia de **DIR**, que solo indica si existió un movimiento significativo hacia arriba o hacia abajo, y de **BAR**, que evalúa si una operación habría alcanzado primero el Take Profit o el Stop Loss, **OPC** combina ambas ideas en una única categoría operativa.

La pregunta que busca responder es:

**¿Qué tipo de evento operativo ocurrió dentro del horizonte evaluado?**

Conceptualmente, **OPC** se calcula a partir de la combinación entre **DIR** y **BAR**. Es decir, primero se identifica la dirección operativa del movimiento futuro mediante **DIR**, y luego se evalúa el resultado operativo de esa dirección mediante **BAR**.

La codificación base considerada es:

DIR = 1 → dirección LONG  
 DIR = -1 → dirección SHORT  
 DIR = 0 → sin dirección operativa clara

BAR = 1 → Take Profit alcanzado primero  
 BAR = -1 → Stop Loss alcanzado primero  
 BAR = 0 → sin resolución operativa

Por lo tanto, las combinaciones conceptuales de OPC son:

| DIR        | BAR        | Etiqueta lógica | Interpretación                                                    |
|------------|------------|-----------------|-------------------------------------------------------------------|
| 1          | 1          | LONG_TP         | Movimiento LONG significativo con Take Profit alcanzado primero.  |
| 1          | -1         | LONG_SL         | Movimiento LONG significativo con Stop Loss alcanzado primero.    |
| -1         | 1          | SHORT_TP        | Movimiento SHORT significativo con Take Profit alcanzado primero. |
| -1         | -1         | SHORT_SL        | Movimiento SHORT significativo con Stop Loss alcanzado primero.   |
| 0          | cualquiera | NO_TRADE        | Sin operación válida por ausencia de dirección operativa clara.   |
| cualquiera | 0          | NO_TRADE        | Sin resolución operativa dentro del horizonte evaluado.           |

Es importante aclarar que no se utilizarán etiquetas lógicas del tipo 11, 10, -11 o -10, ya que podrían generar confusión con la definición original de BAR. En este estudio, BAR = -1 representa Stop Loss alcanzado primero, mientras que BAR = 0 representa ausencia de resolución operativa. Por lo tanto, una etiqueta como 10 podría interpretarse erróneamente como DIR = 1 y BAR = 0, cuando en realidad se quiere representar un LONG perdedor.

Para evitar esta ambigüedad, se utilizarán etiquetas nominales explícitas:

LONG\_TP  
 LONG\_SL  
 SHORT\_TP  
 SHORT\_SL  
 NO\_TRADE

Estas etiquetas pueden codificarse posteriormente como clases numéricas:

| Clase OPC | Etiqueta lógica | Interpretación                      |
|-----------|-----------------|-------------------------------------|
| 0         | NO_TRADE        | No trade / sin resolución operativa |
| 1         | LONG_TP         | LONG ganador                        |
| 2         | LONG_SL         | LONG perdedor                       |
| 3         | SHORT_TP        | SHORT ganador                       |

| Clase OPC | Etiqueta lógica | Interpretación |
|-----------|-----------------|----------------|
| 4         | SHORT_SL        | SHORT perdedor |

La interpretación correcta es que **OPC** no predice una magnitud ni un retorno continuo. Predice una categoría operativa.

Por este motivo, las clases numéricas no deben interpretarse como valores ordinales. Por ejemplo, la clase **4** no es mayor que la clase **1** ; simplemente representa otro tipo de evento. En etapas posteriores, si se entrena un modelo sobre **OPC** , deberá tratarse como un problema de clasificación multiclase nominal.

## 1.4. Objetivo del capítulo

El objetivo de este capítulo es evaluar y comparar las propiedades estadísticas y operativas de los targets **DIR** , **BAR** y **OPC** .

En particular, se busca responder preguntas como:

- ¿Qué tan balanceadas se encuentran las clases de cada target?
- ¿Qué porcentaje de observaciones genera una señal operable?
- ¿Cómo cambian los targets según el horizonte evaluado?
- ¿Qué tan estables son sus distribuciones a través del tiempo?
- ¿Qué diferencias aparecen por régimen intradiario?
- ¿Qué diferencias aparecen por contrato?
- ¿Qué ocurre en períodos extremos o de comportamiento particular, como **2022** , **2025Q2** y **2026** ?
- ¿Qué target parece más claro, más estable y más útil desde una perspectiva operativa?

La respuesta a estas preguntas permitirá identificar targets candidatos para las siguientes etapas del proyecto.

Es importante remarcar que el análisis de períodos posteriores al período de desarrollo, como **2025** y **2026** , deberá interpretarse como una evaluación de validación o estrés fuera de muestra. Estos períodos no deberán utilizarse para diseñar thresholds, ajustar reglas ni seleccionar targets de forma retroactiva.

## 1.5. Metodología general

La investigación se realizará utilizando el dataset intradiario del MNQ previamente preparado y analizado en stages anteriores.

Para cada target se seguirá una metodología común:

1. Definición conceptual del target.
2. Construcción práctica del target.
3. Validación de consistencia.

4. Análisis de distribución de clases.
5. Evaluación de frecuencia operativa.
6. Análisis por horizonte.
7. Análisis por año.
8. Análisis por régimen intradiario.
9. Análisis por contrato.
10. Análisis por combinaciones de contexto.
11. Comparación entre targets.
12. Selección preliminar de targets candidatos.

La lógica será avanzar desde lo más simple hacia lo más operativo. Primero se analizará **DIR**, que permite estudiar la dirección significativa del movimiento futuro. Luego se evaluará **BAR**, que incorpora la lógica de barreras tipo Take Profit y Stop Loss. Finalmente, se analizará **OPC**, que combina dirección y resultado operativo en una única categoría multiclase.

En esta etapa no se evaluará todavía qué variables pueden predecir estos targets. Tampoco se entrenarán modelos ni se seleccionarán features. Ese análisis se realizará posteriormente, una vez que los targets hayan sido definidos, construidos y validados.

## 1.6. Restricción temporal, no cruce de días y prevención de leakage

Un aspecto fundamental de esta investigación es garantizar que la construcción de los targets respete la estructura temporal real de los mercados.

Todos los targets deberán construirse sin cruzar días. Esto significa que cada observación solo podrá utilizar información futura perteneciente a su mismo trading day.

Si el horizonte requerido para calcular un target excede el final de la sesión correspondiente, esa observación será considerada inválida y se marcará como valor faltante.

Este criterio garantiza que:

- No exista contaminación entre sesiones consecutivas.
- Las observaciones sean temporalmente consistentes.
- Los targets respeten una lógica operativa realista.
- No se utilice información futura mal alineada.
- Los futuros modelos no sean entrenados con datos contaminados por leakage.

Además, debe mantenerse la separación temporal definida previamente en el proyecto. El período **2020-2024** corresponde al período de desarrollo, utilizado para diseñar y analizar los targets. En cambio, el período **2025-2026** debe quedar reservado para evaluación posterior fuera de muestra.

Por lo tanto, la construcción y validación inicial de los targets deberá evitar cualquier decisión basada en el comportamiento del período final de test. Esto incluye la selección

de thresholds, la elección de reglas operativas y la selección preliminar de targets candidatos.

En síntesis, este capítulo no busca todavía entrenar modelos ni seleccionar features. Su objetivo es definir, construir y evaluar targets operativos candidatos de forma rigurosa, estadística y contextual, manteniendo una separación clara entre diseño, validación y evaluación final.

## 1.7. Principio metodológico: targets separados y features causales

Antes de continuar con la construcción y análisis de los targets, es importante dejar explícito un principio metodológico central para este estudio.

Todos los modelos predictivos que se desarrollen en etapas posteriores deberán utilizar únicamente información disponible en el instante  $t$ . Esto significa que las variables explicativas deberán construirse a partir de datos históricos o contemporáneos al momento de decisión, como precios OHLCV, contexto temporal, régimen intradiario, contrato, volumen, volatilidad reciente, momentum e indicadores técnicos calculados hasta  $t$ .

Por lo tanto, ningún modelo deberá utilizar información futura ni variables derivadas de la ventana que se intenta predecir.

La regla general es:

Para cualquier modelo predictivo, las features deben estar disponibles en  $t$ .

Nunca se deben usar targets DIR, BAR u OPC como features.

Nunca se debe usar la predicción de un modelo como feature de entrada de otro modelo en esta etapa.

Conceptualmente, cada target deberá evaluarse como un problema predictivo independiente:

Modelo DIR:

$$X_t = \text{OHLCV histórico} + \text{contexto} + \text{indicadores calculados hasta } t$$

$$y = \text{DIR}$$

Modelo BAR:

$$X_t = \text{OHLCV histórico} + \text{contexto} + \text{indicadores calculados hasta } t$$

$$y = \text{BAR}$$

Modelo OPC:

$$X_t = \text{OHLCV histórico} + \text{contexto} + \text{indicadores calculados hasta } t$$

$$y = \text{OPC}$$

En cambio, no se utilizarán esquemas como los siguientes:

Modelo BAR:

$$X_t = \text{features} + \text{DIR real}$$



Modelo BAR:

$X_t = \text{features} + \text{predicción DIR}$

Modelo OPC:

$X_t = \text{features} + \text{DIR} + \text{BAR}$

Modelo OPC:

$X_t = \text{features} + \text{predicción DIR} + \text{predicción BAR}$

La diferencia metodológica es importante. **BAR** está condicionado por **DIR** en su construcción, porque primero se necesita conocer si la operación es **LONG** o **SHORT** para evaluar las barreras de **Take Profit** y **Stop Loss**. Sin embargo, esto no significa que **DIR** pueda utilizarse como variable explicativa para predecir **BAR**.

En otras palabras:

DIR se usa para construir BAR.

DIR no se usa para predecir BAR.

De forma equivalente, **OPC** se construye combinando **DIR** y **BAR**, pero estos targets no deben utilizarse como entradas para predecir **OPC**.

DIR y BAR se usan para construir OPC.

DIR y BAR no se usan como inputs para predecir OPC.

Esto evita introducir información del propio target dentro del modelo y reduce el riesgo de **data leakage**. También evita construir un sistema artificialmente favorecido, donde un modelo se beneficie directa o indirectamente de información que no estaría disponible en una situación real de decisión.

Por lo tanto, en este estudio los targets **DIR**, **BAR** y **OPC** deberán entenderse como alternativas de modelado independientes. En etapas posteriores se podrá evaluar si existe señal predictiva para cada uno de ellos, pero siempre respetando la misma regla: las features deben ser causales y estar disponibles en el instante **t**.

Cualquier arquitectura encadenada, donde una predicción de **DIR** alimente un modelo de **BAR**, o donde predicciones de **DIR** y **BAR** alimenten un modelo de **OPC**, deberá considerarse una investigación posterior y separada. No forma parte de la evaluación inicial de targets de este stage.

En síntesis, el principio metodológico de esta notebook es:

Targets separados.

Features causales.

Sin encadenamiento de modelos.

Sin uso de targets como variables explicativas.

## 2. Construcción práctica de targets

Una vez establecidas las definiciones conceptuales de los targets candidatos, el siguiente paso consiste en describir su construcción práctica a partir de los datos intradiarios del MNQ.

En este capítulo se investigarán únicamente los targets. Todavía no se evaluará la relación entre variables predictivas y targets, ni se entrenarán modelos de Machine Learning. El objetivo de esta etapa es construir, validar y analizar estadísticamente los targets operativos candidatos.

La construcción de los targets se realizará a partir de dos insumos principales: el dataset intradiario base del MNQ y la tabla de thresholds operativos calibrada previamente en el Stage 03. Los thresholds utilizados provienen exclusivamente del período de desarrollo 2020-2024 y permanecerán fijos al aplicarse sobre el resto del dataset.

Los targets podrán construirse sobre todo el período disponible, incluyendo 2025 y 2026. Sin embargo, estos últimos períodos serán utilizados únicamente para evaluación posterior fuera de muestra, no para recalibrar thresholds, ajustar reglas ni seleccionar targets de forma retroactiva.

Los targets evaluados serán:

DIR = Dirección significativa con umbral  
BAR = Resultado por barreras TP/SL  
OPC = Target operativo combinado

## 2.1. Dataset base

La investigación utiliza datos intradiarios del contrato de futuros MNQ, almacenados en el dataset `mnq_intraday`.

Cada observación representa una vela temporal de un minuto e incluye información OHLCV:

open  
high  
low  
close  
volume

Además, el dataset incluye variables de contexto temporal y operativo, tales como:

date  
minute\_of\_day  
regime\_id  
regime\_name  
contract

Estas variables no se utilizarán todavía como features predictivas. En este capítulo se emplearán principalmente para construir, segmentar y analizar la distribución de los targets por año, régimen intradiario, contrato y otros contextos relevantes.

Los targets serán contruidos sobre el dataset intradiario disponible completo. Esto permite generar etiquetas para todo el período analizado, incluyendo tanto el período de desarrollo como el período final de evaluación. Sin embargo, el uso metodológico de cada período será diferente: 2020-2024 corresponde al período de desarrollo, mientras que 2025-2026 quedará reservado para evaluación posterior fuera de muestra.

Los datasets con los targets calculados serán guardados en las siguientes rutas:

```
MNQ_TARGETS_PATH = PROJECT_ROOT / "data" / "04_mnq_targets"
MNQ_TARGETS_PATH.mkdir(parents=True, exist_ok=True)

MNQ_TARGETS_DIR_PARQUET = MNQ_TARGETS_PATH /
"mnq_targets_dir.parquet"
MNQ_TARGETS_BAR_PARQUET = MNQ_TARGETS_PATH /
"mnq_targets_bar.parquet"
MNQ_TARGETS_OPC_PARQUET = MNQ_TARGETS_PATH /
"mnq_targets_opc.parquet"

MNQ_TARGETS_ALL_PARQUET = MNQ_TARGETS_PATH /
"mnq_targets_all.parquet"
```

Los archivos individuales permitirán analizar cada familia de targets por separado.

Adicionalmente, se guardará un dataset consolidado, `mnq_targets_all.parquet`, que integrará los targets `DIR`, `BAR` y `OPC` en una única tabla. Este archivo será útil para realizar comparaciones generales entre targets sin necesidad de unir manualmente los resultados en etapas posteriores.

## 2.2. Dataset de thresholds operativos

Además del dataset intradiario base, la construcción de los targets requiere utilizar los thresholds operativos calibrados previamente en el Stage 03.

Estos thresholds fueron calculados exclusivamente con observaciones del período de desarrollo `2020-2024`. Por lo tanto, representan valores definidos antes de la evaluación final y permanecerán fijos al aplicarse sobre el resto del dataset, incluyendo `2025` y `2026`.

Las tablas finales de thresholds fueron guardadas en las siguientes rutas:

```
MNQ_THRESHOLDS_PATH = PROJECT_ROOT / "data" / "03_mnq_thresholds"

MNQ_THRESHOLDS_GLOBAL_BENCHMARK_PARQUET = (
    MNQ_THRESHOLDS_PATH / "mnq_thresholds_global_benchmark.parquet"
)

MNQ_THRESHOLDS_REGIME_PRIMARY_PARQUET = (
    MNQ_THRESHOLDS_PATH / "mnq_thresholds_regime_primary.parquet"
)

MNQ_THRESHOLDS_REGIME_ROBUST_CHECK_PARQUET = (
    MNQ_THRESHOLDS_PATH /
"mnq_thresholds_regime_robust_check.parquet"
)

MNQ_THRESHOLDS_FINAL_CANDIDATES_PARQUET = (
    MNQ_THRESHOLDS_PATH / "mnq_thresholds_final_candidates.parquet"
)

MNQ_THRESHOLDS_FINAL_CANDIDATES_CSV = (
    MNQ_THRESHOLDS_PATH / "mnq_thresholds_final_candidates.csv"
```

```
)

MNQ_THRESHOLDS_FINAL_SUMMARY_JSON = (
    MNQ_THRESHOLDS_PATH / "mnq_thresholds_final_summary.json"
)
```

La tabla principal para la construcción de targets será:

```
MNQ_THRESHOLDS_REGIME_PRIMARY_PARQUET
```

Esta tabla contiene la metodología principal seleccionada para Stage 04:

```
threshold_method = regime_pooled_2020_2024
threshold_scope  = regime
selection_period  = development_2020_2024
threshold_type    = common
threshold_col     = threshold_common_pts
```

La lógica principal consiste en utilizar thresholds por régimen intradiario. Esto significa que cada target no se construye con un único threshold global, sino con un threshold condicionado por:

```
horizon
percentile
regime_id
```

Por ejemplo, el target `DIR p50 h30` representa una sola definición de target, pero internamente utiliza distintos thresholds según el régimen intradiario correspondiente a cada observación.

La tabla consolidada final, `mnq_thresholds_final_candidates.parquet`, contiene todas las metodologías seleccionadas para Stage 04:

| Método                                    | Rol             | Uso                                           |
|-------------------------------------------|-----------------|-----------------------------------------------|
| <code>regime_pooled_2020_2024</code>      | Principal       | Construcción principal de targets por régimen |
| <code>global_pooled_2020_2024</code>      | Benchmark       | Comparación contra threshold global           |
| <code>regime_median_year_2020_2024</code> | Control robusto | Validación de robustez frente a años extremos |

Para la construcción principal de `DIR`, `BAR` y `OPC`, se utilizará inicialmente la columna:

```
threshold_common_pts
```

Esta columna representa un threshold común de relevancia operativa y será aplicada según la combinación correspondiente de horizonte, percentil y régimen intradiario.

Los horizontes y percentiles principales definidos para Stage 04 son:

```
horizons      = [30, 60, 90]
percentiles    = [40, 50, 60]
```

De esta forma, los thresholds quedan congelados antes de construir los targets sobre el dataset completo. Esto evita que el período final de evaluación influya en la definición de los targets y reduce el riesgo de leakage o ajuste retroactivo.

## 2.2.1 Carga de thersholds operativos

```

In [9]: # 2.2.1 Carga de thersholds operativos
# =====

import json
from pathlib import Path

import pandas as pd

# -----
# Rutas de thresholds generadas en Stage 03
# -----

MNQ_THRESHOLDS_PATH = PROJECT_ROOT / "data" / "03_mnq_thresholds"

MNQ_THRESHOLDS_GLOBAL_BENCHMARK_PARQUET = (
    MNQ_THRESHOLDS_PATH / "mnq_thresholds_global_benchmark.parquet"
)

MNQ_THRESHOLDS_REGIME_PRIMARY_PARQUET = (
    MNQ_THRESHOLDS_PATH / "mnq_thresholds_regime_primary.parquet"
)

MNQ_THRESHOLDS_REGIME_ROBUST_CHECK_PARQUET = (
    MNQ_THRESHOLDS_PATH / "mnq_thresholds_regime_robust_check.parquet"
)

MNQ_THRESHOLDS_FINAL_CANDIDATES_PARQUET = (
    MNQ_THRESHOLDS_PATH / "mnq_thresholds_final_candidates.parquet"
)

MNQ_THRESHOLDS_FINAL_CANDIDATES_CSV = (
    MNQ_THRESHOLDS_PATH / "mnq_thresholds_final_candidates.csv"
)

MNQ_THRESHOLDS_FINAL_SUMMARY_JSON = (
    MNQ_THRESHOLDS_PATH / "mnq_thresholds_final_summary.json"
)

# -----
# Funciones auxiliares de carga
# -----

def load_parquet_checked(path, name):
    """
    Carga un archivo parquet y valida que exista en disco.
    """
    path = Path(path)

    if not path.exists():
        raise FileNotFoundError(f"No se encontró el archivo {name}: {path}")

    print(f"Cargando {name}...")
    print(f"Ruta: {path}")

    df = pd.read_parquet(path)

    print(f"{name} cargado correctamente.")

```

```

print(f"Filas: {df.shape[0]:,}")
print(f"Columnas: {df.shape[1]:,}")
print("-" * 80)

return df

def load_csv_checked(path, name):
    """
    Carga un archivo CSV y valida que exista en disco.
    """
    path = Path(path)

    if not path.exists():
        raise FileNotFoundError(f"No se encontró el archivo {name}: {path}")

    print(f"Cargando {name}...")
    print(f"Ruta: {path}")

    df = pd.read_csv(path)

    print(f"{name} cargado correctamente.")
    print(f"Filas: {df.shape[0]:,}")
    print(f"Columnas: {df.shape[1]:,}")
    print("-" * 80)

    return df

def load_json_checked(path, name):
    """
    Carga un archivo JSON y valida que exista en disco.
    """
    path = Path(path)

    if not path.exists():
        raise FileNotFoundError(f"No se encontró el archivo {name}: {path}")

    print(f"Cargando {name}...")
    print(f"Ruta: {path}")

    with path.open("r", encoding="utf-8") as f:
        data = json.load(f)

    print(f"{name} cargado correctamente.")
    print("-" * 80)

    return data

# -----
# Carga de datasets de thresholds
# -----

df_thresholds_global_benchmark = load_parquet_checked(
    MNQ_THRESHOLDS_GLOBAL_BENCHMARK_PARQUET,
    "df_thresholds_global_benchmark"
)

df_thresholds_regime_primary = load_parquet_checked(

```

```

    MNQ_THRESHOLDS_REGIME_PRIMARY_PARQUET,
    "df_thresholds_regime_primary"
)

df_thresholds_regime_robust_check = load_parquet_checked(
    MNQ_THRESHOLDS_REGIME_ROBUST_CHECK_PARQUET,
    "df_thresholds_regime_robust_check"
)

df_thresholds_final_candidates = load_parquet_checked(
    MNQ_THRESHOLDS_FINAL_CANDIDATES_PARQUET,
    "df_thresholds_final_candidates"
)

df_thresholds_final_candidates_csv = load_csv_checked(
    MNQ_THRESHOLDS_FINAL_CANDIDATES_CSV,
    "df_thresholds_final_candidates_csv"
)

thresholds_final_summary = load_json_checked(
    MNQ_THRESHOLDS_FINAL_SUMMARY_JSON,
    "thresholds_final_summary"
)

# -----
# Validación básica de las tablas cargadas
# -----

print("VALIDACIÓN BÁSICA DE THRESHOLDS")
print("=" * 80)

threshold_tables = {
    "global_benchmark": df_thresholds_global_benchmark,
    "regime_primary": df_thresholds_regime_primary,
    "regime_robust_check": df_thresholds_regime_robust_check,
    "final_candidates": df_thresholds_final_candidates,
    "final_candidates_csv": df_thresholds_final_candidates_csv,
}

for name, df in threshold_tables.items():
    print(f"\nTabla: {name}")
    print(f"Filas: {df.shape[0]:,}")
    print(f"Columnas: {df.shape[1]:,}")

    if "horizon" in df.columns:
        print(f"Horizontes: {sorted(df['horizon'].dropna().astype(int).unique())}")

    if "percentile" in df.columns:
        print(f"Percentiles: {sorted(df['percentile'].dropna().astype(int).unique())}")

    if "threshold_method" in df.columns:
        print(f"Métodos: {df['threshold_method'].dropna().unique().tolist()}")

    if "threshold_scope" in df.columns:
        print(f"Scopes: {df['threshold_scope'].dropna().unique().tolist()}")

    if "threshold_common_pts" in df.columns:
        missing_common = df["threshold_common_pts"].isna().sum()
        non_positive_common = (df["threshold_common_pts"] <= 0).sum()

```

```
print(f"threshold_common_pts faltantes: {missing_common}")  
print(f"threshold_common_pts no positivos: {non_positive_common}")  
  
print("\nValidación básica completada.")
```



```
Cargando df_thresholds_global_benchmark...
Ruta: c:\Users\heguu\OneDrive\Escritorio\neural_profit_local\data\03_mnq_thresholds\mnq_thresholds_global_benchmark.parquet
df_thresholds_global_benchmark cargado correctamente.
Filas: 9
Columnas: 26
-----
Cargando df_thresholds_regime_primary...
Ruta: c:\Users\heguu\OneDrive\Escritorio\neural_profit_local\data\03_mnq_thresholds\mnq_thresholds_regime_primary.parquet
df_thresholds_regime_primary cargado correctamente.
Filas: 39
Columnas: 26
-----
Cargando df_thresholds_regime_robust_check...
Ruta: c:\Users\heguu\OneDrive\Escritorio\neural_profit_local\data\03_mnq_thresholds\mnq_thresholds_regime_robust_check.parquet
df_thresholds_regime_robust_check cargado correctamente.
Filas: 39
Columnas: 26
-----
Cargando df_thresholds_final_candidates...
Ruta: c:\Users\heguu\OneDrive\Escritorio\neural_profit_local\data\03_mnq_thresholds\mnq_thresholds_final_candidates.parquet
df_thresholds_final_candidates cargado correctamente.
Filas: 87
Columnas: 27
-----
Cargando df_thresholds_final_candidates_csv...
Ruta: c:\Users\heguu\OneDrive\Escritorio\neural_profit_local\data\03_mnq_thresholds\mnq_thresholds_final_candidates.csv
df_thresholds_final_candidates_csv cargado correctamente.
Filas: 87
Columnas: 27
-----
Cargando thresholds_final_summary...
Ruta: c:\Users\heguu\OneDrive\Escritorio\neural_profit_local\data\03_mnq_thresholds\mnq_thresholds_final_summary.json
thresholds_final_summary cargado correctamente.
-----
VALIDACIÓN BÁSICA DE THRESHOLDS
=====

Tabla: global_benchmark
Filas: 9
Columnas: 26
Horizontes: [30, 60, 90]
Percentiles: [40, 50, 60]
Métodos: ['global_pooled_2020_2024']
Scopes: ['global']
threshold_common_pts faltantes: 0
threshold_common_pts no positivos: 0

Tabla: regime_primary
Filas: 39
Columnas: 26
Horizontes: [30, 60, 90]
Percentiles: [40, 50, 60]
Métodos: ['regime_pooled_2020_2024']
Scopes: ['regime']
```

threshold\_common\_pts faltantes: 0  
 threshold\_common\_pts no positivos: 0

Tabla: regime\_robust\_check  
 Filas: 39  
 Columnas: 26  
 Horizontes: [30, 60, 90]  
 Percentiles: [40, 50, 60]  
 Métodos: ['regime\_median\_year\_2020\_2024']  
 Scopes: ['regime']  
 threshold\_common\_pts faltantes: 0  
 threshold\_common\_pts no positivos: 0

Tabla: final\_candidates  
 Filas: 87  
 Columnas: 27  
 Horizontes: [30, 60, 90]  
 Percentiles: [40, 50, 60]  
 Métodos: ['regime\_pooled\_2020\_2024', 'global\_pooled\_2020\_2024', 'regime\_median\_year\_2020\_2024']  
 Scopes: ['regime', 'global']  
 threshold\_common\_pts faltantes: 0  
 threshold\_common\_pts no positivos: 0

Tabla: final\_candidates\_csv  
 Filas: 87  
 Columnas: 27  
 Horizontes: [30, 60, 90]  
 Percentiles: [40, 50, 60]  
 Métodos: ['regime\_pooled\_2020\_2024', 'global\_pooled\_2020\_2024', 'regime\_median\_year\_2020\_2024']  
 Scopes: ['regime', 'global']  
 threshold\_common\_pts faltantes: 0  
 threshold\_common\_pts no positivos: 0

Validación básica completada.

## 2.2.2 Selección de thresholds principales para Stage 04

```
In [10]: # 2.2.2. Selección de thresholds principales para Stage 04
# =====

# Configuración principal definida para Stage 04
DIR_HORIZONS = [30, 60, 90]
DIR_PERCENTILES = [40, 50, 60]

# Tabla principal de thresholds:
# - por régimen
# - calibrada con 2020-2024
# - usando threshold_common_pts
df_dir_thresholds_primary = df_thresholds_regime_primary.copy()

# Normalización de tipos
df_dir_thresholds_primary["horizon"] = df_dir_thresholds_primary["horizon"].astype(int)
df_dir_thresholds_primary["percentile"] = df_dir_thresholds_primary["percentile"].astype(int)
df_dir_thresholds_primary["regime_id"] = df_dir_thresholds_primary["regime_id"].astype(int)

# Filtro de horizontes y percentiles principales
df_dir_thresholds_primary = df_dir_thresholds_primary[
```

```

df_dir_thresholds_primary["horizon"].isin(DIR_HORIZONS)
& df_dir_thresholds_primary["percentile"].isin(DIR_PERCENTILES)
].copy()

# Validación de duplicados por clave de construcción
duplicated_keys = df_dir_thresholds_primary.duplicated(
    subset=["horizon", "percentile", "regime_id"]
).sum()

missing_thresholds = df_dir_thresholds_primary["threshold_common_pts"].isna().sum()
non_positive_thresholds = (df_dir_thresholds_primary["threshold_common_pts"] <= 0).sum()

print("THRESHOLDS PRINCIPALES PARA DIR")
print("=" * 80)
print(f"Filas disponibles: {len(df_dir_thresholds_primary):,}")
print(f"Horizontes: {sorted(df_dir_thresholds_primary['horizon'].unique().tolist())}")
print(f"Percentiles: {sorted(df_dir_thresholds_primary['percentile'].unique().tolist())}")
print(f"Regímenes: {sorted(df_dir_thresholds_primary['regime_id'].unique().tolist())}")
print(f"Claves duplicadas horizon-percentile-regime_id: {duplicated_keys}")
print(f"threshold_common_pts faltantes: {missing_thresholds}")
print(f"threshold_common_pts no positivos: {non_positive_thresholds}")

if duplicated_keys > 0:
    raise ValueError("Existen thresholds duplicados por horizon, percentile y regime_id")

if missing_thresholds > 0:
    raise ValueError("Existen thresholds faltantes en threshold_common_pts.")

if non_positive_thresholds > 0:
    raise ValueError("Existen thresholds no positivos en threshold_common_pts.")

print("\nTabla df_dir_thresholds_primary lista para construir DIR.")

```

THRESHOLDS PRINCIPALES PARA DIR

=====

Filas disponibles: 39  
 Horizontes: [30, 60, 90]  
 Percentiles: [40, 50, 60]  
 Regímenes: [0, 1, 2, 3, 4]  
 Claves duplicadas horizon-percentile-regime\_id: 0  
 threshold\_common\_pts faltantes: 0  
 threshold\_common\_pts no positivos: 0

Tabla df\_dir\_thresholds\_primary lista para construir DIR.

In [11]: *# Vista compacta de thresholds principales*

```

cols_preview = [
    "threshold_method",
    "threshold_scope",
    "selection_period",
    "horizon",
    "percentile",
    "regime_id",
    "regime_name",
    "threshold_common_pts",
    "n_valid",
]

```

```
print("Vista previa de df_dir_thresholds_primary:")
print(df_dir_thresholds_primary[cols_preview].head(20).to_string(index=False))
```

Vista previa de df\_dir\_thresholds\_primary:

|                         | threshold_method             | threshold_scope | selection_period | horizon | percentil |
|-------------------------|------------------------------|-----------------|------------------|---------|-----------|
| e regime_id regime_name | threshold_common_pts         | n_valid         |                  |         |           |
| regime_pooled_2020_2024 | regime development_2020_2024 | 30              | 4                |         |           |
| 0 0 Overnight           | 14.00 284400                 |                 |                  |         |           |
| regime_pooled_2020_2024 | regime development_2020_2024 | 30              | 5                |         |           |
| 0 0 Overnight           | 16.50 284400                 |                 |                  |         |           |
| regime_pooled_2020_2024 | regime development_2020_2024 | 30              | 6                |         |           |
| 0 0 Overnight           | 19.25 284400                 |                 |                  |         |           |
| regime_pooled_2020_2024 | regime development_2020_2024 | 30              | 4                |         |           |
| 0 1 Pre-market          | 25.25 71100                  |                 |                  |         |           |
| regime_pooled_2020_2024 | regime development_2020_2024 | 30              | 5                |         |           |
| 0 1 Pre-market          | 31.00 71100                  |                 |                  |         |           |
| regime_pooled_2020_2024 | regime development_2020_2024 | 30              | 6                |         |           |
| 0 1 Pre-market          | 38.25 71100                  |                 |                  |         |           |
| regime_pooled_2020_2024 | regime development_2020_2024 | 30              | 4                |         |           |
| 0 2 Opening             | 42.75 71100                  |                 |                  |         |           |
| regime_pooled_2020_2024 | regime development_2020_2024 | 30              | 5                |         |           |
| 0 2 Opening             | 50.00 71100                  |                 |                  |         |           |
| regime_pooled_2020_2024 | regime development_2020_2024 | 30              | 6                |         |           |
| 0 2 Opening             | 58.50 71100                  |                 |                  |         |           |
| regime_pooled_2020_2024 | regime development_2020_2024 | 30              | 4                |         |           |
| 0 3 Regular             | 25.50 355500                 |                 |                  |         |           |
| regime_pooled_2020_2024 | regime development_2020_2024 | 30              | 5                |         |           |
| 0 3 Regular             | 30.00 355500                 |                 |                  |         |           |
| regime_pooled_2020_2024 | regime development_2020_2024 | 30              | 6                |         |           |
| 0 3 Regular             | 35.50 355500                 |                 |                  |         |           |
| regime_pooled_2020_2024 | regime development_2020_2024 | 30              | 4                |         |           |
| 0 4 Closing             | 30.00 1185                   |                 |                  |         |           |
| regime_pooled_2020_2024 | regime development_2020_2024 | 30              | 5                |         |           |
| 0 4 Closing             | 35.50 1185                   |                 |                  |         |           |
| regime_pooled_2020_2024 | regime development_2020_2024 | 30              | 6                |         |           |
| 0 4 Closing             | 41.75 1185                   |                 |                  |         |           |
| regime_pooled_2020_2024 | regime development_2020_2024 | 60              | 4                |         |           |
| 0 0 Overnight           | 21.00 284400                 |                 |                  |         |           |
| regime_pooled_2020_2024 | regime development_2020_2024 | 60              | 5                |         |           |
| 0 0 Overnight           | 24.75 284400                 |                 |                  |         |           |
| regime_pooled_2020_2024 | regime development_2020_2024 | 60              | 6                |         |           |
| 0 0 Overnight           | 29.25 284400                 |                 |                  |         |           |
| regime_pooled_2020_2024 | regime development_2020_2024 | 60              | 4                |         |           |
| 0 1 Pre-market          | 54.00 71100                  |                 |                  |         |           |
| regime_pooled_2020_2024 | regime development_2020_2024 | 60              | 5                |         |           |
| 0 1 Pre-market          | 63.25 71100                  |                 |                  |         |           |

## 2.3. Horizontes temporales

La construcción de targets se basa en horizontes forward definidos en cantidad de velas futuras.

Sea **H** el horizonte de evaluación. Dado que el dataset intradiario está compuesto por velas de un minuto, un horizonte de **H=30** representa 30 barras futuras, equivalente a 30 minutos.

En este proyecto se analizarán inicialmente los siguientes horizontes:

30 minutos

60 minutos

90 minutos

Para una observación ubicada en el instante  $t$ , la barra actual representa el momento de decisión. Por lo tanto, el target no debe utilizar información de la propia barra  $t$  como resultado futuro, sino únicamente información posterior.

La ventana futura asociada al horizonte  $H$  estará compuesta por las barras:

$t+1, t+2, \dots, t+H$

Es importante destacar que esta ventana deberá construirse siempre dentro del mismo trading day. Si el horizonte requerido excede el final de la sesión correspondiente, la observación será considerada inválida y se marcará como valor faltante.

Los horizontes utilizados en esta etapa coinciden con los horizontes empleados previamente para calibrar los thresholds operativos en el Stage 03:

$H = 30, 60, 90$

De esta manera, cada target será construido utilizando el threshold correspondiente a su misma combinación de horizonte, percentil y régimen intradiario.

Horizontes más cortos, como  $30m$ , suelen capturar movimientos más locales y potencialmente más ruidosos. Horizontes más amplios, como  $60m$  o  $90m$ , pueden capturar movimientos de mayor tamaño, aunque también modifican la naturaleza operativa del problema.

En esta etapa no se elegirá todavía un único horizonte definitivo. El objetivo será comparar cómo cambian las propiedades estadísticas y operativas de los targets según el horizonte utilizado.

## 2.4. Construcción de la ventana futura

Para cada instante  $t$  y horizonte  $H$ , se construye una ventana futura sin cruzar días:

$\text{Future Window}(t, H) = [t+1, t+2, \dots, t+H]$

La barra actual  $t$  representa el momento de decisión. Por lo tanto, no forma parte de la ventana futura utilizada para calcular el target.

A partir de esa ventana se calcularán distintos valores futuros necesarios para construir los targets. En esta etapa, el análisis principal se basará en precios de cierre futuros, manteniendo consistencia con la metodología utilizada en la calibración de thresholds del Stage 03.

Para cada horizonte  $H$ , se calcularán:

$\text{future\_close\_max\_H} = \text{máximo precio close observado entre } t+1 \text{ y } t+H$

$\text{future\_close\_min\_H} = \text{mínimo precio close observado entre } t+1 \text{ y } t+H$

Estos valores permiten medir la máxima excursión alcista y bajista del precio de cierre dentro del horizonte futuro.

Las excursiones se definen como:

$$\begin{aligned} \text{up\_excursion\_pts\_H} &= \text{future\_close\_max\_H} - \text{close\_t} \\ \text{down\_excursion\_pts\_H} &= \text{close\_t} - \text{future\_close\_min\_H} \end{aligned}$$

donde:

$\text{close\_t}$  = precio de cierre en el instante actual  $t$

De esta forma, el target no depende únicamente del precio final observado en  $t+H$ , sino del movimiento máximo alcanzado dentro de toda la ventana futura.

Esta decisión es importante porque, desde una perspectiva operativa, puede ser más relevante saber si el precio llegó a moverse significativamente dentro del horizonte, aunque luego haya retrocedido antes del cierre de la ventana.

Todas las ventanas futuras deberán construirse sin cruzar días. Si una observación no cuenta con suficientes barras futuras dentro del mismo trading day para completar el horizonte  $H$ , será marcada como inválida para ese horizonte.

## 2.5. Construcción del target DIR

El target **DIR** busca clasificar la dirección significativa del movimiento futuro.

Para cada combinación de horizonte  $H$ , percentil  $p$  y régimen intradiario, se utilizará un threshold operativo expresado en puntos. Este threshold permite diferenciar movimientos relevantes de fluctuaciones pequeñas o ruido de mercado.

La lógica general será:

| Condición                                                                 | Valor DIR | Interpretación                        |
|---------------------------------------------------------------------------|-----------|---------------------------------------|
| La excursión alcista supera el threshold y domina a la bajista            | 1         | Movimiento alcista significativo      |
| La excursión bajista supera el threshold y domina a la alcista            | -1        | Movimiento bajista significativo      |
| Ninguna excursión supera el threshold, o no hay dirección dominante clara | 0         | Movimiento no significativo o ambiguo |

En términos simples, **DIR** intenta responder:

¿Dentro del horizonte  $H$ , el precio tuvo un movimiento suficientemente claro hacia arriba, hacia abajo o no tuvo una dirección operable?

La construcción puede expresarse de forma conceptual como:

Si  $\text{up\_excursion\_pts\_H} \geq \text{threshold}$  y  $\text{up\_excursion\_pts\_H} > \text{down\_excursion\_pts\_H}$ :

$$\text{DIR} = 1$$

Si  $\text{down\_excursion\_pts\_H} \geq \text{threshold}$  y  $\text{down\_excursion\_pts\_H} > \text{up\_excursion\_pts\_H}$ :

$$\text{DIR} = -1$$

En caso contrario:

$DIR = 0$

En los casos donde ambas excursiones superen el threshold, se asignará la dirección de la excursión dominante. Si ambas excursiones son iguales o no existe una dirección claramente dominante, la observación será clasificada como  $0$ .

Este target no busca predecir cualquier movimiento mínimo. Su objetivo es filtrar movimientos pequeños y concentrarse en desplazamientos con mayor relevancia operativa.

Los thresholds utilizados no serán recalibrados en esta etapa. Proviene del Stage 03 y serán aplicados según la combinación correspondiente de horizonte, percentil y régimen intradiario. Posteriormente se analizará cómo cambia la distribución de clases de  $DIR$  según el horizonte y el percentil seleccionado.

## 2.6. Construcción del target BAR

El target  $BAR$  busca evaluar el resultado operativo de una posible operación mediante barreras de Take Profit y Stop Loss.

A diferencia de  $DIR$ , que analiza la dirección significativa del movimiento,  $BAR$  incorpora una lógica más cercana a una operación real.

La pregunta principal es:

Si se toma una operación en una dirección determinada,  
¿qué ocurre primero dentro del horizonte H:  
Take Profit, Stop Loss o ninguno?

Para construir  $BAR$  se requiere definir:

dirección de la operación  
take profit  
stop loss  
horizonte máximo de evaluación  
regla de prioridad si TP y SL aparecen en la misma barra

En esta investigación, la dirección de la operación será definida a partir del target  $DIR$ :

$DIR = 1$  → evaluar operación LONG  
 $DIR = -1$  → evaluar operación SHORT  
 $DIR = 0$  → no hay operación direccional clara

La interpretación general de  $BAR$  será:

| Valor<br>$BAR$ | Interpretación                                                                          |
|----------------|-----------------------------------------------------------------------------------------|
| 1              | Se alcanza primero el Take Profit                                                       |
| -1             | Se alcanza primero el Stop Loss                                                         |
| 0              | No se alcanza ni TP ni SL dentro del horizonte                                          |
| NaN            | No corresponde evaluar operación, por ejemplo cuando no existe señal direccional válida |

Para una operación LONG, el Take Profit se ubica por encima del precio de entrada y el Stop Loss por debajo:

```
TP_LONG = close_t + threshold
SL_LONG = close_t - threshold
```

Para una operación SHORT, el Take Profit se ubica por debajo del precio de entrada y el Stop Loss por encima:

```
TP_SHORT = close_t - threshold
SL_SHORT = close_t + threshold
```

En esta etapa, el criterio principal será evaluar las barreras utilizando precios `close`, para mantener consistencia con los thresholds calibrados previamente a partir de excursiones de cierre.

El uso de precios `high/low` puede aproximarse más a una lógica de ejecución intrabar, pero también introduce una mayor sensibilidad a movimientos extremos dentro de cada vela y posibles ambigüedades cuando TP y SL se alcanzan dentro de la misma barra. Por este motivo, podrá considerarse más adelante como análisis alternativo o prueba de sensibilidad.

Cualquier regla adoptada deberá mantenerse constante durante todo el análisis para que los resultados sean comparables entre horizontes, percentiles y regímenes.

## 2.7. Construcción del target OPC

El target `OPC` combina la dirección operable del movimiento con el resultado operativo de esa dirección.

En términos simples:

```
OPC = DIR + BAR
```

Esta expresión no debe interpretarse como una suma matemática, sino como una combinación lógica entre:

```
DIR = dirección significativa del movimiento futuro
BAR = resultado operativo asociado a esa dirección
```

Primero se identifica la dirección operativa mediante `DIR`. Luego, para las observaciones con dirección válida, se evalúa el resultado operativo mediante `BAR`.

La codificación base es:

```
DIR = 1  → dirección LONG
DIR = -1 → dirección SHORT
DIR = 0  → sin dirección operativa clara

BAR = 1  → Take Profit alcanzado primero
BAR = -1 → Stop Loss alcanzado primero
BAR = 0  → sin resolución operativa
```

Por lo tanto, las combinaciones conceptuales de `OPC` serán:



| DIR        | BAR        | Etiqueta lógica | Interpretación                                |
|------------|------------|-----------------|-----------------------------------------------|
| 1          | 1          | LONG_TP         | LONG con Take Profit alcanzado primero        |
| 1          | -1         | LONG_SL         | LONG con Stop Loss alcanzado primero          |
| -1         | 1          | SHORT_TP        | SHORT con Take Profit alcanzado primero       |
| -1         | -1         | SHORT_SL        | SHORT con Stop Loss alcanzado primero         |
| 0          | cualquiera | NO_TRADE        | Sin operación válida                          |
| cualquiera | 0          | NO_TRADE        | Sin resolución operativa dentro del horizonte |

Para evitar ambigüedades, no se utilizarán etiquetas como **11**, **10**, **-11** o **-10**. En este estudio, **BAR = -1** representa Stop Loss alcanzado primero, mientras que **BAR = 0** representa ausencia de resolución operativa. Por lo tanto, una etiqueta como **10** podría confundirse con **DIR = 1** y **BAR = 0**, cuando en realidad se quiere representar un LONG perdedor.

Por este motivo, se utilizarán etiquetas nominales explícitas:

LONG\_TP  
LONG\_SL  
SHORT\_TP  
SHORT\_SL  
NO\_TRADE

Luego, estas etiquetas podrán codificarse como clases numéricas:

| Clase OPC | Etiqueta lógica | Interpretación                      |
|-----------|-----------------|-------------------------------------|
| 0         | NO_TRADE        | No trade / sin resolución operativa |
| 1         | LONG_TP         | LONG ganador                        |
| 2         | LONG_SL         | LONG perdedor                       |
| 3         | SHORT_TP        | SHORT ganador                       |
| 4         | SHORT_SL        | SHORT perdedor                      |

La interpretación correcta es que **OPC** representa una categoría operativa. No es una variable ordinal ni una magnitud continua.

Por lo tanto, la clase **4** no es mayor que la clase **1**. Simplemente representa un tipo de evento diferente. En etapas posteriores, si se utiliza como target de Machine Learning, deberá tratarse como clasificación multiclase nominal.

## 2.8. Regímenes intradiarios

Además del análisis global, todos los targets serán evaluados por régimen intradiario.

Los regímenes considerados serán:

Overnight  
Pre-market  
Opening  
Regular  
Closing

El análisis exploratorio previo mostró que el mercado no se comporta de manera uniforme a lo largo del día. La volatilidad, el volumen, la frecuencia de movimientos significativos y la magnitud de las excursiones futuras cambian según el régimen intradiario.

Por este motivo, cada target será analizado tanto a nivel global como por régimen, con el objetivo de identificar diferencias en:

distribución de clases  
frecuencia operativa  
balance del target  
estabilidad temporal  
proporción de señales LONG y SHORT  
proporción de eventos TP y SL

Este análisis permitirá determinar si determinadas formulaciones de target son más estables o más operables en ciertos períodos específicos de la sesión.

Es importante considerar que no todos los regímenes necesariamente tendrán la misma cantidad de observaciones válidas para todos los horizontes. En particular, los regímenes cercanos al final del día pueden perder observaciones cuando el horizonte futuro requerido excede el cierre de la sesión. En esos casos, dichas observaciones serán marcadas como inválidas para evitar cruces entre trading days.

## 2.9. Análisis por año, contrato y contexto

Además del régimen intradiario, los targets también serán evaluados por otras dimensiones temporales y operativas:

year  
year\_quarter  
contract  
contract\_code  
year + regime  
contract + regime

Esto es necesario porque el análisis exploratorio mostró que el comportamiento del MNQ no fue uniforme a lo largo del tiempo.

En particular, períodos como 2022 , 2025Q2 y el tramo disponible de 2026 presentaron condiciones de mercado más activas y exigentes. Por lo tanto, no será suficiente evaluar un target únicamente mediante su distribución global. También será necesario revisar si sus clases se mantienen razonablemente estables en distintos contextos de mercado.

Un target puede parecer atractivo en promedio, pero ser poco estable por año, régimen o contrato. Esa inestabilidad podría hacerlo más difícil de modelar o menos confiable

desde una perspectiva operativa.

El análisis de 2025 y 2026 deberá interpretarse como una evaluación posterior fuera de muestra. Estos períodos no serán utilizados para recalibrar thresholds, modificar reglas de construcción ni seleccionar targets de forma retroactiva. Su función será evaluar si los targets diseñados con el período de desarrollo mantienen un comportamiento razonable en datos no utilizados para su definición.

## 2.10. Integridad temporal de los targets

La construcción de targets basados en información futura requiere un control estricto de la consistencia temporal de cada observación.

Todos los targets deberán construirse sin cruzar días.

La regla principal será:

Ninguna observación puede utilizar información perteneciente a un trading day diferente al de su instante de origen.

Bajo este criterio, si el horizonte requerido para calcular un target excede el final de la sesión correspondiente, esa observación se considera inválida y se marca como valor faltante.

Como parte del proceso de validación se verificará:

- ausencia de cruces entre trading days
- cantidad de observaciones inválidas por horizonte
- distribución de filas descartadas
- ejemplos representativos de validación manual
- consistencia de las clases generadas
- ausencia de información futura mal alineada

Esta verificación garantiza que todos los análisis estadísticos posteriores se encuentren correctamente alineados con la estructura temporal real del mercado.

Además, permite reducir el riesgo de leakage en etapas posteriores de modelado, ya que los futuros modelos solo deberán utilizar como variables predictivas información disponible hasta el instante de decisión  $t$ , mientras que los targets se construirán exclusivamente a partir de la ventana futura válida correspondiente.

## 2.11. Resumen metodológico

En síntesis, la construcción práctica de targets en este capítulo seguirá una lógica estrictamente temporal y operativa.

Primero se construirá **DIR**, que identifica movimientos futuros significativos por dirección. Luego se construirá **BAR**, que evalúa el resultado de una operación mediante barreras TP/SL. Finalmente, se construirá **OPC**, que combina dirección y resultado en una categoría operativa más completa.

Todos los targets serán contruidos sin cruzar días y serán evaluados estadísticamente por horizonte, percentil, año, régimen intradiario, contrato y combinaciones de contexto.

Los thresholds utilizados para construir los targets provienen del Stage 03 y fueron calibrados únicamente con el período de desarrollo 2020-2024 . Posteriormente, esos thresholds serán aplicados sobre todo el dataset disponible, incluyendo 2025 y 2026 , pero estos últimos períodos se utilizarán solo para evaluación fuera de muestra.

En esta etapa no se buscará todavía predecir los targets. El objetivo será determinar cuáles son estadísticamente razonables, operativamente interpretables y suficientemente estables para ser utilizados en etapas posteriores de modelado.

## 3. DIR — Investigación estadística y operativa

El objetivo de esta sección es construir, validar y analizar el target DIR, que clasifica la dirección significativa del movimiento futuro del precio.

Este target busca responder una pregunta simple:

¿Dentro del horizonte H, el precio tuvo un movimiento relevante hacia arriba, hacia abajo o no presentó una dirección operable clara? El análisis se mantendrá simple y ordenado. No se evaluarán features ni modelos predictivos. El foco estará únicamente en entender si DIR genera clases razonables, balanceadas, estables e interpretables desde una perspectiva operativa.

### 3.1. Construcción práctica de DIR

En este punto se construirá el target DIR para las combinaciones definidas de horizonte y percentil.

Se utilizarán los horizontes:

$H = 30, 60, 90$

y los percentiles principales:

$p = 40, 50, 60$

La construcción se basará en las excursiones futuras del precio de cierre:

$$\begin{aligned} \text{up\_excursion\_pts\_H} &= \text{future\_close\_max\_H} - \text{close\_t} \\ \text{down\_excursion\_pts\_H} &= \text{close\_t} - \text{future\_close\_min\_H} \end{aligned}$$

Para cada observación, el threshold correspondiente se asignará según:

horizon  
percentile  
regime\_id

La lógica del target será:

| Condición                                                                 | Valor DIR | Interpretación                        |
|---------------------------------------------------------------------------|-----------|---------------------------------------|
| La excursión alcista supera el threshold y domina a la bajista            | 1         | Movimiento alcista significativo      |
| La excursión bajista supera el threshold y domina a la alcista            | -1        | Movimiento bajista significativo      |
| Ninguna excursión supera el threshold, o no hay dirección dominante clara | 0         | Movimiento no significativo o ambiguo |

Este punto debe dejar generado el dataset `mnq_targets_dir.parquet`.

## Celda 1 — Configuración general

Esta celda define los parámetros principales que usaremos para construir `DIR`.

Ahí se establecen:

```
DIR_HORIZONS = [30, 60, 90]
DIR_PERCENTILES = [40, 50, 60]
```

Es decir, vamos a construir targets para estas combinaciones:

```
H30 p40, p50, p60
H60 p40, p50, p60
H90 p40, p50, p60
```

También se definen los nombres de columnas importantes:

```
PRICE_COL = "close"
DATE_COL = "date"
REGIME_COL = "regime_id"
THRESHOLD_COL = "threshold_common_pts"
```

Y se crea la ruta donde se guardará el resultado final:

```
data/04_mnq_targets/mnq_targets_dir.parquet
```

En resumen, esta celda **no construye nada todavía**. Solo deja preparada la configuración del análisis.

```
In [12]: # 3.1. Construcción práctica del target DIR
# Celda 1 – Configuración general
# =====

import numpy as np
import pandas as pd
from pathlib import Path

DIR_HORIZONS = [30, 60, 90]
DIR_PERCENTILES = [40, 50, 60]

PRICE_COL = "close"
DATE_COL = "date"
REGIME_COL = "regime_id"
THRESHOLD_COL = "threshold_common_pts"

MNQ_TARGETS_PATH = PROJECT_ROOT / "data" / "04_mnq_targets"
```

```

MNQ_TARGETS_PATH.mkdir(parents=True, exist_ok=True)

MNQ_TARGETS_DIR_PARQUET = MNQ_TARGETS_PATH / "mnq_targets_dir.parquet"

print("Configuración DIR cargada.")
print(f"Horizontes: {DIR_HORIZONS}")
print(f"Percentiles: {DIR_PERCENTILES}")
print(f"Ruta de salida: {MNQ_TARGETS_DIR_PARQUET}")

```

Configuración DIR cargada.

Horizontes: [30, 60, 90]

Percentiles: [40, 50, 60]

Ruta de salida: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_dir.parquet

## Celda 2 — Preparar dataset base y validar thresholds

Esta celda toma el dataset original:

`mnq_intraday`

y crea una copia de trabajo:

`df_targets_dir`

Esto es importante para no modificar directamente el dataset original.

También valida que existan las columnas mínimas necesarias:

```

close
date
regime_id

```

Luego revisa que el índice temporal esté ordenado, porque estamos trabajando con ventanas futuras. Si el índice estuviera desordenado, los targets quedarían mal calculados.

Después prepara la tabla de thresholds:

`df_dir_thresholds_primary`

y convierte a enteros columnas como:

```

horizon
percentile
regime_id

```

Esto evita errores al hacer los cruces entre el dataset base y la tabla de thresholds.

En resumen, esta celda verifica que los dos insumos principales estén listos:

```

mnq_intraday
df_dir_thresholds_primary

```

```

In [13]: # Celda 2 – Preparación del dataset base
# =====

df_targets_dir = mnq_intraday.copy()
df_targets_dir = df_targets_dir.sort_index()

required_base_cols = [PRICE_COL, DATE_COL, REGIME_COL]

```

```

missing_base_cols = [
    col for col in required_base_cols
    if col not in df_targets_dir.columns
]

if missing_base_cols:
    raise KeyError(f"Faltan columnas en mnq_intraday: {missing_base_cols}")

if not df_targets_dir.index.is_monotonic_increasing:
    raise ValueError("El índice temporal no está ordenado.")

df_targets_dir[REGIME_COL] = df_targets_dir[REGIME_COL].astype(int)

df_dir_thresholds_primary = df_dir_thresholds_primary.copy()
df_dir_thresholds_primary["horizon"] = df_dir_thresholds_primary["horizon"].astype(int)
df_dir_thresholds_primary["percentile"] = df_dir_thresholds_primary["percentile"].astype(int)
df_dir_thresholds_primary[REGIME_COL] = df_dir_thresholds_primary[REGIME_COL].astype(int)

print("Dataset base preparado.")
print(f"Filas: {len(df_targets_dir):,}")
print(f"Columnas: {df_targets_dir.shape[1]:,}")
print(f"Inicio: {df_targets_dir.index.min()}")
print(f"Fin: {df_targets_dir.index.max()}")

print("\nThresholds principales cargados:")
print(f"Filas: {len(df_dir_thresholds_primary):,}")
print(f"Horizontes: {sorted(df_dir_thresholds_primary['horizon'].unique().tolist())}")
print(f"Percentiles: {sorted(df_dir_thresholds_primary['percentile'].unique().tolist())}")
print(f"Regímenes: {sorted(df_dir_thresholds_primary[REGIME_COL].unique().tolist())}")

```

Dataset base preparado.

Filas: 1,024,062

Columnas: 9

Inicio: 2020-01-02 04:30:00-05:00

Fin: 2026-04-17 16:00:00-04:00

Thresholds principales cargados:

Filas: 39

Horizontes: [30, 60, 90]

Percentiles: [40, 50, 60]

Regímenes: [0, 1, 2, 3, 4]

## Celda 3 — Calcular excursiones futuras con close

Esta celda calcula las ventanas futuras para cada horizonte.

Para cada instante `t`, y para cada horizonte `H`, calcula:

`future_close_max_H`

`future_close_min_H`

Por ejemplo, para `H=30`, calcula:

`future_close_max_30m` = máximo close entre `t+1` y `t+30`

`future_close_min_30m` = mínimo close entre `t+1` y `t+30`

Luego calcula las excursiones:

`up_excursion_pts_30m` = `future_close_max_30m` - `close_t`

`down_excursion_pts_30m` = `close_t` - `future_close_min_30m`

Esto se repite para:

30m  
60m  
90m

Lo importante es que el cálculo se hace agrupando por `date`, por lo tanto **no cruza días**.

Si una observación está cerca del final del día y no tiene suficientes barras futuras, queda como `NaN`.

En resumen, esta celda calcula la materia prima del target `DIR`: cuánto pudo subir o bajar el precio dentro de la ventana futura.

```
In [14]: # Celda 3 – Cálculo de excursiones futuras sin cruzar días
# =====

def add_future_close_excursions(df, horizons, price_col="close", date_col="date")
    """
    Calcula máximos y mínimos futuros del precio close dentro del mismo trading
    Ventana futura:
        t+1, t+2, ..., t+H
    La barra actual t no se incluye.
    """

    df = df.copy()

    for h in horizons:
        future_max_col = f"future_close_max_{h}m"
        future_min_col = f"future_close_min_{h}m"
        up_col = f"up_excursion_pts_{h}m"
        down_col = f"down_excursion_pts_{h}m"

        print(f"Calculando excursiones futuras para H={h}...")

        df[future_max_col] = (
            df.groupby(date_col)[price_col]
            .transform(
                lambda s: (
                    s.iloc[::-1]
                    .shift(1)
                    .rolling(window=h, min_periods=h)
                    .max()
                    .iloc[::-1]
                )
            )
        )

        df[future_min_col] = (
            df.groupby(date_col)[price_col]
            .transform(
                lambda s: (
                    s.iloc[::-1]
                    .shift(1)
```



```

        .rolling(window=h, min_periods=h)
        .min()
        .iloc[::-1]
    )
)

df[up_col] = df[future_max_col] - df[price_col]
df[down_col] = df[price_col] - df[future_min_col]

print(f"Filas válidas H={h}: {df[up_col].notna().sum():,}")
print(f"Filas inválidas H={h}: {df[up_col].isna().sum():,}")
print("-" * 80)

return df

df_targets_dir = add_future_close_excursions(
    df=df_targets_dir,
    horizons=DIR_HORIZONS,
    price_col=PRICE_COL,
    date_col=DATE_COL
)

```

Calculando excursiones futuras para H=30...

Filas válidas H=30: 979,602

Filas inválidas H=30: 44,460

-----

Calculando excursiones futuras para H=60...

Filas válidas H=60: 935,142

Filas inválidas H=60: 88,920

-----

Calculando excursiones futuras para H=90...

Filas válidas H=90: 890,682

Filas inválidas H=90: 133,380

## Celda 4 — Función para construir un target DIR

Esta celda define una función que construye **un solo target DIR** para una combinación específica de horizonte y percentil.

Por ejemplo:

H = 30

p = 50

La función busca el threshold correspondiente según:

```

horizon
percentile
regime_id

```

Luego aplica la lógica:

Si up\_excursion supera el threshold y domina:

DIR = 1

Si down\_excursion supera el threshold y domina:

DIR = -1

Si no hay movimiento suficiente o no hay dirección clara:

DIR = 0

Si no hay ventana futura válida:

DIR = NaN

Por ejemplo, si estamos construyendo:

dir\_p50\_h30

la función genera dos columnas:

dir\_p50\_h30

dir\_threshold\_pts\_p50\_h30

La primera es el target. La segunda guarda el threshold que se aplicó a cada fila.

En resumen, esta celda no ejecuta todavía todos los targets. Solo crea la función que sabe construir uno.

```
In [15]: # Celda 4 – Función para construir un target DIR
# =====

def build_single_dir_target(
    df,
    thresholds_df,
    horizon,
    percentile,
    regime_col="regime_id",
    threshold_col="threshold_common_pts"
):
    """
    Construye un target DIR para una combinación H/p.

    Reglas:
        DIR = 1 si up_excursion supera el threshold y domina.
        DIR = -1 si down_excursion supera el threshold y domina.
        DIR = 0 si no hay movimiento significativo o no hay dirección clara.
        DIR = NaN si no existe ventana futura válida.
    """

    up_col = f"up_excursion_pts_{horizon}m"
    down_col = f"down_excursion_pts_{horizon}m"

    target_col = f"dir_p{percentile}_h{horizon}"
    threshold_target_col = f"dir_threshold_pts_p{percentile}_h{horizon}"

    th = thresholds_df[
        (thresholds_df["horizon"] == horizon)
        & (thresholds_df["percentile"] == percentile)
    ].copy()

    if th.empty:
        raise ValueError(f"No hay thresholds para H={horizon}, p={percentile}.")

    duplicated_regimes = th.duplicated(subset=[regime_col]).sum()

    if duplicated_regimes > 0:
        raise ValueError(
            f"Hay thresholds duplicados para H={horizon}, p={percentile}."
        )
```

```

    )

    threshold_map = th.set_index(regime_col)[threshold_col].to_dict()

    df[threshold_target_col] = df[regime_col].map(threshold_map)

    valid_future = df[up_col].notna() & df[down_col].notna()
    valid_threshold = df[threshold_target_col].notna()
    valid = valid_future & valid_threshold

    dir_values = pd.Series(pd.NA, index=df.index, dtype="Int64")

    # Si la ventana futura es válida, por defecto es 0.
    dir_values.loc[valid] = 0

    up_signal = (
        valid
        & (df[up_col] >= df[threshold_target_col])
        & (df[up_col] > df[down_col])
    )

    down_signal = (
        valid
        & (df[down_col] >= df[threshold_target_col])
        & (df[down_col] > df[up_col])
    )

    dir_values.loc[up_signal] = 1
    dir_values.loc[down_signal] = -1

    df[target_col] = dir_values

    return df, target_col, threshold_target_col

```

## Celda 5 — Construir los 9 targets DIR

Esta celda usa la función anterior y la ejecuta para todas las combinaciones:

3 horizontes x 3 percentiles = 9 targets

Genera estas columnas:

```

dir_p40_h30
dir_p50_h30
dir_p60_h30
dir_p40_h60
dir_p50_h60
dir_p60_h60
dir_p40_h90
dir_p50_h90
dir_p60_h90

```

También guarda los nombres de esas columnas en:

```
dir_target_cols
```

Esto será útil después para resumir, analizar o guardar los targets.

En resumen, esta celda es la que realmente construye todos los targets **DIR**.

```
In [16]: # Celda 5 – Construcción de todos los targets DIR
# =====

dir_target_cols = []
dir_threshold_cols = []

for h in DIR_HORIZONS:
    for p in DIR_PERCENTILES:
        print(f"Construyendo DIR para H={h}, p={p}...")

        df_targets_dir, target_col, threshold_col = build_single_dir_target(
            df=df_targets_dir,
            thresholds_df=df_dir_thresholds_primary,
            horizon=h,
            percentile=p,
            regime_col=REGIME_COL,
            threshold_col=THRESHOLD_COL
        )

        dir_target_cols.append(target_col)
        dir_threshold_cols.append(threshold_col)

        print(f"Target creado: {target_col}")
        print("-" * 80)

print("Targets DIR contruidos:")
for col in dir_target_cols:
    print(f"- {col}")
```

Construyendo DIR para H=30, p=40...  
Target creado: dir\_p40\_h30

Construyendo DIR para H=30, p=50...  
Target creado: dir\_p50\_h30

Construyendo DIR para H=30, p=60...  
Target creado: dir\_p60\_h30

Construyendo DIR para H=60, p=40...  
Target creado: dir\_p40\_h60

Construyendo DIR para H=60, p=50...  
Target creado: dir\_p50\_h60

Construyendo DIR para H=60, p=60...  
Target creado: dir\_p60\_h60

Construyendo DIR para H=90, p=40...  
Target creado: dir\_p40\_h90

Construyendo DIR para H=90, p=50...  
Target creado: dir\_p50\_h90

Construyendo DIR para H=90, p=60...  
Target creado: dir\_p60\_h90

Targets DIR contruidos:

- dir\_p40\_h30
- dir\_p50\_h30
- dir\_p60\_h30
- dir\_p40\_h60
- dir\_p50\_h60
- dir\_p60\_h60
- dir\_p40\_h90
- dir\_p50\_h90
- dir\_p60\_h90

## Celda 6 — Resumen rápido de distribución

Esta celda analiza cómo quedaron distribuidas las clases de cada target.

Para cada columna **DIR**, calcula:

cantidad de LONG  
cantidad de SHORT  
cantidad de NO SIGNAL  
cantidad de NaN  
porcentaje de LONG  
porcentaje de SHORT  
porcentaje de NO SIGNAL  
frecuencia operativa  
balance LONG/SHORT

La frecuencia operativa es:

$DIR \neq \emptyset$

Es decir, mide qué porcentaje de filas genera una señal operable:

DIR = 1 → LONG  
 DIR = -1 → SHORT

Esto sirve para una primera revisión rápida.

Por ejemplo, si un target tiene demasiados 0, significa que genera pocas señales. Si tiene muchos más LONG que SHORT, puede tener sesgo direccional.

En resumen, esta celda permite ver si los targets contruidos tienen sentido estadístico básico antes de avanzar.

```
In [17]: # Celda 6 – Resumen rápido de targets DIR
# =====

def summarize_dir_targets(df, target_cols):
    rows = []

    for col in target_cols:
        valid = df[col].notna()
        values = df.loc[valid, col]

        total_rows = len(df)
        valid_rows = valid.sum()
        nan_rows = total_rows - valid_rows

        long_count = (values == 1).sum()
        short_count = (values == -1).sum()
        no_signal_count = (values == 0).sum()

        signal_count = long_count + short_count

        rows.append({
            "target": col,
            "total_rows": total_rows,
            "valid_rows": valid_rows,
            "nan_rows": nan_rows,
            "long_count": long_count,
            "short_count": short_count,
            "no_signal_count": no_signal_count,
            "long_ratio": long_count / valid_rows if valid_rows > 0 else np.nan,
            "short_ratio": short_count / valid_rows if valid_rows > 0 else np.nan,
            "no_signal_ratio": no_signal_count / valid_rows if valid_rows > 0 else np.nan,
            "signal_ratio": signal_count / valid_rows if valid_rows > 0 else np.nan,
            "long_share_over_signals": long_count / signal_count if signal_count > 0 else np.nan,
            "short_share_over_signals": short_count / signal_count if signal_count > 0 else np.nan
        })

    return pd.DataFrame(rows)

df_dir_summary = summarize_dir_targets(
    df=df_targets_dir,
    target_cols=dir_target_cols
)

print("Resumen de distribución de targets DIR")
print("=" * 80)
print(df_dir_summary.to_string(index=False))
```

## Resumen de distribución de targets DIR

```

=====
target total_rows valid_rows nan_rows long_count short_count no_signal
_count long_ratio short_ratio no_signal_ratio signal_ratio long_share_over_s
ignals short_share_over_signals
dir_p40_h30 1024062 979602 44460 309481 314255
355866 0.315925 0.320799 0.363276 0.636724 0.
496173 0.503827
dir_p50_h30 1024062 979602 44460 259900 270082
449620 0.265312 0.275706 0.458982 0.541018 0.
490394 0.509606
dir_p60_h30 1024062 979602 44460 210169 224826
544607 0.214545 0.229507 0.555947 0.444053 0.
483153 0.516847
dir_p40_h60 1024062 935142 88920 296160 300864
338118 0.316701 0.321731 0.361569 0.638431 0.
496060 0.503940
dir_p50_h60 1024062 935142 88920 246274 258746
430122 0.263355 0.276692 0.459954 0.540046 0.
487652 0.512348
dir_p60_h60 1024062 935142 88920 197901 215897
521344 0.211627 0.230871 0.557502 0.442498 0.
478255 0.521745
dir_p40_h90 1024062 890682 133380 280594 287376
322712 0.315033 0.322647 0.362320 0.637680 0.
494030 0.505970
dir_p50_h90 1024062 890682 133380 234585 248018
408079 0.263377 0.278459 0.458165 0.541835 0.
486083 0.513917
dir_p60_h90 1024062 890682 133380 187768 206733
496181 0.210814 0.232106 0.557080 0.442920 0.
475963 0.524037

```

## Celda 7 — Agregar split temporal y guardar dataset

Esta celda agrega variables auxiliares para análisis posterior:

```

year
quarter
year_quarter
target_analysis_split

```

La columna más importante es:

```
target_analysis_split
```

que separa:

```

development_2020_2024
final_test_2025_2026

```

Después guarda el dataset en:

```
mnq_targets_dir.parquet
```

Pero antes de guardar, elimina columnas futuras auxiliares como:

```

future_close_max
future_close_min
up_excursion_pts
down_excursion_pts

```

Esto es importante porque esas columnas contienen información futura. Sirven para construir el target, pero no deberían quedar disponibles fácilmente para el modelado posterior, porque podrían generar leakage.

En resumen, esta celda deja guardado el dataset final de **DIR** de forma más limpia y segura.

```
In [18]: # Celda 7 – Guardado del dataset DIR
# =====

if "year" not in df_targets_dir.columns:
    df_targets_dir["year"] = df_targets_dir.index.year

if "quarter" not in df_targets_dir.columns:
    df_targets_dir["quarter"] = df_targets_dir.index.quarter

if "year_quarter" not in df_targets_dir.columns:
    df_targets_dir["year_quarter"] = (
        df_targets_dir["year"].astype(str)
        + "Q"
        + df_targets_dir["quarter"].astype(str)
    )

df_targets_dir["target_analysis_split"] = np.where(
    df_targets_dir["year"].between(2020, 2024),
    "development_2020_2024",
    "final_test_2025_2026"
)

# Para evitar problemas de leakage en etapas posteriores,
# guardamos el dataset principal sin columnas futuras auxiliares.
future_aux_cols = [
    col for col in df_targets_dir.columns
    if col.startswith("future_close_")
    or col.startswith("up_excursion_pts_")
    or col.startswith("down_excursion_pts_")
]

df_targets_dir_export = df_targets_dir.drop(columns=future_aux_cols)

df_targets_dir_export.to_parquet(MNQ_TARGETS_DIR_PARQUET)

print("Dataset DIR guardado correctamente.")
print(f"Ruta: {MNQ_TARGETS_DIR_PARQUET}")
print(f"Filas guardadas: {len(df_targets_dir_export):,}")
print(f"Columnas guardadas: {df_targets_dir_export.shape[1]:,}")

print("\nColumnas DIR guardadas:")
for col in dir_target_cols:
    print(f"- {col}")
```



Dataset DIR guardado correctamente.

Ruta: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets  
\mnq\_targets\_dir.parquet

Filas guardadas: 1,024,062

Columnas guardadas: 31

Columnas DIR guardadas:

- dir\_p40\_h30
- dir\_p50\_h30
- dir\_p60\_h30
- dir\_p40\_h60
- dir\_p50\_h60
- dir\_p60\_h60
- dir\_p40\_h90
- dir\_p50\_h90
- dir\_p60\_h90

La lógica completa queda así:

1. Definir configuración
2. Preparar datos
3. Calcular excursiones futuras
4. Definir cómo se construye un DIR
5. Construir todos los DIR
6. Revisar distribución inicial
7. Guardar dataset final

La celda más delicada es la **Celda 3**, porque ahí se calcula la ventana futura. Si esa parte está bien construida sin cruzar días, el resto del proceso queda mucho más sólido.

## 3.2. Validación de consistencia de DIR

Antes de analizar resultados, se verificará que el target haya sido construido correctamente.

La validación deberá revisar:

- que no existan cruces entre trading days
- que los últimos minutos de cada día queden como NaN cuando no haya ventana futura suficiente
- que las clases generadas sean solamente -1, 0 y 1
- que cada target tenga el threshold correcto según horizonte, percentil y régimen
- que no existan columnas futuras usadas como features

Este punto es importante porque **DIR** se construye usando información futura. Por lo tanto, debe garantizarse que esa información solo se use para crear el target y no contamine etapas posteriores de modelado.

### Celda 1 — Configuración de validación

Esta celda prepara las listas de columnas que vamos a revisar. Esta celda no valida todavía los valores. Solo revisa que existan todas las columnas que deberían haberse creado.

```
In [19]: # Celda 1 – Configuración de validación
# =====

print("Validación de consistencia del target DIR")
print("=" * 80)

# Targets DIR esperados
expected_dir_target_cols = [
    f"dir_p{p}_h{h}"
    for h in DIR_HORIZONS
    for p in DIR_PERCENTILES
]

# Columnas de threshold esperadas
expected_dir_threshold_cols = [
    f"dir_threshold_pts_p{p}_h{h}"
    for h in DIR_HORIZONS
    for p in DIR_PERCENTILES
]

missing_target_cols = [
    col for col in expected_dir_target_cols
    if col not in df_targets_dir.columns
]

missing_threshold_cols = [
    col for col in expected_dir_threshold_cols
    if col not in df_targets_dir.columns
]

if missing_target_cols:
    raise KeyError(f"Faltan columnas target DIR: {missing_target_cols}")

if missing_threshold_cols:
    raise KeyError(f"Faltan columnas de threshold DIR: {missing_threshold_cols}")

print("Columnas target DIR encontradas:")
for col in expected_dir_target_cols:
    print(f"- {col}")

print("\nColumnas threshold DIR encontradas:")
for col in expected_dir_threshold_cols:
    print(f"- {col}")

print("\nConfiguración de validación lista.")
```

## Validación de consistencia del target DIR

=====

Columnas target DIR encontradas:

- dir\_p40\_h30
- dir\_p50\_h30
- dir\_p60\_h30
- dir\_p40\_h60
- dir\_p50\_h60
- dir\_p60\_h60
- dir\_p40\_h90
- dir\_p50\_h90
- dir\_p60\_h90

Columnas threshold DIR encontradas:

- dir\_threshold\_pts\_p40\_h30
- dir\_threshold\_pts\_p50\_h30
- dir\_threshold\_pts\_p60\_h30
- dir\_threshold\_pts\_p40\_h60
- dir\_threshold\_pts\_p50\_h60
- dir\_threshold\_pts\_p60\_h60
- dir\_threshold\_pts\_p40\_h90
- dir\_threshold\_pts\_p50\_h90
- dir\_threshold\_pts\_p60\_h90

Configuración de validación lista.

## Celda 2 — Validar clases generadas

Esta celda revisa que cada target tenga únicamente estas clases:

-1  
0  
1  
NaN

Esto es importante porque DIR debe ser un target de clasificación con tres clases válidas:

1 = LONG  
-1 = SHORT  
0 = NO SIGNAL

Si aparece otro valor, por ejemplo 2, 99 o algún error, significa que la construcción falló.

```
In [20]: # Celda 2 - Validación de clases generadas
# =====

allowed_classes = {-1, 0, 1}

print("Validación de clases DIR")
print("=" * 80)

for target_col in expected_dir_target_cols:
    observed_values = (
        df_targets_dir[target_col]
        .dropna()
        .unique()
        .tolist()
    )

    observed_values = sorted([int(v) for v in observed_values])
```

```

invalid_values = set(observed_values) - allowed_classes

print(f"\nTarget: {target_col}")
print(f"Clases observadas: {observed_values}")

if invalid_values:
    raise ValueError(
        f"{target_col} contiene clases inválidas: {invalid_values}"
    )

print("\nValidación de clases completada correctamente.")

```

Validación de clases DIR

=====

Target: dir\_p40\_h30  
Clases observadas: [-1, 0, 1]

Target: dir\_p50\_h30  
Clases observadas: [-1, 0, 1]

Target: dir\_p60\_h30  
Clases observadas: [-1, 0, 1]

Target: dir\_p40\_h60  
Clases observadas: [-1, 0, 1]

Target: dir\_p50\_h60  
Clases observadas: [-1, 0, 1]

Target: dir\_p60\_h60  
Clases observadas: [-1, 0, 1]

Target: dir\_p40\_h90  
Clases observadas: [-1, 0, 1]

Target: dir\_p50\_h90  
Clases observadas: [-1, 0, 1]

Target: dir\_p60\_h90  
Clases observadas: [-1, 0, 1]

Validación de clases completada correctamente.

## Celda 3 — Validar NaN por falta de ventana futura

Esta celda verifica que los NaN estén donde corresponde.

Como no cruzamos días, para cada horizonte H deben quedar como inválidas las últimas H filas de cada trading day.

Por ejemplo:

H30 → últimos 30 minutos de cada día deben quedar NaN  
H60 → últimos 60 minutos de cada día deben quedar NaN  
H90 → últimos 90 minutos de cada día deben quedar NaN

Esta es una de las validaciones más importantes.

Si una fila cerca del cierre tuviera target válido, sería una señal de error, porque para construirlo habría tenido que usar información del día siguiente.

```
In [21]: # Celda 3 – Validación de NaN por falta de ventana futura
# =====

print("Validación de NaN por horizonte")
print("=" * 80)

# Posición de cada barra dentro del día
df_targets_dir["_bar_number_in_day"] = (
    df_targets_dir
    .groupby(DATE_COL)
    .cumcount()
)

# Cantidad total de barras por día
df_targets_dir["_bars_in_day"] = (
    df_targets_dir
    .groupby(DATE_COL)[PRICE_COL]
    .transform("size")
)

# Cantidad de barras futuras disponibles hasta el cierre del día
df_targets_dir["_future_bars_available"] = (
    df_targets_dir["_bars_in_day"]
    - df_targets_dir["_bar_number_in_day"]
    - 1
)

for h in DIR_HORIZONS:
    expected_invalid = df_targets_dir["_future_bars_available"] < h

    expected_invalid_rows = expected_invalid.sum()
    expected_valid_rows = (~expected_invalid).sum()

    print(f"\nHorizonte H={h}")
    print(f"Filas esperadas válidas: {expected_valid_rows:,}")
    print(f"Filas esperadas inválidas: {expected_invalid_rows:,}")

    for p in DIR_PERCENTILES:
        target_col = f"dir_p{p}_h{h}"

        observed_invalid = df_targets_dir[target_col].isna()

        unexpected_nan = (~expected_invalid & observed_invalid).sum()
        unexpected_valid = (expected_invalid & ~observed_invalid).sum()

        print(f" {target_col}")
        print(f"    NaN inesperados en filas válidas: {unexpected_nan:,}")
        print(f"    Valores no NaN en filas inválidas: {unexpected_valid:,}")

        if unexpected_nan > 0:
            raise ValueError(
                f"{target_col} tiene NaN donde debería existir target."
            )

        if unexpected_valid > 0:
            raise ValueError(
```

```

        f"{target_col} tiene valores donde debería ser NaN."
    )

print("\nValidación de NaN por ventana futura completada correctamente.")

```

Validación de NaN por horizonte

=====

Horizonte H=30

Filas esperadas válidas: 979,602

Filas esperadas inválidas: 44,460

dir\_p40\_h30

NaN inesperados en filas válidas: 0

Valores no NaN en filas inválidas: 0

dir\_p50\_h30

NaN inesperados en filas válidas: 0

Valores no NaN en filas inválidas: 0

dir\_p60\_h30

NaN inesperados en filas válidas: 0

Valores no NaN en filas inválidas: 0

Horizonte H=60

Filas esperadas válidas: 935,142

Filas esperadas inválidas: 88,920

dir\_p40\_h60

NaN inesperados en filas válidas: 0

Valores no NaN en filas inválidas: 0

dir\_p50\_h60

NaN inesperados en filas válidas: 0

Valores no NaN en filas inválidas: 0

dir\_p60\_h60

NaN inesperados en filas válidas: 0

Valores no NaN en filas inválidas: 0

Horizonte H=90

Filas esperadas válidas: 890,682

Filas esperadas inválidas: 133,380

dir\_p40\_h90

NaN inesperados en filas válidas: 0

Valores no NaN en filas inválidas: 0

dir\_p50\_h90

NaN inesperados en filas válidas: 0

Valores no NaN en filas inválidas: 0

dir\_p60\_h90

NaN inesperados en filas válidas: 0

Valores no NaN en filas inválidas: 0

Validación de NaN por ventana futura completada correctamente.

## Celda 4 — Validar columnas de excursión futura

Esta celda revisa que las columnas futuras usadas para construir DIR respeten la misma lógica temporal.

Esta validación confirma que:

future\_close\_max

future\_close\_min

up\_excursion\_pts  
down\_excursion\_pts

también quedan en NaN cuando no existe ventana futura suficiente.

```
In [22]: # Celda 4 – Validación de excursiones futuras
# =====

print("Validación de excursiones futuras")
print("=" * 80)

for h in DIR_HORIZONS:
    future_max_col = f"future_close_max_{h}m"
    future_min_col = f"future_close_min_{h}m"
    up_col = f"up_excursion_pts_{h}m"
    down_col = f"down_excursion_pts_{h}m"

    required_cols = [
        future_max_col,
        future_min_col,
        up_col,
        down_col
    ]

    missing_cols = [
        col for col in required_cols
        if col not in df_targets_dir.columns
    ]

    if missing_cols:
        raise KeyError(
            f"Faltan columnas futuras para H={h}: {missing_cols}"
        )

    expected_invalid = df_targets_dir["_future_bars_available"] < h

    print(f"\nHorizonte H={h}")

    for col in required_cols:
        unexpected_nan = (
            ~expected_invalid
            & df_targets_dir[col].isna()
        ).sum()

        unexpected_valid = (
            expected_invalid
            & df_targets_dir[col].notna()
        ).sum()

        print(f"    {col}")
        print(f"        NaN inesperados en filas válidas: {unexpected_nan:,}")
        print(f"        Valores no NaN en filas inválidas: {unexpected_valid:,}")

        if unexpected_nan > 0:
            raise ValueError(
                f"{col} tiene NaN donde debería tener valor."
            )

        if unexpected_valid > 0:
            raise ValueError(
```

```

        f"{col} tiene valores donde debería ser NaN."
    )

print("\nValidación de excursiones futuras completada correctamente.")

```

Validación de excursiones futuras

=====

Horizonte H=30

```

future_close_max_30m
    NaN inesperados en filas válidas: 0
    Valores no NaN en filas inválidas: 0
future_close_min_30m
    NaN inesperados en filas válidas: 0
    Valores no NaN en filas inválidas: 0
up_excursion_pts_30m
    NaN inesperados en filas válidas: 0
    Valores no NaN en filas inválidas: 0
down_excursion_pts_30m
    NaN inesperados en filas válidas: 0
    Valores no NaN en filas inválidas: 0

```

Horizonte H=60

```

future_close_max_60m
    NaN inesperados en filas válidas: 0
    Valores no NaN en filas inválidas: 0
future_close_min_60m
    NaN inesperados en filas válidas: 0
    Valores no NaN en filas inválidas: 0
up_excursion_pts_60m
    NaN inesperados en filas válidas: 0
    Valores no NaN en filas inválidas: 0
down_excursion_pts_60m
    NaN inesperados en filas válidas: 0
    Valores no NaN en filas inválidas: 0

```

Horizonte H=90

```

future_close_max_90m
    NaN inesperados en filas válidas: 0
    Valores no NaN en filas inválidas: 0
future_close_min_90m
    NaN inesperados en filas válidas: 0
    Valores no NaN en filas inválidas: 0
up_excursion_pts_90m
    NaN inesperados en filas válidas: 0
    Valores no NaN en filas inválidas: 0
down_excursion_pts_90m
    NaN inesperados en filas válidas: 0
    Valores no NaN en filas inválidas: 0

```

Validación de excursiones futuras completada correctamente.

## Celda 5 — Validar threshold aplicado por target

Esta celda verifica que cada target haya usado el threshold correcto según:

```

horizon
percentile
regime_id

```



Esta celda es clave porque confirma que, por ejemplo:

dir\_p50\_h30

usa el threshold p50 , H30 , y no otro.

Además, confirma que el threshold cambia correctamente por régimen intradiario.

```
In [23]: # Celda 5 – Validación de thresholds aplicados
# =====

print("Validación de thresholds aplicados")
print("=" * 80)

for h in DIR_HORIZONS:
    for p in DIR_PERCENTILES:
        target_col = f"dir_p{p}_h{h}"
        threshold_target_col = f"dir_threshold_pts_p{p}_h{h}"

        th = df_dir_thresholds_primary[
            (df_dir_thresholds_primary["horizon"] == h)
            & (df_dir_thresholds_primary["percentile"] == p)
        ].copy()

        if th.empty:
            raise ValueError(f"No hay thresholds para H={h}, p={p}.")

        threshold_map = th.set_index(REGIME_COL)[THRESHOLD_COL].to_dict()

        expected_threshold = df_targets_dir[REGIME_COL].map(threshold_map)
        observed_threshold = df_targets_dir[threshold_target_col]

        comparison_mask = (
            expected_threshold.notna()
            & observed_threshold.notna()
        )

        mismatches = (
            expected_threshold[comparison_mask]
            != observed_threshold[comparison_mask]
        ).sum()

        valid_target_without_threshold = (
            df_targets_dir[target_col].notna()
            & observed_threshold.isna()
        ).sum()

        print(f"\nTarget: {target_col}")
        print(f"Threshold column: {threshold_target_col}")
        print(f"Diferencias contra threshold esperado: {mismatches:,}")
        print(f"Targets válidos sin threshold: {valid_target_without_threshold:,}")

        if mismatches > 0:
            raise ValueError(
                f"{threshold_target_col} no coincide con la tabla de thresholds.
            )

        if valid_target_without_threshold > 0:
            raise ValueError(
                f"{target_col} tiene valores válidos sin threshold asignado."
```

```
)

print("\nValidación de thresholds completada correctamente.")
```

Validación de thresholds aplicados

=====

Target: dir\_p40\_h30  
 Threshold column: dir\_threshold\_pts\_p40\_h30  
 Diferencias contra threshold esperado: 0  
 Targets válidos sin threshold: 0

Target: dir\_p50\_h30  
 Threshold column: dir\_threshold\_pts\_p50\_h30  
 Diferencias contra threshold esperado: 0  
 Targets válidos sin threshold: 0

Target: dir\_p60\_h30  
 Threshold column: dir\_threshold\_pts\_p60\_h30  
 Diferencias contra threshold esperado: 0  
 Targets válidos sin threshold: 0

Target: dir\_p40\_h60  
 Threshold column: dir\_threshold\_pts\_p40\_h60  
 Diferencias contra threshold esperado: 0  
 Targets válidos sin threshold: 0

Target: dir\_p50\_h60  
 Threshold column: dir\_threshold\_pts\_p50\_h60  
 Diferencias contra threshold esperado: 0  
 Targets válidos sin threshold: 0

Target: dir\_p60\_h60  
 Threshold column: dir\_threshold\_pts\_p60\_h60  
 Diferencias contra threshold esperado: 0  
 Targets válidos sin threshold: 0

Target: dir\_p40\_h90  
 Threshold column: dir\_threshold\_pts\_p40\_h90  
 Diferencias contra threshold esperado: 0  
 Targets válidos sin threshold: 0

Target: dir\_p50\_h90  
 Threshold column: dir\_threshold\_pts\_p50\_h90  
 Diferencias contra threshold esperado: 0  
 Targets válidos sin threshold: 0

Target: dir\_p60\_h90  
 Threshold column: dir\_threshold\_pts\_p60\_h90  
 Diferencias contra threshold esperado: 0  
 Targets válidos sin threshold: 0

Validación de thresholds completada correctamente.

## Celda 6 — Validar ausencia de columnas futuras en el dataset exportable

Esta celda revisa que el dataset que se use posteriormente para modelado no conserve columnas futuras.

En memoria, df\_targets\_dir sí puede tener columnas futuras porque las necesitamos para validar. Pero el dataset exportable no debería conservarlas.

La lógica es:

df\_targets\_dir → puede tener columnas futuras para análisis y validación

df\_targets\_dir\_export → no debe tener columnas futuras para evitar leakage

```
In [24]: # Celda 6 – Validación anti-leakage del dataset exportable
# =====

print("Validación anti-leakage")
print("=" * 80)

future_aux_cols = [
    col for col in df_targets_dir.columns
    if col.startswith("future_close_")
    or col.startswith("up_excursion_pts_")
    or col.startswith("down_excursion_pts_")
]

print("Columnas futuras auxiliares detectadas en memoria:")
for col in future_aux_cols:
    print(f"- {col}")

# Dataset limpio para uso posterior
df_targets_dir_export = df_targets_dir.drop(columns=future_aux_cols)

leakage_cols_export = [
    col for col in df_targets_dir_export.columns
    if col.startswith("future_close_")
    or col.startswith("up_excursion_pts_")
    or col.startswith("down_excursion_pts_")
]

print(f"\nColumnas futuras en dataset exportable: {len(leakage_cols_export)}")

if leakage_cols_export:
    raise ValueError(
        f"El dataset exportable contiene columnas futuras: {leakage_cols_export}"
    )

print("\nDataset exportable libre de columnas futuras.")
print(f"Columnas en memoria: {df_targets_dir.shape[1]:,}")
print(f"Columnas exportables: {df_targets_dir_export.shape[1]:,}")
```

## Validación anti-leakage

=====

Columnas futuras auxiliares detectadas en memoria:

- future\_close\_max\_30m
- future\_close\_min\_30m
- up\_excursion\_pts\_30m
- down\_excursion\_pts\_30m
- future\_close\_max\_60m
- future\_close\_min\_60m
- up\_excursion\_pts\_60m
- down\_excursion\_pts\_60m
- future\_close\_max\_90m
- future\_close\_min\_90m
- up\_excursion\_pts\_90m
- down\_excursion\_pts\_90m

Columnas futuras en dataset exportable: 0

Dataset exportable libre de columnas futuras.

Columnas en memoria: 46

Columnas exportables: 34

## Celda 7 — Resumen final de validación

Esta celda deja una tabla resumen simple para documentar que las validaciones fueron superadas.

```
In [25]: # Celda 7 – Resumen final de validación DIR
# =====

validation_rows = []

for h in DIR_HORIZONS:
    expected_invalid = df_targets_dir["_future_bars_available"] < h

    for p in DIR_PERCENTILES:
        target_col = f"dir_p{p}_h{h}"
        threshold_col = f"dir_threshold_pts_p{p}_h{h}"

        valid_rows = df_targets_dir[target_col].notna().sum()
        nan_rows = df_targets_dir[target_col].isna().sum()

        class_values = sorted(
            df_targets_dir[target_col]
                .dropna()
                .astype(int)
                .unique()
                .tolist()
        )

        validation_rows.append({
            "target": target_col,
            "horizon": h,
            "percentile": p,
            "valid_rows": valid_rows,
            "nan_rows": nan_rows,
            "expected_nan_rows": expected_invalid.sum(),
            "classes": class_values,
```

```

        "threshold_column": threshold_col,
        "validation_status": "OK"
    })

df_dir_validation_summary = pd.DataFrame(validation_rows)

print("Resumen final de validación DIR")
print("=" * 80)
print(df_dir_validation_summary.to_string(index=False))

print("\nValidación de consistencia DIR completada correctamente.")

```

Resumen final de validación DIR

```

=====
      target  horizon  percentile  valid_rows  nan_rows  expected_nan_rows  clas
ses      threshold_column validation_status
dir_p40_h30      30         40      979602      44460      44460 [-1, 0,
1] dir_threshold_pts_p40_h30              OK
dir_p50_h30      30         50      979602      44460      44460 [-1, 0,
1] dir_threshold_pts_p50_h30              OK
dir_p60_h30      30         60      979602      44460      44460 [-1, 0,
1] dir_threshold_pts_p60_h30              OK
dir_p40_h60      60         40      935142      88920      88920 [-1, 0,
1] dir_threshold_pts_p40_h60              OK
dir_p50_h60      60         50      935142      88920      88920 [-1, 0,
1] dir_threshold_pts_p50_h60              OK
dir_p60_h60      60         60      935142      88920      88920 [-1, 0,
1] dir_threshold_pts_p60_h60              OK
dir_p40_h90      90         40      890682     133380     133380 [-1, 0,
1] dir_threshold_pts_p40_h90              OK
dir_p50_h90      90         50      890682     133380     133380 [-1, 0,
1] dir_threshold_pts_p50_h90              OK
dir_p60_h90      90         60      890682     133380     133380 [-1, 0,
1] dir_threshold_pts_p60_h90              OK

```

Validación de consistencia DIR completada correctamente.

### 3.3. Distribución global de clases

En este punto se analizará cuántas observaciones pertenecen a cada clase del target.

Para cada variante de `DIR`, se calculará:

- cantidad de LONG
- cantidad de SHORT
- cantidad de NO SIGNAL
- porcentaje de LONG
- porcentaje de SHORT
- porcentaje de NO SIGNAL
- cantidad de filas válidas
- cantidad de filas inválidas

El objetivo es responder:

¿El target está balanceado o domina una clase?

Una distribución demasiado concentrada en `0` indicaría que el target genera pocas señales operables. Una distribución demasiado inclinada hacia `1` o `-1` podría indicar sesgo direccional.

## Celda 1 — Configuración del análisis

Esta celda solo prepara las librerías, etiquetas y valida que existan los targets DIR.

```
In [26]: # Celda 1 – Configuración del análisis
# =====

import re
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# Targets DIR esperados
dir_target_cols = [
    f"dir_p{p}_h{h}"
    for h in DIR_HORIZONS
    for p in DIR_PERCENTILES
]

missing_cols = [
    col for col in dir_target_cols
    if col not in df_targets_dir.columns
]

if missing_cols:
    raise KeyError(f"Faltan columnas DIR en df_targets_dir: {missing_cols}")

DIR_CLASS_LABELS = {
    -1: "SHORT",
    0: "NO_SIGNAL",
    1: "LONG"
}

print("Análisis de distribución de clases DIR")
print("=" * 80)
print(f"Targets a analizar: {len(dir_target_cols)}")
```

```
for col in dir_target_cols:
    print(f"- {col}")
```

Análisis de distribución de clases DIR

=====

Targets a analizar: 9

- dir\_p40\_h30
- dir\_p50\_h30
- dir\_p60\_h30
- dir\_p40\_h60
- dir\_p50\_h60
- dir\_p60\_h60
- dir\_p40\_h90
- dir\_p50\_h90
- dir\_p60\_h90

## Celda 2 — Construcción de tabla resumen

Esta celda calcula la distribución de clases para cada target.

Incluye:

- LONG
- SHORT
- NO SIGNAL
- filas válidas
- filas NaN
- frecuencia operativa
- balance LONG/SHORT
- clase dominante

```
In [27]: # Celda 2 – Tabla resumen de distribución de clases
# =====

def parse_dir_target_name(target_col):
    """
    Extrae percentil y horizonte desde nombres como:
    dir_p50_h30
    """
    pattern = r"dir_p(?P<percentile>\d+)_h(?P<horizon>\d+)"
    match = re.match(pattern, target_col)

    if match is None:
        raise ValueError(f"No se pudo interpretar el nombre del target: {target_col}")

    percentile = int(match.group("percentile"))
    horizon = int(match.group("horizon"))

    return horizon, percentile

def summarize_dir_class_distribution(df, target_cols):
    """
    Calcula la distribución de clases para cada target DIR.
    """
    rows = []

    for target_col in target_cols:
```

```

horizon, percentile = parse_dir_target_name(target_col)

total_rows = len(df)

valid_mask = df[target_col].notna()
valid_values = df.loc[valid_mask, target_col].astype(int)

valid_rows = valid_mask.sum()
nan_rows = total_rows - valid_rows

long_count = (valid_values == 1).sum()
short_count = (valid_values == -1).sum()
no_signal_count = (valid_values == 0).sum()

signal_count = long_count + short_count

long_ratio = long_count / valid_rows if valid_rows > 0 else np.nan
short_ratio = short_count / valid_rows if valid_rows > 0 else np.nan
no_signal_ratio = no_signal_count / valid_rows if valid_rows > 0 else np.nan
signal_ratio = signal_count / valid_rows if valid_rows > 0 else np.nan

long_share_over_signals = (
    long_count / signal_count if signal_count > 0 else np.nan
)

short_share_over_signals = (
    short_count / signal_count if signal_count > 0 else np.nan
)

class_ratios = {
    "LONG": long_ratio,
    "SHORT": short_ratio,
    "NO_SIGNAL": no_signal_ratio,
}

dominant_class = max(class_ratios, key=class_ratios.get)
dominant_ratio = class_ratios[dominant_class]

direction_imbalance = (
    abs(long_share_over_signals - 0.50)
    if signal_count > 0
    else np.nan
)

rows.append({
    "target": target_col,
    "horizon": horizon,
    "percentile": percentile,
    "total_rows": total_rows,
    "valid_rows": valid_rows,
    "nan_rows": nan_rows,
    "valid_ratio": valid_rows / total_rows,
    "nan_ratio": nan_rows / total_rows,
    "long_count": long_count,
    "short_count": short_count,
    "no_signal_count": no_signal_count,
    "long_ratio": long_ratio,
    "short_ratio": short_ratio,
    "no_signal_ratio": no_signal_ratio,
    "signal_count": signal_count,

```



```
        "signal_ratio": signal_ratio,
        "long_share_over_signals": long_share_over_signals,
        "short_share_over_signals": short_share_over_signals,
        "direction_imbalance": direction_imbalance,
        "dominant_class": dominant_class,
        "dominant_ratio": dominant_ratio,
    })

    summary = pd.DataFrame(rows)

    summary = summary.sort_values(
        ["horizon", "percentile"]
    ).reset_index(drop=True)

    return summary

df_dir_class_distribution = summarize_dir_class_distribution(
    df=df_targets_dir,
    target_cols=dir_target_cols
)

print("Resumen de distribución de clases DIR")
print("=" * 80)
print(df_dir_class_distribution.to_string(index=False))
```

## Resumen de distribución de clases DIR

```

=====
target horizon percentile total_rows valid_rows nan_rows valid_ratio
nan_ratio long_count short_count no_signal_count long_ratio short_ratio no_
signal_ratio signal_count signal_ratio long_share_over_signals short_share_ov
er_signals direction_imbalance dominant_class dominant_ratio
dir_p40_h30      30      40      1024062      979602      44460      0.956585
0.043415      309481      314255      355866      0.315925      0.320799
0.363276      623736      0.636724      0.496173      0.
503827      0.003827      NO_SIGNAL      0.363276
dir_p50_h30      30      50      1024062      979602      44460      0.956585
0.043415      259900      270082      449620      0.265312      0.275706
0.458982      529982      0.541018      0.490394      0.
509606      0.009606      NO_SIGNAL      0.458982
dir_p60_h30      30      60      1024062      979602      44460      0.956585
0.043415      210169      224826      544607      0.214545      0.229507
0.555947      434995      0.444053      0.483153      0.
516847      0.016847      NO_SIGNAL      0.555947
dir_p40_h60      60      40      1024062      935142      88920      0.913169
0.086831      296160      300864      338118      0.316701      0.321731
0.361569      597024      0.638431      0.496060      0.
503940      0.003940      NO_SIGNAL      0.361569
dir_p50_h60      60      50      1024062      935142      88920      0.913169
0.086831      246274      258746      430122      0.263355      0.276692
0.459954      505020      0.540046      0.487652      0.
512348      0.012348      NO_SIGNAL      0.459954
dir_p60_h60      60      60      1024062      935142      88920      0.913169
0.086831      197901      215897      521344      0.211627      0.230871
0.557502      413798      0.442498      0.478255      0.
521745      0.021745      NO_SIGNAL      0.557502
dir_p40_h90      90      40      1024062      890682      133380      0.869754
0.130246      280594      287376      322712      0.315033      0.322647
0.362320      567970      0.637680      0.494030      0.
505970      0.005970      NO_SIGNAL      0.362320
dir_p50_h90      90      50      1024062      890682      133380      0.869754
0.130246      234585      248018      408079      0.263377      0.278459
0.458165      482603      0.541835      0.486083      0.
513917      0.013917      NO_SIGNAL      0.458165
dir_p60_h90      90      60      1024062      890682      133380      0.869754
0.130246      187768      206733      496181      0.210814      0.232106
0.557080      394501      0.442920      0.475963      0.
524037      0.024037      NO_SIGNAL      0.557080

```

## Celda 3 — Lectura rápida automática

Esta celda imprime una interpretación rápida de cada target.

Sirve para ver, sin gráficos todavía, cuáles targets generan más señales y cuáles están más balanceados

```

In [28]: # Celda 3 – Lectura rápida de la distribución
# =====

print("Lectura rápida de targets DIR")
print("=" * 80)

for _, row in df_dir_class_distribution.iterrows():
    print(f"\nTarget: {row['target']}")

```

```
print(f" Horizonte: {row['horizon']} minutos")
print(f" Percentil: p{row['percentile']}")
print(f" Filas válidas: {row['valid_rows'],}")
print(f" Filas inválidas / NaN: {row['nan_rows'],}")
print(f" LONG:      {row['long_ratio']:.2%}")
print(f" SHORT:     {row['short_ratio']:.2%}")
print(f" NO SIGNAL: {row['no_signal_ratio']:.2%}")
print(f" Frecuencia operativa: {row['signal_ratio']:.2%}")
print(f" Balance LONG sobre señales: {row['long_share_over_signals']:.2%}")
print(f" Balance SHORT sobre señales: {row['short_share_over_signals']:.2%}")
print(f" Clase dominante: {row['dominant_class']} ({row['dominant_ratio']:.2%})
```

## Lectura rápida de targets DIR

=====

## Target: dir\_p40\_h30

Horizonte: 30 minutos  
Percentil: p40  
Filas válidas: 979,602  
Filas inválidas / NaN: 44,460  
LONG: 31.59%  
SHORT: 32.08%  
NO SIGNAL: 36.33%  
Frecuencia operativa: 63.67%  
Balance LONG sobre señales: 49.62%  
Balance SHORT sobre señales: 50.38%  
Clase dominante: NO\_SIGNAL (36.33%)

## Target: dir\_p50\_h30

Horizonte: 30 minutos  
Percentil: p50  
Filas válidas: 979,602  
Filas inválidas / NaN: 44,460  
LONG: 26.53%  
SHORT: 27.57%  
NO SIGNAL: 45.90%  
Frecuencia operativa: 54.10%  
Balance LONG sobre señales: 49.04%  
Balance SHORT sobre señales: 50.96%  
Clase dominante: NO\_SIGNAL (45.90%)

## Target: dir\_p60\_h30

Horizonte: 30 minutos  
Percentil: p60  
Filas válidas: 979,602  
Filas inválidas / NaN: 44,460  
LONG: 21.45%  
SHORT: 22.95%  
NO SIGNAL: 55.59%  
Frecuencia operativa: 44.41%  
Balance LONG sobre señales: 48.32%  
Balance SHORT sobre señales: 51.68%  
Clase dominante: NO\_SIGNAL (55.59%)

## Target: dir\_p40\_h60

Horizonte: 60 minutos  
Percentil: p40  
Filas válidas: 935,142  
Filas inválidas / NaN: 88,920  
LONG: 31.67%  
SHORT: 32.17%  
NO SIGNAL: 36.16%  
Frecuencia operativa: 63.84%  
Balance LONG sobre señales: 49.61%  
Balance SHORT sobre señales: 50.39%  
Clase dominante: NO\_SIGNAL (36.16%)

## Target: dir\_p50\_h60

Horizonte: 60 minutos  
Percentil: p50  
Filas válidas: 935,142  
Filas inválidas / NaN: 88,920

LONG: 26.34%  
SHORT: 27.67%  
NO SIGNAL: 46.00%  
Frecuencia operativa: 54.00%  
Balance LONG sobre señales: 48.77%  
Balance SHORT sobre señales: 51.23%  
Clase dominante: NO\_SIGNAL (46.00%)

Target: dir\_p60\_h60

Horizonte: 60 minutos  
Percentil: p60  
Filas válidas: 935,142  
Filas inválidas / NaN: 88,920  
LONG: 21.16%  
SHORT: 23.09%  
NO SIGNAL: 55.75%  
Frecuencia operativa: 44.25%  
Balance LONG sobre señales: 47.83%  
Balance SHORT sobre señales: 52.17%  
Clase dominante: NO\_SIGNAL (55.75%)

Target: dir\_p40\_h90

Horizonte: 90 minutos  
Percentil: p40  
Filas válidas: 890,682  
Filas inválidas / NaN: 133,380  
LONG: 31.50%  
SHORT: 32.26%  
NO SIGNAL: 36.23%  
Frecuencia operativa: 63.77%  
Balance LONG sobre señales: 49.40%  
Balance SHORT sobre señales: 50.60%  
Clase dominante: NO\_SIGNAL (36.23%)

Target: dir\_p50\_h90

Horizonte: 90 minutos  
Percentil: p50  
Filas válidas: 890,682  
Filas inválidas / NaN: 133,380  
LONG: 26.34%  
SHORT: 27.85%  
NO SIGNAL: 45.82%  
Frecuencia operativa: 54.18%  
Balance LONG sobre señales: 48.61%  
Balance SHORT sobre señales: 51.39%  
Clase dominante: NO\_SIGNAL (45.82%)

Target: dir\_p60\_h90

Horizonte: 90 minutos  
Percentil: p60  
Filas válidas: 890,682  
Filas inválidas / NaN: 133,380  
LONG: 21.08%  
SHORT: 23.21%  
NO SIGNAL: 55.71%  
Frecuencia operativa: 44.29%  
Balance LONG sobre señales: 47.60%  
Balance SHORT sobre señales: 52.40%  
Clase dominante: NO\_SIGNAL (55.71%)

## Celda 4 — Gráfico de distribución porcentual por target

Este es el gráfico más importante del punto 3.3.

Muestra qué proporción de cada target corresponde a:

SHORT  
NO\_SIGNAL  
LONG

Cómo leer este gráfico

Cada barra representa un target, por ejemplo:

H30 p40  
H30 p50  
H30 p60

La barra completa suma 100% de filas válidas.

Si la zona **NO SIGNAL** es muy grande, significa que el target genera pocas oportunidades operativas.

Si **LONG** o **SHORT** domina demasiado, puede existir sesgo direccional.

```
In [29]: # Celda 4 – Gráfico de distribución porcentual por target
# =====

df_plot = df_dir_class_distribution.copy()

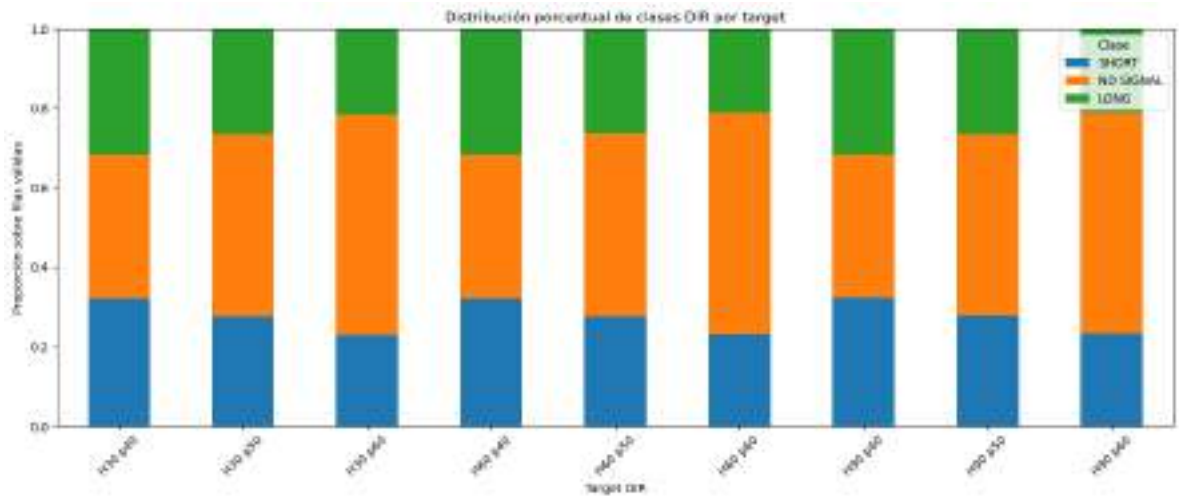
df_plot["target_label"] = (
    "H" + df_plot["horizon"].astype(str)
    + " p" + df_plot["percentile"].astype(str)
)

plot_data = df_plot.set_index("target_label")[
    ["short_ratio", "no_signal_ratio", "long_ratio"]
]

ax = plot_data.plot(
    kind="bar",
    stacked=True,
    figsize=(14, 6)
)

ax.set_title("Distribución porcentual de clases DIR por target")
ax.set_xlabel("Target DIR")
ax.set_ylabel("Proporción sobre filas válidas")
ax.legend(["SHORT", "NO SIGNAL", "LONG"], title="Clase")
ax.set_ylim(0, 1)

plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



## Celda 5 — Gráfico de frecuencia operativa

Este gráfico muestra qué porcentaje de observaciones genera señal:

DIR = 1 o DIR = -1

Es decir, mide:

signal\_ratio = LONG + SHORT

### Cómo leer este gráfico

Si el percentil sube, el threshold es más exigente.

Por eso esperamos que:

p40 tenga más señales  
p50 tenga señales intermedias  
p60 tenga menos señales

Este gráfico nos ayuda a ver si esa relación se cumple.

```
In [30]: # Celda 5 – Frecuencia operativa por horizonte y percentil
# =====

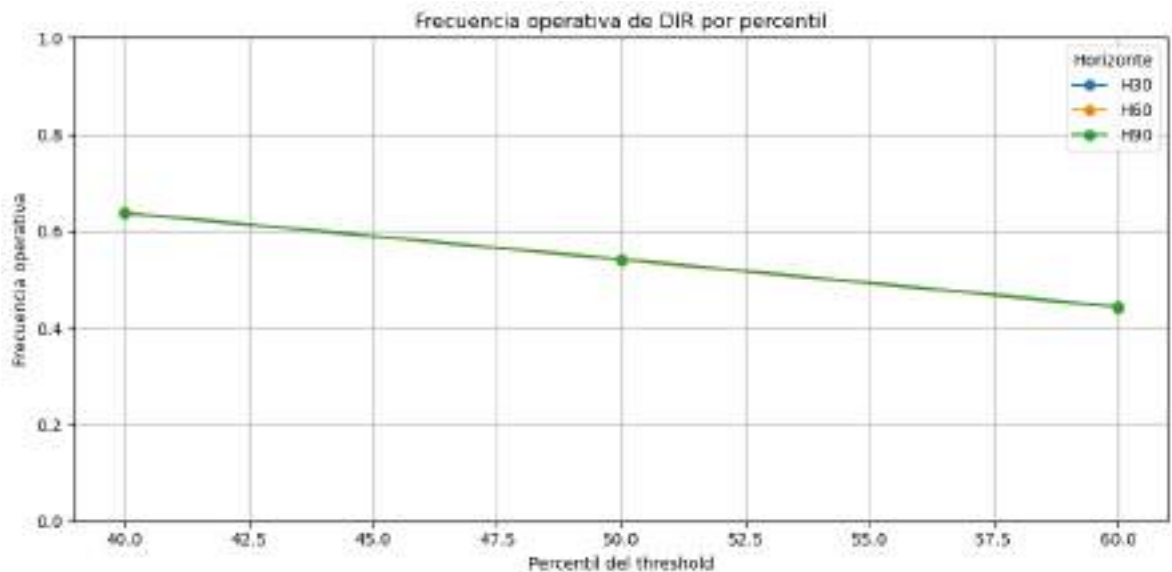
fig, ax = plt.subplots(figsize=(10, 5))

for h in DIR_HORIZONS:
    tmp = df_dir_class_distribution[
        df_dir_class_distribution["horizon"] == h
    ].sort_values("percentile")

    ax.plot(
        tmp["percentile"],
        tmp["signal_ratio"],
        marker="o",
        label=f"H{h}"
    )

ax.set_title("Frecuencia operativa de DIR por percentil")
ax.set_xlabel("Percentil del threshold")
ax.set_ylabel("Frecuencia operativa")
ax.set_ylim(0, 1)
ax.legend(title="Horizonte")
ax.grid(True)
```

```
plt.tight_layout()
plt.show()
```



## Celda 6 — Gráfico de balance LONG/SHORT

Este gráfico analiza solo las señales operativas, es decir, excluye `DIR = 0`.

Muestra qué porcentaje de señales son LONG.

### Cómo leer este gráfico

La línea punteada en 50% representa equilibrio perfecto entre LONG y SHORT.

Si una barra está cerca de 0.50:

LONG y SHORT están balanceados

Si está muy por encima:

predominan señales LONG

Si está muy por debajo:

predominan señales SHORT

```
In [31]: # Celda 6 – Balance LONG/SHORT entre señales operativas
# =====

fig, ax = plt.subplots(figsize=(12, 5))

df_balance = df_dir_class_distribution.copy()

df_balance["target_label"] = (
    "H" + df_balance["horizon"].astype(str)
    + " p" + df_balance["percentile"].astype(str)
)

ax.bar(
    df_balance["target_label"],
    df_balance["long_share_over_signals"]
)
```



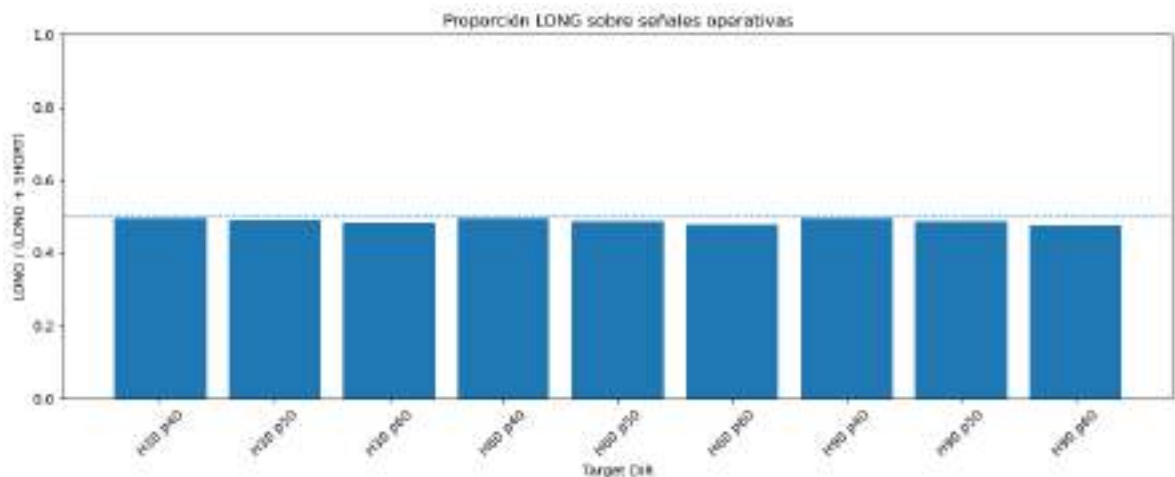
```

ax.axhline(
    0.50,
    linestyle="--",
    linewidth=1
)

ax.set_title("Proporción LONG sobre señales operativas")
ax.set_xlabel("Target DIR")
ax.set_ylabel("LONG / (LONG + SHORT)")
ax.set_ylim(0, 1)

plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

```



## Celda 7 — Gráfico de filas válidas e inválidas

Este gráfico sirve para confirmar visualmente que al aumentar el horizonte se pierden más filas por falta de ventana futura.

```

In [32]: # Celda 7 – Filas válidas e inválidas por horizonte
# =====

df_validity = (
    df_dir_class_distribution
    .groupby("horizon", as_index=False)
    .agg(
        valid_rows=("valid_rows", "first"),
        nan_rows=("nan_rows", "first")
    )
)

df_validity["horizon_label"] = "H" + df_validity["horizon"].astype(str)

plot_data = df_validity.set_index("horizon_label")[
    ["valid_rows", "nan_rows"]
]

ax = plot_data.plot(
    kind="bar",
    stacked=True,
    figsize=(9, 5)
)

```

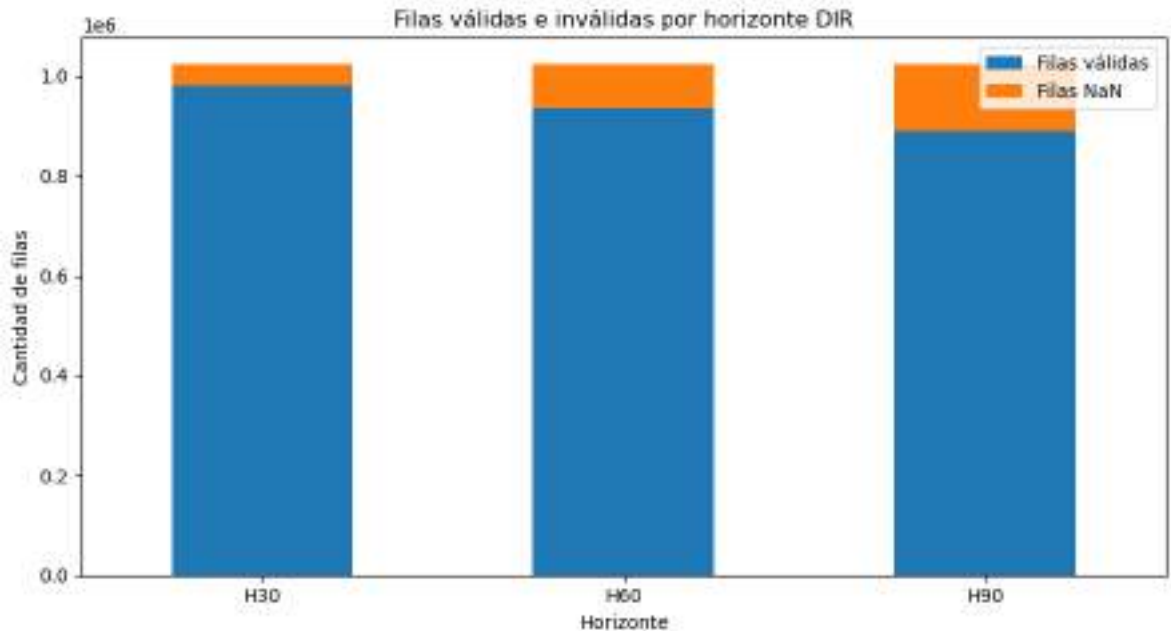
```

)

ax.set_title("Filas válidas e inválidas por horizonte DIR")
ax.set_xlabel("Horizonte")
ax.set_ylabel("Cantidad de filas")
ax.legend(["Filas válidas", "Filas NaN"])

plt.xticks(rotation=0)
plt.tight_layout()
plt.show()

```



## Observaciones preliminares — Distribución de clases DIR

La distribución global de clases muestra un comportamiento coherente para las variantes del target `DIR`.

En primer lugar, la cantidad de filas inválidas aumenta con el horizonte evaluado. Esto es esperado, ya que al usar horizontes más largos se pierden más observaciones al final de cada trading day por falta de ventana futura suficiente. Este resultado confirma que la construcción respeta la restricción de no cruzar días.

El percentil controla correctamente la frecuencia operativa del target. A medida que el percentil aumenta, el threshold se vuelve más exigente y disminuye la proporción de señales `LONG` y `SHORT`. En términos generales, `p40` genera más señales, `p60` es más restrictivo y `p50` representa un punto intermedio.

En todos los targets, la clase individual más frecuente es `NO_SIGNAL`. Sin embargo, para `p40` y `p50`, la suma de señales operativas (`LONG + SHORT`) todavía supera a `NO_SIGNAL`. En cambio, para `p60`, `NO_SIGNAL` pasa a dominar con mayor claridad, lo que indica una formulación más selectiva pero con menor frecuencia operativa.

El balance entre señales `LONG` y `SHORT` es razonablemente bueno. La proporción de señales `LONG` sobre el total de señales operativas se mantiene cercana al 50%, aunque

se observa una leve inclinación hacia señales `SHORT`, especialmente en los percentiles más altos.

También se observa que, para un mismo percentil, los horizontes `H30`, `H60` y `H90` presentan distribuciones muy similares. Esto sugiere que los thresholds por horizonte están calibrados de forma consistente y permiten comparar variantes sin que un horizonte genere señales desproporcionadamente más frecuentes que otro.

De manera preliminar, las variantes `dir_p50_h30`, `dir_p50_h60` y `dir_p50_h90` parecen las más equilibradas a nivel global, ya que combinan una frecuencia operativa razonable con buen balance entre señales `LONG` y `SHORT`.

En síntesis, el target `DIR` quedó bien construido y presenta una distribución global estadísticamente razonable. El siguiente paso será evaluar si este comportamiento se mantiene al segmentar por régimen intradiario, año, contrato y otros contextos relevantes.

### 3.4. Frecuencia operativa DIR

En este punto se analizará qué porcentaje de observaciones genera una señal operable.

La frecuencia operativa se define como:

`frecuencia_operativa = proporción de observaciones donde DIR != 0`

Es decir:

`DIR = 1` → señal `LONG`

`DIR = -1` → señal `SHORT`

`DIR = 0` → sin señal operable

Este análisis permite entender si una combinación de horizonte y percentil genera demasiadas señales, muy pocas señales o una cantidad razonable para ser estudiada posteriormente.

Un target puede estar relativamente balanceado, pero generar señales muy poco frecuentes. En ese caso, podría ser difícil de modelar o poco útil desde una perspectiva operativa.

#### Celda 1 — Preparar tabla de frecuencia operativa

Esta celda toma el resumen ya creado en el punto 3.3 y se queda solo con las columnas necesarias para analizar frecuencia operativa.

```
In [33]: # Celda 1 – Preparación de tabla de frecuencia operativa
# =====

required_cols = [
    "target",
    "horizon",
    "percentile",
    "valid_rows",
```

```
        "nan_rows",
        "long_count",
        "short_count",
        "no_signal_count",
        "signal_count",
        "signal_ratio",
        "no_signal_ratio",
        "long_share_over_signals",
        "short_share_over_signals",
    ]

    missing_cols = [
        col for col in required_cols
        if col not in df_dir_class_distribution.columns
    ]

    if missing_cols:
        raise KeyError(f"Faltan columnas en df_dir_class_distribution: {missing_cols}")

    df_dir_operational_frequency = (
        df_dir_class_distribution[required_cols]
        .copy()
        .sort_values(["horizon", "percentile"])
        .reset_index(drop=True)
    )

    print("Tabla de frecuencia operativa DIR preparada.")
    print("=" * 80)
    print(df_dir_operational_frequency.to_string(index=False))
```

Tabla de frecuencia operativa DIR preparada.

| target          | horizon      | percentile   | valid_rows      | nan_rows                | long_count               | short_count |
|-----------------|--------------|--------------|-----------------|-------------------------|--------------------------|-------------|
| no_signal_count | signal_count | signal_ratio | no_signal_ratio | long_share_over_signals | short_share_over_signals |             |
| dir_p40_h30     | 30           | 40           | 979602          | 44460                   | 309481                   | 314255      |
| 355866          | 623736       | 0.636724     | 0.363276        |                         | 0.496173                 |             |
| 0.503827        |              |              |                 |                         |                          |             |
| dir_p50_h30     | 30           | 50           | 979602          | 44460                   | 259900                   | 270082      |
| 449620          | 529982       | 0.541018     | 0.458982        |                         | 0.490394                 |             |
| 0.509606        |              |              |                 |                         |                          |             |
| dir_p60_h30     | 30           | 60           | 979602          | 44460                   | 210169                   | 224826      |
| 544607          | 434995       | 0.444053     | 0.555947        |                         | 0.483153                 |             |
| 0.516847        |              |              |                 |                         |                          |             |
| dir_p40_h60     | 60           | 40           | 935142          | 88920                   | 296160                   | 300864      |
| 338118          | 597024       | 0.638431     | 0.361569        |                         | 0.496060                 |             |
| 0.503940        |              |              |                 |                         |                          |             |
| dir_p50_h60     | 60           | 50           | 935142          | 88920                   | 246274                   | 258746      |
| 430122          | 505020       | 0.540046     | 0.459954        |                         | 0.487652                 |             |
| 0.512348        |              |              |                 |                         |                          |             |
| dir_p60_h60     | 60           | 60           | 935142          | 88920                   | 197901                   | 215897      |
| 521344          | 413798       | 0.442498     | 0.557502        |                         | 0.478255                 |             |
| 0.521745        |              |              |                 |                         |                          |             |
| dir_p40_h90     | 90           | 40           | 890682          | 133380                  | 280594                   | 287376      |
| 322712          | 567970       | 0.637680     | 0.362320        |                         | 0.494030                 |             |
| 0.505970        |              |              |                 |                         |                          |             |
| dir_p50_h90     | 90           | 50           | 890682          | 133380                  | 234585                   | 248018      |
| 408079          | 482603       | 0.541835     | 0.458165        |                         | 0.486083                 |             |
| 0.513917        |              |              |                 |                         |                          |             |
| dir_p60_h90     | 90           | 60           | 890682          | 133380                  | 187768                   | 206733      |
| 496181          | 394501       | 0.442920     | 0.557080        |                         | 0.475963                 |             |
| 0.524037        |              |              |                 |                         |                          |             |

## Celda 2 — Clasificación simple de frecuencia operativa

Esta celda agrega una lectura rápida: si un target genera muchas, pocas o una cantidad intermedia de señales.

Lectura

Alta → muchas señales, posible mayor ruido  
 Media → frecuencia razonable para seguir estudiando  
 Baja → target más selectivo, menos oportunidades

```
In [34]: # Celda 2 – Clasificación simple de frecuencia operativa
# =====

def classify_operational_frequency(signal_ratio):
    """
    Clasificación simple de la frecuencia operativa.
    Los rangos son criterios exploratorios, no reglas definitivas.
    """
    if signal_ratio >= 0.60:
        return "Alta"
    elif signal_ratio >= 0.45:
        return "Media"
    else:
        return "Baja"
```

```

df_dir_operational_frequency["frequency_category"] = (
    df_dir_operational_frequency["signal_ratio"]
    .apply(classify_operational_frequency)
)

print("Clasificación de frecuencia operativa")
print("=" * 80)

cols_show = [
    "target",
    "horizon",
    "percentile",
    "signal_ratio",
    "no_signal_ratio",
    "frequency_category",
]

print(df_dir_operational_frequency[cols_show].to_string(index=False))

```

Clasificación de frecuencia operativa

```

=====
      target  horizon  percentile  signal_ratio  no_signal_ratio  frequency_categor
y
dir_p40_h30      30        40      0.636724      0.363276           Alt
a
dir_p50_h30      30        50      0.541018      0.458982           Medi
a
dir_p60_h30      30        60      0.444053      0.555947           Baj
a
dir_p40_h60      60        40      0.638431      0.361569           Alt
a
dir_p50_h60      60        50      0.540046      0.459954           Medi
a
dir_p60_h60      60        60      0.442498      0.557502           Baj
a
dir_p40_h90      90        40      0.637680      0.362320           Alt
a
dir_p50_h90      90        50      0.541835      0.458165           Medi
a
dir_p60_h90      90        60      0.442920      0.557080           Baj
a

```

### Celda 3 — Matriz horizonte vs percentil

Esta celda permite ver rápidamente cómo cambia la frecuencia operativa según H y p.

Esta tabla es útil porque resume todo en una matriz simple:

```

filas      = horizontes
columnas   = percentiles
valores    = frecuencia operativa

```

```

In [35]: # Celda 3 – Matriz de frecuencia operativa H x p
# =====

df_signal_pivot = df_dir_operational_frequency.pivot(
    index="horizon",

```

```

        columns="percentile",
        values="signal_ratio"
    )

    print("Matriz de frecuencia operativa DIR")
    print("=" * 80)
    print(df_signal_pivot.to_string(float_format=lambda x: f"{x:.2%}"))

```

Matriz de frecuencia operativa DIR

```

=====
percentile    40    50    60
horizon
30            63.67% 54.10% 44.41%
60            63.84% 54.00% 44.25%
90            63.77% 54.18% 44.29%

```

## Celda 4 — Gráfico de frecuencia operativa por target

Este gráfico muestra directamente qué tan operable es cada variante de `DIR`.

### Cómo leerlo

- Una barra alta significa que el target genera muchas señales `LONG` o `SHORT`.
- Una barra baja significa que el target clasifica más observaciones como `NO_SIGNAL`.

```

In [36]: # Celda 4 – Gráfico de frecuencia operativa por target
# =====

df_plot = df_dir_operational_frequency.copy()

df_plot["target_label"] = (
    "H" + df_plot["horizon"].astype(str)
    + " p" + df_plot["percentile"].astype(str)
)

fig, ax = plt.subplots(figsize=(12, 5))

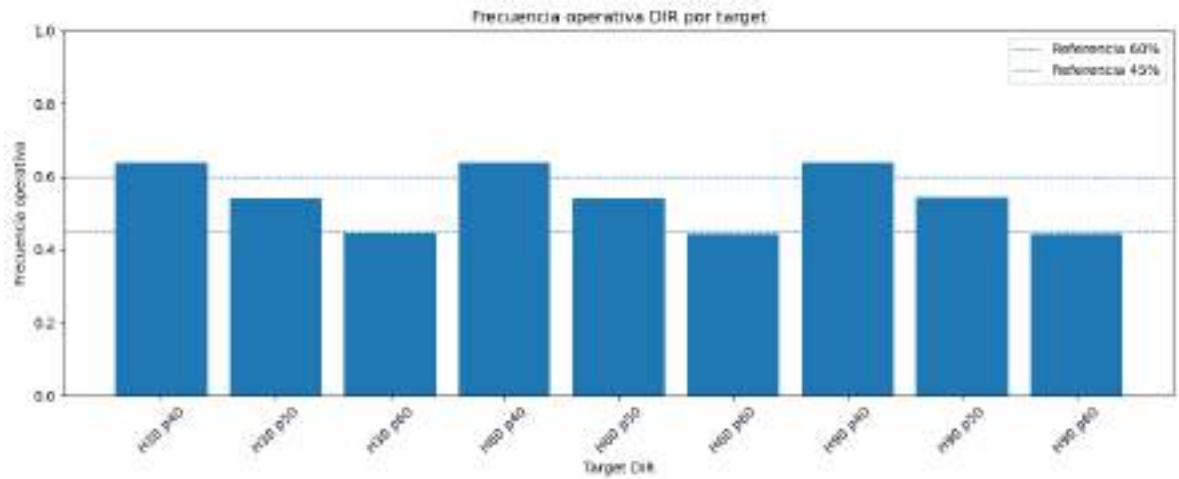
ax.bar(
    df_plot["target_label"],
    df_plot["signal_ratio"]
)

ax.axhline(0.60, linestyle="--", linewidth=1, label="Referencia 60%")
ax.axhline(0.45, linestyle="--", linewidth=1, label="Referencia 45%")

ax.set_title("Frecuencia operativa DIR por target")
ax.set_xlabel("Target DIR")
ax.set_ylabel("Frecuencia operativa")
ax.set_ylim(0, 1)
ax.legend()

plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

```



## Celda 5 — Gráfico por percentil y horizonte

Este gráfico muestra si la frecuencia cae de forma ordenada cuando sube el percentil.

### Cómo leerlo

Lo esperado es que la curva baje cuando sube el percentil:

- p40 → más señales
- p50 → frecuencia intermedia
- p60 → menos señales

Si eso ocurre, significa que el threshold está controlando correctamente la selectividad del target.

```
In [37]: # Celda 5 – Frecuencia operativa por percentil y horizonte
# =====

fig, ax = plt.subplots(figsize=(10, 5))

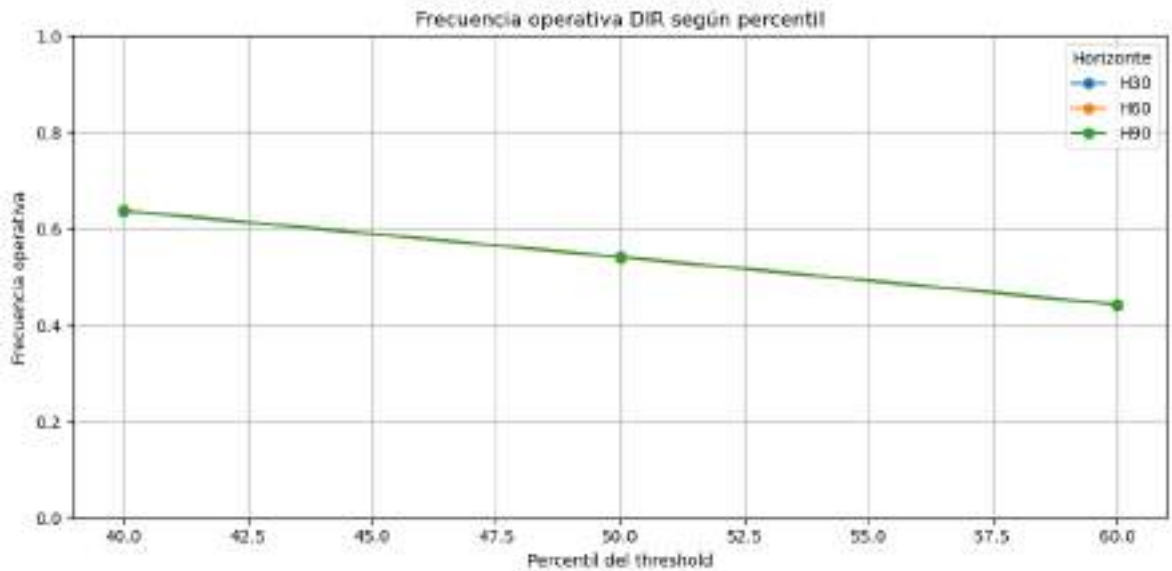
for h in DIR_HORIZONS:
    tmp = (
        df_dir_operational_frequency
        [df_dir_operational_frequency["horizon"] == h]
        .sort_values("percentile")
    )

    ax.plot(
        tmp["percentile"],
        tmp["signal_ratio"],
        marker="o",
        label=f"H{h}"
    )

ax.set_title("Frecuencia operativa DIR según percentil")
ax.set_xlabel("Percentil del threshold")
ax.set_ylabel("Frecuencia operativa")
ax.set_ylim(0, 1)
ax.legend(title="Horizonte")
ax.grid(True)

plt.tight_layout()
plt.show()
```





## Celda 6 — Resumen automático del punto 3.4

Esta celda imprime una conclusión rápida para documentar el análisis.

```
In [38]: # Celda 6 – Resumen automático de frecuencia operativa
# =====

print("Conclusiones preliminares - Frecuencia operativa DIR")
print("=" * 80)

for p in DIR_PERCENTILES:
    tmp = df_dir_operational_frequency[
        df_dir_operational_frequency["percentile"] == p
    ]

    avg_signal_ratio = tmp["signal_ratio"].mean()
    avg_no_signal_ratio = tmp["no_signal_ratio"].mean()

    print(f"\nPercentil {p}:")
    print(f"  Frecuencia operativa promedio: {avg_signal_ratio:.2%}")
    print(f"  NO_SIGNAL promedio: {avg_no_signal_ratio:.2%}")

print("\nLectura por target:")
for _, row in df_dir_operational_frequency.iterrows():
    print(
        f"  {row['target']}: "
        f"signal_ratio={row['signal_ratio']:.2%}, "
        f"NO_SIGNAL={row['no_signal_ratio']:.2%}, "
        f"categoria={row['frequency_category']}"
    )

print("\nInterpretación general:")
print("- p40 genera la mayor frecuencia operativa, pero puede incluir más ruido.")
print("- p50 mantiene una frecuencia intermedia y parece razonable para seguir a")
print("- p60 es más selectivo, pero reduce la cantidad de señales disponibles.")
print("- La frecuencia operativa debe evaluarse luego por año, régimen y contrat
```

## Conclusiones preliminares - Frecuencia operativa DIR

=====

### Percentil p40:

Frecuencia operativa promedio: 63.76%  
NO\_SIGNAL promedio: 36.24%

### Percentil p50:

Frecuencia operativa promedio: 54.10%  
NO\_SIGNAL promedio: 45.90%

### Percentil p60:

Frecuencia operativa promedio: 44.32%  
NO\_SIGNAL promedio: 55.68%

### Lectura por target:

dir\_p40\_h30: signal\_ratio=63.67%, NO\_SIGNAL=36.33%, categoría=Alta  
dir\_p50\_h30: signal\_ratio=54.10%, NO\_SIGNAL=45.90%, categoría=Media  
dir\_p60\_h30: signal\_ratio=44.41%, NO\_SIGNAL=55.59%, categoría=Baja  
dir\_p40\_h60: signal\_ratio=63.84%, NO\_SIGNAL=36.16%, categoría=Alta  
dir\_p50\_h60: signal\_ratio=54.00%, NO\_SIGNAL=46.00%, categoría=Media  
dir\_p60\_h60: signal\_ratio=44.25%, NO\_SIGNAL=55.75%, categoría=Baja  
dir\_p40\_h90: signal\_ratio=63.77%, NO\_SIGNAL=36.23%, categoría=Alta  
dir\_p50\_h90: signal\_ratio=54.18%, NO\_SIGNAL=45.82%, categoría=Media  
dir\_p60\_h90: signal\_ratio=44.29%, NO\_SIGNAL=55.71%, categoría=Baja

### Interpretación general:

- p40 genera la mayor frecuencia operativa, pero puede incluir más ruido.
- p50 mantiene una frecuencia intermedia y parece razonable para seguir analizando.
- p60 es más selectivo, pero reduce la cantidad de señales disponibles.
- La frecuencia operativa debe evaluarse luego por año, régimen y contrato.

## Observaciones premilinares - Frecuencia operativa DIR

La frecuencia operativa se comporta de forma muy ordenada: al aumentar el percentil, baja la cantidad de señales. Esto confirma que el threshold está controlando correctamente la selectividad del target.

p40 → ~63.8% de señales operativas

p50 → ~54.1% de señales operativas

p60 → ~44.3% de señales operativas

p40 genera muchas señales. Puede ser útil si queremos más oportunidades, pero probablemente incluya más ruido.

p50 queda como el punto más equilibrado. Mantiene una frecuencia operativa media, con aproximadamente la mitad de las observaciones generando señal y la otra parte quedando como NO\_SIGNAL.

p60 es más restrictivo. Genera menos señales y deja más del 55% de las observaciones como NO\_SIGNAL. Puede ser más selectivo, pero quizás demasiado limitado para algunos modelos.

Los horizontes H30, H60 y H90 muestran frecuencias casi idénticas para un mismo percentil. Esto es positivo porque indica que los thresholds por horizonte están bien

calibrados y permiten comparar targets de forma consistente.

La frecuencia operativa depende mucho más del **percentil** que del **horizonte**. En esta vista global, cambiar de H30 a H90 casi no modifica la proporción de señales.

Conclusión preliminar: p50 parece la opción más razonable para seguir analizando, mientras que p40 y p60 deben mantenerse como variantes comparativas: p40 por ser más activo y p60 por ser más selectivo.

## 3.5. Sensibilidad por horizonte y percentil

En este punto se evaluará cómo cambia DIR cuando se modifican los parámetros principales:

```
horizon
percentile
```

La pregunta central será:

¿Qué ocurre con la distribución de clases y la frecuencia operativa cuando cambia el horizonte o el threshold?

En general, percentiles más bajos tienden a generar más señales porque el threshold es menor. Percentiles más altos tienden a generar menos señales, pero potencialmente más selectivas.

Del mismo modo, horizontes más amplios pueden capturar movimientos mayores, pero también modifican la naturaleza temporal del problema.

Este análisis permitirá comparar variantes como:

```
dir_p40_h30
dir_p50_h30
dir_p60_h30
dir_p40_h60
dir_p50_h60
dir_p60_h60
dir_p40_h90
dir_p50_h90
dir_p60_h90
```

El objetivo no será elegir todavía un target definitivo, sino identificar cuáles combinaciones parecen más razonables.

## Celda 1 — Preparar tabla de sensibilidad

Esta celda solo prepara la tabla que vamos a usar para comparar. No calcula nada nuevo del precio.

```
In [39]: # Celda 1 – Preparación de tabla base
# =====

required_cols = [
    "target",
```

```

    "horizon",
    "percentile",
    "long_ratio",
    "short_ratio",
    "no_signal_ratio",
    "signal_ratio",
    "long_share_over_signals",
    "short_share_over_signals",
]

missing_cols = [
    col for col in required_cols
    if col not in df_dir_class_distribution.columns
]

if missing_cols:
    raise KeyError(f"Faltan columnas en df_dir_class_distribution: {missing_cols}")

df_dir_sensitivity = (
    df_dir_class_distribution[required_cols]
    .copy()
    .sort_values(["horizon", "percentile"])
    .reset_index(drop=True)
)

df_dir_sensitivity["target_label"] = (
    "H" + df_dir_sensitivity["horizon"].astype(str)
    + " p" + df_dir_sensitivity["percentile"].astype(str)
)

print("Tabla base de sensibilidad DIR")
print("=" * 80)
print(df_dir_sensitivity.to_string(index=False))

```

Tabla base de sensibilidad DIR

```

=====
target  horizon  percentile  long_ratio  short_ratio  no_signal_ratio  signa
l_ratio  long_share_over_signals  short_share_over_signals  target_label
dir_p40_h30      30          40    0.315925    0.320799    0.363276
0.636724          0.496173          0.503827    H30 p40
dir_p50_h30      30          50    0.265312    0.275706    0.458982
0.541018          0.490394          0.509606    H30 p50
dir_p60_h30      30          60    0.214545    0.229507    0.555947
0.444053          0.483153          0.516847    H30 p60
dir_p40_h60      60          40    0.316701    0.321731    0.361569
0.638431          0.496060          0.503940    H60 p40
dir_p50_h60      60          50    0.263355    0.276692    0.459954
0.540046          0.487652          0.512348    H60 p50
dir_p60_h60      60          60    0.211627    0.230871    0.557502
0.442498          0.478255          0.521745    H60 p60
dir_p40_h90      90          40    0.315033    0.322647    0.362320
0.637680          0.494030          0.505970    H90 p40
dir_p50_h90      90          50    0.263377    0.278459    0.458165
0.541835          0.486083          0.513917    H90 p50
dir_p60_h90      90          60    0.210814    0.232106    0.557080
0.442920          0.475963          0.524037    H90 p60

```

## Celda 2 — Matrices de sensibilidad

Estas matrices permiten ver rápidamente cómo cambian las métricas por horizonte y percentil.

Esto nos permite ver, por ejemplo, si `signal_ratio` cambia más por percentil o por horizonte.

```
In [40]: # Celda 2 – Matrices horizon x percentile
# =====

metrics_to_pivot = [
    "signal_ratio",
    "no_signal_ratio",
    "long_ratio",
    "short_ratio",
    "long_share_over_signals",
]

for metric in metrics_to_pivot:
    pivot_table = df_dir_sensitivity.pivot(
        index="horizon",
        columns="percentile",
        values=metric
    )

    print(f"\nMatriz: {metric}")
    print("=" * 80)
    print(pivot_table.to_string(float_format=lambda x: f"{x:.2%}"))
```

Matriz: signal\_ratio

| percentile | 40     | 50     | 60     |
|------------|--------|--------|--------|
| horizon    |        |        |        |
| 30         | 63.67% | 54.10% | 44.41% |
| 60         | 63.84% | 54.00% | 44.25% |
| 90         | 63.77% | 54.18% | 44.29% |

Matriz: no\_signal\_ratio

| percentile | 40     | 50     | 60     |
|------------|--------|--------|--------|
| horizon    |        |        |        |
| 30         | 36.33% | 45.90% | 55.59% |
| 60         | 36.16% | 46.00% | 55.75% |
| 90         | 36.23% | 45.82% | 55.71% |

Matriz: long\_ratio

| percentile | 40     | 50     | 60     |
|------------|--------|--------|--------|
| horizon    |        |        |        |
| 30         | 31.59% | 26.53% | 21.45% |
| 60         | 31.67% | 26.34% | 21.16% |
| 90         | 31.50% | 26.34% | 21.08% |

Matriz: short\_ratio

| percentile | 40     | 50     | 60     |
|------------|--------|--------|--------|
| horizon    |        |        |        |
| 30         | 32.08% | 27.57% | 22.95% |
| 60         | 32.17% | 27.67% | 23.09% |
| 90         | 32.26% | 27.85% | 23.21% |

Matriz: long\_share\_over\_signals

| percentile | 40     | 50     | 60     |
|------------|--------|--------|--------|
| horizon    |        |        |        |
| 30         | 49.62% | 49.04% | 48.32% |
| 60         | 49.61% | 48.77% | 47.83% |
| 90         | 49.40% | 48.61% | 47.60% |

## Celda 3 — Sensibilidad al percentil

Acá medimos cuánto cae la frecuencia operativa cuando pasamos de p40 a p60 .

Lectura simple:

drop\_p40\_to\_p60 = cuánto se reduce la frecuencia operativa al hacer el target más exigente.

```
In [41]: # Celda 3 – Sensibilidad al percentil
# =====

df_percentile_sensitivity = (
    df_dir_sensitivity
    .pivot(
        index="horizon",
        columns="percentile",
        values="signal_ratio"
    )
)
```

```

        .reset_index()
    )

df_percentile_sensitivity["drop_p40_to_p50"] = (
    df_percentile_sensitivity[40] - df_percentile_sensitivity[50]
)

df_percentile_sensitivity["drop_p50_to_p60"] = (
    df_percentile_sensitivity[50] - df_percentile_sensitivity[60]
)

df_percentile_sensitivity["drop_p40_to_p60"] = (
    df_percentile_sensitivity[40] - df_percentile_sensitivity[60]
)

print("Sensibilidad de frecuencia operativa al percentil")
print("=" * 80)
print(df_percentile_sensitivity.to_string(
    index=False,
    float_format=lambda x: f"{x:.2%}" if isinstance(x, float) else str(x)
))

```

Sensibilidad de frecuencia operativa al percentil

```

=====
horizon      40      50      60  drop_p40_to_p50  drop_p50_to_p60  drop_p40_to_p60
30  63.67%  54.10%  44.41%           9.57%           9.70%           19.27%
60  63.84%  54.00%  44.25%           9.84%           9.75%           19.59%
90  63.77%  54.18%  44.29%           9.58%           9.89%           19.48%

```

## Celda 4 — Sensibilidad al horizonte

Acá medimos si cambiar de H30 a H90 altera mucho la frecuencia operativa.

Lectura simple:

signal\_range bajo = el horizonte casi no cambia la frecuencia operativa.  
 signal\_range alto = el horizonte afecta mucho al target.

```

In [42]: # Celda 4 - Sensibilidad al horizonte
# =====

df_horizon_sensitivity = (
    df_dir_sensitivity
    .groupby("percentile")
    .agg(
        signal_min=("signal_ratio", "min"),
        signal_max=("signal_ratio", "max"),
        signal_mean=("signal_ratio", "mean"),
        signal_range=("signal_ratio", lambda x: x.max() - x.min()),
        no_signal_mean=("no_signal_ratio", "mean"),
    )
    .reset_index()
)

print("Sensibilidad de frecuencia operativa al horizonte")
print("=" * 80)
print(df_horizon_sensitivity.to_string(
    index=False,

```

```
float_format=lambda x: f"{x:.2%}"
))
```

Sensibilidad de frecuencia operativa al horizonte

| percentile | signal_min | signal_max | signal_mean | signal_range | no_signal_mean |
|------------|------------|------------|-------------|--------------|----------------|
| 40         | 63.67%     | 63.84%     | 63.76%      | 0.17%        | 36.24%         |
| 50         | 54.00%     | 54.18%     | 54.10%      | 0.18%        | 45.90%         |
| 60         | 44.25%     | 44.41%     | 44.32%      | 0.16%        | 55.68%         |

## Celda 5 — Gráfico de sensibilidad al percentil

Este gráfico muestra si el target responde correctamente al aumento del threshold.

Lo esperado es:

p40 > p50 > p60

```
In [43]: # Celda 5 – Gráfico de frecuencia operativa por percentil
# =====

fig, ax = plt.subplots(figsize=(10, 5))

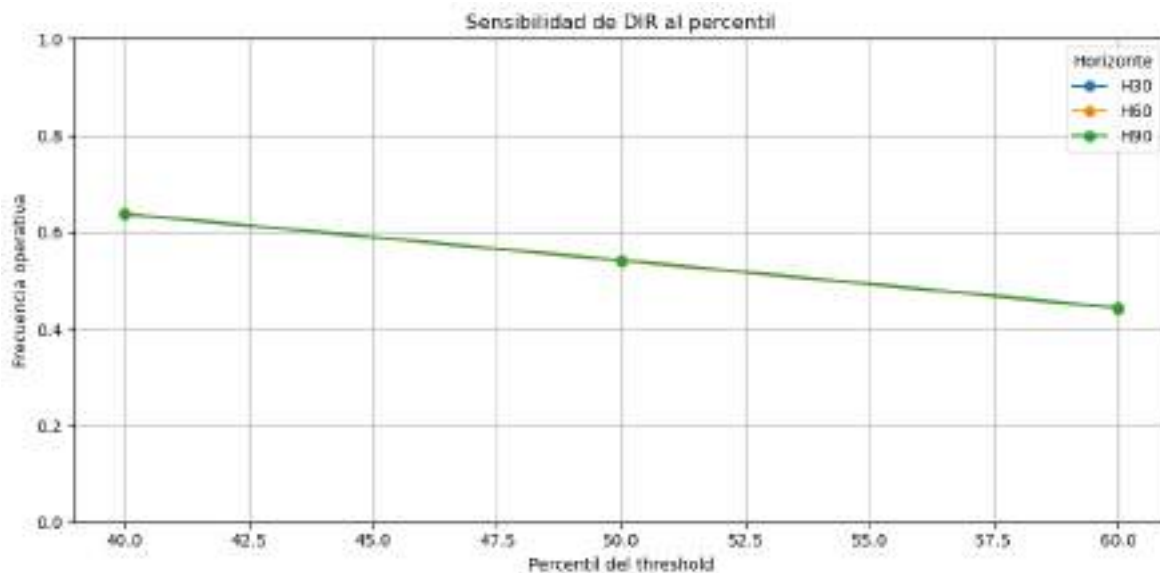
for h in DIR_HORIZONS:
    tmp = (
        df_dir_sensitivity[df_dir_sensitivity["horizon"] == h]
        .sort_values("percentile")
    )

    ax.plot(
        tmp["percentile"],
        tmp["signal_ratio"],
        marker="o",
        label=f"H{h}"
    )

ax.set_title("Sensibilidad de DIR al percentil")
ax.set_xlabel("Percentil del threshold")
ax.set_ylabel("Frecuencia operativa")
ax.set_ylim(0, 1)
ax.legend(title="Horizonte")
ax.grid(True)

plt.tight_layout()
plt.show()
```





## Celda 6 — Gráfico de caída de frecuencia p40 → p60

Este gráfico resume cuánto se vuelve más selectivo el target al pasar de p40 a p60.

```
In [44]: # Celda 6 – Caída de frecuencia operativa p40 a p60
# =====

fig, ax = plt.subplots(figsize=(8, 5))

ax.bar(
    "H" + df_percentile_sensitivity["horizon"].astype(str),
    df_percentile_sensitivity["drop_p40_to_p60"]
)

ax.set_title("Caída de frecuencia operativa al pasar de p40 a p60")
ax.set_xlabel("Horizonte")
ax.set_ylabel("Caída de frecuencia operativa")

plt.tight_layout()
plt.show()
```



## Celda 7 — Conclusión automática simple

```
In [45]: # Celda 7 – Conclusión automática de sensibilidad
# =====

print("Conclusiones preliminares - Sensibilidad DIR")
print("=" * 80)

print("\nSensibilidad al percentil:")
for _, row in df_percentile_sensitivity.iterrows():
    print(
        f"H{int(row['horizon'])}: "
        f"p40={row[40]:.2%}, "
        f"p50={row[50]:.2%}, "
        f"p60={row[60]:.2%}, "
        f"caída p40→p60={row['drop_p40_to_p60']:.2%}"
    )

print("\nSensibilidad al horizonte:")
for _, row in df_horizon_sensitivity.iterrows():
    print(
        f"p{int(row['percentile'])}: "
        f"frecuencia media={row['signal_mean']:.2%}, "
        f"rango entre horizontes={row['signal_range']:.2%}"
    )

print("\nLectura general:")
print("- Si la frecuencia cae claramente de p40 a p60, el percentil está control
print("- Si el rango entre horizontes es bajo, los horizontes son comparables a
print("- Si p50 queda en una zona intermedia, puede ser una variante candidata r
print("- La decisión final no debe tomarse solo con esta vista global; falta rev
```

## Conclusiones preliminares - Sensibilidad DIR

=====

### Sensibilidad al percentil:

H30: p40=63.67%, p50=54.10%, p60=44.41%, caída p40→p60=19.27%

H60: p40=63.84%, p50=54.00%, p60=44.25%, caída p40→p60=19.59%

H90: p40=63.77%, p50=54.18%, p60=44.29%, caída p40→p60=19.48%

### Sensibilidad al horizonte:

p40: frecuencia media=63.76%, rango entre horizontes=0.17%

p50: frecuencia media=54.10%, rango entre horizontes=0.18%

p60: frecuencia media=44.32%, rango entre horizontes=0.16%

### Lectura general:

- Si la frecuencia cae claramente de p40 a p60, el percentil está controlando bien la selectividad.
- Si el rango entre horizontes es bajo, los horizontes son comparables a nivel global.
- Si p50 queda en una zona intermedia, puede ser una variante candidata razonable.
- La decisión final no debe tomarse solo con esta vista global; falta revisar estabilidad temporal, régimen y contrato.

## Observaciones premilinares - Sensibilidad por horizontes y percentil

El análisis muestra que el target **DIR** es mucho más sensible al **percentil del threshold** que al **horizonte temporal**.

Al aumentar el percentil de **p40** a **p60**, la frecuencia operativa cae de forma clara y ordenada en todos los horizontes:

p40 → ~63.8% de señales operativas

p50 → ~54.1% de señales operativas

p60 → ~44.3% de señales operativas

La caída total entre **p40** y **p60** es cercana al 19% para **H30**, **H60** y **H90**. Esto confirma que el percentil controla correctamente la selectividad del target: percentiles más bajos generan más señales, mientras que percentiles más altos filtran más movimientos y aumentan la proporción de **NO\_SIGNAL**.

En cambio, el horizonte tiene muy poco impacto sobre la frecuencia operativa global. Para cada percentil, las diferencias entre **H30**, **H60** y **H90** son mínimas, con rangos inferiores al 0.20%. Esto sugiere que los thresholds por horizonte están calibrados de forma consistente y permiten comparar las variantes de manera razonable.

También se observa que **NO\_SIGNAL** aumenta progresivamente al subir el percentil:

p40 → ~36% **NO\_SIGNAL**

p50 → ~46% **NO\_SIGNAL**

p60 → ~56% **NO\_SIGNAL**

Por lo tanto, **p40** representa una configuración más activa, **p60** una configuración más selectiva, y **p50** queda como una alternativa intermedia.

De forma preliminar, `p50` parece la variante más equilibrada a nivel global, ya que mantiene una frecuencia operativa razonable sin generar señales tan abundantes como `p40` ni tan restrictivas como `p60`.

En síntesis, la sensibilidad principal de `DIR` proviene del percentil seleccionado, no del horizonte. Sin embargo, esta conclusión todavía debe complementarse con análisis por estabilidad temporal, régimen intradiario y contrato antes de seleccionar variantes candidatas definitivas.

## 3.6. Estabilidad temporal

En este punto se evaluará si la distribución del target se mantiene razonablemente estable a lo largo del tiempo.

El análisis se realizará por:

```
year
year_quarter
development_2020_2024 vs final_test_2025_2026
```

También se prestará atención a períodos particulares como:

```
2022
2025Q2
2026
```

La idea es verificar si una variante de `DIR` funciona de manera razonable en distintos contextos de mercado o si depende demasiado de un período específico.

Un target puede verse bien en promedio, pero ser inestable año a año. Esa inestabilidad puede dificultar el entrenamiento de modelos que generalicen correctamente.

## Celda 1 — Configuración temporal

```
In [46]: # Celda 1 – Configuración temporal
# =====

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# Asegurar columnas temporales básicas
if "year" not in df_targets_dir.columns:
    df_targets_dir["year"] = df_targets_dir.index.year

if "quarter" not in df_targets_dir.columns:
    df_targets_dir["quarter"] = df_targets_dir.index.quarter

if "year_quarter" not in df_targets_dir.columns:
    df_targets_dir["year_quarter"] = (
        df_targets_dir["year"].astype(str)
        + "Q"
        + df_targets_dir["quarter"].astype(str)
    )
```

```

if "target_analysis_split" not in df_targets_dir.columns:
    df_targets_dir["target_analysis_split"] = np.where(
        df_targets_dir["year"].between(2020, 2024),
        "development_2020_2024",
        "final_test_2025_2026"
    )

# Targets DIR
dir_target_cols = [
    f"dir_p{p}_h{h}"
    for h in DIR_HORIZONS
    for p in DIR_PERCENTILES
]

# Variante central para gráficos simples
selected_targets_temporal = [
    "dir_p50_h30",
    "dir_p50_h60",
    "dir_p50_h90",
]

print("Configuración temporal lista.")
print("=" * 80)
print(f"Targets DIR a analizar: {len(dir_target_cols)}")
print(f"Targets seleccionados para gráficos: {selected_targets_temporal}")
print(f"Años disponibles: {sorted(df_targets_dir['year'].unique().tolist())}")
print(f"Splits disponibles: {df_targets_dir['target_analysis_split'].unique().to

```

Configuración temporal lista.

```

=====
Targets DIR a analizar: 9
Targets seleccionados para gráficos: ['dir_p50_h30', 'dir_p50_h60', 'dir_p50_h90']
Años disponibles: [2020, 2021, 2022, 2023, 2024, 2025, 2026]
Splits disponibles: ['development_2020_2024', 'final_test_2025_2026']

```

## Celda 2 — Función resumen por período

Esta función nos permite resumir **DIR** por año, trimestre o split temporal.

```

In [47]: # Celda 2 – Función de resumen temporal
# =====

def summarize_dir_by_period(df, target_cols, period_col):
    """
    Resume distribución y frecuencia operativa de DIR por período temporal.
    period_col puede ser: year, year_quarter, target_analysis_split.
    """

    rows = []

    for target_col in target_cols:
        horizon, percentile = parse_dir_target_name(target_col)

        for period_value, group in df.groupby(period_col):
            total_rows = len(group)

            valid_mask = group[target_col].notna()

```

```

valid_values = group.loc[valid_mask, target_col].astype(int)

valid_rows = valid_mask.sum()
nan_rows = total_rows - valid_rows

long_count = (valid_values == 1).sum()
short_count = (valid_values == -1).sum()
no_signal_count = (valid_values == 0).sum()

signal_count = long_count + short_count

rows.append({
    "target": target_col,
    "horizon": horizon,
    "percentile": percentile,
    "period_col": period_value,
    "total_rows": total_rows,
    "valid_rows": valid_rows,
    "nan_rows": nan_rows,
    "long_count": long_count,
    "short_count": short_count,
    "no_signal_count": no_signal_count,
    "signal_count": signal_count,
    "long_ratio": long_count / valid_rows if valid_rows > 0 else np.nan,
    "short_ratio": short_count / valid_rows if valid_rows > 0 else np.nan,
    "no_signal_ratio": no_signal_count / valid_rows if valid_rows > 0 else np.nan,
    "signal_ratio": signal_count / valid_rows if valid_rows > 0 else np.nan,
    "long_share_over_signals": (
        long_count / signal_count if signal_count > 0 else np.nan
    ),
    "short_share_over_signals": (
        short_count / signal_count if signal_count > 0 else np.nan
    ),
})

return pd.DataFrame(rows)

```

## Celda 3 — Comparación development vs final test

Primero vemos si los targets cambian mucho entre el período de desarrollo y el período final.

```

In [48]: # Celda 3 - Comparación development vs final test
# =====

df_dir_split_distribution = summarize_dir_by_period(
    df=df_targets_dir,
    target_cols=dir_target_cols,
    period_col="target_analysis_split"
)

split_signal_pivot = df_dir_split_distribution.pivot(
    index="target",
    columns="target_analysis_split",
    values="signal_ratio"
)

split_signal_pivot["difference_final_minus_dev"] = (

```

```

split_signal_pivot["final_test_2025_2026"]
- split_signal_pivot["development_2020_2024"]
)

print("Frecuencia operativa por split temporal")
print("=" * 80)
print(split_signal_pivot.to_string(float_format=lambda x: f"{x:.2%}"))

```

Frecuencia operativa por split temporal

```

=====
target_analysis_split  development_2020_2024  final_test_2025_2026  difference_fi
nal_minus_dev
target
dir_p40_h30           60.20%                77.54%
17.35%
dir_p40_h60           60.22%                78.28%
18.06%
dir_p40_h90           60.16%                78.14%
17.98%
dir_p50_h30           50.24%                69.51%
19.27%
dir_p50_h60           50.03%                69.86%
19.83%
dir_p50_h90           50.23%                69.96%
19.73%
dir_p60_h30           40.32%                60.71%
20.40%
dir_p60_h60           40.09%                60.84%
20.75%
dir_p60_h90           40.08%                61.08%
21.00%

```

```

In [49]: # Celda 3.1 – Gráfico development vs final test
# =====

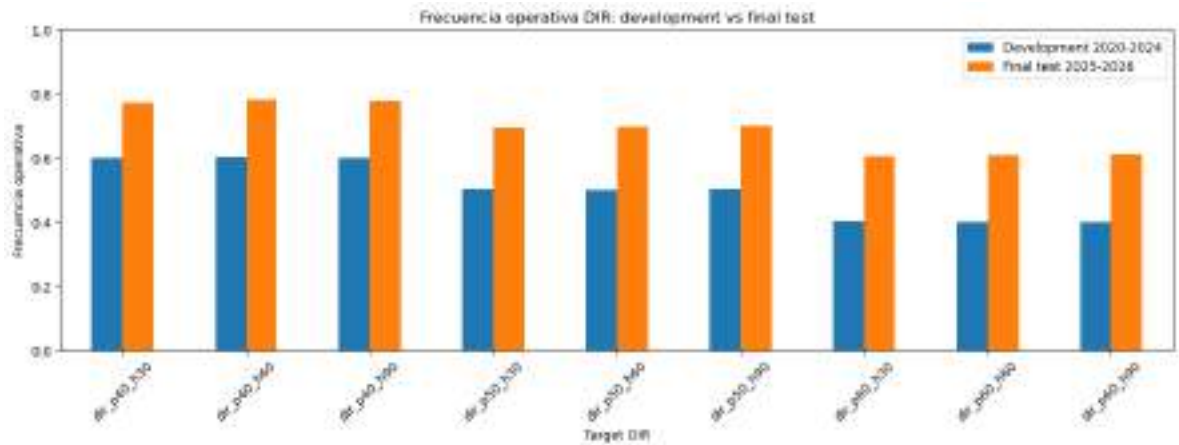
plot_data = split_signal_pivot[
    ["development_2020_2024", "final_test_2025_2026"]
].copy()

ax = plot_data.plot(
    kind="bar",
    figsize=(13, 5)
)

ax.set_title("Frecuencia operativa DIR: development vs final test")
ax.set_xlabel("Target DIR")
ax.set_ylabel("Frecuencia operativa")
ax.set_ylim(0, 1)
ax.legend(["Development 2020-2024", "Final test 2025-2026"])

plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

```



## Celda 4 — Estabilidad por año

Acá revisamos si la frecuencia operativa cambia mucho año a año.

```
In [50]: # Celda 4 – Estabilidad anual
# =====

df_dir_year_distribution = summarize_dir_by_period(
    df=df_targets_dir,
    target_cols=dir_target_cols,
    period_col="year"
)

year_signal_pivot = df_dir_year_distribution.pivot(
    index="year",
    columns="target",
    values="signal_ratio"
)

print("Frecuencia operativa anual por target")
print("=" * 80)
print(year_signal_pivot.to_string(float_format=lambda x: f"{x:.2%}"))
```

Frecuencia operativa anual por target

```
=====
target  dir_p40_h30  dir_p40_h60  dir_p40_h90  dir_p50_h30  dir_p50_h60  dir_p50_
h90  dir_p60_h30  dir_p60_h60  dir_p60_h90
year
2020      51.53%      50.55%      50.23%      42.26%      41.05%      40.
74%      33.57%      32.30%      31.67%
2021      48.98%      48.84%      49.19%      39.53%      39.26%      39.
73%      30.80%      30.21%      30.28%
2022      80.47%      81.41%      81.27%      71.69%      72.19%      72.
19%      61.37%      61.58%      61.63%
2023      55.51%      55.40%      55.29%      43.91%      43.40%      43.
89%      33.06%      32.75%      32.75%
2024      64.09%      64.47%      64.37%      53.43%      53.80%      54.
12%      42.45%      43.21%      43.65%
2025      74.61%      75.20%      74.94%      66.06%      66.19%      66.
12%      56.97%      56.76%      56.91%
2026      87.07%      88.27%      88.54%      80.69%      81.76%      82.
42%      72.84%      74.09%      74.59%
```

```
In [51]: # Celda 4.1 – Gráfico anual para variantes p50
# =====
```



```
fig, axes = plt.subplots(
    nrows=3,
    ncols=1,
    figsize=(11, 10),
    sharex=True
)

for ax, target_col in zip(axes, selected_targets_temporal):
    tmp = df_dir_year_distribution[
        df_dir_year_distribution["target"] == target_col
    ].sort_values("year")

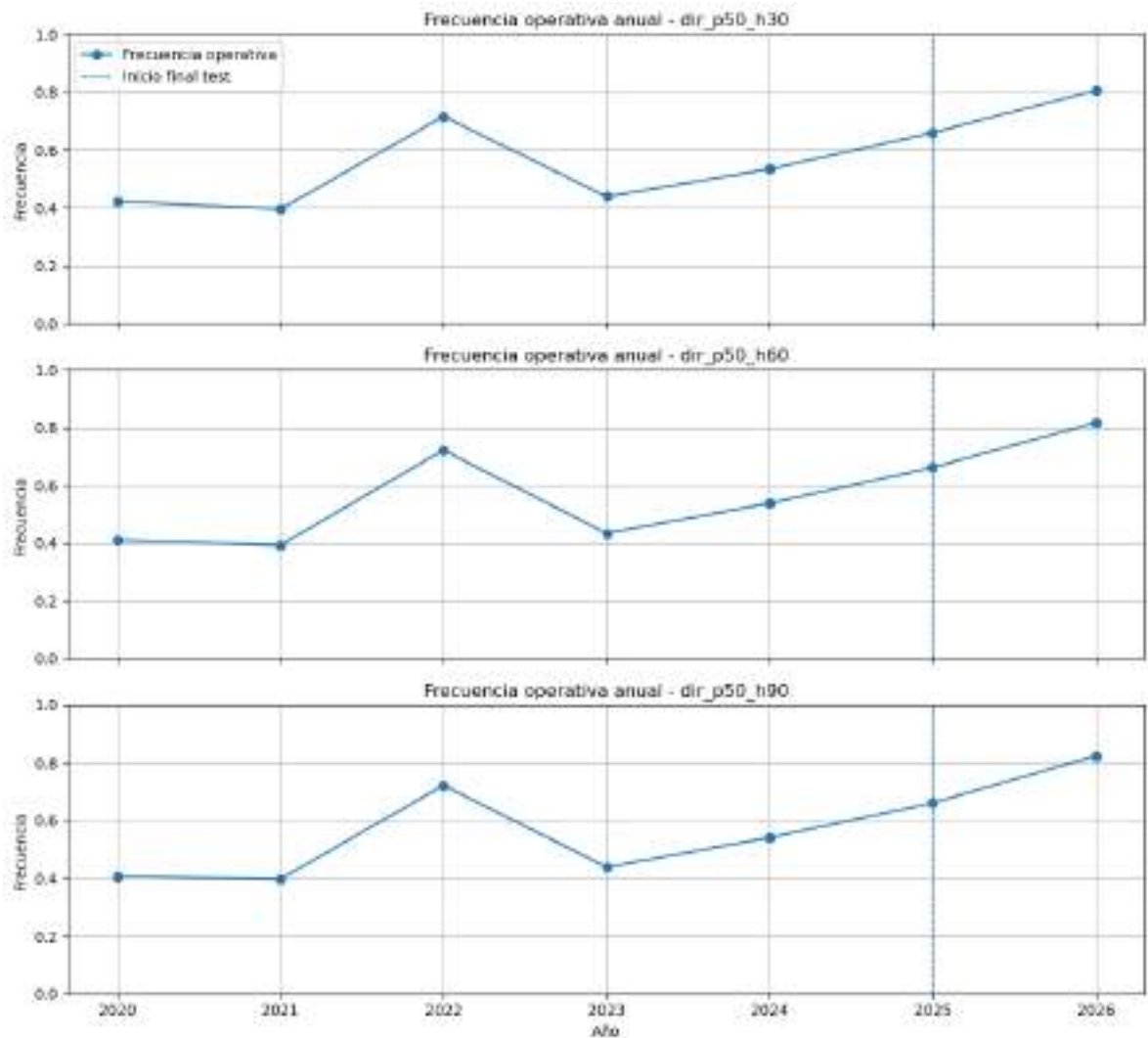
    ax.plot(
        tmp["year"],
        tmp["signal_ratio"],
        marker="o",
        label="Frecuencia operativa"
    )

    ax.axvline(2025, linestyle="--", linewidth=1, label="Inicio final test")

    ax.set_title(f"Frecuencia operativa anual - {target_col}")
    ax.set_ylabel("Frecuencia")
    ax.set_ylim(0, 1)
    ax.grid(True)

axes[0].legend(loc="best")
axes[-1].set_xlabel("Año")

plt.tight_layout()
plt.show()
```



## Celda 5 — Estabilidad por trimestre

Ahora bajamos a `year_quarter` para detectar cambios más finos, como `2025Q2`.

```
In [52]: # Celda 5 – Estabilidad trimestral
# =====

df_dir_quarter_distribution = summarize_dir_by_period(
    df=df_targets_dir,
    target_cols=dir_target_cols,
    period_col="year_quarter"
)

quarter_signal_pivot = df_dir_quarter_distribution.pivot(
    index="year_quarter",
    columns="target",
    values="signal_ratio"
)

print("Frecuencia operativa trimestral por target")
print("=" * 80)
print(quarter_signal_pivot.to_string(float_format=lambda x: f"{x:.2%}"))
```

## Frecuencia operativa trimestral por target

| target       | dir_p40_h30 | dir_p40_h60 | dir_p40_h90 | dir_p50_h30 | dir_p50_h60 | dir_p50_h90 | dir_p60_h30 | dir_p60_h60 | dir_p60_h90 |
|--------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| year_quarter |             |             |             |             |             |             |             |             |             |
| 2020Q1       | 43.44%      | 42.18%      | 41.44%      | 37.57%      | 36.47%      |             |             |             |             |
| 36.05%       | 32.35%      | 31.00%      | 31.07%      |             |             |             |             |             |             |
| 2020Q2       | 54.89%      | 54.15%      | 52.56%      | 43.80%      | 42.19%      |             |             |             |             |
| 40.67%       | 33.11%      | 31.44%      | 30.01%      |             |             |             |             |             |             |
| 2020Q3       | 57.39%      | 56.85%      | 57.93%      | 47.22%      | 46.53%      |             |             |             |             |
| 47.59%       | 37.63%      | 36.77%      | 36.61%      |             |             |             |             |             |             |
| 2020Q4       | 49.43%      | 47.95%      | 47.82%      | 39.68%      | 38.18%      |             |             |             |             |
| 37.70%       | 30.64%      | 29.40%      | 28.40%      |             |             |             |             |             |             |
| 2021Q1       | 61.54%      | 60.27%      | 59.93%      | 51.96%      | 50.76%      |             |             |             |             |
| 51.50%       | 42.79%      | 41.71%      | 42.21%      |             |             |             |             |             |             |
| 2021Q2       | 40.58%      | 39.50%      | 39.78%      | 31.17%      | 30.02%      |             |             |             |             |
| 30.34%       | 22.81%      | 21.75%      | 21.71%      |             |             |             |             |             |             |
| 2021Q3       | 39.82%      | 40.85%      | 42.32%      | 30.94%      | 31.70%      |             |             |             |             |
| 32.73%       | 22.99%      | 22.92%      | 23.21%      |             |             |             |             |             |             |
| 2021Q4       | 54.11%      | 54.83%      | 54.80%      | 44.17%      | 44.67%      |             |             |             |             |
| 44.46%       | 34.73%      | 34.59%      | 34.12%      |             |             |             |             |             |             |
| 2022Q1       | 88.83%      | 90.29%      | 90.00%      | 82.33%      | 83.29%      |             |             |             |             |
| 83.00%       | 73.72%      | 74.13%      | 73.52%      |             |             |             |             |             |             |
| 2022Q2       | 87.06%      | 88.71%      | 88.54%      | 80.24%      | 81.38%      |             |             |             |             |
| 81.18%       | 70.87%      | 71.55%      | 71.90%      |             |             |             |             |             |             |
| 2022Q3       | 73.96%      | 74.60%      | 74.03%      | 62.81%      | 62.88%      |             |             |             |             |
| 62.81%       | 51.01%      | 50.81%      | 51.03%      |             |             |             |             |             |             |
| 2022Q4       | 71.74%      | 71.71%      | 72.19%      | 61.03%      | 60.82%      |             |             |             |             |
| 61.41%       | 49.47%      | 49.43%      | 49.66%      |             |             |             |             |             |             |
| 2023Q1       | 64.11%      | 64.22%      | 64.67%      | 52.51%      | 52.37%      |             |             |             |             |
| 53.25%       | 41.30%      | 41.15%      | 41.17%      |             |             |             |             |             |             |
| 2023Q2       | 47.76%      | 47.51%      | 47.57%      | 36.17%      | 35.63%      |             |             |             |             |
| 35.92%       | 25.78%      | 25.65%      | 25.10%      |             |             |             |             |             |             |
| 2023Q3       | 58.04%      | 57.94%      | 56.13%      | 46.01%      | 44.76%      |             |             |             |             |
| 44.31%       | 34.27%      | 33.08%      | 32.97%      |             |             |             |             |             |             |
| 2023Q4       | 51.25%      | 51.01%      | 51.91%      | 40.05%      | 39.94%      |             |             |             |             |
| 41.17%       | 30.06%      | 30.30%      | 30.89%      |             |             |             |             |             |             |
| 2024Q1       | 57.86%      | 57.30%      | 57.55%      | 45.94%      | 45.84%      |             |             |             |             |
| 46.58%       | 34.56%      | 34.97%      | 35.77%      |             |             |             |             |             |             |
| 2024Q2       | 57.86%      | 58.12%      | 57.51%      | 47.04%      | 47.71%      |             |             |             |             |
| 47.26%       | 36.30%      | 37.38%      | 37.54%      |             |             |             |             |             |             |
| 2024Q3       | 74.51%      | 76.83%      | 77.10%      | 65.66%      | 67.67%      |             |             |             |             |
| 67.86%       | 55.49%      | 57.44%      | 57.21%      |             |             |             |             |             |             |
| 2024Q4       | 66.84%      | 66.47%      | 66.18%      | 55.89%      | 54.90%      |             |             |             |             |
| 55.70%       | 44.30%      | 43.94%      | 44.96%      |             |             |             |             |             |             |
| 2025Q1       | 82.74%      | 83.03%      | 83.37%      | 75.10%      | 75.24%      |             |             |             |             |
| 76.33%       | 66.64%      | 66.93%      | 68.27%      |             |             |             |             |             |             |
| 2025Q2       | 82.50%      | 82.95%      | 83.21%      | 74.83%      | 75.16%      |             |             |             |             |
| 75.47%       | 66.18%      | 66.00%      | 66.51%      |             |             |             |             |             |             |
| 2025Q3       | 58.62%      | 58.83%      | 57.85%      | 48.15%      | 47.17%      |             |             |             |             |
| 46.06%       | 37.64%      | 36.33%      | 35.32%      |             |             |             |             |             |             |
| 2025Q4       | 75.90%      | 77.28%      | 76.71%      | 67.66%      | 68.68%      |             |             |             |             |
| 68.26%       | 59.01%      | 59.39%      | 59.32%      |             |             |             |             |             |             |
| 2026Q1       | 87.43%      | 88.32%      | 88.77%      | 81.19%      | 82.24%      |             |             |             |             |
| 82.85%       | 73.75%      | 75.00%      | 75.15%      |             |             |             |             |             |             |
| 2026Q2       | 85.31%      | 88.02%      | 87.41%      | 78.24%      | 79.44%      |             |             |             |             |
| 80.38%       | 68.43%      | 69.70%      | 71.88%      |             |             |             |             |             |             |

In [53]: # Celda 5.1 – Gráfico trimestral para variantes p50

# =====

```

fig, axes = plt.subplots(
    nrows=3,
    ncols=1,
    figsize=(13, 10),
    sharex=True
)

for ax, target_col in zip(axes, selected_targets_temporal):
    tmp = df_dir_quarter_distribution[
        df_dir_quarter_distribution["target"] == target_col
    ].copy()

    tmp = tmp.sort_values("year_quarter")

    ax.plot(
        tmp["year_quarter"],
        tmp["signal_ratio"],
        marker="o"
    )

    ax.set_title(f"Frecuencia operativa trimestral - {target_col}")
    ax.set_ylabel("Frecuencia")
    ax.set_ylim(0, 1)
    ax.grid(True)

axes[-1].set_xlabel("Trimestre")
plt.xticks(rotation=90)

plt.tight_layout()
plt.show()

```



## Celda 6 — Períodos especiales: 2022, 2025Q2 y 2026

Esta celda extrae directamente los períodos que queremos mirar con más atención.

```
In [54]: # Celda 6 – Revisión de períodos especiales
# =====

special_years = [2022, 2026]
special_quarters = ["2025Q2"]

df_special_years = df_dir_year_distribution[
    df_dir_year_distribution["year"].isin(special_years)
].copy()

df_special_quarters = df_dir_quarter_distribution[
    df_dir_quarter_distribution["year_quarter"].isin(special_quarters)
].copy()

print("Períodos especiales por año")
print("=" * 80)
print(
    df_special_years[
        ["target", "year", "signal_ratio", "no_signal_ratio", "long_share_over_s
    ].to_string(index=False, float_format=lambda x: f"{x:.2%}")
)

print("\nPeríodos especiales por trimestre")
print("=" * 80)
print(
    df_special_quarters[
        ["target", "year_quarter", "signal_ratio", "no_signal_ratio", "long_shar
    ].to_string(index=False, float_format=lambda x: f"{x:.2%}")
)
```

## Períodos especiales por año

| target      | year | signal_ratio | no_signal_ratio | long_share_over_signals |
|-------------|------|--------------|-----------------|-------------------------|
| dir_p40_h30 | 2022 | 80.47%       | 19.53%          | 48.95%                  |
| dir_p40_h30 | 2026 | 87.07%       | 12.93%          | 49.29%                  |
| dir_p50_h30 | 2022 | 71.69%       | 28.31%          | 48.73%                  |
| dir_p50_h30 | 2026 | 80.69%       | 19.31%          | 49.17%                  |
| dir_p60_h30 | 2022 | 61.37%       | 38.63%          | 48.36%                  |
| dir_p60_h30 | 2026 | 72.84%       | 27.16%          | 49.03%                  |
| dir_p40_h60 | 2022 | 81.41%       | 18.59%          | 48.60%                  |
| dir_p40_h60 | 2026 | 88.27%       | 11.73%          | 49.31%                  |
| dir_p50_h60 | 2022 | 72.19%       | 27.81%          | 48.32%                  |
| dir_p50_h60 | 2026 | 81.76%       | 18.24%          | 48.89%                  |
| dir_p60_h60 | 2022 | 61.58%       | 38.42%          | 47.98%                  |
| dir_p60_h60 | 2026 | 74.09%       | 25.91%          | 48.66%                  |
| dir_p40_h90 | 2022 | 81.27%       | 18.73%          | 47.64%                  |
| dir_p40_h90 | 2026 | 88.54%       | 11.46%          | 49.81%                  |
| dir_p50_h90 | 2022 | 72.19%       | 27.81%          | 47.34%                  |
| dir_p50_h90 | 2026 | 82.42%       | 17.58%          | 49.67%                  |
| dir_p60_h90 | 2022 | 61.63%       | 38.37%          | 47.13%                  |
| dir_p60_h90 | 2026 | 74.59%       | 25.41%          | 49.06%                  |

## Períodos especiales por trimestre

| target      | year_quarter | signal_ratio | no_signal_ratio | long_share_over_signals |
|-------------|--------------|--------------|-----------------|-------------------------|
| dir_p40_h30 | 2025Q2       | 82.50%       | 17.50%          | 52.12%                  |
| dir_p50_h30 | 2025Q2       | 74.83%       | 25.17%          | 51.80%                  |
| dir_p60_h30 | 2025Q2       | 66.18%       | 33.82%          | 51.43%                  |
| dir_p40_h60 | 2025Q2       | 82.95%       | 17.05%          | 51.87%                  |
| dir_p50_h60 | 2025Q2       | 75.16%       | 24.84%          | 51.35%                  |
| dir_p60_h60 | 2025Q2       | 66.00%       | 34.00%          | 50.41%                  |
| dir_p40_h90 | 2025Q2       | 83.21%       | 16.79%          | 52.41%                  |
| dir_p50_h90 | 2025Q2       | 75.47%       | 24.53%          | 51.85%                  |
| dir_p60_h90 | 2025Q2       | 66.51%       | 33.49%          | 51.03%                  |

## Celda 7 — Métricas de estabilidad temporal

Esta celda resume qué targets son más o menos estables.

```
In [55]: # Celda 7 - Métricas de estabilidad temporal por target
# =====

df_dir_year_stability = (
    df_dir_year_distribution
    .groupby(["target", "horizon", "percentile"], as_index=False)
    .agg(
        signal_mean=("signal_ratio", "mean"),
        signal_std=("signal_ratio", "std"),
        signal_min=("signal_ratio", "min"),
        signal_max=("signal_ratio", "max"),
        signal_range=("signal_ratio", lambda x: x.max() - x.min()),
        no_signal_mean=("no_signal_ratio", "mean"),
        long_share_mean=("long_share_over_signals", "mean"),
        long_share_range=("long_share_over_signals", lambda x: x.max() - x.min())
    )
    .sort_values(["percentile", "horizon"])
    .reset_index(drop=True)
)
```

```
print("Métricas de estabilidad anual DIR")
print("=" * 80)
print(
    df_dir_year_stability.to_string(
        index=False,
        float_format=lambda x: f"{x:.2%}"
    )
)
```

Métricas de estabilidad anual DIR

```
=====
      target horizon percentile signal_mean signal_std signal_min signal_max
signal_range no_signal_mean long_share_mean long_share_range
dir_p40_h30    30          40      66.04%      14.94%      48.98%      87.07%
38.08%          33.96%          49.62%          2.18%
dir_p40_h60    60          40      66.30%      15.63%      48.84%      88.27%
39.42%          33.70%          49.63%          3.05%
dir_p40_h90    90          40      66.26%      15.65%      49.19%      88.54%
39.35%          33.74%          49.52%          4.87%
dir_p50_h30    30          50      56.79%      16.15%      39.53%      80.69%
41.15%          43.21%          49.03%          2.52%
dir_p50_h60    60          50      56.81%      16.79%      39.26%      81.76%
42.49%          43.19%          48.77%          2.99%
dir_p50_h90    90          50      57.03%      16.85%      39.73%      82.42%
42.70%          42.97%          48.75%          4.56%
dir_p60_h30    30          60      47.29%      16.49%      30.80%      72.84%
42.04%          52.71%          48.28%          2.63%
dir_p60_h60    60          60      47.27%      17.11%      30.21%      74.09%
43.88%          52.73%          47.81%          3.67%
dir_p60_h90    90          60      47.36%      17.33%      30.28%      74.59%
44.31%          52.64%          47.70%          4.92%
```

## Celda 8 — Conclusión automática simple

```
In [56]: # Celda 8 - Conclusión automática de estabilidad temporal
# =====

print("Conclusiones preliminares - Estabilidad temporal DIR")
print("=" * 80)

print("\nTargets con menor rango anual de frecuencia operativa:")
print(
    df_dir_year_stability
    .sort_values("signal_range")
    [["target", "signal_mean", "signal_range", "signal_std"]]
    .head(5)
    .to_string(index=False, float_format=lambda x: f"{x:.2%}")
)

print("\nTargets con mayor rango anual de frecuencia operativa:")
print(
    df_dir_year_stability
    .sort_values("signal_range", ascending=False)
    [["target", "signal_mean", "signal_range", "signal_std"]]
    .head(5)
    .to_string(index=False, float_format=lambda x: f"{x:.2%}")
)

print("\nLectura general:")
```

```
print("- Un target estable debería mantener una frecuencia operativa relativamen
print("- Cambios fuertes en 2022, 2025Q2 o 2026 pueden indicar sensibilidad a re
print("- La comparación development vs final test permite revisar si el target c
print("- p50 sigue siendo una variante importante para observar, pero todavía de
```

## Conclusiones preliminares - Estabilidad temporal DIR

=====

Targets con menor rango anual de frecuencia operativa:

| target      | signal_mean | signal_range | signal_std |
|-------------|-------------|--------------|------------|
| dir_p40_h30 | 66.04%      | 38.08%       | 14.94%     |
| dir_p40_h90 | 66.26%      | 39.35%       | 15.65%     |
| dir_p40_h60 | 66.30%      | 39.42%       | 15.63%     |
| dir_p50_h30 | 56.79%      | 41.15%       | 16.15%     |
| dir_p60_h30 | 47.29%      | 42.04%       | 16.49%     |

Targets con mayor rango anual de frecuencia operativa:

| target      | signal_mean | signal_range | signal_std |
|-------------|-------------|--------------|------------|
| dir_p60_h90 | 47.36%      | 44.31%       | 17.33%     |
| dir_p60_h60 | 47.27%      | 43.88%       | 17.11%     |
| dir_p50_h90 | 57.03%      | 42.70%       | 16.85%     |
| dir_p50_h60 | 56.81%      | 42.49%       | 16.79%     |
| dir_p60_h30 | 47.29%      | 42.04%       | 16.49%     |

Lectura general:

- Un target estable debería mantener una frecuencia operativa relativamente parecida entre años.
- Cambios fuertes en 2022, 2025Q2 o 2026 pueden indicar sensibilidad a regímenes de mercado extremos.
- La comparación development vs final test permite revisar si el target cambia fuera de muestra.
- p50 sigue siendo una variante importante para observar, pero todavía debe validarse temporalmente.

## Observaciones preliminares - Estabilidad Temporal

1. **Hay una diferencia fuerte entre development y final test.** En 2025-2026 la frecuencia operativa sube mucho frente a 2020-2024. Por ejemplo, en dir\_p50\_h30 pasa de 50.24% a 69.51%. Esto indica que el período final tuvo movimientos mucho más frecuentes o intensos respecto al período usado para calibrar thresholds.
2. **El target no es temporalmente estable en frecuencia.** Los rangos anuales son altos, entre 38% y 44% según el target. Eso significa que DIR cambia bastante de comportamiento según el año.
3. **2022 es un año claramente más activo.** En 2022, incluso p60 supera el 61% de frecuencia operativa. Esto confirma que fue un año de alta volatilidad / mayor movimiento relativo.
4. **2026 es todavía más extremo.** En 2026, p50 está alrededor de 80%-82% y p60 alrededor de 73%-75%. Esto es muy alto. Hay que tratarlo como período de estrés, además de recordar que 2026 está incompleto.



5. **2021 fue el período menos activo.** En 2021, `p50` ronda solo **39%–40%** y `p60` cerca de **30%**. Esto muestra que el target puede generar muy pocas señales en años más tranquilos.
6. **Trimestralmente se ven cambios fuertes.** `2022Q1` , `2022Q2` , `2025Q2` y `2026Q1` aparecen como períodos de alta frecuencia operativa. En cambio, `2021Q2` y `2021Q3` son períodos mucho más débiles.
7. **El balance LONG/SHORT se mantiene bastante bien.** Aunque la frecuencia cambia mucho, `long_share_over_signals` se mantiene cerca del **48%–50%**. Esto es positivo: el target no se vuelve extremadamente sesgado hacia LONG o SHORT.
8. **El problema principal no es el balance direccional, sino la estabilidad temporal.** `DIR` parece balanceado en dirección, pero su cantidad de señales depende mucho del régimen de mercado de cada año.
9. **Lectura preliminar:** `p50` sigue siendo una buena variante intermedia, pero no puede considerarse estable todavía. Necesitamos seguir revisando por régimen, contrato y contexto antes de decidir si es un target candidato fuerte.

Conclusión rápida: **DIR funciona bien conceptualmente, pero la frecuencia operativa es sensible al contexto temporal. Esto confirma que el mercado cambió bastante entre 2020-2026 y que el análisis por año/régimen/contrato es necesario antes de modelar.**

## 3.7. Análisis por régimen intradiario

En este punto se analizará el comportamiento de `DIR` por régimen intradiario.

Los regímenes considerados serán:

- Overnight
- Pre-market
- Opening
- Regular
- Closing

Para cada régimen se evaluará:

- distribución de clases
- frecuencia operativa
- proporción LONG
- proporción SHORT
- proporción NO SIGNAL

Este análisis es importante porque el MNQ no se comporta igual durante toda la sesión. Algunos regímenes pueden presentar más volatilidad, más direccionalidad o mayor frecuencia de movimientos significativos.

El objetivo será identificar si `DIR` es más claro, estable u operable en ciertos momentos del día.

## Celda 1 — Configuración del análisis por régimen

Esta celda solo prepara el análisis y asegura que exista `regime_name`.

```
In [57]: # Celda 1 – Configuración
# =====

import re
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

REGIME_ORDER = [
    "Overnight",
    "Pre-market",
    "Opening",
    "Regular",
    "Closing"
]

REGIME_NAME_BY_ID = {
    0: "Overnight",
    1: "Pre-market",
    2: "Opening",
    3: "Regular",
    4: "Closing",
}

# Asegurar columna regime_name
if "regime_name" not in df_targets_dir.columns:
    df_targets_dir["regime_name"] = df_targets_dir["regime_id"].map(REGIME_NAME_BY_ID)

df_targets_dir["regime_name"] = pd.Categorical(
    df_targets_dir["regime_name"],
    categories=REGIME_ORDER,
    ordered=True
)

# Asegurar columna split temporal
if "target_analysis_split" not in df_targets_dir.columns:
    if "year" not in df_targets_dir.columns:
        df_targets_dir["year"] = df_targets_dir.index.year

    df_targets_dir["target_analysis_split"] = np.where(
        df_targets_dir["year"].between(2020, 2024),
        "development_2020_2024",
        "final_test_2025_2026"
    )

# Targets DIR esperados
dir_target_cols = [
    f"dir_p{p}_h{h}"
    for h in DIR_HORIZONS
    for p in DIR_PERCENTILES
]

missing_cols = [
    col for col in dir_target_cols
```

```

    if col not in df_targets_dir.columns
]

if missing_cols:
    raise KeyError(f"Faltan columnas DIR: {missing_cols}")

# Función auxiliar por si no quedó definida antes
def parse_dir_target_name(target_col):
    pattern = r"dir_p(?P<percentile>\d+)_h(?P<horizon>\d+)"
    match = re.match(pattern, target_col)

    if match is None:
        raise ValueError(f"No se pudo interpretar el target: {target_col}")

    return int(match.group("horizon")), int(match.group("percentile"))

print("Configuración del análisis por régimen lista.")
print("=" * 80)
print(f"Targets a analizar: {len(dir_target_cols)}")
print(f"Regímenes: {REGIME_ORDER}")
print(f"Splits: {df_targets_dir['target_analysis_split'].unique().tolist()}")

```

Configuración del análisis por régimen lista.

```

=====
Targets a analizar: 9
Regímenes: ['Overnight', 'Pre-market', 'Opening', 'Regular', 'Closing']
Splits: ['development_2020_2024', 'final_test_2025_2026']

```

## Celda 2 — Función de resumen por régimen

```

In [58]: # Celda 2 – Función para resumir DIR por régimen
# =====

def summarize_dir_by_group(df, target_cols, group_cols):
    """
    Resume la distribución de clases DIR por una o más columnas de agrupación.

    group_cols puede ser:
        ["regime_name"]
        ["regime_name", "target_analysis_split"]
    """

    rows = []

    for target_col in target_cols:
        horizon, percentile = parse_dir_target_name(target_col)

        grouped = df.groupby(group_cols, observed=False)

        for group_key, group in grouped:
            if not isinstance(group_key, tuple):
                group_key = (group_key,)

            group_info = dict(zip(group_cols, group_key))

            total_rows = len(group)

            valid_mask = group[target_col].notna()
            valid_values = group.loc[valid_mask, target_col].astype(int)

```

```

valid_rows = valid_mask.sum()
nan_rows = total_rows - valid_rows

long_count = (valid_values == 1).sum()
short_count = (valid_values == -1).sum()
no_signal_count = (valid_values == 0).sum()

signal_count = long_count + short_count

row = {
    "target": target_col,
    "horizon": horizon,
    "percentile": percentile,
    "total_rows": total_rows,
    "valid_rows": valid_rows,
    "nan_rows": nan_rows,
    "long_count": long_count,
    "short_count": short_count,
    "no_signal_count": no_signal_count,
    "signal_count": signal_count,
    "long_ratio": long_count / valid_rows if valid_rows > 0 else np.nan,
    "short_ratio": short_count / valid_rows if valid_rows > 0 else np.nan,
    "no_signal_ratio": no_signal_count / valid_rows if valid_rows > 0 else np.nan,
    "signal_ratio": signal_count / valid_rows if valid_rows > 0 else np.nan,
    "long_share_over_signals": (
        long_count / signal_count if signal_count > 0 else np.nan
    ),
    "short_share_over_signals": (
        short_count / signal_count if signal_count > 0 else np.nan
    ),
}

row.update(group_info)
rows.append(row)

summary = pd.DataFrame(rows)

if "regime_name" in summary.columns:
    summary["regime_name"] = pd.Categorical(
        summary["regime_name"],
        categories=REGIME_ORDER,
        ordered=True
    )

sort_cols = ["horizon", "percentile"] + group_cols

summary = (
    summary
    .sort_values(sort_cols)
    .reset_index(drop=True)
)

return summary

print("Función de resumen por grupo lista.")

```

Función de resumen por grupo lista.

## Celda 3 — Distribución global de DIR por régimen

```
In [59]: # Celda 3 – Distribución global de DIR por régimen
# =====

df_dir_regime_distribution = summarize_dir_by_group(
    df=df_targets_dir,
    target_cols=dir_target_cols,
    group_cols=["regime_name"]
)

print("Distribución global de DIR por régimen intradiario")
print("=" * 80)

cols_show = [
    "target",
    "horizon",
    "percentile",
    "regime_name",
    "valid_rows",
    "nan_rows",
    "long_ratio",
    "short_ratio",
    "no_signal_ratio",
    "signal_ratio",
    "long_share_over_signals",
]

print(
    df_dir_regime_distribution[cols_show]
    .to_string(index=False, float_format=lambda x: f"{x:.2%}")
)
```

## Distribución global de DIR por régimen intradiario

| target      | horizon         | percentile   | regime_name             | valid_rows | nan_rows | long_ratio | s |
|-------------|-----------------|--------------|-------------------------|------------|----------|------------|---|
| hort_ratio  | no_signal_ratio | signal_ratio | long_share_over_signals |            |          |            |   |
| dir_p40_h30 | 30              | 40           | Overnight               | 355680     | 1482     | 31.87%     |   |
| 32.03%      | 36.11%          | 63.89%       |                         |            | 49.87%   |            |   |
| dir_p40_h30 | 30              | 40           | Pre-market              | 88920      | 0        | 31.27%     |   |
| 32.44%      | 36.29%          | 63.71%       |                         |            | 49.08%   |            |   |
| dir_p40_h30 | 30              | 40           | Opening                 | 88920      | 0        | 31.58%     |   |
| 32.49%      | 35.93%          | 64.07%       |                         |            | 49.29%   |            |   |
| dir_p40_h30 | 30              | 40           | Regular                 | 444600     | 0        | 31.44%     |   |
| 31.97%      | 36.59%          | 63.41%       |                         |            | 49.59%   |            |   |
| dir_p40_h30 | 30              | 40           | Closing                 | 1482       | 42978    | 31.04%     |   |
| 32.66%      | 36.30%          | 63.70%       |                         |            | 48.73%   |            |   |
| dir_p50_h30 | 30              | 50           | Overnight               | 355680     | 1482     | 26.80%     |   |
| 27.45%      | 45.76%          | 54.24%       |                         |            | 49.40%   |            |   |
| dir_p50_h30 | 30              | 50           | Pre-market              | 88920      | 0        | 26.33%     |   |
| 27.77%      | 45.90%          | 54.10%       |                         |            | 48.67%   |            |   |
| dir_p50_h30 | 30              | 50           | Opening                 | 88920      | 0        | 26.33%     |   |
| 28.04%      | 45.63%          | 54.37%       |                         |            | 48.42%   |            |   |
| dir_p50_h30 | 30              | 50           | Regular                 | 444600     | 0        | 26.40%     |   |
| 27.53%      | 46.06%          | 53.94%       |                         |            | 48.95%   |            |   |
| dir_p50_h30 | 30              | 50           | Closing                 | 1482       | 42978    | 25.91%     |   |
| 27.13%      | 46.96%          | 53.04%       |                         |            | 48.85%   |            |   |
| dir_p60_h30 | 30              | 60           | Overnight               | 355680     | 1482     | 21.97%     |   |
| 23.04%      | 54.99%          | 45.01%       |                         |            | 48.82%   |            |   |
| dir_p60_h30 | 30              | 60           | Pre-market              | 88920      | 0        | 21.36%     |   |
| 22.64%      | 56.00%          | 44.00%       |                         |            | 48.54%   |            |   |
| dir_p60_h30 | 30              | 60           | Opening                 | 88920      | 0        | 21.09%     |   |
| 23.48%      | 55.42%          | 44.58%       |                         |            | 47.32%   |            |   |
| dir_p60_h30 | 30              | 60           | Regular                 | 444600     | 0        | 21.13%     |   |
| 22.84%      | 56.03%          | 43.97%       |                         |            | 48.06%   |            |   |
| dir_p60_h30 | 30              | 60           | Closing                 | 1482       | 42978    | 21.46%     |   |
| 21.86%      | 56.68%          | 43.32%       |                         |            | 49.53%   |            |   |
| dir_p40_h60 | 60              | 40           | Overnight               | 355680     | 1482     | 31.91%     |   |
| 32.17%      | 35.92%          | 64.08%       |                         |            | 49.80%   |            |   |
| dir_p40_h60 | 60              | 40           | Pre-market              | 88920      | 0        | 31.64%     |   |
| 32.46%      | 35.90%          | 64.10%       |                         |            | 49.37%   |            |   |
| dir_p40_h60 | 60              | 40           | Opening                 | 88920      | 0        | 31.46%     |   |
| 32.79%      | 35.75%          | 64.25%       |                         |            | 48.96%   |            |   |
| dir_p40_h60 | 60              | 40           | Regular                 | 401622     | 42978    | 31.51%     |   |
| 31.98%      | 36.51%          | 63.49%       |                         |            | 49.63%   |            |   |
| dir_p40_h60 | 60              | 40           | Closing                 | 0          | 44460    | NaN        |   |
| NaN         | NaN             | NaN          |                         |            | NaN      |            |   |
| dir_p50_h60 | 60              | 50           | Overnight               | 355680     | 1482     | 26.74%     |   |
| 27.52%      | 45.74%          | 54.26%       |                         |            | 49.29%   |            |   |
| dir_p50_h60 | 60              | 50           | Pre-market              | 88920      | 0        | 26.70%     |   |
| 27.98%      | 45.32%          | 54.68%       |                         |            | 48.83%   |            |   |
| dir_p50_h60 | 60              | 50           | Opening                 | 88920      | 0        | 26.03%     |   |
| 28.39%      | 45.58%          | 54.42%       |                         |            | 47.84%   |            |   |
| dir_p50_h60 | 60              | 50           | Regular                 | 401622     | 42978    | 25.96%     |   |
| 27.58%      | 46.46%          | 53.54%       |                         |            | 48.49%   |            |   |
| dir_p50_h60 | 60              | 50           | Closing                 | 0          | 44460    | NaN        |   |
| NaN         | NaN             | NaN          |                         |            | NaN      |            |   |
| dir_p60_h60 | 60              | 60           | Overnight               | 355680     | 1482     | 21.42%     |   |
| 22.86%      | 55.72%          | 44.28%       |                         |            | 48.37%   |            |   |
| dir_p60_h60 | 60              | 60           | Pre-market              | 88920      | 0        | 22.13%     |   |
| 22.97%      | 54.90%          | 45.10%       |                         |            | 49.08%   |            |   |
| dir_p60_h60 | 60              | 60           | Opening                 | 88920      | 0        | 20.87%     |   |
| 23.92%      | 55.21%          | 44.79%       |                         |            | 46.60%   |            |   |

|             |        |        |            |        |       |        |
|-------------|--------|--------|------------|--------|-------|--------|
| dir_p60_h60 | 60     | 60     | Regular    | 401622 | 42978 | 20.78% |
| 23.13%      | 56.09% | 43.91% |            | 47.33% |       |        |
| dir_p60_h60 | 60     | 60     | Closing    | 0      | 44460 | NaN    |
| NaN         | NaN    | NaN    |            | NaN    |       |        |
| dir_p40_h90 | 90     | 40     | Overnight  | 355680 | 1482  | 31.58% |
| 32.29%      | 36.13% | 63.87% |            | 49.44% |       |        |
| dir_p40_h90 | 90     | 40     | Pre-market | 88920  | 0     | 31.91% |
| 32.42%      | 35.67% | 64.33% |            | 49.60% |       |        |
| dir_p40_h90 | 90     | 40     | Opening    | 88920  | 0     | 31.15% |
| 33.04%      | 35.81% | 64.19% |            | 48.52% |       |        |
| dir_p40_h90 | 90     | 40     | Regular    | 357162 | 87438 | 31.41% |
| 32.01%      | 36.58% | 63.42% |            | 49.53% |       |        |
| dir_p40_h90 | 90     | 40     | Closing    | 0      | 44460 | NaN    |
| NaN         | NaN    | NaN    |            | NaN    |       |        |
| dir_p50_h90 | 90     | 50     | Overnight  | 355680 | 1482  | 26.57% |
| 27.73%      | 45.70% | 54.30% |            | 48.93% |       |        |
| dir_p50_h90 | 90     | 50     | Pre-market | 88920  | 0     | 27.13% |
| 27.92%      | 44.95% | 55.05% |            | 49.27% |       |        |
| dir_p50_h90 | 90     | 50     | Opening    | 88920  | 0     | 25.79% |
| 28.82%      | 45.39% | 54.61% |            | 47.22% |       |        |
| dir_p50_h90 | 90     | 50     | Regular    | 357162 | 87438 | 26.05% |
| 27.70%      | 46.25% | 53.75% |            | 48.46% |       |        |
| dir_p50_h90 | 90     | 50     | Closing    | 0      | 44460 | NaN    |
| NaN         | NaN    | NaN    |            | NaN    |       |        |
| dir_p60_h90 | 90     | 60     | Overnight  | 355680 | 1482  | 21.39% |
| 22.79%      | 55.81% | 44.19% |            | 48.42% |       |        |
| dir_p60_h90 | 90     | 60     | Pre-market | 88920  | 0     | 22.02% |
| 23.41%      | 54.57% | 45.43% |            | 48.46% |       |        |
| dir_p60_h90 | 90     | 60     | Opening    | 88920  | 0     | 20.49% |
| 24.28%      | 55.23% | 44.77% |            | 45.76% |       |        |
| dir_p60_h90 | 90     | 60     | Regular    | 357162 | 87438 | 20.68% |
| 23.31%      | 56.01% | 43.99% |            | 47.02% |       |        |
| dir_p60_h90 | 90     | 60     | Closing    | 0      | 44460 | NaN    |
| NaN         | NaN    | NaN    |            | NaN    |       |        |

## Celda 4 — Matriz y heatmap de frecuencia operativa por régimen

```
In [60]: # Celda 4 – Frecuencia operativa por régimen y target
# =====

df_regime_signal_matrix = df_dir_regime_distribution.copy()

df_regime_signal_matrix["target_label"] = (
    "H" + df_regime_signal_matrix["horizon"].astype(str)
    + " p" + df_regime_signal_matrix["percentile"].astype(str)
)

signal_pivot_regime = df_regime_signal_matrix.pivot(
    index="regime_name",
    columns="target_label",
    values="signal_ratio"
)

print("Frecuencia operativa por régimen y target")
print("=" * 80)
print(signal_pivot_regime.to_string(float_format=lambda x: f"{x:.2%}"))
```

## Frecuencia operativa por régimen y target

| target_label | H30 p40 | H30 p50 | H30 p60 | H60 p40 | H60 p50 | H60 p60 | H90 p40 | H90 p50 | H90 p60 |
|--------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| regime_name  |         |         |         |         |         |         |         |         |         |
| Overnight    | 63.89%  | 54.24%  | 45.01%  | 64.08%  | 54.26%  | 44.28%  | 63.87%  | 54.30%  | 44.19%  |
| Pre-market   | 63.71%  | 54.10%  | 44.00%  | 64.10%  | 54.68%  | 45.10%  | 64.33%  | 55.05%  | 45.43%  |
| Opening      | 64.07%  | 54.37%  | 44.58%  | 64.25%  | 54.42%  | 44.79%  | 64.19%  | 54.61%  | 44.77%  |
| Regular      | 63.41%  | 53.94%  | 43.97%  | 63.49%  | 53.54%  | 43.91%  | 63.42%  | 53.75%  | 43.99%  |
| Closing      | 63.70%  | 53.04%  | 43.32%  | NaN     | NaN     | NaN     | NaN     | NaN     | NaN     |
| NaN          | NaN     |         |         |         |         |         |         |         |         |

## Celda 5 — Distribución de clases por régimen para p50

Usamos `p50` como variante central para evitar saturar el análisis con demasiados gráficos.

```
In [61]: # Celda 5 – Distribución de clases por régimen para p50
# Gráfico agrupado: 3 filas x 1 columna
# =====

selected_targets_regime = [
    "dir_p50_h30",
    "dir_p50_h60",
    "dir_p50_h90",
]

fig, axes = plt.subplots(
    nrows=3,
    ncols=1,
    figsize=(11, 12),
    sharex=True
)

for ax, target_col in zip(axes, selected_targets_regime):
    tmp = df_dir_regime_distribution[
        df_dir_regime_distribution["target"] == target_col
    ].copy()

    tmp = tmp.sort_values("regime_name")

    plot_data = tmp.set_index("regime_name")[
        ["short_ratio", "no_signal_ratio", "long_ratio"]
    ]

    plot_data.plot(
        kind="bar",
        stacked=True,
        ax=ax,
        legend=False
    )

    ax.set_title(f"Distribución de clases por régimen - {target_col}")
    ax.set_xlabel("")
```



```

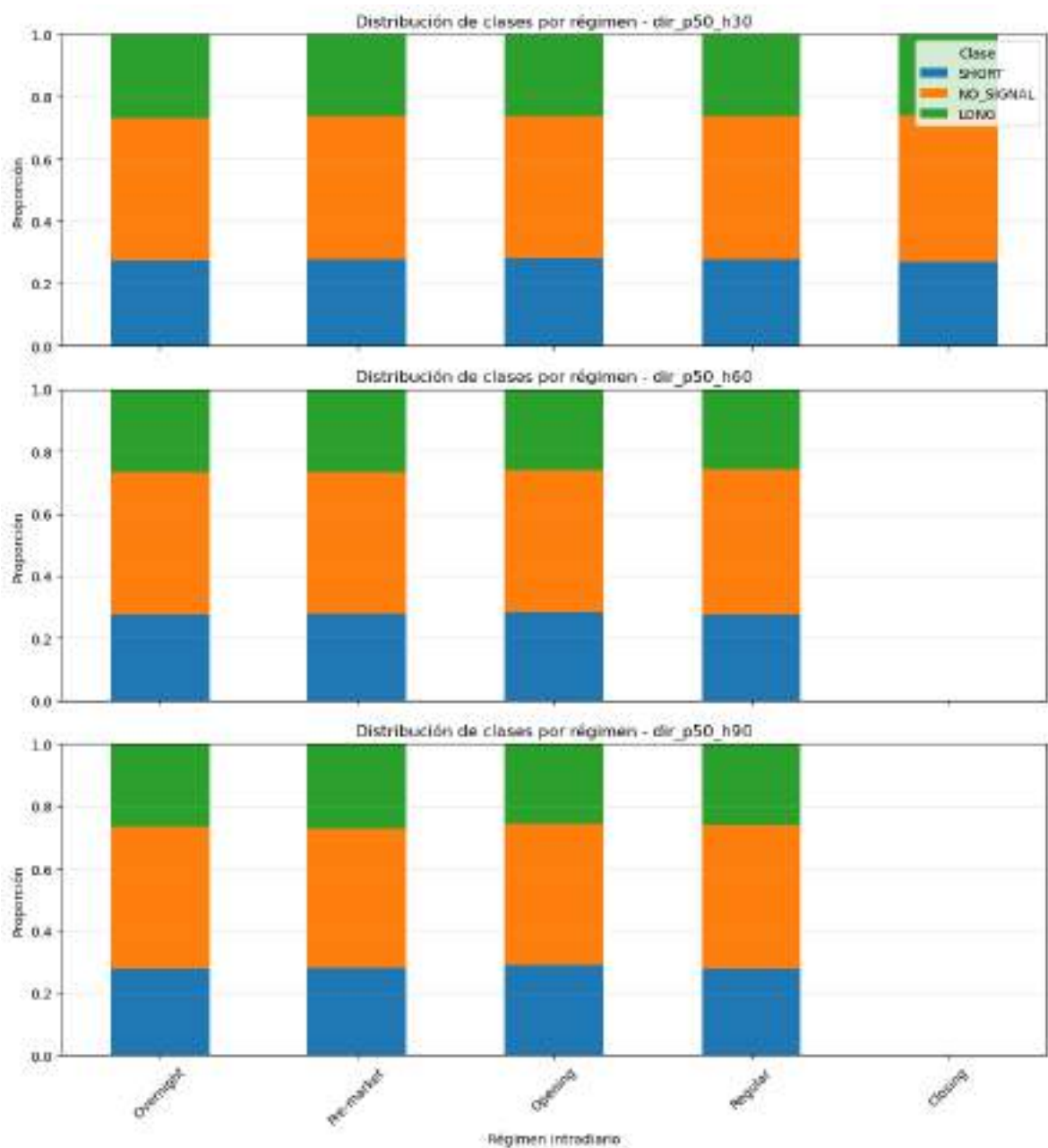
ax.set_ylabel("Proporción")
ax.set_ylim(0, 1)
ax.grid(axis="y", alpha=0.3)

axes[0].legend(
    ["SHORT", "NO_SIGNAL", "LONG"],
    title="Clase",
    loc="upper right"
)

axes[-1].set_xlabel("Régimen intradiario")

plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

```



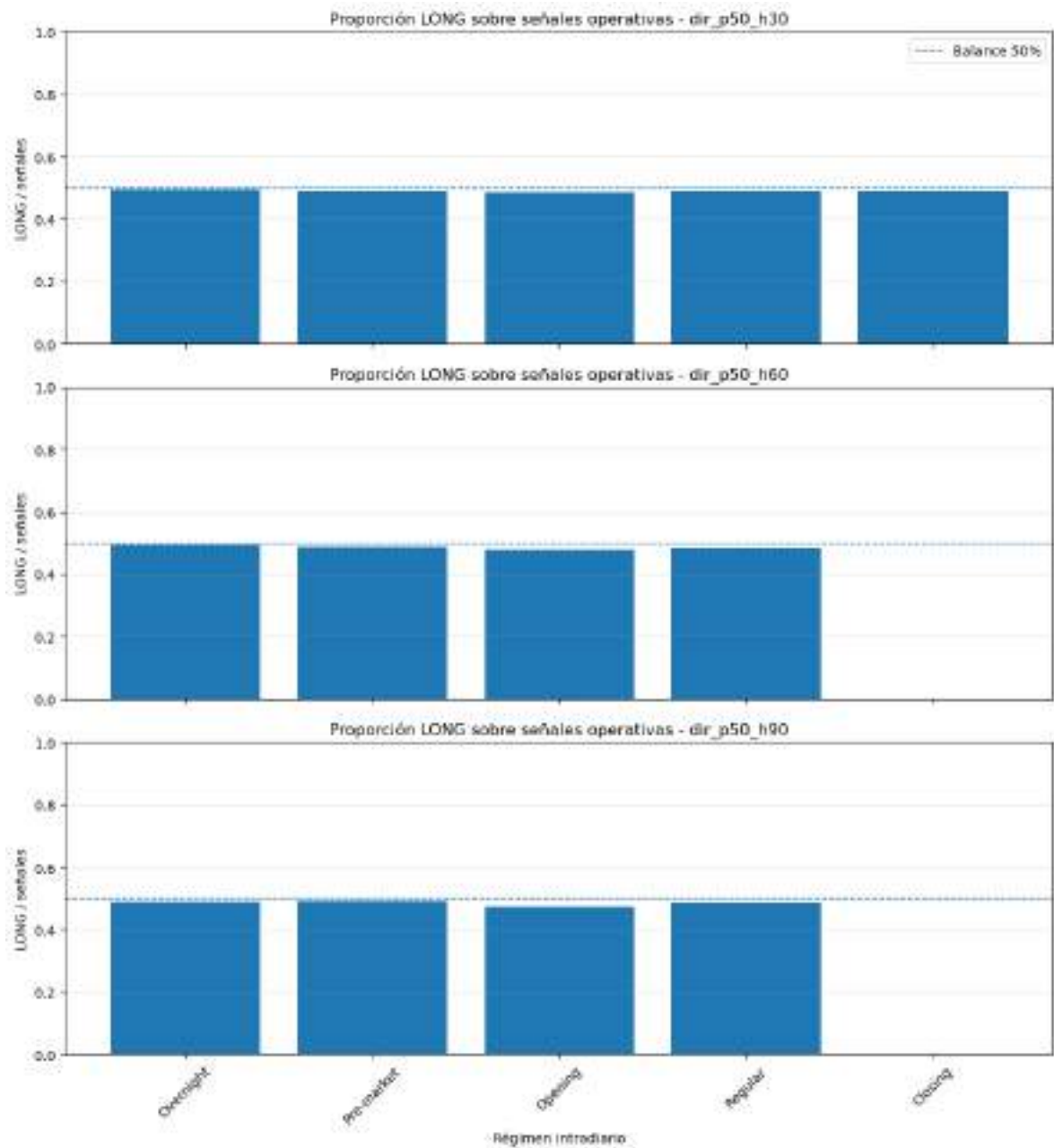
## Celda 6 — Balance LONG/SHORT por régimen para p50

```

In [62]: # Celda 6 – Balance LONG/SHORT por régimen para p50
          # Gráfico agrupado: 3 filas x 1 columna

```

```
# =====  
  
fig, axes = plt.subplots(  
    nrows=3,  
    ncols=1,  
    figsize=(11, 12),  
    sharex=True  
)  
  
for ax, target_col in zip(axes, selected_targets_regime):  
    tmp = df_dir_regime_distribution[  
        df_dir_regime_distribution["target"] == target_col  
    ].copy()  
  
    tmp = tmp.sort_values("regime_name")  
  
    ax.bar(  
        tmp["regime_name"].astype(str),  
        tmp["long_share_over_signals"]  
    )  
  
    ax.axhline(  
        0.50,  
        linestyle="--",  
        linewidth=1,  
        label="Balance 50%"  
    )  
  
    ax.set_title(f"Proporción LONG sobre señales operativas - {target_col}")  
    ax.set_xlabel("")  
    ax.set_ylabel("LONG / señales")  
    ax.set_ylim(0, 1)  
    ax.grid(axis="y", alpha=0.3)  
  
axes[0].legend(loc="upper right")  
axes[-1].set_xlabel("Régimen intradiario")  
  
plt.xticks(rotation=45)  
plt.tight_layout()  
plt.show()
```



## Celda 7 — Régimen + split temporal

Esta es la parte nueva importante. Permite ver si el aumento de frecuencia en 2025-2026 ocurre en todos los regímenes.

```
In [63]: # Celda 7 – Distribución por régimen y split temporal
# =====

df_dir_regime_split_distribution = summarize_dir_by_group(
    df=df_targets_dir,
    target_cols=dir_target_cols,
    group_cols=["regime_name", "target_analysis_split"]
)

print("Distribución de DIR por régimen y split temporal")
print("=" * 80)

cols_show = [
    "target",
```

```
    "horizon",
    "percentile",
    "regime_name",
    "target_analysis_split",
    "valid_rows",
    "nan_rows",
    "signal_ratio",
    "no_signal_ratio",
    "long_share_over_signals",
]

print(
    df_dir_regime_split_distribution[cols_show]
    .to_string(index=False, float_format=lambda x: f"{x:.2%}")
)
```

## Distribución de DIR por régimen y split temporal

| target      | horizon      | percentile      | regime_name             | target_analysis_split | valid_rows | n |
|-------------|--------------|-----------------|-------------------------|-----------------------|------------|---|
| an_rows     | signal_ratio | no_signal_ratio | long_share_over_signals |                       |            |   |
| dir_p40_h30 | 30           | 40              | Overnight               | development_2020_2024 | 284400     |   |
| 1185        | 59.98%       | 40.02%          |                         | 49.80%                |            |   |
| dir_p40_h30 | 30           | 40              | Overnight               | final_test_2025_2026  | 71280      |   |
| 297         | 79.50%       | 20.50%          |                         | 50.10%                |            |   |
| dir_p40_h30 | 30           | 40              | Pre-market              | development_2020_2024 | 71100      |   |
| 0           | 60.37%       | 39.63%          |                         | 49.20%                |            |   |
| dir_p40_h30 | 30           | 40              | Pre-market              | final_test_2025_2026  | 17820      |   |
| 0           | 77.06%       | 22.94%          |                         | 48.70%                |            |   |
| dir_p40_h30 | 30           | 40              | Opening                 | development_2020_2024 | 71100      |   |
| 0           | 60.21%       | 39.79%          |                         | 48.71%                |            |   |
| dir_p40_h30 | 30           | 40              | Opening                 | final_test_2025_2026  | 17820      |   |
| 0           | 79.47%       | 20.53%          |                         | 51.07%                |            |   |
| dir_p40_h30 | 30           | 40              | Regular                 | development_2020_2024 | 355500     |   |
| 0           | 60.33%       | 39.67%          |                         | 49.73%                |            |   |
| dir_p40_h30 | 30           | 40              | Regular                 | final_test_2025_2026  | 89100      |   |
| 0           | 75.69%       | 24.31%          |                         | 49.14%                |            |   |
| dir_p40_h30 | 30           | 40              | Closing                 | development_2020_2024 | 1185       |   |
| 34365       | 60.42%       | 39.58%          |                         | 48.32%                |            |   |
| dir_p40_h30 | 30           | 40              | Closing                 | final_test_2025_2026  | 297        |   |
| 8613        | 76.77%       | 23.23%          |                         | 50.00%                |            |   |
| dir_p50_h30 | 30           | 50              | Overnight               | development_2020_2024 | 284400     |   |
| 1185        | 49.98%       | 50.02%          |                         | 49.27%                |            |   |
| dir_p50_h30 | 30           | 50              | Overnight               | final_test_2025_2026  | 71280      |   |
| 297         | 71.27%       | 28.73%          |                         | 49.76%                |            |   |
| dir_p50_h30 | 30           | 50              | Pre-market              | development_2020_2024 | 71100      |   |
| 0           | 50.38%       | 49.62%          |                         | 48.73%                |            |   |
| dir_p50_h30 | 30           | 50              | Pre-market              | final_test_2025_2026  | 17820      |   |
| 0           | 68.97%       | 31.03%          |                         | 48.50%                |            |   |
| dir_p50_h30 | 30           | 50              | Opening                 | development_2020_2024 | 71100      |   |
| 0           | 50.11%       | 49.89%          |                         | 47.66%                |            |   |
| dir_p50_h30 | 30           | 50              | Opening                 | final_test_2025_2026  | 17820      |   |
| 0           | 71.38%       | 28.62%          |                         | 50.57%                |            |   |
| dir_p50_h30 | 30           | 50              | Regular                 | development_2020_2024 | 355500     |   |
| 0           | 50.45%       | 49.55%          |                         | 48.99%                |            |   |
| dir_p50_h30 | 30           | 50              | Regular                 | final_test_2025_2026  | 89100      |   |
| 0           | 67.85%       | 32.15%          |                         | 48.84%                |            |   |
| dir_p50_h30 | 30           | 50              | Closing                 | development_2020_2024 | 1185       |   |
| 34365       | 50.21%       | 49.79%          |                         | 47.73%                |            |   |
| dir_p50_h30 | 30           | 50              | Closing                 | final_test_2025_2026  | 297        |   |
| 8613        | 64.31%       | 35.69%          |                         | 52.36%                |            |   |
| dir_p60_h30 | 30           | 60              | Overnight               | development_2020_2024 | 284400     |   |
| 1185        | 40.62%       | 59.38%          |                         | 48.62%                |            |   |
| dir_p60_h30 | 30           | 60              | Overnight               | final_test_2025_2026  | 71280      |   |
| 297         | 62.51%       | 37.49%          |                         | 49.33%                |            |   |
| dir_p60_h30 | 30           | 60              | Pre-market              | development_2020_2024 | 71100      |   |
| 0           | 40.09%       | 59.91%          |                         | 48.71%                |            |   |
| dir_p60_h30 | 30           | 60              | Pre-market              | final_test_2025_2026  | 17820      |   |
| 0           | 59.60%       | 40.40%          |                         | 48.09%                |            |   |
| dir_p60_h30 | 30           | 60              | Opening                 | development_2020_2024 | 71100      |   |
| 0           | 39.99%       | 60.01%          |                         | 46.25%                |            |   |
| dir_p60_h30 | 30           | 60              | Opening                 | final_test_2025_2026  | 17820      |   |
| 0           | 62.87%       | 37.13%          |                         | 50.03%                |            |   |
| dir_p60_h30 | 30           | 60              | Regular                 | development_2020_2024 | 355500     |   |
| 0           | 40.18%       | 59.82%          |                         | 47.94%                |            |   |
| dir_p60_h30 | 30           | 60              | Regular                 | final_test_2025_2026  | 89100      |   |
| 0           | 59.09%       | 40.91%          |                         | 48.37%                |            |   |

|             |        |        |            |                       |        |
|-------------|--------|--------|------------|-----------------------|--------|
| dir_p60_h30 | 30     | 60     | Closing    | development_2020_2024 | 1185   |
| 34365       | 40.25% | 59.75% |            | 47.80%                |        |
| dir_p60_h30 | 30     | 60     | Closing    | final_test_2025_2026  | 297    |
| 8613        | 55.56% | 44.44% |            | 54.55%                |        |
| dir_p40_h60 | 60     | 40     | Overnight  | development_2020_2024 | 284400 |
| 1185        | 60.13% | 39.87% |            | 49.70%                |        |
| dir_p40_h60 | 60     | 40     | Overnight  | final_test_2025_2026  | 71280  |
| 297         | 79.84% | 20.16% |            | 50.09%                |        |
| dir_p40_h60 | 60     | 40     | Pre-market | development_2020_2024 | 71100  |
| 0           | 60.11% | 39.89% |            | 49.40%                |        |
| dir_p40_h60 | 60     | 40     | Pre-market | final_test_2025_2026  | 17820  |
| 0           | 80.01% | 19.99% |            | 49.26%                |        |
| dir_p40_h60 | 60     | 40     | Opening    | development_2020_2024 | 71100  |
| 0           | 60.20% | 39.80% |            | 48.63%                |        |
| dir_p40_h60 | 60     | 40     | Opening    | final_test_2025_2026  | 17820  |
| 0           | 80.38% | 19.62% |            | 49.95%                |        |
| dir_p40_h60 | 60     | 40     | Regular    | development_2020_2024 | 321135 |
| 34365       | 60.34% | 39.66% |            | 49.79%                |        |
| dir_p40_h60 | 60     | 40     | Regular    | final_test_2025_2026  | 80487  |
| 8613        | 76.05% | 23.95% |            | 49.13%                |        |
| dir_p40_h60 | 60     | 40     | Closing    | development_2020_2024 | 0      |
| 35550       | NaN    | NaN    |            | NaN                   |        |
| dir_p40_h60 | 60     | 40     | Closing    | final_test_2025_2026  | 0      |
| 8910        | NaN    | NaN    |            | NaN                   |        |
| dir_p50_h60 | 60     | 50     | Overnight  | development_2020_2024 | 284400 |
| 1185        | 50.06% | 49.94% |            | 49.17%                |        |
| dir_p50_h60 | 60     | 50     | Overnight  | final_test_2025_2026  | 71280  |
| 297         | 71.00% | 29.00% |            | 49.61%                |        |
| dir_p50_h60 | 60     | 50     | Pre-market | development_2020_2024 | 71100  |
| 0           | 50.10% | 49.90% |            | 48.79%                |        |
| dir_p50_h60 | 60     | 50     | Pre-market | final_test_2025_2026  | 17820  |
| 0           | 72.93% | 27.07% |            | 48.93%                |        |
| dir_p50_h60 | 60     | 50     | Opening    | development_2020_2024 | 71100  |
| 0           | 50.00% | 50.00% |            | 47.30%                |        |
| dir_p50_h60 | 60     | 50     | Opening    | final_test_2025_2026  | 17820  |
| 0           | 72.05% | 27.95% |            | 49.33%                |        |
| dir_p50_h60 | 60     | 50     | Regular    | development_2020_2024 | 321135 |
| 34365       | 49.99% | 50.01% |            | 48.61%                |        |
| dir_p50_h60 | 60     | 50     | Regular    | final_test_2025_2026  | 80487  |
| 8613        | 67.69% | 32.31% |            | 48.14%                |        |
| dir_p50_h60 | 60     | 50     | Closing    | development_2020_2024 | 0      |
| 35550       | NaN    | NaN    |            | NaN                   |        |
| dir_p50_h60 | 60     | 50     | Closing    | final_test_2025_2026  | 0      |
| 8910        | NaN    | NaN    |            | NaN                   |        |
| dir_p60_h60 | 60     | 60     | Overnight  | development_2020_2024 | 284400 |
| 1185        | 40.05% | 59.95% |            | 48.20%                |        |
| dir_p60_h60 | 60     | 60     | Overnight  | final_test_2025_2026  | 71280  |
| 297         | 61.19% | 38.81% |            | 48.83%                |        |
| dir_p60_h60 | 60     | 60     | Pre-market | development_2020_2024 | 71100  |
| 0           | 40.16% | 59.84% |            | 48.89%                |        |
| dir_p60_h60 | 60     | 60     | Pre-market | final_test_2025_2026  | 17820  |
| 0           | 64.79% | 35.21% |            | 49.55%                |        |
| dir_p60_h60 | 60     | 60     | Opening    | development_2020_2024 | 71100  |
| 0           | 40.18% | 59.82% |            | 45.88%                |        |
| dir_p60_h60 | 60     | 60     | Opening    | final_test_2025_2026  | 17820  |
| 0           | 63.18% | 36.82% |            | 48.45%                |        |
| dir_p60_h60 | 60     | 60     | Regular    | development_2020_2024 | 321135 |
| 34365       | 40.09% | 59.91% |            | 47.38%                |        |
| dir_p60_h60 | 60     | 60     | Regular    | final_test_2025_2026  | 80487  |
| 8613        | 59.15% | 40.85% |            | 47.18%                |        |

|             |        |        |            |                       |        |
|-------------|--------|--------|------------|-----------------------|--------|
| dir_p60_h60 | 60     | 60     | Closing    | development_2020_2024 | 0      |
| 35550       | NaN    | NaN    |            | NaN                   |        |
| dir_p60_h60 | 60     | 60     | Closing    | final_test_2025_2026  | 0      |
| 8910        | NaN    | NaN    |            | NaN                   |        |
| dir_p40_h90 | 90     | 40     | Overnight  | development_2020_2024 | 284400 |
| 1185        | 60.14% | 39.86% |            | 49.38%                |        |
| dir_p40_h90 | 90     | 40     | Overnight  | final_test_2025_2026  | 71280  |
| 297         | 78.76% | 21.24% |            | 49.64%                |        |
| dir_p40_h90 | 90     | 40     | Pre-market | development_2020_2024 | 71100  |
| 0           | 60.04% | 39.96% |            | 49.11%                |        |
| dir_p40_h90 | 90     | 40     | Pre-market | final_test_2025_2026  | 17820  |
| 0           | 81.45% | 18.55% |            | 51.05%                |        |
| dir_p40_h90 | 90     | 40     | Opening    | development_2020_2024 | 71100  |
| 0           | 60.18% | 39.82% |            | 48.22%                |        |
| dir_p40_h90 | 90     | 40     | Opening    | final_test_2025_2026  | 17820  |
| 0           | 80.19% | 19.81% |            | 49.44%                |        |
| dir_p40_h90 | 90     | 40     | Regular    | development_2020_2024 | 285585 |
| 69915       | 60.21% | 39.79% |            | 49.64%                |        |
| dir_p40_h90 | 90     | 40     | Regular    | final_test_2025_2026  | 71577  |
| 17523       | 76.20% | 23.80% |            | 49.18%                |        |
| dir_p40_h90 | 90     | 40     | Closing    | development_2020_2024 | 0      |
| 35550       | NaN    | NaN    |            | NaN                   |        |
| dir_p40_h90 | 90     | 40     | Closing    | final_test_2025_2026  | 0      |
| 8910        | NaN    | NaN    |            | NaN                   |        |
| dir_p50_h90 | 90     | 50     | Overnight  | development_2020_2024 | 284400 |
| 1185        | 50.35% | 49.65% |            | 48.78%                |        |
| dir_p50_h90 | 90     | 50     | Overnight  | final_test_2025_2026  | 71280  |
| 297         | 70.05% | 29.95% |            | 49.37%                |        |
| dir_p50_h90 | 90     | 50     | Pre-market | development_2020_2024 | 71100  |
| 0           | 50.09% | 49.91% |            | 48.53%                |        |
| dir_p50_h90 | 90     | 50     | Pre-market | final_test_2025_2026  | 17820  |
| 0           | 74.84% | 25.16% |            | 51.26%                |        |
| dir_p50_h90 | 90     | 50     | Opening    | development_2020_2024 | 71100  |
| 0           | 50.20% | 49.80% |            | 46.76%                |        |
| dir_p50_h90 | 90     | 50     | Opening    | final_test_2025_2026  | 17820  |
| 0           | 72.17% | 27.83% |            | 48.52%                |        |
| dir_p50_h90 | 90     | 50     | Regular    | development_2020_2024 | 285585 |
| 69915       | 50.15% | 49.85% |            | 48.51%                |        |
| dir_p50_h90 | 90     | 50     | Regular    | final_test_2025_2026  | 71577  |
| 17523       | 68.11% | 31.89% |            | 48.32%                |        |
| dir_p50_h90 | 90     | 50     | Closing    | development_2020_2024 | 0      |
| 35550       | NaN    | NaN    |            | NaN                   |        |
| dir_p50_h90 | 90     | 50     | Closing    | final_test_2025_2026  | 0      |
| 8910        | NaN    | NaN    |            | NaN                   |        |
| dir_p60_h90 | 90     | 60     | Overnight  | development_2020_2024 | 284400 |
| 1185        | 40.05% | 59.95% |            | 48.26%                |        |
| dir_p60_h90 | 90     | 60     | Overnight  | final_test_2025_2026  | 71280  |
| 297         | 60.70% | 39.30% |            | 48.84%                |        |
| dir_p60_h90 | 90     | 60     | Pre-market | development_2020_2024 | 71100  |
| 0           | 40.07% | 59.93% |            | 47.42%                |        |
| dir_p60_h90 | 90     | 60     | Pre-market | final_test_2025_2026  | 17820  |
| 0           | 66.81% | 33.19% |            | 50.96%                |        |
| dir_p60_h90 | 90     | 60     | Opening    | development_2020_2024 | 71100  |
| 0           | 40.13% | 59.87% |            | 45.02%                |        |
| dir_p60_h90 | 90     | 60     | Opening    | final_test_2025_2026  | 17820  |
| 0           | 63.30% | 36.70% |            | 47.64%                |        |
| dir_p60_h90 | 90     | 60     | Regular    | development_2020_2024 | 285585 |
| 69915       | 40.11% | 59.89% |            | 47.02%                |        |
| dir_p60_h90 | 90     | 60     | Regular    | final_test_2025_2026  | 71577  |
| 17523       | 59.48% | 40.52% |            | 47.00%                |        |

|             |     |     |                               |   |
|-------------|-----|-----|-------------------------------|---|
| dir_p60_h90 | 90  | 60  | Closing development_2020_2024 | 0 |
| 35550       | NaN | NaN | NaN                           |   |
| dir_p60_h90 | 90  | 60  | Closing final_test_2025_2026  | 0 |
| 8910        | NaN | NaN | NaN                           |   |

## Celda 8 — Comparación development vs final test por régimen para p50

```
In [64]: # Celda 8 - Comparación régimen + split para p50
# =====

df_p50_regime_split = df_dir_regime_split_distribution[
    df_dir_regime_split_distribution["target"].isin(selected_targets_regime)
].copy()

for target_col in selected_targets_regime:
    tmp = df_p50_regime_split[
        df_p50_regime_split["target"] == target_col
    ].copy()

    pivot_tmp = tmp.pivot(
        index="regime_name",
        columns="target_analysis_split",
        values="signal_ratio"
    )

    if (
        "development_2020_2024" in pivot_tmp.columns
        and "final_test_2025_2026" in pivot_tmp.columns
    ):
        pivot_tmp["difference_final_minus_dev"] = (
            pivot_tmp["final_test_2025_2026"]
            - pivot_tmp["development_2020_2024"]
        )

    print(f"\nFrecuencia operativa por régimen y split - {target_col}")
    print("=" * 80)
    print(pivot_tmp.to_string(float_format=lambda x: f"{x:.2%}"))
```



## Frecuencia operativa por régimen y split - dir\_p50\_h30

| target_analysis_split | development_2020_2024 | final_test_2025_2026 | difference_fi |
|-----------------------|-----------------------|----------------------|---------------|
| regime_name           |                       |                      |               |
| Overnight             | 49.98%                | 71.27%               |               |
| 21.29%                |                       |                      |               |
| Pre-market            | 50.38%                | 68.97%               |               |
| 18.60%                |                       |                      |               |
| Opening               | 50.11%                | 71.38%               |               |
| 21.27%                |                       |                      |               |
| Regular               | 50.45%                | 67.85%               |               |
| 17.40%                |                       |                      |               |
| Closing               | 50.21%                | 64.31%               |               |
| 14.10%                |                       |                      |               |

## Frecuencia operativa por régimen y split - dir\_p50\_h60

| target_analysis_split | development_2020_2024 | final_test_2025_2026 | difference_fi |
|-----------------------|-----------------------|----------------------|---------------|
| regime_name           |                       |                      |               |
| Overnight             | 50.06%                | 71.00%               |               |
| 20.93%                |                       |                      |               |
| Pre-market            | 50.10%                | 72.93%               |               |
| 22.83%                |                       |                      |               |
| Opening               | 50.00%                | 72.05%               |               |
| 22.05%                |                       |                      |               |
| Regular               | 49.99%                | 67.69%               |               |
| 17.70%                |                       |                      |               |
| Closing               | NaN                   | NaN                  |               |
| NaN                   |                       |                      |               |

## Frecuencia operativa por régimen y split - dir\_p50\_h90

| target_analysis_split | development_2020_2024 | final_test_2025_2026 | difference_fi |
|-----------------------|-----------------------|----------------------|---------------|
| regime_name           |                       |                      |               |
| Overnight             | 50.35%                | 70.05%               |               |
| 19.69%                |                       |                      |               |
| Pre-market            | 50.09%                | 74.84%               |               |
| 24.75%                |                       |                      |               |
| Opening               | 50.20%                | 72.17%               |               |
| 21.97%                |                       |                      |               |
| Regular               | 50.15%                | 68.11%               |               |
| 17.96%                |                       |                      |               |
| Closing               | NaN                   | NaN                  |               |
| NaN                   |                       |                      |               |

## Celda 9 — Gráfico development vs final test por régimen para p50

```
In [65]: # Celda 9 – Gráfico régimen + split para p50
# Gráfico agrupado: 3 filas x 1 columna
# =====
```

```
fig, axes = plt.subplots(
    nrows=3,
    ncols=1,
    figsize=(12, 12),
```

```
sharex=True
)

for ax, target_col in zip(axes, selected_targets_regime):
    tmp = df_p50_regime_split[
        df_p50_regime_split["target"] == target_col
    ].copy()

    pivot_tmp = tmp.pivot(
        index="regime_name",
        columns="target_analysis_split",
        values="signal_ratio"
    )

    pivot_tmp = pivot_tmp.reindex(REGIME_ORDER)

    plot_cols = [
        col for col in [
            "development_2020_2024",
            "final_test_2025_2026"
        ]
        if col in pivot_tmp.columns
    ]

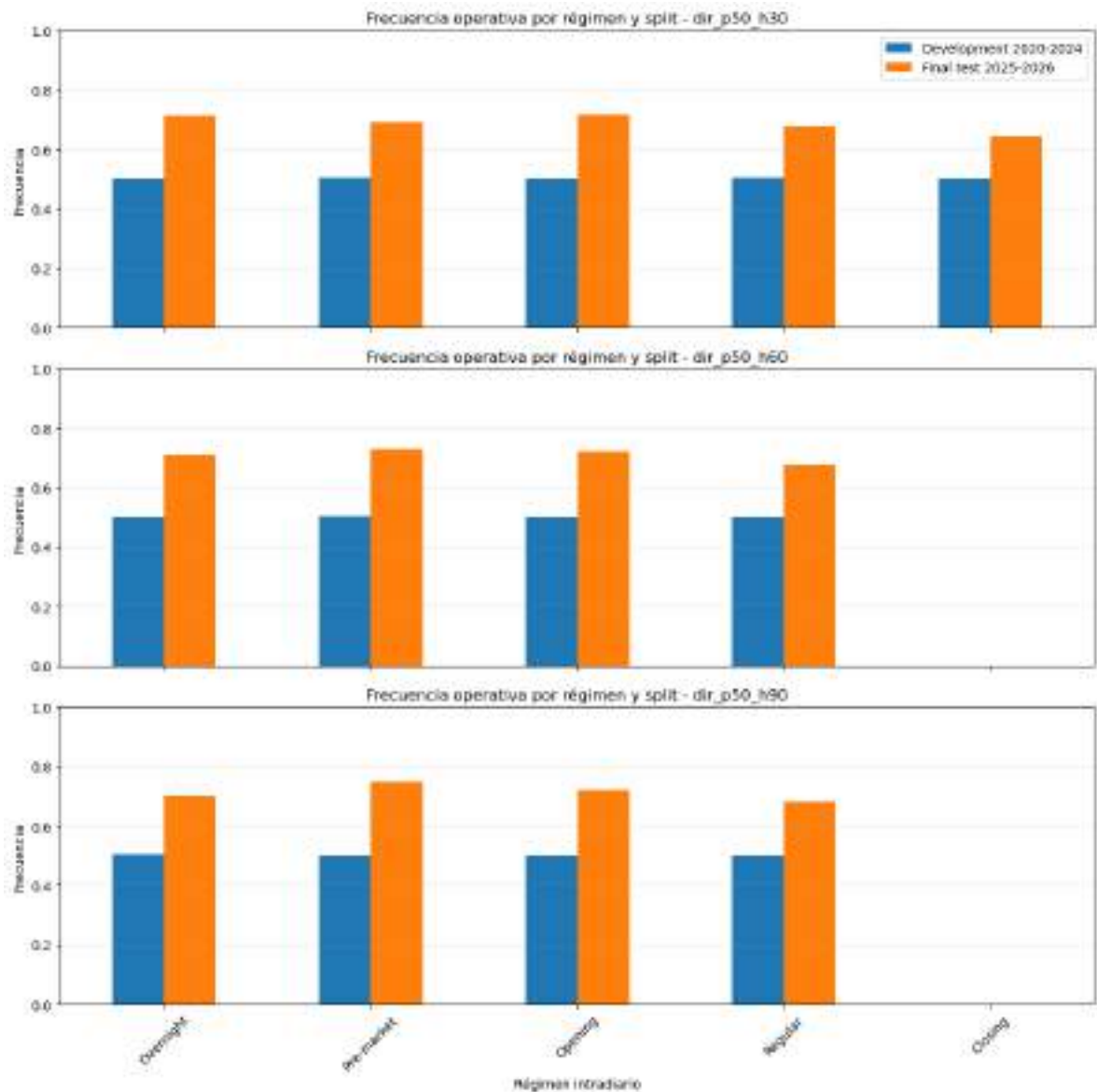
    pivot_tmp[plot_cols].plot(
        kind="bar",
        ax=ax,
        legend=False
    )

    ax.set_title(f"Frecuencia operativa por régimen y split - {target_col}")
    ax.set_xlabel("")
    ax.set_ylabel("Frecuencia")
    ax.set_ylim(0, 1)
    ax.grid(axis="y", alpha=0.3)

axes[0].legend(
    ["Development 2020-2024", "Final test 2025-2026"],
    loc="upper right"
)

axes[-1].set_xlabel("Régimen intradiario")

plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



## Celda 10 — Revisión especial de Closing

```
In [66]: # Celda 10 – Revisión especial del régimen Closing
# =====

df_closing_check = df_dir_regime_distribution[
    df_dir_regime_distribution["regime_name"] == "Closing"
].copy()

print("Revisión especial del régimen Closing")
print("=" * 80)

cols_show = [
    "target",
    "horizon",
    "percentile",
    "total_rows",
    "valid_rows",
    "nan_rows",
    "signal_ratio",
    "no_signal_ratio",
]
```

```
print(
    df_closing_check[cols_show]
    .to_string(index=False, float_format=lambda x: f"{x:.2%}")
)

print("\nLectura:")
print("- Closing puede tener pocas observaciones válidas porque está cerca del c
print("- Para H60 y H90 es esperable que Closing no tenga ventana futura suficiente
print("- Por este motivo, Closing debe interpretarse con cuidado y probablemente
```

#### Revisión especial del régimen Closing

```
=====
target horizon percentile total_rows valid_rows nan_rows signal_ratio
no_signal_ratio
dir_p40_h30 30 40 44460 1482 42978 63.70%
36.30%
dir_p50_h30 30 50 44460 1482 42978 53.04%
46.96%
dir_p60_h30 30 60 44460 1482 42978 43.32%
56.68%
dir_p40_h60 60 40 44460 0 44460 NaN
NaN
dir_p50_h60 60 50 44460 0 44460 NaN
NaN
dir_p60_h60 60 60 44460 0 44460 NaN
NaN
dir_p40_h90 90 40 44460 0 44460 NaN
NaN
dir_p50_h90 90 50 44460 0 44460 NaN
NaN
dir_p60_h90 90 60 44460 0 44460 NaN
NaN
```

#### Lectura:

- Closing puede tener pocas observaciones válidas porque está cerca del cierre de 1 día.
- Para H60 y H90 es esperable que Closing no tenga ventana futura suficiente.
- Por este motivo, Closing debe interpretarse con cuidado y probablemente limitarse a H30.

## Celda 11 — Resumen automático del punto 3.7

```
In [67]: # Celda 11 - Resumen automático por régimen
# =====

print("Resumen por régimen intradiario - DIR")
print("=" * 80)

for regime in REGIME_ORDER:
    tmp = df_dir_regime_distribution[
        df_dir_regime_distribution["regime_name"] == regime
    ].copy()

    avg_signal = tmp["signal_ratio"].mean()
    avg_no_signal = tmp["no_signal_ratio"].mean()
    avg_long_share = tmp["long_share_over_signals"].mean()
    total_valid = tmp["valid_rows"].sum()

    print(f"\nRégimen: {regime}")
```

```

print(f"  Frecuencia operativa promedio: {avg_signal:.2%}")
print(f"  NO_SIGNAL promedio:                {avg_no_signal:.2%}")
print(f"  LONG sobre señales promedio:         {avg_long_share:.2%}")
print(f"  Filas válidas acumuladas:             {total_valid:,}")

print("\nLectura general:")
print("- Se evalúa si DIR genera más o menos señales según el momento del día.")
print("- Se revisa si el balance LONG/SHORT se mantiene cerca de 50%.")
print("- Se compara development vs final test para ver si el aumento de señales")
print("- Closing se analiza por separado porque puede no tener ventana futura vá

```

#### Resumen por régimen intradiario - DIR

=====

##### Régimen: Overnight

Frecuencia operativa promedio: 54.24%  
 NO\_SIGNAL promedio: 45.76%  
 LONG sobre señales promedio: 49.15%  
 Filas válidas acumuladas: 3,201,120

##### Régimen: Pre-market

Frecuencia operativa promedio: 54.50%  
 NO\_SIGNAL promedio: 45.50%  
 LONG sobre señales promedio: 48.99%  
 Filas válidas acumuladas: 800,280

##### Régimen: Opening

Frecuencia operativa promedio: 54.45%  
 NO\_SIGNAL promedio: 45.55%  
 LONG sobre señales promedio: 47.77%  
 Filas válidas acumuladas: 800,280

##### Régimen: Regular

Frecuencia operativa promedio: 53.71%  
 NO\_SIGNAL promedio: 46.29%  
 LONG sobre señales promedio: 48.56%  
 Filas válidas acumuladas: 3,610,152

##### Régimen: Closing

Frecuencia operativa promedio: 53.35%  
 NO\_SIGNAL promedio: 46.65%  
 LONG sobre señales promedio: 49.04%  
 Filas válidas acumuladas: 4,446

##### Lectura general:

- Se evalúa si DIR genera más o menos señales según el momento del día.
- Se revisa si el balance LONG/SHORT se mantiene cerca de 50%.
- Se compara development vs final test para ver si el aumento de señales ocurre en todos los regímenes.
- Closing se analiza por separado porque puede no tener ventana futura válida para H60 y H90.

## Observaciones preliminares - Análisis por régimen intradiario

1. **A nivel global, DIR es bastante estable por régimen.** Las frecuencias operativas son muy parecidas entre Overnight , Pre-market , Opening y Regular .

2. **No hay un régimen claramente dominante.** Para `p50`, la frecuencia ronda aproximadamente el **54%** en casi todos los regímenes. Esto indica que el target no depende excesivamente del momento del día.
3. **Pre-market y Opening son levemente más activos.** Tienen una frecuencia operativa algo superior, especialmente en `H60` y `H90`, pero la diferencia no es muy grande.
4. **Regular es apenas menos activo.** Presenta frecuencias un poco más bajas, aunque sigue estando muy cerca del resto de regímenes.
5. **El balance LONG/SHORT se mantiene razonable.** La proporción LONG sobre señales operativas se mantiene cerca del **50%**, con una leve inclinación hacia `SHORT`.
6. **Opening muestra algo más de sesgo hacia SHORT.** Especialmente en `p60` y horizontes largos, la proporción LONG baja un poco más que en otros regímenes.
7. **El aumento en final test ocurre en todos los regímenes.** En `2025-2026`, la frecuencia operativa sube claramente frente a `2020-2024` en todos los regímenes. Por lo tanto, el cambio no está concentrado en un solo momento del día.
8. **Pre-market y Opening aumentan bastante en final test.** En `p50`, varios casos suben más de **20 puntos porcentuales**, confirmando que estos regímenes fueron mucho más activos fuera de muestra.
9. **Closing debe tratarse aparte.** Para `H60` y `H90` no tiene observaciones válidas, porque no hay suficiente ventana futura antes del cierre del día. Para `H30` sí existe, pero con pocas filas válidas.
10. **Conclusión preliminar:** `DIR` se comporta de forma bastante consistente por régimen intradiario, pero el efecto temporal visto en `2025-2026` impacta a todos los regímenes. El régimen no parece ser la principal fuente de inestabilidad; la mayor diferencia viene del cambio de contexto temporal.

## 3.8. Análisis por contrato y contexto — DIR

Además del análisis por año y régimen, se evaluará el target por contrato y por combinaciones de contexto.

Las dimensiones principales serán:

```
contract
contract_code
year + regime
contract + regime
```

Este análisis permitirá revisar si el target mantiene un comportamiento razonable entre distintos contratos o si aparecen diferencias fuertes asociadas al rolling de contratos.

La idea no es hacer un análisis excesivamente complejo, sino detectar posibles cambios estructurales que puedan afectar la interpretación del target.

## Celda 1 — Configuración del análisis

```
In [68]: # Celda 1 – Configuración
# =====

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

print("Análisis por contrato y contexto - DIR")
print("=" * 80)

# -----
# Asegurar columnas necesarias
# -----

if "year" not in df_targets_dir.columns:
    df_targets_dir["year"] = df_targets_dir.index.year

if "regime_name" not in df_targets_dir.columns:
    REGIME_NAME_BY_ID = {
        0: "Overnight",
        1: "Pre-market",
        2: "Opening",
        3: "Regular",
        4: "Closing",
    }
    df_targets_dir["regime_name"] = df_targets_dir["regime_id"].map(REGIME_NAME_

if "contract" not in df_targets_dir.columns:
    raise KeyError("No existe la columna 'contract' en df_targets_dir.")

# -----
# Crear contract_code como familia del contrato
# -----
# contract      = H20, M20, U20, Z20...
# contract_code = H, M, U, Z
# -----

df_targets_dir["contract_code"] = (
    df_targets_dir["contract"]
    .astype(str)
    .str[0]
)

print("Columna 'contract_code' creada correctamente usando solo la letra del con

# Targets DIR
dir_target_cols = [
    f"dir_p{p}_h{h}"
    for h in DIR_HORIZONS
    for p in DIR_PERCENTILES
]

# Variante central para gráficos
```

```

selected_targets_context = [
    "dir_p50_h30",
    "dir_p50_h60",
    "dir_p50_h90",
]

print(f"Targets DIR a analizar: {len(dir_target_cols)}")
print(f"Targets seleccionados para gráficos: {selected_targets_context}")
print(f"Contratos disponibles: {sorted(df_targets_dir['contract'].dropna().unique())}")
print(f"Contract codes disponibles: {sorted(df_targets_dir['contract_code'].dropna().unique())}")

```

Análisis por contrato y contexto - DIR

```

=====
Columna 'contract_code' creada correctamente usando solo la letra del contrato.
Targets DIR a analizar: 9
Targets seleccionados para gráficos: ['dir_p50_h30', 'dir_p50_h60', 'dir_p50_h90']
Contratos disponibles: ['H20', 'H21', 'H22', 'H23', 'H24', 'H25', 'H26', 'M20', 'M21', 'M22', 'M23', 'M24', 'M25', 'M26', 'U20', 'U21', 'U22', 'U23', 'U24', 'U25', 'Z20', 'Z21', 'Z22', 'Z23', 'Z24', 'Z25']
Contract codes disponibles: ['H', 'M', 'U', 'Z']

```

## Celda 2 — Función resumen por contexto

Esta función permite resumir DIR por contract, contract\_code, year + regime o contract + regime.

```

In [69]: # Celda 2 – Función de resumen por contexto
# =====

def summarize_dir_by_context(df, target_cols, group_cols):
    """
    Resume distribución de clases y frecuencia operativa de DIR
    para una o más columnas de contexto.

    Ejemplos:
        group_cols = ["contract"]
        group_cols = ["contract_code"]
        group_cols = ["year", "regime_name"]
        group_cols = ["contract", "regime_name"]
    """

    rows = []

    for target_col in target_cols:
        horizon, percentile = parse_dir_target_name(target_col)

        grouped = df.groupby(group_cols, observed=True)

        for group_key, group in grouped:
            if not isinstance(group_key, tuple):
                group_key = (group_key,)

            group_info = dict(zip(group_cols, group_key))

            total_rows = len(group)

            valid_mask = group[target_col].notna()
            valid_values = group.loc[valid_mask, target_col].astype(int)

```



```

valid_rows = valid_mask.sum()
nan_rows = total_rows - valid_rows

long_count = (valid_values == 1).sum()
short_count = (valid_values == -1).sum()
no_signal_count = (valid_values == 0).sum()

signal_count = long_count + short_count

row = {
    "target": target_col,
    "horizon": horizon,
    "percentile": percentile,
    "total_rows": total_rows,
    "valid_rows": valid_rows,
    "nan_rows": nan_rows,
    "long_count": long_count,
    "short_count": short_count,
    "no_signal_count": no_signal_count,
    "signal_count": signal_count,
    "long_ratio": long_count / valid_rows if valid_rows > 0 else np.nan,
    "short_ratio": short_count / valid_rows if valid_rows > 0 else np.nan,
    "no_signal_ratio": no_signal_count / valid_rows if valid_rows > 0 else np.nan,
    "signal_ratio": signal_count / valid_rows if valid_rows > 0 else np.nan,
    "long_share_over_signals": (
        long_count / signal_count if signal_count > 0 else np.nan
    ),
    "short_share_over_signals": (
        short_count / signal_count if signal_count > 0 else np.nan
    ),
}

row.update(group_info)
rows.append(row)

summary = pd.DataFrame(rows)

sort_cols = group_cols + ["horizon", "percentile"]
summary = summary.sort_values(sort_cols).reset_index(drop=True)

return summary

print("Función summarize_dir_by_context lista.")

```

Función summarize\_dir\_by\_context lista.

## Celda 3 — Análisis por contrato

```

In [70]: # Celda 3 – Distribución de DIR por contract
# =====

df_dir_contract_distribution = summarize_dir_by_context(
    df=df_targets_dir,
    target_cols=dir_target_cols,
    group_cols=["contract"]
)

```

```
print("Distribución de DIR por contrato")
print("=" * 80)

cols_show = [
    "contract",
    "target",
    "horizon",
    "percentile",
    "valid_rows",
    "nan_rows",
    "signal_ratio",
    "no_signal_ratio",
    "long_share_over_signals",
]

print(
    df_dir_contract_distribution[cols_show]
    .to_string(index=False, float_format=lambda x: f"{x:.2%}")
)
```

## Distribución de DIR por contrato

| contract        | target                  | horizon | percentile | valid_rows | nan_rows | signal_ratio | no |
|-----------------|-------------------------|---------|------------|------------|----------|--------------|----|
| _signal_ratio   | long_share_over_signals |         |            |            |          |              |    |
| H20 dir_p40_h30 | 30                      | 40      | 31728      | 1440       | 37.29%   |              |    |
| 62.71%          | 46.06%                  |         |            |            |          |              |    |
| H20 dir_p50_h30 | 30                      | 50      | 31728      | 1440       | 31.48%   |              |    |
| 68.52%          | 45.88%                  |         |            |            |          |              |    |
| H20 dir_p60_h30 | 30                      | 60      | 31728      | 1440       | 26.62%   |              |    |
| 73.38%          | 45.82%                  |         |            |            |          |              |    |
| H20 dir_p40_h60 | 60                      | 40      | 30288      | 2880       | 36.19%   |              |    |
| 63.81%          | 44.32%                  |         |            |            |          |              |    |
| H20 dir_p50_h60 | 60                      | 50      | 30288      | 2880       | 30.62%   |              |    |
| 69.38%          | 43.48%                  |         |            |            |          |              |    |
| H20 dir_p60_h60 | 60                      | 60      | 30288      | 2880       | 25.38%   |              |    |
| 74.62%          | 42.92%                  |         |            |            |          |              |    |
| H20 dir_p40_h90 | 90                      | 40      | 28848      | 4320       | 35.26%   |              |    |
| 64.74%          | 42.60%                  |         |            |            |          |              |    |
| H20 dir_p50_h90 | 90                      | 50      | 28848      | 4320       | 29.85%   |              |    |
| 70.15%          | 42.03%                  |         |            |            |          |              |    |
| H20 dir_p60_h90 | 90                      | 60      | 28848      | 4320       | 25.12%   |              |    |
| 74.88%          | 41.22%                  |         |            |            |          |              |    |
| H21 dir_p40_h30 | 30                      | 40      | 38999      | 1770       | 55.78%   |              |    |
| 44.22%          | 48.72%                  |         |            |            |          |              |    |
| H21 dir_p50_h30 | 30                      | 50      | 38999      | 1770       | 46.81%   |              |    |
| 53.19%          | 47.72%                  |         |            |            |          |              |    |
| H21 dir_p60_h30 | 30                      | 60      | 38999      | 1770       | 38.65%   |              |    |
| 61.35%          | 46.42%                  |         |            |            |          |              |    |
| H21 dir_p40_h60 | 60                      | 40      | 37229      | 3540       | 54.42%   |              |    |
| 45.58%          | 48.15%                  |         |            |            |          |              |    |
| H21 dir_p50_h60 | 60                      | 50      | 37229      | 3540       | 45.64%   |              |    |
| 54.36%          | 47.05%                  |         |            |            |          |              |    |
| H21 dir_p60_h60 | 60                      | 60      | 37229      | 3540       | 37.43%   |              |    |
| 62.57%          | 45.29%                  |         |            |            |          |              |    |
| H21 dir_p40_h90 | 90                      | 40      | 35459      | 5310       | 53.96%   |              |    |
| 46.04%          | 48.12%                  |         |            |            |          |              |    |
| H21 dir_p50_h90 | 90                      | 50      | 35459      | 5310       | 46.08%   |              |    |
| 53.92%          | 47.14%                  |         |            |            |          |              |    |
| H21 dir_p60_h90 | 90                      | 60      | 35459      | 5310       | 38.05%   |              |    |
| 61.95%          | 46.07%                  |         |            |            |          |              |    |
| H22 dir_p40_h30 | 30                      | 40      | 40321      | 1830       | 84.56%   |              |    |
| 15.44%          | 48.66%                  |         |            |            |          |              |    |
| H22 dir_p50_h30 | 30                      | 50      | 40321      | 1830       | 77.83%   |              |    |
| 22.17%          | 48.49%                  |         |            |            |          |              |    |
| H22 dir_p60_h30 | 30                      | 60      | 40321      | 1830       | 69.54%   |              |    |
| 30.46%          | 48.29%                  |         |            |            |          |              |    |
| H22 dir_p40_h60 | 60                      | 40      | 38491      | 3660       | 85.35%   |              |    |
| 14.65%          | 48.95%                  |         |            |            |          |              |    |
| H22 dir_p50_h60 | 60                      | 50      | 38491      | 3660       | 78.34%   |              |    |
| 21.66%          | 48.35%                  |         |            |            |          |              |    |
| H22 dir_p60_h60 | 60                      | 60      | 38491      | 3660       | 69.63%   |              |    |
| 30.37%          | 47.71%                  |         |            |            |          |              |    |
| H22 dir_p40_h90 | 90                      | 40      | 36661      | 5490       | 85.34%   |              |    |
| 14.66%          | 47.46%                  |         |            |            |          |              |    |
| H22 dir_p50_h90 | 90                      | 50      | 36661      | 5490       | 78.24%   |              |    |
| 21.76%          | 46.92%                  |         |            |            |          |              |    |
| H22 dir_p60_h90 | 90                      | 60      | 36661      | 5490       | 68.99%   |              |    |
| 31.01%          | 46.34%                  |         |            |            |          |              |    |
| H23 dir_p40_h30 | 30                      | 40      | 39660      | 1800       | 65.07%   |              |    |
| 34.93%          | 50.83%                  |         |            |            |          |              |    |

|                 |        |    |       |      |        |
|-----------------|--------|----|-------|------|--------|
| H23 dir_p50_h30 | 30     | 50 | 39660 | 1800 | 53.45% |
| 46.55%          | 50.76% |    |       |      |        |
| H23 dir_p60_h30 | 30     | 60 | 39660 | 1800 | 42.28% |
| 57.72%          | 50.32% |    |       |      |        |
| H23 dir_p40_h60 | 60     | 40 | 37860 | 3600 | 65.95% |
| 34.05%          | 51.51% |    |       |      |        |
| H23 dir_p50_h60 | 60     | 50 | 37860 | 3600 | 54.04% |
| 45.96%          | 50.90% |    |       |      |        |
| H23 dir_p60_h60 | 60     | 60 | 37860 | 3600 | 42.55% |
| 57.45%          | 50.20% |    |       |      |        |
| H23 dir_p40_h90 | 90     | 40 | 36060 | 5400 | 66.58% |
| 33.42%          | 52.08% |    |       |      |        |
| H23 dir_p50_h90 | 90     | 50 | 36060 | 5400 | 55.00% |
| 45.00%          | 51.58% |    |       |      |        |
| H23 dir_p60_h90 | 90     | 60 | 36060 | 5400 | 42.84% |
| 57.16%          | 50.87% |    |       |      |        |
| H24 dir_p40_h30 | 30     | 40 | 39660 | 1800 | 54.19% |
| 45.81%          | 49.51% |    |       |      |        |
| H24 dir_p50_h30 | 30     | 50 | 39660 | 1800 | 42.90% |
| 57.10%          | 48.52% |    |       |      |        |
| H24 dir_p60_h30 | 30     | 60 | 39660 | 1800 | 32.22% |
| 67.78%          | 47.32% |    |       |      |        |
| H24 dir_p40_h60 | 60     | 40 | 37860 | 3600 | 53.57% |
| 46.43%          | 49.87% |    |       |      |        |
| H24 dir_p50_h60 | 60     | 50 | 37860 | 3600 | 42.61% |
| 57.39%          | 48.43% |    |       |      |        |
| H24 dir_p60_h60 | 60     | 60 | 37860 | 3600 | 32.39% |
| 67.61%          | 46.32% |    |       |      |        |
| H24 dir_p40_h90 | 90     | 40 | 36060 | 5400 | 53.75% |
| 46.25%          | 49.82% |    |       |      |        |
| H24 dir_p50_h90 | 90     | 50 | 36060 | 5400 | 43.33% |
| 56.67%          | 48.50% |    |       |      |        |
| H24 dir_p60_h90 | 90     | 60 | 36060 | 5400 | 33.31% |
| 66.69%          | 46.55% |    |       |      |        |
| H25 dir_p40_h30 | 30     | 40 | 37677 | 1710 | 82.67% |
| 17.33%          | 48.19% |    |       |      |        |
| H25 dir_p50_h30 | 30     | 50 | 37677 | 1710 | 75.00% |
| 25.00%          | 48.07% |    |       |      |        |
| H25 dir_p60_h30 | 30     | 60 | 37677 | 1710 | 66.42% |
| 33.58%          | 47.84% |    |       |      |        |
| H25 dir_p40_h60 | 60     | 40 | 35967 | 3420 | 82.93% |
| 17.07%          | 48.59% |    |       |      |        |
| H25 dir_p50_h60 | 60     | 50 | 35967 | 3420 | 74.98% |
| 25.02%          | 47.84% |    |       |      |        |
| H25 dir_p60_h60 | 60     | 60 | 35967 | 3420 | 66.41% |
| 33.59%          | 47.15% |    |       |      |        |
| H25 dir_p40_h90 | 90     | 40 | 34257 | 5130 | 83.15% |
| 16.85%          | 48.41% |    |       |      |        |
| H25 dir_p50_h90 | 90     | 50 | 34257 | 5130 | 75.80% |
| 24.20%          | 47.56% |    |       |      |        |
| H25 dir_p60_h90 | 90     | 60 | 34257 | 5130 | 67.57% |
| 32.43%          | 46.52% |    |       |      |        |
| H26 dir_p40_h30 | 30     | 40 | 37677 | 1710 | 82.07% |
| 17.93%          | 48.24% |    |       |      |        |
| H26 dir_p50_h30 | 30     | 50 | 37677 | 1710 | 75.05% |
| 24.95%          | 48.12% |    |       |      |        |
| H26 dir_p60_h30 | 30     | 60 | 37677 | 1710 | 67.26% |
| 32.74%          | 48.08% |    |       |      |        |
| H26 dir_p40_h60 | 60     | 40 | 35967 | 3420 | 83.10% |
| 16.90%          | 48.97% |    |       |      |        |

|                 |        |    |       |      |        |
|-----------------|--------|----|-------|------|--------|
| H26 dir_p50_h60 | 60     | 50 | 35967 | 3420 | 75.89% |
| 24.11%          | 48.52% |    |       |      |        |
| H26 dir_p60_h60 | 60     | 60 | 35967 | 3420 | 67.85% |
| 32.15%          | 48.25% |    |       |      |        |
| H26 dir_p40_h90 | 90     | 40 | 34257 | 5130 | 83.01% |
| 16.99%          | 49.99% |    |       |      |        |
| H26 dir_p50_h90 | 90     | 50 | 34257 | 5130 | 75.91% |
| 24.09%          | 49.91% |    |       |      |        |
| H26 dir_p60_h90 | 90     | 60 | 34257 | 5130 | 67.61% |
| 32.39%          | 49.37% |    |       |      |        |
| M20 dir_p40_h30 | 30     | 40 | 37016 | 1680 | 59.50% |
| 40.50%          | 51.01% |    |       |      |        |
| M20 dir_p50_h30 | 30     | 50 | 37016 | 1680 | 49.04% |
| 50.96%          | 49.68% |    |       |      |        |
| M20 dir_p60_h30 | 30     | 60 | 37016 | 1680 | 38.57% |
| 61.43%          | 48.22% |    |       |      |        |
| M20 dir_p40_h60 | 60     | 40 | 35336 | 3360 | 58.66% |
| 41.34%          | 51.27% |    |       |      |        |
| M20 dir_p50_h60 | 60     | 50 | 35336 | 3360 | 47.29% |
| 52.71%          | 49.69% |    |       |      |        |
| M20 dir_p60_h60 | 60     | 60 | 35336 | 3360 | 36.81% |
| 63.19%          | 48.99% |    |       |      |        |
| M20 dir_p40_h90 | 90     | 40 | 33656 | 5040 | 56.84% |
| 43.16%          | 51.93% |    |       |      |        |
| M20 dir_p50_h90 | 90     | 50 | 33656 | 5040 | 45.92% |
| 54.08%          | 51.58% |    |       |      |        |
| M20 dir_p60_h90 | 90     | 60 | 33656 | 5040 | 35.89% |
| 64.11%          | 50.40% |    |       |      |        |
| M21 dir_p40_h30 | 30     | 40 | 40321 | 1830 | 46.73% |
| 53.27%          | 49.89% |    |       |      |        |
| M21 dir_p50_h30 | 30     | 50 | 40321 | 1830 | 36.80% |
| 63.20%          | 48.99% |    |       |      |        |
| M21 dir_p60_h30 | 30     | 60 | 40321 | 1830 | 27.57% |
| 72.43%          | 47.40% |    |       |      |        |
| M21 dir_p40_h60 | 60     | 40 | 38491 | 3660 | 45.87% |
| 54.13%          | 50.20% |    |       |      |        |
| M21 dir_p50_h60 | 60     | 50 | 38491 | 3660 | 35.76% |
| 64.24%          | 48.42% |    |       |      |        |
| M21 dir_p60_h60 | 60     | 60 | 38491 | 3660 | 26.72% |
| 73.28%          | 45.87% |    |       |      |        |
| M21 dir_p40_h90 | 90     | 40 | 36661 | 5490 | 46.41% |
| 53.59%          | 48.75% |    |       |      |        |
| M21 dir_p50_h90 | 90     | 50 | 36661 | 5490 | 36.43% |
| 63.57%          | 47.30% |    |       |      |        |
| M21 dir_p60_h90 | 90     | 60 | 36661 | 5490 | 26.89% |
| 73.11%          | 45.59% |    |       |      |        |
| M22 dir_p40_h30 | 30     | 40 | 40321 | 1830 | 87.05% |
| 12.95%          | 47.61% |    |       |      |        |
| M22 dir_p50_h30 | 30     | 50 | 40321 | 1830 | 80.02% |
| 19.98%          | 47.30% |    |       |      |        |
| M22 dir_p60_h30 | 30     | 60 | 40321 | 1830 | 70.41% |
| 29.59%          | 46.81% |    |       |      |        |
| M22 dir_p40_h60 | 60     | 40 | 38491 | 3660 | 89.18% |
| 10.82%          | 46.42% |    |       |      |        |
| M22 dir_p50_h60 | 60     | 50 | 38491 | 3660 | 81.51% |
| 18.49%          | 46.20% |    |       |      |        |
| M22 dir_p60_h60 | 60     | 60 | 38491 | 3660 | 71.33% |
| 28.67%          | 45.92% |    |       |      |        |
| M22 dir_p40_h90 | 90     | 40 | 36661 | 5490 | 88.86% |
| 11.14%          | 46.13% |    |       |      |        |

|                 |        |    |       |      |        |
|-----------------|--------|----|-------|------|--------|
| M22 dir_p50_h90 | 90     | 50 | 36661 | 5490 | 81.42% |
| 18.58%          | 46.01% |    |       |      |        |
| M22 dir_p60_h90 | 90     | 60 | 36661 | 5490 | 72.11% |
| 27.89%          | 45.96% |    |       |      |        |
| M23 dir_p40_h30 | 30     | 40 | 36355 | 1650 | 47.71% |
| 52.29%          | 52.46% |    |       |      |        |
| M23 dir_p50_h30 | 30     | 50 | 36355 | 1650 | 36.20% |
| 63.80%          | 51.93% |    |       |      |        |
| M23 dir_p60_h30 | 30     | 60 | 36355 | 1650 | 25.72% |
| 74.28%          | 50.50% |    |       |      |        |
| M23 dir_p40_h60 | 60     | 40 | 34705 | 3300 | 46.64% |
| 53.36%          | 53.30% |    |       |      |        |
| M23 dir_p50_h60 | 60     | 50 | 34705 | 3300 | 35.02% |
| 64.98%          | 53.09% |    |       |      |        |
| M23 dir_p60_h60 | 60     | 60 | 34705 | 3300 | 25.11% |
| 74.89%          | 53.71% |    |       |      |        |
| M23 dir_p40_h90 | 90     | 40 | 33055 | 4950 | 46.46% |
| 53.54%          | 54.57% |    |       |      |        |
| M23 dir_p50_h90 | 90     | 50 | 33055 | 4950 | 34.63% |
| 65.37%          | 53.87% |    |       |      |        |
| M23 dir_p60_h90 | 90     | 60 | 33055 | 4950 | 23.95% |
| 76.05%          | 53.25% |    |       |      |        |
| M24 dir_p40_h30 | 30     | 40 | 43626 | 1980 | 56.08% |
| 43.92%          | 49.17% |    |       |      |        |
| M24 dir_p50_h30 | 30     | 50 | 43626 | 1980 | 45.11% |
| 54.89%          | 48.19% |    |       |      |        |
| M24 dir_p60_h30 | 30     | 60 | 43626 | 1980 | 34.55% |
| 65.45%          | 47.20% |    |       |      |        |
| M24 dir_p40_h60 | 60     | 40 | 41646 | 3960 | 55.80% |
| 44.20%          | 49.46% |    |       |      |        |
| M24 dir_p50_h60 | 60     | 50 | 41646 | 3960 | 45.54% |
| 54.46%          | 48.34% |    |       |      |        |
| M24 dir_p60_h60 | 60     | 60 | 41646 | 3960 | 35.47% |
| 64.53%          | 46.99% |    |       |      |        |
| M24 dir_p40_h90 | 90     | 40 | 39666 | 5940 | 55.26% |
| 44.74%          | 49.12% |    |       |      |        |
| M24 dir_p50_h90 | 90     | 50 | 39666 | 5940 | 45.18% |
| 54.82%          | 47.65% |    |       |      |        |
| M24 dir_p60_h90 | 90     | 60 | 39666 | 5940 | 35.80% |
| 64.20%          | 45.41% |    |       |      |        |
| M25 dir_p40_h30 | 30     | 40 | 33050 | 1500 | 84.94% |
| 15.06%          | 52.05% |    |       |      |        |
| M25 dir_p50_h30 | 30     | 50 | 33050 | 1500 | 77.63% |
| 22.37%          | 51.83% |    |       |      |        |
| M25 dir_p60_h30 | 30     | 60 | 33050 | 1500 | 69.13% |
| 30.87%          | 51.56% |    |       |      |        |
| M25 dir_p40_h60 | 60     | 40 | 31550 | 3000 | 85.44% |
| 14.56%          | 51.79% |    |       |      |        |
| M25 dir_p50_h60 | 60     | 50 | 31550 | 3000 | 77.93% |
| 22.07%          | 51.39% |    |       |      |        |
| M25 dir_p60_h60 | 60     | 60 | 31550 | 3000 | 68.91% |
| 31.09%          | 50.57% |    |       |      |        |
| M25 dir_p40_h90 | 90     | 40 | 30050 | 4500 | 85.81% |
| 14.19%          | 52.36% |    |       |      |        |
| M25 dir_p50_h90 | 90     | 50 | 30050 | 4500 | 78.39% |
| 21.61%          | 51.83% |    |       |      |        |
| M25 dir_p60_h90 | 90     | 60 | 30050 | 4500 | 69.46% |
| 30.54%          | 51.10% |    |       |      |        |
| M26 dir_p40_h30 | 30     | 40 | 12559 | 570  | 89.25% |
| 10.75%          | 52.10% |    |       |      |        |

|                 |        |    |       |      |        |
|-----------------|--------|----|-------|------|--------|
| M26 dir_p50_h30 | 30     | 50 | 12559 | 570  | 83.37% |
| 16.63%          | 51.87% |    |       |      |        |
| M26 dir_p60_h30 | 30     | 60 | 12559 | 570  | 74.84% |
| 25.16%          | 51.51% |    |       |      |        |
| M26 dir_p40_h60 | 60     | 40 | 11989 | 1140 | 91.28% |
| 8.72%           | 50.72% |    |       |      |        |
| M26 dir_p50_h60 | 60     | 50 | 11989 | 1140 | 84.84% |
| 15.16%          | 49.98% |    |       |      |        |
| M26 dir_p60_h60 | 60     | 60 | 11989 | 1140 | 77.05% |
| 22.95%          | 49.36% |    |       |      |        |
| M26 dir_p40_h90 | 90     | 40 | 11419 | 1710 | 91.79% |
| 8.21%           | 49.40% |    |       |      |        |
| M26 dir_p50_h90 | 90     | 50 | 11419 | 1710 | 86.45% |
| 13.55%          | 49.17% |    |       |      |        |
| M26 dir_p60_h90 | 90     | 60 | 11419 | 1710 | 79.24% |
| 20.76%          | 48.46% |    |       |      |        |
| U20 dir_p40_h30 | 30     | 40 | 40321 | 1830 | 54.07% |
| 45.93%          | 48.63% |    |       |      |        |
| U20 dir_p50_h30 | 30     | 50 | 40321 | 1830 | 43.56% |
| 56.44%          | 46.71% |    |       |      |        |
| U20 dir_p60_h30 | 30     | 60 | 40321 | 1830 | 34.04% |
| 65.96%          | 44.81% |    |       |      |        |
| U20 dir_p40_h60 | 60     | 40 | 38491 | 3660 | 53.24% |
| 46.76%          | 47.98% |    |       |      |        |
| U20 dir_p50_h60 | 60     | 50 | 38491 | 3660 | 42.66% |
| 57.34%          | 45.61% |    |       |      |        |
| U20 dir_p60_h60 | 60     | 60 | 38491 | 3660 | 33.00% |
| 67.00%          | 43.02% |    |       |      |        |
| U20 dir_p40_h90 | 90     | 40 | 36661 | 5490 | 54.61% |
| 45.39%          | 46.99% |    |       |      |        |
| U20 dir_p50_h90 | 90     | 50 | 36661 | 5490 | 43.77% |
| 56.23%          | 44.66% |    |       |      |        |
| U20 dir_p60_h90 | 90     | 60 | 36661 | 5490 | 32.70% |
| 67.30%          | 41.79% |    |       |      |        |
| U21 dir_p40_h30 | 30     | 40 | 39660 | 1800 | 33.27% |
| 66.73%          | 49.01% |    |       |      |        |
| U21 dir_p50_h30 | 30     | 50 | 39660 | 1800 | 24.89% |
| 75.11%          | 47.76% |    |       |      |        |
| U21 dir_p60_h30 | 30     | 60 | 39660 | 1800 | 17.59% |
| 82.41%          | 45.64% |    |       |      |        |
| U21 dir_p40_h60 | 60     | 40 | 37860 | 3600 | 33.81% |
| 66.19%          | 47.52% |    |       |      |        |
| U21 dir_p50_h60 | 60     | 50 | 37860 | 3600 | 25.37% |
| 74.63%          | 45.30% |    |       |      |        |
| U21 dir_p60_h60 | 60     | 60 | 37860 | 3600 | 17.76% |
| 82.24%          | 43.90% |    |       |      |        |
| U21 dir_p40_h90 | 90     | 40 | 36060 | 5400 | 35.46% |
| 64.54%          | 46.21% |    |       |      |        |
| U21 dir_p50_h90 | 90     | 50 | 36060 | 5400 | 26.54% |
| 73.46%          | 43.74% |    |       |      |        |
| U21 dir_p60_h90 | 90     | 60 | 36060 | 5400 | 17.95% |
| 82.05%          | 42.51% |    |       |      |        |
| U22 dir_p40_h30 | 30     | 40 | 38999 | 1770 | 73.20% |
| 26.80%          | 50.20% |    |       |      |        |
| U22 dir_p50_h30 | 30     | 50 | 38999 | 1770 | 62.05% |
| 37.95%          | 49.79% |    |       |      |        |
| U22 dir_p60_h30 | 30     | 60 | 38999 | 1770 | 50.32% |
| 49.68%          | 49.30% |    |       |      |        |
| U22 dir_p40_h60 | 60     | 40 | 37229 | 3540 | 74.27% |
| 25.73%          | 50.26% |    |       |      |        |

|                 |        |    |       |      |        |
|-----------------|--------|----|-------|------|--------|
| U22 dir_p50_h60 | 60     | 50 | 37229 | 3540 | 62.43% |
| 37.57%          | 50.22% |    |       |      |        |
| U22 dir_p60_h60 | 60     | 60 | 37229 | 3540 | 50.32% |
| 49.68%          | 50.27% |    |       |      |        |
| U22 dir_p40_h90 | 90     | 40 | 35459 | 5310 | 74.01% |
| 25.99%          | 50.21% |    |       |      |        |
| U22 dir_p50_h90 | 90     | 50 | 35459 | 5310 | 62.58% |
| 37.42%          | 50.19% |    |       |      |        |
| U22 dir_p60_h90 | 90     | 60 | 35459 | 5310 | 50.51% |
| 49.49%          | 50.15% |    |       |      |        |
| U23 dir_p40_h30 | 30     | 40 | 39660 | 1800 | 57.10% |
| 42.90%          | 49.82% |    |       |      |        |
| U23 dir_p50_h30 | 30     | 50 | 39660 | 1800 | 44.81% |
| 55.19%          | 48.53% |    |       |      |        |
| U23 dir_p60_h30 | 30     | 60 | 39660 | 1800 | 33.18% |
| 66.82%          | 46.85% |    |       |      |        |
| U23 dir_p40_h60 | 60     | 40 | 37860 | 3600 | 57.78% |
| 42.22%          | 50.04% |    |       |      |        |
| U23 dir_p50_h60 | 60     | 50 | 37860 | 3600 | 44.27% |
| 55.73%          | 48.35% |    |       |      |        |
| U23 dir_p60_h60 | 60     | 60 | 37860 | 3600 | 32.58% |
| 67.42%          | 45.94% |    |       |      |        |
| U23 dir_p40_h90 | 90     | 40 | 36060 | 5400 | 56.11% |
| 43.89%          | 50.50% |    |       |      |        |
| U23 dir_p50_h90 | 90     | 50 | 36060 | 5400 | 44.22% |
| 55.78%          | 49.09% |    |       |      |        |
| U23 dir_p60_h90 | 90     | 60 | 36060 | 5400 | 32.76% |
| 67.24%          | 46.83% |    |       |      |        |
| U24 dir_p40_h30 | 30     | 40 | 37016 | 1680 | 74.75% |
| 25.25%          | 50.11% |    |       |      |        |
| U24 dir_p50_h30 | 30     | 50 | 37016 | 1680 | 66.06% |
| 33.94%          | 49.53% |    |       |      |        |
| U24 dir_p60_h30 | 30     | 60 | 37016 | 1680 | 55.92% |
| 44.08%          | 48.84% |    |       |      |        |
| U24 dir_p40_h60 | 60     | 40 | 35336 | 3360 | 77.09% |
| 22.91%          | 49.73% |    |       |      |        |
| U24 dir_p50_h60 | 60     | 50 | 35336 | 3360 | 68.06% |
| 31.94%          | 49.13% |    |       |      |        |
| U24 dir_p60_h60 | 60     | 60 | 35336 | 3360 | 58.01% |
| 41.99%          | 47.92% |    |       |      |        |
| U24 dir_p40_h90 | 90     | 40 | 33656 | 5040 | 77.41% |
| 22.59%          | 49.69% |    |       |      |        |
| U24 dir_p50_h90 | 90     | 50 | 33656 | 5040 | 68.54% |
| 31.46%          | 48.65% |    |       |      |        |
| U24 dir_p60_h90 | 90     | 60 | 33656 | 5040 | 57.82% |
| 42.18%          | 47.26% |    |       |      |        |
| U25 dir_p40_h30 | 30     | 40 | 37677 | 1710 | 57.97% |
| 42.03%          | 48.63% |    |       |      |        |
| U25 dir_p50_h30 | 30     | 50 | 37677 | 1710 | 47.43% |
| 52.57%          | 47.42% |    |       |      |        |
| U25 dir_p60_h30 | 30     | 60 | 37677 | 1710 | 36.97% |
| 63.03%          | 45.14% |    |       |      |        |
| U25 dir_p40_h60 | 60     | 40 | 35967 | 3420 | 57.92% |
| 42.08%          | 46.90% |    |       |      |        |
| U25 dir_p50_h60 | 60     | 50 | 35967 | 3420 | 46.41% |
| 53.59%          | 45.15% |    |       |      |        |
| U25 dir_p60_h60 | 60     | 60 | 35967 | 3420 | 35.56% |
| 64.44%          | 43.34% |    |       |      |        |
| U25 dir_p40_h90 | 90     | 40 | 34257 | 5130 | 57.13% |
| 42.87%          | 46.10% |    |       |      |        |



|                 |        |    |       |      |        |
|-----------------|--------|----|-------|------|--------|
| U25 dir_p50_h90 | 90     | 50 | 34257 | 5130 | 45.29% |
| 54.71%          | 44.83% |    |       |      |        |
| U25 dir_p60_h90 | 90     | 60 | 34257 | 5130 | 34.72% |
| 65.28%          | 43.39% |    |       |      |        |
| Z20 dir_p40_h30 | 30     | 40 | 35033 | 1590 | 56.69% |
| 43.31%          | 49.67% |    |       |      |        |
| Z20 dir_p50_h30 | 30     | 50 | 35033 | 1590 | 46.64% |
| 53.36%          | 48.96% |    |       |      |        |
| Z20 dir_p60_h30 | 30     | 60 | 35033 | 1590 | 36.81% |
| 63.19%          | 49.03% |    |       |      |        |
| Z20 dir_p40_h60 | 60     | 40 | 33443 | 3180 | 55.53% |
| 44.47%          | 51.30% |    |       |      |        |
| Z20 dir_p50_h60 | 60     | 50 | 33443 | 3180 | 45.26% |
| 54.74%          | 51.17% |    |       |      |        |
| Z20 dir_p60_h60 | 60     | 60 | 33443 | 3180 | 35.74% |
| 64.26%          | 50.90% |    |       |      |        |
| Z20 dir_p40_h90 | 90     | 40 | 31853 | 4770 | 55.41% |
| 44.59%          | 52.06% |    |       |      |        |
| Z20 dir_p50_h90 | 90     | 50 | 31853 | 4770 | 44.90% |
| 55.10%          | 51.27% |    |       |      |        |
| Z20 dir_p60_h90 | 90     | 60 | 31853 | 4770 | 34.47% |
| 65.53%          | 50.32% |    |       |      |        |
| Z21 dir_p40_h30 | 30     | 40 | 40982 | 1860 | 56.90% |
| 43.10%          | 48.66% |    |       |      |        |
| Z21 dir_p50_h30 | 30     | 50 | 40982 | 1860 | 47.02% |
| 52.98%          | 47.57% |    |       |      |        |
| Z21 dir_p60_h30 | 30     | 60 | 40982 | 1860 | 37.43% |
| 62.57%          | 46.74% |    |       |      |        |
| Z21 dir_p40_h60 | 60     | 40 | 39122 | 3720 | 57.72% |
| 42.28%          | 48.73% |    |       |      |        |
| Z21 dir_p50_h60 | 60     | 50 | 39122 | 3720 | 47.46% |
| 52.54%          | 47.58% |    |       |      |        |
| Z21 dir_p60_h60 | 60     | 60 | 39122 | 3720 | 36.81% |
| 63.19%          | 46.34% |    |       |      |        |
| Z21 dir_p40_h90 | 90     | 40 | 37262 | 5580 | 57.14% |
| 42.86%          | 48.20% |    |       |      |        |
| Z21 dir_p50_h90 | 90     | 50 | 37262 | 5580 | 46.68% |
| 53.32%          | 47.51% |    |       |      |        |
| Z21 dir_p60_h90 | 90     | 60 | 37262 | 5580 | 35.99% |
| 64.01%          | 45.91% |    |       |      |        |
| Z22 dir_p40_h30 | 30     | 40 | 38999 | 1770 | 75.24% |
| 24.76%          | 50.80% |    |       |      |        |
| Z22 dir_p50_h30 | 30     | 50 | 38999 | 1770 | 65.00% |
| 35.00%          | 50.82% |    |       |      |        |
| Z22 dir_p60_h30 | 30     | 60 | 38999 | 1770 | 53.43% |
| 46.57%          | 50.66% |    |       |      |        |
| Z22 dir_p40_h60 | 60     | 40 | 37229 | 3540 | 74.55% |
| 25.45%          | 50.88% |    |       |      |        |
| Z22 dir_p50_h60 | 60     | 50 | 37229 | 3540 | 64.36% |
| 35.64%          | 50.73% |    |       |      |        |
| Z22 dir_p60_h60 | 60     | 60 | 37229 | 3540 | 53.16% |
| 46.84%          | 50.62% |    |       |      |        |
| Z22 dir_p40_h90 | 90     | 40 | 35459 | 5310 | 74.94% |
| 25.06%          | 49.51% |    |       |      |        |
| Z22 dir_p50_h90 | 90     | 50 | 35459 | 5310 | 64.82% |
| 35.18%          | 49.32% |    |       |      |        |
| Z22 dir_p60_h90 | 90     | 60 | 35459 | 5310 | 53.25% |
| 46.75%          | 49.45% |    |       |      |        |
| Z23 dir_p40_h30 | 30     | 40 | 40982 | 1860 | 56.78% |
| 43.22%          | 50.66% |    |       |      |        |

|                 |        |    |       |      |        |
|-----------------|--------|----|-------|------|--------|
| Z23 dir_p50_h30 | 30     | 50 | 40982 | 1860 | 45.10% |
| 54.90%          | 50.37% |    |       |      |        |
| Z23 dir_p60_h30 | 30     | 60 | 40982 | 1860 | 34.21% |
| 65.79%          | 49.74% |    |       |      |        |
| Z23 dir_p40_h60 | 60     | 40 | 39122 | 3720 | 56.82% |
| 43.18%          | 50.69% |    |       |      |        |
| Z23 dir_p50_h60 | 60     | 50 | 39122 | 3720 | 44.95% |
| 55.05%          | 49.57% |    |       |      |        |
| Z23 dir_p60_h60 | 60     | 60 | 39122 | 3720 | 34.33% |
| 65.67%          | 48.36% |    |       |      |        |
| Z23 dir_p40_h90 | 90     | 40 | 37262 | 5580 | 57.47% |
| 42.53%          | 50.89% |    |       |      |        |
| Z23 dir_p50_h90 | 90     | 50 | 37262 | 5580 | 46.00% |
| 54.00%          | 50.25% |    |       |      |        |
| Z23 dir_p60_h90 | 90     | 60 | 37262 | 5580 | 34.72% |
| 65.28%          | 49.98% |    |       |      |        |
| Z24 dir_p40_h30 | 30     | 40 | 40982 | 1860 | 65.78% |
| 34.22%          | 50.55% |    |       |      |        |
| Z24 dir_p50_h30 | 30     | 50 | 40982 | 1860 | 54.53% |
| 45.47%          | 50.47% |    |       |      |        |
| Z24 dir_p60_h30 | 30     | 60 | 40982 | 1860 | 42.77% |
| 57.23%          | 49.74% |    |       |      |        |
| Z24 dir_p40_h60 | 60     | 40 | 39122 | 3720 | 65.45% |
| 34.55%          | 50.92% |    |       |      |        |
| Z24 dir_p50_h60 | 60     | 50 | 39122 | 3720 | 53.65% |
| 46.35%          | 50.95% |    |       |      |        |
| Z24 dir_p60_h60 | 60     | 60 | 39122 | 3720 | 42.63% |
| 57.37%          | 49.75% |    |       |      |        |
| Z24 dir_p40_h90 | 90     | 40 | 37262 | 5580 | 65.26% |
| 34.74%          | 51.40% |    |       |      |        |
| Z24 dir_p50_h90 | 90     | 50 | 37262 | 5580 | 54.49% |
| 45.51%          | 50.41% |    |       |      |        |
| Z24 dir_p60_h90 | 90     | 60 | 37262 | 5580 | 43.58% |
| 56.42%          | 49.07% |    |       |      |        |
| Z25 dir_p40_h30 | 30     | 40 | 40321 | 1830 | 77.37% |
| 22.63%          | 49.81% |    |       |      |        |
| Z25 dir_p50_h30 | 30     | 50 | 40321 | 1830 | 69.13% |
| 30.87%          | 49.40% |    |       |      |        |
| Z25 dir_p60_h30 | 30     | 60 | 40321 | 1830 | 60.34% |
| 39.66%          | 48.92% |    |       |      |        |
| Z25 dir_p40_h60 | 60     | 40 | 38491 | 3660 | 78.77% |
| 21.23%          | 50.36% |    |       |      |        |
| Z25 dir_p50_h60 | 60     | 50 | 38491 | 3660 | 70.19% |
| 29.81%          | 49.54% |    |       |      |        |
| Z25 dir_p60_h60 | 60     | 60 | 38491 | 3660 | 60.98% |
| 39.02%          | 48.55% |    |       |      |        |
| Z25 dir_p40_h90 | 90     | 40 | 36661 | 5490 | 78.15% |
| 21.85%          | 49.83% |    |       |      |        |
| Z25 dir_p50_h90 | 90     | 50 | 36661 | 5490 | 69.87% |
| 30.13%          | 49.17% |    |       |      |        |
| Z25 dir_p60_h90 | 90     | 60 | 36661 | 5490 | 60.84% |
| 39.16%          | 47.79% |    |       |      |        |

## Celda 4 — Matriz de frecuencia operativa por contrato

```
In [71]: # Celda 4 - Matriz de frecuencia operativa por contrato
# =====

df_contract_matrix = df_dir_contract_distribution.copy()
```

```
df_contract_matrix["target_label"] = (  
    "H" + df_contract_matrix["horizon"].astype(str)  
    + " p" + df_contract_matrix["percentile"].astype(str)  
)  
  
contract_signal_pivot = df_contract_matrix.pivot(  
    index="contract",  
    columns="target_label",  
    values="signal_ratio"  
)  
  
print("Frecuencia operativa por contrato y target")  
print("=" * 80)  
print(contract_signal_pivot.to_string(float_format=lambda x: f"{x:.2%}"))
```

## Frecuencia operativa por contrato y target

| target_label | H30 p40 | H30 p50 | H30 p60 | H60 p40 | H60 p50 | H60 p60 | H90 p40 | H90 p50 | H90 p60 |
|--------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| contract     |         |         |         |         |         |         |         |         |         |
| H20          | 37.29%  | 31.48%  | 26.62%  | 36.19%  | 30.62%  | 25.38%  | 35.26%  | 29.85%  | 25.12%  |
| H21          | 55.78%  | 46.81%  | 38.65%  | 54.42%  | 45.64%  | 37.43%  | 53.96%  | 46.08%  | 38.05%  |
| H22          | 84.56%  | 77.83%  | 69.54%  | 85.35%  | 78.34%  | 69.63%  | 85.34%  | 78.24%  | 68.99%  |
| H23          | 65.07%  | 53.45%  | 42.28%  | 65.95%  | 54.04%  | 42.55%  | 66.58%  | 55.00%  | 42.84%  |
| H24          | 54.19%  | 42.90%  | 32.22%  | 53.57%  | 42.61%  | 32.39%  | 53.75%  | 43.33%  | 33.31%  |
| H25          | 82.67%  | 75.00%  | 66.42%  | 82.93%  | 74.98%  | 66.41%  | 83.15%  | 75.80%  | 67.57%  |
| H26          | 82.07%  | 75.05%  | 67.26%  | 83.10%  | 75.89%  | 67.85%  | 83.01%  | 75.91%  | 67.61%  |
| M20          | 59.50%  | 49.04%  | 38.57%  | 58.66%  | 47.29%  | 36.81%  | 56.84%  | 45.92%  | 35.89%  |
| M21          | 46.73%  | 36.80%  | 27.57%  | 45.87%  | 35.76%  | 26.72%  | 46.41%  | 36.43%  | 26.89%  |
| M22          | 87.05%  | 80.02%  | 70.41%  | 89.18%  | 81.51%  | 71.33%  | 88.86%  | 81.42%  | 72.11%  |
| M23          | 47.71%  | 36.20%  | 25.72%  | 46.64%  | 35.02%  | 25.11%  | 46.46%  | 34.63%  | 23.95%  |
| M24          | 56.08%  | 45.11%  | 34.55%  | 55.80%  | 45.54%  | 35.47%  | 55.26%  | 45.18%  | 35.80%  |
| M25          | 84.94%  | 77.63%  | 69.13%  | 85.44%  | 77.93%  | 68.91%  | 85.81%  | 78.39%  | 69.46%  |
| M26          | 89.25%  | 83.37%  | 74.84%  | 91.28%  | 84.84%  | 77.05%  | 91.79%  | 86.45%  | 79.24%  |
| U20          | 54.07%  | 43.56%  | 34.04%  | 53.24%  | 42.66%  | 33.00%  | 54.61%  | 43.77%  | 32.70%  |
| U21          | 33.27%  | 24.89%  | 17.59%  | 33.81%  | 25.37%  | 17.76%  | 35.46%  | 26.54%  | 17.95%  |
| U22          | 73.20%  | 62.05%  | 50.32%  | 74.27%  | 62.43%  | 50.32%  | 74.01%  | 62.58%  | 50.51%  |
| U23          | 57.10%  | 44.81%  | 33.18%  | 57.78%  | 44.27%  | 32.58%  | 56.11%  | 44.22%  | 32.76%  |
| U24          | 74.75%  | 66.06%  | 55.92%  | 77.09%  | 68.06%  | 58.01%  | 77.41%  | 68.54%  | 57.82%  |
| U25          | 57.97%  | 47.43%  | 36.97%  | 57.92%  | 46.41%  | 35.56%  | 57.13%  | 45.29%  | 34.72%  |
| Z20          | 56.69%  | 46.64%  | 36.81%  | 55.53%  | 45.26%  | 35.74%  | 55.41%  | 44.90%  | 34.47%  |
| Z21          | 56.90%  | 47.02%  | 37.43%  | 57.72%  | 47.46%  | 36.81%  | 57.14%  | 46.68%  | 35.99%  |
| Z22          | 75.24%  | 65.00%  | 53.43%  | 74.55%  | 64.36%  | 53.16%  | 74.94%  | 64.82%  | 53.25%  |
| Z23          | 56.78%  | 45.10%  | 34.21%  | 56.82%  | 44.95%  | 34.33%  | 57.47%  | 46.00%  | 34.72%  |
| Z24          | 65.78%  | 54.53%  | 42.77%  | 65.45%  | 53.65%  | 42.63%  | 65.26%  | 54.49%  | 43.58%  |
| Z25          | 77.37%  | 69.13%  | 60.34%  | 78.77%  | 70.19%  | 60.98%  | 78.15%  | 69.87%  | 60.84%  |

## Celda 5 — Heatmap simple por contrato

```
In [72]: # Celda 5 – Heatmap frecuencia operativa por contrato
# =====

plot_data = contract_signal_pivot.copy()

fig, ax = plt.subplots(figsize=(14, 5))

im = ax.imshow(plot_data.values, aspect="auto")

ax.set_title("Frecuencia operativa DIR por contrato")
ax.set_xlabel("Target DIR")
ax.set_ylabel("Contrato")

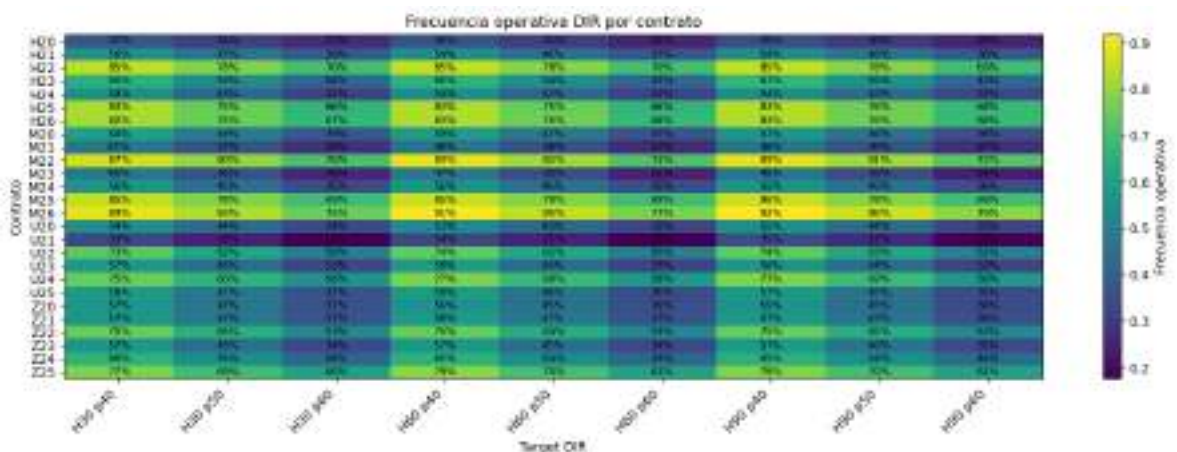
ax.set_xticks(np.arange(len(plot_data.columns)))
ax.set_xticklabels(plot_data.columns, rotation=45, ha="right")

ax.set_yticks(np.arange(len(plot_data.index)))
ax.set_yticklabels(plot_data.index)

for i in range(plot_data.shape[0]):
    for j in range(plot_data.shape[1]):
        value = plot_data.iloc[i, j]
        if pd.notna(value):
            ax.text(
                j,
                i,
                f"{value:.0%}",
                ha="center",
                va="center",
                fontsize=8
            )

fig.colorbar(im, ax=ax, label="Frecuencia operativa")

plt.tight_layout()
plt.show()
```



## Celda 6 — Gráfico por contrato para p50

Este gráfico se enfoca en p50, que venimos usando como variante central.

```
In [73]: # Celda 6 – Frecuencia operativa por contrato para p50
# Gráfico agrupado: 3 filas x 1 columna
# =====
```

```
fig, axes = plt.subplots(
    nrows=3,
    ncols=1,
    figsize=(12, 11),
    sharex=True
)

for ax, target_col in zip(axes, selected_targets_context):
    tmp = df_dir_contract_distribution[
        df_dir_contract_distribution["target"] == target_col
    ].copy()

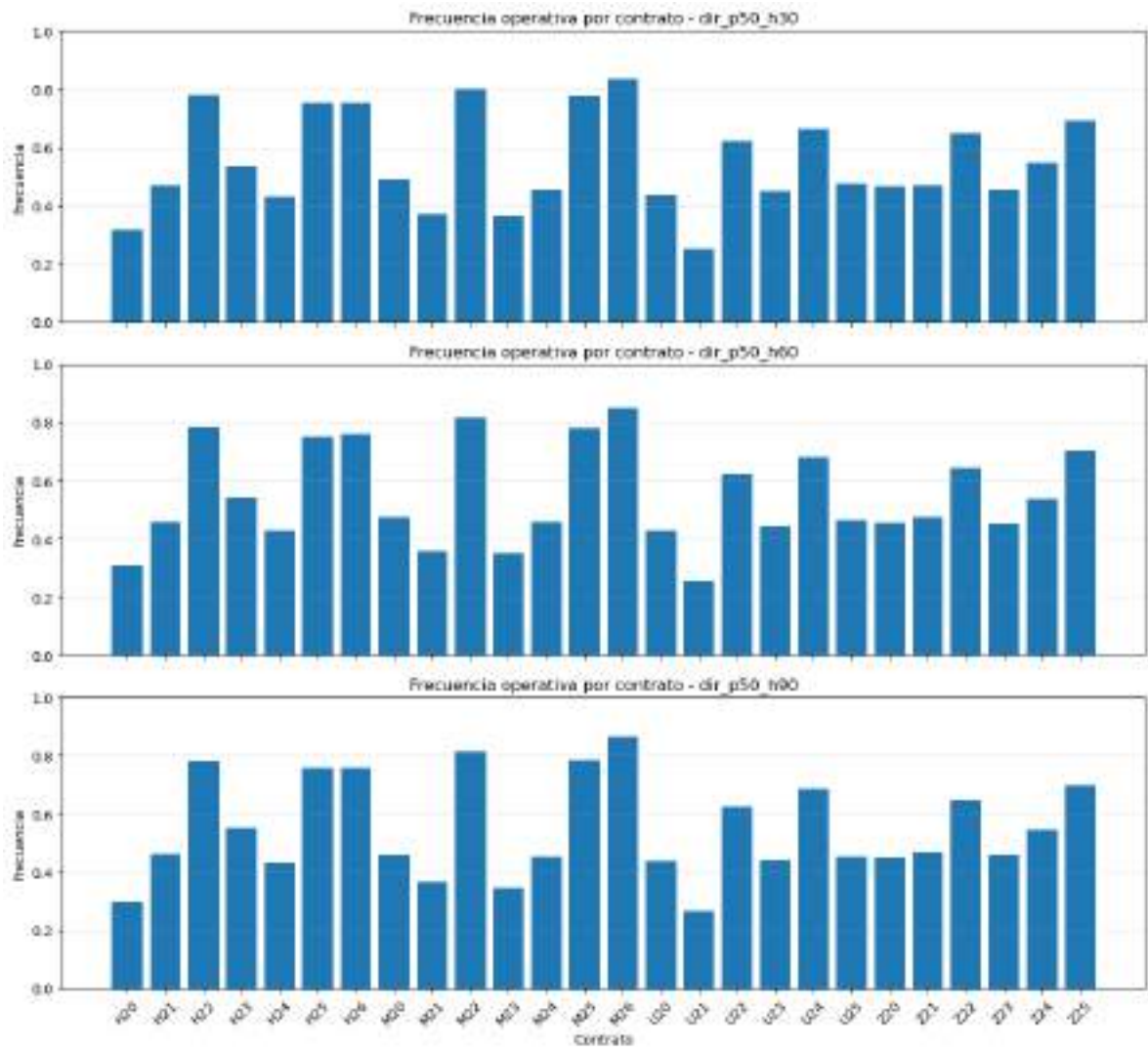
    tmp = tmp.sort_values("contract")

    ax.bar(
        tmp["contract"].astype(str),
        tmp["signal_ratio"]
    )

    ax.set_title(f"Frecuencia operativa por contrato - {target_col}")
    ax.set_xlabel("")
    ax.set_ylabel("Frecuencia")
    ax.set_ylim(0, 1)
    ax.grid(axis="y", alpha=0.3)

axes[-1].set_xlabel("Contrato")

plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



## Celda 7 — Análisis por contract\_code

Usamos contract\_code para revisar si hay diferencias más específicas asociadas al rolling de contratos.

```
In [74]: # Celda 7 – Distribución de DIR por contract_code
# =====

df_dir_contract_code_distribution = summarize_dir_by_context(
    df=df_targets_dir,
    target_cols=dir_target_cols,
    group_cols=["contract_code"]
)

print("Distribución de DIR por contract_code")
print("=" * 80)

cols_show = [
    "contract_code",
    "target",
    "horizon",
    "percentile",
    "valid_rows",
    "nan_rows",
    "signal_ratio",
```

```
    "no_signal_ratio",  
    "long_share_over_signals",  
]  
  
print(  
    df_dir_contract_code_distribution[cols_show]  
    .to_string(index=False, float_format=lambda x: f"{x:.2%}")  
)
```



## Distribución de DIR por contract\_code

| =====             |               |              |            |            |          |              |
|-------------------|---------------|--------------|------------|------------|----------|--------------|
| contract_code     | target        | horizon      | percentile | valid_rows | nan_rows | signal_ratio |
| o no_signal_ratio | long_share    | over_signals |            |            |          |              |
|                   | H dir_p40_h30 | 30           | 40         | 265722     | 12060    | 66.6         |
| 3%                | 33.37%        |              | 48.76%     |            |          |              |
|                   | H dir_p50_h30 | 30           | 50         | 265722     | 12060    | 58.0         |
| 9%                | 41.91%        |              | 48.40%     |            |          |              |
|                   | H dir_p60_h30 | 30           | 60         | 265722     | 12060    | 49.4         |
| 8%                | 50.52%        |              | 47.96%     |            |          |              |
|                   | H dir_p40_h60 | 60           | 40         | 253662     | 24120    | 66.6         |
| 4%                | 33.36%        |              | 48.98%     |            |          |              |
|                   | H dir_p50_h60 | 60           | 50         | 253662     | 24120    | 58.0         |
| 6%                | 41.94%        |              | 48.19%     |            |          |              |
|                   | H dir_p60_h60 | 60           | 60         | 253662     | 24120    | 49.3         |
| 1%                | 50.69%        |              | 47.33%     |            |          |              |
|                   | H dir_p40_h90 | 90           | 40         | 241602     | 36180    | 66.6         |
| 0%                | 33.40%        |              | 48.82%     |            |          |              |
|                   | H dir_p50_h90 | 90           | 50         | 241602     | 36180    | 58.3         |
| 9%                | 41.61%        |              | 48.15%     |            |          |              |
|                   | H dir_p60_h90 | 90           | 60         | 241602     | 36180    | 49.5         |
| 9%                | 50.41%        |              | 47.23%     |            |          |              |
|                   | M dir_p40_h30 | 30           | 40         | 243248     | 11040    | 64.5         |
| 7%                | 35.43%        |              | 50.25%     |            |          |              |
|                   | M dir_p50_h30 | 30           | 50         | 243248     | 11040    | 55.1         |
| 8%                | 44.82%        |              | 49.62%     |            |          |              |
|                   | M dir_p60_h30 | 30           | 60         | 243248     | 11040    | 45.4         |
| 1%                | 54.59%        |              | 48.80%     |            |          |              |
|                   | M dir_p40_h60 | 60           | 40         | 232208     | 22080    | 64.6         |
| 1%                | 35.39%        |              | 50.03%     |            |          |              |
|                   | M dir_p50_h60 | 60           | 50         | 232208     | 22080    | 55.0         |
| 0%                | 45.00%        |              | 49.17%     |            |          |              |
|                   | M dir_p60_h60 | 60           | 60         | 232208     | 22080    | 45.3         |
| 1%                | 54.69%        |              | 48.35%     |            |          |              |
|                   | M dir_p40_h90 | 90           | 40         | 221168     | 33120    | 64.3         |
| 2%                | 35.68%        |              | 49.97%     |            |          |              |
|                   | M dir_p50_h90 | 90           | 50         | 221168     | 33120    | 54.9         |
| 2%                | 45.08%        |              | 49.23%     |            |          |              |
|                   | M dir_p60_h90 | 90           | 60         | 221168     | 33120    | 45.4         |
| 0%                | 54.60%        |              | 48.25%     |            |          |              |
|                   | U dir_p40_h30 | 30           | 40         | 233333     | 10590    | 58.1         |
| 6%                | 41.84%        |              | 49.50%     |            |          |              |
|                   | U dir_p50_h30 | 30           | 50         | 233333     | 10590    | 47.8         |
| 8%                | 52.12%        |              | 48.49%     |            |          |              |
|                   | U dir_p60_h30 | 30           | 60         | 233333     | 10590    | 37.7         |
| 6%                | 62.24%        |              | 47.18%     |            |          |              |
|                   | U dir_p40_h60 | 60           | 40         | 222743     | 21180    | 58.7         |
| 6%                | 41.24%        |              | 48.95%     |            |          |              |
|                   | U dir_p50_h60 | 60           | 50         | 222743     | 21180    | 47.9         |
| 3%                | 52.07%        |              | 47.74%     |            |          |              |
|                   | U dir_p60_h60 | 60           | 60         | 222743     | 21180    | 37.6         |
| 1%                | 62.39%        |              | 46.39%     |            |          |              |
|                   | U dir_p40_h90 | 90           | 40         | 212153     | 31770    | 58.8         |
| 8%                | 41.12%        |              | 48.58%     |            |          |              |
|                   | U dir_p50_h90 | 90           | 50         | 212153     | 31770    | 48.2         |
| 4%                | 51.76%        |              | 47.39%     |            |          |              |
|                   | U dir_p60_h90 | 90           | 60         | 212153     | 31770    | 37.4         |
| 9%                | 62.51%        |              | 46.06%     |            |          |              |
|                   | Z dir_p40_h30 | 30           | 40         | 237299     | 10770    | 64.8         |
| 7%                | 35.13%        |              | 50.06%     |            |          |              |

|    |               |    |        |        |       |      |
|----|---------------|----|--------|--------|-------|------|
|    | Z dir_p50_h30 | 30 | 50     | 237299 | 10770 | 54.6 |
| 4% | 45.36%        |    | 49.67% |        |       |      |
|    | Z dir_p60_h30 | 30 | 60     | 237299 | 10770 | 44.2 |
| 3% | 55.77%        |    | 49.21% |        |       |      |
|    | Z dir_p40_h60 | 60 | 40     | 226529 | 21540 | 64.9 |
| 2% | 35.08%        |    | 50.47% |        |       |      |
|    | Z dir_p50_h60 | 60 | 50     | 226529 | 21540 | 54.4 |
| 1% | 45.59%        |    | 49.92% |        |       |      |
|    | Z dir_p60_h60 | 60 | 60     | 226529 | 21540 | 44.0 |
| 2% | 55.98%        |    | 49.10% |        |       |      |
|    | Z dir_p40_h90 | 90 | 40     | 215759 | 32310 | 64.8 |
| 4% | 35.16%        |    | 50.24% |        |       |      |
|    | Z dir_p50_h90 | 90 | 50     | 215759 | 32310 | 54.5 |
| 7% | 45.43%        |    | 49.58% |        |       |      |
|    | Z dir_p60_h90 | 90 | 60     | 215759 | 32310 | 43.9 |
| 2% | 56.08%        |    | 48.67% |        |       |      |

## Celda 8 — Estabilidad por contract\_code

```
In [75]: # Celda 8 – Métricas de estabilidad por contract_code
# =====

df_contract_code_stability = (
    df_dir_contract_code_distribution
    .groupby(["target", "horizon", "percentile"], as_index=False)
    .agg(
        signal_mean=("signal_ratio", "mean"),
        signal_std=("signal_ratio", "std"),
        signal_min=("signal_ratio", "min"),
        signal_max=("signal_ratio", "max"),
        signal_range=("signal_ratio", lambda x: x.max() - x.min()),
        long_share_mean=("long_share_over_signals", "mean"),
        long_share_range=("long_share_over_signals", lambda x: x.max() - x.min())
    )
    .sort_values(["percentile", "horizon"])
    .reset_index(drop=True)
)

print("Estabilidad de DIR por contract_code")
print("=" * 80)
print(
    df_contract_code_stability.to_string(
        index=False,
        float_format=lambda x: f"{x:.2%}"
    )
)
```

## Estabilidad de DIR por contract\_code

| target       | horizon         | percentile       | signal_mean | signal_std | signal_min | signal_max |
|--------------|-----------------|------------------|-------------|------------|------------|------------|
| signal_range | long_share_mean | long_share_range |             |            |            |            |
| dir_p40_h30  | 30              | 40               | 63.56%      | 3.71%      | 58.16%     | 66.63%     |
| 8.47%        | 49.64%          |                  | 1.49%       |            |            |            |
| dir_p40_h60  | 60              | 40               | 63.73%      | 3.43%      | 58.76%     | 66.64%     |
| 7.87%        | 49.61%          |                  | 1.52%       |            |            |            |
| dir_p40_h90  | 90              | 40               | 63.66%      | 3.33%      | 58.88%     | 66.60%     |
| 7.72%        | 49.40%          |                  | 1.66%       |            |            |            |
| dir_p50_h30  | 30              | 50               | 53.95%      | 4.32%      | 47.88%     | 58.09%     |
| 10.21%       | 49.05%          |                  | 1.27%       |            |            |            |
| dir_p50_h60  | 60              | 50               | 53.85%      | 4.26%      | 47.93%     | 58.06%     |
| 10.13%       | 48.75%          |                  | 2.18%       |            |            |            |
| dir_p50_h90  | 90              | 50               | 54.03%      | 4.23%      | 48.24%     | 58.39%     |
| 10.15%       | 48.59%          |                  | 2.19%       |            |            |            |
| dir_p60_h30  | 30              | 60               | 44.22%      | 4.86%      | 37.76%     | 49.48%     |
| 11.71%       | 48.29%          |                  | 2.03%       |            |            |            |
| dir_p60_h60  | 60              | 60               | 44.06%      | 4.85%      | 37.61%     | 49.31%     |
| 11.70%       | 47.79%          |                  | 2.71%       |            |            |            |
| dir_p60_h90  | 90              | 60               | 44.10%      | 5.02%      | 37.49%     | 49.59%     |
| 12.09%       | 47.55%          |                  | 2.61%       |            |            |            |

## Celda 9 — Análisis year + regime

Esta celda ayuda a detectar si ciertos años tienen más señales en algunos regímenes específicos.

```
In [76]: # Celda 9 – Análisis DIR por year + regime
# =====

df_dir_year_regime_distribution = summarize_dir_by_context(
    df=df_targets_dir,
    target_cols=dir_target_cols,
    group_cols=["year", "regime_name"]
)

print("Distribución de DIR por year + regime")
print("=" * 80)

cols_show = [
    "year",
    "regime_name",
    "target",
    "valid_rows",
    "nan_rows",
    "signal_ratio",
    "no_signal_ratio",
    "long_share_over_signals",
]

print(
    df_dir_year_regime_distribution[cols_show]
    .to_string(index=False, float_format=lambda x: f"{x:.2%}")
)
```

## Distribución de DIR por year + regime

| =====                     |             |             |            |          |              |                 |
|---------------------------|-------------|-------------|------------|----------|--------------|-----------------|
| year                      | regime_name | target      | valid_rows | nan_rows | signal_ratio | no_signal_ratio |
| o long_share_over_signals |             |             |            |          |              |                 |
| 2020                      | Closing     | dir_p40_h30 | 225        | 6525     | 54.67%       | 45.3            |
| 3%                        |             | 47.97%      |            |          |              |                 |
| 2020                      | Closing     | dir_p50_h30 | 225        | 6525     | 45.33%       | 54.6            |
| 7%                        |             | 49.02%      |            |          |              |                 |
| 2020                      | Closing     | dir_p60_h30 | 225        | 6525     | 38.67%       | 61.3            |
| 3%                        |             | 50.57%      |            |          |              |                 |
| 2020                      | Closing     | dir_p40_h60 | 0          | 6750     | NaN          | Na              |
| N                         |             | NaN         |            |          |              |                 |
| 2020                      | Closing     | dir_p50_h60 | 0          | 6750     | NaN          | Na              |
| N                         |             | NaN         |            |          |              |                 |
| 2020                      | Closing     | dir_p60_h60 | 0          | 6750     | NaN          | Na              |
| N                         |             | NaN         |            |          |              |                 |
| 2020                      | Closing     | dir_p40_h90 | 0          | 6750     | NaN          | Na              |
| N                         |             | NaN         |            |          |              |                 |
| 2020                      | Closing     | dir_p50_h90 | 0          | 6750     | NaN          | Na              |
| N                         |             | NaN         |            |          |              |                 |
| 2020                      | Closing     | dir_p60_h90 | 0          | 6750     | NaN          | Na              |
| N                         |             | NaN         |            |          |              |                 |
| 2020                      | Opening     | dir_p40_h30 | 13500      | 0        | 45.37%       | 54.6            |
| 3%                        |             | 47.43%      |            |          |              |                 |
| 2020                      | Opening     | dir_p50_h30 | 13500      | 0        | 35.46%       | 64.5            |
| 4%                        |             | 45.27%      |            |          |              |                 |
| 2020                      | Opening     | dir_p60_h30 | 13500      | 0        | 26.47%       | 73.5            |
| 3%                        |             | 43.21%      |            |          |              |                 |
| 2020                      | Opening     | dir_p40_h60 | 13500      | 0        | 45.21%       | 54.7            |
| 9%                        |             | 46.61%      |            |          |              |                 |
| 2020                      | Opening     | dir_p50_h60 | 13500      | 0        | 35.16%       | 64.8            |
| 4%                        |             | 44.43%      |            |          |              |                 |
| 2020                      | Opening     | dir_p60_h60 | 13500      | 0        | 27.16%       | 72.8            |
| 4%                        |             | 42.85%      |            |          |              |                 |
| 2020                      | Opening     | dir_p40_h90 | 13500      | 0        | 46.04%       | 53.9            |
| 6%                        |             | 46.30%      |            |          |              |                 |
| 2020                      | Opening     | dir_p50_h90 | 13500      | 0        | 36.48%       | 63.5            |
| 2%                        |             | 43.86%      |            |          |              |                 |
| 2020                      | Opening     | dir_p60_h90 | 13500      | 0        | 28.24%       | 71.7            |
| 6%                        |             | 42.05%      |            |          |              |                 |
| 2020                      | Overnight   | dir_p40_h30 | 54000      | 225      | 54.36%       | 45.6            |
| 4%                        |             | 49.57%      |            |          |              |                 |
| 2020                      | Overnight   | dir_p50_h30 | 54000      | 225      | 45.21%       | 54.7            |
| 9%                        |             | 49.05%      |            |          |              |                 |
| 2020                      | Overnight   | dir_p60_h30 | 54000      | 225      | 36.93%       | 63.0            |
| 7%                        |             | 48.65%      |            |          |              |                 |
| 2020                      | Overnight   | dir_p40_h60 | 54000      | 225      | 53.64%       | 46.3            |
| 6%                        |             | 50.00%      |            |          |              |                 |
| 2020                      | Overnight   | dir_p50_h60 | 54000      | 225      | 44.55%       | 55.4            |
| 5%                        |             | 49.48%      |            |          |              |                 |
| 2020                      | Overnight   | dir_p60_h60 | 54000      | 225      | 35.56%       | 64.4            |
| 4%                        |             | 48.56%      |            |          |              |                 |
| 2020                      | Overnight   | dir_p40_h90 | 54000      | 225      | 53.57%       | 46.4            |
| 3%                        |             | 49.19%      |            |          |              |                 |
| 2020                      | Overnight   | dir_p50_h90 | 54000      | 225      | 44.37%       | 55.6            |
| 3%                        |             | 48.92%      |            |          |              |                 |
| 2020                      | Overnight   | dir_p60_h90 | 54000      | 225      | 34.93%       | 65.0            |
| 7%                        |             | 48.45%      |            |          |              |                 |
| 2020                      | Pre-market  | dir_p40_h30 | 13500      | 0        | 50.19%       | 49.8            |
| 1%                        |             | 49.51%      |            |          |              |                 |

|            |            |                       |       |       |        |      |
|------------|------------|-----------------------|-------|-------|--------|------|
| 2020<br>3% | Pre-market | dir_p50_h30<br>49.19% | 13500 | 0     | 41.07% | 58.9 |
| 2020<br>7% | Pre-market | dir_p60_h30<br>49.67% | 13500 | 0     | 32.63% | 67.3 |
| 2020<br>7% | Pre-market | dir_p40_h60<br>49.83% | 13500 | 0     | 47.73% | 52.2 |
| 2020<br>0% | Pre-market | dir_p50_h60<br>46.72% | 13500 | 0     | 37.40% | 62.6 |
| 2020<br>0% | Pre-market | dir_p60_h60<br>47.63% | 13500 | 0     | 28.10% | 71.9 |
| 2020<br>3% | Pre-market | dir_p40_h90<br>49.91% | 13500 | 0     | 44.67% | 55.3 |
| 2020<br>8% | Pre-market | dir_p50_h90<br>51.22% | 13500 | 0     | 34.62% | 65.3 |
| 2020<br>9% | Pre-market | dir_p60_h90<br>49.41% | 13500 | 0     | 25.71% | 74.2 |
| 2020<br>4% | Regular    | dir_p40_h30<br>49.02% | 67500 | 0     | 50.76% | 49.2 |
| 2020<br>1% | Regular    | dir_p50_h30<br>47.33% | 67500 | 0     | 41.49% | 58.5 |
| 2020<br>2% | Regular    | dir_p60_h30<br>45.73% | 67500 | 0     | 32.48% | 67.5 |
| 2020<br>8% | Regular    | dir_p40_h60<br>48.77% | 60975 | 6525  | 49.62% | 50.3 |
| 2020<br>4% | Regular    | dir_p50_h60<br>47.20% | 60975 | 6525  | 40.06% | 59.9 |
| 2020<br>3% | Regular    | dir_p60_h60<br>45.58% | 60975 | 6525  | 31.47% | 68.5 |
| 2020<br>8% | Regular    | dir_p40_h90<br>49.19% | 54225 | 13275 | 49.32% | 50.6 |
| 2020<br>0% | Regular    | dir_p50_h90<br>46.93% | 54225 | 13275 | 39.70% | 60.3 |
| 2020<br>3% | Regular    | dir_p60_h90<br>44.40% | 54225 | 13275 | 30.77% | 69.2 |
| 2021<br>7% | Closing    | dir_p40_h30<br>48.80% | 244   | 7076  | 51.23% | 48.7 |
| 2021<br>3% | Closing    | dir_p50_h30<br>45.45% | 244   | 7076  | 40.57% | 59.4 |
| 2021<br>7% | Closing    | dir_p60_h30<br>43.24% | 244   | 7076  | 30.33% | 69.6 |
| 2021<br>N  | Closing    | dir_p40_h60<br>NaN    | 0     | 7320  | NaN    | Na   |
| 2021<br>N  | Closing    | dir_p50_h60<br>NaN    | 0     | 7320  | NaN    | Na   |
| 2021<br>N  | Closing    | dir_p60_h60<br>NaN    | 0     | 7320  | NaN    | Na   |
| 2021<br>N  | Closing    | dir_p40_h90<br>NaN    | 0     | 7320  | NaN    | Na   |
| 2021<br>N  | Closing    | dir_p50_h90<br>NaN    | 0     | 7320  | NaN    | Na   |
| 2021<br>N  | Closing    | dir_p60_h90<br>NaN    | 0     | 7320  | NaN    | Na   |
| 2021<br>3% | Opening    | dir_p40_h30<br>48.53% | 14640 | 0     | 49.97% | 50.0 |
| 2021<br>7% | Opening    | dir_p50_h30<br>46.82% | 14640 | 0     | 39.83% | 60.1 |
| 2021<br>2% | Opening    | dir_p60_h30<br>44.40% | 14640 | 0     | 30.58% | 69.4 |
| 2021<br>4% | Opening    | dir_p40_h60<br>47.24% | 14640 | 0     | 49.56% | 50.4 |

|            |                                  |       |       |        |      |
|------------|----------------------------------|-------|-------|--------|------|
| 2021<br>3% | Opening dir_p50_h60<br>44.52%    | 14640 | 0     | 39.57% | 60.4 |
| 2021<br>5% | Opening dir_p60_h60<br>41.84%    | 14640 | 0     | 30.15% | 69.8 |
| 2021<br>4% | Opening dir_p40_h90<br>46.45%    | 14640 | 0     | 50.86% | 49.1 |
| 2021<br>6% | Opening dir_p50_h90<br>43.69%    | 14640 | 0     | 40.74% | 59.2 |
| 2021<br>2% | Opening dir_p60_h90<br>40.63%    | 14640 | 0     | 30.78% | 69.2 |
| 2021<br>3% | Overnight dir_p40_h30<br>49.92%  | 58560 | 244   | 49.37% | 50.6 |
| 2021<br>6% | Overnight dir_p50_h30<br>48.77%  | 58560 | 244   | 40.04% | 59.9 |
| 2021<br>1% | Overnight dir_p60_h30<br>47.34%  | 58560 | 244   | 31.89% | 68.1 |
| 2021<br>7% | Overnight dir_p40_h60<br>48.21%  | 58560 | 244   | 49.83% | 50.1 |
| 2021<br>2% | Overnight dir_p50_h60<br>47.27%  | 58560 | 244   | 40.08% | 59.9 |
| 2021<br>6% | Overnight dir_p60_h60<br>46.08%  | 58560 | 244   | 30.84% | 69.1 |
| 2021<br>8% | Overnight dir_p40_h90<br>48.36%  | 58560 | 244   | 50.72% | 49.2 |
| 2021<br>3% | Overnight dir_p50_h90<br>47.58%  | 58560 | 244   | 41.07% | 58.9 |
| 2021<br>6% | Overnight dir_p60_h90<br>47.02%  | 58560 | 244   | 31.44% | 68.5 |
| 2021<br>2% | Pre-market dir_p40_h30<br>50.41% | 14640 | 0     | 53.78% | 46.2 |
| 2021<br>5% | Pre-market dir_p50_h30<br>49.90% | 14640 | 0     | 43.55% | 56.4 |
| 2021<br>3% | Pre-market dir_p60_h30<br>49.53% | 14640 | 0     | 33.37% | 66.6 |
| 2021<br>8% | Pre-market dir_p40_h60<br>50.53% | 14640 | 0     | 52.52% | 47.4 |
| 2021<br>5% | Pre-market dir_p50_h60<br>49.84% | 14640 | 0     | 42.75% | 57.2 |
| 2021<br>4% | Pre-market dir_p60_h60<br>49.26% | 14640 | 0     | 32.16% | 67.8 |
| 2021<br>8% | Pre-market dir_p40_h90<br>48.37% | 14640 | 0     | 51.82% | 48.1 |
| 2021<br>5% | Pre-market dir_p50_h90<br>49.14% | 14640 | 0     | 40.85% | 59.1 |
| 2021<br>7% | Pre-market dir_p60_h90<br>48.97% | 14640 | 0     | 30.03% | 69.9 |
| 2021<br>9% | Regular dir_p40_h30<br>48.69%    | 73200 | 0     | 47.51% | 52.4 |
| 2021<br>4% | Regular dir_p50_h30<br>47.73%    | 73200 | 0     | 38.26% | 61.7 |
| 2021<br>4% | Regular dir_p60_h30<br>46.41%    | 73200 | 0     | 29.46% | 70.5 |
| 2021<br>0% | Regular dir_p40_h60<br>49.86%    | 66124 | 7076  | 47.00% | 53.0 |
| 2021<br>0% | Regular dir_p50_h60<br>48.10%    | 66124 | 7076  | 37.70% | 62.3 |
| 2021<br>6% | Regular dir_p60_h60<br>45.87%    | 66124 | 7076  | 29.24% | 70.7 |
| 2021<br>1% | Regular dir_p40_h90<br>48.76%    | 58804 | 14396 | 46.59% | 53.4 |

|            |            |                       |       |       |        |      |
|------------|------------|-----------------------|-------|-------|--------|------|
| 2021<br>4% | Regular    | dir_p50_h90<br>47.11% | 58804 | 14396 | 37.86% | 62.1 |
| 2021<br>4% | Regular    | dir_p60_h90<br>45.07% | 58804 | 14396 | 29.06% | 70.9 |
| 2022<br>2% | Closing    | dir_p40_h30<br>43.46% | 240   | 6960  | 79.58% | 20.4 |
| 2022<br>0% | Closing    | dir_p50_h30<br>43.10% | 240   | 6960  | 72.50% | 27.5 |
| 2022<br>8% | Closing    | dir_p60_h30<br>41.06% | 240   | 6960  | 62.92% | 37.0 |
| 2022<br>N  | Closing    | dir_p40_h60<br>NaN    | 0     | 7200  | NaN    | Na   |
| 2022<br>N  | Closing    | dir_p50_h60<br>NaN    | 0     | 7200  | NaN    | Na   |
| 2022<br>N  | Closing    | dir_p60_h60<br>NaN    | 0     | 7200  | NaN    | Na   |
| 2022<br>N  | Closing    | dir_p40_h90<br>NaN    | 0     | 7200  | NaN    | Na   |
| 2022<br>N  | Closing    | dir_p50_h90<br>NaN    | 0     | 7200  | NaN    | Na   |
| 2022<br>N  | Closing    | dir_p60_h90<br>NaN    | 0     | 7200  | NaN    | Na   |
| 2022<br>7% | Opening    | dir_p40_h30<br>49.30% | 14400 | 0     | 81.53% | 18.4 |
| 2022<br>6% | Opening    | dir_p50_h30<br>49.11% | 14400 | 0     | 72.94% | 27.0 |
| 2022<br>0% | Opening    | dir_p60_h30<br>48.63% | 14400 | 0     | 62.90% | 37.1 |
| 2022<br>1% | Opening    | dir_p40_h60<br>49.53% | 14400 | 0     | 81.89% | 18.1 |
| 2022<br>4% | Opening    | dir_p50_h60<br>49.62% | 14400 | 0     | 73.26% | 26.7 |
| 2022<br>0% | Opening    | dir_p60_h60<br>49.69% | 14400 | 0     | 62.90% | 37.1 |
| 2022<br>8% | Opening    | dir_p40_h90<br>48.23% | 14400 | 0     | 81.92% | 18.0 |
| 2022<br>2% | Opening    | dir_p50_h90<br>48.27% | 14400 | 0     | 73.38% | 26.6 |
| 2022<br>1% | Opening    | dir_p60_h90<br>47.83% | 14400 | 0     | 63.09% | 36.9 |
| 2022<br>6% | Overnight  | dir_p40_h30<br>48.72% | 57600 | 240   | 80.44% | 19.5 |
| 2022<br>4% | Overnight  | dir_p50_h30<br>48.43% | 57600 | 240   | 71.56% | 28.4 |
| 2022<br>6% | Overnight  | dir_p60_h30<br>48.17% | 57600 | 240   | 61.44% | 38.5 |
| 2022<br>0% | Overnight  | dir_p40_h60<br>48.34% | 57600 | 240   | 81.10% | 18.9 |
| 2022<br>2% | Overnight  | dir_p50_h60<br>48.07% | 57600 | 240   | 71.98% | 28.0 |
| 2022<br>8% | Overnight  | dir_p60_h60<br>47.48% | 57600 | 240   | 61.22% | 38.7 |
| 2022<br>7% | Overnight  | dir_p40_h90<br>46.81% | 57600 | 240   | 80.33% | 19.6 |
| 2022<br>8% | Overnight  | dir_p50_h90<br>46.50% | 57600 | 240   | 71.12% | 28.8 |
| 2022<br>8% | Overnight  | dir_p60_h90<br>46.56% | 57600 | 240   | 60.42% | 39.5 |
| 2022<br>4% | Pre-market | dir_p40_h30<br>46.07% | 14400 | 0     | 76.26% | 23.7 |

|            |            |                       |       |       |        |      |
|------------|------------|-----------------------|-------|-------|--------|------|
| 2022<br>7% | Pre-market | dir_p50_h30<br>45.98% | 14400 | 0     | 68.03% | 31.9 |
| 2022<br>7% | Pre-market | dir_p60_h30<br>46.25% | 14400 | 0     | 58.53% | 41.4 |
| 2022<br>5% | Pre-market | dir_p40_h60<br>46.40% | 14400 | 0     | 82.35% | 17.6 |
| 2022<br>1% | Pre-market | dir_p50_h60<br>46.83% | 14400 | 0     | 73.69% | 26.3 |
| 2022<br>9% | Pre-market | dir_p60_h60<br>47.59% | 14400 | 0     | 63.51% | 36.4 |
| 2022<br>5% | Pre-market | dir_p40_h90<br>45.64% | 14400 | 0     | 84.35% | 15.6 |
| 2022<br>3% | Pre-market | dir_p50_h90<br>45.41% | 14400 | 0     | 76.17% | 23.8 |
| 2022<br>1% | Pre-market | dir_p60_h90<br>45.57% | 14400 | 0     | 65.89% | 34.1 |
| 2022<br>7% | Regular    | dir_p40_h30<br>49.61% | 72000 | 0     | 81.13% | 18.8 |
| 2022<br>2% | Regular    | dir_p50_h30<br>49.43% | 72000 | 0     | 72.28% | 27.7 |
| 2022<br>3% | Regular    | dir_p60_h30<br>48.89% | 72000 | 0     | 61.57% | 38.4 |
| 2022<br>4% | Regular    | dir_p40_h60<br>49.12% | 65040 | 6960  | 81.36% | 18.6 |
| 2022<br>1% | Regular    | dir_p50_h60<br>48.59% | 65040 | 6960  | 71.79% | 28.2 |
| 2022<br>1% | Regular    | dir_p60_h60<br>48.12% | 65040 | 6960  | 61.19% | 38.8 |
| 2022<br>3% | Regular    | dir_p40_h90<br>48.82% | 57840 | 14160 | 81.27% | 18.7 |
| 2022<br>3% | Regular    | dir_p50_h90<br>48.45% | 57840 | 14160 | 71.97% | 28.0 |
| 2022<br>0% | Regular    | dir_p60_h90<br>47.93% | 57840 | 14160 | 61.40% | 38.6 |
| 2023<br>5% | Closing    | dir_p40_h30<br>49.59% | 237   | 6873  | 51.05% | 48.9 |
| 2023<br>0% | Closing    | dir_p50_h30<br>49.45% | 237   | 6873  | 38.40% | 61.6 |
| 2023<br>0% | Closing    | dir_p60_h30<br>56.25% | 237   | 6873  | 27.00% | 73.0 |
| 2023<br>N  | Closing    | dir_p40_h60<br>NaN    | 0     | 7110  | NaN    | Na   |
| 2023<br>N  | Closing    | dir_p50_h60<br>NaN    | 0     | 7110  | NaN    | Na   |
| 2023<br>N  | Closing    | dir_p60_h60<br>NaN    | 0     | 7110  | NaN    | Na   |
| 2023<br>N  | Closing    | dir_p40_h90<br>NaN    | 0     | 7110  | NaN    | Na   |
| 2023<br>N  | Closing    | dir_p50_h90<br>NaN    | 0     | 7110  | NaN    | Na   |
| 2023<br>N  | Closing    | dir_p60_h90<br>NaN    | 0     | 7110  | NaN    | Na   |
| 2023<br>4% | Opening    | dir_p40_h30<br>49.20% | 14220 | 0     | 57.76% | 42.2 |
| 2023<br>6% | Opening    | dir_p50_h30<br>48.15% | 14220 | 0     | 46.24% | 53.7 |
| 2023<br>0% | Opening    | dir_p60_h30<br>46.86% | 14220 | 0     | 34.80% | 65.2 |
| 2023<br>9% | Opening    | dir_p40_h60<br>49.35% | 14220 | 0     | 57.01% | 42.9 |



|            |            |             |       |       |        |      |
|------------|------------|-------------|-------|-------|--------|------|
| 2023<br>0% | Opening    | dir_p50_h60 | 14220 | 0     | 44.80% | 55.2 |
|            |            | 48.41%      |       |       |        |      |
| 2023<br>1% | Opening    | dir_p60_h60 | 14220 | 0     | 34.09% | 65.9 |
|            |            | 47.33%      |       |       |        |      |
| 2023<br>7% | Opening    | dir_p40_h90 | 14220 | 0     | 56.13% | 43.8 |
|            |            | 50.45%      |       |       |        |      |
| 2023<br>5% | Opening    | dir_p50_h90 | 14220 | 0     | 44.55% | 55.4 |
|            |            | 49.44%      |       |       |        |      |
| 2023<br>4% | Opening    | dir_p60_h90 | 14220 | 0     | 33.16% | 66.8 |
|            |            | 48.03%      |       |       |        |      |
| 2023<br>4% | Overnight  | dir_p40_h30 | 56880 | 237   | 53.06% | 46.9 |
|            |            | 49.49%      |       |       |        |      |
| 2023<br>3% | Overnight  | dir_p50_h30 | 56880 | 237   | 41.47% | 58.5 |
|            |            | 48.86%      |       |       |        |      |
| 2023<br>1% | Overnight  | dir_p60_h30 | 56880 | 237   | 31.59% | 68.4 |
|            |            | 47.65%      |       |       |        |      |
| 2023<br>1% | Overnight  | dir_p40_h60 | 56880 | 237   | 52.89% | 47.1 |
|            |            | 49.96%      |       |       |        |      |
| 2023<br>2% | Overnight  | dir_p50_h60 | 56880 | 237   | 41.38% | 58.6 |
|            |            | 49.22%      |       |       |        |      |
| 2023<br>4% | Overnight  | dir_p60_h60 | 56880 | 237   | 31.36% | 68.6 |
|            |            | 48.07%      |       |       |        |      |
| 2023<br>9% | Overnight  | dir_p40_h90 | 56880 | 237   | 52.91% | 47.0 |
|            |            | 51.32%      |       |       |        |      |
| 2023<br>8% | Overnight  | dir_p50_h90 | 56880 | 237   | 42.52% | 57.4 |
|            |            | 51.35%      |       |       |        |      |
| 2023<br>1% | Overnight  | dir_p60_h90 | 56880 | 237   | 31.79% | 68.2 |
|            |            | 51.77%      |       |       |        |      |
| 2023<br>5% | Pre-market | dir_p40_h30 | 14220 | 0     | 56.15% | 43.8 |
|            |            | 56.53%      |       |       |        |      |
| 2023<br>9% | Pre-market | dir_p50_h30 | 14220 | 0     | 44.81% | 55.1 |
|            |            | 57.42%      |       |       |        |      |
| 2023<br>5% | Pre-market | dir_p60_h30 | 14220 | 0     | 33.85% | 66.1 |
|            |            | 58.38%      |       |       |        |      |
| 2023<br>1% | Pre-market | dir_p40_h60 | 14220 | 0     | 54.29% | 45.7 |
|            |            | 57.19%      |       |       |        |      |
| 2023<br>9% | Pre-market | dir_p50_h60 | 14220 | 0     | 42.41% | 57.5 |
|            |            | 57.31%      |       |       |        |      |
| 2023<br>8% | Pre-market | dir_p60_h60 | 14220 | 0     | 32.12% | 67.8 |
|            |            | 57.44%      |       |       |        |      |
| 2023<br>4% | Pre-market | dir_p40_h90 | 14220 | 0     | 54.16% | 45.8 |
|            |            | 56.96%      |       |       |        |      |
| 2023<br>6% | Pre-market | dir_p50_h90 | 14220 | 0     | 42.34% | 57.6 |
|            |            | 55.01%      |       |       |        |      |
| 2023<br>1% | Pre-market | dir_p60_h90 | 14220 | 0     | 31.39% | 68.6 |
|            |            | 52.69%      |       |       |        |      |
| 2023<br>8% | Regular    | dir_p40_h30 | 71100 | 0     | 56.92% | 43.0 |
|            |            | 51.67%      |       |       |        |      |
| 2023<br>7% | Regular    | dir_p50_h30 | 71100 | 0     | 45.23% | 54.7 |
|            |            | 50.88%      |       |       |        |      |
| 2023<br>5% | Regular    | dir_p60_h30 | 71100 | 0     | 33.75% | 66.2 |
|            |            | 49.66%      |       |       |        |      |
| 2023<br>0% | Regular    | dir_p40_h60 | 64227 | 6873  | 57.50% | 42.5 |
|            |            | 52.38%      |       |       |        |      |
| 2023<br>0% | Regular    | dir_p50_h60 | 64227 | 6873  | 45.10% | 54.9 |
|            |            | 50.82%      |       |       |        |      |
| 2023<br>8% | Regular    | dir_p60_h60 | 64227 | 6873  | 33.82% | 66.1 |
|            |            | 49.68%      |       |       |        |      |
| 2023<br>7% | Regular    | dir_p40_h90 | 57117 | 13983 | 57.73% | 42.2 |
|            |            | 53.04%      |       |       |        |      |

|            |            |             |       |       |        |      |
|------------|------------|-------------|-------|-------|--------|------|
| 2023<br>2% | Regular    | dir_p50_h90 | 57117 | 13983 | 45.48% | 54.5 |
|            |            | 51.87%      |       |       |        |      |
| 2023<br>5% | Regular    | dir_p60_h90 | 57117 | 13983 | 33.95% | 66.0 |
|            |            | 49.97%      |       |       |        |      |
| 2024<br>3% | Closing    | dir_p40_h30 | 239   | 6931  | 65.27% | 34.7 |
|            |            | 53.21%      |       |       |        |      |
| 2024<br>3% | Closing    | dir_p50_h30 | 239   | 6931  | 53.97% | 46.0 |
|            |            | 53.49%      |       |       |        |      |
| 2024<br>4% | Closing    | dir_p60_h30 | 239   | 6931  | 42.26% | 57.7 |
|            |            | 53.47%      |       |       |        |      |
| 2024<br>N  | Closing    | dir_p40_h60 | 0     | 7170  | NaN    | Na   |
|            |            | NaN         |       |       |        |      |
| 2024<br>N  | Closing    | dir_p50_h60 | 0     | 7170  | NaN    | Na   |
|            |            | NaN         |       |       |        |      |
| 2024<br>N  | Closing    | dir_p60_h60 | 0     | 7170  | NaN    | Na   |
|            |            | NaN         |       |       |        |      |
| 2024<br>N  | Closing    | dir_p40_h90 | 0     | 7170  | NaN    | Na   |
|            |            | NaN         |       |       |        |      |
| 2024<br>N  | Closing    | dir_p50_h90 | 0     | 7170  | NaN    | Na   |
|            |            | NaN         |       |       |        |      |
| 2024<br>N  | Closing    | dir_p60_h90 | 0     | 7170  | NaN    | Na   |
|            |            | NaN         |       |       |        |      |
| 2024<br>5% | Opening    | dir_p40_h30 | 14340 | 0     | 65.65% | 34.3 |
|            |            | 48.50%      |       |       |        |      |
| 2024<br>9% | Opening    | dir_p50_h30 | 14340 | 0     | 55.31% | 44.6 |
|            |            | 47.38%      |       |       |        |      |
| 2024<br>2% | Opening    | dir_p60_h30 | 14340 | 0     | 44.48% | 55.5 |
|            |            | 45.41%      |       |       |        |      |
| 2024<br>2% | Opening    | dir_p40_h60 | 14340 | 0     | 66.58% | 33.4 |
|            |            | 49.27%      |       |       |        |      |
| 2024<br>9% | Opening    | dir_p50_h60 | 14340 | 0     | 56.41% | 43.5 |
|            |            | 47.08%      |       |       |        |      |
| 2024<br>7% | Opening    | dir_p60_h60 | 14340 | 0     | 45.93% | 54.0 |
|            |            | 43.95%      |       |       |        |      |
| 2024<br>1% | Opening    | dir_p40_h90 | 14340 | 0     | 65.19% | 34.8 |
|            |            | 48.97%      |       |       |        |      |
| 2024<br>9% | Opening    | dir_p50_h90 | 14340 | 0     | 55.11% | 44.8 |
|            |            | 46.72%      |       |       |        |      |
| 2024<br>9% | Opening    | dir_p60_h90 | 14340 | 0     | 44.71% | 55.2 |
|            |            | 43.67%      |       |       |        |      |
| 2024<br>7% | Overnight  | dir_p40_h30 | 57360 | 239   | 62.43% | 37.5 |
|            |            | 51.55%      |       |       |        |      |
| 2024<br>2% | Overnight  | dir_p50_h30 | 57360 | 239   | 51.38% | 48.6 |
|            |            | 51.34%      |       |       |        |      |
| 2024<br>3% | Overnight  | dir_p60_h30 | 57360 | 239   | 41.07% | 58.9 |
|            |            | 51.01%      |       |       |        |      |
| 2024<br>1% | Overnight  | dir_p40_h60 | 57360 | 239   | 62.89% | 37.1 |
|            |            | 52.23%      |       |       |        |      |
| 2024<br>6% | Overnight  | dir_p50_h60 | 57360 | 239   | 52.04% | 47.9 |
|            |            | 51.92%      |       |       |        |      |
| 2024<br>8% | Overnight  | dir_p60_h60 | 57360 | 239   | 41.02% | 58.9 |
|            |            | 50.68%      |       |       |        |      |
| 2024<br>6% | Overnight  | dir_p40_h90 | 57360 | 239   | 62.84% | 37.1 |
|            |            | 52.06%      |       |       |        |      |
| 2024<br>2% | Overnight  | dir_p50_h90 | 57360 | 239   | 52.38% | 47.6 |
|            |            | 50.68%      |       |       |        |      |
| 2024<br>1% | Overnight  | dir_p60_h90 | 57360 | 239   | 41.39% | 58.6 |
|            |            | 48.89%      |       |       |        |      |
| 2024<br>0% | Pre-market | dir_p40_h30 | 14340 | 0     | 64.90% | 35.1 |
|            |            | 45.36%      |       |       |        |      |

|            |            |                       |       |       |        |      |
|------------|------------|-----------------------|-------|-------|--------|------|
| 2024<br>9% | Pre-market | dir_p50_h30<br>43.75% | 14340 | 0     | 53.91% | 46.0 |
| 2024<br>5% | Pre-market | dir_p60_h30<br>42.98% | 14340 | 0     | 41.65% | 58.3 |
| 2024<br>6% | Pre-market | dir_p40_h60<br>45.44% | 14340 | 0     | 62.94% | 37.0 |
| 2024<br>7% | Pre-market | dir_p50_h60<br>45.31% | 14340 | 0     | 53.53% | 46.4 |
| 2024<br>9% | Pre-market | dir_p60_h60<br>45.06% | 14340 | 0     | 44.21% | 55.7 |
| 2024<br>8% | Pre-market | dir_p40_h90<br>47.21% | 14340 | 0     | 64.32% | 35.6 |
| 2024<br>3% | Pre-market | dir_p50_h90<br>45.90% | 14340 | 0     | 55.57% | 44.4 |
| 2024<br>9% | Pre-market | dir_p60_h90<br>44.47% | 14340 | 0     | 46.51% | 53.4 |
| 2024<br>7% | Regular    | dir_p40_h30<br>49.47% | 71700 | 0     | 64.93% | 35.0 |
| 2024<br>1% | Regular    | dir_p50_h30<br>48.93% | 71700 | 0     | 54.59% | 45.4 |
| 2024<br>0% | Regular    | dir_p60_h30<br>47.90% | 71700 | 0     | 43.30% | 56.7 |
| 2024<br>5% | Regular    | dir_p40_h60<br>49.03% | 64769 | 6931  | 65.75% | 34.2 |
| 2024<br>5% | Regular    | dir_p50_h60<br>48.16% | 64769 | 6931  | 54.85% | 45.1 |
| 2024<br>8% | Regular    | dir_p60_h60<br>46.85% | 64769 | 6931  | 44.32% | 55.6 |
| 2024<br>0% | Regular    | dir_p40_h90<br>48.67% | 57599 | 14101 | 65.70% | 34.3 |
| 2024<br>5% | Regular    | dir_p50_h90<br>47.88% | 57599 | 14101 | 55.25% | 44.7 |
| 2024<br>7% | Regular    | dir_p60_h90<br>46.54% | 57599 | 14101 | 44.93% | 55.0 |
| 2025<br>9% | Closing    | dir_p40_h30<br>47.98% | 227   | 6583  | 76.21% | 23.7 |
| 2025<br>6% | Closing    | dir_p50_h30<br>49.31% | 227   | 6583  | 63.44% | 36.5 |
| 2025<br>6% | Closing    | dir_p60_h30<br>51.64% | 227   | 6583  | 53.74% | 46.2 |
| 2025<br>N  | Closing    | dir_p40_h60<br>NaN    | 0     | 6810  | NaN    | Na   |
| 2025<br>N  | Closing    | dir_p50_h60<br>NaN    | 0     | 6810  | NaN    | Na   |
| 2025<br>N  | Closing    | dir_p60_h60<br>NaN    | 0     | 6810  | NaN    | Na   |
| 2025<br>N  | Closing    | dir_p40_h90<br>NaN    | 0     | 6810  | NaN    | Na   |
| 2025<br>N  | Closing    | dir_p50_h90<br>NaN    | 0     | 6810  | NaN    | Na   |
| 2025<br>N  | Closing    | dir_p60_h90<br>NaN    | 0     | 6810  | NaN    | Na   |
| 2025<br>1% | Opening    | dir_p40_h30<br>50.66% | 13620 | 0     | 75.99% | 24.0 |
| 2025<br>9% | Opening    | dir_p50_h30<br>50.09% | 13620 | 0     | 67.31% | 32.6 |
| 2025<br>8% | Opening    | dir_p60_h30<br>49.46% | 13620 | 0     | 58.62% | 41.3 |
| 2025<br>4% | Opening    | dir_p40_h60<br>49.24% | 13620 | 0     | 77.06% | 22.9 |

|            |            |             |       |       |        |      |
|------------|------------|-------------|-------|-------|--------|------|
| 2025<br>1% | Opening    | dir_p50_h60 | 13620 | 0     | 67.59% | 32.4 |
|            |            | 48.33%      |       |       |        |      |
| 2025<br>6% | Opening    | dir_p60_h60 | 13620 | 0     | 58.24% | 41.7 |
|            |            | 47.06%      |       |       |        |      |
| 2025<br>6% | Opening    | dir_p40_h90 | 13620 | 0     | 76.64% | 23.3 |
|            |            | 47.87%      |       |       |        |      |
| 2025<br>5% | Opening    | dir_p50_h90 | 13620 | 0     | 67.55% | 32.4 |
|            |            | 46.50%      |       |       |        |      |
| 2025<br>2% | Opening    | dir_p60_h90 | 13620 | 0     | 58.08% | 41.9 |
|            |            | 44.99%      |       |       |        |      |
| 2025<br>4% | Overnight  | dir_p40_h30 | 54480 | 227   | 76.36% | 23.6 |
|            |            | 50.56%      |       |       |        |      |
| 2025<br>6% | Overnight  | dir_p50_h30 | 54480 | 227   | 67.44% | 32.5 |
|            |            | 50.16%      |       |       |        |      |
| 2025<br>6% | Overnight  | dir_p60_h30 | 54480 | 227   | 58.24% | 41.7 |
|            |            | 49.80%      |       |       |        |      |
| 2025<br>1% | Overnight  | dir_p40_h60 | 54480 | 227   | 76.89% | 23.1 |
|            |            | 50.46%      |       |       |        |      |
| 2025<br>0% | Overnight  | dir_p50_h60 | 54480 | 227   | 67.30% | 32.7 |
|            |            | 49.96%      |       |       |        |      |
| 2025<br>1% | Overnight  | dir_p60_h60 | 54480 | 227   | 56.69% | 43.3 |
|            |            | 49.10%      |       |       |        |      |
| 2025<br>8% | Overnight  | dir_p40_h90 | 54480 | 227   | 75.52% | 24.4 |
|            |            | 49.51%      |       |       |        |      |
| 2025<br>9% | Overnight  | dir_p50_h90 | 54480 | 227   | 66.01% | 33.9 |
|            |            | 49.25%      |       |       |        |      |
| 2025<br>6% | Overnight  | dir_p60_h90 | 54480 | 227   | 56.14% | 43.8 |
|            |            | 48.77%      |       |       |        |      |
| 2025<br>2% | Pre-market | dir_p40_h30 | 13620 | 0     | 74.88% | 25.1 |
|            |            | 46.86%      |       |       |        |      |
| 2025<br>9% | Pre-market | dir_p50_h30 | 13620 | 0     | 66.31% | 33.6 |
|            |            | 46.38%      |       |       |        |      |
| 2025<br>1% | Pre-market | dir_p60_h30 | 13620 | 0     | 56.59% | 43.4 |
|            |            | 45.39%      |       |       |        |      |
| 2025<br>1% | Pre-market | dir_p40_h60 | 13620 | 0     | 76.19% | 23.8 |
|            |            | 46.59%      |       |       |        |      |
| 2025<br>6% | Pre-market | dir_p50_h60 | 13620 | 0     | 68.74% | 31.2 |
|            |            | 46.22%      |       |       |        |      |
| 2025<br>2% | Pre-market | dir_p60_h60 | 13620 | 0     | 59.88% | 40.1 |
|            |            | 46.93%      |       |       |        |      |
| 2025<br>6% | Pre-market | dir_p40_h90 | 13620 | 0     | 77.14% | 22.8 |
|            |            | 48.98%      |       |       |        |      |
| 2025<br>9% | Pre-market | dir_p50_h90 | 13620 | 0     | 69.71% | 30.2 |
|            |            | 49.06%      |       |       |        |      |
| 2025<br>9% | Pre-market | dir_p60_h90 | 13620 | 0     | 61.81% | 38.1 |
|            |            | 49.00%      |       |       |        |      |
| 2025<br>4% | Regular    | dir_p40_h30 | 68100 | 0     | 72.86% | 27.1 |
|            |            | 49.51%      |       |       |        |      |
| 2025<br>3% | Regular    | dir_p50_h30 | 68100 | 0     | 64.67% | 35.3 |
|            |            | 49.18%      |       |       |        |      |
| 2025<br>8% | Regular    | dir_p60_h30 | 68100 | 0     | 55.72% | 44.2 |
|            |            | 48.51%      |       |       |        |      |
| 2025<br>2% | Regular    | dir_p40_h60 | 61517 | 6583  | 73.08% | 26.9 |
|            |            | 49.82%      |       |       |        |      |
| 2025<br>6% | Regular    | dir_p50_h60 | 61517 | 6583  | 64.34% | 35.6 |
|            |            | 48.70%      |       |       |        |      |
| 2025<br>0% | Regular    | dir_p60_h60 | 61517 | 6583  | 55.80% | 44.2 |
|            |            | 47.44%      |       |       |        |      |
| 2025<br>2% | Regular    | dir_p40_h90 | 54707 | 13393 | 73.38% | 26.6 |
|            |            | 50.06%      |       |       |        |      |

|            |            |                       |       |       |        |      |
|------------|------------|-----------------------|-------|-------|--------|------|
| 2025<br>3% | Regular    | dir_p50_h90<br>49.00% | 54707 | 13393 | 64.97% | 35.0 |
| 2025<br>2% | Regular    | dir_p60_h90<br>47.47% | 54707 | 13393 | 56.18% | 43.8 |
| 2026<br>3% | Closing    | dir_p40_h30<br>56.36% | 70    | 2030  | 78.57% | 21.4 |
| 2026<br>6% | Closing    | dir_p50_h30<br>61.70% | 70    | 2030  | 67.14% | 32.8 |
| 2026<br>7% | Closing    | dir_p60_h30<br>62.79% | 70    | 2030  | 61.43% | 38.5 |
| 2026<br>N  | Closing    | dir_p40_h60<br>NaN    | 0     | 2100  | NaN    | Na   |
| 2026<br>N  | Closing    | dir_p50_h60<br>NaN    | 0     | 2100  | NaN    | Na   |
| 2026<br>N  | Closing    | dir_p60_h60<br>NaN    | 0     | 2100  | NaN    | Na   |
| 2026<br>N  | Closing    | dir_p40_h90<br>NaN    | 0     | 2100  | NaN    | Na   |
| 2026<br>N  | Closing    | dir_p50_h90<br>NaN    | 0     | 2100  | NaN    | Na   |
| 2026<br>N  | Closing    | dir_p60_h90<br>NaN    | 0     | 2100  | NaN    | Na   |
| 2026<br>6% | Opening    | dir_p40_h30<br>52.19% | 4200  | 0     | 90.74% | 9.2  |
| 2026<br>0% | Opening    | dir_p50_h30<br>51.82% | 4200  | 0     | 84.60% | 15.4 |
| 2026<br>6% | Opening    | dir_p60_h30<br>51.44% | 4200  | 0     | 76.64% | 23.3 |
| 2026<br>6% | Opening    | dir_p40_h60<br>51.91% | 4200  | 0     | 91.14% | 8.8  |
| 2026<br>8% | Opening    | dir_p50_h60<br>51.87% | 4200  | 0     | 86.52% | 13.4 |
| 2026<br>1% | Opening    | dir_p60_h60<br>51.77% | 4200  | 0     | 79.19% | 20.8 |
| 2026<br>1% | Opening    | dir_p40_h90<br>53.67% | 4200  | 0     | 91.69% | 8.3  |
| 2026<br>3% | Opening    | dir_p50_h90<br>53.59% | 4200  | 0     | 87.17% | 12.8 |
| 2026<br>6% | Opening    | dir_p60_h90<br>53.86% | 4200  | 0     | 80.24% | 19.7 |
| 2026<br>2% | Overnight  | dir_p40_h30<br>48.83% | 16800 | 70    | 89.68% | 10.3 |
| 2026<br>2% | Overnight  | dir_p50_h30<br>48.70% | 16800 | 70    | 83.68% | 16.3 |
| 2026<br>5% | Overnight  | dir_p60_h30<br>48.16% | 16800 | 70    | 76.35% | 23.6 |
| 2026<br>7% | Overnight  | dir_p40_h60<br>49.08% | 16800 | 70    | 89.43% | 10.5 |
| 2026<br>1% | Overnight  | dir_p50_h60<br>48.67% | 16800 | 70    | 82.99% | 17.0 |
| 2026<br>5% | Overnight  | dir_p60_h60<br>48.18% | 16800 | 70    | 75.75% | 24.2 |
| 2026<br>4% | Overnight  | dir_p40_h90<br>50.00% | 16800 | 70    | 89.26% | 10.7 |
| 2026<br>7% | Overnight  | dir_p50_h90<br>49.69% | 16800 | 70    | 83.13% | 16.8 |
| 2026<br>0% | Overnight  | dir_p60_h90<br>49.01% | 16800 | 70    | 75.50% | 24.5 |
| 2026<br>6% | Pre-market | dir_p40_h30<br>53.99% | 4200  | 0     | 84.14% | 15.8 |

|            |            |             |       |      |        |      |
|------------|------------|-------------|-------|------|--------|------|
| 2026<br>0% | Pre-market | dir_p50_h30 | 4200  | 0    | 77.60% | 22.4 |
|            |            | 54.37%      |       |      |        |      |
| 2026<br>7% | Pre-market | dir_p60_h30 | 4200  | 0    | 69.33% | 30.6 |
|            |            | 55.22%      |       |      |        |      |
| 2026<br>0% | Pre-market | dir_p40_h60 | 4200  | 0    | 92.40% | 7.6  |
|            |            | 56.38%      |       |      |        |      |
| 2026<br>8% | Pre-market | dir_p50_h60 | 4200  | 0    | 86.52% | 13.4 |
|            |            | 55.92%      |       |      |        |      |
| 2026<br>9% | Pre-market | dir_p60_h60 | 4200  | 0    | 80.71% | 19.2 |
|            |            | 55.84%      |       |      |        |      |
| 2026<br>0% | Pre-market | dir_p40_h90 | 4200  | 0    | 95.40% | 4.6  |
|            |            | 56.50%      |       |      |        |      |
| 2026<br>5% | Pre-market | dir_p50_h90 | 4200  | 0    | 91.45% | 8.5  |
|            |            | 56.70%      |       |      |        |      |
| 2026<br>0% | Pre-market | dir_p60_h90 | 4200  | 0    | 83.00% | 17.0 |
|            |            | 55.71%      |       |      |        |      |
| 2026<br>5% | Regular    | dir_p40_h30 | 21000 | 0    | 84.85% | 15.1 |
|            |            | 48.11%      |       |      |        |      |
| 2026<br>3% | Regular    | dir_p50_h30 | 21000 | 0    | 78.17% | 21.8 |
|            |            | 47.93%      |       |      |        |      |
| 2026<br>9% | Regular    | dir_p60_h30 | 21000 | 0    | 70.01% | 29.9 |
|            |            | 48.00%      |       |      |        |      |
| 2026<br>2% | Regular    | dir_p40_h60 | 18970 | 2030 | 85.68% | 14.3 |
|            |            | 47.23%      |       |      |        |      |
| 2026<br>5% | Regular    | dir_p50_h60 | 18970 | 2030 | 78.55% | 21.4 |
|            |            | 46.65%      |       |      |        |      |
| 2026<br>7% | Regular    | dir_p60_h60 | 18970 | 2030 | 70.03% | 29.9 |
|            |            | 46.50%      |       |      |        |      |
| 2026<br>8% | Regular    | dir_p40_h90 | 16870 | 4130 | 85.32% | 14.6 |
|            |            | 46.71%      |       |      |        |      |
| 2026<br>1% | Regular    | dir_p50_h90 | 16870 | 4130 | 78.29% | 21.7 |
|            |            | 46.50%      |       |      |        |      |
| 2026<br>1% | Regular    | dir_p60_h90 | 16870 | 4130 | 70.19% | 29.8 |
|            |            | 45.79%      |       |      |        |      |

## Celda 10 — Heatmap year + regime para p50

```
In [77]: # Celda 10 - Heatmap year + regime para p50
# Gráfico agrupado: 3 filas x 1 columna + colorbar separado
# =====

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd

fig = plt.figure(figsize=(12, 12))

gs = fig.add_gridspec(
    nrows=3,
    ncols=2,
    width_ratios=[20, 1],
    height_ratios=[1, 1, 1],
    wspace=0.08,
    hspace=0.35
)

axes = [fig.add_subplot(gs[i, 0]) for i in range(3)]
cax = fig.add_subplot(gs[:, 1])
```

```

vmin = 0.30
vmax = 0.95

last_im = None

for ax, target_col in zip(axes, selected_targets_context):
    tmp = df_dir_year_regime_distribution[
        df_dir_year_regime_distribution["target"] == target_col
    ].copy()

    pivot_tmp = tmp.pivot(
        index="year",
        columns="regime_name",
        values="signal_ratio"
    )

    pivot_tmp = pivot_tmp.reindex(columns=REGIME_ORDER)

    data = np.ma.masked_invalid(pivot_tmp.values)

    last_im = ax.imshow(
        data,
        aspect="auto",
        vmin=vmin,
        vmax=vmax
    )

    ax.set_title(f"Frecuencia operativa year + regime - {target_col}")
    ax.set_ylabel("Año")

    ax.set_yticks(np.arange(len(pivot_tmp.index)))
    ax.set_yticklabels(pivot_tmp.index)

    ax.set_xticks(np.arange(len(pivot_tmp.columns)))
    ax.set_xticklabels(pivot_tmp.columns, rotation=45, ha="right")

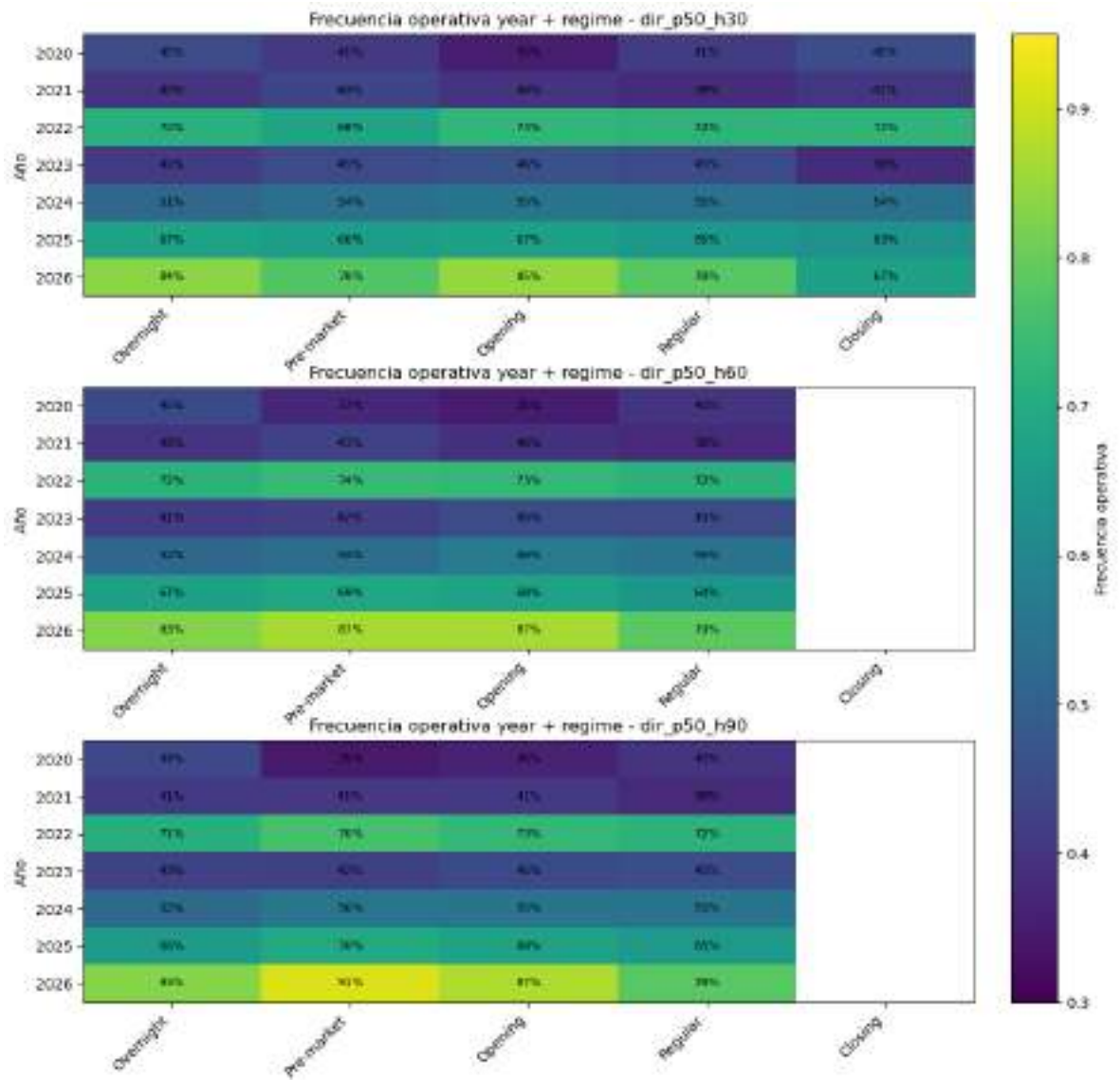
    for i in range(pivot_tmp.shape[0]):
        for j in range(pivot_tmp.shape[1]):
            value = pivot_tmp.iloc[i, j]

            if pd.notna(value):
                ax.text(
                    j,
                    i,
                    f"{value:.0%}",
                    ha="center",
                    va="center",
                    fontsize=8
                )

fig.colorbar(
    last_im,
    cax=cax,
    label="Frecuencia operativa"
)

plt.show()

```



## Celda 11 — Análisis contract + regime

Esta celda revisa si ciertos contratos se comportan distinto en ciertos regímenes.

```
In [78]: # Celda 11 - Análisis DIR por contract + regime
# =====

df_dir_contract_regime_distribution = summarize_dir_by_context(
    df=df_targets_dir,
    target_cols=dir_target_cols,
    group_cols=["contract", "regime_name"]
)

print("Distribución de DIR por contract + regime")
print("=" * 80)

cols_show = [
    "contract",
    "regime_name",
    "target",
    "valid_rows",
    "nan_rows",
    "signal_ratio",
    "no_signal_ratio",
```



```
    "long_share_over_signals",  
]  
  
print(  
    df_dir_contract_regime_distribution[cols_show]  
    .to_string(index=False, float_format=lambda x: f"{x:.2%}")  
)
```

## Distribución de DIR por contract + regime

| =====    |                         |             |            |          |              |             |
|----------|-------------------------|-------------|------------|----------|--------------|-------------|
| contract | regime_name             | target      | valid_rows | nan_rows | signal_ratio | no_signal_r |
| atio     | long_share_over_signals |             |            |          |              |             |
| H20      | Closing                 | dir_p40_h30 | 48         | 1392     | 33.33%       | 6           |
| 6.67%    |                         | 56.25%      |            |          |              |             |
| H20      | Closing                 | dir_p50_h30 | 48         | 1392     | 29.17%       | 7           |
| 0.83%    |                         | 57.14%      |            |          |              |             |
| H20      | Closing                 | dir_p60_h30 | 48         | 1392     | 27.08%       | 7           |
| 2.92%    |                         | 53.85%      |            |          |              |             |
| H20      | Closing                 | dir_p40_h60 | 0          | 1440     | NaN          |             |
| NaN      |                         | NaN         |            |          |              |             |
| H20      | Closing                 | dir_p50_h60 | 0          | 1440     | NaN          |             |
| NaN      |                         | NaN         |            |          |              |             |
| H20      | Closing                 | dir_p60_h60 | 0          | 1440     | NaN          |             |
| NaN      |                         | NaN         |            |          |              |             |
| H20      | Closing                 | dir_p40_h90 | 0          | 1440     | NaN          |             |
| NaN      |                         | NaN         |            |          |              |             |
| H20      | Closing                 | dir_p50_h90 | 0          | 1440     | NaN          |             |
| NaN      |                         | NaN         |            |          |              |             |
| H20      | Closing                 | dir_p60_h90 | 0          | 1440     | NaN          |             |
| NaN      |                         | NaN         |            |          |              |             |
| H20      | Opening                 | dir_p40_h30 | 2880       | 0        | 29.20%       | 7           |
| 0.80%    |                         | 44.23%      |            |          |              |             |
| H20      | Opening                 | dir_p50_h30 | 2880       | 0        | 24.34%       | 7           |
| 5.66%    |                         | 46.50%      |            |          |              |             |
| H20      | Opening                 | dir_p60_h30 | 2880       | 0        | 20.49%       | 7           |
| 9.51%    |                         | 47.63%      |            |          |              |             |
| H20      | Opening                 | dir_p40_h60 | 2880       | 0        | 31.42%       | 6           |
| 8.58%    |                         | 43.87%      |            |          |              |             |
| H20      | Opening                 | dir_p50_h60 | 2880       | 0        | 26.04%       | 7           |
| 3.96%    |                         | 45.20%      |            |          |              |             |
| H20      | Opening                 | dir_p60_h60 | 2880       | 0        | 20.87%       | 7           |
| 9.13%    |                         | 48.75%      |            |          |              |             |
| H20      | Opening                 | dir_p40_h90 | 2880       | 0        | 30.76%       | 6           |
| 9.24%    |                         | 44.81%      |            |          |              |             |
| H20      | Opening                 | dir_p50_h90 | 2880       | 0        | 25.62%       | 7           |
| 4.38%    |                         | 46.07%      |            |          |              |             |
| H20      | Opening                 | dir_p60_h90 | 2880       | 0        | 21.60%       | 7           |
| 8.40%    |                         | 46.14%      |            |          |              |             |
| H20      | Overnight               | dir_p40_h30 | 11520      | 48       | 40.26%       | 5           |
| 9.74%    |                         | 47.50%      |            |          |              |             |
| H20      | Overnight               | dir_p50_h30 | 11520      | 48       | 34.18%       | 6           |
| 5.82%    |                         | 46.14%      |            |          |              |             |
| H20      | Overnight               | dir_p60_h30 | 11520      | 48       | 29.42%       | 7           |
| 0.58%    |                         | 44.50%      |            |          |              |             |
| H20      | Overnight               | dir_p40_h60 | 11520      | 48       | 38.87%       | 6           |
| 1.13%    |                         | 46.23%      |            |          |              |             |
| H20      | Overnight               | dir_p50_h60 | 11520      | 48       | 32.68%       | 6           |
| 7.32%    |                         | 44.54%      |            |          |              |             |
| H20      | Overnight               | dir_p60_h60 | 11520      | 48       | 26.95%       | 7           |
| 3.05%    |                         | 42.61%      |            |          |              |             |
| H20      | Overnight               | dir_p40_h90 | 11520      | 48       | 37.73%       | 6           |
| 2.27%    |                         | 43.79%      |            |          |              |             |
| H20      | Overnight               | dir_p50_h90 | 11520      | 48       | 31.67%       | 6           |
| 8.33%    |                         | 41.34%      |            |          |              |             |
| H20      | Overnight               | dir_p60_h90 | 11520      | 48       | 26.95%       | 7           |
| 3.05%    |                         | 39.74%      |            |          |              |             |
| H20      | Pre-market              | dir_p40_h30 | 2880       | 0        | 38.37%       | 6           |
| 1.63%    |                         | 46.97%      |            |          |              |             |

|       |            |             |       |      |        |   |
|-------|------------|-------------|-------|------|--------|---|
| H20   | Pre-market | dir_p50_h30 | 2880  | 0    | 31.77% | 6 |
| 8.23% |            | 47.87%      |       |      |        |   |
| H20   | Pre-market | dir_p60_h30 | 2880  | 0    | 25.94% | 7 |
| 4.06% |            | 51.54%      |       |      |        |   |
| H20   | Pre-market | dir_p40_h60 | 2880  | 0    | 33.58% | 6 |
| 6.42% |            | 48.81%      |       |      |        |   |
| H20   | Pre-market | dir_p50_h60 | 2880  | 0    | 28.89% | 7 |
| 1.11% |            | 47.12%      |       |      |        |   |
| H20   | Pre-market | dir_p60_h60 | 2880  | 0    | 23.61% | 7 |
| 6.39% |            | 46.76%      |       |      |        |   |
| H20   | Pre-market | dir_p40_h90 | 2880  | 0    | 33.09% | 6 |
| 6.91% |            | 50.16%      |       |      |        |   |
| H20   | Pre-market | dir_p50_h90 | 2880  | 0    | 28.72% | 7 |
| 1.28% |            | 52.24%      |       |      |        |   |
| H20   | Pre-market | dir_p60_h90 | 2880  | 0    | 22.26% | 7 |
| 7.74% |            | 52.89%      |       |      |        |   |
| H20   | Regular    | dir_p40_h30 | 14400 | 0    | 36.32% | 6 |
| 3.68% |            | 44.86%      |       |      |        |   |
| H20   | Regular    | dir_p50_h30 | 14400 | 0    | 30.69% | 6 |
| 9.31% |            | 45.09%      |       |      |        |   |
| H20   | Regular    | dir_p60_h30 | 14400 | 0    | 25.75% | 7 |
| 4.25% |            | 45.55%      |       |      |        |   |
| H20   | Regular    | dir_p40_h60 | 13008 | 1392 | 35.45% | 6 |
| 4.55% |            | 41.62%      |       |      |        |   |
| H20   | Regular    | dir_p50_h60 | 13008 | 1392 | 30.19% | 6 |
| 9.81% |            | 41.35%      |       |      |        |   |
| H20   | Regular    | dir_p60_h60 | 13008 | 1392 | 25.38% | 7 |
| 4.62% |            | 41.37%      |       |      |        |   |
| H20   | Regular    | dir_p40_h90 | 11568 | 2832 | 34.47% | 6 |
| 5.53% |            | 39.00%      |       |      |        |   |
| H20   | Regular    | dir_p50_h90 | 11568 | 2832 | 29.38% | 7 |
| 0.62% |            | 39.42%      |       |      |        |   |
| H20   | Regular    | dir_p60_h90 | 11568 | 2832 | 24.89% | 7 |
| 5.11% |            | 39.15%      |       |      |        |   |
| H21   | Closing    | dir_p40_h30 | 59    | 1711 | 64.41% | 3 |
| 5.59% |            | 52.63%      |       |      |        |   |
| H21   | Closing    | dir_p50_h30 | 59    | 1711 | 47.46% | 5 |
| 2.54% |            | 53.57%      |       |      |        |   |
| H21   | Closing    | dir_p60_h30 | 59    | 1711 | 35.59% | 6 |
| 4.41% |            | 47.62%      |       |      |        |   |
| H21   | Closing    | dir_p40_h60 | 0     | 1770 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H21   | Closing    | dir_p50_h60 | 0     | 1770 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H21   | Closing    | dir_p60_h60 | 0     | 1770 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H21   | Closing    | dir_p40_h90 | 0     | 1770 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H21   | Closing    | dir_p50_h90 | 0     | 1770 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H21   | Closing    | dir_p60_h90 | 0     | 1770 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H21   | Opening    | dir_p40_h30 | 3540  | 0    | 57.23% | 4 |
| 2.77% |            | 46.59%      |       |      |        |   |
| H21   | Opening    | dir_p50_h30 | 3540  | 0    | 46.55% | 5 |
| 3.45% |            | 44.90%      |       |      |        |   |
| H21   | Opening    | dir_p60_h30 | 3540  | 0    | 37.34% | 6 |
| 2.66% |            | 42.51%      |       |      |        |   |
| H21   | Opening    | dir_p40_h60 | 3540  | 0    | 55.31% | 4 |
| 4.69% |            | 44.43%      |       |      |        |   |

|       |            |             |       |      |        |   |
|-------|------------|-------------|-------|------|--------|---|
| H21   | Opening    | dir_p50_h60 | 3540  | 0    | 46.55% | 5 |
| 3.45% |            | 43.26%      |       |      |        |   |
| H21   | Opening    | dir_p60_h60 | 3540  | 0    | 38.22% | 6 |
| 1.78% |            | 42.50%      |       |      |        |   |
| H21   | Opening    | dir_p40_h90 | 3540  | 0    | 56.05% | 4 |
| 3.95% |            | 42.99%      |       |      |        |   |
| H21   | Opening    | dir_p50_h90 | 3540  | 0    | 47.91% | 5 |
| 2.09% |            | 42.10%      |       |      |        |   |
| H21   | Opening    | dir_p60_h90 | 3540  | 0    | 38.70% | 6 |
| 1.30% |            | 41.75%      |       |      |        |   |
| H21   | Overnight  | dir_p40_h30 | 14160 | 59   | 54.67% | 4 |
| 5.33% |            | 49.39%      |       |      |        |   |
| H21   | Overnight  | dir_p50_h30 | 14160 | 59   | 46.84% | 5 |
| 3.16% |            | 47.95%      |       |      |        |   |
| H21   | Overnight  | dir_p60_h30 | 14160 | 59   | 39.75% | 6 |
| 0.25% |            | 46.83%      |       |      |        |   |
| H21   | Overnight  | dir_p40_h60 | 14160 | 59   | 53.31% | 4 |
| 6.69% |            | 47.54%      |       |      |        |   |
| H21   | Overnight  | dir_p50_h60 | 14160 | 59   | 45.90% | 5 |
| 4.10% |            | 46.58%      |       |      |        |   |
| H21   | Overnight  | dir_p60_h60 | 14160 | 59   | 38.62% | 6 |
| 1.38% |            | 45.90%      |       |      |        |   |
| H21   | Overnight  | dir_p40_h90 | 14160 | 59   | 53.93% | 4 |
| 6.07% |            | 48.71%      |       |      |        |   |
| H21   | Overnight  | dir_p50_h90 | 14160 | 59   | 46.92% | 5 |
| 3.08% |            | 48.37%      |       |      |        |   |
| H21   | Overnight  | dir_p60_h90 | 14160 | 59   | 38.95% | 6 |
| 1.05% |            | 47.99%      |       |      |        |   |
| H21   | Pre-market | dir_p40_h30 | 3540  | 0    | 59.97% | 4 |
| 0.03% |            | 48.00%      |       |      |        |   |
| H21   | Pre-market | dir_p50_h30 | 3540  | 0    | 48.64% | 5 |
| 1.36% |            | 48.95%      |       |      |        |   |
| H21   | Pre-market | dir_p60_h30 | 3540  | 0    | 38.70% | 6 |
| 1.30% |            | 50.51%      |       |      |        |   |
| H21   | Pre-market | dir_p40_h60 | 3540  | 0    | 54.55% | 4 |
| 5.45% |            | 46.61%      |       |      |        |   |
| H21   | Pre-market | dir_p50_h60 | 3540  | 0    | 42.68% | 5 |
| 7.32% |            | 45.73%      |       |      |        |   |
| H21   | Pre-market | dir_p60_h60 | 3540  | 0    | 32.12% | 6 |
| 7.88% |            | 45.91%      |       |      |        |   |
| H21   | Pre-market | dir_p40_h90 | 3540  | 0    | 51.16% | 4 |
| 8.84% |            | 42.68%      |       |      |        |   |
| H21   | Pre-market | dir_p50_h90 | 3540  | 0    | 39.69% | 6 |
| 0.31% |            | 43.99%      |       |      |        |   |
| H21   | Pre-market | dir_p60_h90 | 3540  | 0    | 32.97% | 6 |
| 7.03% |            | 43.36%      |       |      |        |   |
| H21   | Regular    | dir_p40_h30 | 17700 | 0    | 55.51% | 4 |
| 4.49% |            | 48.78%      |       |      |        |   |
| H21   | Regular    | dir_p50_h30 | 17700 | 0    | 46.47% | 5 |
| 3.53% |            | 47.82%      |       |      |        |   |
| H21   | Regular    | dir_p60_h30 | 17700 | 0    | 38.03% | 6 |
| 1.97% |            | 46.01%      |       |      |        |   |
| H21   | Regular    | dir_p40_h60 | 15989 | 1711 | 55.18% | 4 |
| 4.82% |            | 49.82%      |       |      |        |   |
| H21   | Regular    | dir_p50_h60 | 15989 | 1711 | 45.86% | 5 |
| 4.14% |            | 48.58%      |       |      |        |   |
| H21   | Regular    | dir_p60_h60 | 15989 | 1711 | 37.39% | 6 |
| 2.61% |            | 45.25%      |       |      |        |   |
| H21   | Regular    | dir_p40_h90 | 14219 | 3481 | 54.16% | 4 |
| 5.84% |            | 50.14%      |       |      |        |   |

|       |            |             |       |      |        |   |
|-------|------------|-------------|-------|------|--------|---|
| H21   | Regular    | dir_p50_h90 | 14219 | 3481 | 46.37% | 5 |
| 3.63% |            | 47.88%      |       |      |        |   |
| H21   | Regular    | dir_p60_h90 | 14219 | 3481 | 38.24% | 6 |
| 1.76% |            | 45.79%      |       |      |        |   |
| H22   | Closing    | dir_p40_h30 | 61    | 1769 | 90.16% |   |
| 9.84% |            | 45.45%      |       |      |        |   |
| H22   | Closing    | dir_p50_h30 | 61    | 1769 | 83.61% | 1 |
| 6.39% |            | 45.10%      |       |      |        |   |
| H22   | Closing    | dir_p60_h30 | 61    | 1769 | 72.13% | 2 |
| 7.87% |            | 38.64%      |       |      |        |   |
| H22   | Closing    | dir_p40_h60 | 0     | 1830 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H22   | Closing    | dir_p50_h60 | 0     | 1830 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H22   | Closing    | dir_p60_h60 | 0     | 1830 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H22   | Closing    | dir_p40_h90 | 0     | 1830 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H22   | Closing    | dir_p50_h90 | 0     | 1830 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H22   | Closing    | dir_p60_h90 | 0     | 1830 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H22   | Opening    | dir_p40_h30 | 3660  | 0    | 88.99% | 1 |
| 1.01% |            | 46.58%      |       |      |        |   |
| H22   | Opening    | dir_p50_h30 | 3660  | 0    | 82.16% | 1 |
| 7.84% |            | 46.39%      |       |      |        |   |
| H22   | Opening    | dir_p60_h30 | 3660  | 0    | 73.20% | 2 |
| 6.80% |            | 45.76%      |       |      |        |   |
| H22   | Opening    | dir_p40_h60 | 3660  | 0    | 88.93% | 1 |
| 1.07% |            | 47.93%      |       |      |        |   |
| H22   | Opening    | dir_p50_h60 | 3660  | 0    | 81.20% | 1 |
| 8.80% |            | 46.67%      |       |      |        |   |
| H22   | Opening    | dir_p60_h60 | 3660  | 0    | 71.99% | 2 |
| 8.01% |            | 46.49%      |       |      |        |   |
| H22   | Opening    | dir_p40_h90 | 3660  | 0    | 88.11% | 1 |
| 1.89% |            | 45.52%      |       |      |        |   |
| H22   | Opening    | dir_p50_h90 | 3660  | 0    | 82.05% | 1 |
| 7.95% |            | 45.05%      |       |      |        |   |
| H22   | Opening    | dir_p60_h90 | 3660  | 0    | 73.61% | 2 |
| 6.39% |            | 43.69%      |       |      |        |   |
| H22   | Overnight  | dir_p40_h30 | 14640 | 61   | 85.01% | 1 |
| 4.99% |            | 48.36%      |       |      |        |   |
| H22   | Overnight  | dir_p50_h30 | 14640 | 61   | 79.13% | 2 |
| 0.87% |            | 48.04%      |       |      |        |   |
| H22   | Overnight  | dir_p60_h30 | 14640 | 61   | 71.97% | 2 |
| 8.03% |            | 48.11%      |       |      |        |   |
| H22   | Overnight  | dir_p40_h60 | 14640 | 61   | 86.87% | 1 |
| 3.13% |            | 49.02%      |       |      |        |   |
| H22   | Overnight  | dir_p50_h60 | 14640 | 61   | 80.33% | 1 |
| 9.67% |            | 48.32%      |       |      |        |   |
| H22   | Overnight  | dir_p60_h60 | 14640 | 61   | 71.92% | 2 |
| 8.08% |            | 47.32%      |       |      |        |   |
| H22   | Overnight  | dir_p40_h90 | 14640 | 61   | 85.73% | 1 |
| 4.27% |            | 47.81%      |       |      |        |   |
| H22   | Overnight  | dir_p50_h90 | 14640 | 61   | 78.92% | 2 |
| 1.08% |            | 47.51%      |       |      |        |   |
| H22   | Overnight  | dir_p60_h90 | 14640 | 61   | 69.93% | 3 |
| 0.07% |            | 47.30%      |       |      |        |   |
| H22   | Pre-market | dir_p40_h30 | 3660  | 0    | 80.74% | 1 |
| 9.26% |            | 49.17%      |       |      |        |   |

|       |            |             |       |      |        |   |
|-------|------------|-------------|-------|------|--------|---|
| H22   | Pre-market | dir_p50_h30 | 3660  | 0    | 69.89% | 3 |
| 0.11% |            | 48.55%      |       |      |        |   |
| H22   | Pre-market | dir_p60_h30 | 3660  | 0    | 60.03% | 3 |
| 9.97% |            | 47.56%      |       |      |        |   |
| H22   | Pre-market | dir_p40_h60 | 3660  | 0    | 82.35% | 1 |
| 7.65% |            | 48.81%      |       |      |        |   |
| H22   | Pre-market | dir_p50_h60 | 3660  | 0    | 74.13% | 2 |
| 5.87% |            | 48.47%      |       |      |        |   |
| H22   | Pre-market | dir_p60_h60 | 3660  | 0    | 64.73% | 3 |
| 5.27% |            | 47.83%      |       |      |        |   |
| H22   | Pre-market | dir_p40_h90 | 3660  | 0    | 84.34% | 1 |
| 5.66% |            | 44.02%      |       |      |        |   |
| H22   | Pre-market | dir_p50_h90 | 3660  | 0    | 76.86% | 2 |
| 3.14% |            | 42.66%      |       |      |        |   |
| H22   | Pre-market | dir_p60_h90 | 3660  | 0    | 66.26% | 3 |
| 3.74% |            | 41.98%      |       |      |        |   |
| H22   | Regular    | dir_p40_h30 | 18300 | 0    | 84.07% | 1 |
| 5.93% |            | 49.26%      |       |      |        |   |
| H22   | Regular    | dir_p50_h30 | 18300 | 0    | 77.49% | 2 |
| 2.51% |            | 49.31%      |       |      |        |   |
| H22   | Regular    | dir_p60_h30 | 18300 | 0    | 68.77% | 3 |
| 1.23% |            | 49.14%      |       |      |        |   |
| H22   | Regular    | dir_p40_h60 | 16531 | 1769 | 83.87% | 1 |
| 6.13% |            | 49.16%      |       |      |        |   |
| H22   | Regular    | dir_p50_h60 | 16531 | 1769 | 76.87% | 2 |
| 3.13% |            | 48.75%      |       |      |        |   |
| H22   | Regular    | dir_p60_h60 | 16531 | 1769 | 68.16% | 3 |
| 1.84% |            | 48.33%      |       |      |        |   |
| H22   | Regular    | dir_p40_h90 | 14701 | 3599 | 84.50% | 1 |
| 5.50% |            | 48.46%      |       |      |        |   |
| H22   | Regular    | dir_p50_h90 | 14701 | 3599 | 76.95% | 2 |
| 3.05% |            | 47.89%      |       |      |        |   |
| H22   | Regular    | dir_p60_h90 | 14701 | 3599 | 67.59% | 3 |
| 2.41% |            | 47.13%      |       |      |        |   |
| H23   | Closing    | dir_p40_h30 | 60    | 1740 | 53.33% | 4 |
| 6.67% |            | 59.38%      |       |      |        |   |
| H23   | Closing    | dir_p50_h30 | 60    | 1740 | 43.33% | 5 |
| 6.67% |            | 61.54%      |       |      |        |   |
| H23   | Closing    | dir_p60_h30 | 60    | 1740 | 40.00% | 6 |
| 0.00% |            | 62.50%      |       |      |        |   |
| H23   | Closing    | dir_p40_h60 | 0     | 1800 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H23   | Closing    | dir_p50_h60 | 0     | 1800 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H23   | Closing    | dir_p60_h60 | 0     | 1800 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H23   | Closing    | dir_p40_h90 | 0     | 1800 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H23   | Closing    | dir_p50_h90 | 0     | 1800 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H23   | Closing    | dir_p60_h90 | 0     | 1800 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H23   | Opening    | dir_p40_h30 | 3600  | 0    | 69.69% | 3 |
| 0.31% |            | 55.92%      |       |      |        |   |
| H23   | Opening    | dir_p50_h30 | 3600  | 0    | 58.56% | 4 |
| 1.44% |            | 55.74%      |       |      |        |   |
| H23   | Opening    | dir_p60_h30 | 3600  | 0    | 46.39% | 5 |
| 3.61% |            | 53.59%      |       |      |        |   |
| H23   | Opening    | dir_p40_h60 | 3600  | 0    | 71.11% | 2 |
| 8.89% |            | 55.20%      |       |      |        |   |

|       |            |             |       |      |        |   |
|-------|------------|-------------|-------|------|--------|---|
| H23   | Opening    | dir_p50_h60 | 3600  | 0    | 58.78% | 4 |
| 1.22% |            | 54.21%      |       |      |        |   |
| H23   | Opening    | dir_p60_h60 | 3600  | 0    | 46.03% | 5 |
| 3.97% |            | 53.59%      |       |      |        |   |
| H23   | Opening    | dir_p40_h90 | 3600  | 0    | 71.39% | 2 |
| 8.61% |            | 54.59%      |       |      |        |   |
| H23   | Opening    | dir_p50_h90 | 3600  | 0    | 59.69% | 4 |
| 0.31% |            | 54.12%      |       |      |        |   |
| H23   | Opening    | dir_p60_h90 | 3600  | 0    | 47.03% | 5 |
| 2.97% |            | 53.51%      |       |      |        |   |
| H23   | Overnight  | dir_p40_h30 | 14400 | 60   | 65.71% | 3 |
| 4.29% |            | 46.69%      |       |      |        |   |
| H23   | Overnight  | dir_p50_h30 | 14400 | 60   | 54.06% | 4 |
| 5.94% |            | 46.51%      |       |      |        |   |
| H23   | Overnight  | dir_p60_h30 | 14400 | 60   | 43.55% | 5 |
| 6.45% |            | 45.35%      |       |      |        |   |
| H23   | Overnight  | dir_p40_h60 | 14400 | 60   | 67.20% | 3 |
| 2.80% |            | 47.37%      |       |      |        |   |
| H23   | Overnight  | dir_p50_h60 | 14400 | 60   | 55.35% | 4 |
| 4.65% |            | 46.49%      |       |      |        |   |
| H23   | Overnight  | dir_p60_h60 | 14400 | 60   | 44.01% | 5 |
| 5.99% |            | 45.82%      |       |      |        |   |
| H23   | Overnight  | dir_p40_h90 | 14400 | 60   | 67.12% | 3 |
| 2.88% |            | 48.06%      |       |      |        |   |
| H23   | Overnight  | dir_p50_h90 | 14400 | 60   | 55.77% | 4 |
| 4.23% |            | 48.24%      |       |      |        |   |
| H23   | Overnight  | dir_p60_h90 | 14400 | 60   | 43.29% | 5 |
| 6.71% |            | 48.01%      |       |      |        |   |
| H23   | Pre-market | dir_p40_h30 | 3600  | 0    | 64.06% | 3 |
| 5.94% |            | 53.82%      |       |      |        |   |
| H23   | Pre-market | dir_p50_h30 | 3600  | 0    | 54.39% | 4 |
| 5.61% |            | 54.14%      |       |      |        |   |
| H23   | Pre-market | dir_p60_h30 | 3600  | 0    | 43.69% | 5 |
| 6.31% |            | 55.31%      |       |      |        |   |
| H23   | Pre-market | dir_p40_h60 | 3600  | 0    | 64.81% | 3 |
| 5.19% |            | 55.47%      |       |      |        |   |
| H23   | Pre-market | dir_p50_h60 | 3600  | 0    | 54.19% | 4 |
| 5.81% |            | 54.59%      |       |      |        |   |
| H23   | Pre-market | dir_p60_h60 | 3600  | 0    | 42.36% | 5 |
| 7.64% |            | 52.39%      |       |      |        |   |
| H23   | Pre-market | dir_p40_h90 | 3600  | 0    | 66.17% | 3 |
| 3.83% |            | 55.58%      |       |      |        |   |
| H23   | Pre-market | dir_p50_h90 | 3600  | 0    | 56.22% | 4 |
| 3.78% |            | 53.36%      |       |      |        |   |
| H23   | Pre-market | dir_p60_h90 | 3600  | 0    | 44.50% | 5 |
| 5.50% |            | 53.25%      |       |      |        |   |
| H23   | Regular    | dir_p40_h30 | 18000 | 0    | 63.87% | 3 |
| 6.13% |            | 52.51%      |       |      |        |   |
| H23   | Regular    | dir_p50_h30 | 18000 | 0    | 51.79% | 4 |
| 8.21% |            | 52.45%      |       |      |        |   |
| H23   | Regular    | dir_p60_h30 | 18000 | 0    | 40.17% | 5 |
| 9.83% |            | 52.75%      |       |      |        |   |
| H23   | Regular    | dir_p40_h60 | 16260 | 1740 | 63.95% | 3 |
| 6.05% |            | 53.56%      |       |      |        |   |
| H23   | Regular    | dir_p50_h60 | 16260 | 1740 | 51.80% | 4 |
| 8.20% |            | 53.40%      |       |      |        |   |
| H23   | Regular    | dir_p60_h60 | 16260 | 1740 | 40.53% | 5 |
| 9.47% |            | 53.07%      |       |      |        |   |
| H23   | Regular    | dir_p40_h90 | 14460 | 3540 | 64.94% | 3 |
| 5.06% |            | 54.64%      |       |      |        |   |

|       |            |             |       |      |        |   |
|-------|------------|-------------|-------|------|--------|---|
| H23   | Regular    | dir_p50_h90 | 14460 | 3540 | 52.76% | 4 |
| 7.24% |            | 53.91%      |       |      |        |   |
| H23   | Regular    | dir_p60_h90 | 14460 | 3540 | 40.94% | 5 |
| 9.06% |            | 52.48%      |       |      |        |   |
| H24   | Closing    | dir_p40_h30 | 60    | 1740 | 56.67% | 4 |
| 3.33% |            | 61.76%      |       |      |        |   |
| H24   | Closing    | dir_p50_h30 | 60    | 1740 | 38.33% | 6 |
| 1.67% |            | 60.87%      |       |      |        |   |
| H24   | Closing    | dir_p60_h30 | 60    | 1740 | 25.00% | 7 |
| 5.00% |            | 66.67%      |       |      |        |   |
| H24   | Closing    | dir_p40_h60 | 0     | 1800 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H24   | Closing    | dir_p50_h60 | 0     | 1800 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H24   | Closing    | dir_p60_h60 | 0     | 1800 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H24   | Closing    | dir_p40_h90 | 0     | 1800 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H24   | Closing    | dir_p50_h90 | 0     | 1800 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H24   | Closing    | dir_p60_h90 | 0     | 1800 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H24   | Opening    | dir_p40_h30 | 3600  | 0    | 51.31% | 4 |
| 8.69% |            | 47.64%      |       |      |        |   |
| H24   | Opening    | dir_p50_h30 | 3600  | 0    | 39.72% | 6 |
| 0.28% |            | 46.71%      |       |      |        |   |
| H24   | Opening    | dir_p60_h30 | 3600  | 0    | 29.22% | 7 |
| 0.78% |            | 46.01%      |       |      |        |   |
| H24   | Opening    | dir_p40_h60 | 3600  | 0    | 51.67% | 4 |
| 8.33% |            | 53.39%      |       |      |        |   |
| H24   | Opening    | dir_p50_h60 | 3600  | 0    | 40.69% | 5 |
| 9.31% |            | 51.26%      |       |      |        |   |
| H24   | Opening    | dir_p60_h60 | 3600  | 0    | 30.83% | 6 |
| 9.17% |            | 47.57%      |       |      |        |   |
| H24   | Opening    | dir_p40_h90 | 3600  | 0    | 50.33% | 4 |
| 9.67% |            | 54.58%      |       |      |        |   |
| H24   | Opening    | dir_p50_h90 | 3600  | 0    | 41.14% | 5 |
| 8.86% |            | 54.49%      |       |      |        |   |
| H24   | Opening    | dir_p60_h90 | 3600  | 0    | 30.31% | 6 |
| 9.69% |            | 52.98%      |       |      |        |   |
| H24   | Overnight  | dir_p40_h30 | 14400 | 60   | 51.67% | 4 |
| 8.33% |            | 49.55%      |       |      |        |   |
| H24   | Overnight  | dir_p50_h30 | 14400 | 60   | 40.54% | 5 |
| 9.46% |            | 49.02%      |       |      |        |   |
| H24   | Overnight  | dir_p60_h30 | 14400 | 60   | 30.84% | 6 |
| 9.16% |            | 48.75%      |       |      |        |   |
| H24   | Overnight  | dir_p40_h60 | 14400 | 60   | 51.32% | 4 |
| 8.68% |            | 50.34%      |       |      |        |   |
| H24   | Overnight  | dir_p50_h60 | 14400 | 60   | 41.06% | 5 |
| 8.94% |            | 50.08%      |       |      |        |   |
| H24   | Overnight  | dir_p60_h60 | 14400 | 60   | 31.75% | 6 |
| 8.25% |            | 49.19%      |       |      |        |   |
| H24   | Overnight  | dir_p40_h90 | 14400 | 60   | 52.45% | 4 |
| 7.55% |            | 50.30%      |       |      |        |   |
| H24   | Overnight  | dir_p50_h90 | 14400 | 60   | 43.01% | 5 |
| 6.99% |            | 48.87%      |       |      |        |   |
| H24   | Overnight  | dir_p60_h90 | 14400 | 60   | 33.35% | 6 |
| 6.65% |            | 47.70%      |       |      |        |   |
| H24   | Pre-market | dir_p40_h30 | 3600  | 0    | 54.97% | 4 |
| 5.03% |            | 50.28%      |       |      |        |   |



|       |            |             |       |      |        |   |
|-------|------------|-------------|-------|------|--------|---|
| H24   | Pre-market | dir_p50_h30 | 3600  | 0    | 43.14% | 5 |
| 6.86% |            | 49.00%      |       |      |        |   |
| H24   | Pre-market | dir_p60_h30 | 3600  | 0    | 30.94% | 6 |
| 9.06% |            | 47.67%      |       |      |        |   |
| H24   | Pre-market | dir_p40_h60 | 3600  | 0    | 43.08% | 5 |
| 6.92% |            | 42.49%      |       |      |        |   |
| H24   | Pre-market | dir_p50_h60 | 3600  | 0    | 33.78% | 6 |
| 6.22% |            | 37.99%      |       |      |        |   |
| H24   | Pre-market | dir_p60_h60 | 3600  | 0    | 25.78% | 7 |
| 4.22% |            | 35.02%      |       |      |        |   |
| H24   | Pre-market | dir_p40_h90 | 3600  | 0    | 44.42% | 5 |
| 5.58% |            | 40.34%      |       |      |        |   |
| H24   | Pre-market | dir_p50_h90 | 3600  | 0    | 36.03% | 6 |
| 3.97% |            | 37.86%      |       |      |        |   |
| H24   | Pre-market | dir_p60_h90 | 3600  | 0    | 29.00% | 7 |
| 1.00% |            | 34.87%      |       |      |        |   |
| H24   | Regular    | dir_p40_h30 | 18000 | 0    | 56.61% | 4 |
| 3.39% |            | 49.63%      |       |      |        |   |
| H24   | Regular    | dir_p50_h30 | 18000 | 0    | 45.39% | 5 |
| 4.61% |            | 48.36%      |       |      |        |   |
| H24   | Regular    | dir_p60_h30 | 18000 | 0    | 34.19% | 6 |
| 5.81% |            | 46.40%      |       |      |        |   |
| H24   | Regular    | dir_p40_h60 | 16260 | 1740 | 58.31% | 4 |
| 1.69% |            | 50.03%      |       |      |        |   |
| H24   | Regular    | dir_p50_h60 | 16260 | 1740 | 46.37% | 5 |
| 3.63% |            | 48.28%      |       |      |        |   |
| H24   | Regular    | dir_p60_h60 | 16260 | 1740 | 34.75% | 6 |
| 5.25% |            | 45.60%      |       |      |        |   |
| H24   | Regular    | dir_p40_h90 | 14460 | 3540 | 58.21% | 4 |
| 1.79% |            | 50.16%      |       |      |        |   |
| H24   | Regular    | dir_p50_h90 | 14460 | 3540 | 46.02% | 5 |
| 3.98% |            | 48.90%      |       |      |        |   |
| H24   | Regular    | dir_p60_h90 | 14460 | 3540 | 35.08% | 6 |
| 4.92% |            | 46.48%      |       |      |        |   |
| H25   | Closing    | dir_p40_h30 | 57    | 1653 | 94.74% |   |
| 5.26% |            | 42.59%      |       |      |        |   |
| H25   | Closing    | dir_p50_h30 | 57    | 1653 | 84.21% | 1 |
| 5.79% |            | 39.58%      |       |      |        |   |
| H25   | Closing    | dir_p60_h30 | 57    | 1653 | 71.93% | 2 |
| 8.07% |            | 43.90%      |       |      |        |   |
| H25   | Closing    | dir_p40_h60 | 0     | 1710 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H25   | Closing    | dir_p50_h60 | 0     | 1710 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H25   | Closing    | dir_p60_h60 | 0     | 1710 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H25   | Closing    | dir_p40_h90 | 0     | 1710 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H25   | Closing    | dir_p50_h90 | 0     | 1710 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H25   | Closing    | dir_p60_h90 | 0     | 1710 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H25   | Opening    | dir_p40_h30 | 3420  | 0    | 86.55% | 1 |
| 3.45% |            | 46.52%      |       |      |        |   |
| H25   | Opening    | dir_p50_h30 | 3420  | 0    | 80.61% | 1 |
| 9.39% |            | 45.56%      |       |      |        |   |
| H25   | Opening    | dir_p60_h30 | 3420  | 0    | 74.33% | 2 |
| 5.67% |            | 44.89%      |       |      |        |   |
| H25   | Opening    | dir_p40_h60 | 3420  | 0    | 85.53% | 1 |
| 4.47% |            | 46.97%      |       |      |        |   |

|       |            |             |       |      |        |   |
|-------|------------|-------------|-------|------|--------|---|
| H25   | Opening    | dir_p50_h60 | 3420  | 0    | 78.13% | 2 |
| 1.87% |            | 45.96%      |       |      |        |   |
| H25   | Opening    | dir_p60_h60 | 3420  | 0    | 69.62% | 3 |
| 0.38% |            | 44.69%      |       |      |        |   |
| H25   | Opening    | dir_p40_h90 | 3420  | 0    | 84.27% | 1 |
| 5.73% |            | 47.26%      |       |      |        |   |
| H25   | Opening    | dir_p50_h90 | 3420  | 0    | 76.40% | 2 |
| 3.60% |            | 45.77%      |       |      |        |   |
| H25   | Opening    | dir_p60_h90 | 3420  | 0    | 66.40% | 3 |
| 3.60% |            | 44.39%      |       |      |        |   |
| H25   | Overnight  | dir_p40_h30 | 13680 | 57   | 80.75% | 1 |
| 9.25% |            | 48.84%      |       |      |        |   |
| H25   | Overnight  | dir_p50_h30 | 13680 | 57   | 71.83% | 2 |
| 8.17% |            | 48.98%      |       |      |        |   |
| H25   | Overnight  | dir_p60_h30 | 13680 | 57   | 62.51% | 3 |
| 7.49% |            | 49.08%      |       |      |        |   |
| H25   | Overnight  | dir_p40_h60 | 13680 | 57   | 79.88% | 2 |
| 0.12% |            | 49.40%      |       |      |        |   |
| H25   | Overnight  | dir_p50_h60 | 13680 | 57   | 70.96% | 2 |
| 9.04% |            | 48.62%      |       |      |        |   |
| H25   | Overnight  | dir_p60_h60 | 13680 | 57   | 61.62% | 3 |
| 8.38% |            | 47.89%      |       |      |        |   |
| H25   | Overnight  | dir_p40_h90 | 13680 | 57   | 79.44% | 2 |
| 0.56% |            | 48.53%      |       |      |        |   |
| H25   | Overnight  | dir_p50_h90 | 13680 | 57   | 71.71% | 2 |
| 8.29% |            | 47.00%      |       |      |        |   |
| H25   | Overnight  | dir_p60_h90 | 13680 | 57   | 62.68% | 3 |
| 7.32% |            | 45.99%      |       |      |        |   |
| H25   | Pre-market | dir_p40_h30 | 3420  | 0    | 80.41% | 1 |
| 9.59% |            | 42.25%      |       |      |        |   |
| H25   | Pre-market | dir_p50_h30 | 3420  | 0    | 72.49% | 2 |
| 7.51% |            | 41.87%      |       |      |        |   |
| H25   | Pre-market | dir_p60_h30 | 3420  | 0    | 63.80% | 3 |
| 6.20% |            | 41.25%      |       |      |        |   |
| H25   | Pre-market | dir_p40_h60 | 3420  | 0    | 83.80% | 1 |
| 6.20% |            | 41.17%      |       |      |        |   |
| H25   | Pre-market | dir_p50_h60 | 3420  | 0    | 78.54% | 2 |
| 1.46% |            | 40.06%      |       |      |        |   |
| H25   | Pre-market | dir_p60_h60 | 3420  | 0    | 71.23% | 2 |
| 8.77% |            | 39.86%      |       |      |        |   |
| H25   | Pre-market | dir_p40_h90 | 3420  | 0    | 88.54% | 1 |
| 1.46% |            | 41.58%      |       |      |        |   |
| H25   | Pre-market | dir_p50_h90 | 3420  | 0    | 83.95% | 1 |
| 6.05% |            | 41.66%      |       |      |        |   |
| H25   | Pre-market | dir_p60_h90 | 3420  | 0    | 78.63% | 2 |
| 1.37% |            | 40.54%      |       |      |        |   |
| H25   | Regular    | dir_p40_h30 | 17100 | 0    | 83.83% | 1 |
| 6.17% |            | 49.19%      |       |      |        |   |
| H25   | Regular    | dir_p50_h30 | 17100 | 0    | 76.88% | 2 |
| 3.12% |            | 49.13%      |       |      |        |   |
| H25   | Regular    | dir_p60_h30 | 17100 | 0    | 68.48% | 3 |
| 1.52% |            | 48.81%      |       |      |        |   |
| H25   | Regular    | dir_p40_h60 | 15447 | 1653 | 84.86% | 1 |
| 5.14% |            | 49.90%      |       |      |        |   |
| H25   | Regular    | dir_p50_h60 | 15447 | 1653 | 77.06% | 2 |
| 2.94% |            | 49.37%      |       |      |        |   |
| H25   | Regular    | dir_p60_h60 | 15447 | 1653 | 68.87% | 3 |
| 1.13% |            | 48.78%      |       |      |        |   |
| H25   | Regular    | dir_p40_h90 | 13737 | 3363 | 85.21% | 1 |
| 4.79% |            | 50.35%      |       |      |        |   |

|       |            |             |       |      |        |   |
|-------|------------|-------------|-------|------|--------|---|
| H25   | Regular    | dir_p50_h90 | 13737 | 3363 | 77.70% | 2 |
| 2.30% |            | 50.10%      |       |      |        |   |
| H25   | Regular    | dir_p60_h90 | 13737 | 3363 | 69.98% | 3 |
| 0.02% |            | 49.18%      |       |      |        |   |
| H26   | Closing    | dir_p40_h30 | 57    | 1653 | 70.18% | 2 |
| 9.82% |            | 55.00%      |       |      |        |   |
| H26   | Closing    | dir_p50_h30 | 57    | 1653 | 57.89% | 4 |
| 2.11% |            | 60.61%      |       |      |        |   |
| H26   | Closing    | dir_p60_h30 | 57    | 1653 | 52.63% | 4 |
| 7.37% |            | 60.00%      |       |      |        |   |
| H26   | Closing    | dir_p40_h60 | 0     | 1710 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H26   | Closing    | dir_p50_h60 | 0     | 1710 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H26   | Closing    | dir_p60_h60 | 0     | 1710 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H26   | Closing    | dir_p40_h90 | 0     | 1710 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H26   | Closing    | dir_p50_h90 | 0     | 1710 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H26   | Closing    | dir_p60_h90 | 0     | 1710 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| H26   | Opening    | dir_p40_h30 | 3420  | 0    | 89.65% | 1 |
| 0.35% |            | 48.63%      |       |      |        |   |
| H26   | Opening    | dir_p50_h30 | 3420  | 0    | 83.39% | 1 |
| 6.61% |            | 48.46%      |       |      |        |   |
| H26   | Opening    | dir_p60_h30 | 3420  | 0    | 75.70% | 2 |
| 4.30% |            | 48.17%      |       |      |        |   |
| H26   | Opening    | dir_p40_h60 | 3420  | 0    | 88.01% | 1 |
| 1.99% |            | 48.97%      |       |      |        |   |
| H26   | Opening    | dir_p50_h60 | 3420  | 0    | 81.87% | 1 |
| 8.13% |            | 49.29%      |       |      |        |   |
| H26   | Opening    | dir_p60_h60 | 3420  | 0    | 74.47% | 2 |
| 5.53% |            | 49.59%      |       |      |        |   |
| H26   | Opening    | dir_p40_h90 | 3420  | 0    | 86.35% | 1 |
| 3.65% |            | 50.59%      |       |      |        |   |
| H26   | Opening    | dir_p50_h90 | 3420  | 0    | 80.99% | 1 |
| 9.01% |            | 50.94%      |       |      |        |   |
| H26   | Opening    | dir_p60_h90 | 3420  | 0    | 73.57% | 2 |
| 6.43% |            | 51.27%      |       |      |        |   |
| H26   | Overnight  | dir_p40_h30 | 13680 | 57   | 84.42% | 1 |
| 5.58% |            | 48.70%      |       |      |        |   |
| H26   | Overnight  | dir_p50_h30 | 13680 | 57   | 77.44% | 2 |
| 2.56% |            | 48.76%      |       |      |        |   |
| H26   | Overnight  | dir_p60_h30 | 13680 | 57   | 69.85% | 3 |
| 0.15% |            | 48.47%      |       |      |        |   |
| H26   | Overnight  | dir_p40_h60 | 13680 | 57   | 85.47% | 1 |
| 4.53% |            | 49.99%      |       |      |        |   |
| H26   | Overnight  | dir_p50_h60 | 13680 | 57   | 77.79% | 2 |
| 2.21% |            | 49.28%      |       |      |        |   |
| H26   | Overnight  | dir_p60_h60 | 13680 | 57   | 69.58% | 3 |
| 0.42% |            | 48.23%      |       |      |        |   |
| H26   | Overnight  | dir_p40_h90 | 13680 | 57   | 85.62% | 1 |
| 4.38% |            | 50.92%      |       |      |        |   |
| H26   | Overnight  | dir_p50_h90 | 13680 | 57   | 78.14% | 2 |
| 1.86% |            | 50.44%      |       |      |        |   |
| H26   | Overnight  | dir_p60_h90 | 13680 | 57   | 69.02% | 3 |
| 0.98% |            | 49.85%      |       |      |        |   |
| H26   | Pre-market | dir_p40_h30 | 3420  | 0    | 83.39% | 1 |
| 6.61% |            | 52.49%      |       |      |        |   |

|       |            |             |       |      |        |   |
|-------|------------|-------------|-------|------|--------|---|
| H26   | Pre-market | dir_p50_h30 | 3420  | 0    | 75.70% | 2 |
| 4.30% |            | 52.92%      |       |      |        |   |
| H26   | Pre-market | dir_p60_h30 | 3420  | 0    | 67.28% | 3 |
| 2.72% |            | 53.76%      |       |      |        |   |
| H26   | Pre-market | dir_p40_h60 | 3420  | 0    | 91.40% |   |
| 8.60% |            | 56.40%      |       |      |        |   |
| H26   | Pre-market | dir_p50_h60 | 3420  | 0    | 86.32% | 1 |
| 3.68% |            | 56.37%      |       |      |        |   |
| H26   | Pre-market | dir_p60_h60 | 3420  | 0    | 80.35% | 1 |
| 9.65% |            | 56.84%      |       |      |        |   |
| H26   | Pre-market | dir_p40_h90 | 3420  | 0    | 92.22% |   |
| 7.78% |            | 57.58%      |       |      |        |   |
| H26   | Pre-market | dir_p50_h90 | 3420  | 0    | 87.89% | 1 |
| 2.11% |            | 58.15%      |       |      |        |   |
| H26   | Pre-market | dir_p60_h90 | 3420  | 0    | 80.29% | 1 |
| 9.71% |            | 57.32%      |       |      |        |   |
| H26   | Regular    | dir_p40_h30 | 17100 | 0    | 78.46% | 2 |
| 1.54% |            | 46.83%      |       |      |        |   |
| H26   | Regular    | dir_p50_h30 | 17100 | 0    | 71.39% | 2 |
| 8.61% |            | 46.43%      |       |      |        |   |
| H26   | Regular    | dir_p60_h30 | 17100 | 0    | 63.54% | 3 |
| 6.46% |            | 46.49%      |       |      |        |   |
| H26   | Regular    | dir_p40_h60 | 15447 | 1653 | 78.07% | 2 |
| 1.93% |            | 46.07%      |       |      |        |   |
| H26   | Regular    | dir_p50_h60 | 15447 | 1653 | 70.58% | 2 |
| 9.42% |            | 45.45%      |       |      |        |   |
| H26   | Regular    | dir_p60_h60 | 15447 | 1653 | 62.08% | 3 |
| 7.92% |            | 45.44%      |       |      |        |   |
| H26   | Regular    | dir_p40_h90 | 13737 | 3363 | 77.29% | 2 |
| 2.71% |            | 46.55%      |       |      |        |   |
| H26   | Regular    | dir_p50_h90 | 13737 | 3363 | 69.45% | 3 |
| 0.55% |            | 46.44%      |       |      |        |   |
| H26   | Regular    | dir_p60_h90 | 13737 | 3363 | 61.57% | 3 |
| 8.43% |            | 45.68%      |       |      |        |   |
| M20   | Closing    | dir_p40_h30 | 56    | 1624 | 67.86% | 3 |
| 2.14% |            | 50.00%      |       |      |        |   |
| M20   | Closing    | dir_p50_h30 | 56    | 1624 | 62.50% | 3 |
| 7.50% |            | 51.43%      |       |      |        |   |
| M20   | Closing    | dir_p60_h30 | 56    | 1624 | 55.36% | 4 |
| 4.64% |            | 54.84%      |       |      |        |   |
| M20   | Closing    | dir_p40_h60 | 0     | 1680 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M20   | Closing    | dir_p50_h60 | 0     | 1680 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M20   | Closing    | dir_p60_h60 | 0     | 1680 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M20   | Closing    | dir_p40_h90 | 0     | 1680 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M20   | Closing    | dir_p50_h90 | 0     | 1680 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M20   | Closing    | dir_p60_h90 | 0     | 1680 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M20   | Opening    | dir_p40_h30 | 3360  | 0    | 48.54% | 5 |
| 1.46% |            | 53.71%      |       |      |        |   |
| M20   | Opening    | dir_p50_h30 | 3360  | 0    | 36.49% | 6 |
| 3.51% |            | 51.22%      |       |      |        |   |
| M20   | Opening    | dir_p60_h30 | 3360  | 0    | 26.34% | 7 |
| 3.66% |            | 48.02%      |       |      |        |   |
| M20   | Opening    | dir_p40_h60 | 3360  | 0    | 48.36% | 5 |
| 1.64% |            | 52.18%      |       |      |        |   |

|       |            |             |       |      |        |   |
|-------|------------|-------------|-------|------|--------|---|
| M20   | Opening    | dir_p50_h60 | 3360  | 0    | 37.44% | 6 |
| 2.56% |            | 49.21%      |       |      |        |   |
| M20   | Opening    | dir_p60_h60 | 3360  | 0    | 27.86% | 7 |
| 2.14% |            | 47.65%      |       |      |        |   |
| M20   | Opening    | dir_p40_h90 | 3360  | 0    | 49.23% | 5 |
| 0.77% |            | 50.42%      |       |      |        |   |
| M20   | Opening    | dir_p50_h90 | 3360  | 0    | 37.86% | 6 |
| 2.14% |            | 48.27%      |       |      |        |   |
| M20   | Opening    | dir_p60_h90 | 3360  | 0    | 27.77% | 7 |
| 2.23% |            | 45.02%      |       |      |        |   |
| M20   | Overnight  | dir_p40_h30 | 13440 | 56   | 67.91% | 3 |
| 2.09% |            | 49.60%      |       |      |        |   |
| M20   | Overnight  | dir_p50_h30 | 13440 | 56   | 57.22% | 4 |
| 2.78% |            | 49.55%      |       |      |        |   |
| M20   | Overnight  | dir_p60_h30 | 13440 | 56   | 46.56% | 5 |
| 3.44% |            | 49.29%      |       |      |        |   |
| M20   | Overnight  | dir_p40_h60 | 13440 | 56   | 67.83% | 3 |
| 2.17% |            | 50.52%      |       |      |        |   |
| M20   | Overnight  | dir_p50_h60 | 13440 | 56   | 56.43% | 4 |
| 3.57% |            | 50.13%      |       |      |        |   |
| M20   | Overnight  | dir_p60_h60 | 13440 | 56   | 44.71% | 5 |
| 5.29% |            | 49.51%      |       |      |        |   |
| M20   | Overnight  | dir_p40_h90 | 13440 | 56   | 65.42% | 3 |
| 4.58% |            | 50.80%      |       |      |        |   |
| M20   | Overnight  | dir_p50_h90 | 13440 | 56   | 54.59% | 4 |
| 5.41% |            | 51.32%      |       |      |        |   |
| M20   | Overnight  | dir_p60_h90 | 13440 | 56   | 43.56% | 5 |
| 6.44% |            | 51.33%      |       |      |        |   |
| M20   | Pre-market | dir_p40_h30 | 3360  | 0    | 60.51% | 3 |
| 9.49% |            | 55.93%      |       |      |        |   |
| M20   | Pre-market | dir_p50_h30 | 3360  | 0    | 49.88% | 5 |
| 0.12% |            | 55.25%      |       |      |        |   |
| M20   | Pre-market | dir_p60_h30 | 3360  | 0    | 39.79% | 6 |
| 0.21% |            | 54.60%      |       |      |        |   |
| M20   | Pre-market | dir_p40_h60 | 3360  | 0    | 55.06% | 4 |
| 4.94% |            | 56.00%      |       |      |        |   |
| M20   | Pre-market | dir_p50_h60 | 3360  | 0    | 41.31% | 5 |
| 8.69% |            | 50.65%      |       |      |        |   |
| M20   | Pre-market | dir_p60_h60 | 3360  | 0    | 29.38% | 7 |
| 0.62% |            | 53.09%      |       |      |        |   |
| M20   | Pre-market | dir_p40_h90 | 3360  | 0    | 47.77% | 5 |
| 2.23% |            | 57.38%      |       |      |        |   |
| M20   | Pre-market | dir_p50_h90 | 3360  | 0    | 35.95% | 6 |
| 4.05% |            | 63.66%      |       |      |        |   |
| M20   | Pre-market | dir_p60_h90 | 3360  | 0    | 26.22% | 7 |
| 3.78% |            | 61.98%      |       |      |        |   |
| M20   | Regular    | dir_p40_h30 | 16800 | 0    | 54.74% | 4 |
| 5.26% |            | 50.84%      |       |      |        |   |
| M20   | Regular    | dir_p50_h30 | 16800 | 0    | 44.78% | 5 |
| 5.22% |            | 48.31%      |       |      |        |   |
| M20   | Regular    | dir_p60_h30 | 16800 | 0    | 34.33% | 6 |
| 5.67% |            | 45.58%      |       |      |        |   |
| M20   | Regular    | dir_p40_h60 | 15176 | 1624 | 53.61% | 4 |
| 6.39% |            | 50.86%      |       |      |        |   |
| M20   | Regular    | dir_p50_h60 | 15176 | 1624 | 42.70% | 5 |
| 7.30% |            | 49.07%      |       |      |        |   |
| M20   | Regular    | dir_p60_h60 | 15176 | 1624 | 33.43% | 6 |
| 6.57% |            | 47.81%      |       |      |        |   |
| M20   | Regular    | dir_p40_h90 | 13496 | 3304 | 52.45% | 4 |
| 7.55% |            | 52.47%      |       |      |        |   |

|       |            |             |       |      |        |   |
|-------|------------|-------------|-------|------|--------|---|
| M20   | Regular    | dir_p50_h90 | 13496 | 3304 | 41.78% | 5 |
| 8.22% |            | 50.08%      |       |      |        |   |
| M20   | Regular    | dir_p60_h90 | 13496 | 3304 | 32.68% | 6 |
| 7.32% |            | 47.98%      |       |      |        |   |
| M21   | Closing    | dir_p40_h30 | 61    | 1769 | 55.74% | 4 |
| 4.26% |            | 47.06%      |       |      |        |   |
| M21   | Closing    | dir_p50_h30 | 61    | 1769 | 45.90% | 5 |
| 4.10% |            | 39.29%      |       |      |        |   |
| M21   | Closing    | dir_p60_h30 | 61    | 1769 | 29.51% | 7 |
| 0.49% |            | 38.89%      |       |      |        |   |
| M21   | Closing    | dir_p40_h60 | 0     | 1830 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M21   | Closing    | dir_p50_h60 | 0     | 1830 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M21   | Closing    | dir_p60_h60 | 0     | 1830 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M21   | Closing    | dir_p40_h90 | 0     | 1830 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M21   | Closing    | dir_p50_h90 | 0     | 1830 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M21   | Closing    | dir_p60_h90 | 0     | 1830 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M21   | Opening    | dir_p40_h30 | 3660  | 0    | 45.38% | 5 |
| 4.62% |            | 51.48%      |       |      |        |   |
| M21   | Opening    | dir_p50_h30 | 3660  | 0    | 35.71% | 6 |
| 4.29% |            | 50.11%      |       |      |        |   |
| M21   | Opening    | dir_p60_h30 | 3660  | 0    | 25.44% | 7 |
| 4.56% |            | 47.15%      |       |      |        |   |
| M21   | Opening    | dir_p40_h60 | 3660  | 0    | 44.86% | 5 |
| 5.14% |            | 50.18%      |       |      |        |   |
| M21   | Opening    | dir_p50_h60 | 3660  | 0    | 32.51% | 6 |
| 7.49% |            | 44.87%      |       |      |        |   |
| M21   | Opening    | dir_p60_h60 | 3660  | 0    | 23.58% | 7 |
| 6.42% |            | 39.75%      |       |      |        |   |
| M21   | Opening    | dir_p40_h90 | 3660  | 0    | 46.37% | 5 |
| 3.63% |            | 49.44%      |       |      |        |   |
| M21   | Opening    | dir_p50_h90 | 3660  | 0    | 35.33% | 6 |
| 4.67% |            | 44.62%      |       |      |        |   |
| M21   | Opening    | dir_p60_h90 | 3660  | 0    | 25.77% | 7 |
| 4.23% |            | 40.08%      |       |      |        |   |
| M21   | Overnight  | dir_p40_h30 | 14640 | 61   | 49.89% | 5 |
| 0.11% |            | 51.94%      |       |      |        |   |
| M21   | Overnight  | dir_p50_h30 | 14640 | 61   | 39.54% | 6 |
| 0.46% |            | 51.20%      |       |      |        |   |
| M21   | Overnight  | dir_p60_h30 | 14640 | 61   | 30.29% | 6 |
| 9.71% |            | 48.49%      |       |      |        |   |
| M21   | Overnight  | dir_p40_h60 | 14640 | 61   | 50.30% | 4 |
| 9.70% |            | 50.42%      |       |      |        |   |
| M21   | Overnight  | dir_p50_h60 | 14640 | 61   | 39.20% | 6 |
| 0.80% |            | 49.10%      |       |      |        |   |
| M21   | Overnight  | dir_p60_h60 | 14640 | 61   | 28.51% | 7 |
| 1.49% |            | 46.98%      |       |      |        |   |
| M21   | Overnight  | dir_p40_h90 | 14640 | 61   | 50.83% | 4 |
| 9.17% |            | 50.39%      |       |      |        |   |
| M21   | Overnight  | dir_p50_h90 | 14640 | 61   | 39.25% | 6 |
| 0.75% |            | 49.83%      |       |      |        |   |
| M21   | Overnight  | dir_p60_h90 | 14640 | 61   | 28.50% | 7 |
| 1.50% |            | 49.15%      |       |      |        |   |
| M21   | Pre-market | dir_p40_h30 | 3660  | 0    | 50.68% | 4 |
| 9.32% |            | 54.66%      |       |      |        |   |

|       |            |             |       |      |        |   |
|-------|------------|-------------|-------|------|--------|---|
| M21   | Pre-market | dir_p50_h30 | 3660  | 0    | 39.97% | 6 |
| 0.03% |            | 54.00%      |       |      |        |   |
| M21   | Pre-market | dir_p60_h30 | 3660  | 0    | 30.68% | 6 |
| 9.32% |            | 54.85%      |       |      |        |   |
| M21   | Pre-market | dir_p40_h60 | 3660  | 0    | 46.83% | 5 |
| 3.17% |            | 58.05%      |       |      |        |   |
| M21   | Pre-market | dir_p50_h60 | 3660  | 0    | 36.89% | 6 |
| 3.11% |            | 59.04%      |       |      |        |   |
| M21   | Pre-market | dir_p60_h60 | 3660  | 0    | 29.40% | 7 |
| 0.60% |            | 58.83%      |       |      |        |   |
| M21   | Pre-market | dir_p40_h90 | 3660  | 0    | 49.18% | 5 |
| 0.82% |            | 55.83%      |       |      |        |   |
| M21   | Pre-market | dir_p50_h90 | 3660  | 0    | 38.33% | 6 |
| 1.67% |            | 57.09%      |       |      |        |   |
| M21   | Pre-market | dir_p60_h90 | 3660  | 0    | 28.06% | 7 |
| 1.94% |            | 62.03%      |       |      |        |   |
| M21   | Regular    | dir_p40_h30 | 18300 | 0    | 43.66% | 5 |
| 6.34% |            | 46.58%      |       |      |        |   |
| M21   | Regular    | dir_p50_h30 | 18300 | 0    | 34.15% | 6 |
| 5.85% |            | 45.58%      |       |      |        |   |
| M21   | Regular    | dir_p60_h30 | 18300 | 0    | 25.20% | 7 |
| 4.80% |            | 44.62%      |       |      |        |   |
| M21   | Regular    | dir_p40_h60 | 16531 | 1769 | 41.95% | 5 |
| 8.05% |            | 48.02%      |       |      |        |   |
| M21   | Regular    | dir_p50_h60 | 16531 | 1769 | 33.18% | 6 |
| 6.82% |            | 45.85%      |       |      |        |   |
| M21   | Regular    | dir_p60_h60 | 16531 | 1769 | 25.23% | 7 |
| 4.77% |            | 42.69%      |       |      |        |   |
| M21   | Regular    | dir_p40_h90 | 14701 | 3599 | 41.33% | 5 |
| 8.67% |            | 44.45%      |       |      |        |   |
| M21   | Regular    | dir_p50_h90 | 14701 | 3599 | 33.42% | 6 |
| 6.58% |            | 42.26%      |       |      |        |   |
| M21   | Regular    | dir_p60_h90 | 14701 | 3599 | 25.26% | 7 |
| 4.74% |            | 38.45%      |       |      |        |   |
| M22   | Closing    | dir_p40_h30 | 61    | 1769 | 93.44% |   |
| 6.56% |            | 43.86%      |       |      |        |   |
| M22   | Closing    | dir_p50_h30 | 61    | 1769 | 88.52% | 1 |
| 1.48% |            | 46.30%      |       |      |        |   |
| M22   | Closing    | dir_p60_h30 | 61    | 1769 | 83.61% | 1 |
| 6.39% |            | 43.14%      |       |      |        |   |
| M22   | Closing    | dir_p40_h60 | 0     | 1830 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M22   | Closing    | dir_p50_h60 | 0     | 1830 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M22   | Closing    | dir_p60_h60 | 0     | 1830 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M22   | Closing    | dir_p40_h90 | 0     | 1830 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M22   | Closing    | dir_p50_h90 | 0     | 1830 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M22   | Closing    | dir_p60_h90 | 0     | 1830 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M22   | Opening    | dir_p40_h30 | 3660  | 0    | 87.54% | 1 |
| 2.46% |            | 45.13%      |       |      |        |   |
| M22   | Opening    | dir_p50_h30 | 3660  | 0    | 80.87% | 1 |
| 9.13% |            | 45.24%      |       |      |        |   |
| M22   | Opening    | dir_p60_h30 | 3660  | 0    | 72.35% | 2 |
| 7.65% |            | 44.98%      |       |      |        |   |
| M22   | Opening    | dir_p40_h60 | 3660  | 0    | 88.36% | 1 |
| 1.64% |            | 44.93%      |       |      |        |   |

|       |            |             |       |      |        |   |
|-------|------------|-------------|-------|------|--------|---|
| M22   | Opening    | dir_p50_h60 | 3660  | 0    | 81.48% | 1 |
| 8.52% |            | 44.77%      |       |      |        |   |
| M22   | Opening    | dir_p60_h60 | 3660  | 0    | 72.16% | 2 |
| 7.84% |            | 43.88%      |       |      |        |   |
| M22   | Opening    | dir_p40_h90 | 3660  | 0    | 88.58% | 1 |
| 1.42% |            | 44.17%      |       |      |        |   |
| M22   | Opening    | dir_p50_h90 | 3660  | 0    | 81.94% | 1 |
| 8.06% |            | 43.95%      |       |      |        |   |
| M22   | Opening    | dir_p60_h90 | 3660  | 0    | 72.95% | 2 |
| 7.05% |            | 42.96%      |       |      |        |   |
| M22   | Overnight  | dir_p40_h30 | 14640 | 61   | 86.66% | 1 |
| 3.34% |            | 46.92%      |       |      |        |   |
| M22   | Overnight  | dir_p50_h30 | 14640 | 61   | 78.85% | 2 |
| 1.15% |            | 46.48%      |       |      |        |   |
| M22   | Overnight  | dir_p60_h30 | 14640 | 61   | 68.55% | 3 |
| 1.45% |            | 46.13%      |       |      |        |   |
| M22   | Overnight  | dir_p40_h60 | 14640 | 61   | 88.07% | 1 |
| 1.93% |            | 44.82%      |       |      |        |   |
| M22   | Overnight  | dir_p50_h60 | 14640 | 61   | 79.43% | 2 |
| 0.57% |            | 44.63%      |       |      |        |   |
| M22   | Overnight  | dir_p60_h60 | 14640 | 61   | 67.90% | 3 |
| 2.10% |            | 44.55%      |       |      |        |   |
| M22   | Overnight  | dir_p40_h90 | 14640 | 61   | 86.84% | 1 |
| 3.16% |            | 43.66%      |       |      |        |   |
| M22   | Overnight  | dir_p50_h90 | 14640 | 61   | 77.96% | 2 |
| 2.04% |            | 43.91%      |       |      |        |   |
| M22   | Overnight  | dir_p60_h90 | 14640 | 61   | 67.34% | 3 |
| 2.66% |            | 44.12%      |       |      |        |   |
| M22   | Pre-market | dir_p40_h30 | 3660  | 0    | 76.99% | 2 |
| 3.01% |            | 44.32%      |       |      |        |   |
| M22   | Pre-market | dir_p50_h30 | 3660  | 0    | 69.78% | 3 |
| 0.22% |            | 44.28%      |       |      |        |   |
| M22   | Pre-market | dir_p60_h30 | 3660  | 0    | 61.45% | 3 |
| 8.55% |            | 43.98%      |       |      |        |   |
| M22   | Pre-market | dir_p40_h60 | 3660  | 0    | 87.73% | 1 |
| 2.27% |            | 44.81%      |       |      |        |   |
| M22   | Pre-market | dir_p50_h60 | 3660  | 0    | 79.92% | 2 |
| 0.08% |            | 44.89%      |       |      |        |   |
| M22   | Pre-market | dir_p60_h60 | 3660  | 0    | 71.42% | 2 |
| 8.58% |            | 45.07%      |       |      |        |   |
| M22   | Pre-market | dir_p40_h90 | 3660  | 0    | 92.57% |   |
| 7.43% |            | 43.71%      |       |      |        |   |
| M22   | Pre-market | dir_p50_h90 | 3660  | 0    | 87.73% | 1 |
| 2.27% |            | 42.63%      |       |      |        |   |
| M22   | Pre-market | dir_p60_h90 | 3660  | 0    | 79.97% | 2 |
| 0.03% |            | 43.56%      |       |      |        |   |
| M22   | Regular    | dir_p40_h30 | 18300 | 0    | 89.25% | 1 |
| 0.75% |            | 49.20%      |       |      |        |   |
| M22   | Regular    | dir_p50_h30 | 18300 | 0    | 82.81% | 1 |
| 7.19% |            | 48.84%      |       |      |        |   |
| M22   | Regular    | dir_p60_h30 | 18300 | 0    | 73.26% | 2 |
| 6.74% |            | 48.16%      |       |      |        |   |
| M22   | Regular    | dir_p40_h60 | 16531 | 1769 | 90.66% |   |
| 9.34% |            | 48.46%      |       |      |        |   |
| M22   | Regular    | dir_p50_h60 | 16531 | 1769 | 83.71% | 1 |
| 6.29% |            | 48.09%      |       |      |        |   |
| M22   | Regular    | dir_p60_h60 | 16531 | 1769 | 74.15% | 2 |
| 5.85% |            | 47.64%      |       |      |        |   |
| M22   | Regular    | dir_p40_h90 | 14701 | 3599 | 90.01% |   |
| 9.99% |            | 49.59%      |       |      |        |   |



|       |            |             |       |      |        |   |
|-------|------------|-------------|-------|------|--------|---|
| M22   | Regular    | dir_p50_h90 | 14701 | 3599 | 83.16% | 1 |
| 6.84% |            | 49.35%      |       |      |        |   |
| M22   | Regular    | dir_p60_h90 | 14701 | 3599 | 74.68% | 2 |
| 5.32% |            | 48.98%      |       |      |        |   |
| M23   | Closing    | dir_p40_h30 | 55    | 1595 | 45.45% | 5 |
| 4.55% |            | 52.00%      |       |      |        |   |
| M23   | Closing    | dir_p50_h30 | 55    | 1595 | 34.55% | 6 |
| 5.45% |            | 52.63%      |       |      |        |   |
| M23   | Closing    | dir_p60_h30 | 55    | 1595 | 20.00% | 8 |
| 0.00% |            | 54.55%      |       |      |        |   |
| M23   | Closing    | dir_p40_h60 | 0     | 1650 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M23   | Closing    | dir_p50_h60 | 0     | 1650 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M23   | Closing    | dir_p60_h60 | 0     | 1650 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M23   | Closing    | dir_p40_h90 | 0     | 1650 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M23   | Closing    | dir_p50_h90 | 0     | 1650 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M23   | Closing    | dir_p60_h90 | 0     | 1650 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M23   | Opening    | dir_p40_h30 | 3300  | 0    | 45.55% | 5 |
| 4.45% |            | 48.84%      |       |      |        |   |
| M23   | Opening    | dir_p50_h30 | 3300  | 0    | 33.48% | 6 |
| 6.52% |            | 46.52%      |       |      |        |   |
| M23   | Opening    | dir_p60_h30 | 3300  | 0    | 23.15% | 7 |
| 6.85% |            | 44.24%      |       |      |        |   |
| M23   | Opening    | dir_p40_h60 | 3300  | 0    | 46.67% | 5 |
| 3.33% |            | 52.14%      |       |      |        |   |
| M23   | Opening    | dir_p50_h60 | 3300  | 0    | 35.30% | 6 |
| 4.70% |            | 51.33%      |       |      |        |   |
| M23   | Opening    | dir_p60_h60 | 3300  | 0    | 26.09% | 7 |
| 3.91% |            | 49.01%      |       |      |        |   |
| M23   | Opening    | dir_p40_h90 | 3300  | 0    | 48.73% | 5 |
| 1.27% |            | 54.91%      |       |      |        |   |
| M23   | Opening    | dir_p50_h90 | 3300  | 0    | 36.24% | 6 |
| 3.76% |            | 53.26%      |       |      |        |   |
| M23   | Opening    | dir_p60_h90 | 3300  | 0    | 26.09% | 7 |
| 3.91% |            | 51.22%      |       |      |        |   |
| M23   | Overnight  | dir_p40_h30 | 13200 | 55   | 45.32% | 5 |
| 4.68% |            | 50.15%      |       |      |        |   |
| M23   | Overnight  | dir_p50_h30 | 13200 | 55   | 34.30% | 6 |
| 5.70% |            | 50.09%      |       |      |        |   |
| M23   | Overnight  | dir_p60_h30 | 13200 | 55   | 24.75% | 7 |
| 5.25% |            | 50.11%      |       |      |        |   |
| M23   | Overnight  | dir_p40_h60 | 13200 | 55   | 43.40% | 5 |
| 6.60% |            | 50.86%      |       |      |        |   |
| M23   | Overnight  | dir_p50_h60 | 13200 | 55   | 32.52% | 6 |
| 7.48% |            | 52.15%      |       |      |        |   |
| M23   | Overnight  | dir_p60_h60 | 13200 | 55   | 23.45% | 7 |
| 6.55% |            | 52.92%      |       |      |        |   |
| M23   | Overnight  | dir_p40_h90 | 13200 | 55   | 42.91% | 5 |
| 7.09% |            | 53.11%      |       |      |        |   |
| M23   | Overnight  | dir_p50_h90 | 13200 | 55   | 32.99% | 6 |
| 7.01% |            | 53.69%      |       |      |        |   |
| M23   | Overnight  | dir_p60_h90 | 13200 | 55   | 22.41% | 7 |
| 7.59% |            | 55.78%      |       |      |        |   |
| M23   | Pre-market | dir_p40_h30 | 3300  | 0    | 43.82% | 5 |
| 6.18% |            | 59.20%      |       |      |        |   |

|       |            |             |       |      |        |   |
|-------|------------|-------------|-------|------|--------|---|
| M23   | Pre-market | dir_p50_h30 | 3300  | 0    | 34.06% | 6 |
| 5.94% |            | 60.50%      |       |      |        |   |
| M23   | Pre-market | dir_p60_h30 | 3300  | 0    | 24.97% | 7 |
| 5.03% |            | 59.83%      |       |      |        |   |
| M23   | Pre-market | dir_p40_h60 | 3300  | 0    | 41.21% | 5 |
| 8.79% |            | 60.07%      |       |      |        |   |
| M23   | Pre-market | dir_p50_h60 | 3300  | 0    | 31.39% | 6 |
| 8.61% |            | 64.86%      |       |      |        |   |
| M23   | Pre-market | dir_p60_h60 | 3300  | 0    | 23.76% | 7 |
| 6.24% |            | 69.01%      |       |      |        |   |
| M23   | Pre-market | dir_p40_h90 | 3300  | 0    | 40.24% | 5 |
| 9.76% |            | 63.93%      |       |      |        |   |
| M23   | Pre-market | dir_p50_h90 | 3300  | 0    | 28.15% | 7 |
| 1.85% |            | 61.36%      |       |      |        |   |
| M23   | Pre-market | dir_p60_h90 | 3300  | 0    | 20.79% | 7 |
| 9.21% |            | 54.23%      |       |      |        |   |
| M23   | Regular    | dir_p40_h30 | 16500 | 0    | 50.85% | 4 |
| 9.15% |            | 53.59%      |       |      |        |   |
| M23   | Regular    | dir_p50_h30 | 16500 | 0    | 38.68% | 6 |
| 1.32% |            | 52.67%      |       |      |        |   |
| M23   | Regular    | dir_p60_h30 | 16500 | 0    | 27.18% | 7 |
| 2.82% |            | 50.13%      |       |      |        |   |
| M23   | Regular    | dir_p40_h60 | 14905 | 1595 | 50.71% | 4 |
| 9.29% |            | 54.17%      |       |      |        |   |
| M23   | Regular    | dir_p50_h60 | 14905 | 1595 | 37.97% | 6 |
| 2.03% |            | 52.01%      |       |      |        |   |
| M23   | Regular    | dir_p60_h60 | 14905 | 1595 | 26.68% | 7 |
| 3.32% |            | 52.31%      |       |      |        |   |
| M23   | Regular    | dir_p40_h90 | 13255 | 3245 | 50.97% | 4 |
| 9.03% |            | 53.88%      |       |      |        |   |
| M23   | Regular    | dir_p50_h90 | 13255 | 3245 | 37.48% | 6 |
| 2.52% |            | 52.78%      |       |      |        |   |
| M23   | Regular    | dir_p60_h90 | 13255 | 3245 | 25.75% | 7 |
| 4.25% |            | 51.36%      |       |      |        |   |
| M24   | Closing    | dir_p40_h30 | 66    | 1914 | 66.67% | 3 |
| 3.33% |            | 54.55%      |       |      |        |   |
| M24   | Closing    | dir_p50_h30 | 66    | 1914 | 59.09% | 4 |
| 0.91% |            | 51.28%      |       |      |        |   |
| M24   | Closing    | dir_p60_h30 | 66    | 1914 | 43.94% | 5 |
| 6.06% |            | 55.17%      |       |      |        |   |
| M24   | Closing    | dir_p40_h60 | 0     | 1980 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M24   | Closing    | dir_p50_h60 | 0     | 1980 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M24   | Closing    | dir_p60_h60 | 0     | 1980 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M24   | Closing    | dir_p40_h90 | 0     | 1980 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M24   | Closing    | dir_p50_h90 | 0     | 1980 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M24   | Closing    | dir_p60_h90 | 0     | 1980 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M24   | Opening    | dir_p40_h30 | 3960  | 0    | 52.35% | 4 |
| 7.65% |            | 48.96%      |       |      |        |   |
| M24   | Opening    | dir_p50_h30 | 3960  | 0    | 40.78% | 5 |
| 9.22% |            | 46.87%      |       |      |        |   |
| M24   | Opening    | dir_p60_h30 | 3960  | 0    | 30.15% | 6 |
| 9.85% |            | 42.63%      |       |      |        |   |
| M24   | Opening    | dir_p40_h60 | 3960  | 0    | 55.98% | 4 |
| 4.02% |            | 50.07%      |       |      |        |   |

|       |            |             |       |      |        |   |
|-------|------------|-------------|-------|------|--------|---|
| M24   | Opening    | dir_p50_h60 | 3960  | 0    | 44.87% | 5 |
| 5.13% |            | 46.99%      |       |      |        |   |
| M24   | Opening    | dir_p60_h60 | 3960  | 0    | 33.61% | 6 |
| 6.39% |            | 41.92%      |       |      |        |   |
| M24   | Opening    | dir_p40_h90 | 3960  | 0    | 52.27% | 4 |
| 7.73% |            | 49.32%      |       |      |        |   |
| M24   | Opening    | dir_p50_h90 | 3960  | 0    | 41.79% | 5 |
| 8.21% |            | 44.05%      |       |      |        |   |
| M24   | Opening    | dir_p60_h90 | 3960  | 0    | 32.80% | 6 |
| 7.20% |            | 38.72%      |       |      |        |   |
| M24   | Overnight  | dir_p40_h30 | 15840 | 66   | 56.41% | 4 |
| 3.59% |            | 48.98%      |       |      |        |   |
| M24   | Overnight  | dir_p50_h30 | 15840 | 66   | 45.08% | 5 |
| 4.92% |            | 48.14%      |       |      |        |   |
| M24   | Overnight  | dir_p60_h30 | 15840 | 66   | 34.91% | 6 |
| 5.09% |            | 47.83%      |       |      |        |   |
| M24   | Overnight  | dir_p40_h60 | 15840 | 66   | 57.08% | 4 |
| 2.92% |            | 50.10%      |       |      |        |   |
| M24   | Overnight  | dir_p50_h60 | 15840 | 66   | 47.09% | 5 |
| 2.91% |            | 49.07%      |       |      |        |   |
| M24   | Overnight  | dir_p60_h60 | 15840 | 66   | 36.75% | 6 |
| 3.25% |            | 47.48%      |       |      |        |   |
| M24   | Overnight  | dir_p40_h90 | 15840 | 66   | 57.75% | 4 |
| 2.25% |            | 48.81%      |       |      |        |   |
| M24   | Overnight  | dir_p50_h90 | 15840 | 66   | 48.08% | 5 |
| 1.92% |            | 47.06%      |       |      |        |   |
| M24   | Overnight  | dir_p60_h90 | 15840 | 66   | 38.34% | 6 |
| 1.66% |            | 45.02%      |       |      |        |   |
| M24   | Pre-market | dir_p40_h30 | 3960  | 0    | 57.98% | 4 |
| 2.02% |            | 45.60%      |       |      |        |   |
| M24   | Pre-market | dir_p50_h30 | 3960  | 0    | 46.69% | 5 |
| 3.31% |            | 42.56%      |       |      |        |   |
| M24   | Pre-market | dir_p60_h30 | 3960  | 0    | 33.94% | 6 |
| 6.06% |            | 40.33%      |       |      |        |   |
| M24   | Pre-market | dir_p40_h60 | 3960  | 0    | 52.02% | 4 |
| 7.98% |            | 43.45%      |       |      |        |   |
| M24   | Pre-market | dir_p50_h60 | 3960  | 0    | 42.42% | 5 |
| 7.58% |            | 44.94%      |       |      |        |   |
| M24   | Pre-market | dir_p60_h60 | 3960  | 0    | 32.37% | 6 |
| 7.63% |            | 45.79%      |       |      |        |   |
| M24   | Pre-market | dir_p40_h90 | 3960  | 0    | 51.67% | 4 |
| 8.33% |            | 47.26%      |       |      |        |   |
| M24   | Pre-market | dir_p50_h90 | 3960  | 0    | 39.24% | 6 |
| 0.76% |            | 46.53%      |       |      |        |   |
| M24   | Pre-market | dir_p60_h90 | 3960  | 0    | 28.46% | 7 |
| 1.54% |            | 40.02%      |       |      |        |   |
| M24   | Regular    | dir_p40_h30 | 19800 | 0    | 56.14% | 4 |
| 3.86% |            | 50.09%      |       |      |        |   |
| M24   | Regular    | dir_p50_h30 | 19800 | 0    | 45.64% | 5 |
| 4.36% |            | 49.61%      |       |      |        |   |
| M24   | Regular    | dir_p60_h30 | 19800 | 0    | 35.23% | 6 |
| 4.77% |            | 48.78%      |       |      |        |   |
| M24   | Regular    | dir_p40_h60 | 17886 | 1914 | 55.46% | 4 |
| 4.54% |            | 49.99%      |       |      |        |   |
| M24   | Regular    | dir_p50_h60 | 17886 | 1914 | 45.00% | 5 |
| 5.00% |            | 48.68%      |       |      |        |   |
| M24   | Regular    | dir_p60_h60 | 17886 | 1914 | 35.43% | 6 |
| 4.57% |            | 47.85%      |       |      |        |   |
| M24   | Regular    | dir_p40_h90 | 15906 | 3894 | 54.43% | 4 |
| 5.57% |            | 49.84%      |       |      |        |   |

|       |            |             |       |      |        |   |
|-------|------------|-------------|-------|------|--------|---|
| M24   | Regular    | dir_p50_h90 | 15906 | 3894 | 44.61% | 5 |
| 5.39% |            | 49.37%      |       |      |        |   |
| M24   | Regular    | dir_p60_h90 | 15906 | 3894 | 35.84% | 6 |
| 4.16% |            | 48.42%      |       |      |        |   |
| M25   | Closing    | dir_p40_h30 | 50    | 1450 | 86.00% | 1 |
| 4.00% |            | 58.14%      |       |      |        |   |
| M25   | Closing    | dir_p50_h30 | 50    | 1450 | 80.00% | 2 |
| 0.00% |            | 62.50%      |       |      |        |   |
| M25   | Closing    | dir_p60_h30 | 50    | 1450 | 70.00% | 3 |
| 0.00% |            | 62.86%      |       |      |        |   |
| M25   | Closing    | dir_p40_h60 | 0     | 1500 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M25   | Closing    | dir_p50_h60 | 0     | 1500 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M25   | Closing    | dir_p60_h60 | 0     | 1500 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M25   | Closing    | dir_p40_h90 | 0     | 1500 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M25   | Closing    | dir_p50_h90 | 0     | 1500 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M25   | Closing    | dir_p60_h90 | 0     | 1500 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M25   | Opening    | dir_p40_h30 | 3000  | 0    | 78.83% | 2 |
| 1.17% |            | 57.76%      |       |      |        |   |
| M25   | Opening    | dir_p50_h30 | 3000  | 0    | 70.57% | 2 |
| 9.43% |            | 57.68%      |       |      |        |   |
| M25   | Opening    | dir_p60_h30 | 3000  | 0    | 63.03% | 3 |
| 6.97% |            | 58.33%      |       |      |        |   |
| M25   | Opening    | dir_p40_h60 | 3000  | 0    | 81.07% | 1 |
| 8.93% |            | 58.55%      |       |      |        |   |
| M25   | Opening    | dir_p50_h60 | 3000  | 0    | 72.87% | 2 |
| 7.13% |            | 57.59%      |       |      |        |   |
| M25   | Opening    | dir_p60_h60 | 3000  | 0    | 64.30% | 3 |
| 5.70% |            | 56.04%      |       |      |        |   |
| M25   | Opening    | dir_p40_h90 | 3000  | 0    | 82.07% | 1 |
| 7.93% |            | 55.61%      |       |      |        |   |
| M25   | Opening    | dir_p50_h90 | 3000  | 0    | 74.83% | 2 |
| 5.17% |            | 54.57%      |       |      |        |   |
| M25   | Opening    | dir_p60_h90 | 3000  | 0    | 65.67% | 3 |
| 4.33% |            | 53.81%      |       |      |        |   |
| M25   | Overnight  | dir_p40_h30 | 12000 | 50   | 90.33% |   |
| 9.67% |            | 52.32%      |       |      |        |   |
| M25   | Overnight  | dir_p50_h30 | 12000 | 50   | 83.88% | 1 |
| 6.12% |            | 52.19%      |       |      |        |   |
| M25   | Overnight  | dir_p60_h30 | 12000 | 50   | 75.93% | 2 |
| 4.07% |            | 51.94%      |       |      |        |   |
| M25   | Overnight  | dir_p40_h60 | 12000 | 50   | 90.92% |   |
| 9.08% |            | 51.37%      |       |      |        |   |
| M25   | Overnight  | dir_p50_h60 | 12000 | 50   | 84.00% | 1 |
| 6.00% |            | 51.32%      |       |      |        |   |
| M25   | Overnight  | dir_p60_h60 | 12000 | 50   | 74.50% | 2 |
| 5.50% |            | 50.64%      |       |      |        |   |
| M25   | Overnight  | dir_p40_h90 | 12000 | 50   | 89.94% | 1 |
| 0.06% |            | 50.92%      |       |      |        |   |
| M25   | Overnight  | dir_p50_h90 | 12000 | 50   | 82.93% | 1 |
| 7.08% |            | 50.98%      |       |      |        |   |
| M25   | Overnight  | dir_p60_h90 | 12000 | 50   | 73.85% | 2 |
| 6.15% |            | 50.44%      |       |      |        |   |
| M25   | Pre-market | dir_p40_h30 | 3000  | 0    | 83.27% | 1 |
| 6.73% |            | 52.88%      |       |      |        |   |

|       |            |             |       |      |        |   |
|-------|------------|-------------|-------|------|--------|---|
| M25   | Pre-market | dir_p50_h30 | 3000  | 0    | 74.30% | 2 |
| 5.70% |            | 52.98%      |       |      |        |   |
| M25   | Pre-market | dir_p60_h30 | 3000  | 0    | 64.77% | 3 |
| 5.23% |            | 52.50%      |       |      |        |   |
| M25   | Pre-market | dir_p40_h60 | 3000  | 0    | 81.93% | 1 |
| 8.07% |            | 54.68%      |       |      |        |   |
| M25   | Pre-market | dir_p50_h60 | 3000  | 0    | 74.17% | 2 |
| 5.83% |            | 55.15%      |       |      |        |   |
| M25   | Pre-market | dir_p60_h60 | 3000  | 0    | 65.73% | 3 |
| 4.27% |            | 55.98%      |       |      |        |   |
| M25   | Pre-market | dir_p40_h90 | 3000  | 0    | 82.80% | 1 |
| 7.20% |            | 58.21%      |       |      |        |   |
| M25   | Pre-market | dir_p50_h90 | 3000  | 0    | 74.13% | 2 |
| 5.87% |            | 58.09%      |       |      |        |   |
| M25   | Pre-market | dir_p60_h90 | 3000  | 0    | 63.67% | 3 |
| 6.33% |            | 58.17%      |       |      |        |   |
| M25   | Regular    | dir_p40_h30 | 15000 | 0    | 82.19% | 1 |
| 7.81% |            | 50.53%      |       |      |        |   |
| M25   | Regular    | dir_p50_h30 | 15000 | 0    | 74.70% | 2 |
| 5.30% |            | 50.15%      |       |      |        |   |
| M25   | Regular    | dir_p60_h30 | 15000 | 0    | 65.77% | 3 |
| 4.23% |            | 49.69%      |       |      |        |   |
| M25   | Regular    | dir_p40_h60 | 13550 | 1450 | 82.32% | 1 |
| 7.68% |            | 50.09%      |       |      |        |   |
| M25   | Regular    | dir_p50_h60 | 13550 | 1450 | 74.51% | 2 |
| 5.49% |            | 49.29%      |       |      |        |   |
| M25   | Regular    | dir_p60_h60 | 13550 | 1450 | 65.69% | 3 |
| 4.31% |            | 48.13%      |       |      |        |   |
| M25   | Regular    | dir_p40_h90 | 12050 | 2950 | 83.39% | 1 |
| 6.61% |            | 51.66%      |       |      |        |   |
| M25   | Regular    | dir_p50_h90 | 12050 | 2950 | 75.82% | 2 |
| 4.18% |            | 50.55%      |       |      |        |   |
| M25   | Regular    | dir_p60_h90 | 12050 | 2950 | 67.47% | 3 |
| 2.53% |            | 49.51%      |       |      |        |   |
| M26   | Closing    | dir_p40_h30 | 19    | 551  | 84.21% | 1 |
| 5.79% |            | 62.50%      |       |      |        |   |
| M26   | Closing    | dir_p50_h30 | 19    | 551  | 73.68% | 2 |
| 6.32% |            | 64.29%      |       |      |        |   |
| M26   | Closing    | dir_p60_h30 | 19    | 551  | 68.42% | 3 |
| 1.58% |            | 69.23%      |       |      |        |   |
| M26   | Closing    | dir_p40_h60 | 0     | 570  | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M26   | Closing    | dir_p50_h60 | 0     | 570  | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M26   | Closing    | dir_p60_h60 | 0     | 570  | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M26   | Closing    | dir_p40_h90 | 0     | 570  | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M26   | Closing    | dir_p50_h90 | 0     | 570  | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M26   | Closing    | dir_p60_h90 | 0     | 570  | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| M26   | Opening    | dir_p40_h30 | 1140  | 0    | 87.72% | 1 |
| 2.28% |            | 58.60%      |       |      |        |   |
| M26   | Opening    | dir_p50_h30 | 1140  | 0    | 79.39% | 2 |
| 0.61% |            | 58.67%      |       |      |        |   |
| M26   | Opening    | dir_p60_h30 | 1140  | 0    | 68.42% | 3 |
| 1.58% |            | 58.46%      |       |      |        |   |
| M26   | Opening    | dir_p40_h60 | 1140  | 0    | 92.46% |   |
| 7.54% |            | 57.59%      |       |      |        |   |

|       |            |             |      |      |        |   |
|-------|------------|-------------|------|------|--------|---|
| M26   | Opening    | dir_p50_h60 | 1140 | 0    | 86.75% | 1 |
| 3.25% |            | 57.53%      |      |      |        |   |
| M26   | Opening    | dir_p60_h60 | 1140 | 0    | 77.63% | 2 |
| 2.37% |            | 56.84%      |      |      |        |   |
| M26   | Opening    | dir_p40_h90 | 1140 | 0    | 94.65% |   |
| 5.35% |            | 60.61%      |      |      |        |   |
| M26   | Opening    | dir_p50_h90 | 1140 | 0    | 89.47% | 1 |
| 0.53% |            | 60.10%      |      |      |        |   |
| M26   | Opening    | dir_p60_h90 | 1140 | 0    | 81.49% | 1 |
| 8.51% |            | 60.82%      |      |      |        |   |
| M26   | Overnight  | dir_p40_h30 | 4560 | 19   | 92.32% |   |
| 7.68% |            | 49.26%      |      |      |        |   |
| M26   | Overnight  | dir_p50_h30 | 4560 | 19   | 87.79% | 1 |
| 2.21% |            | 48.79%      |      |      |        |   |
| M26   | Overnight  | dir_p60_h30 | 4560 | 19   | 80.75% | 1 |
| 9.25% |            | 47.91%      |      |      |        |   |
| M26   | Overnight  | dir_p40_h60 | 4560 | 19   | 91.43% |   |
| 8.57% |            | 47.93%      |      |      |        |   |
| M26   | Overnight  | dir_p50_h60 | 4560 | 19   | 85.83% | 1 |
| 4.17% |            | 47.55%      |      |      |        |   |
| M26   | Overnight  | dir_p60_h60 | 4560 | 19   | 78.86% | 2 |
| 1.14% |            | 47.33%      |      |      |        |   |
| M26   | Overnight  | dir_p40_h90 | 4560 | 19   | 91.54% |   |
| 8.46% |            | 47.56%      |      |      |        |   |
| M26   | Overnight  | dir_p50_h90 | 4560 | 19   | 85.79% | 1 |
| 4.21% |            | 47.44%      |      |      |        |   |
| M26   | Overnight  | dir_p60_h90 | 4560 | 19   | 79.89% | 2 |
| 0.11% |            | 46.72%      |      |      |        |   |
| M26   | Pre-market | dir_p40_h30 | 1140 | 0    | 82.02% | 1 |
| 7.98% |            | 57.97%      |      |      |        |   |
| M26   | Pre-market | dir_p50_h30 | 1140 | 0    | 75.70% | 2 |
| 4.30% |            | 57.82%      |      |      |        |   |
| M26   | Pre-market | dir_p60_h30 | 1140 | 0    | 67.72% | 3 |
| 2.28% |            | 58.29%      |      |      |        |   |
| M26   | Pre-market | dir_p40_h60 | 1140 | 0    | 90.61% |   |
| 9.39% |            | 54.99%      |      |      |        |   |
| M26   | Pre-market | dir_p50_h60 | 1140 | 0    | 79.39% | 2 |
| 0.61% |            | 53.37%      |      |      |        |   |
| M26   | Pre-market | dir_p60_h60 | 1140 | 0    | 71.84% | 2 |
| 8.16% |            | 52.26%      |      |      |        |   |
| M26   | Pre-market | dir_p40_h90 | 1140 | 0    | 92.63% |   |
| 7.37% |            | 52.56%      |      |      |        |   |
| M26   | Pre-market | dir_p50_h90 | 1140 | 0    | 87.89% | 1 |
| 2.11% |            | 52.50%      |      |      |        |   |
| M26   | Pre-market | dir_p60_h90 | 1140 | 0    | 78.86% | 2 |
| 1.14% |            | 51.39%      |      |      |        |   |
| M26   | Regular    | dir_p40_h30 | 5700 | 0    | 88.56% | 1 |
| 1.44% |            | 52.06%      |      |      |        |   |
| M26   | Regular    | dir_p50_h30 | 5700 | 0    | 82.21% | 1 |
| 7.79% |            | 52.05%      |      |      |        |   |
| M26   | Regular    | dir_p60_h30 | 5700 | 0    | 72.84% | 2 |
| 7.16% |            | 52.07%      |      |      |        |   |
| M26   | Regular    | dir_p40_h60 | 5149 | 551  | 91.03% |   |
| 8.97% |            | 50.71%      |      |      |        |   |
| M26   | Regular    | dir_p50_h60 | 5149 | 551  | 84.75% | 1 |
| 5.25% |            | 49.75%      |      |      |        |   |
| M26   | Regular    | dir_p60_h60 | 5149 | 551  | 76.46% | 2 |
| 3.54% |            | 48.92%      |      |      |        |   |
| M26   | Regular    | dir_p40_h90 | 4579 | 1121 | 91.13% |   |
| 8.87% |            | 47.54%      |      |      |        |   |

|       |            |             |       |      |        |   |
|-------|------------|-------------|-------|------|--------|---|
| M26   | Regular    | dir_p50_h90 | 4579  | 1121 | 86.00% | 1 |
| 4.00% |            | 47.21%      |       |      |        |   |
| M26   | Regular    | dir_p60_h90 | 4579  | 1121 | 78.12% | 2 |
| 1.88% |            | 46.30%      |       |      |        |   |
| U20   | Closing    | dir_p40_h30 | 61    | 1769 | 59.02% | 4 |
| 0.98% |            | 44.44%      |       |      |        |   |
| U20   | Closing    | dir_p50_h30 | 61    | 1769 | 40.98% | 5 |
| 9.02% |            | 48.00%      |       |      |        |   |
| U20   | Closing    | dir_p60_h30 | 61    | 1769 | 36.07% | 6 |
| 3.93% |            | 50.00%      |       |      |        |   |
| U20   | Closing    | dir_p40_h60 | 0     | 1830 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U20   | Closing    | dir_p50_h60 | 0     | 1830 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U20   | Closing    | dir_p60_h60 | 0     | 1830 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U20   | Closing    | dir_p40_h90 | 0     | 1830 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U20   | Closing    | dir_p50_h90 | 0     | 1830 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U20   | Closing    | dir_p60_h90 | 0     | 1830 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U20   | Opening    | dir_p40_h30 | 3660  | 0    | 51.64% | 4 |
| 8.36% |            | 45.08%      |       |      |        |   |
| U20   | Opening    | dir_p50_h30 | 3660  | 0    | 40.63% | 5 |
| 9.37% |            | 40.75%      |       |      |        |   |
| U20   | Opening    | dir_p60_h30 | 3660  | 0    | 29.84% | 7 |
| 0.16% |            | 37.27%      |       |      |        |   |
| U20   | Opening    | dir_p40_h60 | 3660  | 0    | 51.56% | 4 |
| 8.44% |            | 44.73%      |       |      |        |   |
| U20   | Opening    | dir_p50_h60 | 3660  | 0    | 39.78% | 6 |
| 0.22% |            | 39.84%      |       |      |        |   |
| U20   | Opening    | dir_p60_h60 | 3660  | 0    | 31.23% | 6 |
| 8.77% |            | 35.43%      |       |      |        |   |
| U20   | Opening    | dir_p40_h90 | 3660  | 0    | 54.29% | 4 |
| 5.71% |            | 45.60%      |       |      |        |   |
| U20   | Opening    | dir_p50_h90 | 3660  | 0    | 42.27% | 5 |
| 7.73% |            | 39.82%      |       |      |        |   |
| U20   | Opening    | dir_p60_h90 | 3660  | 0    | 32.79% | 6 |
| 7.21% |            | 36.50%      |       |      |        |   |
| U20   | Overnight  | dir_p40_h30 | 14640 | 61   | 52.31% | 4 |
| 7.69% |            | 49.97%      |       |      |        |   |
| U20   | Overnight  | dir_p50_h30 | 14640 | 61   | 41.79% | 5 |
| 8.21% |            | 49.22%      |       |      |        |   |
| U20   | Overnight  | dir_p60_h30 | 14640 | 61   | 32.87% | 6 |
| 7.13% |            | 48.21%      |       |      |        |   |
| U20   | Overnight  | dir_p40_h60 | 14640 | 61   | 52.13% | 4 |
| 7.87% |            | 49.71%      |       |      |        |   |
| U20   | Overnight  | dir_p50_h60 | 14640 | 61   | 42.09% | 5 |
| 7.91% |            | 48.83%      |       |      |        |   |
| U20   | Overnight  | dir_p60_h60 | 14640 | 61   | 32.10% | 6 |
| 7.90% |            | 47.80%      |       |      |        |   |
| U20   | Overnight  | dir_p40_h90 | 14640 | 61   | 53.62% | 4 |
| 6.38% |            | 48.55%      |       |      |        |   |
| U20   | Overnight  | dir_p50_h90 | 14640 | 61   | 43.27% | 5 |
| 6.73% |            | 48.95%      |       |      |        |   |
| U20   | Overnight  | dir_p60_h90 | 14640 | 61   | 31.67% | 6 |
| 8.33% |            | 48.93%      |       |      |        |   |
| U20   | Pre-market | dir_p40_h30 | 3660  | 0    | 52.76% | 4 |
| 7.24% |            | 49.92%      |       |      |        |   |

|       |            |             |       |      |        |   |
|-------|------------|-------------|-------|------|--------|---|
| U20   | Pre-market | dir_p50_h30 | 3660  | 0    | 42.60% | 5 |
| 7.40% |            | 48.88%      |       |      |        |   |
| U20   | Pre-market | dir_p60_h30 | 3660  | 0    | 34.18% | 6 |
| 5.82% |            | 46.68%      |       |      |        |   |
| U20   | Pre-market | dir_p40_h60 | 3660  | 0    | 50.33% | 4 |
| 9.67% |            | 45.06%      |       |      |        |   |
| U20   | Pre-market | dir_p50_h60 | 3660  | 0    | 38.74% | 6 |
| 1.26% |            | 37.94%      |       |      |        |   |
| U20   | Pre-market | dir_p60_h60 | 3660  | 0    | 29.67% | 7 |
| 0.33% |            | 35.17%      |       |      |        |   |
| U20   | Pre-market | dir_p40_h90 | 3660  | 0    | 50.60% | 4 |
| 9.40% |            | 40.23%      |       |      |        |   |
| U20   | Pre-market | dir_p50_h90 | 3660  | 0    | 38.14% | 6 |
| 1.86% |            | 34.53%      |       |      |        |   |
| U20   | Pre-market | dir_p60_h90 | 3660  | 0    | 28.09% | 7 |
| 1.91% |            | 26.95%      |       |      |        |   |
| U20   | Regular    | dir_p40_h30 | 18300 | 0    | 56.21% | 4 |
| 3.79% |            | 48.06%      |       |      |        |   |
| U20   | Regular    | dir_p50_h30 | 18300 | 0    | 45.75% | 5 |
| 4.25% |            | 45.53%      |       |      |        |   |
| U20   | Regular    | dir_p60_h30 | 18300 | 0    | 35.79% | 6 |
| 4.21% |            | 43.19%      |       |      |        |   |
| U20   | Regular    | dir_p40_h60 | 16531 | 1769 | 55.25% | 4 |
| 4.75% |            | 47.79%      |       |      |        |   |
| U20   | Regular    | dir_p50_h60 | 16531 | 1769 | 44.66% | 5 |
| 5.34% |            | 45.52%      |       |      |        |   |
| U20   | Regular    | dir_p60_h60 | 16531 | 1769 | 34.93% | 6 |
| 5.07% |            | 42.11%      |       |      |        |   |
| U20   | Regular    | dir_p40_h90 | 14701 | 3599 | 56.67% | 4 |
| 3.33% |            | 47.37%      |       |      |        |   |
| U20   | Regular    | dir_p50_h90 | 14701 | 3599 | 46.05% | 5 |
| 3.95% |            | 43.84%      |       |      |        |   |
| U20   | Regular    | dir_p60_h90 | 14701 | 3599 | 34.85% | 6 |
| 5.15% |            | 39.54%      |       |      |        |   |
| U21   | Closing    | dir_p40_h30 | 60    | 1740 | 16.67% | 8 |
| 3.33% |            | 40.00%      |       |      |        |   |
| U21   | Closing    | dir_p50_h30 | 60    | 1740 | 13.33% | 8 |
| 6.67% |            | 25.00%      |       |      |        |   |
| U21   | Closing    | dir_p60_h30 | 60    | 1740 | 8.33%  | 9 |
| 1.67% |            | 20.00%      |       |      |        |   |
| U21   | Closing    | dir_p40_h60 | 0     | 1800 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U21   | Closing    | dir_p50_h60 | 0     | 1800 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U21   | Closing    | dir_p60_h60 | 0     | 1800 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U21   | Closing    | dir_p40_h90 | 0     | 1800 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U21   | Closing    | dir_p50_h90 | 0     | 1800 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U21   | Closing    | dir_p60_h90 | 0     | 1800 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U21   | Opening    | dir_p40_h30 | 3600  | 0    | 36.11% | 6 |
| 3.89% |            | 45.54%      |       |      |        |   |
| U21   | Opening    | dir_p50_h30 | 3600  | 0    | 26.61% | 7 |
| 3.39% |            | 43.84%      |       |      |        |   |
| U21   | Opening    | dir_p60_h30 | 3600  | 0    | 19.58% | 8 |
| 0.42% |            | 41.56%      |       |      |        |   |
| U21   | Opening    | dir_p40_h60 | 3600  | 0    | 37.19% | 6 |
| 2.81% |            | 46.90%      |       |      |        |   |



|       |            |             |       |      |        |   |
|-------|------------|-------------|-------|------|--------|---|
| U21   | Opening    | dir_p50_h60 | 3600  | 0    | 28.69% | 7 |
| 1.31% |            | 44.14%      |       |      |        |   |
| U21   | Opening    | dir_p60_h60 | 3600  | 0    | 19.44% | 8 |
| 0.56% |            | 39.29%      |       |      |        |   |
| U21   | Opening    | dir_p40_h90 | 3600  | 0    | 40.33% | 5 |
| 9.67% |            | 47.11%      |       |      |        |   |
| U21   | Opening    | dir_p50_h90 | 3600  | 0    | 30.58% | 6 |
| 9.42% |            | 44.78%      |       |      |        |   |
| U21   | Opening    | dir_p60_h90 | 3600  | 0    | 19.58% | 8 |
| 0.42% |            | 41.13%      |       |      |        |   |
| U21   | Overnight  | dir_p40_h30 | 14400 | 60   | 33.72% | 6 |
| 6.28% |            | 50.39%      |       |      |        |   |
| U21   | Overnight  | dir_p50_h30 | 14400 | 60   | 25.50% | 7 |
| 4.50% |            | 49.26%      |       |      |        |   |
| U21   | Overnight  | dir_p60_h30 | 14400 | 60   | 18.61% | 8 |
| 1.39% |            | 47.31%      |       |      |        |   |
| U21   | Overnight  | dir_p40_h60 | 14400 | 60   | 34.59% | 6 |
| 5.41% |            | 47.06%      |       |      |        |   |
| U21   | Overnight  | dir_p50_h60 | 14400 | 60   | 25.78% | 7 |
| 4.22% |            | 45.88%      |       |      |        |   |
| U21   | Overnight  | dir_p60_h60 | 14400 | 60   | 18.67% | 8 |
| 1.33% |            | 43.32%      |       |      |        |   |
| U21   | Overnight  | dir_p40_h90 | 14400 | 60   | 37.46% | 6 |
| 2.54% |            | 45.79%      |       |      |        |   |
| U21   | Overnight  | dir_p50_h90 | 14400 | 60   | 28.17% | 7 |
| 1.83% |            | 41.83%      |       |      |        |   |
| U21   | Overnight  | dir_p60_h90 | 14400 | 60   | 19.66% | 8 |
| 0.34% |            | 39.77%      |       |      |        |   |
| U21   | Pre-market | dir_p40_h30 | 3600  | 0    | 39.69% | 6 |
| 0.31% |            | 51.01%      |       |      |        |   |
| U21   | Pre-market | dir_p50_h30 | 3600  | 0    | 31.69% | 6 |
| 8.31% |            | 48.55%      |       |      |        |   |
| U21   | Pre-market | dir_p60_h30 | 3600  | 0    | 23.42% | 7 |
| 6.58% |            | 41.76%      |       |      |        |   |
| U21   | Pre-market | dir_p40_h60 | 3600  | 0    | 38.72% | 6 |
| 1.28% |            | 40.82%      |       |      |        |   |
| U21   | Pre-market | dir_p50_h60 | 3600  | 0    | 31.17% | 6 |
| 8.83% |            | 38.32%      |       |      |        |   |
| U21   | Pre-market | dir_p60_h60 | 3600  | 0    | 21.44% | 7 |
| 8.56% |            | 41.58%      |       |      |        |   |
| U21   | Pre-market | dir_p40_h90 | 3600  | 0    | 39.08% | 6 |
| 0.92% |            | 39.30%      |       |      |        |   |
| U21   | Pre-market | dir_p50_h90 | 3600  | 0    | 30.83% | 6 |
| 9.17% |            | 42.61%      |       |      |        |   |
| U21   | Pre-market | dir_p60_h90 | 3600  | 0    | 20.00% | 8 |
| 0.00% |            | 44.44%      |       |      |        |   |
| U21   | Regular    | dir_p40_h30 | 18000 | 0    | 31.11% | 6 |
| 8.89% |            | 48.12%      |       |      |        |   |
| U21   | Regular    | dir_p50_h30 | 18000 | 0    | 22.74% | 7 |
| 7.26% |            | 47.14%      |       |      |        |   |
| U21   | Regular    | dir_p60_h30 | 18000 | 0    | 15.25% | 8 |
| 4.75% |            | 46.30%      |       |      |        |   |
| U21   | Regular    | dir_p40_h60 | 16260 | 1740 | 31.29% | 6 |
| 8.71% |            | 49.97%      |       |      |        |   |
| U21   | Regular    | dir_p50_h60 | 16260 | 1740 | 22.98% | 7 |
| 7.02% |            | 47.15%      |       |      |        |   |
| U21   | Regular    | dir_p60_h60 | 16260 | 1740 | 15.76% | 8 |
| 4.24% |            | 46.47%      |       |      |        |   |
| U21   | Regular    | dir_p40_h90 | 14460 | 3540 | 31.36% | 6 |
| 8.64% |            | 48.56%      |       |      |        |   |

|       |            |             |       |      |        |   |
|-------|------------|-------------|-------|------|--------|---|
| U21   | Regular    | dir_p50_h90 | 14460 | 3540 | 22.84% | 7 |
| 7.16% |            | 46.11%      |       |      |        |   |
| U21   | Regular    | dir_p60_h90 | 14460 | 3540 | 15.34% | 8 |
| 4.66% |            | 45.81%      |       |      |        |   |
| U22   | Closing    | dir_p40_h30 | 59    | 1711 | 69.49% | 3 |
| 0.51% |            | 31.71%      |       |      |        |   |
| U22   | Closing    | dir_p50_h30 | 59    | 1711 | 55.93% | 4 |
| 4.07% |            | 30.30%      |       |      |        |   |
| U22   | Closing    | dir_p60_h30 | 59    | 1711 | 44.07% | 5 |
| 5.93% |            | 34.62%      |       |      |        |   |
| U22   | Closing    | dir_p40_h60 | 0     | 1770 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U22   | Closing    | dir_p50_h60 | 0     | 1770 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U22   | Closing    | dir_p60_h60 | 0     | 1770 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U22   | Closing    | dir_p40_h90 | 0     | 1770 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U22   | Closing    | dir_p50_h90 | 0     | 1770 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U22   | Closing    | dir_p60_h90 | 0     | 1770 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U22   | Opening    | dir_p40_h30 | 3540  | 0    | 80.00% | 2 |
| 0.00% |            | 53.25%      |       |      |        |   |
| U22   | Opening    | dir_p50_h30 | 3540  | 0    | 68.90% | 3 |
| 1.10% |            | 53.05%      |       |      |        |   |
| U22   | Opening    | dir_p60_h30 | 3540  | 0    | 56.67% | 4 |
| 3.33% |            | 52.79%      |       |      |        |   |
| U22   | Opening    | dir_p40_h60 | 3540  | 0    | 80.25% | 1 |
| 9.75% |            | 53.33%      |       |      |        |   |
| U22   | Opening    | dir_p50_h60 | 3540  | 0    | 69.44% | 3 |
| 0.56% |            | 54.31%      |       |      |        |   |
| U22   | Opening    | dir_p60_h60 | 3540  | 0    | 56.92% | 4 |
| 3.08% |            | 55.98%      |       |      |        |   |
| U22   | Opening    | dir_p40_h90 | 3540  | 0    | 81.24% | 1 |
| 8.76% |            | 53.13%      |       |      |        |   |
| U22   | Opening    | dir_p50_h90 | 3540  | 0    | 70.37% | 2 |
| 9.63% |            | 53.55%      |       |      |        |   |
| U22   | Opening    | dir_p60_h90 | 3540  | 0    | 57.51% | 4 |
| 2.49% |            | 54.52%      |       |      |        |   |
| U22   | Overnight  | dir_p40_h30 | 14160 | 59   | 71.52% | 2 |
| 8.48% |            | 49.51%      |       |      |        |   |
| U22   | Overnight  | dir_p50_h30 | 14160 | 59   | 59.75% | 4 |
| 0.25% |            | 49.27%      |       |      |        |   |
| U22   | Overnight  | dir_p60_h30 | 14160 | 59   | 47.94% | 5 |
| 2.06% |            | 48.82%      |       |      |        |   |
| U22   | Overnight  | dir_p40_h60 | 14160 | 59   | 72.79% | 2 |
| 7.21% |            | 48.62%      |       |      |        |   |
| U22   | Overnight  | dir_p50_h60 | 14160 | 59   | 60.83% | 3 |
| 9.17% |            | 49.10%      |       |      |        |   |
| U22   | Overnight  | dir_p60_h60 | 14160 | 59   | 48.47% | 5 |
| 1.53% |            | 48.87%      |       |      |        |   |
| U22   | Overnight  | dir_p40_h90 | 14160 | 59   | 72.49% | 2 |
| 7.51% |            | 48.19%      |       |      |        |   |
| U22   | Overnight  | dir_p50_h90 | 14160 | 59   | 60.31% | 3 |
| 9.69% |            | 47.81%      |       |      |        |   |
| U22   | Overnight  | dir_p60_h90 | 14160 | 59   | 48.07% | 5 |
| 1.93% |            | 47.98%      |       |      |        |   |
| U22   | Pre-market | dir_p40_h30 | 3540  | 0    | 72.01% | 2 |
| 7.99% |            | 48.29%      |       |      |        |   |

|       |            |             |       |      |        |   |
|-------|------------|-------------|-------|------|--------|---|
| U22   | Pre-market | dir_p50_h30 | 3540  | 0    | 62.77% | 3 |
| 7.23% |            | 48.38%      |       |      |        |   |
| U22   | Pre-market | dir_p60_h30 | 3540  | 0    | 53.19% | 4 |
| 6.81% |            | 49.50%      |       |      |        |   |
| U22   | Pre-market | dir_p40_h60 | 3540  | 0    | 79.89% | 2 |
| 0.11% |            | 50.39%      |       |      |        |   |
| U22   | Pre-market | dir_p50_h60 | 3540  | 0    | 70.20% | 2 |
| 9.80% |            | 50.99%      |       |      |        |   |
| U22   | Pre-market | dir_p60_h60 | 3540  | 0    | 58.81% | 4 |
| 1.19% |            | 54.47%      |       |      |        |   |
| U22   | Pre-market | dir_p40_h90 | 3540  | 0    | 81.41% | 1 |
| 8.59% |            | 51.53%      |       |      |        |   |
| U22   | Pre-market | dir_p50_h90 | 3540  | 0    | 70.56% | 2 |
| 9.44% |            | 52.84%      |       |      |        |   |
| U22   | Pre-market | dir_p60_h90 | 3540  | 0    | 57.18% | 4 |
| 2.82% |            | 53.36%      |       |      |        |   |
| U22   | Regular    | dir_p40_h30 | 17700 | 0    | 73.43% | 2 |
| 6.57% |            | 50.50%      |       |      |        |   |
| U22   | Regular    | dir_p50_h30 | 17700 | 0    | 62.40% | 3 |
| 7.60% |            | 49.81%      |       |      |        |   |
| U22   | Regular    | dir_p60_h30 | 17700 | 0    | 50.40% | 4 |
| 9.60% |            | 48.88%      |       |      |        |   |
| U22   | Regular    | dir_p40_h60 | 15989 | 1711 | 73.02% | 2 |
| 6.98% |            | 50.94%      |       |      |        |   |
| U22   | Regular    | dir_p50_h60 | 15989 | 1711 | 60.59% | 3 |
| 9.41% |            | 49.97%      |       |      |        |   |
| U22   | Regular    | dir_p60_h60 | 15989 | 1711 | 48.61% | 5 |
| 1.39% |            | 48.91%      |       |      |        |   |
| U22   | Regular    | dir_p40_h90 | 14219 | 3481 | 71.87% | 2 |
| 8.13% |            | 51.04%      |       |      |        |   |
| U22   | Regular    | dir_p50_h90 | 14219 | 3481 | 60.93% | 3 |
| 9.07% |            | 50.81%      |       |      |        |   |
| U22   | Regular    | dir_p60_h90 | 14219 | 3481 | 49.54% | 5 |
| 0.46% |            | 50.07%      |       |      |        |   |
| U23   | Closing    | dir_p40_h30 | 60    | 1740 | 55.00% | 4 |
| 5.00% |            | 48.48%      |       |      |        |   |
| U23   | Closing    | dir_p50_h30 | 60    | 1740 | 38.33% | 6 |
| 1.67% |            | 52.17%      |       |      |        |   |
| U23   | Closing    | dir_p60_h30 | 60    | 1740 | 26.67% | 7 |
| 3.33% |            | 56.25%      |       |      |        |   |
| U23   | Closing    | dir_p40_h60 | 0     | 1800 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U23   | Closing    | dir_p50_h60 | 0     | 1800 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U23   | Closing    | dir_p60_h60 | 0     | 1800 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U23   | Closing    | dir_p40_h90 | 0     | 1800 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U23   | Closing    | dir_p50_h90 | 0     | 1800 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U23   | Closing    | dir_p60_h90 | 0     | 1800 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U23   | Opening    | dir_p40_h30 | 3600  | 0    | 64.19% | 3 |
| 5.81% |            | 45.22%      |       |      |        |   |
| U23   | Opening    | dir_p50_h30 | 3600  | 0    | 52.56% | 4 |
| 7.44% |            | 42.76%      |       |      |        |   |
| U23   | Opening    | dir_p60_h30 | 3600  | 0    | 40.42% | 5 |
| 9.58% |            | 40.76%      |       |      |        |   |
| U23   | Opening    | dir_p40_h60 | 3600  | 0    | 61.78% | 3 |
| 8.22% |            | 40.47%      |       |      |        |   |

|       |            |             |       |      |        |   |
|-------|------------|-------------|-------|------|--------|---|
| U23   | Opening    | dir_p50_h60 | 3600  | 0    | 50.06% | 4 |
| 9.94% |            | 39.35%      |       |      |        |   |
| U23   | Opening    | dir_p60_h60 | 3600  | 0    | 40.08% | 5 |
| 9.92% |            | 38.95%      |       |      |        |   |
| U23   | Opening    | dir_p40_h90 | 3600  | 0    | 58.44% | 4 |
| 1.56% |            | 39.73%      |       |      |        |   |
| U23   | Opening    | dir_p50_h90 | 3600  | 0    | 47.86% | 5 |
| 2.14% |            | 38.13%      |       |      |        |   |
| U23   | Opening    | dir_p60_h90 | 3600  | 0    | 36.00% | 6 |
| 4.00% |            | 35.96%      |       |      |        |   |
| U23   | Overnight  | dir_p40_h30 | 14400 | 60   | 52.19% | 4 |
| 7.81% |            | 50.15%      |       |      |        |   |
| U23   | Overnight  | dir_p50_h30 | 14400 | 60   | 39.62% | 6 |
| 0.38% |            | 48.83%      |       |      |        |   |
| U23   | Overnight  | dir_p60_h30 | 14400 | 60   | 29.12% | 7 |
| 0.88% |            | 46.84%      |       |      |        |   |
| U23   | Overnight  | dir_p40_h60 | 14400 | 60   | 52.38% | 4 |
| 7.62% |            | 50.69%      |       |      |        |   |
| U23   | Overnight  | dir_p50_h60 | 14400 | 60   | 38.82% | 6 |
| 1.18% |            | 49.11%      |       |      |        |   |
| U23   | Overnight  | dir_p60_h60 | 14400 | 60   | 28.19% | 7 |
| 1.81% |            | 45.75%      |       |      |        |   |
| U23   | Overnight  | dir_p40_h90 | 14400 | 60   | 51.47% | 4 |
| 8.53% |            | 50.78%      |       |      |        |   |
| U23   | Overnight  | dir_p50_h90 | 14400 | 60   | 40.38% | 5 |
| 9.62% |            | 49.24%      |       |      |        |   |
| U23   | Overnight  | dir_p60_h90 | 14400 | 60   | 29.72% | 7 |
| 0.28% |            | 48.19%      |       |      |        |   |
| U23   | Pre-market | dir_p40_h30 | 3600  | 0    | 58.64% | 4 |
| 1.36% |            | 54.52%      |       |      |        |   |
| U23   | Pre-market | dir_p50_h30 | 3600  | 0    | 46.89% | 5 |
| 3.11% |            | 55.98%      |       |      |        |   |
| U23   | Pre-market | dir_p60_h30 | 3600  | 0    | 35.31% | 6 |
| 4.69% |            | 57.83%      |       |      |        |   |
| U23   | Pre-market | dir_p40_h60 | 3600  | 0    | 64.17% | 3 |
| 5.83% |            | 52.77%      |       |      |        |   |
| U23   | Pre-market | dir_p50_h60 | 3600  | 0    | 51.53% | 4 |
| 8.47% |            | 50.24%      |       |      |        |   |
| U23   | Pre-market | dir_p60_h60 | 3600  | 0    | 39.11% | 6 |
| 0.89% |            | 50.85%      |       |      |        |   |
| U23   | Pre-market | dir_p40_h90 | 3600  | 0    | 64.28% | 3 |
| 5.72% |            | 49.83%      |       |      |        |   |
| U23   | Pre-market | dir_p50_h90 | 3600  | 0    | 50.64% | 4 |
| 9.36% |            | 50.19%      |       |      |        |   |
| U23   | Pre-market | dir_p60_h90 | 3600  | 0    | 38.81% | 6 |
| 1.19% |            | 47.82%      |       |      |        |   |
| U23   | Regular    | dir_p40_h30 | 18000 | 0    | 59.31% | 4 |
| 0.69% |            | 49.65%      |       |      |        |   |
| U23   | Regular    | dir_p50_h30 | 18000 | 0    | 47.02% | 5 |
| 2.98% |            | 48.12%      |       |      |        |   |
| U23   | Regular    | dir_p60_h30 | 18000 | 0    | 34.57% | 6 |
| 5.43% |            | 46.02%      |       |      |        |   |
| U23   | Regular    | dir_p40_h60 | 16260 | 1740 | 60.28% | 3 |
| 9.72% |            | 51.07%      |       |      |        |   |
| U23   | Regular    | dir_p50_h60 | 16260 | 1740 | 46.21% | 5 |
| 3.79% |            | 49.49%      |       |      |        |   |
| U23   | Regular    | dir_p60_h60 | 16260 | 1740 | 33.37% | 6 |
| 6.63% |            | 46.66%      |       |      |        |   |
| U23   | Regular    | dir_p40_h90 | 14460 | 3540 | 58.11% | 4 |
| 1.89% |            | 53.14%      |       |      |        |   |

|       |            |             |       |      |        |   |
|-------|------------|-------------|-------|------|--------|---|
| U23   | Regular    | dir_p50_h90 | 14460 | 3540 | 45.54% | 5 |
| 4.46% |            | 51.51%      |       |      |        |   |
| U23   | Regular    | dir_p60_h90 | 14460 | 3540 | 33.47% | 6 |
| 6.53% |            | 48.24%      |       |      |        |   |
| U24   | Closing    | dir_p40_h30 | 56    | 1624 | 82.14% | 1 |
| 7.86% |            | 52.17%      |       |      |        |   |
| U24   | Closing    | dir_p50_h30 | 56    | 1624 | 66.07% | 3 |
| 3.93% |            | 56.76%      |       |      |        |   |
| U24   | Closing    | dir_p60_h30 | 56    | 1624 | 58.93% | 4 |
| 1.07% |            | 54.55%      |       |      |        |   |
| U24   | Closing    | dir_p40_h60 | 0     | 1680 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U24   | Closing    | dir_p50_h60 | 0     | 1680 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U24   | Closing    | dir_p60_h60 | 0     | 1680 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U24   | Closing    | dir_p40_h90 | 0     | 1680 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U24   | Closing    | dir_p50_h90 | 0     | 1680 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U24   | Closing    | dir_p60_h90 | 0     | 1680 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U24   | Opening    | dir_p40_h30 | 3360  | 0    | 82.53% | 1 |
| 7.47% |            | 47.13%      |       |      |        |   |
| U24   | Opening    | dir_p50_h30 | 3360  | 0    | 75.24% | 2 |
| 4.76% |            | 46.28%      |       |      |        |   |
| U24   | Opening    | dir_p60_h30 | 3360  | 0    | 65.80% | 3 |
| 4.20% |            | 44.64%      |       |      |        |   |
| U24   | Opening    | dir_p40_h60 | 3360  | 0    | 85.00% | 1 |
| 5.00% |            | 45.06%      |       |      |        |   |
| U24   | Opening    | dir_p50_h60 | 3360  | 0    | 76.31% | 2 |
| 3.69% |            | 43.56%      |       |      |        |   |
| U24   | Opening    | dir_p60_h60 | 3360  | 0    | 66.31% | 3 |
| 3.69% |            | 42.01%      |       |      |        |   |
| U24   | Opening    | dir_p40_h90 | 3360  | 0    | 85.60% | 1 |
| 4.40% |            | 44.51%      |       |      |        |   |
| U24   | Opening    | dir_p50_h90 | 3360  | 0    | 76.76% | 2 |
| 3.24% |            | 43.70%      |       |      |        |   |
| U24   | Opening    | dir_p60_h90 | 3360  | 0    | 65.86% | 3 |
| 4.14% |            | 41.35%      |       |      |        |   |
| U24   | Overnight  | dir_p40_h30 | 13440 | 56   | 69.43% | 3 |
| 0.57% |            | 52.34%      |       |      |        |   |
| U24   | Overnight  | dir_p50_h30 | 13440 | 56   | 60.27% | 3 |
| 9.73% |            | 52.05%      |       |      |        |   |
| U24   | Overnight  | dir_p60_h30 | 13440 | 56   | 50.83% | 4 |
| 9.17% |            | 51.30%      |       |      |        |   |
| U24   | Overnight  | dir_p40_h60 | 13440 | 56   | 70.60% | 2 |
| 9.40% |            | 52.52%      |       |      |        |   |
| U24   | Overnight  | dir_p50_h60 | 13440 | 56   | 60.72% | 3 |
| 9.28% |            | 52.20%      |       |      |        |   |
| U24   | Overnight  | dir_p60_h60 | 13440 | 56   | 49.72% | 5 |
| 0.28% |            | 50.47%      |       |      |        |   |
| U24   | Overnight  | dir_p40_h90 | 13440 | 56   | 69.84% | 3 |
| 0.16% |            | 52.03%      |       |      |        |   |
| U24   | Overnight  | dir_p50_h90 | 13440 | 56   | 59.58% | 4 |
| 0.42% |            | 50.69%      |       |      |        |   |
| U24   | Overnight  | dir_p60_h90 | 13440 | 56   | 47.42% | 5 |
| 2.58% |            | 49.24%      |       |      |        |   |
| U24   | Pre-market | dir_p40_h30 | 3360  | 0    | 70.92% | 2 |
| 9.08% |            | 47.67%      |       |      |        |   |

|       |            |             |       |      |        |   |
|-------|------------|-------------|-------|------|--------|---|
| U24   | Pre-market | dir_p50_h30 | 3360  | 0    | 60.68% | 3 |
| 9.32% |            | 46.64%      |       |      |        |   |
| U24   | Pre-market | dir_p60_h30 | 3360  | 0    | 47.44% | 5 |
| 2.56% |            | 49.06%      |       |      |        |   |
| U24   | Pre-market | dir_p40_h60 | 3360  | 0    | 76.46% | 2 |
| 3.54% |            | 54.50%      |       |      |        |   |
| U24   | Pre-market | dir_p50_h60 | 3360  | 0    | 66.79% | 3 |
| 3.21% |            | 53.30%      |       |      |        |   |
| U24   | Pre-market | dir_p60_h60 | 3360  | 0    | 57.20% | 4 |
| 2.80% |            | 52.39%      |       |      |        |   |
| U24   | Pre-market | dir_p40_h90 | 3360  | 0    | 80.21% | 1 |
| 9.79% |            | 55.10%      |       |      |        |   |
| U24   | Pre-market | dir_p50_h90 | 3360  | 0    | 73.57% | 2 |
| 6.43% |            | 54.25%      |       |      |        |   |
| U24   | Pre-market | dir_p60_h90 | 3360  | 0    | 63.42% | 3 |
| 6.58% |            | 53.87%      |       |      |        |   |
| U24   | Regular    | dir_p40_h30 | 16800 | 0    | 78.18% | 2 |
| 1.82% |            | 49.59%      |       |      |        |   |
| U24   | Regular    | dir_p50_h30 | 16800 | 0    | 69.94% | 3 |
| 0.06% |            | 48.98%      |       |      |        |   |
| U24   | Regular    | dir_p60_h30 | 16800 | 0    | 59.71% | 4 |
| 0.29% |            | 48.03%      |       |      |        |   |
| U24   | Regular    | dir_p40_h60 | 15176 | 1624 | 81.23% | 1 |
| 8.77% |            | 47.67%      |       |      |        |   |
| U24   | Regular    | dir_p50_h60 | 15176 | 1624 | 73.02% | 2 |
| 6.98% |            | 47.32%      |       |      |        |   |
| U24   | Regular    | dir_p60_h60 | 15176 | 1624 | 63.68% | 3 |
| 6.32% |            | 46.64%      |       |      |        |   |
| U24   | Regular    | dir_p40_h90 | 13496 | 3304 | 82.22% | 1 |
| 7.78% |            | 47.74%      |       |      |        |   |
| U24   | Regular    | dir_p50_h90 | 13496 | 3304 | 74.17% | 2 |
| 5.83% |            | 46.91%      |       |      |        |   |
| U24   | Regular    | dir_p60_h90 | 13496 | 3304 | 64.78% | 3 |
| 5.22% |            | 45.71%      |       |      |        |   |
| U25   | Closing    | dir_p40_h30 | 57    | 1653 | 56.14% | 4 |
| 3.86% |            | 50.00%      |       |      |        |   |
| U25   | Closing    | dir_p50_h30 | 57    | 1653 | 35.09% | 6 |
| 4.91% |            | 60.00%      |       |      |        |   |
| U25   | Closing    | dir_p60_h30 | 57    | 1653 | 24.56% | 7 |
| 5.44% |            | 64.29%      |       |      |        |   |
| U25   | Closing    | dir_p40_h60 | 0     | 1710 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U25   | Closing    | dir_p50_h60 | 0     | 1710 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U25   | Closing    | dir_p60_h60 | 0     | 1710 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U25   | Closing    | dir_p40_h90 | 0     | 1710 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U25   | Closing    | dir_p50_h90 | 0     | 1710 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U25   | Closing    | dir_p60_h90 | 0     | 1710 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| U25   | Opening    | dir_p40_h30 | 3420  | 0    | 64.21% | 3 |
| 5.79% |            | 51.64%      |       |      |        |   |
| U25   | Opening    | dir_p50_h30 | 3420  | 0    | 52.51% | 4 |
| 7.49% |            | 50.17%      |       |      |        |   |
| U25   | Opening    | dir_p60_h30 | 3420  | 0    | 41.14% | 5 |
| 8.86% |            | 48.05%      |       |      |        |   |
| U25   | Opening    | dir_p40_h60 | 3420  | 0    | 63.92% | 3 |
| 6.08% |            | 48.08%      |       |      |        |   |

|       |            |             |       |      |        |   |
|-------|------------|-------------|-------|------|--------|---|
| U25   | Opening    | dir_p50_h60 | 3420  | 0    | 52.40% | 4 |
| 7.60% |            | 47.10%      |       |      |        |   |
| U25   | Opening    | dir_p60_h60 | 3420  | 0    | 41.23% | 5 |
| 8.77% |            | 45.39%      |       |      |        |   |
| U25   | Opening    | dir_p40_h90 | 3420  | 0    | 63.60% | 3 |
| 6.40% |            | 47.40%      |       |      |        |   |
| U25   | Opening    | dir_p50_h90 | 3420  | 0    | 51.35% | 4 |
| 8.65% |            | 44.70%      |       |      |        |   |
| U25   | Opening    | dir_p60_h90 | 3420  | 0    | 41.43% | 5 |
| 8.57% |            | 41.21%      |       |      |        |   |
| U25   | Overnight  | dir_p40_h30 | 13680 | 57   | 57.54% | 4 |
| 2.46% |            | 48.34%      |       |      |        |   |
| U25   | Overnight  | dir_p50_h30 | 13680 | 57   | 46.45% | 5 |
| 3.55% |            | 46.72%      |       |      |        |   |
| U25   | Overnight  | dir_p60_h30 | 13680 | 57   | 36.36% | 6 |
| 3.64% |            | 44.45%      |       |      |        |   |
| U25   | Overnight  | dir_p40_h60 | 13680 | 57   | 57.92% | 4 |
| 2.08% |            | 46.04%      |       |      |        |   |
| U25   | Overnight  | dir_p50_h60 | 13680 | 57   | 45.96% | 5 |
| 4.04% |            | 44.45%      |       |      |        |   |
| U25   | Overnight  | dir_p60_h60 | 13680 | 57   | 33.87% | 6 |
| 6.13% |            | 42.84%      |       |      |        |   |
| U25   | Overnight  | dir_p40_h90 | 13680 | 57   | 56.67% | 4 |
| 3.33% |            | 44.48%      |       |      |        |   |
| U25   | Overnight  | dir_p50_h90 | 13680 | 57   | 43.69% | 5 |
| 6.31% |            | 44.24%      |       |      |        |   |
| U25   | Overnight  | dir_p60_h90 | 13680 | 57   | 32.71% | 6 |
| 7.29% |            | 43.84%      |       |      |        |   |
| U25   | Pre-market | dir_p40_h30 | 3420  | 0    | 59.74% | 4 |
| 0.26% |            | 46.11%      |       |      |        |   |
| U25   | Pre-market | dir_p50_h30 | 3420  | 0    | 51.49% | 4 |
| 8.51% |            | 45.03%      |       |      |        |   |
| U25   | Pre-market | dir_p60_h30 | 3420  | 0    | 40.88% | 5 |
| 9.12% |            | 42.85%      |       |      |        |   |
| U25   | Pre-market | dir_p40_h60 | 3420  | 0    | 57.98% | 4 |
| 2.02% |            | 46.14%      |       |      |        |   |
| U25   | Pre-market | dir_p50_h60 | 3420  | 0    | 49.21% | 5 |
| 0.79% |            | 45.81%      |       |      |        |   |
| U25   | Pre-market | dir_p60_h60 | 3420  | 0    | 40.12% | 5 |
| 9.88% |            | 46.65%      |       |      |        |   |
| U25   | Pre-market | dir_p40_h90 | 3420  | 0    | 60.56% | 3 |
| 9.44% |            | 47.80%      |       |      |        |   |
| U25   | Pre-market | dir_p50_h90 | 3420  | 0    | 52.16% | 4 |
| 7.84% |            | 48.77%      |       |      |        |   |
| U25   | Pre-market | dir_p60_h90 | 3420  | 0    | 43.95% | 5 |
| 6.05% |            | 50.03%      |       |      |        |   |
| U25   | Regular    | dir_p40_h30 | 17100 | 0    | 56.71% | 4 |
| 3.29% |            | 48.71%      |       |      |        |   |
| U25   | Regular    | dir_p50_h30 | 17100 | 0    | 46.42% | 5 |
| 3.58% |            | 47.86%      |       |      |        |   |
| U25   | Regular    | dir_p60_h30 | 17100 | 0    | 35.88% | 6 |
| 4.12% |            | 45.52%      |       |      |        |   |
| U25   | Regular    | dir_p40_h60 | 15447 | 1653 | 56.57% | 4 |
| 3.43% |            | 47.55%      |       |      |        |   |
| U25   | Regular    | dir_p50_h60 | 15447 | 1653 | 44.86% | 5 |
| 5.14% |            | 45.13%      |       |      |        |   |
| U25   | Regular    | dir_p60_h60 | 15447 | 1653 | 34.79% | 6 |
| 5.21% |            | 42.39%      |       |      |        |   |
| U25   | Regular    | dir_p40_h90 | 13737 | 3363 | 55.12% | 4 |
| 4.88% |            | 46.91%      |       |      |        |   |

|       |            |             |       |      |        |   |
|-------|------------|-------------|-------|------|--------|---|
| U25   | Regular    | dir_p50_h90 | 13737 | 3363 | 43.67% | 5 |
| 6.33% |            | 44.29%      |       |      |        |   |
| U25   | Regular    | dir_p60_h90 | 13737 | 3363 | 32.76% | 6 |
| 7.24% |            | 41.40%      |       |      |        |   |
| Z20   | Closing    | dir_p40_h30 | 53    | 1537 | 56.60% | 4 |
| 3.40% |            | 46.67%      |       |      |        |   |
| Z20   | Closing    | dir_p50_h30 | 53    | 1537 | 49.06% | 5 |
| 0.94% |            | 42.31%      |       |      |        |   |
| Z20   | Closing    | dir_p60_h30 | 53    | 1537 | 35.85% | 6 |
| 4.15% |            | 42.11%      |       |      |        |   |
| Z20   | Closing    | dir_p40_h60 | 0     | 1590 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| Z20   | Closing    | dir_p50_h60 | 0     | 1590 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| Z20   | Closing    | dir_p60_h60 | 0     | 1590 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| Z20   | Closing    | dir_p40_h90 | 0     | 1590 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| Z20   | Closing    | dir_p50_h90 | 0     | 1590 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| Z20   | Closing    | dir_p60_h90 | 0     | 1590 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| Z20   | Opening    | dir_p40_h30 | 3180  | 0    | 51.13% | 4 |
| 8.87% |            | 47.66%      |       |      |        |   |
| Z20   | Opening    | dir_p50_h30 | 3180  | 0    | 40.25% | 5 |
| 9.75% |            | 46.02%      |       |      |        |   |
| Z20   | Opening    | dir_p60_h30 | 3180  | 0    | 29.69% | 7 |
| 0.31% |            | 44.70%      |       |      |        |   |
| Z20   | Opening    | dir_p40_h60 | 3180  | 0    | 49.75% | 5 |
| 0.25% |            | 46.78%      |       |      |        |   |
| Z20   | Opening    | dir_p50_h60 | 3180  | 0    | 38.11% | 6 |
| 1.89% |            | 46.78%      |       |      |        |   |
| Z20   | Opening    | dir_p60_h60 | 3180  | 0    | 29.18% | 7 |
| 0.82% |            | 46.01%      |       |      |        |   |
| Z20   | Opening    | dir_p40_h90 | 3180  | 0    | 49.25% | 5 |
| 0.75% |            | 45.98%      |       |      |        |   |
| Z20   | Opening    | dir_p50_h90 | 3180  | 0    | 39.81% | 6 |
| 0.19% |            | 45.58%      |       |      |        |   |
| Z20   | Opening    | dir_p60_h90 | 3180  | 0    | 31.04% | 6 |
| 8.96% |            | 45.69%      |       |      |        |   |
| Z20   | Overnight  | dir_p40_h30 | 12720 | 53   | 59.53% | 4 |
| 0.47% |            | 50.04%      |       |      |        |   |
| Z20   | Overnight  | dir_p50_h30 | 12720 | 53   | 50.21% | 4 |
| 9.79% |            | 49.99%      |       |      |        |   |
| Z20   | Overnight  | dir_p60_h30 | 12720 | 53   | 41.23% | 5 |
| 8.77% |            | 51.13%      |       |      |        |   |
| Z20   | Overnight  | dir_p40_h60 | 12720 | 53   | 58.44% | 4 |
| 1.56% |            | 51.89%      |       |      |        |   |
| Z20   | Overnight  | dir_p50_h60 | 12720 | 53   | 49.55% | 5 |
| 0.45% |            | 52.40%      |       |      |        |   |
| Z20   | Overnight  | dir_p60_h60 | 12720 | 53   | 40.79% | 5 |
| 9.21% |            | 51.98%      |       |      |        |   |
| Z20   | Overnight  | dir_p40_h90 | 12720 | 53   | 59.69% | 4 |
| 0.31% |            | 51.15%      |       |      |        |   |
| Z20   | Overnight  | dir_p50_h90 | 12720 | 53   | 50.13% | 4 |
| 9.87% |            | 50.85%      |       |      |        |   |
| Z20   | Overnight  | dir_p60_h90 | 12720 | 53   | 39.60% | 6 |
| 0.40% |            | 50.27%      |       |      |        |   |
| Z20   | Pre-market | dir_p40_h30 | 3180  | 0    | 48.90% | 5 |
| 1.10% |            | 44.44%      |       |      |        |   |



|       |            |             |       |      |        |   |
|-------|------------|-------------|-------|------|--------|---|
| Z20   | Pre-market | dir_p50_h30 | 3180  | 0    | 40.06% | 5 |
| 9.94% |            | 44.82%      |       |      |        |   |
| Z20   | Pre-market | dir_p60_h30 | 3180  | 0    | 30.97% | 6 |
| 9.03% |            | 46.90%      |       |      |        |   |
| Z20   | Pre-market | dir_p40_h60 | 3180  | 0    | 50.35% | 4 |
| 9.65% |            | 52.09%      |       |      |        |   |
| Z20   | Pre-market | dir_p50_h60 | 3180  | 0    | 40.66% | 5 |
| 9.34% |            | 53.52%      |       |      |        |   |
| Z20   | Pre-market | dir_p60_h60 | 3180  | 0    | 31.86% | 6 |
| 8.14% |            | 56.96%      |       |      |        |   |
| Z20   | Pre-market | dir_p40_h90 | 3180  | 0    | 47.17% | 5 |
| 2.83% |            | 57.27%      |       |      |        |   |
| Z20   | Pre-market | dir_p50_h90 | 3180  | 0    | 37.86% | 6 |
| 2.14% |            | 59.05%      |       |      |        |   |
| Z20   | Pre-market | dir_p60_h90 | 3180  | 0    | 28.27% | 7 |
| 1.73% |            | 61.51%      |       |      |        |   |
| Z20   | Regular    | dir_p40_h30 | 15900 | 0    | 57.08% | 4 |
| 2.92% |            | 50.62%      |       |      |        |   |
| Z20   | Regular    | dir_p50_h30 | 15900 | 0    | 46.36% | 5 |
| 3.64% |            | 49.32%      |       |      |        |   |
| Z20   | Regular    | dir_p60_h30 | 15900 | 0    | 35.86% | 6 |
| 4.14% |            | 48.21%      |       |      |        |   |
| Z20   | Regular    | dir_p40_h60 | 14363 | 1537 | 55.38% | 4 |
| 4.62% |            | 51.48%      |       |      |        |   |
| Z20   | Regular    | dir_p50_h60 | 14363 | 1537 | 44.05% | 5 |
| 5.95% |            | 50.29%      |       |      |        |   |
| Z20   | Regular    | dir_p60_h60 | 14363 | 1537 | 33.57% | 6 |
| 6.43% |            | 49.41%      |       |      |        |   |
| Z20   | Regular    | dir_p40_h90 | 12773 | 3127 | 54.74% | 4 |
| 5.26% |            | 53.29%      |       |      |        |   |
| Z20   | Regular    | dir_p50_h90 | 12773 | 3127 | 42.72% | 5 |
| 7.28% |            | 51.36%      |       |      |        |   |
| Z20   | Regular    | dir_p60_h90 | 12773 | 3127 | 31.77% | 6 |
| 8.23% |            | 49.04%      |       |      |        |   |
| Z21   | Closing    | dir_p40_h30 | 62    | 1798 | 62.90% | 3 |
| 7.10% |            | 46.15%      |       |      |        |   |
| Z21   | Closing    | dir_p50_h30 | 62    | 1798 | 51.61% | 4 |
| 8.39% |            | 46.88%      |       |      |        |   |
| Z21   | Closing    | dir_p60_h30 | 62    | 1798 | 46.77% | 5 |
| 3.23% |            | 48.28%      |       |      |        |   |
| Z21   | Closing    | dir_p40_h60 | 0     | 1860 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| Z21   | Closing    | dir_p50_h60 | 0     | 1860 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| Z21   | Closing    | dir_p60_h60 | 0     | 1860 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| Z21   | Closing    | dir_p40_h90 | 0     | 1860 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| Z21   | Closing    | dir_p50_h90 | 0     | 1860 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| Z21   | Closing    | dir_p60_h90 | 0     | 1860 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| Z21   | Opening    | dir_p40_h30 | 3720  | 0    | 56.29% | 4 |
| 3.71% |            | 47.99%      |       |      |        |   |
| Z21   | Opening    | dir_p50_h30 | 3720  | 0    | 45.89% | 5 |
| 4.11% |            | 45.99%      |       |      |        |   |
| Z21   | Opening    | dir_p60_h30 | 3720  | 0    | 36.56% | 6 |
| 3.44% |            | 44.49%      |       |      |        |   |
| Z21   | Opening    | dir_p40_h60 | 3720  | 0    | 54.84% | 4 |
| 5.16% |            | 45.00%      |       |      |        |   |

|       |     |            |             |       |      |        |   |
|-------|-----|------------|-------------|-------|------|--------|---|
| 4.30% | Z21 | Opening    | dir_p50_h60 | 3720  | 0    | 45.70% | 5 |
|       |     |            | 43.59%      |       |      |        |   |
| 4.14% | Z21 | Opening    | dir_p60_h60 | 3720  | 0    | 35.86% | 6 |
|       |     |            | 41.23%      |       |      |        |   |
| 4.09% | Z21 | Opening    | dir_p40_h90 | 3720  | 0    | 55.91% | 4 |
|       |     |            | 43.75%      |       |      |        |   |
| 4.81% | Z21 | Opening    | dir_p50_h90 | 3720  | 0    | 45.19% | 5 |
|       |     |            | 39.98%      |       |      |        |   |
| 3.92% | Z21 | Opening    | dir_p60_h90 | 3720  | 0    | 36.08% | 6 |
|       |     |            | 35.77%      |       |      |        |   |
| 4.42% | Z21 | Overnight  | dir_p40_h30 | 14880 | 62   | 55.58% | 4 |
|       |     |            | 48.59%      |       |      |        |   |
| 4.49% | Z21 | Overnight  | dir_p50_h30 | 14880 | 62   | 45.51% | 5 |
|       |     |            | 47.18%      |       |      |        |   |
| 3.12% | Z21 | Overnight  | dir_p60_h30 | 14880 | 62   | 36.88% | 6 |
|       |     |            | 46.37%      |       |      |        |   |
| 3.47% | Z21 | Overnight  | dir_p40_h60 | 14880 | 62   | 56.53% | 4 |
|       |     |            | 47.25%      |       |      |        |   |
| 3.86% | Z21 | Overnight  | dir_p50_h60 | 14880 | 62   | 46.14% | 5 |
|       |     |            | 46.60%      |       |      |        |   |
| 4.56% | Z21 | Overnight  | dir_p60_h60 | 14880 | 62   | 35.44% | 6 |
|       |     |            | 46.25%      |       |      |        |   |
| 3.95% | Z21 | Overnight  | dir_p40_h90 | 14880 | 62   | 56.05% | 4 |
|       |     |            | 47.58%      |       |      |        |   |
| 3.68% | Z21 | Overnight  | dir_p50_h90 | 14880 | 62   | 46.32% | 5 |
|       |     |            | 47.29%      |       |      |        |   |
| 3.73% | Z21 | Overnight  | dir_p60_h90 | 14880 | 62   | 36.27% | 6 |
|       |     |            | 47.06%      |       |      |        |   |
| 9.06% | Z21 | Pre-market | dir_p40_h30 | 3720  | 0    | 60.94% | 3 |
|       |     |            | 45.74%      |       |      |        |   |
| 8.36% | Z21 | Pre-market | dir_p50_h30 | 3720  | 0    | 51.64% | 4 |
|       |     |            | 45.71%      |       |      |        |   |
| 0.67% | Z21 | Pre-market | dir_p60_h30 | 3720  | 0    | 39.33% | 6 |
|       |     |            | 48.12%      |       |      |        |   |
| 3.17% | Z21 | Pre-market | dir_p40_h60 | 3720  | 0    | 66.83% | 3 |
|       |     |            | 51.69%      |       |      |        |   |
| 3.15% | Z21 | Pre-market | dir_p50_h60 | 3720  | 0    | 56.85% | 4 |
|       |     |            | 51.68%      |       |      |        |   |
| 8.47% | Z21 | Pre-market | dir_p60_h60 | 3720  | 0    | 41.53% | 5 |
|       |     |            | 48.93%      |       |      |        |   |
| 6.99% | Z21 | Pre-market | dir_p40_h90 | 3720  | 0    | 63.01% | 3 |
|       |     |            | 49.91%      |       |      |        |   |
| 1.59% | Z21 | Pre-market | dir_p50_h90 | 3720  | 0    | 48.41% | 5 |
|       |     |            | 50.47%      |       |      |        |   |
| 6.16% | Z21 | Pre-market | dir_p60_h90 | 3720  | 0    | 33.84% | 6 |
|       |     |            | 47.10%      |       |      |        |   |
| 2.76% | Z21 | Regular    | dir_p40_h30 | 18600 | 0    | 57.24% | 4 |
|       |     |            | 49.47%      |       |      |        |   |
| 2.49% | Z21 | Regular    | dir_p50_h30 | 18600 | 0    | 47.51% | 5 |
|       |     |            | 48.57%      |       |      |        |   |
| 2.36% | Z21 | Regular    | dir_p60_h30 | 18600 | 0    | 37.64% | 6 |
|       |     |            | 47.18%      |       |      |        |   |
| 2.61% | Z21 | Regular    | dir_p40_h60 | 16802 | 1798 | 57.39% | 4 |
|       |     |            | 50.04%      |       |      |        |   |
| 3.05% | Z21 | Regular    | dir_p50_h60 | 16802 | 1798 | 46.95% | 5 |
|       |     |            | 48.19%      |       |      |        |   |
| 2.80% | Z21 | Regular    | dir_p60_h60 | 16802 | 1798 | 37.20% | 6 |
|       |     |            | 46.86%      |       |      |        |   |
| 2.93% | Z21 | Regular    | dir_p40_h90 | 14942 | 3658 | 57.07% | 4 |
|       |     |            | 49.43%      |       |      |        |   |

|       |     |            |             |       |      |        |   |
|-------|-----|------------|-------------|-------|------|--------|---|
| 3.01% | Z21 | Regular    | dir_p50_h90 | 14942 | 3658 | 46.99% | 5 |
|       |     |            | 48.78%      |       |      |        |   |
| 3.77% | Z21 | Regular    | dir_p60_h90 | 14942 | 3658 | 36.23% | 6 |
|       |     |            | 46.99%      |       |      |        |   |
| 8.81% | Z22 | Closing    | dir_p40_h30 | 59    | 1711 | 71.19% | 2 |
|       |     |            | 52.38%      |       |      |        |   |
| 5.59% | Z22 | Closing    | dir_p50_h30 | 59    | 1711 | 64.41% | 3 |
|       |     |            | 47.37%      |       |      |        |   |
| 9.15% | Z22 | Closing    | dir_p60_h30 | 59    | 1711 | 50.85% | 4 |
|       |     |            | 43.33%      |       |      |        |   |
| NaN   | Z22 | Closing    | dir_p40_h60 | 0     | 1770 | NaN    |   |
|       |     |            | NaN         |       |      |        |   |
| NaN   | Z22 | Closing    | dir_p50_h60 | 0     | 1770 | NaN    |   |
|       |     |            | NaN         |       |      |        |   |
| NaN   | Z22 | Closing    | dir_p60_h60 | 0     | 1770 | NaN    |   |
|       |     |            | NaN         |       |      |        |   |
| NaN   | Z22 | Closing    | dir_p40_h90 | 0     | 1770 | NaN    |   |
|       |     |            | NaN         |       |      |        |   |
| NaN   | Z22 | Closing    | dir_p50_h90 | 0     | 1770 | NaN    |   |
|       |     |            | NaN         |       |      |        |   |
| NaN   | Z22 | Closing    | dir_p60_h90 | 0     | 1770 | NaN    |   |
|       |     |            | NaN         |       |      |        |   |
| 2.03% | Z22 | Opening    | dir_p40_h30 | 3540  | 0    | 67.97% | 3 |
|       |     |            | 52.00%      |       |      |        |   |
| 2.23% | Z22 | Opening    | dir_p50_h30 | 3540  | 0    | 57.77% | 4 |
|       |     |            | 51.88%      |       |      |        |   |
| 2.91% | Z22 | Opening    | dir_p60_h30 | 3540  | 0    | 47.09% | 5 |
|       |     |            | 51.77%      |       |      |        |   |
| 1.81% | Z22 | Opening    | dir_p40_h60 | 3540  | 0    | 68.19% | 3 |
|       |     |            | 51.78%      |       |      |        |   |
| 1.64% | Z22 | Opening    | dir_p50_h60 | 3540  | 0    | 58.36% | 4 |
|       |     |            | 52.90%      |       |      |        |   |
| 2.06% | Z22 | Opening    | dir_p60_h60 | 3540  | 0    | 47.94% | 5 |
|       |     |            | 54.10%      |       |      |        |   |
| 2.57% | Z22 | Opening    | dir_p40_h90 | 3540  | 0    | 67.43% | 3 |
|       |     |            | 51.15%      |       |      |        |   |
| 2.51% | Z22 | Opening    | dir_p50_h90 | 3540  | 0    | 57.49% | 4 |
|       |     |            | 52.78%      |       |      |        |   |
| 3.50% | Z22 | Opening    | dir_p60_h90 | 3540  | 0    | 46.50% | 5 |
|       |     |            | 54.50%      |       |      |        |   |
| 3.62% | Z22 | Overnight  | dir_p40_h30 | 14160 | 59   | 76.38% | 2 |
|       |     |            | 51.55%      |       |      |        |   |
| 3.71% | Z22 | Overnight  | dir_p50_h30 | 14160 | 59   | 66.29% | 3 |
|       |     |            | 51.44%      |       |      |        |   |
| 4.62% | Z22 | Overnight  | dir_p60_h30 | 14160 | 59   | 55.38% | 4 |
|       |     |            | 51.30%      |       |      |        |   |
| 5.18% | Z22 | Overnight  | dir_p40_h60 | 14160 | 59   | 74.82% | 2 |
|       |     |            | 53.44%      |       |      |        |   |
| 4.48% | Z22 | Overnight  | dir_p50_h60 | 14160 | 59   | 65.52% | 3 |
|       |     |            | 53.03%      |       |      |        |   |
| 5.28% | Z22 | Overnight  | dir_p60_h60 | 14160 | 59   | 54.72% | 4 |
|       |     |            | 52.15%      |       |      |        |   |
| 5.35% | Z22 | Overnight  | dir_p40_h90 | 14160 | 59   | 74.65% | 2 |
|       |     |            | 51.14%      |       |      |        |   |
| 4.43% | Z22 | Overnight  | dir_p50_h90 | 14160 | 59   | 65.57% | 3 |
|       |     |            | 50.52%      |       |      |        |   |
| 5.37% | Z22 | Overnight  | dir_p60_h90 | 14160 | 59   | 54.63% | 4 |
|       |     |            | 50.74%      |       |      |        |   |
| 3.95% | Z22 | Pre-market | dir_p40_h30 | 3540  | 0    | 76.05% | 2 |
|       |     |            | 46.58%      |       |      |        |   |

|       |            |             |       |      |        |   |
|-------|------------|-------------|-------|------|--------|---|
| Z22   | Pre-market | dir_p50_h30 | 3540  | 0    | 68.39% | 3 |
| 1.61% |            | 46.51%      |       |      |        |   |
| Z22   | Pre-market | dir_p60_h30 | 3540  | 0    | 57.18% | 4 |
| 2.82% |            | 46.84%      |       |      |        |   |
| Z22   | Pre-market | dir_p40_h60 | 3540  | 0    | 76.67% | 2 |
| 3.33% |            | 43.81%      |       |      |        |   |
| Z22   | Pre-market | dir_p50_h60 | 3540  | 0    | 67.03% | 3 |
| 2.97% |            | 45.17%      |       |      |        |   |
| Z22   | Pre-market | dir_p60_h60 | 3540  | 0    | 56.41% | 4 |
| 3.59% |            | 46.37%      |       |      |        |   |
| Z22   | Pre-market | dir_p40_h90 | 3540  | 0    | 77.54% | 2 |
| 2.46% |            | 45.68%      |       |      |        |   |
| Z22   | Pre-market | dir_p50_h90 | 3540  | 0    | 67.43% | 3 |
| 2.57% |            | 46.17%      |       |      |        |   |
| Z22   | Pre-market | dir_p60_h90 | 3540  | 0    | 57.09% | 4 |
| 2.91% |            | 46.46%      |       |      |        |   |
| Z22   | Regular    | dir_p40_h30 | 17700 | 0    | 75.62% | 2 |
| 4.38% |            | 50.82%      |       |      |        |   |
| Z22   | Regular    | dir_p50_h30 | 17700 | 0    | 64.74% | 3 |
| 5.26% |            | 51.05%      |       |      |        |   |
| Z22   | Regular    | dir_p60_h30 | 17700 | 0    | 52.41% | 4 |
| 7.59% |            | 50.79%      |       |      |        |   |
| Z22   | Regular    | dir_p40_h60 | 15989 | 1711 | 75.25% | 2 |
| 4.75% |            | 50.05%      |       |      |        |   |
| Z22   | Regular    | dir_p50_h60 | 15989 | 1711 | 64.06% | 3 |
| 5.94% |            | 49.50%      |       |      |        |   |
| Z22   | Regular    | dir_p60_h60 | 15989 | 1711 | 52.20% | 4 |
| 7.80% |            | 49.50%      |       |      |        |   |
| Z22   | Regular    | dir_p40_h90 | 14219 | 3481 | 76.45% | 2 |
| 3.55% |            | 48.54%      |       |      |        |   |
| Z22   | Regular    | dir_p50_h90 | 14219 | 3481 | 65.25% | 3 |
| 4.75% |            | 48.17%      |       |      |        |   |
| Z22   | Regular    | dir_p60_h90 | 14219 | 3481 | 52.60% | 4 |
| 7.40% |            | 47.81%      |       |      |        |   |
| Z23   | Closing    | dir_p40_h30 | 62    | 1798 | 48.39% | 5 |
| 1.61% |            | 40.00%      |       |      |        |   |
| Z23   | Closing    | dir_p50_h30 | 62    | 1798 | 40.32% | 5 |
| 9.68% |            | 36.00%      |       |      |        |   |
| Z23   | Closing    | dir_p60_h30 | 62    | 1798 | 25.81% | 7 |
| 4.19% |            | 50.00%      |       |      |        |   |
| Z23   | Closing    | dir_p40_h60 | 0     | 1860 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| Z23   | Closing    | dir_p50_h60 | 0     | 1860 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| Z23   | Closing    | dir_p60_h60 | 0     | 1860 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| Z23   | Closing    | dir_p40_h90 | 0     | 1860 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| Z23   | Closing    | dir_p50_h90 | 0     | 1860 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| Z23   | Closing    | dir_p60_h90 | 0     | 1860 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| Z23   | Opening    | dir_p40_h30 | 3720  | 0    | 59.35% | 4 |
| 0.65% |            | 49.14%      |       |      |        |   |
| Z23   | Opening    | dir_p50_h30 | 3720  | 0    | 47.39% | 5 |
| 2.61% |            | 49.57%      |       |      |        |   |
| Z23   | Opening    | dir_p60_h30 | 3720  | 0    | 34.84% | 6 |
| 5.16% |            | 50.15%      |       |      |        |   |
| Z23   | Opening    | dir_p40_h60 | 3720  | 0    | 56.80% | 4 |
| 3.20% |            | 53.38%      |       |      |        |   |

|       |            |             |       |      |        |   |
|-------|------------|-------------|-------|------|--------|---|
| Z23   | Opening    | dir_p50_h60 | 3720  | 0    | 42.53% | 5 |
| 7.47% |            | 53.22%      |       |      |        |   |
| Z23   | Opening    | dir_p60_h60 | 3720  | 0    | 30.30% | 6 |
| 9.70% |            | 52.62%      |       |      |        |   |
| Z23   | Opening    | dir_p40_h90 | 3720  | 0    | 54.65% | 4 |
| 5.35% |            | 55.39%      |       |      |        |   |
| Z23   | Opening    | dir_p50_h90 | 3720  | 0    | 41.16% | 5 |
| 8.84% |            | 55.78%      |       |      |        |   |
| Z23   | Opening    | dir_p60_h90 | 3720  | 0    | 28.74% | 7 |
| 1.26% |            | 54.63%      |       |      |        |   |
| Z23   | Overnight  | dir_p40_h30 | 14880 | 62   | 54.68% | 4 |
| 5.32% |            | 49.80%      |       |      |        |   |
| Z23   | Overnight  | dir_p50_h30 | 14880 | 62   | 42.98% | 5 |
| 7.02% |            | 49.31%      |       |      |        |   |
| Z23   | Overnight  | dir_p60_h30 | 14880 | 62   | 33.22% | 6 |
| 6.78% |            | 48.33%      |       |      |        |   |
| Z23   | Overnight  | dir_p40_h60 | 14880 | 62   | 54.90% | 4 |
| 5.10% |            | 48.65%      |       |      |        |   |
| Z23   | Overnight  | dir_p50_h60 | 14880 | 62   | 44.15% | 5 |
| 5.85% |            | 47.48%      |       |      |        |   |
| Z23   | Overnight  | dir_p60_h60 | 14880 | 62   | 33.84% | 6 |
| 6.16% |            | 46.18%      |       |      |        |   |
| Z23   | Overnight  | dir_p40_h90 | 14880 | 62   | 56.24% | 4 |
| 3.76% |            | 49.15%      |       |      |        |   |
| Z23   | Overnight  | dir_p50_h90 | 14880 | 62   | 45.93% | 5 |
| 4.07% |            | 49.54%      |       |      |        |   |
| Z23   | Overnight  | dir_p60_h90 | 14880 | 62   | 35.46% | 6 |
| 4.54% |            | 50.72%      |       |      |        |   |
| Z23   | Pre-market | dir_p40_h30 | 3720  | 0    | 61.51% | 3 |
| 8.49% |            | 53.41%      |       |      |        |   |
| Z23   | Pre-market | dir_p50_h30 | 3720  | 0    | 48.36% | 5 |
| 1.64% |            | 53.59%      |       |      |        |   |
| Z23   | Pre-market | dir_p60_h30 | 3720  | 0    | 36.24% | 6 |
| 3.76% |            | 54.53%      |       |      |        |   |
| Z23   | Pre-market | dir_p40_h60 | 3720  | 0    | 55.86% | 4 |
| 4.14% |            | 56.98%      |       |      |        |   |
| Z23   | Pre-market | dir_p50_h60 | 3720  | 0    | 41.40% | 5 |
| 8.60% |            | 59.29%      |       |      |        |   |
| Z23   | Pre-market | dir_p60_h60 | 3720  | 0    | 30.22% | 6 |
| 9.78% |            | 56.85%      |       |      |        |   |
| Z23   | Pre-market | dir_p40_h90 | 3720  | 0    | 53.04% | 4 |
| 6.96% |            | 58.84%      |       |      |        |   |
| Z23   | Pre-market | dir_p50_h90 | 3720  | 0    | 41.61% | 5 |
| 8.39% |            | 54.46%      |       |      |        |   |
| Z23   | Pre-market | dir_p60_h90 | 3720  | 0    | 29.76% | 7 |
| 0.24% |            | 51.22%      |       |      |        |   |
| Z23   | Regular    | dir_p40_h30 | 18600 | 0    | 57.03% | 4 |
| 2.97% |            | 51.07%      |       |      |        |   |
| Z23   | Regular    | dir_p50_h30 | 18600 | 0    | 45.70% | 5 |
| 4.30% |            | 50.69%      |       |      |        |   |
| Z23   | Regular    | dir_p60_h30 | 18600 | 0    | 34.50% | 6 |
| 5.50% |            | 49.74%      |       |      |        |   |
| Z23   | Regular    | dir_p40_h60 | 16802 | 1798 | 58.74% | 4 |
| 1.26% |            | 50.47%      |       |      |        |   |
| Z23   | Regular    | dir_p50_h60 | 16802 | 1798 | 46.98% | 5 |
| 3.02% |            | 48.68%      |       |      |        |   |
| Z23   | Regular    | dir_p60_h60 | 16802 | 1798 | 36.57% | 6 |
| 3.43% |            | 47.82%      |       |      |        |   |
| Z23   | Regular    | dir_p40_h90 | 14942 | 3658 | 60.49% | 3 |
| 9.51% |            | 49.77%      |       |      |        |   |

|       |            |             |       |      |        |   |
|-------|------------|-------------|-------|------|--------|---|
| Z23   | Regular    | dir_p50_h90 | 14942 | 3658 | 48.37% | 5 |
| 1.63% |            | 48.84%      |       |      |        |   |
| Z23   | Regular    | dir_p60_h90 | 14942 | 3658 | 36.72% | 6 |
| 3.28% |            | 48.11%      |       |      |        |   |
| Z24   | Closing    | dir_p40_h30 | 62    | 1798 | 51.61% | 4 |
| 8.39% |            | 46.88%      |       |      |        |   |
| Z24   | Closing    | dir_p50_h30 | 62    | 1798 | 43.55% | 5 |
| 6.45% |            | 48.15%      |       |      |        |   |
| Z24   | Closing    | dir_p60_h30 | 62    | 1798 | 33.87% | 6 |
| 6.13% |            | 42.86%      |       |      |        |   |
| Z24   | Closing    | dir_p40_h60 | 0     | 1860 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| Z24   | Closing    | dir_p50_h60 | 0     | 1860 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| Z24   | Closing    | dir_p60_h60 | 0     | 1860 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| Z24   | Closing    | dir_p40_h90 | 0     | 1860 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| Z24   | Closing    | dir_p50_h90 | 0     | 1860 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| Z24   | Closing    | dir_p60_h90 | 0     | 1860 | NaN    |   |
| NaN   |            | NaN         |       |      |        |   |
| Z24   | Opening    | dir_p40_h30 | 3720  | 0    | 69.73% | 3 |
| 0.27% |            | 50.69%      |       |      |        |   |
| Z24   | Opening    | dir_p50_h30 | 3720  | 0    | 59.44% | 4 |
| 0.56% |            | 50.07%      |       |      |        |   |
| Z24   | Opening    | dir_p60_h30 | 3720  | 0    | 47.80% | 5 |
| 2.20% |            | 48.82%      |       |      |        |   |
| Z24   | Opening    | dir_p40_h60 | 3720  | 0    | 67.15% | 3 |
| 2.85% |            | 50.16%      |       |      |        |   |
| Z24   | Opening    | dir_p50_h60 | 3720  | 0    | 58.17% | 4 |
| 1.83% |            | 48.52%      |       |      |        |   |
| Z24   | Opening    | dir_p60_h60 | 3720  | 0    | 48.71% | 5 |
| 1.29% |            | 45.75%      |       |      |        |   |
| Z24   | Opening    | dir_p40_h90 | 3720  | 0    | 66.08% | 3 |
| 3.92% |            | 49.15%      |       |      |        |   |
| Z24   | Opening    | dir_p50_h90 | 3720  | 0    | 55.48% | 4 |
| 4.52% |            | 46.46%      |       |      |        |   |
| Z24   | Opening    | dir_p60_h90 | 3720  | 0    | 45.83% | 5 |
| 4.17% |            | 44.11%      |       |      |        |   |
| Z24   | Overnight  | dir_p40_h30 | 14880 | 62   | 66.16% | 3 |
| 3.84% |            | 55.72%      |       |      |        |   |
| Z24   | Overnight  | dir_p50_h30 | 14880 | 62   | 54.17% | 4 |
| 5.83% |            | 56.22%      |       |      |        |   |
| Z24   | Overnight  | dir_p60_h30 | 14880 | 62   | 43.03% | 5 |
| 6.97% |            | 56.27%      |       |      |        |   |
| Z24   | Overnight  | dir_p40_h60 | 14880 | 62   | 66.01% | 3 |
| 3.99% |            | 56.72%      |       |      |        |   |
| Z24   | Overnight  | dir_p50_h60 | 14880 | 62   | 53.34% | 4 |
| 6.66% |            | 57.38%      |       |      |        |   |
| Z24   | Overnight  | dir_p60_h60 | 14880 | 62   | 41.19% | 5 |
| 8.81% |            | 57.27%      |       |      |        |   |
| Z24   | Overnight  | dir_p40_h90 | 14880 | 62   | 64.99% | 3 |
| 5.01% |            | 58.48%      |       |      |        |   |
| Z24   | Overnight  | dir_p50_h90 | 14880 | 62   | 53.40% | 4 |
| 6.60% |            | 58.18%      |       |      |        |   |
| Z24   | Overnight  | dir_p60_h90 | 14880 | 62   | 41.96% | 5 |
| 8.04% |            | 56.71%      |       |      |        |   |
| Z24   | Pre-market | dir_p40_h30 | 3720  | 0    | 70.27% | 2 |
| 9.73% |            | 43.27%      |       |      |        |   |

|       |     |            |             |       |      |        |   |
|-------|-----|------------|-------------|-------|------|--------|---|
| 0.54% | Z24 | Pre-market | dir_p50_h30 | 3720  | 0    | 59.46% | 4 |
|       |     |            | 42.27%      |       |      |        |   |
| 1.21% | Z24 | Pre-market | dir_p60_h30 | 3720  | 0    | 48.79% | 5 |
|       |     |            | 40.22%      |       |      |        |   |
| 7.34% | Z24 | Pre-market | dir_p40_h60 | 3720  | 0    | 72.66% | 2 |
|       |     |            | 43.95%      |       |      |        |   |
| 6.24% | Z24 | Pre-market | dir_p50_h60 | 3720  | 0    | 63.76% | 3 |
|       |     |            | 45.53%      |       |      |        |   |
| 4.81% | Z24 | Pre-market | dir_p60_h60 | 3720  | 0    | 55.19% | 4 |
|       |     |            | 45.88%      |       |      |        |   |
| 5.54% | Z24 | Pre-market | dir_p40_h90 | 3720  | 0    | 74.46% | 2 |
|       |     |            | 46.79%      |       |      |        |   |
| 2.15% | Z24 | Pre-market | dir_p50_h90 | 3720  | 0    | 67.85% | 3 |
|       |     |            | 44.97%      |       |      |        |   |
| 0.86% | Z24 | Pre-market | dir_p60_h90 | 3720  | 0    | 59.14% | 4 |
|       |     |            | 45.64%      |       |      |        |   |
| 6.17% | Z24 | Regular    | dir_p40_h30 | 18600 | 0    | 63.83% | 3 |
|       |     |            | 47.85%      |       |      |        |   |
| 7.11% | Z24 | Regular    | dir_p50_h30 | 18600 | 0    | 52.89% | 4 |
|       |     |            | 47.71%      |       |      |        |   |
| 9.62% | Z24 | Regular    | dir_p60_h30 | 18600 | 0    | 40.38% | 5 |
|       |     |            | 46.70%      |       |      |        |   |
| 7.01% | Z24 | Regular    | dir_p40_h60 | 16802 | 1798 | 62.99% | 3 |
|       |     |            | 47.50%      |       |      |        |   |
| 9.32% | Z24 | Regular    | dir_p50_h60 | 16802 | 1798 | 50.68% | 4 |
|       |     |            | 47.09%      |       |      |        |   |
| 0.23% | Z24 | Regular    | dir_p60_h60 | 16802 | 1798 | 39.77% | 6 |
|       |     |            | 45.12%      |       |      |        |   |
| 6.97% | Z24 | Regular    | dir_p40_h90 | 14942 | 3658 | 63.03% | 3 |
|       |     |            | 46.06%      |       |      |        |   |
| 8.00% | Z24 | Regular    | dir_p50_h90 | 14942 | 3658 | 52.00% | 4 |
|       |     |            | 45.29%      |       |      |        |   |
| 9.26% | Z24 | Regular    | dir_p60_h90 | 14942 | 3658 | 40.74% | 5 |
|       |     |            | 43.86%      |       |      |        |   |
| 2.95% | Z25 | Closing    | dir_p40_h30 | 61    | 1769 | 77.05% | 2 |
|       |     |            | 40.43%      |       |      |        |   |
| 4.43% | Z25 | Closing    | dir_p50_h30 | 61    | 1769 | 65.57% | 3 |
|       |     |            | 40.00%      |       |      |        |   |
| 2.62% | Z25 | Closing    | dir_p60_h30 | 61    | 1769 | 57.38% | 4 |
|       |     |            | 42.86%      |       |      |        |   |
| NaN   | Z25 | Closing    | dir_p40_h60 | 0     | 1830 | NaN    |   |
|       |     |            | NaN         |       |      |        |   |
| NaN   | Z25 | Closing    | dir_p50_h60 | 0     | 1830 | NaN    |   |
|       |     |            | NaN         |       |      |        |   |
| NaN   | Z25 | Closing    | dir_p60_h60 | 0     | 1830 | NaN    |   |
|       |     |            | NaN         |       |      |        |   |
| NaN   | Z25 | Closing    | dir_p40_h90 | 0     | 1830 | NaN    |   |
|       |     |            | NaN         |       |      |        |   |
| NaN   | Z25 | Closing    | dir_p50_h90 | 0     | 1830 | NaN    |   |
|       |     |            | NaN         |       |      |        |   |
| NaN   | Z25 | Closing    | dir_p60_h90 | 0     | 1830 | NaN    |   |
|       |     |            | NaN         |       |      |        |   |
| 3.63% | Z25 | Opening    | dir_p40_h30 | 3660  | 0    | 76.37% | 2 |
|       |     |            | 48.34%      |       |      |        |   |
| 1.58% | Z25 | Opening    | dir_p50_h30 | 3660  | 0    | 68.42% | 3 |
|       |     |            | 48.28%      |       |      |        |   |
| 0.44% | Z25 | Opening    | dir_p60_h30 | 3660  | 0    | 59.56% | 4 |
|       |     |            | 47.57%      |       |      |        |   |
| 9.92% | Z25 | Opening    | dir_p40_h60 | 3660  | 0    | 80.08% | 1 |
|       |     |            | 44.63%      |       |      |        |   |

|       |     |            |             |       |      |        |   |
|-------|-----|------------|-------------|-------|------|--------|---|
| 9.21% | Z25 | Opening    | dir_p50_h60 | 3660  | 0    | 70.79% | 2 |
|       |     |            | 43.27%      |       |      |        |   |
| 8.20% | Z25 | Opening    | dir_p60_h60 | 3660  | 0    | 61.80% | 3 |
|       |     |            | 42.04%      |       |      |        |   |
| 9.15% | Z25 | Opening    | dir_p40_h90 | 3660  | 0    | 80.85% | 1 |
|       |     |            | 42.14%      |       |      |        |   |
| 7.57% | Z25 | Opening    | dir_p50_h90 | 3660  | 0    | 72.43% | 2 |
|       |     |            | 40.93%      |       |      |        |   |
| 6.26% | Z25 | Opening    | dir_p60_h90 | 3660  | 0    | 63.74% | 3 |
|       |     |            | 39.39%      |       |      |        |   |
| 9.05% | Z25 | Overnight  | dir_p40_h30 | 14640 | 61   | 80.95% | 1 |
|       |     |            | 51.57%      |       |      |        |   |
| 7.83% | Z25 | Overnight  | dir_p50_h30 | 14640 | 61   | 72.17% | 2 |
|       |     |            | 50.80%      |       |      |        |   |
| 7.17% | Z25 | Overnight  | dir_p60_h30 | 14640 | 61   | 62.83% | 3 |
|       |     |            | 50.37%      |       |      |        |   |
| 8.27% | Z25 | Overnight  | dir_p40_h60 | 14640 | 61   | 81.73% | 1 |
|       |     |            | 52.52%      |       |      |        |   |
| 7.90% | Z25 | Overnight  | dir_p50_h60 | 14640 | 61   | 72.10% | 2 |
|       |     |            | 52.32%      |       |      |        |   |
| 8.72% | Z25 | Overnight  | dir_p60_h60 | 14640 | 61   | 61.28% | 3 |
|       |     |            | 51.33%      |       |      |        |   |
| 1.30% | Z25 | Overnight  | dir_p40_h90 | 14640 | 61   | 78.70% | 2 |
|       |     |            | 51.68%      |       |      |        |   |
| 0.55% | Z25 | Overnight  | dir_p50_h90 | 14640 | 61   | 69.45% | 3 |
|       |     |            | 51.69%      |       |      |        |   |
| 0.31% | Z25 | Overnight  | dir_p60_h90 | 14640 | 61   | 59.69% | 4 |
|       |     |            | 50.81%      |       |      |        |   |
| 2.38% | Z25 | Pre-market | dir_p40_h30 | 3660  | 0    | 77.62% | 2 |
|       |     |            | 44.49%      |       |      |        |   |
| 0.60% | Z25 | Pre-market | dir_p50_h30 | 3660  | 0    | 69.40% | 3 |
|       |     |            | 43.90%      |       |      |        |   |
| 0.63% | Z25 | Pre-market | dir_p60_h30 | 3660  | 0    | 59.37% | 4 |
|       |     |            | 42.66%      |       |      |        |   |
| 7.65% | Z25 | Pre-market | dir_p40_h60 | 3660  | 0    | 82.35% | 1 |
|       |     |            | 42.00%      |       |      |        |   |
| 4.84% | Z25 | Pre-market | dir_p50_h60 | 3660  | 0    | 75.16% | 2 |
|       |     |            | 41.51%      |       |      |        |   |
| 5.22% | Z25 | Pre-market | dir_p60_h60 | 3660  | 0    | 64.78% | 3 |
|       |     |            | 42.85%      |       |      |        |   |
| 0.00% | Z25 | Pre-market | dir_p40_h90 | 3660  | 0    | 80.00% | 2 |
|       |     |            | 45.90%      |       |      |        |   |
| 8.17% | Z25 | Pre-market | dir_p50_h90 | 3660  | 0    | 71.83% | 2 |
|       |     |            | 45.80%      |       |      |        |   |
| 6.20% | Z25 | Pre-market | dir_p60_h90 | 3660  | 0    | 63.80% | 3 |
|       |     |            | 46.17%      |       |      |        |   |
| 5.35% | Z25 | Regular    | dir_p40_h30 | 18300 | 0    | 74.65% | 2 |
|       |     |            | 49.73%      |       |      |        |   |
| 3.20% | Z25 | Regular    | dir_p50_h30 | 18300 | 0    | 66.80% | 3 |
|       |     |            | 49.59%      |       |      |        |   |
| 1.30% | Z25 | Regular    | dir_p60_h30 | 18300 | 0    | 58.70% | 4 |
|       |     |            | 49.25%      |       |      |        |   |
| 4.94% | Z25 | Regular    | dir_p40_h60 | 16531 | 1769 | 75.06% | 2 |
|       |     |            | 51.68%      |       |      |        |   |
| 2.73% | Z25 | Regular    | dir_p50_h60 | 16531 | 1769 | 67.27% | 3 |
|       |     |            | 50.34%      |       |      |        |   |
| 0.30% | Z25 | Regular    | dir_p60_h60 | 16531 | 1769 | 59.70% | 4 |
|       |     |            | 48.89%      |       |      |        |   |
| 3.52% | Z25 | Regular    | dir_p40_h90 | 14701 | 3599 | 76.48% | 2 |
|       |     |            | 50.97%      |       |      |        |   |



|       |     |         |             |       |      |        |   |
|-------|-----|---------|-------------|-------|------|--------|---|
|       | Z25 | Regular | dir_p50_h90 | 14701 | 3599 | 69.17% | 3 |
| 0.83% |     |         | 49.67%      |       |      |        |   |
|       | Z25 | Regular | dir_p60_h90 | 14701 | 3599 | 60.52% | 3 |
| 9.48% |     |         | 47.44%      |       |      |        |   |

## Celda 12 — Heatmap contract + regime para p50

```
In [79]: # Celda 12 - Heatmap contract + regime para p50
# Gráfico agrupado: 3 filas x 1 columna + colorbar separado
# =====

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd

def contract_sort_key(contract):
    """
    Ordena contratos del tipo H20, M20, U20, Z20, H21...
    """
    month_order = {"H": 1, "M": 2, "U": 3, "Z": 4}

    contract = str(contract)
    letter = contract[0]
    year = int(contract[1:]) if contract[1:].isdigit() else 999

    return year, month_order.get(letter, 99)

fig = plt.figure(figsize=(13, 22))

gs = fig.add_gridspec(
    nrows=3,
    ncols=2,
    width_ratios=[20, 1],
    height_ratios=[1, 1, 1],
    wspace=0.08,
    hspace=0.30
)

axes = [fig.add_subplot(gs[i, 0]) for i in range(3)]
cax = fig.add_subplot(gs[:, 1])

vmin = 0.20
vmax = 0.95

last_im = None

for ax, target_col in zip(axes, selected_targets_context):
    tmp = df_dir_contract_regime_distribution[
        df_dir_contract_regime_distribution["target"] == target_col
    ].copy()

    pivot_tmp = tmp.pivot(
        index="contract",
        columns="regime_name",
        values="signal_ratio"
    )

    ordered_contracts = sorted(
```

```
        pivot_tmp.index.astype(str).tolist(),
        key=contract_sort_key
    )

    pivot_tmp = pivot_tmp.reindex(index=ordered_contracts)
    pivot_tmp = pivot_tmp.reindex(columns=REGIME_ORDER)

    data = np.ma.masked_invalid(pivot_tmp.values)

    last_im = ax.imshow(
        data,
        aspect="auto",
        vmin=vmin,
        vmax=vmax
    )

    ax.set_title(f"Frecuencia operativa contract + regime - {target_col}")
    ax.set_ylabel("Contrato")

    ax.set_yticks(np.arange(len(pivot_tmp.index)))
    ax.set_yticklabels(pivot_tmp.index, fontsize=8)

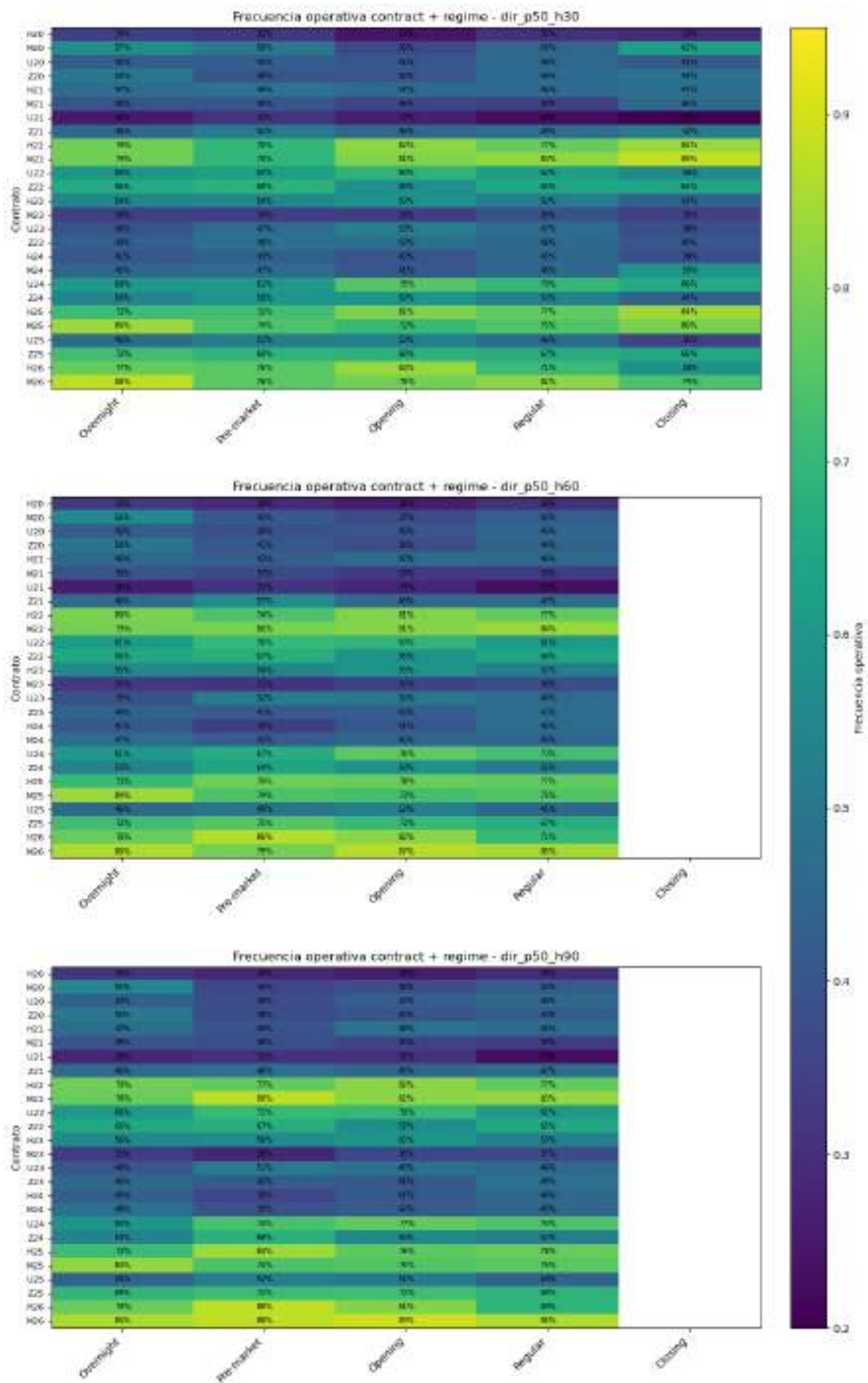
    ax.set_xticks(np.arange(len(pivot_tmp.columns)))
    ax.set_xticklabels(pivot_tmp.columns, rotation=45, ha="right")

    for i in range(pivot_tmp.shape[0]):
        for j in range(pivot_tmp.shape[1]):
            value = pivot_tmp.iloc[i, j]

            if pd.notna(value):
                ax.text(
                    j,
                    i,
                    f"{value:.0%}",
                    ha="center",
                    va="center",
                    fontsize=7
                )

    fig.colorbar(
        last_im,
        cax=cax,
        label="Frecuencia operativa"
    )

    plt.show()
```



## Celda 13 — Resumen automático de contrato y contexto

```
In [80]: # Celda 13 – Resumen automático del análisis por contrato y contexto
# =====

print("Resumen por contrato y contexto - DIR")
print("=" * 80)

print("\nEstabilidad por contract_code:")
print(
    df_contract_code_stability[
        ["target", "signal_mean", "signal_std", "signal_range", "long_share_mean"]
    ].sort_values("signal_range")
    .to_string(index=False, float_format=lambda x: f"{x:.2%}")
)

print("\nLectura general:")
print("- contract permite revisar si la familia de contrato H/M/U/Z cambia la fr")
print("- contract_code permite revisar diferencias más específicas asociadas al")
print("- year + regime ayuda a ver si ciertos años fueron más activos en momento")
print("- contract + regime ayuda a detectar si ciertos contratos presentan difer")
print("- El objetivo no es elegir aún un target final, sino detectar posibles ca
```

Resumen por contrato y contexto - DIR

=====

Estabilidad por contract\_code:

|             | target | signal_mean | signal_std | signal_range | long_share_mean |
|-------------|--------|-------------|------------|--------------|-----------------|
| dir_p40_h90 | 63.66% | 3.33%       | 7.72%      | 49.40%       |                 |
| dir_p40_h60 | 63.73% | 3.43%       | 7.87%      | 49.61%       |                 |
| dir_p40_h30 | 63.56% | 3.71%       | 8.47%      | 49.64%       |                 |
| dir_p50_h60 | 53.85% | 4.26%       | 10.13%     | 48.75%       |                 |
| dir_p50_h90 | 54.03% | 4.23%       | 10.15%     | 48.59%       |                 |
| dir_p50_h30 | 53.95% | 4.32%       | 10.21%     | 49.05%       |                 |
| dir_p60_h60 | 44.06% | 4.85%       | 11.70%     | 47.79%       |                 |
| dir_p60_h30 | 44.22% | 4.86%       | 11.71%     | 48.29%       |                 |
| dir_p60_h90 | 44.10% | 5.02%       | 12.09%     | 47.55%       |                 |

Lectura general:

- contract permite revisar si la familia de contrato H/M/U/Z cambia la frecuencia operativa.
- contract\_code permite revisar diferencias más específicas asociadas al rolling de contratos.
- year + regime ayuda a ver si ciertos años fueron más activos en momentos específicos del día.
- contract + regime ayuda a detectar si ciertos contratos presentan diferencias por régimen.
- El objetivo no es elegir aún un target final, sino detectar posibles cambios estructurales.

## Observaciones preliminares - Análisis por contrato y contexto

El análisis por contrato muestra que la frecuencia operativa de DIR cambia de forma importante entre contratos individuales. Contratos como H22, M22, H25, M25, M26 y H26 presentan frecuencias operativas elevadas, mientras que contratos como H20, U21 o M23 muestran valores mucho más bajos.

Esto sugiere que las diferencias observadas no dependen únicamente del target, sino también del contexto temporal del mercado en el que se ubica cada contrato. En otras palabras, ciertos contratos coinciden con períodos de mayor volatilidad o mayor actividad, como 2022 , 2025 y 2026 .

Al agrupar por contract\_code , es decir, por familia de contrato ( H , M , U , Z ), las diferencias son más moderadas. Esto indica que la letra del contrato parece tener menos impacto que el contrato individual específico. Por lo tanto, la principal variación parece estar más asociada al período histórico que a la familia trimestral del contrato.

El análisis year + regime permite estudiar el contexto combinado entre año y régimen intradiario. Este análisis confirma que los años más activos elevan la frecuencia operativa en varios regímenes al mismo tiempo. Por ejemplo, 2022 , 2025 y 2026 muestran frecuencias altas en Overnight , Pre-market , Opening y Regular , no solamente en un único momento del día.

El análisis contract + regime permite revisar si ciertos contratos presentan comportamientos diferentes según el régimen intradiario. Los resultados muestran que los contratos asociados a períodos más activos tienden a presentar mayor frecuencia operativa en varios regímenes simultáneamente. Esto refuerza la idea de que la principal fuente de variación proviene del contexto temporal del mercado.

El régimen Closing debe interpretarse con cuidado. Para horizontes H60 y H90 , aparecen valores faltantes porque no existe suficiente ventana futura dentro del mismo trading day. Esto es esperable y no representa un error de construcción.

En síntesis, DIR mantiene una lógica razonable por contrato y contexto, pero su frecuencia operativa cambia bastante según el período histórico. La variación parece estar más relacionada con el contexto temporal del mercado que con el régimen intradiario o la familia del contrato de forma aislada.

Para etapas posteriores, esta observación será importante porque el modelo no debería aprender simplemente patrones asociados a un contrato o año específico. En cambio, deberá recibir features que describan el contexto de mercado disponible en el instante  $t$  , como volatilidad reciente, rango intradiario, volumen, momentum y variables de sesión.

### 3.9. Observaciones e interpretación

El análisis del target DIR permite extraer varias conclusiones preliminares sobre su comportamiento estadístico y operativo.

En primer lugar, la construcción del target resultó consistente. Las clases generadas son válidas, la cantidad de observaciones inválidas aumenta correctamente con el horizonte y no se observaron señales de cruce entre trading days. Esto confirma que DIR fue construido respetando la lógica temporal definida para el proyecto.

Desde el punto de vista de distribución global de clases, **DIR** presenta un comportamiento razonable. Las señales **LONG** y **SHORT** se mantienen bastante balanceadas, con una leve inclinación hacia **SHORT**, especialmente en percentiles más altos. La clase **NO\_SIGNAL** aumenta de forma progresiva a medida que el threshold se vuelve más exigente.

La frecuencia operativa está controlada principalmente por el percentil seleccionado. Las variantes **p40** generan la mayor cantidad de señales, con una frecuencia operativa cercana al 64%. Las variantes **p50** quedan en un punto intermedio, alrededor del 54%. Las variantes **p60** son más restrictivas, con una frecuencia cercana al 44%.

Esto sugiere la siguiente lectura:

- p40** → variante más activa, pero potencialmente más ruidosa
- p50** → variante intermedia y más equilibrada
- p60** → variante más selectiva, pero con menor frecuencia operativa

El horizonte temporal, en cambio, no modifica demasiado la frecuencia operativa global. Para un mismo percentil, los horizontes **H30**, **H60** y **H90** presentan distribuciones muy similares. Esto indica que los thresholds por horizonte están calibrados de forma consistente.

Sin embargo, el análisis de estabilidad temporal muestra una conclusión importante:

**DIR** no es completamente estable a lo largo del tiempo. La frecuencia operativa cambia de forma significativa entre años. Períodos como **2022**, **2025** y **2026** presentan una frecuencia de señales mucho mayor que años más tranquilos como **2021**. Esto indica que el target es sensible al contexto temporal del mercado.

La comparación entre **development\_2020\_2024** y **final\_test\_2025\_2026** también muestra un aumento claro de la frecuencia operativa fuera de muestra. Por lo tanto, el período final debe interpretarse como un escenario de validación exigente o de estrés, no como un período para recalibrar el target.

El análisis por régimen intradiario muestra que **DIR** se comporta de manera bastante consistente entre **Overnight**, **Pre-market**, **Opening** y **Regular**. No aparece un régimen claramente dominante ni completamente desbalanceado. Sin embargo, **Closing** debe tratarse con cuidado, ya que para horizontes **H60** y **H90** no cuenta con suficiente ventana futura dentro del mismo trading day.

El análisis por contrato y contexto confirma que las diferencias más fuertes no parecen depender exclusivamente de la familia del contrato **H**, **M**, **U** o **Z**, sino del período histórico en el que se ubica cada contrato. Contratos correspondientes a años más activos presentan frecuencias operativas más altas, lo que refuerza la idea de que el contexto temporal del mercado es una variable importante para etapas posteriores.

En función de estos resultados, las variantes más razonables para pasar a la siguiente etapa son:

- dir\_p50\_h30**
- dir\_p50\_h60**
- dir\_p50\_h90**

Estas variantes mantienen una frecuencia operativa intermedia, un balance razonable entre `LONG` y `SHORT`, y una interpretación operativa clara.

Adicionalmente, conviene conservar como variantes comparativas:

`dir_p40_h30`, `dir_p40_h60`, `dir_p40_h90` → alternativas más activas  
`dir_p60_h30`, `dir_p60_h60`, `dir_p60_h90` → alternativas más selectivas

La selección definitiva todavía no debe realizarse en este punto. `DIR` debe ser comparado posteriormente con `BAR` y `OPC`, ya que estos targets incorporan una lógica operativa más cercana a una operación real mediante barreras de Take Profit y Stop Loss.

En síntesis, `DIR` es un target bien construido, interpretable y estadísticamente razonable. Sus principales fortalezas son el buen balance direccional y la facilidad de interpretación. Su principal debilidad es la sensibilidad temporal de la frecuencia operativa, lo que indica que futuras etapas deberán incorporar features capaces de describir el contexto de mercado en el instante `t`, como volatilidad reciente, rango intradiario, volumen, momentum y variables de sesión.

## 4. BAR - Resultado operativo por barreras TP/SL

En esta sección se construirá, validará y analizará el target `BAR`, cuyo objetivo es evaluar el resultado operativo de una señal direccional mediante una lógica de barreras de Take Profit y Stop Loss.

A diferencia de `DIR`, que identifica si el precio tuvo un movimiento significativo hacia arriba, hacia abajo o sin dirección clara, `BAR` incorpora la trayectoria futura del precio y permite analizar qué ocurre primero después de una posible entrada:

### Take Profit o Stop Loss

En esta primera etapa, `BAR` no se utilizará para definir la dirección de la operación. La dirección será tomada desde `DIR`. Por lo tanto, `BAR` se construirá como una extensión operativa de las señales direccionales previamente identificadas.

El análisis se mantendrá alineado con las mismas restricciones metodológicas utilizadas en `DIR`:

No cruzar trading days  
Usar ventanas futuras de H minutos  
Usar thresholds congelados de `development_2020_2024`  
No recalibrar reglas con `final_test_2025_2026`

El objetivo de esta sección no es seleccionar todavía una estrategia final de trading, sino evaluar si las señales `DIR` tienen suficiente calidad operativa bajo distintas configuraciones de Take Profit y Stop Loss.

### 4.1. Objetivo del target BAR

El objetivo del target `BAR` es evaluar si una señal direccional puede transformarse en una operación favorable bajo reglas explícitas de Take Profit y Stop Loss.

Mientras que `DIR` responde una pregunta direccional:

¿El precio tuvo un movimiento significativo hacia arriba, hacia abajo o sin dirección clara?

`BAR` responde una pregunta más operativa:

Dada una dirección operativa, ¿el precio alcanza primero el Take Profit o el Stop Loss?

Por lo tanto, `BAR` no busca reemplazar a `DIR`, sino complementarlo. `DIR` identifica la dirección potencial de la operación, mientras que `BAR` evalúa la calidad de esa dirección desde el punto de vista de ejecución.

La lógica general será:

```
DIR = 1  → se evalúa una operación LONG
DIR = -1 → se evalúa una operación SHORT
DIR = 0  → no se evalúa operación
DIR = NaN → no existe ventana futura válida
```

La codificación general de `BAR` será:

```
BAR = 1  → Take Profit alcanzado primero
BAR = -1 → Stop Loss alcanzado primero
BAR = 0  → no se alcanza ni TP ni SL dentro del horizonte
BAR = NaN → no hay operación válida o no existe ventana futura
```

suficiente

En síntesis, `BAR` representa el paso siguiente después de `DIR`: pasa de identificar movimientos direccionales significativos a evaluar si esos movimientos pueden sostener una estructura operativa de riesgo y beneficio.

## 4.2. Lógica metodológica de BAR

El target `BAR` será construido de forma condicionada por `DIR`. Esto significa que solo se evaluarán barreras de Take Profit y Stop Loss cuando `DIR` indique una dirección operativa válida.

Cuando `DIR = 0`, la observación representa un escenario sin dirección clara. En ese caso, no corresponde evaluar una operación y el valor de `BAR` será considerado no operable.

Una decisión importante es que el threshold utilizado para construir `DIR` no será interpretado directamente como el Take Profit definitivo. En cambio, será utilizado como una unidad base de movimiento significativo.

Esto es necesario porque, si se usara exactamente el mismo threshold de `DIR` como Take Profit, entonces en los casos donde `DIR != 0` el TP estaría prácticamente garantizado por construcción. En ese escenario, `BAR` mediría únicamente si el TP ocurrió antes que el SL, pero no permitiría evaluar objetivos operativos más exigentes.



Por este motivo, **BAR** utilizará el threshold de **DIR** como referencia base y construirá las barreras mediante múltiplos:

```
threshold DIR → unidad base de movimiento significativo
TP           → múltiplo del threshold
SL           → múltiplo del threshold
```

De esta manera, se podrán comparar distintas relaciones de riesgo/beneficio, por ejemplo:

```
TP 1.0 / SL 1.0
TP 1.5 / SL 1.0
TP 2.0 / SL 1.0
```

Estas combinaciones permitirán evaluar si las señales **DIR** solo alcanzan movimientos mínimos o si también pueden sostener objetivos de Take Profit más exigentes.

Para mantener consistencia con la construcción de **DIR**, esta primera versión de **BAR** evaluará las barreras usando la columna **close**. No se utilizarán inicialmente **high** ni **low**, ya que eso introduciría mayor sensibilidad intrabar y requeriría reglas adicionales para resolver casos ambiguos.

La ventana futura se mantiene igual que en **DIR**:

```
Future Window(t, H) = [t+1, ..., t+H]
```

Dentro de esa ventana se revisará, barra por barra, qué barrera se alcanza primero.

En síntesis, la metodología de **BAR** será:

1. Tomar la dirección desde **DIR**.
2. Usar el threshold de **DIR** como unidad base.
3. Construir barreras TP/SL mediante múltiplos del threshold.
4. Evaluar qué barrera se alcanza primero dentro del horizonte futuro.
5. Analizar los resultados por horizonte, percentil, año, régimen y contrato.

La definición formal de las barreras para operaciones LONG y SHORT se desarrollará en el punto 4.3.

## 4.3. Reglas de TP/SL para LONG y SHORT

El target **BAR** será construido a partir de las señales generadas por **DIR**. Para cada variante de **DIR**, se construirá una familia de targets **BAR** usando distintas configuraciones de Take Profit y Stop Loss.

Las variantes de **DIR** consideradas serán las mismas nueve utilizadas en el análisis anterior:

```
dir_p40_h30
dir_p50_h30
dir_p60_h30

dir_p40_h60
dir_p50_h60
```

dir\_p60\_h60

dir\_p40\_h90

dir\_p50\_h90

dir\_p60\_h90

Para cada una de estas variantes, se utilizará el threshold correspondiente como unidad base de movimiento significativo. Ese threshold depende de:

horizon

percentile

regime\_id

Sin embargo, el threshold no será interpretado directamente como el Take Profit definitivo. En cambio, se utilizará como una unidad base para construir diferentes relaciones de riesgo/beneficio.

La lógica general será:

TP = tp\_mult × threshold

SL = sl\_mult × threshold

donde:

tp\_mult = múltiplo aplicado al Take Profit

sl\_mult = múltiplo aplicado al Stop Loss

Las configuraciones iniciales a evaluar serán:

TP 1.0 / SL 1.0

TP 1.5 / SL 1.0

TP 2.0 / SL 1.0

TP 2.5 / SL 1.0

TP 1.0 / SL 0.5

TP 1.5 / SL 0.5

TP 2.0 / SL 0.5

Estas combinaciones permiten evaluar distintos escenarios operativos.

Las configuraciones con **SL = 1.0** mantienen el Stop Loss igual a la unidad base del movimiento significativo y aumentan progresivamente la exigencia del Take Profit.

TP 1.0 / SL 1.0 → benchmark 1:1

TP 1.5 / SL 1.0 → objetivo más exigente

TP 2.0 / SL 1.0 → objetivo amplio

TP 2.5 / SL 1.0 → objetivo muy exigente

Las configuraciones con **SL = 0.5** evalúan escenarios con Stop Loss más ajustado. Esto permite analizar si las señales de **DIR** pueden sostener operaciones con menor tolerancia al movimiento adverso.

TP 1.0 / SL 0.5

TP 1.5 / SL 0.5

TP 2.0 / SL 0.5

Para una operación **LONG**, el precio de entrada será el cierre actual:

entry\_price = close\_t

Las barreras se definen como:

```
TP_LONG = close_t + (tp_mult × threshold)
SL_LONG = close_t - (sl_mult × threshold)
```

Para una operación **SHORT**, el precio de entrada también será el cierre actual:

```
entry_price = close_t
```

Las barreras se definen de forma inversa:

```
TP_SHORT = close_t - (tp_mult × threshold)
SL_SHORT = close_t + (sl_mult × threshold)
```

Luego, dentro de la ventana futura:

```
Future Window(t, H) = [t+1, ..., t+H]
```

se revisa barra por barra qué barrera se alcanza primero.

La codificación del target será:

```
BAR = 1 → Take Profit alcanzado primero
BAR = -1 → Stop Loss alcanzado primero
BAR = 0 → no se alcanza ni TP ni SL dentro del horizonte
BAR = NaN → no hay operación válida o no existe ventana futura
```

suficiente

Cuando **DIR = 0**, no se evalúa operación y **BAR** será considerado no operable.

Cuando **DIR = NaN**, tampoco se evalúa **BAR**, ya que no existe una ventana futura válida.

Para evitar nombres ambiguos, las columnas **BAR** seguirán una nomenclatura explícita.

Por ejemplo:

```
bar_p50_h30_tp10_sl10
bar_p50_h30_tp15_sl10
bar_p50_h30_tp20_sl10
bar_p50_h30_tp25_sl10
```

```
bar_p50_h30_tp10_sl05
bar_p50_h30_tp15_sl05
bar_p50_h30_tp20_sl05
```

donde:

```
tp10 → TP = 1.0 × threshold
tp15 → TP = 1.5 × threshold
tp20 → TP = 2.0 × threshold
tp25 → TP = 2.5 × threshold
```

```
sl10 → SL = 1.0 × threshold
sl05 → SL = 0.5 × threshold
```

En total, se construirán:

9 variantes DIR × 7 configuraciones TP/SL = 63 variantes BAR

Estas variantes no implican que todas deban ser utilizadas posteriormente para entrenar modelos. Su objetivo inicial es permitir un análisis estadístico y operativo amplio, para identificar cuáles combinaciones de dirección, horizonte, percentil y relación TP/SL merecen pasar a las siguientes etapas.

Desde el punto de vista de equilibrio teórico, cada relación TP/SL requiere una tasa mínima de acierto distinta antes de considerar costos.

En este contexto, estar en equilibrio significa no ganar ni perder dinero en promedio. Es decir, después de muchas operaciones, las ganancias compensan exactamente las pérdidas y el resultado total queda aproximadamente en cero.

Por ejemplo, si una operación tiene:

TP = +10 puntos

SL = -10 puntos

entonces una estrategia con 50 operaciones ganadoras y 50 operaciones perdedoras tendría:

50 ganadoras × +10 puntos = +500 puntos

50 perdedoras × -10 puntos = -500 puntos

Resultado total = 0 puntos

Ese sería el punto de equilibrio para una relación TP 1.0 / SL 1.0 . No se gana ni se pierde, sin considerar costos.

En cambio, si una operación tiene:

TP = +20 puntos

SL = -10 puntos

cada operación ganadora vale el doble que una operación perdedora. En ese caso, no hace falta acertar el 50% de las veces para estar cerca del equilibrio.

Por ejemplo:

34 ganadoras × +20 puntos = +680 puntos

66 perdedoras × -10 puntos = -660 puntos

Resultado total = +20 puntos

Por eso, una relación TP 2.0 / SL 1.0 necesita una tasa de acierto aproximada de solo 33% para estar cerca del equilibrio antes de costos.

La fórmula general del punto de equilibrio es:

$$\text{win\_rate\_mínimo} = \text{SL} / (\text{TP} + \text{SL})$$

Aplicando esa fórmula a las configuraciones iniciales:

TP 1.0 / SL 1.0 → necesita superar aproximadamente 50%

TP 1.5 / SL 1.0 → necesita superar aproximadamente 40%

TP 2.0 / SL 1.0 → necesita superar aproximadamente 33%

TP 2.5 / SL 1.0 → necesita superar aproximadamente 29%

TP 1.0 / SL 0.5 → necesita superar aproximadamente 33%

TP 1.5 / SL 0.5 → necesita superar aproximadamente 25%

TP 2.0 / SL 0.5 → necesita superar aproximadamente 20%

Estos valores son solo referencias teóricas, ya que todavía no incluyen comisiones, spread, slippage ni otras restricciones operativas.

Además, una relación TP/SL más alta no significa automáticamente que sea mejor. Por ejemplo, un `TP 2.0 / SL 1.0` necesita menor tasa de acierto que un `TP 1.0 / SL 1.0`, pero también será más difícil de alcanzar porque el Take Profit está más lejos.

Por lo tanto, el análisis de `BAR` permitirá evaluar empíricamente qué configuraciones TP/SL tienen un comportamiento más razonable en los datos, considerando no solo la tasa de TP y SL, sino también su estabilidad por horizonte, percentil, régimen, año y contrato.

En síntesis, `BAR` permitirá evaluar si las señales `DIR` no solo indican una dirección futura significativa, sino si esa dirección puede sostener diferentes estructuras de riesgo y beneficio dentro del horizonte definido.

## 4.4. Construcción práctica de BAR

En este punto se construyen los targets `BAR` a partir de las señales `DIR`, los thresholds calibrados previamente y las configuraciones TP/SL definidas.

Para cada una de las nueve variantes de `DIR`, se evaluarán distintas relaciones de Take Profit y Stop Loss dentro de la ventana futura correspondiente. El resultado indicará si primero se alcanza el TP, si primero se alcanza el SL, si no ocurre ningún evento o si la fila no es operable.

Esta construcción mantiene la misma lógica temporal utilizada en `DIR`: no cruzar días, usar solo la ventana futura válida y conservar los thresholds congelados del período `development_2020_2024`.

- Celda 1 — Configuración general de BAR
- Celda 2 — Validación de inputs
- Celda 3 — Función para parsear nombres DIR
- Celda 4 — Función para construir un target BAR
- Celda 5 — Construcción de los 63 targets BAR
- Celda 6 — Resumen rápido de targets contruidos
- Celda 7 — Guardado del dataset BAR

### Celda 1 — Configuración general de BAR.

Esta celda solo deja preparada la configuración. En la Celda 2 validamos que existan los inputs necesarios: `df_targets_dir`, los 9 `DIR`, `df_dir_thresholds_primary`, `close`, `date`, `regime_id`, etc.

```
In [19]: # Celda 1 – Configuración general
# =====

import numpy as np
import pandas as pd
from pathlib import Path

print("Construcción práctica de BAR")
```

```

print("=" * 80)

# -----
# Configuración base
# -----

BAR_HORIZONS = [30, 60, 90]
BAR_PERCENTILES = [40, 50, 60]

PRICE_COL = "close"
DATE_COL = "date"
REGIME_COL = "regime_id"
THRESHOLD_COL = "threshold_common_pts"

# -----
# Configuraciones TP/SL a evaluar
# -----
# tp_mult y sl_mult son múltiplos del threshold base
# -----

BAR_TP_SL_CONFIGS = [
    {"tp_mult": 1.0, "sl_mult": 1.0, "tp_label": "tp10", "sl_label": "sl10"},
    {"tp_mult": 1.5, "sl_mult": 1.0, "tp_label": "tp15", "sl_label": "sl10"},
    {"tp_mult": 2.0, "sl_mult": 1.0, "tp_label": "tp20", "sl_label": "sl10"},
    {"tp_mult": 2.5, "sl_mult": 1.0, "tp_label": "tp25", "sl_label": "sl10"},

    {"tp_mult": 1.0, "sl_mult": 0.5, "tp_label": "tp10", "sl_label": "sl05"},
    {"tp_mult": 1.5, "sl_mult": 0.5, "tp_label": "tp15", "sl_label": "sl05"},
    {"tp_mult": 2.0, "sl_mult": 0.5, "tp_label": "tp20", "sl_label": "sl05"},
]

# -----
# Targets DIR base
# -----

dir_target_cols = [
    f"dir_p{p}_h{h}"
    for h in BAR_HORIZONS
    for p in BAR_PERCENTILES
]

# Variantes centrales para análisis y gráficos posteriores
selected_dir_targets_bar = [
    "dir_p50_h30",
    "dir_p50_h60",
    "dir_p50_h90",
]

# -----
# Nombres esperados de targets BAR
# -----

bar_target_cols = []

for h in BAR_HORIZONS:
    for p in BAR_PERCENTILES:
        for cfg in BAR_TP_SL_CONFIGS:
            bar_col = (
                f"bar_p{p}_h{h}_"
                f"{cfg['tp_label']}_{cfg['sl_label']}"
            )

```

```

    )
    bar_target_cols.append(bar_col)

# Columnas auxiliares para tiempo hasta evento
bar_event_step_cols = [
    f"{bar_col}_event_step"
    for bar_col in bar_target_cols
]

# -----
# Rutas de guardado
# -----

MNQ_TARGETS_PATH = PROJECT_ROOT / "data" / "04_mnq_targets"
MNQ_TARGETS_PATH.mkdir(parents=True, exist_ok=True)

MNQ_TARGETS_BAR_PARQUET = MNQ_TARGETS_PATH / "mnq_targets_bar.parquet"
MNQ_TARGETS_BAR_SUMMARY_CSV = MNQ_TARGETS_PATH / "mnq_targets_bar_summary.csv"

# -----
# Resumen de configuración
# -----

print(f"Horizontes BAR: {BAR_HORIZONS}")
print(f"Percentiles BAR: {BAR_PERCENTILES}")
print(f"Configuraciones TP/SL: {len(BAR_TP_SL_CONFIGS)}")
print(f"Targets DIR base: {len(dir_target_cols)}")
print(f"Targets BAR a construir: {len(bar_target_cols)}")
print(f"Columnas event_step a construir: {len(bar_event_step_cols)}")

print("\nConfiguraciones TP/SL:")
for cfg in BAR_TP_SL_CONFIGS:
    print(
        f"    {cfg['tp_label']}_{cfg['sl_label']} "
        f"→ TP={cfg['tp_mult']}x threshold, "
        f"SL={cfg['sl_mult']}x threshold"
    )

print("\nEjemplos de targets BAR:")
for col in bar_target_cols[:7]:
    print(f"    {col}")

print(f"\nArchivo de salida BAR: {MNQ_TARGETS_BAR_PARQUET}")

```

### Construcción práctica de BAR

```
=====
Horizontes BAR: [30, 60, 90]
Percentiles BAR: [40, 50, 60]
Configuraciones TP/SL: 7
Targets DIR base: 9
Targets BAR a construir: 63
Columnas event_step a construir: 63
```

#### Configuraciones TP/SL:

```
tp10_sl10 → TP=1.0x threshold, SL=1.0x threshold
tp15_sl10 → TP=1.5x threshold, SL=1.0x threshold
tp20_sl10 → TP=2.0x threshold, SL=1.0x threshold
tp25_sl10 → TP=2.5x threshold, SL=1.0x threshold
tp10_sl05 → TP=1.0x threshold, SL=0.5x threshold
tp15_sl05 → TP=1.5x threshold, SL=0.5x threshold
tp20_sl05 → TP=2.0x threshold, SL=0.5x threshold
```

#### Ejemplos de targets BAR:

```
bar_p40_h30_tp10_sl10
bar_p40_h30_tp15_sl10
bar_p40_h30_tp20_sl10
bar_p40_h30_tp25_sl10
bar_p40_h30_tp10_sl05
bar_p40_h30_tp15_sl05
bar_p40_h30_tp20_sl05
```

Archivo de salida BAR: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar.parquet

## Celda 2 — Validación de inputs para BAR.

Esta celda deja listo `df_targets_bar` y `df_bar_thresholds`, que serán la base para construir los 63 targets `BAR`.

```
In [20]: # Celda 2 - Validación de inputs
# =====

print("Validación de inputs para construir BAR")
print("=" * 80)

# -----
# 1. Validar existencia de datasets base
# -----

if "df_targets_dir" not in globals():
    raise NameError("No existe df_targets_dir. Primero debes construir o cargar")

if "df_dir_thresholds_primary" not in globals():
    raise NameError("No existe df_dir_thresholds_primary. Primero debes cargar 1")

# Copia de trabajo para BAR
df_targets_bar = df_targets_dir.copy()

# -----
# 2. Validar índice temporal
# -----
```



```

if not isinstance(df_targets_bar.index, pd.DatetimeIndex):
    raise TypeError("El índice de df_targets_bar debe ser un DatetimeIndex.")

if not df_targets_bar.index.is_monotonic_increasing:
    print("El índice no estaba ordenado. Se ordena por datetime.")
    df_targets_bar = df_targets_bar.sort_index()

if df_targets_bar.index.has_duplicates:
    raise ValueError("El índice contiene timestamps duplicados. Revisar antes de

print("Índice temporal validado correctamente.")

# -----
# 3. Validar columnas mínimas necesarias
# -----

required_base_cols = [
    PRICE_COL,
    DATE_COL,
    REGIME_COL,
]

missing_base_cols = [
    col for col in required_base_cols
    if col not in df_targets_bar.columns
]

if missing_base_cols:
    raise KeyError(f"Faltan columnas base necesarias: {missing_base_cols}")

print("Columnas base validadas correctamente.")

# -----
# 4. Validar targets DIR necesarios
# -----

missing_dir_cols = [
    col for col in dir_target_cols
    if col not in df_targets_bar.columns
]

if missing_dir_cols:
    raise KeyError(f"Faltan targets DIR necesarios para construir BAR: {missing_

print(f"Targets DIR disponibles: {len(dir_target_cols)}")

# Validar valores permitidos en DIR
valid_dir_values = {-1, 0, 1}

for col in dir_target_cols:
    values = set(df_targets_bar[col].dropna().unique())

    invalid_values = values - valid_dir_values

    if invalid_values:
        raise ValueError(
            f"El target {col} contiene valores inválidos: {invalid_values}. "
            "Solo se permiten -1, 0, 1 o NaN."
        )

```

```

print("Valores de targets DIR validados correctamente.")

# -----
# 5. Validar tabla de thresholds
# -----

required_threshold_cols = [
    "horizon",
    "percentile",
    REGIME_COL,
    THRESHOLD_COL,
]

missing_threshold_cols = [
    col for col in required_threshold_cols
    if col not in df_dir_thresholds_primary.columns
]

if missing_threshold_cols:
    raise KeyError(
        f"Faltan columnas necesarias en df_dir_thresholds_primary: {missing_thre
    )

# Nos quedamos solo con los thresholds necesarios para BAR
df_bar_thresholds = df_dir_thresholds_primary[
    df_dir_thresholds_primary["horizon"].isin(BAR_HORIZONS)
    & df_dir_thresholds_primary["percentile"].isin(BAR_PERCENTILES)
].copy()

# Validar duplicados en claves
threshold_key_cols = ["horizon", "percentile", REGIME_COL]

duplicated_thresholds = df_bar_thresholds.duplicated(
    subset=threshold_key_cols,
    keep=False
)

if duplicated_thresholds.any():
    duplicated_rows = df_bar_thresholds.loc[
        duplicated_thresholds,
        threshold_key_cols + [THRESHOLD_COL]
    ]

    raise ValueError(
        "Existen thresholds duplicados para la misma combinación "
        "(horizon, percentile, regime_id). Revisar df_bar_thresholds.\n"
        f"{duplicated_rows}"
    )

print("Tabla de thresholds validada correctamente.")

# -----
# 6. Validar que los thresholds sean positivos
# -----

invalid_threshold_values = df_bar_thresholds[
    df_bar_thresholds[THRESHOLD_COL].isna()
    | (df_bar_thresholds[THRESHOLD_COL] <= 0)
]

```

```

if len(invalid_threshold_values) > 0:
    raise ValueError(
        "Existen thresholds nulos o menores/iguales a cero. "
        "Revisar df_bar_thresholds."
    )

print("Valores de thresholds validados correctamente.")

# -----
# 7. Validar cobertura de thresholds para filas operables
# -----
# No exigimos thresholds para combinaciones sin filas operables.
# Por ejemplo, Closing en H60/H90 puede no tener ventanas válidas.
# -----

threshold_keys_available = set(
    df_bar_thresholds[threshold_key_cols]
    .apply(tuple, axis=1)
    .tolist()
)

missing_required_thresholds = []

for h in BAR_HORIZONS:
    for p in BAR_PERCENTILES:
        dir_col = f"dir_p{p}_h{h}"

        # Solo revisamos filas donde DIR es operable: LONG o SHORT
        operable_mask = df_targets_bar[dir_col].isin([1, -1])

        required_keys = set(
            df_targets_bar.loc[
                operable_mask,
                ["regime_id"]
            ]
            .assign(horizon=h, percentile=p)
            .loc[["horizon", "percentile", "regime_id"]]
            .apply(tuple, axis=1)
            .tolist()
        )

        missing_keys = required_keys - threshold_keys_available

        if missing_keys:
            missing_required_thresholds.extend(
                [
                    {
                        "target_dir": dir_col,
                        "horizon": key[0],
                        "percentile": key[1],
                        "regime_id": key[2],
                    }
                    for key in missing_keys
                ]
            )

if missing_required_thresholds:
    df_missing_required_thresholds = pd.DataFrame(missing_required_thresholds)
    raise ValueError(
        "Faltan thresholds para algunas filas operables de DIR. "

```

```

        "Revisar df_missing_required_thresholds."
    )

print("Cobertura de thresholds para filas operables validada correctamente.")

# -----
# 8. Resumen final de validación
# -----

print("\nResumen:")
print(f"Filas en df_targets_bar: {len(df_targets_bar):,}")
print(f"Columnas actuales: {df_targets_bar.shape[1]}")
print(f"Targets DIR validados: {len(dir_target_cols)}")
print(f"Thresholds disponibles para BAR: {len(df_bar_thresholds)}")
print(f"Targets BAR esperados: {len(bar_target_cols)}")
print(f"Columnas event_step esperadas: {len(bar_event_step_cols)}")

print("\nValidación finalizada correctamente.")

```

Validación de inputs para construir BAR

```

=====
Índice temporal validado correctamente.
Columnas base validadas correctamente.
Targets DIR disponibles: 9
Valores de targets DIR validados correctamente.
Tabla de thresholds validada correctamente.
Valores de thresholds validados correctamente.
Cobertura de thresholds para filas operables validada correctamente.

```

Resumen:

```

Filas en df_targets_bar: 1,024,062
Columnas actuales: 43
Targets DIR validados: 9
Thresholds disponibles para BAR: 39
Targets BAR esperados: 63
Columnas event_step esperadas: 63

```

Validación finalizada correctamente.

## Celda 3 — Funciones auxiliares para BAR.

Esta celda deja preparado el parseo de nombres y el `threshold_lookup`. En la Celda 4 ya podemos construir la función principal que evalúa TP/SL dentro de la ventana futura.

```

In [21]: # Celda 3 – Funciones auxiliares
# =====

import re

print("Funciones auxiliares para construir BAR")
print("=" * 80)

# -----
# 1. Parsear nombre de target DIR
# -----
# Ejemplo:
# dir_p50_h30 -> percentile=50, horizon=30
# -----

```

```

def parse_dir_target_name(dir_col):
    """
    Extrae percentile y horizon desde un nombre de target DIR.

    Ejemplo:
        dir_p50_h30 -> percentile=50, horizon=30
    """

    pattern = r"^dir_p(?:P<percentile>\d+)_h(?:P<horizon>\d+)$"
    match = re.match(pattern, dir_col)

    if match is None:
        raise ValueError(f"Nombre de target DIR inválido: {dir_col}")

    percentile = int(match.group("percentile"))
    horizon = int(match.group("horizon"))

    return percentile, horizon

# -----
# 2. Crear nombre de target BAR
# -----
# Ejemplo:
# dir_p50_h30 + tp15/sl10 -> bar_p50_h30_tp15_sl10
# -----

def make_bar_target_name(dir_col, tp_label, sl_label):
    """
    Crea el nombre del target BAR a partir de un target DIR
    y una configuración TP/SL.
    """

    percentile, horizon = parse_dir_target_name(dir_col)

    bar_col = (
        f"bar_p{percentile}_h{horizon}_"
        f"{tp_label}_{sl_label}"
    )

    return bar_col

# -----
# 3. Crear nombre de columna event_step
# -----
# Ejemplo:
# bar_p50_h30_tp15_sl10 -> bar_p50_h30_tp15_sl10_event_step
# -----

def make_bar_event_step_name(bar_col):
    """
    Crea el nombre de la columna que guarda el tiempo hasta evento.
    """

    return f"{bar_col}_event_step"

# -----
# 4. Crear lookup de thresholds

```

```

# -----
# Clave:
# (horizon, percentile, regime_id)
# -----

threshold_lookup = {
    (
        int(row["horizon"]),
        int(row["percentile"]),
        int(row[REGIME_COL]),
    ): float(row[THRESHOLD_COL])
    for _, row in df_bar_thresholds.iterrows()
}

print(f"Thresholds en lookup: {len(threshold_lookup)}")

# -----
# 5. Validación rápida de nombres esperados
# -----

parsed_dir_targets = []

for dir_col in dir_target_cols:
    percentile, horizon = parse_dir_target_name(dir_col)

    parsed_dir_targets.append(
        {
            "dir_col": dir_col,
            "percentile": percentile,
            "horizon": horizon,
        }
    )

df_parsed_dir_targets = pd.DataFrame(parsed_dir_targets)

print("\nTargets DIR parseados:")
print(df_parsed_dir_targets.to_string(index=False))

# -----
# 6. Validar nombres BAR esperados
# -----

generated_bar_cols_check = []

for dir_col in dir_target_cols:
    for cfg in BAR_TP_SL_CONFIGS:
        bar_col = make_bar_target_name(
            dir_col=dir_col,
            tp_label=cfg["tp_label"],
            sl_label=cfg["sl_label"],
        )

        generated_bar_cols_check.append(bar_col)

if len(generated_bar_cols_check) != len(set(generated_bar_cols_check)):
    raise ValueError("Existen nombres duplicados en los targets BAR generados.")

print("\nValidación de nombres BAR:")

```

```
print(f"Targets BAR generados: {len(generated_bar_cols_check)}")
print(f"Targets BAR únicos: {len(set(generated_bar_cols_check))}")

print("\nEjemplos:")
for col in generated_bar_cols_check[:10]:
    print(f" {col}")

print("\nFunciones auxiliares listas.")
```

Funciones auxiliares para construir BAR

=====

Thresholds en lookup: 39

Targets DIR parseados:

| dir_col     | percentile | horizon |
|-------------|------------|---------|
| dir_p40_h30 | 40         | 30      |
| dir_p50_h30 | 50         | 30      |
| dir_p60_h30 | 60         | 30      |
| dir_p40_h60 | 40         | 60      |
| dir_p50_h60 | 50         | 60      |
| dir_p60_h60 | 60         | 60      |
| dir_p40_h90 | 40         | 90      |
| dir_p50_h90 | 50         | 90      |
| dir_p60_h90 | 60         | 90      |

Validación de nombres BAR:

Targets BAR generados: 63

Targets BAR únicos: 63

Ejemplos:

```
bar_p40_h30_tp10_sl10
bar_p40_h30_tp15_sl10
bar_p40_h30_tp20_sl10
bar_p40_h30_tp25_sl10
bar_p40_h30_tp10_sl05
bar_p40_h30_tp15_sl05
bar_p40_h30_tp20_sl05
bar_p50_h30_tp10_sl10
bar_p50_h30_tp15_sl10
bar_p50_h30_tp20_sl10
```

Funciones auxiliares listas.

## Celda 4 — Función para construir un target BAR.

Esta función todavía no construye los 63 targets. Solo deja preparada la lógica para construir un BAR específico, por ejemplo:

dir\_p50\_h30 + TP 1.5 / SL 1.0

En la Celda 5 hacemos el loop completo para construir las 63 variantes.

```
In [22]: # Celda 4 – Función para construir un target BAR
# =====

print("Función para construir un target BAR")
print("=" * 80)

# -----
# Función auxiliar: primer paso donde ocurre una condición
```

```

# -----

def first_true_step(hit_matrix):
    """
    Devuelve el primer paso donde una condición es True.

    Parámetros
    -----
    hit_matrix : np.ndarray
        Matriz booleana de forma (n_observaciones, horizon).

    Retorna
    -----
    first_step : np.ndarray
        Paso del primer True, usando numeración 1, 2, ..., H.
        Si nunca ocurre True, devuelve NaN.
    """

    has_hit = hit_matrix.any(axis=1)

    first_step = np.full(hit_matrix.shape[0], np.nan)

    if has_hit.any():
        first_step[has_hit] = np.argmax(hit_matrix[has_hit], axis=1) + 1

    return first_step

# -----
# Función principal: construir un BAR para una variante DIR
# -----

def build_single_bar_target(
    df,
    dir_col,
    tp_mult,
    sl_mult,
    tp_label,
    sl_label,
    threshold_lookup,
    price_col=PRICE_COL,
    date_col=DATE_COL,
    regime_col=REGIME_COL,
):
    """
    Construye un target BAR a partir de un target DIR y una configuración TP/SL.

    BAR = 1 -> TP alcanzado primero
    BAR = -1 -> SL alcanzado primero
    BAR = 0 -> no se alcanza TP ni SL dentro del horizonte
    BAR = NaN -> no operable o sin ventana futura válida

    Además construye una columna event_step que indica en qué barra futura
    ocurrió el evento TP o SL.
    """

    # -----
    # 1. Identificar percentil, horizonte y nombres de columnas
    # -----

```



```

percentile, horizon = parse_dir_target_name(dir_col)

bar_col = make_bar_target_name(
    dir_col=dir_col,
    tp_label=tp_label,
    sl_label=sl_label,
)

event_step_col = make_bar_event_step_name(bar_col)

# -----
# 2. Inicializar arrays de salida
# -----

n_rows = len(df)

bar_values = np.full(n_rows, np.nan)
event_steps = np.full(n_rows, np.nan)

# Copia mínima de trabajo con posición numérica
work = df[[price_col, date_col, regime_col, dir_col]].copy()
work["_pos"] = np.arange(n_rows)

# -----
# 3. Procesar día por día para no cruzar trading days
# -----

for _, day_df in work.groupby(date_col, sort=False):
    n_day = len(day_df)

    # Si el día no tiene suficientes barras para el horizonte, se omite
    if n_day <= horizon:
        continue

    close_arr = day_df[price_col].to_numpy(dtype=float)
    regime_arr = day_df[regime_col].to_numpy()
    dir_arr = day_df[dir_col].to_numpy(dtype=float)
    pos_arr = day_df["_pos"].to_numpy()

    # Número de filas con ventana futura completa
    n_valid = n_day - horizon

    entry_price = close_arr[:n_valid]
    regime_valid = regime_arr[:n_valid]
    dir_valid = dir_arr[:n_valid]
    pos_valid = pos_arr[:n_valid]

    # -----
    # 4. Construir matriz de precios futuros
    # -----
    # Cada fila contiene:
    # [close_{t+1}, close_{t+2}, ..., close_{t+H}]
    # -----

    future_windows = np.lib.stride_tricks.sliding_window_view(
        close_arr,
        window_shape=horizon + 1,
    )[:, 1:]

    # -----

```

```

# 5. Asignar threshold por régimen
# -----

thresholds = np.full(n_valid, np.nan)

for regime_id in np.unique(regime_valid):
    key = (horizon, percentile, int(regime_id))
    threshold_value = threshold_lookup.get(key, np.nan)

    thresholds[regime_valid == regime_id] = threshold_value

# -----
# 6. Identificar filas operables
# -----

long_mask = (dir_valid == 1) & np.isfinite(thresholds)
short_mask = (dir_valid == -1) & np.isfinite(thresholds)

operable_mask = long_mask | short_mask

# Las filas operables empiezan como 0:
# no se alcanzó ni TP ni SL
bar_day = np.full(n_valid, np.nan)
step_day = np.full(n_valid, np.nan)

bar_day[operable_mask] = 0

# -----
# 7. Calcular barreras
# -----

tp_distance = tp_mult * thresholds
sl_distance = sl_mult * thresholds

# -----
# 8. Evaluar operaciones LONG
# -----

if long_mask.any():
    long_idx = np.where(long_mask)[0]

    long_tp_level = entry_price[long_idx] + tp_distance[long_idx]
    long_sl_level = entry_price[long_idx] - sl_distance[long_idx]

    long_future = future_windows[long_idx]

    long_tp_hits = long_future >= long_tp_level[:, None]
    long_sl_hits = long_future <= long_sl_level[:, None]

    long_tp_step = first_true_step(long_tp_hits)
    long_sl_step = first_true_step(long_sl_hits)

    tp_first = (
        np.isfinite(long_tp_step)
        & (
            ~np.isfinite(long_sl_step)
            | (long_tp_step < long_sl_step)
        )
    )

```

```

sl_first = (
    np.isfinite(long_sl_step)
    & (
        ~np.isfinite(long_tp_step)
        | (long_sl_step < long_tp_step)
    )
)

bar_day[long_idx[tp_first]] = 1
step_day[long_idx[tp_first]] = long_tp_step[tp_first]

bar_day[long_idx[sl_first]] = -1
step_day[long_idx[sl_first]] = long_sl_step[sl_first]

# -----
# 9. Evaluar operaciones SHORT
# -----

if short_mask.any():
    short_idx = np.where(short_mask)[0]

    short_tp_level = entry_price[short_idx] - tp_distance[short_idx]
    short_sl_level = entry_price[short_idx] + sl_distance[short_idx]

    short_future = future_windows[short_idx]

    short_tp_hits = short_future <= short_tp_level[:, None]
    short_sl_hits = short_future >= short_sl_level[:, None]

    short_tp_step = first_true_step(short_tp_hits)
    short_sl_step = first_true_step(short_sl_hits)

    tp_first = (
        np.isfinite(short_tp_step)
        & (
            ~np.isfinite(short_sl_step)
            | (short_tp_step < short_sl_step)
        )
    )

    sl_first = (
        np.isfinite(short_sl_step)
        & (
            ~np.isfinite(short_tp_step)
            | (short_sl_step < short_tp_step)
        )
    )

    bar_day[short_idx[tp_first]] = 1
    step_day[short_idx[tp_first]] = short_tp_step[tp_first]

    bar_day[short_idx[sl_first]] = -1
    step_day[short_idx[sl_first]] = short_sl_step[sl_first]

# -----
# 10. Guardar resultados del día en los arrays globales
# -----

bar_values[pos_valid] = bar_day
event_steps[pos_valid] = step_day

```

```

    return bar_col, event_step_col, bar_values, event_steps

print("Función build_single_bar_target lista.")

```

Función para construir un target BAR

=====

Función build\_single\_bar\_target lista.

## Celda 5 — Construcción de los 63 targets BAR.

Esta celda crea:

```

63 columnas BAR
63 columnas event_step

```

Después seguimos con la Celda 6, que sería un resumen rápido para revisar si los targets quedaron razonables antes de guardar.

```

In [23]: # Celda 5 – Construcción de Los targets BAR
# =====

import time
import gc

print("Construcción de targets BAR")
print("=" * 80)

start_time = time.time()

# -----
# 1. Eliminar columnas BAR previas si la celda se vuelve a correr
# -----

existing_bar_cols = [
    col for col in df_targets_bar.columns
    if col.startswith("bar_")
]

if existing_bar_cols:
    df_targets_bar = df_targets_bar.drop(columns=existing_bar_cols)
    print(f"Se eliminaron columnas BAR previas: {len(existing_bar_cols)}")

# -----
# 2. Construcción de targets BAR
# -----

bar_new_cols = {}
bar_metadata_rows = []

total_targets = len(dir_target_cols) * len(BAR_TP_SL_CONFIGS)
target_counter = 0

for dir_col in dir_target_cols:
    percentile, horizon = parse_dir_target_name(dir_col)

    for cfg in BAR_TP_SL_CONFIGS:
        target_counter += 1

```

```

tp_mult = cfg["tp_mult"]
sl_mult = cfg["sl_mult"]
tp_label = cfg["tp_label"]
sl_label = cfg["sl_label"]

bar_col, event_step_col, bar_values, event_steps = build_single_bar_target(
    df=df_targets_bar,
    dir_col=dir_col,
    tp_mult=tp_mult,
    sl_mult=sl_mult,
    tp_label=tp_label,
    sl_label=sl_label,
    threshold_lookup=threshold_lookup,
    price_col=PRICE_COL,
    date_col=DATE_COL,
    regime_col=REGIME_COL,
)

# Guardamos como tipos livianos:
# BAR: Int8 permite -1, 0, 1 y NaN
# event_step: Int16 permite pasos 1...90 y NaN
bar_new_cols[bar_col] = (
    pd.Series(bar_values, index=df_targets_bar.index)
    .astype("Int8")
)

bar_new_cols[event_step_col] = (
    pd.Series(event_steps, index=df_targets_bar.index)
    .astype("Int16")
)

bar_metadata_rows.append(
    {
        "dir_col": dir_col,
        "bar_col": bar_col,
        "event_step_col": event_step_col,
        "horizon": horizon,
        "percentile": percentile,
        "tp_mult": tp_mult,
        "sl_mult": sl_mult,
        "tp_label": tp_label,
        "sl_label": sl_label,
    }
)

print(
    f"[{target_counter:02d}/{total_targets}] "
    f"{bar_col} construido"
)

# -----
# 3. Unir nuevas columnas al dataset BAR
# -----

df_bar_new_cols = pd.DataFrame(bar_new_cols, index=df_targets_bar.index)

df_targets_bar = pd.concat(
    [df_targets_bar, df_bar_new_cols],
    axis=1
)

```

```
)

df_bar_metadata = pd.DataFrame(bar_metadata_rows)

# Liberar memoria auxiliar
del df_bar_new_cols
gc.collect()

# -----
# 4. Resumen final
# -----

elapsed_time = time.time() - start_time

bar_cols_created = [
    col for col in df_targets_bar.columns
    if col.startswith("bar_") and not col.endswith("_event_step")
]

bar_event_step_cols_created = [
    col for col in df_targets_bar.columns
    if col.startswith("bar_") and col.endswith("_event_step")
]

print("\nResumen de construcción BAR:")
print(f"Targets BAR contruidos: {len(bar_cols_created)}")
print(f"Columnas event_step construidas: {len(bar_event_step_cols_created)}")
print(f"Filas del dataset: {len(df_targets_bar):,}")
print(f"Columnas totales del dataset: {df_targets_bar.shape[1]}")
print(f"Tiempo total: {elapsed_time:.2f} segundos")

print("\nEjemplos de columnas BAR:")
for col in bar_cols_created[:10]:
    print(f" {col}")

print("\nConstrucción de targets BAR finalizada correctamente.")
```

## Construcción de targets BAR

```
=====
[01/63] bar_p40_h30_tp10_sl10 construido
[02/63] bar_p40_h30_tp15_sl10 construido
[03/63] bar_p40_h30_tp20_sl10 construido
[04/63] bar_p40_h30_tp25_sl10 construido
[05/63] bar_p40_h30_tp10_sl05 construido
[06/63] bar_p40_h30_tp15_sl05 construido
[07/63] bar_p40_h30_tp20_sl05 construido
[08/63] bar_p50_h30_tp10_sl10 construido
[09/63] bar_p50_h30_tp15_sl10 construido
[10/63] bar_p50_h30_tp20_sl10 construido
[11/63] bar_p50_h30_tp25_sl10 construido
[12/63] bar_p50_h30_tp10_sl05 construido
[13/63] bar_p50_h30_tp15_sl05 construido
[14/63] bar_p50_h30_tp20_sl05 construido
[15/63] bar_p60_h30_tp10_sl10 construido
[16/63] bar_p60_h30_tp15_sl10 construido
[17/63] bar_p60_h30_tp20_sl10 construido
[18/63] bar_p60_h30_tp25_sl10 construido
[19/63] bar_p60_h30_tp10_sl05 construido
[20/63] bar_p60_h30_tp15_sl05 construido
[21/63] bar_p60_h30_tp20_sl05 construido
[22/63] bar_p40_h60_tp10_sl10 construido
[23/63] bar_p40_h60_tp15_sl10 construido
[24/63] bar_p40_h60_tp20_sl10 construido
[25/63] bar_p40_h60_tp25_sl10 construido
[26/63] bar_p40_h60_tp10_sl05 construido
[27/63] bar_p40_h60_tp15_sl05 construido
[28/63] bar_p40_h60_tp20_sl05 construido
[29/63] bar_p50_h60_tp10_sl10 construido
[30/63] bar_p50_h60_tp15_sl10 construido
[31/63] bar_p50_h60_tp20_sl10 construido
[32/63] bar_p50_h60_tp25_sl10 construido
[33/63] bar_p50_h60_tp10_sl05 construido
[34/63] bar_p50_h60_tp15_sl05 construido
[35/63] bar_p50_h60_tp20_sl05 construido
[36/63] bar_p60_h60_tp10_sl10 construido
[37/63] bar_p60_h60_tp15_sl10 construido
[38/63] bar_p60_h60_tp20_sl10 construido
[39/63] bar_p60_h60_tp25_sl10 construido
[40/63] bar_p60_h60_tp10_sl05 construido
[41/63] bar_p60_h60_tp15_sl05 construido
[42/63] bar_p60_h60_tp20_sl05 construido
[43/63] bar_p40_h90_tp10_sl10 construido
[44/63] bar_p40_h90_tp15_sl10 construido
[45/63] bar_p40_h90_tp20_sl10 construido
[46/63] bar_p40_h90_tp25_sl10 construido
[47/63] bar_p40_h90_tp10_sl05 construido
[48/63] bar_p40_h90_tp15_sl05 construido
[49/63] bar_p40_h90_tp20_sl05 construido
[50/63] bar_p50_h90_tp10_sl10 construido
[51/63] bar_p50_h90_tp15_sl10 construido
[52/63] bar_p50_h90_tp20_sl10 construido
[53/63] bar_p50_h90_tp25_sl10 construido
[54/63] bar_p50_h90_tp10_sl05 construido
[55/63] bar_p50_h90_tp15_sl05 construido
[56/63] bar_p50_h90_tp20_sl05 construido
[57/63] bar_p60_h90_tp10_sl10 construido
[58/63] bar_p60_h90_tp15_sl10 construido
```

```
[59/63] bar_p60_h90_tp20_sl10 construido
[60/63] bar_p60_h90_tp25_sl10 construido
[61/63] bar_p60_h90_tp10_sl05 construido
[62/63] bar_p60_h90_tp15_sl05 construido
[63/63] bar_p60_h90_tp20_sl05 construido
```

Resumen de construcción BAR:  
 Targets BAR contruidos: 63  
 Columnas event\_step construidas: 63  
 Filas del dataset: 1,024,062  
 Columnas totales del dataset: 169  
 Tiempo total: 26.32 segundos

Ejemplos de columnas BAR:

```
bar_p40_h30_tp10_sl10
bar_p40_h30_tp15_sl10
bar_p40_h30_tp20_sl10
bar_p40_h30_tp25_sl10
bar_p40_h30_tp10_sl05
bar_p40_h30_tp15_sl05
bar_p40_h30_tp20_sl05
bar_p50_h30_tp10_sl10
bar_p50_h30_tp15_sl10
bar_p50_h30_tp20_sl10
```

Construcción de targets BAR finalizada correctamente.

## Celda 6 — Resumen rápido de targets BAR

```
In [24]: # Celda 6 - Resumen rápido de targets BAR
# =====

print("Resumen rápido de targets BAR")
print("=" * 80)

# -----
# 1. Identificar columnas BAR construidas
# -----

bar_cols_created = [
    col for col in df_targets_bar.columns
    if col.startswith("bar_") and not col.endswith("_event_step")
]

bar_event_step_cols_created = [
    col for col in df_targets_bar.columns
    if col.startswith("bar_") and col.endswith("_event_step")
]

print(f"Targets BAR encontrados: {len(bar_cols_created)}")
print(f"Columnas event_step encontradas: {len(bar_event_step_cols_created)}")

if len(bar_cols_created) == 0:
    raise ValueError("No se encontraron columnas BAR. Revisar la Celda 5.")

# -----
# 2. Función para resumir targets BAR
# -----
```



```

def summarize_bar_targets(df, bar_cols, metadata_df):
    """
    Resume la distribución de los targets BAR.

    BAR = 1 -> TP primero
    BAR = -1 -> SL primero
    BAR = 0 -> no se alcanza TP ni SL
    BAR = NaN -> no operable
    """

    rows = []

    metadata_lookup = (
        metadata_df
        .set_index("bar_col")
        .to_dict(orient="index")
    )

    for bar_col in bar_cols:
        event_step_col = f"{bar_col}_event_step"

        meta = metadata_lookup.get(bar_col, {})

        total_rows = len(df)

        valid_mask = df[bar_col].notna()
        valid_rows = int(valid_mask.sum())
        nan_rows = int(total_rows - valid_rows)

        valid_values = df.loc[valid_mask, bar_col].astype(int)

        tp_count = int((valid_values == 1).sum())
        sl_count = int((valid_values == -1).sum())
        no_event_count = int((valid_values == 0).sum())

        resolved_count = tp_count + sl_count

        if event_step_col in df.columns:
            event_steps = df.loc[df[event_step_col].notna(), event_step_col].ast

            event_step_mean = event_steps.mean() if len(event_steps) > 0 else np
            event_step_median = event_steps.median() if len(event_steps) > 0 els
            event_step_min = event_steps.min() if len(event_steps) > 0 else np.n
            event_step_max = event_steps.max() if len(event_steps) > 0 else np.n
        else:
            event_step_mean = np.nan
            event_step_median = np.nan
            event_step_min = np.nan
            event_step_max = np.nan

        row = {
            "bar_col": bar_col,
            "dir_col": meta.get("dir_col", np.nan),
            "horizon": meta.get("horizon", np.nan),
            "percentile": meta.get("percentile", np.nan),
            "tp_mult": meta.get("tp_mult", np.nan),
            "sl_mult": meta.get("sl_mult", np.nan),
            "tp_label": meta.get("tp_label", np.nan),
            "sl_label": meta.get("sl_label", np.nan),

```

```

        "total_rows": total_rows,
        "valid_rows": valid_rows,
        "nan_rows": nan_rows,

        "tp_count": tp_count,
        "sl_count": sl_count,
        "no_event_count": no_event_count,
        "resolved_count": resolved_count,

        "tp_rate": tp_count / valid_rows if valid_rows > 0 else np.nan,
        "sl_rate": sl_count / valid_rows if valid_rows > 0 else np.nan,
        "no_event_rate": no_event_count / valid_rows if valid_rows > 0 else

        "tp_share_over_resolved": tp_count / resolved_count if resolved_count
        "sl_share_over_resolved": sl_count / resolved_count if resolved_count

        "event_step_mean": event_step_mean,
        "event_step_median": event_step_median,
        "event_step_min": event_step_min,
        "event_step_max": event_step_max,
    }

    rows.append(row)

summary = pd.DataFrame(rows)

summary = summary.sort_values(
    ["percentile", "horizon", "sl_mult", "tp_mult"]
).reset_index(drop=True)

return summary

# -----
# 3. Construir resumen
# -----

df_bar_summary = summarize_bar_targets(
    df=df_targets_bar,
    bar_cols=bar_cols_created,
    metadata_df=df_bar_metadata,
)

# -----
# 4. Mostrar resumen completo por bloques
# -----

summary_view_cols = [
    "bar_col",
    "horizon",
    "percentile",
    "tp_mult",
    "sl_mult",
    "valid_rows",
    "nan_rows",
    "tp_rate",
    "sl_rate",
    "no_event_rate",
    "tp_share_over_resolved",
    "event_step_mean",

```

```

        "event_step_median",
    ]

    formatters_bar_summary = {
        "tp_rate": "{:.2%}".format,
        "sl_rate": "{:.2%}".format,
        "no_event_rate": "{:.2%}".format,
        "tp_share_over_resolved": "{:.2%}".format,
        "event_step_mean": "{:.2f}".format,
        "event_step_median": "{:.2f}".format,
    }

    print("\nResumen completo de targets BAR por horizonte y percentil:")
    print("=" * 100)

    for p in BAR_PERCENTILES:
        for h in BAR_HORIZONS:
            tmp = df_bar_summary[
                (df_bar_summary["percentile"] == p)
                & (df_bar_summary["horizon"] == h)
            ].copy()

            tmp = tmp.sort_values(["sl_mult", "tp_mult"]).reset_index(drop=True)

            print("\n" + "-" * 100)
            print(f"Percentil p{p} | Horizonte H{h}")
            print("-" * 100)

            print(
                tmp[summary_view_cols]
                .to_string(
                    index=False,
                    formatters=formatters_bar_summary
                )
            )

# -----
# 5. Resumen agregado por configuración TP/SL
# -----

df_bar_summary_by_config = (
    df_bar_summary
    .groupby(["tp_mult", "sl_mult"], as_index=False)
    .agg(
        n_targets=("bar_col", "count"),
        tp_rate_mean=("tp_rate", "mean"),
        sl_rate_mean=("sl_rate", "mean"),
        no_event_rate_mean=("no_event_rate", "mean"),
        tp_share_over_resolved_mean=("tp_share_over_resolved", "mean"),
        event_step_mean=("event_step_mean", "mean"),
    )
    .sort_values(["sl_mult", "tp_mult"])
    .reset_index(drop=True)
)

print("\n" + "=" * 100)
print("Resumen promedio por configuración TP/SL")
print("=" * 100)

print(

```

```
df_bar_summary_by_config.to_string(  
    index=False,  
    formatters={  
        "tp_rate_mean": "{:.2%}".format,  
        "sl_rate_mean": "{:.2%}".format,  
        "no_event_rate_mean": "{:.2%}".format,  
        "tp_share_over_resolved_mean": "{:.2%}".format,  
        "event_step_mean": "{:.2f}".format,  
    }  
)  
)  
  
print("\nResumen rápido de BAR finalizado correctamente.")
```

## Resumen rápido de targets BAR

Targets BAR encontrados: 63

Columnas event\_step encontradas: 63

## Resumen completo de targets BAR por horizonte y percentil:

## Percentil p40 | Horizonte H30

|                       | bar_col | horizon | percentile    | tp_mult                | sl_mult         | valid_rows | nan_row |
|-----------------------|---------|---------|---------------|------------------------|-----------------|------------|---------|
| s                     | tp_rate | sl_rate | no_event_rate | tp_share_over_resolved | event_step_mean | event_step | median  |
| bar_p40_h30_tp10_sl05 | 30      | 40      | 1.0           | 0.5                    | 623736          | 40032      |         |
| 6                     | 78.98%  | 21.02%  | 0.00%         | 78.98%                 | 11.18           |            |         |
| 9.00                  |         |         |               |                        |                 |            |         |
| bar_p40_h30_tp15_sl05 | 30      | 40      | 1.5           | 0.5                    | 623736          | 40032      |         |
| 6                     | 48.14%  | 25.07%  | 26.79%        | 65.76%                 | 12.46           |            |         |
| 11.00                 |         |         |               |                        |                 |            |         |
| bar_p40_h30_tp20_sl05 | 30      | 40      | 2.0           | 0.5                    | 623736          | 40032      |         |
| 6                     | 29.68%  | 26.94%  | 43.38%        | 52.42%                 | 12.70           |            |         |
| 11.00                 |         |         |               |                        |                 |            |         |
| bar_p40_h30_tp10_sl10 | 30      | 40      | 1.0           | 1.0                    | 623736          | 40032      |         |
| 6                     | 93.84%  | 6.16%   | 0.00%         | 93.84%                 | 13.20           |            |         |
| 12.00                 |         |         |               |                        |                 |            |         |
| bar_p40_h30_tp15_sl10 | 30      | 40      | 1.5           | 1.0                    | 623736          | 40032      |         |
| 6                     | 57.52%  | 7.35%   | 35.13%        | 88.67%                 | 14.83           |            |         |
| 14.00                 |         |         |               |                        |                 |            |         |
| bar_p40_h30_tp20_sl10 | 30      | 40      | 2.0           | 1.0                    | 623736          | 40032      |         |
| 6                     | 35.63%  | 8.35%   | 56.03%        | 81.02%                 | 15.66           |            |         |
| 15.00                 |         |         |               |                        |                 |            |         |
| bar_p40_h30_tp25_sl10 | 30      | 40      | 2.5           | 1.0                    | 623736          | 40032      |         |
| 6                     | 22.59%  | 8.92%   | 68.49%        | 71.69%                 | 15.97           |            |         |
| 16.00                 |         |         |               |                        |                 |            |         |

## Percentil p40 | Horizonte H60

|                       | bar_col | horizon | percentile    | tp_mult                | sl_mult         | valid_rows | nan_row |
|-----------------------|---------|---------|---------------|------------------------|-----------------|------------|---------|
| s                     | tp_rate | sl_rate | no_event_rate | tp_share_over_resolved | event_step_mean | event_step | median  |
| bar_p40_h60_tp10_sl05 | 60      | 40      | 1.0           | 0.5                    | 597024          | 42703      |         |
| 8                     | 77.94%  | 22.06%  | 0.00%         | 77.94%                 | 21.95           |            |         |
| 18.00                 |         |         |               |                        |                 |            |         |
| bar_p40_h60_tp15_sl05 | 60      | 40      | 1.5           | 0.5                    | 597024          | 42703      |         |
| 8                     | 46.45%  | 26.43%  | 27.12%        | 63.74%                 | 24.35           |            |         |
| 21.00                 |         |         |               |                        |                 |            |         |
| bar_p40_h60_tp20_sl05 | 60      | 40      | 2.0           | 0.5                    | 597024          | 42703      |         |
| 8                     | 27.84%  | 28.31%  | 43.85%        | 49.58%                 | 24.57           |            |         |
| 21.00                 |         |         |               |                        |                 |            |         |
| bar_p40_h60_tp10_sl10 | 60      | 40      | 1.0           | 1.0                    | 597024          | 42703      |         |
| 8                     | 94.00%  | 6.00%   | 0.00%         | 94.00%                 | 26.19           |            |         |
| 24.00                 |         |         |               |                        |                 |            |         |
| bar_p40_h60_tp15_sl10 | 60      | 40      | 1.5           | 1.0                    | 597024          | 42703      |         |

|                       |        |       |        |        |        |       |
|-----------------------|--------|-------|--------|--------|--------|-------|
| 8                     | 56.30% | 7.31% | 36.39% | 88.51% | 29.41  |       |
| 28.00                 |        |       |        |        |        |       |
| bar_p40_h60_tp20_sl10 |        |       | 60     | 40     | 2.0    | 1.0   |
| 8                     | 34.08% | 8.30% | 57.61% | 80.41% | 597024 | 42703 |
| 30.00                 |        |       |        |        | 31.05  |       |
| bar_p40_h60_tp25_sl10 |        |       | 60     | 40     | 2.5    | 1.0   |
| 8                     | 21.05% | 8.84% | 70.11% | 70.42% | 597024 | 42703 |
| 31.00                 |        |       |        |        | 31.48  |       |

-----

Percentil p40 | Horizonte H90

-----

|                       | bar_col | horizon | percentile    | tp_mult                | sl_mult         | valid_rows | nan_row |
|-----------------------|---------|---------|---------------|------------------------|-----------------|------------|---------|
| s                     | tp_rate | sl_rate | no_event_rate | tp_share_over_resolved | event_step_mean | event_step | median  |
| bar_p40_h90_tp10_sl05 |         |         | 90            | 40                     | 1.0             | 0.5        | 567970  |
| 2                     | 77.10%  | 22.90%  | 0.00%         | 77.10%                 |                 | 32.93      | 45609   |
| 27.00                 |         |         |               |                        |                 |            |         |
| bar_p40_h90_tp15_sl05 |         |         | 90            | 40                     | 1.5             | 0.5        | 567970  |
| 2                     | 45.52%  | 27.32%  | 27.16%        | 62.49%                 |                 | 36.63      | 45609   |
| 32.00                 |         |         |               |                        |                 |            |         |
| bar_p40_h90_tp20_sl05 |         |         | 90            | 40                     | 2.0             | 0.5        | 567970  |
| 2                     | 27.45%  | 29.21%  | 43.34%        | 48.45%                 |                 | 37.08      | 45609   |
| 32.00                 |         |         |               |                        |                 |            |         |
| bar_p40_h90_tp10_sl10 |         |         | 90            | 40                     | 1.0             | 1.0        | 567970  |
| 2                     | 93.57%  | 6.43%   | 0.00%         | 93.57%                 |                 | 39.52      | 45609   |
| 36.00                 |         |         |               |                        |                 |            |         |
| bar_p40_h90_tp15_sl10 |         |         | 90            | 40                     | 1.5             | 1.0        | 567970  |
| 2                     | 55.98%  | 7.76%   | 36.26%        | 87.83%                 |                 | 44.77      | 45609   |
| 43.00                 |         |         |               |                        |                 |            |         |
| bar_p40_h90_tp20_sl10 |         |         | 90            | 40                     | 2.0             | 1.0        | 567970  |
| 2                     | 34.09%  | 8.83%   | 57.08%        | 79.43%                 |                 | 47.51      | 45609   |
| 47.00                 |         |         |               |                        |                 |            |         |
| bar_p40_h90_tp25_sl10 |         |         | 90            | 40                     | 2.5             | 1.0        | 567970  |
| 2                     | 21.19%  | 9.43%   | 69.38%        | 69.19%                 |                 | 48.34      | 45609   |
| 49.00                 |         |         |               |                        |                 |            |         |

-----

Percentil p50 | Horizonte H30

-----

|                       | bar_col | horizon | percentile    | tp_mult                | sl_mult         | valid_rows | nan_row |
|-----------------------|---------|---------|---------------|------------------------|-----------------|------------|---------|
| s                     | tp_rate | sl_rate | no_event_rate | tp_share_over_resolved | event_step_mean | event_step | median  |
| bar_p50_h30_tp10_sl05 |         |         | 30            | 50                     | 1.0             | 0.5        | 529982  |
| 0                     | 81.93%  | 18.07%  | 0.00%         | 81.93%                 |                 | 12.37      | 49408   |
| 11.00                 |         |         |               |                        |                 |            |         |
| bar_p50_h30_tp15_sl05 |         |         | 30            | 50                     | 1.5             | 0.5        | 529982  |
| 0                     | 45.74%  | 21.67%  | 32.60%        | 67.85%                 |                 | 13.40      | 49408   |
| 12.00                 |         |         |               |                        |                 |            |         |
| bar_p50_h30_tp20_sl05 |         |         | 30            | 50                     | 2.0             | 0.5        | 529982  |
| 0                     | 26.31%  | 23.19%  | 50.50%        | 53.15%                 |                 | 13.39      | 49408   |
| 12.00                 |         |         |               |                        |                 |            |         |
| bar_p50_h30_tp10_sl10 |         |         | 30            | 50                     | 1.0             | 1.0        | 529982  |
| 0                     | 95.26%  | 4.74%   | 0.00%         | 95.26%                 |                 | 14.21      | 49408   |
| 13.00                 |         |         |               |                        |                 |            |         |
| bar_p50_h30_tp15_sl10 |         |         | 30            | 50                     | 1.5             | 1.0        | 529982  |
|                       |         |         |               |                        |                 |            | 49408   |

|                       |        |       |        |        |        |       |
|-----------------------|--------|-------|--------|--------|--------|-------|
| 0                     | 53.60% | 5.71% | 40.70% | 90.38% | 15.74  |       |
| 15.00                 |        |       |        |        |        |       |
| bar_p50_h30_tp20_sl10 |        |       | 30     | 50     | 2.0    | 1.0   |
| 0                     | 31.07% | 6.49% | 62.44% | 82.73% | 529982 | 49408 |
| 17.00                 |        |       |        |        | 16.50  |       |
| bar_p50_h30_tp25_sl10 |        |       | 30     | 50     | 2.5    | 1.0   |
| 0                     | 18.63% | 6.90% | 74.47% | 72.97% | 529982 | 49408 |
| 17.00                 |        |       |        |        | 16.70  |       |

-----

Percentil p50 | Horizonte H60

-----

|                       | bar_col | horizon | percentile    | tp_mult                | sl_mult         | valid_rows | nan_row |
|-----------------------|---------|---------|---------------|------------------------|-----------------|------------|---------|
| s                     | tp_rate | sl_rate | no_event_rate | tp_share_over_resolved | event_step_mean | event_step | median  |
| bar_p50_h60_tp10_sl05 |         |         | 60            | 50                     | 1.0             | 0.5        | 505020  |
| 2                     | 81.32%  | 18.68%  | 0.00%         | 81.32%                 |                 | 24.40      | 51904   |
| 21.00                 |         |         |               |                        |                 |            |         |
| bar_p50_h60_tp15_sl05 |         |         | 60            | 50                     | 1.5             | 0.5        | 505020  |
| 2                     | 44.00%  | 22.52%  | 33.48%        | 66.15%                 |                 | 26.28      | 51904   |
| 24.00                 |         |         |               |                        |                 |            |         |
| bar_p50_h60_tp20_sl05 |         |         | 60            | 50                     | 2.0             | 0.5        | 505020  |
| 2                     | 24.58%  | 23.99%  | 51.44%        | 50.61%                 |                 | 25.95      | 51904   |
| 23.00                 |         |         |               |                        |                 |            |         |
| bar_p50_h60_tp10_sl10 |         |         | 60            | 50                     | 1.0             | 1.0        | 505020  |
| 2                     | 95.45%  | 4.55%   | 0.00%         | 95.45%                 |                 | 28.22      | 51904   |
| 26.00                 |         |         |               |                        |                 |            |         |
| bar_p50_h60_tp15_sl10 |         |         | 60            | 50                     | 1.5             | 1.0        | 505020  |
| 2                     | 52.23%  | 5.54%   | 42.23%        | 90.41%                 |                 | 31.28      | 51904   |
| 31.00                 |         |         |               |                        |                 |            |         |
| bar_p50_h60_tp20_sl10 |         |         | 60            | 50                     | 2.0             | 1.0        | 505020  |
| 2                     | 29.49%  | 6.29%   | 64.22%        | 82.42%                 |                 | 32.68      | 51904   |
| 33.00                 |         |         |               |                        |                 |            |         |
| bar_p50_h60_tp25_sl10 |         |         | 60            | 50                     | 2.5             | 1.0        | 505020  |
| 2                     | 17.33%  | 6.69%   | 75.98%        | 72.16%                 |                 | 32.87      | 51904   |
| 34.00                 |         |         |               |                        |                 |            |         |

-----

Percentil p50 | Horizonte H90

-----

|                       | bar_col | horizon | percentile    | tp_mult                | sl_mult         | valid_rows | nan_row |
|-----------------------|---------|---------|---------------|------------------------|-----------------|------------|---------|
| s                     | tp_rate | sl_rate | no_event_rate | tp_share_over_resolved | event_step_mean | event_step | median  |
| bar_p50_h90_tp10_sl05 |         |         | 90            | 50                     | 1.0             | 0.5        | 482603  |
| 9                     | 80.23%  | 19.77%  | 0.00%         | 80.23%                 |                 | 36.58      | 54145   |
| 32.00                 |         |         |               |                        |                 |            |         |
| bar_p50_h90_tp15_sl05 |         |         | 90            | 50                     | 1.5             | 0.5        | 482603  |
| 9                     | 43.60%  | 23.71%  | 32.70%        | 64.78%                 |                 | 39.72      | 54145   |
| 36.00                 |         |         |               |                        |                 |            |         |
| bar_p50_h90_tp20_sl05 |         |         | 90            | 50                     | 2.0             | 0.5        | 482603  |
| 9                     | 24.52%  | 25.25%  | 50.23%        | 49.26%                 |                 | 39.31      | 54145   |
| 35.00                 |         |         |               |                        |                 |            |         |
| bar_p50_h90_tp10_sl10 |         |         | 90            | 50                     | 1.0             | 1.0        | 482603  |
| 9                     | 95.11%  | 4.89%   | 0.00%         | 95.11%                 |                 | 42.70      | 54145   |
| 40.00                 |         |         |               |                        |                 |            |         |
| bar_p50_h90_tp15_sl10 |         |         | 90            | 50                     | 1.5             | 1.0        | 482603  |
|                       |         |         |               |                        |                 |            | 54145   |

```

9 52.38% 5.93% 41.69% 89.83% 47.81
48.00
bar_p50_h90_tp20_sl10 90 50 2.0 1.0 482603 54145
9 29.78% 6.77% 63.45% 81.48% 50.14
51.00
bar_p50_h90_tp25_sl10 90 50 2.5 1.0 482603 54145
9 17.65% 7.17% 75.18% 71.12% 50.79
53.00

```

-----

Percentil p60 | Horizonte H30

-----

```

          bar_col horizon percentile tp_mult sl_mult valid_rows nan_row
s tp_rate sl_rate no_event_rate tp_share_over_resolved event_step_mean event_step
_median
bar_p60_h30_tp10_sl05 30 60 1.0 0.5 434995 58906
7 84.78% 15.22% 0.00% 84.78% 13.51
12.00
bar_p60_h30_tp15_sl05 30 60 1.5 0.5 434995 58906
7 43.07% 18.43% 38.51% 70.04% 14.34
14.00
bar_p60_h30_tp20_sl05 30 60 2.0 0.5 434995 58906
7 23.11% 19.59% 57.29% 54.12% 14.07
13.00
bar_p60_h30_tp10_sl10 30 60 1.0 1.0 434995 58906
7 96.41% 3.59% 0.00% 96.41% 15.14
15.00
bar_p60_h30_tp15_sl10 30 60 1.5 1.0 434995 58906
7 49.53% 4.37% 46.11% 91.90% 16.62
17.00
bar_p60_h30_tp20_sl10 30 60 2.0 1.0 434995 58906
7 26.78% 4.94% 68.28% 84.42% 17.26
18.00
bar_p60_h30_tp25_sl10 30 60 2.5 1.0 434995 58906
7 15.06% 5.21% 79.73% 74.30% 17.32
18.00

```

-----

Percentil p60 | Horizonte H60

-----

```

          bar_col horizon percentile tp_mult sl_mult valid_rows nan_row
s tp_rate sl_rate no_event_rate tp_share_over_resolved event_step_mean event_step
_median
bar_p60_h60_tp10_sl05 60 60 1.0 0.5 413798 61026
4 84.19% 15.81% 0.00% 84.19% 26.62
24.00
bar_p60_h60_tp15_sl05 60 60 1.5 0.5 413798 61026
4 41.45% 19.09% 39.46% 68.46% 28.02
27.00
bar_p60_h60_tp20_sl05 60 60 2.0 0.5 413798 61026
4 21.61% 20.25% 58.13% 51.63% 27.12
25.00
bar_p60_h60_tp10_sl10 60 60 1.0 1.0 413798 61026
4 96.62% 3.38% 0.00% 96.62% 30.03
29.00
bar_p60_h60_tp15_sl10 60 60 1.5 1.0 413798 61026

```



|                       |        |       |        |        |        |       |
|-----------------------|--------|-------|--------|--------|--------|-------|
| 4                     | 48.27% | 4.14% | 47.58% | 92.10% | 32.97  |       |
| 33.00                 |        |       |        |        |        |       |
| bar_p60_h60_tp20_sl10 |        |       | 60     | 60     | 2.0    | 1.0   |
| 4                     | 25.48% | 4.72% | 69.80% | 84.38% | 413798 | 61026 |
| 35.00                 |        |       |        |        | 34.12  |       |
| bar_p60_h60_tp25_sl10 |        |       | 60     | 60     | 2.5    | 1.0   |
| 4                     | 14.23% | 4.97% | 80.80% | 74.13% | 413798 | 61026 |
| 35.00                 |        |       |        |        | 34.09  |       |

-----  
 -----  
 Percentil p60 | Horizonte H90  
 -----  
 -----

|                       | bar_col | horizon | percentile    | tp_mult                | sl_mult         | valid_rows | nan_row |
|-----------------------|---------|---------|---------------|------------------------|-----------------|------------|---------|
| s                     | tp_rate | sl_rate | no_event_rate | tp_share_over_resolved | event_step_mean | event_step | median  |
| bar_p60_h90_tp10_sl05 |         |         | 90            | 60                     | 1.0             | 0.5        | 394501  |
| 1                     | 83.19%  | 16.81%  | 0.00%         | 83.19%                 |                 | 40.14      | 62956   |
| 37.00                 |         |         |               |                        |                 |            |         |
| bar_p60_h90_tp15_sl05 |         |         | 90            | 60                     | 1.5             | 0.5        | 394501  |
| 1                     | 41.58%  | 20.19%  | 38.23%        | 67.31%                 |                 | 42.67      | 62956   |
| 41.00                 |         |         |               |                        |                 |            |         |
| bar_p60_h90_tp20_sl05 |         |         | 90            | 60                     | 2.0             | 0.5        | 394501  |
| 1                     | 21.82%  | 21.44%  | 56.74%        | 50.45%                 |                 | 41.59      | 62956   |
| 39.00                 |         |         |               |                        |                 |            |         |
| bar_p60_h90_tp10_sl10 |         |         | 90            | 60                     | 1.0             | 1.0        | 394501  |
| 1                     | 96.37%  | 3.63%   | 0.00%         | 96.37%                 |                 | 45.68      | 62956   |
| 44.00                 |         |         |               |                        |                 |            |         |
| bar_p60_h90_tp15_sl10 |         |         | 90            | 60                     | 1.5             | 1.0        | 394501  |
| 1                     | 48.78%  | 4.52%   | 46.70%        | 91.52%                 |                 | 50.61      | 62956   |
| 51.00                 |         |         |               |                        |                 |            |         |
| bar_p60_h90_tp20_sl10 |         |         | 90            | 60                     | 2.0             | 1.0        | 394501  |
| 1                     | 25.89%  | 5.12%   | 68.98%        | 83.48%                 |                 | 52.77      | 62956   |
| 55.00                 |         |         |               |                        |                 |            |         |
| bar_p60_h90_tp25_sl10 |         |         | 90            | 60                     | 2.5             | 1.0        | 394501  |
| 1                     | 14.40%  | 5.42%   | 80.18%        | 72.67%                 |                 | 53.00      | 62956   |
| 56.00                 |         |         |               |                        |                 |            |         |

=====  
 =====  
 Resumen promedio por configuración TP/SL  
 =====  
 =====

| tp_mult | sl_mult | n_targets | tp_rate_mean | sl_rate_mean | no_event_rate_mean | tp_share_over_resolved_mean | event_step_mean |
|---------|---------|-----------|--------------|--------------|--------------------|-----------------------------|-----------------|
| 1.0     | 0.5     | 9         | 81.07%       | 18.93%       | 0.00%              |                             |                 |
| 81.07%  |         | 24.41     |              |              |                    |                             |                 |
| 1.5     | 0.5     | 9         | 44.39%       | 22.71%       | 32.89%             |                             |                 |
| 66.29%  |         | 26.43     |              |              |                    |                             |                 |
| 2.0     | 0.5     | 9         | 25.21%       | 24.24%       | 50.55%             |                             |                 |
| 51.08%  |         | 26.20     |              |              |                    |                             |                 |
| 1.0     | 1.0     | 9         | 95.18%       | 4.82%        | 0.00%              |                             |                 |
| 95.18%  |         | 28.32     |              |              |                    |                             |                 |
| 1.5     | 1.0     | 9         | 52.73%       | 5.85%        | 41.42%             |                             |                 |
| 90.13%  |         | 31.56     |              |              |                    |                             |                 |
| 2.0     | 1.0     | 9         | 30.26%       | 6.65%        | 63.10%             |                             |                 |
| 82.20%  |         | 33.08     |              |              |                    |                             |                 |
| 2.5     | 1.0     | 9         | 18.02%       | 7.06%        | 74.92%             |                             |                 |
| 72.07%  |         | 33.39     |              |              |                    |                             |                 |

Resumen rápido de BAR finalizado correctamente.

## Celda 7 — Guardado del dataset BAR

```
In [25]: # Celda 7 — Guardado del dataset BAR
# =====

print("Guardado del dataset BAR")
print("=" * 80)

# -----
# 1. Definir rutas adicionales de guardado
# -----

MNQ_TARGETS_BAR_METADATA_CSV = MNQ_TARGETS_PATH / "mnq_targets_bar_metadata.csv"
MNQ_TARGETS_BAR_METADATA_PARQUET = MNQ_TARGETS_PATH / "mnq_targets_bar_metadata."

# -----
# 2. Validaciones antes de guardar
# -----

if "df_targets_bar" not in globals():
    raise NameError("No existe df_targets_bar. Primero debes construir los targe

if "df_bar_summary" not in globals():
    raise NameError("No existe df_bar_summary. Primero debes correr la Celda 6."

if "df_bar_metadata" not in globals():
    raise NameError("No existe df_bar_metadata. Primero debes correr la Celda 5.

bar_cols_to_save = [
    col for col in df_targets_bar.columns
    if col.startswith("bar_") and not col.endswith("_event_step")
]

bar_event_step_cols_to_save = [
    col for col in df_targets_bar.columns
    if col.startswith("bar_") and col.endswith("_event_step")
]

if len(bar_cols_to_save) != 63:
    print(f"Advertencia: se esperaban 63 targets BAR, pero se encontraron {len(b

if len(bar_event_step_cols_to_save) != 63:
    print(f"Advertencia: se esperaban 63 columnas event_step, pero se encontraro

print(f"Targets BAR a guardar: {len(bar_cols_to_save)}")
print(f"Columnas event_step a guardar: {len(bar_event_step_cols_to_save)}")
print(f"Filas del dataset: {len(df_targets_bar):,}")
print(f"Columnas totales: {df_targets_bar.shape[1]}")

# -----
# 3. Guardar dataset completo BAR
# -----
# Nota:
# Las columnas event_step se guardan para análisis del target.
# No deben usarse como features de modelos predictivos.
# -----
```

```

df_targets_bar.to_parquet(MNQ_TARGETS_BAR_PARQUET)

# -----
# 4. Guardar resumen y metadata
# -----

df_bar_summary.to_csv(MNQ_TARGETS_BAR_SUMMARY_CSV, index=False)

df_bar_metadata.to_csv(MNQ_TARGETS_BAR_METADATA_CSV, index=False)
df_bar_metadata.to_parquet(MNQ_TARGETS_BAR_METADATA_PARQUET)

# -----
# 5. Confirmación final
# -----

print("\nArchivos guardados:")
print(f"Dataset BAR:      {MNQ_TARGETS_BAR_PARQUET}")
print(f"Resumen BAR:      {MNQ_TARGETS_BAR_SUMMARY_CSV}")
print(f"Metadata BAR CSV: {MNQ_TARGETS_BAR_METADATA_CSV}")
print(f"Metadata BAR PQ: {MNQ_TARGETS_BAR_METADATA_PARQUET}")

print("\nGuardado finalizado correctamente.")

```

Guardado del dataset BAR

=====

Targets BAR a guardar: 63

Columnas event\_step a guardar: 63

Filas del dataset: 1,024,062

Columnas totales: 169

Archivos guardados:

Dataset BAR: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar.parquet

Resumen BAR: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_summary.csv

Metadata BAR CSV: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_metadata.csv

Metadata BAR PQ: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_metadata.parquet

Guardado finalizado correctamente.

## Observaciones breves

Primero, las variantes TP 1.0 / SL 1.0 y TP 1.0 / SL 0.5 tienen

no\_event\_rate = 0%. Esto es esperable porque DIR ya exige que exista una excursión favorable igual o superior al threshold. Por eso, cuando el TP es  $1.0 \times \text{threshold}$ , el TP se alcanza en algún momento de la ventana. La diferencia está en si ocurre antes o después del SL.

Segundo, TP 1.0 / SL 1.0 es el caso más favorable: en promedio alcanza TP primero el 95.18% de las veces y SL primero solo el 4.82%. Pero este resultado está muy condicionado por la lógica de construcción de DIR, así que debe interpretarse como **benchmark**, no como una señal de trading definitiva.

Tercero, al aumentar el TP, cae la tasa de TP y sube mucho el `no_event_rate`. Por ejemplo, en promedio:

TP 1.5 / SL 1.0 → TP 52.73%, no\_event 41.42%  
 TP 2.0 / SL 1.0 → TP 30.26%, no\_event 63.10%  
 TP 2.5 / SL 1.0 → TP 18.02%, no\_event 74.92%

Esto indica que objetivos más ambiciosos son mucho más difíciles de alcanzar dentro del horizonte.

Cuarto, las configuraciones con `SL = 0.5` son más exigentes porque el Stop Loss está más cerca. Por eso aumentan bastante los SL. Por ejemplo:

TP 1.0 / SL 0.5 → SL 18.93%  
 TP 1.5 / SL 0.5 → SL 22.71%  
 TP 2.0 / SL 0.5 → SL 24.24%

Esto tiene sentido: cuanto más ajustado el SL, más fácil es que el precio lo toque antes de llegar al TP.

Quinto, las variantes `TP 2.0 / SL 0.5` parecen poco atractivas inicialmente, porque el TP y el SL quedan casi equilibrados entre los eventos resueltos:

`tp_share_over_resolved = 51.08%`, pero con `no_event_rate = 50.55%`. Es decir, cuando resuelve, no muestra una ventaja clara.

Sexto, las variantes `TP 1.5 / SL 1.0` parecen interesantes para seguir analizando.

Tienen un TP promedio de `52.73%`, SL bajo de `5.85%` y un

`tp_share_over_resolved` de `90.13%`. El problema es que dejan un `41.42%` como no resuelto, así que habrá que ver qué pasa con esos casos en análisis posteriores.

Séptimo, las variantes `TP 2.0 / SL 1.0` también podrían analizarse, pero son más exigentes. El TP promedio cae a `30.26%` y el `no_event_rate` sube a `63.10%`.

Podrían ser útiles solo si los eventos TP tienen suficiente valor operativo y los no resueltos no son perjudiciales.

## Lectura preliminar

Todavía no descartaremos ninguna familia, pero para análisis posteriores me enfocaré en:

Benchmark:

TP 1.0 / SL 1.0

Candidata principal:

TP 1.5 / SL 1.0

Candidata exigente:

TP 2.0 / SL 1.0

Candidata con SL ajustado:

TP 1.5 / SL 0.5

Las variantes `TP 2.5 / SL 1.0` parecen demasiado exigentes por ahora, porque dejan cerca del `75%` de casos sin evento. Y `TP 2.0 / SL 0.5` parece demasiado agresiva

porque combina TP lejano con SL muy ajustado.

La conclusión importante es: **BAR está funcionando como esperábamos**. A medida que exigimos más TP o ajustamos más el SL, el target se vuelve más difícil, menos frecuente y más selectivo.

## 4.5. Validación de consistencia de BAR

Antes de analizar los resultados operativos de `BAR`, es necesario validar que los targets fueron construidos correctamente.

El objetivo de este punto es comprobar que las columnas `BAR` respeten la lógica definida previamente: solo deben tomar valores válidos, deben estar correctamente condicionadas por `DIR`, no deben cruzar trading days y las columnas auxiliares de tiempo hasta evento deben ser consistentes con el resultado del target.

En esta etapa no se busca determinar si una configuración TP/SL es buena o mala. El objetivo es verificar que la construcción del target sea temporalmente correcta, lógica y libre de errores estructurales.

Las validaciones principales serán:

1. Confirmar que existan los 63 targets BAR y sus 63 columnas `event_step`.
2. Verificar que BAR solo contenga valores -1, 0, 1 o NaN.
3. Confirmar que BAR sea NaN cuando DIR sea 0 o NaN.
4. Confirmar que BAR solo tenga valor cuando DIR sea LONG o SHORT.
5. Verificar que `event_step` solo exista cuando BAR sea 1 o -1.
6. Confirmar que `event_step` esté dentro del horizonte correspondiente.
7. Validar que no exista cruce entre trading days.
8. Revisar casos esperados, como la ausencia de `no_event` cuando  $TP = 1.0 \times \text{threshold}$ .

Si estas validaciones son correctas, se podrá avanzar con mayor confianza al análisis de distribución global, frecuencia de TP/SL, tiempo hasta evento, estabilidad temporal, régimen y contrato.

### 4.5.1. Validación de columnas construidas

Esta celda solo confirma que la estructura está completa.

```
In [88]: # 4.5.1 - Validación de columnas construidas
# =====

print("4.5.1 Validación de columnas construidas - BAR")
print("=" * 80)

# -----
# 1. Validar existencia de objetos necesarios
# -----

if "df_targets_bar" not in globals():
```

```

        raise NameError("No existe df_targets_bar. Primero debes construir los targe

if "df_bar_metadata" not in globals():
    raise NameError("No existe df_bar_metadata. Primero debes construir df_bar_m

# -----
# 2. Identificar columnas BAR reales en el dataset
# -----

bar_cols_found = [
    col for col in df_targets_bar.columns
    if col.startswith("bar_") and not col.endswith("_event_step")
]

bar_event_step_cols_found = [
    col for col in df_targets_bar.columns
    if col.startswith("bar_") and col.endswith("_event_step")
]

# -----
# 3. Identificar columnas BAR esperadas según metadata
# -----

bar_cols_expected = df_bar_metadata["bar_col"].tolist()
bar_event_step_cols_expected = df_bar_metadata["event_step_col"].tolist()

# -----
# 4. Validar cantidad esperada
# -----

expected_n_bar = 63
expected_n_event_step = 63

print(f"Targets BAR encontrados: {len(bar_cols_found)}")
print(f"Targets BAR esperados: {expected_n_bar}")

print(f"Columnas event_step encontradas: {len(bar_event_step_cols_found)}")
print(f"Columnas event_step esperadas: {expected_n_event_step}")

if len(bar_cols_found) != expected_n_bar:
    raise ValueError(
        f"Cantidad incorrecta de targets BAR. "
        f"Esperados: {expected_n_bar}, encontrados: {len(bar_cols_found)}"
    )

if len(bar_event_step_cols_found) != expected_n_event_step:
    raise ValueError(
        f"Cantidad incorrecta de columnas event_step. "
        f"Esperadas: {expected_n_event_step}, encontradas: {len(bar_event_step_c

# -----
# 5. Validar columnas faltantes o inesperadas
# -----

missing_bar_cols = sorted(set(bar_cols_expected) - set(bar_cols_found))
unexpected_bar_cols = sorted(set(bar_cols_found) - set(bar_cols_expected))

missing_event_step_cols = sorted(set(bar_event_step_cols_expected) - set(bar_eve
unexpected_event_step_cols = sorted(set(bar_event_step_cols_found) - set(bar_eve

```

```

if missing_bar_cols:
    raise ValueError(f"Faltan targets BAR esperados: {missing_bar_cols}")

if unexpected_bar_cols:
    raise ValueError(f"Existen targets BAR no esperados: {unexpected_bar_cols}")

if missing_event_step_cols:
    raise ValueError(f"Faltan columnas event_step esperadas: {missing_event_step_cols}")

if unexpected_event_step_cols:
    raise ValueError(f"Existen columnas event_step no esperadas: {unexpected_event_step_cols}")

# -----
# 6. Validar duplicados en metadata
# -----

duplicated_bar_metadata = df_bar_metadata["bar_col"].duplicated().sum()
duplicated_event_metadata = df_bar_metadata["event_step_col"].duplicated().sum()

if duplicated_bar_metadata > 0:
    raise ValueError("Existen nombres duplicados en df_bar_metadata['bar_col'].")

if duplicated_event_metadata > 0:
    raise ValueError("Existen nombres duplicados en df_bar_metadata['event_step_col'].")

# -----
# 7. Resumen final
# -----

print("\nValidación de columnas construidas:")
print("OK - Existen los 63 targets BAR.")
print("OK - Existen las 63 columnas event_step.")
print("OK - No faltan columnas esperadas.")
print("OK - No existen columnas inesperadas.")
print("OK - No hay duplicados en metadata.")

print("\nEjemplos de targets BAR:")
for col in bar_cols_found[:10]:
    print(f"  {col}")

print("\nValidación 4.5.1 finalizada correctamente.")

```

#### 4.5.1 Validación de columnas construidas - BAR

```
=====
Targets BAR encontrados: 63
Targets BAR esperados:   63
Columnas event_step encontradas: 63
Columnas event_step esperadas:  63
```

Validación de columnas construidas:

- OK - Existen los 63 targets BAR.
- OK - Existen las 63 columnas event\_step.
- OK - No faltan columnas esperadas.
- OK - No existen columnas inesperadas.
- OK - No hay duplicados en metadata.

Ejemplos de targets BAR:

```
bar_p40_h30_tp10_sl10
bar_p40_h30_tp15_sl10
bar_p40_h30_tp20_sl10
bar_p40_h30_tp25_sl10
bar_p40_h30_tp10_sl05
bar_p40_h30_tp15_sl05
bar_p40_h30_tp20_sl05
bar_p50_h30_tp10_sl10
bar_p50_h30_tp15_sl10
bar_p50_h30_tp20_sl10
```

Validación 4.5.1 finalizada correctamente.

### 4.5.2. Validación de valores permitidos

Esta celda confirma que ningún target BAR tenga valores raros.

```
In [89]: # 4.5.2 - Validación de valores permitidos
# =====

print("4.5.2 Validación de valores permitidos - BAR")
print("=" * 80)

# -----
# 1. Valores permitidos
# -----

valid_bar_values = {-1, 0, 1}

validation_rows = []
invalid_value_rows = []

# -----
# 2. Revisar cada target BAR
# -----

for bar_col in bar_cols_found:
    series = df_targets_bar[bar_col]

    non_na_values = series.dropna()

    unique_values = sorted(non_na_values.astype(int).unique().tolist())

    invalid_values = set(unique_values) - valid_bar_values
```



```

value_counts = series.value_counts(dropna=False)

count_tp = int((series == 1).sum())
count_sl = int((series == -1).sum())
count_no_event = int((series == 0).sum())
count_nan = int(series.isna().sum())

validation_rows.append(
    {
        "bar_col": bar_col,
        "unique_values": unique_values,
        "count_tp_1": count_tp,
        "count_sl_minus_1": count_sl,
        "count_no_event_0": count_no_event,
        "count_nan": count_nan,
        "invalid_values": sorted(list(invalid_values)),
        "has_invalid_values": len(invalid_values) > 0,
    }
)

if invalid_values:
    invalid_value_rows.append(
        {
            "bar_col": bar_col,
            "invalid_values": sorted(list(invalid_values)),
        }
    )

# -----
# 3. Crear resumen de validación
# -----

df_bar_value_validation = pd.DataFrame(validation_rows)

n_targets_checked = len(df_bar_value_validation)
n_targets_with_invalid_values = int(df_bar_value_validation["has_invalid_values"])

print(f"Targets BAR revisados: {n_targets_checked}")
print(f"Targets BAR con valores inválidos: {n_targets_with_invalid_values}")

# -----
# 4. Detener si existen valores inválidos
# -----

if n_targets_with_invalid_values > 0:
    df_invalid_bar_values = pd.DataFrame(invalid_value_rows)

    print("\nTargets con valores inválidos:")
    print(df_invalid_bar_values.to_string(index=False))

    raise ValueError(
        "Existen targets BAR con valores fuera de {-1, 0, 1, NaN}. "
        "Revisar df_invalid_bar_values."
    )

# -----
# 5. Mostrar vista de resumen
# -----

```

```
print("\nVista previa de validación de valores:")
print(
    df_bar_value_validation[
        [
            "bar_col",
            "unique_values",
            "count_tp_1",
            "count_sl_minus_1",
            "count_no_event_0",
            "count_nan",
        ]
    ]
    .head(15)
    .to_string(index=False)
)

# -----
# 6. Validación agregada
# -----

total_tp = df_bar_value_validation["count_tp_1"].sum()
total_sl = df_bar_value_validation["count_sl_minus_1"].sum()
total_no_event = df_bar_value_validation["count_no_event_0"].sum()
total_nan = df_bar_value_validation["count_nan"].sum()

print("\nResumen agregado de valores BAR:")
print(f"Total TP primero      (1): {total_tp:,}")
print(f"Total SL primero      (-1): {total_sl:,}")
print(f"Total no_event         (0): {total_no_event:,}")
print(f"Total NaN / no operable: {total_nan:,}")

print("\nValidación de valores permitidos:")
print("OK - Todos los targets BAR contienen únicamente -1, 0, 1 o NaN.")

print("\nValidación 4.5.2 finalizada correctamente.")
```

## 4.5.2 Validación de valores permitidos - BAR

```
=====
```

Targets BAR revisados: 63

Targets BAR con valores inválidos: 0

Vista previa de validación de valores:

|                       | bar_col   | unique_values | count_tp_1 | count_sl_minus_1 | count_no_event |
|-----------------------|-----------|---------------|------------|------------------|----------------|
| _0                    | count_nan |               |            |                  |                |
| bar_p40_h30_tp10_sl10 |           | [-1, 1]       | 585334     | 38402            |                |
| 0                     | 400326    |               |            |                  |                |
| bar_p40_h30_tp15_sl10 |           | [-1, 0, 1]    | 358774     | 45854            | 2191           |
| 08                    | 400326    |               |            |                  |                |
| bar_p40_h30_tp20_sl10 |           | [-1, 0, 1]    | 222218     | 52061            | 3494           |
| 57                    | 400326    |               |            |                  |                |
| bar_p40_h30_tp25_sl10 |           | [-1, 0, 1]    | 140906     | 55648            | 4271           |
| 82                    | 400326    |               |            |                  |                |
| bar_p40_h30_tp10_sl05 |           | [-1, 1]       | 492645     | 131091           |                |
| 0                     | 400326    |               |            |                  |                |
| bar_p40_h30_tp15_sl05 |           | [-1, 0, 1]    | 300268     | 156341           | 1671           |
| 27                    | 400326    |               |            |                  |                |
| bar_p40_h30_tp20_sl05 |           | [-1, 0, 1]    | 185138     | 168011           | 2705           |
| 87                    | 400326    |               |            |                  |                |
| bar_p50_h30_tp10_sl10 |           | [-1, 1]       | 504861     | 25121            |                |
| 0                     | 494080    |               |            |                  |                |
| bar_p50_h30_tp15_sl10 |           | [-1, 0, 1]    | 284054     | 30244            | 2156           |
| 84                    | 494080    |               |            |                  |                |
| bar_p50_h30_tp20_sl10 |           | [-1, 0, 1]    | 164678     | 34387            | 3309           |
| 17                    | 494080    |               |            |                  |                |
| bar_p50_h30_tp25_sl10 |           | [-1, 0, 1]    | 98753      | 36576            | 3946           |
| 53                    | 494080    |               |            |                  |                |
| bar_p50_h30_tp10_sl05 |           | [-1, 1]       | 434203     | 95779            |                |
| 0                     | 494080    |               |            |                  |                |
| bar_p50_h30_tp15_sl05 |           | [-1, 0, 1]    | 242389     | 114830           | 1727           |
| 63                    | 494080    |               |            |                  |                |
| bar_p50_h30_tp20_sl05 |           | [-1, 0, 1]    | 139429     | 122917           | 2676           |
| 36                    | 494080    |               |            |                  |                |
| bar_p60_h30_tp10_sl10 |           | [-1, 1]       | 419380     | 15615            |                |
| 0                     | 589067    |               |            |                  |                |

Resumen agregado de valores BAR:

Total TP primero (1): 15,857,316

Total SL primero (-1): 4,193,918

Total no\_event (0): 11,796,169

Total NaN / no operable: 32,668,503

Validación de valores permitidos:

OK - Todos los targets BAR contienen únicamente -1, 0, 1 o NaN.

Validación 4.5.2 finalizada correctamente.

## 4.5.3. Consistencia entre DIR y BAR

```
In [90]: # 4.5.3 - Consistencia entre DIR y BAR
# =====

print("4.5.3 Consistencia entre DIR y BAR")
print("=" * 80)

# -----
```

```

# Objetivo:
# Validar que BAR solo exista cuando DIR sea operable.
#
# Reglas esperadas:
# DIR = 1 -> BAR debe tener valor: 1, -1 o 0
# DIR = -1 -> BAR debe tener valor: 1, -1 o 0
# DIR = 0 -> BAR debe ser NaN
# DIR = NaN -> BAR debe ser NaN
# -----

dir_bar_consistency_rows = []
dir_bar_error_rows = []

# -----
# 1. Revisar cada target BAR contra su DIR fuente
# -----

for _, meta in df_bar_metadata.iterrows():
    dir_col = meta["dir_col"]
    bar_col = meta["bar_col"]

    if dir_col not in df_targets_bar.columns:
        raise KeyError(f"No existe el target DIR requerido: {dir_col}")

    if bar_col not in df_targets_bar.columns:
        raise KeyError(f"No existe el target BAR requerido: {bar_col}")

    dir_series = df_targets_bar[dir_col]
    bar_series = df_targets_bar[bar_col]

    # DIR operable: LONG o SHORT
    dir_operable_mask = dir_series.isin([1, -1])

    # BAR válido: cualquier valor no NaN
    bar_valid_mask = bar_series.notna()

    # Casos correctos esperados
    dir_operable_count = int(dir_operable_mask.sum())
    bar_valid_count = int(bar_valid_mask.sum())

    # Error 1:
    # BAR tiene valor, pero DIR no era operable
    bar_valid_when_dir_not_operable = bar_valid_mask & (~dir_operable_mask)

    # Error 2:
    # DIR era operable, pero BAR quedó NaN
    bar_nan_when_dir_operable = dir_operable_mask & (~bar_valid_mask)

    n_error_bar_valid_when_dir_not_operable = int(bar_valid_when_dir_not_operable.sum())
    n_error_bar_nan_when_dir_operable = int(bar_nan_when_dir_operable.sum())

    has_error = (
        n_error_bar_valid_when_dir_not_operable > 0
        or n_error_bar_nan_when_dir_operable > 0
    )

    dir_bar_consistency_rows.append(
        {
            "dir_col": dir_col,
            "bar_col": bar_col,

```

```

        "horizon": meta["horizon"],
        "percentile": meta["percentile"],
        "tp_mult": meta["tp_mult"],
        "sl_mult": meta["sl_mult"],
        "dir_operable_count": dir_operable_count,
        "bar_valid_count": bar_valid_count,
        "diff_bar_vs_dir_operable": bar_valid_count - dir_operable_count,
        "bar_valid_when_dir_not_operable": n_error_bar_valid_when_dir_not_op
        "bar_nan_when_dir_operable": n_error_bar_nan_when_dir_operable,
        "has_error": has_error,
    }
)

if has_error:
    dir_bar_error_rows.append(
        {
            "dir_col": dir_col,
            "bar_col": bar_col,
            "bar_valid_when_dir_not_operable": n_error_bar_valid_when_dir_no
            "bar_nan_when_dir_operable": n_error_bar_nan_when_dir_operable,
        }
    )

# -----
# 2. Crear resumen de consistencia
# -----

df_bar_dir_consistency = pd.DataFrame(dir_bar_consistency_rows)

n_targets_checked = len(df_bar_dir_consistency)
n_targets_with_errors = int(df_bar_dir_consistency["has_error"].sum())

print(f"Targets BAR revisados: {n_targets_checked}")
print(f"Targets BAR con errores de consistencia DIR/BAR: {n_targets_with_errors}")

# -----
# 3. Detener si existen errores
# -----

if n_targets_with_errors > 0:
    df_bar_dir_errors = pd.DataFrame(dir_bar_error_rows)

    print("\nErrores encontrados:")
    print(df_bar_dir_errors.to_string(index=False))

    raise ValueError(
        "Existen inconsistencias entre DIR y BAR. "
        "Revisar df_bar_dir_errors."
    )

# -----
# 4. Mostrar vista de validación
# -----

view_cols = [
    "dir_col",
    "bar_col",
    "horizon",
    "percentile",
    "tp_mult",

```

```

        "sl_mult",
        "dir_operable_count",
        "bar_valid_count",
        "diff_bar_vs_dir_operable",
    ]

    print("\nVista previa de consistencia DIR/BAR:")
    print(
        df_bar_dir_consistency[view_cols]
        .head(20)
        .to_string(index=False)
    )

    # -----
    # 5. Resumen agregado por DIR
    # -----

    df_bar_dir_consistency_by_dir = (
        df_bar_dir_consistency
        .groupby(["dir_col", "horizon", "percentile"], as_index=False)
        .agg(
            n_bar_variants=("bar_col", "count"),
            dir_operable_count=("dir_operable_count", "first"),
            min_bar_valid_count=("bar_valid_count", "min"),
            max_bar_valid_count=("bar_valid_count", "max"),
            max_abs_diff=("diff_bar_vs_dir_operable", lambda x: int(np.abs(x).max()))
        )
        .sort_values(["horizon", "percentile"])
        .reset_index(drop=True)
    )

    print("\nResumen por target DIR:")
    print(
        df_bar_dir_consistency_by_dir
        .to_string(index=False)
    )

    # -----
    # 6. Confirmación final
    # -----

    print("\nValidación de consistencia DIR/BAR:")
    print("OK - BAR es NaN cuando DIR = 0 o DIR = NaN.")
    print("OK - BAR solo tiene valor cuando DIR = 1 o DIR = -1.")
    print("OK - La cantidad de filas operables en DIR coincide con la cantidad de fi")
    print("OK - Todas las variantes TP/SL de un mismo DIR tienen la misma base opera")

    print("\nValidación 4.5.3 finalizada correctamente.")

```

## 4.5.3 Consistencia entre DIR y BAR

=====

Targets BAR revisados: 63

Targets BAR con errores de consistencia DIR/BAR: 0

Vista previa de consistencia DIR/BAR:

| dir_col     | bar_col               | horizon                  | percentile | tp_mult | sl_mult | dir_ope |
|-------------|-----------------------|--------------------------|------------|---------|---------|---------|
| table_count | bar_valid_count       | diff_bar_vs_dir_operable |            |         |         |         |
| dir_p40_h30 | bar_p40_h30_tp10_sl10 | 30                       | 40         | 1.0     | 1.0     |         |
| 623736      | 623736                |                          | 0          |         |         |         |
| dir_p40_h30 | bar_p40_h30_tp15_sl10 | 30                       | 40         | 1.5     | 1.0     |         |
| 623736      | 623736                |                          | 0          |         |         |         |
| dir_p40_h30 | bar_p40_h30_tp20_sl10 | 30                       | 40         | 2.0     | 1.0     |         |
| 623736      | 623736                |                          | 0          |         |         |         |
| dir_p40_h30 | bar_p40_h30_tp25_sl10 | 30                       | 40         | 2.5     | 1.0     |         |
| 623736      | 623736                |                          | 0          |         |         |         |
| dir_p40_h30 | bar_p40_h30_tp10_sl05 | 30                       | 40         | 1.0     | 0.5     |         |
| 623736      | 623736                |                          | 0          |         |         |         |
| dir_p40_h30 | bar_p40_h30_tp15_sl05 | 30                       | 40         | 1.5     | 0.5     |         |
| 623736      | 623736                |                          | 0          |         |         |         |
| dir_p40_h30 | bar_p40_h30_tp20_sl05 | 30                       | 40         | 2.0     | 0.5     |         |
| 623736      | 623736                |                          | 0          |         |         |         |
| dir_p50_h30 | bar_p50_h30_tp10_sl10 | 30                       | 50         | 1.0     | 1.0     |         |
| 529982      | 529982                |                          | 0          |         |         |         |
| dir_p50_h30 | bar_p50_h30_tp15_sl10 | 30                       | 50         | 1.5     | 1.0     |         |
| 529982      | 529982                |                          | 0          |         |         |         |
| dir_p50_h30 | bar_p50_h30_tp20_sl10 | 30                       | 50         | 2.0     | 1.0     |         |
| 529982      | 529982                |                          | 0          |         |         |         |
| dir_p50_h30 | bar_p50_h30_tp25_sl10 | 30                       | 50         | 2.5     | 1.0     |         |
| 529982      | 529982                |                          | 0          |         |         |         |
| dir_p50_h30 | bar_p50_h30_tp10_sl05 | 30                       | 50         | 1.0     | 0.5     |         |
| 529982      | 529982                |                          | 0          |         |         |         |
| dir_p50_h30 | bar_p50_h30_tp15_sl05 | 30                       | 50         | 1.5     | 0.5     |         |
| 529982      | 529982                |                          | 0          |         |         |         |
| dir_p50_h30 | bar_p50_h30_tp20_sl05 | 30                       | 50         | 2.0     | 0.5     |         |
| 529982      | 529982                |                          | 0          |         |         |         |
| dir_p60_h30 | bar_p60_h30_tp10_sl10 | 30                       | 60         | 1.0     | 1.0     |         |
| 434995      | 434995                |                          | 0          |         |         |         |
| dir_p60_h30 | bar_p60_h30_tp15_sl10 | 30                       | 60         | 1.5     | 1.0     |         |
| 434995      | 434995                |                          | 0          |         |         |         |
| dir_p60_h30 | bar_p60_h30_tp20_sl10 | 30                       | 60         | 2.0     | 1.0     |         |
| 434995      | 434995                |                          | 0          |         |         |         |
| dir_p60_h30 | bar_p60_h30_tp25_sl10 | 30                       | 60         | 2.5     | 1.0     |         |
| 434995      | 434995                |                          | 0          |         |         |         |
| dir_p60_h30 | bar_p60_h30_tp10_sl05 | 30                       | 60         | 1.0     | 0.5     |         |
| 434995      | 434995                |                          | 0          |         |         |         |
| dir_p60_h30 | bar_p60_h30_tp15_sl05 | 30                       | 60         | 1.5     | 0.5     |         |
| 434995      | 434995                |                          | 0          |         |         |         |

Resumen por target DIR:

| dir_col     | horizon             | percentile   | n_bar_variants | dir_operable_count | min_bar_val |
|-------------|---------------------|--------------|----------------|--------------------|-------------|
| id_count    | max_bar_valid_count | max_abs_diff |                |                    |             |
| dir_p40_h30 | 30                  | 40           | 7              | 623736             |             |
| 623736      | 623736              | 0            |                |                    |             |
| dir_p50_h30 | 30                  | 50           | 7              | 529982             |             |
| 529982      | 529982              | 0            |                |                    |             |
| dir_p60_h30 | 30                  | 60           | 7              | 434995             |             |
| 434995      | 434995              | 0            |                |                    |             |
| dir_p40_h60 | 60                  | 40           | 7              | 597024             |             |
| 597024      | 597024              | 0            |                |                    |             |

|             |        |    |   |        |
|-------------|--------|----|---|--------|
| dir_p50_h60 | 60     | 50 | 7 | 505020 |
| 505020      | 505020 | 0  |   |        |
| dir_p60_h60 | 60     | 60 | 7 | 413798 |
| 413798      | 413798 | 0  |   |        |
| dir_p40_h90 | 90     | 40 | 7 | 567970 |
| 567970      | 567970 | 0  |   |        |
| dir_p50_h90 | 90     | 50 | 7 | 482603 |
| 482603      | 482603 | 0  |   |        |
| dir_p60_h90 | 90     | 60 | 7 | 394501 |
| 394501      | 394501 | 0  |   |        |

Validación de consistencia DIR/BAR:

OK - BAR es NaN cuando DIR = 0 o DIR = NaN.

OK - BAR solo tiene valor cuando DIR = 1 o DIR = -1.

OK - La cantidad de filas operables en DIR coincide con la cantidad de filas válidas en BAR.

OK - Todas las variantes TP/SL de un mismo DIR tienen la misma base operable.

Validación 4.5.3 finalizada correctamente.

#### 4.5.4. Consistencia de event\_step

Esta validación revisa que la columna `event_step` tenga sentido:

BAR = 1 → debe existir event\_step

BAR = -1 → debe existir event\_step

BAR = 0 → event\_step debe ser NaN

BAR = NaN → event\_step debe ser NaN

Además, `event_step` debe estar entre 1 y el horizonte H.

```
In [91]: # 4.5.4 - Consistencia de event_step
# =====

print("4.5.4 Consistencia de event_step - BAR")
print("=" * 80)

# -----
# Objetivo:
# Validar que event_step sea coherente con BAR.
#
# Reglas esperadas:
# BAR = 1 -> event_step debe existir
# BAR = -1 -> event_step debe existir
# BAR = 0 -> event_step debe ser NaN
# BAR = NaN -> event_step debe ser NaN
#
# Además:
# event_step debe estar entre 1 y horizon
# -----

event_step_validation_rows = []
event_step_error_rows = []

# -----
# 1. Revisar cada target BAR y su event_step
# -----

for _, meta in df_bar_metadata.iterrows():
```



```

bar_col = meta["bar_col"]
event_step_col = meta["event_step_col"]
horizon = int(meta["horizon"])

if bar_col not in df_targets_bar.columns:
    raise KeyError(f"No existe el target BAR requerido: {bar_col}")

if event_step_col not in df_targets_bar.columns:
    raise KeyError(f"No existe la columna event_step requerida: {event_step_col}")

bar_series = df_targets_bar[bar_col]
step_series = df_targets_bar[event_step_col]

# Máscaras principales
bar_tp_or_sl_mask = bar_series.isin([1, -1])
bar_no_event_mask = bar_series == 0
bar_nan_mask = bar_series.isna()

step_valid_mask = step_series.notna()

# -----
# Errores posibles
# -----

# Error 1:
# Hay TP o SL, pero event_step está NaN
missing_step_when_event = bar_tp_or_sl_mask & (~step_valid_mask)

# Error 2:
# BAR = 0, pero event_step tiene valor
step_when_no_event = bar_no_event_mask & step_valid_mask

# Error 3:
# BAR = NaN, pero event_step tiene valor
step_when_bar_nan = bar_nan_mask & step_valid_mask

# Error 4:
# event_step menor a 1
step_less_than_1 = step_valid_mask & (step_series.astype(float) < 1)

# Error 5:
# event_step mayor al horizonte
step_greater_than_horizon = step_valid_mask & (step_series.astype(float) > h

n_missing_step_when_event = int(missing_step_when_event.sum())
n_step_when_no_event = int(step_when_no_event.sum())
n_step_when_bar_nan = int(step_when_bar_nan.sum())
n_step_less_than_1 = int(step_less_than_1.sum())
n_step_greater_than_horizon = int(step_greater_than_horizon.sum())

has_error = (
    n_missing_step_when_event > 0
    or n_step_when_no_event > 0
    or n_step_when_bar_nan > 0
    or n_step_less_than_1 > 0
    or n_step_greater_than_horizon > 0
)

# Estadísticas de event_step
if step_valid_mask.sum() > 0:

```

```

step_values = step_series.loc[step_valid_mask].astype(float)

step_min = step_values.min()
step_max = step_values.max()
step_mean = step_values.mean()
step_median = step_values.median()
else:
    step_min = np.nan
    step_max = np.nan
    step_mean = np.nan
    step_median = np.nan

event_step_validation_rows.append(
    {
        "bar_col": bar_col,
        "event_step_col": event_step_col,
        "horizon": horizon,
        "percentile": meta["percentile"],
        "tp_mult": meta["tp_mult"],
        "sl_mult": meta["sl_mult"],

        "bar_event_count": int(bar_tp_or_sl_mask.sum()),
        "event_step_valid_count": int(step_valid_mask.sum()),

        "step_min": step_min,
        "step_max": step_max,
        "step_mean": step_mean,
        "step_median": step_median,

        "missing_step_when_event": n_missing_step_when_event,
        "step_when_no_event": n_step_when_no_event,
        "step_when_bar_nan": n_step_when_bar_nan,
        "step_less_than_1": n_step_less_than_1,
        "step_greater_than_horizon": n_step_greater_than_horizon,

        "has_error": has_error,
    }
)

if has_error:
    event_step_error_rows.append(
        {
            "bar_col": bar_col,
            "event_step_col": event_step_col,
            "horizon": horizon,
            "missing_step_when_event": n_missing_step_when_event,
            "step_when_no_event": n_step_when_no_event,
            "step_when_bar_nan": n_step_when_bar_nan,
            "step_less_than_1": n_step_less_than_1,
            "step_greater_than_horizon": n_step_greater_than_horizon,
        }
    )

# -----
# 2. Crear resumen de validación
# -----

df_bar_event_step_validation = pd.DataFrame(event_step_validation_rows)

n_targets_checked = len(df_bar_event_step_validation)

```

```

n_targets_with_errors = int(df_bar_event_step_validation["has_error"].sum())

print(f"Targets BAR revisados: {n_targets_checked}")
print(f"Targets BAR con errores en event_step: {n_targets_with_errors}")

# -----
# 3. Detener si existen errores
# -----

if n_targets_with_errors > 0:
    df_bar_event_step_errors = pd.DataFrame(event_step_error_rows)

    print("\nErrores encontrados:")
    print(df_bar_event_step_errors.to_string(index=False))

    raise ValueError(
        "Existen inconsistencias entre BAR y event_step. "
        "Revisar df_bar_event_step_errors."
    )

# -----
# 4. Vista previa de validación
# -----

view_cols = [
    "bar_col",
    "horizon",
    "percentile",
    "tp_mult",
    "sl_mult",
    "bar_event_count",
    "event_step_valid_count",
    "step_min",
    "step_max",
    "step_mean",
    "step_median",
]

print("\nVista previa de consistencia event_step:")
print(
    df_bar_event_step_validation[view_cols]
    .head(20)
    .to_string(
        index=False,
        formatters={
            "step_min": "{:.0f}".format,
            "step_max": "{:.0f}".format,
            "step_mean": "{:.2f}".format,
            "step_median": "{:.2f}".format,
        }
    )
)

# -----
# 5. Resumen agregado por horizonte y configuración TP/SL
# -----

df_bar_event_step_by_config = (
    df_bar_event_step_validation
    .groupby(["horizon", "tp_mult", "sl_mult"], as_index=False)

```

```
.agg(  
    n_targets=("bar_col", "count"),  
    event_step_mean=("step_mean", "mean"),  
    event_step_median=("step_median", "mean"),  
    min_step_observed=("step_min", "min"),  
    max_step_observed=("step_max", "max"),  
)  
.sort_values(["horizon", "sl_mult", "tp_mult"])  
.reset_index(drop=True)  
)  
  
print("\nResumen de event_step por horizonte y configuración TP/SL:")  
print(  
    df_bar_event_step_by_config.to_string(  
        index=False,  
        formatters={  
            "event_step_mean": "{:.2f}".format,  
            "event_step_median": "{:.2f}".format,  
            "min_step_observed": "{:.0f}".format,  
            "max_step_observed": "{:.0f}".format,  
        }  
    )  
)  
  
# -----  
# 6. Confirmación final  
# -----  
  
print("\nValidación de event_step:")  
print("OK - event_step existe cuando BAR = 1 o BAR = -1.")  
print("OK - event_step es NaN cuando BAR = 0.")  
print("OK - event_step es NaN cuando BAR = NaN.")  
print("OK - event_step siempre está entre 1 y H.")  
  
print("\nValidación 4.5.4 finalizada correctamente.")
```

## 4.5.4 Consistencia de event\_step - BAR

=====

Targets BAR revisados: 63

Targets BAR con errores en event\_step: 0

Vista previa de consistencia event\_step:

|                       | bar_col  | horizon  | percentile | tp_mult     | sl_mult | bar_event_count | ev |
|-----------------------|----------|----------|------------|-------------|---------|-----------------|----|
| ent_step_valid_count  | step_min | step_max | step_mean  | step_median |         |                 |    |
| bar_p40_h30_tp10_sl10 | 30       | 40       | 1.0        | 1.0         | 623736  |                 |    |
| 623736                | 1        | 30       | 13.20      | 12.00       |         |                 |    |
| bar_p40_h30_tp15_sl10 | 30       | 40       | 1.5        | 1.0         | 404628  |                 |    |
| 404628                | 1        | 30       | 14.83      | 14.00       |         |                 |    |
| bar_p40_h30_tp20_sl10 | 30       | 40       | 2.0        | 1.0         | 274279  |                 |    |
| 274279                | 1        | 30       | 15.66      | 15.00       |         |                 |    |
| bar_p40_h30_tp25_sl10 | 30       | 40       | 2.5        | 1.0         | 196554  |                 |    |
| 196554                | 1        | 30       | 15.97      | 16.00       |         |                 |    |
| bar_p40_h30_tp10_sl05 | 30       | 40       | 1.0        | 0.5         | 623736  |                 |    |
| 623736                | 1        | 30       | 11.18      | 9.00        |         |                 |    |
| bar_p40_h30_tp15_sl05 | 30       | 40       | 1.5        | 0.5         | 456609  |                 |    |
| 456609                | 1        | 30       | 12.46      | 11.00       |         |                 |    |
| bar_p40_h30_tp20_sl05 | 30       | 40       | 2.0        | 0.5         | 353149  |                 |    |
| 353149                | 1        | 30       | 12.70      | 11.00       |         |                 |    |
| bar_p50_h30_tp10_sl10 | 30       | 50       | 1.0        | 1.0         | 529982  |                 |    |
| 529982                | 1        | 30       | 14.21      | 13.00       |         |                 |    |
| bar_p50_h30_tp15_sl10 | 30       | 50       | 1.5        | 1.0         | 314298  |                 |    |
| 314298                | 1        | 30       | 15.74      | 15.00       |         |                 |    |
| bar_p50_h30_tp20_sl10 | 30       | 50       | 2.0        | 1.0         | 199065  |                 |    |
| 199065                | 1        | 30       | 16.50      | 17.00       |         |                 |    |
| bar_p50_h30_tp25_sl10 | 30       | 50       | 2.5        | 1.0         | 135329  |                 |    |
| 135329                | 1        | 30       | 16.70      | 17.00       |         |                 |    |
| bar_p50_h30_tp10_sl05 | 30       | 50       | 1.0        | 0.5         | 529982  |                 |    |
| 529982                | 1        | 30       | 12.37      | 11.00       |         |                 |    |
| bar_p50_h30_tp15_sl05 | 30       | 50       | 1.5        | 0.5         | 357219  |                 |    |
| 357219                | 1        | 30       | 13.40      | 12.00       |         |                 |    |
| bar_p50_h30_tp20_sl05 | 30       | 50       | 2.0        | 0.5         | 262346  |                 |    |
| 262346                | 1        | 30       | 13.39      | 12.00       |         |                 |    |
| bar_p60_h30_tp10_sl10 | 30       | 60       | 1.0        | 1.0         | 434995  |                 |    |
| 434995                | 1        | 30       | 15.14      | 15.00       |         |                 |    |
| bar_p60_h30_tp15_sl10 | 30       | 60       | 1.5        | 1.0         | 234426  |                 |    |
| 234426                | 1        | 30       | 16.62      | 17.00       |         |                 |    |
| bar_p60_h30_tp20_sl10 | 30       | 60       | 2.0        | 1.0         | 137975  |                 |    |
| 137975                | 1        | 30       | 17.26      | 18.00       |         |                 |    |
| bar_p60_h30_tp25_sl10 | 30       | 60       | 2.5        | 1.0         | 88176   |                 |    |
| 88176                 | 1        | 30       | 17.32      | 18.00       |         |                 |    |
| bar_p60_h30_tp10_sl05 | 30       | 60       | 1.0        | 0.5         | 434995  |                 |    |
| 434995                | 1        | 30       | 13.51      | 12.00       |         |                 |    |
| bar_p60_h30_tp15_sl05 | 30       | 60       | 1.5        | 0.5         | 267487  |                 |    |
| 267487                | 1        | 30       | 14.34      | 14.00       |         |                 |    |

Resumen de event\_step por horizonte y configuración TP/SL:

| horizon | tp_mult | sl_mult | n_targets | event_step_mean | event_step_median | min_step_observed | max_step_observed |
|---------|---------|---------|-----------|-----------------|-------------------|-------------------|-------------------|
| 30      | 1.0     | 0.5     | 3         | 12.35           | 10.67             |                   |                   |
| 1       | 30      |         |           |                 |                   |                   |                   |
| 30      | 1.5     | 0.5     | 3         | 13.40           | 12.33             |                   |                   |
| 1       | 30      |         |           |                 |                   |                   |                   |
| 30      | 2.0     | 0.5     | 3         | 13.39           | 12.00             |                   |                   |
| 1       | 30      |         |           |                 |                   |                   |                   |
| 30      | 1.0     | 1.0     | 3         | 14.18           | 13.33             |                   |                   |
| 1       | 30      |         |           |                 |                   |                   |                   |

|   |    |     |     |   |       |       |
|---|----|-----|-----|---|-------|-------|
|   | 30 | 1.5 | 1.0 | 3 | 15.73 | 15.33 |
| 1 |    | 30  |     |   |       |       |
|   | 30 | 2.0 | 1.0 | 3 | 16.47 | 16.67 |
| 1 |    | 30  |     |   |       |       |
|   | 30 | 2.5 | 1.0 | 3 | 16.66 | 17.00 |
| 1 |    | 30  |     |   |       |       |
|   | 60 | 1.0 | 0.5 | 3 | 24.32 | 21.00 |
| 1 |    | 60  |     |   |       |       |
|   | 60 | 1.5 | 0.5 | 3 | 26.22 | 24.00 |
| 1 |    | 60  |     |   |       |       |
|   | 60 | 2.0 | 0.5 | 3 | 25.88 | 23.00 |
| 1 |    | 60  |     |   |       |       |
|   | 60 | 1.0 | 1.0 | 3 | 28.15 | 26.33 |
| 1 |    | 60  |     |   |       |       |
|   | 60 | 1.5 | 1.0 | 3 | 31.22 | 30.67 |
| 1 |    | 60  |     |   |       |       |
|   | 60 | 2.0 | 1.0 | 3 | 32.62 | 32.67 |
| 1 |    | 60  |     |   |       |       |
|   | 60 | 2.5 | 1.0 | 3 | 32.81 | 33.33 |
| 1 |    | 60  |     |   |       |       |
|   | 90 | 1.0 | 0.5 | 3 | 36.55 | 32.00 |
| 1 |    | 90  |     |   |       |       |
|   | 90 | 1.5 | 0.5 | 3 | 39.67 | 36.33 |
| 1 |    | 90  |     |   |       |       |
|   | 90 | 2.0 | 0.5 | 3 | 39.33 | 35.33 |
| 1 |    | 90  |     |   |       |       |
|   | 90 | 1.0 | 1.0 | 3 | 42.63 | 40.00 |
| 1 |    | 90  |     |   |       |       |
|   | 90 | 1.5 | 1.0 | 3 | 47.73 | 47.33 |
| 1 |    | 90  |     |   |       |       |
|   | 90 | 2.0 | 1.0 | 3 | 50.14 | 51.00 |
| 1 |    | 90  |     |   |       |       |
|   | 90 | 2.5 | 1.0 | 3 | 50.71 | 52.67 |
| 1 |    | 90  |     |   |       |       |

Validación de event\_step:

OK - event\_step existe cuando BAR = 1 o BAR = -1.

OK - event\_step es NaN cuando BAR = 0.

OK - event\_step es NaN cuando BAR = NaN.

OK - event\_step siempre está entre 1 y H.

Validación 4.5.4 finalizada correctamente.

### 4.5.5. Validación de ventanas futuras y no cruce de días

```
In [92]: # 4.5.5 – Validación de ventanas futuras y no cruce de días
# =====

print("4.5.5 Validación de ventanas futuras y no cruce de días - BAR")
print("=" * 80)

# -----
# Objetivo:
# Validar que BAR solo tenga valores cuando existe una ventana
# futura completa dentro del mismo trading day.
#
# Reglas esperadas:
# - Si una fila no tiene al menos H barras futuras dentro del día,
#   BAR debe ser NaN.
```

```

# - Si BAR tiene valor, entonces debe existir ventana futura completa.
# - event_step no puede superar la cantidad de barras futuras disponibles.
# -----

# -----
# 1. Calcular posición intradía y barras futuras disponibles
# -----

df_bar_temporal_check = df_targets_bar[[DATE_COL]].copy()

df_bar_temporal_check["_bar_position_in_day"] = (
    df_bar_temporal_check
    .groupby(DATE_COL)
    .cumcount()
)

df_bar_temporal_check["_bars_in_day"] = (
    df_bar_temporal_check
    .groupby(DATE_COL)[DATE_COL]
    .transform("size")
)

df_bar_temporal_check["_future_bars_available"] = (
    df_bar_temporal_check["_bars_in_day"]
    - df_bar_temporal_check["_bar_position_in_day"]
    - 1
)

# -----
# 2. Validar cada target BAR
# -----

temporal_validation_rows = []
temporal_error_rows = []

for _, meta in df_bar_metadata.iterrows():
    bar_col = meta["bar_col"]
    event_step_col = meta["event_step_col"]
    horizon = int(meta["horizon"])

    bar_series = df_targets_bar[bar_col]
    step_series = df_targets_bar[event_step_col]

    future_bars_available = df_bar_temporal_check["_future_bars_available"]

    full_window_mask = future_bars_available >= horizon
    insufficient_window_mask = ~full_window_mask

    bar_valid_mask = bar_series.notna()
    step_valid_mask = step_series.notna()

    # Error 1:
    # BAR tiene valor aunque no existe ventana futura completa
    bar_valid_without_full_window = bar_valid_mask & insufficient_window_mask

    # Error 2:
    # event_step tiene valor aunque no existe ventana futura completa
    step_valid_without_full_window = step_valid_mask & insufficient_window_mask

    # Error 3:

```

```

# event_step supera las barras futuras disponibles
step_greater_than_available = (
    step_valid_mask
    & (
        step_series.astype(float)
        > future_bars_available.astype(float)
    )
)

n_bar_valid_without_full_window = int(bar_valid_without_full_window.sum())
n_step_valid_without_full_window = int(step_valid_without_full_window.sum())
n_step_greater_than_available = int(step_greater_than_available.sum())

total_rows = len(df_targets_bar)
full_window_rows = int(full_window_mask.sum())
insufficient_window_rows = int(insufficient_window_mask.sum())

bar_valid_rows = int(bar_valid_mask.sum())
bar_nan_insufficient_window = int(
    bar_series.loc[insufficient_window_mask].isna().sum()
)

has_error = (
    n_bar_valid_without_full_window > 0
    or n_step_valid_without_full_window > 0
    or n_step_greater_than_available > 0
)

temporal_validation_rows.append(
    {
        "bar_col": bar_col,
        "event_step_col": event_step_col,
        "horizon": horizon,
        "percentile": meta["percentile"],
        "tp_mult": meta["tp_mult"],
        "sl_mult": meta["sl_mult"],

        "total_rows": total_rows,
        "full_window_rows": full_window_rows,
        "insufficient_window_rows": insufficient_window_rows,
        "bar_valid_rows": bar_valid_rows,
        "bar_nan_insufficient_window": bar_nan_insufficient_window,

        "bar_valid_without_full_window": n_bar_valid_without_full_window,
        "step_valid_without_full_window": n_step_valid_without_full_window,
        "step_greater_than_available": n_step_greater_than_available,

        "has_error": has_error,
    }
)

if has_error:
    temporal_error_rows.append(
        {
            "bar_col": bar_col,
            "event_step_col": event_step_col,
            "horizon": horizon,
            "bar_valid_without_full_window": n_bar_valid_without_full_window,
            "step_valid_without_full_window": n_step_valid_without_full_window,
            "step_greater_than_available": n_step_greater_than_available,
        }
    )

```



```

    }
)

# -----
# 3. Crear resumen de validación
# -----

df_bar_temporal_validation = pd.DataFrame(temporal_validation_rows)

n_targets_checked = len(df_bar_temporal_validation)
n_targets_with_errors = int(df_bar_temporal_validation["has_error"].sum())

print(f"Targets BAR revisados: {n_targets_checked}")
print(f"Targets BAR con errores temporales: {n_targets_with_errors}")

# -----
# 4. Detener si existen errores
# -----

if n_targets_with_errors > 0:
    df_bar_temporal_errors = pd.DataFrame(temporal_error_rows)

    print("\nErrores encontrados:")
    print(df_bar_temporal_errors.to_string(index=False))

    raise ValueError(
        "Existen inconsistencias temporales en BAR. "
        "Revisar df_bar_temporal_errors."
    )

# -----
# 5. Vista previa de validación temporal
# -----

view_cols = [
    "bar_col",
    "horizon",
    "percentile",
    "tp_mult",
    "sl_mult",
    "full_window_rows",
    "insufficient_window_rows",
    "bar_valid_rows",
    "bar_nan_insufficient_window",
]

print("\nVista previa de validación temporal:")
print(
    df_bar_temporal_validation[view_cols]
    .head(20)
    .to_string(index=False)
)

# -----
# 6. Resumen por horizonte
# -----

df_bar_temporal_by_horizon = (
    df_bar_temporal_validation
    .groupby("horizon", as_index=False)

```

```
.agg(  
    n_targets=("bar_col", "count"),  
    full_window_rows=("full_window_rows", "first"),  
    insufficient_window_rows=("insufficient_window_rows", "first"),  
    min_bar_valid_rows=("bar_valid_rows", "min"),  
    max_bar_valid_rows=("bar_valid_rows", "max"),  
)  
.sort_values("horizon")  
.reset_index(drop=True)  
)  
  
print("\nResumen temporal por horizonte:")  
print(df_bar_temporal_by_horizon.to_string(index=False))  
  
# -----  
# 7. Confirmación final  
# -----  
  
print("\nValidación temporal de BAR:")  
print("OK - BAR no tiene valores cuando no existe ventana futura completa.")  
print("OK - event_step no aparece fuera de ventanas futuras válidas.")  
print("OK - event_step no supera la cantidad de barras futuras disponibles.")  
print("OK - La construcción respeta la regla de no cruzar trading days.")  
  
print("\nValidación 4.5.5 finalizada correctamente.")
```

## 4.5.5 Validación de ventanas futuras y no cruce de días - BAR

=====

Targets BAR revisados: 63

Targets BAR con errores temporales: 0

Vista previa de validación temporal:

| bar_col                 | horizon        | percentile                  | tp_mult | sl_mult | full_window_rows | i |
|-------------------------|----------------|-----------------------------|---------|---------|------------------|---|
| nsufficient_window_rows | bar_valid_rows | bar_nan_insufficient_window |         |         |                  |   |
| bar_p40_h30_tp10_sl10   | 30             | 40                          | 1.0     | 1.0     | 979602           |   |
| 44460                   | 623736         | 44460                       |         |         |                  |   |
| bar_p40_h30_tp15_sl10   | 30             | 40                          | 1.5     | 1.0     | 979602           |   |
| 44460                   | 623736         | 44460                       |         |         |                  |   |
| bar_p40_h30_tp20_sl10   | 30             | 40                          | 2.0     | 1.0     | 979602           |   |
| 44460                   | 623736         | 44460                       |         |         |                  |   |
| bar_p40_h30_tp25_sl10   | 30             | 40                          | 2.5     | 1.0     | 979602           |   |
| 44460                   | 623736         | 44460                       |         |         |                  |   |
| bar_p40_h30_tp10_sl05   | 30             | 40                          | 1.0     | 0.5     | 979602           |   |
| 44460                   | 623736         | 44460                       |         |         |                  |   |
| bar_p40_h30_tp15_sl05   | 30             | 40                          | 1.5     | 0.5     | 979602           |   |
| 44460                   | 623736         | 44460                       |         |         |                  |   |
| bar_p40_h30_tp20_sl05   | 30             | 40                          | 2.0     | 0.5     | 979602           |   |
| 44460                   | 623736         | 44460                       |         |         |                  |   |
| bar_p50_h30_tp10_sl10   | 30             | 50                          | 1.0     | 1.0     | 979602           |   |
| 44460                   | 529982         | 44460                       |         |         |                  |   |
| bar_p50_h30_tp15_sl10   | 30             | 50                          | 1.5     | 1.0     | 979602           |   |
| 44460                   | 529982         | 44460                       |         |         |                  |   |
| bar_p50_h30_tp20_sl10   | 30             | 50                          | 2.0     | 1.0     | 979602           |   |
| 44460                   | 529982         | 44460                       |         |         |                  |   |
| bar_p50_h30_tp25_sl10   | 30             | 50                          | 2.5     | 1.0     | 979602           |   |
| 44460                   | 529982         | 44460                       |         |         |                  |   |
| bar_p50_h30_tp10_sl05   | 30             | 50                          | 1.0     | 0.5     | 979602           |   |
| 44460                   | 529982         | 44460                       |         |         |                  |   |
| bar_p50_h30_tp15_sl05   | 30             | 50                          | 1.5     | 0.5     | 979602           |   |
| 44460                   | 529982         | 44460                       |         |         |                  |   |
| bar_p50_h30_tp20_sl05   | 30             | 50                          | 2.0     | 0.5     | 979602           |   |
| 44460                   | 529982         | 44460                       |         |         |                  |   |
| bar_p60_h30_tp10_sl10   | 30             | 60                          | 1.0     | 1.0     | 979602           |   |
| 44460                   | 434995         | 44460                       |         |         |                  |   |
| bar_p60_h30_tp15_sl10   | 30             | 60                          | 1.5     | 1.0     | 979602           |   |
| 44460                   | 434995         | 44460                       |         |         |                  |   |
| bar_p60_h30_tp20_sl10   | 30             | 60                          | 2.0     | 1.0     | 979602           |   |
| 44460                   | 434995         | 44460                       |         |         |                  |   |
| bar_p60_h30_tp25_sl10   | 30             | 60                          | 2.5     | 1.0     | 979602           |   |
| 44460                   | 434995         | 44460                       |         |         |                  |   |
| bar_p60_h30_tp10_sl05   | 30             | 60                          | 1.0     | 0.5     | 979602           |   |
| 44460                   | 434995         | 44460                       |         |         |                  |   |
| bar_p60_h30_tp15_sl05   | 30             | 60                          | 1.5     | 0.5     | 979602           |   |
| 44460                   | 434995         | 44460                       |         |         |                  |   |

Resumen temporal por horizonte:

| horizon | n_targets | full_window_rows | insufficient_window_rows | min_bar_valid_rows |
|---------|-----------|------------------|--------------------------|--------------------|
| 30      | 21        | 979602           | 44460                    | 4349               |
| 95      | 623736    |                  |                          |                    |
| 60      | 21        | 935142           | 88920                    | 4137               |
| 98      | 597024    |                  |                          |                    |
| 90      | 21        | 890682           | 133380                   | 3945               |
| 01      | 567970    |                  |                          |                    |

Validación temporal de BAR:

OK - BAR no tiene valores cuando no existe ventana futura completa.  
 OK - event\_step no aparece fuera de ventanas futuras válidas.  
 OK - event\_step no supera la cantidad de barras futuras disponibles.  
 OK - La construcción respeta la regla de no cruzar trading days.

Validación 4.5.5 finalizada correctamente.

## 4.5.6. Validación específica de TP base

```
In [93]: # 4.5.6 – Validación específica de TP base
# =====

print("4.5.6 Validación específica de TP base - BAR")
print("=" * 80)

# -----
# Objetivo:
# Validar que cuando  $TP = 1.0 \times threshold$ , no existan casos
# no_event entre las filas operables.
#
# Motivo:
# Si  $DIR = 1$  o  $DIR = -1$ , significa que dentro del horizonte
# el precio alcanzó una excursión favorable  $\geq threshold$ .
#
# Por lo tanto, si  $TP = 1.0 \times threshold$ , el TP debería ser
# alcanzado en algún momento de la ventana futura.
#
# Entonces:
# BAR debe resolver como TP primero o SL primero.
# No debería quedar  $BAR = 0$ .
# -----

tp_base_validation_rows = []
tp_base_error_rows = []

# -----
# 1. Filtrar metadata de configuraciones con TP base
# -----

df_tp_base_metadata = df_bar_metadata[
    df_bar_metadata["tp_mult"] == 1.0
].copy()

print(f"Targets BAR con TP base encontrados: {len(df_tp_base_metadata)}")

expected_tp_base_targets = len(BAR_HORIZONS) * len(BAR_PERCENTILES) * 2
print(f"Targets BAR con TP base esperados: {expected_tp_base_targets}")

if len(df_tp_base_metadata) != expected_tp_base_targets:
    raise ValueError(
        "La cantidad de targets BAR con TP base no coincide con lo esperado."
    )

# -----
# 2. Validar cada target con  $TP = 1.0 \times threshold$ 
# -----

for _, meta in df_tp_base_metadata.iterrows():
    dir_col = meta["dir_col"]
```

```

bar_col = meta["bar_col"]
event_step_col = meta["event_step_col"]

horizon = int(meta["horizon"])
percentile = int(meta["percentile"])
tp_mult = float(meta["tp_mult"])
sl_mult = float(meta["sl_mult"])

dir_series = df_targets_bar[dir_col]
bar_series = df_targets_bar[bar_col]
step_series = df_targets_bar[event_step_col]

dir_operable_mask = dir_series.isin([1, -1])
bar_valid_mask = bar_series.notna()

no_event_mask = bar_series == 0
tp_mask = bar_series == 1
sl_mask = bar_series == -1

operable_count = int(dir_operable_mask.sum())
bar_valid_count = int(bar_valid_mask.sum())

tp_count = int(tp_mask.sum())
sl_count = int(sl_mask.sum())
no_event_count = int(no_event_mask.sum())

resolved_count = tp_count + sl_count

# Errores esperados para TP base
error_no_event = no_event_count > 0
error_not_fully_resolved = resolved_count != bar_valid_count
error_bar_valid_vs_operable = bar_valid_count != operable_count

has_error = (
    error_no_event
    or error_not_fully_resolved
    or error_bar_valid_vs_operable
)

tp_base_validation_rows.append(
    {
        "dir_col": dir_col,
        "bar_col": bar_col,
        "horizon": horizon,
        "percentile": percentile,
        "tp_mult": tp_mult,
        "sl_mult": sl_mult,

        "operable_count": operable_count,
        "bar_valid_count": bar_valid_count,
        "tp_count": tp_count,
        "sl_count": sl_count,
        "resolved_count": resolved_count,
        "no_event_count": no_event_count,

        "tp_rate": tp_count / bar_valid_count if bar_valid_count > 0 else np
        "sl_rate": sl_count / bar_valid_count if bar_valid_count > 0 else np
        "no_event_rate": no_event_count / bar_valid_count if bar_valid_count

        "error_no_event": error_no_event,

```

```

        "error_not_fully_resolved": error_not_fully_resolved,
        "error_bar_valid_vs_operable": error_bar_valid_vs_operable,
        "has_error": has_error,
    }
)

if has_error:
    tp_base_error_rows.append(
        {
            "dir_col": dir_col,
            "bar_col": bar_col,
            "horizon": horizon,
            "percentile": percentile,
            "tp_mult": tp_mult,
            "sl_mult": sl_mult,
            "operable_count": operable_count,
            "bar_valid_count": bar_valid_count,
            "tp_count": tp_count,
            "sl_count": sl_count,
            "resolved_count": resolved_count,
            "no_event_count": no_event_count,
        }
    )

# -----
# 3. Crear resumen de validación
# -----

df_bar_tp_base_validation = pd.DataFrame(tp_base_validation_rows)

n_targets_checked = len(df_bar_tp_base_validation)
n_targets_with_errors = int(df_bar_tp_base_validation["has_error"].sum())

print(f"\nTargets TP base revisados: {n_targets_checked}")
print(f"Targets TP base con errores: {n_targets_with_errors}")

# -----
# 4. Detener si existen errores
# -----

if n_targets_with_errors > 0:
    df_bar_tp_base_errors = pd.DataFrame(tp_base_error_rows)

    print("\nErrores encontrados:")
    print(df_bar_tp_base_errors.to_string(index=False))

    raise ValueError(
        "Existen inconsistencias en la validación de TP base. "
        "Revisar df_bar_tp_base_errors."
    )

# -----
# 5. Mostrar tabla de validación
# -----

view_cols = [
    "bar_col",
    "horizon",
    "percentile",
    "tp_mult",

```

```

        "sl_mult",
        "bar_valid_count",
        "tp_rate",
        "sl_rate",
        "no_event_rate",
    ]

print("\nValidación de targets con TP base:")
print(
    df_bar_tp_base_validation[view_cols]
    .sort_values(["percentile", "horizon", "sl_mult"])
    .to_string(
        index=False,
        formatters={
            "tp_rate": "{:.2%}".format,
            "sl_rate": "{:.2%}".format,
            "no_event_rate": "{:.2%}".format,
        }
    )
)

# -----
# 6. Resumen agregado por SL
# -----

df_bar_tp_base_by_sl = (
    df_bar_tp_base_validation
    .groupby("sl_mult", as_index=False)
    .agg(
        n_targets=("bar_col", "count"),
        tp_rate_mean=("tp_rate", "mean"),
        sl_rate_mean=("sl_rate", "mean"),
        no_event_rate_mean=("no_event_rate", "mean"),
        min_no_event_count=("no_event_count", "min"),
        max_no_event_count=("no_event_count", "max"),
    )
    .sort_values("sl_mult")
    .reset_index(drop=True)
)

print("\nResumen agregado de TP base por SL:")
print(
    df_bar_tp_base_by_sl.to_string(
        index=False,
        formatters={
            "tp_rate_mean": "{:.2%}".format,
            "sl_rate_mean": "{:.2%}".format,
            "no_event_rate_mean": "{:.2%}".format,
        }
    )
)

# -----
# 7. Confirmación final
# -----

print("\nValidación específica de TP base:")
print("OK - Cuando TP = 1.0 × threshold, no existen casos no_event.")
print("OK - Todos los casos operables resuelven como TP primero o SL primero.")
print("OK - La lógica de BAR es consistente con la construcción previa de DIR.")

```

```
print("\nValidación 4.5.6 finalizada correctamente.")
```



## 4.5.6 Validación específica de TP base - BAR

=====

Targets BAR con TP base encontrados: 18

Targets BAR con TP base esperados: 18

Targets TP base revisados: 18

Targets TP base con errores: 0

Validación de targets con TP base:

| bar_col               | horizon | percentile | tp_mult | sl_mult | bar_valid_count | tp_rate | sl_rate | no_event_rate |
|-----------------------|---------|------------|---------|---------|-----------------|---------|---------|---------------|
| bar_p40_h30_tp10_sl05 | 30      | 40         | 1.0     | 0.5     | 623736          | 7       | 8.98%   | 21.02%        |
|                       |         |            |         |         |                 |         |         | 0.00%         |
| bar_p40_h30_tp10_sl10 | 30      | 40         | 1.0     | 1.0     | 623736          | 9       | 3.84%   | 6.16%         |
|                       |         |            |         |         |                 |         |         | 0.00%         |
| bar_p40_h60_tp10_sl05 | 60      | 40         | 1.0     | 0.5     | 597024          | 7       | 7.94%   | 22.06%        |
|                       |         |            |         |         |                 |         |         | 0.00%         |
| bar_p40_h60_tp10_sl10 | 60      | 40         | 1.0     | 1.0     | 597024          | 9       | 4.00%   | 6.00%         |
|                       |         |            |         |         |                 |         |         | 0.00%         |
| bar_p40_h90_tp10_sl05 | 90      | 40         | 1.0     | 0.5     | 567970          | 7       | 7.10%   | 22.90%        |
|                       |         |            |         |         |                 |         |         | 0.00%         |
| bar_p40_h90_tp10_sl10 | 90      | 40         | 1.0     | 1.0     | 567970          | 9       | 3.57%   | 6.43%         |
|                       |         |            |         |         |                 |         |         | 0.00%         |
| bar_p50_h30_tp10_sl05 | 30      | 50         | 1.0     | 0.5     | 529982          | 8       | 1.93%   | 18.07%        |
|                       |         |            |         |         |                 |         |         | 0.00%         |
| bar_p50_h30_tp10_sl10 | 30      | 50         | 1.0     | 1.0     | 529982          | 9       | 5.26%   | 4.74%         |
|                       |         |            |         |         |                 |         |         | 0.00%         |
| bar_p50_h60_tp10_sl05 | 60      | 50         | 1.0     | 0.5     | 505020          | 8       | 1.32%   | 18.68%        |
|                       |         |            |         |         |                 |         |         | 0.00%         |
| bar_p50_h60_tp10_sl10 | 60      | 50         | 1.0     | 1.0     | 505020          | 9       | 5.45%   | 4.55%         |
|                       |         |            |         |         |                 |         |         | 0.00%         |
| bar_p50_h90_tp10_sl05 | 90      | 50         | 1.0     | 0.5     | 482603          | 8       | 0.23%   | 19.77%        |
|                       |         |            |         |         |                 |         |         | 0.00%         |
| bar_p50_h90_tp10_sl10 | 90      | 50         | 1.0     | 1.0     | 482603          | 9       | 5.11%   | 4.89%         |
|                       |         |            |         |         |                 |         |         | 0.00%         |
| bar_p60_h30_tp10_sl05 | 30      | 60         | 1.0     | 0.5     | 434995          | 8       | 4.78%   | 15.22%        |
|                       |         |            |         |         |                 |         |         | 0.00%         |
| bar_p60_h30_tp10_sl10 | 30      | 60         | 1.0     | 1.0     | 434995          | 9       | 6.41%   | 3.59%         |
|                       |         |            |         |         |                 |         |         | 0.00%         |
| bar_p60_h60_tp10_sl05 | 60      | 60         | 1.0     | 0.5     | 413798          | 8       | 4.19%   | 15.81%        |
|                       |         |            |         |         |                 |         |         | 0.00%         |
| bar_p60_h60_tp10_sl10 | 60      | 60         | 1.0     | 1.0     | 413798          | 9       | 6.62%   | 3.38%         |
|                       |         |            |         |         |                 |         |         | 0.00%         |
| bar_p60_h90_tp10_sl05 | 90      | 60         | 1.0     | 0.5     | 394501          | 8       | 3.19%   | 16.81%        |
|                       |         |            |         |         |                 |         |         | 0.00%         |
| bar_p60_h90_tp10_sl10 | 90      | 60         | 1.0     | 1.0     | 394501          | 9       | 6.37%   | 3.63%         |
|                       |         |            |         |         |                 |         |         | 0.00%         |

Resumen agregado de TP base por SL:

| sl_mult | n_targets | tp_rate_mean | sl_rate_mean | no_event_rate_mean | min_no_event_count | max_no_event_count |
|---------|-----------|--------------|--------------|--------------------|--------------------|--------------------|
| 0.5     | 9         | 81.07%       | 18.93%       | 0.00%              | 0                  | 0                  |
| 1.0     | 9         | 95.18%       | 4.82%        | 0.00%              | 0                  | 0                  |

Validación específica de TP base:

OK - Cuando TP = 1.0 × threshold, no existen casos no\_event.

OK - Todos los casos operables resuelven como TP primero o SL primero.

OK - La lógica de BAR es consistente con la construcción previa de DIR.

Validación 4.5.6 finalizada correctamente.

## 4.5.7. Resumen final de validación

```
In [94]: # 4.5.7 – Resumen final de validación
# =====

print("4.5.7 Resumen final de validación - BAR")
print("=" * 80)

# -----
# 1. Validar que existan los objetos de validación previos
# -----

required_validation_objects = [
    "df_bar_value_validation",
    "df_bar_dir_consistency",
    "df_bar_event_step_validation",
    "df_bar_temporal_validation",
    "df_bar_tp_base_validation",
]

missing_objects = [
    obj for obj in required_validation_objects
    if obj not in globals()
]

if missing_objects:
    raise NameError(
        "Faltan objetos de validación previos. "
        f"Debes correr primero: {missing_objects}"
    )

# -----
# 2. Construir resumen general de validaciones
# -----

validation_summary_rows = []

def add_validation_result(validation_name, checked_items, error_count, status_ok):
    """
    Agrega una fila al resumen final de validaciones.
    """

    status = "OK" if error_count == 0 else "ERROR"

    validation_summary_rows.append(
        {
            "validation": validation_name,
            "checked_items": checked_items,
            "error_count": error_count,
            "status": status,
            "message": status_ok_message if status == "OK" else "Revisar errores"
        }
    )

# -----
```

```

# 3. Validación 4.5.1 – Columnas construidas
# -----

n_bar_cols = len(bar_cols_found)
n_event_step_cols = len(bar_event_step_cols_found)

columns_error_count = 0

if n_bar_cols != 63:
    columns_error_count += abs(n_bar_cols - 63)

if n_event_step_cols != 63:
    columns_error_count += abs(n_event_step_cols - 63)

add_validation_result(
    validation_name="4.5.1 Columnas construidas",
    checked_items=n_bar_cols + n_event_step_cols,
    error_count=columns_error_count,
    status_ok_message="Existen 63 targets BAR y 63 columnas event_step.",
)

# -----
# 4. Validación 4.5.2 – Valores permitidos
# -----

value_error_count = int(df_bar_value_validation["has_invalid_values"].sum())

add_validation_result(
    validation_name="4.5.2 Valores permitidos",
    checked_items=len(df_bar_value_validation),
    error_count=value_error_count,
    status_ok_message="Todos los targets BAR contienen solo -1, 0, 1 o NaN.",
)

# -----
# 5. Validación 4.5.3 – Consistencia DIR/BAR
# -----

dir_bar_error_count = int(df_bar_dir_consistency["has_error"].sum())

add_validation_result(
    validation_name="4.5.3 Consistencia DIR/BAR",
    checked_items=len(df_bar_dir_consistency),
    error_count=dir_bar_error_count,
    status_ok_message="BAR solo tiene valor cuando DIR es operable.",
)

# -----
# 6. Validación 4.5.4 – Consistencia event_step
# -----

event_step_error_count = int(df_bar_event_step_validation["has_error"].sum())

add_validation_result(
    validation_name="4.5.4 Consistencia event_step",
    checked_items=len(df_bar_event_step_validation),
    error_count=event_step_error_count,
    status_ok_message="event_step es coherente con TP, SL, no_event y NaN.",
)

```

```

# -----
# 7. Validación 4.5.5 – Ventanas futuras y no cruce de días
# -----

temporal_error_count = int(df_bar_temporal_validation["has_error"].sum())

add_validation_result(
    validation_name="4.5.5 Ventanas futuras",
    checked_items=len(df_bar_temporal_validation),
    error_count=temporal_error_count,
    status_ok_message="BAR respeta ventanas futuras completas y no cruza días.",
)

# -----
# 8. Validación 4.5.6 – TP base
# -----

tp_base_error_count = int(df_bar_tp_base_validation["has_error"].sum())

add_validation_result(
    validation_name="4.5.6 TP base",
    checked_items=len(df_bar_tp_base_validation),
    error_count=tp_base_error_count,
    status_ok_message="Cuando TP = 1.0 × threshold, no existen casos no_event.",
)

# -----
# 9. Crear DataFrame final de validación
# -----

df_bar_validation_summary = pd.DataFrame(validation_summary_rows)

total_errors = int(df_bar_validation_summary["error_count"].sum())

print("\nResumen final de validaciones BAR:")
print(df_bar_validation_summary.to_string(index=False))

# -----
# 10. Resumen estructural de targets BAR
# -----

print("\nResumen estructural:")
print(f"Targets BAR contruidos: {len(bar_cols_found)}")
print(f"Columnas event_step contruidas: {len(bar_event_step_cols_found)}")
print(f"Filas del dataset BAR: {len(df_targets_bar):,}")
print(f"Columnas totales dataset BAR: {df_targets_bar.shape[1]}")

# -----
# 11. Resumen rápido de distribución general
# -----

if "df_bar_summary" in globals():
    print("\nResumen promedio por configuración TP/SL:")

    summary_by_config_view = (
        df_bar_summary
        .groupby(["tp_mult", "sl_mult"], as_index=False)
        .agg(
            n_targets=("bar_col", "count"),
            tp_rate_mean=("tp_rate", "mean"),

```

```

        sl_rate_mean=("sl_rate", "mean"),
        no_event_rate_mean=("no_event_rate", "mean"),
        tp_share_over_resolved_mean=("tp_share_over_resolved", "mean"),
        event_step_mean=("event_step_mean", "mean"),
    )
    .sort_values(["sl_mult", "tp_mult"])
    .reset_index(drop=True)
)

print(
    summary_by_config_view.to_string(
        index=False,
        formatters={
            "tp_rate_mean": "{:.2%}".format,
            "sl_rate_mean": "{:.2%}".format,
            "no_event_rate_mean": "{:.2%}".format,
            "tp_share_over_resolved_mean": "{:.2%}".format,
            "event_step_mean": "{:.2f}".format,
        }
    )
)

# -----
# 12. Guardar resumen final de validación
# -----

MNQ_TARGETS_BAR_VALIDATION_SUMMARY_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_validation_summary.csv"
)

df_bar_validation_summary.to_csv(
    MNQ_TARGETS_BAR_VALIDATION_SUMMARY_CSV,
    index=False
)

print(f"\nResumen de validación guardado en:")
print(MNQ_TARGETS_BAR_VALIDATION_SUMMARY_CSV)

# -----
# 13. Resultado final
# -----

if total_errors > 0:
    raise ValueError(
        "La validación final de BAR encontró errores. "
        "Revisar df_bar_validation_summary."
    )

print("\nResultado final:")
print("OK - Todas las validaciones de consistencia de BAR fueron superadas.")
print("OK - Los targets BAR están listos para análisis de distribución, TP/SL, t

print("\nValidación 4.5 finalizada correctamente.")

```

## 4.5.7 Resumen final de validación - BAR

```

=====
Resumen final de validaciones BAR:
      validation  checked_items  error_count status
message
  4.5.1 Columnas construidas          126          0    OK      Existen 6
3 targets BAR y 63 columnas event_step.
  4.5.2 Valores permitidos           63          0    OK      Todos los tar
gets BAR contienen solo -1, 0, 1 o NaN.
  4.5.3 Consistencia DIR/BAR          63          0    OK      BAR s
olo tiene valor cuando DIR es operable.
  4.5.4 Consistencia event_step       63          0    OK      event_step e
s coherente con TP, SL, no_event y NaN.
  4.5.5 Ventanas futuras             63          0    OK      BAR respeta vent
anas futuras completas y no cruza días.
  4.5.6 TP base                      18          0    OK      Cuando TP = 1.0
x threshold, no existen casos no_event.

```

## Resumen estructural:

```

Targets BAR contruidos:      63
Columnas event_step construidas:63
Filas del dataset BAR:      1,024,062
Columnas totales dataset BAR: 174

```

## Resumen promedio por configuración TP/SL:

| tp_mult | sl_mult | n_targets | tp_rate_mean | sl_rate_mean | no_event_rate_mean | tp_share_over_resolved_mean | event_step_mean |
|---------|---------|-----------|--------------|--------------|--------------------|-----------------------------|-----------------|
| 1.0     | 0.5     | 9         | 81.07%       | 18.93%       | 0.00%              |                             |                 |
| 81.07%  | 24.41   |           |              |              |                    |                             |                 |
| 1.5     | 0.5     | 9         | 44.39%       | 22.71%       | 32.89%             |                             |                 |
| 66.29%  | 26.43   |           |              |              |                    |                             |                 |
| 2.0     | 0.5     | 9         | 25.21%       | 24.24%       | 50.55%             |                             |                 |
| 51.08%  | 26.20   |           |              |              |                    |                             |                 |
| 1.0     | 1.0     | 9         | 95.18%       | 4.82%        | 0.00%              |                             |                 |
| 95.18%  | 28.32   |           |              |              |                    |                             |                 |
| 1.5     | 1.0     | 9         | 52.73%       | 5.85%        | 41.42%             |                             |                 |
| 90.13%  | 31.56   |           |              |              |                    |                             |                 |
| 2.0     | 1.0     | 9         | 30.26%       | 6.65%        | 63.10%             |                             |                 |
| 82.20%  | 33.08   |           |              |              |                    |                             |                 |
| 2.5     | 1.0     | 9         | 18.02%       | 7.06%        | 74.92%             |                             |                 |
| 72.07%  | 33.39   |           |              |              |                    |                             |                 |

## Resumen de validación guardado en:

```

c:\Users\heguu\OneDrive\Escritorio\neural_profit_local\data\04_mnq_targets\mnq_targets_bar_validation_summary.csv

```

## Resultado final:

OK - Todas las validaciones de consistencia de BAR fueron superadas.  
 OK - Los targets BAR están listos para análisis de distribución, TP/SL, tiempo hasta evento, estabilidad, régimen y contrato.

Validación 4.5 finalizada correctamente.

## 4.6. Distribución global de resultados

En este punto se analiza la distribución global de los targets **BAR**.

El objetivo es revisar, para cada combinación de horizonte, percentil y relación TP/SL, qué proporción de casos termina en Take Profit, Stop Loss, no\_event o NaN.

Este análisis permite entender qué configuraciones son demasiado fáciles, cuáles son demasiado exigentes y cuáles podrían tener un comportamiento operativo razonable para continuar hacia análisis más detallados.

## Celda 1 — Preparación del resumen global de BAR

Esta celda deja preparado df\_bar\_global\_distribution, que será la tabla base para analizar la distribución global de BAR.

```
In [95]: # Celda 1 – Preparación del resumen global de BAR
# =====

print("4.6. Distribución global de resultados - BAR")
print("=" * 80)

# -----
# Validar objetos necesarios
# -----

if "df_bar_summary" not in globals():
    raise NameError("No existe df_bar_summary. Primero debes correr el resumen d

if "df_bar_metadata" not in globals():
    raise NameError("No existe df_bar_metadata. Primero debes construir los targ

# -----
# Copia de trabajo
# -----

df_bar_global_distribution = df_bar_summary.copy()

# -----
# Agregar etiquetas útiles
# -----

df_bar_global_distribution["target_base"] = (
    "p" + df_bar_global_distribution["percentile"].astype(str)
    + "_h" + df_bar_global_distribution["horizon"].astype(str)
)

df_bar_global_distribution["tp_sl_config"] = (
    "TP " + df_bar_global_distribution["tp_mult"].astype(str)
    + " / SL " + df_bar_global_distribution["sl_mult"].astype(str)
)

df_bar_global_distribution["risk_reward_ratio"] = (
    df_bar_global_distribution["tp_mult"]
    / df_bar_global_distribution["sl_mult"]
)

# Punto de equilibrio teórico sin costos
df_bar_global_distribution["breakeven_win_rate"] = (
    df_bar_global_distribution["sl_mult"]
    / (
```

```

        df_bar_global_distribution["tp_mult"]
        + df_bar_global_distribution["sl_mult"]
    )
)

# Expectativa simple en unidades de threshold
# Supone:
# TP = +tp_mult
# SL = -sl_mult
# no_event = 0
df_bar_global_distribution["expectancy_units"] = (
    df_bar_global_distribution["tp_rate"] * df_bar_global_distribution["tp_mult"]
    - df_bar_global_distribution["sl_rate"] * df_bar_global_distribution["sl_mult"]
)

# Diferencia entre TP rate observado y punto de equilibrio teórico
df_bar_global_distribution["tp_rate_vs_breakeven"] = (
    df_bar_global_distribution["tp_rate"]
    - df_bar_global_distribution["breakeven_win_rate"]
)

# -----
# Ordenar columnas principales
# -----

global_distribution_cols = [
    "bar_col",
    "target_base",
    "horizon",
    "percentile",
    "tp_mult",
    "sl_mult",
    "tp_sl_config",
    "risk_reward_ratio",
    "valid_rows",
    "nan_rows",
    "tp_rate",
    "sl_rate",
    "no_event_rate",
    "tp_share_over_resolved",
    "breakeven_win_rate",
    "tp_rate_vs_breakeven",
    "expectancy_units",
    "event_step_mean",
    "event_step_median",
]

df_bar_global_distribution = (
    df_bar_global_distribution[global_distribution_cols]
    .sort_values(["percentile", "horizon", "sl_mult", "tp_mult"])
    .reset_index(drop=True)
)

# -----
# Mostrar vista inicial
# -----

print("Targets BAR analizados:", len(df_bar_global_distribution))

print("\nVista inicial de distribución global BAR:")

```



```
print(  
    df_bar_global_distribution  
    .head(21)  
    .to_string(  
        index=False,  
        formatters={  
            "risk_reward_ratio": "{:.2f}".format,  
            "tp_rate": "{:.2%}".format,  
            "sl_rate": "{:.2%}".format,  
            "no_event_rate": "{:.2%}".format,  
            "tp_share_over_resolved": "{:.2%}".format,  
            "breakeven_win_rate": "{:.2%}".format,  
            "tp_rate_vs_breakeven": "{:.2%}".format,  
            "expectancy_units": "{:.3f}".format,  
            "event_step_mean": "{:.2f}".format,  
            "event_step_median": "{:.2f}".format,  
        }  
    )  
)  
  
print("\nPreparación del resumen global finalizada correctamente.")
```

## 4.6. Distribución global de resultados - BAR

=====

Targets BAR analizados: 63

Vista inicial de distribución global BAR:

| bar_col                   | target_base        | horizon              | percentile       | tp_mult | sl_mult       | tp_sl    |
|---------------------------|--------------------|----------------------|------------------|---------|---------------|----------|
| _config risk_reward_ratio | valid_rows         | nan_rows             | tp_rate          | sl_rate | no_event_rate | tp_      |
| share_over_resolved       | breakeven_win_rate | tp_rate_vs_breakeven | expectancy_units | even    |               |          |
| t_step_mean               | event_step_median  |                      |                  |         |               |          |
| bar_p40_h30_tp10_sl05     | p40_h30            | 30                   | 40               | 1.0     | 0.5           | TP 1.0 / |
| SL 0.5                    | 2.00               | 623736               | 400326           | 78.98%  | 21.02%        | 0.00%    |
| 78.98%                    | 33.33%             |                      | 45.65%           |         | 0.685         | 11.18    |
| 9.00                      |                    |                      |                  |         |               |          |
| bar_p40_h30_tp15_sl05     | p40_h30            | 30                   | 40               | 1.5     | 0.5           | TP 1.5 / |
| SL 0.5                    | 3.00               | 623736               | 400326           | 48.14%  | 25.07%        | 26.79%   |
| 65.76%                    | 25.00%             |                      | 23.14%           |         | 0.597         | 12.46    |
| 11.00                     |                    |                      |                  |         |               |          |
| bar_p40_h30_tp20_sl05     | p40_h30            | 30                   | 40               | 2.0     | 0.5           | TP 2.0 / |
| SL 0.5                    | 4.00               | 623736               | 400326           | 29.68%  | 26.94%        | 43.38%   |
| 52.42%                    | 20.00%             |                      | 9.68%            |         | 0.459         | 12.70    |
| 11.00                     |                    |                      |                  |         |               |          |
| bar_p40_h30_tp10_sl10     | p40_h30            | 30                   | 40               | 1.0     | 1.0           | TP 1.0 / |
| SL 1.0                    | 1.00               | 623736               | 400326           | 93.84%  | 6.16%         | 0.00%    |
| 93.84%                    | 50.00%             |                      | 43.84%           |         | 0.877         | 13.20    |
| 12.00                     |                    |                      |                  |         |               |          |
| bar_p40_h30_tp15_sl10     | p40_h30            | 30                   | 40               | 1.5     | 1.0           | TP 1.5 / |
| SL 1.0                    | 1.50               | 623736               | 400326           | 57.52%  | 7.35%         | 35.13%   |
| 88.67%                    | 40.00%             |                      | 17.52%           |         | 0.789         | 14.83    |
| 14.00                     |                    |                      |                  |         |               |          |
| bar_p40_h30_tp20_sl10     | p40_h30            | 30                   | 40               | 2.0     | 1.0           | TP 2.0 / |
| SL 1.0                    | 2.00               | 623736               | 400326           | 35.63%  | 8.35%         | 56.03%   |
| 81.02%                    | 33.33%             |                      | 2.29%            |         | 0.629         | 15.66    |
| 15.00                     |                    |                      |                  |         |               |          |
| bar_p40_h30_tp25_sl10     | p40_h30            | 30                   | 40               | 2.5     | 1.0           | TP 2.5 / |
| SL 1.0                    | 2.50               | 623736               | 400326           | 22.59%  | 8.92%         | 68.49%   |
| 71.69%                    | 28.57%             |                      | -5.98%           |         | 0.476         | 15.97    |
| 16.00                     |                    |                      |                  |         |               |          |
| bar_p40_h60_tp10_sl05     | p40_h60            | 60                   | 40               | 1.0     | 0.5           | TP 1.0 / |
| SL 0.5                    | 2.00               | 597024               | 427038           | 77.94%  | 22.06%        | 0.00%    |
| 77.94%                    | 33.33%             |                      | 44.61%           |         | 0.669         | 21.95    |
| 18.00                     |                    |                      |                  |         |               |          |
| bar_p40_h60_tp15_sl05     | p40_h60            | 60                   | 40               | 1.5     | 0.5           | TP 1.5 / |
| SL 0.5                    | 3.00               | 597024               | 427038           | 46.45%  | 26.43%        | 27.12%   |
| 63.74%                    | 25.00%             |                      | 21.45%           |         | 0.565         | 24.35    |
| 21.00                     |                    |                      |                  |         |               |          |
| bar_p40_h60_tp20_sl05     | p40_h60            | 60                   | 40               | 2.0     | 0.5           | TP 2.0 / |
| SL 0.5                    | 4.00               | 597024               | 427038           | 27.84%  | 28.31%        | 43.85%   |
| 49.58%                    | 20.00%             |                      | 7.84%            |         | 0.415         | 24.57    |
| 21.00                     |                    |                      |                  |         |               |          |
| bar_p40_h60_tp10_sl10     | p40_h60            | 60                   | 40               | 1.0     | 1.0           | TP 1.0 / |
| SL 1.0                    | 1.00               | 597024               | 427038           | 94.00%  | 6.00%         | 0.00%    |
| 94.00%                    | 50.00%             |                      | 44.00%           |         | 0.880         | 26.19    |
| 24.00                     |                    |                      |                  |         |               |          |
| bar_p40_h60_tp15_sl10     | p40_h60            | 60                   | 40               | 1.5     | 1.0           | TP 1.5 / |
| SL 1.0                    | 1.50               | 597024               | 427038           | 56.30%  | 7.31%         | 36.39%   |
| 88.51%                    | 40.00%             |                      | 16.30%           |         | 0.771         | 29.41    |
| 28.00                     |                    |                      |                  |         |               |          |
| bar_p40_h60_tp20_sl10     | p40_h60            | 60                   | 40               | 2.0     | 1.0           | TP 2.0 / |
| SL 1.0                    | 2.00               | 597024               | 427038           | 34.08%  | 8.30%         | 57.61%   |
| 80.41%                    | 33.33%             |                      | 0.75%            |         | 0.599         | 31.05    |

|                       |         |        |        |        |              |        |
|-----------------------|---------|--------|--------|--------|--------------|--------|
| 30.00                 |         |        |        |        |              |        |
| bar_p40_h60_tp25_sl10 | p40_h60 | 60     | 40     | 2.5    | 1.0 TP 2.5 / |        |
| SL 1.0                | 2.50    | 597024 | 427038 | 21.05% | 8.84%        | 70.11% |
| 70.42%                | 28.57%  |        | -7.52% |        | 0.438        | 31.48  |
| 31.00                 |         |        |        |        |              |        |
| bar_p40_h90_tp10_sl05 | p40_h90 | 90     | 40     | 1.0    | 0.5 TP 1.0 / |        |
| SL 0.5                | 2.00    | 567970 | 456092 | 77.10% | 22.90%       | 0.00%  |
| 77.10%                | 33.33%  |        | 43.77% |        | 0.656        | 32.93  |
| 27.00                 |         |        |        |        |              |        |
| bar_p40_h90_tp15_sl05 | p40_h90 | 90     | 40     | 1.5    | 0.5 TP 1.5 / |        |
| SL 0.5                | 3.00    | 567970 | 456092 | 45.52% | 27.32%       | 27.16% |
| 62.49%                | 25.00%  |        | 20.52% |        | 0.546        | 36.63  |
| 32.00                 |         |        |        |        |              |        |
| bar_p40_h90_tp20_sl05 | p40_h90 | 90     | 40     | 2.0    | 0.5 TP 2.0 / |        |
| SL 0.5                | 4.00    | 567970 | 456092 | 27.45% | 29.21%       | 43.34% |
| 48.45%                | 20.00%  |        | 7.45%  |        | 0.403        | 37.08  |
| 32.00                 |         |        |        |        |              |        |
| bar_p40_h90_tp10_sl10 | p40_h90 | 90     | 40     | 1.0    | 1.0 TP 1.0 / |        |
| SL 1.0                | 1.00    | 567970 | 456092 | 93.57% | 6.43%        | 0.00%  |
| 93.57%                | 50.00%  |        | 43.57% |        | 0.871        | 39.52  |
| 36.00                 |         |        |        |        |              |        |
| bar_p40_h90_tp15_sl10 | p40_h90 | 90     | 40     | 1.5    | 1.0 TP 1.5 / |        |
| SL 1.0                | 1.50    | 567970 | 456092 | 55.98% | 7.76%        | 36.26% |
| 87.83%                | 40.00%  |        | 15.98% |        | 0.762        | 44.77  |
| 43.00                 |         |        |        |        |              |        |
| bar_p40_h90_tp20_sl10 | p40_h90 | 90     | 40     | 2.0    | 1.0 TP 2.0 / |        |
| SL 1.0                | 2.00    | 567970 | 456092 | 34.09% | 8.83%        | 57.08% |
| 79.43%                | 33.33%  |        | 0.76%  |        | 0.594        | 47.51  |
| 47.00                 |         |        |        |        |              |        |
| bar_p40_h90_tp25_sl10 | p40_h90 | 90     | 40     | 2.5    | 1.0 TP 2.5 / |        |
| SL 1.0                | 2.50    | 567970 | 456092 | 21.19% | 9.43%        | 69.38% |
| 69.19%                | 28.57%  |        | -7.39% |        | 0.435        | 48.34  |
| 49.00                 |         |        |        |        |              |        |

Preparación del resumen global finalizada correctamente.

## Celda 2 — Tablas agregadas de distribución global

```
In [96]: # Celda 2 – Tablas agregadas de distribución global
# =====

print("Tablas agregadas de distribución global - BAR")
print("=" * 80)

# -----
# Validar tabla base
# -----

if "df_bar_global_distribution" not in globals():
    raise NameError("No existe df_bar_global_distribution. Primero debes correr

# -----
# 1. Resumen promedio por configuración TP/SL
# -----

df_bar_distribution_by_config = (
    df_bar_global_distribution
    .groupby(["tp_mult", "sl_mult", "tp_sl_config"], as_index=False)
    .agg(
```

```

        n_targets=("bar_col", "count"),
        tp_rate_mean=("tp_rate", "mean"),
        sl_rate_mean=("sl_rate", "mean"),
        no_event_rate_mean=("no_event_rate", "mean"),
        tp_share_over_resolved_mean=("tp_share_over_resolved", "mean"),
        breakeven_win_rate=("breakeven_win_rate", "first"),
        tp_rate_vs_breakeven_mean=("tp_rate_vs_breakeven", "mean"),
        expectancy_units_mean=("expectancy_units", "mean"),
        expectancy_units_min=("expectancy_units", "min"),
        expectancy_units_max=("expectancy_units", "max"),
        event_step_mean=("event_step_mean", "mean"),
        event_step_median=("event_step_median", "mean"),
    )
    .sort_values(["sl_mult", "tp_mult"])
    .reset_index(drop=True)
)

print("\nResumen promedio por configuración TP/SL:")
print(
    df_bar_distribution_by_config.to_string(
        index=False,
        formatters={
            "tp_rate_mean": "{:.2%}".format,
            "sl_rate_mean": "{:.2%}".format,
            "no_event_rate_mean": "{:.2%}".format,
            "tp_share_over_resolved_mean": "{:.2%}".format,
            "breakeven_win_rate": "{:.2%}".format,
            "tp_rate_vs_breakeven_mean": "{:.2%}".format,
            "expectancy_units_mean": "{:.3f}".format,
            "expectancy_units_min": "{:.3f}".format,
            "expectancy_units_max": "{:.3f}".format,
            "event_step_mean": "{:.2f}".format,
            "event_step_median": "{:.2f}".format,
        }
    )
)

# -----
# 2. Resumen por percentil y horizonte
# -----

df_bar_distribution_by_target_base = (
    df_bar_global_distribution
    .groupby(["percentile", "horizon", "target_base"], as_index=False)
    .agg(
        n_configs=("bar_col", "count"),
        valid_rows_mean=("valid_rows", "mean"),
        tp_rate_mean=("tp_rate", "mean"),
        sl_rate_mean=("sl_rate", "mean"),
        no_event_rate_mean=("no_event_rate", "mean"),
        tp_share_over_resolved_mean=("tp_share_over_resolved", "mean"),
        expectancy_units_mean=("expectancy_units", "mean"),
        event_step_mean=("event_step_mean", "mean"),
    )
    .sort_values(["percentile", "horizon"])
    .reset_index(drop=True)
)

print("\nResumen promedio por percentil y horizonte:")
print(

```

```

df_bar_distribution_by_target_base.to_string(
    index=False,
    formatters={
        "valid_rows_mean": "{:.0f}".format,
        "tp_rate_mean": "{:.2%}".format,
        "sl_rate_mean": "{:.2%}".format,
        "no_event_rate_mean": "{:.2%}".format,
        "tp_share_over_resolved_mean": "{:.2%}".format,
        "expectancy_units_mean": "{:.3f}".format,
        "event_step_mean": "{:.2f}".format,
    }
)

# -----
# 3. Vista enfocada en variantes centrales p50
# -----

df_bar_distribution_p50 = (
    df_bar_global_distribution[
        df_bar_global_distribution["percentile"] == 50
    ]
    .copy()
    .sort_values(["horizon", "sl_mult", "tp_mult"])
    .reset_index(drop=True)
)

print("\nDetalle de variantes centrales p50:")
print(
    df_bar_distribution_p50[
        [
            "bar_col",
            "horizon",
            "tp_mult",
            "sl_mult",
            "tp_rate",
            "sl_rate",
            "no_event_rate",
            "tp_share_over_resolved",
            "breakeven_win_rate",
            "tp_rate_vs_breakeven",
            "expectancy_units",
            "event_step_mean",
            "event_step_median",
        ]
    ].to_string(
        index=False,
        formatters={
            "tp_rate": "{:.2%}".format,
            "sl_rate": "{:.2%}".format,
            "no_event_rate": "{:.2%}".format,
            "tp_share_over_resolved": "{:.2%}".format,
            "breakeven_win_rate": "{:.2%}".format,
            "tp_rate_vs_breakeven": "{:.2%}".format,
            "expectancy_units": "{:.3f}".format,
            "event_step_mean": "{:.2f}".format,
            "event_step_median": "{:.2f}".format,
        }
    )
)

```

```

# -----
# 4. Ranking preliminar por expectativa simple
# -----
# Nota:
# expectancy_units asume:
# TP = +tp_mult
# SL = -sl_mult
# no_event = 0
#
# Es una métrica diagnóstica, no una rentabilidad real.
# -----

df_bar_distribution_ranking = (
    df_bar_global_distribution
    .copy()
    .sort_values(
        ["expectancy_units", "tp_rate", "no_event_rate"],
        ascending=[False, False, True]
    )
    .reset_index(drop=True)
)

print("\nTop 15 variantes por expectativa simple:")
print(
    df_bar_distribution_ranking[
        [
            "bar_col",
            "horizon",
            "percentile",
            "tp_mult",
            "sl_mult",
            "tp_rate",
            "sl_rate",
            "no_event_rate",
            "tp_share_over_resolved",
            "breakeven_win_rate",
            "expectancy_units",
        ]
    ]
    .head(15)
    .to_string(
        index=False,
        formatters={
            "tp_rate": "{:.2%}".format,
            "sl_rate": "{:.2%}".format,
            "no_event_rate": "{:.2%}".format,
            "tp_share_over_resolved": "{:.2%}".format,
            "breakeven_win_rate": "{:.2%}".format,
            "expectancy_units": "{:.3f}".format,
        }
    )
)

print("\nCelda 2 finalizada correctamente.")

```

## Tablas agregadas de distribución global - BAR

=====

## Resumen promedio por configuración TP/SL:

| tp_mult         | sl_mult                     | tp_sl_config       | n_targets                 | tp_rate_mean          | sl_rate_mean         | no_event_r           |
|-----------------|-----------------------------|--------------------|---------------------------|-----------------------|----------------------|----------------------|
| ate_mean        | tp_share_over_resolved_mean | breakeven_win_rate | tp_rate_vs_breakeven_mean | expectancy_units_mean | expectancy_units_min | expectancy_units_max |
| event_step_mean | event_step_median           |                    |                           |                       |                      |                      |
| 1.0             | 0.5                         | TP 1.0 / SL 0.5    | 9                         | 81.07%                | 18.93%               |                      |
| 0.00%           |                             | 81.07%             |                           | 33.33%                |                      | 47.74%               |
| 0.716           |                             | 0.656              | 0.772                     | 24.41                 |                      | 21.22                |
| 1.5             | 0.5                         | TP 1.5 / SL 0.5    | 9                         | 44.39%                | 22.71%               |                      |
| 32.89%          |                             | 66.29%             |                           | 25.00%                |                      | 19.39%               |
| 0.552           |                             | 0.523              | 0.597                     | 26.43                 |                      | 24.22                |
| 2.0             | 0.5                         | TP 2.0 / SL 0.5    | 9                         | 25.21%                | 24.24%               |                      |
| 50.55%          |                             | 51.08%             |                           | 20.00%                |                      | 5.21%                |
| 0.383           |                             | 0.329              | 0.459                     | 26.20                 |                      | 23.44                |
| 1.0             | 1.0                         | TP 1.0 / SL 1.0    | 9                         | 95.18%                | 4.82%                |                      |
| 0.00%           |                             | 95.18%             |                           | 50.00%                |                      | 45.18%               |
| 0.904           |                             | 0.871              | 0.932                     | 28.32                 |                      | 26.56                |
| 1.5             | 1.0                         | TP 1.5 / SL 1.0    | 9                         | 52.73%                | 5.85%                |                      |
| 41.42%          |                             | 90.13%             |                           | 40.00%                |                      | 12.73%               |
| 0.733           |                             | 0.683              | 0.789                     | 31.56                 |                      | 31.11                |
| 2.0             | 1.0                         | TP 2.0 / SL 1.0    | 9                         | 30.26%                | 6.65%                |                      |
| 63.10%          |                             | 82.20%             |                           | 33.33%                |                      | -3.08%               |
| 0.539           |                             | 0.462              | 0.629                     | 33.08                 |                      | 33.44                |
| 2.5             | 1.0                         | TP 2.5 / SL 1.0    | 9                         | 18.02%                | 7.06%                |                      |
| 74.92%          |                             | 72.07%             |                           | 28.57%                |                      | -10.56%              |
| 0.380           |                             | 0.306              | 0.476                     | 33.39                 |                      | 34.33                |

## Resumen promedio por percentil y horizonte:

| percentile         | horizon                     | target_base           | n_configs | valid_rows_mean | tp_rate_mean | sl_rate_mean |
|--------------------|-----------------------------|-----------------------|-----------|-----------------|--------------|--------------|
| no_event_rate_mean | tp_share_over_resolved_mean | expectancy_units_mean | event_s   | tep_mean        |              |              |
| 40                 | 30                          | p40_h30               | 7         | 623736          | 52.34%       | 1            |
| 4.83%              |                             | 32.83%                |           | 76.06%          |              | 0.644        |
| 13.71              |                             |                       |           |                 |              |              |
| 40                 | 60                          | p40_h60               | 7         | 597024          | 51.09%       | 1            |
| 5.32%              |                             | 33.58%                |           | 74.94%          |              | 0.620        |
| 27.00              |                             |                       |           |                 |              |              |
| 40                 | 90                          | p40_h90               | 7         | 567970          | 50.70%       | 1            |
| 5.98%              |                             | 33.32%                |           | 74.01%          |              | 0.610        |
| 40.97              |                             |                       |           |                 |              |              |
| 50                 | 30                          | p50_h30               | 7         | 529982          | 50.36%       | 1            |
| 2.40%              |                             | 37.24%                |           | 77.75%          |              | 0.617        |
| 14.62              |                             |                       |           |                 |              |              |
| 50                 | 60                          | p50_h60               | 7         | 505020          | 49.20%       | 1            |
| 2.61%              |                             | 38.19%                |           | 76.93%          |              | 0.596        |
| 28.81              |                             |                       |           |                 |              |              |
| 50                 | 90                          | p50_h90               | 7         | 482603          | 49.04%       | 1            |
| 3.36%              |                             | 37.61%                |           | 75.97%          |              | 0.590        |
| 43.86              |                             |                       |           |                 |              |              |
| 60                 | 30                          | p60_h30               | 7         | 434995          | 48.39%       | 1            |
| 0.19%              |                             | 41.42%                |           | 79.43%          |              | 0.590        |
| 15.47              |                             |                       |           |                 |              |              |
| 60                 | 60                          | p60_h60               | 7         | 413798          | 47.41%       | 1            |
| 0.34%              |                             | 42.25%                |           | 78.79%          |              | 0.572        |
| 30.42              |                             |                       |           |                 |              |              |
| 60                 | 90                          | p60_h90               | 7         | 394501          | 47.43%       | 1            |
| 1.02%              |                             | 41.55%                |           | 77.86%          |              | 0.569        |

46.64

Detalle de variantes centrales p50:

| bar_col               | horizon            | tp_mult              | sl_mult          | tp_rate | sl_rate      | no_event_rate     | tp |
|-----------------------|--------------------|----------------------|------------------|---------|--------------|-------------------|----|
| _share_over_resolved  | breakeven_win_rate | tp_rate_vs_breakeven | expectancy_units | event   | nt_step_mean | event_step_median |    |
| bar_p50_h30_tp10_sl05 | 30                 | 1.0                  | 0.5              | 81.93%  | 18.07%       | 0.00%             |    |
| 81.93%                | 33.33%             | 48.59%               |                  | 0.729   | 12.37        |                   |    |
| 11.00                 |                    |                      |                  |         |              |                   |    |
| bar_p50_h30_tp15_sl05 | 30                 | 1.5                  | 0.5              | 45.74%  | 21.67%       | 32.60%            |    |
| 67.85%                | 25.00%             | 20.74%               |                  | 0.578   | 13.40        |                   |    |
| 12.00                 |                    |                      |                  |         |              |                   |    |
| bar_p50_h30_tp20_sl05 | 30                 | 2.0                  | 0.5              | 26.31%  | 23.19%       | 50.50%            |    |
| 53.15%                | 20.00%             | 6.31%                |                  | 0.410   | 13.39        |                   |    |
| 12.00                 |                    |                      |                  |         |              |                   |    |
| bar_p50_h30_tp10_sl10 | 30                 | 1.0                  | 1.0              | 95.26%  | 4.74%        | 0.00%             |    |
| 95.26%                | 50.00%             | 45.26%               |                  | 0.905   | 14.21        |                   |    |
| 13.00                 |                    |                      |                  |         |              |                   |    |
| bar_p50_h30_tp15_sl10 | 30                 | 1.5                  | 1.0              | 53.60%  | 5.71%        | 40.70%            |    |
| 90.38%                | 40.00%             | 13.60%               |                  | 0.747   | 15.74        |                   |    |
| 15.00                 |                    |                      |                  |         |              |                   |    |
| bar_p50_h30_tp20_sl10 | 30                 | 2.0                  | 1.0              | 31.07%  | 6.49%        | 62.44%            |    |
| 82.73%                | 33.33%             | -2.26%               |                  | 0.557   | 16.50        |                   |    |
| 17.00                 |                    |                      |                  |         |              |                   |    |
| bar_p50_h30_tp25_sl10 | 30                 | 2.5                  | 1.0              | 18.63%  | 6.90%        | 74.47%            |    |
| 72.97%                | 28.57%             | -9.94%               |                  | 0.397   | 16.70        |                   |    |
| 17.00                 |                    |                      |                  |         |              |                   |    |
| bar_p50_h60_tp10_sl05 | 60                 | 1.0                  | 0.5              | 81.32%  | 18.68%       | 0.00%             |    |
| 81.32%                | 33.33%             | 47.99%               |                  | 0.720   | 24.40        |                   |    |
| 21.00                 |                    |                      |                  |         |              |                   |    |
| bar_p50_h60_tp15_sl05 | 60                 | 1.5                  | 0.5              | 44.00%  | 22.52%       | 33.48%            |    |
| 66.15%                | 25.00%             | 19.00%               |                  | 0.547   | 26.28        |                   |    |
| 24.00                 |                    |                      |                  |         |              |                   |    |
| bar_p50_h60_tp20_sl05 | 60                 | 2.0                  | 0.5              | 24.58%  | 23.99%       | 51.44%            |    |
| 50.61%                | 20.00%             | 4.58%                |                  | 0.372   | 25.95        |                   |    |
| 23.00                 |                    |                      |                  |         |              |                   |    |
| bar_p50_h60_tp10_sl10 | 60                 | 1.0                  | 1.0              | 95.45%  | 4.55%        | 0.00%             |    |
| 95.45%                | 50.00%             | 45.45%               |                  | 0.909   | 28.22        |                   |    |
| 26.00                 |                    |                      |                  |         |              |                   |    |
| bar_p50_h60_tp15_sl10 | 60                 | 1.5                  | 1.0              | 52.23%  | 5.54%        | 42.23%            |    |
| 90.41%                | 40.00%             | 12.23%               |                  | 0.728   | 31.28        |                   |    |
| 31.00                 |                    |                      |                  |         |              |                   |    |
| bar_p50_h60_tp20_sl10 | 60                 | 2.0                  | 1.0              | 29.49%  | 6.29%        | 64.22%            |    |
| 82.42%                | 33.33%             | -3.84%               |                  | 0.527   | 32.68        |                   |    |
| 33.00                 |                    |                      |                  |         |              |                   |    |
| bar_p50_h60_tp25_sl10 | 60                 | 2.5                  | 1.0              | 17.33%  | 6.69%        | 75.98%            |    |
| 72.16%                | 28.57%             | -11.24%              |                  | 0.366   | 32.87        |                   |    |
| 34.00                 |                    |                      |                  |         |              |                   |    |
| bar_p50_h90_tp10_sl05 | 90                 | 1.0                  | 0.5              | 80.23%  | 19.77%       | 0.00%             |    |
| 80.23%                | 33.33%             | 46.89%               |                  | 0.703   | 36.58        |                   |    |
| 32.00                 |                    |                      |                  |         |              |                   |    |
| bar_p50_h90_tp15_sl05 | 90                 | 1.5                  | 0.5              | 43.60%  | 23.71%       | 32.70%            |    |
| 64.78%                | 25.00%             | 18.60%               |                  | 0.535   | 39.72        |                   |    |
| 36.00                 |                    |                      |                  |         |              |                   |    |
| bar_p50_h90_tp20_sl05 | 90                 | 2.0                  | 0.5              | 24.52%  | 25.25%       | 50.23%            |    |
| 49.26%                | 20.00%             | 4.52%                |                  | 0.364   | 39.31        |                   |    |
| 35.00                 |                    |                      |                  |         |              |                   |    |
| bar_p50_h90_tp10_sl10 | 90                 | 1.0                  | 1.0              | 95.11%  | 4.89%        | 0.00%             |    |
| 95.11%                | 50.00%             | 45.11%               |                  | 0.902   | 42.70        |                   |    |
| 40.00                 |                    |                      |                  |         |              |                   |    |



|                       |        |     |     |         |       |        |
|-----------------------|--------|-----|-----|---------|-------|--------|
| bar_p50_h90_tp15_sl10 | 90     | 1.5 | 1.0 | 52.38%  | 5.93% | 41.69% |
| 89.83%                | 40.00% |     |     | 12.38%  | 0.726 | 47.81  |
| 48.00                 |        |     |     |         |       |        |
| bar_p50_h90_tp20_sl10 | 90     | 2.0 | 1.0 | 29.78%  | 6.77% | 63.45% |
| 81.48%                | 33.33% |     |     | -3.56%  | 0.528 | 50.14  |
| 51.00                 |        |     |     |         |       |        |
| bar_p50_h90_tp25_sl10 | 90     | 2.5 | 1.0 | 17.65%  | 7.17% | 75.18% |
| 71.12%                | 28.57% |     |     | -10.92% | 0.370 | 50.79  |
| 53.00                 |        |     |     |         |       |        |

Top 15 variantes por expectativa simple:

| bar_col               | horizon                | percentile         | tp_mult          | sl_mult | tp_rate | sl_rate | no_e |
|-----------------------|------------------------|--------------------|------------------|---------|---------|---------|------|
| vent_rate             | tp_share_over_resolved | breakeven_win_rate | expectancy_units |         |         |         |      |
| bar_p60_h60_tp10_sl10 | 60                     | 60                 | 1.0              | 1.0     | 96.62%  | 3.38%   |      |
| 0.00%                 | 96.62%                 | 50.00%             |                  |         | 0.932   |         |      |
| bar_p60_h30_tp10_sl10 | 30                     | 60                 | 1.0              | 1.0     | 96.41%  | 3.59%   |      |
| 0.00%                 | 96.41%                 | 50.00%             |                  |         | 0.928   |         |      |
| bar_p60_h90_tp10_sl10 | 90                     | 60                 | 1.0              | 1.0     | 96.37%  | 3.63%   |      |
| 0.00%                 | 96.37%                 | 50.00%             |                  |         | 0.927   |         |      |
| bar_p50_h60_tp10_sl10 | 60                     | 50                 | 1.0              | 1.0     | 95.45%  | 4.55%   |      |
| 0.00%                 | 95.45%                 | 50.00%             |                  |         | 0.909   |         |      |
| bar_p50_h30_tp10_sl10 | 30                     | 50                 | 1.0              | 1.0     | 95.26%  | 4.74%   |      |
| 0.00%                 | 95.26%                 | 50.00%             |                  |         | 0.905   |         |      |
| bar_p50_h90_tp10_sl10 | 90                     | 50                 | 1.0              | 1.0     | 95.11%  | 4.89%   |      |
| 0.00%                 | 95.11%                 | 50.00%             |                  |         | 0.902   |         |      |
| bar_p40_h60_tp10_sl10 | 60                     | 40                 | 1.0              | 1.0     | 94.00%  | 6.00%   |      |
| 0.00%                 | 94.00%                 | 50.00%             |                  |         | 0.880   |         |      |
| bar_p40_h30_tp10_sl10 | 30                     | 40                 | 1.0              | 1.0     | 93.84%  | 6.16%   |      |
| 0.00%                 | 93.84%                 | 50.00%             |                  |         | 0.877   |         |      |
| bar_p40_h90_tp10_sl10 | 90                     | 40                 | 1.0              | 1.0     | 93.57%  | 6.43%   |      |
| 0.00%                 | 93.57%                 | 50.00%             |                  |         | 0.871   |         |      |
| bar_p40_h30_tp15_sl10 | 30                     | 40                 | 1.5              | 1.0     | 57.52%  | 7.35%   |      |
| 35.13%                | 88.67%                 | 40.00%             |                  |         | 0.789   |         |      |
| bar_p60_h30_tp10_sl05 | 30                     | 60                 | 1.0              | 0.5     | 84.78%  | 15.22%  |      |
| 0.00%                 | 84.78%                 | 33.33%             |                  |         | 0.772   |         |      |
| bar_p40_h60_tp15_sl10 | 60                     | 40                 | 1.5              | 1.0     | 56.30%  | 7.31%   |      |
| 36.39%                | 88.51%                 | 40.00%             |                  |         | 0.771   |         |      |
| bar_p60_h60_tp10_sl05 | 60                     | 60                 | 1.0              | 0.5     | 84.19%  | 15.81%  |      |
| 0.00%                 | 84.19%                 | 33.33%             |                  |         | 0.763   |         |      |
| bar_p40_h90_tp15_sl10 | 90                     | 40                 | 1.5              | 1.0     | 55.98%  | 7.76%   |      |
| 36.26%                | 87.83%                 | 40.00%             |                  |         | 0.762   |         |      |
| bar_p60_h90_tp10_sl05 | 90                     | 60                 | 1.0              | 0.5     | 83.19%  | 16.81%  |      |
| 0.00%                 | 83.19%                 | 33.33%             |                  |         | 0.748   |         |      |

Celda 2 finalizada correctamente.

## Celda 3 — Gráficos agrupados

Esto genera 3 figuras agrupadas:

1. Distribución promedio TP / SL / no\_event por configuración.
2. Heatmaps p50 por horizonte y configuración.
3. Expectativa simple promedio por configuración.

```
In [97]: # Celda 3 — Gráficos agrupados
# =====

print("Gráficos agrupados de distribución global - BAR")
print("=" * 80)
```

```

# -----
# Validar objetos necesarios
# -----

if "df_bar_distribution_by_config" not in globals():
    raise NameError("No existe df_bar_distribution_by_config. Primero debes corr

if "df_bar_global_distribution" not in globals():
    raise NameError("No existe df_bar_global_distribution. Primero debes correr

# -----
# Orden de configuraciones TP/SL
# -----

config_order = [
    (1.0, 1.0),
    (1.5, 1.0),
    (2.0, 1.0),
    (2.5, 1.0),
    (1.0, 0.5),
    (1.5, 0.5),
    (2.0, 0.5),
]

config_labels = [
    f"TP {tp} / SL {sl}"
    for tp, sl in config_order
]

df_config_plot = df_bar_distribution_by_config.copy()

df_config_plot["config_order"] = df_config_plot.apply(
    lambda row: config_order.index((row["tp_mult"], row["sl_mult"]))
    if (row["tp_mult"], row["sl_mult"]) in config_order
    else 999,
    axis=1
)

df_config_plot = (
    df_config_plot
    .sort_values("config_order")
    .reset_index(drop=True)
)

# =====
# Figura 1 – Distribución promedio por configuración TP/SL
# =====

x = np.arange(len(df_config_plot))
bar_width = 0.25

fig, ax = plt.subplots(figsize=(13, 6))

ax.bar(
    x - bar_width,
    df_config_plot["tp_rate_mean"],
    width=bar_width,
    label="TP primero"
)

```

```

ax.bar(
    x,
    df_config_plot["sl_rate_mean"],
    width=bar_width,
    label="SL primero"
)

ax.bar(
    x + bar_width,
    df_config_plot["no_event_rate_mean"],
    width=bar_width,
    label="No event"
)

ax.set_title("Distribución promedio de resultados BAR por configuración TP/SL")
ax.set_ylabel("Proporción promedio")
ax.set_xticks(x)
ax.set_xticklabels(config_labels, rotation=45, ha="right")
ax.legend()

ax.yaxis.set_major_formatter(
    plt.FuncFormatter(lambda y, _: f"{y:.0%}")
)

for i, row in df_config_plot.iterrows():
    ax.text(
        i - bar_width,
        row["tp_rate_mean"],
        f"{row['tp_rate_mean']:.0%}",
        ha="center",
        va="bottom",
        fontsize=8
    )
    ax.text(
        i,
        row["sl_rate_mean"],
        f"{row['sl_rate_mean']:.0%}",
        ha="center",
        va="bottom",
        fontsize=8
    )
    ax.text(
        i + bar_width,
        row["no_event_rate_mean"],
        f"{row['no_event_rate_mean']:.0%}",
        ha="center",
        va="bottom",
        fontsize=8
    )

plt.tight_layout()
plt.show()

# =====
# Figura 2 – Heatmaps p50 por horizonte y configuración TP/SL
# =====

df_p50_plot = df_bar_global_distribution[
    df_bar_global_distribution["percentile"] == 50

```

```

].copy()

df_p50_plot["config_label"] = df_p50_plot.apply(
    lambda row: f"TP {row['tp_mult']} / SL {row['sl_mult']}",
    axis=1
)

df_p50_plot["config_order"] = df_p50_plot.apply(
    lambda row: config_order.index((row["tp_mult"], row["sl_mult"]))
    if (row["tp_mult"], row["sl_mult"]) in config_order
    else 999,
    axis=1
)

metrics_to_plot = [
    ("tp_rate", "TP primero"),
    ("sl_rate", "SL primero"),
    ("no_event_rate", "No event"),
]

fig = plt.figure(figsize=(13, 10))

gs = fig.add_gridspec(
    nrows=3,
    ncols=2,
    width_ratios=[20, 1],
    hspace=0.45,
    wspace=0.08
)

axes = [fig.add_subplot(gs[i, 0]) for i in range(3)]
caxes = [fig.add_subplot(gs[i, 1]) for i in range(3)]

for ax, cax, (metric_col, metric_title) in zip(axes, caxes, metrics_to_plot):
    tmp = df_p50_plot.copy()

    pivot_tmp = tmp.pivot(
        index="horizon",
        columns="config_label",
        values=metric_col
    )

    ordered_columns = [
        label for label in config_labels
        if label in pivot_tmp.columns
    ]

    pivot_tmp = pivot_tmp.reindex(index=BAR_HORIZONS)
    pivot_tmp = pivot_tmp.reindex(columns=ordered_columns)

    data = np.ma.masked_invalid(pivot_tmp.values)

    im = ax.imshow(
        data,
        aspect="auto",
        vmin=0,
        vmax=1
    )

    ax.set_title(f"{metric_title} - variantes p50")

```

```

ax.set_ylabel("Horizonte")

ax.set_yticks(np.arange(len(pivot_tmp.index)))
ax.set_yticklabels([f"H{h}" for h in pivot_tmp.index])

ax.set_xticks(np.arange(len(pivot_tmp.columns)))
ax.set_xticklabels(pivot_tmp.columns, rotation=45, ha="right")

for i in range(pivot_tmp.shape[0]):
    for j in range(pivot_tmp.shape[1]):
        value = pivot_tmp.iloc[i, j]

        if pd.notna(value):
            ax.text(
                j,
                i,
                f"{value:.0%}",
                ha="center",
                va="center",
                fontsize=8
            )

fig.colorbar(
    im,
    cax=cax,
    label="Proporción"
)

plt.show()

# =====
# Figura 3 – Expectativa simple por configuración TP/SL
# =====

fig, ax = plt.subplots(figsize=(13, 5))

ax.bar(
    x,
    df_config_plot["expectancy_units_mean"],
    width=0.6
)

ax.axhline(0, linestyle="--", linewidth=1)

ax.set_title("Expectativa simple promedio por configuración TP/SL")
ax.set_ylabel("Expectativa en unidades de threshold")
ax.set_xticks(x)
ax.set_xticklabels(config_labels, rotation=45, ha="right")

for i, row in df_config_plot.iterrows():
    ax.text(
        i,
        row["expectancy_units_mean"],
        f"{row['expectancy_units_mean']:.2f}",
        ha="center",
        va="bottom" if row["expectancy_units_mean"] >= 0 else "top",
        fontsize=8
    )

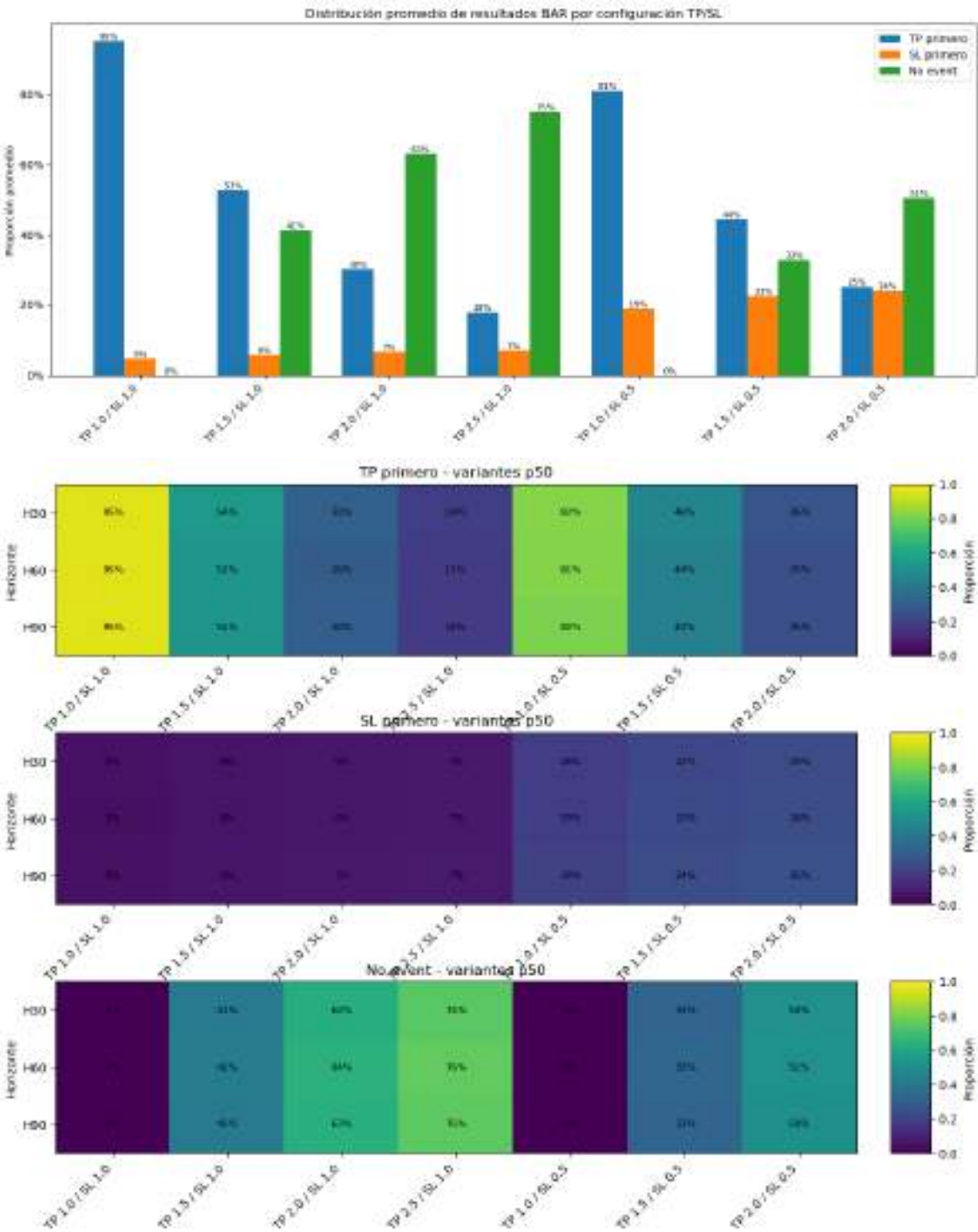
plt.tight_layout()

```

```
plt.show()

print("\nCelda 3 finalizada correctamente.")
```

Gráficos agrupados de distribución global - BAR





Celda 3 finalizada correctamente.

## Celda 4 — Resumen guardado / tabla final para observaciones

```
In [98]: # Celda 4 – Guardado y tabla final para observaciones
# =====

print("Guardado y tabla final de distribución global - BAR")
print("=" * 80)

# -----
# Validar objetos necesarios
# -----

required_objects = [
    "df_bar_global_distribution",
    "df_bar_distribution_by_config",
    "df_bar_distribution_by_target_base",
    "df_bar_distribution_p50",
    "df_bar_distribution_ranking",
]

missing_objects = [
    obj for obj in required_objects
    if obj not in globals()
]

if missing_objects:
    raise NameError(
        "Faltan objetos necesarios. "
        f"Debes correr las celdas anteriores: {missing_objects}"
    )

# -----
# Definir rutas de guardado
# -----

MNQ_TARGETS_BAR_GLOBAL_DISTRIBUTION_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_global_distribution.csv"
)

MNQ_TARGETS_BAR_DISTRIBUTION_BY_CONFIG_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_distribution_by_config.csv"
)
```

```

MNQ_TARGETS_BAR_DISTRIBUTION_BY_TARGET_BASE_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_distribution_by_target_base.csv"
)

MNQ_TARGETS_BAR_DISTRIBUTION_RANKING_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_distribution_ranking.csv"
)

MNQ_TARGETS_BAR_DISTRIBUTION_REPORT_TXT = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_distribution_report.txt"
)

# -----
# Guardar tablas principales
# -----

df_bar_global_distribution.to_csv(
    MNQ_TARGETS_BAR_GLOBAL_DISTRIBUTION_CSV,
    index=False
)

df_bar_distribution_by_config.to_csv(
    MNQ_TARGETS_BAR_DISTRIBUTION_BY_CONFIG_CSV,
    index=False
)

df_bar_distribution_by_target_base.to_csv(
    MNQ_TARGETS_BAR_DISTRIBUTION_BY_TARGET_BASE_CSV,
    index=False
)

df_bar_distribution_ranking.to_csv(
    MNQ_TARGETS_BAR_DISTRIBUTION_RANKING_CSV,
    index=False
)

# -----
# Crear tabla final compacta para observaciones
# -----

df_bar_distribution_observation_table = (
    df_bar_distribution_by_config
    .copy()
    .sort_values(["sl_mult", "tp_mult"])
    .reset_index(drop=True)
)

observation_cols = [
    "tp_sl_config",
    "tp_mult",
    "sl_mult",
    "n_targets",
    "tp_rate_mean",
    "sl_rate_mean",
    "no_event_rate_mean",
    "tp_share_over_resolved_mean",
    "breakeven_win_rate",
    "tp_rate_vs_breakeven_mean",
    "expectancy_units_mean",

```



```

        "event_step_mean",
    ]

print("\nTabla compacta para observaciones:")
print(
    df_bar_distribution_observation_table[observation_cols]
    .to_string(
        index=False,
        formatters={
            "tp_rate_mean": "{:.2%}".format,
            "sl_rate_mean": "{:.2%}".format,
            "no_event_rate_mean": "{:.2%}".format,
            "tp_share_over_resolved_mean": "{:.2%}".format,
            "breakeven_win_rate": "{:.2%}".format,
            "tp_rate_vs_breakeven_mean": "{:.2%}".format,
            "expectancy_units_mean": "{:.3f}".format,
            "event_step_mean": "{:.2f}".format,
        }
    )
)

# -----
# Tabla p50 compacta
# -----

p50_cols = [
    "bar_col",
    "horizon",
    "tp_mult",
    "sl_mult",
    "tp_rate",
    "sl_rate",
    "no_event_rate",
    "tp_share_over_resolved",
    "breakeven_win_rate",
    "tp_rate_vs_breakeven",
    "expectancy_units",
    "event_step_mean",
    "event_step_median",
]

print("\nTabla compacta de variantes p50:")
print(
    df_bar_distribution_p50[p50_cols]
    .to_string(
        index=False,
        formatters={
            "tp_rate": "{:.2%}".format,
            "sl_rate": "{:.2%}".format,
            "no_event_rate": "{:.2%}".format,
            "tp_share_over_resolved": "{:.2%}".format,
            "breakeven_win_rate": "{:.2%}".format,
            "tp_rate_vs_breakeven": "{:.2%}".format,
            "expectancy_units": "{:.3f}".format,
            "event_step_mean": "{:.2f}".format,
            "event_step_median": "{:.2f}".format,
        }
    )
)

```

```

# -----
# Top variantes por expectativa simple
# -----

print("\nTop 20 variantes por expectativa simple:")
print(
    df_bar_distribution_ranking[
        [
            "bar_col",
            "horizon",
            "percentile",
            "tp_mult",
            "sl_mult",
            "tp_rate",
            "sl_rate",
            "no_event_rate",
            "tp_share_over_resolved",
            "breakeven_win_rate",
            "expectancy_units",
            "event_step_mean",
        ]
    ].head(20)
    .to_string(
        index=False,
        formatters={
            "tp_rate": "{:.2%}".format,
            "sl_rate": "{:.2%}".format,
            "no_event_rate": "{:.2%}".format,
            "tp_share_over_resolved": "{:.2%}".format,
            "breakeven_win_rate": "{:.2%}".format,
            "expectancy_units": "{:.3f}".format,
            "event_step_mean": "{:.2f}".format,
        }
    )
)

# -----
# Guardar reporte TXT completo
# -----

with MNQ_TARGETS_BAR_DISTRIBUTION_REPORT_TXT.open("w", encoding="utf-8") as f:
    f.write("4.6. Distribución global de resultados - BAR\n")
    f.write("=" * 100 + "\n\n")

    f.write("Tabla compacta por configuración TP/SL\n")
    f.write("-" * 100 + "\n")
    f.write(
        df_bar_distribution_observation_table[observation_cols]
        .to_string(
            index=False,
            formatters={
                "tp_rate_mean": "{:.2%}".format,
                "sl_rate_mean": "{:.2%}".format,
                "no_event_rate_mean": "{:.2%}".format,
                "tp_share_over_resolved_mean": "{:.2%}".format,
                "breakeven_win_rate": "{:.2%}".format,
                "tp_rate_vs_breakeven_mean": "{:.2%}".format,
                "expectancy_units_mean": "{:.3f}".format,
                "event_step_mean": "{:.2f}".format,
            }
        )
    )

```

```

    }
  )
)

f.write("\n\n")
f.write("Tabla compacta de variantes p50\n")
f.write("-" * 100 + "\n")
f.write(
    df_bar_distribution_p50[p50_cols]
    .to_string(
        index=False,
        formatters={
            "tp_rate": "{:.2%}".format,
            "sl_rate": "{:.2%}".format,
            "no_event_rate": "{:.2%}".format,
            "tp_share_over_resolved": "{:.2%}".format,
            "breakeven_win_rate": "{:.2%}".format,
            "tp_rate_vs_breakeven": "{:.2%}".format,
            "expectancy_units": "{:.3f}".format,
            "event_step_mean": "{:.2f}".format,
            "event_step_median": "{:.2f}".format,
        },
    )
)

f.write("\n\n")
f.write("Top 20 variantes por expectativa simple\n")
f.write("-" * 100 + "\n")
f.write(
    df_bar_distribution_ranking[
        [
            "bar_col",
            "horizon",
            "percentile",
            "tp_mult",
            "sl_mult",
            "tp_rate",
            "sl_rate",
            "no_event_rate",
            "tp_share_over_resolved",
            "breakeven_win_rate",
            "expectancy_units",
            "event_step_mean",
        ]
    ].head(20)
    .to_string(
        index=False,
        formatters={
            "tp_rate": "{:.2%}".format,
            "sl_rate": "{:.2%}".format,
            "no_event_rate": "{:.2%}".format,
            "tp_share_over_resolved": "{:.2%}".format,
            "breakeven_win_rate": "{:.2%}".format,
            "expectancy_units": "{:.3f}".format,
            "event_step_mean": "{:.2f}".format,
        },
    )
)

```

```
# -----  
# Confirmación final  
# -----  
  
print("\nArchivos guardados:")  
print(f"Distribución global BAR:          {MNQ_TARGETS_BAR_GLOBAL_DISTRIBUTION_CSV}")  
print(f"Resumen por configuración:        {MNQ_TARGETS_BAR_DISTRIBUTION_BY_CONFIG_C}")  
print(f"Resumen por target base:            {MNQ_TARGETS_BAR_DISTRIBUTION_BY_TARGET_B}")  
print(f"Ranking BAR:                        {MNQ_TARGETS_BAR_DISTRIBUTION_RANKING_CSV}")  
print(f"Reporte TXT:                        {MNQ_TARGETS_BAR_DISTRIBUTION_REPORT_TXT}")  
  
print("\nCelda 4 finalizada correctamente.")
```

## Guardado y tabla final de distribución global - BAR

=====

Tabla compacta para observaciones:

| tp_sl_config    | tp_mult                     | sl_mult            | n_targets                 | tp_rate_mean          | sl_rate_mean    | no_event_r |
|-----------------|-----------------------------|--------------------|---------------------------|-----------------------|-----------------|------------|
| ate_mean        | tp_share_over_resolved_mean | breakeven_win_rate | tp_rate_vs_breakeven_mean | expectancy_units_mean | event_step_mean |            |
| TP 1.0 / SL 0.5 | 1.0                         | 0.5                | 9                         | 81.07%                | 18.93%          |            |
| 0.00%           |                             | 81.07%             |                           | 33.33%                |                 | 47.74%     |
| 0.716           | 24.41                       |                    |                           |                       |                 |            |
| TP 1.5 / SL 0.5 | 1.5                         | 0.5                | 9                         | 44.39%                | 22.71%          |            |
| 32.89%          |                             | 66.29%             |                           | 25.00%                |                 | 19.39%     |
| 0.552           | 26.43                       |                    |                           |                       |                 |            |
| TP 2.0 / SL 0.5 | 2.0                         | 0.5                | 9                         | 25.21%                | 24.24%          |            |
| 50.55%          |                             | 51.08%             |                           | 20.00%                |                 | 5.21%      |
| 0.383           | 26.20                       |                    |                           |                       |                 |            |
| TP 1.0 / SL 1.0 | 1.0                         | 1.0                | 9                         | 95.18%                | 4.82%           |            |
| 0.00%           |                             | 95.18%             |                           | 50.00%                |                 | 45.18%     |
| 0.904           | 28.32                       |                    |                           |                       |                 |            |
| TP 1.5 / SL 1.0 | 1.5                         | 1.0                | 9                         | 52.73%                | 5.85%           |            |
| 41.42%          |                             | 90.13%             |                           | 40.00%                |                 | 12.73%     |
| 0.733           | 31.56                       |                    |                           |                       |                 |            |
| TP 2.0 / SL 1.0 | 2.0                         | 1.0                | 9                         | 30.26%                | 6.65%           |            |
| 63.10%          |                             | 82.20%             |                           | 33.33%                |                 | -3.08%     |
| 0.539           | 33.08                       |                    |                           |                       |                 |            |
| TP 2.5 / SL 1.0 | 2.5                         | 1.0                | 9                         | 18.02%                | 7.06%           |            |
| 74.92%          |                             | 72.07%             |                           | 28.57%                |                 | -10.56%    |
| 0.380           | 33.39                       |                    |                           |                       |                 |            |

Tabla compacta de variantes p50:

| bar_col               | horizon            | tp_mult              | sl_mult          | tp_rate         | sl_rate           | no_event_rate | tp |
|-----------------------|--------------------|----------------------|------------------|-----------------|-------------------|---------------|----|
| _share_over_resolved  | breakeven_win_rate | tp_rate_vs_breakeven | expectancy_units | event_step_mean | event_step_median |               |    |
| bar_p50_h30_tp10_sl05 | 30                 | 1.0                  | 0.5              | 81.93%          | 18.07%            | 0.00%         |    |
| 81.93%                | 33.33%             | 48.59%               |                  | 0.729           |                   | 12.37         |    |
| 11.00                 |                    |                      |                  |                 |                   |               |    |
| bar_p50_h30_tp15_sl05 | 30                 | 1.5                  | 0.5              | 45.74%          | 21.67%            | 32.60%        |    |
| 67.85%                | 25.00%             | 20.74%               |                  | 0.578           |                   | 13.40         |    |
| 12.00                 |                    |                      |                  |                 |                   |               |    |
| bar_p50_h30_tp20_sl05 | 30                 | 2.0                  | 0.5              | 26.31%          | 23.19%            | 50.50%        |    |
| 53.15%                | 20.00%             | 6.31%                |                  | 0.410           |                   | 13.39         |    |
| 12.00                 |                    |                      |                  |                 |                   |               |    |
| bar_p50_h30_tp10_sl10 | 30                 | 1.0                  | 1.0              | 95.26%          | 4.74%             | 0.00%         |    |
| 95.26%                | 50.00%             | 45.26%               |                  | 0.905           |                   | 14.21         |    |
| 13.00                 |                    |                      |                  |                 |                   |               |    |
| bar_p50_h30_tp15_sl10 | 30                 | 1.5                  | 1.0              | 53.60%          | 5.71%             | 40.70%        |    |
| 90.38%                | 40.00%             | 13.60%               |                  | 0.747           |                   | 15.74         |    |
| 15.00                 |                    |                      |                  |                 |                   |               |    |
| bar_p50_h30_tp20_sl10 | 30                 | 2.0                  | 1.0              | 31.07%          | 6.49%             | 62.44%        |    |
| 82.73%                | 33.33%             | -2.26%               |                  | 0.557           |                   | 16.50         |    |
| 17.00                 |                    |                      |                  |                 |                   |               |    |
| bar_p50_h30_tp25_sl10 | 30                 | 2.5                  | 1.0              | 18.63%          | 6.90%             | 74.47%        |    |
| 72.97%                | 28.57%             | -9.94%               |                  | 0.397           |                   | 16.70         |    |
| 17.00                 |                    |                      |                  |                 |                   |               |    |
| bar_p50_h60_tp10_sl05 | 60                 | 1.0                  | 0.5              | 81.32%          | 18.68%            | 0.00%         |    |
| 81.32%                | 33.33%             | 47.99%               |                  | 0.720           |                   | 24.40         |    |
| 21.00                 |                    |                      |                  |                 |                   |               |    |
| bar_p50_h60_tp15_sl05 | 60                 | 1.5                  | 0.5              | 44.00%          | 22.52%            | 33.48%        |    |
| 66.15%                | 25.00%             | 19.00%               |                  | 0.547           |                   | 26.28         |    |
| 24.00                 |                    |                      |                  |                 |                   |               |    |

|                       |        |     |         |        |        |        |
|-----------------------|--------|-----|---------|--------|--------|--------|
| bar_p50_h60_tp20_sl05 | 60     | 2.0 | 0.5     | 24.58% | 23.99% | 51.44% |
| 50.61%                | 20.00% |     | 4.58%   |        | 0.372  | 25.95  |
| 23.00                 |        |     |         |        |        |        |
| bar_p50_h60_tp10_sl10 | 60     | 1.0 | 1.0     | 95.45% | 4.55%  | 0.00%  |
| 95.45%                | 50.00% |     | 45.45%  |        | 0.909  | 28.22  |
| 26.00                 |        |     |         |        |        |        |
| bar_p50_h60_tp15_sl10 | 60     | 1.5 | 1.0     | 52.23% | 5.54%  | 42.23% |
| 90.41%                | 40.00% |     | 12.23%  |        | 0.728  | 31.28  |
| 31.00                 |        |     |         |        |        |        |
| bar_p50_h60_tp20_sl10 | 60     | 2.0 | 1.0     | 29.49% | 6.29%  | 64.22% |
| 82.42%                | 33.33% |     | -3.84%  |        | 0.527  | 32.68  |
| 33.00                 |        |     |         |        |        |        |
| bar_p50_h60_tp25_sl10 | 60     | 2.5 | 1.0     | 17.33% | 6.69%  | 75.98% |
| 72.16%                | 28.57% |     | -11.24% |        | 0.366  | 32.87  |
| 34.00                 |        |     |         |        |        |        |
| bar_p50_h90_tp10_sl05 | 90     | 1.0 | 0.5     | 80.23% | 19.77% | 0.00%  |
| 80.23%                | 33.33% |     | 46.89%  |        | 0.703  | 36.58  |
| 32.00                 |        |     |         |        |        |        |
| bar_p50_h90_tp15_sl05 | 90     | 1.5 | 0.5     | 43.60% | 23.71% | 32.70% |
| 64.78%                | 25.00% |     | 18.60%  |        | 0.535  | 39.72  |
| 36.00                 |        |     |         |        |        |        |
| bar_p50_h90_tp20_sl05 | 90     | 2.0 | 0.5     | 24.52% | 25.25% | 50.23% |
| 49.26%                | 20.00% |     | 4.52%   |        | 0.364  | 39.31  |
| 35.00                 |        |     |         |        |        |        |
| bar_p50_h90_tp10_sl10 | 90     | 1.0 | 1.0     | 95.11% | 4.89%  | 0.00%  |
| 95.11%                | 50.00% |     | 45.11%  |        | 0.902  | 42.70  |
| 40.00                 |        |     |         |        |        |        |
| bar_p50_h90_tp15_sl10 | 90     | 1.5 | 1.0     | 52.38% | 5.93%  | 41.69% |
| 89.83%                | 40.00% |     | 12.38%  |        | 0.726  | 47.81  |
| 48.00                 |        |     |         |        |        |        |
| bar_p50_h90_tp20_sl10 | 90     | 2.0 | 1.0     | 29.78% | 6.77%  | 63.45% |
| 81.48%                | 33.33% |     | -3.56%  |        | 0.528  | 50.14  |
| 51.00                 |        |     |         |        |        |        |
| bar_p50_h90_tp25_sl10 | 90     | 2.5 | 1.0     | 17.65% | 7.17%  | 75.18% |
| 71.12%                | 28.57% |     | -10.92% |        | 0.370  | 50.79  |
| 53.00                 |        |     |         |        |        |        |

Top 20 variantes por expectativa simple:

| bar_col               | horizon                | percentile         | tp_mult          | sl_mult      | tp_rate | sl_rate | no_e |
|-----------------------|------------------------|--------------------|------------------|--------------|---------|---------|------|
| vent_rate             | tp_share_over_resolved | breakeven_win_rate | expectancy_units | event_step_m | ean     |         |      |
| bar_p60_h60_tp10_sl10 | 60                     | 60                 | 1.0              | 1.0          | 96.62%  | 3.38%   |      |
| 0.00%                 | 96.62%                 | 50.00%             |                  | 0.932        |         | 30.03   |      |
| bar_p60_h30_tp10_sl10 | 30                     | 60                 | 1.0              | 1.0          | 96.41%  | 3.59%   |      |
| 0.00%                 | 96.41%                 | 50.00%             |                  | 0.928        |         | 15.14   |      |
| bar_p60_h90_tp10_sl10 | 90                     | 60                 | 1.0              | 1.0          | 96.37%  | 3.63%   |      |
| 0.00%                 | 96.37%                 | 50.00%             |                  | 0.927        |         | 45.68   |      |
| bar_p50_h60_tp10_sl10 | 60                     | 50                 | 1.0              | 1.0          | 95.45%  | 4.55%   |      |
| 0.00%                 | 95.45%                 | 50.00%             |                  | 0.909        |         | 28.22   |      |
| bar_p50_h30_tp10_sl10 | 30                     | 50                 | 1.0              | 1.0          | 95.26%  | 4.74%   |      |
| 0.00%                 | 95.26%                 | 50.00%             |                  | 0.905        |         | 14.21   |      |
| bar_p50_h90_tp10_sl10 | 90                     | 50                 | 1.0              | 1.0          | 95.11%  | 4.89%   |      |
| 0.00%                 | 95.11%                 | 50.00%             |                  | 0.902        |         | 42.70   |      |
| bar_p40_h60_tp10_sl10 | 60                     | 40                 | 1.0              | 1.0          | 94.00%  | 6.00%   |      |
| 0.00%                 | 94.00%                 | 50.00%             |                  | 0.880        |         | 26.19   |      |
| bar_p40_h30_tp10_sl10 | 30                     | 40                 | 1.0              | 1.0          | 93.84%  | 6.16%   |      |
| 0.00%                 | 93.84%                 | 50.00%             |                  | 0.877        |         | 13.20   |      |
| bar_p40_h90_tp10_sl10 | 90                     | 40                 | 1.0              | 1.0          | 93.57%  | 6.43%   |      |
| 0.00%                 | 93.57%                 | 50.00%             |                  | 0.871        |         | 39.52   |      |
| bar_p40_h30_tp15_sl10 | 30                     | 40                 | 1.5              | 1.0          | 57.52%  | 7.35%   |      |

|                       |        |        |            |        |
|-----------------------|--------|--------|------------|--------|
| 35.13%                | 88.67% | 40.00% | 0.789      | 14.83  |
| bar_p60_h30_tp10_sl05 | 30     | 60 1.0 | 0.5 84.78% | 15.22% |
| 0.00%                 | 84.78% | 33.33% | 0.772      | 13.51  |
| bar_p40_h60_tp15_sl10 | 60     | 40 1.5 | 1.0 56.30% | 7.31%  |
| 36.39%                | 88.51% | 40.00% | 0.771      | 29.41  |
| bar_p60_h60_tp10_sl05 | 60     | 60 1.0 | 0.5 84.19% | 15.81% |
| 0.00%                 | 84.19% | 33.33% | 0.763      | 26.62  |
| bar_p40_h90_tp15_sl10 | 90     | 40 1.5 | 1.0 55.98% | 7.76%  |
| 36.26%                | 87.83% | 40.00% | 0.762      | 44.77  |
| bar_p60_h90_tp10_sl05 | 90     | 60 1.0 | 0.5 83.19% | 16.81% |
| 0.00%                 | 83.19% | 33.33% | 0.748      | 40.14  |
| bar_p50_h30_tp15_sl10 | 30     | 50 1.5 | 1.0 53.60% | 5.71%  |
| 40.70%                | 90.38% | 40.00% | 0.747      | 15.74  |
| bar_p50_h30_tp10_sl05 | 30     | 50 1.0 | 0.5 81.93% | 18.07% |
| 0.00%                 | 81.93% | 33.33% | 0.729      | 12.37  |
| bar_p50_h60_tp15_sl10 | 60     | 50 1.5 | 1.0 52.23% | 5.54%  |
| 42.23%                | 90.41% | 40.00% | 0.728      | 31.28  |
| bar_p50_h90_tp15_sl10 | 90     | 50 1.5 | 1.0 52.38% | 5.93%  |
| 41.69%                | 89.83% | 40.00% | 0.726      | 47.81  |
| bar_p50_h60_tp10_sl05 | 60     | 50 1.0 | 0.5 81.32% | 18.68% |
| 0.00%                 | 81.32% | 33.33% | 0.720      | 24.40  |

Archivos guardados:

Distribución global BAR: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_global\_distribution.csv  
 Resumen por configuración: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_distribution\_by\_config.csv  
 Resumen por target base: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_distribution\_by\_target\_base.csv  
 Ranking BAR: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_distribution\_ranking.csv  
 Reporte TXT: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_distribution\_report.txt

Celda 4 finalizada correctamente.

## Observaciones preliminares — Distribución global de resultados BAR

La distribución global de **BAR** muestra un comportamiento coherente con la lógica definida para las barreras TP/SL.

En primer lugar, las configuraciones con **TP = 1.0 × threshold** presentan **no\_event = 0%**. Esto es esperable, porque **DIR** ya exige que exista una excursión favorable igual o superior al threshold base. Por lo tanto, cuando el TP se ubica exactamente en esa distancia, el evento necesariamente se resuelve como TP primero o SL primero.

La configuración **TP 1.0 / SL 1.0** muestra la mayor tasa de TP primero, con un promedio cercano al 95%. Sin embargo, este resultado debe interpretarse como un benchmark operativo, no como una señal definitiva de calidad, ya que está muy favorecido por la propia construcción de **DIR**.

Al aumentar la distancia del Take Profit, el target se vuelve más exigente. Esto se observa claramente en las configuraciones con **SL = 1.0**:

TP 1.0 / SL 1.0 → TP alto y no\_event nulo  
 TP 1.5 / SL 1.0 → TP intermedio y no\_event moderado  
 TP 2.0 / SL 1.0 → TP bajo y no\_event alto  
 TP 2.5 / SL 1.0 → TP muy bajo y no\_event muy alto

Esto indica que objetivos demasiado amplios no suelen alcanzarse dentro del horizonte definido. En particular, TP 2.0 / SL 1.0 y TP 2.5 / SL 1.0 dejan una gran proporción de casos sin resolución, por lo que parecen configuraciones demasiado exigentes para esta primera etapa.

Las configuraciones con SL = 0.5 muestran una lógica diferente. Al tener el Stop Loss más cerca, aumenta la proporción de SL primero. Esto es esperable: cuanto más ajustado está el stop, más fácil es que el precio lo toque antes de alcanzar el TP.

Entre las configuraciones evaluadas, TP 1.5 / SL 1.0 aparece como una de las alternativas más interesantes para continuar el análisis. No es tan fácil como TP 1.0 / SL 1.0, pero todavía mantiene una proporción razonable de TP primero, un SL bajo y una proporción de eventos resueltos aceptable.

Las variantes p50 muestran un comportamiento bastante consistente entre horizontes H30, H60 y H90. Para TP 1.5 / SL 1.0, las tasas de TP se mantienen alrededor del 52%–54%, con SL cercano al 5%–6% y no\_event alrededor del 41%–42%. Esto sugiere que la configuración es relativamente estable entre horizontes.

El ranking por expectativa simple favorece principalmente las variantes TP 1.0 / SL 1.0, especialmente en p60. Sin embargo, esto no debe interpretarse automáticamente como la mejor opción, porque esa configuración está directamente relacionada con el threshold base usado para construir DIR. Por eso, su utilidad principal es servir como referencia.

En síntesis, la distribución global de BAR confirma que el target está funcionando de forma lógica: TP cercanos se alcanzan con mayor frecuencia, TP lejanos generan más casos sin evento y stops más ajustados aumentan la probabilidad de SL. Para los próximos análisis, conviene mantener como benchmark TP 1.0 / SL 1.0, pero prestar especial atención a TP 1.5 / SL 1.0 como candidata operativa principal.

La lectura simple: **seguimos mirando todas**, pero el foco práctico debería estar en:

Benchmark: TP 1.0 / SL 1.0  
 Candidata fuerte: TP 1.5 / SL 1.0  
 Candidata dura: TP 2.0 / SL 1.0

TP 2.5 / SL 1.0 parece demasiado exigente, y las de SL 0.5 son útiles como comparación, pero más agresivas.

## 4.7. Frecuencia de TP y SL

En este punto se analiza específicamente la frecuencia con la que los targets BAR alcanzan primero el Take Profit o el Stop Loss.



A diferencia del punto anterior, donde se revisó la distribución global incluyendo `no_event`, aquí el foco estará en comparar directamente la relación entre TP y SL.

El objetivo es identificar qué configuraciones generan una proporción favorable de TP primero, cuáles producen demasiados SL y cuáles mantienen una relación TP/SL razonable para continuar hacia análisis más detallados.

## Celda 1 — Preparación de métricas TP/SL

```
In [99]: # Celda 1 — Preparación de métricas TP/SL
# =====

print("4.7. Frecuencia de TP y SL - BAR")
print("=" * 80)

# -----
# Validar tabla base
# -----

if "df_bar_global_distribution" not in globals():
    raise NameError("No existe df_bar_global_distribution. Primero debes correr

# -----
# Copia de trabajo
# -----

df_bar_tp_sl_frequency = df_bar_global_distribution.copy()

# -----
# Métricas específicas TP/SL
# -----

# Casos que resolvieron: TP o SL
df_bar_tp_sl_frequency["resolved_rate"] = (
    df_bar_tp_sl_frequency["tp_rate"]
    + df_bar_tp_sl_frequency["sl_rate"]
)

# Diferencia directa entre TP y SL
df_bar_tp_sl_frequency["tp_minus_sl_rate"] = (
    df_bar_tp_sl_frequency["tp_rate"]
    - df_bar_tp_sl_frequency["sl_rate"]
)

# Relación TP / SL.
# Ejemplo: 10 significa que hubo 10 veces más TP que SL.
df_bar_tp_sl_frequency["tp_to_sl_ratio"] = np.where(
    df_bar_tp_sl_frequency["sl_rate"] > 0,
    df_bar_tp_sl_frequency["tp_rate"] / df_bar_tp_sl_frequency["sl_rate"],
    np.nan
)

# TP entre eventos resueltos.
# Esto ya existía como tp_share_over_resolved,
# pero lo renombramos conceptualmente para esta sección.
df_bar_tp_sl_frequency["tp_rate_resolved_only"] = (
    df_bar_tp_sl_frequency["tp_share_over_resolved"]
```

```

)

df_bar_tp_sl_frequency["sl_rate_resolved_only"] = (
    1 - df_bar_tp_sl_frequency["tp_share_over_resolved"]
)

# -----
# Clasificación simple del balance TP/SL
# -----

def classify_tp_sl_quality(row):
    """
    Clasificación simple y preliminar de la relación TP/SL.
    No selecciona target final, solo ayuda a leer la tabla.
    """

    tp_resolved = row["tp_rate_resolved_only"]
    no_event = row["no_event_rate"]

    if pd.isna(tp_resolved):
        return "sin_eventos"

    if tp_resolved >= 0.85 and no_event <= 0.45:
        return "muy_favorable"

    if tp_resolved >= 0.70 and no_event <= 0.65:
        return "favorable"

    if tp_resolved >= 0.55:
        return "moderada"

    return "débil"

df_bar_tp_sl_frequency["tp_sl_quality"] = df_bar_tp_sl_frequency.apply(
    classify_tp_sl_quality,
    axis=1
)

# -----
# Orden de columnas para análisis
# -----

tp_sl_frequency_cols = [
    "bar_col",
    "target_base",
    "horizon",
    "percentile",
    "tp_mult",
    "sl_mult",
    "tp_sl_config",
    "valid_rows",
    "tp_rate",
    "sl_rate",
    "no_event_rate",
    "resolved_rate",
    "tp_minus_sl_rate",
    "tp_to_sl_ratio",
    "tp_rate_resolved_only",
    "sl_rate_resolved_only",

```

```

        "breakeven_win_rate",
        "tp_rate_vs_breakeven",
        "expectancy_units",
        "tp_sl_quality",
    ]

df_bar_tp_sl_frequency = (
    df_bar_tp_sl_frequency[tp_sl_frequency_cols]
    .sort_values(["percentile", "horizon", "sl_mult", "tp_mult"])
    .reset_index(drop=True)
)

# -----
# Vista inicial
# -----

print(f"Targets BAR analizados: {len(df_bar_tp_sl_frequency)}")

print("\nVista inicial de frecuencia TP/SL:")
print(
    df_bar_tp_sl_frequency
    .head(21)
    .to_string(
        index=False,
        formatters={
            "tp_rate": "{:.2%}".format,
            "sl_rate": "{:.2%}".format,
            "no_event_rate": "{:.2%}".format,
            "resolved_rate": "{:.2%}".format,
            "tp_minus_sl_rate": "{:.2%}".format,
            "tp_to_sl_ratio": "{:.2f}".format,
            "tp_rate_resolved_only": "{:.2%}".format,
            "sl_rate_resolved_only": "{:.2%}".format,
            "breakeven_win_rate": "{:.2%}".format,
            "tp_rate_vs_breakeven": "{:.2%}".format,
            "expectancy_units": "{:.3f}".format,
        },
    )
)

print("\nPreparación de métricas TP/SL finalizada correctamente.")

```

## 4.7. Frecuencia de TP y SL - BAR

=====

Targets BAR analizados: 63

Vista inicial de frecuencia TP/SL:

| bar_col               | target_base           | horizon               | percentile         | tp_mult              | sl_mult          | tp_sl            |
|-----------------------|-----------------------|-----------------------|--------------------|----------------------|------------------|------------------|
| _config               | valid_rows            | tp_rate               | sl_rate            | no_event_rate        | resolved_rate    | tp_minus_sl_rate |
| tp_to_sl_ratio        | tp_rate_resolved_only | sl_rate_resolved_only | breakeven_win_rate | tp_rate_vs_breakeven | expectancy_units | tp_sl_quality    |
| bar_p40_h30_tp10_sl05 | p40_h30               | 30                    | 40                 | 1.0                  | 0.5              | TP 1.0 /         |
| SL 0.5                | 623736                | 78.98%                | 21.02%             | 0.00%                | 100.00%          | 57.97%           |
| 3.76                  | 78.98%                | 21.02%                | 33.33%             |                      |                  |                  |
| 45.65%                | 0.685                 | favorable             |                    |                      |                  |                  |
| bar_p40_h30_tp15_sl05 | p40_h30               | 30                    | 40                 | 1.5                  | 0.5              | TP 1.5 /         |
| SL 0.5                | 623736                | 48.14%                | 25.07%             | 26.79%               | 73.21%           | 23.07%           |
| 1.92                  | 65.76%                | 34.24%                | 25.00%             |                      |                  |                  |
| 23.14%                | 0.597                 | moderada              |                    |                      |                  |                  |
| bar_p40_h30_tp20_sl05 | p40_h30               | 30                    | 40                 | 2.0                  | 0.5              | TP 2.0 /         |
| SL 0.5                | 623736                | 29.68%                | 26.94%             | 43.38%               | 56.62%           | 2.75%            |
| 1.10                  | 52.42%                | 47.58%                | 20.00%             |                      |                  |                  |
| 9.68%                 | 0.459                 | débil                 |                    |                      |                  |                  |
| bar_p40_h30_tp10_sl10 | p40_h30               | 30                    | 40                 | 1.0                  | 1.0              | TP 1.0 /         |
| SL 1.0                | 623736                | 93.84%                | 6.16%              | 0.00%                | 100.00%          | 87.69%           |
| 15.24                 | 93.84%                | 6.16%                 | 50.00%             |                      |                  |                  |
| 43.84%                | 0.877                 | muy_favorable         |                    |                      |                  |                  |
| bar_p40_h30_tp15_sl10 | p40_h30               | 30                    | 40                 | 1.5                  | 1.0              | TP 1.5 /         |
| SL 1.0                | 623736                | 57.52%                | 7.35%              | 35.13%               | 64.87%           | 50.17%           |
| 7.82                  | 88.67%                | 11.33%                | 40.00%             |                      |                  |                  |
| 17.52%                | 0.789                 | muy_favorable         |                    |                      |                  |                  |
| bar_p40_h30_tp20_sl10 | p40_h30               | 30                    | 40                 | 2.0                  | 1.0              | TP 2.0 /         |
| SL 1.0                | 623736                | 35.63%                | 8.35%              | 56.03%               | 43.97%           | 27.28%           |
| 4.27                  | 81.02%                | 18.98%                | 33.33%             |                      |                  |                  |
| 2.29%                 | 0.629                 | favorable             |                    |                      |                  |                  |
| bar_p40_h30_tp25_sl10 | p40_h30               | 30                    | 40                 | 2.5                  | 1.0              | TP 2.5 /         |
| SL 1.0                | 623736                | 22.59%                | 8.92%              | 68.49%               | 31.51%           | 13.67%           |
| 2.53                  | 71.69%                | 28.31%                | 28.57%             |                      |                  |                  |
| -5.98%                | 0.476                 | moderada              |                    |                      |                  |                  |
| bar_p40_h60_tp10_sl05 | p40_h60               | 60                    | 40                 | 1.0                  | 0.5              | TP 1.0 /         |
| SL 0.5                | 597024                | 77.94%                | 22.06%             | 0.00%                | 100.00%          | 55.88%           |
| 3.53                  | 77.94%                | 22.06%                | 33.33%             |                      |                  |                  |
| 44.61%                | 0.669                 | favorable             |                    |                      |                  |                  |
| bar_p40_h60_tp15_sl05 | p40_h60               | 60                    | 40                 | 1.5                  | 0.5              | TP 1.5 /         |
| SL 0.5                | 597024                | 46.45%                | 26.43%             | 27.12%               | 72.88%           | 20.02%           |
| 1.76                  | 63.74%                | 36.26%                | 25.00%             |                      |                  |                  |
| 21.45%                | 0.565                 | moderada              |                    |                      |                  |                  |
| bar_p40_h60_tp20_sl05 | p40_h60               | 60                    | 40                 | 2.0                  | 0.5              | TP 2.0 /         |
| SL 0.5                | 597024                | 27.84%                | 28.31%             | 43.85%               | 56.15%           | -0.47%           |
| 0.98                  | 49.58%                | 50.42%                | 20.00%             |                      |                  |                  |
| 7.84%                 | 0.415                 | débil                 |                    |                      |                  |                  |
| bar_p40_h60_tp10_sl10 | p40_h60               | 60                    | 40                 | 1.0                  | 1.0              | TP 1.0 /         |
| SL 1.0                | 597024                | 94.00%                | 6.00%              | 0.00%                | 100.00%          | 88.00%           |
| 15.66                 | 94.00%                | 6.00%                 | 50.00%             |                      |                  |                  |
| 44.00%                | 0.880                 | muy_favorable         |                    |                      |                  |                  |
| bar_p40_h60_tp15_sl10 | p40_h60               | 60                    | 40                 | 1.5                  | 1.0              | TP 1.5 /         |
| SL 1.0                | 597024                | 56.30%                | 7.31%              | 36.39%               | 63.61%           | 48.99%           |
| 7.70                  | 88.51%                | 11.49%                | 40.00%             |                      |                  |                  |
| 16.30%                | 0.771                 | muy_favorable         |                    |                      |                  |                  |
| bar_p40_h60_tp20_sl10 | p40_h60               | 60                    | 40                 | 2.0                  | 1.0              | TP 2.0 /         |
| SL 1.0                | 597024                | 34.08%                | 8.30%              | 57.61%               | 42.39%           | 25.78%           |
| 4.10                  | 80.41%                | 19.59%                | 33.33%             |                      |                  |                  |

|                       |         |               |        |        |         |          |  |
|-----------------------|---------|---------------|--------|--------|---------|----------|--|
| 0.75%                 | 0.599   | favorable     |        |        |         |          |  |
| bar_p40_h60_tp25_sl10 | p40_h60 | 60            | 40     | 2.5    | 1.0     | TP 2.5 / |  |
| SL 1.0                | 597024  | 21.05%        | 8.84%  | 70.11% | 29.89%  | 12.21%   |  |
| 2.38                  | 70.42%  |               | 29.58% | 28.57% |         |          |  |
| -7.52%                | 0.438   | moderada      |        |        |         |          |  |
| bar_p40_h90_tp10_sl05 | p40_h90 | 90            | 40     | 1.0    | 0.5     | TP 1.0 / |  |
| SL 0.5                | 567970  | 77.10%        | 22.90% | 0.00%  | 100.00% | 54.20%   |  |
| 3.37                  | 77.10%  |               | 22.90% | 33.33% |         |          |  |
| 43.77%                | 0.656   | favorable     |        |        |         |          |  |
| bar_p40_h90_tp15_sl05 | p40_h90 | 90            | 40     | 1.5    | 0.5     | TP 1.5 / |  |
| SL 0.5                | 567970  | 45.52%        | 27.32% | 27.16% | 72.84%  | 18.20%   |  |
| 1.67                  | 62.49%  |               | 37.51% | 25.00% |         |          |  |
| 20.52%                | 0.546   | moderada      |        |        |         |          |  |
| bar_p40_h90_tp20_sl05 | p40_h90 | 90            | 40     | 2.0    | 0.5     | TP 2.0 / |  |
| SL 0.5                | 567970  | 27.45%        | 29.21% | 43.34% | 56.66%  | -1.76%   |  |
| 0.94                  | 48.45%  |               | 51.55% | 20.00% |         |          |  |
| 7.45%                 | 0.403   | débil         |        |        |         |          |  |
| bar_p40_h90_tp10_sl10 | p40_h90 | 90            | 40     | 1.0    | 1.0     | TP 1.0 / |  |
| SL 1.0                | 567970  | 93.57%        | 6.43%  | 0.00%  | 100.00% | 87.13%   |  |
| 14.54                 | 93.57%  |               | 6.43%  | 50.00% |         |          |  |
| 43.57%                | 0.871   | muy_favorable |        |        |         |          |  |
| bar_p40_h90_tp15_sl10 | p40_h90 | 90            | 40     | 1.5    | 1.0     | TP 1.5 / |  |
| SL 1.0                | 567970  | 55.98%        | 7.76%  | 36.26% | 63.74%  | 48.22%   |  |
| 7.22                  | 87.83%  |               | 12.17% | 40.00% |         |          |  |
| 15.98%                | 0.762   | muy_favorable |        |        |         |          |  |
| bar_p40_h90_tp20_sl10 | p40_h90 | 90            | 40     | 2.0    | 1.0     | TP 2.0 / |  |
| SL 1.0                | 567970  | 34.09%        | 8.83%  | 57.08% | 42.92%  | 25.26%   |  |
| 3.86                  | 79.43%  |               | 20.57% | 33.33% |         |          |  |
| 0.76%                 | 0.594   | favorable     |        |        |         |          |  |
| bar_p40_h90_tp25_sl10 | p40_h90 | 90            | 40     | 2.5    | 1.0     | TP 2.5 / |  |
| SL 1.0                | 567970  | 21.19%        | 9.43%  | 69.38% | 30.62%  | 11.75%   |  |
| 2.25                  | 69.19%  |               | 30.81% | 28.57% |         |          |  |
| -7.39%                | 0.435   | moderada      |        |        |         |          |  |

Preparación de métricas TP/SL finalizada correctamente.

## Celda 2 — Tablas agregadas por configuración, percentil y horizonte

In [100...

```
# Celda 2 - Tablas agregadas de frecuencia TP/SL
# =====

print("Tablas agregadas de frecuencia TP/SL - BAR")
print("=" * 80)

# -----
# Validar tabla base
# -----

if "df_bar_tp_sl_frequency" not in globals():
    raise NameError("No existe df_bar_tp_sl_frequency. Primero debes correr la C")

# -----
# 1. Resumen por configuración TP/SL
# -----

df_bar_tp_sl_by_config = (
    df_bar_tp_sl_frequency
    .groupby(["tp_mult", "sl_mult", "tp_sl_config"], as_index=False)
```

```

    .agg(
        n_targets=("bar_col", "count"),
        tp_rate_mean=("tp_rate", "mean"),
        sl_rate_mean=("sl_rate", "mean"),
        no_event_rate_mean=("no_event_rate", "mean"),
        resolved_rate_mean=("resolved_rate", "mean"),
        tp_minus_sl_rate_mean=("tp_minus_sl_rate", "mean"),
        tp_to_sl_ratio_mean=("tp_to_sl_ratio", "mean"),
        tp_rate_resolved_only_mean=("tp_rate_resolved_only", "mean"),
        sl_rate_resolved_only_mean=("sl_rate_resolved_only", "mean"),
        expectancy_units_mean=("expectancy_units", "mean"),
    )
    .sort_values(["sl_mult", "tp_mult"])
    .reset_index(drop=True)
)

print("\nResumen por configuración TP/SL:")
print(
    df_bar_tp_sl_by_config.to_string(
        index=False,
        formatters={
            "tp_rate_mean": "{:.2%}".format,
            "sl_rate_mean": "{:.2%}".format,
            "no_event_rate_mean": "{:.2%}".format,
            "resolved_rate_mean": "{:.2%}".format,
            "tp_minus_sl_rate_mean": "{:.2%}".format,
            "tp_to_sl_ratio_mean": "{:.2f}".format,
            "tp_rate_resolved_only_mean": "{:.2%}".format,
            "sl_rate_resolved_only_mean": "{:.2%}".format,
            "expectancy_units_mean": "{:.3f}".format,
        }
    )
)

# -----
# 2. Resumen por percentil y horizonte
# -----

df_bar_tp_sl_by_target_base = (
    df_bar_tp_sl_frequency
    .groupby(["percentile", "horizon", "target_base"], as_index=False)
    .agg(
        n_configs=("bar_col", "count"),
        tp_rate_mean=("tp_rate", "mean"),
        sl_rate_mean=("sl_rate", "mean"),
        no_event_rate_mean=("no_event_rate", "mean"),
        resolved_rate_mean=("resolved_rate", "mean"),
        tp_minus_sl_rate_mean=("tp_minus_sl_rate", "mean"),
        tp_to_sl_ratio_mean=("tp_to_sl_ratio", "mean"),
        tp_rate_resolved_only_mean=("tp_rate_resolved_only", "mean"),
        expectancy_units_mean=("expectancy_units", "mean"),
    )
    .sort_values(["percentile", "horizon"])
    .reset_index(drop=True)
)

print("\nResumen por percentil y horizonte:")
print(
    df_bar_tp_sl_by_target_base.to_string(
        index=False,

```

```

        formatters={
            "tp_rate_mean": "{:.2%}".format,
            "sl_rate_mean": "{:.2%}".format,
            "no_event_rate_mean": "{:.2%}".format,
            "resolved_rate_mean": "{:.2%}".format,
            "tp_minus_sl_rate_mean": "{:.2%}".format,
            "tp_to_sl_ratio_mean": "{:.2f}".format,
            "tp_rate_resolved_only_mean": "{:.2%}".format,
            "expectancy_units_mean": "{:.3f}".format,
        }
    )
)

# -----
# 3. Detalle de variantes centrales p50
# -----

df_bar_tp_sl_p50 = (
    df_bar_tp_sl_frequency[
        df_bar_tp_sl_frequency["percentile"] == 50
    ]
    .copy()
    .sort_values(["horizon", "sl_mult", "tp_mult"])
    .reset_index(drop=True)
)

print("\nDetalle de frecuencia TP/SL para variantes p50:")
print(
    df_bar_tp_sl_p50[
        [
            "bar_col",
            "horizon",
            "tp_mult",
            "sl_mult",
            "tp_rate",
            "sl_rate",
            "no_event_rate",
            "resolved_rate",
            "tp_minus_sl_rate",
            "tp_to_sl_ratio",
            "tp_rate_resolved_only",
            "sl_rate_resolved_only",
            "expectancy_units",
            "tp_sl_quality",
        ]
    ].to_string(
        index=False,
        formatters={
            "tp_rate": "{:.2%}".format,
            "sl_rate": "{:.2%}".format,
            "no_event_rate": "{:.2%}".format,
            "resolved_rate": "{:.2%}".format,
            "tp_minus_sl_rate": "{:.2%}".format,
            "tp_to_sl_ratio": "{:.2f}".format,
            "tp_rate_resolved_only": "{:.2%}".format,
            "sl_rate_resolved_only": "{:.2%}".format,
            "expectancy_units": "{:.3f}".format,
        }
    )
)

```

```

# -----
# 4. Ranking por relación TP/SL resuelta
# -----
# Este ranking mira solamente los casos que resolvieron como TP o SL.
# No reemplaza el análisis global, porque puede ocultar muchos no_event.
# -----

df_bar_tp_sl_resolved_ranking = (
    df_bar_tp_sl_frequency
    .copy()
    .sort_values(
        ["tp_rate_resolved_only", "tp_minus_sl_rate", "resolved_rate"],
        ascending=[False, False, False]
    )
    .reset_index(drop=True)
)

print("\nTop 20 por proporción de TP entre eventos resueltos:")
print(
    df_bar_tp_sl_resolved_ranking[
        [
            "bar_col",
            "horizon",
            "percentile",
            "tp_mult",
            "sl_mult",
            "tp_rate",
            "sl_rate",
            "no_event_rate",
            "resolved_rate",
            "tp_rate_resolved_only",
            "tp_to_sl_ratio",
            "tp_sl_quality",
        ]
    ]
    .head(20)
    .to_string(
        index=False,
        formatters={
            "tp_rate": "{:.2%}".format,
            "sl_rate": "{:.2%}".format,
            "no_event_rate": "{:.2%}".format,
            "resolved_rate": "{:.2%}".format,
            "tp_rate_resolved_only": "{:.2%}".format,
            "tp_to_sl_ratio": "{:.2f}".format,
        }
    )
)

# -----
# 5. Conteo de calidad preliminar
# -----

df_bar_tp_sl_quality_counts = (
    df_bar_tp_sl_frequency
    .groupby(["tp_sl_quality"], as_index=False)
    .agg(
        n_targets=("bar_col", "count"),
        tp_rate_mean=("tp_rate", "mean"),
    )
)

```



```
sl_rate_mean=("sl_rate", "mean"),
no_event_rate_mean=("no_event_rate", "mean"),
tp_rate_resolved_only_mean=("tp_rate_resolved_only", "mean"),
expectancy_units_mean=("expectancy_units", "mean"),
)
.sort_values("n_targets", ascending=False)
.reset_index(drop=True)
)

print("\nConteo por clasificación preliminar TP/SL:")
print(
    df_bar_tp_sl_quality_counts.to_string(
        index=False,
        formatters={
            "tp_rate_mean": "{:.2%}".format,
            "sl_rate_mean": "{:.2%}".format,
            "no_event_rate_mean": "{:.2%}".format,
            "tp_rate_resolved_only_mean": "{:.2%}".format,
            "expectancy_units_mean": "{:.3f}".format,
        }
    )
)

print("\nCelda 2 finalizada correctamente.")
```

## Tablas agregadas de frecuencia TP/SL - BAR

=====

## Resumen por configuración TP/SL:

| tp_mult | sl_mult | tp_sl_config    | n_targets | tp_rate_mean | sl_rate_mean | no_event_rate_mean | resolved_rate_mean | tp_minus_sl_rate_mean | tp_to_sl_ratio_mean | tp_rate_resolved_only_mean | sl_rate_resolved_only_mean | expectancy_units_mean |
|---------|---------|-----------------|-----------|--------------|--------------|--------------------|--------------------|-----------------------|---------------------|----------------------------|----------------------------|-----------------------|
| 1.0     | 0.5     | TP 1.0 / SL 0.5 | 9         | 81.07%       | 18.93%       | 0.00%              | 100.00%            | 62.15%                | 4.38                | 0.716                      | 0.552                      | 1.98                  |
| 1.5     | 0.5     | TP 1.5 / SL 0.5 | 9         | 44.39%       | 22.71%       | 32.89%             | 67.11%             | 21.68%                | 1.98                | 0.552                      | 0.383                      | 1.05                  |
| 2.0     | 0.5     | TP 2.0 / SL 0.5 | 9         | 25.21%       | 24.24%       | 50.55%             | 49.45%             | 0.97%                 | 1.05                | 0.383                      | 0.539                      | 2.59                  |
| 1.0     | 1.0     | TP 1.0 / SL 1.0 | 9         | 95.18%       | 4.82%        | 0.00%              | 100.00%            | 90.36%                | 20.89               | 0.904                      | 0.539                      | 2.59                  |
| 1.5     | 1.0     | TP 1.5 / SL 1.0 | 9         | 52.73%       | 5.85%        | 41.42%             | 58.58%             | 46.89%                | 9.36                | 0.733                      | 0.539                      | 2.59                  |
| 2.0     | 1.0     | TP 2.0 / SL 1.0 | 9         | 30.26%       | 6.65%        | 63.10%             | 36.90%             | 23.61%                | 4.66                | 0.539                      | 0.380                      | 2.59                  |
| 2.5     | 1.0     | TP 2.5 / SL 1.0 | 9         | 18.02%       | 7.06%        | 74.92%             | 25.08%             | 10.96%                | 2.59                | 0.380                      |                            |                       |
|         |         |                 |           |              |              | 72.07%             | 27.93%             |                       |                     |                            |                            |                       |

## Resumen por percentil y horizonte:

| percentile | horizon | target_base | n_configs | tp_rate_mean | sl_rate_mean | no_event_rate_mean | resolved_rate_mean | tp_minus_sl_rate_mean | tp_to_sl_ratio_mean | tp_rate_resolved_only_mean | sl_rate_resolved_only_mean | expectancy_units_mean |
|------------|---------|-------------|-----------|--------------|--------------|--------------------|--------------------|-----------------------|---------------------|----------------------------|----------------------------|-----------------------|
| 40         | 30      | p40_h30     | 7         | 52.34%       | 14.83%       | 32.83%             | 67.17%             | 37.51%                | 5.24                | 0.644                      | 0.620                      | 1.98                  |
| 40         | 60      | p40_h60     | 7         | 51.09%       | 15.32%       | 33.58%             | 66.42%             | 35.77%                | 5.16                | 0.620                      | 0.610                      | 1.98                  |
| 40         | 90      | p40_h90     | 7         | 50.70%       | 15.98%       | 33.32%             | 66.68%             | 34.72%                | 4.83                | 0.610                      | 0.610                      | 1.98                  |
| 50         | 30      | p50_h30     | 7         | 50.36%       | 12.40%       | 37.24%             | 62.76%             | 37.97%                | 6.39                | 0.617                      | 0.596                      | 1.98                  |
| 50         | 60      | p50_h60     | 7         | 49.20%       | 12.61%       | 38.19%             | 61.81%             | 36.60%                | 6.43                | 0.596                      | 0.590                      | 1.98                  |
| 50         | 90      | p50_h90     | 7         | 49.04%       | 13.36%       | 37.61%             | 62.39%             | 35.68%                | 6.00                | 0.590                      | 0.572                      | 1.98                  |
| 60         | 30      | p60_h30     | 7         | 48.39%       | 10.19%       | 41.42%             | 58.58%             | 38.20%                | 7.94                | 0.590                      | 0.572                      | 1.98                  |
| 60         | 60      | p60_h60     | 7         | 47.41%       | 10.34%       | 42.25%             | 57.75%             | 37.07%                | 8.15                | 0.572                      | 0.569                      | 1.98                  |
| 60         | 90      | p60_h90     | 7         | 47.43%       | 11.02%       | 41.55%             | 58.45%             | 36.42%                | 7.58                | 0.569                      |                            |                       |
|            |         |             |           |              |              | 77.86%             |                    |                       |                     |                            |                            |                       |

Detalle de frecuencia TP/SL para variantes p50:

| bar_col               | horizon | tp_mult | sl_mult | tp_rate | sl_rate | no_event_rate | re_solved_rate | tp_minus_sl_rate | tp_to_sl_ratio | tp_rate_resolved_only | sl_rate_resolved_only | expectancy_units | tp_sl_quality       |
|-----------------------|---------|---------|---------|---------|---------|---------------|----------------|------------------|----------------|-----------------------|-----------------------|------------------|---------------------|
| bar_p50_h30_tp10_sl05 | 30      | 1.0     | 0.5     | 81.93%  | 18.07%  | 0.00%         | 100.00%        | 63.86%           | 4.53           | 81.93%                | 18.0                  | 7%               | 0.729 favorable     |
| bar_p50_h30_tp15_sl05 | 30      | 1.5     | 0.5     | 45.74%  | 21.67%  | 32.60%        | 67.40%         | 24.07%           | 2.11           | 67.85%                | 32.1                  | 5%               | 0.578 moderada      |
| bar_p50_h30_tp20_sl05 | 30      | 2.0     | 0.5     | 26.31%  | 23.19%  | 50.50%        | 49.50%         | 3.12%            | 1.13           | 53.15%                | 46.8                  | 5%               | 0.410 débil         |
| bar_p50_h30_tp10_sl10 | 30      | 1.0     | 1.0     | 95.26%  | 4.74%   | 0.00%         | 100.00%        | 90.52%           | 20.10          | 95.26%                | 4.7                   | 4%               | 0.905 muy_favorable |
| bar_p50_h30_tp15_sl10 | 30      | 1.5     | 1.0     | 53.60%  | 5.71%   | 40.70%        | 59.30%         | 47.89%           | 9.39           | 90.38%                | 9.6                   | 2%               | 0.747 muy_favorable |
| bar_p50_h30_tp20_sl10 | 30      | 2.0     | 1.0     | 31.07%  | 6.49%   | 62.44%        | 37.56%         | 24.58%           | 4.79           | 82.73%                | 17.2                  | 7%               | 0.557 favorable     |
| bar_p50_h30_tp25_sl10 | 30      | 2.5     | 1.0     | 18.63%  | 6.90%   | 74.47%        | 25.53%         | 11.73%           | 2.70           | 72.97%                | 27.0                  | 3%               | 0.397 moderada      |
| bar_p50_h60_tp10_sl05 | 60      | 1.0     | 0.5     | 81.32%  | 18.68%  | 0.00%         | 100.00%        | 62.64%           | 4.35           | 81.32%                | 18.6                  | 8%               | 0.720 favorable     |
| bar_p50_h60_tp15_sl05 | 60      | 1.5     | 0.5     | 44.00%  | 22.52%  | 33.48%        | 66.52%         | 21.48%           | 1.95           | 66.15%                | 33.8                  | 5%               | 0.547 moderada      |
| bar_p50_h60_tp20_sl05 | 60      | 2.0     | 0.5     | 24.58%  | 23.99%  | 51.44%        | 48.56%         | 0.59%            | 1.02           | 50.61%                | 49.3                  | 9%               | 0.372 débil         |
| bar_p50_h60_tp10_sl10 | 60      | 1.0     | 1.0     | 95.45%  | 4.55%   | 0.00%         | 100.00%        | 90.91%           | 21.00          | 95.45%                | 4.5                   | 5%               | 0.909 muy_favorable |
| bar_p50_h60_tp15_sl10 | 60      | 1.5     | 1.0     | 52.23%  | 5.54%   | 42.23%        | 57.77%         | 46.70%           | 9.43           | 90.41%                | 9.5                   | 9%               | 0.728 muy_favorable |
| bar_p50_h60_tp20_sl10 | 60      | 2.0     | 1.0     | 29.49%  | 6.29%   | 64.22%        | 35.78%         | 23.20%           | 4.69           | 82.42%                | 17.5                  | 8%               | 0.527 favorable     |
| bar_p50_h60_tp25_sl10 | 60      | 2.5     | 1.0     | 17.33%  | 6.69%   | 75.98%        | 24.02%         | 10.65%           | 2.59           | 72.16%                | 27.8                  | 4%               | 0.366 moderada      |
| bar_p50_h90_tp10_sl05 | 90      | 1.0     | 0.5     | 80.23%  | 19.77%  | 0.00%         | 100.00%        | 60.45%           | 4.06           | 80.23%                | 19.7                  | 7%               | 0.703 favorable     |
| bar_p50_h90_tp15_sl05 | 90      | 1.5     | 0.5     | 43.60%  | 23.71%  | 32.70%        | 67.30%         | 19.89%           | 1.84           | 64.78%                | 35.2                  | 2%               | 0.535 moderada      |
| bar_p50_h90_tp20_sl05 | 90      | 2.0     | 0.5     | 24.52%  | 25.25%  | 50.23%        | 49.77%         | -0.74%           | 0.97           | 49.26%                | 50.7                  | 4%               | 0.364 débil         |
| bar_p50_h90_tp10_sl10 | 90      | 1.0     | 1.0     | 95.11%  | 4.89%   | 0.00%         | 100.00%        | 90.23%           | 19.46          | 95.11%                | 4.8                   | 9%               | 0.902 muy_favorable |
| bar_p50_h90_tp15_sl10 | 90      | 1.5     | 1.0     | 52.38%  | 5.93%   | 41.69%        |                |                  |                |                       |                       |                  |                     |

|                       |                     |      |                  |        |
|-----------------------|---------------------|------|------------------|--------|
| 58.31%                | 46.45%              | 8.83 | 89.83%           | 10.1   |
| 7%                    | 0.726 muy_favorable |      |                  |        |
| bar_p50_h90_tp20_sl10 | 90                  | 2.0  | 1.0 29.78% 6.77% | 63.45% |
| 36.55%                | 23.01%              | 4.40 | 81.48%           | 18.5   |
| 2%                    | 0.528 favorable     |      |                  |        |
| bar_p50_h90_tp25_sl10 | 90                  | 2.5  | 1.0 17.65% 7.17% | 75.18% |
| 24.82%                | 10.48%              | 2.46 | 71.12%           | 28.8   |
| 8%                    | 0.370 moderada      |      |                  |        |

Top 20 por proporción de TP entre eventos resueltos:

| bar_col               | horizon       | percentile            | tp_mult        | sl_mult       | tp_rate | sl_rate | no_e |
|-----------------------|---------------|-----------------------|----------------|---------------|---------|---------|------|
| vent_rate             | resolved_rate | tp_rate_resolved_only | tp_to_sl_ratio | tp_sl_quality |         |         |      |
| bar_p60_h60_tp10_sl10 | 60            | 60                    | 1.0            | 1.0           | 96.62%  | 3.38%   |      |
| 0.00%                 | 100.00%       | 96.62%                | 28.60          | muy_favorable |         |         |      |
| bar_p60_h30_tp10_sl10 | 30            | 60                    | 1.0            | 1.0           | 96.41%  | 3.59%   |      |
| 0.00%                 | 100.00%       | 96.41%                | 26.86          | muy_favorable |         |         |      |
| bar_p60_h90_tp10_sl10 | 90            | 60                    | 1.0            | 1.0           | 96.37%  | 3.63%   |      |
| 0.00%                 | 100.00%       | 96.37%                | 26.52          | muy_favorable |         |         |      |
| bar_p50_h60_tp10_sl10 | 60            | 50                    | 1.0            | 1.0           | 95.45%  | 4.55%   |      |
| 0.00%                 | 100.00%       | 95.45%                | 21.00          | muy_favorable |         |         |      |
| bar_p50_h30_tp10_sl10 | 30            | 50                    | 1.0            | 1.0           | 95.26%  | 4.74%   |      |
| 0.00%                 | 100.00%       | 95.26%                | 20.10          | muy_favorable |         |         |      |
| bar_p50_h90_tp10_sl10 | 90            | 50                    | 1.0            | 1.0           | 95.11%  | 4.89%   |      |
| 0.00%                 | 100.00%       | 95.11%                | 19.46          | muy_favorable |         |         |      |
| bar_p40_h60_tp10_sl10 | 60            | 40                    | 1.0            | 1.0           | 94.00%  | 6.00%   |      |
| 0.00%                 | 100.00%       | 94.00%                | 15.66          | muy_favorable |         |         |      |
| bar_p40_h30_tp10_sl10 | 30            | 40                    | 1.0            | 1.0           | 93.84%  | 6.16%   |      |
| 0.00%                 | 100.00%       | 93.84%                | 15.24          | muy_favorable |         |         |      |
| bar_p40_h90_tp10_sl10 | 90            | 40                    | 1.0            | 1.0           | 93.57%  | 6.43%   |      |
| 0.00%                 | 100.00%       | 93.57%                | 14.54          | muy_favorable |         |         |      |
| bar_p60_h60_tp15_sl10 | 60            | 60                    | 1.5            | 1.0           | 48.27%  | 4.14%   |      |
| 47.58%                | 52.42%        | 92.10%                | 11.66          | favorable     |         |         |      |
| bar_p60_h30_tp15_sl10 | 30            | 60                    | 1.5            | 1.0           | 49.53%  | 4.37%   |      |
| 46.11%                | 53.89%        | 91.90%                | 11.34          | favorable     |         |         |      |
| bar_p60_h90_tp15_sl10 | 90            | 60                    | 1.5            | 1.0           | 48.78%  | 4.52%   |      |
| 46.70%                | 53.30%        | 91.52%                | 10.80          | favorable     |         |         |      |
| bar_p50_h60_tp15_sl10 | 60            | 50                    | 1.5            | 1.0           | 52.23%  | 5.54%   |      |
| 42.23%                | 57.77%        | 90.41%                | 9.43           | muy_favorable |         |         |      |
| bar_p50_h30_tp15_sl10 | 30            | 50                    | 1.5            | 1.0           | 53.60%  | 5.71%   |      |
| 40.70%                | 59.30%        | 90.38%                | 9.39           | muy_favorable |         |         |      |
| bar_p50_h90_tp15_sl10 | 90            | 50                    | 1.5            | 1.0           | 52.38%  | 5.93%   |      |
| 41.69%                | 58.31%        | 89.83%                | 8.83           | muy_favorable |         |         |      |
| bar_p40_h30_tp15_sl10 | 30            | 40                    | 1.5            | 1.0           | 57.52%  | 7.35%   |      |
| 35.13%                | 64.87%        | 88.67%                | 7.82           | muy_favorable |         |         |      |
| bar_p40_h60_tp15_sl10 | 60            | 40                    | 1.5            | 1.0           | 56.30%  | 7.31%   |      |
| 36.39%                | 63.61%        | 88.51%                | 7.70           | muy_favorable |         |         |      |
| bar_p40_h90_tp15_sl10 | 90            | 40                    | 1.5            | 1.0           | 55.98%  | 7.76%   |      |
| 36.26%                | 63.74%        | 87.83%                | 7.22           | muy_favorable |         |         |      |
| bar_p60_h30_tp10_sl05 | 30            | 60                    | 1.0            | 0.5           | 84.78%  | 15.22%  |      |
| 0.00%                 | 100.00%       | 84.78%                | 5.57           | favorable     |         |         |      |
| bar_p60_h30_tp20_sl10 | 30            | 60                    | 2.0            | 1.0           | 26.78%  | 4.94%   |      |
| 68.28%                | 31.72%        | 84.42%                | 5.42           | moderada      |         |         |      |

Conteo por clasificación preliminar TP/SL:

| tp_sl_quality | n_targets | tp_rate_mean | sl_rate_mean | no_event_rate_mean | tp_rate_resolved_only_mean | expectancy_units_mean |
|---------------|-----------|--------------|--------------|--------------------|----------------------------|-----------------------|
| moderada      | 20        | 29.84%       | 13.22%       | 56.95%             |                            |                       |
| 71.37%        | 0.463     |              |              |                    |                            |                       |
| favorable     | 19        | 58.60%       | 12.99%       | 28.41%             |                            |                       |
| 82.25%        | 0.658     |              |              |                    |                            |                       |

|               |       |        |        |        |
|---------------|-------|--------|--------|--------|
| muy_favorable | 15    | 78.98% | 5.53%  | 15.49% |
| 92.82%        | 0.844 |        |        |        |
| débil         | 9     | 25.21% | 24.24% | 50.55% |
| 51.08%        | 0.383 |        |        |        |

Celda 2 finalizada correctamente.

## Celda 3 — Gráficos agrupados

In [101...

```
# Celda 3 - Gráficos agrupados
# =====

print("Gráficos agrupados de frecuencia TP/SL - BAR")
print("=" * 80)

# -----
# Validar objetos necesarios
# -----

if "df_bar_tp_sl_by_config" not in globals():
    raise NameError("No existe df_bar_tp_sl_by_config. Primero debes correr la C")

if "df_bar_tp_sl_frequency" not in globals():
    raise NameError("No existe df_bar_tp_sl_frequency. Primero debes correr la C")

if "df_bar_tp_sl_p50" not in globals():
    raise NameError("No existe df_bar_tp_sl_p50. Primero debes correr la Celda 2")

# -----
# Orden de configuraciones TP/SL
# -----

config_order = [
    (1.0, 1.0),
    (1.5, 1.0),
    (2.0, 1.0),
    (2.5, 1.0),
    (1.0, 0.5),
    (1.5, 0.5),
    (2.0, 0.5),
]

config_labels = [
    f"TP {tp} / SL {sl}"
    for tp, sl in config_order
]

df_config_plot = df_bar_tp_sl_by_config.copy()

df_config_plot["config_order"] = df_config_plot.apply(
    lambda row: config_order.index((row["tp_mult"], row["sl_mult"])),
    if (row["tp_mult"], row["sl_mult"]) in config_order
    else 999,
    axis=1
)

df_config_plot = (
    df_config_plot
    .sort_values("config_order")

```

```

        .reset_index(drop=True)
    )

    # =====
    # Figura 1 – TP vs SL vs eventos resueltos por configuración
    # =====

    x = np.arange(len(df_config_plot))
    bar_width = 0.25

    fig, axes = plt.subplots(
        nrows=2,
        ncols=1,
        figsize=(13, 10),
        sharex=True
    )

    # -----
    # Gráfico 1A – TP y SL globales
    # -----

    axes[0].bar(
        x - bar_width / 2,
        df_config_plot["tp_rate_mean"],
        width=bar_width,
        label="TP primero"
    )

    axes[0].bar(
        x + bar_width / 2,
        df_config_plot["sl_rate_mean"],
        width=bar_width,
        label="SL primero"
    )

    axes[0].set_title("Frecuencia promedio de TP y SL por configuración TP/SL")
    axes[0].set_ylabel("Proporción promedio")
    axes[0].legend()

    axes[0].yaxis.set_major_formatter(
        plt.FuncFormatter(lambda y, _: f"{y:.0%}")
    )

    for i, row in df_config_plot.iterrows():
        axes[0].text(
            i - bar_width / 2,
            row["tp_rate_mean"],
            f"{row['tp_rate_mean']:.0%}",
            ha="center",
            va="bottom",
            fontsize=8
        )

        axes[0].text(
            i + bar_width / 2,
            row["sl_rate_mean"],
            f"{row['sl_rate_mean']:.0%}",
            ha="center",
            va="bottom",
            fontsize=8
        )

```

```

    )

# -----
# Gráfico 1B – TP sobre eventos resueltos
# -----

axes[1].bar(
    x,
    df_config_plot["tp_rate_resolved_only_mean"],
    width=0.55,
    label="TP sobre eventos resueltos"
)

axes[1].axhline(
    0.5,
    linestyle="--",
    linewidth=1,
    label="50%"
)

axes[1].set_title("Proporción de TP entre eventos resueltos")
axes[1].set_ylabel("TP / (TP + SL)")
axes[1].set_xticks(x)
axes[1].set_xticklabels(config_labels, rotation=45, ha="right")
axes[1].legend()

axes[1].yaxis.set_major_formatter(
    plt.FuncFormatter(lambda y, _: f"{y:.0%}")
)

for i, row in df_config_plot.iterrows():
    axes[1].text(
        i,
        row["tp_rate_resolved_only_mean"],
        f"{row['tp_rate_resolved_only_mean']:.0%}",
        ha="center",
        va="bottom",
        fontsize=8
    )

plt.tight_layout()
plt.show()

# =====
# Figura 2 – Heatmaps p50 por horizonte y configuración
# =====

df_p50_plot = df_bar_tp_sl_p50.copy()

df_p50_plot["config_label"] = df_p50_plot.apply(
    lambda row: f"TP {row['tp_mult']} / SL {row['sl_mult']}",
    axis=1
)

metrics_to_plot = [
    ("tp_rate", "TP primero"),
    ("sl_rate", "SL primero"),
    ("tp_rate_resolved_only", "TP entre eventos resueltos"),
]

```

```

fig = plt.figure(figsize=(13, 10))

gs = fig.add_gridspec(
    nrows=3,
    ncols=2,
    width_ratios=[20, 1],
    hspace=0.45,
    wspace=0.08
)

axes = [fig.add_subplot(gs[i, 0]) for i in range(3)]
caxes = [fig.add_subplot(gs[i, 1]) for i in range(3)]

for ax, cax, (metric_col, metric_title) in zip(axes, caxes, metrics_to_plot):
    pivot_tmp = df_p50_plot.pivot(
        index="horizon",
        columns="config_label",
        values=metric_col
    )

    ordered_columns = [
        label for label in config_labels
        if label in pivot_tmp.columns
    ]

    pivot_tmp = pivot_tmp.reindex(index=BAR_HORIZONS)
    pivot_tmp = pivot_tmp.reindex(columns=ordered_columns)

    data = np.ma.masked_invalid(pivot_tmp.values)

    im = ax.imshow(
        data,
        aspect="auto",
        vmin=0,
        vmax=1
    )

    ax.set_title(f"{metric_title} - variantes p50")
    ax.set_ylabel("Horizonte")

    ax.set_yticks(np.arange(len(pivot_tmp.index)))
    ax.set_yticklabels([f"H{h}" for h in pivot_tmp.index])

    ax.set_xticks(np.arange(len(pivot_tmp.columns)))
    ax.set_xticklabels(pivot_tmp.columns, rotation=45, ha="right")

    for i in range(pivot_tmp.shape[0]):
        for j in range(pivot_tmp.shape[1]):
            value = pivot_tmp.iloc[i, j]

            if pd.notna(value):
                ax.text(
                    j,
                    i,
                    f"{value:.0%}",
                    ha="center",
                    va="center",
                    fontsize=8
                )

```



```

fig.colorbar(
    im,
    cax=cax,
    label="Proporción"
)

plt.show()

# =====
# Figura 3 – Relación TP/SL y diferencia TP - SL
# =====

fig, axes = plt.subplots(
    nrows=2,
    ncols=1,
    figsize=(13, 9),
    sharex=True
)

# -----
# Gráfico 3A – Ratio TP/SL observado
# -----

axes[0].bar(
    x,
    df_config_plot["tp_to_sl_ratio_mean"],
    width=0.55
)

axes[0].axhline(
    1,
    linestyle="--",
    linewidth=1
)

axes[0].set_title("Relación promedio TP / SL por configuración")
axes[0].set_ylabel("TP / SL")

for i, row in df_config_plot.iterrows():
    axes[0].text(
        i,
        row["tp_to_sl_ratio_mean"],
        f"{row['tp_to_sl_ratio_mean']:.1f}",
        ha="center",
        va="bottom",
        fontsize=8
    )

# -----
# Gráfico 3B – Diferencia TP - SL
# -----

axes[1].bar(
    x,
    df_config_plot["tp_minus_sl_rate_mean"],
    width=0.55
)

axes[1].axhline(
    0,

```

```

linestyle="--",
linewidth=1
)

axes[1].set_title("Diferencia promedio entre TP y SL")
axes[1].set_ylabel("TP rate - SL rate")
axes[1].set_xticks(x)
axes[1].set_xticklabels(config_labels, rotation=45, ha="right")

axes[1].yaxis.set_major_formatter(
    plt.FuncFormatter(lambda y, _: f"{y:.0%}")
)

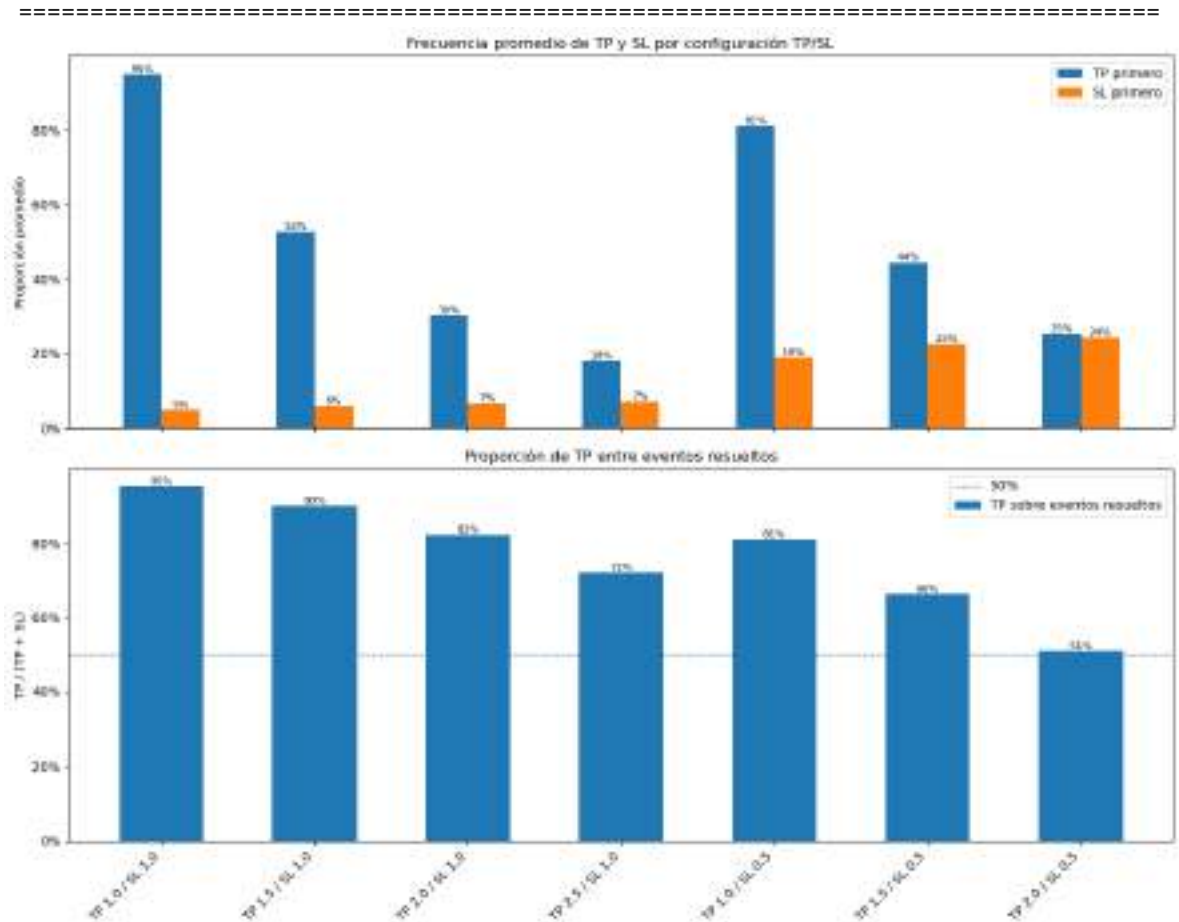
for i, row in df_config_plot.iterrows():
    axes[1].text(
        i,
        row["tp_minus_sl_rate_mean"],
        f"{row['tp_minus_sl_rate_mean']:.0%}",
        ha="center",
        va="bottom" if row["tp_minus_sl_rate_mean"] >= 0 else "top",
        fontsize=8
    )

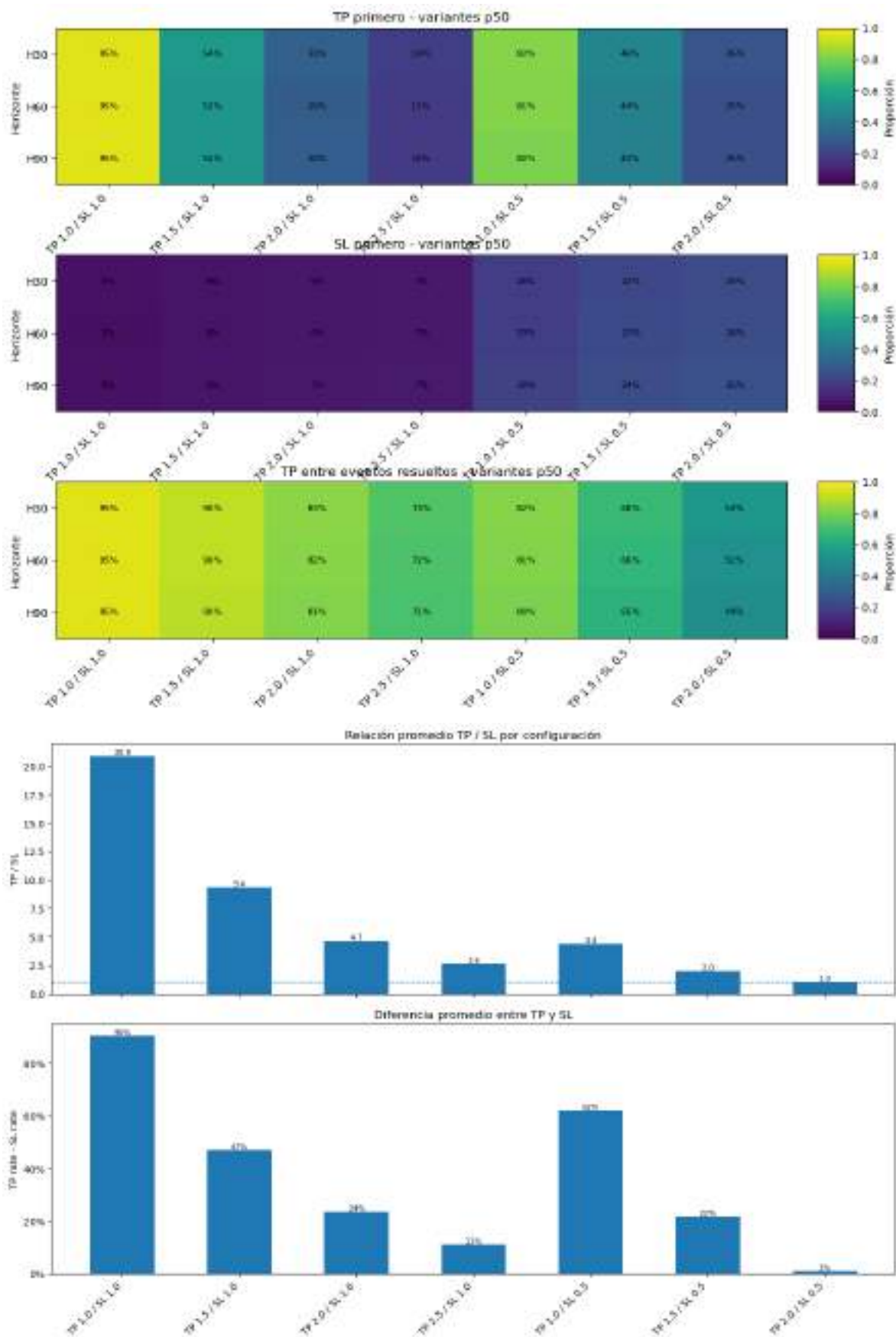
plt.tight_layout()
plt.show()

print("\nCelda 3 finalizada correctamente.")

```

Gráficos agrupados de frecuencia TP/SL - BAR





Celda 3 finalizada correctamente.

## Celda 4 — Tabla final para observaciones

In [102...

```
# Celda 4 - Guardado y tabla final para observaciones
# =====
```

```

print("Guardado y tabla final de frecuencia TP/SL - BAR")
print("=" * 80)

# -----
# Validar objetos necesarios
# -----

required_objects = [
    "df_bar_tp_sl_frequency",
    "df_bar_tp_sl_by_config",
    "df_bar_tp_sl_by_target_base",
    "df_bar_tp_sl_p50",
    "df_bar_tp_sl_resolved_ranking",
    "df_bar_tp_sl_quality_counts",
]

missing_objects = [
    obj for obj in required_objects
    if obj not in globals()
]

if missing_objects:
    raise NameError(
        "Faltan objetos necesarios. "
        f"Debes correr las celdas anteriores: {missing_objects}"
    )

# -----
# Definir rutas de guardado
# -----

MNQ_TARGETS_BAR_TP_SL_FREQUENCY_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_tp_sl_frequency.csv"
)

MNQ_TARGETS_BAR_TP_SL_BY_CONFIG_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_tp_sl_by_config.csv"
)

MNQ_TARGETS_BAR_TP_SL_BY_TARGET_BASE_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_tp_sl_by_target_base.csv"
)

MNQ_TARGETS_BAR_TP_SL_RESOLVED_RANKING_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_tp_sl_resolved_ranking.csv"
)

MNQ_TARGETS_BAR_TP_SL_REPORT_TXT = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_tp_sl_report.txt"
)

# -----
# Guardar tablas principales
# -----

df_bar_tp_sl_frequency.to_csv(
    MNQ_TARGETS_BAR_TP_SL_FREQUENCY_CSV,
    index=False
)

```

```

df_bar_tp_sl_by_config.to_csv(
    MNQ_TARGETS_BAR_TP_SL_BY_CONFIG_CSV,
    index=False
)

df_bar_tp_sl_by_target_base.to_csv(
    MNQ_TARGETS_BAR_TP_SL_BY_TARGET_BASE_CSV,
    index=False
)

df_bar_tp_sl_resolved_ranking.to_csv(
    MNQ_TARGETS_BAR_TP_SL_RESOLVED_RANKING_CSV,
    index=False
)

# -----
# Tabla compacta por configuración TP/SL
# -----

tp_sl_config_cols = [
    "tp_sl_config",
    "tp_mult",
    "sl_mult",
    "n_targets",
    "tp_rate_mean",
    "sl_rate_mean",
    "no_event_rate_mean",
    "resolved_rate_mean",
    "tp_minus_sl_rate_mean",
    "tp_to_sl_ratio_mean",
    "tp_rate_resolved_only_mean",
    "sl_rate_resolved_only_mean",
    "expectancy_units_mean",
]

print("\nTabla compacta por configuración TP/SL:")
print(
    df_bar_tp_sl_by_config[tp_sl_config_cols]
    .to_string(
        index=False,
        formatters={
            "tp_rate_mean": "{:.2%}".format,
            "sl_rate_mean": "{:.2%}".format,
            "no_event_rate_mean": "{:.2%}".format,
            "resolved_rate_mean": "{:.2%}".format,
            "tp_minus_sl_rate_mean": "{:.2%}".format,
            "tp_to_sl_ratio_mean": "{:.2f}".format,
            "tp_rate_resolved_only_mean": "{:.2%}".format,
            "sl_rate_resolved_only_mean": "{:.2%}".format,
            "expectancy_units_mean": "{:.3f}".format,
        }
    )
)

# -----
# Tabla compacta de variantes p50
# -----

p50_tp_sl_cols = [
    "bar_col",

```

```

    "horizon",
    "tp_mult",
    "sl_mult",
    "tp_rate",
    "sl_rate",
    "no_event_rate",
    "resolved_rate",
    "tp_minus_sl_rate",
    "tp_to_sl_ratio",
    "tp_rate_resolved_only",
    "sl_rate_resolved_only",
    "expectancy_units",
    "tp_sl_quality",
]

print("\nTabla compacta de frecuencia TP/SL para variantes p50:")
print(
    df_bar_tp_sl_p50[p50_tp_sl_cols]
    .to_string(
        index=False,
        formatters={
            "tp_rate": "{:.2%}".format,
            "sl_rate": "{:.2%}".format,
            "no_event_rate": "{:.2%}".format,
            "resolved_rate": "{:.2%}".format,
            "tp_minus_sl_rate": "{:.2%}".format,
            "tp_to_sl_ratio": "{:.2f}".format,
            "tp_rate_resolved_only": "{:.2%}".format,
            "sl_rate_resolved_only": "{:.2%}".format,
            "expectancy_units": "{:.3f}".format,
        }
    )
)

# -----
# Top 20 por proporción TP entre eventos resueltos
# -----

resolved_ranking_cols = [
    "bar_col",
    "horizon",
    "percentile",
    "tp_mult",
    "sl_mult",
    "tp_rate",
    "sl_rate",
    "no_event_rate",
    "resolved_rate",
    "tp_rate_resolved_only",
    "tp_to_sl_ratio",
    "expectancy_units",
    "tp_sl_quality",
]

print("\nTop 20 por proporción de TP entre eventos resueltos:")
print(
    df_bar_tp_sl_resolved_ranking[resolved_ranking_cols]
    .head(20)
    .to_string(
        index=False,

```

```

        formatters={
            "tp_rate": "{:.2%}".format,
            "sl_rate": "{:.2%}".format,
            "no_event_rate": "{:.2%}".format,
            "resolved_rate": "{:.2%}".format,
            "tp_rate_resolved_only": "{:.2%}".format,
            "tp_to_sl_ratio": "{:.2f}".format,
            "expectancy_units": "{:.3f}".format,
        }
    )
)

# -----
# Conteo por clasificación preliminar
# -----

print("\nConteo por clasificación preliminar TP/SL:")
print(
    df_bar_tp_sl_quality_counts
    .to_string(
        index=False,
        formatters={
            "tp_rate_mean": "{:.2%}".format,
            "sl_rate_mean": "{:.2%}".format,
            "no_event_rate_mean": "{:.2%}".format,
            "tp_rate_resolved_only_mean": "{:.2%}".format,
            "expectancy_units_mean": "{:.3f}".format,
        }
    )
)

# -----
# Guardar reporte TXT completo
# -----

with MNQ_TARGETS_BAR_TP_SL_REPORT_TXT.open("w", encoding="utf-8") as f:
    f.write("4.7. Frecuencia de TP y SL - BAR\n")
    f.write("=" * 100 + "\n\n")

    f.write("Tabla compacta por configuración TP/SL\n")
    f.write("-" * 100 + "\n")
    f.write(
        df_bar_tp_sl_by_config[tp_sl_config_cols]
        .to_string(
            index=False,
            formatters={
                "tp_rate_mean": "{:.2%}".format,
                "sl_rate_mean": "{:.2%}".format,
                "no_event_rate_mean": "{:.2%}".format,
                "resolved_rate_mean": "{:.2%}".format,
                "tp_minus_sl_rate_mean": "{:.2%}".format,
                "tp_to_sl_ratio_mean": "{:.2f}".format,
                "tp_rate_resolved_only_mean": "{:.2%}".format,
                "sl_rate_resolved_only_mean": "{:.2%}".format,
                "expectancy_units_mean": "{:.3f}".format,
            }
        )
    )

    f.write("\n\n")

```

```

f.write("Tabla compacta de frecuencia TP/SL para variantes p50\n")
f.write("-" * 100 + "\n")
f.write(
    df_bar_tp_sl_p50[p50_tp_sl_cols]
    .to_string(
        index=False,
        formatters={
            "tp_rate": "{:.2%}".format,
            "sl_rate": "{:.2%}".format,
            "no_event_rate": "{:.2%}".format,
            "resolved_rate": "{:.2%}".format,
            "tp_minus_sl_rate": "{:.2%}".format,
            "tp_to_sl_ratio": "{:.2f}".format,
            "tp_rate_resolved_only": "{:.2%}".format,
            "sl_rate_resolved_only": "{:.2%}".format,
            "expectancy_units": "{:.3f}".format,
        }
    )
)

f.write("\n\n")
f.write("Top 20 por proporción de TP entre eventos resueltos\n")
f.write("-" * 100 + "\n")
f.write(
    df_bar_tp_sl_resolved_ranking[resolved_ranking_cols]
    .head(20)
    .to_string(
        index=False,
        formatters={
            "tp_rate": "{:.2%}".format,
            "sl_rate": "{:.2%}".format,
            "no_event_rate": "{:.2%}".format,
            "resolved_rate": "{:.2%}".format,
            "tp_rate_resolved_only": "{:.2%}".format,
            "tp_to_sl_ratio": "{:.2f}".format,
            "expectancy_units": "{:.3f}".format,
        }
    )
)

f.write("\n\n")
f.write("Conteo por clasificación preliminar TP/SL\n")
f.write("-" * 100 + "\n")
f.write(
    df_bar_tp_sl_quality_counts
    .to_string(
        index=False,
        formatters={
            "tp_rate_mean": "{:.2%}".format,
            "sl_rate_mean": "{:.2%}".format,
            "no_event_rate_mean": "{:.2%}".format,
            "tp_rate_resolved_only_mean": "{:.2%}".format,
            "expectancy_units_mean": "{:.3f}".format,
        }
    )
)

# -----
# Confirmación final
# -----

```



```
print("\nArchivos guardados:")
print(f"Frecuencia TP/SL BAR: {MNQ_TARGETS_BAR_TP_SL_FREQUENCY_CSV}")
print(f"Resumen TP/SL por configuración: {MNQ_TARGETS_BAR_TP_SL_BY_CONFIG_CSV}")
print(f"Resumen TP/SL por target base: {MNQ_TARGETS_BAR_TP_SL_BY_TARGET_BASE_CS")
print(f"Ranking TP/SL resuelto: {MNQ_TARGETS_BAR_TP_SL_RESOLVED_RANKING_")
print(f"Reporte TXT: {MNQ_TARGETS_BAR_TP_SL_REPORT_TXT}")

print("\nCelda 4 finalizada correctamente.")
```

## Guardado y tabla final de frecuencia TP/SL - BAR

=====

## Tabla compacta por configuración TP/SL:

| tp_sl_config    | tp_mult | sl_mult | n_targets | tp_rate_mean | sl_rate_mean | no_event_rate_mean | resolved_rate_mean | tp_minus_sl_rate_mean | tp_to_sl_ratio_mean | tp_rate_resolved_only_mean | sl_rate_resolved_only_mean | expectancy_units_mean |
|-----------------|---------|---------|-----------|--------------|--------------|--------------------|--------------------|-----------------------|---------------------|----------------------------|----------------------------|-----------------------|
| TP 1.0 / SL 0.5 | 1.0     | 0.5     | 9         | 81.07%       | 18.93%       | 0.00%              | 100.00%            | 62.15%                | 4.38                | 0.716                      | 0.552                      | 0.383                 |
| TP 1.5 / SL 0.5 | 1.5     | 0.5     | 9         | 44.39%       | 22.71%       | 32.89%             | 67.11%             | 21.68%                | 1.98                | 0.552                      | 0.383                      | 0.383                 |
| TP 2.0 / SL 0.5 | 2.0     | 0.5     | 9         | 25.21%       | 24.24%       | 50.55%             | 49.45%             | 0.97%                 | 1.05                | 0.383                      | 0.383                      | 0.383                 |
| TP 1.0 / SL 1.0 | 1.0     | 1.0     | 9         | 95.18%       | 4.82%        | 0.00%              | 100.00%            | 90.36%                | 20.89               | 0.904                      | 0.539                      | 0.380                 |
| TP 1.5 / SL 1.0 | 1.5     | 1.0     | 9         | 52.73%       | 5.85%        | 41.42%             | 58.58%             | 46.89%                | 9.36                | 0.733                      | 0.539                      | 0.380                 |
| TP 2.0 / SL 1.0 | 2.0     | 1.0     | 9         | 30.26%       | 6.65%        | 63.10%             | 36.90%             | 23.61%                | 4.66                | 0.539                      | 0.539                      | 0.380                 |
| TP 2.5 / SL 1.0 | 2.5     | 1.0     | 9         | 18.02%       | 7.06%        | 74.92%             | 25.08%             | 10.96%                | 2.59                | 0.380                      | 0.539                      | 0.380                 |

## Tabla compacta de frecuencia TP/SL para variantes p50:

| bar_col               | horizon | tp_mult | sl_mult | tp_rate | sl_rate | no_event_rate | resolved_rate | tp_minus_sl_rate | tp_to_sl_ratio | tp_rate_resolved_only | sl_rate_resolved_only | expectancy_units | tp_sl_quality |
|-----------------------|---------|---------|---------|---------|---------|---------------|---------------|------------------|----------------|-----------------------|-----------------------|------------------|---------------|
| bar_p50_h30_tp10_sl05 | 30      | 1.0     | 0.5     | 81.93%  | 18.07%  | 0.00%         | 100.00%       | 63.86%           | 4.53           | 81.93%                | 18.07%                | 0.00%            | favorable     |
| bar_p50_h30_tp15_sl05 | 30      | 1.5     | 0.5     | 45.74%  | 21.67%  | 32.60%        | 67.40%        | 24.07%           | 2.11           | 67.85%                | 21.67%                | 32.1             | moderada      |
| bar_p50_h30_tp20_sl05 | 30      | 2.0     | 0.5     | 26.31%  | 23.19%  | 50.50%        | 49.50%        | 3.12%            | 1.13           | 53.15%                | 23.19%                | 46.8             | débil         |
| bar_p50_h30_tp10_sl10 | 30      | 1.0     | 1.0     | 95.26%  | 4.74%   | 0.00%         | 100.00%       | 90.52%           | 20.10          | 95.26%                | 4.74%                 | 4.7              | muy_favorable |
| bar_p50_h30_tp15_sl10 | 30      | 1.5     | 1.0     | 53.60%  | 5.71%   | 40.70%        | 59.30%        | 47.89%           | 9.39           | 90.38%                | 5.71%                 | 9.6              | muy_favorable |
| bar_p50_h30_tp20_sl10 | 30      | 2.0     | 1.0     | 31.07%  | 6.49%   | 62.44%        | 37.56%        | 24.58%           | 4.79           | 82.73%                | 6.49%                 | 17.2             | favorable     |
| bar_p50_h30_tp25_sl10 | 30      | 2.5     | 1.0     | 18.63%  | 6.90%   | 74.47%        | 25.53%        | 11.73%           | 2.70           | 72.97%                | 6.90%                 | 27.0             | moderada      |
| bar_p50_h60_tp10_sl05 | 60      | 1.0     | 0.5     | 81.32%  | 18.68%  | 0.00%         | 100.00%       | 62.64%           | 4.35           | 81.32%                | 18.68%                | 18.6             | favorable     |
| bar_p50_h60_tp15_sl05 | 60      | 1.5     | 0.5     | 44.00%  | 22.52%  | 33.48%        | 66.52%        | 21.48%           | 1.95           | 66.15%                | 22.52%                | 33.8             | moderada      |

|                       |        |               |     |        |        |        |
|-----------------------|--------|---------------|-----|--------|--------|--------|
| bar_p50_h60_tp20_sl05 | 60     | 2.0           | 0.5 | 24.58% | 23.99% | 51.44% |
| 48.56%                | 0.59%  | 1.02          |     | 50.61% |        | 49.3   |
| 9%                    | 0.372  | débil         |     |        |        |        |
| bar_p50_h60_tp10_sl10 | 60     | 1.0           | 1.0 | 95.45% | 4.55%  | 0.00%  |
| 100.00%               | 90.91% | 21.00         |     | 95.45% |        | 4.5    |
| 5%                    | 0.909  | muy_favorable |     |        |        |        |
| bar_p50_h60_tp15_sl10 | 60     | 1.5           | 1.0 | 52.23% | 5.54%  | 42.23% |
| 57.77%                | 46.70% | 9.43          |     | 90.41% |        | 9.5    |
| 9%                    | 0.728  | muy_favorable |     |        |        |        |
| bar_p50_h60_tp20_sl10 | 60     | 2.0           | 1.0 | 29.49% | 6.29%  | 64.22% |
| 35.78%                | 23.20% | 4.69          |     | 82.42% |        | 17.5   |
| 8%                    | 0.527  | favorable     |     |        |        |        |
| bar_p50_h60_tp25_sl10 | 60     | 2.5           | 1.0 | 17.33% | 6.69%  | 75.98% |
| 24.02%                | 10.65% | 2.59          |     | 72.16% |        | 27.8   |
| 4%                    | 0.366  | moderada      |     |        |        |        |
| bar_p50_h90_tp10_sl05 | 90     | 1.0           | 0.5 | 80.23% | 19.77% | 0.00%  |
| 100.00%               | 60.45% | 4.06          |     | 80.23% |        | 19.7   |
| 7%                    | 0.703  | favorable     |     |        |        |        |
| bar_p50_h90_tp15_sl05 | 90     | 1.5           | 0.5 | 43.60% | 23.71% | 32.70% |
| 67.30%                | 19.89% | 1.84          |     | 64.78% |        | 35.2   |
| 2%                    | 0.535  | moderada      |     |        |        |        |
| bar_p50_h90_tp20_sl05 | 90     | 2.0           | 0.5 | 24.52% | 25.25% | 50.23% |
| 49.77%                | -0.74% | 0.97          |     | 49.26% |        | 50.7   |
| 4%                    | 0.364  | débil         |     |        |        |        |
| bar_p50_h90_tp10_sl10 | 90     | 1.0           | 1.0 | 95.11% | 4.89%  | 0.00%  |
| 100.00%               | 90.23% | 19.46         |     | 95.11% |        | 4.8    |
| 9%                    | 0.902  | muy_favorable |     |        |        |        |
| bar_p50_h90_tp15_sl10 | 90     | 1.5           | 1.0 | 52.38% | 5.93%  | 41.69% |
| 58.31%                | 46.45% | 8.83          |     | 89.83% |        | 10.1   |
| 7%                    | 0.726  | muy_favorable |     |        |        |        |
| bar_p50_h90_tp20_sl10 | 90     | 2.0           | 1.0 | 29.78% | 6.77%  | 63.45% |
| 36.55%                | 23.01% | 4.40          |     | 81.48% |        | 18.5   |
| 2%                    | 0.528  | favorable     |     |        |        |        |
| bar_p50_h90_tp25_sl10 | 90     | 2.5           | 1.0 | 17.65% | 7.17%  | 75.18% |
| 24.82%                | 10.48% | 2.46          |     | 71.12% |        | 28.8   |
| 8%                    | 0.370  | moderada      |     |        |        |        |

Top 20 por proporción de TP entre eventos resueltos:

| bar_col               | horizon       | percentile            | tp_mult        | sl_mult          | tp_rate       | sl_rate | no_e |
|-----------------------|---------------|-----------------------|----------------|------------------|---------------|---------|------|
| vent_rate             | resolved_rate | tp_rate_resolved_only | tp_to_sl_ratio | expectancy_units | tp_sl_quality |         |      |
| bar_p60_h60_tp10_sl10 | 60            | 60                    | 1.0            | 1.0              | 96.62%        | 3.38%   |      |
| 0.00%                 | 100.00%       | 96.62%                | 28.60          |                  | 0.932         | muy_fav |      |
| orable                |               |                       |                |                  |               |         |      |
| bar_p60_h30_tp10_sl10 | 30            | 60                    | 1.0            | 1.0              | 96.41%        | 3.59%   |      |
| 0.00%                 | 100.00%       | 96.41%                | 26.86          |                  | 0.928         | muy_fav |      |
| orable                |               |                       |                |                  |               |         |      |
| bar_p60_h90_tp10_sl10 | 90            | 60                    | 1.0            | 1.0              | 96.37%        | 3.63%   |      |
| 0.00%                 | 100.00%       | 96.37%                | 26.52          |                  | 0.927         | muy_fav |      |
| orable                |               |                       |                |                  |               |         |      |
| bar_p50_h60_tp10_sl10 | 60            | 50                    | 1.0            | 1.0              | 95.45%        | 4.55%   |      |
| 0.00%                 | 100.00%       | 95.45%                | 21.00          |                  | 0.909         | muy_fav |      |
| orable                |               |                       |                |                  |               |         |      |
| bar_p50_h30_tp10_sl10 | 30            | 50                    | 1.0            | 1.0              | 95.26%        | 4.74%   |      |
| 0.00%                 | 100.00%       | 95.26%                | 20.10          |                  | 0.905         | muy_fav |      |
| orable                |               |                       |                |                  |               |         |      |
| bar_p50_h90_tp10_sl10 | 90            | 50                    | 1.0            | 1.0              | 95.11%        | 4.89%   |      |
| 0.00%                 | 100.00%       | 95.11%                | 19.46          |                  | 0.902         | muy_fav |      |
| orable                |               |                       |                |                  |               |         |      |
| bar_p40_h60_tp10_sl10 | 60            | 40                    | 1.0            | 1.0              | 94.00%        | 6.00%   |      |

|                       |         |        |       |       |               |
|-----------------------|---------|--------|-------|-------|---------------|
| 0.00%                 | 100.00% | 94.00% | 15.66 | 0.880 | muy_fav       |
| orable                |         |        |       |       |               |
| bar_p40_h30_tp10_sl10 | 30      | 40     | 1.0   | 1.0   | 93.84% 6.16%  |
| 0.00%                 | 100.00% | 93.84% | 15.24 | 0.877 | muy_fav       |
| orable                |         |        |       |       |               |
| bar_p40_h90_tp10_sl10 | 90      | 40     | 1.0   | 1.0   | 93.57% 6.43%  |
| 0.00%                 | 100.00% | 93.57% | 14.54 | 0.871 | muy_fav       |
| orable                |         |        |       |       |               |
| bar_p60_h60_tp15_sl10 | 60      | 60     | 1.5   | 1.0   | 48.27% 4.14%  |
| 47.58%                | 52.42%  | 92.10% | 11.66 | 0.683 | fa            |
| vorable               |         |        |       |       |               |
| bar_p60_h30_tp15_sl10 | 30      | 60     | 1.5   | 1.0   | 49.53% 4.37%  |
| 46.11%                | 53.89%  | 91.90% | 11.34 | 0.699 | fa            |
| vorable               |         |        |       |       |               |
| bar_p60_h90_tp15_sl10 | 90      | 60     | 1.5   | 1.0   | 48.78% 4.52%  |
| 46.70%                | 53.30%  | 91.52% | 10.80 | 0.687 | fa            |
| vorable               |         |        |       |       |               |
| bar_p50_h60_tp15_sl10 | 60      | 50     | 1.5   | 1.0   | 52.23% 5.54%  |
| 42.23%                | 57.77%  | 90.41% | 9.43  | 0.728 | muy_fa        |
| vorable               |         |        |       |       |               |
| bar_p50_h30_tp15_sl10 | 30      | 50     | 1.5   | 1.0   | 53.60% 5.71%  |
| 40.70%                | 59.30%  | 90.38% | 9.39  | 0.747 | muy_fa        |
| vorable               |         |        |       |       |               |
| bar_p50_h90_tp15_sl10 | 90      | 50     | 1.5   | 1.0   | 52.38% 5.93%  |
| 41.69%                | 58.31%  | 89.83% | 8.83  | 0.726 | muy_fa        |
| vorable               |         |        |       |       |               |
| bar_p40_h30_tp15_sl10 | 30      | 40     | 1.5   | 1.0   | 57.52% 7.35%  |
| 35.13%                | 64.87%  | 88.67% | 7.82  | 0.789 | muy_fa        |
| vorable               |         |        |       |       |               |
| bar_p40_h60_tp15_sl10 | 60      | 40     | 1.5   | 1.0   | 56.30% 7.31%  |
| 36.39%                | 63.61%  | 88.51% | 7.70  | 0.771 | muy_fa        |
| vorable               |         |        |       |       |               |
| bar_p40_h90_tp15_sl10 | 90      | 40     | 1.5   | 1.0   | 55.98% 7.76%  |
| 36.26%                | 63.74%  | 87.83% | 7.22  | 0.762 | muy_fa        |
| vorable               |         |        |       |       |               |
| bar_p60_h30_tp10_sl05 | 30      | 60     | 1.0   | 0.5   | 84.78% 15.22% |
| 0.00%                 | 100.00% | 84.78% | 5.57  | 0.772 | fav           |
| orable                |         |        |       |       |               |
| bar_p60_h30_tp20_sl10 | 30      | 60     | 2.0   | 1.0   | 26.78% 4.94%  |
| 68.28%                | 31.72%  | 84.42% | 5.42  | 0.486 | m             |
| oderada               |         |        |       |       |               |

Conteo por clasificación preliminar TP/SL:

| tp_sl_quality | n_targets | tp_rate_mean | sl_rate_mean | no_event_rate_mean | tp_rate_resolved_only_mean | expectancy_units_mean |
|---------------|-----------|--------------|--------------|--------------------|----------------------------|-----------------------|
| moderada      | 20        | 29.84%       | 13.22%       | 56.95%             |                            |                       |
| 71.37%        | 0.463     |              |              |                    |                            |                       |
| favorable     | 19        | 58.60%       | 12.99%       | 28.41%             |                            |                       |
| 82.25%        | 0.658     |              |              |                    |                            |                       |
| muy_favorable | 15        | 78.98%       | 5.53%        | 15.49%             |                            |                       |
| 92.82%        | 0.844     |              |              |                    |                            |                       |
| débil         | 9         | 25.21%       | 24.24%       | 50.55%             |                            |                       |
| 51.08%        | 0.383     |              |              |                    |                            |                       |

Archivos guardados:

Frecuencia TP/SL BAR: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_tp\_sl\_frequency.csv  
 Resumen TP/SL por configuración: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_tp\_sl\_by\_config.csv  
 Resumen TP/SL por target base: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_

local\data\04\_mnq\_targets\mnq\_targets\_bar\_tp\_sl\_by\_target\_base.csv  
 Ranking TP/SL resuelto: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_  
 local\data\04\_mnq\_targets\mnq\_targets\_bar\_tp\_sl\_resolved\_ranking.csv  
 Reporte TXT: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_  
 local\data\04\_mnq\_targets\mnq\_targets\_bar\_tp\_sl\_report.txt

Celda 4 finalizada correctamente.

## Observaciones preliminares — Frecuencia de TP y SL

El análisis de frecuencia de TP y SL muestra que las configuraciones evaluadas presentan comportamientos muy distintos según la distancia del Take Profit y del Stop Loss.

En primer lugar, TP 1.0 / SL 1.0 vuelve a aparecer como la configuración más favorable en términos brutos. En promedio, alcanza TP primero en el 95.18% de los casos, con solo 4.82% de SL y sin casos no\_event. Sin embargo, esta configuración debe interpretarse principalmente como benchmark, ya que el TP coincide con el threshold base utilizado para construir DIR.

La configuración TP 1.0 / SL 0.5 también resuelve el 100% de los casos, pero presenta una proporción de SL mucho mayor. Esto ocurre porque el Stop Loss está más cerca del precio de entrada. Aunque mantiene un TP promedio alto, cercano al 81.07%, el SL aumenta hasta 18.93%, por lo que es más agresiva desde el punto de vista operativo.

La configuración TP 1.5 / SL 1.0 aparece como una alternativa especialmente interesante. Aunque no resuelve todos los casos, mantiene una proporción favorable entre TP y SL. En promedio, alcanza TP primero en el 52.73% de los casos y SL primero en solo 5.85%. Entre los eventos que sí resuelven como TP o SL, el TP representa alrededor del 90.13%. Esto sugiere que, cuando la operación resuelve, generalmente lo hace a favor.

Las configuraciones más exigentes, como TP 2.0 / SL 1.0 y TP 2.5 / SL 1.0, mantienen una proporción favorable entre TP y SL cuando el evento ocurre, pero generan demasiados casos sin resolución. En particular, TP 2.0 / SL 1.0 deja cerca de 63.10% como no\_event, mientras que TP 2.5 / SL 1.0 deja cerca de 74.92%. Por lo tanto, aunque no necesariamente son descartables, parecen menos prácticas para una primera selección operativa.

Las configuraciones con SL = 0.5 muestran mayor sensibilidad al movimiento adverso. El caso TP 2.0 / SL 0.5 es el más débil, ya que el TP y el SL quedan prácticamente equilibrados entre los eventos resueltos. Su tp\_rate\_resolved\_only es cercano al 51.08%, lo que indica que, cuando ocurre un evento, casi no hay ventaja clara entre TP y SL.

En las variantes p50, el comportamiento es bastante consistente entre H30, H60 y H90. La configuración TP 1.5 / SL 1.0 mantiene tasas de TP cercanas al 52%-54%, SL alrededor del 5%-6%, y una proporción de TP entre eventos resueltos cercana al 90%. Esto refuerza su papel como candidata principal para análisis posteriores.

El ranking por proporción de TP entre eventos resueltos favorece principalmente a las configuraciones TP 1.0 / SL 1.0 y luego a TP 1.5 / SL 1.0. Esto confirma que las señales DIR tienen una trayectoria favorable cuando se evalúan con barreras no excesivamente lejanas.

En síntesis, el análisis de frecuencia TP/SL sugiere mantener TP 1.0 / SL 1.0 como benchmark, pero enfocar la atención principal en TP 1.5 / SL 1.0, ya que ofrece un equilibrio más realista entre exigencia, baja frecuencia de SL y alta proporción de TP entre eventos resueltos.

De forma preliminar, las configuraciones más relevantes para continuar son:

Benchmark:

TP 1.0 / SL 1.0

Candidata principal:

TP 1.5 / SL 1.0

Candidata exigente:

TP 2.0 / SL 1.0

Comparación agresiva:

TP 1.0 / SL 0.5

En simple: TP 1.5 / SL 1.0 sigue siendo la mejor candidata práctica. TP 1.0 / SL 1.0 se ve mejor en números, pero está demasiado pegada a la forma en que construimos DIR. TP 2.0 / SL 1.0 podría seguir en análisis, pero genera demasiados casos sin resolución.

## 4.8. Tiempo hasta evento TP/SL

En este punto se analiza cuánto tarda cada target BAR en alcanzar un evento operativo.

El objetivo es estudiar, para cada configuración TP/SL, en qué barra futura ocurre primero el Take Profit o el Stop Loss. Esto permite diferenciar configuraciones que resuelven rápidamente de aquellas que requieren esperar gran parte del horizonte.

Este análisis es importante porque dos configuraciones pueden tener frecuencias similares de TP y SL, pero comportamientos operativos distintos si una resuelve rápido y otra tarda demasiado en alcanzar el evento.

Las columnas event\_step serán utilizadas únicamente como diagnóstico del target. No deben ser utilizadas posteriormente como features de modelos predictivos, ya que contienen información futura.

### Celda 1 — Preparación de métricas de tiempo hasta evento

In [103...

```
# Celda 1 – Preparación de métricas de tiempo hasta evento
# =====

print("4.8. Tiempo hasta evento TP/SL - BAR")
```

```

print("=" * 80)

# -----
# Validar objetos necesarios
# -----

if "df_targets_bar" not in globals():
    raise NameError("No existe df_targets_bar. Primero debes construir los targe

if "df_bar_metadata" not in globals():
    raise NameError("No existe df_bar_metadata. Primero debes construir df_bar_m

# -----
# Función auxiliar para resumir event_step
# -----

def summarize_step_values(step_values):
    """
    Resume una serie de event_step.
    """

    step_values = pd.Series(step_values).dropna().astype(float)

    if len(step_values) == 0:
        return {
            "step_count": 0,
            "step_mean": np.nan,
            "step_median": np.nan,
            "step_q25": np.nan,
            "step_q75": np.nan,
            "step_min": np.nan,
            "step_max": np.nan,
        }

    return {
        "step_count": int(len(step_values)),
        "step_mean": step_values.mean(),
        "step_median": step_values.median(),
        "step_q25": step_values.quantile(0.25),
        "step_q75": step_values.quantile(0.75),
        "step_min": step_values.min(),
        "step_max": step_values.max(),
    }

# -----
# Construir resumen de tiempo hasta evento
# -----

event_time_rows = []

for _, meta in df_bar_metadata.iterrows():
    bar_col = meta["bar_col"]
    event_step_col = meta["event_step_col"]

    horizon = int(meta["horizon"])
    percentile = int(meta["percentile"])
    tp_mult = float(meta["tp_mult"])
    sl_mult = float(meta["sl_mult"])

```

```

if bar_col not in df_targets_bar.columns:
    raise KeyError(f"No existe la columna BAR: {bar_col}")

if event_step_col not in df_targets_bar.columns:
    raise KeyError(f"No existe la columna event_step: {event_step_col}")

bar_series = df_targets_bar[bar_col]
step_series = df_targets_bar[event_step_col]

tp_mask = bar_series == 1
sl_mask = bar_series == -1
no_event_mask = bar_series == 0
valid_mask = bar_series.notna()

tp_steps = step_series.loc[tp_mask]
sl_steps = step_series.loc[sl_mask]
all_event_steps = step_series.loc[tp_mask | sl_mask]

tp_summary = summarize_step_values(tp_steps)
sl_summary = summarize_step_values(sl_steps)
all_summary = summarize_step_values(all_event_steps)

valid_rows = int(valid_mask.sum())
tp_count = int(tp_mask.sum())
sl_count = int(sl_mask.sum())
no_event_count = int(no_event_mask.sum())
event_count = tp_count + sl_count

# Partes del horizonte
early_limit = horizon / 3
late_limit = 2 * horizon / 3

if event_count > 0:
    all_steps_float = all_event_steps.dropna().astype(float)

    early_event_rate = (all_steps_float <= early_limit).mean()
    middle_event_rate = (
        (all_steps_float > early_limit)
        & (all_steps_float <= late_limit)
    ).mean()
    late_event_rate = (all_steps_float > late_limit).mean()
else:
    early_event_rate = np.nan
    middle_event_rate = np.nan
    late_event_rate = np.nan

event_time_rows.append(
    {
        "bar_col": bar_col,
        "event_step_col": event_step_col,
        "dir_col": meta["dir_col"],
        "horizon": horizon,
        "percentile": percentile,
        "tp_mult": tp_mult,
        "sl_mult": sl_mult,
        "tp_sl_config": f"TP {tp_mult} / SL {sl_mult}",

        "valid_rows": valid_rows,
        "tp_count": tp_count,
        "sl_count": sl_count,
    }
)

```



```

        "no_event_count": no_event_count,
        "event_count": event_count,

        "tp_rate": tp_count / valid_rows if valid_rows > 0 else np.nan,
        "sl_rate": sl_count / valid_rows if valid_rows > 0 else np.nan,
        "no_event_rate": no_event_count / valid_rows if valid_rows > 0 else np.nan,
        "event_rate": event_count / valid_rows if valid_rows > 0 else np.nan

        "event_step_mean": all_summary["step_mean"],
        "event_step_median": all_summary["step_median"],
        "event_step_q25": all_summary["step_q25"],
        "event_step_q75": all_summary["step_q75"],
        "event_step_min": all_summary["step_min"],
        "event_step_max": all_summary["step_max"],

        "tp_step_mean": tp_summary["step_mean"],
        "tp_step_median": tp_summary["step_median"],
        "tp_step_q25": tp_summary["step_q25"],
        "tp_step_q75": tp_summary["step_q75"],

        "sl_step_mean": sl_summary["step_mean"],
        "sl_step_median": sl_summary["step_median"],
        "sl_step_q25": sl_summary["step_q25"],
        "sl_step_q75": sl_summary["step_q75"],

        "early_event_rate": early_event_rate,
        "middle_event_rate": middle_event_rate,
        "late_event_rate": late_event_rate,
    }
)

df_bar_event_time = pd.DataFrame(event_time_rows)

df_bar_event_time = (
    df_bar_event_time
    .sort_values(["percentile", "horizon", "sl_mult", "tp_mult"])
    .reset_index(drop=True)
)

# -----
# Vista inicial
# -----

print(f"Targets BAR analizados: {len(df_bar_event_time)}")

print("\nVista inicial de tiempo hasta evento:")
print(
    df_bar_event_time[
        [
            "bar_col",
            "horizon",
            "percentile",
            "tp_mult",
            "sl_mult",
            "event_rate",
            "tp_rate",
            "sl_rate",
            "no_event_rate",
            "event_step_mean",
            "event_step_median",

```

```
        "tp_step_mean",
        "sl_step_mean",
        "early_event_rate",
        "middle_event_rate",
        "late_event_rate",
    ]
]
.head(21)
.to_string(
    index=False,
    formatters={
        "event_rate": "{:.2%}".format,
        "tp_rate": "{:.2%}".format,
        "sl_rate": "{:.2%}".format,
        "no_event_rate": "{:.2%}".format,
        "event_step_mean": "{:.2f}".format,
        "event_step_median": "{:.2f}".format,
        "tp_step_mean": "{:.2f}".format,
        "sl_step_mean": "{:.2f}".format,
        "early_event_rate": "{:.2%}".format,
        "middle_event_rate": "{:.2%}".format,
        "late_event_rate": "{:.2%}".format,
    }
)
)

print("\nPreparación de métricas de tiempo hasta evento finalizada correctamente")
```

## 4.8. Tiempo hasta evento TP/SL - BAR

=====

Targets BAR analizados: 63

Vista inicial de tiempo hasta evento:

| bar_col               | horizon         | percentile        | tp_mult      | sl_mult      | event_rate       | tp_rate           | sl_rate         |
|-----------------------|-----------------|-------------------|--------------|--------------|------------------|-------------------|-----------------|
| no_event_rate         | event_step_mean | event_step_median | tp_step_mean | sl_step_mean | early_event_rate | middle_event_rate | late_event_rate |
| bar_p40_h30_tp10_sl05 | 30              | 40                | 1.0          | 0.5          | 100.00%          | 78.98%            |                 |
| 21.02%                | 0.00%           | 11.18             | 9.00         | 12.72        | 5.39             |                   |                 |
| 55.06%                | 29.29%          | 15.65%            |              |              |                  |                   |                 |
| bar_p40_h30_tp15_sl05 | 30              | 40                | 1.5          | 0.5          | 73.21%           | 48.14%            |                 |
| 25.07%                | 26.79%          | 12.46             | 11.00        | 14.86        | 7.85             |                   |                 |
| 48.61%                | 30.00%          | 21.39%            |              |              |                  |                   |                 |
| bar_p40_h30_tp20_sl05 | 30              | 40                | 2.0          | 0.5          | 56.62%           | 29.68%            |                 |
| 26.94%                | 43.38%          | 12.70             | 11.00        | 16.26        | 8.78             |                   |                 |
| 48.01%                | 28.26%          | 23.74%            |              |              |                  |                   |                 |
| bar_p40_h30_tp10_sl10 | 30              | 40                | 1.0          | 1.0          | 100.00%          | 93.84%            |                 |
| 6.16%                 | 0.00%           | 13.20             | 12.00        | 13.61        | 6.99             |                   |                 |
| 44.23%                | 33.99%          | 21.78%            |              |              |                  |                   |                 |
| bar_p40_h30_tp15_sl10 | 30              | 40                | 1.5          | 1.0          | 64.87%           | 57.52%            |                 |
| 7.35%                 | 35.13%          | 14.83             | 14.00        | 15.55        | 9.23             |                   |                 |
| 35.76%                | 36.12%          | 28.12%            |              |              |                  |                   |                 |
| bar_p40_h30_tp20_sl10 | 30              | 40                | 2.0          | 1.0          | 43.97%           | 35.63%            |                 |
| 8.35%                 | 56.03%          | 15.66             | 15.00        | 16.81        | 10.78            |                   |                 |
| 32.03%                | 35.73%          | 32.24%            |              |              |                  |                   |                 |
| bar_p40_h30_tp25_sl10 | 30              | 40                | 2.5          | 1.0          | 31.51%           | 22.59%            |                 |
| 8.92%                 | 68.49%          | 15.97             | 16.00        | 17.71        | 11.56            |                   |                 |
| 31.32%                | 34.13%          | 34.54%            |              |              |                  |                   |                 |
| bar_p40_h60_tp10_sl05 | 60              | 40                | 1.0          | 0.5          | 100.00%          | 77.94%            |                 |
| 22.06%                | 0.00%           | 21.95             | 18.00        | 25.27        | 10.22            |                   |                 |
| 54.96%                | 28.94%          | 16.10%            |              |              |                  |                   |                 |
| bar_p40_h60_tp15_sl05 | 60              | 40                | 1.5          | 0.5          | 72.88%           | 46.45%            |                 |
| 26.43%                | 27.12%          | 24.35             | 21.00        | 29.59        | 15.15            |                   |                 |
| 48.90%                | 29.53%          | 21.57%            |              |              |                  |                   |                 |
| bar_p40_h60_tp20_sl05 | 60              | 40                | 2.0          | 0.5          | 56.15%           | 27.84%            |                 |
| 28.31%                | 43.85%          | 24.57             | 21.00        | 32.35        | 16.92            |                   |                 |
| 49.08%                | 27.40%          | 23.52%            |              |              |                  |                   |                 |
| bar_p40_h60_tp10_sl10 | 60              | 40                | 1.0          | 1.0          | 100.00%          | 94.00%            |                 |
| 6.00%                 | 0.00%           | 26.19             | 24.00        | 27.01        | 13.34            |                   |                 |
| 43.64%                | 33.89%          | 22.47%            |              |              |                  |                   |                 |
| bar_p40_h60_tp15_sl10 | 60              | 40                | 1.5          | 1.0          | 63.61%           | 56.30%            |                 |
| 7.31%                 | 36.39%          | 29.41             | 28.00        | 30.86        | 18.22            |                   |                 |
| 35.18%                | 36.11%          | 28.71%            |              |              |                  |                   |                 |
| bar_p40_h60_tp20_sl10 | 60              | 40                | 2.0          | 1.0          | 42.39%           | 34.08%            |                 |
| 8.30%                 | 57.61%          | 31.05             | 30.00        | 33.41        | 21.37            |                   |                 |
| 31.74%                | 35.36%          | 32.90%            |              |              |                  |                   |                 |
| bar_p40_h60_tp25_sl10 | 60              | 40                | 2.5          | 1.0          | 29.89%           | 21.05%            |                 |
| 8.84%                 | 70.11%          | 31.48             | 31.00        | 35.13        | 22.78            |                   |                 |
| 31.37%                | 33.80%          | 34.83%            |              |              |                  |                   |                 |
| bar_p40_h90_tp10_sl05 | 90              | 40                | 1.0          | 0.5          | 100.00%          | 77.10%            |                 |
| 22.90%                | 0.00%           | 32.93             | 27.00        | 38.07        | 15.60            |                   |                 |
| 54.24%                | 29.44%          | 16.32%            |              |              |                  |                   |                 |
| bar_p40_h90_tp15_sl05 | 90              | 40                | 1.5          | 0.5          | 72.84%           | 45.52%            |                 |
| 27.32%                | 27.16%          | 36.63             | 32.00        | 44.92        | 22.83            |                   |                 |
| 48.55%                | 29.16%          | 22.30%            |              |              |                  |                   |                 |
| bar_p40_h90_tp20_sl05 | 90              | 40                | 2.0          | 0.5          | 56.66%           | 27.45%            |                 |
| 29.21%                | 43.34%          | 37.08             | 32.00        | 49.45        | 25.44            |                   |                 |
| 48.52%                | 26.84%          | 24.64%            |              |              |                  |                   |                 |
| bar_p40_h90_tp10_sl10 | 90              | 40                | 1.0          | 1.0          | 100.00%          | 93.57%            |                 |

|                       |        |        |       |       |       |
|-----------------------|--------|--------|-------|-------|-------|
| 6.43%                 | 0.00%  | 39.52  | 36.00 | 40.74 | 21.79 |
| 42.61%                | 34.26% | 23.13% |       |       |       |
| bar_p40_h90_tp15_sl10 |        | 90     | 40    | 1.5   | 1.0   |
| 7.76%                 | 36.26% | 44.77  | 43.00 | 47.01 | 28.64 |
| 34.15%                | 35.43% | 30.42% |       |       |       |
| bar_p40_h90_tp20_sl10 |        | 90     | 40    | 2.0   | 1.0   |
| 8.83%                 | 57.08% | 47.51  | 47.00 | 51.17 | 33.35 |
| 30.35%                | 34.28% | 35.37% |       |       |       |
| bar_p40_h90_tp25_sl10 |        | 90     | 40    | 2.5   | 1.0   |
| 9.43%                 | 69.38% | 48.34  | 49.00 | 54.00 | 35.62 |
| 30.03%                | 32.41% | 37.56% |       |       |       |

Preparación de métricas de tiempo hasta evento finalizada correctamente.

## Celda 2 — Tablas agregadas

In [104...

```
# Celda 2 - Tablas agregadas
# =====

print("Tablas agregadas de tiempo hasta evento - BAR")
print("=" * 80)

# -----
# Validar tabla base
# -----

if "df_bar_event_time" not in globals():
    raise NameError("No existe df_bar_event_time. Primero debes correr la Celda

# -----
# 1. Resumen por configuración TP/SL
# -----

df_bar_event_time_by_config = (
    df_bar_event_time
    .groupby(["tp_mult", "sl_mult", "tp_sl_config"], as_index=False)
    .agg(
        n_targets=("bar_col", "count"),

        event_rate_mean=("event_rate", "mean"),
        tp_rate_mean=("tp_rate", "mean"),
        sl_rate_mean=("sl_rate", "mean"),
        no_event_rate_mean=("no_event_rate", "mean"),

        event_step_mean=("event_step_mean", "mean"),
        event_step_median=("event_step_median", "mean"),
        event_step_q25=("event_step_q25", "mean"),
        event_step_q75=("event_step_q75", "mean"),

        tp_step_mean=("tp_step_mean", "mean"),
        tp_step_median=("tp_step_median", "mean"),

        sl_step_mean=("sl_step_mean", "mean"),
        sl_step_median=("sl_step_median", "mean"),

        early_event_rate_mean=("early_event_rate", "mean"),
        middle_event_rate_mean=("middle_event_rate", "mean"),
        late_event_rate_mean=("late_event_rate", "mean"),
    )
)
```

```

        .sort_values(["sl_mult", "tp_mult"])
        .reset_index(drop=True)
    )

print("\nResumen por configuración TP/SL:")
print(
    df_bar_event_time_by_config.to_string(
        index=False,
        formatters={
            "event_rate_mean": "{:.2%}".format,
            "tp_rate_mean": "{:.2%}".format,
            "sl_rate_mean": "{:.2%}".format,
            "no_event_rate_mean": "{:.2%}".format,
            "event_step_mean": "{:.2f}".format,
            "event_step_median": "{:.2f}".format,
            "event_step_q25": "{:.2f}".format,
            "event_step_q75": "{:.2f}".format,
            "tp_step_mean": "{:.2f}".format,
            "tp_step_median": "{:.2f}".format,
            "sl_step_mean": "{:.2f}".format,
            "sl_step_median": "{:.2f}".format,
            "early_event_rate_mean": "{:.2%}".format,
            "middle_event_rate_mean": "{:.2%}".format,
            "late_event_rate_mean": "{:.2%}".format,
        }
    )
)

# -----
# 2. Resumen por percentil y horizonte
# -----

df_bar_event_time_by_target_base = (
    df_bar_event_time
    .groupby(["percentile", "horizon"], as_index=False)
    .agg(
        n_configs=("bar_col", "count"),

        event_rate_mean=("event_rate", "mean"),
        tp_rate_mean=("tp_rate", "mean"),
        sl_rate_mean=("sl_rate", "mean"),
        no_event_rate_mean=("no_event_rate", "mean"),

        event_step_mean=("event_step_mean", "mean"),
        event_step_median=("event_step_median", "mean"),

        tp_step_mean=("tp_step_mean", "mean"),
        sl_step_mean=("sl_step_mean", "mean"),

        early_event_rate_mean=("early_event_rate", "mean"),
        middle_event_rate_mean=("middle_event_rate", "mean"),
        late_event_rate_mean=("late_event_rate", "mean"),
    )
    .sort_values(["percentile", "horizon"])
    .reset_index(drop=True)
)

print("\nResumen por percentil y horizonte:")
print(
    df_bar_event_time_by_target_base.to_string(

```

```

        index=False,
        formatters={
            "event_rate_mean": "{:.2%}".format,
            "tp_rate_mean": "{:.2%}".format,
            "sl_rate_mean": "{:.2%}".format,
            "no_event_rate_mean": "{:.2%}".format,
            "event_step_mean": "{:.2f}".format,
            "event_step_median": "{:.2f}".format,
            "tp_step_mean": "{:.2f}".format,
            "sl_step_mean": "{:.2f}".format,
            "early_event_rate_mean": "{:.2%}".format,
            "middle_event_rate_mean": "{:.2%}".format,
            "late_event_rate_mean": "{:.2%}".format,
        }
    )
)

# -----
# 3. Detalle de variantes centrales p50
# -----

df_bar_event_time_p50 = (
    df_bar_event_time[
        df_bar_event_time["percentile"] == 50
    ]
    .copy()
    .sort_values(["horizon", "sl_mult", "tp_mult"])
    .reset_index(drop=True)
)

print("\nDetalle de tiempo hasta evento para variantes p50:")
print(
    df_bar_event_time_p50[
        [
            "bar_col",
            "horizon",
            "tp_mult",
            "sl_mult",

            "event_rate",
            "tp_rate",
            "sl_rate",
            "no_event_rate",

            "event_step_mean",
            "event_step_median",
            "event_step_q25",
            "event_step_q75",

            "tp_step_mean",
            "tp_step_median",
            "sl_step_mean",
            "sl_step_median",

            "early_event_rate",
            "middle_event_rate",
            "late_event_rate",
        ]
    ]
    .to_string(

```

```

        index=False,
        formatters={
            "event_rate": "{:.2%}".format,
            "tp_rate": "{:.2%}".format,
            "sl_rate": "{:.2%}".format,
            "no_event_rate": "{:.2%}".format,
            "event_step_mean": "{:.2f}".format,
            "event_step_median": "{:.2f}".format,
            "event_step_q25": "{:.2f}".format,
            "event_step_q75": "{:.2f}".format,
            "tp_step_mean": "{:.2f}".format,
            "tp_step_median": "{:.2f}".format,
            "sl_step_mean": "{:.2f}".format,
            "sl_step_median": "{:.2f}".format,
            "early_event_rate": "{:.2%}".format,
            "middle_event_rate": "{:.2%}".format,
            "late_event_rate": "{:.2%}".format,
        }
    )
)

# -----
# 4. Ranking por resolución rápida
# -----
# early_event_rate mide qué proporción de eventos ocurre
# dentro del primer tercio del horizonte.
# -----

df_bar_event_time_fast_ranking = (
    df_bar_event_time
    .copy()
    .sort_values(
        ["early_event_rate", "event_rate", "tp_rate"],
        ascending=[False, False, False]
    )
    .reset_index(drop=True)
)

print("\nTop 20 variantes con eventos más tempranos:")
print(
    df_bar_event_time_fast_ranking[
        [
            "bar_col",
            "horizon",
            "percentile",
            "tp_mult",
            "sl_mult",
            "event_rate",
            "tp_rate",
            "sl_rate",
            "no_event_rate",
            "event_step_mean",
            "event_step_median",
            "early_event_rate",
            "middle_event_rate",
            "late_event_rate",
        ]
    ]
    .head(20)
    .to_string(

```

```

        index=False,
        formatters={
            "event_rate": "{:.2%}".format,
            "tp_rate": "{:.2%}".format,
            "sl_rate": "{:.2%}".format,
            "no_event_rate": "{:.2%}".format,
            "event_step_mean": "{:.2f}".format,
            "event_step_median": "{:.2f}".format,
            "early_event_rate": "{:.2%}".format,
            "middle_event_rate": "{:.2%}".format,
            "late_event_rate": "{:.2%}".format,
        }
    )
)

# -----
# 5. Ranking por eventos más lentos
# -----

df_bar_event_time_slow_ranking = (
    df_bar_event_time
    .copy()
    .sort_values(
        ["late_event_rate", "event_step_mean", "event_rate"],
        ascending=[False, False, False]
    )
    .reset_index(drop=True)
)

print("\nTop 20 variantes con eventos más tardíos:")
print(
    df_bar_event_time_slow_ranking[
        [
            "bar_col",
            "horizon",
            "percentile",
            "tp_mult",
            "sl_mult",
            "event_rate",
            "tp_rate",
            "sl_rate",
            "no_event_rate",
            "event_step_mean",
            "event_step_median",
            "early_event_rate",
            "middle_event_rate",
            "late_event_rate",
        ]
    ]
    .head(20)
    .to_string(
        index=False,
        formatters={
            "event_rate": "{:.2%}".format,
            "tp_rate": "{:.2%}".format,
            "sl_rate": "{:.2%}".format,
            "no_event_rate": "{:.2%}".format,
            "event_step_mean": "{:.2f}".format,
            "event_step_median": "{:.2f}".format,
            "early_event_rate": "{:.2%}".format,

```



```
        "middle_event_rate": "{:.2%}".format,  
        "late_event_rate": "{:.2%}".format,  
    }  
)  
)  
  
print("\nCelda 2 finalizada correctamente.")
```

## Tablas agregadas de tiempo hasta evento - BAR

=====

## Resumen por configuración TP/SL:

| tp_mult | sl_mult | tp_sl_config    | n_targets | event_rate_mean | tp_rate_mean | sl_rate_mean | no_event_rate_mean | event_step_mean | event_step_median | event_step_q25 | event_step_q75 | tp_step_mean | tp_step_median | sl_step_mean | sl_step_median | early_event_rate_mean | middle_event_rate_mean | late_event_rate_mean |        |
|---------|---------|-----------------|-----------|-----------------|--------------|--------------|--------------------|-----------------|-------------------|----------------|----------------|--------------|----------------|--------------|----------------|-----------------------|------------------------|----------------------|--------|
| 1.0     | 0.5     | TP 1.0 / SL 0.5 | 9         | 100.00%         | 81.07%       | 1            | 8.93%              | 0.00%           | 24.41             | 21.22          | 10.78          | 36.67        | 27.54          | 25.44        | 11.10          | 8.00                  | 48.43%                 | 31.55%               | 20.02% |
| 1.5     | 0.5     | TP 1.5 / SL 0.5 | 9         | 67.11%          | 44.39%       | 2            | 2.71%              | 32.89%          | 26.43             | 24.22          | 11.11          | 40.89        | 31.72          | 31.11        | 16.23          | 10.22                 | 43.88%                 | 30.75%               | 25.37% |
| 2.0     | 0.5     | TP 2.0 / SL 0.5 | 9         | 49.45%          | 25.21%       | 2            | 4.24%              | 50.55%          | 26.20             | 23.44          | 9.56           | 41.89        | 34.35          | 34.56        | 17.90          | 11.11                 | 45.29%                 | 28.03%               | 26.68% |
| 1.0     | 1.0     | TP 1.0 / SL 1.0 | 9         | 100.00%         | 95.18%       |              | 4.82%              | 0.00%           | 28.32             | 26.56          | 14.33          | 41.44        | 28.99          | 27.67        | 14.94          | 12.00                 | 38.24%                 | 35.56%               | 26.20% |
| 1.5     | 1.0     | TP 1.5 / SL 1.0 | 9         | 58.58%          | 52.73%       |              | 5.85%              | 41.42%          | 31.56             | 31.11          | 18.11          | 45.31        | 32.84          | 32.56        | 19.93          | 14.78                 | 30.43%                 | 36.55%               | 33.02% |
| 2.0     | 1.0     | TP 2.0 / SL 1.0 | 9         | 36.90%          | 30.26%       |              | 6.65%              | 63.10%          | 33.08             | 33.44          | 19.44          | 47.33        | 35.27          | 36.00        | 23.05          | 17.00                 | 27.52%                 | 35.33%               | 37.15% |
| 2.5     | 1.0     | TP 2.5 / SL 1.0 | 9         | 25.08%          | 18.02%       |              | 7.06%              | 74.92%          | 33.39             | 34.33          | 19.00          | 48.44        | 36.93          | 38.56        | 24.40          | 18.89                 | 27.88%                 | 33.11%               | 39.02% |

## Resumen por percentil y horizonte:

| percentile | horizon | n_configs | event_rate_mean | tp_rate_mean | sl_rate_mean | no_event_rate_mean | event_step_mean | event_step_median | tp_step_mean | sl_step_mean | early_event_rate_mean | middle_event_rate_mean | late_event_rate_mean |
|------------|---------|-----------|-----------------|--------------|--------------|--------------------|-----------------|-------------------|--------------|--------------|-----------------------|------------------------|----------------------|
| 40         | 30      | 7         | 67.17%          | 52.34%       | 14.83%       | 32.83%             | 13.71           | 12.57             | 15.36        | 8.65         | 42.15%                | 32.50%                 | 25.35%               |
| 40         | 60      | 7         | 66.42%          | 51.09%       | 15.32%       | 33.58%             | 27.00           | 24.71             | 30.52        | 16.86        | 42.12%                | 32.15%                 | 25.73%               |
| 40         | 90      | 7         | 66.68%          | 50.70%       | 15.98%       | 33.32%             | 40.97           | 38.00             | 46.48        | 26.18        | 41.21%                | 31.69%                 | 27.10%               |
| 50         | 30      | 7         | 62.76%          | 50.36%       | 12.40%       | 37.24%             | 14.62           | 13.86             | 16.24        | 9.10         | 37.57%                | 33.63%                 | 28.80%               |
| 50         | 60      | 7         | 61.81%          | 49.20%       | 12.61%       | 38.19%             | 28.81           | 27.43             | 32.26        | 17.69        | 37.54%                | 33.32%                 | 29.14%               |
| 50         | 90      | 7         | 62.39%          | 49.04%       | 13.36%       | 37.61%             | 43.86           | 42.14             | 49.24        | 27.85        | 36.55%                | 32.48%                 | 30.97%               |
| 60         | 30      | 7         | 58.58%          | 48.39%       | 10.19%       |                    |                 |                   |              |              |                       |                        |                      |

|        |       |        |        |        |
|--------|-------|--------|--------|--------|
| 41.42% | 15.47 | 15.29  | 17.05  | 9.51   |
| 33.40% |       | 34.32% | 32.28% |        |
| 60     | 60    | 7      | 57.75% | 47.41% |
| 42.25% | 30.42 | 29.71  | 33.76  | 18.47  |
| 33.57% |       | 33.99% | 32.43% |        |
| 60     | 90    | 7      | 58.45% | 47.43% |
| 41.55% | 46.64 | 46.14  | 51.78  | 29.67  |
| 32.30% |       | 32.77% | 34.93% |        |

Detalle de tiempo hasta evento para variantes p50:

| bar_col               | horizon          | tp_mult           | sl_mult         | event_rate     | tp_rate      | sl_rate        | no_ev        |
|-----------------------|------------------|-------------------|-----------------|----------------|--------------|----------------|--------------|
| ent_rate              | event_step_mean  | event_step_median | event_step_q25  | event_step_q75 | tp_step_mean | tp_step_median | sl_step_mean |
| sl_step_median        | early_event_rate | middle_event_rate | late_event_rate |                |              |                |              |
| bar_p50_h30_tp10_sl05 | 30               | 1.0               | 0.5             | 100.00%        | 81.93%       | 18.07%         |              |
| 0.00%                 | 12.37            | 11.00             | 6.00            | 18.00          | 13.8         |                |              |
| 4                     | 13.00            | 5.70              | 4.00            | 48.62%         |              | 31.90%         |              |
| 19.49%                |                  |                   |                 |                |              |                |              |
| bar_p50_h30_tp15_sl05 | 30               | 1.5               | 0.5             | 67.40%         | 45.74%       | 21.67%         |              |
| 32.60%                | 13.40            | 12.00             | 6.00            | 20.00          | 15.          |                |              |
| 82                    | 15.00            | 8.28              | 5.00            | 43.77%         |              | 31.52%         |              |
| 24.70%                |                  |                   |                 |                |              |                |              |
| bar_p50_h30_tp20_sl05 | 30               | 2.0               | 0.5             | 49.50%         | 26.31%       | 23.19%         |              |
| 50.50%                | 13.39            | 12.00             | 5.00            | 21.00          | 17.          |                |              |
| 11                    | 17.00            | 9.16              | 6.00            | 44.78%         |              | 28.79%         |              |
| 26.43%                |                  |                   |                 |                |              |                |              |
| bar_p50_h30_tp10_sl10 | 30               | 1.0               | 1.0             | 100.00%        | 95.26%       | 4.74%          |              |
| 0.00%                 | 14.21            | 13.00             | 7.00            | 21.00          | 14.5         |                |              |
| 5                     | 14.00            | 7.38              | 6.00            | 38.74%         |              | 35.86%         |              |
| 25.41%                |                  |                   |                 |                |              |                |              |
| bar_p50_h30_tp15_sl10 | 30               | 1.5               | 1.0             | 59.30%         | 53.60%       | 5.71%          |              |
| 40.70%                | 15.74            | 15.00             | 9.00            | 22.00          | 16.          |                |              |
| 37                    | 16.00            | 9.77              | 7.00            | 31.05%         |              | 37.21%         |              |
| 31.75%                |                  |                   |                 |                |              |                |              |
| bar_p50_h30_tp20_sl10 | 30               | 2.0               | 1.0             | 37.56%         | 31.07%       | 6.49%          |              |
| 62.44%                | 16.50            | 17.00             | 10.00           | 24.00          | 17.          |                |              |
| 58                    | 18.00            | 11.34             | 8.00            | 28.03%         |              | 36.02%         |              |
| 35.95%                |                  |                   |                 |                |              |                |              |
| bar_p50_h30_tp25_sl10 | 30               | 2.5               | 1.0             | 25.53%         | 18.63%       | 6.90%          |              |
| 74.47%                | 16.70            | 17.00             | 10.00           | 24.00          | 18.          |                |              |
| 42                    | 19.00            | 12.05             | 9.00            | 28.01%         |              | 34.11%         |              |
| 37.88%                |                  |                   |                 |                |              |                |              |
| bar_p50_h60_tp10_sl05 | 60               | 1.0               | 0.5             | 100.00%        | 81.32%       | 18.68%         |              |
| 0.00%                 | 24.40            | 21.00             | 10.00           | 37.00          | 27.5         |                |              |
| 3                     | 25.00            | 10.80             | 8.00            | 48.36%         |              | 31.58%         |              |
| 20.06%                |                  |                   |                 |                |              |                |              |
| bar_p50_h60_tp15_sl05 | 60               | 1.5               | 0.5             | 66.52%         | 44.00%       | 22.52%         |              |
| 33.48%                | 26.28            | 24.00             | 11.00           | 41.00          | 31.          |                |              |
| 51                    | 31.00            | 16.05             | 10.00           | 44.09%         |              | 30.86%         |              |
| 25.04%                |                  |                   |                 |                |              |                |              |
| bar_p50_h60_tp20_sl05 | 60               | 2.0               | 0.5             | 48.56%         | 24.58%       | 23.99%         |              |
| 51.44%                | 25.95            | 23.00             | 9.00            | 41.00          | 34.          |                |              |
| 01                    | 34.00            | 17.68             | 11.00           | 45.68%         |              | 28.34%         |              |
| 25.98%                |                  |                   |                 |                |              |                |              |
| bar_p50_h60_tp10_sl10 | 60               | 1.0               | 1.0             | 100.00%        | 95.45%       | 4.55%          |              |
| 0.00%                 | 28.22            | 26.00             | 14.00           | 41.00          | 28.8         |                |              |
| 9                     | 27.00            | 14.12             | 11.00           | 38.19%         |              | 35.70%         |              |
| 26.11%                |                  |                   |                 |                |              |                |              |
| bar_p50_h60_tp15_sl10 | 60               | 1.5               | 1.0             | 57.77%         | 52.23%       | 5.54%          |              |
| 42.23%                | 31.28            | 31.00             | 18.00           | 45.00          | 32.          |                |              |

|                       |       |       |       |         |        |
|-----------------------|-------|-------|-------|---------|--------|
| 56                    | 32.00 | 19.20 | 14.00 | 30.53%  | 36.95% |
| 32.52%                |       |       |       |         |        |
| bar_p50_h60_tp20_sl10 | 60    | 2.0   | 1.0   | 35.78%  | 29.49% |
| 64.22%                | 32.68 | 33.00 | 19.00 | 47.00   | 34.    |
| 89                    | 35.00 | 22.33 | 16.00 | 27.63%  | 36.08% |
| 36.29%                |       |       |       |         |        |
| bar_p50_h60_tp25_sl10 | 60    | 2.5   | 1.0   | 24.02%  | 17.33% |
| 75.98%                | 32.87 | 34.00 | 18.00 | 48.00   | 36.    |
| 42                    | 38.00 | 23.68 | 18.00 | 28.31%  | 33.73% |
| 37.95%                |       |       |       |         |        |
| bar_p50_h90_tp10_sl05 | 90    | 1.0   | 0.5   | 100.00% | 80.23% |
| 0.00%                 | 36.58 | 32.00 | 16.00 | 55.00   | 41.4   |
| 5                     | 39.00 | 16.80 | 12.00 | 47.89%  | 31.67% |
| 20.45%                |       |       |       |         |        |
| bar_p50_h90_tp15_sl05 | 90    | 1.5   | 0.5   | 67.30%  | 43.60% |
| 32.70%                | 39.72 | 36.00 | 16.00 | 62.00   | 48.    |
| 03                    | 48.00 | 24.45 | 15.00 | 43.51%  | 30.21% |
| 26.28%                |       |       |       |         |        |
| bar_p50_h90_tp20_sl05 | 90    | 2.0   | 0.5   | 49.77%  | 24.52% |
| 50.23%                | 39.31 | 35.00 | 14.00 | 64.00   | 52.    |
| 06                    | 53.00 | 26.93 | 17.00 | 45.08%  | 27.39% |
| 27.52%                |       |       |       |         |        |
| bar_p50_h90_tp10_sl10 | 90    | 1.0   | 1.0   | 100.00% | 95.11% |
| 0.00%                 | 42.70 | 40.00 | 22.00 | 63.00   | 43.6   |
| 9                     | 42.00 | 23.37 | 19.00 | 37.20%  | 35.60% |
| 27.20%                |       |       |       |         |        |
| bar_p50_h90_tp15_sl10 | 90    | 1.5   | 1.0   | 58.31%  | 52.38% |
| 41.69%                | 47.81 | 48.00 | 27.00 | 69.00   | 49.    |
| 75                    | 50.00 | 30.65 | 23.00 | 29.23%  | 35.98% |
| 34.79%                |       |       |       |         |        |
| bar_p50_h90_tp20_sl10 | 90    | 2.0   | 1.0   | 36.55%  | 29.78% |
| 63.45%                | 50.14 | 51.00 | 29.00 | 72.00   | 53.    |
| 49                    | 55.00 | 35.38 | 27.00 | 26.24%  | 34.57% |
| 39.19%                |       |       |       |         |        |
| bar_p50_h90_tp25_sl10 | 90    | 2.5   | 1.0   | 24.82%  | 17.65% |
| 75.18%                | 50.79 | 53.00 | 29.00 | 74.00   | 56.    |
| 23                    | 59.00 | 37.39 | 29.00 | 26.72%  | 31.90% |
| 41.38%                |       |       |       |         |        |

Top 20 variantes con eventos más tempranos:

| bar_col               | horizon       | percentile      | tp_mult           | sl_mult          | event_rate | tp_rate  | s               |
|-----------------------|---------------|-----------------|-------------------|------------------|------------|----------|-----------------|
| l_rate                | no_event_rate | event_step_mean | event_step_median | early_event_rate | middle_ev  | ent_rate | late_event_rate |
| bar_p40_h30_tp10_sl05 | 30            | 40              | 1.0               | 0.5              | 100.00%    | 78.98%   |                 |
| 21.02%                | 0.00%         | 11.18           | 9.00              | 55.06%           |            |          |                 |
| 29.29%                | 15.65%        |                 |                   |                  |            |          |                 |
| bar_p40_h60_tp10_sl05 | 60            | 40              | 1.0               | 0.5              | 100.00%    | 77.94%   |                 |
| 22.06%                | 0.00%         | 21.95           | 18.00             | 54.96%           |            |          |                 |
| 28.94%                | 16.10%        |                 |                   |                  |            |          |                 |
| bar_p40_h90_tp10_sl05 | 90            | 40              | 1.0               | 0.5              | 100.00%    | 77.10%   |                 |
| 22.90%                | 0.00%         | 32.93           | 27.00             | 54.24%           |            |          |                 |
| 29.44%                | 16.32%        |                 |                   |                  |            |          |                 |
| bar_p40_h60_tp20_sl05 | 60            | 40              | 2.0               | 0.5              | 56.15%     | 27.84%   |                 |
| 28.31%                | 43.85%        | 24.57           | 21.00             | 49.08%           |            |          |                 |
| 27.40%                | 23.52%        |                 |                   |                  |            |          |                 |
| bar_p40_h60_tp15_sl05 | 60            | 40              | 1.5               | 0.5              | 72.88%     | 46.45%   |                 |
| 26.43%                | 27.12%        | 24.35           | 21.00             | 48.90%           |            |          |                 |
| 29.53%                | 21.57%        |                 |                   |                  |            |          |                 |
| bar_p50_h30_tp10_sl05 | 30            | 50              | 1.0               | 0.5              | 100.00%    | 81.93%   |                 |
| 18.07%                | 0.00%         | 12.37           | 11.00             | 48.62%           |            |          |                 |

|                       |        |       |    |       |     |         |        |
|-----------------------|--------|-------|----|-------|-----|---------|--------|
| 31.90%                | 19.49% |       |    |       |     |         |        |
| bar_p40_h30_tp15_sl05 |        | 30    | 40 | 1.5   | 0.5 | 73.21%  | 48.14% |
| 25.07%                | 26.79% | 12.46 |    | 11.00 |     | 48.61%  |        |
| 30.00%                | 21.39% |       |    |       |     |         |        |
| bar_p40_h90_tp15_sl05 |        | 90    | 40 | 1.5   | 0.5 | 72.84%  | 45.52% |
| 27.32%                | 27.16% | 36.63 |    | 32.00 |     | 48.55%  |        |
| 29.16%                | 22.30% |       |    |       |     |         |        |
| bar_p40_h90_tp20_sl05 |        | 90    | 40 | 2.0   | 0.5 | 56.66%  | 27.45% |
| 29.21%                | 43.34% | 37.08 |    | 32.00 |     | 48.52%  |        |
| 26.84%                | 24.64% |       |    |       |     |         |        |
| bar_p50_h60_tp10_sl05 |        | 60    | 50 | 1.0   | 0.5 | 100.00% | 81.32% |
| 18.68%                | 0.00%  | 24.40 |    | 21.00 |     | 48.36%  |        |
| 31.58%                | 20.06% |       |    |       |     |         |        |
| bar_p40_h30_tp20_sl05 |        | 30    | 40 | 2.0   | 0.5 | 56.62%  | 29.68% |
| 26.94%                | 43.38% | 12.70 |    | 11.00 |     | 48.01%  |        |
| 28.26%                | 23.74% |       |    |       |     |         |        |
| bar_p50_h90_tp10_sl05 |        | 90    | 50 | 1.0   | 0.5 | 100.00% | 80.23% |
| 19.77%                | 0.00%  | 36.58 |    | 32.00 |     | 47.89%  |        |
| 31.67%                | 20.45% |       |    |       |     |         |        |
| bar_p50_h60_tp20_sl05 |        | 60    | 50 | 2.0   | 0.5 | 48.56%  | 24.58% |
| 23.99%                | 51.44% | 25.95 |    | 23.00 |     | 45.68%  |        |
| 28.34%                | 25.98% |       |    |       |     |         |        |
| bar_p50_h90_tp20_sl05 |        | 90    | 50 | 2.0   | 0.5 | 49.77%  | 24.52% |
| 25.25%                | 50.23% | 39.31 |    | 35.00 |     | 45.08%  |        |
| 27.39%                | 27.52% |       |    |       |     |         |        |
| bar_p50_h30_tp20_sl05 |        | 30    | 50 | 2.0   | 0.5 | 49.50%  | 26.31% |
| 23.19%                | 50.50% | 13.39 |    | 12.00 |     | 44.78%  |        |
| 28.79%                | 26.43% |       |    |       |     |         |        |
| bar_p40_h30_tp10_sl10 |        | 30    | 40 | 1.0   | 1.0 | 100.00% | 93.84% |
| 6.16%                 | 0.00%  | 13.20 |    | 12.00 |     | 44.23%  |        |
| 33.99%                | 21.78% |       |    |       |     |         |        |
| bar_p50_h60_tp15_sl05 |        | 60    | 50 | 1.5   | 0.5 | 66.52%  | 44.00% |
| 22.52%                | 33.48% | 26.28 |    | 24.00 |     | 44.09%  |        |
| 30.86%                | 25.04% |       |    |       |     |         |        |
| bar_p50_h30_tp15_sl05 |        | 30    | 50 | 1.5   | 0.5 | 67.40%  | 45.74% |
| 21.67%                | 32.60% | 13.40 |    | 12.00 |     | 43.77%  |        |
| 31.52%                | 24.70% |       |    |       |     |         |        |
| bar_p40_h60_tp10_sl10 |        | 60    | 40 | 1.0   | 1.0 | 100.00% | 94.00% |
| 6.00%                 | 0.00%  | 26.19 |    | 24.00 |     | 43.64%  |        |
| 33.89%                | 22.47% |       |    |       |     |         |        |
| bar_p50_h90_tp15_sl05 |        | 90    | 50 | 1.5   | 0.5 | 67.30%  | 43.60% |
| 23.71%                | 32.70% | 39.72 |    | 36.00 |     | 43.51%  |        |
| 30.21%                | 26.28% |       |    |       |     |         |        |

Top 20 variantes con eventos más tardíos:

| bar_col               | horizon       | percentile      | tp_mult           | sl_mult          | event_rate | tp_rate  | s               |
|-----------------------|---------------|-----------------|-------------------|------------------|------------|----------|-----------------|
| l_rate                | no_event_rate | event_step_mean | event_step_median | early_event_rate | middle_ev  | ent_rate | late_event_rate |
| bar_p60_h90_tp25_sl10 |               | 90              | 60                | 2.5              | 1.0        | 19.82%   | 14.40%          |
| 5.42%                 | 80.18%        | 53.00           |                   | 56.00            |            | 23.89%   |                 |
| 31.10%                | 45.01%        |                 |                   |                  |            |          |                 |
| bar_p60_h90_tp20_sl10 |               | 90              | 60                | 2.0              | 1.0        | 31.02%   | 25.89%          |
| 5.12%                 | 68.98%        | 52.77           |                   | 55.00            |            | 22.71%   |                 |
| 33.95%                | 43.34%        |                 |                   |                  |            |          |                 |
| bar_p50_h90_tp25_sl10 |               | 90              | 50                | 2.5              | 1.0        | 24.82%   | 17.65%          |
| 7.17%                 | 75.18%        | 50.79           |                   | 53.00            |            | 26.72%   |                 |
| 31.90%                | 41.38%        |                 |                   |                  |            |          |                 |
| bar_p60_h30_tp25_sl10 |               | 30              | 60                | 2.5              | 1.0        | 20.27%   | 15.06%          |
| 5.21%                 | 79.73%        | 17.32           |                   | 18.00            |            | 25.50%   |                 |
| 33.50%                | 41.00%        |                 |                   |                  |            |          |                 |

|                       |        |       |       |     |        |        |
|-----------------------|--------|-------|-------|-----|--------|--------|
| bar_p60_h60_tp25_sl10 | 60     | 60    | 2.5   | 1.0 | 19.20% | 14.23% |
| 4.97%                 | 80.80% | 34.09 | 35.00 |     | 25.73% |        |
| 33.29%                | 40.99% |       |       |     |        |        |
| bar_p60_h60_tp20_sl10 | 60     | 60    | 2.0   | 1.0 | 30.20% | 25.48% |
| 4.72%                 | 69.80% | 34.12 | 35.00 |     | 24.39% |        |
| 35.88%                | 39.72% |       |       |     |        |        |
| bar_p60_h30_tp20_sl10 | 30     | 60    | 2.0   | 1.0 | 31.72% | 26.78% |
| 4.94%                 | 68.28% | 17.26 | 18.00 |     | 24.52% |        |
| 36.10%                | 39.38% |       |       |     |        |        |
| bar_p50_h90_tp20_sl10 | 90     | 50    | 2.0   | 1.0 | 36.55% | 29.78% |
| 6.77%                 | 63.45% | 50.14 | 51.00 |     | 26.24% |        |
| 34.57%                | 39.19% |       |       |     |        |        |
| bar_p60_h90_tp15_sl10 | 90     | 60    | 1.5   | 1.0 | 53.30% | 48.78% |
| 4.52%                 | 46.70% | 50.61 | 51.00 |     | 24.77% |        |
| 36.09%                | 39.13% |       |       |     |        |        |
| bar_p50_h60_tp25_sl10 | 60     | 50    | 2.5   | 1.0 | 24.02% | 17.33% |
| 6.69%                 | 75.98% | 32.87 | 34.00 |     | 28.31% |        |
| 33.73%                | 37.95% |       |       |     |        |        |
| bar_p50_h30_tp25_sl10 | 30     | 50    | 2.5   | 1.0 | 25.53% | 18.63% |
| 6.90%                 | 74.47% | 16.70 | 17.00 |     | 28.01% |        |
| 34.11%                | 37.88% |       |       |     |        |        |
| bar_p40_h90_tp25_sl10 | 90     | 40    | 2.5   | 1.0 | 30.62% | 21.19% |
| 9.43%                 | 69.38% | 48.34 | 49.00 |     | 30.03% |        |
| 32.41%                | 37.56% |       |       |     |        |        |
| bar_p50_h60_tp20_sl10 | 60     | 50    | 2.0   | 1.0 | 35.78% | 29.49% |
| 6.29%                 | 64.22% | 32.68 | 33.00 |     | 27.63% |        |
| 36.08%                | 36.29% |       |       |     |        |        |
| bar_p60_h60_tp15_sl10 | 60     | 60    | 1.5   | 1.0 | 52.42% | 48.27% |
| 4.14%                 | 47.58% | 32.97 | 33.00 |     | 26.35% |        |
| 37.51%                | 36.14% |       |       |     |        |        |
| bar_p50_h30_tp20_sl10 | 30     | 50    | 2.0   | 1.0 | 37.56% | 31.07% |
| 6.49%                 | 62.44% | 16.50 | 17.00 |     | 28.03% |        |
| 36.02%                | 35.95% |       |       |     |        |        |
| bar_p60_h30_tp15_sl10 | 30     | 60    | 1.5   | 1.0 | 53.89% | 49.53% |
| 4.37%                 | 46.11% | 16.62 | 17.00 |     | 26.82% |        |
| 37.55%                | 35.63% |       |       |     |        |        |
| bar_p40_h90_tp20_sl10 | 90     | 40    | 2.0   | 1.0 | 42.92% | 34.09% |
| 8.83%                 | 57.08% | 47.51 | 47.00 |     | 30.35% |        |
| 34.28%                | 35.37% |       |       |     |        |        |
| bar_p40_h60_tp25_sl10 | 60     | 40    | 2.5   | 1.0 | 29.89% | 21.05% |
| 8.84%                 | 70.11% | 31.48 | 31.00 |     | 31.37% |        |
| 33.80%                | 34.83% |       |       |     |        |        |
| bar_p50_h90_tp15_sl10 | 90     | 50    | 1.5   | 1.0 | 58.31% | 52.38% |
| 5.93%                 | 41.69% | 47.81 | 48.00 |     | 29.23% |        |
| 35.98%                | 34.79% |       |       |     |        |        |
| bar_p40_h30_tp25_sl10 | 30     | 40    | 2.5   | 1.0 | 31.51% | 22.59% |
| 8.92%                 | 68.49% | 15.97 | 16.00 |     | 31.32% |        |
| 34.13%                | 34.54% |       |       |     |        |        |

Celda 2 finalizada correctamente.

## Celda 3 — Gráficos agrupados

In [105...

```
# Celda 3 – Gráficos agrupados por percentil y horizonte
# =====

print("Gráficos agrupados por percentil y horizonte - Tiempo hasta evento BAR")
print("=" * 80)
```

```

# -----
# Validar objetos necesarios
# -----

if "df_bar_event_time" not in globals():
    raise NameError("No existe df_bar_event_time. Primero debes correr la Celda

# -----
# Orden de configuraciones TP/SL
# -----

config_order = [
    (1.0, 1.0),
    (1.5, 1.0),
    (2.0, 1.0),
    (2.5, 1.0),
    (1.0, 0.5),
    (1.5, 0.5),
    (2.0, 0.5),
]

config_labels = [
    f"TP {tp} / SL {sl}"
    for tp, sl in config_order
]

df_event_plot = df_bar_event_time.copy()

df_event_plot["config_label"] = df_event_plot.apply(
    lambda row: f"TP {row['tp_mult']} / SL {row['sl_mult']}",
    axis=1
)

df_event_plot["config_order"] = df_event_plot.apply(
    lambda row: config_order.index((row["tp_mult"], row["sl_mult"]))
    if (row["tp_mult"], row["sl_mult"]) in config_order
    else 999,
    axis=1
)

df_event_plot = (
    df_event_plot
    .sort_values(["percentile", "horizon", "config_order"])
    .reset_index(drop=True)
)

# =====
# Figura 1 – Tiempo promedio hasta evento
# Filas: percentil | Columnas: horizonte
# =====

fig, axes = plt.subplots(
    nrows=len(BAR_PERCENTILES),
    ncols=len(BAR_HORIZONS),
    figsize=(18, 12),
    sharey=True
)

global_max_step = df_event_plot["event_step_mean"].max()

```

```

for i, percentile in enumerate(BAR_PERCENTILES):
    for j, horizon in enumerate(BAR_HORIZONS):
        ax = axes[i, j]

        tmp = df_event_plot[
            (df_event_plot["percentile"] == percentile)
            & (df_event_plot["horizon"] == horizon)
        ].copy()

        tmp = tmp.sort_values("config_order")

        x = np.arange(len(tmp))

        ax.bar(
            x,
            tmp["event_step_mean"],
            width=0.6
        )

        ax.set_title(f"p{percentile} | H{horizon}")
        ax.set_ylim(0, global_max_step * 1.15)

        if j == 0:
            ax.set_ylabel("Barras hasta evento")

        if i == len(BAR_PERCENTILES) - 1:
            ax.set_xticks(x)
            ax.set_xticklabels(config_labels, rotation=45, ha="right")
        else:
            ax.set_xticks(x)
            ax.set_xticklabels([])

        for k, value in enumerate(tmp["event_step_mean"]):
            ax.text(
                k,
                value,
                f"{value:.1f}",
                ha="center",
                va="bottom",
                fontsize=7
            )

fig.suptitle(
    "Tiempo promedio hasta evento por percentil, horizonte y configuración TP/SL",
    fontsize=14
)

plt.tight_layout()
plt.show()

# =====
# Figura 2 – Tiempo promedio hasta TP vs SL
# Filas: percentil | Columnas: horizonte
# =====

fig, axes = plt.subplots(
    nrows=len(BAR_PERCENTILES),
    ncols=len(BAR_HORIZONS),
    figsize=(18, 12),
    sharey=True

```



```

)

bar_width = 0.35

global_max_tp_sl_step = max(
    df_event_plot["tp_step_mean"].max(),
    df_event_plot["sl_step_mean"].max()
)

for i, percentile in enumerate(BAR_PERCENTILES):
    for j, horizon in enumerate(BAR_HORIZONS):
        ax = axes[i, j]

        tmp = df_event_plot[
            (df_event_plot["percentile"] == percentile)
            & (df_event_plot["horizon"] == horizon)
        ].copy()

        tmp = tmp.sort_values("config_order")

        x = np.arange(len(tmp))

        ax.bar(
            x - bar_width / 2,
            tmp["tp_step_mean"],
            width=bar_width,
            label="TP"
        )

        ax.bar(
            x + bar_width / 2,
            tmp["sl_step_mean"],
            width=bar_width,
            label="SL"
        )

        ax.set_title(f"p{percentile} | H{horizon}")
        ax.set_ylim(0, global_max_tp_sl_step * 1.15)

        if j == 0:
            ax.set_ylabel("Barras hasta evento")

        if i == len(BAR_PERCENTILES) - 1:
            ax.set_xticks(x)
            ax.set_xticklabels(config_labels, rotation=45, ha="right")
        else:
            ax.set_xticks(x)
            ax.set_xticklabels([])

        if i == 0 and j == 0:
            ax.legend()

        for k, row in enumerate(tmp.itertuples()):
            ax.text(
                k - bar_width / 2,
                row.tp_step_mean,
                f"{row.tp_step_mean:.1f}",
                ha="center",
                va="bottom",
                fontsize=6
            )

```

```

    )

    ax.text(
        k + bar_width / 2,
        row.sl_step_mean,
        f"{row.sl_step_mean:.1f}",
        ha="center",
        va="bottom",
        fontsize=6
    )

fig.suptitle(
    "Tiempo promedio hasta TP vs SL por percentil, horizonte y configuración TP/"
    fontsize=14
)

plt.tight_layout()
plt.show()

# =====
# Figura 3 – Eventos tempranos, medios y tardíos
# Filas: percentil | Columnas: horizonte
# =====

fig, axes = plt.subplots(
    nrows=len(BAR_PERCENTILES),
    ncols=len(BAR_HORIZONS),
    figsize=(18, 12),
    sharey=True
)

for i, percentile in enumerate(BAR_PERCENTILES):
    for j, horizon in enumerate(BAR_HORIZONS):
        ax = axes[i, j]

        tmp = df_event_plot[
            (df_event_plot["percentile"] == percentile)
            & (df_event_plot["horizon"] == horizon)
        ].copy()

        tmp = tmp.sort_values("config_order")

        x = np.arange(len(tmp))

        early = tmp["early_event_rate"].to_numpy()
        middle = tmp["middle_event_rate"].to_numpy()
        late = tmp["late_event_rate"].to_numpy()

        ax.bar(
            x,
            early,
            width=0.6,
            label="Temprano"
        )

        ax.bar(
            x,
            middle,
            width=0.6,
            bottom=early,

```

```

        label="Medio"
    )

    ax.bar(
        x,
        late,
        width=0.6,
        bottom=early + middle,
        label="Tardío"
    )

    ax.set_title(f"p{percentile} | H{horizon}")
    ax.set_ylim(0, 1)

    if j == 0:
        ax.set_ylabel("Proporción de eventos")

    if i == len(BAR_PERCENTILES) - 1:
        ax.set_xticks(x)
        ax.set_xticklabels(config_labels, rotation=45, ha="right")
    else:
        ax.set_xticks(x)
        ax.set_xticklabels([])

    ax.yaxis.set_major_formatter(
        plt.FuncFormatter(lambda y, _: f"{y:.0%}")
    )

    if i == 0 and j == 0:
        ax.legend()

fig.suptitle(
    "Distribución temporal de eventos por percentil, horizonte y configuración T
    fontsize=14
)

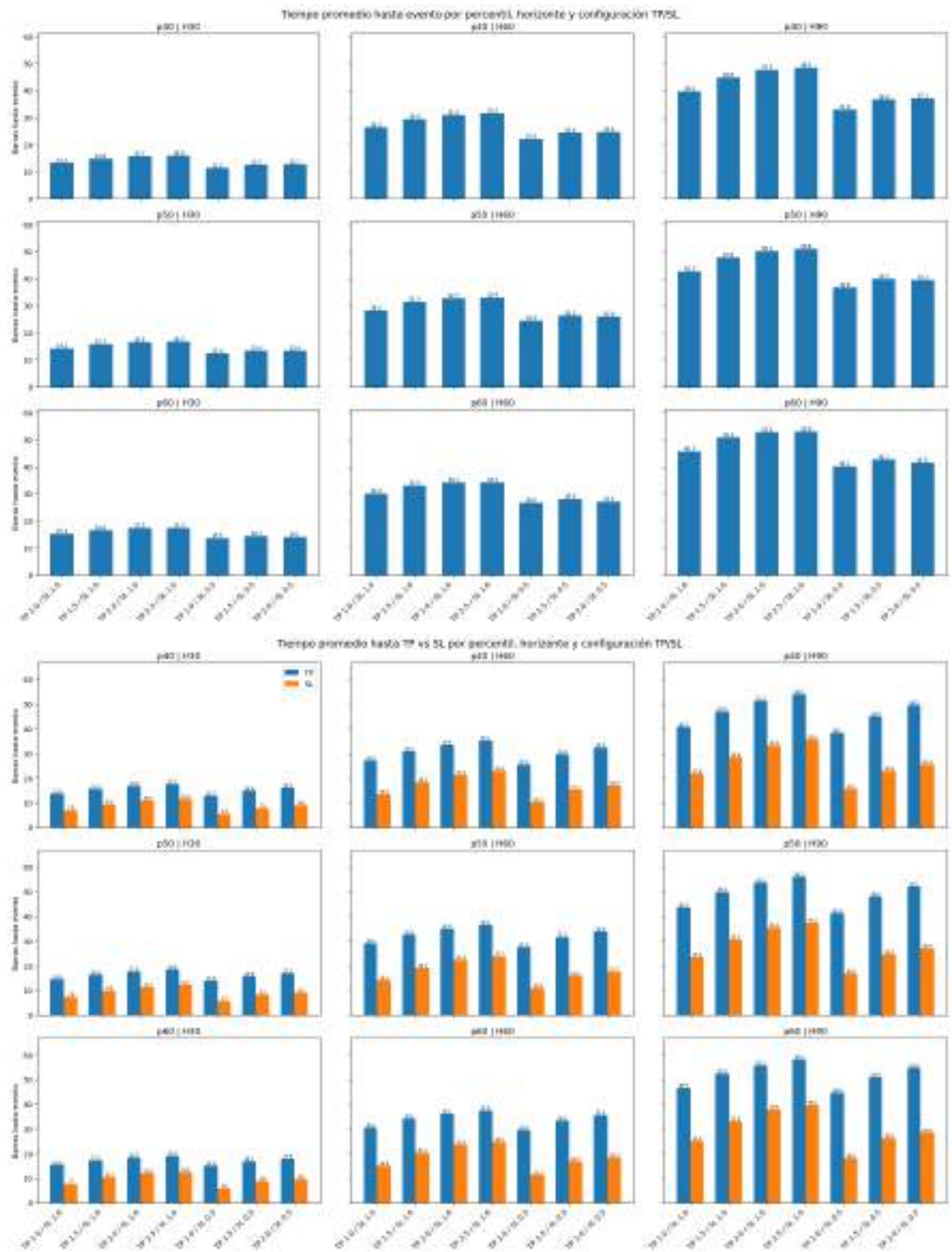
plt.tight_layout()
plt.show()

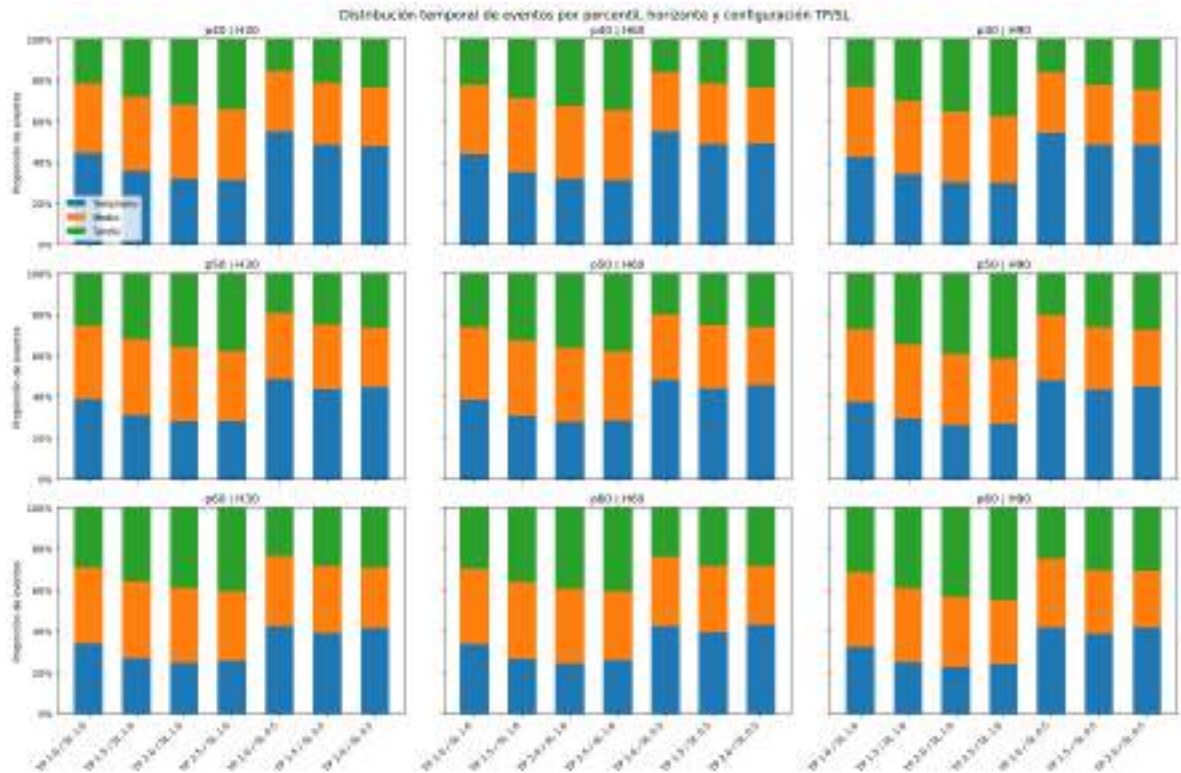
print("\nCelda 3 finalizada correctamente.")

```

Gráficos agrupados por percentil y horizonte - Tiempo hasta evento BAR

=====





Celda 3 finalizada correctamente.

## Celda 4 — Guardado y tabla final para observaciones

In [106...

```
# Celda 4 - Guardado y tabla final para observaciones
# =====

print("Guardado y tabla final de tiempo hasta evento TP/SL - BAR")
print("=" * 80)

# -----
# Validar objetos necesarios
# -----

required_objects = [
    "df_bar_event_time",
    "df_bar_event_time_by_config",
    "df_bar_event_time_by_target_base",
    "df_bar_event_time_fast_ranking",
    "df_bar_event_time_slow_ranking",
    "MNQ_TARGETS_PATH",
]

missing_objects = [
    obj for obj in required_objects
    if obj not in globals()
]

if missing_objects:
    raise NameError(
        "Faltan objetos necesarios. "
        f"Debes correr las celdas anteriores: {missing_objects}"
    )

# -----
# Definir rutas de guardado
```

```

# -----
MNQ_TARGETS_BAR_EVENT_TIME_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_event_time.csv"
)

MNQ_TARGETS_BAR_EVENT_TIME_BY_CONFIG_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_event_time_by_config.csv"
)

MNQ_TARGETS_BAR_EVENT_TIME_BY_TARGET_BASE_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_event_time_by_target_base.csv"
)

MNQ_TARGETS_BAR_EVENT_TIME_FAST_RANKING_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_event_time_fast_ranking.csv"
)

MNQ_TARGETS_BAR_EVENT_TIME_SLOW_RANKING_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_event_time_slow_ranking.csv"
)

MNQ_TARGETS_BAR_EVENT_TIME_FINAL_TABLE_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_event_time_final_table.csv"
)

MNQ_TARGETS_BAR_EVENT_TIME_REPORT_TXT = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_event_time_report.txt"
)

# -----
# Crear tabla final para observaciones
# -----

df_bar_event_time_final_table = df_bar_event_time.copy()

df_bar_event_time_final_table["target_base"] = (
    "p"
    + df_bar_event_time_final_table["percentile"].astype(str)
    + "_h"
    + df_bar_event_time_final_table["horizon"].astype(str)
)

# Clasificación simple del tiempo de resolución
def classify_event_timing(row):
    """
    Clasificación descriptiva del tiempo hasta evento.
    No selecciona targets finales.
    Solo ayuda a interpretar si los eventos tienden a ocurrir temprano,
    en la parte media o tarde del horizonte.
    """

    early = row["early_event_rate"]
    late = row["late_event_rate"]
    event_rate = row["event_rate"]

    if pd.isna(event_rate) or event_rate == 0:
        return "sin_eventos"

    if early >= 0.50:

```

```

        return "resolución_temprana"

    if late >= 0.50:
        return "resolución_tardía"

    return "resolución_intermedia"

df_bar_event_time_final_table["event_timing_profile"] = (
    df_bar_event_time_final_table.apply(
        classify_event_timing,
        axis=1
    )
)

# Ordenar columnas principales
event_time_final_cols = [
    "bar_col",
    "target_base",
    "horizon",
    "percentile",
    "tp_mult",
    "sl_mult",
    "tp_sl_config",

    "valid_rows",
    "event_count",
    "tp_count",
    "sl_count",
    "no_event_count",

    "event_rate",
    "tp_rate",
    "sl_rate",
    "no_event_rate",

    "event_step_mean",
    "event_step_median",
    "event_step_q25",
    "event_step_q75",
    "event_step_min",
    "event_step_max",

    "tp_step_mean",
    "tp_step_median",
    "tp_step_q25",
    "tp_step_q75",

    "sl_step_mean",
    "sl_step_median",
    "sl_step_q25",
    "sl_step_q75",

    "early_event_rate",
    "middle_event_rate",
    "late_event_rate",
    "event_timing_profile",
]

df_bar_event_time_final_table = (

```

```

df_bar_event_time_final_table[event_time_final_cols]
.sort_values(["percentile", "horizon", "sl_mult", "tp_mult"])
.reset_index(drop=True)
)

# -----
# Guardar tablas principales
# -----

df_bar_event_time.to_csv(
    MNQ_TARGETS_BAR_EVENT_TIME_CSV,
    index=False
)

df_bar_event_time_by_config.to_csv(
    MNQ_TARGETS_BAR_EVENT_TIME_BY_CONFIG_CSV,
    index=False
)

df_bar_event_time_by_target_base.to_csv(
    MNQ_TARGETS_BAR_EVENT_TIME_BY_TARGET_BASE_CSV,
    index=False
)

df_bar_event_time_fast_ranking.to_csv(
    MNQ_TARGETS_BAR_EVENT_TIME_FAST_RANKING_CSV,
    index=False
)

df_bar_event_time_slow_ranking.to_csv(
    MNQ_TARGETS_BAR_EVENT_TIME_SLOW_RANKING_CSV,
    index=False
)

df_bar_event_time_final_table.to_csv(
    MNQ_TARGETS_BAR_EVENT_TIME_FINAL_TABLE_CSV,
    index=False
)

# -----
# Tabla compacta por configuración TP/SL
# -----

event_time_config_cols = [
    "tp_sl_config",
    "tp_mult",
    "sl_mult",
    "n_targets",
    "event_rate_mean",
    "tp_rate_mean",
    "sl_rate_mean",
    "no_event_rate_mean",
    "event_step_mean",
    "event_step_median",
    "tp_step_mean",
    "tp_step_median",
    "sl_step_mean",
    "sl_step_median",
    "early_event_rate_mean",
    "middle_event_rate_mean",

```



```

        "late_event_rate_mean",
    ]

print("\nTabla compacta por configuración TP/SL:")
print(
    df_bar_event_time_by_config[event_time_config_cols]
    .to_string(
        index=False,
        formatters={
            "event_rate_mean": "{:.2%}".format,
            "tp_rate_mean": "{:.2%}".format,
            "sl_rate_mean": "{:.2%}".format,
            "no_event_rate_mean": "{:.2%}".format,
            "event_step_mean": "{:.2f}".format,
            "event_step_median": "{:.2f}".format,
            "tp_step_mean": "{:.2f}".format,
            "tp_step_median": "{:.2f}".format,
            "sl_step_mean": "{:.2f}".format,
            "sl_step_median": "{:.2f}".format,
            "early_event_rate_mean": "{:.2%}".format,
            "middle_event_rate_mean": "{:.2%}".format,
            "late_event_rate_mean": "{:.2%}".format,
        }
    )
)

# -----
# Tabla compacta por percentil y horizonte
# -----

event_time_target_base_cols = [
    "percentile",
    "horizon",
    "n_configs",
    "event_rate_mean",
    "tp_rate_mean",
    "sl_rate_mean",
    "no_event_rate_mean",
    "event_step_mean",
    "event_step_median",
    "tp_step_mean",
    "sl_step_mean",
    "early_event_rate_mean",
    "middle_event_rate_mean",
    "late_event_rate_mean",
]

print("\nTabla compacta por percentil y horizonte:")
print(
    df_bar_event_time_by_target_base[event_time_target_base_cols]
    .to_string(
        index=False,
        formatters={
            "event_rate_mean": "{:.2%}".format,
            "tp_rate_mean": "{:.2%}".format,
            "sl_rate_mean": "{:.2%}".format,
            "no_event_rate_mean": "{:.2%}".format,
            "event_step_mean": "{:.2f}".format,
            "event_step_median": "{:.2f}".format,
            "tp_step_mean": "{:.2f}".format,

```

```

        "sl_step_mean": "{:.2f}".format,
        "early_event_rate_mean": "{:.2%}".format,
        "middle_event_rate_mean": "{:.2%}".format,
        "late_event_rate_mean": "{:.2%}".format,
    }
    )
)

# -----
# Tabla final completa: todos los percentiles, horizontes y TP/SL
# -----

final_print_cols = [
    "bar_col",
    "target_base",
    "tp_sl_config",
    "event_rate",
    "tp_rate",
    "sl_rate",
    "no_event_rate",
    "event_step_mean",
    "event_step_median",
    "tp_step_mean",
    "sl_step_mean",
    "early_event_rate",
    "middle_event_rate",
    "late_event_rate",
    "event_timing_profile",
]

print("\nTabla final de tiempo hasta evento – todos los percentiles y horizontes")
print(
    df_bar_event_time_final_table[final_print_cols]
    .to_string(
        index=False,
        formatters={
            "event_rate": "{:.2%}".format,
            "tp_rate": "{:.2%}".format,
            "sl_rate": "{:.2%}".format,
            "no_event_rate": "{:.2%}".format,
            "event_step_mean": "{:.2f}".format,
            "event_step_median": "{:.2f}".format,
            "tp_step_mean": "{:.2f}".format,
            "sl_step_mean": "{:.2f}".format,
            "early_event_rate": "{:.2%}".format,
            "middle_event_rate": "{:.2%}".format,
            "late_event_rate": "{:.2%}".format,
        }
    )
)

# -----
# Top 20 eventos más tempranos
# -----

ranking_cols = [
    "bar_col",
    "horizon",
    "percentile",
    "tp_mult",

```

```

    "sl_mult",
    "event_rate",
    "tp_rate",
    "sl_rate",
    "no_event_rate",
    "event_step_mean",
    "event_step_median",
    "early_event_rate",
    "middle_event_rate",
    "late_event_rate",
]

print("\nTop 20 variantes con eventos más tempranos:")
print(
    df_bar_event_time_fast_ranking[ranking_cols]
    .head(20)
    .to_string(
        index=False,
        formatters={
            "event_rate": "{:.2%}".format,
            "tp_rate": "{:.2%}".format,
            "sl_rate": "{:.2%}".format,
            "no_event_rate": "{:.2%}".format,
            "event_step_mean": "{:.2f}".format,
            "event_step_median": "{:.2f}".format,
            "early_event_rate": "{:.2%}".format,
            "middle_event_rate": "{:.2%}".format,
            "late_event_rate": "{:.2%}".format,
        }
    )
)

# -----
# Top 20 eventos más tardíos
# -----

print("\nTop 20 variantes con eventos más tardíos:")
print(
    df_bar_event_time_slow_ranking[ranking_cols]
    .head(20)
    .to_string(
        index=False,
        formatters={
            "event_rate": "{:.2%}".format,
            "tp_rate": "{:.2%}".format,
            "sl_rate": "{:.2%}".format,
            "no_event_rate": "{:.2%}".format,
            "event_step_mean": "{:.2f}".format,
            "event_step_median": "{:.2f}".format,
            "early_event_rate": "{:.2%}".format,
            "middle_event_rate": "{:.2%}".format,
            "late_event_rate": "{:.2%}".format,
        }
    )
)

# -----
# Conteo por perfil temporal
# -----

```

```

df_bar_event_timing_profile_counts = (
    df_bar_event_time_final_table
    .groupby("event_timing_profile", as_index=False)
    .agg(
        n_targets=("bar_col", "count"),
        event_rate_mean=("event_rate", "mean"),
        tp_rate_mean=("tp_rate", "mean"),
        sl_rate_mean=("sl_rate", "mean"),
        no_event_rate_mean=("no_event_rate", "mean"),
        event_step_mean=("event_step_mean", "mean"),
        event_step_median=("event_step_median", "mean"),
        early_event_rate_mean=("early_event_rate", "mean"),
        middle_event_rate_mean=("middle_event_rate", "mean"),
        late_event_rate_mean=("late_event_rate", "mean"),
    )
    .sort_values("n_targets", ascending=False)
    .reset_index(drop=True)
)

print("\nConteo por perfil temporal:")
print(
    df_bar_event_timing_profile_counts
    .to_string(
        index=False,
        formatters={
            "event_rate_mean": "{:.2%}".format,
            "tp_rate_mean": "{:.2%}".format,
            "sl_rate_mean": "{:.2%}".format,
            "no_event_rate_mean": "{:.2%}".format,
            "event_step_mean": "{:.2f}".format,
            "event_step_median": "{:.2f}".format,
            "early_event_rate_mean": "{:.2%}".format,
            "middle_event_rate_mean": "{:.2%}".format,
            "late_event_rate_mean": "{:.2%}".format,
        }
    )
)

# -----
# Guardar reporte TXT completo
# -----

with MNQ_TARGETS_BAR_EVENT_TIME_REPORT_TXT.open("w", encoding="utf-8") as f:
    f.write("4.8. Tiempo hasta evento TP/SL - BAR\n")
    f.write("=" * 100 + "\n\n")

    f.write("Tabla compacta por configuración TP/SL\n")
    f.write("-" * 100 + "\n")
    f.write(
        df_bar_event_time_by_config[event_time_config_cols]
        .to_string(
            index=False,
            formatters={
                "event_rate_mean": "{:.2%}".format,
                "tp_rate_mean": "{:.2%}".format,
                "sl_rate_mean": "{:.2%}".format,
                "no_event_rate_mean": "{:.2%}".format,
                "event_step_mean": "{:.2f}".format,
                "event_step_median": "{:.2f}".format,
                "tp_step_mean": "{:.2f}".format,
            }
        )
    )

```

```

        "tp_step_median": "{:.2f}".format,
        "sl_step_mean": "{:.2f}".format,
        "sl_step_median": "{:.2f}".format,
        "early_event_rate_mean": "{:.2%}".format,
        "middle_event_rate_mean": "{:.2%}".format,
        "late_event_rate_mean": "{:.2%}".format,
    }
    )
)

f.write("\n\n")
f.write("Tabla compacta por percentil y horizonte\n")
f.write("-" * 100 + "\n")
f.write(
    df_bar_event_time_by_target_base[event_time_target_base_cols]
    .to_string(
        index=False,
        formatters={
            "event_rate_mean": "{:.2%}".format,
            "tp_rate_mean": "{:.2%}".format,
            "sl_rate_mean": "{:.2%}".format,
            "no_event_rate_mean": "{:.2%}".format,
            "event_step_mean": "{:.2f}".format,
            "event_step_median": "{:.2f}".format,
            "tp_step_mean": "{:.2f}".format,
            "sl_step_mean": "{:.2f}".format,
            "early_event_rate_mean": "{:.2%}".format,
            "middle_event_rate_mean": "{:.2%}".format,
            "late_event_rate_mean": "{:.2%}".format,
        }
    )
)

f.write("\n\n")
f.write("Tabla final completa: todos los percentiles, horizontes y configura")
f.write("-" * 100 + "\n")
f.write(
    df_bar_event_time_final_table[final_print_cols]
    .to_string(
        index=False,
        formatters={
            "event_rate": "{:.2%}".format,
            "tp_rate": "{:.2%}".format,
            "sl_rate": "{:.2%}".format,
            "no_event_rate": "{:.2%}".format,
            "event_step_mean": "{:.2f}".format,
            "event_step_median": "{:.2f}".format,
            "tp_step_mean": "{:.2f}".format,
            "sl_step_mean": "{:.2f}".format,
            "early_event_rate": "{:.2%}".format,
            "middle_event_rate": "{:.2%}".format,
            "late_event_rate": "{:.2%}".format,
        }
    )
)

f.write("\n\n")
f.write("Top 20 variantes con eventos más tempranos\n")
f.write("-" * 100 + "\n")
f.write(

```

```

df_bar_event_time_fast_ranking[ranking_cols]
.head(20)
.to_string(
    index=False,
    formatters={
        "event_rate": "{:.2%}".format,
        "tp_rate": "{:.2%}".format,
        "sl_rate": "{:.2%}".format,
        "no_event_rate": "{:.2%}".format,
        "event_step_mean": "{:.2f}".format,
        "event_step_median": "{:.2f}".format,
        "early_event_rate": "{:.2%}".format,
        "middle_event_rate": "{:.2%}".format,
        "late_event_rate": "{:.2%}".format,
    }
)

f.write("\n\n")
f.write("Top 20 variantes con eventos más tardíos\n")
f.write("-" * 100 + "\n")
f.write(
    df_bar_event_time_slow_ranking[ranking_cols]
    .head(20)
    .to_string(
        index=False,
        formatters={
            "event_rate": "{:.2%}".format,
            "tp_rate": "{:.2%}".format,
            "sl_rate": "{:.2%}".format,
            "no_event_rate": "{:.2%}".format,
            "event_step_mean": "{:.2f}".format,
            "event_step_median": "{:.2f}".format,
            "early_event_rate": "{:.2%}".format,
            "middle_event_rate": "{:.2%}".format,
            "late_event_rate": "{:.2%}".format,
        }
    )
)

f.write("\n\n")
f.write("Conteo por perfil temporal\n")
f.write("-" * 100 + "\n")
f.write(
    df_bar_event_timing_profile_counts
    .to_string(
        index=False,
        formatters={
            "event_rate_mean": "{:.2%}".format,
            "tp_rate_mean": "{:.2%}".format,
            "sl_rate_mean": "{:.2%}".format,
            "no_event_rate_mean": "{:.2%}".format,
            "event_step_mean": "{:.2f}".format,
            "event_step_median": "{:.2f}".format,
            "early_event_rate_mean": "{:.2%}".format,
            "middle_event_rate_mean": "{:.2%}".format,
            "late_event_rate_mean": "{:.2%}".format,
        }
    )
)

```

```
# -----  
# Confirmación final  
# -----  
  
print("\nArchivos guardados:")  
print(f"Tiempo hasta evento BAR: {MNQ_TARGETS_BAR_EVENT_TIME_CSV}")  
print(f"Tiempo por configuración TP/SL: {MNQ_TARGETS_BAR_EVENT_TIME_BY_CON}")  
print(f"Tiempo por percentil y horizonte: {MNQ_TARGETS_BAR_EVENT_TIME_BY_TAR}")  
print(f"Ranking eventos tempranos: {MNQ_TARGETS_BAR_EVENT_TIME_FAST_R}")  
print(f"Ranking eventos tardíos: {MNQ_TARGETS_BAR_EVENT_TIME_SLOW_R}")  
print(f"Tabla final tiempo hasta evento: {MNQ_TARGETS_BAR_EVENT_TIME_FINAL}")  
print(f"Reporte TXT: {MNQ_TARGETS_BAR_EVENT_TIME_REPORT}")  
  
print("\nCelda 4 finalizada correctamente.")
```

## Guardado y tabla final de tiempo hasta evento TP/SL - BAR

=====

## Tabla compacta por configuración TP/SL:

| tp_sl_config    | tp_mult | sl_mult | n_targets | event_rate_mean | tp_rate_mean | sl_rate_mean | no_event_rate_mean | event_step_mean | event_step_median | tp_step_mean | tp_step_median | sl_step_mean | sl_step_median | early_event_rate_mean | middle_event_rate_mean | late_event_rate_mean |
|-----------------|---------|---------|-----------|-----------------|--------------|--------------|--------------------|-----------------|-------------------|--------------|----------------|--------------|----------------|-----------------------|------------------------|----------------------|
| TP 1.0 / SL 0.5 | 1.0     | 0.5     | 9         | 100.00%         | 81.07%       | 1            | 8.93%              | 0.00%           | 24.41             | 21.22        | 27.54          | 25.44        | 11.10          | 8.00                  | 48.43%                 | 31.55%               |
|                 |         |         |           |                 |              |              | 20.02%             |                 |                   |              |                |              |                |                       |                        |                      |
| TP 1.5 / SL 0.5 | 1.5     | 0.5     | 9         | 67.11%          | 44.39%       | 2            | 2.71%              | 32.89%          | 26.43             | 24.22        | 31.72          | 31.11        | 16.23          | 10.22                 | 43.88%                 | 30.75%               |
|                 |         |         |           |                 |              |              | 25.37%             |                 |                   |              |                |              |                |                       |                        |                      |
| TP 2.0 / SL 0.5 | 2.0     | 0.5     | 9         | 49.45%          | 25.21%       | 2            | 4.24%              | 50.55%          | 26.20             | 23.44        | 34.35          | 34.56        | 17.90          | 11.11                 | 45.29%                 | 28.03%               |
|                 |         |         |           |                 |              |              | 26.68%             |                 |                   |              |                |              |                |                       |                        |                      |
| TP 1.0 / SL 1.0 | 1.0     | 1.0     | 9         | 100.00%         | 95.18%       |              | 4.82%              | 0.00%           | 28.32             | 26.56        | 28.99          | 27.67        | 14.94          | 12.00                 | 38.24%                 | 35.56%               |
|                 |         |         |           |                 |              |              | 26.20%             |                 |                   |              |                |              |                |                       |                        |                      |
| TP 1.5 / SL 1.0 | 1.5     | 1.0     | 9         | 58.58%          | 52.73%       |              | 5.85%              | 41.42%          | 31.56             | 31.11        | 32.84          | 32.56        | 19.93          | 14.78                 | 30.43%                 | 36.55%               |
|                 |         |         |           |                 |              |              | 33.02%             |                 |                   |              |                |              |                |                       |                        |                      |
| TP 2.0 / SL 1.0 | 2.0     | 1.0     | 9         | 36.90%          | 30.26%       |              | 6.65%              | 63.10%          | 33.08             | 33.44        | 35.27          | 36.00        | 23.05          | 17.00                 | 27.52%                 | 35.33%               |
|                 |         |         |           |                 |              |              | 37.15%             |                 |                   |              |                |              |                |                       |                        |                      |
| TP 2.5 / SL 1.0 | 2.5     | 1.0     | 9         | 25.08%          | 18.02%       |              | 7.06%              | 74.92%          | 33.39             | 34.33        | 36.93          | 38.56        | 24.40          | 18.89                 | 27.88%                 | 33.11%               |
|                 |         |         |           |                 |              |              | 39.02%             |                 |                   |              |                |              |                |                       |                        |                      |

## Tabla compacta por percentil y horizonte:

| percentile | horizon | n_configs | event_rate_mean | tp_rate_mean | sl_rate_mean | no_event_rate_mean | event_step_mean | event_step_median | tp_step_mean | sl_step_mean | early_event_rate_mean | middle_event_rate_mean | late_event_rate_mean |
|------------|---------|-----------|-----------------|--------------|--------------|--------------------|-----------------|-------------------|--------------|--------------|-----------------------|------------------------|----------------------|
| 40         | 30      | 7         | 67.17%          | 52.34%       | 14.83%       | 32.83%             | 13.71           | 12.57             | 15.36        | 8.65         | 42.15%                | 32.50%                 | 25.35%               |
| 40         | 60      | 7         | 66.42%          | 51.09%       | 15.32%       | 33.58%             | 27.00           | 24.71             | 30.52        | 16.86        | 42.12%                | 32.15%                 | 25.73%               |
| 40         | 90      | 7         | 66.68%          | 50.70%       | 15.98%       | 33.32%             | 40.97           | 38.00             | 46.48        | 26.18        | 41.21%                | 31.69%                 | 27.10%               |
| 50         | 30      | 7         | 62.76%          | 50.36%       | 12.40%       | 37.24%             | 14.62           | 13.86             | 16.24        | 9.10         | 37.57%                | 33.63%                 | 28.80%               |
| 50         | 60      | 7         | 61.81%          | 49.20%       | 12.61%       | 38.19%             | 28.81           | 27.43             | 32.26        | 17.69        | 37.54%                | 33.32%                 | 29.14%               |
| 50         | 90      | 7         | 62.39%          | 49.04%       | 13.36%       | 37.61%             | 43.86           | 42.14             | 49.24        | 27.85        | 36.55%                | 32.48%                 | 30.97%               |
| 60         | 30      | 7         | 58.58%          | 48.39%       | 10.19%       |                    |                 |                   |              |              |                       |                        |                      |



|        |       |        |        |        |
|--------|-------|--------|--------|--------|
| 41.42% | 15.47 | 15.29  | 17.05  | 9.51   |
| 33.40% |       | 34.32% | 32.28% |        |
| 60     | 60    | 7      | 57.75% | 47.41% |
| 42.25% | 30.42 | 29.71  | 33.76  | 18.47  |
| 33.57% |       | 33.99% | 32.43% |        |
| 60     | 90    | 7      | 58.45% | 47.43% |
| 41.55% | 46.64 | 46.14  | 51.78  | 29.67  |
| 32.30% |       | 32.77% | 34.93% |        |

Tabla final de tiempo hasta evento – todos los percentiles y horizontes:

| bar_col               | target_base       | tp_sl_config      | event_rate            | tp_rate      | sl_rate     | no_e |
|-----------------------|-------------------|-------------------|-----------------------|--------------|-------------|------|
| vent_rate             | event_step_mean   | event_step_median | tp_step_mean          | sl_step_mean | early_event |      |
| _rate                 | middle_event_rate | late_event_rate   | event_timing_profile  |              |             |      |
| bar_p40_h30_tp10_sl05 | p40_h30           | TP 1.0 / SL 0.5   | 100.00%               | 78.98%       | 21.02%      |      |
| 0.00%                 | 11.18             | 9.00              | 12.72                 | 5.39         | 55.0        |      |
| 6%                    | 29.29%            | 15.65%            | resolución_temprana   |              |             |      |
| bar_p40_h30_tp15_sl05 | p40_h30           | TP 1.5 / SL 0.5   | 73.21%                | 48.14%       | 25.07%      |      |
| 26.79%                | 12.46             | 11.00             | 14.86                 | 7.85         | 48.6        |      |
| 1%                    | 30.00%            | 21.39%            | resolución_intermedia |              |             |      |
| bar_p40_h30_tp20_sl05 | p40_h30           | TP 2.0 / SL 0.5   | 56.62%                | 29.68%       | 26.94%      |      |
| 43.38%                | 12.70             | 11.00             | 16.26                 | 8.78         | 48.0        |      |
| 1%                    | 28.26%            | 23.74%            | resolución_intermedia |              |             |      |
| bar_p40_h30_tp10_sl10 | p40_h30           | TP 1.0 / SL 1.0   | 100.00%               | 93.84%       | 6.16%       |      |
| 0.00%                 | 13.20             | 12.00             | 13.61                 | 6.99         | 44.2        |      |
| 3%                    | 33.99%            | 21.78%            | resolución_intermedia |              |             |      |
| bar_p40_h30_tp15_sl10 | p40_h30           | TP 1.5 / SL 1.0   | 64.87%                | 57.52%       | 7.35%       |      |
| 35.13%                | 14.83             | 14.00             | 15.55                 | 9.23         | 35.7        |      |
| 6%                    | 36.12%            | 28.12%            | resolución_intermedia |              |             |      |
| bar_p40_h30_tp20_sl10 | p40_h30           | TP 2.0 / SL 1.0   | 43.97%                | 35.63%       | 8.35%       |      |
| 56.03%                | 15.66             | 15.00             | 16.81                 | 10.78        | 32.0        |      |
| 3%                    | 35.73%            | 32.24%            | resolución_intermedia |              |             |      |
| bar_p40_h30_tp25_sl10 | p40_h30           | TP 2.5 / SL 1.0   | 31.51%                | 22.59%       | 8.92%       |      |
| 68.49%                | 15.97             | 16.00             | 17.71                 | 11.56        | 31.3        |      |
| 2%                    | 34.13%            | 34.54%            | resolución_intermedia |              |             |      |
| bar_p40_h60_tp10_sl05 | p40_h60           | TP 1.0 / SL 0.5   | 100.00%               | 77.94%       | 22.06%      |      |
| 0.00%                 | 21.95             | 18.00             | 25.27                 | 10.22        | 54.9        |      |
| 6%                    | 28.94%            | 16.10%            | resolución_temprana   |              |             |      |
| bar_p40_h60_tp15_sl05 | p40_h60           | TP 1.5 / SL 0.5   | 72.88%                | 46.45%       | 26.43%      |      |
| 27.12%                | 24.35             | 21.00             | 29.59                 | 15.15        | 48.9        |      |
| 0%                    | 29.53%            | 21.57%            | resolución_intermedia |              |             |      |
| bar_p40_h60_tp20_sl05 | p40_h60           | TP 2.0 / SL 0.5   | 56.15%                | 27.84%       | 28.31%      |      |
| 43.85%                | 24.57             | 21.00             | 32.35                 | 16.92        | 49.0        |      |
| 8%                    | 27.40%            | 23.52%            | resolución_intermedia |              |             |      |
| bar_p40_h60_tp10_sl10 | p40_h60           | TP 1.0 / SL 1.0   | 100.00%               | 94.00%       | 6.00%       |      |
| 0.00%                 | 26.19             | 24.00             | 27.01                 | 13.34        | 43.6        |      |
| 4%                    | 33.89%            | 22.47%            | resolución_intermedia |              |             |      |
| bar_p40_h60_tp15_sl10 | p40_h60           | TP 1.5 / SL 1.0   | 63.61%                | 56.30%       | 7.31%       |      |
| 36.39%                | 29.41             | 28.00             | 30.86                 | 18.22        | 35.1        |      |
| 8%                    | 36.11%            | 28.71%            | resolución_intermedia |              |             |      |
| bar_p40_h60_tp20_sl10 | p40_h60           | TP 2.0 / SL 1.0   | 42.39%                | 34.08%       | 8.30%       |      |
| 57.61%                | 31.05             | 30.00             | 33.41                 | 21.37        | 31.7        |      |
| 4%                    | 35.36%            | 32.90%            | resolución_intermedia |              |             |      |
| bar_p40_h60_tp25_sl10 | p40_h60           | TP 2.5 / SL 1.0   | 29.89%                | 21.05%       | 8.84%       |      |
| 70.11%                | 31.48             | 31.00             | 35.13                 | 22.78        | 31.3        |      |
| 7%                    | 33.80%            | 34.83%            | resolución_intermedia |              |             |      |
| bar_p40_h90_tp10_sl05 | p40_h90           | TP 1.0 / SL 0.5   | 100.00%               | 77.10%       | 22.90%      |      |
| 0.00%                 | 32.93             | 27.00             | 38.07                 | 15.60        | 54.2        |      |
| 4%                    | 29.44%            | 16.32%            | resolución_temprana   |              |             |      |
| bar_p40_h90_tp15_sl05 | p40_h90           | TP 1.5 / SL 0.5   | 72.84%                | 45.52%       | 27.32%      |      |
| 27.16%                | 36.63             | 32.00             | 44.92                 | 22.83        | 48.5        |      |

|                       |        |                         |                       |        |        |      |  |
|-----------------------|--------|-------------------------|-----------------------|--------|--------|------|--|
| 5%                    | 29.16% | 22.30%                  | resolución_intermedia |        |        |      |  |
| bar_p40_h90_tp20_sl05 |        | p40_h90 TP 2.0 / SL 0.5 | 56.66%                | 27.45% | 29.21% |      |  |
| 43.34%                | 37.08  | 32.00                   | 49.45                 | 25.44  |        | 48.5 |  |
| 2%                    | 26.84% | 24.64%                  | resolución_intermedia |        |        |      |  |
| bar_p40_h90_tp10_sl10 |        | p40_h90 TP 1.0 / SL 1.0 | 100.00%               | 93.57% | 6.43%  |      |  |
| 0.00%                 | 39.52  | 36.00                   | 40.74                 | 21.79  |        | 42.6 |  |
| 1%                    | 34.26% | 23.13%                  | resolución_intermedia |        |        |      |  |
| bar_p40_h90_tp15_sl10 |        | p40_h90 TP 1.5 / SL 1.0 | 63.74%                | 55.98% | 7.76%  |      |  |
| 36.26%                | 44.77  | 43.00                   | 47.01                 | 28.64  |        | 34.1 |  |
| 5%                    | 35.43% | 30.42%                  | resolución_intermedia |        |        |      |  |
| bar_p40_h90_tp20_sl10 |        | p40_h90 TP 2.0 / SL 1.0 | 42.92%                | 34.09% | 8.83%  |      |  |
| 57.08%                | 47.51  | 47.00                   | 51.17                 | 33.35  |        | 30.3 |  |
| 5%                    | 34.28% | 35.37%                  | resolución_intermedia |        |        |      |  |
| bar_p40_h90_tp25_sl10 |        | p40_h90 TP 2.5 / SL 1.0 | 30.62%                | 21.19% | 9.43%  |      |  |
| 69.38%                | 48.34  | 49.00                   | 54.00                 | 35.62  |        | 30.0 |  |
| 3%                    | 32.41% | 37.56%                  | resolución_intermedia |        |        |      |  |
| bar_p50_h30_tp10_sl05 |        | p50_h30 TP 1.0 / SL 0.5 | 100.00%               | 81.93% | 18.07% |      |  |
| 0.00%                 | 12.37  | 11.00                   | 13.84                 | 5.70   |        | 48.6 |  |
| 2%                    | 31.90% | 19.49%                  | resolución_intermedia |        |        |      |  |
| bar_p50_h30_tp15_sl05 |        | p50_h30 TP 1.5 / SL 0.5 | 67.40%                | 45.74% | 21.67% |      |  |
| 32.60%                | 13.40  | 12.00                   | 15.82                 | 8.28   |        | 43.7 |  |
| 7%                    | 31.52% | 24.70%                  | resolución_intermedia |        |        |      |  |
| bar_p50_h30_tp20_sl05 |        | p50_h30 TP 2.0 / SL 0.5 | 49.50%                | 26.31% | 23.19% |      |  |
| 50.50%                | 13.39  | 12.00                   | 17.11                 | 9.16   |        | 44.7 |  |
| 8%                    | 28.79% | 26.43%                  | resolución_intermedia |        |        |      |  |
| bar_p50_h30_tp10_sl10 |        | p50_h30 TP 1.0 / SL 1.0 | 100.00%               | 95.26% | 4.74%  |      |  |
| 0.00%                 | 14.21  | 13.00                   | 14.55                 | 7.38   |        | 38.7 |  |
| 4%                    | 35.86% | 25.41%                  | resolución_intermedia |        |        |      |  |
| bar_p50_h30_tp15_sl10 |        | p50_h30 TP 1.5 / SL 1.0 | 59.30%                | 53.60% | 5.71%  |      |  |
| 40.70%                | 15.74  | 15.00                   | 16.37                 | 9.77   |        | 31.0 |  |
| 5%                    | 37.21% | 31.75%                  | resolución_intermedia |        |        |      |  |
| bar_p50_h30_tp20_sl10 |        | p50_h30 TP 2.0 / SL 1.0 | 37.56%                | 31.07% | 6.49%  |      |  |
| 62.44%                | 16.50  | 17.00                   | 17.58                 | 11.34  |        | 28.0 |  |
| 3%                    | 36.02% | 35.95%                  | resolución_intermedia |        |        |      |  |
| bar_p50_h30_tp25_sl10 |        | p50_h30 TP 2.5 / SL 1.0 | 25.53%                | 18.63% | 6.90%  |      |  |
| 74.47%                | 16.70  | 17.00                   | 18.42                 | 12.05  |        | 28.0 |  |
| 1%                    | 34.11% | 37.88%                  | resolución_intermedia |        |        |      |  |
| bar_p50_h60_tp10_sl05 |        | p50_h60 TP 1.0 / SL 0.5 | 100.00%               | 81.32% | 18.68% |      |  |
| 0.00%                 | 24.40  | 21.00                   | 27.53                 | 10.80  |        | 48.3 |  |
| 6%                    | 31.58% | 20.06%                  | resolución_intermedia |        |        |      |  |
| bar_p50_h60_tp15_sl05 |        | p50_h60 TP 1.5 / SL 0.5 | 66.52%                | 44.00% | 22.52% |      |  |
| 33.48%                | 26.28  | 24.00                   | 31.51                 | 16.05  |        | 44.0 |  |
| 9%                    | 30.86% | 25.04%                  | resolución_intermedia |        |        |      |  |
| bar_p50_h60_tp20_sl05 |        | p50_h60 TP 2.0 / SL 0.5 | 48.56%                | 24.58% | 23.99% |      |  |
| 51.44%                | 25.95  | 23.00                   | 34.01                 | 17.68  |        | 45.6 |  |
| 8%                    | 28.34% | 25.98%                  | resolución_intermedia |        |        |      |  |
| bar_p50_h60_tp10_sl10 |        | p50_h60 TP 1.0 / SL 1.0 | 100.00%               | 95.45% | 4.55%  |      |  |
| 0.00%                 | 28.22  | 26.00                   | 28.89                 | 14.12  |        | 38.1 |  |
| 9%                    | 35.70% | 26.11%                  | resolución_intermedia |        |        |      |  |
| bar_p50_h60_tp15_sl10 |        | p50_h60 TP 1.5 / SL 1.0 | 57.77%                | 52.23% | 5.54%  |      |  |
| 42.23%                | 31.28  | 31.00                   | 32.56                 | 19.20  |        | 30.5 |  |
| 3%                    | 36.95% | 32.52%                  | resolución_intermedia |        |        |      |  |
| bar_p50_h60_tp20_sl10 |        | p50_h60 TP 2.0 / SL 1.0 | 35.78%                | 29.49% | 6.29%  |      |  |
| 64.22%                | 32.68  | 33.00                   | 34.89                 | 22.33  |        | 27.6 |  |
| 3%                    | 36.08% | 36.29%                  | resolución_intermedia |        |        |      |  |
| bar_p50_h60_tp25_sl10 |        | p50_h60 TP 2.5 / SL 1.0 | 24.02%                | 17.33% | 6.69%  |      |  |
| 75.98%                | 32.87  | 34.00                   | 36.42                 | 23.68  |        | 28.3 |  |
| 1%                    | 33.73% | 37.95%                  | resolución_intermedia |        |        |      |  |
| bar_p50_h90_tp10_sl05 |        | p50_h90 TP 1.0 / SL 0.5 | 100.00%               | 80.23% | 19.77% |      |  |
| 0.00%                 | 36.58  | 32.00                   | 41.45                 | 16.80  |        | 47.8 |  |

|                       |        |                              |         |        |        |
|-----------------------|--------|------------------------------|---------|--------|--------|
| 9%                    | 31.67% | 20.45% resolución_intermedia |         |        |        |
| bar_p50_h90_tp15_sl05 |        | p50_h90 TP 1.5 / SL 0.5      | 67.30%  | 43.60% | 23.71% |
| 32.70%                | 39.72  | 36.00                        | 48.03   | 24.45  | 43.5   |
| 1%                    | 30.21% | 26.28% resolución_intermedia |         |        |        |
| bar_p50_h90_tp20_sl05 |        | p50_h90 TP 2.0 / SL 0.5      | 49.77%  | 24.52% | 25.25% |
| 50.23%                | 39.31  | 35.00                        | 52.06   | 26.93  | 45.0   |
| 8%                    | 27.39% | 27.52% resolución_intermedia |         |        |        |
| bar_p50_h90_tp10_sl10 |        | p50_h90 TP 1.0 / SL 1.0      | 100.00% | 95.11% | 4.89%  |
| 0.00%                 | 42.70  | 40.00                        | 43.69   | 23.37  | 37.2   |
| 0%                    | 35.60% | 27.20% resolución_intermedia |         |        |        |
| bar_p50_h90_tp15_sl10 |        | p50_h90 TP 1.5 / SL 1.0      | 58.31%  | 52.38% | 5.93%  |
| 41.69%                | 47.81  | 48.00                        | 49.75   | 30.65  | 29.2   |
| 3%                    | 35.98% | 34.79% resolución_intermedia |         |        |        |
| bar_p50_h90_tp20_sl10 |        | p50_h90 TP 2.0 / SL 1.0      | 36.55%  | 29.78% | 6.77%  |
| 63.45%                | 50.14  | 51.00                        | 53.49   | 35.38  | 26.2   |
| 4%                    | 34.57% | 39.19% resolución_intermedia |         |        |        |
| bar_p50_h90_tp25_sl10 |        | p50_h90 TP 2.5 / SL 1.0      | 24.82%  | 17.65% | 7.17%  |
| 75.18%                | 50.79  | 53.00                        | 56.23   | 37.39  | 26.7   |
| 2%                    | 31.90% | 41.38% resolución_intermedia |         |        |        |
| bar_p60_h30_tp10_sl05 |        | p60_h30 TP 1.0 / SL 0.5      | 100.00% | 84.78% | 15.22% |
| 0.00%                 | 13.51  | 12.00                        | 14.86   | 6.02   | 42.5   |
| 0%                    | 34.05% | 23.45% resolución_intermedia |         |        |        |
| bar_p60_h30_tp15_sl05 |        | p60_h30 TP 1.5 / SL 0.5      | 61.49%  | 43.07% | 18.43% |
| 38.51%                | 14.34  | 14.00                        | 16.73   | 8.76   | 39.1   |
| 0%                    | 32.49% | 28.40% resolución_intermedia |         |        |        |
| bar_p60_h30_tp20_sl05 |        | p60_h30 TP 2.0 / SL 0.5      | 42.71%  | 23.11% | 19.59% |
| 57.29%                | 14.07  | 13.00                        | 17.89   | 9.57   | 41.5   |
| 2%                    | 29.33% | 29.15% resolución_intermedia |         |        |        |
| bar_p60_h30_tp10_sl10 |        | p60_h30 TP 1.0 / SL 1.0      | 100.00% | 96.41% | 3.59%  |
| 0.00%                 | 15.14  | 15.00                        | 15.42   | 7.69   | 33.8   |
| 7%                    | 37.20% | 28.93% resolución_intermedia |         |        |        |
| bar_p60_h30_tp15_sl10 |        | p60_h30 TP 1.5 / SL 1.0      | 53.89%  | 49.53% | 4.37%  |
| 46.11%                | 16.62  | 17.00                        | 17.18   | 10.28  | 26.8   |
| 2%                    | 37.55% | 35.63% resolución_intermedia |         |        |        |
| bar_p60_h30_tp20_sl10 |        | p60_h30 TP 2.0 / SL 1.0      | 31.72%  | 26.78% | 4.94%  |
| 68.28%                | 17.26  | 18.00                        | 18.26   | 11.81  | 24.5   |
| 2%                    | 36.10% | 39.38% resolución_intermedia |         |        |        |
| bar_p60_h30_tp25_sl10 |        | p60_h30 TP 2.5 / SL 1.0      | 20.27%  | 15.06% | 5.21%  |
| 79.73%                | 17.32  | 18.00                        | 19.02   | 12.42  | 25.5   |
| 0%                    | 33.50% | 41.00% resolución_intermedia |         |        |        |
| bar_p60_h60_tp10_sl05 |        | p60_h60 TP 1.0 / SL 0.5      | 100.00% | 84.19% | 15.81% |
| 0.00%                 | 26.62  | 24.00                        | 29.49   | 11.32  | 42.4   |
| 4%                    | 33.68% | 23.88% resolución_intermedia |         |        |        |
| bar_p60_h60_tp15_sl05 |        | p60_h60 TP 1.5 / SL 0.5      | 60.54%  | 41.45% | 19.09% |
| 39.46%                | 28.02  | 27.00                        | 33.21   | 16.74  | 39.7   |
| 1%                    | 31.97% | 28.33% resolución_intermedia |         |        |        |
| bar_p60_h60_tp20_sl05 |        | p60_h60 TP 2.0 / SL 0.5      | 41.87%  | 21.61% | 20.25% |
| 58.13%                | 27.12  | 25.00                        | 35.41   | 18.27  | 43.0   |
| 0%                    | 28.59% | 28.41% resolución_intermedia |         |        |        |
| bar_p60_h60_tp10_sl10 |        | p60_h60 TP 1.0 / SL 1.0      | 100.00% | 96.62% | 3.38%  |
| 0.00%                 | 30.03  | 29.00                        | 30.56   | 14.85  | 33.3   |
| 7%                    | 37.06% | 29.57% resolución_intermedia |         |        |        |
| bar_p60_h60_tp15_sl10 |        | p60_h60 TP 1.5 / SL 1.0      | 52.42%  | 48.27% | 4.14%  |
| 47.58%                | 32.97  | 33.00                        | 34.06   | 20.25  | 26.3   |
| 5%                    | 37.51% | 36.14% resolución_intermedia |         |        |        |
| bar_p60_h60_tp20_sl10 |        | p60_h60 TP 2.0 / SL 1.0      | 30.20%  | 25.48% | 4.72%  |
| 69.80%                | 34.12  | 35.00                        | 36.11   | 23.37  | 24.3   |
| 9%                    | 35.88% | 39.72% resolución_intermedia |         |        |        |
| bar_p60_h60_tp25_sl10 |        | p60_h60 TP 2.5 / SL 1.0      | 19.20%  | 14.23% | 4.97%  |
| 80.80%                | 34.09  | 35.00                        | 37.44   | 24.48  | 25.7   |

|                       |        |                              |         |        |        |
|-----------------------|--------|------------------------------|---------|--------|--------|
| 3%                    | 33.29% | 40.99% resolución_intermedia |         |        |        |
| bar_p60_h90_tp10_sl05 |        | p60_h90 TP 1.0 / SL 0.5      | 100.00% | 83.19% | 16.81% |
| 0.00%                 | 40.14  | 37.00                        | 44.61   | 18.02  | 41.8   |
| 1%                    | 33.41% | 24.78% resolución_intermedia |         |        |        |
| bar_p60_h90_tp15_sl05 |        | p60_h90 TP 1.5 / SL 0.5      | 61.77%  | 41.58% | 20.19% |
| 38.23%                | 42.67  | 41.00                        | 50.80   | 25.93  | 38.7   |
| 0%                    | 31.02% | 30.28% resolución_intermedia |         |        |        |
| bar_p60_h90_tp20_sl05 |        | p60_h90 TP 2.0 / SL 0.5      | 43.26%  | 21.82% | 21.44% |
| 56.74%                | 41.59  | 39.00                        | 54.58   | 28.36  | 41.9   |
| 6%                    | 27.35% | 30.70% resolución_intermedia |         |        |        |
| bar_p60_h90_tp10_sl10 |        | p60_h90 TP 1.0 / SL 1.0      | 100.00% | 96.37% | 3.63%  |
| 0.00%                 | 45.68  | 44.00                        | 46.46   | 24.89  | 32.2   |
| 9%                    | 36.48% | 31.24% resolución_intermedia |         |        |        |
| bar_p60_h90_tp15_sl10 |        | p60_h90 TP 1.5 / SL 1.0      | 53.30%  | 48.78% | 4.52%  |
| 46.70%                | 50.61  | 51.00                        | 52.22   | 33.17  | 24.7   |
| 7%                    | 36.09% | 39.13% resolución_intermedia |         |        |        |
| bar_p60_h90_tp20_sl10 |        | p60_h90 TP 2.0 / SL 1.0      | 31.02%  | 25.89% | 5.12%  |
| 68.98%                | 52.77  | 55.00                        | 55.73   | 37.76  | 22.7   |
| 1%                    | 33.95% | 43.34% resolución_intermedia |         |        |        |
| bar_p60_h90_tp25_sl10 |        | p60_h90 TP 2.5 / SL 1.0      | 19.82%  | 14.40% | 5.42%  |
| 80.18%                | 53.00  | 56.00                        | 58.05   | 39.58  | 23.8   |
| 9%                    | 31.10% | 45.01% resolución_intermedia |         |        |        |

Top 20 variantes con eventos más tempranos:

|                       | bar_col | horizon       | percentile      | tp_mult           | sl_mult          | event_rate | tp_rate  | s               |
|-----------------------|---------|---------------|-----------------|-------------------|------------------|------------|----------|-----------------|
|                       | l_rate  | no_event_rate | event_step_mean | event_step_median | early_event_rate | middle_ev  | ent_rate | late_event_rate |
| bar_p40_h30_tp10_sl05 |         | 30            | 40              | 1.0               | 0.5              | 100.00%    | 78.98%   |                 |
| 21.02%                | 0.00%   | 11.18         |                 | 9.00              |                  | 55.06%     |          |                 |
| 29.29%                | 15.65%  |               |                 |                   |                  |            |          |                 |
| bar_p40_h60_tp10_sl05 |         | 60            | 40              | 1.0               | 0.5              | 100.00%    | 77.94%   |                 |
| 22.06%                | 0.00%   | 21.95         |                 | 18.00             |                  | 54.96%     |          |                 |
| 28.94%                | 16.10%  |               |                 |                   |                  |            |          |                 |
| bar_p40_h90_tp10_sl05 |         | 90            | 40              | 1.0               | 0.5              | 100.00%    | 77.10%   |                 |
| 22.90%                | 0.00%   | 32.93         |                 | 27.00             |                  | 54.24%     |          |                 |
| 29.44%                | 16.32%  |               |                 |                   |                  |            |          |                 |
| bar_p40_h60_tp20_sl05 |         | 60            | 40              | 2.0               | 0.5              | 56.15%     | 27.84%   |                 |
| 28.31%                | 43.85%  | 24.57         |                 | 21.00             |                  | 49.08%     |          |                 |
| 27.40%                | 23.52%  |               |                 |                   |                  |            |          |                 |
| bar_p40_h60_tp15_sl05 |         | 60            | 40              | 1.5               | 0.5              | 72.88%     | 46.45%   |                 |
| 26.43%                | 27.12%  | 24.35         |                 | 21.00             |                  | 48.90%     |          |                 |
| 29.53%                | 21.57%  |               |                 |                   |                  |            |          |                 |
| bar_p50_h30_tp10_sl05 |         | 30            | 50              | 1.0               | 0.5              | 100.00%    | 81.93%   |                 |
| 18.07%                | 0.00%   | 12.37         |                 | 11.00             |                  | 48.62%     |          |                 |
| 31.90%                | 19.49%  |               |                 |                   |                  |            |          |                 |
| bar_p40_h30_tp15_sl05 |         | 30            | 40              | 1.5               | 0.5              | 73.21%     | 48.14%   |                 |
| 25.07%                | 26.79%  | 12.46         |                 | 11.00             |                  | 48.61%     |          |                 |
| 30.00%                | 21.39%  |               |                 |                   |                  |            |          |                 |
| bar_p40_h90_tp15_sl05 |         | 90            | 40              | 1.5               | 0.5              | 72.84%     | 45.52%   |                 |
| 27.32%                | 27.16%  | 36.63         |                 | 32.00             |                  | 48.55%     |          |                 |
| 29.16%                | 22.30%  |               |                 |                   |                  |            |          |                 |
| bar_p40_h90_tp20_sl05 |         | 90            | 40              | 2.0               | 0.5              | 56.66%     | 27.45%   |                 |
| 29.21%                | 43.34%  | 37.08         |                 | 32.00             |                  | 48.52%     |          |                 |
| 26.84%                | 24.64%  |               |                 |                   |                  |            |          |                 |
| bar_p50_h60_tp10_sl05 |         | 60            | 50              | 1.0               | 0.5              | 100.00%    | 81.32%   |                 |
| 18.68%                | 0.00%   | 24.40         |                 | 21.00             |                  | 48.36%     |          |                 |
| 31.58%                | 20.06%  |               |                 |                   |                  |            |          |                 |
| bar_p40_h30_tp20_sl05 |         | 30            | 40              | 2.0               | 0.5              | 56.62%     | 29.68%   |                 |
| 26.94%                | 43.38%  | 12.70         |                 | 11.00             |                  | 48.01%     |          |                 |
| 28.26%                | 23.74%  |               |                 |                   |                  |            |          |                 |

|                       |        |       |       |        |         |        |
|-----------------------|--------|-------|-------|--------|---------|--------|
| bar_p50_h90_tp10_sl05 | 90     | 50    | 1.0   | 0.5    | 100.00% | 80.23% |
| 19.77%                | 0.00%  | 36.58 | 32.00 | 47.89% |         |        |
| 31.67%                | 20.45% |       |       |        |         |        |
| bar_p50_h60_tp20_sl05 | 60     | 50    | 2.0   | 0.5    | 48.56%  | 24.58% |
| 23.99%                | 51.44% | 25.95 | 23.00 | 45.68% |         |        |
| 28.34%                | 25.98% |       |       |        |         |        |
| bar_p50_h90_tp20_sl05 | 90     | 50    | 2.0   | 0.5    | 49.77%  | 24.52% |
| 25.25%                | 50.23% | 39.31 | 35.00 | 45.08% |         |        |
| 27.39%                | 27.52% |       |       |        |         |        |
| bar_p50_h30_tp20_sl05 | 30     | 50    | 2.0   | 0.5    | 49.50%  | 26.31% |
| 23.19%                | 50.50% | 13.39 | 12.00 | 44.78% |         |        |
| 28.79%                | 26.43% |       |       |        |         |        |
| bar_p40_h30_tp10_sl10 | 30     | 40    | 1.0   | 1.0    | 100.00% | 93.84% |
| 6.16%                 | 0.00%  | 13.20 | 12.00 | 44.23% |         |        |
| 33.99%                | 21.78% |       |       |        |         |        |
| bar_p50_h60_tp15_sl05 | 60     | 50    | 1.5   | 0.5    | 66.52%  | 44.00% |
| 22.52%                | 33.48% | 26.28 | 24.00 | 44.09% |         |        |
| 30.86%                | 25.04% |       |       |        |         |        |
| bar_p50_h30_tp15_sl05 | 30     | 50    | 1.5   | 0.5    | 67.40%  | 45.74% |
| 21.67%                | 32.60% | 13.40 | 12.00 | 43.77% |         |        |
| 31.52%                | 24.70% |       |       |        |         |        |
| bar_p40_h60_tp10_sl10 | 60     | 40    | 1.0   | 1.0    | 100.00% | 94.00% |
| 6.00%                 | 0.00%  | 26.19 | 24.00 | 43.64% |         |        |
| 33.89%                | 22.47% |       |       |        |         |        |
| bar_p50_h90_tp15_sl05 | 90     | 50    | 1.5   | 0.5    | 67.30%  | 43.60% |
| 23.71%                | 32.70% | 39.72 | 36.00 | 43.51% |         |        |
| 30.21%                | 26.28% |       |       |        |         |        |

Top 20 variantes con eventos más tardíos:

| bar_col               | horizon       | percentile      | tp_mult           | sl_mult          | event_rate | tp_rate  | s               |
|-----------------------|---------------|-----------------|-------------------|------------------|------------|----------|-----------------|
| l_rate                | no_event_rate | event_step_mean | event_step_median | early_event_rate | middle_ev  | ent_rate | late_event_rate |
| bar_p60_h90_tp25_sl10 | 90            | 60              | 2.5               | 1.0              | 19.82%     | 14.40%   |                 |
| 5.42%                 | 80.18%        | 53.00           | 56.00             | 23.89%           |            |          |                 |
| 31.10%                | 45.01%        |                 |                   |                  |            |          |                 |
| bar_p60_h90_tp20_sl10 | 90            | 60              | 2.0               | 1.0              | 31.02%     | 25.89%   |                 |
| 5.12%                 | 68.98%        | 52.77           | 55.00             | 22.71%           |            |          |                 |
| 33.95%                | 43.34%        |                 |                   |                  |            |          |                 |
| bar_p50_h90_tp25_sl10 | 90            | 50              | 2.5               | 1.0              | 24.82%     | 17.65%   |                 |
| 7.17%                 | 75.18%        | 50.79           | 53.00             | 26.72%           |            |          |                 |
| 31.90%                | 41.38%        |                 |                   |                  |            |          |                 |
| bar_p60_h30_tp25_sl10 | 30            | 60              | 2.5               | 1.0              | 20.27%     | 15.06%   |                 |
| 5.21%                 | 79.73%        | 17.32           | 18.00             | 25.50%           |            |          |                 |
| 33.50%                | 41.00%        |                 |                   |                  |            |          |                 |
| bar_p60_h60_tp25_sl10 | 60            | 60              | 2.5               | 1.0              | 19.20%     | 14.23%   |                 |
| 4.97%                 | 80.80%        | 34.09           | 35.00             | 25.73%           |            |          |                 |
| 33.29%                | 40.99%        |                 |                   |                  |            |          |                 |
| bar_p60_h60_tp20_sl10 | 60            | 60              | 2.0               | 1.0              | 30.20%     | 25.48%   |                 |
| 4.72%                 | 69.80%        | 34.12           | 35.00             | 24.39%           |            |          |                 |
| 35.88%                | 39.72%        |                 |                   |                  |            |          |                 |
| bar_p60_h30_tp20_sl10 | 30            | 60              | 2.0               | 1.0              | 31.72%     | 26.78%   |                 |
| 4.94%                 | 68.28%        | 17.26           | 18.00             | 24.52%           |            |          |                 |
| 36.10%                | 39.38%        |                 |                   |                  |            |          |                 |
| bar_p50_h90_tp20_sl10 | 90            | 50              | 2.0               | 1.0              | 36.55%     | 29.78%   |                 |
| 6.77%                 | 63.45%        | 50.14           | 51.00             | 26.24%           |            |          |                 |
| 34.57%                | 39.19%        |                 |                   |                  |            |          |                 |
| bar_p60_h90_tp15_sl10 | 90            | 60              | 1.5               | 1.0              | 53.30%     | 48.78%   |                 |
| 4.52%                 | 46.70%        | 50.61           | 51.00             | 24.77%           |            |          |                 |
| 36.09%                | 39.13%        |                 |                   |                  |            |          |                 |
| bar_p50_h60_tp25_sl10 | 60            | 50              | 2.5               | 1.0              | 24.02%     | 17.33%   |                 |

|                       |        |       |       |        |     |        |        |
|-----------------------|--------|-------|-------|--------|-----|--------|--------|
| 6.69%                 | 75.98% | 32.87 | 34.00 | 28.31% |     |        |        |
| 33.73%                | 37.95% |       |       |        |     |        |        |
| bar_p50_h30_tp25_sl10 |        | 30    | 50    | 2.5    | 1.0 | 25.53% | 18.63% |
| 6.90%                 | 74.47% | 16.70 | 17.00 | 28.01% |     |        |        |
| 34.11%                | 37.88% |       |       |        |     |        |        |
| bar_p40_h90_tp25_sl10 |        | 90    | 40    | 2.5    | 1.0 | 30.62% | 21.19% |
| 9.43%                 | 69.38% | 48.34 | 49.00 | 30.03% |     |        |        |
| 32.41%                | 37.56% |       |       |        |     |        |        |
| bar_p50_h60_tp20_sl10 |        | 60    | 50    | 2.0    | 1.0 | 35.78% | 29.49% |
| 6.29%                 | 64.22% | 32.68 | 33.00 | 27.63% |     |        |        |
| 36.08%                | 36.29% |       |       |        |     |        |        |
| bar_p60_h60_tp15_sl10 |        | 60    | 60    | 1.5    | 1.0 | 52.42% | 48.27% |
| 4.14%                 | 47.58% | 32.97 | 33.00 | 26.35% |     |        |        |
| 37.51%                | 36.14% |       |       |        |     |        |        |
| bar_p50_h30_tp20_sl10 |        | 30    | 50    | 2.0    | 1.0 | 37.56% | 31.07% |
| 6.49%                 | 62.44% | 16.50 | 17.00 | 28.03% |     |        |        |
| 36.02%                | 35.95% |       |       |        |     |        |        |
| bar_p60_h30_tp15_sl10 |        | 30    | 60    | 1.5    | 1.0 | 53.89% | 49.53% |
| 4.37%                 | 46.11% | 16.62 | 17.00 | 26.82% |     |        |        |
| 37.55%                | 35.63% |       |       |        |     |        |        |
| bar_p40_h90_tp20_sl10 |        | 90    | 40    | 2.0    | 1.0 | 42.92% | 34.09% |
| 8.83%                 | 57.08% | 47.51 | 47.00 | 30.35% |     |        |        |
| 34.28%                | 35.37% |       |       |        |     |        |        |
| bar_p40_h60_tp25_sl10 |        | 60    | 40    | 2.5    | 1.0 | 29.89% | 21.05% |
| 8.84%                 | 70.11% | 31.48 | 31.00 | 31.37% |     |        |        |
| 33.80%                | 34.83% |       |       |        |     |        |        |
| bar_p50_h90_tp15_sl10 |        | 90    | 50    | 1.5    | 1.0 | 58.31% | 52.38% |
| 5.93%                 | 41.69% | 47.81 | 48.00 | 29.23% |     |        |        |
| 35.98%                | 34.79% |       |       |        |     |        |        |
| bar_p40_h30_tp25_sl10 |        | 30    | 40    | 2.5    | 1.0 | 31.51% | 22.59% |
| 8.92%                 | 68.49% | 15.97 | 16.00 | 31.32% |     |        |        |
| 34.13%                | 34.54% |       |       |        |     |        |        |

#### Conteo por perfil temporal:

|                       |           |                 |              |              |                    |                 |                   |                       |                        |                      |
|-----------------------|-----------|-----------------|--------------|--------------|--------------------|-----------------|-------------------|-----------------------|------------------------|----------------------|
| event_timing_profile  | n_targets | event_rate_mean | tp_rate_mean | sl_rate_mean | no_event_rate_mean | event_step_mean | event_step_median | early_event_rate_mean | middle_event_rate_mean | late_event_rate_mean |
| resolución_intermedia | 60        | 60.57%          | 48.13%       | 12.44%       |                    |                 |                   |                       |                        |                      |
| 39.43%                | 29.41     | 28.25           | 36.51%       | 3            |                    |                 |                   |                       |                        |                      |
| 3.17%                 | 30.32%    |                 |              |              |                    |                 |                   |                       |                        |                      |
| resolución_temprana   | 3         | 100.00%         | 78.01%       | 21.99%       |                    |                 |                   |                       |                        |                      |
| 0.00%                 | 22.02     | 18.00           | 54.75%       | 29.22%       |                    |                 |                   |                       |                        |                      |
|                       | 16.02%    |                 |              |              |                    |                 |                   |                       |                        |                      |

#### Archivos guardados:

Tiempo hasta evento BAR: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_event\_time.csv

Tiempo por configuración TP/SL: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_event\_time\_by\_config.csv

Tiempo por percentil y horizonte: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_event\_time\_by\_target\_base.csv

Ranking eventos tempranos: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_event\_time\_fast\_ranking.csv

Ranking eventos tardíos: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_event\_time\_slow\_ranking.csv

Tabla final tiempo hasta evento: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_event\_time\_final\_table.csv

Reporte TXT: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_event\_time\_report.txt

Celda 4 finalizada correctamente.

## Observaciones preliminares — Tiempo hasta evento TP/SL

El análisis de tiempo hasta evento muestra que los targets `BAR` presentan una dinámica temporal coherente con la lógica de las barreras.

En primer lugar, el tiempo promedio hasta evento aumenta naturalmente con el horizonte. Para `H30`, los eventos suelen ocurrir alrededor de las barras 13 a 16; para `H60`, alrededor de las barras 27 a 31; y para `H90`, alrededor de las barras 41 a 47. Esto indica que, en términos relativos, muchas configuraciones tienden a resolverse aproximadamente hacia la mitad del horizonte disponible.

También se observa que, a medida que aumenta el percentil, los eventos tienden a tardar más. Esto es esperable, porque percentiles más altos implican thresholds más exigentes. Por ejemplo, al pasar de `p40` a `p60`, el tiempo promedio hasta evento aumenta de forma consistente en `H30`, `H60` y `H90`.

Una observación importante es que los eventos de Stop Loss suelen ocurrir antes que los eventos de Take Profit. Esto aparece de forma consistente en casi todas las configuraciones. Por ejemplo, en configuraciones con `SL = 0.5`, el SL ocurre muy rápido porque la barrera adversa está más cerca del precio de entrada. Esto confirma que los stops más ajustados aumentan la velocidad de resolución, pero también aumentan la sensibilidad al ruido.

La configuración `TP 1.0 / SL 0.5` es la que resuelve más temprano. De hecho, las únicas variantes clasificadas como `resolución_temprana` corresponden a `p40` con `TP 1.0 / SL 0.5` en los tres horizontes. Sin embargo, esta configuración no necesariamente es la más interesante operativamente, porque tiene una frecuencia de SL relativamente alta.

La configuración `TP 1.0 / SL 1.0` mantiene una resolución completa, con `event_rate = 100%`, y tiempos de resolución moderados. Sin embargo, al igual que en los puntos anteriores, debe seguir interpretándose como benchmark, ya que el TP coincide con el threshold base usado para construir `DIR`.

La configuración `TP 1.5 / SL 1.0` mantiene un perfil temporal más equilibrado. No resuelve tan rápido como las configuraciones más fáciles, pero tampoco queda excesivamente tardía. En las variantes `p50`, por ejemplo, el evento promedio ocurre aproximadamente en la barra 16 para `H30`, 31 para `H60` y 48 para `H90`. Esto sugiere que la operación necesita tiempo para desarrollarse, pero no depende exclusivamente del final de la ventana.

Las configuraciones más exigentes, especialmente `TP 2.0 / SL 1.0` y `TP 2.5 / SL 1.0`, muestran una mayor proporción de eventos tardíos y muchos casos `no_event`. Esto indica que, aunque algunas operaciones alcanzan objetivos grandes, muchas necesitan demasiado tiempo o directamente no resuelven dentro del horizonte definido.

El caso más extremo aparece en variantes como `p60_h90` con `TP 2.5 / SL 1.0`, donde el `no_event` supera el 80% y una proporción importante de los eventos ocurre tarde. Esto refuerza la idea de que esa configuración es demasiado exigente para una primera selección operativa.

En términos generales, la mayoría de los targets quedan clasificados como `resolución_intermedia`. Esto es positivo, porque indica que las operaciones no dependen exclusivamente de movimientos inmediatos ni tampoco solamente del cierre del horizonte. La dinámica parece repartirse entre eventos tempranos, medios y tardíos.

En síntesis, el análisis de tiempo hasta evento confirma que `TP 1.5 / SL 1.0` sigue siendo una candidata razonable. Ofrece una frecuencia de TP favorable, baja frecuencia de SL y un tiempo de resolución intermedio. Por el contrario, `TP 2.0 / SL 1.0` y `TP 2.5 / SL 1.0` parecen más exigentes y lentas, por lo que deberían mantenerse solo como referencias comparativas o candidatas más agresivas.

La lectura simple sería esta:

`TP 1.0 / SL 1.0` → benchmark, resuelve siempre, pero favorecido por construcción.

`TP 1.5 / SL 1.0` → mejor equilibrio entre calidad y tiempo.

`TP 2.0 / SL 1.0` → más exigente, más lenta, muchos `no_event`.

`TP 2.5 / SL 1.0` → demasiado exigente para esta etapa.

`SL 0.5` → resuelve rápido, pero aumenta mucho la sensibilidad al ruido.

Para mí, después de 4.8, la candidata principal sigue siendo:

`bar_p50_h30_tp15_sl10`

`bar_p50_h60_tp15_sl10`

`bar_p50_h90_tp15_sl10`

Y como comparación más selectiva:

`bar_p60_h30_tp15_sl10`

`bar_p60_h60_tp15_sl10`

`bar_p60_h90_tp15_sl10`

## Aclaración sobre el tiempo hasta TP y SL

Una lectura importante del análisis de tiempo hasta evento es que el hecho de que el `SL` tenga un menor tiempo promedio que el `TP` no significa que el Stop Loss se alcance antes en todas las operaciones.

La codificación de `BAR` ya resuelve esa lógica fila por fila:

`BAR = 1` → el Take Profit fue alcanzado primero

`BAR = -1` → el Stop Loss fue alcanzado primero

`BAR = 0` → no se alcanzó ni TP ni SL dentro del horizonte

Por lo tanto, cuando una observación tiene `BAR = 1`, significa que el `TP` ocurrió antes que el `SL`. Cuando una observación tiene `BAR = -1`, significa que el `SL` ocurrió antes que el `TP`.



Lo que muestran las métricas `tp_step_mean` y `sl_step_mean` es distinto: son promedios condicionados al tipo de evento. Es decir, miden cuánto tardan, en promedio, las operaciones que terminan en TP y cuánto tardan, en promedio, las operaciones que terminan en SL.

En la configuración `TP 1.5 / SL 1.0`, por ejemplo, el TP está más lejos que el SL:

$$\text{TP} = 1.5 \times \text{threshold}$$

$$\text{SL} = 1.0 \times \text{threshold}$$

Por esta razón, cuando la operación sale bien, normalmente necesita más barras para alcanzar el Take Profit. En cambio, cuando la operación sale mal, el Stop Loss puede aparecer antes porque está más cerca del precio de entrada.

La interpretación correcta es la siguiente:

El SL suele ocurrir más rápido cuando ocurre una pérdida, pero eso no significa que el SL ocurra antes en las operaciones ganadoras.

En términos operativos, esto es razonable. Las pérdidas tienden a resolverse más rápido porque la barrera adversa está más cerca, mientras que las ganancias requieren más tiempo para desarrollar un movimiento favorable más amplio.

Por eso, una configuración como `TP 1.5 / SL 1.0` puede seguir siendo interesante: no porque el TP llegue más rápido, sino porque cuando el evento se resuelve, la mayoría de las veces resuelve a favor del TP, aunque tarde más en alcanzarlo.

## 4.9. Estabilidad temporal

En este punto se analiza la estabilidad temporal de los targets `BAR`.

El objetivo es revisar si las métricas operativas de cada target se mantienen relativamente consistentes a lo largo del tiempo o si cambian de forma significativa entre períodos.

Este análisis es importante porque un target puede mostrar buenos resultados globales, pero ser poco confiable si su comportamiento depende demasiado de un año específico o de un período particular del mercado.

La evaluación se realizará sobre los 63 targets `BAR`, considerando todos los percentiles, horizontes y configuraciones TP/SL.

Se analizarán principalmente:

- Frecuencia de TP

- Frecuencia de SL

- Frecuencia de `no_event`

- Eventos resueltos

- Proporción de TP entre eventos resueltos

- Expectativa simple en unidades de `threshold`

- Tiempo promedio hasta evento

Además, se mantendrá la separación metodológica definida previamente:

development\_2020\_2024 → período usado para diseño y calibración  
 final\_test\_2025\_2026 → período reservado para evaluación fuera de muestra

## Celda 1 — Preparación de métricas temporales para los 63 targets BAR

Esta celda deja listo el dataframe principal:

df\_bar\_temporal\_stability

Ese dataframe contiene las métricas temporales de los **63 targets BAR** por:

dataset\_split  
 year  
 quarter

In [107...

```
# Celda 1 – Preparación de métricas temporales para los 63 targets BAR
# =====

print("4.9. Estabilidad temporal - BAR")
print("=" * 80)

# -----
# Validar objetos necesarios
# -----

if "df_targets_bar" not in globals():
    raise NameError("No existe df_targets_bar. Primero debes construir los targe

if "df_bar_metadata" not in globals():
    raise NameError("No existe df_bar_metadata. Primero debes construir df_bar_m

# -----
# Copia de trabajo
# -----

df_bar_temporal_base = df_targets_bar.copy()

# -----
# Crear columnas temporales auxiliares
# -----
# Importante:
# Recalculamos year y year_quarter explícitamente para evitar usar
# columnas antiguas o columnas quarter con valores 1, 2, 3, 4.
# -----

if isinstance(df_bar_temporal_base.index, pd.DatetimeIndex):
    date_series_tmp = pd.Series(
        df_bar_temporal_base.index,
        index=df_bar_temporal_base.index
    )

elif "date" in df_bar_temporal_base.columns:
    date_series_tmp = pd.to_datetime(df_bar_temporal_base["date"])

else:
    raise KeyError(
        "No se pudieron crear columnas temporales. "
```

```

        "Se requiere DatetimeIndex o columna date."
    )

# Año
df_bar_temporal_base["year"] = date_series_tmp.dt.year

# Trimestre temporal correcto: 2020Q1, 2020Q2, ..., 2026Q2
df_bar_temporal_base["year_quarter"] = (
    date_series_tmp.dt.year.astype(str)
    + "Q"
    + date_series_tmp.dt.quarter.astype(str)
)

# Split metodológico
df_bar_temporal_base["dataset_split"] = np.where(
    df_bar_temporal_base["year"].between(2020, 2024),
    "development_2020_2024",
    np.where(
        df_bar_temporal_base["year"].between(2025, 2026),
        "final_test_2025_2026",
        "other"
    )
)

# -----
# Función auxiliar para resumir un target BAR en un período
# -----

def summarize_bar_target_for_period(df, bar_col, event_step_col, tp_mult, sl_mult):
    """
    Resume un target BAR dentro de un período temporal específico.

    BAR:
        1 = TP primero
        -1 = SL primero
        0 = no_event
        NaN = no operable / sin ventana válida
    """

    bar_series = df[bar_col]
    step_series = df[event_step_col]

    valid_mask = bar_series.notna()
    tp_mask = bar_series == 1
    sl_mask = bar_series == -1
    no_event_mask = bar_series == 0
    resolved_mask = tp_mask | sl_mask

    total_rows = len(df)
    valid_rows = int(valid_mask.sum())
    nan_rows = int(bar_series.isna().sum())

    tp_count = int(tp_mask.sum())
    sl_count = int(sl_mask.sum())
    no_event_count = int(no_event_mask.sum())
    resolved_count = tp_count + sl_count

    if valid_rows > 0:
        tp_rate = tp_count / valid_rows
        sl_rate = sl_count / valid_rows

```

```

no_event_rate = no_event_count / valid_rows
resolved_rate = resolved_count / valid_rows
else:
    tp_rate = np.nan
    sl_rate = np.nan
    no_event_rate = np.nan
    resolved_rate = np.nan

if resolved_count > 0:
    tp_share_over_resolved = tp_count / resolved_count
    sl_share_over_resolved = sl_count / resolved_count
else:
    tp_share_over_resolved = np.nan
    sl_share_over_resolved = np.nan

expectancy_units = (
    tp_rate * tp_mult
    - sl_rate * sl_mult
    if valid_rows > 0
    else np.nan
)

event_steps = step_series.loc[resolved_mask].dropna()

if len(event_steps) > 0:
    event_step_mean = event_steps.mean()
    event_step_median = event_steps.median()
else:
    event_step_mean = np.nan
    event_step_median = np.nan

return {
    "total_rows": total_rows,
    "valid_rows": valid_rows,
    "nan_rows": nan_rows,

    "tp_count": tp_count,
    "sl_count": sl_count,
    "no_event_count": no_event_count,
    "resolved_count": resolved_count,

    "tp_rate": tp_rate,
    "sl_rate": sl_rate,
    "no_event_rate": no_event_rate,
    "resolved_rate": resolved_rate,

    "tp_share_over_resolved": tp_share_over_resolved,
    "sl_share_over_resolved": sl_share_over_resolved,

    "expectancy_units": expectancy_units,

    "event_step_mean": event_step_mean,
    "event_step_median": event_step_median,
}

```

```

# -----
# Definición de periodos temporales
# -----
# dataset_split: development vs final_test
# year: año contra año

```

```

# quarter: trimestre temporal real, por ejemplo 2020Q1, 2020Q2, ...
# -----

period_definitions = [
    {
        "period_type": "dataset_split",
        "period_col": "dataset_split",
    },
    {
        "period_type": "year",
        "period_col": "year",
    },
    {
        "period_type": "quarter",
        "period_col": "year_quarter",
    },
]

# -----
# Construir métricas temporales por split, año y trimestre temporal
# -----

temporal_rows = []

for _, meta in df_bar_metadata.iterrows():

    bar_col = meta["bar_col"]
    event_step_col = meta["event_step_col"]

    horizon = int(meta["horizon"])
    percentile = int(meta["percentile"])
    tp_mult = float(meta["tp_mult"])
    sl_mult = float(meta["sl_mult"])

    tp_sl_config = f"TP {tp_mult} / SL {sl_mult}"
    target_base = f"p{percentile}_h{horizon}"

    if bar_col not in df_bar_temporal_base.columns:
        raise KeyError(f"No existe la columna BAR: {bar_col}")

    if event_step_col not in df_bar_temporal_base.columns:
        raise KeyError(f"No existe la columna event_step: {event_step_col}")

    for period_info in period_definitions:

        period_type = period_info["period_type"]
        period_col = period_info["period_col"]

        period_values = (
            df_bar_temporal_base[period_col]
            .dropna()
            .unique()
        )

        for period_value in sorted(period_values):

            if period_value == "other":
                continue

            period_mask = df_bar_temporal_base[period_col] == period_value

```

```

df_period = df_bar_temporal_base.loc[period_mask]

summary = summarize_bar_target_for_period(
    df=df_period,
    bar_col=bar_col,
    event_step_col=event_step_col,
    tp_mult=tp_mult,
    sl_mult=sl_mult,
)

temporal_rows.append(
    {
        "bar_col": bar_col,
        "event_step_col": event_step_col,
        "dir_col": meta["dir_col"],

        "target_base": target_base,
        "horizon": horizon,
        "percentile": percentile,
        "tp_mult": tp_mult,
        "sl_mult": sl_mult,
        "tp_sl_config": tp_sl_config,

        "period_type": period_type,
        "period": str(period_value),

        **summary,
    }
)

df_bar_temporal_stability = pd.DataFrame(temporal_rows)

df_bar_temporal_stability = (
    df_bar_temporal_stability
    .sort_values(
        [
            "period_type",
            "period",
            "percentile",
            "horizon",
            "sl_mult",
            "tp_mult",
        ]
    )
    .reset_index(drop=True)
)

# -----
# Vista inicial
# -----

print(f"Targets BAR analizados: {df_bar_metadata['bar_col'].unique()}")
print(f"Filas de estabilidad temporal generadas: {len(df_bar_temporal_stability)}")

print("\nPeriodos analizados:")
print(
    df_bar_temporal_stability[
        ["period_type", "period"]
    ]
    .drop_duplicates()
)

```

```

        .sort_values(["period_type", "period"])
        .to_string(index=False)
    )

print("\nVista inicial de estabilidad temporal BAR:")
print(
    df_bar_temporal_stability[
        [
            "bar_col",
            "period_type",
            "period",
            "horizon",
            "percentile",
            "tp_mult",
            "sl_mult",
            "valid_rows",
            "tp_rate",
            "sl_rate",
            "no_event_rate",
            "resolved_rate",
            "tp_share_over_resolved",
            "expectancy_units",
            "event_step_mean",
        ]
    ]
    .head(30)
    .to_string(
        index=False,
        formatters={
            "tp_rate": "{:.2%}".format,
            "sl_rate": "{:.2%}".format,
            "no_event_rate": "{:.2%}".format,
            "resolved_rate": "{:.2%}".format,
            "tp_share_over_resolved": "{:.2%}".format,
            "expectancy_units": "{:.3f}".format,
            "event_step_mean": "{:.2f}".format,
        }
    )
)

print("\nValidación de trimestres temporales:")
print(
    df_bar_temporal_stability[
        df_bar_temporal_stability["period_type"] == "quarter"
    ][["period_type", "period"]]
    .drop_duplicates()
    .sort_values("period")
    .head(12)
    .to_string(index=False)
)

print("\nPreparación de métricas temporales finalizada correctamente.")

```

## 4.9. Estabilidad temporal - BAR

=====

Targets BAR analizados: 63

Filas de estabilidad temporal generadas: 2205

Periodos analizados:

| period_type   | period                |
|---------------|-----------------------|
| dataset_split | development_2020_2024 |
| dataset_split | final_test_2025_2026  |
| quarter       | 2020Q1                |
| quarter       | 2020Q2                |
| quarter       | 2020Q3                |
| quarter       | 2020Q4                |
| quarter       | 2021Q1                |
| quarter       | 2021Q2                |
| quarter       | 2021Q3                |
| quarter       | 2021Q4                |
| quarter       | 2022Q1                |
| quarter       | 2022Q2                |
| quarter       | 2022Q3                |
| quarter       | 2022Q4                |
| quarter       | 2023Q1                |
| quarter       | 2023Q2                |
| quarter       | 2023Q3                |
| quarter       | 2023Q4                |
| quarter       | 2024Q1                |
| quarter       | 2024Q2                |
| quarter       | 2024Q3                |
| quarter       | 2024Q4                |
| quarter       | 2025Q1                |
| quarter       | 2025Q2                |
| quarter       | 2025Q3                |
| quarter       | 2025Q4                |
| quarter       | 2026Q1                |
| quarter       | 2026Q2                |
| year          | 2020                  |
| year          | 2021                  |
| year          | 2022                  |
| year          | 2023                  |
| year          | 2024                  |
| year          | 2025                  |
| year          | 2026                  |

Vista inicial de estabilidad temporal BAR:

| bar_col               | period_type      | period                | horizon | percentile | t             |               |          |
|-----------------------|------------------|-----------------------|---------|------------|---------------|---------------|----------|
| p_mult                | sl_mult          | valid_rows            | tp_rate | sl_rate    | no_event_rate | resolved_rate | tp_share |
| over_resolved         | expectancy_units | event_step_mean       |         |            |               |               |          |
| bar_p40_h30_tp10_sl05 | dataset_split    | development_2020_2024 | 30      | 40         |               |               |          |
| 1.0                   | 0.5              | 471508                | 80.81%  | 19.19%     | 0.00%         | 100.00%       |          |
| 80.81%                |                  | 0.712                 |         | 11.92      |               |               |          |
| bar_p40_h30_tp15_sl05 | dataset_split    | development_2020_2024 | 30      | 40         |               |               |          |
| 1.5                   | 0.5              | 471508                | 47.11%  | 23.00%     | 29.88%        | 70.12%        |          |
| 67.19%                |                  | 0.592                 |         | 13.24      |               |               |          |
| bar_p40_h30_tp20_sl05 | dataset_split    | development_2020_2024 | 30      | 40         |               |               |          |
| 2.0                   | 0.5              | 471508                | 27.75%  | 24.59%     | 47.67%        | 52.33%        |          |
| 53.01%                |                  | 0.432                 |         | 13.41      |               |               |          |
| bar_p40_h30_tp10_sl10 | dataset_split    | development_2020_2024 | 30      | 40         |               |               |          |
| 1.0                   | 1.0              | 471508                | 95.09%  | 4.91%      | 0.00%         | 100.00%       |          |
| 95.09%                |                  | 0.902                 |         | 13.90      |               |               |          |
| bar_p40_h30_tp15_sl10 | dataset_split    | development_2020_2024 | 30      | 40         |               |               |          |



|                       |               |                       |        |        |        |         |    |
|-----------------------|---------------|-----------------------|--------|--------|--------|---------|----|
| 1.5                   | 1.0           | 471508                | 55.62% | 5.95%  | 38.42% | 61.58%  |    |
| 90.33%                |               | 0.775                 |        | 15.67  |        |         |    |
| bar_p40_h30_tp20_sl10 | dataset_split | development_2020_2024 |        |        |        | 30      | 40 |
| 2.0                   | 1.0           | 471508                | 32.82% | 6.76%  | 60.41% | 39.59%  |    |
| 82.92%                |               | 0.589                 |        | 16.58  |        |         |    |
| bar_p40_h30_tp25_sl10 | dataset_split | development_2020_2024 |        |        |        | 30      | 40 |
| 2.5                   | 1.0           | 471508                | 19.75% | 7.18%  | 73.08% | 26.92%  |    |
| 73.34%                |               | 0.422                 |        | 16.88  |        |         |    |
| bar_p40_h60_tp10_sl05 | dataset_split | development_2020_2024 |        |        |        | 60      | 40 |
| 1.0                   | 0.5           | 450321                | 80.13% | 19.87% | 0.00%  | 100.00% |    |
| 80.13%                |               | 0.702                 |        | 23.45  |        |         |    |
| bar_p40_h60_tp15_sl05 | dataset_split | development_2020_2024 |        |        |        | 60      | 40 |
| 1.5                   | 0.5           | 450321                | 45.72% | 23.99% | 30.28% | 69.72%  |    |
| 65.58%                |               | 0.566                 |        | 26.04  |        |         |    |
| bar_p40_h60_tp20_sl05 | dataset_split | development_2020_2024 |        |        |        | 60      | 40 |
| 2.0                   | 0.5           | 450321                | 26.03% | 25.55% | 48.42% | 51.58%  |    |
| 50.47%                |               | 0.393                 |        | 26.11  |        |         |    |
| bar_p40_h60_tp10_sl10 | dataset_split | development_2020_2024 |        |        |        | 60      | 40 |
| 1.0                   | 1.0           | 450321                | 95.34% | 4.66%  | 0.00%  | 100.00% |    |
| 95.34%                |               | 0.907                 |        | 27.58  |        |         |    |
| bar_p40_h60_tp15_sl10 | dataset_split | development_2020_2024 |        |        |        | 60      | 40 |
| 1.5                   | 1.0           | 450321                | 54.44% | 5.78%  | 39.78% | 60.22%  |    |
| 90.40%                |               | 0.759                 |        | 31.17  |        |         |    |
| bar_p40_h60_tp20_sl10 | dataset_split | development_2020_2024 |        |        |        | 60      | 40 |
| 2.0                   | 1.0           | 450321                | 31.18% | 6.54%  | 62.27% | 37.73%  |    |
| 82.66%                |               | 0.558                 |        | 33.01  |        |         |    |
| bar_p40_h60_tp25_sl10 | dataset_split | development_2020_2024 |        |        |        | 60      | 40 |
| 2.5                   | 1.0           | 450321                | 18.07% | 6.92%  | 75.02% | 24.98%  |    |
| 72.31%                |               | 0.382                 |        | 33.38  |        |         |    |
| bar_p40_h90_tp10_sl05 | dataset_split | development_2020_2024 |        |        |        | 90      | 40 |
| 1.0                   | 0.5           | 428486                | 79.41% | 20.59% | 0.00%  | 100.00% |    |
| 79.41%                |               | 0.691                 |        | 35.31  |        |         |    |
| bar_p40_h90_tp15_sl05 | dataset_split | development_2020_2024 |        |        |        | 90      | 40 |
| 1.5                   | 0.5           | 428486                | 44.83% | 24.77% | 30.40% | 69.60%  |    |
| 64.41%                |               | 0.549                 |        | 39.35  |        |         |    |
| bar_p40_h90_tp20_sl05 | dataset_split | development_2020_2024 |        |        |        | 90      | 40 |
| 2.0                   | 0.5           | 428486                | 25.64% | 26.36% | 48.00% | 52.00%  |    |
| 49.30%                |               | 0.381                 |        | 39.62  |        |         |    |
| bar_p40_h90_tp10_sl10 | dataset_split | development_2020_2024 |        |        |        | 90      | 40 |
| 1.0                   | 1.0           | 428486                | 94.92% | 5.08%  | 0.00%  | 100.00% |    |
| 94.92%                |               | 0.898                 |        | 41.73  |        |         |    |
| bar_p40_h90_tp15_sl10 | dataset_split | development_2020_2024 |        |        |        | 90      | 40 |
| 1.5                   | 1.0           | 428486                | 54.07% | 6.18%  | 39.75% | 60.25%  |    |
| 89.74%                |               | 0.749                 |        | 47.60  |        |         |    |
| bar_p40_h90_tp20_sl10 | dataset_split | development_2020_2024 |        |        |        | 90      | 40 |
| 2.0                   | 1.0           | 428486                | 31.10% | 7.03%  | 61.87% | 38.13%  |    |
| 81.56%                |               | 0.552                 |        | 50.66  |        |         |    |
| bar_p40_h90_tp25_sl10 | dataset_split | development_2020_2024 |        |        |        | 90      | 40 |
| 2.5                   | 1.0           | 428486                | 18.23% | 7.44%  | 74.32% | 25.68%  |    |
| 71.01%                |               | 0.381                 |        | 51.31  |        |         |    |
| bar_p50_h30_tp10_sl05 | dataset_split | development_2020_2024 |        |        |        | 30      | 50 |
| 1.0                   | 0.5           | 393525                | 83.91% | 16.09% | 0.00%  | 100.00% |    |
| 83.91%                |               | 0.759                 |        | 13.17  |        |         |    |
| bar_p50_h30_tp15_sl05 | dataset_split | development_2020_2024 |        |        |        | 30      | 50 |
| 1.5                   | 0.5           | 393525                | 44.36% | 19.40% | 36.24% | 63.76%  |    |
| 69.57%                |               | 0.568                 |        | 14.26  |        |         |    |
| bar_p50_h30_tp20_sl05 | dataset_split | development_2020_2024 |        |        |        | 30      | 50 |
| 2.0                   | 0.5           | 393525                | 24.07% | 20.60% | 55.32% | 44.68%  |    |
| 53.88%                |               | 0.378                 |        | 14.13  |        |         |    |
| bar_p50_h30_tp10_sl10 | dataset_split | development_2020_2024 |        |        |        | 30      | 50 |

|                       |               |                       |        |        |        |         |    |
|-----------------------|---------------|-----------------------|--------|--------|--------|---------|----|
| 1.0                   | 1.0           | 393525                | 96.38% | 3.62%  | 0.00%  | 100.00% |    |
| 96.38%                |               | 0.928                 |        | 14.94  |        |         |    |
| bar_p50_h30_tp15_sl10 | dataset_split | development_2020_2024 |        |        |        | 30      | 50 |
| 1.5                   | 1.0           | 393525                | 51.18% | 4.44%  | 44.37% | 55.63%  |    |
| 92.01%                |               | 0.723                 |        | 16.62  |        |         |    |
| bar_p50_h30_tp20_sl10 | dataset_split | development_2020_2024 |        |        |        | 30      | 50 |
| 2.0                   | 1.0           | 393525                | 27.93% | 5.02%  | 67.04% | 32.96%  |    |
| 84.76%                |               | 0.508                 |        | 17.45  |        |         |    |
| bar_p50_h30_tp25_sl10 | dataset_split | development_2020_2024 |        |        |        | 30      | 50 |
| 2.5                   | 1.0           | 393525                | 15.76% | 5.30%  | 78.94% | 21.06%  |    |
| 74.83%                |               | 0.341                 |        | 17.62  |        |         |    |
| bar_p50_h60_tp10_sl05 | dataset_split | development_2020_2024 |        |        |        | 60      | 50 |
| 1.0                   | 0.5           | 374094                | 83.68% | 16.32% | 0.00%  | 100.00% |    |
| 83.68%                |               | 0.755                 |        | 26.09  |        |         |    |
| bar_p50_h60_tp15_sl05 | dataset_split | development_2020_2024 |        |        |        | 60      | 50 |
| 1.5                   | 0.5           | 374094                | 42.79% | 19.78% | 37.43% | 62.57%  |    |
| 68.39%                |               | 0.543                 |        | 28.16  |        |         |    |

Validación de trimestres temporales:

```
period_type period
quarter 2020Q1
quarter 2020Q2
quarter 2020Q3
quarter 2020Q4
quarter 2021Q1
quarter 2021Q2
quarter 2021Q3
quarter 2021Q4
quarter 2022Q1
quarter 2022Q2
quarter 2022Q3
quarter 2022Q4
```

Preparación de métricas temporales finalizada correctamente.

## Celda 2 — Tablas agregadas de estabilidad temporal

In [108...

```
# Celda 2 - Tablas agregadas de estabilidad temporal
# =====

print("Tablas agregadas de estabilidad temporal - BAR")
print("=" * 80)

# -----
# Validar tabla base
# -----

if "df_bar_temporal_stability" not in globals():
    raise NameError("No existe df_bar_temporal_stability. Primero debes correr 1

# -----
# Métricas principales para evaluar estabilidad
# -----

stability_metrics = [
    "tp_rate",
    "sl_rate",
    "no_event_rate",
    "resolved_rate",
```

```

        "tp_share_over_resolved",
        "expectancy_units",
        "event_step_mean",
    ]

# -----
# Separar tablas por tipo de período
# -----

df_bar_stability_split = (
    df_bar_temporal_stability[
        df_bar_temporal_stability["period_type"] == "dataset_split"
    ]
    .copy()
    .reset_index(drop=True)
)

df_bar_stability_year = (
    df_bar_temporal_stability[
        df_bar_temporal_stability["period_type"] == "year"
    ]
    .copy()
    .reset_index(drop=True)
)

df_bar_stability_quarter = (
    df_bar_temporal_stability[
        df_bar_temporal_stability["period_type"] == "quarter"
    ]
    .copy()
    .reset_index(drop=True)
)

print(f"Filas por dataset_split: {len(df_bar_stability_split)}")
print(f"Filas por año: {len(df_bar_stability_year)}")
print(f"Filas por trimestre: {len(df_bar_stability_quarter)}")

print("\nPrimeros trimestres temporales detectados:")
print(
    df_bar_stability_quarter[["period_type", "period"]]
    .drop_duplicates()
    .sort_values("period")
    .head(12)
    .to_string(index=False)
)

# =====
# 1. Comparación development vs final_test
# =====

id_cols = [
    "bar_col",
    "target_base",
    "horizon",
    "percentile",
    "tp_mult",
    "sl_mult",
    "tp_sl_config",
]

```

```

df_bar_split_comparison = (
    df_bar_stability_split
    .pivot_table(
        index=id_cols,
        columns="period",
        values=stability_metrics,
        aggfunc="first"
    )
)

df_bar_split_comparison.columns = [
    f"{metric}_{period}"
    for metric, period in df_bar_split_comparison.columns
]

df_bar_split_comparison = (
    df_bar_split_comparison
    .reset_index()
)

dev_label = "development_2020_2024"
test_label = "final_test_2025_2026"

for metric in stability_metrics:
    dev_col = f"{metric}_{dev_label}"
    test_col = f"{metric}_{test_label}"

    if dev_col in df_bar_split_comparison.columns and test_col in df_bar_split_c
        df_bar_split_comparison[f"{metric}_diff_final_minus_dev"] = (
            df_bar_split_comparison[test_col]
            - df_bar_split_comparison[dev_col]
        )

    df_bar_split_comparison[f"{metric}_abs_diff_final_vs_dev"] = (
        df_bar_split_comparison[f"{metric}_diff_final_minus_dev"].abs()
    )

split_score_cols = [
    "tp_rate_abs_diff_final_vs_dev",
    "sl_rate_abs_diff_final_vs_dev",
    "no_event_rate_abs_diff_final_vs_dev",
    "tp_share_over_resolved_abs_diff_final_vs_dev",
]

available_split_score_cols = [
    col for col in split_score_cols
    if col in df_bar_split_comparison.columns
]

df_bar_split_comparison["split_shift_score"] = (
    df_bar_split_comparison[available_split_score_cols]
    .sum(axis=1)
)

df_bar_split_comparison = (
    df_bar_split_comparison
    .sort_values(
        ["split_shift_score", "percentile", "horizon", "sl_mult", "tp_mult"],
        ascending=[False, True, True, True, True]
    )
)

```

```

        .reset_index(drop=True)
    )

print("\nComparación development vs final_test - Top 20 con mayor cambio:")
print(
    df_bar_split_comparison[
        [
            "bar_col",
            "target_base",
            "tp_sl_config",

            f"tp_rate_{dev_label}",
            f"tp_rate_{test_label}",
            "tp_rate_diff_final_minus_dev",

            f"sl_rate_{dev_label}",
            f"sl_rate_{test_label}",
            "sl_rate_diff_final_minus_dev",

            f"no_event_rate_{dev_label}",
            f"no_event_rate_{test_label}",
            "no_event_rate_diff_final_minus_dev",

            f"tp_share_over_resolved_{dev_label}",
            f"tp_share_over_resolved_{test_label}",
            "tp_share_over_resolved_diff_final_minus_dev",

            "split_shift_score",
        ]
    ]
    .head(20)
    .to_string(
        index=False,
        formatters={
            f"tp_rate_{dev_label}": "{:.2%}".format,
            f"tp_rate_{test_label}": "{:.2%}".format,
            "tp_rate_diff_final_minus_dev": "{:.2%}".format,

            f"sl_rate_{dev_label}": "{:.2%}".format,
            f"sl_rate_{test_label}": "{:.2%}".format,
            "sl_rate_diff_final_minus_dev": "{:.2%}".format,

            f"no_event_rate_{dev_label}": "{:.2%}".format,
            f"no_event_rate_{test_label}": "{:.2%}".format,
            "no_event_rate_diff_final_minus_dev": "{:.2%}".format,

            f"tp_share_over_resolved_{dev_label}": "{:.2%}".format,
            f"tp_share_over_resolved_{test_label}": "{:.2%}".format,
            "tp_share_over_resolved_diff_final_minus_dev": "{:.2%}".format,

            "split_shift_score": "{:.3f}".format,
        }
    )
)

print("\nComparación development vs final_test - Top 20 más estables:")
print(
    df_bar_split_comparison[
        [
            "bar_col",

```

```

        "target_base",
        "tp_sl_config",

        f"tp_rate_{dev_label}",
        f"tp_rate_{test_label}",
        "tp_rate_diff_final_minus_dev",

        f"sl_rate_{dev_label}",
        f"sl_rate_{test_label}",
        "sl_rate_diff_final_minus_dev",

        f"no_event_rate_{dev_label}",
        f"no_event_rate_{test_label}",
        "no_event_rate_diff_final_minus_dev",

        f"tp_share_over_resolved_{dev_label}",
        f"tp_share_over_resolved_{test_label}",
        "tp_share_over_resolved_diff_final_minus_dev",

        "split_shift_score",
    ]
]
.tail(20)
.sort_values("split_shift_score")
.to_string(
    index=False,
    formatters={
        f"tp_rate_{dev_label}": "{:.2%}".format,
        f"tp_rate_{test_label}": "{:.2%}".format,
        "tp_rate_diff_final_minus_dev": "{:.2%}".format,

        f"sl_rate_{dev_label}": "{:.2%}".format,
        f"sl_rate_{test_label}": "{:.2%}".format,
        "sl_rate_diff_final_minus_dev": "{:.2%}".format,

        f"no_event_rate_{dev_label}": "{:.2%}".format,
        f"no_event_rate_{test_label}": "{:.2%}".format,
        "no_event_rate_diff_final_minus_dev": "{:.2%}".format,

        f"tp_share_over_resolved_{dev_label}": "{:.2%}".format,
        f"tp_share_over_resolved_{test_label}": "{:.2%}".format,
        "tp_share_over_resolved_diff_final_minus_dev": "{:.2%}".format,

        "split_shift_score": "{:.3f}".format,
    }
)

# =====
# 2. Resumen del cambio por configuración TP/SL
# =====

df_bar_split_change_by_config = (
    df_bar_split_comparison
    .groupby(["tp_mult", "sl_mult", "tp_sl_config"], as_index=False)
    .agg(
        n_targets=("bar_col", "count"),

        tp_rate_diff_mean=("tp_rate_diff_final_minus_dev", "mean"),
        tp_rate_abs_diff_mean=("tp_rate_abs_diff_final_vs_dev", "mean"),

```

```

sl_rate_diff_mean=("sl_rate_diff_final_minus_dev", "mean"),
sl_rate_abs_diff_mean=("sl_rate_abs_diff_final_vs_dev", "mean"),

no_event_rate_diff_mean=("no_event_rate_diff_final_minus_dev", "mean"),
no_event_rate_abs_diff_mean=("no_event_rate_abs_diff_final_vs_dev", "mea

tp_share_diff_mean=("tp_share_over_resolved_diff_final_minus_dev", "mean
tp_share_abs_diff_mean=("tp_share_over_resolved_abs_diff_final_vs_dev",

split_shift_score_mean=("split_shift_score", "mean"),
split_shift_score_max=("split_shift_score", "max"),
)
.sort_values(["sl_mult", "tp_mult"])
.reset_index(drop=True)
)

print("\nResumen de cambio temporal por configuración TP/SL:")
print(
    df_bar_split_change_by_config.to_string(
        index=False,
        formatters={
            "tp_rate_diff_mean": "{:.2%}".format,
            "tp_rate_abs_diff_mean": "{:.2%}".format,
            "sl_rate_diff_mean": "{:.2%}".format,
            "sl_rate_abs_diff_mean": "{:.2%}".format,
            "no_event_rate_diff_mean": "{:.2%}".format,
            "no_event_rate_abs_diff_mean": "{:.2%}".format,
            "tp_share_diff_mean": "{:.2%}".format,
            "tp_share_abs_diff_mean": "{:.2%}".format,
            "split_shift_score_mean": "{:.3f}".format,
            "split_shift_score_max": "{:.3f}".format,
        }
    )
)

# =====
# 3. Resumen del cambio por percentil y horizonte
# =====

df_bar_split_change_by_target_base = (
    df_bar_split_comparison
    .groupby(["percentile", "horizon", "target_base"], as_index=False)
    .agg(
        n_configs=("bar_col", "count"),

        tp_rate_diff_mean=("tp_rate_diff_final_minus_dev", "mean"),
        tp_rate_abs_diff_mean=("tp_rate_abs_diff_final_vs_dev", "mean"),

        sl_rate_diff_mean=("sl_rate_diff_final_minus_dev", "mean"),
        sl_rate_abs_diff_mean=("sl_rate_abs_diff_final_vs_dev", "mean"),

        no_event_rate_diff_mean=("no_event_rate_diff_final_minus_dev", "mean"),
        no_event_rate_abs_diff_mean=("no_event_rate_abs_diff_final_vs_dev", "mea

        tp_share_diff_mean=("tp_share_over_resolved_diff_final_minus_dev", "mean
        tp_share_abs_diff_mean=("tp_share_over_resolved_abs_diff_final_vs_dev",

        split_shift_score_mean=("split_shift_score", "mean"),
        split_shift_score_max=("split_shift_score", "max"),

```

```

    )
    .sort_values(["percentile", "horizon"])
    .reset_index(drop=True)
)

print("\nResumen de cambio temporal por percentil y horizonte:")
print(
    df_bar_split_change_by_target_base.to_string(
        index=False,
        formatters={
            "tp_rate_diff_mean": "{:.2%}".format,
            "tp_rate_abs_diff_mean": "{:.2%}".format,
            "sl_rate_diff_mean": "{:.2%}".format,
            "sl_rate_abs_diff_mean": "{:.2%}".format,
            "no_event_rate_diff_mean": "{:.2%}".format,
            "no_event_rate_abs_diff_mean": "{:.2%}".format,
            "tp_share_diff_mean": "{:.2%}".format,
            "tp_share_abs_diff_mean": "{:.2%}".format,
            "split_shift_score_mean": "{:.3f}".format,
            "split_shift_score_max": "{:.3f}".format,
        }
    )
)

# =====
# 4. Estabilidad anual por target
# =====

def value_range(series):
    return series.max() - series.min()

df_bar_yearly_stability_summary = (
    df_bar_stability_year
    .groupby(id_cols, as_index=False)
    .agg(
        n_years=("period", "nunique"),

        tp_rate_mean=("tp_rate", "mean"),
        tp_rate_std=("tp_rate", "std"),
        tp_rate_min=("tp_rate", "min"),
        tp_rate_max=("tp_rate", "max"),
        tp_rate_range=("tp_rate", value_range),

        sl_rate_mean=("sl_rate", "mean"),
        sl_rate_std=("sl_rate", "std"),
        sl_rate_min=("sl_rate", "min"),
        sl_rate_max=("sl_rate", "max"),
        sl_rate_range=("sl_rate", value_range),

        no_event_rate_mean=("no_event_rate", "mean"),
        no_event_rate_std=("no_event_rate", "std"),
        no_event_rate_min=("no_event_rate", "min"),
        no_event_rate_max=("no_event_rate", "max"),
        no_event_rate_range=("no_event_rate", value_range),

        tp_share_mean=("tp_share_over_resolved", "mean"),
        tp_share_std=("tp_share_over_resolved", "std"),
        tp_share_min=("tp_share_over_resolved", "min"),
        tp_share_max=("tp_share_over_resolved", "max"),
        tp_share_range=("tp_share_over_resolved", value_range),
    )
)

```



```

    expectancy_mean=("expectancy_units", "mean"),
    expectancy_std=("expectancy_units", "std"),
    expectancy_min=("expectancy_units", "min"),
    expectancy_max=("expectancy_units", "max"),
    expectancy_range=("expectancy_units", value_range),

    event_step_mean_mean=("event_step_mean", "mean"),
    event_step_mean_std=("event_step_mean", "std"),
    event_step_mean_min=("event_step_mean", "min"),
    event_step_mean_max=("event_step_mean", "max"),
    event_step_mean_range=("event_step_mean", value_range),
)
.reset_index(drop=True)
)

year_score_cols = [
    "tp_rate_range",
    "sl_rate_range",
    "no_event_rate_range",
    "tp_share_range",
]

df_bar_yearly_stability_summary["yearly_instability_score"] = (
    df_bar_yearly_stability_summary[year_score_cols]
    .sum(axis=1)
)

df_bar_yearly_stability_summary = (
    df_bar_yearly_stability_summary
    .sort_values(
        ["yearly_instability_score", "percentile", "horizon", "sl_mult", "tp_mult"],
        ascending=[False, True, True, True, True]
    )
    .reset_index(drop=True)
)

print("\nTop 20 targets con mayor variación anual:")
print(
    df_bar_yearly_stability_summary[
        [
            "bar_col",
            "target_base",
            "tp_sl_config",
            "n_years",
            "tp_rate_mean",
            "tp_rate_range",
            "sl_rate_mean",
            "sl_rate_range",
            "no_event_rate_mean",
            "no_event_rate_range",
            "tp_share_mean",
            "tp_share_range",
            "yearly_instability_score",
        ]
    ]
    .head(20)
    .to_string(
        index=False,
        formatters={

```

```

        "tp_rate_mean": "{:.2%}".format,
        "tp_rate_range": "{:.2%}".format,
        "sl_rate_mean": "{:.2%}".format,
        "sl_rate_range": "{:.2%}".format,
        "no_event_rate_mean": "{:.2%}".format,
        "no_event_rate_range": "{:.2%}".format,
        "tp_share_mean": "{:.2%}".format,
        "tp_share_range": "{:.2%}".format,
        "yearly_instability_score": "{:.3f}".format,
    }
)
)

print("\nTop 20 targets con menor variación anual:")
print(
    df_bar_yearly_stability_summary[
        [
            "bar_col",
            "target_base",
            "tp_sl_config",
            "n_years",
            "tp_rate_mean",
            "tp_rate_range",
            "sl_rate_mean",
            "sl_rate_range",
            "no_event_rate_mean",
            "no_event_rate_range",
            "tp_share_mean",
            "tp_share_range",
            "yearly_instability_score",
        ]
    ]
    .tail(20)
    .sort_values("yearly_instability_score")
    .to_string(
        index=False,
        formatters={
            "tp_rate_mean": "{:.2%}".format,
            "tp_rate_range": "{:.2%}".format,
            "sl_rate_mean": "{:.2%}".format,
            "sl_rate_range": "{:.2%}".format,
            "no_event_rate_mean": "{:.2%}".format,
            "no_event_rate_range": "{:.2%}".format,
            "tp_share_mean": "{:.2%}".format,
            "tp_share_range": "{:.2%}".format,
            "yearly_instability_score": "{:.3f}".format,
        }
    )
)

# =====
# 5. Estabilidad anual agregada por configuración TP/SL
# =====

df_bar_yearly_stability_by_config = (
    df_bar_yearly_stability_summary
    .groupby(["tp_mult", "sl_mult", "tp_sl_config"], as_index=False)
    .agg(
        n_targets=("bar_col", "count"),

```

```

    tp_rate_range_mean=("tp_rate_range", "mean"),
    sl_rate_range_mean=("sl_rate_range", "mean"),
    no_event_rate_range_mean=("no_event_rate_range", "mean"),
    tp_share_range_mean=("tp_share_range", "mean"),

    expectancy_range_mean=("expectancy_range", "mean"),
    event_step_range_mean=("event_step_mean_range", "mean"),

    yearly_instability_score_mean=("yearly_instability_score", "mean"),
    yearly_instability_score_max=("yearly_instability_score", "max"),
)
.sort_values(["sl_mult", "tp_mult"])
.reset_index(drop=True)
)

print("\nEstabilidad anual agregada por configuración TP/SL:")
print(
    df_bar_yearly_stability_by_config.to_string(
        index=False,
        formatters={
            "tp_rate_range_mean": "{:.2%}".format,
            "sl_rate_range_mean": "{:.2%}".format,
            "no_event_rate_range_mean": "{:.2%}".format,
            "tp_share_range_mean": "{:.2%}".format,
            "expectancy_range_mean": "{:.3f}".format,
            "event_step_range_mean": "{:.2f}".format,
            "yearly_instability_score_mean": "{:.3f}".format,
            "yearly_instability_score_max": "{:.3f}".format,
        }
    )
)

# =====
# 6. Estabilidad trimestral temporal por target
# =====
# Ahora quarter significa year_quarter:
# 2020Q1, 2020Q2, ..., 2026Q2
# =====

df_bar_quarterly_stability_summary = (
    df_bar_stability_quarter
    .groupby(id_cols, as_index=False)
    .agg(
        n_quarters=("period", "nunique"),

        tp_rate_mean=("tp_rate", "mean"),
        tp_rate_std=("tp_rate", "std"),
        tp_rate_min=("tp_rate", "min"),
        tp_rate_max=("tp_rate", "max"),
        tp_rate_range=("tp_rate", value_range),

        sl_rate_mean=("sl_rate", "mean"),
        sl_rate_std=("sl_rate", "std"),
        sl_rate_min=("sl_rate", "min"),
        sl_rate_max=("sl_rate", "max"),
        sl_rate_range=("sl_rate", value_range),

        no_event_rate_mean=("no_event_rate", "mean"),
        no_event_rate_std=("no_event_rate", "std"),
        no_event_rate_min=("no_event_rate", "min"),

```

```

no_event_rate_max=("no_event_rate", "max"),
no_event_rate_range=("no_event_rate", value_range),

tp_share_mean=("tp_share_over_resolved", "mean"),
tp_share_std=("tp_share_over_resolved", "std"),
tp_share_min=("tp_share_over_resolved", "min"),
tp_share_max=("tp_share_over_resolved", "max"),
tp_share_range=("tp_share_over_resolved", value_range),

expectancy_mean=("expectancy_units", "mean"),
expectancy_std=("expectancy_units", "std"),
expectancy_min=("expectancy_units", "min"),
expectancy_max=("expectancy_units", "max"),
expectancy_range=("expectancy_units", value_range),

event_step_mean_mean=("event_step_mean", "mean"),
event_step_mean_std=("event_step_mean", "std"),
event_step_mean_min=("event_step_mean", "min"),
event_step_mean_max=("event_step_mean", "max"),
event_step_mean_range=("event_step_mean", value_range),
)
.reset_index(drop=True)
)

df_bar_quarterly_stability_summary["quarterly_instability_score"] = (
    df_bar_quarterly_stability_summary[
        [
            "tp_rate_range",
            "sl_rate_range",
            "no_event_rate_range",
            "tp_share_range",
        ]
    ]
    .sum(axis=1)
)

df_bar_quarterly_stability_summary = (
    df_bar_quarterly_stability_summary
    .sort_values(
        ["quarterly_instability_score", "percentile", "horizon", "sl_mult", "tp_
        ascending=[False, True, True, True, True]
    )
    .reset_index(drop=True)
)

print("\nTop 20 targets con mayor variación trimestral temporal:")
print(
    df_bar_quarterly_stability_summary[
        [
            "bar_col",
            "target_base",
            "tp_sl_config",
            "n_quarters",
            "tp_rate_mean",
            "tp_rate_range",
            "sl_rate_mean",
            "sl_rate_range",
            "no_event_rate_mean",
            "no_event_rate_range",
            "tp_share_mean",

```

```

        "tp_share_range",
        "quarterly_instability_score",
    ]
]
.head(20)
.to_string(
    index=False,
    formatters={
        "tp_rate_mean": "{:.2%}".format,
        "tp_rate_range": "{:.2%}".format,
        "sl_rate_mean": "{:.2%}".format,
        "sl_rate_range": "{:.2%}".format,
        "no_event_rate_mean": "{:.2%}".format,
        "no_event_rate_range": "{:.2%}".format,
        "tp_share_mean": "{:.2%}".format,
        "tp_share_range": "{:.2%}".format,
        "quarterly_instability_score": "{:.3f}".format,
    }
)
)

print("\nTop 20 targets con menor variación trimestral temporal:")
print(
    df_bar_quarterly_stability_summary[
        [
            "bar_col",
            "target_base",
            "tp_sl_config",
            "n_quarters",
            "tp_rate_mean",
            "tp_rate_range",
            "sl_rate_mean",
            "sl_rate_range",
            "no_event_rate_mean",
            "no_event_rate_range",
            "tp_share_mean",
            "tp_share_range",
            "quarterly_instability_score",
        ]
    ]
    .tail(20)
    .sort_values("quarterly_instability_score")
    .to_string(
        index=False,
        formatters={
            "tp_rate_mean": "{:.2%}".format,
            "tp_rate_range": "{:.2%}".format,
            "sl_rate_mean": "{:.2%}".format,
            "sl_rate_range": "{:.2%}".format,
            "no_event_rate_mean": "{:.2%}".format,
            "no_event_rate_range": "{:.2%}".format,
            "tp_share_mean": "{:.2%}".format,
            "tp_share_range": "{:.2%}".format,
            "quarterly_instability_score": "{:.3f}".format,
        }
    )
)

print("\nCelda 2 finalizada correctamente.")

```

## Tablas agregadas de estabilidad temporal - BAR

=====

Filas por dataset\_split: 126

Filas por año: 441

Filas por trimestre: 1638

Primeros trimestres temporales detectados:

```

period_type period
quarter 2020Q1
quarter 2020Q2
quarter 2020Q3
quarter 2020Q4
quarter 2021Q1
quarter 2021Q2
quarter 2021Q3
quarter 2021Q4
quarter 2022Q1
quarter 2022Q2
quarter 2022Q3
quarter 2022Q4

```

Comparación development vs final\_test - Top 20 con mayor cambio:

| bar_col               | target_base             | tp_sl_config | tp_rate_development_2020_2024 | tp_rate_final_test_2025_2026 | tp_rate_diff_final_minus_dev | sl_rate_development_2020_2024 | sl_rate_final_test_2025_2026 | sl_rate_diff_final_minus_dev | no_event_rate_development_2020_2024 | no_event_rate_final_test_2025_2026 | no_event_rate_diff_final_minus_dev | tp_share_over_resolved_development_2020_2024 | tp_share_over_resolved_final_test_2025_2026 | tp_share_over_resolved_diff_final_minus_dev | split_shift_score |
|-----------------------|-------------------------|--------------|-------------------------------|------------------------------|------------------------------|-------------------------------|------------------------------|------------------------------|-------------------------------------|------------------------------------|------------------------------------|----------------------------------------------|---------------------------------------------|---------------------------------------------|-------------------|
| bar_p60_h60_tp20_sl05 | p60_h60 TP 2.0 / SL 0.5 |              |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 18.96%            |
| 28.58%                | 9.62%                   |              |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 16.95%            |
| 28.92%                | 11.97%                  |              |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 64.08%            |
| 42.49%                | -21.59%                 |              |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 5                 |
| 2.80%                 |                         |              |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 49.70%            |
| -3.10%                | 0.463                   |              |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             |                   |
| bar_p60_h90_tp20_sl05 | p60_h90 TP 2.0 / SL 0.5 |              |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 19.33%            |
| 28.37%                | 9.04%                   |              |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 18.07%            |
| 30.23%                | 12.16%                  |              |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 62.60%            |
| 41.40%                | -21.20%                 |              |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 5                 |
| 1.67%                 |                         |              |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 48.41%            |
| -3.26%                | 0.457                   |              |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             |                   |
| bar_p40_h90_tp20_sl10 | p40_h90 TP 2.0 / SL 1.0 |              |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 31.10%            |
| 43.30%                | 12.20%                  |              |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 7.03%             |
| 14.36%                | 7.33%                   |              |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 61.87%            |
| 42.34%                | -19.53%                 |              |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 8                 |
| 1.56%                 |                         |              |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 75.09%            |
| -6.47%                | 0.455                   |              |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             |                   |
| bar_p50_h60_tp20_sl10 | p50_h60 TP 2.0 / SL 1.0 |              |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 26.10%            |
| 39.19%                | 13.10%                  |              |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 4.66%             |
| 10.96%                | 6.30%                   |              |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 69.25%            |
| 49.85%                | -19.40%                 |              |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 8                 |
| 4.85%                 |                         |              |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 78.15%            |
| -6.70%                | 0.455                   |              |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             |                   |
| bar_p40_h90_tp25_sl10 | p40_h90 TP 2.5 / SL 1.0 |              |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 18.23%            |
| 30.26%                | 12.03%                  |              |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 7.44%             |
| 15.54%                | 8.10%                   |              |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 74.32%            |
| 54.19%                | -20.13%                 |              |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 7                 |
| 1.01%                 |                         |              |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 66.06%            |
| -4.94%                | 0.452                   |              |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             |                   |
| bar_p50_h90_tp20_sl10 | p50_h90 TP 2.0 / SL 1.0 |              |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 26.35%            |
| 39.60%                | 13.25%                  |              |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 5.15%             |

|                       |                         |        |   |
|-----------------------|-------------------------|--------|---|
| 11.42%                | 6.27%                   | 68.50% |   |
| 48.98%                | -19.53%                 |        | 8 |
| 3.66%                 | 77.62%                  |        |   |
| -6.04%                | 0.451                   |        |   |
| bar_p40_h60_tp25_sl10 | p40_h60 TP 2.5 / SL 1.0 | 18.07% |   |
| 30.21%                | 12.14%                  | 6.92%  |   |
| 14.75%                | 7.83%                   | 75.02% |   |
| 55.04%                | -19.98%                 |        | 7 |
| 2.31%                 | 67.20%                  |        |   |
| -5.12%                | 0.451                   |        |   |
| bar_p50_h90_tp20_sl05 | p50_h90 TP 2.0 / SL 0.5 | 22.26% |   |
| 30.97%                | 8.70%                   | 22.07% |   |
| 34.38%                | 12.31%                  | 55.67% |   |
| 34.66%                | -21.01%                 |        | 5 |
| 0.22%                 | 47.39%                  |        |   |
| -2.83%                | 0.449                   |        |   |
| bar_p40_h60_tp20_sl10 | p40_h60 TP 2.0 / SL 1.0 | 31.18% |   |
| 42.99%                | 11.80%                  | 6.54%  |   |
| 13.71%                | 7.17%                   | 62.27% |   |
| 43.31%                | -18.97%                 |        | 8 |
| 2.66%                 | 75.82%                  |        |   |
| -6.84%                | 0.448                   |        |   |
| bar_p50_h60_tp20_sl05 | p50_h60 TP 2.0 / SL 0.5 | 22.27% |   |
| 31.16%                | 8.89%                   | 20.93% |   |
| 32.71%                | 11.78%                  | 56.80% |   |
| 36.13%                | -20.67%                 |        | 5 |
| 1.55%                 | 48.79%                  |        |   |
| -2.76%                | 0.441                   |        |   |
| bar_p60_h60_tp15_sl05 | p60_h60 TP 1.5 / SL 0.5 | 39.59% |   |
| 46.32%                | 6.72%                   | 16.09% |   |
| 27.00%                | 10.91%                  | 44.32% |   |
| 26.68%                | -17.64%                 |        | 7 |
| 1.11%                 | 63.17%                  |        |   |
| -7.94%                | 0.432                   |        |   |
| bar_p50_h60_tp25_sl10 | p50_h60 TP 2.5 / SL 1.0 | 14.28% |   |
| 26.05%                | 11.76%                  | 4.91%  |   |
| 11.76%                | 6.84%                   | 80.80% |   |
| 62.20%                | -18.60%                 |        | 7 |
| 4.41%                 | 68.90%                  |        |   |
| -5.50%                | 0.427                   |        |   |
| bar_p60_h90_tp15_sl05 | p60_h90 TP 1.5 / SL 0.5 | 39.74% |   |
| 46.39%                | 6.65%                   | 17.18% |   |
| 28.08%                | 10.90%                  | 43.08% |   |
| 25.53%                | -17.54%                 |        | 6 |
| 9.82%                 | 62.29%                  |        |   |
| -7.52%                | 0.426                   |        |   |
| bar_p40_h30_tp25_sl10 | p40_h30 TP 2.5 / SL 1.0 | 19.75% |   |
| 31.40%                | 11.66%                  | 7.18%  |   |
| 14.33%                | 7.15%                   | 73.08% |   |
| 54.27%                | -18.81%                 |        | 7 |
| 3.34%                 | 68.67%                  |        |   |
| -4.67%                | 0.423                   |        |   |
| bar_p60_h60_tp20_sl10 | p60_h60 TP 2.0 / SL 1.0 | 21.84% |   |
| 35.07%                | 13.23%                  | 3.37%  |   |
| 8.26%                 | 4.88%                   | 74.79% |   |
| 56.68%                | -18.11%                 |        | 8 |
| 6.62%                 | 80.94%                  |        |   |
| -5.68%                | 0.419                   |        |   |
| bar_p40_h30_tp20_sl10 | p40_h30 TP 2.0 / SL 1.0 | 32.82% |   |
| 44.31%                | 11.48%                  | 6.76%  |   |

|                       |       |                         |        |        |
|-----------------------|-------|-------------------------|--------|--------|
| 13.26%                |       | 6.49%                   | 60.41% |        |
| 42.44%                |       | -17.98%                 |        | 8      |
| 2.92%                 |       |                         | 76.97% |        |
| -5.95%                | 0.419 |                         |        |        |
| bar_p60_h30_tp20_sl05 |       | p60_h30 TP 2.0 / SL 0.5 |        | 20.58% |
| 29.82%                |       | 9.23%                   | 16.78% |        |
| 27.04%                |       | 10.25%                  | 62.63% |        |
| 43.15%                |       | -19.49%                 |        | 5      |
| 5.09%                 |       |                         | 52.44% |        |
| -2.64%                | 0.416 |                         |        |        |
| bar_p50_h30_tp20_sl10 |       | p50_h30 TP 2.0 / SL 1.0 |        | 27.93% |
| 40.12%                |       | 12.19%                  | 5.02%  |        |
| 10.71%                |       | 5.69%                   | 67.04% |        |
| 49.16%                |       | -17.88%                 |        | 8      |
| 4.76%                 |       |                         | 78.92% |        |
| -5.83%                | 0.416 |                         |        |        |
| bar_p50_h90_tp25_sl10 |       | p50_h90 TP 2.5 / SL 1.0 |        | 14.66% |
| 26.21%                |       | 11.55%                  | 5.43%  |        |
| 12.15%                |       | 6.73%                   | 79.91% |        |
| 61.63%                |       | -18.28%                 |        | 7      |
| 2.98%                 |       |                         | 68.32% |        |
| -4.66%                | 0.412 |                         |        |        |
| bar_p60_h90_tp20_sl10 |       | p60_h90 TP 2.0 / SL 1.0 |        | 22.27% |
| 35.38%                |       | 13.11%                  | 3.78%  |        |
| 8.64%                 |       | 4.86%                   | 73.95% |        |
| 55.98%                |       | -17.96%                 |        | 8      |
| 5.49%                 |       |                         | 80.38% |        |
| -5.11%                | 0.410 |                         |        |        |

Comparación development vs final\_test - Top 20 más estables:

|                       |             |                         |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             |                   |
|-----------------------|-------------|-------------------------|-------------------------------|------------------------------|------------------------------|-------------------------------|------------------------------|------------------------------|-------------------------------------|------------------------------------|------------------------------------|----------------------------------------------|---------------------------------------------|---------------------------------------------|-------------------|
| bar_col               | target_base | tp_sl_config            | tp_rate_development_2020_2024 | tp_rate_final_test_2025_2026 | tp_rate_diff_final_minus_dev | sl_rate_development_2020_2024 | sl_rate_final_test_2025_2026 | sl_rate_diff_final_minus_dev | no_event_rate_development_2020_2024 | no_event_rate_final_test_2025_2026 | no_event_rate_diff_final_minus_dev | tp_share_over_resolved_development_2020_2024 | tp_share_over_resolved_final_test_2025_2026 | tp_share_over_resolved_diff_final_minus_dev | split_shift_score |
| bar_p60_h90_tp10_sl10 |             | p60_h90 TP 1.0 / SL 1.0 |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 97.33%            |
| 93.83%                |             | -3.50%                  |                               | 2.67%                        |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             |                   |
| 6.17%                 |             | 3.50%                   |                               | 0.00%                        |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             |                   |
| 0.00%                 |             | 0.00%                   |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 9                 |
| 7.33%                 |             |                         | 93.83%                        |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             |                   |
| -3.50%                | 0.105       |                         |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             |                   |
| bar_p60_h60_tp10_sl10 |             | p60_h60 TP 1.0 / SL 1.0 |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 97.60%            |
| 94.05%                |             | -3.55%                  |                               | 2.40%                        |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             |                   |
| 5.95%                 |             | 3.55%                   |                               | 0.00%                        |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             |                   |
| 0.00%                 |             | 0.00%                   |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 9                 |
| 7.60%                 |             |                         | 94.05%                        |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             |                   |
| -3.55%                | 0.107       |                         |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             |                   |
| bar_p60_h30_tp10_sl10 |             | p60_h30 TP 1.0 / SL 1.0 |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 97.39%            |
| 93.82%                |             | -3.56%                  |                               | 2.61%                        |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             |                   |
| 6.18%                 |             | 3.56%                   |                               | 0.00%                        |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             |                   |
| 0.00%                 |             | 0.00%                   |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 9                 |
| 7.39%                 |             |                         | 93.82%                        |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             |                   |
| -3.56%                | 0.107       |                         |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             |                   |
| bar_p50_h30_tp10_sl10 |             | p50_h30 TP 1.0 / SL 1.0 |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 96.38%            |
| 92.04%                |             | -4.34%                  |                               | 3.62%                        |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             |                   |
| 7.96%                 |             | 4.34%                   |                               | 0.00%                        |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             |                   |
| 0.00%                 |             | 0.00%                   |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             | 9                 |
| 6.38%                 |             |                         | 92.04%                        |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             |                   |
| -4.34%                | 0.130       |                         |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             |                   |



|                       |                         |        |        |
|-----------------------|-------------------------|--------|--------|
| bar_p50_h90_tp10_sl10 | p50_h90 TP 1.0 / SL 1.0 |        | 96.31% |
| 91.68%                | -4.63%                  | 3.69%  |        |
| 8.32%                 | 4.63%                   | 0.00%  |        |
| 0.00%                 | 0.00%                   |        | 9      |
| 6.31%                 |                         | 91.68% |        |
| -4.63%                | 0.139                   |        |        |
| bar_p50_h60_tp10_sl10 | p50_h60 TP 1.0 / SL 1.0 |        | 96.67% |
| 91.99%                | -4.67%                  | 3.33%  |        |
| 8.01%                 | 4.67%                   | 0.00%  |        |
| 0.00%                 | 0.00%                   |        | 9      |
| 6.67%                 |                         | 91.99% |        |
| -4.67%                | 0.140                   |        |        |
| bar_p40_h30_tp10_sl10 | p40_h30 TP 1.0 / SL 1.0 |        | 95.09% |
| 89.97%                | -5.12%                  | 4.91%  |        |
| 10.03%                | 5.12%                   | 0.00%  |        |
| 0.00%                 | 0.00%                   |        | 9      |
| 5.09%                 |                         | 89.97% |        |
| -5.12%                | 0.154                   |        |        |
| bar_p40_h60_tp10_sl10 | p40_h60 TP 1.0 / SL 1.0 |        | 95.34% |
| 89.90%                | -5.44%                  | 4.66%  |        |
| 10.10%                | 5.44%                   | 0.00%  |        |
| 0.00%                 | 0.00%                   |        | 9      |
| 5.34%                 |                         | 89.90% |        |
| -5.44%                | 0.163                   |        |        |
| bar_p40_h90_tp10_sl10 | p40_h90 TP 1.0 / SL 1.0 |        | 94.92% |
| 89.40%                | -5.52%                  | 5.08%  |        |
| 10.60%                | 5.52%                   | 0.00%  |        |
| 0.00%                 | 0.00%                   |        | 9      |
| 4.92%                 |                         | 89.40% |        |
| -5.52%                | 0.166                   |        |        |
| bar_p40_h30_tp10_sl05 | p40_h30 TP 1.0 / SL 0.5 |        | 80.81% |
| 73.31%                | -7.50%                  | 19.19% |        |
| 26.69%                | 7.50%                   | 0.00%  |        |
| 0.00%                 | 0.00%                   |        | 8      |
| 0.81%                 |                         | 73.31% |        |
| -7.50%                | 0.225                   |        |        |
| bar_p50_h30_tp10_sl05 | p50_h30 TP 1.0 / SL 0.5 |        | 83.91% |
| 76.22%                | -7.69%                  | 16.09% |        |
| 23.78%                | 7.69%                   | 0.00%  |        |
| 0.00%                 | 0.00%                   |        | 8      |
| 3.91%                 |                         | 76.22% |        |
| -7.69%                | 0.231                   |        |        |
| bar_p60_h30_tp10_sl05 | p60_h30 TP 1.0 / SL 0.5 |        | 86.92% |
| 79.12%                | -7.79%                  | 13.08% |        |
| 20.88%                | 7.79%                   | 0.00%  |        |
| 0.00%                 | 0.00%                   |        | 8      |
| 6.92%                 |                         | 79.12% |        |
| -7.79%                | 0.234                   |        |        |
| bar_p40_h60_tp10_sl05 | p40_h60 TP 1.0 / SL 0.5 |        | 80.13% |
| 71.23%                | -8.90%                  | 19.87% |        |
| 28.77%                | 8.90%                   | 0.00%  |        |
| 0.00%                 | 0.00%                   |        | 8      |
| 0.13%                 |                         | 71.23% |        |
| -8.90%                | 0.267                   |        |        |
| bar_p50_h60_tp10_sl05 | p50_h60 TP 1.0 / SL 0.5 |        | 83.68% |
| 74.57%                | -9.11%                  | 16.32% |        |
| 25.43%                | 9.11%                   | 0.00%  |        |
| 0.00%                 | 0.00%                   |        | 8      |
| 3.68%                 |                         | 74.57% |        |
| -9.11%                | 0.273                   |        |        |

|                       |                         |        |
|-----------------------|-------------------------|--------|
| bar_p60_h60_tp10_sl05 | p60_h60 TP 1.0 / SL 0.5 | 86.72% |
| 77.53%                | -9.19%                  | 13.28% |
| 22.47%                | 9.19%                   | 0.00%  |
| 0.00%                 | 0.00%                   | 8      |
| 6.72%                 | 77.53%                  |        |
| -9.19%                | 0.276                   |        |
| bar_p60_h90_tp10_sl05 | p60_h90 TP 1.0 / SL 0.5 | 85.74% |
| 76.53%                | -9.21%                  | 14.26% |
| 23.47%                | 9.21%                   | 0.00%  |
| 0.00%                 | 0.00%                   | 8      |
| 5.74%                 | 76.53%                  |        |
| -9.21%                | 0.276                   |        |
| bar_p40_h90_tp10_sl05 | p40_h90 TP 1.0 / SL 0.5 | 79.41% |
| 70.00%                | -9.41%                  | 20.59% |
| 30.00%                | 9.41%                   | 0.00%  |
| 0.00%                 | 0.00%                   | 7      |
| 9.41%                 | 70.00%                  |        |
| -9.41%                | 0.282                   |        |
| bar_p50_h90_tp10_sl05 | p50_h90 TP 1.0 / SL 0.5 | 82.68% |
| 73.19%                | -9.50%                  | 17.32% |
| 26.81%                | 9.50%                   | 0.00%  |
| 0.00%                 | 0.00%                   | 8      |
| 2.68%                 | 73.19%                  |        |
| -9.50%                | 0.285                   |        |
| bar_p40_h30_tp15_sl05 | p40_h30 TP 1.5 / SL 0.5 | 47.11% |
| 51.32%                | 4.21%                   | 23.00% |
| 31.45%                | 8.44%                   | 29.88% |
| 17.23%                | -12.65%                 | 6      |
| 7.19%                 | 62.01%                  |        |
| -5.19%                | 0.305                   |        |
| bar_p40_h60_tp15_sl05 | p40_h60 TP 1.5 / SL 0.5 | 45.72% |
| 48.69%                | 2.97%                   | 23.99% |
| 33.90%                | 9.91%                   | 30.28% |
| 17.40%                | -12.88%                 | 6      |
| 5.58%                 | 58.95%                  |        |
| -6.63%                | 0.324                   |        |

## Resumen de cambio temporal por configuración TP/SL:

| tp_mult | sl_mult | tp_sl_config    | n_targets | tp_rate_diff_mean | tp_rate_abs_diff_m | ean sl_rate_diff_mean | sl_rate_abs_diff_mean | no_event_rate_diff_mean | no_event_rate | _abs_diff_mean | tp_share_diff_mean | tp_share_abs_diff_mean | split_shift_score_mean | s |
|---------|---------|-----------------|-----------|-------------------|--------------------|-----------------------|-----------------------|-------------------------|---------------|----------------|--------------------|------------------------|------------------------|---|
| 1.0     | 0.5     | TP 1.0 / SL 0.5 | 9         | -8.70%            | 8.70%              | 0.00%                 | -8.70%                | 8.70%                   | 0.261         | 0.285          |                    |                        |                        |   |
| 1.5     | 0.5     | TP 1.5 / SL 0.5 | 9         | 4.96%             | 9.99%              | -14.95%               | -6.79%                | 6.79%                   | 0.367         | 0.432          |                    |                        |                        |   |
| 2.0     | 0.5     | TP 2.0 / SL 0.5 | 9         | 8.54%             | 11.22%             | -19.76%               | -2.71%                | 2.71%                   | 0.422         | 0.463          |                    |                        |                        |   |
| 1.0     | 1.0     | TP 1.0 / SL 1.0 | 9         | -4.48%            | 4.48%              | 0.00%                 | -4.48%                | 4.48%                   | 0.134         | 0.166          |                    |                        |                        |   |
| 1.5     | 1.0     | TP 1.5 / SL 1.0 | 9         | 9.68%             | 5.20%              | -14.88%               |                       |                         |               |                |                    |                        |                        |   |

|        |                     |       |         |     |
|--------|---------------------|-------|---------|-----|
| 14.88% | -5.72%              | 5.72% | 0.355   |     |
| 0.374  |                     |       |         |     |
| 2.0    | 1.0 TP 2.0 / SL 1.0 | 9     | 12.52%  | 12. |
| 52%    | 5.98%               | 5.98% | -18.50% |     |
| 18.50% | -6.03%              | 6.03% | 0.430   |     |
| 0.455  |                     |       |         |     |
| 2.5    | 1.0 TP 2.5 / SL 1.0 | 9     | 11.24%  | 11. |
| 24%    | 6.50%               | 6.50% | -17.74% |     |
| 17.74% | -4.85%              | 4.85% | 0.403   |     |
| 0.452  |                     |       |         |     |

Resumen de cambio temporal por percentil y horizonte:

percentile horizon target\_base n\_configs tp\_rate\_diff\_mean tp\_rate\_abs\_diff\_mean sl\_rate\_diff\_mean sl\_rate\_abs\_diff\_mean no\_event\_rate\_diff\_mean no\_event\_rate\_abs\_diff\_mean tp\_share\_diff\_mean tp\_share\_abs\_diff\_mean split\_shift\_score\_mean split\_shift\_score\_max

|        |        |         |         |       |     |
|--------|--------|---------|---------|-------|-----|
| 40     | 30     | p40_h30 | 7       | 4.35% | 7.9 |
| 5%     | 7.15%  | 7.15%   | -11.50% |       |     |
| 11.50% | -5.18% | 5.18%   | 0.318   |       |     |
| 0.423  |        |         |         |       |     |
| 40     | 60     | p40_h60 | 7       | 3.93% | 8.0 |
| 3%     | 8.10%  | 8.10%   | -12.03% |       |     |
| 12.03% | -6.06% | 6.06%   | 0.342   |       |     |
| 0.451  |        |         |         |       |     |
| 40     | 90     | p40_h90 | 7       | 3.90% | 8.1 |
| 6%     | 8.39%  | 8.39%   | -12.29% |       |     |
| 12.29% | -6.09% | 6.09%   | 0.349   |       |     |
| 0.455  |        |         |         |       |     |
| 50     | 30     | p50_h30 | 7       | 4.96% | 8.4 |
| 0%     | 6.81%  | 6.81%   | -11.77% |       |     |
| 11.77% | -5.15% | 5.15%   | 0.321   |       |     |
| 0.416  |        |         |         |       |     |
| 50     | 60     | p50_h60 | 7       | 4.92% | 8.8 |
| 6%     | 7.82%  | 7.82%   | -12.74% |       |     |
| 12.74% | -6.05% | 6.05%   | 0.355   |       |     |
| 0.455  |        |         |         |       |     |
| 50     | 90     | p50_h90 | 7       | 4.92% | 8.9 |
| 6%     | 7.95%  | 7.95%   | -12.87% |       |     |
| 12.87% | -5.82% | 5.82%   | 0.356   |       |     |
| 0.451  |        |         |         |       |     |
| 60     | 30     | p60_h30 | 7       | 5.40% | 8.6 |
| 4%     | 6.42%  | 6.42%   | -11.81% |       |     |
| 11.81% | -5.13% | 5.13%   | 0.320   |       |     |
| 0.416  |        |         |         |       |     |
| 60     | 60     | p60_h60 | 7       | 5.56% | 9.2 |
| 1%     | 7.14%  | 7.14%   | -12.71% |       |     |
| 12.71% | -5.66% | 5.66%   | 0.347   |       |     |
| 0.463  |        |         |         |       |     |
| 60     | 90     | p60_h90 | 7       | 5.46% | 9.1 |
| 0%     | 7.15%  | 7.15%   | -12.62% |       |     |
| 12.62% | -5.37% | 5.37%   | 0.342   |       |     |
| 0.457  |        |         |         |       |     |

Top 20 targets con mayor variación anual:

|                       |                         |              |         |              |               |              |               |                    |                     |               |                |                          |
|-----------------------|-------------------------|--------------|---------|--------------|---------------|--------------|---------------|--------------------|---------------------|---------------|----------------|--------------------------|
| bar_col               | target_base             | tp_sl_config | n_years | tp_rate_mean | tp_rate_range | sl_rate_mean | sl_rate_range | no_event_rate_mean | no_event_rate_range | tp_share_mean | tp_share_range | yearly_instability_score |
| bar_p40_h60_tp20_sl10 | p40_h60 TP 2.0 / SL 1.0 | 7            | 34.42%  | 2            |               |              |               |                    |                     |               |                |                          |
| 6.14%                 | 8.61%                   | 13.42%       | 56.97%  | 39.56%       | 8             |              |               |                    |                     |               |                |                          |
| 1.20%                 | 12.71%                  | 0.918        |         |              |               |              |               |                    |                     |               |                |                          |

|                       |                         |        |        |   |
|-----------------------|-------------------------|--------|--------|---|
| bar_p50_h60_tp20_sl05 | p50_h60 TP 2.0 / SL 0.5 | 7      | 24.57% | 1 |
| 8.04%                 | 24.05%                  | 23.94% | 51.39% | 5 |
| 1.14%                 | 6.64%                   | 0.906  |        |   |
| bar_p50_h90_tp20_sl05 | p50_h90 TP 2.0 / SL 0.5 | 7      | 24.40% | 1 |
| 5.77%                 | 25.40%                  | 25.06% | 50.20% | 4 |
| 9.64%                 | 8.42%                   | 0.901  |        |   |
| bar_p40_h90_tp20_sl10 | p40_h90 TP 2.0 / SL 1.0 | 7      | 34.36% | 2 |
| 4.96%                 | 9.15%                   | 13.43% | 56.50% | 8 |
| 0.10%                 | 12.01%                  | 0.888  |        |   |
| bar_p60_h60_tp20_sl05 | p60_h60 TP 2.0 / SL 0.5 | 7      | 21.26% | 1 |
| 6.40%                 | 20.07%                  | 23.18% | 58.67% | 5 |
| 2.33%                 | 9.61%                   | 0.888  |        |   |
| bar_p60_h60_tp15_sl05 | p60_h60 TP 1.5 / SL 0.5 | 7      | 41.08% | 1 |
| 5.51%                 | 18.86%                  | 20.93% | 40.07% | 6 |
| 9.55%                 | 15.76%                  | 0.886  |        |   |
| bar_p40_h60_tp25_sl10 | p40_h60 TP 2.5 / SL 1.0 | 7      | 21.43% | 2 |
| 3.18%                 | 9.22%                   | 14.84% | 69.35% | 7 |
| 1.31%                 | 11.45%                  | 0.875  |        |   |
| bar_p40_h90_tp20_sl05 | p40_h90 TP 2.0 / SL 0.5 | 7      | 27.55% | 1 |
| 5.51%                 | 29.52%                  | 24.41% | 42.93% | 4 |
| 8.75%                 | 6.64%                   | 0.865  |        |   |
| bar_p60_h90_tp20_sl05 | p60_h90 TP 2.0 / SL 0.5 | 7      | 21.42% | 1 |
| 4.81%                 | 21.28%                  | 23.21% | 57.30% | 5 |
| 0.99%                 | 10.33%                  | 0.864  |        |   |
| bar_p50_h60_tp20_sl10 | p50_h60 TP 2.0 / SL 1.0 | 7      | 29.55% | 2 |
| 5.94%                 | 6.44%                   | 11.04% | 64.01% | 8 |
| 3.43%                 | 12.11%                  | 0.861  |        |   |
| bar_p40_h60_tp20_sl05 | p40_h60 TP 2.0 / SL 0.5 | 7      | 27.97% | 1 |
| 6.96%                 | 28.59%                  | 23.38% | 43.43% | 4 |
| 9.88%                 | 5.10%                   | 0.858  |        |   |
| bar_p40_h30_tp20_sl10 | p40_h30 TP 2.0 / SL 1.0 | 7      | 35.99% | 2 |
| 4.36%                 | 8.65%                   | 12.54% | 55.36% | 8 |
| 1.65%                 | 11.86%                  | 0.857  |        |   |
| bar_p40_h30_tp25_sl10 | p40_h30 TP 2.5 / SL 1.0 | 7      | 23.08% | 2 |
| 3.48%                 | 9.26%                   | 13.61% | 67.66% | 7 |
| 2.54%                 | 9.80%                   | 0.840  |        |   |
| bar_p60_h90_tp15_sl05 | p60_h90 TP 1.5 / SL 0.5 | 7      | 41.12% | 1 |
| 3.17%                 | 19.98%                  | 20.83% | 38.90% | 6 |
| 8.21%                 | 15.91%                  | 0.839  |        |   |
| bar_p40_h90_tp25_sl10 | p40_h90 TP 2.5 / SL 1.0 | 7      | 21.48% | 2 |
| 1.51%                 | 9.81%                   | 14.91% | 68.71% | 6 |
| 9.90%                 | 10.99%                  | 0.838  |        |   |
| bar_p60_h30_tp20_sl05 | p60_h30 TP 2.0 / SL 0.5 | 7      | 22.93% | 1 |
| 7.46%                 | 19.45%                  | 20.36% | 57.62% | 5 |
| 4.79%                 | 7.54%                   | 0.832  |        |   |
| bar_p50_h60_tp15_sl05 | p50_h60 TP 1.5 / SL 0.5 | 7      | 43.82% | 1 |
| 2.09%                 | 22.54%                  | 21.69% | 33.64% | 6 |
| 6.85%                 | 15.26%                  | 0.828  |        |   |
| bar_p50_h30_tp20_sl05 | p50_h30 TP 2.0 / SL 0.5 | 7      | 26.34% | 1 |
| 7.84%                 | 23.25%                  | 20.54% | 50.41% | 5 |
| 3.55%                 | 5.33%                   | 0.821  |        |   |
| bar_p50_h30_tp20_sl10 | p50_h30 TP 2.0 / SL 1.0 | 7      | 31.21% | 2 |
| 4.81%                 | 6.61%                   | 10.15% | 62.17% | 8 |
| 3.58%                 | 10.64%                  | 0.806  |        |   |
| bar_p50_h90_tp15_sl05 | p50_h90 TP 1.5 / SL 0.5 | 7      | 43.41% | 1 |
| 0.81%                 | 23.75%                  | 21.84% | 32.85% | 6 |
| 5.42%                 | 15.18%                  | 0.805  |        |   |

Top 20 targets con menor variación anual:

bar\_col target\_base tp\_sl\_config n\_years tp\_rate\_mean tp\_rate\_r

| ange sl_rate_mean     | sl_rate_range            | no_event_rate_mean | no_event_rate_range | tp_share_m |
|-----------------------|--------------------------|--------------------|---------------------|------------|
| ean tp_share_range    | yearly_instability_score |                    |                     |            |
| bar_p60_h90_tp10_sl10 | p60_h90 TP 1.0 / SL 1.0  | 7                  | 96.39%              |            |
| 5.52%                 | 3.61%                    | 5.52%              | 0.00%               | 9          |
| 6.39%                 | 5.52%                    | 0.166              |                     |            |
| bar_p60_h60_tp10_sl10 | p60_h60 TP 1.0 / SL 1.0  | 7                  | 96.65%              |            |
| 5.57%                 | 3.35%                    | 5.57%              | 0.00%               | 9          |
| 6.65%                 | 5.57%                    | 0.167              |                     |            |
| bar_p60_h30_tp10_sl10 | p60_h30 TP 1.0 / SL 1.0  | 7                  | 96.42%              |            |
| 5.76%                 | 3.58%                    | 5.76%              | 0.00%               | 9          |
| 6.42%                 | 5.76%                    | 0.173              |                     |            |
| bar_p50_h30_tp10_sl10 | p50_h30 TP 1.0 / SL 1.0  | 7                  | 95.15%              |            |
| 7.74%                 | 4.85%                    | 7.74%              | 0.00%               | 9          |
| 5.15%                 | 7.74%                    | 0.232              |                     |            |
| bar_p50_h90_tp10_sl10 | p50_h90 TP 1.0 / SL 1.0  | 7                  | 95.04%              |            |
| 7.81%                 | 4.96%                    | 7.81%              | 0.00%               | 9          |
| 5.04%                 | 7.81%                    | 0.234              |                     |            |
| bar_p50_h60_tp10_sl10 | p50_h60 TP 1.0 / SL 1.0  | 7                  | 95.36%              |            |
| 7.94%                 | 4.64%                    | 7.94%              | 0.00%               | 9          |
| 5.36%                 | 7.94%                    | 0.238              |                     |            |
| bar_p40_h30_tp10_sl10 | p40_h30 TP 1.0 / SL 1.0  | 7                  | 93.59%              |            |
| 9.72%                 | 6.41%                    | 9.72%              | 0.00%               | 9          |
| 3.59%                 | 9.72%                    | 0.291              |                     |            |
| bar_p40_h90_tp10_sl10 | p40_h90 TP 1.0 / SL 1.0  | 7                  | 93.36%              |            |
| 9.94%                 | 6.64%                    | 9.94%              | 0.00%               | 9          |
| 3.36%                 | 9.94%                    | 0.298              |                     |            |
| bar_p40_h60_tp10_sl10 | p40_h60 TP 1.0 / SL 1.0  | 7                  | 93.76%              |            |
| 9.99%                 | 6.24%                    | 9.99%              | 0.00%               | 9          |
| 3.76%                 | 9.99%                    | 0.300              |                     |            |
| bar_p60_h30_tp10_sl05 | p60_h30 TP 1.0 / SL 0.5  | 7                  | 84.90%              | 1          |
| 5.47%                 | 15.10%                   | 15.47%             | 0.00%               | 8          |
| 4.90%                 | 15.47%                   | 0.464              |                     |            |
| bar_p40_h30_tp10_sl05 | p40_h30 TP 1.0 / SL 0.5  | 7                  | 78.74%              | 1          |
| 5.61%                 | 21.26%                   | 15.61%             | 0.00%               | 7          |
| 8.74%                 | 15.61%                   | 0.468              |                     |            |
| bar_p50_h30_tp10_sl05 | p50_h30 TP 1.0 / SL 0.5  | 7                  | 81.86%              | 1          |
| 5.77%                 | 18.14%                   | 15.77%             | 0.00%               | 8          |
| 1.86%                 | 15.77%                   | 0.473              |                     |            |
| bar_p60_h90_tp10_sl05 | p60_h90 TP 1.0 / SL 0.5  | 7                  | 83.43%              | 1          |
| 7.15%                 | 16.57%                   | 17.15%             | 0.00%               | 8          |
| 3.43%                 | 17.15%                   | 0.515              |                     |            |
| bar_p60_h60_tp10_sl05 | p60_h60 TP 1.0 / SL 0.5  | 7                  | 84.38%              | 1          |
| 7.85%                 | 15.62%                   | 17.85%             | 0.00%               | 8          |
| 4.38%                 | 17.85%                   | 0.535              |                     |            |
| bar_p40_h60_tp10_sl05 | p40_h60 TP 1.0 / SL 0.5  | 7                  | 77.68%              | 1          |
| 8.64%                 | 22.32%                   | 18.64%             | 0.00%               | 7          |
| 7.68%                 | 18.64%                   | 0.559              |                     |            |
| bar_p50_h60_tp10_sl05 | p50_h60 TP 1.0 / SL 0.5  | 7                  | 81.25%              | 1          |
| 8.76%                 | 18.75%                   | 18.76%             | 0.00%               | 8          |
| 1.25%                 | 18.76%                   | 0.563              |                     |            |
| bar_p50_h90_tp10_sl05 | p50_h90 TP 1.0 / SL 0.5  | 7                  | 80.19%              | 1          |
| 8.91%                 | 19.81%                   | 18.91%             | 0.00%               | 8          |
| 0.19%                 | 18.91%                   | 0.567              |                     |            |
| bar_p60_h90_tp25_sl10 | p60_h90 TP 2.5 / SL 1.0  | 7                  | 14.05%              | 1          |
| 4.42%                 | 5.43%                    | 8.36%              | 80.52%              | 7          |
| 3.29%                 | 12.15%                   | 0.577              |                     |            |
| bar_p40_h90_tp10_sl05 | p40_h90 TP 1.0 / SL 0.5  | 7                  | 76.84%              | 1          |
| 9.57%                 | 23.16%                   | 19.57%             | 0.00%               | 7          |
| 6.84%                 | 19.57%                   | 0.587              |                     |            |
| bar_p60_h60_tp25_sl10 | p60_h60 TP 2.5 / SL 1.0  | 7                  | 13.95%              | 1          |

|       |        |       |        |        |   |
|-------|--------|-------|--------|--------|---|
| 5.35% | 5.01%  | 8.62% | 81.04% | 23.97% | 7 |
| 5.12% | 12.90% |       | 0.608  |        |   |

Estabilidad anual agregada por configuración TP/SL:

| tp_mult | sl_mult | tp_sl_config    | n_targets | tp_rate_range_mean | sl_rate_range_mean | no_event_rate_range_mean | tp_share_range_mean | expectancy_range_mean | event_step_range_mean | yearly_instability_score_mean | yearly_instability_score_max |
|---------|---------|-----------------|-----------|--------------------|--------------------|--------------------------|---------------------|-----------------------|-----------------------|-------------------------------|------------------------------|
| 1.0     | 0.5     | TP 1.0 / SL 0.5 | 9         | 17.53%             | 0.263              | 17.5                     |                     |                       |                       |                               |                              |
| 3%      |         |                 |           | 0.00%              |                    |                          |                     |                       |                       |                               |                              |
| 13.25   |         |                 |           | 0.526              | 0.587              |                          |                     |                       |                       |                               |                              |
| 1.5     | 0.5     | TP 1.5 / SL 0.5 | 9         | 11.64%             | 0.086              | 20.2                     |                     |                       |                       |                               |                              |
| 8%      |         |                 |           | 31.88%             | 14.18%             |                          |                     |                       |                       |                               |                              |
| 11.59   |         |                 |           | 0.780              | 0.886              |                          |                     |                       |                       |                               |                              |
| 2.0     | 0.5     | TP 2.0 / SL 0.5 | 9         | 16.63%             | 0.219              | 22.7                     |                     |                       |                       |                               |                              |
| 0%      |         |                 |           | 39.32%             | 7.09%              |                          |                     |                       |                       |                               |                              |
| 8.16    |         |                 |           | 0.857              | 0.906              |                          |                     |                       |                       |                               |                              |
| 1.0     | 1.0     | TP 1.0 / SL 1.0 | 9         | 7.78%              | 0.156              | 7.7                      |                     |                       |                       |                               |                              |
| 8%      |         |                 |           | 0.00%              | 7.78%              |                          |                     |                       |                       |                               |                              |
| 11.20   |         |                 |           | 0.233              | 0.300              |                          |                     |                       |                       |                               |                              |
| 1.5     | 1.0     | TP 1.5 / SL 1.0 | 9         | 21.27%             | 0.228              | 9.1                      |                     |                       |                       |                               |                              |
| 2%      |         |                 |           | 30.39%             | 9.82%              |                          |                     |                       |                       |                               |                              |
| 10.56   |         |                 |           | 0.706              | 0.751              |                          |                     |                       |                       |                               |                              |
| 2.0     | 1.0     | TP 2.0 / SL 1.0 | 9         | 24.07%             | 0.376              | 10.5                     |                     |                       |                       |                               |                              |
| 7%      |         |                 |           | 34.64%             | 11.08%             |                          |                     |                       |                       |                               |                              |
| 8.79    |         |                 |           | 0.804              | 0.918              |                          |                     |                       |                       |                               |                              |
| 2.5     | 1.0     | TP 2.5 / SL 1.0 | 9         | 19.40%             | 0.371              | 11.5                     |                     |                       |                       |                               |                              |
| 0%      |         |                 |           | 30.90%             | 11.17%             |                          |                     |                       |                       |                               |                              |
| 7.39    |         |                 |           | 0.730              | 0.875              |                          |                     |                       |                       |                               |                              |

Top 20 targets con mayor variación trimestral temporal:

| bar_col               | target_base | tp_sl_config    | n_quarters | tp_rate_mean | tp_rate_range | sl_rate_mean | sl_rate_range | no_event_rate_mean | no_event_rate_range | tp_share_mean | tp_share_range | quarterly_instability_score |
|-----------------------|-------------|-----------------|------------|--------------|---------------|--------------|---------------|--------------------|---------------------|---------------|----------------|-----------------------------|
| bar_p60_h60_tp15_sl05 | p60_h60     | TP 1.5 / SL 0.5 | 26         | 39.81%       |               |              |               |                    |                     |               |                |                             |
| 24.02%                | 16.97%      | 26.77%          |            | 43.22%       | 49.90%        | 7            |               |                    |                     |               |                |                             |
| 1.60%                 | 21.77%      |                 |            | 1.225        |               |              |               |                    |                     |               |                |                             |
| bar_p50_h60_tp20_sl05 | p50_h60     | TP 2.0 / SL 0.5 | 26         | 23.23%       |               |              |               |                    |                     |               |                |                             |
| 24.21%                | 22.17%      | 30.56%          |            | 54.60%       | 54.77%        | 5            |               |                    |                     |               |                |                             |
| 1.98%                 | 10.82%      |                 |            | 1.203        |               |              |               |                    |                     |               |                |                             |
| bar_p50_h90_tp20_sl05 | p50_h90     | TP 2.0 / SL 0.5 | 26         | 23.21%       |               |              |               |                    |                     |               |                |                             |
| 21.84%                | 23.48%      | 31.01%          |            | 53.31%       | 52.85%        | 5            |               |                    |                     |               |                |                             |
| 0.44%                 | 13.53%      |                 |            | 1.192        |               |              |               |                    |                     |               |                |                             |
| bar_p60_h60_tp20_sl05 | p60_h60     | TP 2.0 / SL 0.5 | 26         | 19.80%       |               |              |               |                    |                     |               |                |                             |
| 22.58%                | 17.97%      | 29.67%          |            | 62.23%       | 52.25%        | 5            |               |                    |                     |               |                |                             |
| 3.73%                 | 14.43%      |                 |            | 1.189        |               |              |               |                    |                     |               |                |                             |
| bar_p60_h90_tp20_sl05 | p60_h90     | TP 2.0 / SL 0.5 | 26         | 19.98%       |               |              |               |                    |                     |               |                |                             |
| 21.39%                | 19.24%      | 28.69%          |            | 60.78%       | 49.44%        | 5            |               |                    |                     |               |                |                             |
| 1.79%                 | 19.14%      |                 |            | 1.187        |               |              |               |                    |                     |               |                |                             |
| bar_p60_h90_tp15_sl05 | p60_h90     | TP 1.5 / SL 0.5 | 26         | 40.02%       |               |              |               |                    |                     |               |                |                             |
| 22.11%                | 18.16%      | 25.54%          |            | 41.82%       | 46.52%        | 6            |               |                    |                     |               |                |                             |
| 9.96%                 | 22.69%      |                 |            | 1.169        |               |              |               |                    |                     |               |                |                             |
| bar_p40_h60_tp20_sl10 | p40_h60     | TP 2.0 / SL 1.0 | 26         | 32.70%       |               |              |               |                    |                     |               |                |                             |
| 33.56%                | 7.61%       | 16.32%          |            | 59.69%       | 49.88%        | 8            |               |                    |                     |               |                |                             |
| 2.81%                 | 17.08%      |                 |            | 1.168        |               |              |               |                    |                     |               |                |                             |
| bar_p40_h90_tp20_sl05 | p40_h90     | TP 2.0 / SL 0.5 | 26         | 26.59%       |               |              |               |                    |                     |               |                |                             |
| 23.03%                | 27.82%      | 30.70%          |            | 45.60%       | 51.90%        | 4            |               |                    |                     |               |                |                             |
| 9.36%                 | 11.03%      |                 |            | 1.167        |               |              |               |                    |                     |               |                |                             |
| bar_p40_h90_tp20_sl10 | p40_h90     | TP 2.0 / SL 1.0 | 26         | 32.76%       |               |              |               |                    |                     |               |                |                             |
| 32.53%                | 8.13%       | 16.37%          |            | 59.11%       | 48.89%        | 8            |               |                    |                     |               |                |                             |
| 1.57%                 | 16.48%      |                 |            | 1.143        |               |              |               |                    |                     |               |                |                             |

|                       |                         |        |        |        |
|-----------------------|-------------------------|--------|--------|--------|
| bar_p50_h60_tp15_sl05 | p50_h60 TP 1.5 / SL 0.5 | 26     | 42.98% |        |
| 19.88%                | 20.86%                  | 27.57% | 36.15% | 44.98% |
| 8.46%                 | 21.09%                  |        | 1.135  |        |
| bar_p40_h60_tp20_sl05 | p40_h60 TP 2.0 / SL 0.5 | 26     | 26.89% |        |
| 22.25%                | 26.99%                  | 30.28% | 46.13% | 52.53% |
| 0.42%                 | 7.58%                   |        | 1.126  |        |
| bar_p40_h60_tp25_sl10 | p40_h60 TP 2.5 / SL 1.0 | 26     | 19.84% |        |
| 29.22%                | 8.10%                   | 18.06% | 72.06% | 47.28% |
| 3.02%                 | 16.49%                  |        | 1.110  |        |
| bar_p50_h90_tp15_sl05 | p50_h90 TP 1.5 / SL 0.5 | 26     | 42.67% |        |
| 21.23%                | 22.06%                  | 27.08% | 35.27% | 43.23% |
| 6.92%                 | 19.08%                  |        | 1.106  |        |
| bar_p50_h60_tp20_sl10 | p50_h60 TP 2.0 / SL 1.0 | 26     | 27.59% |        |
| 33.74%                | 5.52%                   | 13.38% | 66.89% | 47.12% |
| 5.17%                 | 16.16%                  |        | 1.104  |        |
| bar_p60_h30_tp20_sl05 | p60_h30 TP 2.0 / SL 0.5 | 26     | 21.43% |        |
| 23.30%                | 17.63%                  | 26.11% | 60.94% | 49.41% |
| 5.74%                 | 11.43%                  |        | 1.103  |        |
| bar_p60_h30_tp15_sl05 | p60_h30 TP 1.5 / SL 0.5 | 26     | 41.55% |        |
| 19.89%                | 16.63%                  | 23.94% | 41.82% | 43.84% |
| 2.50%                 | 20.09%                  |        | 1.078  |        |
| bar_p40_h90_tp25_sl10 | p40_h90 TP 2.5 / SL 1.0 | 26     | 19.94% |        |
| 27.58%                | 8.70%                   | 18.15% | 71.36% | 45.73% |
| 1.23%                 | 16.28%                  |        | 1.077  |        |
| bar_p50_h30_tp20_sl05 | p50_h30 TP 2.0 / SL 0.5 | 26     | 25.00% |        |
| 23.01%                | 21.63%                  | 26.54% | 53.37% | 49.55% |
| 4.24%                 | 8.24%                   |        | 1.073  |        |
| bar_p40_h30_tp20_sl10 | p40_h30 TP 2.0 / SL 1.0 | 26     | 34.38% |        |
| 30.70%                | 7.71%                   | 15.19% | 57.91% | 45.89% |
| 3.03%                 | 15.43%                  |        | 1.072  |        |
| bar_p50_h90_tp20_sl10 | p50_h90 TP 2.0 / SL 1.0 | 26     | 27.86% |        |
| 31.49%                | 6.02%                   | 13.25% | 66.12% | 44.74% |
| 3.83%                 | 16.27%                  |        | 1.057  |        |

Top 20 targets con menor variación trimestral temporal:

| bar_col               | target_base             | tp_sl_config | n_quarters | tp_rate_mean | tp_rate_range | sl_rate_mean | sl_rate_range | no_event_rate_mean | no_event_rate_range | tp_share_mean | tp_share_range | quarterly_instability_score |
|-----------------------|-------------------------|--------------|------------|--------------|---------------|--------------|---------------|--------------------|---------------------|---------------|----------------|-----------------------------|
| bar_p60_h60_tp10_sl10 | p60_h60 TP 1.0 / SL 1.0 | 26           | 97.19%     |              |               |              |               |                    |                     |               |                |                             |
| 7.07%                 | 2.81%                   | 7.07%        | 0.00%      | 0.00%        |               |              |               |                    |                     |               |                |                             |
| 7.19%                 | 7.07%                   |              | 0.212      |              |               |              |               |                    |                     |               |                |                             |
| bar_p60_h30_tp10_sl10 | p60_h30 TP 1.0 / SL 1.0 | 26           | 96.98%     |              |               |              |               |                    |                     |               |                |                             |
| 7.25%                 | 3.02%                   | 7.25%        | 0.00%      | 0.00%        |               |              |               |                    |                     |               |                |                             |
| 6.98%                 | 7.25%                   |              | 0.217      |              |               |              |               |                    |                     |               |                |                             |
| bar_p60_h90_tp10_sl10 | p60_h90 TP 1.0 / SL 1.0 | 26           | 96.96%     |              |               |              |               |                    |                     |               |                |                             |
| 7.33%                 | 3.04%                   | 7.33%        | 0.00%      | 0.00%        |               |              |               |                    |                     |               |                |                             |
| 6.96%                 | 7.33%                   |              | 0.220      |              |               |              |               |                    |                     |               |                |                             |
| bar_p50_h90_tp10_sl10 | p50_h90 TP 1.0 / SL 1.0 | 26           | 95.70%     |              |               |              |               |                    |                     |               |                |                             |
| 9.36%                 | 4.30%                   | 9.36%        | 0.00%      | 0.00%        |               |              |               |                    |                     |               |                |                             |
| 5.70%                 | 9.36%                   |              | 0.281      |              |               |              |               |                    |                     |               |                |                             |
| bar_p50_h30_tp10_sl10 | p50_h30 TP 1.0 / SL 1.0 | 26           | 95.81%     |              |               |              |               |                    |                     |               |                |                             |
| 9.52%                 | 4.19%                   | 9.52%        | 0.00%      | 0.00%        |               |              |               |                    |                     |               |                |                             |
| 5.81%                 | 9.52%                   |              | 0.286      |              |               |              |               |                    |                     |               |                |                             |
| bar_p50_h60_tp10_sl10 | p50_h60 TP 1.0 / SL 1.0 | 26           | 96.03%     |              |               |              |               |                    |                     |               |                |                             |
| 9.62%                 | 3.97%                   | 9.62%        | 0.00%      | 0.00%        |               |              |               |                    |                     |               |                |                             |
| 6.03%                 | 9.62%                   |              | 0.289      |              |               |              |               |                    |                     |               |                |                             |
| bar_p40_h30_tp10_sl10 | p40_h30 TP 1.0 / SL 1.0 | 26           | 94.32%     |              |               |              |               |                    |                     |               |                |                             |
| 11.76%                | 5.68%                   | 11.76%       | 0.00%      | 0.00%        |               |              |               |                    |                     |               |                |                             |
| 4.32%                 | 11.76%                  |              | 0.353      |              |               |              |               |                    |                     |               |                |                             |
| bar_p40_h90_tp10_sl10 | p40_h90 TP 1.0 / SL 1.0 | 26           | 94.11%     |              |               |              |               |                    |                     |               |                |                             |

|                       |        |                         |        |        |   |
|-----------------------|--------|-------------------------|--------|--------|---|
| 11.83%                | 5.89%  | 11.83%                  | 0.00%  | 0.00%  | 9 |
| 4.11%                 | 11.83% |                         | 0.355  |        |   |
| bar_p40_h60_tp10_sl10 |        | p40_h60 TP 1.0 / SL 1.0 | 26     | 94.51% |   |
| 12.17%                | 5.49%  | 12.17%                  | 0.00%  | 0.00%  | 9 |
| 4.51%                 | 12.17% |                         | 0.365  |        |   |
| bar_p60_h30_tp10_sl05 |        | p60_h30 TP 1.0 / SL 0.5 | 26     | 86.27% |   |
| 19.91%                | 13.73% | 19.91%                  | 0.00%  | 0.00%  | 8 |
| 6.27%                 | 19.91% |                         | 0.597  |        |   |
| bar_p60_h90_tp10_sl05 |        | p60_h90 TP 1.0 / SL 0.5 | 26     | 84.92% |   |
| 20.20%                | 15.08% | 20.20%                  | 0.00%  | 0.00%  | 8 |
| 4.92%                 | 20.20% |                         | 0.606  |        |   |
| bar_p40_h30_tp10_sl05 |        | p40_h30 TP 1.0 / SL 0.5 | 26     | 79.85% |   |
| 20.24%                | 20.15% | 20.24%                  | 0.00%  | 0.00%  | 7 |
| 9.85%                 | 20.24% |                         | 0.607  |        |   |
| bar_p50_h30_tp10_sl05 |        | p50_h30 TP 1.0 / SL 0.5 | 26     | 83.11% |   |
| 20.38%                | 16.89% | 20.38%                  | 0.00%  | 0.00%  | 8 |
| 3.11%                 | 20.38% |                         | 0.611  |        |   |
| bar_p60_h60_tp10_sl05 |        | p60_h60 TP 1.0 / SL 0.5 | 26     | 85.97% |   |
| 22.83%                | 14.03% | 22.83%                  | 0.00%  | 0.00%  | 8 |
| 5.97%                 | 22.83% |                         | 0.685  |        |   |
| bar_p50_h90_tp10_sl05 |        | p50_h90 TP 1.0 / SL 0.5 | 26     | 81.63% |   |
| 23.15%                | 18.37% | 23.15%                  | 0.00%  | 0.00%  | 8 |
| 1.63%                 | 23.15% |                         | 0.694  |        |   |
| bar_p50_h60_tp10_sl05 |        | p50_h60 TP 1.0 / SL 0.5 | 26     | 82.72% |   |
| 24.02%                | 17.28% | 24.02%                  | 0.00%  | 0.00%  | 8 |
| 2.72%                 | 24.02% |                         | 0.721  |        |   |
| bar_p40_h60_tp10_sl05 |        | p40_h60 TP 1.0 / SL 0.5 | 26     | 78.98% |   |
| 24.22%                | 21.02% | 24.22%                  | 0.00%  | 0.00%  | 7 |
| 8.98%                 | 24.22% |                         | 0.727  |        |   |
| bar_p40_h90_tp10_sl05 |        | p40_h90 TP 1.0 / SL 0.5 | 26     | 78.20% |   |
| 24.50%                | 21.80% | 24.50%                  | 0.00%  | 0.00%  | 7 |
| 8.20%                 | 24.50% |                         | 0.735  |        |   |
| bar_p60_h30_tp25_sl10 |        | p60_h30 TP 2.5 / SL 1.0 | 26     | 13.47% |   |
| 22.41%                | 4.39%  | 10.31%                  | 82.14% | 32.71% | 7 |
| 7.52%                 | 17.45% |                         | 0.829  |        |   |
| bar_p60_h60_tp25_sl10 |        | p60_h60 TP 2.5 / SL 1.0 | 26     | 12.50% |   |
| 20.52%                | 4.18%  | 11.10%                  | 83.32% | 30.98% | 7 |
| 7.22%                 | 22.37% |                         | 0.850  |        |   |

Celda 2 finalizada correctamente.

### Celda 3 — Gráficos agrupados por año, percentil, horizonte y configuración

Esta celda te nos deja ver la estabilidad desde tres ángulos:

- 1. Cambio entre development\_2020\_2024 y final\_test\_2025\_2026.
- 2. Evolución anual de TP rate y no\_event rate.
- 3. Variación anual acumulada para los 63 targets BAR.

In [109...

```
# Celda 3 - Gráficos agrupados de estabilidad temporal
# =====

print("Gráficos agrupados de estabilidad temporal - BAR")
print("=" * 80)

# -----
# Validar objetos necesarios
```



```

# -----
required_objects = [
    "df_bar_split_comparison",
    "df_bar_split_change_by_config",
    "df_bar_split_change_by_target_base",
    "df_bar_yearly_stability_summary",
    "df_bar_yearly_stability_by_config",
    "df_bar_stability_year",
    "BAR_PERCENTILES",
    "BAR_HORIZONS",
]

missing_objects = [
    obj for obj in required_objects
    if obj not in globals()
]

if missing_objects:
    raise NameError(
        "Faltan objetos necesarios. "
        f"Debes correr las celdas anteriores: {missing_objects}"
    )

# -----
# Orden de configuraciones TP/SL
# -----

config_order = [
    (1.0, 1.0),
    (1.5, 1.0),
    (2.0, 1.0),
    (2.5, 1.0),
    (1.0, 0.5),
    (1.5, 0.5),
    (2.0, 0.5),
]

config_labels = [
    f"TP {tp} / SL {sl}"
    for tp, sl in config_order
]

target_base_order = [
    f"p{p}_h{h}"
    for p in BAR_PERCENTILES
    for h in BAR_HORIZONS
]

# -----
# Copias de trabajo
# -----

df_split_plot = df_bar_split_comparison.copy()
df_year_plot = df_bar_stability_year.copy()
df_yearly_summary_plot = df_bar_yearly_stability_summary.copy()

for df_tmp in [df_split_plot, df_year_plot, df_yearly_summary_plot]:
    df_tmp["config_label"] = df_tmp.apply(
        lambda row: f"TP {row['tp_mult']} / SL {row['sl_mult']}",

```

```

        axis=1
    )

    df_tmp["config_order"] = df_tmp.apply(
        lambda row: config_order.index((row["tp_mult"], row["sl_mult"]))
        if (row["tp_mult"], row["sl_mult"]) in config_order
        else 999,
        axis=1
    )

# -----
# Función auxiliar para heatmaps
# -----

def plot_heatmap_with_values(pivot_data, title, value_format="{:.2f}", figsize=(
    """
    Grafica un heatmap simple con valores anotados.
    """

    fig, ax = plt.subplots(figsize=figsize)

    data = np.ma.masked_invalid(pivot_data.values)

    im = ax.imshow(
        data,
        aspect="auto"
    )

    ax.set_title(title)
    ax.set_yticks(np.arange(len(pivot_data.index)))
    ax.set_yticklabels(pivot_data.index)

    ax.set_xticks(np.arange(len(pivot_data.columns)))
    ax.set_xticklabels(pivot_data.columns, rotation=45, ha="right")

    for i in range(pivot_data.shape[0]):
        for j in range(pivot_data.shape[1]):
            value = pivot_data.iloc[i, j]

            if pd.notna(value):
                ax.text(
                    j,
                    i,
                    value_format.format(value),
                    ha="center",
                    va="center",
                    fontsize=8
                )

    fig.colorbar(im, ax=ax)
    plt.tight_layout()
    plt.show()

# =====
# Figura 1 – Cambio total development vs final_test
# Todos los percentiles, horizontes y configuraciones
# =====

pivot_split_score = (

```

```

df_split_plot
.pivot_table(
    index="target_base",
    columns="config_label",
    values="split_shift_score",
    aggfunc="first"
)
.reindex(index=target_base_order)
.reindex(columns=config_labels)
)

plot_heatmap_with_values(
    pivot_data=pivot_split_score,
    title="Cambio total development vs final_test - split_shift_score",
    value_format="{:.2f}",
    figsize=(15, 7)
)

# =====
# Figura 2 – Diferencias final_test - development
# TP rate, SL rate, no_event y TP sobre resueltos
# =====

diff_metrics = [
    ("tp_rate_diff_final_minus_dev", "Diferencia TP rate final_test - developmen
    ("sl_rate_diff_final_minus_dev", "Diferencia SL rate final_test - developmen
    ("no_event_rate_diff_final_minus_dev", "Diferencia no_event final_test - dev
    ("tp_share_over_resolved_diff_final_minus_dev", "Diferencia TP sobre eventos
]

fig = plt.figure(figsize=(16, 20))

gs = fig.add_gridspec(
    nrows=len(diff_metrics),
    ncols=2,
    width_ratios=[20, 1],
    hspace=0.35,
    wspace=0.08
)

axes = [fig.add_subplot(gs[i, 0]) for i in range(len(diff_metrics))]
caxes = [fig.add_subplot(gs[i, 1]) for i in range(len(diff_metrics))]

for ax, cax, (metric_col, metric_title) in zip(axes, caxes, diff_metrics):

    pivot_tmp = (
        df_split_plot
        .pivot_table(
            index="target_base",
            columns="config_label",
            values=metric_col,
            aggfunc="first"
        )
        .reindex(index=target_base_order)
        .reindex(columns=config_labels)
    )

    data = np.ma.masked_invalid(pivot_tmp.values)

    im = ax.imshow(

```

```

        data,
        aspect="auto"
    )

    ax.set_title(metric_title)
    ax.set_yticks(np.arange(len(pivot_tmp.index)))
    ax.set_yticklabels(pivot_tmp.index)

    ax.set_xticks(np.arange(len(pivot_tmp.columns)))
    ax.set_xticklabels(pivot_tmp.columns, rotation=45, ha="right")

    for i in range(pivot_tmp.shape[0]):
        for j in range(pivot_tmp.shape[1]):
            value = pivot_tmp.iloc[i, j]

            if pd.notna(value):
                ax.text(
                    j,
                    i,
                    f"{value:.1%}",
                    ha="center",
                    va="center",
                    fontsize=7
                )

    fig.colorbar(
        im,
        cax=cax,
        label="Diferencia"
    )

plt.show()

# =====
# Figura 3 – Evolución anual de TP rate
# Filas: percentil | Columnas: horizonte
# Líneas: configuraciones TP/SL
# =====

fig, axes = plt.subplots(
    nrows=len(BAR_PERCENTILES),
    ncols=len(BAR_HORIZONS),
    figsize=(18, 12),
    sharex=True,
    sharey=True
)

for i, percentile in enumerate(BAR_PERCENTILES):
    for j, horizon in enumerate(BAR_HORIZONS):

        ax = axes[i, j]

        tmp_base = df_year_plot[
            (df_year_plot["percentile"] == percentile)
            & (df_year_plot["horizon"] == horizon)
        ].copy()

        for tp_mult, sl_mult in config_order:
            tmp = tmp_base[
                (tmp_base["tp_mult"] == tp_mult)

```

```

        & (tmp_base["sl_mult"] == sl_mult)
    ].copy()

    if tmp.empty:
        continue

    tmp["year"] = tmp["period"].astype(int)
    tmp = tmp.sort_values("year")

    ax.plot(
        tmp["year"],
        tmp["tp_rate"],
        marker="o",
        linewidth=1,
        label=f"TP {tp_mult} / SL {sl_mult}"
    )

    ax.set_title(f"p{percentile} | H{horizon}")

    if j == 0:
        ax.set_ylabel("TP rate")

    if i == len(BAR_PERCENTILES) - 1:
        ax.set_xlabel("Año")

    ax.yaxis.set_major_formatter(
        plt.FuncFormatter(lambda y, _: f"{y:.0%}")
    )

    ax.grid(True, alpha=0.3)

handles, labels = axes[0, 0].get_legend_handles_labels()

fig.legend(
    handles,
    labels,
    loc="upper center",
    ncol=4,
    bbox_to_anchor=(0.5, 1.03)
)

fig.suptitle(
    "Evolución anual del TP rate – todos los percentiles, horizontes y configura
    fontsize=14,
    y=1.08
)

plt.tight_layout()
plt.show()

# =====
# Figura 4 – Evolución anual de no_event rate
# Filas: percentil | Columnas: horizonte
# Líneas: configuraciones TP/SL
# =====

fig, axes = plt.subplots(
    nrows=len(BAR_PERCENTILES),
    ncols=len(BAR_HORIZONS),
    figsize=(18, 12),

```

```

sharex=True,
sharey=True
)

for i, percentile in enumerate(BAR_PERCENTILES):
    for j, horizon in enumerate(BAR_HORIZONS):

        ax = axes[i, j]

        tmp_base = df_year_plot[
            (df_year_plot["percentile"] == percentile)
            & (df_year_plot["horizon"] == horizon)
        ].copy()

        for tp_mult, sl_mult in config_order:
            tmp = tmp_base[
                (tmp_base["tp_mult"] == tp_mult)
                & (tmp_base["sl_mult"] == sl_mult)
            ].copy()

            if tmp.empty:
                continue

            tmp["year"] = tmp["period"].astype(int)
            tmp = tmp.sort_values("year")

            ax.plot(
                tmp["year"],
                tmp["no_event_rate"],
                marker="o",
                linewidth=1,
                label=f"TP {tp_mult} / SL {sl_mult}"
            )

        ax.set_title(f"p{percentile} | H{horizon}")

        if j == 0:
            ax.set_ylabel("no_event rate")

        if i == len(BAR_PERCENTILES) - 1:
            ax.set_xlabel("Año")

        ax.yaxis.set_major_formatter(
            plt.FuncFormatter(lambda y, _: f"{y:.0%}")
        )

        ax.grid(True, alpha=0.3)

handles, labels = axes[0, 0].get_legend_handles_labels()

fig.legend(
    handles,
    labels,
    loc="upper center",
    ncol=4,
    bbox_to_anchor=(0.5, 1.03)
)

fig.suptitle(
    "Evolución anual del no_event rate – todos los percentiles, horizontes y con

```

```

        fontsize=14,
        y=1.08
    )

plt.tight_layout()
plt.show()

# =====
# Figura 5 – Variación anual agregada por target
# Heatmap: target_base x configuración
# =====

pivot_yearly_instability = (
    df_yearly_summary_plot
    .pivot_table(
        index="target_base",
        columns="config_label",
        values="yearly_instability_score",
        aggfunc="first"
    )
    .reindex(index=target_base_order)
    .reindex(columns=config_labels)
)

plot_heatmap_with_values(
    pivot_data=pivot_yearly_instability,
    title="Inestabilidad anual por target – yearly_instability_score",
    value_format="{:.2f}",
    figsize=(15, 7)
)

# =====
# Figura 6 – Rangos anuales por métrica
# TP range, SL range, no_event range y TP share range
# =====

range_metrics = [
    ("tp_rate_range", "Rango anual TP rate"),
    ("sl_rate_range", "Rango anual SL rate"),
    ("no_event_rate_range", "Rango anual no_event rate"),
    ("tp_share_range", "Rango anual TP sobre eventos resueltos"),
]

fig = plt.figure(figsize=(16, 20))

gs = fig.add_gridspec(
    nrows=len(range_metrics),
    ncols=2,
    width_ratios=[20, 1],
    hspace=0.35,
    wspace=0.08
)

axes = [fig.add_subplot(gs[i, 0]) for i in range(len(range_metrics))]
caxes = [fig.add_subplot(gs[i, 1]) for i in range(len(range_metrics))]

for ax, cax, (metric_col, metric_title) in zip(axes, caxes, range_metrics):

    pivot_tmp = (
        df_yearly_summary_plot

```

```

        .pivot_table(
            index="target_base",
            columns="config_label",
            values=metric_col,
            aggfunc="first"
        )
        .reindex(index=target_base_order)
        .reindex(columns=config_labels)
    )

    data = np.ma.masked_invalid(pivot_tmp.values)

    im = ax.imshow(
        data,
        aspect="auto"
    )

    ax.set_title(metric_title)
    ax.set_yticks(np.arange(len(pivot_tmp.index)))
    ax.set_yticklabels(pivot_tmp.index)

    ax.set_xticks(np.arange(len(pivot_tmp.columns)))
    ax.set_xticklabels(pivot_tmp.columns, rotation=45, ha="right")

    for i in range(pivot_tmp.shape[0]):
        for j in range(pivot_tmp.shape[1]):
            value = pivot_tmp.iloc[i, j]

            if pd.notna(value):
                ax.text(
                    j,
                    i,
                    f"{value:.1%}",
                    ha="center",
                    va="center",
                    fontsize=7
                )

    fig.colorbar(
        im,
        cax=cax,
        label="Rango anual"
    )

plt.show()

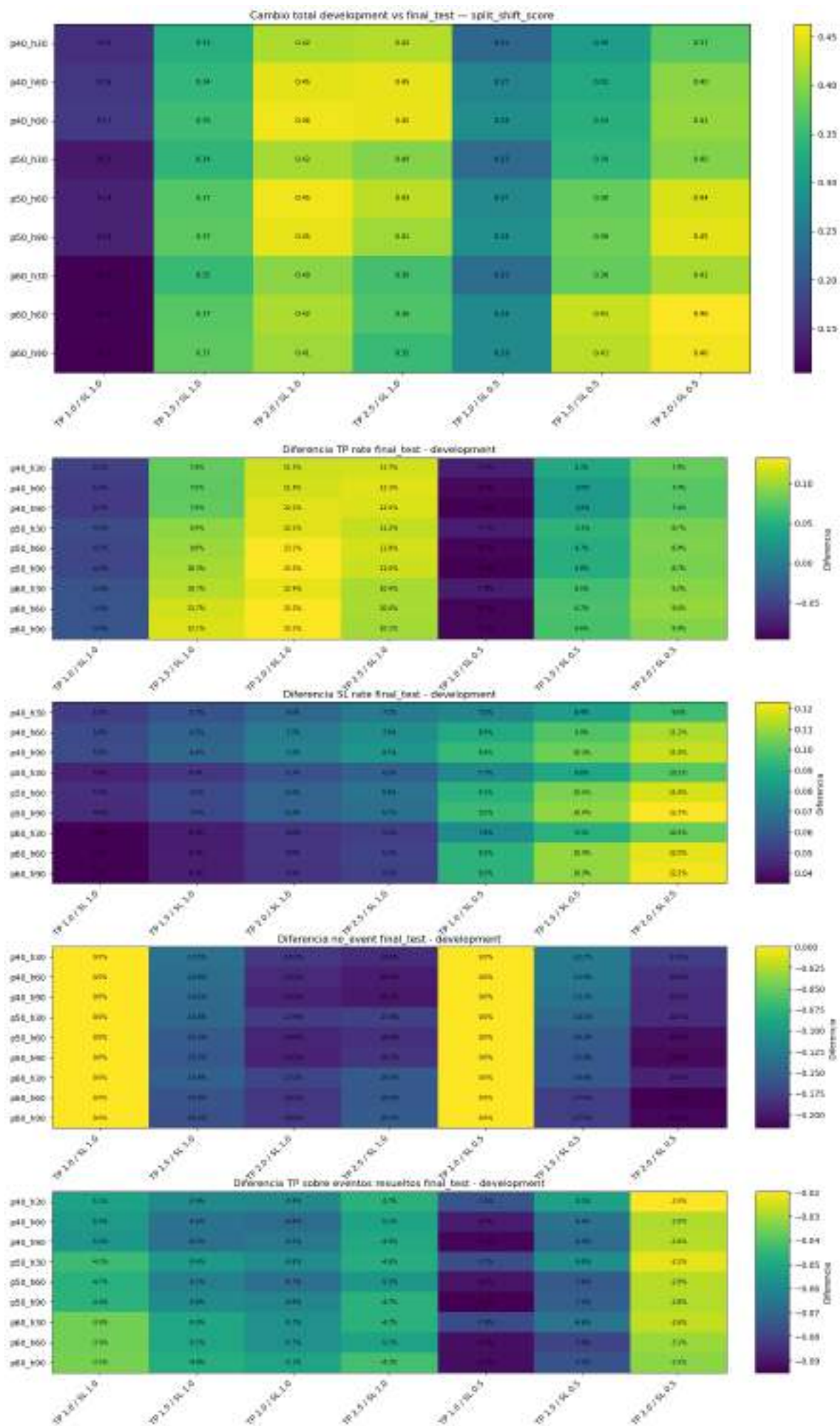
print("\nCelda 3 finalizada correctamente.")

```

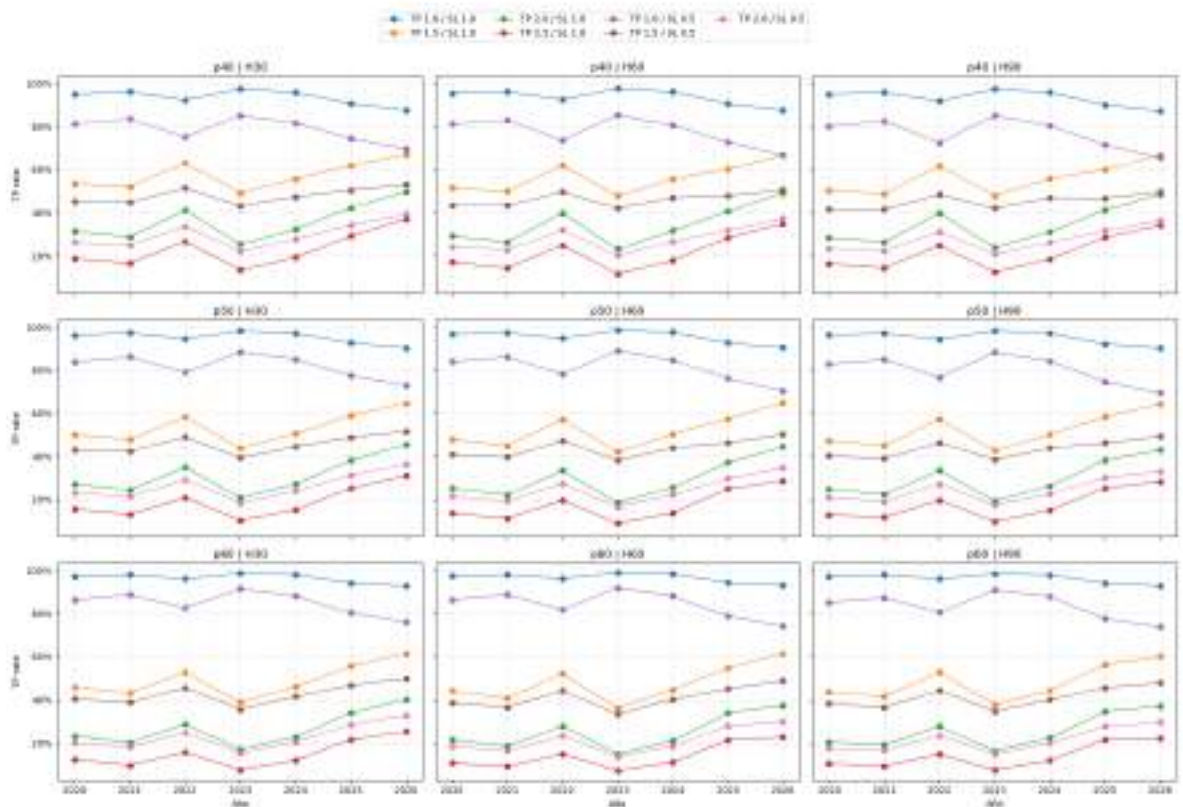
Gráficos agrupados de estabilidad temporal - BAR

=====

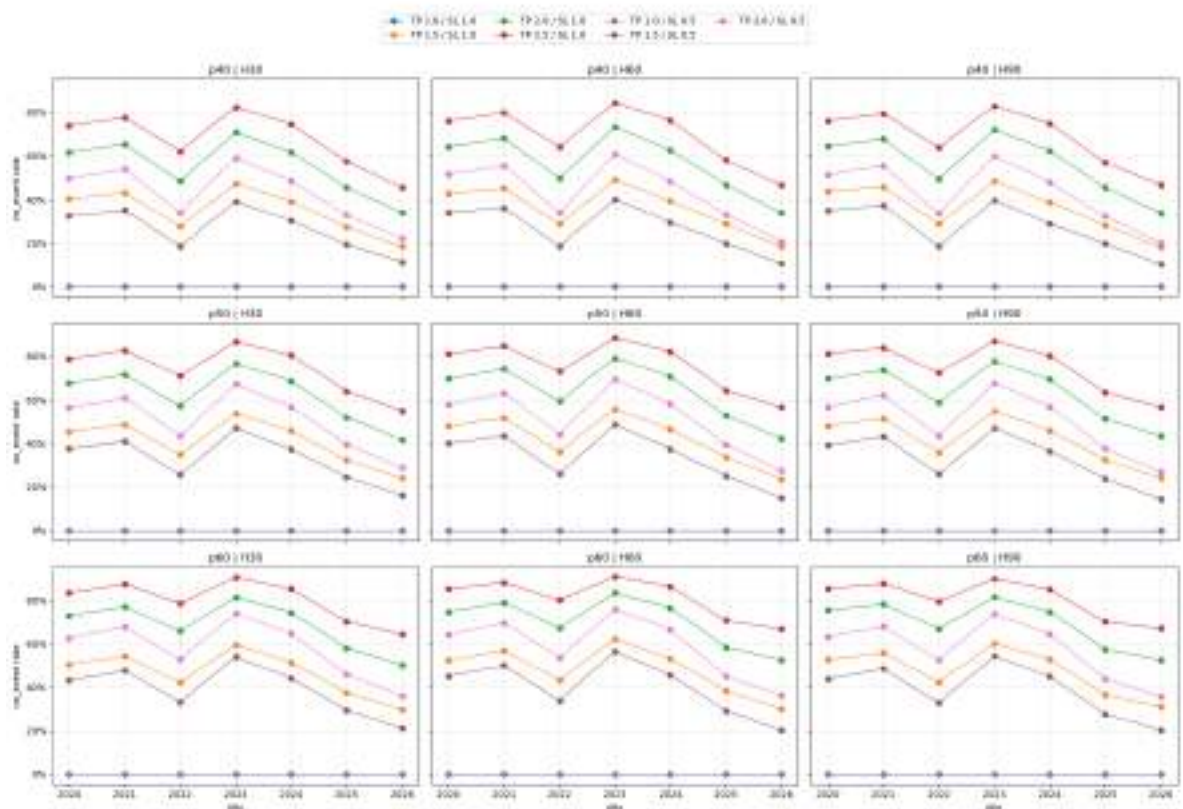




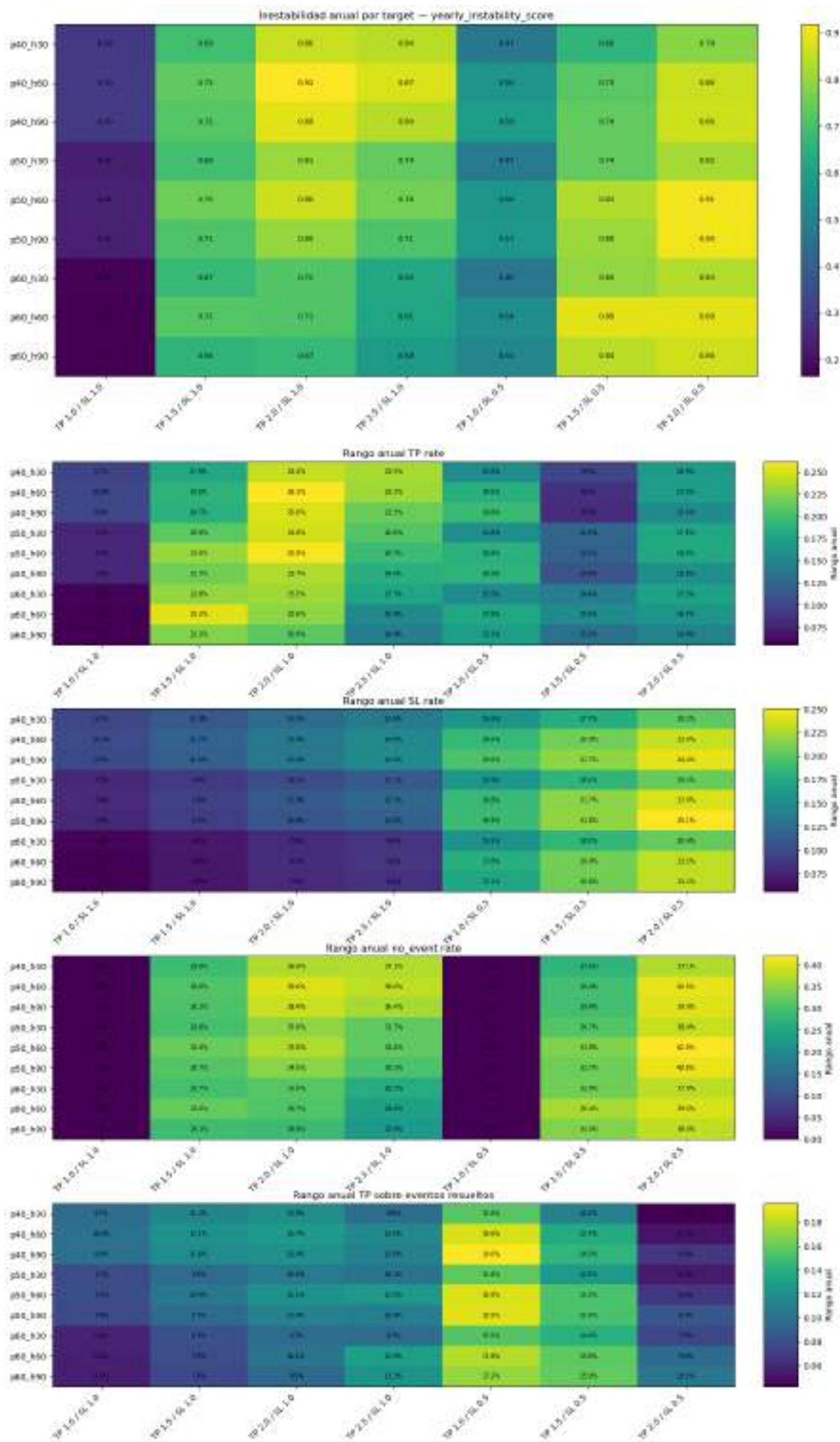
Evolución anual del TP rate -- todos los percentiles, horizontes y configuraciones



Evolución anual del tp\_event rate -- todos los percentiles, horizontes y configuraciones







Celda 3 finalizada correctamente.

## Celda 4 — Guardado y tabla final para observaciones

Esta celda te deja una tabla final lista para interpretar:

df\_bar\_temporal\_stability\_final\_table

y resume la estabilidad temporal de los **63 targets BAR** cruzando tres niveles:

development vs final\_test

variación anual

variación trimestral

```
In [110... # Celda 4 – Guardado y tabla final para observaciones
# =====

print("Guardado y tabla final de estabilidad temporal - BAR")
print("=" * 80)

# -----
# Validar objetos necesarios
# -----

required_objects = [
    "df_bar_temporal_stability",
    "df_bar_stability_split",
    "df_bar_stability_year",
    "df_bar_stability_quarter",
    "df_bar_split_comparison",
    "df_bar_split_change_by_config",
    "df_bar_split_change_by_target_base",
    "df_bar_yearly_stability_summary",
    "df_bar_yearly_stability_by_config",
    "df_bar_quarterly_stability_summary",
    "MNQ_TARGETS_PATH",
]

missing_objects = [
    obj for obj in required_objects
    if obj not in globals()
]

if missing_objects:
    raise NameError(
        "Faltan objetos necesarios. "
        f"Debes correr las celdas anteriores: {missing_objects}"
    )

# -----
# Definir rutas de guardado
# -----

MNQ_TARGETS_BAR_TEMPORAL_STABILITY_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_temporal_stability.csv"
)

MNQ_TARGETS_BAR_STABILITY_SPLIT_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_stability_split.csv"
```

```

)

MNQ_TARGETS_BAR_STABILITY_YEAR_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_stability_year.csv"
)

MNQ_TARGETS_BAR_STABILITY_QUARTER_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_stability_quarter.csv"
)

MNQ_TARGETS_BAR_SPLIT_COMPARISON_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_split_comparison.csv"
)

MNQ_TARGETS_BAR_SPLIT_CHANGE_BY_CONFIG_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_split_change_by_config.csv"
)

MNQ_TARGETS_BAR_SPLIT_CHANGE_BY_TARGET_BASE_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_split_change_by_target_base.csv"
)

MNQ_TARGETS_BAR_YEARLY_STABILITY_SUMMARY_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_yearly_stability_summary.csv"
)

MNQ_TARGETS_BAR_YEARLY_STABILITY_BY_CONFIG_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_yearly_stability_by_config.csv"
)

MNQ_TARGETS_BAR_QUARTERLY_STABILITY_SUMMARY_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_quarterly_stability_summary.csv"
)

MNQ_TARGETS_BAR_TEMPORAL_STABILITY_FINAL_TABLE_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_temporal_stability_final_table.csv"
)

MNQ_TARGETS_BAR_TEMPORAL_STABILITY_REPORT_TXT = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_temporal_stability_report.txt"
)

# -----
# Columnas identificadoras
# -----

id_cols = [
    "bar_col",
    "target_base",
    "horizon",
    "percentile",
    "tp_mult",
    "sl_mult",
    "tp_sl_config",
]

dev_label = "development_2020_2024"
test_label = "final_test_2025_2026"

# -----

```

```

# Construir tabla final de estabilidad temporal
# -----

split_cols = [
    "bar_col",
    "target_base",
    "horizon",
    "percentile",
    "tp_mult",
    "sl_mult",
    "tp_sl_config",

    f"tp_rate_{dev_label}",
    f"tp_rate_{test_label}",
    "tp_rate_diff_final_minus_dev",
    "tp_rate_abs_diff_final_vs_dev",

    f"sl_rate_{dev_label}",
    f"sl_rate_{test_label}",
    "sl_rate_diff_final_minus_dev",
    "sl_rate_abs_diff_final_vs_dev",

    f"no_event_rate_{dev_label}",
    f"no_event_rate_{test_label}",
    "no_event_rate_diff_final_minus_dev",
    "no_event_rate_abs_diff_final_vs_dev",

    f"resolved_rate_{dev_label}",
    f"resolved_rate_{test_label}",
    "resolved_rate_diff_final_minus_dev",
    "resolved_rate_abs_diff_final_vs_dev",

    f"tp_share_over_resolved_{dev_label}",
    f"tp_share_over_resolved_{test_label}",
    "tp_share_over_resolved_diff_final_minus_dev",
    "tp_share_over_resolved_abs_diff_final_vs_dev",

    f"expectancy_units_{dev_label}",
    f"expectancy_units_{test_label}",
    "expectancy_units_diff_final_minus_dev",
    "expectancy_units_abs_diff_final_vs_dev",

    f"event_step_mean_{dev_label}",
    f"event_step_mean_{test_label}",
    "event_step_mean_diff_final_minus_dev",
    "event_step_mean_abs_diff_final_vs_dev",

    "split_shift_score",
]

available_split_cols = [
    col for col in split_cols
    if col in df_bar_split_comparison.columns
]

df_bar_temporal_stability_final_table = (
    df_bar_split_comparison[available_split_cols]
    .copy()
)

```

```
# -----  
# Añadir resumen anual  
# -----  
  
yearly_cols = [  
    "bar_col",  
    "target_base",  
    "horizon",  
    "percentile",  
    "tp_mult",  
    "sl_mult",  
    "tp_sl_config",  
  
    "n_years",  
  
    "tp_rate_mean",  
    "tp_rate_std",  
    "tp_rate_min",  
    "tp_rate_max",  
    "tp_rate_range",  
  
    "sl_rate_mean",  
    "sl_rate_std",  
    "sl_rate_min",  
    "sl_rate_max",  
    "sl_rate_range",  
  
    "no_event_rate_mean",  
    "no_event_rate_std",  
    "no_event_rate_min",  
    "no_event_rate_max",  
    "no_event_rate_range",  
  
    "tp_share_mean",  
    "tp_share_std",  
    "tp_share_min",  
    "tp_share_max",  
    "tp_share_range",  
  
    "expectancy_mean",  
    "expectancy_std",  
    "expectancy_min",  
    "expectancy_max",  
    "expectancy_range",  
  
    "event_step_mean_mean",  
    "event_step_mean_std",  
    "event_step_mean_min",  
    "event_step_mean_max",  
    "event_step_mean_range",  
  
    "yearly_instability_score",  
]  
  
available_yearly_cols = [  
    col for col in yearly_cols  
    if col in df_bar_yearly_stability_summary.columns  
]  
  
df_yearly_merge = (
```

```

df_bar_yearly_stability_summary[available_yearly_cols]
    .copy()
)

rename_yearly_cols = {
    col: f"yearly_{col}"
    for col in df_yearly_merge.columns
    if col not in id_cols
}

df_yearly_merge = df_yearly_merge.rename(columns=rename_yearly_cols)

df_bar_temporal_stability_final_table = (
    df_bar_temporal_stability_final_table
    .merge(
        df_yearly_merge,
        on=id_cols,
        how="left"
    )
)

# -----
# Añadir resumen trimestral
# -----

quarterly_cols = [
    "bar_col",
    "target_base",
    "horizon",
    "percentile",
    "tp_mult",
    "sl_mult",
    "tp_sl_config",

    "n_quarters",

    "tp_rate_mean",
    "tp_rate_std",
    "tp_rate_range",

    "sl_rate_mean",
    "sl_rate_std",
    "sl_rate_range",

    "no_event_rate_mean",
    "no_event_rate_std",
    "no_event_rate_range",

    "tp_share_mean",
    "tp_share_std",
    "tp_share_range",

    "expectancy_mean",
    "expectancy_std",
    "expectancy_range",

    "event_step_mean_mean",
    "event_step_mean_std",
    "event_step_mean_range",

```



```

        "quarterly_instability_score",
    ]

    available_quarterly_cols = [
        col for col in quarterly_cols
        if col in df_bar_quarterly_stability_summary.columns
    ]

    df_quarterly_merge = (
        df_bar_quarterly_stability_summary[available_quarterly_cols]
        .copy()
    )

    rename_quarterly_cols = {
        col: f"quarterly_{col}"
        for col in df_quarterly_merge.columns
        if col not in id_cols
    }

    df_quarterly_merge = df_quarterly_merge.rename(columns=rename_quarterly_cols)

    df_bar_temporal_stability_final_table = (
        df_bar_temporal_stability_final_table
        .merge(
            df_quarterly_merge,
            on=id_cols,
            how="left"
        )
    )

    # -----
    # Clasificación descriptiva de estabilidad
    # -----
    # Esta clasificación es relativa al conjunto de 63 targets.
    # No selecciona el target final.
    # -----

    df_bar_temporal_stability_final_table["temporal_instability_score"] = (
        df_bar_temporal_stability_final_table["split_shift_score"].fillna(0)
        + df_bar_temporal_stability_final_table["yearly_yearly_instability_score"].f
        + df_bar_temporal_stability_final_table["quarterly_quarterly_instability_sco
    )

    q33 = df_bar_temporal_stability_final_table["temporal_instability_score"].quanti
    q66 = df_bar_temporal_stability_final_table["temporal_instability_score"].quanti

    def classify_temporal_stability(score):
        """
        Clasificación relativa de estabilidad temporal.
        Menor score = más estable.
        Mayor score = más variable temporalmente.
        """

        if pd.isna(score):
            return "sin_clasificar"

        if score <= q33:
            return "más_estable"

        if score <= q66:

```

```

        return "estabilidad_intermedia"

    return "más_inestable"

df_bar_temporal_stability_final_table["temporal_stability_profile"] = (
    df_bar_temporal_stability_final_table["temporal_instability_score"]
    .apply(classify_temporal_stability)
)

df_bar_temporal_stability_final_table = (
    df_bar_temporal_stability_final_table
    .sort_values(
        [
            "temporal_instability_score",
            "percentile",
            "horizon",
            "sl_mult",
            "tp_mult",
        ],
        ascending=[True, True, True, True, True]
    )
    .reset_index(drop=True)
)

# -----
# Guardar tablas principales
# -----

df_bar_temporal_stability.to_csv(
    MNQ_TARGETS_BAR_TEMPORAL_STABILITY_CSV,
    index=False
)

df_bar_stability_split.to_csv(
    MNQ_TARGETS_BAR_STABILITY_SPLIT_CSV,
    index=False
)

df_bar_stability_year.to_csv(
    MNQ_TARGETS_BAR_STABILITY_YEAR_CSV,
    index=False
)

df_bar_stability_quarter.to_csv(
    MNQ_TARGETS_BAR_STABILITY_QUARTER_CSV,
    index=False
)

df_bar_split_comparison.to_csv(
    MNQ_TARGETS_BAR_SPLIT_COMPARISON_CSV,
    index=False
)

df_bar_split_change_by_config.to_csv(
    MNQ_TARGETS_BAR_SPLIT_CHANGE_BY_CONFIG_CSV,
    index=False
)

df_bar_split_change_by_target_base.to_csv(

```

```

    MNQ_TARGETS_BAR_SPLIT_CHANGE_BY_TARGET_BASE_CSV,
    index=False
)

df_bar_yearly_stability_summary.to_csv(
    MNQ_TARGETS_BAR_YEARLY_STABILITY_SUMMARY_CSV,
    index=False
)

df_bar_yearly_stability_by_config.to_csv(
    MNQ_TARGETS_BAR_YEARLY_STABILITY_BY_CONFIG_CSV,
    index=False
)

df_bar_quarterly_stability_summary.to_csv(
    MNQ_TARGETS_BAR_QUARTERLY_STABILITY_SUMMARY_CSV,
    index=False
)

df_bar_temporal_stability_final_table.to_csv(
    MNQ_TARGETS_BAR_TEMPORAL_STABILITY_FINAL_TABLE_CSV,
    index=False
)

# -----
# Tablas compactas para imprimir
# -----

split_change_config_cols = [
    "tp_sl_config",
    "tp_mult",
    "sl_mult",
    "n_targets",

    "tp_rate_diff_mean",
    "tp_rate_abs_diff_mean",

    "sl_rate_diff_mean",
    "sl_rate_abs_diff_mean",

    "no_event_rate_diff_mean",
    "no_event_rate_abs_diff_mean",

    "tp_share_diff_mean",
    "tp_share_abs_diff_mean",

    "split_shift_score_mean",
    "split_shift_score_max",
]

split_change_target_cols = [
    "target_base",
    "percentile",
    "horizon",
    "n_configs",

    "tp_rate_diff_mean",
    "tp_rate_abs_diff_mean",

    "sl_rate_diff_mean",

```

```

        "sl_rate_abs_diff_mean",

        "no_event_rate_diff_mean",
        "no_event_rate_abs_diff_mean",

        "tp_share_diff_mean",
        "tp_share_abs_diff_mean",

        "split_shift_score_mean",
        "split_shift_score_max",
    ]

    yearly_config_cols = [
        "tp_sl_config",
        "tp_mult",
        "sl_mult",
        "n_targets",

        "tp_rate_range_mean",
        "sl_rate_range_mean",
        "no_event_rate_range_mean",
        "tp_share_range_mean",

        "expectancy_range_mean",
        "event_step_range_mean",

        "yearly_instability_score_mean",
        "yearly_instability_score_max",
    ]

    final_obs_cols = [
        "bar_col",
        "target_base",
        "tp_sl_config",

        f"tp_rate_{dev_label}",
        f"tp_rate_{test_label}",
        "tp_rate_diff_final_minus_dev",

        f"sl_rate_{dev_label}",
        f"sl_rate_{test_label}",
        "sl_rate_diff_final_minus_dev",

        f"no_event_rate_{dev_label}",
        f"no_event_rate_{test_label}",
        "no_event_rate_diff_final_minus_dev",

        f"tp_share_over_resolved_{dev_label}",
        f"tp_share_over_resolved_{test_label}",
        "tp_share_over_resolved_diff_final_minus_dev",

        "split_shift_score",
        "yearly_yearly_instability_score",
        "quarterly_quarterly_instability_score",
        "temporal_instability_score",
        "temporal_stability_profile",
    ]

    available_final_obs_cols = [
        col for col in final_obs_cols

```

```

    if col in df_bar_temporal_stability_final_table.columns
]

# -----
# Impresiones principales
# -----

print("\nResumen de cambio development vs final_test por configuración TP/SL:")
print(
    df_bar_split_change_by_config[split_change_config_cols]
    .to_string(
        index=False,
        formatters={
            "tp_rate_diff_mean": "{:.2%}".format,
            "tp_rate_abs_diff_mean": "{:.2%}".format,
            "sl_rate_diff_mean": "{:.2%}".format,
            "sl_rate_abs_diff_mean": "{:.2%}".format,
            "no_event_rate_diff_mean": "{:.2%}".format,
            "no_event_rate_abs_diff_mean": "{:.2%}".format,
            "tp_share_diff_mean": "{:.2%}".format,
            "tp_share_abs_diff_mean": "{:.2%}".format,
            "split_shift_score_mean": "{:.3f}".format,
            "split_shift_score_max": "{:.3f}".format,
        }
    )
)

print("\nResumen de cambio development vs final_test por percentil y horizonte:")
print(
    df_bar_split_change_by_target_base[split_change_target_cols]
    .to_string(
        index=False,
        formatters={
            "tp_rate_diff_mean": "{:.2%}".format,
            "tp_rate_abs_diff_mean": "{:.2%}".format,
            "sl_rate_diff_mean": "{:.2%}".format,
            "sl_rate_abs_diff_mean": "{:.2%}".format,
            "no_event_rate_diff_mean": "{:.2%}".format,
            "no_event_rate_abs_diff_mean": "{:.2%}".format,
            "tp_share_diff_mean": "{:.2%}".format,
            "tp_share_abs_diff_mean": "{:.2%}".format,
            "split_shift_score_mean": "{:.3f}".format,
            "split_shift_score_max": "{:.3f}".format,
        }
    )
)

print("\nEstabilidad anual agregada por configuración TP/SL:")
print(
    df_bar_yearly_stability_by_config[yearly_config_cols]
    .to_string(
        index=False,
        formatters={
            "tp_rate_range_mean": "{:.2%}".format,
            "sl_rate_range_mean": "{:.2%}".format,
            "no_event_rate_range_mean": "{:.2%}".format,
            "tp_share_range_mean": "{:.2%}".format,
            "expectancy_range_mean": "{:.3f}".format,
            "event_step_range_mean": "{:.2f}".format,
            "yearly_instability_score_mean": "{:.3f}".format,
        }
    )
)

```

```

        "yearly_instability_score_max": "{:.3f}".format,
    }
)
)

print("\nTabla final de estabilidad temporal – 63 targets BAR:")
print(
    df_bar_temporal_stability_final_table[available_final_obs_cols]
    .to_string(
        index=False,
        formatters={
            f"tp_rate_{dev_label}": "{:.2%}".format,
            f"tp_rate_{test_label}": "{:.2%}".format,
            "tp_rate_diff_final_minus_dev": "{:.2%}".format,

            f"sl_rate_{dev_label}": "{:.2%}".format,
            f"sl_rate_{test_label}": "{:.2%}".format,
            "sl_rate_diff_final_minus_dev": "{:.2%}".format,

            f"no_event_rate_{dev_label}": "{:.2%}".format,
            f"no_event_rate_{test_label}": "{:.2%}".format,
            "no_event_rate_diff_final_minus_dev": "{:.2%}".format,

            f"tp_share_over_resolved_{dev_label}": "{:.2%}".format,
            f"tp_share_over_resolved_{test_label}": "{:.2%}".format,
            "tp_share_over_resolved_diff_final_minus_dev": "{:.2%}".format,

            "split_shift_score": "{:.3f}".format,
            "yearly_yearly_instability_score": "{:.3f}".format,
            "quarterly_quarterly_instability_score": "{:.3f}".format,
            "temporal_instability_score": "{:.3f}".format,
        }
    )
)

# -----
# Top 15 más estables y más inestables
# -----

print("\nTop 15 targets temporalmente más estables:")
print(
    df_bar_temporal_stability_final_table[available_final_obs_cols]
    .head(15)
    .to_string(
        index=False,
        formatters={
            f"tp_rate_{dev_label}": "{:.2%}".format,
            f"tp_rate_{test_label}": "{:.2%}".format,
            "tp_rate_diff_final_minus_dev": "{:.2%}".format,

            f"sl_rate_{dev_label}": "{:.2%}".format,
            f"sl_rate_{test_label}": "{:.2%}".format,
            "sl_rate_diff_final_minus_dev": "{:.2%}".format,

            f"no_event_rate_{dev_label}": "{:.2%}".format,
            f"no_event_rate_{test_label}": "{:.2%}".format,
            "no_event_rate_diff_final_minus_dev": "{:.2%}".format,

            f"tp_share_over_resolved_{dev_label}": "{:.2%}".format,
            f"tp_share_over_resolved_{test_label}": "{:.2%}".format,

```

```

        "tp_share_over_resolved_diff_final_minus_dev": "{:.2%}".format,
        "split_shift_score": "{:.3f}".format,
        "yearly_yearly_instability_score": "{:.3f}".format,
        "quarterly_quarterly_instability_score": "{:.3f}".format,
        "temporal_instability_score": "{:.3f}".format,
    }
    )
)

print("\nTop 15 targets temporalmente más inestables:")
print(
    df_bar_temporal_stability_final_table[available_final_obs_cols]
    .tail(15)
    .sort_values("temporal_instability_score", ascending=False)
    .to_string(
        index=False,
        formatters={
            f"tp_rate_{dev_label}": "{:.2%}".format,
            f"tp_rate_{test_label}": "{:.2%}".format,
            "tp_rate_diff_final_minus_dev": "{:.2%}".format,

            f"sl_rate_{dev_label}": "{:.2%}".format,
            f"sl_rate_{test_label}": "{:.2%}".format,
            "sl_rate_diff_final_minus_dev": "{:.2%}".format,

            f"no_event_rate_{dev_label}": "{:.2%}".format,
            f"no_event_rate_{test_label}": "{:.2%}".format,
            "no_event_rate_diff_final_minus_dev": "{:.2%}".format,

            f"tp_share_over_resolved_{dev_label}": "{:.2%}".format,
            f"tp_share_over_resolved_{test_label}": "{:.2%}".format,
            "tp_share_over_resolved_diff_final_minus_dev": "{:.2%}".format,

            "split_shift_score": "{:.3f}".format,
            "yearly_yearly_instability_score": "{:.3f}".format,
            "quarterly_quarterly_instability_score": "{:.3f}".format,
            "temporal_instability_score": "{:.3f}".format,
        }
    )
)

# -----
# Conteo por perfil de estabilidad temporal
# -----

df_bar_temporal_stability_profile_counts = (
    df_bar_temporal_stability_final_table
    .groupby("temporal_stability_profile", as_index=False)
    .agg(
        n_targets=("bar_col", "count"),

        split_shift_score_mean=("split_shift_score", "mean"),
        yearly_instability_score_mean=("yearly_yearly_instability_score", "mean"),
        quarterly_instability_score_mean=("quarterly_quarterly_instability_score", "mean"),
        temporal_instability_score_mean=("temporal_instability_score", "mean"),

        tp_rate_diff_abs_mean=("tp_rate_abs_diff_final_vs_dev", "mean"),
        sl_rate_diff_abs_mean=("sl_rate_abs_diff_final_vs_dev", "mean"),
        no_event_rate_diff_abs_mean=("no_event_rate_abs_diff_final_vs_dev", "mea

```

```

        tp_share_diff_abs_mean=("tp_share_over_resolved_abs_diff_final_vs_dev",
    )
    .sort_values("temporal_instability_score_mean")
    .reset_index(drop=True)
)

print("\nConteo por perfil de estabilidad temporal:")
print(
    df_bar_temporal_stability_profile_counts
    .to_string(
        index=False,
        formatters={
            "split_shift_score_mean": "{:.3f}".format,
            "yearly_instability_score_mean": "{:.3f}".format,
            "quarterly_instability_score_mean": "{:.3f}".format,
            "temporal_instability_score_mean": "{:.3f}".format,
            "tp_rate_diff_abs_mean": "{:.2%}".format,
            "sl_rate_diff_abs_mean": "{:.2%}".format,
            "no_event_rate_diff_abs_mean": "{:.2%}".format,
            "tp_share_diff_abs_mean": "{:.2%}".format,
        }
    )
)

# -----
# Guardar reporte TXT
# -----

with MNQ_TARGETS_BAR_TEMPORAL_STABILITY_REPORT_TXT.open("w", encoding="utf-8") as f:
    f.write("4.9. Estabilidad temporal - BAR\n")
    f.write("=" * 100 + "\n\n")

    f.write("Resumen de cambio development vs final_test por configuración TP/SL")
    f.write("-" * 100 + "\n")
    f.write(
        df_bar_split_change_by_config[split_change_config_cols]
        .to_string(
            index=False,
            formatters={
                "tp_rate_diff_mean": "{:.2%}".format,
                "tp_rate_abs_diff_mean": "{:.2%}".format,
                "sl_rate_diff_mean": "{:.2%}".format,
                "sl_rate_abs_diff_mean": "{:.2%}".format,
                "no_event_rate_diff_mean": "{:.2%}".format,
                "no_event_rate_abs_diff_mean": "{:.2%}".format,
                "tp_share_diff_mean": "{:.2%}".format,
                "tp_share_abs_diff_mean": "{:.2%}".format,
                "split_shift_score_mean": "{:.3f}".format,
                "split_shift_score_max": "{:.3f}".format,
            }
        )
    )

    f.write("\n\n")
    f.write("Resumen de cambio development vs final_test por percentil y horizon")
    f.write("-" * 100 + "\n")
    f.write(
        df_bar_split_change_by_target_base[split_change_target_cols]
        .to_string(
            index=False,

```



```

        formatters={
            "tp_rate_diff_mean": "{:.2%}".format,
            "tp_rate_abs_diff_mean": "{:.2%}".format,
            "sl_rate_diff_mean": "{:.2%}".format,
            "sl_rate_abs_diff_mean": "{:.2%}".format,
            "no_event_rate_diff_mean": "{:.2%}".format,
            "no_event_rate_abs_diff_mean": "{:.2%}".format,
            "tp_share_diff_mean": "{:.2%}".format,
            "tp_share_abs_diff_mean": "{:.2%}".format,
            "split_shift_score_mean": "{:.3f}".format,
            "split_shift_score_max": "{:.3f}".format,
        }
    )
)

f.write("\n\n")
f.write("Estabilidad anual agregada por configuración TP/SL\n")
f.write("-" * 100 + "\n")
f.write(
    df_bar_yearly_stability_by_config[yearly_config_cols]
    .to_string(
        index=False,
        formatters={
            "tp_rate_range_mean": "{:.2%}".format,
            "sl_rate_range_mean": "{:.2%}".format,
            "no_event_rate_range_mean": "{:.2%}".format,
            "tp_share_range_mean": "{:.2%}".format,
            "expectancy_range_mean": "{:.3f}".format,
            "event_step_range_mean": "{:.2f}".format,
            "yearly_instability_score_mean": "{:.3f}".format,
            "yearly_instability_score_max": "{:.3f}".format,
        }
    )
)

f.write("\n\n")
f.write("Tabla final de estabilidad temporal – 63 targets BAR\n")
f.write("-" * 100 + "\n")
f.write(
    df_bar_temporal_stability_final_table[available_final_obs_cols]
    .to_string(
        index=False,
        formatters={
            f"tp_rate_{dev_label}": "{:.2%}".format,
            f"tp_rate_{test_label}": "{:.2%}".format,
            "tp_rate_diff_final_minus_dev": "{:.2%}".format,

            f"sl_rate_{dev_label}": "{:.2%}".format,
            f"sl_rate_{test_label}": "{:.2%}".format,
            "sl_rate_diff_final_minus_dev": "{:.2%}".format,

            f"no_event_rate_{dev_label}": "{:.2%}".format,
            f"no_event_rate_{test_label}": "{:.2%}".format,
            "no_event_rate_diff_final_minus_dev": "{:.2%}".format,

            f"tp_share_over_resolved_{dev_label}": "{:.2%}".format,
            f"tp_share_over_resolved_{test_label}": "{:.2%}".format,
            "tp_share_over_resolved_diff_final_minus_dev": "{:.2%}".format,

            "split_shift_score": "{:.3f}".format,

```

```

        "yearly_yearly_instability_score": "{:.3f}".format,
        "quarterly_quarterly_instability_score": "{:.3f}".format,
        "temporal_instability_score": "{:.3f}".format,
    }
    )
)

f.write("\n\n")
f.write("Conteo por perfil de estabilidad temporal\n")
f.write("-" * 100 + "\n")
f.write(
    df_bar_temporal_stability_profile_counts
    .to_string(
        index=False,
        formatters={
            "split_shift_score_mean": "{:.3f}".format,
            "yearly_instability_score_mean": "{:.3f}".format,
            "quarterly_instability_score_mean": "{:.3f}".format,
            "temporal_instability_score_mean": "{:.3f}".format,
            "tp_rate_diff_abs_mean": "{:.2%}".format,
            "sl_rate_diff_abs_mean": "{:.2%}".format,
            "no_event_rate_diff_abs_mean": "{:.2%}".format,
            "tp_share_diff_abs_mean": "{:.2%}".format,
        }
    )
)

# -----
# Confirmación final
# -----

print("\nArchivos guardados:")
print(f"Estabilidad temporal completa: {MNQ_TARGETS_BAR_TEMPORAL_STABILI")
print(f"Estabilidad por split: {MNQ_TARGETS_BAR_STABILITY_SPLIT_")
print(f"Estabilidad por año: {MNQ_TARGETS_BAR_STABILITY_YEAR_C")
print(f"Estabilidad por trimestre: {MNQ_TARGETS_BAR_STABILITY_QUARTE")
print(f"Comparación development vs final_test: {MNQ_TARGETS_BAR_SPLIT_COMPARISON")
print(f"Cambio por configuración TP/SL: {MNQ_TARGETS_BAR_SPLIT_CHANGE_BY_")
print(f"Cambio por percentil y horizonte: {MNQ_TARGETS_BAR_SPLIT_CHANGE_BY_")
print(f"Estabilidad anual por target: {MNQ_TARGETS_BAR_YEARLY_STABILITY")
print(f"Estabilidad anual por configuración: {MNQ_TARGETS_BAR_YEARLY_STABILITY")
print(f"Estabilidad trimestral por target: {MNQ_TARGETS_BAR_QUARTERLY_STABIL")
print(f"Tabla final estabilidad temporal: {MNQ_TARGETS_BAR_TEMPORAL_STABILI")
print(f"Reporte TXT: {MNQ_TARGETS_BAR_TEMPORAL_STABILI")

print("\nCelda 4 finalizada correctamente.")

```

## Guardado y tabla final de estabilidad temporal - BAR

=====

Resumen de cambio development vs final\_test por configuración TP/SL:

| tp_sl_config    | tp_mult | sl_mult | n_targets | tp_rate_diff_mean | tp_rate_abs_diff_mean | sl_rate_diff_mean | sl_rate_abs_diff_mean | no_event_rate_diff_mean | no_event_rate_abs_diff_mean | tp_share_diff_mean | tp_share_abs_diff_mean | split_shift_score_mean | split_shift_score_max |
|-----------------|---------|---------|-----------|-------------------|-----------------------|-------------------|-----------------------|-------------------------|-----------------------------|--------------------|------------------------|------------------------|-----------------------|
| TP 1.0 / SL 0.5 | 1.0     | 0.5     | 9         | -8.70%            | 8.70%                 | 0.00%             | 0.00%                 | 0.261                   | 0.285                       |                    |                        |                        |                       |
| 70%             | 8.70%   |         | 8.70%     | 0.00%             |                       |                   |                       |                         |                             |                    |                        |                        |                       |
| 0.00%           | -8.70%  |         | 8.70%     |                   |                       |                   |                       |                         |                             |                    |                        |                        |                       |
| TP 1.5 / SL 0.5 | 1.5     | 0.5     | 9         | 4.96%             | 9.99%                 | -14.95%           | 0.367                 | 0.432                   |                             |                    |                        |                        |                       |
| 96%             | 9.99%   |         | 9.99%     |                   |                       |                   |                       |                         |                             |                    |                        |                        |                       |
| 14.95%          | -6.79%  |         | 6.79%     |                   |                       |                   |                       |                         |                             |                    |                        |                        |                       |
| TP 2.0 / SL 0.5 | 2.0     | 0.5     | 9         | 8.54%             | 11.22%                | -19.76%           | 0.422                 | 0.463                   |                             |                    |                        |                        |                       |
| 54%             | 11.22%  |         | 11.22%    |                   |                       |                   |                       |                         |                             |                    |                        |                        |                       |
| 19.76%          | -2.71%  |         | 2.71%     |                   |                       |                   |                       |                         |                             |                    |                        |                        |                       |
| TP 1.0 / SL 1.0 | 1.0     | 1.0     | 9         | -4.48%            | 4.48%                 | 0.00%             | 0.134                 | 0.166                   |                             |                    |                        |                        |                       |
| 48%             | 4.48%   |         | 4.48%     |                   |                       |                   |                       |                         |                             |                    |                        |                        |                       |
| 0.00%           | -4.48%  |         | 4.48%     |                   |                       |                   |                       |                         |                             |                    |                        |                        |                       |
| TP 1.5 / SL 1.0 | 1.5     | 1.0     | 9         | 9.68%             | 5.20%                 | -14.88%           | 0.355                 | 0.374                   |                             |                    |                        |                        |                       |
| 68%             | 5.20%   |         | 5.20%     |                   |                       |                   |                       |                         |                             |                    |                        |                        |                       |
| 14.88%          | -5.72%  |         | 5.72%     |                   |                       |                   |                       |                         |                             |                    |                        |                        |                       |
| TP 2.0 / SL 1.0 | 2.0     | 1.0     | 9         | 12.52%            | 5.98%                 | -18.50%           | 0.430                 | 0.455                   |                             |                    |                        |                        |                       |
| 52%             | 5.98%   |         | 5.98%     |                   |                       |                   |                       |                         |                             |                    |                        |                        |                       |
| 18.50%          | -6.03%  |         | 6.03%     |                   |                       |                   |                       |                         |                             |                    |                        |                        |                       |
| TP 2.5 / SL 1.0 | 2.5     | 1.0     | 9         | 11.24%            | 6.50%                 | -17.74%           | 0.403                 | 0.452                   |                             |                    |                        |                        |                       |
| 24%             | 6.50%   |         | 6.50%     |                   |                       |                   |                       |                         |                             |                    |                        |                        |                       |
| 17.74%          | -4.85%  |         | 4.85%     |                   |                       |                   |                       |                         |                             |                    |                        |                        |                       |

Resumen de cambio development vs final\_test por percentil y horizonte:

| target_base | percentile | horizon | n_configs | tp_rate_diff_mean | tp_rate_abs_diff_mean | sl_rate_diff_mean | sl_rate_abs_diff_mean | no_event_rate_diff_mean | no_event_rate_abs_diff_mean | tp_share_diff_mean | tp_share_abs_diff_mean | split_shift_score_mean | split_shift_score_max |
|-------------|------------|---------|-----------|-------------------|-----------------------|-------------------|-----------------------|-------------------------|-----------------------------|--------------------|------------------------|------------------------|-----------------------|
| p40_h30     | 40         | 30      | 7         | 4.35%             | 7.15%                 | -11.50%           | 0.318                 | 0.423                   |                             |                    |                        |                        |                       |
| 5%          | 7.15%      |         | 7.15%     |                   |                       |                   |                       |                         |                             |                    |                        |                        |                       |
| 11.50%      | -5.18%     |         | 5.18%     |                   |                       |                   |                       |                         |                             |                    |                        |                        |                       |
| p40_h60     | 40         | 60      | 7         | 3.93%             | 8.10%                 | -12.03%           | 0.342                 | 0.451                   |                             |                    |                        |                        |                       |
| 3%          | 8.10%      |         | 8.10%     |                   |                       |                   |                       |                         |                             |                    |                        |                        |                       |
| 12.03%      | -6.06%     |         | 6.06%     |                   |                       |                   |                       |                         |                             |                    |                        |                        |                       |
| p40_h90     | 40         | 90      | 7         | 3.90%             | 8.39%                 | -12.29%           | 0.349                 | 0.455                   |                             |                    |                        |                        |                       |
| 6%          | 8.39%      |         | 8.39%     |                   |                       |                   |                       |                         |                             |                    |                        |                        |                       |
| 12.29%      | -6.09%     |         | 6.09%     |                   |                       |                   |                       |                         |                             |                    |                        |                        |                       |
| p50_h30     | 50         | 30      | 7         | 4.96%             | 6.81%                 | -11.77%           | 0.321                 | 0.416                   |                             |                    |                        |                        |                       |
| 0%          | 6.81%      |         | 6.81%     |                   |                       |                   |                       |                         |                             |                    |                        |                        |                       |
| 11.77%      | -5.15%     |         | 5.15%     |                   |                       |                   |                       |                         |                             |                    |                        |                        |                       |
| p50_h60     | 50         | 60      | 7         | 4.92%             | 7.82%                 | -12.74%           |                       |                         |                             |                    |                        |                        |                       |
| 6%          | 7.82%      |         | 7.82%     |                   |                       |                   |                       |                         |                             |                    |                        |                        |                       |

|         |        |    |       |         |     |
|---------|--------|----|-------|---------|-----|
| 12.74%  | -6.05% |    | 6.05% | 0.355   |     |
| 0.455   |        |    |       |         |     |
| p50_h90 | 50     | 90 | 7     | 4.92%   | 8.9 |
| 6%      | 7.95%  |    | 7.95% | -12.87% |     |
| 12.87%  | -5.82% |    | 5.82% | 0.356   |     |
| 0.451   |        |    |       |         |     |
| p60_h30 | 60     | 30 | 7     | 5.40%   | 8.6 |
| 4%      | 6.42%  |    | 6.42% | -11.81% |     |
| 11.81%  | -5.13% |    | 5.13% | 0.320   |     |
| 0.416   |        |    |       |         |     |
| p60_h60 | 60     | 60 | 7     | 5.56%   | 9.2 |
| 1%      | 7.14%  |    | 7.14% | -12.71% |     |
| 12.71%  | -5.66% |    | 5.66% | 0.347   |     |
| 0.463   |        |    |       |         |     |
| p60_h90 | 60     | 90 | 7     | 5.46%   | 9.1 |
| 0%      | 7.15%  |    | 7.15% | -12.62% |     |
| 12.62%  | -5.37% |    | 5.37% | 0.342   |     |
| 0.457   |        |    |       |         |     |

Estabilidad anual agregada por configuración TP/SL:

| tp_sl_config    | tp_mult | sl_mult | n_targets | tp_rate_range_mean | sl_rate_range_mean | n    | no_event_rate_range_mean | tp_share_range_mean | expectancy_range_mean | event_step_range_mean | yearly_instability_score_mean | yearly_instability_score_max |
|-----------------|---------|---------|-----------|--------------------|--------------------|------|--------------------------|---------------------|-----------------------|-----------------------|-------------------------------|------------------------------|
| TP 1.0 / SL 0.5 | 1.0     | 0.5     | 9         | 17.53%             |                    | 17.5 |                          |                     |                       |                       |                               |                              |
| 3%              | 0.00%   |         | 17.53%    |                    | 0.263              |      |                          |                     |                       |                       |                               |                              |
| 13.25           |         | 0.526   |           |                    | 0.587              |      |                          |                     |                       |                       |                               |                              |
| TP 1.5 / SL 0.5 | 1.5     | 0.5     | 9         | 11.64%             |                    | 20.2 |                          |                     |                       |                       |                               |                              |
| 8%              | 31.88%  |         | 14.18%    |                    | 0.086              |      |                          |                     |                       |                       |                               |                              |
| 11.59           |         | 0.780   |           |                    | 0.886              |      |                          |                     |                       |                       |                               |                              |
| TP 2.0 / SL 0.5 | 2.0     | 0.5     | 9         | 16.63%             |                    | 22.7 |                          |                     |                       |                       |                               |                              |
| 0%              | 39.32%  |         | 7.09%     |                    | 0.219              |      |                          |                     |                       |                       |                               |                              |
| 8.16            |         | 0.857   |           |                    | 0.906              |      |                          |                     |                       |                       |                               |                              |
| TP 1.0 / SL 1.0 | 1.0     | 1.0     | 9         | 7.78%              |                    | 7.7  |                          |                     |                       |                       |                               |                              |
| 8%              | 0.00%   |         | 7.78%     |                    | 0.156              |      |                          |                     |                       |                       |                               |                              |
| 11.20           |         | 0.233   |           |                    | 0.300              |      |                          |                     |                       |                       |                               |                              |
| TP 1.5 / SL 1.0 | 1.5     | 1.0     | 9         | 21.27%             |                    | 9.1  |                          |                     |                       |                       |                               |                              |
| 2%              | 30.39%  |         | 9.82%     |                    | 0.228              |      |                          |                     |                       |                       |                               |                              |
| 10.56           |         | 0.706   |           |                    | 0.751              |      |                          |                     |                       |                       |                               |                              |
| TP 2.0 / SL 1.0 | 2.0     | 1.0     | 9         | 24.07%             |                    | 10.5 |                          |                     |                       |                       |                               |                              |
| 7%              | 34.64%  |         | 11.08%    |                    | 0.376              |      |                          |                     |                       |                       |                               |                              |
| 8.79            |         | 0.804   |           |                    | 0.918              |      |                          |                     |                       |                       |                               |                              |
| TP 2.5 / SL 1.0 | 2.5     | 1.0     | 9         | 19.40%             |                    | 11.5 |                          |                     |                       |                       |                               |                              |
| 0%              | 30.90%  |         | 11.17%    |                    | 0.371              |      |                          |                     |                       |                       |                               |                              |
| 7.39            |         | 0.730   |           |                    | 0.875              |      |                          |                     |                       |                       |                               |                              |

Tabla final de estabilidad temporal – 63 targets BAR:

| bar_col               | target_base | tp_sl_config    | tp_rate_development_2020_2024 | tp_rate_final_test_2025_2026 | tp_rate_diff_final_minus_dev | sl_rate_development_2020_2024 | sl_rate_final_test_2025_2026 | sl_rate_diff_final_minus_dev | no_event_rate_development_2020_2024 | no_event_rate_final_test_2025_2026 | no_event_rate_diff_final_minus_dev | tp_share_over_resolved_development_2020_2024 | tp_share_over_resolved_final_test_2025_2026 | tp_share_over_resolved_diff_final_minus_dev | split_shift_score | yearly_yearly_instability_score | quarterly_quarterly_instability_score | temporal_instability_score | temporal_stability_profile |
|-----------------------|-------------|-----------------|-------------------------------|------------------------------|------------------------------|-------------------------------|------------------------------|------------------------------|-------------------------------------|------------------------------------|------------------------------------|----------------------------------------------|---------------------------------------------|---------------------------------------------|-------------------|---------------------------------|---------------------------------------|----------------------------|----------------------------|
| bar_p60_h60_tp10_sl10 | p60_h60     | TP 1.0 / SL 1.0 |                               |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             |                   |                                 |                                       |                            | 97.60%                     |
| 94.05%                |             | -3.55%          |                               |                              |                              | 2.40%                         |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             |                   |                                 |                                       |                            |                            |
| 5.95%                 |             | 3.55%           |                               |                              |                              |                               |                              | 0.00%                        |                                     |                                    |                                    |                                              |                                             |                                             |                   |                                 |                                       |                            |                            |
| 0.00%                 |             |                 | 0.00%                         |                              |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             |                   |                                 |                                       |                            | 9                          |
| 7.60%                 |             |                 |                               | 94.05%                       |                              |                               |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             |                   |                                 |                                       |                            |                            |
| -3.55%                | 0.107       |                 |                               |                              |                              | 0.167                         |                              |                              |                                     |                                    |                                    |                                              |                                             |                                             |                   |                                 |                                       |                            |                            |

|                       |                         |             |        |
|-----------------------|-------------------------|-------------|--------|
| 0.212                 | 0.486                   | más_estable |        |
| bar_p60_h90_tp10_sl10 | p60_h90 TP 1.0 / SL 1.0 |             | 97.33% |
| 93.83%                | -3.50%                  | 2.67%       |        |
| 6.17%                 | 3.50%                   | 0.00%       |        |
| 0.00%                 | 0.00%                   |             | 9      |
| 7.33%                 |                         | 93.83%      |        |
| -3.50%                | 0.105                   | 0.166       |        |
| 0.220                 | 0.491                   | más_estable |        |
| bar_p60_h30_tp10_sl10 | p60_h30 TP 1.0 / SL 1.0 |             | 97.39% |
| 93.82%                | -3.56%                  | 2.61%       |        |
| 6.18%                 | 3.56%                   | 0.00%       |        |
| 0.00%                 | 0.00%                   |             | 9      |
| 7.39%                 |                         | 93.82%      |        |
| -3.56%                | 0.107                   | 0.173       |        |
| 0.217                 | 0.497                   | más_estable |        |
| bar_p50_h30_tp10_sl10 | p50_h30 TP 1.0 / SL 1.0 |             | 96.38% |
| 92.04%                | -4.34%                  | 3.62%       |        |
| 7.96%                 | 4.34%                   | 0.00%       |        |
| 0.00%                 | 0.00%                   |             | 9      |
| 6.38%                 |                         | 92.04%      |        |
| -4.34%                | 0.130                   | 0.232       |        |
| 0.286                 | 0.648                   | más_estable |        |
| bar_p50_h90_tp10_sl10 | p50_h90 TP 1.0 / SL 1.0 |             | 96.31% |
| 91.68%                | -4.63%                  | 3.69%       |        |
| 8.32%                 | 4.63%                   | 0.00%       |        |
| 0.00%                 | 0.00%                   |             | 9      |
| 6.31%                 |                         | 91.68%      |        |
| -4.63%                | 0.139                   | 0.234       |        |
| 0.281                 | 0.654                   | más_estable |        |
| bar_p50_h60_tp10_sl10 | p50_h60 TP 1.0 / SL 1.0 |             | 96.67% |
| 91.99%                | -4.67%                  | 3.33%       |        |
| 8.01%                 | 4.67%                   | 0.00%       |        |
| 0.00%                 | 0.00%                   |             | 9      |
| 6.67%                 |                         | 91.99%      |        |
| -4.67%                | 0.140                   | 0.238       |        |
| 0.289                 | 0.667                   | más_estable |        |
| bar_p40_h30_tp10_sl10 | p40_h30 TP 1.0 / SL 1.0 |             | 95.09% |
| 89.97%                | -5.12%                  | 4.91%       |        |
| 10.03%                | 5.12%                   | 0.00%       |        |
| 0.00%                 | 0.00%                   |             | 9      |
| 5.09%                 |                         | 89.97%      |        |
| -5.12%                | 0.154                   | 0.291       |        |
| 0.353                 | 0.798                   | más_estable |        |
| bar_p40_h90_tp10_sl10 | p40_h90 TP 1.0 / SL 1.0 |             | 94.92% |
| 89.40%                | -5.52%                  | 5.08%       |        |
| 10.60%                | 5.52%                   | 0.00%       |        |
| 0.00%                 | 0.00%                   |             | 9      |
| 4.92%                 |                         | 89.40%      |        |
| -5.52%                | 0.166                   | 0.298       |        |
| 0.355                 | 0.819                   | más_estable |        |
| bar_p40_h60_tp10_sl10 | p40_h60 TP 1.0 / SL 1.0 |             | 95.34% |
| 89.90%                | -5.44%                  | 4.66%       |        |
| 10.10%                | 5.44%                   | 0.00%       |        |
| 0.00%                 | 0.00%                   |             | 9      |
| 5.34%                 |                         | 89.90%      |        |
| -5.44%                | 0.163                   | 0.300       |        |
| 0.365                 | 0.828                   | más_estable |        |
| bar_p60_h30_tp10_sl05 | p60_h30 TP 1.0 / SL 0.5 |             | 86.92% |
| 79.12%                | -7.79%                  | 13.08%      |        |
| 20.88%                | 7.79%                   | 0.00%       |        |

|                       |       |                         |             |        |
|-----------------------|-------|-------------------------|-------------|--------|
| 0.00%                 |       | 0.00%                   |             | 8      |
| 6.92%                 |       |                         | 79.12%      |        |
| -7.79%                | 0.234 |                         | 0.464       |        |
| 0.597                 |       | 1.295                   | más_estable |        |
| bar_p40_h30_tp10_sl05 |       | p40_h30 TP 1.0 / SL 0.5 |             | 80.81% |
| 73.31%                |       | -7.50%                  | 19.19%      |        |
| 26.69%                |       | 7.50%                   | 0.00%       |        |
| 0.00%                 |       | 0.00%                   |             | 8      |
| 0.81%                 |       |                         | 73.31%      |        |
| -7.50%                | 0.225 |                         | 0.468       |        |
| 0.607                 |       | 1.301                   | más_estable |        |
| bar_p50_h30_tp10_sl05 |       | p50_h30 TP 1.0 / SL 0.5 |             | 83.91% |
| 76.22%                |       | -7.69%                  | 16.09%      |        |
| 23.78%                |       | 7.69%                   | 0.00%       |        |
| 0.00%                 |       | 0.00%                   |             | 8      |
| 3.91%                 |       |                         | 76.22%      |        |
| -7.69%                | 0.231 |                         | 0.473       |        |
| 0.611                 |       | 1.315                   | más_estable |        |
| bar_p60_h90_tp10_sl05 |       | p60_h90 TP 1.0 / SL 0.5 |             | 85.74% |
| 76.53%                |       | -9.21%                  | 14.26%      |        |
| 23.47%                |       | 9.21%                   | 0.00%       |        |
| 0.00%                 |       | 0.00%                   |             | 8      |
| 5.74%                 |       |                         | 76.53%      |        |
| -9.21%                | 0.276 |                         | 0.515       |        |
| 0.606                 |       | 1.397                   | más_estable |        |
| bar_p60_h60_tp10_sl05 |       | p60_h60 TP 1.0 / SL 0.5 |             | 86.72% |
| 77.53%                |       | -9.19%                  | 13.28%      |        |
| 22.47%                |       | 9.19%                   | 0.00%       |        |
| 0.00%                 |       | 0.00%                   |             | 8      |
| 6.72%                 |       |                         | 77.53%      |        |
| -9.19%                | 0.276 |                         | 0.535       |        |
| 0.685                 |       | 1.496                   | más_estable |        |
| bar_p50_h90_tp10_sl05 |       | p50_h90 TP 1.0 / SL 0.5 |             | 82.68% |
| 73.19%                |       | -9.50%                  | 17.32%      |        |
| 26.81%                |       | 9.50%                   | 0.00%       |        |
| 0.00%                 |       | 0.00%                   |             | 8      |
| 2.68%                 |       |                         | 73.19%      |        |
| -9.50%                | 0.285 |                         | 0.567       |        |
| 0.694                 |       | 1.547                   | más_estable |        |
| bar_p40_h60_tp10_sl05 |       | p40_h60 TP 1.0 / SL 0.5 |             | 80.13% |
| 71.23%                |       | -8.90%                  | 19.87%      |        |
| 28.77%                |       | 8.90%                   | 0.00%       |        |
| 0.00%                 |       | 0.00%                   |             | 8      |
| 0.13%                 |       |                         | 71.23%      |        |
| -8.90%                | 0.267 |                         | 0.559       |        |
| 0.727                 |       | 1.553                   | más_estable |        |
| bar_p50_h60_tp10_sl05 |       | p50_h60 TP 1.0 / SL 0.5 |             | 83.68% |
| 74.57%                |       | -9.11%                  | 16.32%      |        |
| 25.43%                |       | 9.11%                   | 0.00%       |        |
| 0.00%                 |       | 0.00%                   |             | 8      |
| 3.68%                 |       |                         | 74.57%      |        |
| -9.11%                | 0.273 |                         | 0.563       |        |
| 0.721                 |       | 1.557                   | más_estable |        |
| bar_p40_h90_tp10_sl05 |       | p40_h90 TP 1.0 / SL 0.5 |             | 79.41% |
| 70.00%                |       | -9.41%                  | 20.59%      |        |
| 30.00%                |       | 9.41%                   | 0.00%       |        |
| 0.00%                 |       | 0.00%                   |             | 7      |
| 9.41%                 |       |                         | 70.00%      |        |
| -9.41%                | 0.282 |                         | 0.587       |        |
| 0.735                 |       | 1.605                   | más_estable |        |

|                       |                         |                        |        |
|-----------------------|-------------------------|------------------------|--------|
| bar_p60_h90_tp25_sl10 | p60_h90 TP 2.5 / SL 1.0 |                        | 11.61% |
| 21.71%                | 10.10%                  | 3.98%                  |        |
| 9.18%                 | 5.20%                   | 84.41%                 |        |
| 69.11%                | -15.30%                 |                        | 7      |
| 4.48%                 |                         | 70.29%                 |        |
| -4.19%                | 0.348                   | 0.577                  |        |
| 0.869                 | 1.794                   | más_estable            |        |
| bar_p60_h30_tp25_sl10 | p60_h30 TP 2.5 / SL 1.0 |                        | 12.22% |
| 22.59%                | 10.37%                  | 3.80%                  |        |
| 8.94%                 | 5.14%                   | 83.98%                 |        |
| 68.47%                | -15.51%                 |                        | 7      |
| 6.29%                 |                         | 71.64%                 |        |
| -4.65%                | 0.357                   | 0.621                  |        |
| 0.829                 | 1.806                   | más_estable            |        |
| bar_p60_h60_tp25_sl10 | p60_h60 TP 2.5 / SL 1.0 |                        | 11.38% |
| 21.75%                | 10.37%                  | 3.52%                  |        |
| 8.78%                 | 5.26%                   | 85.10%                 |        |
| 69.48%                | -15.63%                 |                        | 7      |
| 6.38%                 |                         | 71.25%                 |        |
| -5.13%                | 0.364                   | 0.608                  |        |
| 0.850                 | 1.822                   | más_estable            |        |
| bar_p40_h30_tp15_sl05 | p40_h30 TP 1.5 / SL 0.5 |                        | 47.11% |
| 51.32%                | 4.21%                   | 23.00%                 |        |
| 31.45%                | 8.44%                   | 29.88%                 |        |
| 17.23%                | -12.65%                 |                        | 6      |
| 7.19%                 |                         | 62.01%                 |        |
| -5.19%                | 0.305                   | 0.662                  |        |
| 0.896                 | 1.863                   | estabilidad_intermedia |        |
| bar_p40_h30_tp15_sl10 | p40_h30 TP 1.5 / SL 1.0 |                        | 55.62% |
| 63.39%                | 7.77%                   | 5.95%                  |        |
| 11.69%                | 5.73%                   | 38.42%                 |        |
| 24.92%                | -13.50%                 |                        | 9      |
| 0.33%                 |                         | 84.43%                 |        |
| -5.90%                | 0.329                   | 0.692                  |        |
| 0.877                 | 1.898                   | estabilidad_intermedia |        |
| bar_p60_h30_tp15_sl10 | p60_h30 TP 1.5 / SL 1.0 |                        | 46.59% |
| 57.29%                | 10.70%                  | 3.22%                  |        |
| 7.41%                 | 4.19%                   | 50.19%                 |        |
| 35.30%                | -14.89%                 |                        | 9      |
| 3.54%                 |                         | 88.55%                 |        |
| -4.99%                | 0.348                   | 0.674                  |        |
| 0.888                 | 1.910                   | estabilidad_intermedia |        |
| bar_p50_h30_tp15_sl10 | p50_h30 TP 1.5 / SL 1.0 |                        | 51.18% |
| 60.55%                | 9.37%                   | 4.44%                  |        |
| 9.35%                 | 4.91%                   | 44.37%                 |        |
| 30.10%                | -14.28%                 |                        | 9      |
| 2.01%                 |                         | 86.63%                 |        |
| -5.39%                | 0.339                   | 0.692                  |        |
| 0.890                 | 1.921                   | estabilidad_intermedia |        |
| bar_p60_h90_tp15_sl10 | p60_h90 TP 1.5 / SL 1.0 |                        | 45.45% |
| 57.50%                | 12.06%                  | 3.34%                  |        |
| 7.59%                 | 4.24%                   | 51.21%                 |        |
| 34.91%                | -16.30%                 |                        | 9      |
| 3.14%                 |                         | 88.34%                 |        |
| -4.80%                | 0.374                   | 0.659                  |        |
| 0.936                 | 1.969                   | estabilidad_intermedia |        |
| bar_p40_h60_tp15_sl10 | p40_h60 TP 1.5 / SL 1.0 |                        | 54.44% |
| 62.01%                | 7.57%                   | 5.78%                  |        |
| 12.01%                | 6.22%                   | 39.78%                 |        |
| 25.99%                | -13.79%                 |                        | 9      |

|                       |       |                         |                        |        |
|-----------------------|-------|-------------------------|------------------------|--------|
| 0.40%                 |       |                         | 83.78%                 |        |
| -6.62%                | 0.342 |                         | 0.734                  |        |
| 0.936                 |       | 2.013                   | estabilidad_intermedia |        |
| bar_p50_h90_tp15_sl10 |       | p50_h90 TP 1.5 / SL 1.0 |                        | 49.72% |
| 60.00%                |       | 10.28%                  | 4.53%                  |        |
| 9.93%                 |       | 5.40%                   | 45.74%                 |        |
| 30.07%                |       | -15.68%                 |                        | 9      |
| 1.65%                 |       |                         | 85.80%                 |        |
| -5.85%                | 0.372 |                         | 0.712                  |        |
| 0.937                 |       | 2.021                   | estabilidad_intermedia |        |
| bar_p40_h90_tp15_sl10 |       | p40_h90 TP 1.5 / SL 1.0 |                        | 54.07% |
| 61.86%                |       | 7.79%                   | 6.18%                  |        |
| 12.61%                |       | 6.43%                   | 39.75%                 |        |
| 25.53%                |       | -14.22%                 |                        | 8      |
| 9.74%                 |       |                         | 83.06%                 |        |
| -6.68%                | 0.351 |                         | 0.721                  |        |
| 0.952                 |       | 2.023                   | estabilidad_intermedia |        |
| bar_p60_h90_tp20_sl10 |       | p60_h90 TP 2.0 / SL 1.0 |                        | 22.27% |
| 35.38%                |       | 13.11%                  | 3.78%                  |        |
| 8.64%                 |       | 4.86%                   | 73.95%                 |        |
| 55.98%                |       | -17.96%                 |                        | 8      |
| 5.49%                 |       |                         | 80.38%                 |        |
| -5.11%                | 0.410 |                         | 0.671                  |        |
| 0.956                 |       | 2.037                   | estabilidad_intermedia |        |
| bar_p40_h60_tp15_sl05 |       | p40_h60 TP 1.5 / SL 0.5 |                        | 45.72% |
| 48.69%                |       | 2.97%                   | 23.99%                 |        |
| 33.90%                |       | 9.91%                   | 30.28%                 |        |
| 17.40%                |       | -12.88%                 |                        | 6      |
| 5.58%                 |       |                         | 58.95%                 |        |
| -6.63%                | 0.324 |                         | 0.725                  |        |
| 1.002                 |       | 2.051                   | estabilidad_intermedia |        |
| bar_p60_h30_tp20_sl10 |       | p60_h30 TP 2.0 / SL 1.0 |                        | 23.39% |
| 35.76%                |       | 12.37%                  | 3.62%                  |        |
| 8.44%                 |       | 4.82%                   | 72.99%                 |        |
| 55.81%                |       | -17.18%                 |                        | 8      |
| 6.59%                 |       |                         | 80.91%                 |        |
| -5.68%                | 0.401 |                         | 0.718                  |        |
| 0.941                 |       | 2.060                   | estabilidad_intermedia |        |
| bar_p50_h30_tp15_sl05 |       | p50_h30 TP 1.5 / SL 0.5 |                        | 44.36% |
| 49.70%                |       | 5.34%                   | 19.40%                 |        |
| 28.20%                |       | 8.80%                   | 36.24%                 |        |
| 22.10%                |       | -14.14%                 |                        | 6      |
| 9.57%                 |       |                         | 63.80%                 |        |
| -5.77%                | 0.340 |                         | 0.739                  |        |
| 0.983                 |       | 2.062                   | estabilidad_intermedia |        |
| bar_p60_h60_tp15_sl10 |       | p60_h60 TP 1.5 / SL 1.0 |                        | 45.04% |
| 56.79%                |       | 11.75%                  | 2.98%                  |        |
| 7.21%                 |       | 4.23%                   | 51.99%                 |        |
| 36.01%                |       | -15.98%                 |                        | 9      |
| 3.80%                 |       |                         | 88.74%                 |        |
| -5.07%                | 0.370 |                         | 0.719                  |        |
| 0.979                 |       | 2.069                   | estabilidad_intermedia |        |
| bar_p50_h90_tp25_sl10 |       | p50_h90 TP 2.5 / SL 1.0 |                        | 14.66% |
| 26.21%                |       | 11.55%                  | 5.43%                  |        |
| 12.15%                |       | 6.73%                   | 79.91%                 |        |
| 61.63%                |       | -18.28%                 |                        | 7      |
| 2.98%                 |       |                         | 68.32%                 |        |
| -4.66%                | 0.412 |                         | 0.711                  |        |
| 0.952                 |       | 2.076                   | estabilidad_intermedia |        |
| bar_p40_h90_tp15_sl05 |       | p40_h90 TP 1.5 / SL 0.5 |                        | 44.83% |



|                       |       |                         |                        |        |   |
|-----------------------|-------|-------------------------|------------------------|--------|---|
| 47.64%                |       | 2.81%                   |                        | 24.77% |   |
| 35.15%                |       | 10.38%                  |                        | 30.40% |   |
| 17.21%                |       | -13.19%                 |                        |        | 6 |
| 4.41%                 |       |                         | 57.55%                 |        |   |
| -6.86%                | 0.332 |                         | 0.737                  |        |   |
| 1.007                 |       | 2.076                   | estabilidad_intermedia |        |   |
| bar_p50_h30_tp25_sl10 |       | p50_h30 TP 2.5 / SL 1.0 |                        | 15.76% |   |
| 26.93%                |       | 11.17%                  |                        | 5.30%  |   |
| 11.52%                |       | 6.22%                   |                        | 78.94% |   |
| 61.55%                |       | -17.39%                 |                        |        | 7 |
| 4.83%                 |       |                         | 70.04%                 |        |   |
| -4.80%                | 0.396 |                         | 0.736                  |        |   |
| 0.952                 |       | 2.084                   | estabilidad_intermedia |        |   |
| bar_p50_h60_tp15_sl10 |       | p50_h60 TP 1.5 / SL 1.0 |                        | 49.69% |   |
| 59.50%                |       | 9.80%                   |                        | 4.12%  |   |
| 9.59%                 |       | 5.47%                   |                        | 46.19% |   |
| 30.91%                |       | -15.28%                 |                        |        | 9 |
| 2.34%                 |       |                         | 86.12%                 |        |   |
| -6.23%                | 0.368 |                         | 0.751                  |        |   |
| 0.972                 |       | 2.091                   | estabilidad_intermedia |        |   |
| bar_p60_h60_tp20_sl10 |       | p60_h60 TP 2.0 / SL 1.0 |                        | 21.84% |   |
| 35.07%                |       | 13.23%                  |                        | 3.37%  |   |
| 8.26%                 |       | 4.88%                   |                        | 74.79% |   |
| 56.68%                |       | -18.11%                 |                        |        | 8 |
| 6.62%                 |       |                         | 80.94%                 |        |   |
| -5.68%                | 0.419 |                         | 0.715                  |        |   |
| 0.967                 |       | 2.101                   | estabilidad_intermedia |        |   |
| bar_p50_h60_tp25_sl10 |       | p50_h60 TP 2.5 / SL 1.0 |                        | 14.28% |   |
| 26.05%                |       | 11.76%                  |                        | 4.91%  |   |
| 11.76%                |       | 6.84%                   |                        | 80.80% |   |
| 62.20%                |       | -18.60%                 |                        |        | 7 |
| 4.41%                 |       |                         | 68.90%                 |        |   |
| -5.50%                | 0.427 |                         | 0.760                  |        |   |
| 0.985                 |       | 2.172                   | estabilidad_intermedia |        |   |
| bar_p40_h30_tp20_sl05 |       | p40_h30 TP 2.0 / SL 0.5 |                        | 27.75% |   |
| 35.68%                |       | 7.94%                   |                        | 24.59% |   |
| 34.20%                |       | 9.61%                   |                        | 47.67% |   |
| 30.11%                |       | -17.55%                 |                        |        | 5 |
| 3.01%                 |       |                         | 51.06%                 |        |   |
| -1.96%                | 0.371 |                         | 0.783                  |        |   |
| 1.021                 |       | 2.175                   | estabilidad_intermedia |        |   |
| bar_p60_h30_tp15_sl05 |       | p60_h30 TP 1.5 / SL 0.5 |                        | 41.29% |   |
| 47.77%                |       | 6.48%                   |                        | 15.92% |   |
| 25.07%                |       | 9.15%                   |                        | 42.79% |   |
| 27.16%                |       | -15.63%                 |                        |        | 7 |
| 2.17%                 |       |                         | 65.59%                 |        |   |
| -6.59%                | 0.379 |                         | 0.797                  |        |   |
| 1.078                 |       | 2.253                   | más_inestable          |        |   |
| bar_p50_h30_tp20_sl10 |       | p50_h30 TP 2.0 / SL 1.0 |                        | 27.93% |   |
| 40.12%                |       | 12.19%                  |                        | 5.02%  |   |
| 10.71%                |       | 5.69%                   |                        | 67.04% |   |
| 49.16%                |       | -17.88%                 |                        |        | 8 |
| 4.76%                 |       |                         | 78.92%                 |        |   |
| -5.83%                | 0.416 |                         | 0.806                  |        |   |
| 1.034                 |       | 2.256                   | más_inestable          |        |   |
| bar_p50_h30_tp20_sl05 |       | p50_h30 TP 2.0 / SL 0.5 |                        | 24.07% |   |
| 32.75%                |       | 8.68%                   |                        | 20.60% |   |
| 30.66%                |       | 10.05%                  |                        | 55.32% |   |
| 36.59%                |       | -18.73%                 |                        |        | 5 |
| 3.88%                 |       |                         | 51.65%                 |        |   |

|                       |                         |               |        |   |
|-----------------------|-------------------------|---------------|--------|---|
| -2.23%                | 0.397                   | 0.821         |        |   |
| 1.073                 | 2.291                   | más_inestable |        |   |
| bar_p50_h90_tp15_sl05 | p50_h90 TP 1.5 / SL 0.5 |               | 42.36% |   |
| 47.13%                | 4.77%                   | 20.90%        |        |   |
| 31.74%                | 10.84%                  | 36.74%        |        |   |
| 21.12%                | -15.61%                 |               |        | 6 |
| 6.96%                 |                         | 59.75%        |        |   |
| -7.21%                | 0.384                   | 0.805         |        |   |
| 1.106                 | 2.295                   | más_inestable |        |   |
| bar_p50_h90_tp20_sl10 | p50_h90 TP 2.0 / SL 1.0 |               | 26.35% |   |
| 39.60%                | 13.25%                  | 5.15%         |        |   |
| 11.42%                | 6.27%                   | 68.50%        |        |   |
| 48.98%                | -19.53%                 |               |        | 8 |
| 3.66%                 |                         | 77.62%        |        |   |
| -6.04%                | 0.451                   | 0.799         |        |   |
| 1.057                 | 2.308                   | más_inestable |        |   |
| bar_p40_h30_tp25_sl10 | p40_h30 TP 2.5 / SL 1.0 |               | 19.75% |   |
| 31.40%                | 11.66%                  | 7.18%         |        |   |
| 14.33%                | 7.15%                   | 73.08%        |        |   |
| 54.27%                | -18.81%                 |               |        | 7 |
| 3.34%                 |                         | 68.67%        |        |   |
| -4.67%                | 0.423                   | 0.840         |        |   |
| 1.055                 | 2.318                   | más_inestable |        |   |
| bar_p50_h60_tp15_sl05 | p50_h60 TP 1.5 / SL 0.5 |               | 42.79% |   |
| 47.47%                | 4.68%                   | 19.78%        |        |   |
| 30.34%                | 10.56%                  | 37.43%        |        |   |
| 22.19%                | -15.24%                 |               |        | 6 |
| 8.39%                 |                         | 61.01%        |        |   |
| -7.38%                | 0.379                   | 0.828         |        |   |
| 1.135                 | 2.342                   | más_inestable |        |   |
| bar_p40_h30_tp20_sl10 | p40_h30 TP 2.0 / SL 1.0 |               | 32.82% |   |
| 44.31%                | 11.48%                  | 6.76%         |        |   |
| 13.26%                | 6.49%                   | 60.41%        |        |   |
| 42.44%                | -17.98%                 |               |        | 8 |
| 2.92%                 |                         | 76.97%        |        |   |
| -5.95%                | 0.419                   | 0.857         |        |   |
| 1.072                 | 2.348                   | más_inestable |        |   |
| bar_p60_h30_tp20_sl05 | p60_h30 TP 2.0 / SL 0.5 |               | 20.58% |   |
| 29.82%                | 9.23%                   | 16.78%        |        |   |
| 27.04%                | 10.25%                  | 62.63%        |        |   |
| 43.15%                | -19.49%                 |               |        | 5 |
| 5.09%                 |                         | 52.44%        |        |   |
| -2.64%                | 0.416                   | 0.832         |        |   |
| 1.103                 | 2.351                   | más_inestable |        |   |
| bar_p40_h90_tp25_sl10 | p40_h90 TP 2.5 / SL 1.0 |               | 18.23% |   |
| 30.26%                | 12.03%                  | 7.44%         |        |   |
| 15.54%                | 8.10%                   | 74.32%        |        |   |
| 54.19%                | -20.13%                 |               |        | 7 |
| 1.01%                 |                         | 66.06%        |        |   |
| -4.94%                | 0.452                   | 0.838         |        |   |
| 1.077                 | 2.368                   | más_inestable |        |   |
| bar_p40_h60_tp20_sl05 | p40_h60 TP 2.0 / SL 0.5 |               | 26.03% |   |
| 33.41%                | 7.38%                   | 25.55%        |        |   |
| 36.78%                | 11.23%                  | 48.42%        |        |   |
| 29.81%                | -18.61%                 |               |        | 5 |
| 0.47%                 |                         | 47.60%        |        |   |
| -2.87%                | 0.401                   | 0.858         |        |   |
| 1.126                 | 2.385                   | más_inestable |        |   |
| bar_p50_h60_tp20_sl10 | p50_h60 TP 2.0 / SL 1.0 |               | 26.10% |   |
| 39.19%                | 13.10%                  | 4.66%         |        |   |

|                       |       |                         |               |        |   |
|-----------------------|-------|-------------------------|---------------|--------|---|
| 10.96%                |       | 6.30%                   |               | 69.25% |   |
| 49.85%                |       | -19.40%                 |               |        | 8 |
| 4.85%                 |       |                         | 78.15%        |        |   |
| -6.70%                | 0.455 |                         | 0.861         |        |   |
| 1.104                 |       | 2.420                   | más_inestable |        |   |
| bar_p60_h90_tp15_sl05 |       | p60_h90 TP 1.5 / SL 0.5 |               | 39.74% |   |
| 46.39%                |       | 6.65%                   |               | 17.18% |   |
| 28.08%                |       | 10.90%                  |               | 43.08% |   |
| 25.53%                |       | -17.54%                 |               |        | 6 |
| 9.82%                 |       |                         | 62.29%        |        |   |
| -7.52%                | 0.426 |                         | 0.839         |        |   |
| 1.169                 |       | 2.434                   | más_inestable |        |   |
| bar_p40_h60_tp25_sl10 |       | p40_h60 TP 2.5 / SL 1.0 |               | 18.07% |   |
| 30.21%                |       | 12.14%                  |               | 6.92%  |   |
| 14.75%                |       | 7.83%                   |               | 75.02% |   |
| 55.04%                |       | -19.98%                 |               |        | 7 |
| 2.31%                 |       |                         | 67.20%        |        |   |
| -5.12%                | 0.451 |                         | 0.875         |        |   |
| 1.110                 |       | 2.436                   | más_inestable |        |   |
| bar_p40_h90_tp20_sl05 |       | p40_h90 TP 2.0 / SL 0.5 |               | 25.64% |   |
| 33.02%                |       | 7.38%                   |               | 26.36% |   |
| 37.94%                |       | 11.57%                  |               | 48.00% |   |
| 29.04%                |       | -18.95%                 |               |        | 4 |
| 9.30%                 |       |                         | 46.53%        |        |   |
| -2.77%                | 0.407 |                         | 0.865         |        |   |
| 1.167                 |       | 2.438                   | más_inestable |        |   |
| bar_p40_h90_tp20_sl10 |       | p40_h90 TP 2.0 / SL 1.0 |               | 31.10% |   |
| 43.30%                |       | 12.20%                  |               | 7.03%  |   |
| 14.36%                |       | 7.33%                   |               | 61.87% |   |
| 42.34%                |       | -19.53%                 |               |        | 8 |
| 1.56%                 |       |                         | 75.09%        |        |   |
| -6.47%                | 0.455 |                         | 0.888         |        |   |
| 1.143                 |       | 2.486                   | más_inestable |        |   |
| bar_p60_h90_tp20_sl05 |       | p60_h90 TP 2.0 / SL 0.5 |               | 19.33% |   |
| 28.37%                |       | 9.04%                   |               | 18.07% |   |
| 30.23%                |       | 12.16%                  |               | 62.60% |   |
| 41.40%                |       | -21.20%                 |               |        | 5 |
| 1.67%                 |       |                         | 48.41%        |        |   |
| -3.26%                | 0.457 |                         | 0.864         |        |   |
| 1.187                 |       | 2.507                   | más_inestable |        |   |
| bar_p40_h60_tp20_sl10 |       | p40_h60 TP 2.0 / SL 1.0 |               | 31.18% |   |
| 42.99%                |       | 11.80%                  |               | 6.54%  |   |
| 13.71%                |       | 7.17%                   |               | 62.27% |   |
| 43.31%                |       | -18.97%                 |               |        | 8 |
| 2.66%                 |       |                         | 75.82%        |        |   |
| -6.84%                | 0.448 |                         | 0.918         |        |   |
| 1.168                 |       | 2.534                   | más_inestable |        |   |
| bar_p60_h60_tp20_sl05 |       | p60_h60 TP 2.0 / SL 0.5 |               | 18.96% |   |
| 28.58%                |       | 9.62%                   |               | 16.95% |   |
| 28.92%                |       | 11.97%                  |               | 64.08% |   |
| 42.49%                |       | -21.59%                 |               |        | 5 |
| 2.80%                 |       |                         | 49.70%        |        |   |
| -3.10%                | 0.463 |                         | 0.888         |        |   |
| 1.189                 |       | 2.540                   | más_inestable |        |   |
| bar_p50_h90_tp20_sl05 |       | p50_h90 TP 2.0 / SL 0.5 |               | 22.26% |   |
| 30.97%                |       | 8.70%                   |               | 22.07% |   |
| 34.38%                |       | 12.31%                  |               | 55.67% |   |
| 34.66%                |       | -21.01%                 |               |        | 5 |
| 0.22%                 |       |                         | 47.39%        |        |   |
| -2.83%                | 0.449 |                         | 0.901         |        |   |

|                       |                         |               |        |
|-----------------------|-------------------------|---------------|--------|
| 1.192                 | 2.542                   | más_inestable |        |
| bar_p60_h60_tp15_sl05 | p60_h60 TP 1.5 / SL 0.5 |               | 39.59% |
| 46.32%                | 6.72%                   | 16.09%        |        |
| 27.00%                | 10.91%                  | 44.32%        |        |
| 26.68%                | -17.64%                 |               | 7      |
| 1.11%                 |                         | 63.17%        |        |
| -7.94%                | 0.432                   | 0.886         |        |
| 1.225                 | 2.543                   | más_inestable |        |
| bar_p50_h60_tp20_sl05 | p50_h60 TP 2.0 / SL 0.5 |               | 22.27% |
| 31.16%                | 8.89%                   | 20.93%        |        |
| 32.71%                | 11.78%                  | 56.80%        |        |
| 36.13%                | -20.67%                 |               | 5      |
| 1.55%                 |                         | 48.79%        |        |
| -2.76%                | 0.441                   | 0.906         |        |
| 1.203                 | 2.550                   | más_inestable |        |

Top 15 targets temporalmente más estables:

|                                              |                                             |                                             |                                    |
|----------------------------------------------|---------------------------------------------|---------------------------------------------|------------------------------------|
| bar_col target_base                          | tp_sl_config                                | tp_rate_development_2020_2024               | t                                  |
| p_rate_final_test_2025_2026                  | tp_rate_diff_final_minus_dev                | sl_rate_development_2020_2024               | sl_rate_final_test_2025_2026       |
| sl_rate_diff_final_minus_dev                 | no_event_rate_development_2020_2024         | no_event_rate_final_test_2025_2026          | no_event_rate_diff_final_minus_dev |
| tp_share_over_resolved_development_2020_2024 | tp_share_over_resolved_final_test_2025_2026 | tp_share_over_resolved_diff_final_minus_dev | split_shift_score                  |
| yearly_yearly_instability_score              | quarterly_quarterly_instability_score       | temporal_instability_score                  | temporal_stability_profile         |
| bar_p60_h60_tp10_sl10                        | p60_h60 TP 1.0 / SL 1.0                     |                                             | 97.60%                             |
| 94.05%                                       | -3.55%                                      | 2.40%                                       |                                    |
| 5.95%                                        | 3.55%                                       | 0.00%                                       |                                    |
| 0.00%                                        | 0.00%                                       |                                             | 9                                  |
| 7.60%                                        |                                             | 94.05%                                      |                                    |
| -3.55%                                       | 0.107                                       | 0.167                                       |                                    |
| 0.212                                        | 0.486                                       | más_estable                                 |                                    |
| bar_p60_h90_tp10_sl10                        | p60_h90 TP 1.0 / SL 1.0                     |                                             | 97.33%                             |
| 93.83%                                       | -3.50%                                      | 2.67%                                       |                                    |
| 6.17%                                        | 3.50%                                       | 0.00%                                       |                                    |
| 0.00%                                        | 0.00%                                       |                                             | 9                                  |
| 7.33%                                        |                                             | 93.83%                                      |                                    |
| -3.50%                                       | 0.105                                       | 0.166                                       |                                    |
| 0.220                                        | 0.491                                       | más_estable                                 |                                    |
| bar_p60_h30_tp10_sl10                        | p60_h30 TP 1.0 / SL 1.0                     |                                             | 97.39%                             |
| 93.82%                                       | -3.56%                                      | 2.61%                                       |                                    |
| 6.18%                                        | 3.56%                                       | 0.00%                                       |                                    |
| 0.00%                                        | 0.00%                                       |                                             | 9                                  |
| 7.39%                                        |                                             | 93.82%                                      |                                    |
| -3.56%                                       | 0.107                                       | 0.173                                       |                                    |
| 0.217                                        | 0.497                                       | más_estable                                 |                                    |
| bar_p50_h30_tp10_sl10                        | p50_h30 TP 1.0 / SL 1.0                     |                                             | 96.38%                             |
| 92.04%                                       | -4.34%                                      | 3.62%                                       |                                    |
| 7.96%                                        | 4.34%                                       | 0.00%                                       |                                    |
| 0.00%                                        | 0.00%                                       |                                             | 9                                  |
| 6.38%                                        |                                             | 92.04%                                      |                                    |
| -4.34%                                       | 0.130                                       | 0.232                                       |                                    |
| 0.286                                        | 0.648                                       | más_estable                                 |                                    |
| bar_p50_h90_tp10_sl10                        | p50_h90 TP 1.0 / SL 1.0                     |                                             | 96.31%                             |
| 91.68%                                       | -4.63%                                      | 3.69%                                       |                                    |
| 8.32%                                        | 4.63%                                       | 0.00%                                       |                                    |
| 0.00%                                        | 0.00%                                       |                                             | 9                                  |
| 6.31%                                        |                                             | 91.68%                                      |                                    |
| -4.63%                                       | 0.139                                       | 0.234                                       |                                    |
| 0.281                                        | 0.654                                       | más_estable                                 |                                    |

|                       |                         |             |  |        |
|-----------------------|-------------------------|-------------|--|--------|
| bar_p50_h60_tp10_sl10 | p50_h60 TP 1.0 / SL 1.0 |             |  | 96.67% |
| 91.99%                | -4.67%                  | 3.33%       |  |        |
| 8.01%                 | 4.67%                   | 0.00%       |  |        |
| 0.00%                 | 0.00%                   |             |  | 9      |
| 6.67%                 |                         | 91.99%      |  |        |
| -4.67%                | 0.140                   | 0.238       |  |        |
| 0.289                 | 0.667                   | más_estable |  |        |
| bar_p40_h30_tp10_sl10 | p40_h30 TP 1.0 / SL 1.0 |             |  | 95.09% |
| 89.97%                | -5.12%                  | 4.91%       |  |        |
| 10.03%                | 5.12%                   | 0.00%       |  |        |
| 0.00%                 | 0.00%                   |             |  | 9      |
| 5.09%                 |                         | 89.97%      |  |        |
| -5.12%                | 0.154                   | 0.291       |  |        |
| 0.353                 | 0.798                   | más_estable |  |        |
| bar_p40_h90_tp10_sl10 | p40_h90 TP 1.0 / SL 1.0 |             |  | 94.92% |
| 89.40%                | -5.52%                  | 5.08%       |  |        |
| 10.60%                | 5.52%                   | 0.00%       |  |        |
| 0.00%                 | 0.00%                   |             |  | 9      |
| 4.92%                 |                         | 89.40%      |  |        |
| -5.52%                | 0.166                   | 0.298       |  |        |
| 0.355                 | 0.819                   | más_estable |  |        |
| bar_p40_h60_tp10_sl10 | p40_h60 TP 1.0 / SL 1.0 |             |  | 95.34% |
| 89.90%                | -5.44%                  | 4.66%       |  |        |
| 10.10%                | 5.44%                   | 0.00%       |  |        |
| 0.00%                 | 0.00%                   |             |  | 9      |
| 5.34%                 |                         | 89.90%      |  |        |
| -5.44%                | 0.163                   | 0.300       |  |        |
| 0.365                 | 0.828                   | más_estable |  |        |
| bar_p60_h30_tp10_sl05 | p60_h30 TP 1.0 / SL 0.5 |             |  | 86.92% |
| 79.12%                | -7.79%                  | 13.08%      |  |        |
| 20.88%                | 7.79%                   | 0.00%       |  |        |
| 0.00%                 | 0.00%                   |             |  | 8      |
| 6.92%                 |                         | 79.12%      |  |        |
| -7.79%                | 0.234                   | 0.464       |  |        |
| 0.597                 | 1.295                   | más_estable |  |        |
| bar_p40_h30_tp10_sl05 | p40_h30 TP 1.0 / SL 0.5 |             |  | 80.81% |
| 73.31%                | -7.50%                  | 19.19%      |  |        |
| 26.69%                | 7.50%                   | 0.00%       |  |        |
| 0.00%                 | 0.00%                   |             |  | 8      |
| 0.81%                 |                         | 73.31%      |  |        |
| -7.50%                | 0.225                   | 0.468       |  |        |
| 0.607                 | 1.301                   | más_estable |  |        |
| bar_p50_h30_tp10_sl05 | p50_h30 TP 1.0 / SL 0.5 |             |  | 83.91% |
| 76.22%                | -7.69%                  | 16.09%      |  |        |
| 23.78%                | 7.69%                   | 0.00%       |  |        |
| 0.00%                 | 0.00%                   |             |  | 8      |
| 3.91%                 |                         | 76.22%      |  |        |
| -7.69%                | 0.231                   | 0.473       |  |        |
| 0.611                 | 1.315                   | más_estable |  |        |
| bar_p60_h90_tp10_sl05 | p60_h90 TP 1.0 / SL 0.5 |             |  | 85.74% |
| 76.53%                | -9.21%                  | 14.26%      |  |        |
| 23.47%                | 9.21%                   | 0.00%       |  |        |
| 0.00%                 | 0.00%                   |             |  | 8      |
| 5.74%                 |                         | 76.53%      |  |        |
| -9.21%                | 0.276                   | 0.515       |  |        |
| 0.606                 | 1.397                   | más_estable |  |        |
| bar_p60_h60_tp10_sl05 | p60_h60 TP 1.0 / SL 0.5 |             |  | 86.72% |
| 77.53%                | -9.19%                  | 13.28%      |  |        |
| 22.47%                | 9.19%                   | 0.00%       |  |        |
| 0.00%                 | 0.00%                   |             |  | 8      |

|                       |       |                         |             |        |
|-----------------------|-------|-------------------------|-------------|--------|
| 6.72%                 |       |                         | 77.53%      |        |
| -9.19%                | 0.276 |                         | 0.535       |        |
| 0.685                 |       | 1.496                   | más_estable |        |
| bar_p50_h90_tp10_sl05 |       | p50_h90 TP 1.0 / SL 0.5 |             | 82.68% |
| 73.19%                |       | -9.50%                  | 17.32%      |        |
| 26.81%                |       | 9.50%                   | 0.00%       |        |
| 0.00%                 |       | 0.00%                   |             | 8      |
| 2.68%                 |       |                         | 73.19%      |        |
| -9.50%                | 0.285 |                         | 0.567       |        |
| 0.694                 |       | 1.547                   | más_estable |        |

Top 15 targets temporalmente más inestables:

|                                                                                   |       |                         |               |        |
|-----------------------------------------------------------------------------------|-------|-------------------------|---------------|--------|
| bar_col target_base tp_sl_config tp_rate_development_2020_2024 t                  |       |                         |               |        |
| p_rate_final_test_2025_2026 tp_rate_diff_final_minus_dev sl_rate_development_2020 |       |                         |               |        |
| _2024 sl_rate_final_test_2025_2026 sl_rate_diff_final_minus_dev no_event_rate_dev |       |                         |               |        |
| elopment_2020_2024 no_event_rate_final_test_2025_2026 no_event_rate_diff_final_mi |       |                         |               |        |
| nus_dev tp_share_over_resolved_development_2020_2024 tp_share_over_resolved_final |       |                         |               |        |
| _test_2025_2026 tp_share_over_resolved_diff_final_minus_dev split_shift_score yea |       |                         |               |        |
| rly_yearly_instability_score quarterly_quarterly_instability_score temporal_insta |       |                         |               |        |
| bility_score temporal_stability_profile                                           |       |                         |               |        |
| bar_p50_h60_tp20_sl05                                                             |       | p50_h60 TP 2.0 / SL 0.5 |               | 22.27% |
| 31.16%                                                                            |       | 8.89%                   | 20.93%        |        |
| 32.71%                                                                            |       | 11.78%                  | 56.80%        |        |
| 36.13%                                                                            |       | -20.67%                 |               | 5      |
| 1.55%                                                                             |       |                         | 48.79%        |        |
| -2.76%                                                                            | 0.441 |                         | 0.906         |        |
| 1.203                                                                             |       | 2.550                   | más_inestable |        |
| bar_p60_h60_tp15_sl05                                                             |       | p60_h60 TP 1.5 / SL 0.5 |               | 39.59% |
| 46.32%                                                                            |       | 6.72%                   | 16.09%        |        |
| 27.00%                                                                            |       | 10.91%                  | 44.32%        |        |
| 26.68%                                                                            |       | -17.64%                 |               | 7      |
| 1.11%                                                                             |       |                         | 63.17%        |        |
| -7.94%                                                                            | 0.432 |                         | 0.886         |        |
| 1.225                                                                             |       | 2.543                   | más_inestable |        |
| bar_p50_h90_tp20_sl05                                                             |       | p50_h90 TP 2.0 / SL 0.5 |               | 22.26% |
| 30.97%                                                                            |       | 8.70%                   | 22.07%        |        |
| 34.38%                                                                            |       | 12.31%                  | 55.67%        |        |
| 34.66%                                                                            |       | -21.01%                 |               | 5      |
| 0.22%                                                                             |       |                         | 47.39%        |        |
| -2.83%                                                                            | 0.449 |                         | 0.901         |        |
| 1.192                                                                             |       | 2.542                   | más_inestable |        |
| bar_p60_h60_tp20_sl05                                                             |       | p60_h60 TP 2.0 / SL 0.5 |               | 18.96% |
| 28.58%                                                                            |       | 9.62%                   | 16.95%        |        |
| 28.92%                                                                            |       | 11.97%                  | 64.08%        |        |
| 42.49%                                                                            |       | -21.59%                 |               | 5      |
| 2.80%                                                                             |       |                         | 49.70%        |        |
| -3.10%                                                                            | 0.463 |                         | 0.888         |        |
| 1.189                                                                             |       | 2.540                   | más_inestable |        |
| bar_p40_h60_tp20_sl10                                                             |       | p40_h60 TP 2.0 / SL 1.0 |               | 31.18% |
| 42.99%                                                                            |       | 11.80%                  | 6.54%         |        |
| 13.71%                                                                            |       | 7.17%                   | 62.27%        |        |
| 43.31%                                                                            |       | -18.97%                 |               | 8      |
| 2.66%                                                                             |       |                         | 75.82%        |        |
| -6.84%                                                                            | 0.448 |                         | 0.918         |        |
| 1.168                                                                             |       | 2.534                   | más_inestable |        |
| bar_p60_h90_tp20_sl05                                                             |       | p60_h90 TP 2.0 / SL 0.5 |               | 19.33% |
| 28.37%                                                                            |       | 9.04%                   | 18.07%        |        |
| 30.23%                                                                            |       | 12.16%                  | 62.60%        |        |
| 41.40%                                                                            |       | -21.20%                 |               | 5      |
| 1.67%                                                                             |       |                         | 48.41%        |        |

|                       |                         |               |        |   |
|-----------------------|-------------------------|---------------|--------|---|
| -3.26%                | 0.457                   | 0.864         |        |   |
| 1.187                 | 2.507                   | más_inestable |        |   |
| bar_p40_h90_tp20_sl10 | p40_h90 TP 2.0 / SL 1.0 |               | 31.10% |   |
| 43.30%                | 12.20%                  | 7.03%         |        |   |
| 14.36%                | 7.33%                   | 61.87%        |        |   |
| 42.34%                | -19.53%                 |               |        | 8 |
| 1.56%                 |                         | 75.09%        |        |   |
| -6.47%                | 0.455                   | 0.888         |        |   |
| 1.143                 | 2.486                   | más_inestable |        |   |
| bar_p40_h90_tp20_sl05 | p40_h90 TP 2.0 / SL 0.5 |               | 25.64% |   |
| 33.02%                | 7.38%                   | 26.36%        |        |   |
| 37.94%                | 11.57%                  | 48.00%        |        |   |
| 29.04%                | -18.95%                 |               |        | 4 |
| 9.30%                 |                         | 46.53%        |        |   |
| -2.77%                | 0.407                   | 0.865         |        |   |
| 1.167                 | 2.438                   | más_inestable |        |   |
| bar_p40_h60_tp25_sl10 | p40_h60 TP 2.5 / SL 1.0 |               | 18.07% |   |
| 30.21%                | 12.14%                  | 6.92%         |        |   |
| 14.75%                | 7.83%                   | 75.02%        |        |   |
| 55.04%                | -19.98%                 |               |        | 7 |
| 2.31%                 |                         | 67.20%        |        |   |
| -5.12%                | 0.451                   | 0.875         |        |   |
| 1.110                 | 2.436                   | más_inestable |        |   |
| bar_p60_h90_tp15_sl05 | p60_h90 TP 1.5 / SL 0.5 |               | 39.74% |   |
| 46.39%                | 6.65%                   | 17.18%        |        |   |
| 28.08%                | 10.90%                  | 43.08%        |        |   |
| 25.53%                | -17.54%                 |               |        | 6 |
| 9.82%                 |                         | 62.29%        |        |   |
| -7.52%                | 0.426                   | 0.839         |        |   |
| 1.169                 | 2.434                   | más_inestable |        |   |
| bar_p50_h60_tp20_sl10 | p50_h60 TP 2.0 / SL 1.0 |               | 26.10% |   |
| 39.19%                | 13.10%                  | 4.66%         |        |   |
| 10.96%                | 6.30%                   | 69.25%        |        |   |
| 49.85%                | -19.40%                 |               |        | 8 |
| 4.85%                 |                         | 78.15%        |        |   |
| -6.70%                | 0.455                   | 0.861         |        |   |
| 1.104                 | 2.420                   | más_inestable |        |   |
| bar_p40_h60_tp20_sl05 | p40_h60 TP 2.0 / SL 0.5 |               | 26.03% |   |
| 33.41%                | 7.38%                   | 25.55%        |        |   |
| 36.78%                | 11.23%                  | 48.42%        |        |   |
| 29.81%                | -18.61%                 |               |        | 5 |
| 0.47%                 |                         | 47.60%        |        |   |
| -2.87%                | 0.401                   | 0.858         |        |   |
| 1.126                 | 2.385                   | más_inestable |        |   |
| bar_p40_h90_tp25_sl10 | p40_h90 TP 2.5 / SL 1.0 |               | 18.23% |   |
| 30.26%                | 12.03%                  | 7.44%         |        |   |
| 15.54%                | 8.10%                   | 74.32%        |        |   |
| 54.19%                | -20.13%                 |               |        | 7 |
| 1.01%                 |                         | 66.06%        |        |   |
| -4.94%                | 0.452                   | 0.838         |        |   |
| 1.077                 | 2.368                   | más_inestable |        |   |
| bar_p60_h30_tp20_sl05 | p60_h30 TP 2.0 / SL 0.5 |               | 20.58% |   |
| 29.82%                | 9.23%                   | 16.78%        |        |   |
| 27.04%                | 10.25%                  | 62.63%        |        |   |
| 43.15%                | -19.49%                 |               |        | 5 |
| 5.09%                 |                         | 52.44%        |        |   |
| -2.64%                | 0.416                   | 0.832         |        |   |
| 1.103                 | 2.351                   | más_inestable |        |   |
| bar_p40_h30_tp20_sl10 | p40_h30 TP 2.0 / SL 1.0 |               | 32.82% |   |
| 44.31%                | 11.48%                  | 6.76%         |        |   |

|        |       |         |               |        |   |
|--------|-------|---------|---------------|--------|---|
| 13.26% |       | 6.49%   |               | 60.41% |   |
| 42.44% |       | -17.98% |               |        | 8 |
| 2.92%  |       |         | 76.97%        |        |   |
| -5.95% | 0.419 |         | 0.857         |        |   |
| 1.072  |       | 2.348   | más_inestable |        |   |

Conteo por perfil de estabilidad temporal:

| temporal_stability_profile | n_targets | split_shift_score_mean | yearly_instability_score_mean | quarterly_instability_score_mean | temporal_instability_score_mean | tp_rate_diff_abs_mean | sl_rate_diff_abs_mean | no_event_rate_diff_abs_mean | tp_share_diff_abs_mean |
|----------------------------|-----------|------------------------|-------------------------------|----------------------------------|---------------------------------|-----------------------|-----------------------|-----------------------------|------------------------|
|----------------------------|-----------|------------------------|-------------------------------|----------------------------------|---------------------------------|-----------------------|-----------------------|-----------------------------|------------------------|

|       |                        |       |        |       |  |       |  |  |  |
|-------|------------------------|-------|--------|-------|--|-------|--|--|--|
|       | más_estable            | 21    | 0.220  |       |  |       |  |  |  |
| 0.411 |                        | 0.529 |        | 1.161 |  |       |  |  |  |
| 7.12% | 6.39%                  |       | 2.21%  |       |  | 6.32% |  |  |  |
|       | estabilidad_intermedia | 20    | 0.367  |       |  |       |  |  |  |
| 0.716 |                        | 0.951 |        | 2.034 |  |       |  |  |  |
| 9.18% | 6.42%                  |       | 15.59% |       |  | 5.47% |  |  |  |
|       | más_inestable          | 22    | 0.427  |       |  |       |  |  |  |
| 0.853 |                        | 1.126 |        | 2.407 |  |       |  |  |  |
| 9.46% | 9.36%                  |       | 18.82% |       |  | 5.08% |  |  |  |

Archivos guardados:

Estabilidad temporal completa: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_temporal\_stability.csv

Estabilidad por split: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_stability\_split.csv

Estabilidad por año: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_stability\_year.csv

Estabilidad por trimestre: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_stability\_quarter.csv

Comparación development vs final\_test: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_split\_comparison.csv

Cambio por configuración TP/SL: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_split\_change\_by\_config.csv

Cambio por percentil y horizonte: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_split\_change\_by\_target\_base.csv

Estabilidad anual por target: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_yearly\_stability\_summary.csv

Estabilidad anual por configuración: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_yearly\_stability\_by\_config.csv

Estabilidad trimestral por target: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_quarterly\_stability\_summary.csv

Tabla final estabilidad temporal: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_temporal\_stability\_final\_table.csv

Reporte TXT: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_temporal\_stability\_report.txt

Celda 4 finalizada correctamente.

## Observaciones preliminares — Estabilidad temporal BAR

Luego de corregir la segmentación trimestral, el análisis de estabilidad temporal queda correctamente calculado trimestre a trimestre. Es decir, los períodos trimestrales ya no agrupan simplemente Q1, Q2, Q3 y Q4, sino que ahora distinguen períodos temporales específicos como 2020Q1, 2020Q2, ..., 2026Q2.



Esto aumenta la cantidad de observaciones temporales analizadas y permite evaluar mejor la variación de los targets `BAR` a lo largo del tiempo. La tabla final considera los 63 targets `BAR`, evaluados por `dataset_split`, año y trimestre temporal.

El primer resultado importante es que el patrón general se mantiene: el período `final_test_2025_2026` presenta más actividad que el período `development_2020_2024`. En la mayoría de las configuraciones disminuye el `no_event`, aumentan los TP y también aumentan los SL. Esto indica que las barreras calibradas con datos de 2020–2024 fueron alcanzadas con mayor frecuencia en 2025–2026.

Por configuración TP/SL, el comportamiento es bastante claro. `TP 1.0 / SL 1.0` es la configuración más estable, con el menor `split_shift_score_mean`, igual a `0.134`. Sin embargo, debe seguir interpretándose como benchmark, no como target operativo principal, porque el TP coincide con el threshold base utilizado para construir `DIR`.

La configuración `TP 1.0 / SL 0.5` también aparece relativamente estable, pero su interpretación es distinta. No tiene `no_event`, porque el TP base se alcanza por construcción, pero el stop es más ajustado. En `final_test_2025_2026`, el TP cae en promedio `-8.70%` y el SL sube `+8.70%`. Esto muestra que, aunque resuelve siempre, es sensible al ruido y al movimiento adverso.

La configuración `TP 1.5 / SL 1.0` mantiene una posición intermedia. En promedio, el TP aumenta `+9.68%`, el SL aumenta `+5.20%` y el `no_event` baja `-14.88%` en `final_test_2025_2026`. Esto significa que en el período final hubo más operaciones resueltas, pero también más riesgo de tocar SL. La proporción de TP entre eventos resueltos cae `-5.72%`, lo que indica que el entorno fue más activo, pero no necesariamente más limpio.

Las configuraciones más inestables son principalmente `TP 2.0 / SL 0.5`, `TP 2.0 / SL 1.0` y algunas variantes de `TP 2.5 / SL 1.0`. Esto es esperable, porque objetivos más lejanos son más dependientes del contexto de volatilidad. Cuando el mercado se mueve más, muchos casos que antes quedaban como `no_event` pasan a resolverse como TP o SL, generando cambios temporales más fuertes.

El análisis por percentil y horizonte muestra que la fuente principal de inestabilidad no parece ser el horizonte por sí solo. Los `split_shift_score_mean` por combinación p/h se mantienen en un rango relativamente acotado, aproximadamente entre `0.318` y `0.356`. Esto sugiere que la configuración TP/SL influye más que elegir `H30`, `H60` o `H90`.

El análisis anual confirma la misma lectura. `TP 1.0 / SL 1.0` tiene la menor inestabilidad anual promedio, mientras que `TP 2.0 / SL 0.5` y `TP 2.0 / SL 1.0` tienen las mayores variaciones. En particular, `TP 2.0 / SL 0.5` presenta un `yearly_instability_score_mean` de `0.857`, y `TP 2.0 / SL 1.0` un valor de `0.804`.

El análisis trimestral corregido refuerza esta conclusión. Al evaluar trimestre a trimestre, las configuraciones más sensibles vuelven a ser las que combinan TP exigente con SL ajustado, especialmente **TP 2.0 / SL 0.5**. Esto confirma que esas variantes dependen demasiado del contexto temporal y pueden cambiar de comportamiento de forma importante entre períodos.

En cambio, **TP 1.5 / SL 1.0** no aparece como la configuración más estable, pero tampoco como la más inestable. Su perfil es intermedio: tiene cambios temporales relevantes, pero conserva una estructura operativa razonable. Por eso sigue siendo una candidata práctica para continuar el análisis, siempre teniendo presente que el modelo deberá aprender a distinguir contextos de mercado más activos de contextos más laterales.

En síntesis, el análisis de estabilidad temporal muestra que los targets **BAR** no son estacionarios. El comportamiento cambia entre años y trimestres, especialmente cuando las barreras son más exigentes o cuando el stop es demasiado ajustado. Esto no invalida los targets, pero confirma que la etapa de modelado deberá considerar explícitamente variables de contexto, volatilidad, régimen intradiario y comportamiento reciente del mercado.

De forma preliminar:

- TP 1.0 / SL 1.0 → más estable, pero benchmark por construcción.
- TP 1.0 / SL 0.5 → estable en resolución, pero sensible al ruido y al SL.
- TP 1.5 / SL 1.0 → estabilidad intermedia y candidata operativa principal.
- TP 2.0 / SL 1.0 → más exigente y más dependiente del contexto.
- TP 2.5 / SL 1.0 → muy exigente, con alta dependencia temporal.
- SL 0.5 → aumenta la sensibilidad al ruido y la inestabilidad.

La conclusión operativa se mantiene: **TP 1.5 / SL 1.0** sigue siendo una de las configuraciones más razonables para continuar el análisis, no porque sea la más estable, sino porque ofrece un equilibrio aceptable entre exigencia, frecuencia de TP, baja frecuencia relativa de SL y estabilidad temporal intermedia.

## 4.10. Análisis por régimen intradiario

En este punto se analiza el comportamiento de los targets **BAR** según el régimen intradiario.

El objetivo es evaluar si las métricas operativas de **BAR** cambian entre las distintas etapas de la sesión. Esto es importante porque el mercado no se comporta igual durante el **Overnight**, el **Pre-market**, la apertura, la sesión regular o el cierre.

El análisis se realizará sobre los 63 targets **BAR**, considerando todos los percentiles, horizontes y configuraciones TP/SL.

Se analizarán principalmente:

Tasa de operaciones válidas  
 Frecuencia de TP  
 Frecuencia de SL  
 Frecuencia de no\_event  
 Eventos resueltos  
 Proporción de TP entre eventos resueltos  
 Expectativa simple en unidades de threshold  
 Tiempo promedio hasta evento

Una consideración importante es que el régimen **Closing** puede tener menos observaciones válidas, especialmente para horizontes largos, porque no siempre existe suficiente ventana futura sin cruzar el final del trading day.

## Celda 1 — Preparación de métricas BAR por régimen

In [111...

```
# Celda 1 – Preparación de métricas BAR por régimen
# =====

print("4.10. Análisis por régimen intradiario - BAR")
print("=" * 80)

# -----
# Validar objetos necesarios
# -----

if "df_targets_bar" not in globals():
    raise NameError("No existe df_targets_bar. Primero debes construir los targe

if "df_bar_metadata" not in globals():
    raise NameError("No existe df_bar_metadata. Primero debes construir df_bar_m

# -----
# Copia de trabajo
# -----

df_bar_regime_base = df_targets_bar.copy()

# -----
# Validar columna de régimen
# -----

REGIME_COL = "regime_id"

if REGIME_COL not in df_bar_regime_base.columns:
    raise KeyError(
        f"No existe la columna {REGIME_COL}. "
        "Se requiere regime_id para analizar BAR por régimen intradiario."
    )

# -----
# Diccionario de nombres de régimen
# -----

REGIME_NAME_MAP = {
    0: "Overnight",
    1: "Pre-market",
    2: "Opening",
    3: "Regular",
```

```

    4: "Closing",
}

df_bar_regime_base["regime_name"] = (
    df_bar_regime_base[REGIME_COL]
    .map(REGIME_NAME_MAP)
    .fillna("Unknown")
)

# -----
# Función auxiliar para resumir un target BAR por régimen
# -----

def summarize_bar_target_for_regime(df, bar_col, event_step_col, tp_mult, sl_mult):
    """
    Resume un target BAR dentro de un régimen intradiario.

    BAR:
        1 = TP primero
        -1 = SL primero
        0 = no_event
        NaN = no operable / sin ventana válida
    """

    bar_series = df[bar_col]
    step_series = df[event_step_col]

    valid_mask = bar_series.notna()
    tp_mask = bar_series == 1
    sl_mask = bar_series == -1
    no_event_mask = bar_series == 0
    resolved_mask = tp_mask | sl_mask

    total_rows = len(df)
    valid_rows = int(valid_mask.sum())
    nan_rows = int(bar_series.isna().sum())

    tp_count = int(tp_mask.sum())
    sl_count = int(sl_mask.sum())
    no_event_count = int(no_event_mask.sum())
    resolved_count = tp_count + sl_count

    # Tasa de filas operables dentro del régimen.
    # En BAR, una fila válida significa que DIR generó una operación LONG o SHOR
    operable_rate = valid_rows / total_rows if total_rows > 0 else np.nan

    if valid_rows > 0:
        tp_rate = tp_count / valid_rows
        sl_rate = sl_count / valid_rows
        no_event_rate = no_event_count / valid_rows
        resolved_rate = resolved_count / valid_rows
    else:
        tp_rate = np.nan
        sl_rate = np.nan
        no_event_rate = np.nan
        resolved_rate = np.nan

    if resolved_count > 0:
        tp_share_over_resolved = tp_count / resolved_count
        sl_share_over_resolved = sl_count / resolved_count

```

```

else:
    tp_share_over_resolved = np.nan
    sl_share_over_resolved = np.nan

expectancy_units = (
    tp_rate * tp_mult
    - sl_rate * sl_mult
    if valid_rows > 0
    else np.nan
)

event_steps = step_series.loc[resolved_mask].dropna()

if len(event_steps) > 0:
    event_step_mean = event_steps.mean()
    event_step_median = event_steps.median()
else:
    event_step_mean = np.nan
    event_step_median = np.nan

return {
    "total_rows": total_rows,
    "valid_rows": valid_rows,
    "nan_rows": nan_rows,

    "operable_rate": operable_rate,

    "tp_count": tp_count,
    "sl_count": sl_count,
    "no_event_count": no_event_count,
    "resolved_count": resolved_count,

    "tp_rate": tp_rate,
    "sl_rate": sl_rate,
    "no_event_rate": no_event_rate,
    "resolved_rate": resolved_rate,

    "tp_share_over_resolved": tp_share_over_resolved,
    "sl_share_over_resolved": sl_share_over_resolved,

    "expectancy_units": expectancy_units,

    "event_step_mean": event_step_mean,
    "event_step_median": event_step_median,
}

# -----
# Construir métricas por régimen para los 63 targets BAR
# -----

regime_rows = []

regime_values = (
    df_bar_regime_base[[REGIME_COL, "regime_name"]]
    .drop_duplicates()
    .sort_values(REGIME_COL)
)

for _, meta in df_bar_metadata.iterrows():

```

```

bar_col = meta["bar_col"]
event_step_col = meta["event_step_col"]

horizon = int(meta["horizon"])
percentile = int(meta["percentile"])
tp_mult = float(meta["tp_mult"])
sl_mult = float(meta["sl_mult"])

tp_sl_config = f"TP {tp_mult} / SL {sl_mult}"
target_base = f"p{percentile}_h{horizon}"

if bar_col not in df_bar_regime_base.columns:
    raise KeyError(f"No existe la columna BAR: {bar_col}")

if event_step_col not in df_bar_regime_base.columns:
    raise KeyError(f"No existe la columna event_step: {event_step_col}")

for _, regime_info in regime_values.iterrows():

    regime_id = int(regime_info[REGIME_COL])
    regime_name = regime_info["regime_name"]

    df_regime = df_bar_regime_base[
        df_bar_regime_base[REGIME_COL] == regime_id
    ]

    summary = summarize_bar_target_for_regime(
        df=df_regime,
        bar_col=bar_col,
        event_step_col=event_step_col,
        tp_mult=tp_mult,
        sl_mult=sl_mult,
    )

    regime_rows.append(
        {
            "bar_col": bar_col,
            "event_step_col": event_step_col,
            "dir_col": meta["dir_col"],

            "target_base": target_base,
            "horizon": horizon,
            "percentile": percentile,
            "tp_mult": tp_mult,
            "sl_mult": sl_mult,
            "tp_sl_config": tp_sl_config,

            "regime_id": regime_id,
            "regime_name": regime_name,

            **summary,
        }
    )

df_bar_regime_analysis = pd.DataFrame(regime_rows)

df_bar_regime_analysis = (
    df_bar_regime_analysis
    .sort_values(
        [

```

```

        "regime_id",
        "percentile",
        "horizon",
        "sl_mult",
        "tp_mult",
    ]
)
.reset_index(drop=True)
)

# -----
# Vista inicial
# -----

print(f"Targets BAR analizados: {df_bar_metadata['bar_col'].nunique()}")
print(f"Regímenes analizados: {df_bar_regime_analysis['regime_id'].nunique()}")
print(f"Filas generadas: {len(df_bar_regime_analysis)}")

print("\nRegímenes detectados:")
print(
    df_bar_regime_analysis[
        ["regime_id", "regime_name"]
    ]
    .drop_duplicates()
    .sort_values("regime_id")
    .to_string(index=False)
)

print("\nVista inicial de BAR por régimen:")
print(
    df_bar_regime_analysis[
        [
            "bar_col",
            "regime_id",
            "regime_name",
            "horizon",
            "percentile",
            "tp_mult",
            "sl_mult",
            "total_rows",
            "valid_rows",
            "operable_rate",
            "tp_rate",
            "sl_rate",
            "no_event_rate",
            "resolved_rate",
            "tp_share_over_resolved",
            "expectancy_units",
            "event_step_mean",
        ]
    ]
    .head(35)
    .to_string(
        index=False,
        formatters={
            "operable_rate": "{:.2%}".format,
            "tp_rate": "{:.2%}".format,
            "sl_rate": "{:.2%}".format,
            "no_event_rate": "{:.2%}".format,
            "resolved_rate": "{:.2%}".format,
        },
    )
)

```

```
        "tp_share_over_resolved": "{:.2%}".format,  
        "expectancy_units": "{:.3f}".format,  
        "event_step_mean": "{:.2f}".format,  
    }  
)  
)  
  
print("\nPreparación de métricas BAR por régimen finalizada correctamente.")
```



## 4.10. Análisis por régimen intradiario - BAR

Targets BAR analizados: 63

Regímenes analizados: 5

Filas generadas: 315

Regímenes detectados:

| regime_id | regime_name |
|-----------|-------------|
| 0         | Overnight   |
| 1         | Pre-market  |
| 2         | Opening     |
| 3         | Regular     |
| 4         | Closing     |

Vista inicial de BAR por régimen:

| bar_col               | regime_id  | regime_name | horizon       | percentile | tp_mult | sl_mu         |               |                        |                  |                 |
|-----------------------|------------|-------------|---------------|------------|---------|---------------|---------------|------------------------|------------------|-----------------|
| lt                    | total_rows | valid_rows  | operable_rate | tp_rate    | sl_rate | no_event_rate | resolved_rate | tp_share_over_resolved | expectancy_units | event_step_mean |
| bar_p40_h30_tp10_sl05 | 0          | Overnight   | 30            | 40         | 1.0     |               |               |                        |                  |                 |
| 0.5                   | 357162     | 227256      | 63.63%        | 78.94%     | 21.06%  | 0.00%         | 10            |                        |                  |                 |
| 0.00%                 |            | 78.94%      |               | 0.684      | 11.18   |               |               |                        |                  |                 |
| bar_p40_h30_tp15_sl05 | 0          | Overnight   | 30            | 40         | 1.5     |               |               |                        |                  |                 |
| 0.5                   | 357162     | 227256      | 63.63%        | 47.69%     | 25.10%  | 27.21%        | 7             |                        |                  |                 |
| 2.79%                 |            | 65.51%      |               | 0.590      | 12.44   |               |               |                        |                  |                 |
| bar_p40_h30_tp20_sl05 | 0          | Overnight   | 30            | 40         | 2.0     |               |               |                        |                  |                 |
| 0.5                   | 357162     | 227256      | 63.63%        | 29.20%     | 26.88%  | 43.92%        | 5             |                        |                  |                 |
| 6.08%                 |            | 52.07%      |               | 0.450      | 12.59   |               |               |                        |                  |                 |
| bar_p40_h30_tp10_sl10 | 0          | Overnight   | 30            | 40         | 1.0     |               |               |                        |                  |                 |
| 1.0                   | 357162     | 227256      | 63.63%        | 93.84%     | 6.16%   | 0.00%         | 10            |                        |                  |                 |
| 0.00%                 |            | 93.84%      |               | 0.877      | 13.23   |               |               |                        |                  |                 |
| bar_p40_h30_tp15_sl10 | 0          | Overnight   | 30            | 40         | 1.5     |               |               |                        |                  |                 |
| 1.0                   | 357162     | 227256      | 63.63%        | 57.02%     | 7.34%   | 35.64%        | 6             |                        |                  |                 |
| 4.36%                 |            | 88.60%      |               | 0.782      | 14.83   |               |               |                        |                  |                 |
| bar_p40_h30_tp20_sl10 | 0          | Overnight   | 30            | 40         | 2.0     |               |               |                        |                  |                 |
| 1.0                   | 357162     | 227256      | 63.63%        | 35.18%     | 8.30%   | 56.52%        | 4             |                        |                  |                 |
| 3.48%                 |            | 80.91%      |               | 0.621      | 15.57   |               |               |                        |                  |                 |
| bar_p40_h30_tp25_sl10 | 0          | Overnight   | 30            | 40         | 2.5     |               |               |                        |                  |                 |
| 1.0                   | 357162     | 227256      | 63.63%        | 22.48%     | 8.83%   | 68.69%        | 3             |                        |                  |                 |
| 1.31%                 |            | 71.80%      |               | 0.474      | 15.79   |               |               |                        |                  |                 |
| bar_p40_h60_tp10_sl05 | 0          | Overnight   | 60            | 40         | 1.0     |               |               |                        |                  |                 |
| 0.5                   | 357162     | 227928      | 63.82%        | 77.15%     | 22.85%  | 0.00%         | 10            |                        |                  |                 |
| 0.00%                 |            | 77.15%      |               | 0.657      | 21.84   |               |               |                        |                  |                 |
| bar_p40_h60_tp15_sl05 | 0          | Overnight   | 60            | 40         | 1.5     |               |               |                        |                  |                 |
| 0.5                   | 357162     | 227928      | 63.82%        | 46.31%     | 27.11%  | 26.58%        | 7             |                        |                  |                 |
| 3.42%                 |            | 63.08%      |               | 0.559      | 24.14   |               |               |                        |                  |                 |
| bar_p40_h60_tp20_sl05 | 0          | Overnight   | 60            | 40         | 2.0     |               |               |                        |                  |                 |
| 0.5                   | 357162     | 227928      | 63.82%        | 28.11%     | 29.02%  | 42.87%        | 5             |                        |                  |                 |
| 7.13%                 |            | 49.21%      |               | 0.417      | 24.20   |               |               |                        |                  |                 |
| bar_p40_h60_tp10_sl10 | 0          | Overnight   | 60            | 40         | 1.0     |               |               |                        |                  |                 |
| 1.0                   | 357162     | 227928      | 63.82%        | 93.55%     | 6.45%   | 0.00%         | 10            |                        |                  |                 |
| 0.00%                 |            | 93.55%      |               | 0.871      | 26.24   |               |               |                        |                  |                 |
| bar_p40_h60_tp15_sl10 | 0          | Overnight   | 60            | 40         | 1.5     |               |               |                        |                  |                 |
| 1.0                   | 357162     | 227928      | 63.82%        | 56.64%     | 7.74%   | 35.62%        | 6             |                        |                  |                 |
| 4.38%                 |            | 87.98%      |               | 0.772      | 29.31   |               |               |                        |                  |                 |
| bar_p40_h60_tp20_sl10 | 0          | Overnight   | 60            | 40         | 2.0     |               |               |                        |                  |                 |
| 1.0                   | 357162     | 227928      | 63.82%        | 34.90%     | 8.75%   | 56.35%        | 4             |                        |                  |                 |
| 3.65%                 |            | 79.95%      |               | 0.610      | 30.69   |               |               |                        |                  |                 |
| bar_p40_h60_tp25_sl10 | 0          | Overnight   | 60            | 40         | 2.5     |               |               |                        |                  |                 |
| 1.0                   | 357162     | 227928      | 63.82%        | 22.41%     | 9.27%   | 68.31%        | 3             |                        |                  |                 |
| 1.69%                 |            | 70.73%      |               | 0.468      | 31.02   |               |               |                        |                  |                 |

|                       |   |               |        |        |     |  |
|-----------------------|---|---------------|--------|--------|-----|--|
| bar_p40_h90_tp10_sl05 | 0 | Overnight     | 90     | 40     | 1.0 |  |
| 0.5 357162 227186     |   | 63.61% 75.13% | 24.87% | 0.00%  | 10  |  |
| 0.00% 75.13%          |   | 0.627         | 34.38  |        |     |  |
| bar_p40_h90_tp15_sl05 | 0 | Overnight     | 90     | 40     | 1.5 |  |
| 0.5 357162 227186     |   | 63.61% 44.74% | 29.26% | 26.00% | 7   |  |
| 4.00% 60.46%          |   | 0.525         | 37.68  |        |     |  |
| bar_p40_h90_tp20_sl05 | 0 | Overnight     | 90     | 40     | 2.0 |  |
| 0.5 357162 227186     |   | 63.61% 28.08% | 31.16% | 40.76% | 5   |  |
| 9.24% 47.40%          |   | 0.406         | 38.11  |        |     |  |
| bar_p40_h90_tp10_sl10 | 0 | Overnight     | 90     | 40     | 1.0 |  |
| 1.0 357162 227186     |   | 63.61% 92.39% | 7.61%  | 0.00%  | 10  |  |
| 0.00% 92.39%          |   | 0.848         | 41.65  |        |     |  |
| bar_p40_h90_tp15_sl10 | 0 | Overnight     | 90     | 40     | 1.5 |  |
| 1.0 357162 227186     |   | 63.61% 56.13% | 9.05%  | 34.82% | 6   |  |
| 5.18% 86.11%          |   | 0.751         | 46.46  |        |     |  |
| bar_p40_h90_tp20_sl10 | 0 | Overnight     | 90     | 40     | 2.0 |  |
| 1.0 357162 227186     |   | 63.61% 35.88% | 10.20% | 53.92% | 4   |  |
| 6.08% 77.87%          |   | 0.616         | 48.98  |        |     |  |
| bar_p40_h90_tp25_sl10 | 0 | Overnight     | 90     | 40     | 2.5 |  |
| 1.0 357162 227186     |   | 63.61% 23.85% | 10.93% | 65.23% | 3   |  |
| 4.77% 68.58%          |   | 0.487         | 49.83  |        |     |  |
| bar_p50_h30_tp10_sl05 | 0 | Overnight     | 30     | 50     | 1.0 |  |
| 0.5 357162 192938     |   | 54.02% 81.88% | 18.12% | 0.00%  | 10  |  |
| 0.00% 81.88%          |   | 0.728         | 12.39  |        |     |  |
| bar_p50_h30_tp15_sl05 | 0 | Overnight     | 30     | 50     | 1.5 |  |
| 0.5 357162 192938     |   | 54.02% 45.27% | 21.59% | 33.13% | 6   |  |
| 6.87% 67.71%          |   | 0.571         | 13.33  |        |     |  |
| bar_p50_h30_tp20_sl05 | 0 | Overnight     | 30     | 50     | 2.0 |  |
| 0.5 357162 192938     |   | 54.02% 26.09% | 23.02% | 50.88% | 4   |  |
| 9.12% 53.12%          |   | 0.407         | 13.22  |        |     |  |
| bar_p50_h30_tp10_sl10 | 0 | Overnight     | 30     | 50     | 1.0 |  |
| 1.0 357162 192938     |   | 54.02% 95.12% | 4.88%  | 0.00%  | 10  |  |
| 0.00% 95.12%          |   | 0.902         | 14.25  |        |     |  |
| bar_p50_h30_tp15_sl10 | 0 | Overnight     | 30     | 50     | 1.5 |  |
| 1.0 357162 192938     |   | 54.02% 53.11% | 5.80%  | 41.08% | 5   |  |
| 8.92% 90.15%          |   | 0.739         | 15.68  |        |     |  |
| bar_p50_h30_tp20_sl10 | 0 | Overnight     | 30     | 50     | 2.0 |  |
| 1.0 357162 192938     |   | 54.02% 30.99% | 6.55%  | 62.46% | 3   |  |
| 7.54% 82.54%          |   | 0.554         | 16.33  |        |     |  |
| bar_p50_h30_tp25_sl10 | 0 | Overnight     | 30     | 50     | 2.5 |  |
| 1.0 357162 192938     |   | 54.02% 18.98% | 6.93%  | 74.09% | 2   |  |
| 5.91% 73.25%          |   | 0.405         | 16.39  |        |     |  |
| bar_p50_h60_tp10_sl05 | 0 | Overnight     | 60     | 50     | 1.0 |  |
| 0.5 357162 192987     |   | 54.03% 80.61% | 19.39% | 0.00%  | 10  |  |
| 0.00% 80.61%          |   | 0.709         | 24.34  |        |     |  |
| bar_p50_h60_tp15_sl05 | 0 | Overnight     | 60     | 50     | 1.5 |  |
| 0.5 357162 192987     |   | 54.03% 44.15% | 23.09% | 32.76% | 6   |  |
| 7.24% 65.66%          |   | 0.547         | 26.01  |        |     |  |
| bar_p50_h60_tp20_sl05 | 0 | Overnight     | 60     | 50     | 2.0 |  |
| 0.5 357162 192987     |   | 54.03% 25.45% | 24.55% | 50.00% | 5   |  |
| 0.00% 50.90%          |   | 0.386         | 25.59  |        |     |  |
| bar_p50_h60_tp10_sl10 | 0 | Overnight     | 60     | 50     | 1.0 |  |
| 1.0 357162 192987     |   | 54.03% 94.96% | 5.04%  | 0.00%  | 10  |  |
| 0.00% 94.96%          |   | 0.899         | 28.22  |        |     |  |
| bar_p50_h60_tp15_sl10 | 0 | Overnight     | 60     | 50     | 1.5 |  |
| 1.0 357162 192987     |   | 54.03% 52.81% | 6.02%  | 41.17% | 5   |  |
| 8.83% 89.77%          |   | 0.732         | 31.04  |        |     |  |
| bar_p50_h60_tp20_sl10 | 0 | Overnight     | 60     | 50     | 2.0 |  |
| 1.0 357162 192987     |   | 54.03% 30.97% | 6.77%  | 62.26% | 3   |  |
| 7.74% 82.07%          |   | 0.552         | 32.15  |        |     |  |

|                       |        |           |        |        |       |        |
|-----------------------|--------|-----------|--------|--------|-------|--------|
| bar_p50_h60_tp25_sl10 | 0      | Overnight | 60     | 50     | 2.5   |        |
| 1.0                   | 357162 | 192987    | 54.03% | 19.31% | 7.19% | 73.49% |
| 6.51%                 |        | 72.87%    | 0.411  |        | 32.22 | 2      |

Preparación de métricas BAR por régimen finalizada correctamente.

## Celda 2 — Tablas agregadas por régimen, configuración, percentil y horizonte

In [112...

```
# Celda 2 – Tablas agregadas por régimen, configuración, percentil y horizonte
# =====

print("Tablas agregadas por régimen intradiario - BAR")
print("=" * 80)

# -----
# Validar tabla base
# -----

if "df_bar_regime_analysis" not in globals():
    raise NameError("No existe df_bar_regime_analysis. Primero debes correr la C

# -----
# Orden auxiliar de configuraciones TP/SL
# -----

config_order = [
    (1.0, 1.0),
    (1.5, 1.0),
    (2.0, 1.0),
    (2.5, 1.0),
    (1.0, 0.5),
    (1.5, 0.5),
    (2.0, 0.5),
]

target_base_order = (
    df_bar_regime_analysis[["percentile", "horizon", "target_base"]]
    .drop_duplicates()
    .sort_values(["percentile", "horizon"])
    ["target_base"]
    .tolist()
)

# -----
# Función auxiliar para rango
# -----

def value_range(series):
    return series.max() - series.min()

# =====
# 1. Resumen general por régimen
# =====

df_bar_regime_by_regime = (
    df_bar_regime_analysis
    .groupby(["regime_id", "regime_name"], as_index=False)
    .agg(
```

```

n_targets=("bar_col", "count"),

total_rows_mean=("total_rows", "mean"),
valid_rows_mean=("valid_rows", "mean"),
operable_rate_mean=("operable_rate", "mean"),

tp_rate_mean=("tp_rate", "mean"),
sl_rate_mean=("sl_rate", "mean"),
no_event_rate_mean=("no_event_rate", "mean"),
resolved_rate_mean=("resolved_rate", "mean"),

tp_share_over_resolved_mean=("tp_share_over_resolved", "mean"),
expectancy_units_mean=("expectancy_units", "mean"),

event_step_mean=("event_step_mean", "mean"),
event_step_median=("event_step_median", "mean"),
)
.sort_values("regime_id")
.reset_index(drop=True)
)

print("\nResumen general por régimen:")
print(
    df_bar_regime_by_regime.to_string(
        index=False,
        formatters={
            "total_rows_mean": "{:.0f}".format,
            "valid_rows_mean": "{:.0f}".format,
            "operable_rate_mean": "{:.2%}".format,
            "tp_rate_mean": "{:.2%}".format,
            "sl_rate_mean": "{:.2%}".format,
            "no_event_rate_mean": "{:.2%}".format,
            "resolved_rate_mean": "{:.2%}".format,
            "tp_share_over_resolved_mean": "{:.2%}".format,
            "expectancy_units_mean": "{:.3f}".format,
            "event_step_mean": "{:.2f}".format,
            "event_step_median": "{:.2f}".format,
        }
    )
)

# =====
# 2. Resumen por régimen y configuración TP/SL
# =====

df_bar_regime_by_config = (
    df_bar_regime_analysis
    .groupby(
        [
            "regime_id",
            "regime_name",
            "tp_mult",
            "sl_mult",
            "tp_sl_config",
        ],
        as_index=False
    )
    .agg(
        n_targets=("bar_col", "count"),

```

```

operable_rate_mean=("operable_rate", "mean"),

tp_rate_mean=("tp_rate", "mean"),
sl_rate_mean=("sl_rate", "mean"),
no_event_rate_mean=("no_event_rate", "mean"),
resolved_rate_mean=("resolved_rate", "mean"),

tp_share_over_resolved_mean=("tp_share_over_resolved", "mean"),
expectancy_units_mean=("expectancy_units", "mean"),

event_step_mean=("event_step_mean", "mean"),
event_step_median=("event_step_median", "mean"),
)
.reset_index(drop=True)
)

df_bar_regime_by_config["config_order"] = df_bar_regime_by_config.apply(
    lambda row: config_order.index((row["tp_mult"], row["sl_mult"]))
    if (row["tp_mult"], row["sl_mult"]) in config_order
    else 999,
    axis=1
)

df_bar_regime_by_config = (
    df_bar_regime_by_config
    .sort_values(["regime_id", "config_order"])
    .reset_index(drop=True)
)

print("\nResumen por régimen y configuración TP/SL:")
print(
    df_bar_regime_by_config[
        [
            "regime_id",
            "regime_name",
            "tp_sl_config",
            "n_targets",
            "operable_rate_mean",
            "tp_rate_mean",
            "sl_rate_mean",
            "no_event_rate_mean",
            "resolved_rate_mean",
            "tp_share_over_resolved_mean",
            "expectancy_units_mean",
            "event_step_mean",
        ]
    ]
    .to_string(
        index=False,
        formatters={
            "operable_rate_mean": "{:.2%}".format,
            "tp_rate_mean": "{:.2%}".format,
            "sl_rate_mean": "{:.2%}".format,
            "no_event_rate_mean": "{:.2%}".format,
            "resolved_rate_mean": "{:.2%}".format,
            "tp_share_over_resolved_mean": "{:.2%}".format,
            "expectancy_units_mean": "{:.3f}".format,
            "event_step_mean": "{:.2f}".format,
        }
    )
)

```

```

)

# =====
# 3. Resumen por régimen, percentil y horizonte
# =====

df_bar_regime_by_target_base = (
    df_bar_regime_analysis
    .groupby(
        [
            "regime_id",
            "regime_name",
            "percentile",
            "horizon",
            "target_base",
        ],
        as_index=False
    )
    .agg(
        n_configs=("bar_col", "count"),

        operable_rate_mean=("operable_rate", "mean"),

        tp_rate_mean=("tp_rate", "mean"),
        sl_rate_mean=("sl_rate", "mean"),
        no_event_rate_mean=("no_event_rate", "mean"),
        resolved_rate_mean=("resolved_rate", "mean"),

        tp_share_over_resolved_mean=("tp_share_over_resolved", "mean"),
        expectancy_units_mean=("expectancy_units", "mean"),

        event_step_mean=("event_step_mean", "mean"),
        event_step_median=("event_step_median", "mean"),
    )
    .sort_values(["regime_id", "percentile", "horizon"])
    .reset_index(drop=True)
)

print("\nResumen por régimen, percentil y horizonte:")
print(
    df_bar_regime_by_target_base[
        [
            "regime_id",
            "regime_name",
            "target_base",
            "n_configs",
            "operable_rate_mean",
            "tp_rate_mean",
            "sl_rate_mean",
            "no_event_rate_mean",
            "resolved_rate_mean",
            "tp_share_over_resolved_mean",
            "expectancy_units_mean",
            "event_step_mean",
        ]
    ]
    .to_string(
        index=False,
        formatters={
            "operable_rate_mean": "{:.2%}".format,

```

```

        "tp_rate_mean": "{:.2%}".format,
        "sl_rate_mean": "{:.2%}".format,
        "no_event_rate_mean": "{:.2%}".format,
        "resolved_rate_mean": "{:.2%}".format,
        "tp_share_over_resolved_mean": "{:.2%}".format,
        "expectancy_units_mean": "{:.3f}".format,
        "event_step_mean": "{:.2f}".format,
    }
)
)

# =====
# 4. Tabla específica para candidata principal TP 1.5 / SL 1.0
# =====

df_bar_regime_candidate_tp15_sl10 = (
    df_bar_regime_analysis[
        (df_bar_regime_analysis["tp_mult"] == 1.5)
        & (df_bar_regime_analysis["sl_mult"] == 1.0)
    ]
    .copy()
    .sort_values(["percentile", "horizon", "regime_id"])
    .reset_index(drop=True)
)

print("\nDetalle por régimen para configuración candidata TP 1.5 / SL 1.0:")
print(
    df_bar_regime_candidate_tp15_sl10[
        [
            "bar_col",
            "target_base",
            "regime_id",
            "regime_name",
            "total_rows",
            "valid_rows",
            "operable_rate",
            "tp_rate",
            "sl_rate",
            "no_event_rate",
            "resolved_rate",
            "tp_share_over_resolved",
            "expectancy_units",
            "event_step_mean",
        ]
    ]
    .to_string(
        index=False,
        formatters={
            "operable_rate": "{:.2%}".format,
            "tp_rate": "{:.2%}".format,
            "sl_rate": "{:.2%}".format,
            "no_event_rate": "{:.2%}".format,
            "resolved_rate": "{:.2%}".format,
            "tp_share_over_resolved": "{:.2%}".format,
            "expectancy_units": "{:.3f}".format,
            "event_step_mean": "{:.2f}".format,
        }
    )
)

```

```

# =====
# 5. Estabilidad entre regímenes por target BAR
# =====
# Mide cuánto cambia cada target entre los distintos regímenes.
# No selecciona el target final, solo identifica sensibilidad al régimen.
# =====

id_cols = [
    "bar_col",
    "target_base",
    "horizon",
    "percentile",
    "tp_mult",
    "sl_mult",
    "tp_sl_config",
]

df_bar_regime_stability_by_target = (
    df_bar_regime_analysis
    .groupby(id_cols, as_index=False)
    .agg(
        n_regimes=("regime_id", "nunique"),

        operable_rate_mean=("operable_rate", "mean"),
        operable_rate_range=("operable_rate", value_range),

        tp_rate_mean=("tp_rate", "mean"),
        tp_rate_range=("tp_rate", value_range),

        sl_rate_mean=("sl_rate", "mean"),
        sl_rate_range=("sl_rate", value_range),

        no_event_rate_mean=("no_event_rate", "mean"),
        no_event_rate_range=("no_event_rate", value_range),

        resolved_rate_mean=("resolved_rate", "mean"),
        resolved_rate_range=("resolved_rate", value_range),

        tp_share_over_resolved_mean=("tp_share_over_resolved", "mean"),
        tp_share_over_resolved_range=("tp_share_over_resolved", value_range),

        expectancy_units_mean=("expectancy_units", "mean"),
        expectancy_units_range=("expectancy_units", value_range),

        event_step_mean_mean=("event_step_mean", "mean"),
        event_step_mean_range=("event_step_mean", value_range),
    )
    .reset_index(drop=True)
)

# Score simple de sensibilidad al régimen
df_bar_regime_stability_by_target["regime_sensitivity_score"] = (
    df_bar_regime_stability_by_target[
        [
            "operable_rate_range",
            "tp_rate_range",
            "sl_rate_range",
            "no_event_rate_range",
            "tp_share_over_resolved_range",
        ]
    ]

```



```

    ]
    .sum(axis=1)
)

df_bar_regime_stability_by_target = (
    df_bar_regime_stability_by_target
    .sort_values(
        [
            "regime_sensitivity_score",
            "percentile",
            "horizon",
            "sl_mult",
            "tp_mult",
        ],
        ascending=[False, True, True, True, True]
    )
    .reset_index(drop=True)
)

print("\nTop 20 targets con mayor sensibilidad al régimen:")
print(
    df_bar_regime_stability_by_target[
        [
            "bar_col",
            "target_base",
            "tp_sl_config",
            "n_regimes",
            "operable_rate_range",
            "tp_rate_range",
            "sl_rate_range",
            "no_event_rate_range",
            "tp_share_over_resolved_range",
            "expectancy_units_range",
            "regime_sensitivity_score",
        ]
    ]
    .head(20)
    .to_string(
        index=False,
        formatters={
            "operable_rate_range": "{:.2%}".format,
            "tp_rate_range": "{:.2%}".format,
            "sl_rate_range": "{:.2%}".format,
            "no_event_rate_range": "{:.2%}".format,
            "tp_share_over_resolved_range": "{:.2%}".format,
            "expectancy_units_range": "{:.3f}".format,
            "regime_sensitivity_score": "{:.3f}".format,
        }
    )
)

print("\nTop 20 targets con menor sensibilidad al régimen:")
print(
    df_bar_regime_stability_by_target[
        [
            "bar_col",
            "target_base",
            "tp_sl_config",
            "n_regimes",
            "operable_rate_range",

```

```

        "tp_rate_range",
        "sl_rate_range",
        "no_event_rate_range",
        "tp_share_over_resolved_range",
        "expectancy_units_range",
        "regime_sensitivity_score",
    ]
]
.tail(20)
.sort_values("regime_sensitivity_score")
.to_string(
    index=False,
    formatters={
        "operable_rate_range": "{:.2%}".format,
        "tp_rate_range": "{:.2%}".format,
        "sl_rate_range": "{:.2%}".format,
        "no_event_rate_range": "{:.2%}".format,
        "tp_share_over_resolved_range": "{:.2%}".format,
        "expectancy_units_range": "{:.3f}".format,
        "regime_sensitivity_score": "{:.3f}".format,
    }
)
)

# =====
# 6. Sensibilidad agregada al régimen por configuración TP/SL
# =====

df_bar_regime_sensitivity_by_config = (
    df_bar_regime_stability_by_target
    .groupby(["tp_mult", "sl_mult", "tp_sl_config"], as_index=False)
    .agg(
        n_targets=("bar_col", "count"),

        operable_rate_range_mean=("operable_rate_range", "mean"),
        tp_rate_range_mean=("tp_rate_range", "mean"),
        sl_rate_range_mean=("sl_rate_range", "mean"),
        no_event_rate_range_mean=("no_event_rate_range", "mean"),
        tp_share_range_mean=("tp_share_over_resolved_range", "mean"),
        expectancy_range_mean=("expectancy_units_range", "mean"),

        regime_sensitivity_score_mean=("regime_sensitivity_score", "mean"),
        regime_sensitivity_score_max=("regime_sensitivity_score", "max"),
    )
    .reset_index(drop=True)
)

df_bar_regime_sensitivity_by_config["config_order"] = df_bar_regime_sensitivity_
    lambda row: config_order.index((row["tp_mult"], row["sl_mult"]))
    if (row["tp_mult"], row["sl_mult"]) in config_order
    else 999,
    axis=1
)

df_bar_regime_sensitivity_by_config = (
    df_bar_regime_sensitivity_by_config
    .sort_values("config_order")
    .reset_index(drop=True)
)

```

```

print("\nSensibilidad agregada al régimen por configuración TP/SL:")
print(
    df_bar_regime_sensitivity_by_config[
        [
            "tp_sl_config",
            "n_targets",
            "operable_rate_range_mean",
            "tp_rate_range_mean",
            "sl_rate_range_mean",
            "no_event_rate_range_mean",
            "tp_share_range_mean",
            "expectancy_range_mean",
            "regime_sensitivity_score_mean",
            "regime_sensitivity_score_max",
        ]
    ].to_string(
        index=False,
        formatters={
            "operable_rate_range_mean": "{:.2%}".format,
            "tp_rate_range_mean": "{:.2%}".format,
            "sl_rate_range_mean": "{:.2%}".format,
            "no_event_rate_range_mean": "{:.2%}".format,
            "tp_share_range_mean": "{:.2%}".format,
            "expectancy_range_mean": "{:.3f}".format,
            "regime_sensitivity_score_mean": "{:.3f}".format,
            "regime_sensitivity_score_max": "{:.3f}".format,
        }
    )
)

# =====
# 7. Sensibilidad agregada al régimen por percentil y horizonte
# =====

df_bar_regime_sensitivity_by_target_base = (
    df_bar_regime_stability_by_target
    .groupby(["percentile", "horizon", "target_base"], as_index=False)
    .agg(
        n_configs=("bar_col", "count"),

        operable_rate_range_mean=("operable_rate_range", "mean"),
        tp_rate_range_mean=("tp_rate_range", "mean"),
        sl_rate_range_mean=("sl_rate_range", "mean"),
        no_event_rate_range_mean=("no_event_rate_range", "mean"),
        tp_share_range_mean=("tp_share_over_resolved_range", "mean"),
        expectancy_range_mean=("expectancy_units_range", "mean"),

        regime_sensitivity_score_mean=("regime_sensitivity_score", "mean"),
        regime_sensitivity_score_max=("regime_sensitivity_score", "max"),
    )
    .sort_values(["percentile", "horizon"])
    .reset_index(drop=True)
)

print("\nSensibilidad agregada al régimen por percentil y horizonte:")
print(
    df_bar_regime_sensitivity_by_target_base.to_string(
        index=False,
        formatters={

```

```
        "operable_rate_range_mean": "{:.2%}".format,  
        "tp_rate_range_mean": "{:.2%}".format,  
        "sl_rate_range_mean": "{:.2%}".format,  
        "no_event_rate_range_mean": "{:.2%}".format,  
        "tp_share_range_mean": "{:.2%}".format,  
        "expectancy_range_mean": "{:.3f}".format,  
        "regime_sensitivity_score_mean": "{:.3f}".format,  
        "regime_sensitivity_score_max": "{:.3f}".format,  
    }  
)  
)  
  
print("\nCelda 2 finalizada correctamente.")
```

## Tablas agregadas por régimen intradiario - BAR

=====

## Resumen general por régimen:

|        | regime_id | regime_name | n_targets | total_rows_mean | valid_rows_mean | operable_rate_mean | tp_rate_mean | sl_rate_mean | no_event_rate_mean | resolved_rate_mean | tp_share_over_resolved_mean | expectancy_units_mean | event_step_mean | event_step_median |
|--------|-----------|-------------|-----------|-----------------|-----------------|--------------------|--------------|--------------|--------------------|--------------------|-----------------------------|-----------------------|-----------------|-------------------|
|        | 0         | Overnight   | 63        | 357162          | 192910          | 54.01%             | 94.60%       | 0.00%        | 100.00%            | 94.60%             |                             |                       |                 |                   |
| 01%    |           |             |           |                 |                 |                    |              |              |                    |                    |                             |                       |                 |                   |
| 76.30% |           |             |           |                 |                 |                    |              |              |                    |                    |                             |                       |                 |                   |
|        | 1         | Pre-market  | 63        | 88920           | 48461           | 54.01%             | 53.21%       | 40.32%       | 59.68%             | 89.26%             |                             |                       |                 |                   |
| 50%    |           |             |           |                 |                 |                    |              |              |                    |                    |                             |                       |                 |                   |
| 76.06% |           |             |           |                 |                 |                    |              |              |                    |                    |                             |                       |                 |                   |
|        | 2         | Opening     | 63        | 88920           | 48416           | 54.01%             | 31.72%       | 60.98%       | 39.02%             | 81.50%             |                             |                       |                 |                   |
| 45%    |           |             |           |                 |                 |                    |              |              |                    |                    |                             |                       |                 |                   |
| 78.70% |           |             |           |                 |                 |                    |              |              |                    |                    |                             |                       |                 |                   |
|        | 3         | Regular     | 63        | 444600          | 215464          | 48.01%             | 80.15%       | 32.11%       | 67.89%             | 65.43%             |                             |                       |                 |                   |
| 46%    |           |             |           |                 |                 |                    |              |              |                    |                    |                             |                       |                 |                   |
| 77.20% |           |             |           |                 |                 |                    |              |              |                    |                    |                             |                       |                 |                   |
|        | 4         | Closing     | 63        | 44460           | 264             | 0.00%              | 16.73%       | 76.01%       | 23.99%             | 69.81%             |                             |                       |                 |                   |
| 59%    |           |             |           |                 |                 |                    |              |              |                    |                    |                             |                       |                 |                   |
| 79.16% |           |             |           |                 |                 |                    |              |              |                    |                    |                             |                       |                 |                   |

## Resumen por régimen y configuración TP/SL:

|        | regime_id | regime_name | tp_sl_config    | n_targets | operable_rate_mean | tp_rate_mean | sl_rate_mean | no_event_rate_mean | resolved_rate_mean | tp_share_over_resolved_mean | expectancy_units_mean | event_step_mean |
|--------|-----------|-------------|-----------------|-----------|--------------------|--------------|--------------|--------------------|--------------------|-----------------------------|-----------------------|-----------------|
|        | 0         | Overnight   | TP 1.0 / SL 1.0 | 9         | 54.01%             | 94.60%       | 0.00%        | 100.00%            | 94.60%             |                             |                       |                 |
| 5.40%  |           |             |                 |           |                    |              |              |                    |                    |                             |                       |                 |
| 0.892  |           |             |                 |           |                    |              |              |                    |                    |                             |                       |                 |
|        | 0         | Overnight   | TP 1.5 / SL 1.0 | 9         | 54.01%             | 53.21%       | 40.32%       | 59.68%             | 89.26%             |                             |                       |                 |
| 6.47%  |           |             |                 |           |                    |              |              |                    |                    |                             |                       |                 |
| 0.733  |           |             |                 |           |                    |              |              |                    |                    |                             |                       |                 |
|        | 0         | Overnight   | TP 2.0 / SL 1.0 | 9         | 54.01%             | 31.72%       | 60.98%       | 39.02%             | 81.50%             |                             |                       |                 |
| 7.30%  |           |             |                 |           |                    |              |              |                    |                    |                             |                       |                 |
| 0.561  |           |             |                 |           |                    |              |              |                    |                    |                             |                       |                 |
|        | 0         | Overnight   | TP 2.5 / SL 1.0 | 9         | 54.01%             | 19.78%       | 72.46%       | 27.54%             | 72.10%             |                             |                       |                 |
| 7.76%  |           |             |                 |           |                    |              |              |                    |                    |                             |                       |                 |
| 0.417  |           |             |                 |           |                    |              |              |                    |                    |                             |                       |                 |
|        | 0         | Overnight   | TP 1.0 / SL 0.5 | 9         | 54.01%             | 80.15%       | 0.00%        | 100.00%            | 80.15%             |                             |                       |                 |
| 19.85% |           |             |                 |           |                    |              |              |                    |                    |                             |                       |                 |
| 0.702  |           |             |                 |           |                    |              |              |                    |                    |                             |                       |                 |
|        | 0         | Overnight   | TP 1.5 / SL 0.5 | 9         | 54.01%             | 44.32%       | 32.11%       | 67.89%             | 65.43%             |                             |                       |                 |
| 23.57% |           |             |                 |           |                    |              |              |                    |                    |                             |                       |                 |
| 0.547  |           |             |                 |           |                    |              |              |                    |                    |                             |                       |                 |
|        | 0         | Overnight   | TP 2.0 / SL 0.5 | 9         | 54.01%             | 26.00%       | 48.89%       | 51.11%             | 51.05%             |                             |                       |                 |
| 25.11% |           |             |                 |           |                    |              |              |                    |                    |                             |                       |                 |
| 0.394  |           |             |                 |           |                    |              |              |                    |                    |                             |                       |                 |
|        | 1         | Pre-market  | TP 1.0 / SL 1.0 | 9         | 54.50%             | 95.17%       | 0.00%        | 100.00%            | 95.17%             |                             |                       |                 |
| 4.83%  |           |             |                 |           |                    |              |              |                    |                    |                             |                       |                 |
| 0.903  |           |             |                 |           |                    |              |              |                    |                    |                             |                       |                 |
|        | 1         | Pre-market  | TP 1.5 / SL 1.0 | 9         | 54.50%             | 53.07%       | 40.98%       | 59.02%             | 90.20%             |                             |                       |                 |
| 5.95%  |           |             |                 |           |                    |              |              |                    |                    |                             |                       |                 |
| 0.737  |           |             |                 |           |                    |              |              |                    |                    |                             |                       |                 |
|        | 1         | Pre-market  | TP 2.0 / SL 1.0 | 9         | 54.50%             | 29.54%       | 63.64%       | 36.36%             | 81.71%             |                             |                       |                 |
| 6.82%  |           |             |                 |           |                    |              |              |                    |                    |                             |                       |                 |
| 0.523  |           |             |                 |           |                    |              |              |                    |                    |                             |                       |                 |
|        | 1         | Pre-market  | TP 2.5 / SL 1.0 | 9         | 54.50%             | 16.73%       | 76.01%       | 23.99%             | 69.81%             |                             |                       |                 |
| 7.26%  |           |             |                 |           |                    |              |              |                    |                    |                             |                       |                 |
| 0.346  |           |             |                 |           |                    |              |              |                    |                    |                             |                       |                 |

|        |   |                            |   |        |        |
|--------|---|----------------------------|---|--------|--------|
| 19.44% | 1 | Pre-market TP 1.0 / SL 0.5 | 9 | 54.50% | 80.56% |
| 0.708  |   | 0.00% 100.00%              |   | 80.56% |        |
|        |   | 30.17                      |   |        |        |
| 23.50% | 1 | Pre-market TP 1.5 / SL 0.5 | 9 | 54.50% | 44.59% |
| 0.551  |   | 31.91% 68.09%              |   | 65.64% |        |
|        |   | 32.00                      |   |        |        |
| 25.16% | 1 | Pre-market TP 2.0 / SL 0.5 | 9 | 54.50% | 24.68% |
| 0.368  |   | 50.15% 49.85%              |   | 49.31% |        |
|        |   | 31.40                      |   |        |        |
| 3.62%  | 2 | Opening TP 1.0 / SL 1.0    | 9 | 54.45% | 96.38% |
| 0.928  |   | 0.00% 100.00%              |   | 96.38% |        |
|        |   | 25.39                      |   |        |        |
| 4.46%  | 2 | Opening TP 1.5 / SL 1.0    | 9 | 54.45% | 51.49% |
| 0.728  |   | 44.05% 55.95%              |   | 92.17% |        |
|        |   | 29.59                      |   |        |        |
| 5.10%  | 2 | Opening TP 2.0 / SL 1.0    | 9 | 54.45% | 27.17% |
| 0.492  |   | 67.74% 32.26%              |   | 84.47% |        |
|        |   | 31.82                      |   |        |        |
| 5.39%  | 2 | Opening TP 2.5 / SL 1.0    | 9 | 54.45% | 14.34% |
| 0.304  |   | 80.28% 19.72%              |   | 72.88% |        |
|        |   | 32.31                      |   |        |        |
| 16.45% | 2 | Opening TP 1.0 / SL 0.5    | 9 | 54.45% | 83.55% |
| 0.753  |   | 0.00% 100.00%              |   | 83.55% |        |
|        |   | 21.82                      |   |        |        |
| 19.96% | 2 | Opening TP 1.5 / SL 0.5    | 9 | 54.45% | 44.39% |
| 0.566  |   | 35.65% 64.35%              |   | 69.13% |        |
|        |   | 24.61                      |   |        |        |
| 21.31% | 2 | Opening TP 2.0 / SL 0.5    | 9 | 54.45% | 23.37% |
| 0.361  |   | 55.32% 44.68%              |   | 52.30% |        |
|        |   | 24.43                      |   |        |        |
| 4.56%  | 3 | Regular TP 1.0 / SL 1.0    | 9 | 48.46% | 95.44% |
| 0.909  |   | 0.00% 100.00%              |   | 95.44% |        |
|        |   | 27.00                      |   |        |        |
| 5.58%  | 3 | Regular TP 1.5 / SL 1.0    | 9 | 48.46% | 52.51% |
| 0.732  |   | 41.91% 58.09%              |   | 90.51% |        |
|        |   | 30.31                      |   |        |        |
| 6.36%  | 3 | Regular TP 2.0 / SL 1.0    | 9 | 48.46% | 29.80% |
| 0.532  |   | 63.84% 36.16%              |   | 82.61% |        |
|        |   | 31.93                      |   |        |        |
| 6.76%  | 3 | Regular TP 2.5 / SL 1.0    | 9 | 48.46% | 17.56% |
| 0.371  |   | 75.68% 24.32%              |   | 72.44% |        |
|        |   | 32.33                      |   |        |        |
| 18.53% | 3 | Regular TP 1.0 / SL 0.5    | 9 | 48.46% | 81.47% |
| 0.722  |   | 0.00% 100.00%              |   | 81.47% |        |
|        |   | 23.14                      |   |        |        |
| 22.37% | 3 | Regular TP 1.5 / SL 0.5    | 9 | 48.46% | 44.43% |
| 0.555  |   | 33.21% 66.79%              |   | 66.64% |        |
|        |   | 25.23                      |   |        |        |
| 23.90% | 3 | Regular TP 2.0 / SL 0.5    | 9 | 48.46% | 25.05% |
| 0.382  |   | 51.05% 48.95%              |   | 51.25% |        |
|        |   | 25.05                      |   |        |        |
| 3.95%  | 4 | Closing TP 1.0 / SL 1.0    | 9 | 0.59%  | 96.05% |
| 0.921  |   | 0.00% 100.00%              |   | 96.05% |        |
|        |   | 16.45                      |   |        |        |
| 4.83%  | 4 | Closing TP 1.5 / SL 1.0    | 9 | 0.59%  | 52.78% |
| 0.743  |   | 42.38% 57.62%              |   | 91.76% |        |
|        |   | 18.25                      |   |        |        |
| 5.29%  | 4 | Closing TP 2.0 / SL 1.0    | 9 | 0.59%  | 29.83% |
| 0.544  |   | 64.88% 35.12%              |   | 85.37% |        |
|        |   | 18.80                      |   |        |        |

|        |   |                         |   |        |        |
|--------|---|-------------------------|---|--------|--------|
| 5.62%  | 4 | Closing TP 2.5 / SL 1.0 | 9 | 0.59%  | 17.77% |
| 0.388  |   | 76.61%                  |   | 76.55% |        |
|        |   | 18.71                   |   |        |        |
| 17.02% | 4 | Closing TP 1.0 / SL 0.5 | 9 | 0.59%  | 82.98% |
| 0.745  |   | 0.00%                   |   | 82.98% |        |
|        |   | 14.54                   |   |        |        |
| 20.96% | 4 | Closing TP 1.5 / SL 0.5 | 9 | 0.59%  | 44.87% |
| 0.568  |   | 34.17%                  |   | 68.28% |        |
|        |   | 15.63                   |   |        |        |
| 22.26% | 4 | Closing TP 2.0 / SL 0.5 | 9 | 0.59%  | 25.09% |
| 0.391  |   | 52.65%                  |   | 53.13% |        |
|        |   | 15.31                   |   |        |        |

Resumen por régimen, percentil y horizonte:

|        | regime_id | regime_name | target_base | n_configs | operable_rate_mean | tp_rate_mean | sl_rate_mean | no_event_rate_mean | resolved_rate_mean | tp_share_over_resolved_mean | expectancy_units_mean | event_step_mean |
|--------|-----------|-------------|-------------|-----------|--------------------|--------------|--------------|--------------------|--------------------|-----------------------------|-----------------------|-----------------|
| 14.81% | 0         | Overnight   | p40_h30     | 7         | 63.63%             | 52.05%       |              |                    |                    |                             |                       |                 |
| 0.640  |           |             |             |           |                    | 75.95%       |              |                    |                    |                             |                       |                 |
|        |           |             |             |           |                    |              |              |                    |                    |                             |                       |                 |
| 15.88% | 0         | Overnight   | p40_h60     | 7         | 63.82%             | 51.30%       |              |                    |                    |                             |                       |                 |
| 0.622  |           |             |             |           |                    | 74.52%       |              |                    |                    |                             |                       |                 |
|        |           |             |             |           |                    |              |              |                    |                    |                             |                       |                 |
| 17.58% | 0         | Overnight   | p40_h90     | 7         | 63.61%             | 50.88%       |              |                    |                    |                             |                       |                 |
| 0.608  |           |             |             |           |                    | 72.56%       |              |                    |                    |                             |                       |                 |
|        |           |             |             |           |                    |              |              |                    |                    |                             |                       |                 |
| 12.42% | 0         | Overnight   | p50_h30     | 7         | 54.02%             | 50.21%       |              |                    |                    |                             |                       |                 |
| 0.615  |           |             |             |           |                    | 77.68%       |              |                    |                    |                             |                       |                 |
|        |           |             |             |           |                    |              |              |                    |                    |                             |                       |                 |
| 13.15% | 0         | Overnight   | p50_h60     | 7         | 54.03%             | 49.75%       |              |                    |                    |                             |                       |                 |
| 0.605  |           |             |             |           |                    | 76.69%       |              |                    |                    |                             |                       |                 |
|        |           |             |             |           |                    |              |              |                    |                    |                             |                       |                 |
| 15.02% | 0         | Overnight   | p50_h90     | 7         | 54.07%             | 49.79%       |              |                    |                    |                             |                       |                 |
| 0.599  |           |             |             |           |                    | 74.58%       |              |                    |                    |                             |                       |                 |
|        |           |             |             |           |                    |              |              |                    |                    |                             |                       |                 |
| 10.37% | 0         | Overnight   | p60_h30     | 7         | 44.82%             | 48.58%       |              |                    |                    |                             |                       |                 |
| 0.593  |           |             |             |           |                    | 79.30%       |              |                    |                    |                             |                       |                 |
|        |           |             |             |           |                    |              |              |                    |                    |                             |                       |                 |
| 10.90% | 0         | Overnight   | p60_h60     | 7         | 44.10%             | 48.31%       |              |                    |                    |                             |                       |                 |
| 0.588  |           |             |             |           |                    | 78.68%       |              |                    |                    |                             |                       |                 |
|        |           |             |             |           |                    |              |              |                    |                    |                             |                       |                 |
| 12.60% | 0         | Overnight   | p60_h90     | 7         | 44.00%             | 48.85%       |              |                    |                    |                             |                       |                 |
| 0.590  |           |             |             |           |                    | 76.73%       |              |                    |                    |                             |                       |                 |
|        |           |             |             |           |                    |              |              |                    |                    |                             |                       |                 |
| 18.30% | 1         | Pre-market  | p40_h30     | 7         | 63.71%             | 54.37%       |              |                    |                    |                             |                       |                 |
| 0.669  |           |             |             |           |                    | 73.53%       |              |                    |                    |                             |                       |                 |
|        |           |             |             |           |                    |              |              |                    |                    |                             |                       |                 |
| 15.24% | 1         | Pre-market  | p40_h60     | 7         | 64.10%             | 50.85%       |              |                    |                    |                             |                       |                 |
| 0.614  |           |             |             |           |                    | 74.79%       |              |                    |                    |                             |                       |                 |
|        |           |             |             |           |                    |              |              |                    |                    |                             |                       |                 |
| 14.55% | 1         | Pre-market  | p40_h90     | 7         | 64.33%             | 49.59%       |              |                    |                    |                             |                       |                 |
| 0.591  |           |             |             |           |                    | 74.61%       |              |                    |                    |                             |                       |                 |
|        |           |             |             |           |                    |              |              |                    |                    |                             |                       |                 |
| 15.48% | 1         | Pre-market  | p50_h30     | 7         | 54.10%             | 52.42%       |              |                    |                    |                             |                       |                 |
| 0.643  |           |             |             |           |                    | 75.32%       |              |                    |                    |                             |                       |                 |
|        |           |             |             |           |                    |              |              |                    |                    |                             |                       |                 |
| 12.44% | 1         | Pre-market  | p50_h60     | 7         | 54.68%             | 48.29%       |              |                    |                    |                             |                       |                 |
| 0.577  |           |             |             |           |                    | 76.32%       |              |                    |                    |                             |                       |                 |
|        |           |             |             |           |                    |              |              |                    |                    |                             |                       |                 |
|        | 1         | Pre-market  | p50_h90     | 7         | 55.05%             | 47.01%       |              |                    |                    |                             |                       |                 |

|        |   |            |         |        |   |        |        |        |
|--------|---|------------|---------|--------|---|--------|--------|--------|
| 11.82% |   |            |         | 41.17% |   | 58.83% |        | 76.49% |
| 0.556  |   |            |         | 51.44  |   |        |        |        |
|        | 1 | Pre-market | p60_h30 |        | 7 |        | 44.00% | 50.01% |
| 12.48% |   |            |         | 37.51% |   | 62.49% |        | 77.37% |
| 0.610  |   |            |         | 17.17  |   |        |        |        |
|        | 1 | Pre-market | p60_h60 |        | 7 |        | 45.10% | 45.75% |
| 10.02% |   |            |         | 44.23% |   | 55.77% |        | 77.82% |
| 0.540  |   |            |         | 36.87  |   |        |        |        |
|        | 1 | Pre-market | p60_h90 |        | 7 |        | 45.43% | 44.44% |
| 9.19%  |   |            |         | 46.37% |   | 53.63% |        | 78.26% |
| 0.519  |   |            |         | 53.69  |   |        |        |        |
|        | 2 | Opening    | p40_h30 |        | 7 |        | 64.07% | 51.77% |
| 12.61% |   |            |         | 35.62% |   | 64.38% |        | 78.16% |
| 0.641  |   |            |         | 13.68  |   |        |        |        |
|        | 2 | Opening    | p40_h60 |        | 7 |        | 64.25% | 51.09% |
| 13.51% |   |            |         | 35.40% |   | 64.60% |        | 76.87% |
| 0.625  |   |            |         | 25.55  |   |        |        |        |
|        | 2 | Opening    | p40_h90 |        | 7 |        | 64.19% | 50.72% |
| 13.81% |   |            |         | 35.46% |   | 64.54% |        | 76.28% |
| 0.617  |   |            |         | 36.17  |   |        |        |        |
|        | 2 | Opening    | p50_h30 |        | 7 |        | 54.37% | 49.32% |
| 10.40% |   |            |         | 40.28% |   | 59.72% |        | 79.50% |
| 0.604  |   |            |         | 14.73  |   |        |        |        |
|        | 2 | Opening    | p50_h60 |        | 7 |        | 54.42% | 48.56% |
| 10.89% |   |            |         | 40.55% |   | 59.45% |        | 78.55% |
| 0.588  |   |            |         | 27.55  |   |        |        |        |
|        | 2 | Opening    | p50_h90 |        | 7 |        | 54.61% | 48.24% |
| 11.10% |   |            |         | 40.67% |   | 59.33% |        | 78.17% |
| 0.582  |   |            |         | 39.28  |   |        |        |        |
|        | 2 | Opening    | p60_h30 |        | 7 |        | 44.58% | 46.70% |
| 8.32%  |   |            |         | 44.98% |   | 55.02% |        | 80.95% |
| 0.564  |   |            |         | 15.73  |   |        |        |        |
|        | 2 | Opening    | p60_h60 |        | 7 |        | 44.79% | 45.97% |
| 8.56%  |   |            |         | 45.47% |   | 54.53% |        | 80.24% |
| 0.550  |   |            |         | 29.44  |   |        |        |        |
|        | 2 | Opening    | p60_h90 |        | 7 |        | 44.77% | 45.66% |
| 8.87%  |   |            |         | 45.47% |   | 54.53% |        | 79.56% |
| 0.543  |   |            |         | 42.13  |   |        |        |        |
|        | 3 | Regular    | p40_h30 |        | 7 |        | 63.41% | 52.28% |
| 14.60% |   |            |         | 33.12% |   | 66.88% |        | 76.30% |
| 0.644  |   |            |         | 13.44  |   |        |        |        |
|        | 3 | Regular    | p40_h60 |        | 7 |        | 57.35% | 50.97% |
| 15.24% |   |            |         | 33.79% |   | 66.21% |        | 74.94% |
| 0.617  |   |            |         | 25.98  |   |        |        |        |
|        | 3 | Regular    | p40_h90 |        | 7 |        | 50.95% | 50.79% |
| 15.29% |   |            |         | 33.92% |   | 66.08% |        | 74.80% |
| 0.614  |   |            |         | 38.63  |   |        |        |        |
|        | 3 | Regular    | p50_h30 |        | 7 |        | 53.94% | 50.29% |
| 12.17% |   |            |         | 37.55% |   | 62.45% |        | 78.02% |
| 0.617  |   |            |         | 14.33  |   |        |        |        |
|        | 3 | Regular    | p50_h60 |        | 7 |        | 48.36% | 49.06% |
| 12.54% |   |            |         | 38.40% |   | 61.60% |        | 76.92% |
| 0.593  |   |            |         | 27.91  |   |        |        |        |
|        | 3 | Regular    | p50_h90 |        | 7 |        | 43.18% | 49.00% |
| 12.64% |   |            |         | 38.35% |   | 61.65% |        | 76.80% |
| 0.592  |   |            |         | 41.52  |   |        |        |        |
|        | 3 | Regular    | p60_h30 |        | 7 |        | 43.97% | 48.25% |
| 9.97%  |   |            |         | 41.78% |   | 58.22% |        | 79.70% |
| 0.588  |   |            |         | 15.20  |   |        |        |        |
|        | 3 | Regular    | p60_h60 |        | 7 |        | 39.67% | 47.31% |



|        |   |         |         |   |        |        |        |
|--------|---|---------|---------|---|--------|--------|--------|
| 10.31% |   |         | 42.38%  |   | 57.62% |        | 78.74% |
| 0.570  |   |         | 29.57   |   |        |        |        |
|        | 3 | Regular | p60_h90 | 7 |        | 35.34% | 47.24% |
| 10.45% |   |         | 42.31%  |   | 57.69% |        | 78.56% |
| 0.568  |   |         | 44.12   |   |        |        |        |
|        | 4 | Closing | p40_h30 | 7 |        | 2.12%  | 51.33% |
| 14.10% |   |         | 34.56%  |   | 65.44% |        | 76.35% |
| 0.629  |   |         | 15.78   |   |        |        |        |
|        | 4 | Closing | p40_h60 | 7 |        | 0.00%  | NaN    |
| NaN    |   |         | NaN     |   | NaN    |        | NaN    |
| NaN    |   |         | NaN     |   |        |        |        |
|        | 4 | Closing | p40_h90 | 7 |        | 0.00%  | NaN    |
| NaN    |   |         | NaN     |   | NaN    |        | NaN    |
| NaN    |   |         | NaN     |   |        |        |        |
|        | 4 | Closing | p50_h30 | 7 |        | 1.77%  | 50.47% |
| 11.16% |   |         | 38.37%  |   | 61.63% |        | 79.64% |
| 0.625  |   |         | 16.97   |   |        |        |        |
|        | 4 | Closing | p50_h60 | 7 |        | 0.00%  | NaN    |
| NaN    |   |         | NaN     |   | NaN    |        | NaN    |
| NaN    |   |         | NaN     |   |        |        |        |
|        | 4 | Closing | p50_h90 | 7 |        | 0.00%  | NaN    |
| NaN    |   |         | NaN     |   | NaN    |        | NaN    |
| NaN    |   |         | NaN     |   |        |        |        |
|        | 4 | Closing | p60_h30 | 7 |        | 1.44%  | 47.93% |
| 8.99%  |   |         | 43.08%  |   | 56.92% |        | 81.50% |
| 0.589  |   |         | 17.69   |   |        |        |        |
|        | 4 | Closing | p60_h60 | 7 |        | 0.00%  | NaN    |
| NaN    |   |         | NaN     |   | NaN    |        | NaN    |
| NaN    |   |         | NaN     |   |        |        |        |
|        | 4 | Closing | p60_h90 | 7 |        | 0.00%  | NaN    |
| NaN    |   |         | NaN     |   | NaN    |        | NaN    |
| NaN    |   |         | NaN     |   |        |        |        |

Detalle por régimen para configuración candidata TP 1.5 / SL 1.0:

| bar_col               | target_base     | regime_id | regime_name   | total_rows    | valid_rows             |
|-----------------------|-----------------|-----------|---------------|---------------|------------------------|
| operable_rate         | tp_rate         | sl_rate   | no_event_rate | resolved_rate | tp_share_over_resolved |
| expectancy_units      | event_step_mean |           |               |               |                        |
| bar_p40_h30_tp15_sl10 |                 | p40_h30   | 0 Overnight   | 357162        | 227256                 |
| 63.63%                | 57.02%          | 7.34%     | 35.64%        | 64.36%        | 88.60%                 |
| 0.782                 |                 | 14.83     |               |               |                        |
| bar_p40_h30_tp15_sl10 |                 | p40_h30   | 1 Pre-market  | 88920         | 56654                  |
| 63.71%                | 61.19%          | 10.11%    | 28.70%        | 71.30%        | 85.82%                 |
| 0.817                 |                 | 16.17     |               |               |                        |
| bar_p40_h30_tp15_sl10 |                 | p40_h30   | 2 Opening     | 88920         | 56968                  |
| 64.07%                | 56.45%          | 5.81%     | 37.74%        | 62.26%        | 90.66%                 |
| 0.789                 |                 | 14.75     |               |               |                        |
| bar_p40_h30_tp15_sl10 |                 | p40_h30   | 3 Regular     | 444600        | 281914                 |
| 63.41%                | 57.41%          | 7.12%     | 35.47%        | 64.53%        | 88.97%                 |
| 0.790                 |                 | 14.54     |               |               |                        |
| bar_p40_h30_tp15_sl10 |                 | p40_h30   | 4 Closing     | 44460         | 944                    |
| 2.12%                 | 56.25%          | 6.67%     | 37.08%        | 62.92%        | 89.39%                 |
| 0.777                 |                 | 17.17     |               |               |                        |
| bar_p40_h60_tp15_sl10 |                 | p40_h60   | 0 Overnight   | 357162        | 227928                 |
| 63.82%                | 56.64%          | 7.74%     | 35.62%        | 64.38%        | 87.98%                 |
| 0.772                 |                 | 29.31     |               |               |                        |
| bar_p40_h60_tp15_sl10 |                 | p40_h60   | 1 Pre-market  | 88920         | 56996                  |
| 64.10%                | 56.79%          | 6.95%     | 36.25%        | 63.75%        | 89.09%                 |
| 0.782                 |                 | 36.21     |               |               |                        |
| bar_p40_h60_tp15_sl10 |                 | p40_h60   | 2 Opening     | 88920         | 57128                  |
| 64.25%                | 56.26%          | 5.96%     | 37.78%        | 62.22%        | 90.43%                 |

|                       |        |         |        |            |        |        |
|-----------------------|--------|---------|--------|------------|--------|--------|
| 0.784                 | 27.78  |         |        |            |        |        |
| bar_p40_h60_tp15_sl10 |        | p40_h60 | 3      | Regular    | 444600 | 254972 |
| 57.35%                | 55.89% | 7.31%   | 36.80% | 63.20%     | 88.43% |        |
| 0.765                 | 28.32  |         |        |            |        |        |
| bar_p40_h60_tp15_sl10 |        | p40_h60 | 4      | Closing    | 44460  | 0      |
| 0.00%                 | NaN    | NaN     | NaN    | NaN        | NaN    |        |
| NaN                   | NaN    |         |        |            |        |        |
| bar_p40_h90_tp15_sl10 |        | p40_h90 | 0      | Overnight  | 357162 | 227186 |
| 63.61%                | 56.13% | 9.05%   | 34.82% | 65.18%     | 86.11% |        |
| 0.751                 | 46.46  |         |        |            |        |        |
| bar_p40_h90_tp15_sl10 |        | p40_h90 | 1      | Pre-market | 88920  | 57202  |
| 64.33%                | 55.47% | 6.57%   | 37.96% | 62.04%     | 89.41% |        |
| 0.766                 | 52.67  |         |        |            |        |        |
| bar_p40_h90_tp15_sl10 |        | p40_h90 | 2      | Opening    | 88920  | 57077  |
| 64.19%                | 56.18% | 6.21%   | 37.62% | 62.38%     | 90.05% |        |
| 0.781                 | 39.61  |         |        |            |        |        |
| bar_p40_h90_tp15_sl10 |        | p40_h90 | 3      | Regular    | 444600 | 226505 |
| 50.95%                | 55.92% | 7.15%   | 36.93% | 63.07%     | 88.66% |        |
| 0.767                 | 42.35  |         |        |            |        |        |
| bar_p40_h90_tp15_sl10 |        | p40_h90 | 4      | Closing    | 44460  | 0      |
| 0.00%                 | NaN    | NaN     | NaN    | NaN        | NaN    |        |
| NaN                   | NaN    |         |        |            |        |        |
| bar_p50_h30_tp15_sl10 |        | p50_h30 | 0      | Overnight  | 357162 | 192938 |
| 54.02%                | 53.11% | 5.80%   | 41.08% | 58.92%     | 90.15% |        |
| 0.739                 | 15.68  |         |        |            |        |        |
| bar_p50_h30_tp15_sl10 |        | p50_h30 | 1      | Pre-market | 88920  | 48109  |
| 54.10%                | 57.96% | 7.88%   | 34.16% | 65.84%     | 88.03% |        |
| 0.791                 | 17.19  |         |        |            |        |        |
| bar_p50_h30_tp15_sl10 |        | p50_h30 | 2      | Opening    | 88920  | 48348  |
| 54.37%                | 52.04% | 4.37%   | 43.59% | 56.41%     | 92.25% |        |
| 0.737                 | 15.84  |         |        |            |        |        |
| bar_p50_h30_tp15_sl10 |        | p50_h30 | 3      | Regular    | 444600 | 239801 |
| 53.94%                | 53.42% | 5.46%   | 41.11% | 58.89%     | 90.72% |        |
| 0.747                 | 15.44  |         |        |            |        |        |
| bar_p50_h30_tp15_sl10 |        | p50_h30 | 4      | Closing    | 44460  | 786    |
| 1.77%                 | 53.82% | 4.71%   | 41.48% | 58.52%     | 91.96% |        |
| 0.760                 | 18.32  |         |        |            |        |        |
| bar_p50_h60_tp15_sl10 |        | p50_h60 | 0      | Overnight  | 357162 | 192987 |
| 54.03%                | 52.81% | 6.02%   | 41.17% | 58.83%     | 89.77% |        |
| 0.732                 | 31.04  |         |        |            |        |        |
| bar_p50_h60_tp15_sl10 |        | p50_h60 | 1      | Pre-market | 88920  | 48621  |
| 54.68%                | 51.78% | 5.22%   | 43.00% | 57.00%     | 90.84% |        |
| 0.725                 | 37.80  |         |        |            |        |        |
| bar_p50_h60_tp15_sl10 |        | p50_h60 | 2      | Opening    | 88920  | 48390  |
| 54.42%                | 51.69% | 4.33%   | 43.99% | 56.01%     | 92.28% |        |
| 0.732                 | 30.08  |         |        |            |        |        |
| bar_p50_h60_tp15_sl10 |        | p50_h60 | 3      | Regular    | 444600 | 215022 |
| 48.36%                | 51.95% | 5.45%   | 42.60% | 57.40%     | 90.50% |        |
| 0.725                 | 30.29  |         |        |            |        |        |
| bar_p50_h60_tp15_sl10 |        | p50_h60 | 4      | Closing    | 44460  | 0      |
| 0.00%                 | NaN    | NaN     | NaN    | NaN        | NaN    |        |
| NaN                   | NaN    |         |        |            |        |        |
| bar_p50_h90_tp15_sl10 |        | p50_h90 | 0      | Overnight  | 357162 | 193131 |
| 54.07%                | 53.39% | 7.25%   | 39.36% | 60.64%     | 88.05% |        |
| 0.728                 | 49.43  |         |        |            |        |        |
| bar_p50_h90_tp15_sl10 |        | p50_h90 | 1      | Pre-market | 88920  | 48950  |
| 55.05%                | 50.27% | 4.42%   | 45.31% | 54.69%     | 91.92% |        |
| 0.710                 | 55.42  |         |        |            |        |        |
| bar_p50_h90_tp15_sl10 |        | p50_h90 | 2      | Opening    | 88920  | 48555  |
| 54.61%                | 51.28% | 4.39%   | 44.33% | 55.67%     | 92.12% |        |

|                       |         |        |            |        |        |  |
|-----------------------|---------|--------|------------|--------|--------|--|
| 0.725                 | 43.09   |        |            |        |        |  |
| bar_p50_h90_tp15_sl10 | p50_h90 | 3      | Regular    | 444600 | 191967 |  |
| 43.18% 52.19% 5.38%   | 42.43%  | 57.57% |            | 90.66% |        |  |
| 0.729                 | 45.40   |        |            |        |        |  |
| bar_p50_h90_tp15_sl10 | p50_h90 | 4      | Closing    | 44460  | 0      |  |
| 0.00% NaN NaN         | NaN     | NaN    |            | NaN    |        |  |
| NaN                   | NaN     |        |            |        |        |  |
| bar_p60_h30_tp15_sl10 | p60_h30 | 0      | Overnight  | 357162 | 160088 |  |
| 44.82% 49.41% 4.70%   | 45.89%  | 54.11% |            | 91.32% |        |  |
| 0.694                 | 16.45   |        |            |        |        |  |
| bar_p60_h30_tp15_sl10 | p60_h30 | 1      | Pre-market | 88920  | 39124  |  |
| 44.00% 53.23% 5.74%   | 41.03%  | 58.97% |            | 90.27% |        |  |
| 0.741                 | 18.14   |        |            |        |        |  |
| bar_p60_h30_tp15_sl10 | p60_h30 | 2      | Opening    | 88920  | 39638  |  |
| 44.58% 46.94% 3.02%   | 50.04%  | 49.96% |            | 93.96% |        |  |
| 0.674                 | 16.88   |        |            |        |        |  |
| bar_p60_h30_tp15_sl10 | p60_h30 | 3      | Regular    | 444600 | 195503 |  |
| 43.97% 49.40% 4.10%   | 46.50%  | 53.50% |            | 92.34% |        |  |
| 0.700                 | 16.36   |        |            |        |        |  |
| bar_p60_h30_tp15_sl10 | p60_h30 | 4      | Closing    | 44460  | 642    |  |
| 1.44% 48.29% 3.12%    | 48.60%  | 51.40% |            | 93.94% |        |  |
| 0.693                 | 19.25   |        |            |        |        |  |
| bar_p60_h60_tp15_sl10 | p60_h60 | 0      | Overnight  | 357162 | 157509 |  |
| 44.10% 49.21% 4.62%   | 46.17%  | 53.83% |            | 91.42% |        |  |
| 0.692                 | 32.61   |        |            |        |        |  |
| bar_p60_h60_tp15_sl10 | p60_h60 | 1      | Pre-market | 88920  | 40102  |  |
| 45.10% 46.89% 3.65%   | 49.46%  | 50.54% |            | 92.78% |        |  |
| 0.667                 | 39.28   |        |            |        |        |  |
| bar_p60_h60_tp15_sl10 | p60_h60 | 2      | Opening    | 88920  | 39829  |  |
| 44.79% 46.36% 2.97%   | 50.67%  | 49.33% |            | 93.98% |        |  |
| 0.666                 | 32.22   |        |            |        |        |  |
| bar_p60_h60_tp15_sl10 | p60_h60 | 3      | Regular    | 444600 | 176358 |  |
| 39.67% 48.19% 4.09%   | 47.72%  | 52.28% |            | 92.17% |        |  |
| 0.682                 | 32.06   |        |            |        |        |  |
| bar_p60_h60_tp15_sl10 | p60_h60 | 4      | Closing    | 44460  | 0      |  |
| 0.00% NaN NaN         | NaN     | NaN    |            | NaN    |        |  |
| NaN                   | NaN     |        |            |        |        |  |
| bar_p60_h90_tp15_sl10 | p60_h90 | 0      | Overnight  | 357162 | 157163 |  |
| 44.00% 51.16% 5.69%   | 43.15%  | 56.85% |            | 89.99% |        |  |
| 0.710                 | 52.44   |        |            |        |        |  |
| bar_p60_h90_tp15_sl10 | p60_h90 | 1      | Pre-market | 88920  | 40395  |  |
| 45.43% 44.09% 2.98%   | 52.93%  | 47.07% |            | 93.67% |        |  |
| 0.632                 | 57.69   |        |            |        |        |  |
| bar_p60_h90_tp15_sl10 | p60_h90 | 2      | Opening    | 88920  | 39811  |  |
| 44.77% 46.23% 3.05%   | 50.71%  | 49.29% |            | 93.81% |        |  |
| 0.663                 | 46.10   |        |            |        |        |  |
| bar_p60_h90_tp15_sl10 | p60_h90 | 3      | Regular    | 444600 | 157132 |  |
| 35.34% 48.26% 4.11%   | 47.64%  | 52.36% |            | 92.15% |        |  |
| 0.683                 | 48.06   |        |            |        |        |  |
| bar_p60_h90_tp15_sl10 | p60_h90 | 4      | Closing    | 44460  | 0      |  |
| 0.00% NaN NaN         | NaN     | NaN    |            | NaN    |        |  |
| NaN                   | NaN     |        |            |        |        |  |

Top 20 targets con mayor sensibilidad al régimen:

|                       |                          |                     |                              |                     |  |  |
|-----------------------|--------------------------|---------------------|------------------------------|---------------------|--|--|
| bar_col               | target_base              | tp_sl_config        | n_regimes                    | operable_rate_range |  |  |
| tp_rate_range         | sl_rate_range            | no_event_rate_range | tp_share_over_resolved_range | expe                |  |  |
| ctancy_units_range    | regime_sensitivity_score |                     |                              |                     |  |  |
| bar_p40_h30_tp25_sl10 | p40_h30                  | TP 2.5 / SL 1.0     | 5                            | 61.94%              |  |  |
| 8.98%                 | 5.34%                    | 14.32%              | 3.66%                        |                     |  |  |
| 0.171                 | 0.942                    |                     |                              |                     |  |  |

|                       |                         |   |        |        |
|-----------------------|-------------------------|---|--------|--------|
| bar_p40_h30_tp20_sl10 | p40_h30 TP 2.0 / SL 1.0 | 5 |        | 61.94% |
| 8.82%                 | 4.86%                   |   | 13.68% | 4.81%  |
| 0.128                 | 0.941                   |   |        |        |
| bar_p40_h30_tp20_sl05 | p40_h30 TP 2.0 / SL 0.5 | 5 |        | 61.94% |
| 5.83%                 | 7.72%                   |   | 12.96% | 3.36%  |
| 0.092                 | 0.918                   |   |        |        |
| bar_p40_h90_tp25_sl10 | p40_h90 TP 2.5 / SL 1.0 | 5 |        | 64.33% |
| 7.54%                 | 3.44%                   |   | 10.52% | 3.47%  |
| 0.159                 | 0.893                   |   |        |        |
| bar_p40_h30_tp15_sl05 | p40_h30 TP 1.5 / SL 0.5 | 5 |        | 61.94% |
| 2.86%                 | 7.04%                   |   | 8.02%  | 5.78%  |
| 0.040                 | 0.856                   |   |        |        |
| bar_p40_h90_tp20_sl10 | p40_h90 TP 2.0 / SL 1.0 | 5 |        | 64.33% |
| 5.54%                 | 3.13%                   |   | 8.19%  | 4.19%  |
| 0.084                 | 0.854                   |   |        |        |
| bar_p40_h30_tp15_sl10 | p40_h30 TP 1.5 / SL 1.0 | 5 |        | 61.94% |
| 4.94%                 | 4.30%                   |   | 9.05%  | 4.85%  |
| 0.040                 | 0.851                   |   |        |        |
| bar_p50_h30_tp25_sl10 | p50_h30 TP 2.5 / SL 1.0 | 5 |        | 52.60% |
| 8.52%                 | 4.37%                   |   | 12.85% | 6.18%  |
| 0.170                 | 0.845                   |   |        |        |
| bar_p50_h30_tp20_sl10 | p50_h30 TP 2.0 / SL 1.0 | 5 |        | 52.60% |
| 8.58%                 | 4.04%                   |   | 12.53% | 5.78%  |
| 0.132                 | 0.835                   |   |        |        |
| bar_p50_h90_tp20_sl10 | p50_h90 TP 2.0 / SL 1.0 | 5 |        | 55.05% |
| 8.28%                 | 3.28%                   |   | 11.46% | 4.43%  |
| 0.134                 | 0.825                   |   |        |        |
| bar_p40_h30_tp10_sl05 | p40_h30 TP 1.0 / SL 0.5 | 5 |        | 61.94% |
| 6.67%                 | 6.67%                   |   | 0.00%  | 6.67%  |
| 0.100                 | 0.820                   |   |        |        |
| bar_p40_h90_tp20_sl05 | p40_h90 TP 2.0 / SL 0.5 | 5 |        | 64.33% |
| 2.75%                 | 5.05%                   |   | 6.24%  | 3.34%  |
| 0.039                 | 0.817                   |   |        |        |
| bar_p50_h90_tp25_sl10 | p50_h90 TP 2.5 / SL 1.0 | 5 |        | 55.05% |
| 8.51%                 | 3.53%                   |   | 11.98% | 2.30%  |
| 0.178                 | 0.814                   |   |        |        |
| bar_p50_h30_tp20_sl05 | p50_h30 TP 2.0 / SL 0.5 | 5 |        | 52.60% |
| 5.44%                 | 7.51%                   |   | 12.95% | 2.83%  |
| 0.071                 | 0.813                   |   |        |        |
| bar_p40_h90_tp15_sl05 | p40_h90 TP 1.5 / SL 0.5 | 5 |        | 64.33% |
| 2.20%                 | 4.81%                   |   | 2.61%  | 5.29%  |
| 0.057                 | 0.792                   |   |        |        |
| bar_p50_h90_tp20_sl05 | p50_h90 TP 2.0 / SL 0.5 | 5 |        | 55.05% |
| 5.11%                 | 5.44%                   |   | 8.62%  | 4.30%  |
| 0.085                 | 0.785                   |   |        |        |
| bar_p40_h60_tp25_sl10 | p40_h60 TP 2.5 / SL 1.0 | 5 |        | 64.25% |
| 3.77%                 | 1.95%                   |   | 5.71%  | 2.72%  |
| 0.075                 | 0.784                   |   |        |        |
| bar_p40_h90_tp10_sl05 | p40_h90 TP 1.0 / SL 0.5 | 5 |        | 64.33% |
| 4.57%                 | 4.57%                   |   | 0.00%  | 4.57%  |
| 0.069                 | 0.780                   |   |        |        |
| bar_p50_h30_tp15_sl05 | p50_h30 TP 1.5 / SL 0.5 | 5 |        | 52.60% |
| 2.58%                 | 6.92%                   |   | 9.49%  | 5.63%  |
| 0.017                 | 0.772                   |   |        |        |
| bar_p60_h90_tp20_sl10 | p60_h90 TP 2.0 / SL 1.0 | 5 |        | 45.43% |
| 10.23%                | 3.07%                   |   | 13.30% | 3.73%  |
| 0.174                 | 0.758                   |   |        |        |

Top 20 targets con menor sensibilidad al régimen:

bar\_col target\_base tp\_sl\_config n\_regimes operable\_rate\_range

| tp_rate_range         | sl_rate_range            | no_event_rate_range | tp_share_over_resolved_range | expe   |
|-----------------------|--------------------------|---------------------|------------------------------|--------|
| ctancy_units_range    | regime_sensitivity_score |                     |                              |        |
| bar_p60_h60_tp10_sl10 | p60_h60                  | TP 1.0 / SL 1.0     | 5                            | 45.10% |
| 1.45%                 | 1.45%                    | 0.00%               | 1.45%                        |        |
| 0.029                 | 0.494                    |                     |                              |        |
| bar_p60_h30_tp10_sl10 | p60_h30                  | TP 1.0 / SL 1.0     | 5                            | 43.38% |
| 2.38%                 | 2.38%                    | 0.00%               | 2.38%                        |        |
| 0.048                 | 0.505                    |                     |                              |        |
| bar_p60_h90_tp10_sl10 | p60_h90                  | TP 1.0 / SL 1.0     | 5                            | 45.43% |
| 2.33%                 | 2.33%                    | 0.00%               | 2.33%                        |        |
| 0.047                 | 0.524                    |                     |                              |        |
| bar_p60_h60_tp10_sl05 | p60_h60                  | TP 1.0 / SL 0.5     | 5                            | 45.10% |
| 2.96%                 | 2.96%                    | 0.00%               | 2.96%                        |        |
| 0.044                 | 0.540                    |                     |                              |        |
| bar_p60_h60_tp15_sl10 | p60_h60                  | TP 1.5 / SL 1.0     | 5                            | 45.10% |
| 2.85%                 | 1.65%                    | 4.50%               | 2.56%                        |        |
| 0.026                 | 0.567                    |                     |                              |        |
| bar_p60_h60_tp15_sl05 | p60_h60                  | TP 1.5 / SL 0.5     | 5                            | 45.10% |
| 1.44%                 | 3.15%                    | 4.11%               | 3.32%                        |        |
| 0.020                 | 0.571                    |                     |                              |        |
| bar_p60_h30_tp10_sl05 | p60_h30                  | TP 1.0 / SL 0.5     | 5                            | 43.38% |
| 5.19%                 | 5.19%                    | 0.00%               | 5.19%                        |        |
| 0.078                 | 0.589                    |                     |                              |        |
| bar_p60_h90_tp10_sl05 | p60_h90                  | TP 1.0 / SL 0.5     | 5                            | 45.43% |
| 4.62%                 | 4.62%                    | 0.00%               | 4.62%                        |        |
| 0.069                 | 0.593                    |                     |                              |        |
| bar_p50_h60_tp10_sl10 | p50_h60                  | TP 1.0 / SL 1.0     | 5                            | 54.68% |
| 1.55%                 | 1.55%                    | 0.00%               | 1.55%                        |        |
| 0.031                 | 0.593                    |                     |                              |        |
| bar_p50_h30_tp10_sl10 | p50_h30                  | TP 1.0 / SL 1.0     | 5                            | 52.60% |
| 3.06%                 | 3.06%                    | 0.00%               | 3.06%                        |        |
| 0.061                 | 0.618                    |                     |                              |        |
| bar_p50_h60_tp15_sl10 | p50_h60                  | TP 1.5 / SL 1.0     | 5                            | 54.68% |
| 1.12%                 | 1.70%                    | 2.82%               | 2.51%                        |        |
| 0.008                 | 0.628                    |                     |                              |        |
| bar_p50_h90_tp10_sl10 | p50_h90                  | TP 1.0 / SL 1.0     | 5                            | 55.05% |
| 2.63%                 | 2.63%                    | 0.00%               | 2.63%                        |        |
| 0.053                 | 0.629                    |                     |                              |        |
| bar_p50_h60_tp10_sl05 | p50_h60                  | TP 1.0 / SL 0.5     | 5                            | 54.68% |
| 2.84%                 | 2.84%                    | 0.00%               | 2.84%                        |        |
| 0.043                 | 0.632                    |                     |                              |        |
| bar_p50_h60_tp15_sl05 | p50_h60                  | TP 1.5 / SL 0.5     | 5                            | 54.68% |
| 0.65%                 | 2.92%                    | 2.61%               | 3.14%                        |        |
| 0.022                 | 0.640                    |                     |                              |        |
| bar_p60_h60_tp20_sl10 | p60_h60                  | TP 2.0 / SL 1.0     | 5                            | 45.10% |
| 6.22%                 | 1.85%                    | 8.07%               | 2.84%                        |        |
| 0.109                 | 0.641                    |                     |                              |        |
| bar_p60_h60_tp20_sl05 | p60_h60                  | TP 2.0 / SL 0.5     | 5                            | 45.10% |
| 4.32%                 | 3.25%                    | 7.35%               | 4.58%                        |        |
| 0.084                 | 0.646                    |                     |                              |        |
| bar_p60_h30_tp15_sl10 | p60_h30                  | TP 1.5 / SL 1.0     | 5                            | 43.38% |
| 6.29%                 | 2.72%                    | 9.00%               | 3.68%                        |        |
| 0.067                 | 0.651                    |                     |                              |        |
| bar_p60_h90_tp15_sl05 | p60_h90                  | TP 1.5 / SL 0.5     | 5                            | 45.43% |
| 3.79%                 | 4.93%                    | 7.68%               | 4.43%                        |        |
| 0.037                 | 0.663                    |                     |                              |        |
| bar_p60_h60_tp25_sl10 | p60_h60                  | TP 2.5 / SL 1.0     | 5                            | 45.10% |
| 6.63%                 | 2.06%                    | 8.69%               | 4.20%                        |        |
| 0.145                 | 0.667                    |                     |                              |        |
| bar_p60_h30_tp15_sl05 | p60_h30                  | TP 1.5 / SL 0.5     | 5                            | 43.38% |

|       |       |       |       |
|-------|-------|-------|-------|
| 3.49% | 5.97% | 9.47% | 5.02% |
| 0.023 |       | 0.673 |       |

Sensibilidad agregada al régimen por configuración TP/SL:

| tp_sl_config    | n_targets | operable_rate_range_mean | tp_rate_range_mean | sl_rate_range_mean | no_event_rate_range_mean | tp_share_range_mean | expectancy_range_mean | regime_sensitivity_score_mean | regime_sensitivity_score_max |
|-----------------|-----------|--------------------------|--------------------|--------------------|--------------------------|---------------------|-----------------------|-------------------------------|------------------------------|
| TP 1.0 / SL 1.0 | 9         |                          | 54.08%             |                    | 2.38%                    |                     |                       |                               |                              |
|                 |           | 0.00%                    |                    | 2.38%              |                          |                     |                       | 0.048                         |                              |
|                 |           | 0.612                    |                    | 0.737              |                          |                     |                       |                               |                              |
| TP 1.5 / SL 1.0 | 9         |                          | 54.08%             |                    | 3.66%                    |                     |                       |                               |                              |
|                 |           | 6.20%                    |                    | 3.57%              |                          |                     |                       | 0.038                         |                              |
|                 |           | 0.702                    |                    | 0.851              |                          |                     |                       |                               |                              |
| TP 2.0 / SL 1.0 | 9         |                          | 54.08%             |                    | 6.92%                    |                     |                       |                               |                              |
|                 |           | 9.84%                    |                    | 4.11%              |                          |                     |                       | 0.109                         |                              |
|                 |           | 0.780                    |                    | 0.941              |                          |                     |                       |                               |                              |
| TP 2.5 / SL 1.0 | 9         |                          | 54.08%             |                    | 7.27%                    |                     |                       |                               |                              |
|                 |           | 10.48%                   |                    | 4.27%              |                          |                     |                       | 0.150                         |                              |
|                 |           | 0.794                    |                    | 0.942              |                          |                     |                       |                               |                              |
| TP 1.0 / SL 0.5 | 9         |                          | 54.08%             |                    | 4.54%                    |                     |                       |                               |                              |
|                 |           | 0.00%                    |                    | 4.54%              |                          |                     |                       | 0.068                         |                              |
|                 |           | 0.677                    |                    | 0.820              |                          |                     |                       |                               |                              |
| TP 1.5 / SL 0.5 | 9         |                          | 54.08%             |                    | 2.13%                    |                     |                       |                               |                              |
|                 |           | 5.64%                    |                    | 4.57%              |                          |                     |                       | 0.031                         |                              |
|                 |           | 0.713                    |                    | 0.856              |                          |                     |                       |                               |                              |
| TP 2.0 / SL 0.5 | 9         |                          | 54.08%             |                    | 4.40%                    |                     |                       |                               |                              |
|                 |           | 9.04%                    |                    | 3.27%              |                          |                     |                       | 0.071                         |                              |
|                 |           | 0.760                    |                    | 0.918              |                          |                     |                       |                               |                              |

Sensibilidad agregada al régimen por percentil y horizonte:

| percentile | horizon | target_base | n_configs | operable_rate_range_mean | tp_rate_range_mean | sl_rate_range_mean | no_event_rate_range_mean | tp_share_range_mean | expectancy_range_mean | regime_sensitivity_score_mean | regime_sensitivity_score_max |
|------------|---------|-------------|-----------|--------------------------|--------------------|--------------------|--------------------------|---------------------|-----------------------|-------------------------------|------------------------------|
|            | 40      | 30          | p40_h30   | 7                        |                    |                    |                          |                     |                       | 61.94%                        |                              |
|            |         |             |           |                          | 8.29%              |                    |                          |                     |                       | 4.72%                         |                              |
|            |         |             |           |                          |                    | 0.866              |                          |                     |                       | 0.942                         |                              |
|            | 40      | 60          | p40_h60   | 7                        |                    |                    |                          |                     |                       | 64.25%                        |                              |
|            |         |             |           |                          | 2.61%              |                    |                          |                     |                       | 2.66%                         |                              |
|            |         |             |           |                          |                    | 0.738              |                          |                     |                       | 0.784                         |                              |
|            | 40      | 90          | p40_h90   | 7                        |                    |                    |                          |                     |                       | 64.33%                        |                              |
|            |         |             |           |                          | 4.38%              |                    |                          |                     |                       | 3.91%                         |                              |
|            |         |             |           |                          |                    | 0.801              |                          |                     |                       | 0.893                         |                              |
|            | 50      | 30          | p50_h30   | 7                        |                    |                    |                          |                     |                       | 52.60%                        |                              |
|            |         |             |           |                          | 8.18%              |                    |                          |                     |                       | 4.85%                         |                              |
|            |         |             |           |                          |                    | 0.765              |                          |                     |                       | 0.845                         |                              |
|            | 50      | 60          | p50_h60   | 7                        |                    |                    |                          |                     |                       | 54.68%                        |                              |
|            |         |             |           |                          | 3.45%              |                    |                          |                     |                       | 2.72%                         |                              |
|            |         |             |           |                          |                    | 0.657              |                          |                     |                       | 0.734                         |                              |
|            | 50      | 90          | p50_h90   | 7                        |                    |                    |                          |                     |                       | 55.05%                        |                              |
|            |         |             |           |                          | 6.09%              |                    |                          |                     |                       | 3.95%                         |                              |
|            |         |             |           |                          |                    | 0.738              |                          |                     |                       | 0.825                         |                              |
|            | 60      | 30          | p60_h30   | 7                        |                    |                    |                          |                     |                       | 43.38%                        |                              |
|            |         |             |           |                          | 7.48%              |                    |                          |                     |                       | 4.73%                         |                              |
|            |         |             |           |                          |                    | 0.653              |                          |                     |                       | 0.726                         |                              |
|            | 60      | 60          | p60_h60   | 7                        |                    |                    |                          |                     |                       | 45.10%                        |                              |
|            |         |             |           |                          | 4.67%              |                    |                          |                     |                       | 3.13%                         |                              |
|            |         |             |           |                          |                    | 0.589              |                          |                     |                       | 0.667                         |                              |
|            | 60      | 90          | p60_h90   | 7                        |                    |                    |                          |                     |                       | 45.43%                        |                              |
|            |         |             |           |                          | 7.82%              |                    |                          |                     |                       | 3.66%                         |                              |
|            |         |             |           |                          |                    | 0.670              |                          |                     |                       | 0.758                         |                              |

Celda 2 finalizada correctamente.

## Celda 3 — Gráficos agrupados por régimen

```
In [113... # Celda 3 – Gráficos agrupados por régimen
# =====

print("Gráficos agrupados por régimen intradiario - BAR")
print("=" * 80)

# -----
# Validar objetos necesarios
# -----

required_objects = [
    "df_bar_regime_analysis",
    "df_bar_regime_by_regime",
    "df_bar_regime_by_config",
    "df_bar_regime_by_target_base",
    "df_bar_regime_stability_by_target",
    "df_bar_regime_sensitivity_by_config",
    "df_bar_regime_sensitivity_by_target_base",
    "BAR_PERCENTILES",
    "BAR_HORIZONS",
]

missing_objects = [
    obj for obj in required_objects
    if obj not in globals()
]

if missing_objects:
    raise NameError(
        "Faltan objetos necesarios. "
        f"Debes correr las celdas anteriores: {missing_objects}"
    )

# -----
# Orden de configuraciones TP/SL
# -----

config_order = [
    (1.0, 1.0),
    (1.5, 1.0),
    (2.0, 1.0),
    (2.5, 1.0),
    (1.0, 0.5),
    (1.5, 0.5),
    (2.0, 0.5),
]

config_labels = [
    f"TP {tp} / SL {sl}"
    for tp, sl in config_order
]

target_base_order = [
    f"p{p}_h{h}"
```

```

    for p in BAR_PERCENTILES
    for h in BAR_HORIZONS
]

regime_order = (
    df_bar_regime_analysis[["regime_id", "regime_name"]]
    .drop_duplicates()
    .sort_values("regime_id")
    .reset_index(drop=True)
)

regime_labels = regime_order["regime_name"].tolist()

# -----
# Copias de trabajo
# -----

df_regime_plot = df_bar_regime_analysis.copy()
df_regime_config_plot = df_bar_regime_by_config.copy()
df_regime_sensitivity_plot = df_bar_regime_stability_by_target.copy()

for df_tmp in [df_regime_plot, df_regime_config_plot, df_regime_sensitivity_plot]:
    df_tmp["config_label"] = df_tmp.apply(
        lambda row: f"TP {row['tp_mult']} / SL {row['sl_mult']}",
        axis=1
    )

    df_tmp["config_order"] = df_tmp.apply(
        lambda row: config_order.index((row["tp_mult"], row["sl_mult"]))
        if (row["tp_mult"], row["sl_mult"]) in config_order
        else 999,
        axis=1
    )

# =====
# Figura 1 – Resumen global por régimen
# =====

df_summary_plot = df_bar_regime_by_regime.copy()
df_summary_plot = df_summary_plot.sort_values("regime_id").reset_index(drop=True)

x = np.arange(len(df_summary_plot))
bar_width = 0.22

fig, ax = plt.subplots(figsize=(13, 6))

ax.bar(
    x - bar_width,
    df_summary_plot["tp_rate_mean"],
    width=bar_width,
    label="TP primero"
)

ax.bar(
    x,
    df_summary_plot["sl_rate_mean"],
    width=bar_width,
    label="SL primero"
)

```



```

ax.bar(
    x + bar_width,
    df_summary_plot["no_event_rate_mean"],
    width=bar_width,
    label="no_event"
)

ax.set_title("Distribución promedio de BAR por régimen intradiario")
ax.set_ylabel("Proporción promedio")
ax.set_xticks(x)
ax.set_xticklabels(df_summary_plot["regime_name"], rotation=30, ha="right")
ax.legend()

ax.yaxis.set_major_formatter(
    plt.FuncFormatter(lambda y, _: f"{y:.0%}")
)

for i, row in df_summary_plot.iterrows():
    ax.text(
        i - bar_width,
        row["tp_rate_mean"],
        f"{row['tp_rate_mean']:.0%}",
        ha="center",
        va="bottom",
        fontsize=8
    )

    ax.text(
        i,
        row["sl_rate_mean"],
        f"{row['sl_rate_mean']:.0%}",
        ha="center",
        va="bottom",
        fontsize=8
    )

    ax.text(
        i + bar_width,
        row["no_event_rate_mean"],
        f"{row['no_event_rate_mean']:.0%}",
        ha="center",
        va="bottom",
        fontsize=8
    )

plt.tight_layout()
plt.show()

# =====
# Función auxiliar para heatmaps por régimen
# =====

def plot_metric_heatmaps_by_regime(
    df,
    metric_col,
    title,
    value_format="{:.0%}",
    vmin=0,
    vmax=1,
    colorbar_label="Proporción",

```

```

figsize=(17, 18)
):
    """
    Grafica heatmaps separados por régimen.

    Cada heatmap contiene:
    - Filas: target_base, por ejemplo p40_h30, p50_h60, p60_h90
    - Columnas: configuraciones TP/SL

    La barra de color se ubica en una columna separada para evitar
    superposición con los heatmaps.
    """

    n_regimes = len(regime_order)

    fig = plt.figure(figsize=figsize)

    gs = fig.add_gridspec(
        nrows=n_regimes,
        ncols=2,
        width_ratios=[30, 1],
        hspace=0.35,
        wspace=0.08
    )

    axes = [
        fig.add_subplot(gs[i, 0])
        for i in range(n_regimes)
    ]

    cax = fig.add_subplot(gs[:, 1])

    last_im = None

    for ax, regime_info in zip(axes, regime_order.itertuples(index=False)):

        regime_id = regime_info.regime_id
        regime_name = regime_info.regime_name

        tmp = df[df["regime_id"] == regime_id].copy()

        pivot_tmp = (
            tmp
            .pivot_table(
                index="target_base",
                columns="config_label",
                values=metric_col,
                aggfunc="first"
            )
            .reindex(index=target_base_order)
            .reindex(columns=config_labels)
        )

        data = np.ma.masked_invalid(pivot_tmp.values)

        last_im = ax.imshow(
            data,
            aspect="auto",
            vmin=vmin,
            vmax=vmax

```

```

    )

    ax.set_title(regime_name)

    ax.set_yticks(np.arange(len(pivot_tmp.index)))
    ax.set_yticklabels(pivot_tmp.index)

    ax.set_xticks(np.arange(len(pivot_tmp.columns)))

    # Solo mostrar etiquetas del eje X en el último heatmap
    if ax == axes[-1]:
        ax.set_xticklabels(
            pivot_tmp.columns,
            rotation=45,
            ha="right"
        )
    else:
        ax.set_xticklabels([])

    for i in range(pivot_tmp.shape[0]):
        for j in range(pivot_tmp.shape[1]):
            value = pivot_tmp.iloc[i, j]

            if pd.notna(value):
                ax.text(
                    j,
                    i,
                    value_format.format(value),
                    ha="center",
                    va="center",
                    fontsize=7
                )

    fig.colorbar(
        last_im,
        cax=cax,
        label=colorbar_label
    )

    fig.suptitle(
        title,
        fontsize=14,
        y=0.995
    )

    plt.show()

# =====
# Figura 2 – TP rate por régimen para Los 63 targets BAR
# =====

plot_metric_heatmaps_by_regime(
    df=df_regime_plot,
    metric_col="tp_rate",
    title="TP rate por régimen – 63 targets BAR",
    value_format="{:.0%}",
    vmin=0,
    vmax=1,
    colorbar_label="TP rate"
)

```

```

# =====
# Figura 3 – SL rate por régimen para Los 63 targets BAR
# =====

plot_metric_heatmaps_by_regime(
    df=df_regime_plot,
    metric_col="sl_rate",
    title="SL rate por régimen – 63 targets BAR",
    value_format="{:.0%}",
    vmin=0,
    vmax=1,
    colorbar_label="SL rate"
)

# =====
# Figura 4 – no_event rate por régimen para Los 63 targets BAR
# =====

plot_metric_heatmaps_by_regime(
    df=df_regime_plot,
    metric_col="no_event_rate",
    title="no_event rate por régimen – 63 targets BAR",
    value_format="{:.0%}",
    vmin=0,
    vmax=1,
    colorbar_label="no_event rate"
)

# =====
# Figura 5 – TP sobre eventos resueltos por régimen
# =====

plot_metric_heatmaps_by_regime(
    df=df_regime_plot,
    metric_col="tp_share_over_resolved",
    title="TP sobre eventos resueltos por régimen – 63 targets BAR",
    value_format="{:.0%}",
    vmin=0,
    vmax=1,
    colorbar_label="TP / (TP + SL)"
)

# =====
# Figura 6 – Tiempo promedio hasta evento por régimen
# =====

global_max_event_step = df_regime_plot["event_step_mean"].max()

plot_metric_heatmaps_by_regime(
    df=df_regime_plot,
    metric_col="event_step_mean",
    title="Tiempo promedio hasta evento por régimen – 63 targets BAR",
    value_format="{:.1f}",
    vmin=0,
    vmax=global_max_event_step,
    colorbar_label="Barras hasta evento"
)

# =====

```

```

# Figura 7 – Sensibilidad al régimen por target
# =====

pivot_sensitivity = (
    df_regime_sensitivity_plot
    .pivot_table(
        index="target_base",
        columns="config_label",
        values="regime_sensitivity_score",
        aggfunc="first"
    )
    .reindex(index=target_base_order)
    .reindex(columns=config_labels)
)

fig, ax = plt.subplots(figsize=(15, 7))

data = np.ma.masked_invalid(pivot_sensitivity.values)

im = ax.imshow(
    data,
    aspect="auto"
)

ax.set_title("Sensibilidad al régimen por target – regime_sensitivity_score")
ax.set_ylabel("Target base")
ax.set_yticks(np.arange(len(pivot_sensitivity.index)))
ax.set_yticklabels(pivot_sensitivity.index)

ax.set_xticks(np.arange(len(pivot_sensitivity.columns)))
ax.set_xticklabels(pivot_sensitivity.columns, rotation=45, ha="right")

for i in range(pivot_sensitivity.shape[0]):
    for j in range(pivot_sensitivity.shape[1]):
        value = pivot_sensitivity.iloc[i, j]

        if pd.notna(value):
            ax.text(
                j,
                i,
                f"{value:.2f}",
                ha="center",
                va="center",
                fontsize=8
            )

fig.colorbar(
    im,
    ax=ax,
    label="Sensibilidad al régimen"
)

plt.tight_layout()
plt.show()

# =====
# Figura 8 – Sensibilidad promedio al régimen por configuración
# =====

df_config_sensitivity_plot = df_bar_regime_sensitivity_by_config.copy()

```

```

df_config_sensitivity_plot["config_order"] = df_config_sensitivity_plot.apply(
    lambda row: config_order.index((row["tp_mult"], row["sl_mult"]))
    if (row["tp_mult"], row["sl_mult"]) in config_order
    else 999,
    axis=1
)

df_config_sensitivity_plot = (
    df_config_sensitivity_plot
    .sort_values("config_order")
    .reset_index(drop=True)
)

x = np.arange(len(df_config_sensitivity_plot))

fig, ax = plt.subplots(figsize=(13, 6))

ax.bar(
    x,
    df_config_sensitivity_plot["regime_sensitivity_score_mean"],
    width=0.6
)

ax.set_title("Sensibilidad promedio al régimen por configuración TP/SL")
ax.set_ylabel("regime_sensitivity_score promedio")
ax.set_xticks(x)
ax.set_xticklabels(df_config_sensitivity_plot["tp_sl_config"], rotation=45, ha="left")

for i, row in df_config_sensitivity_plot.iterrows():
    ax.text(
        i,
        row["regime_sensitivity_score_mean"],
        f"{row['regime_sensitivity_score_mean']:.2f}",
        ha="center",
        va="bottom",
        fontsize=8
    )

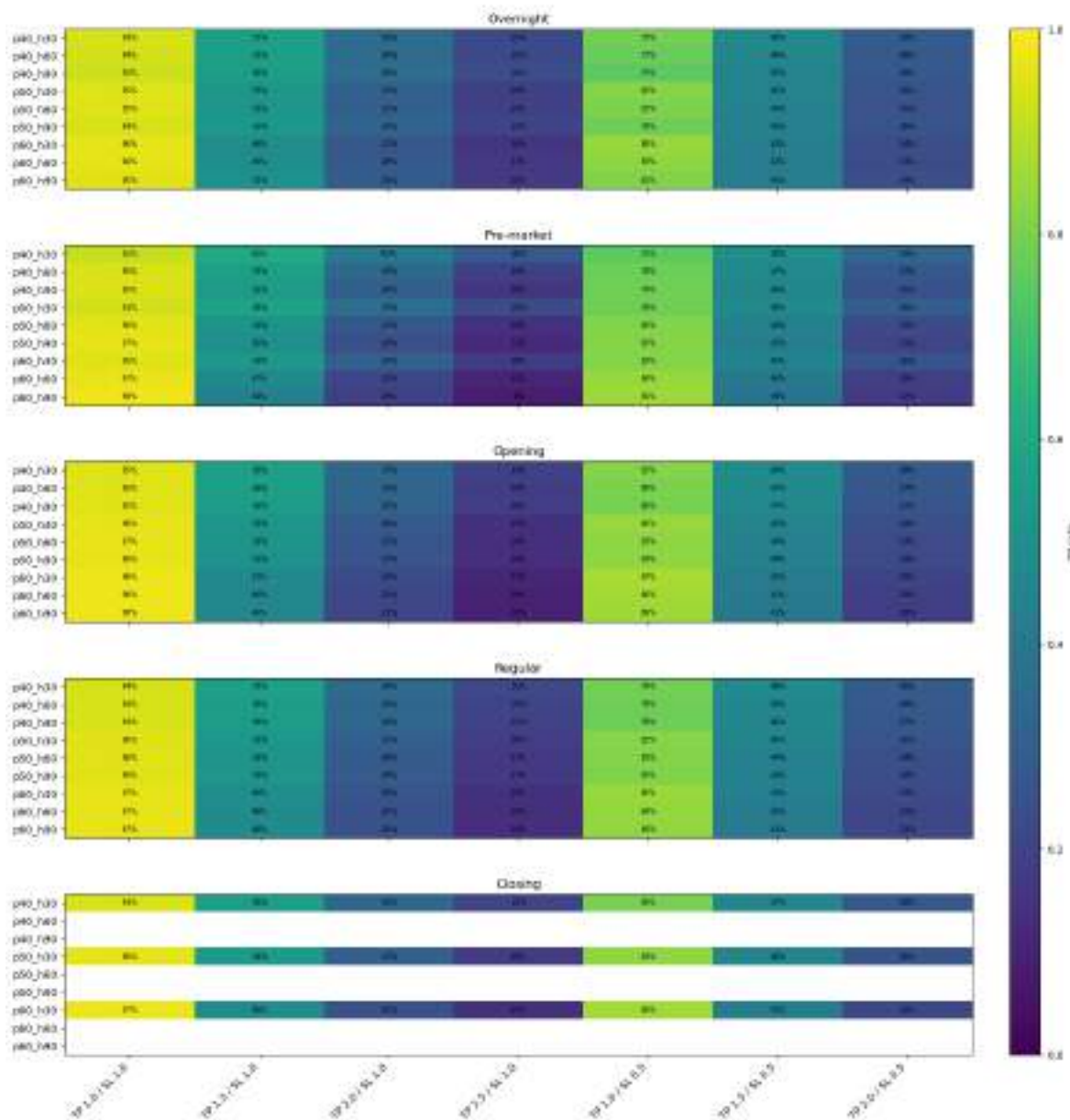
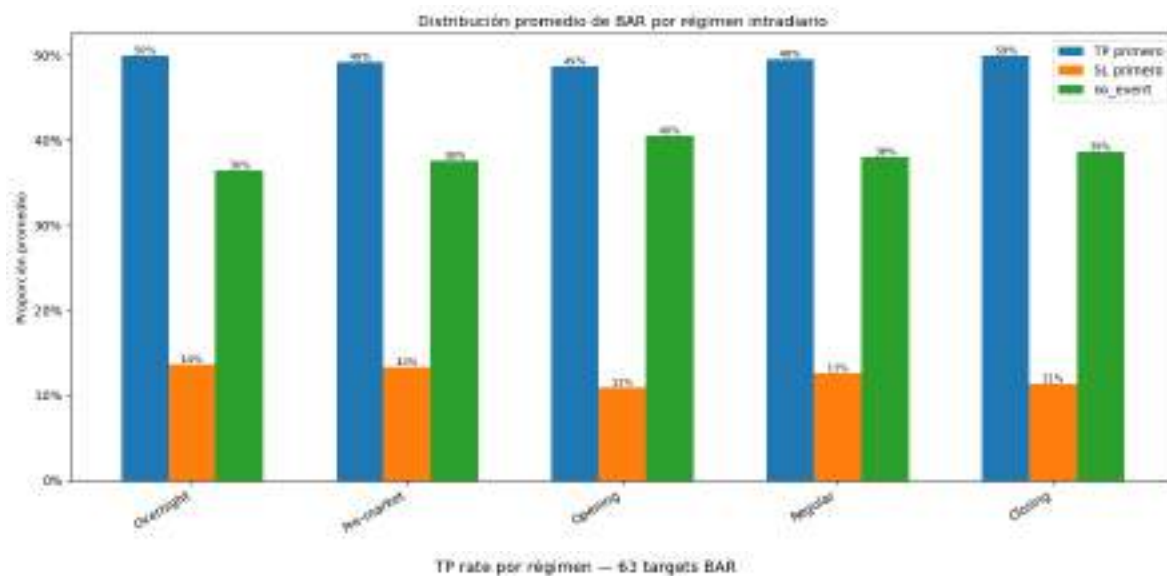
plt.tight_layout()
plt.show()

print("\nCelda 3 finalizada correctamente.")

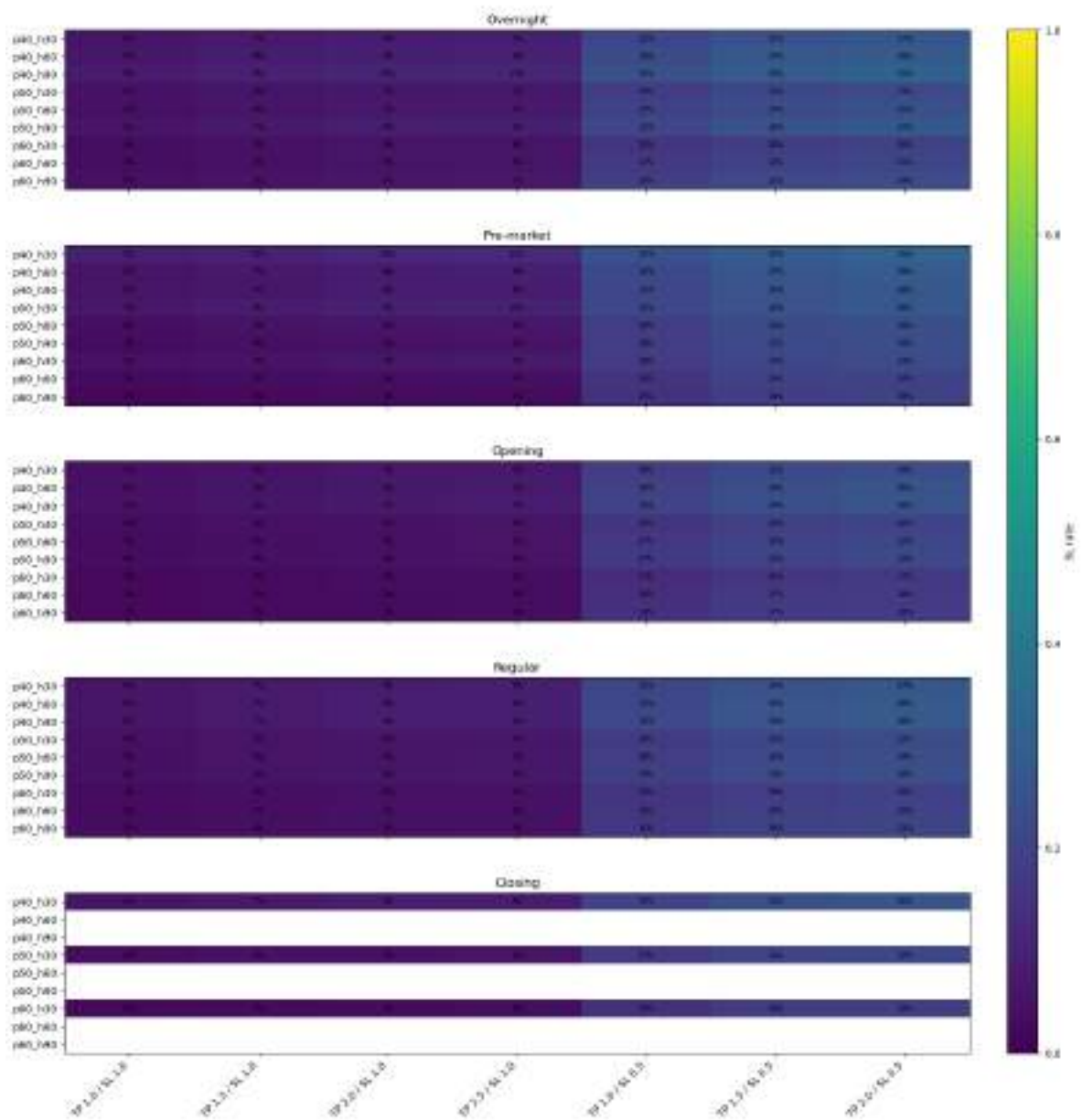
```

Gráficos agrupados por régimen intradiario - BAR

=====

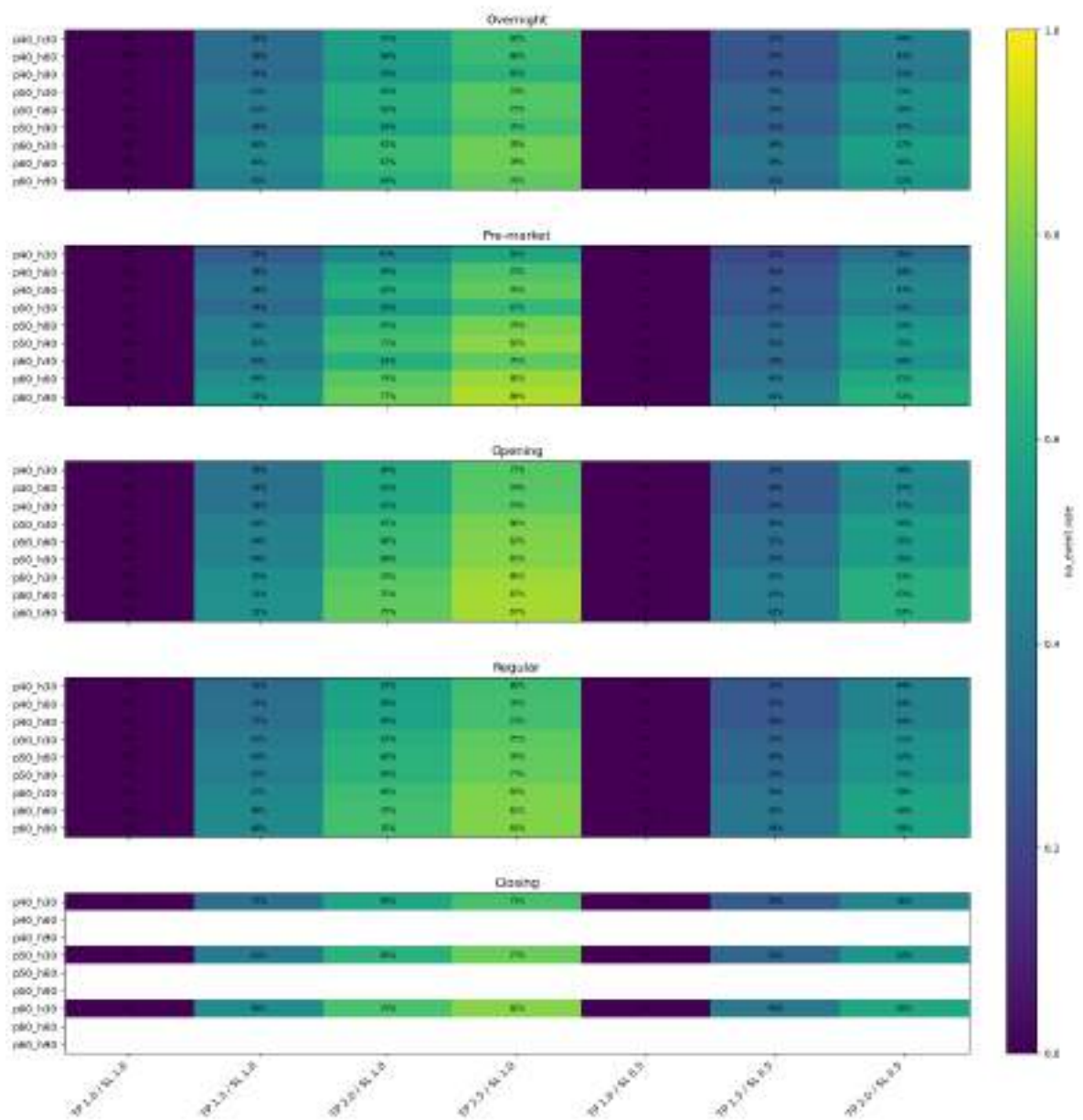


SL rate per regimen — 63 targets BAR

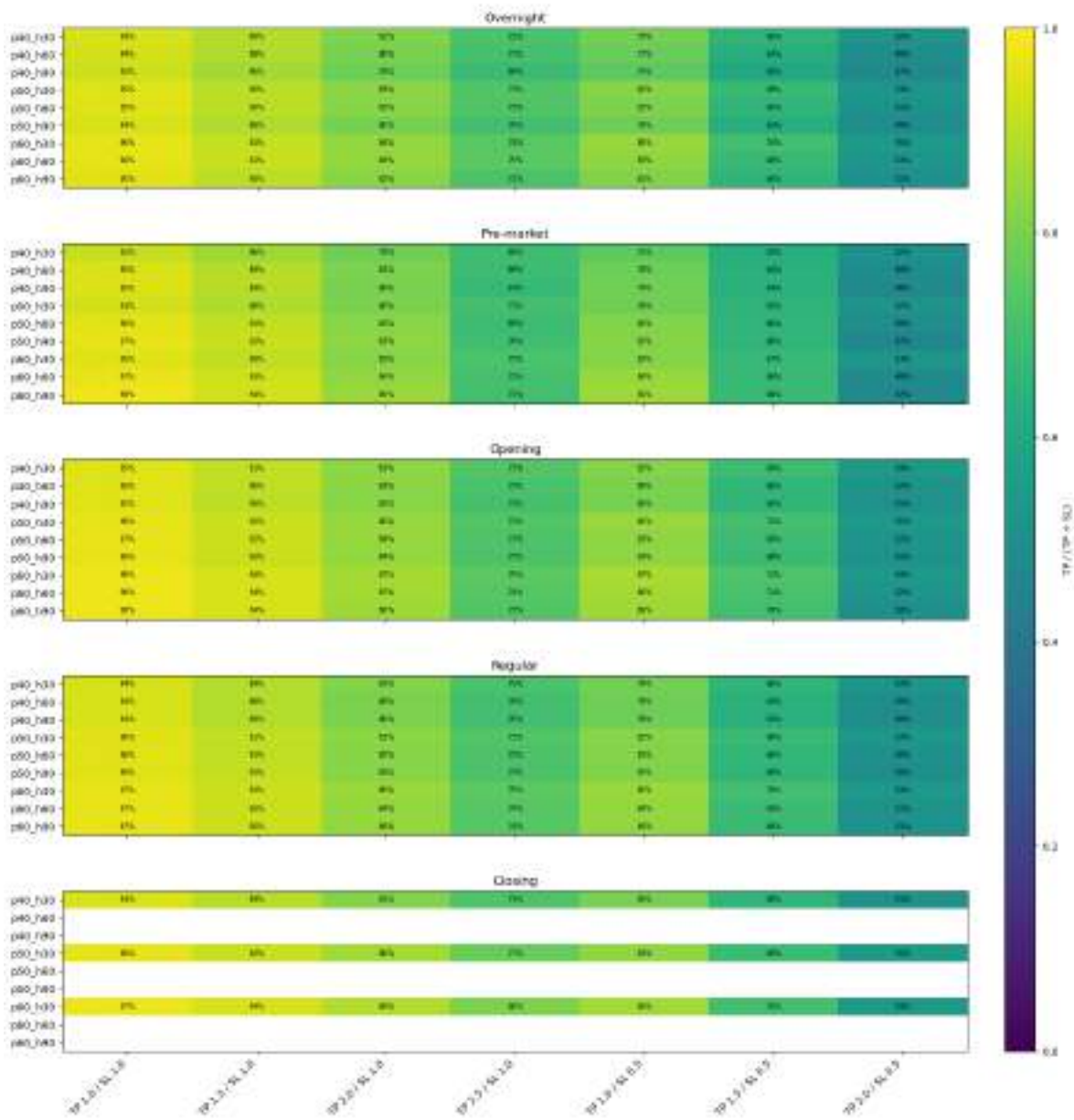




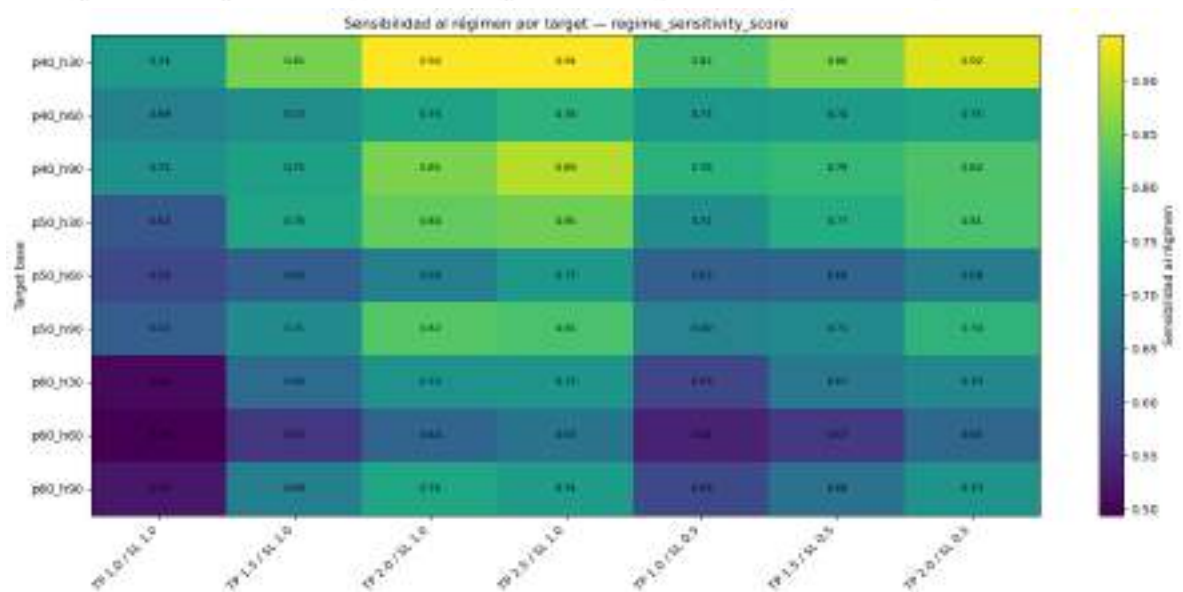
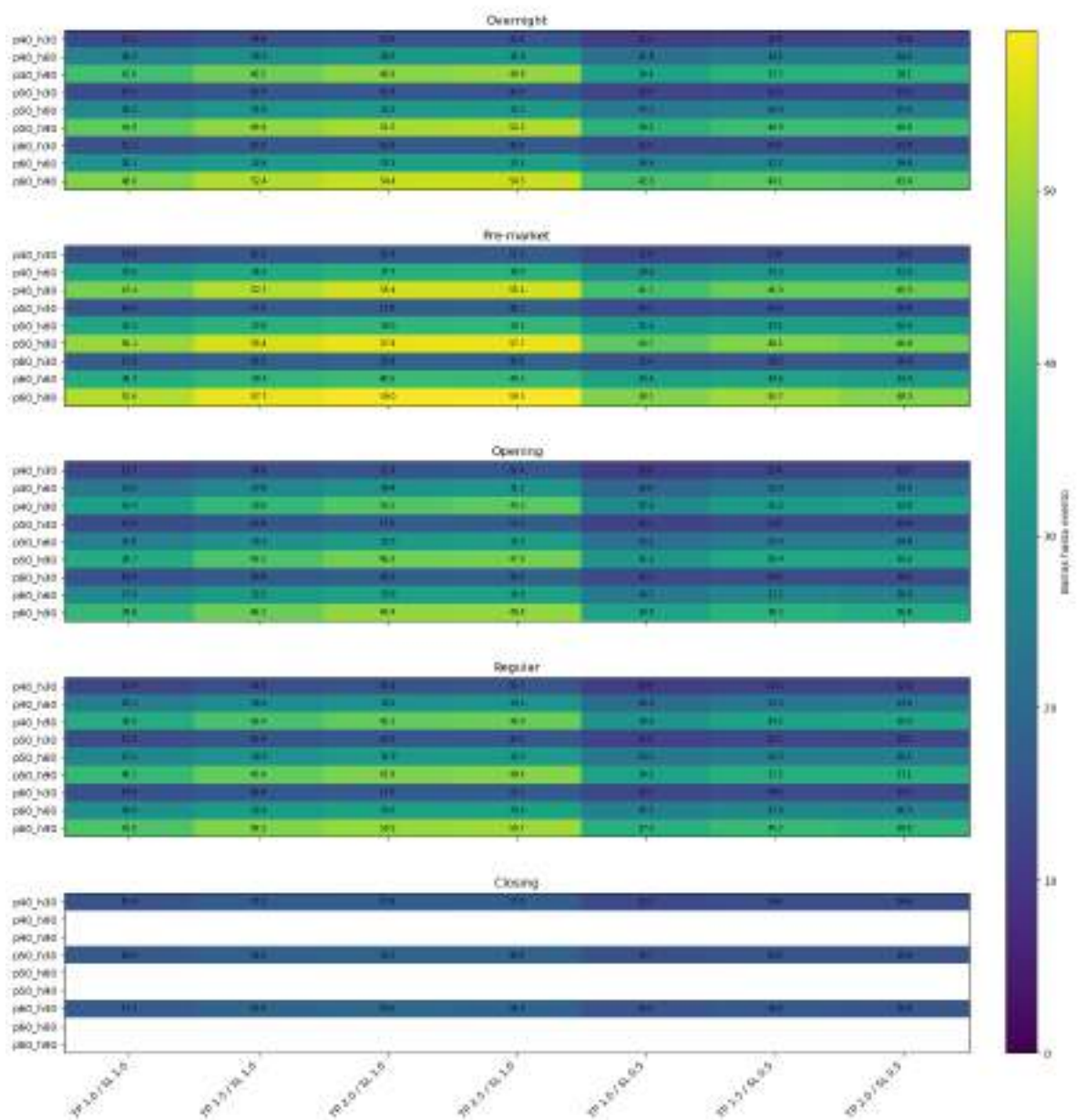
no\_event rate per regimen — 63 targets B&amp;B

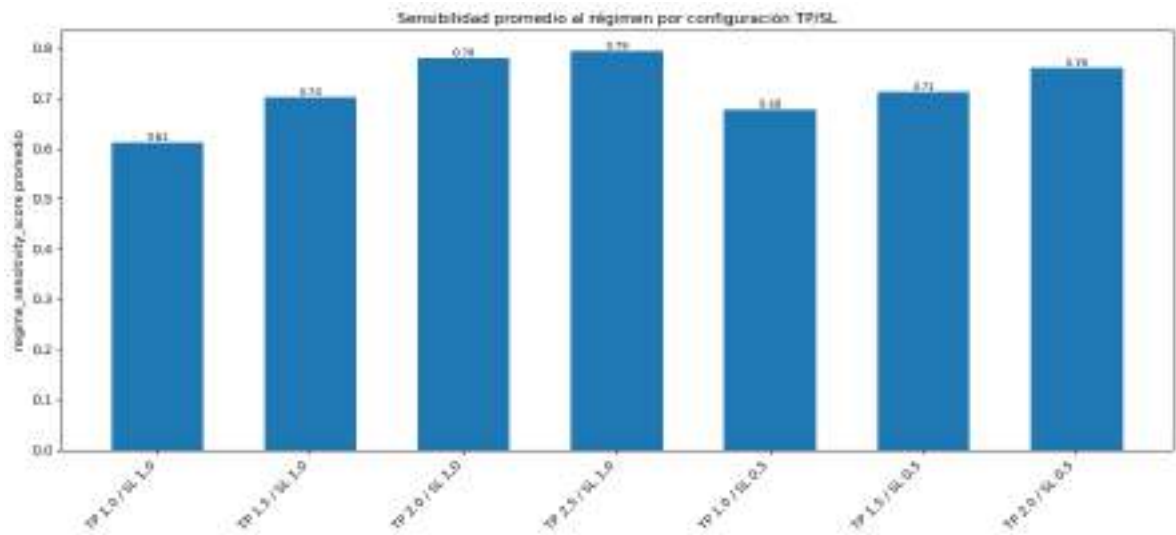


TP sobre eventos resueltos por régimen — 63 targets BARR



Tiempo promedio hasta evento por régimen — 63 targets BAI





Celda 3 finalizada correctamente.

## Celda 4 — Guardado y tabla final para observaciones

In [114...

```
# Celda 4 – Guardado y tabla final para observaciones
# =====

print("Guardado y tabla final de análisis por régimen intradiario - BAR")
print("=" * 80)

# -----
# Validar objetos necesarios
# -----

required_objects = [
    "df_bar_regime_analysis",
    "df_bar_regime_by_regime",
    "df_bar_regime_by_config",
    "df_bar_regime_by_target_base",
    "df_bar_regime_candidate_tp15_sl10",
    "df_bar_regime_stability_by_target",
    "df_bar_regime_sensitivity_by_config",
    "df_bar_regime_sensitivity_by_target_base",
    "MNQ_TARGETS_PATH",
]

missing_objects = [
    obj for obj in required_objects
    if obj not in globals()
]

if missing_objects:
    raise NameError(
        "Faltan objetos necesarios. "
        f"Debes correr las celdas anteriores: {missing_objects}"
    )

# -----
# Definir rutas de guardado
# -----

MNQ_TARGETS_BAR_REGIME_ANALYSIS_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_regime_analysis.csv"
```

```

)

MNQ_TARGETS_BAR_REGIME_BY_REGIME_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_regime_by_regime.csv"
)

MNQ_TARGETS_BAR_REGIME_BY_CONFIG_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_regime_by_config.csv"
)

MNQ_TARGETS_BAR_REGIME_BY_TARGET_BASE_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_regime_by_target_base.csv"
)

MNQ_TARGETS_BAR_REGIME_CANDIDATE_TP15_SL10_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_regime_candidate_tp15_sl10.csv"
)

MNQ_TARGETS_BAR_REGIME_STABILITY_BY_TARGET_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_regime_stability_by_target.csv"
)

MNQ_TARGETS_BAR_REGIME_SENSITIVITY_BY_CONFIG_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_regime_sensitivity_by_config.csv"
)

MNQ_TARGETS_BAR_REGIME_SENSITIVITY_BY_TARGET_BASE_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_regime_sensitivity_by_target_base.csv"
)

MNQ_TARGETS_BAR_REGIME_FINAL_TABLE_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_regime_final_table.csv"
)

MNQ_TARGETS_BAR_REGIME_REPORT_TXT = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_regime_report.txt"
)

# -----
# Crear tabla final de sensibilidad por régimen
# -----

df_bar_regime_final_table = df_bar_regime_stability_by_target.copy()

# -----
# Clasificación relativa de sensibilidad al régimen
# -----
# Menor score = más estable entre regímenes.
# Mayor score = más dependiente del régimen intradiario.
# -----

q33 = df_bar_regime_final_table["regime_sensitivity_score"].quantile(0.33)
q66 = df_bar_regime_final_table["regime_sensitivity_score"].quantile(0.66)

def classify_regime_sensitivity(score):
    """
    Clasificación descriptiva de sensibilidad al régimen.
    Es relativa al conjunto de 63 targets BAR.
    """

```

```
    if pd.isna(score):
        return "sin_clasificar"

    if score <= q33:
        return "menor_sensibilidad"

    if score <= q66:
        return "sensibilidad_intermedia"

    return "mayor_sensibilidad"

df_bar_regime_final_table["regime_sensitivity_profile"] = (
    df_bar_regime_final_table["regime_sensitivity_score"]
    .apply(classify_regime_sensitivity)
)

df_bar_regime_final_table = (
    df_bar_regime_final_table
    .sort_values(
        [
            "regime_sensitivity_score",
            "percentile",
            "horizon",
            "sl_mult",
            "tp_mult",
        ],
        ascending=[True, True, True, True, True]
    )
    .reset_index(drop=True)
)

# -----
# Guardar tablas principales
# -----

df_bar_regime_analysis.to_csv(
    MNQ_TARGETS_BAR_REGIME_ANALYSIS_CSV,
    index=False
)

df_bar_regime_by_regime.to_csv(
    MNQ_TARGETS_BAR_REGIME_BY_REGIME_CSV,
    index=False
)

df_bar_regime_by_config.to_csv(
    MNQ_TARGETS_BAR_REGIME_BY_CONFIG_CSV,
    index=False
)

df_bar_regime_by_target_base.to_csv(
    MNQ_TARGETS_BAR_REGIME_BY_TARGET_BASE_CSV,
    index=False
)

df_bar_regime_candidate_tp15_sl10.to_csv(
    MNQ_TARGETS_BAR_REGIME_CANDIDATE_TP15_SL10_CSV,
    index=False
)
```

```
df_bar_regime_stability_by_target.to_csv(
    MNQ_TARGETS_BAR_REGIME_STABILITY_BY_TARGET_CSV,
    index=False
)

df_bar_regime_sensitivity_by_config.to_csv(
    MNQ_TARGETS_BAR_REGIME_SENSITIVITY_BY_CONFIG_CSV,
    index=False
)

df_bar_regime_sensitivity_by_target_base.to_csv(
    MNQ_TARGETS_BAR_REGIME_SENSITIVITY_BY_TARGET_BASE_CSV,
    index=False
)

df_bar_regime_final_table.to_csv(
    MNQ_TARGETS_BAR_REGIME_FINAL_TABLE_CSV,
    index=False
)

# -----
# Columnas para impresión
# -----

regime_summary_cols = [
    "regime_id",
    "regime_name",
    "n_targets",
    "operable_rate_mean",
    "tp_rate_mean",
    "sl_rate_mean",
    "no_event_rate_mean",
    "resolved_rate_mean",
    "tp_share_over_resolved_mean",
    "expectancy_units_mean",
    "event_step_mean",
]

config_summary_cols = [
    "regime_id",
    "regime_name",
    "tp_sl_config",
    "n_targets",
    "operable_rate_mean",
    "tp_rate_mean",
    "sl_rate_mean",
    "no_event_rate_mean",
    "resolved_rate_mean",
    "tp_share_over_resolved_mean",
    "expectancy_units_mean",
    "event_step_mean",
]

candidate_cols = [
    "bar_col",
    "target_base",
    "regime_id",
    "regime_name",
    "total_rows",
```



```
"valid_rows",
"operable_rate",
"tp_rate",
"sl_rate",
"no_event_rate",
"resolved_rate",
"tp_share_over_resolved",
"expectancy_units",
"event_step_mean",
]

sensitivity_config_cols = [
    "tp_sl_config",
    "n_targets",
    "operable_rate_range_mean",
    "tp_rate_range_mean",
    "sl_rate_range_mean",
    "no_event_rate_range_mean",
    "tp_share_range_mean",
    "expectancy_range_mean",
    "regime_sensitivity_score_mean",
    "regime_sensitivity_score_max",
]

sensitivity_target_base_cols = [
    "target_base",
    "percentile",
    "horizon",
    "n_configs",
    "operable_rate_range_mean",
    "tp_rate_range_mean",
    "sl_rate_range_mean",
    "no_event_rate_range_mean",
    "tp_share_range_mean",
    "expectancy_range_mean",
    "regime_sensitivity_score_mean",
    "regime_sensitivity_score_max",
]

final_table_cols = [
    "bar_col",
    "target_base",
    "tp_sl_config",
    "n_regimes",
    "operable_rate_mean",
    "operable_rate_range",
    "tp_rate_mean",
    "tp_rate_range",
    "sl_rate_mean",
    "sl_rate_range",
    "no_event_rate_mean",
    "no_event_rate_range",
    "resolved_rate_mean",
    "resolved_rate_range",
    "tp_share_over_resolved_mean",
    "tp_share_over_resolved_range",
    "expectancy_units_mean",
    "expectancy_units_range",
    "event_step_mean_mean",
    "event_step_mean_range",
```



```

    "regime_sensitivity_score",
    "regime_sensitivity_profile",
]

# -----
# Impresiones principales
# -----

print("\nResumen general por régimen:")
print(
    df_bar_regime_by_regime[regime_summary_cols]
    .to_string(
        index=False,
        formatters={
            "operable_rate_mean": "{:.2%}".format,
            "tp_rate_mean": "{:.2%}".format,
            "sl_rate_mean": "{:.2%}".format,
            "no_event_rate_mean": "{:.2%}".format,
            "resolved_rate_mean": "{:.2%}".format,
            "tp_share_over_resolved_mean": "{:.2%}".format,
            "expectancy_units_mean": "{:.3f}".format,
            "event_step_mean": "{:.2f}".format,
        }
    )
)

print("\nResumen por régimen y configuración TP/SL:")
print(
    df_bar_regime_by_config[config_summary_cols]
    .to_string(
        index=False,
        formatters={
            "operable_rate_mean": "{:.2%}".format,
            "tp_rate_mean": "{:.2%}".format,
            "sl_rate_mean": "{:.2%}".format,
            "no_event_rate_mean": "{:.2%}".format,
            "resolved_rate_mean": "{:.2%}".format,
            "tp_share_over_resolved_mean": "{:.2%}".format,
            "expectancy_units_mean": "{:.3f}".format,
            "event_step_mean": "{:.2f}".format,
        }
    )
)

print("\nDetalle por régimen para configuración candidata TP 1.5 / SL 1.0:")
print(
    df_bar_regime_candidate_tp15_sl10[candidate_cols]
    .to_string(
        index=False,
        formatters={
            "operable_rate": "{:.2%}".format,
            "tp_rate": "{:.2%}".format,
            "sl_rate": "{:.2%}".format,
            "no_event_rate": "{:.2%}".format,
            "resolved_rate": "{:.2%}".format,
            "tp_share_over_resolved": "{:.2%}".format,
            "expectancy_units": "{:.3f}".format,
            "event_step_mean": "{:.2f}".format,
        }
    )
)

```

```

)

print("\nSensibilidad agregada al régimen por configuración TP/SL:")
print(
    df_bar_regime_sensitivity_by_config[sensitivity_config_cols]
    .to_string(
        index=False,
        formatters={
            "operable_rate_range_mean": "{:.2%}".format,
            "tp_rate_range_mean": "{:.2%}".format,
            "sl_rate_range_mean": "{:.2%}".format,
            "no_event_rate_range_mean": "{:.2%}".format,
            "tp_share_range_mean": "{:.2%}".format,
            "expectancy_range_mean": "{:.3f}".format,
            "regime_sensitivity_score_mean": "{:.3f}".format,
            "regime_sensitivity_score_max": "{:.3f}".format,
        }
    )
)

print("\nSensibilidad agregada al régimen por percentil y horizonte:")
print(
    df_bar_regime_sensitivity_by_target_base[sensitivity_target_base_cols]
    .to_string(
        index=False,
        formatters={
            "operable_rate_range_mean": "{:.2%}".format,
            "tp_rate_range_mean": "{:.2%}".format,
            "sl_rate_range_mean": "{:.2%}".format,
            "no_event_rate_range_mean": "{:.2%}".format,
            "tp_share_range_mean": "{:.2%}".format,
            "expectancy_range_mean": "{:.3f}".format,
            "regime_sensitivity_score_mean": "{:.3f}".format,
            "regime_sensitivity_score_max": "{:.3f}".format,
        }
    )
)

print("\nTabla final de sensibilidad al régimen – 63 targets BAR:")
print(
    df_bar_regime_final_table[final_table_cols]
    .to_string(
        index=False,
        formatters={
            "operable_rate_mean": "{:.2%}".format,
            "operable_rate_range": "{:.2%}".format,
            "tp_rate_mean": "{:.2%}".format,
            "tp_rate_range": "{:.2%}".format,
            "sl_rate_mean": "{:.2%}".format,
            "sl_rate_range": "{:.2%}".format,
            "no_event_rate_mean": "{:.2%}".format,
            "no_event_rate_range": "{:.2%}".format,
            "resolved_rate_mean": "{:.2%}".format,
            "resolved_rate_range": "{:.2%}".format,
            "tp_share_over_resolved_mean": "{:.2%}".format,
            "tp_share_over_resolved_range": "{:.2%}".format,
            "expectancy_units_mean": "{:.3f}".format,
            "expectancy_units_range": "{:.3f}".format,
            "event_step_mean_mean": "{:.2f}".format,
            "event_step_mean_range": "{:.2f}".format,
        }
    )
)

```

```

        "regime_sensitivity_score": "{:.3f}".format,
    }
)
)

# -----
# Top 15 menor y mayor sensibilidad al régimen
# -----

print("\nTop 15 targets con menor sensibilidad al régimen:")
print(
    df_bar_regime_final_table[final_table_cols]
    .head(15)
    .to_string(
        index=False,
        formatters={
            "operable_rate_mean": "{:.2%}".format,
            "operable_rate_range": "{:.2%}".format,
            "tp_rate_mean": "{:.2%}".format,
            "tp_rate_range": "{:.2%}".format,
            "sl_rate_mean": "{:.2%}".format,
            "sl_rate_range": "{:.2%}".format,
            "no_event_rate_mean": "{:.2%}".format,
            "no_event_rate_range": "{:.2%}".format,
            "resolved_rate_mean": "{:.2%}".format,
            "resolved_rate_range": "{:.2%}".format,
            "tp_share_over_resolved_mean": "{:.2%}".format,
            "tp_share_over_resolved_range": "{:.2%}".format,
            "expectancy_units_mean": "{:.3f}".format,
            "expectancy_units_range": "{:.3f}".format,
            "event_step_mean_mean": "{:.2f}".format,
            "event_step_mean_range": "{:.2f}".format,
            "regime_sensitivity_score": "{:.3f}".format,
        }
    )
)

print("\nTop 15 targets con mayor sensibilidad al régimen:")
print(
    df_bar_regime_final_table[final_table_cols]
    .tail(15)
    .sort_values("regime_sensitivity_score", ascending=False)
    .to_string(
        index=False,
        formatters={
            "operable_rate_mean": "{:.2%}".format,
            "operable_rate_range": "{:.2%}".format,
            "tp_rate_mean": "{:.2%}".format,
            "tp_rate_range": "{:.2%}".format,
            "sl_rate_mean": "{:.2%}".format,
            "sl_rate_range": "{:.2%}".format,
            "no_event_rate_mean": "{:.2%}".format,
            "no_event_rate_range": "{:.2%}".format,
            "resolved_rate_mean": "{:.2%}".format,
            "resolved_rate_range": "{:.2%}".format,
            "tp_share_over_resolved_mean": "{:.2%}".format,
            "tp_share_over_resolved_range": "{:.2%}".format,
            "expectancy_units_mean": "{:.3f}".format,
            "expectancy_units_range": "{:.3f}".format,
            "event_step_mean_mean": "{:.2f}".format,
        }
    )
)

```

```

        "event_step_mean_range": "{:.2f}".format,
        "regime_sensitivity_score": "{:.3f}".format,
    }
)
)

# -----
# Conteo por perfil de sensibilidad al régimen
# -----

df_bar_regime_sensitivity_profile_counts = (
    df_bar_regime_final_table
    .groupby("regime_sensitivity_profile", as_index=False)
    .agg(
        n_targets=("bar_col", "count"),

        regime_sensitivity_score_mean=("regime_sensitivity_score", "mean"),

        operable_rate_range_mean=("operable_rate_range", "mean"),
        tp_rate_range_mean=("tp_rate_range", "mean"),
        sl_rate_range_mean=("sl_rate_range", "mean"),
        no_event_rate_range_mean=("no_event_rate_range", "mean"),
        tp_share_range_mean=("tp_share_over_resolved_range", "mean"),

        expectancy_range_mean=("expectancy_units_range", "mean"),
        event_step_range_mean=("event_step_mean_range", "mean"),
    )
    .sort_values("regime_sensitivity_score_mean")
    .reset_index(drop=True)
)

print("\nConteo por perfil de sensibilidad al régimen:")
print(
    df_bar_regime_sensitivity_profile_counts
    .to_string(
        index=False,
        formatters={
            "regime_sensitivity_score_mean": "{:.3f}".format,
            "operable_rate_range_mean": "{:.2%}".format,
            "tp_rate_range_mean": "{:.2%}".format,
            "sl_rate_range_mean": "{:.2%}".format,
            "no_event_rate_range_mean": "{:.2%}".format,
            "tp_share_range_mean": "{:.2%}".format,
            "expectancy_range_mean": "{:.3f}".format,
            "event_step_range_mean": "{:.2f}".format,
        }
    )
)

# -----
# Guardar reporte TXT
# -----

with MNQ_TARGETS_BAR_REGIME_REPORT_TXT.open("w", encoding="utf-8") as f:
    f.write("4.10. Análisis por régimen intradiario - BAR\n")
    f.write("=" * 100 + "\n\n")

    f.write("Resumen general por régimen\n")
    f.write("-" * 100 + "\n")
    f.write(

```

```

df_bar_regime_by_regime[regime_summary_cols]
.to_string(
    index=False,
    formatters={
        "operable_rate_mean": "{:.2%}".format,
        "tp_rate_mean": "{:.2%}".format,
        "sl_rate_mean": "{:.2%}".format,
        "no_event_rate_mean": "{:.2%}".format,
        "resolved_rate_mean": "{:.2%}".format,
        "tp_share_over_resolved_mean": "{:.2%}".format,
        "expectancy_units_mean": "{:.3f}".format,
        "event_step_mean": "{:.2f}".format,
    }
)

f.write("\n\n")
f.write("Resumen por régimen y configuración TP/SL\n")
f.write("-" * 100 + "\n")
f.write(
    df_bar_regime_by_config[config_summary_cols]
    .to_string(
        index=False,
        formatters={
            "operable_rate_mean": "{:.2%}".format,
            "tp_rate_mean": "{:.2%}".format,
            "sl_rate_mean": "{:.2%}".format,
            "no_event_rate_mean": "{:.2%}".format,
            "resolved_rate_mean": "{:.2%}".format,
            "tp_share_over_resolved_mean": "{:.2%}".format,
            "expectancy_units_mean": "{:.3f}".format,
            "event_step_mean": "{:.2f}".format,
        }
    )
)

f.write("\n\n")
f.write("Detalle por régimen para configuración candidata TP 1.5 / SL 1.0\n")
f.write("-" * 100 + "\n")
f.write(
    df_bar_regime_candidate_tp15_sl10[candidate_cols]
    .to_string(
        index=False,
        formatters={
            "operable_rate": "{:.2%}".format,
            "tp_rate": "{:.2%}".format,
            "sl_rate": "{:.2%}".format,
            "no_event_rate": "{:.2%}".format,
            "resolved_rate": "{:.2%}".format,
            "tp_share_over_resolved": "{:.2%}".format,
            "expectancy_units": "{:.3f}".format,
            "event_step_mean": "{:.2f}".format,
        }
    )
)

f.write("\n\n")
f.write("Sensibilidad agregada al régimen por configuración TP/SL\n")
f.write("-" * 100 + "\n")
f.write(

```

```

df_bar_regime_sensitivity_by_config[sensitivity_config_cols]
.to_string(
    index=False,
    formatters={
        "operable_rate_range_mean": "{:.2%}".format,
        "tp_rate_range_mean": "{:.2%}".format,
        "sl_rate_range_mean": "{:.2%}".format,
        "no_event_rate_range_mean": "{:.2%}".format,
        "tp_share_range_mean": "{:.2%}".format,
        "expectancy_range_mean": "{:.3f}".format,
        "regime_sensitivity_score_mean": "{:.3f}".format,
        "regime_sensitivity_score_max": "{:.3f}".format,
    }
)

f.write("\n\n")
f.write("Sensibilidad agregada al régimen por percentil y horizonte\n")
f.write("-" * 100 + "\n")
f.write(
    df_bar_regime_sensitivity_by_target_base[sensitivity_target_base_cols]
    .to_string(
        index=False,
        formatters={
            "operable_rate_range_mean": "{:.2%}".format,
            "tp_rate_range_mean": "{:.2%}".format,
            "sl_rate_range_mean": "{:.2%}".format,
            "no_event_rate_range_mean": "{:.2%}".format,
            "tp_share_range_mean": "{:.2%}".format,
            "expectancy_range_mean": "{:.3f}".format,
            "regime_sensitivity_score_mean": "{:.3f}".format,
            "regime_sensitivity_score_max": "{:.3f}".format,
        }
    )
)

f.write("\n\n")
f.write("Tabla final de sensibilidad al régimen – 63 targets BAR\n")
f.write("-" * 100 + "\n")
f.write(
    df_bar_regime_final_table[final_table_cols]
    .to_string(
        index=False,
        formatters={
            "operable_rate_mean": "{:.2%}".format,
            "operable_rate_range": "{:.2%}".format,
            "tp_rate_mean": "{:.2%}".format,
            "tp_rate_range": "{:.2%}".format,
            "sl_rate_mean": "{:.2%}".format,
            "sl_rate_range": "{:.2%}".format,
            "no_event_rate_mean": "{:.2%}".format,
            "no_event_rate_range": "{:.2%}".format,
            "resolved_rate_mean": "{:.2%}".format,
            "resolved_rate_range": "{:.2%}".format,
            "tp_share_over_resolved_mean": "{:.2%}".format,
            "tp_share_over_resolved_range": "{:.2%}".format,
            "expectancy_units_mean": "{:.3f}".format,
            "expectancy_units_range": "{:.3f}".format,
            "event_step_mean_mean": "{:.2f}".format,
            "event_step_mean_range": "{:.2f}".format,
        }
    )
)

```

```

        "regime_sensitivity_score": "{:.3f}".format,
    }
    )
)

f.write("\n\n")
f.write("Conteo por perfil de sensibilidad al régimen\n")
f.write("-" * 100 + "\n")
f.write(
    df_bar_regime_sensitivity_profile_counts
    .to_string(
        index=False,
        formatters={
            "regime_sensitivity_score_mean": "{:.3f}".format,
            "operable_rate_range_mean": "{:.2%}".format,
            "tp_rate_range_mean": "{:.2%}".format,
            "sl_rate_range_mean": "{:.2%}".format,
            "no_event_rate_range_mean": "{:.2%}".format,
            "tp_share_range_mean": "{:.2%}".format,
            "expectancy_range_mean": "{:.3f}".format,
            "event_step_range_mean": "{:.2f}".format,
        }
    )
)

# -----
# Confirmación final
# -----

print("\nArchivos guardados:")
print(f"Análisis BAR por régimen: {MNQ_TARGETS_BAR_REGIME_ANALYSIS_")
print(f"Resumen general por régimen: {MNQ_TARGETS_BAR_REGIME_BY_REGIME")
print(f"Resumen por régimen y configuración: {MNQ_TARGETS_BAR_REGIME_BY_CONFIG")
print(f"Resumen por régimen y target base: {MNQ_TARGETS_BAR_REGIME_BY_TARGET")
print(f"Candidata TP 1.5 / SL 1.0 por régimen: {MNQ_TARGETS_BAR_REGIME_CANDIDATE")
print(f"Sensibilidad por target: {MNQ_TARGETS_BAR_REGIME_STABILITY")
print(f"Sensibilidad por configuración: {MNQ_TARGETS_BAR_REGIME_SENSITIVI")
print(f"Sensibilidad por percentil/horizonte: {MNQ_TARGETS_BAR_REGIME_SENSITIVI")
print(f"Tabla final por régimen: {MNQ_TARGETS_BAR_REGIME_FINAL_TAB")
print(f"Reporte TXT: {MNQ_TARGETS_BAR_REGIME_REPORT_TX")

print("\nCelda 4 finalizada correctamente.")

```

## Guardado y tabla final de análisis por régimen intradiario - BAR

=====

## Resumen general por régimen:

| regime_id | regime_name | n_targets | operable_rate_mean | tp_rate_mean | sl_rate_mean | no_event_rate_mean | resolved_rate_mean | tp_share_over_resolved_mean | expectancy_units_mean | event_step_mean |
|-----------|-------------|-----------|--------------------|--------------|--------------|--------------------|--------------------|-----------------------------|-----------------------|-----------------|
| 0         | Overnight   | 63        | 54.01%             | 49.97%       | 13.64%       | 36.39%             | 63.61%             | 76.30%                      | 0.607                 | 29.45           |
| 1         | Pre-market  | 63        | 54.50%             | 49.19%       | 13.28%       | 37.53%             | 62.47%             | 76.06%                      | 0.591                 | 34.33           |
| 2         | Opening     | 63        | 54.45%             | 48.67%       | 10.90%       | 40.43%             | 59.57%             | 78.70%                      | 0.590                 | 27.14           |
| 3         | Regular     | 63        | 48.46%             | 49.47%       | 12.58%       | 37.96%             | 62.04%             | 77.20%                      | 0.600                 | 27.86           |
| 4         | Closing     | 63        | 0.59%              | 49.91%       | 11.42%       | 38.67%             | 61.33%             | 79.16%                      | 0.614                 | 16.81           |

## Resumen por régimen y configuración TP/SL:

| regime_id | regime_name | tp_sl_config    | n_targets | operable_rate_mean | tp_rate_mean | sl_rate_mean | no_event_rate_mean | resolved_rate_mean | tp_share_over_resolved_mean | expectancy_units_mean | event_step_mean |
|-----------|-------------|-----------------|-----------|--------------------|--------------|--------------|--------------------|--------------------|-----------------------------|-----------------------|-----------------|
| 0         | Overnight   | TP 1.0 / SL 1.0 | 9         | 54.01%             | 94.60%       | 5.40%        | 0.00%              | 100.00%            | 94.60%                      | 0.892                 | 29.08           |
| 0         | Overnight   | TP 1.5 / SL 1.0 | 9         | 54.01%             | 53.21%       | 6.47%        | 40.32%             | 59.68%             | 89.26%                      | 0.733                 | 32.03           |
| 0         | Overnight   | TP 2.0 / SL 1.0 | 9         | 54.01%             | 31.72%       | 7.30%        | 60.98%             | 39.02%             | 81.50%                      | 0.561                 | 33.34           |
| 0         | Overnight   | TP 2.5 / SL 1.0 | 9         | 54.01%             | 19.78%       | 7.76%        | 72.46%             | 27.54%             | 72.10%                      | 0.417                 | 33.58           |
| 0         | Overnight   | TP 1.0 / SL 0.5 | 9         | 54.01%             | 80.15%       | 19.85%       | 0.00%              | 100.00%            | 80.15%                      | 0.702                 | 24.93           |
| 0         | Overnight   | TP 1.5 / SL 0.5 | 9         | 54.01%             | 44.32%       | 23.57%       | 32.11%             | 67.89%             | 65.43%                      | 0.547                 | 26.72           |
| 0         | Overnight   | TP 2.0 / SL 0.5 | 9         | 54.01%             | 26.00%       | 25.11%       | 48.89%             | 51.11%             | 51.05%                      | 0.394                 | 26.49           |
| 1         | Pre-market  | TP 1.0 / SL 1.0 | 9         | 54.50%             | 95.17%       | 4.83%        | 0.00%              | 100.00%            | 95.17%                      | 0.903                 | 33.70           |
| 1         | Pre-market  | TP 1.5 / SL 1.0 | 9         | 54.50%             | 53.07%       | 5.95%        | 40.98%             | 59.02%             | 90.20%                      | 0.737                 | 36.73           |
| 1         | Pre-market  | TP 2.0 / SL 1.0 | 9         | 54.50%             | 29.54%       | 6.82%        | 63.64%             | 36.36%             | 81.71%                      | 0.523                 | 38.06           |
| 1         | Pre-market  | TP 2.5 / SL 1.0 | 9         | 54.50%             | 16.73%       | 7.26%        | 76.01%             | 23.99%             | 69.81%                      | 0.346                 | 38.24           |



|        |   |                            |   |        |        |
|--------|---|----------------------------|---|--------|--------|
| 19.44% | 1 | Pre-market TP 1.0 / SL 0.5 | 9 | 54.50% | 80.56% |
| 0.708  |   | 0.00% 100.00%              |   | 80.56% |        |
|        |   | 30.17                      |   |        |        |
| 23.50% | 1 | Pre-market TP 1.5 / SL 0.5 | 9 | 54.50% | 44.59% |
| 0.551  |   | 31.91% 68.09%              |   | 65.64% |        |
|        |   | 32.00                      |   |        |        |
| 25.16% | 1 | Pre-market TP 2.0 / SL 0.5 | 9 | 54.50% | 24.68% |
| 0.368  |   | 50.15% 49.85%              |   | 49.31% |        |
|        |   | 31.40                      |   |        |        |
| 3.62%  | 2 | Opening TP 1.0 / SL 1.0    | 9 | 54.45% | 96.38% |
| 0.928  |   | 0.00% 100.00%              |   | 96.38% |        |
|        |   | 25.39                      |   |        |        |
| 4.46%  | 2 | Opening TP 1.5 / SL 1.0    | 9 | 54.45% | 51.49% |
| 0.728  |   | 44.05% 55.95%              |   | 92.17% |        |
|        |   | 29.59                      |   |        |        |
| 5.10%  | 2 | Opening TP 2.0 / SL 1.0    | 9 | 54.45% | 27.17% |
| 0.492  |   | 67.74% 32.26%              |   | 84.47% |        |
|        |   | 31.82                      |   |        |        |
| 5.39%  | 2 | Opening TP 2.5 / SL 1.0    | 9 | 54.45% | 14.34% |
| 0.304  |   | 80.28% 19.72%              |   | 72.88% |        |
|        |   | 32.31                      |   |        |        |
| 16.45% | 2 | Opening TP 1.0 / SL 0.5    | 9 | 54.45% | 83.55% |
| 0.753  |   | 0.00% 100.00%              |   | 83.55% |        |
|        |   | 21.82                      |   |        |        |
| 19.96% | 2 | Opening TP 1.5 / SL 0.5    | 9 | 54.45% | 44.39% |
| 0.566  |   | 35.65% 64.35%              |   | 69.13% |        |
|        |   | 24.61                      |   |        |        |
| 21.31% | 2 | Opening TP 2.0 / SL 0.5    | 9 | 54.45% | 23.37% |
| 0.361  |   | 55.32% 44.68%              |   | 52.30% |        |
|        |   | 24.43                      |   |        |        |
| 4.56%  | 3 | Regular TP 1.0 / SL 1.0    | 9 | 48.46% | 95.44% |
| 0.909  |   | 0.00% 100.00%              |   | 95.44% |        |
|        |   | 27.00                      |   |        |        |
| 5.58%  | 3 | Regular TP 1.5 / SL 1.0    | 9 | 48.46% | 52.51% |
| 0.732  |   | 41.91% 58.09%              |   | 90.51% |        |
|        |   | 30.31                      |   |        |        |
| 6.36%  | 3 | Regular TP 2.0 / SL 1.0    | 9 | 48.46% | 29.80% |
| 0.532  |   | 63.84% 36.16%              |   | 82.61% |        |
|        |   | 31.93                      |   |        |        |
| 6.76%  | 3 | Regular TP 2.5 / SL 1.0    | 9 | 48.46% | 17.56% |
| 0.371  |   | 75.68% 24.32%              |   | 72.44% |        |
|        |   | 32.33                      |   |        |        |
| 18.53% | 3 | Regular TP 1.0 / SL 0.5    | 9 | 48.46% | 81.47% |
| 0.722  |   | 0.00% 100.00%              |   | 81.47% |        |
|        |   | 23.14                      |   |        |        |
| 22.37% | 3 | Regular TP 1.5 / SL 0.5    | 9 | 48.46% | 44.43% |
| 0.555  |   | 33.21% 66.79%              |   | 66.64% |        |
|        |   | 25.23                      |   |        |        |
| 23.90% | 3 | Regular TP 2.0 / SL 0.5    | 9 | 48.46% | 25.05% |
| 0.382  |   | 51.05% 48.95%              |   | 51.25% |        |
|        |   | 25.05                      |   |        |        |
| 3.95%  | 4 | Closing TP 1.0 / SL 1.0    | 9 | 0.59%  | 96.05% |
| 0.921  |   | 0.00% 100.00%              |   | 96.05% |        |
|        |   | 16.45                      |   |        |        |
| 4.83%  | 4 | Closing TP 1.5 / SL 1.0    | 9 | 0.59%  | 52.78% |
| 0.743  |   | 42.38% 57.62%              |   | 91.76% |        |
|        |   | 18.25                      |   |        |        |
| 5.29%  | 4 | Closing TP 2.0 / SL 1.0    | 9 | 0.59%  | 29.83% |
| 0.544  |   | 64.88% 35.12%              |   | 85.37% |        |
|        |   | 18.80                      |   |        |        |

|        |   |                         |   |        |        |
|--------|---|-------------------------|---|--------|--------|
| 5.62%  | 4 | Closing TP 2.5 / SL 1.0 | 9 | 0.59%  | 17.77% |
| 0.388  |   | 76.61% 23.39%           |   | 76.55% |        |
|        |   | 18.71                   |   |        |        |
| 17.02% | 4 | Closing TP 1.0 / SL 0.5 | 9 | 0.59%  | 82.98% |
| 0.745  |   | 0.00% 100.00%           |   | 82.98% |        |
|        |   | 14.54                   |   |        |        |
| 20.96% | 4 | Closing TP 1.5 / SL 0.5 | 9 | 0.59%  | 44.87% |
| 0.568  |   | 34.17% 65.83%           |   | 68.28% |        |
|        |   | 15.63                   |   |        |        |
| 22.26% | 4 | Closing TP 2.0 / SL 0.5 | 9 | 0.59%  | 25.09% |
| 0.391  |   | 52.65% 47.35%           |   | 53.13% |        |
|        |   | 15.31                   |   |        |        |

Detalle por régimen para configuración candidata TP 1.5 / SL 1.0:

| operable_rate         | tp_rate     | sl_rate   | no_event_rate | resolved_rate | tp_share_over_resolved | expectancy_units | event_step_mean |
|-----------------------|-------------|-----------|---------------|---------------|------------------------|------------------|-----------------|
| bar_col               | target_base | regime_id | regime_name   | total_rows    | valid_rows             |                  |                 |
| bar_p40_h30_tp15_sl10 | p40_h30     | 0         | Overnight     | 357162        | 227256                 |                  |                 |
| 63.63%                | 57.02%      | 7.34%     | 35.64%        | 64.36%        | 88.60%                 |                  |                 |
| 0.782                 |             | 14.83     |               |               |                        |                  |                 |
| bar_p40_h30_tp15_sl10 | p40_h30     | 1         | Pre-market    | 88920         | 56654                  |                  |                 |
| 63.71%                | 61.19%      | 10.11%    | 28.70%        | 71.30%        | 85.82%                 |                  |                 |
| 0.817                 |             | 16.17     |               |               |                        |                  |                 |
| bar_p40_h30_tp15_sl10 | p40_h30     | 2         | Opening       | 88920         | 56968                  |                  |                 |
| 64.07%                | 56.45%      | 5.81%     | 37.74%        | 62.26%        | 90.66%                 |                  |                 |
| 0.789                 |             | 14.75     |               |               |                        |                  |                 |
| bar_p40_h30_tp15_sl10 | p40_h30     | 3         | Regular       | 444600        | 281914                 |                  |                 |
| 63.41%                | 57.41%      | 7.12%     | 35.47%        | 64.53%        | 88.97%                 |                  |                 |
| 0.790                 |             | 14.54     |               |               |                        |                  |                 |
| bar_p40_h30_tp15_sl10 | p40_h30     | 4         | Closing       | 44460         | 944                    |                  |                 |
| 2.12%                 | 56.25%      | 6.67%     | 37.08%        | 62.92%        | 89.39%                 |                  |                 |
| 0.777                 |             | 17.17     |               |               |                        |                  |                 |
| bar_p40_h60_tp15_sl10 | p40_h60     | 0         | Overnight     | 357162        | 227928                 |                  |                 |
| 63.82%                | 56.64%      | 7.74%     | 35.62%        | 64.38%        | 87.98%                 |                  |                 |
| 0.772                 |             | 29.31     |               |               |                        |                  |                 |
| bar_p40_h60_tp15_sl10 | p40_h60     | 1         | Pre-market    | 88920         | 56996                  |                  |                 |
| 64.10%                | 56.79%      | 6.95%     | 36.25%        | 63.75%        | 89.09%                 |                  |                 |
| 0.782                 |             | 36.21     |               |               |                        |                  |                 |
| bar_p40_h60_tp15_sl10 | p40_h60     | 2         | Opening       | 88920         | 57128                  |                  |                 |
| 64.25%                | 56.26%      | 5.96%     | 37.78%        | 62.22%        | 90.43%                 |                  |                 |
| 0.784                 |             | 27.78     |               |               |                        |                  |                 |
| bar_p40_h60_tp15_sl10 | p40_h60     | 3         | Regular       | 444600        | 254972                 |                  |                 |
| 57.35%                | 55.89%      | 7.31%     | 36.80%        | 63.20%        | 88.43%                 |                  |                 |
| 0.765                 |             | 28.32     |               |               |                        |                  |                 |
| bar_p40_h60_tp15_sl10 | p40_h60     | 4         | Closing       | 44460         | 0                      |                  |                 |
| 0.00%                 | NaN         | NaN       | NaN           | NaN           | NaN                    |                  |                 |
| NaN                   |             | NaN       |               |               |                        |                  |                 |
| bar_p40_h90_tp15_sl10 | p40_h90     | 0         | Overnight     | 357162        | 227186                 |                  |                 |
| 63.61%                | 56.13%      | 9.05%     | 34.82%        | 65.18%        | 86.11%                 |                  |                 |
| 0.751                 |             | 46.46     |               |               |                        |                  |                 |
| bar_p40_h90_tp15_sl10 | p40_h90     | 1         | Pre-market    | 88920         | 57202                  |                  |                 |
| 64.33%                | 55.47%      | 6.57%     | 37.96%        | 62.04%        | 89.41%                 |                  |                 |
| 0.766                 |             | 52.67     |               |               |                        |                  |                 |
| bar_p40_h90_tp15_sl10 | p40_h90     | 2         | Opening       | 88920         | 57077                  |                  |                 |
| 64.19%                | 56.18%      | 6.21%     | 37.62%        | 62.38%        | 90.05%                 |                  |                 |
| 0.781                 |             | 39.61     |               |               |                        |                  |                 |
| bar_p40_h90_tp15_sl10 | p40_h90     | 3         | Regular       | 444600        | 226505                 |                  |                 |
| 50.95%                | 55.92%      | 7.15%     | 36.93%        | 63.07%        | 88.66%                 |                  |                 |
| 0.767                 |             | 42.35     |               |               |                        |                  |                 |
| bar_p40_h90_tp15_sl10 | p40_h90     | 4         | Closing       | 44460         | 0                      |                  |                 |

|                       |        |       |         |              |        |        |
|-----------------------|--------|-------|---------|--------------|--------|--------|
| 0.00%                 | NaN    | NaN   | NaN     | NaN          | NaN    | NaN    |
| NaN                   |        | NaN   |         |              |        |        |
| bar_p50_h30_tp15_sl10 |        |       | p50_h30 | 0 Overnight  | 357162 | 192938 |
| 54.02%                | 53.11% | 5.80% | 41.08%  | 58.92%       | 90.15% |        |
| 0.739                 |        | 15.68 |         |              |        |        |
| bar_p50_h30_tp15_sl10 |        |       | p50_h30 | 1 Pre-market | 88920  | 48109  |
| 54.10%                | 57.96% | 7.88% | 34.16%  | 65.84%       | 88.03% |        |
| 0.791                 |        | 17.19 |         |              |        |        |
| bar_p50_h30_tp15_sl10 |        |       | p50_h30 | 2 Opening    | 88920  | 48348  |
| 54.37%                | 52.04% | 4.37% | 43.59%  | 56.41%       | 92.25% |        |
| 0.737                 |        | 15.84 |         |              |        |        |
| bar_p50_h30_tp15_sl10 |        |       | p50_h30 | 3 Regular    | 444600 | 239801 |
| 53.94%                | 53.42% | 5.46% | 41.11%  | 58.89%       | 90.72% |        |
| 0.747                 |        | 15.44 |         |              |        |        |
| bar_p50_h30_tp15_sl10 |        |       | p50_h30 | 4 Closing    | 44460  | 786    |
| 1.77%                 | 53.82% | 4.71% | 41.48%  | 58.52%       | 91.96% |        |
| 0.760                 |        | 18.32 |         |              |        |        |
| bar_p50_h60_tp15_sl10 |        |       | p50_h60 | 0 Overnight  | 357162 | 192987 |
| 54.03%                | 52.81% | 6.02% | 41.17%  | 58.83%       | 89.77% |        |
| 0.732                 |        | 31.04 |         |              |        |        |
| bar_p50_h60_tp15_sl10 |        |       | p50_h60 | 1 Pre-market | 88920  | 48621  |
| 54.68%                | 51.78% | 5.22% | 43.00%  | 57.00%       | 90.84% |        |
| 0.725                 |        | 37.80 |         |              |        |        |
| bar_p50_h60_tp15_sl10 |        |       | p50_h60 | 2 Opening    | 88920  | 48390  |
| 54.42%                | 51.69% | 4.33% | 43.99%  | 56.01%       | 92.28% |        |
| 0.732                 |        | 30.08 |         |              |        |        |
| bar_p50_h60_tp15_sl10 |        |       | p50_h60 | 3 Regular    | 444600 | 215022 |
| 48.36%                | 51.95% | 5.45% | 42.60%  | 57.40%       | 90.50% |        |
| 0.725                 |        | 30.29 |         |              |        |        |
| bar_p50_h60_tp15_sl10 |        |       | p50_h60 | 4 Closing    | 44460  | 0      |
| 0.00%                 | NaN    | NaN   | NaN     | NaN          | NaN    |        |
| NaN                   |        | NaN   |         |              |        |        |
| bar_p50_h90_tp15_sl10 |        |       | p50_h90 | 0 Overnight  | 357162 | 193131 |
| 54.07%                | 53.39% | 7.25% | 39.36%  | 60.64%       | 88.05% |        |
| 0.728                 |        | 49.43 |         |              |        |        |
| bar_p50_h90_tp15_sl10 |        |       | p50_h90 | 1 Pre-market | 88920  | 48950  |
| 55.05%                | 50.27% | 4.42% | 45.31%  | 54.69%       | 91.92% |        |
| 0.710                 |        | 55.42 |         |              |        |        |
| bar_p50_h90_tp15_sl10 |        |       | p50_h90 | 2 Opening    | 88920  | 48555  |
| 54.61%                | 51.28% | 4.39% | 44.33%  | 55.67%       | 92.12% |        |
| 0.725                 |        | 43.09 |         |              |        |        |
| bar_p50_h90_tp15_sl10 |        |       | p50_h90 | 3 Regular    | 444600 | 191967 |
| 43.18%                | 52.19% | 5.38% | 42.43%  | 57.57%       | 90.66% |        |
| 0.729                 |        | 45.40 |         |              |        |        |
| bar_p50_h90_tp15_sl10 |        |       | p50_h90 | 4 Closing    | 44460  | 0      |
| 0.00%                 | NaN    | NaN   | NaN     | NaN          | NaN    |        |
| NaN                   |        | NaN   |         |              |        |        |
| bar_p60_h30_tp15_sl10 |        |       | p60_h30 | 0 Overnight  | 357162 | 160088 |
| 44.82%                | 49.41% | 4.70% | 45.89%  | 54.11%       | 91.32% |        |
| 0.694                 |        | 16.45 |         |              |        |        |
| bar_p60_h30_tp15_sl10 |        |       | p60_h30 | 1 Pre-market | 88920  | 39124  |
| 44.00%                | 53.23% | 5.74% | 41.03%  | 58.97%       | 90.27% |        |
| 0.741                 |        | 18.14 |         |              |        |        |
| bar_p60_h30_tp15_sl10 |        |       | p60_h30 | 2 Opening    | 88920  | 39638  |
| 44.58%                | 46.94% | 3.02% | 50.04%  | 49.96%       | 93.96% |        |
| 0.674                 |        | 16.88 |         |              |        |        |
| bar_p60_h30_tp15_sl10 |        |       | p60_h30 | 3 Regular    | 444600 | 195503 |
| 43.97%                | 49.40% | 4.10% | 46.50%  | 53.50%       | 92.34% |        |
| 0.700                 |        | 16.36 |         |              |        |        |
| bar_p60_h30_tp15_sl10 |        |       | p60_h30 | 4 Closing    | 44460  | 642    |

|                       |        |       |         |              |        |        |
|-----------------------|--------|-------|---------|--------------|--------|--------|
| 1.44%                 | 48.29% | 3.12% | 48.60%  | 51.40%       | 93.94% |        |
| 0.693                 |        | 19.25 |         |              |        |        |
| bar_p60_h60_tp15_sl10 |        |       | p60_h60 | 0 Overnight  | 357162 | 157509 |
| 44.10%                | 49.21% | 4.62% | 46.17%  | 53.83%       | 91.42% |        |
| 0.692                 |        | 32.61 |         |              |        |        |
| bar_p60_h60_tp15_sl10 |        |       | p60_h60 | 1 Pre-market | 88920  | 40102  |
| 45.10%                | 46.89% | 3.65% | 49.46%  | 50.54%       | 92.78% |        |
| 0.667                 |        | 39.28 |         |              |        |        |
| bar_p60_h60_tp15_sl10 |        |       | p60_h60 | 2 Opening    | 88920  | 39829  |
| 44.79%                | 46.36% | 2.97% | 50.67%  | 49.33%       | 93.98% |        |
| 0.666                 |        | 32.22 |         |              |        |        |
| bar_p60_h60_tp15_sl10 |        |       | p60_h60 | 3 Regular    | 444600 | 176358 |
| 39.67%                | 48.19% | 4.09% | 47.72%  | 52.28%       | 92.17% |        |
| 0.682                 |        | 32.06 |         |              |        |        |
| bar_p60_h60_tp15_sl10 |        |       | p60_h60 | 4 Closing    | 44460  | 0      |
| 0.00%                 | NaN    | NaN   | NaN     | NaN          | NaN    |        |
| NaN                   |        | NaN   |         |              |        |        |
| bar_p60_h90_tp15_sl10 |        |       | p60_h90 | 0 Overnight  | 357162 | 157163 |
| 44.00%                | 51.16% | 5.69% | 43.15%  | 56.85%       | 89.99% |        |
| 0.710                 |        | 52.44 |         |              |        |        |
| bar_p60_h90_tp15_sl10 |        |       | p60_h90 | 1 Pre-market | 88920  | 40395  |
| 45.43%                | 44.09% | 2.98% | 52.93%  | 47.07%       | 93.67% |        |
| 0.632                 |        | 57.69 |         |              |        |        |
| bar_p60_h90_tp15_sl10 |        |       | p60_h90 | 2 Opening    | 88920  | 39811  |
| 44.77%                | 46.23% | 3.05% | 50.71%  | 49.29%       | 93.81% |        |
| 0.663                 |        | 46.10 |         |              |        |        |
| bar_p60_h90_tp15_sl10 |        |       | p60_h90 | 3 Regular    | 444600 | 157132 |
| 35.34%                | 48.26% | 4.11% | 47.64%  | 52.36%       | 92.15% |        |
| 0.683                 |        | 48.06 |         |              |        |        |
| bar_p60_h90_tp15_sl10 |        |       | p60_h90 | 4 Closing    | 44460  | 0      |
| 0.00%                 | NaN    | NaN   | NaN     | NaN          | NaN    |        |
| NaN                   |        | NaN   |         |              |        |        |

Sensibilidad agregada al régimen por configuración TP/SL:

| tp_sl_config    | n_targets | operable_rate_range_mean | tp_rate_range_mean | sl_rate_range_mean | no_event_rate_range_mean | tp_share_range_mean | expectancy_range_mean | regime_sensitivity_score_mean | regime_sensitivity_score_max |
|-----------------|-----------|--------------------------|--------------------|--------------------|--------------------------|---------------------|-----------------------|-------------------------------|------------------------------|
| TP 1.0 / SL 1.0 | 9         |                          | 54.08%             | 2.38%              |                          |                     |                       |                               |                              |
| 2.38%           |           | 0.00%                    | 2.38%              | 0.048              |                          |                     |                       |                               |                              |
| 0.612           |           | 0.737                    |                    |                    |                          |                     |                       |                               |                              |
| TP 1.5 / SL 1.0 | 9         |                          | 54.08%             | 3.66%              |                          |                     |                       |                               |                              |
| 2.67%           |           | 6.20%                    | 3.57%              | 0.038              |                          |                     |                       |                               |                              |
| 0.702           |           | 0.851                    |                    |                    |                          |                     |                       |                               |                              |
| TP 2.0 / SL 1.0 | 9         |                          | 54.08%             | 6.92%              |                          |                     |                       |                               |                              |
| 3.03%           |           | 9.84%                    | 4.11%              | 0.109              |                          |                     |                       |                               |                              |
| 0.780           |           | 0.941                    |                    |                    |                          |                     |                       |                               |                              |
| TP 2.5 / SL 1.0 | 9         |                          | 54.08%             | 7.27%              |                          |                     |                       |                               |                              |
| 3.28%           |           | 10.48%                   | 4.27%              | 0.150              |                          |                     |                       |                               |                              |
| 0.794           |           | 0.942                    |                    |                    |                          |                     |                       |                               |                              |
| TP 1.0 / SL 0.5 | 9         |                          | 54.08%             | 4.54%              |                          |                     |                       |                               |                              |
| 4.54%           |           | 0.00%                    | 4.54%              | 0.068              |                          |                     |                       |                               |                              |
| 0.677           |           | 0.820                    |                    |                    |                          |                     |                       |                               |                              |
| TP 1.5 / SL 0.5 | 9         |                          | 54.08%             | 2.13%              |                          |                     |                       |                               |                              |
| 4.88%           |           | 5.64%                    | 4.57%              | 0.031              |                          |                     |                       |                               |                              |
| 0.713           |           | 0.856                    |                    |                    |                          |                     |                       |                               |                              |
| TP 2.0 / SL 0.5 | 9         |                          | 54.08%             | 4.40%              |                          |                     |                       |                               |                              |
| 5.24%           |           | 9.04%                    | 3.27%              | 0.071              |                          |                     |                       |                               |                              |
| 0.760           |           | 0.918                    |                    |                    |                          |                     |                       |                               |                              |

Sensibilidad agregada al régimen por percentil y horizonte:

| target_base | percentile | horizon | n_configs | operable_rate_range_mean | tp_rate_range_mean | sl_rate_range_mean | no_event_rate_range_mean | tp_share_range_mean | expectancy_range_mean | regime_sensitivity_score_mean | regime_sensitivity_score_max |
|-------------|------------|---------|-----------|--------------------------|--------------------|--------------------|--------------------------|---------------------|-----------------------|-------------------------------|------------------------------|
| p40_h30     | 40         | 30      | 7         |                          | 61.94%             |                    |                          |                     |                       |                               |                              |
| 6.00%       | 5.69%      |         |           | 8.29%                    | 4.72%              |                    |                          |                     |                       |                               |                              |
| 0.093       |            | 0.866   |           |                          | 0.942              |                    |                          |                     |                       |                               |                              |
| p40_h60     | 40         | 60      | 7         |                          | 64.25%             |                    |                          |                     |                       |                               |                              |
| 1.94%       | 2.38%      |         |           | 2.61%                    | 2.66%              |                    |                          |                     |                       |                               |                              |
| 0.035       |            | 0.738   |           |                          | 0.784              |                    |                          |                     |                       |                               |                              |
| p40_h90     | 40         | 90      | 7         |                          | 64.33%             |                    |                          |                     |                       |                               |                              |
| 3.69%       | 3.77%      |         |           | 4.38%                    | 3.91%              |                    |                          |                     |                       |                               |                              |
| 0.070       |            | 0.801   |           |                          | 0.893              |                    |                          |                     |                       |                               |                              |
| p50_h30     | 50         | 30      | 7         |                          | 52.60%             |                    |                          |                     |                       |                               |                              |
| 5.77%       | 5.10%      |         |           | 8.18%                    | 4.85%              |                    |                          |                     |                       |                               |                              |
| 0.086       |            | 0.765   |           |                          | 0.845              |                    |                          |                     |                       |                               |                              |
| p50_h60     | 50         | 60      | 7         |                          | 54.68%             |                    |                          |                     |                       |                               |                              |
| 2.58%       | 2.26%      |         |           | 3.45%                    | 2.72%              |                    |                          |                     |                       |                               |                              |
| 0.047       |            | 0.657   |           |                          | 0.734              |                    |                          |                     |                       |                               |                              |
| p50_h90     | 50         | 90      | 7         |                          | 55.05%             |                    |                          |                     |                       |                               |                              |
| 4.77%       | 3.93%      |         |           | 6.09%                    | 3.95%              |                    |                          |                     |                       |                               |                              |
| 0.082       |            | 0.738   |           |                          | 0.825              |                    |                          |                     |                       |                               |                              |
| p60_h30     | 60         | 30      | 7         |                          | 43.38%             |                    |                          |                     |                       |                               |                              |
| 5.48%       | 4.21%      |         |           | 7.48%                    | 4.73%              |                    |                          |                     |                       |                               |                              |
| 0.082       |            | 0.653   |           |                          | 0.726              |                    |                          |                     |                       |                               |                              |
| p60_h60     | 60         | 60      | 7         |                          | 45.10%             |                    |                          |                     |                       |                               |                              |
| 3.70%       | 2.34%      |         |           | 4.67%                    | 3.13%              |                    |                          |                     |                       |                               |                              |
| 0.065       |            | 0.589   |           |                          | 0.667              |                    |                          |                     |                       |                               |                              |
| p60_h90     | 60         | 90      | 7         |                          | 45.43%             |                    |                          |                     |                       |                               |                              |
| 6.32%       | 3.78%      |         |           | 7.82%                    | 3.66%              |                    |                          |                     |                       |                               |                              |
| 0.103       |            | 0.670   |           |                          | 0.758              |                    |                          |                     |                       |                               |                              |

Tabla final de sensibilidad al régimen – 63 targets BAR:

| bar_col               | target_base | tp_sl_config       | n_regimes | operable_rate_mean | operable_rate_range | tp_rate_mean | tp_rate_range | sl_rate_mean | sl_rate_range | no_event_rate_mean | no_event_rate_range | resolved_rate_mean | resolved_rate_range | tp_share_over_resolved_mean | tp_share_over_resolved_range | expectancy_units_mean | expectancy_units_range | event_step_mean_mean | event_step_mean_range | regime_sensitivity_score | regime_sensitivity_profile |
|-----------------------|-------------|--------------------|-----------|--------------------|---------------------|--------------|---------------|--------------|---------------|--------------------|---------------------|--------------------|---------------------|-----------------------------|------------------------------|-----------------------|------------------------|----------------------|-----------------------|--------------------------|----------------------------|
| bar_p60_h60_tp10_sl10 | p60_h60     | TP 1.0 / SL 1.0    | 5         |                    |                     |              |               |              |               |                    |                     |                    |                     |                             |                              |                       |                        |                      |                       |                          |                            |
| 45.10%                | 96.92%      | 1.45%              |           | 3.08%              | 1.45%               |              |               |              |               |                    |                     |                    |                     |                             |                              | 34.73%                |                        |                      |                       |                          |                            |
| 0.00%                 | 100.00%     |                    |           | 0.00%              |                     |              |               |              |               |                    |                     |                    |                     |                             |                              | 96.92%                |                        |                      |                       |                          |                            |
| 1.45%                 | 0.938       |                    |           | 0.029              |                     |              |               |              |               |                    |                     |                    |                     |                             |                              | 30.91                 |                        |                      |                       |                          |                            |
| 8.73                  | 0.494       | menor_sensibilidad |           |                    |                     |              |               |              |               |                    |                     |                    |                     |                             |                              |                       |                        |                      |                       |                          |                            |
| bar_p60_h30_tp10_sl10 | p60_h30     | TP 1.0 / SL 1.0    | 5         |                    |                     |              |               |              |               |                    |                     |                    |                     |                             |                              |                       |                        |                      |                       |                          |                            |
| 43.38%                | 96.59%      | 2.38%              |           | 3.41%              | 2.38%               |              |               |              |               |                    |                     |                    |                     |                             |                              | 35.76%                |                        |                      |                       |                          |                            |
| 0.00%                 | 100.00%     |                    |           | 0.00%              |                     |              |               |              |               |                    |                     |                    |                     |                             |                              | 96.59%                |                        |                      |                       |                          |                            |
| 2.38%                 | 0.932       |                    |           | 0.048              |                     |              |               |              |               |                    |                     |                    |                     |                             |                              | 15.82                 |                        |                      |                       |                          |                            |
| 2.45                  | 0.505       | menor_sensibilidad |           |                    |                     |              |               |              |               |                    |                     |                    |                     |                             |                              |                       |                        |                      |                       |                          |                            |
| bar_p60_h90_tp10_sl10 | p60_h90     | TP 1.0 / SL 1.0    | 5         |                    |                     |              |               |              |               |                    |                     |                    |                     |                             |                              |                       |                        |                      |                       |                          |                            |
| 45.43%                | 96.83%      | 2.33%              |           | 3.17%              | 2.33%               |              |               |              |               |                    |                     |                    |                     |                             |                              | 33.91%                |                        |                      |                       |                          |                            |
| 0.00%                 | 100.00%     |                    |           | 0.00%              |                     |              |               |              |               |                    |                     |                    |                     |                             |                              | 96.83%                |                        |                      |                       |                          |                            |
| 2.33%                 | 0.937       |                    |           | 0.047              |                     |              |               |              |               |                    |                     |                    |                     |                             |                              | 45.86                 |                        |                      |                       |                          |                            |
| 12.88                 | 0.524       | menor_sensibilidad |           |                    |                     |              |               |              |               |                    |                     |                    |                     |                             |                              |                       |                        |                      |                       |                          |                            |
| bar_p60_h60_tp10_sl05 | p60_h60     | TP 1.0 / SL 0.5    | 5         |                    |                     |              |               |              |               |                    |                     |                    |                     |                             |                              |                       |                        |                      |                       |                          |                            |
| 45.10%                | 84.66%      | 2.96%              |           | 15.34%             | 2.96%               |              |               |              |               |                    |                     |                    |                     |                             |                              | 34.73%                |                        |                      |                       |                          |                            |
| 0.00%                 | 100.00%     |                    |           | 0.00%              |                     |              |               |              |               |                    |                     |                    |                     |                             |                              | 84.66%                |                        |                      |                       |                          |                            |
| 2.96%                 | 0.770       |                    |           | 0.044              |                     |              |               |              |               |                    |                     |                    |                     |                             |                              | 27.60                 |                        |                      |                       |                          |                            |
| 8.83                  | 0.540       | menor_sensibilidad |           |                    |                     |              |               |              |               |                    |                     |                    |                     |                             |                              |                       |                        |                      |                       |                          |                            |
| bar_p60_h60_tp15_sl10 | p60_h60     | TP 1.5 / SL 1.0    | 5         |                    |                     |              |               |              |               |                    |                     |                    |                     |                             |                              |                       |                        |                      |                       |                          |                            |
| 45.10%                | 47.66%      | 2.85%              |           | 3.83%              | 1.65%               |              |               |              |               |                    |                     |                    |                     |                             |                              | 34.73%                |                        |                      |                       |                          |                            |
|                       |             |                    |           |                    |                     |              |               |              |               |                    |                     |                    |                     |                             |                              | 48.51%                |                        |                      |                       |                          |                            |

|                       |                         |                    |        |
|-----------------------|-------------------------|--------------------|--------|
| 4.50%                 | 51.49%                  | 4.50%              | 92.59% |
| 2.56%                 | 0.677                   | 0.026              | 34.04  |
| 7.22                  | 0.567                   | menor_sensibilidad |        |
| bar_p60_h60_tp15_sl05 | p60_h60 TP 1.5 / SL 0.5 | 5                  | 34.73% |
| 45.10%                | 41.13%                  | 1.44%              | 18.64% |
|                       |                         | 3.15%              | 40.23% |
| 4.11%                 | 59.77%                  | 4.11%              | 68.84% |
| 3.32%                 | 0.524                   | 0.020              | 29.13  |
| 7.59                  | 0.571                   | menor_sensibilidad |        |
| bar_p60_h30_tp10_sl05 | p60_h30 TP 1.0 / SL 0.5 | 5                  | 35.76% |
| 43.38%                | 84.95%                  | 5.19%              | 15.05% |
|                       |                         | 5.19%              | 0.00%  |
| 0.00%                 | 100.00%                 | 0.00%              | 84.95% |
| 5.19%                 | 0.774                   | 0.078              | 14.19  |
| 2.44                  | 0.589                   | menor_sensibilidad |        |
| bar_p60_h90_tp10_sl05 | p60_h90 TP 1.0 / SL 0.5 | 5                  | 33.91% |
| 45.43%                | 84.10%                  | 4.62%              | 15.90% |
|                       |                         | 4.62%              | 0.00%  |
| 0.00%                 | 100.00%                 | 0.00%              | 84.10% |
| 4.62%                 | 0.761                   | 0.069              | 40.62  |
| 13.36                 | 0.593                   | menor_sensibilidad |        |
| bar_p50_h60_tp10_sl10 | p50_h60 TP 1.0 / SL 1.0 | 5                  | 42.30% |
| 54.68%                | 95.74%                  | 1.55%              | 4.26%  |
|                       |                         | 1.55%              | 0.00%  |
| 0.00%                 | 100.00%                 | 0.00%              | 95.74% |
| 1.55%                 | 0.915                   | 0.031              | 29.09  |
| 9.34                  | 0.593                   | menor_sensibilidad |        |
| bar_p50_h30_tp10_sl10 | p50_h30 TP 1.0 / SL 1.0 | 5                  | 43.64% |
| 52.60%                | 95.35%                  | 3.06%              | 4.65%  |
|                       |                         | 3.06%              | 0.00%  |
| 0.00%                 | 100.00%                 | 0.00%              | 95.35% |
| 3.06%                 | 0.907                   | 0.061              | 14.91  |
| 2.81                  | 0.618                   | menor_sensibilidad |        |
| bar_p50_h60_tp15_sl10 | p50_h60 TP 1.5 / SL 1.0 | 5                  | 42.30% |
| 54.68%                | 52.06%                  | 1.12%              | 5.25%  |
|                       |                         | 1.70%              | 42.69% |
| 2.82%                 | 57.31%                  | 2.82%              | 90.85% |
| 2.51%                 | 0.728                   | 0.008              | 32.30  |
| 7.72                  | 0.628                   | menor_sensibilidad |        |
| bar_p50_h90_tp10_sl10 | p50_h90 TP 1.0 / SL 1.0 | 5                  | 41.38% |
| 55.05%                | 95.62%                  | 2.63%              | 4.38%  |
|                       |                         | 2.63%              | 0.00%  |
| 0.00%                 | 100.00%                 | 0.00%              | 95.62% |
| 2.63%                 | 0.912                   | 0.053              | 42.94  |
| 13.40                 | 0.629                   | menor_sensibilidad |        |
| bar_p50_h60_tp10_sl05 | p50_h60 TP 1.0 / SL 0.5 | 5                  | 42.30% |
| 54.68%                | 81.75%                  | 2.84%              | 18.25% |
|                       |                         | 2.84%              | 0.00%  |
| 0.00%                 | 100.00%                 | 0.00%              | 81.75% |
| 2.84%                 | 0.726                   | 0.043              | 25.38  |
| 9.41                  | 0.632                   | menor_sensibilidad |        |
| bar_p50_h60_tp15_sl05 | p50_h60 TP 1.5 / SL 0.5 | 5                  | 42.30% |
| 54.68%                | 44.06%                  | 0.65%              | 22.12% |
|                       |                         | 2.92%              | 33.81% |
| 2.61%                 | 66.19%                  | 2.61%              | 66.59% |
| 3.14%                 | 0.550                   | 0.022              | 27.35  |
| 8.05                  | 0.640                   | menor_sensibilidad |        |
| bar_p60_h60_tp20_sl10 | p60_h60 TP 2.0 / SL 1.0 | 5                  | 34.73% |
| 45.10%                | 24.08%                  | 6.22%              | 4.38%  |
|                       |                         | 1.85%              | 71.54% |
| 8.07%                 | 28.46%                  | 8.07%              | 84.67% |
| 2.84%                 | 0.438                   | 0.109              | 35.35  |
| 6.96                  | 0.641                   | menor_sensibilidad |        |
| bar_p60_h60_tp20_sl05 | p60_h60 TP 2.0 / SL 0.5 | 5                  | 34.73% |
| 45.10%                | 20.62%                  | 4.32%              | 19.78% |
|                       |                         | 3.25%              | 59.61% |
| 7.35%                 | 40.39%                  | 7.35%              | 51.00% |
| 4.58%                 | 0.313                   | 0.084              | 28.12  |
| 7.34                  | 0.646                   | menor_sensibilidad |        |
| bar_p60_h30_tp15_sl10 | p60_h30 TP 1.5 / SL 1.0 | 5                  | 35.76% |
| 43.38%                | 49.46%                  | 6.29%              | 4.13%  |
|                       |                         | 2.72%              | 46.41% |

|                       |                         |                         |        |
|-----------------------|-------------------------|-------------------------|--------|
| 9.00%                 | 53.59%                  | 9.00%                   | 92.37% |
| 3.68%                 | 0.701                   | 0.067                   | 17.42  |
| 2.89                  | 0.651                   | menor_sensibilidad      |        |
| bar_p60_h90_tp15_sl05 | p60_h90 TP 1.5 / SL 0.5 | 5                       | 33.91% |
| 45.43%                | 40.89%                  | 3.79%                   | 19.28% |
| 7.68%                 | 60.17%                  | 7.68%                   | 4.93%  |
| 4.43%                 | 0.517                   | 0.037                   | 68.02% |
| 12.46                 | 0.663                   | menor_sensibilidad      | 43.35  |
| bar_p60_h60_tp25_sl10 | p60_h60 TP 2.5 / SL 1.0 | 5                       | 34.73% |
| 45.10%                | 12.77%                  | 6.63%                   | 4.58%  |
| 8.69%                 | 17.36%                  | 8.69%                   | 2.06%  |
| 4.20%                 | 0.274                   | 0.145                   | 82.64% |
| 6.73                  | 0.667                   | menor_sensibilidad      | 73.43% |
| bar_p60_h30_tp15_sl05 | p60_h30 TP 1.5 / SL 0.5 | 5                       | 35.76% |
| 43.38%                | 42.96%                  | 3.49%                   | 18.43% |
| 9.47%                 | 61.39%                  | 9.47%                   | 5.97%  |
| 5.02%                 | 0.552                   | 0.023                   | 38.61% |
| 2.83                  | 0.673                   | menor_sensibilidad      | 70.06% |
| bar_p50_h60_tp20_sl05 | p50_h60 TP 2.0 / SL 0.5 | 5                       | 15.13  |
| 54.68%                | 24.03%                  | 2.35%                   | 23.56% |
| 5.37%                 | 47.59%                  | 5.37%                   | 3.02%  |
| 2.89%                 | 0.363                   | 0.045                   | 50.51% |
| 7.58                  | 0.683                   | menor_sensibilidad      | 26.98  |
| bar_p60_h90_tp15_sl10 | p60_h90 TP 1.5 / SL 1.0 | 5                       | 33.91% |
| 45.43%                | 47.43%                  | 7.07%                   | 3.96%  |
| 9.78%                 | 51.39%                  | 9.78%                   | 2.71%  |
| 3.82%                 | 0.672                   | 0.079                   | 48.61% |
| 11.59                 | 0.688                   | sensibilidad_intermedia | 92.40% |
| bar_p50_h60_tp20_sl10 | p50_h60 TP 2.0 / SL 1.0 | 5                       | 51.07  |
| 54.68%                | 28.63%                  | 4.08%                   | 5.98%  |
| 5.90%                 | 34.62%                  | 5.90%                   | 1.81%  |
| 2.38%                 | 0.513                   | 0.063                   | 42.30% |
| 7.13                  | 0.688                   | sensibilidad_intermedia | 65.38% |
| bar_p40_h60_tp10_sl10 | p40_h60 TP 1.0 / SL 1.0 | 5                       | 82.76% |
| 64.25%                | 94.32%                  | 1.59%                   | 5.98%  |
| 0.00%                 | 100.00%                 | 0.00%                   | 1.81%  |
| 1.59%                 | 0.886                   | 0.032                   | 82.76% |
| 9.99                  | 0.690                   | sensibilidad_intermedia | 33.85  |
| bar_p50_h90_tp10_sl05 | p50_h90 TP 1.0 / SL 0.5 | 5                       | 49.90% |
| 55.05%                | 81.03%                  | 4.73%                   | 18.97% |
| 0.00%                 | 100.00%                 | 0.00%                   | 4.73%  |
| 4.73%                 | 0.715                   | 0.071                   | 0.00%  |
| 13.43                 | 0.692                   | sensibilidad_intermedia | 81.03% |
| bar_p60_h30_tp20_sl05 | p60_h30 TP 2.0 / SL 0.5 | 5                       | 37.09  |
| 43.38%                | 22.84%                  | 6.07%                   | 19.56% |
| 12.50%                | 42.40%                  | 12.50%                  | 6.43%  |
| 1.81%                 | 0.359                   | 0.089                   | 53.91% |
| 2.25                  | 0.702                   | sensibilidad_intermedia | 14.76  |
| bar_p50_h90_tp15_sl05 | p50_h90 TP 1.5 / SL 0.5 | 5                       | 41.38% |
| 55.05%                | 43.53%                  | 0.98%                   | 22.89% |
| 4.61%                 | 66.42%                  | 4.61%                   | 5.06%  |
| 5.18%                 | 0.539                   | 0.032                   | 33.58% |
| 12.65                 | 0.709                   | sensibilidad_intermedia | 65.58% |
| bar_p50_h90_tp15_sl10 | p50_h90 TP 1.5 / SL 1.0 | 5                       | 40.47  |
| 55.05%                | 51.78%                  | 3.11%                   | 5.36%  |
| 5.95%                 | 57.14%                  | 5.95%                   | 2.86%  |
| 4.07%                 | 0.723                   | 0.019                   | 90.69% |
| 12.33                 | 0.710                   | sensibilidad_intermedia | 48.34  |
| bar_p50_h30_tp10_sl05 | p50_h30 TP 1.0 / SL 0.5 | 5                       | 43.64% |
| 52.60%                | 82.00%                  | 6.28%                   | 18.00% |
|                       |                         | 6.28%                   | 0.00%  |

|                       |                         |                         |        |
|-----------------------|-------------------------|-------------------------|--------|
| 0.00%                 | 100.00%                 | 0.00%                   | 82.00% |
| 6.28%                 | 0.730                   | 0.094                   | 13.06  |
| 2.63                  | 0.714                   | sensibilidad_intermedia |        |
| bar_p40_h60_tp15_sl10 | p40_h60 TP 1.5 / SL 1.0 | 5                       | 49.90% |
| 64.25%                | 56.40%                  | 0.90%                   | 6.99%  |
| 2.16%                 | 63.39%                  | 2.16%                   | 88.98% |
| 2.45%                 | 0.776                   | 0.019                   | 30.40  |
| 8.43                  | 0.715                   | sensibilidad_intermedia |        |
| bar_p40_h90_tp10_sl10 | p40_h90 TP 1.0 / SL 1.0 | 5                       | 48.61% |
| 64.33%                | 94.05%                  | 2.54%                   | 5.95%  |
| 0.00%                 | 100.00%                 | 0.00%                   | 94.05% |
| 2.54%                 | 0.881                   | 0.051                   | 39.85  |
| 14.02                 | 0.720                   | sensibilidad_intermedia |        |
| bar_p60_h30_tp20_sl10 | p60_h30 TP 2.0 / SL 1.0 | 5                       | 35.76% |
| 43.38%                | 26.52%                  | 8.17%                   | 4.65%  |
| 11.22%                | 31.17%                  | 11.22%                  | 85.29% |
| 6.17%                 | 0.484                   | 0.133                   | 18.06  |
| 2.57                  | 0.723                   | sensibilidad_intermedia |        |
| bar_p60_h30_tp25_sl10 | p60_h30 TP 2.5 / SL 1.0 | 5                       | 35.76% |
| 43.38%                | 14.77%                  | 6.77%                   | 4.94%  |
| 10.13%                | 19.71%                  | 10.13%                  | 75.17% |
| 8.83%                 | 0.320                   | 0.136                   | 18.11  |
| 2.49                  | 0.726                   | sensibilidad_intermedia |        |
| bar_p60_h90_tp20_sl05 | p60_h90 TP 2.0 / SL 0.5 | 5                       | 33.91% |
| 45.43%                | 20.36%                  | 6.96%                   | 20.42% |
| 11.36%                | 40.78%                  | 11.36%                  | 49.84% |
| 3.40%                 | 0.305                   | 0.117                   | 41.90  |
| 11.52                 | 0.726                   | sensibilidad_intermedia |        |
| bar_p40_h60_tp10_sl05 | p40_h60 TP 1.0 / SL 0.5 | 5                       | 49.90% |
| 64.25%                | 78.33%                  | 3.02%                   | 21.67% |
| 0.00%                 | 100.00%                 | 0.00%                   | 78.33% |
| 3.02%                 | 0.675                   | 0.045                   | 22.97  |
| 9.94                  | 0.733                   | sensibilidad_intermedia |        |
| bar_p50_h60_tp25_sl10 | p50_h60 TP 2.5 / SL 1.0 | 5                       | 42.30% |
| 54.68%                | 16.13%                  | 5.50%                   | 6.36%  |
| 7.48%                 | 22.49%                  | 7.48%                   | 71.64% |
| 3.75%                 | 0.340                   | 0.118                   | 34.08  |
| 7.02                  | 0.734                   | sensibilidad_intermedia |        |
| bar_p40_h30_tp10_sl10 | p40_h30 TP 1.0 / SL 1.0 | 5                       | 51.39% |
| 61.94%                | 93.78%                  | 3.91%                   | 6.22%  |
| 0.00%                 | 100.00%                 | 0.00%                   | 93.78% |
| 3.91%                 | 0.876                   | 0.078                   | 13.83  |
| 2.71                  | 0.737                   | sensibilidad_intermedia |        |
| bar_p60_h90_tp25_sl10 | p60_h90 TP 2.5 / SL 1.0 | 5                       | 33.91% |
| 45.43%                | 12.43%                  | 9.24%                   | 4.72%  |
| 12.62%                | 17.14%                  | 12.62%                  | 72.41% |
| 3.32%                 | 0.264                   | 0.197                   | 53.56  |
| 9.54                  | 0.740                   | sensibilidad_intermedia |        |
| bar_p40_h60_tp15_sl05 | p40_h60 TP 1.5 / SL 0.5 | 5                       | 49.90% |
| 64.25%                | 46.73%                  | 1.18%                   | 26.02% |
| 2.17%                 | 72.75%                  | 2.17%                   | 64.25% |
| 3.32%                 | 0.571                   | 0.032                   | 25.40  |
| 8.42                  | 0.740                   | sensibilidad_intermedia |        |
| bar_p40_h90_tp15_sl10 | p40_h90 TP 1.5 / SL 1.0 | 5                       | 48.61% |
| 64.33%                | 55.92%                  | 0.71%                   | 7.25%  |
| 3.14%                 | 63.17%                  | 3.14%                   | 88.56% |
| 3.94%                 | 0.766                   | 0.029                   | 45.27  |
| 13.07                 | 0.750                   | sensibilidad_intermedia |        |
| bar_p40_h60_tp20_sl05 | p40_h60 TP 2.0 / SL 0.5 | 5                       | 49.90% |
| 64.25%                | 27.66%                  | 0.81%                   | 27.92% |



|                       |                         |                         |        |
|-----------------------|-------------------------|-------------------------|--------|
| 3.99%                 | 55.58%                  | 3.99%                   | 49.80% |
| 2.91%                 | 0.414                   | 0.018                   | 25.65  |
| 7.74                  | 0.753                   | sensibilidad_intermedia |        |
| bar_p40_h60_tp20_sl10 | p40_h60 TP 2.0 / SL 1.0 | 5                       | 49.90% |
| 64.25%                | 33.67%                  | 2.32%                   | 58.34% |
| 4.21%                 | 41.66%                  | 4.21%                   | 80.86% |
| 2.64%                 | 0.594                   | 0.028                   | 32.21  |
| 7.67                  | 0.753                   | mayor_sensibilidad      |        |
| bar_p50_h30_tp15_sl10 | p50_h30 TP 1.5 / SL 1.0 | 5                       | 43.64% |
| 52.60%                | 54.07%                  | 5.92%                   | 40.28% |
| 9.43%                 | 59.72%                  | 9.43%                   | 90.62% |
| 4.22%                 | 0.755                   | 0.054                   | 16.49  |
| 2.88                  | 0.757                   | mayor_sensibilidad      |        |
| bar_p60_h90_tp20_sl10 | p60_h90 TP 2.0 / SL 1.0 | 5                       | 33.91% |
| 45.43%                | 23.79%                  | 10.23%                  | 71.71% |
| 13.30%                | 28.29%                  | 13.30%                  | 84.35% |
| 3.73%                 | 0.431                   | 0.174                   | 53.26  |
| 9.63                  | 0.758                   | mayor_sensibilidad      |        |
| bar_p50_h30_tp15_sl05 | p50_h30 TP 1.5 / SL 0.5 | 5                       | 43.64% |
| 52.60%                | 45.94%                  | 2.58%                   | 32.38% |
| 9.49%                 | 67.62%                  | 9.49%                   | 68.02% |
| 5.63%                 | 0.581                   | 0.017                   | 14.09  |
| 2.52                  | 0.772                   | mayor_sensibilidad      |        |
| bar_p40_h90_tp10_sl05 | p40_h90 TP 1.0 / SL 0.5 | 5                       | 48.61% |
| 64.33%                | 77.92%                  | 4.57%                   | 0.00%  |
| 0.00%                 | 100.00%                 | 0.00%                   | 77.92% |
| 4.57%                 | 0.669                   | 0.069                   | 33.57  |
| 14.13                 | 0.780                   | mayor_sensibilidad      |        |
| bar_p40_h60_tp25_sl10 | p40_h60 TP 2.5 / SL 1.0 | 5                       | 49.90% |
| 64.25%                | 20.26%                  | 3.77%                   | 71.22% |
| 5.71%                 | 28.78%                  | 5.71%                   | 70.43% |
| 2.72%                 | 0.421                   | 0.075                   | 32.68  |
| 7.39                  | 0.784                   | mayor_sensibilidad      |        |
| bar_p50_h90_tp20_sl05 | p50_h90 TP 2.0 / SL 0.5 | 5                       | 41.38% |
| 55.05%                | 23.56%                  | 5.11%                   | 52.08% |
| 8.62%                 | 47.92%                  | 8.62%                   | 49.17% |
| 4.30%                 | 0.349                   | 0.085                   | 39.92  |
| 11.56                 | 0.785                   | mayor_sensibilidad      |        |
| bar_p40_h90_tp15_sl05 | p40_h90 TP 1.5 / SL 0.5 | 5                       | 48.61% |
| 64.33%                | 45.89%                  | 2.20%                   | 27.62% |
| 2.61%                 | 72.38%                  | 2.61%                   | 63.42% |
| 5.29%                 | 0.556                   | 0.057                   | 37.43  |
| 13.11                 | 0.792                   | mayor_sensibilidad      |        |
| bar_p50_h30_tp20_sl05 | p50_h30 TP 2.0 / SL 0.5 | 5                       | 43.64% |
| 52.60%                | 26.47%                  | 5.44%                   | 50.39% |
| 12.95%                | 49.61%                  | 12.95%                  | 53.43% |
| 2.83%                 | 0.414                   | 0.071                   | 14.10  |
| 2.49                  | 0.813                   | mayor_sensibilidad      |        |
| bar_p50_h90_tp25_sl10 | p50_h90 TP 2.5 / SL 1.0 | 5                       | 41.38% |
| 55.05%                | 15.87%                  | 8.51%                   | 77.68% |
| 11.98%                | 22.32%                  | 11.98%                  | 71.13% |
| 2.30%                 | 0.332                   | 0.178                   | 51.34  |
| 10.70                 | 0.814                   | mayor_sensibilidad      |        |
| bar_p40_h90_tp20_sl05 | p40_h90 TP 2.0 / SL 0.5 | 5                       | 48.61% |
| 64.33%                | 26.95%                  | 2.75%                   | 44.71% |
| 6.24%                 | 55.29%                  | 6.24%                   | 48.78% |
| 3.34%                 | 0.397                   | 0.039                   | 37.84  |
| 12.41                 | 0.817                   | mayor_sensibilidad      |        |
| bar_p40_h30_tp10_sl05 | p40_h30 TP 1.0 / SL 0.5 | 5                       | 51.39% |
| 61.94%                | 78.95%                  | 6.67%                   | 0.00%  |

|                       |                         |                    |        |       |        |
|-----------------------|-------------------------|--------------------|--------|-------|--------|
| 0.00%                 | 100.00%                 | 0.00%              | 78.95% |       |        |
| 6.67%                 | 0.684                   | 0.100              | 11.81  |       |        |
| 2.49                  | 0.820                   | mayor_sensibilidad |        |       |        |
| bar_p50_h90_tp20_sl10 | p50_h90 TP 2.0 / SL 1.0 | 5                  | 41.38% |       |        |
| 55.05%                | 28.17%                  | 8.28%              | 6.11%  | 3.28% | 65.72% |
| 11.46%                | 34.28%                  | 11.46%             | 82.36% |       |        |
| 4.43%                 | 0.502                   | 0.134              | 50.76  |       |        |
| 11.07                 | 0.825                   | mayor_sensibilidad |        |       |        |
| bar_p50_h30_tp20_sl10 | p50_h30 TP 2.0 / SL 1.0 | 5                  | 43.64% |       |        |
| 52.60%                | 31.34%                  | 8.58%              | 6.36%  | 4.04% | 62.31% |
| 12.53%                | 37.69%                  | 12.53%             | 83.33% |       |        |
| 5.78%                 | 0.563                   | 0.132              | 17.30  |       |        |
| 2.90                  | 0.835                   | mayor_sensibilidad |        |       |        |
| bar_p50_h30_tp25_sl10 | p50_h30 TP 2.5 / SL 1.0 | 5                  | 43.64% |       |        |
| 52.60%                | 18.60%                  | 8.52%              | 6.80%  | 4.37% | 74.60% |
| 12.85%                | 25.40%                  | 12.85%             | 73.47% |       |        |
| 6.18%                 | 0.397                   | 0.170              | 17.44  |       |        |
| 2.46                  | 0.845                   | mayor_sensibilidad |        |       |        |
| bar_p40_h30_tp15_sl10 | p40_h30 TP 1.5 / SL 1.0 | 5                  | 51.39% |       |        |
| 61.94%                | 57.66%                  | 4.94%              | 7.41%  | 4.30% | 34.93% |
| 9.05%                 | 65.07%                  | 9.05%              | 88.69% |       |        |
| 4.85%                 | 0.791                   | 0.040              | 15.49  |       |        |
| 2.63                  | 0.851                   | mayor_sensibilidad |        |       |        |
| bar_p40_h90_tp20_sl10 | p40_h90 TP 2.0 / SL 1.0 | 5                  | 48.61% |       |        |
| 64.33%                | 33.06%                  | 5.54%              | 8.26%  | 3.13% | 58.68% |
| 8.19%                 | 41.32%                  | 8.19%              | 80.10% |       |        |
| 4.19%                 | 0.579                   | 0.084              | 48.17  |       |        |
| 12.21                 | 0.854                   | mayor_sensibilidad |        |       |        |
| bar_p40_h30_tp15_sl05 | p40_h30 TP 1.5 / SL 0.5 | 5                  | 51.39% |       |        |
| 61.94%                | 48.14%                  | 2.86%              | 25.03% | 7.04% | 26.83% |
| 8.02%                 | 73.17%                  | 8.02%              | 65.85% |       |        |
| 5.78%                 | 0.597                   | 0.040              | 13.06  |       |        |
| 2.22                  | 0.856                   | mayor_sensibilidad |        |       |        |
| bar_p40_h90_tp25_sl10 | p40_h90 TP 2.5 / SL 1.0 | 5                  | 48.61% |       |        |
| 64.33%                | 19.69%                  | 7.54%              | 8.79%  | 3.44% | 71.52% |
| 10.52%                | 28.48%                  | 10.52%             | 69.13% |       |        |
| 3.47%                 | 0.404                   | 0.159              | 48.89  |       |        |
| 11.15                 | 0.893                   | mayor_sensibilidad |        |       |        |
| bar_p40_h30_tp20_sl05 | p40_h30 TP 2.0 / SL 0.5 | 5                  | 51.39% |       |        |
| 61.94%                | 29.58%                  | 5.83%              | 27.03% | 7.72% | 43.39% |
| 12.96%                | 56.61%                  | 12.96%             | 52.30% |       |        |
| 3.36%                 | 0.456                   | 0.092              | 13.26  |       |        |
| 1.96                  | 0.918                   | mayor_sensibilidad |        |       |        |
| bar_p40_h30_tp20_sl10 | p40_h30 TP 2.0 / SL 1.0 | 5                  | 51.39% |       |        |
| 61.94%                | 35.74%                  | 8.82%              | 8.44%  | 4.86% | 55.82% |
| 13.68%                | 44.18%                  | 13.68%             | 81.07% |       |        |
| 4.81%                 | 0.631                   | 0.128              | 16.31  |       |        |
| 2.42                  | 0.941                   | mayor_sensibilidad |        |       |        |
| bar_p40_h30_tp25_sl10 | p40_h30 TP 2.5 / SL 1.0 | 5                  | 51.39% |       |        |
| 61.94%                | 22.69%                  | 8.98%              | 9.01%  | 5.34% | 68.30% |
| 14.32%                | 31.70%                  | 14.32%             | 71.78% |       |        |
| 3.66%                 | 0.477                   | 0.171              | 16.62  |       |        |
| 2.25                  | 0.942                   | mayor_sensibilidad |        |       |        |

Top 15 targets con menor sensibilidad al régimen:

bar\_col target\_base tp\_sl\_config n\_regimes operable\_rate\_mean o  
perable\_rate\_range tp\_rate\_mean tp\_rate\_range sl\_rate\_mean sl\_rate\_range no\_event  
\_rate\_mean no\_event\_rate\_range resolved\_rate\_mean resolved\_rate\_range tp\_share\_ov  
er\_resolved\_mean tp\_share\_over\_resolved\_range expectancy\_units\_mean expectancy\_un  
its\_range event\_step\_mean\_mean event\_step\_mean\_range regime\_sensitivity\_score reg

## ime\_sensitivity\_profile

|                       |         |                    |        |        |        |
|-----------------------|---------|--------------------|--------|--------|--------|
| bar_p60_h60_tp10_sl10 | p60_h60 | TP 1.0 / SL 1.0    | 5      | 34.73% |        |
| 45.10%                | 96.92%  | 1.45%              | 3.08%  | 1.45%  | 0.00%  |
| 0.00%                 | 100.00% | 0.00%              |        | 96.92% |        |
| 1.45%                 | 0.938   | 0.029              |        | 30.91  |        |
| 8.73                  | 0.494   | menor_sensibilidad |        |        |        |
| bar_p60_h30_tp10_sl10 | p60_h30 | TP 1.0 / SL 1.0    | 5      | 35.76% |        |
| 43.38%                | 96.59%  | 2.38%              | 3.41%  | 2.38%  | 0.00%  |
| 0.00%                 | 100.00% | 0.00%              |        | 96.59% |        |
| 2.38%                 | 0.932   | 0.048              |        | 15.82  |        |
| 2.45                  | 0.505   | menor_sensibilidad |        |        |        |
| bar_p60_h90_tp10_sl10 | p60_h90 | TP 1.0 / SL 1.0    | 5      | 33.91% |        |
| 45.43%                | 96.83%  | 2.33%              | 3.17%  | 2.33%  | 0.00%  |
| 0.00%                 | 100.00% | 0.00%              |        | 96.83% |        |
| 2.33%                 | 0.937   | 0.047              |        | 45.86  |        |
| 12.88                 | 0.524   | menor_sensibilidad |        |        |        |
| bar_p60_h60_tp10_sl05 | p60_h60 | TP 1.0 / SL 0.5    | 5      | 34.73% |        |
| 45.10%                | 84.66%  | 2.96%              | 15.34% | 2.96%  | 0.00%  |
| 0.00%                 | 100.00% | 0.00%              |        | 84.66% |        |
| 2.96%                 | 0.770   | 0.044              |        | 27.60  |        |
| 8.83                  | 0.540   | menor_sensibilidad |        |        |        |
| bar_p60_h60_tp15_sl10 | p60_h60 | TP 1.5 / SL 1.0    | 5      | 34.73% |        |
| 45.10%                | 47.66%  | 2.85%              | 3.83%  | 1.65%  | 48.51% |
| 4.50%                 | 51.49%  | 4.50%              |        | 92.59% |        |
| 2.56%                 | 0.677   | 0.026              |        | 34.04  |        |
| 7.22                  | 0.567   | menor_sensibilidad |        |        |        |
| bar_p60_h60_tp15_sl05 | p60_h60 | TP 1.5 / SL 0.5    | 5      | 34.73% |        |
| 45.10%                | 41.13%  | 1.44%              | 18.64% | 3.15%  | 40.23% |
| 4.11%                 | 59.77%  | 4.11%              |        | 68.84% |        |
| 3.32%                 | 0.524   | 0.020              |        | 29.13  |        |
| 7.59                  | 0.571   | menor_sensibilidad |        |        |        |
| bar_p60_h30_tp10_sl05 | p60_h30 | TP 1.0 / SL 0.5    | 5      | 35.76% |        |
| 43.38%                | 84.95%  | 5.19%              | 15.05% | 5.19%  | 0.00%  |
| 0.00%                 | 100.00% | 0.00%              |        | 84.95% |        |
| 5.19%                 | 0.774   | 0.078              |        | 14.19  |        |
| 2.44                  | 0.589   | menor_sensibilidad |        |        |        |
| bar_p60_h90_tp10_sl05 | p60_h90 | TP 1.0 / SL 0.5    | 5      | 33.91% |        |
| 45.43%                | 84.10%  | 4.62%              | 15.90% | 4.62%  | 0.00%  |
| 0.00%                 | 100.00% | 0.00%              |        | 84.10% |        |
| 4.62%                 | 0.761   | 0.069              |        | 40.62  |        |
| 13.36                 | 0.593   | menor_sensibilidad |        |        |        |
| bar_p50_h60_tp10_sl10 | p50_h60 | TP 1.0 / SL 1.0    | 5      | 42.30% |        |
| 54.68%                | 95.74%  | 1.55%              | 4.26%  | 1.55%  | 0.00%  |
| 0.00%                 | 100.00% | 0.00%              |        | 95.74% |        |
| 1.55%                 | 0.915   | 0.031              |        | 29.09  |        |
| 9.34                  | 0.593   | menor_sensibilidad |        |        |        |
| bar_p50_h30_tp10_sl10 | p50_h30 | TP 1.0 / SL 1.0    | 5      | 43.64% |        |
| 52.60%                | 95.35%  | 3.06%              | 4.65%  | 3.06%  | 0.00%  |
| 0.00%                 | 100.00% | 0.00%              |        | 95.35% |        |
| 3.06%                 | 0.907   | 0.061              |        | 14.91  |        |
| 2.81                  | 0.618   | menor_sensibilidad |        |        |        |
| bar_p50_h60_tp15_sl10 | p50_h60 | TP 1.5 / SL 1.0    | 5      | 42.30% |        |
| 54.68%                | 52.06%  | 1.12%              | 5.25%  | 1.70%  | 42.69% |
| 2.82%                 | 57.31%  | 2.82%              |        | 90.85% |        |
| 2.51%                 | 0.728   | 0.008              |        | 32.30  |        |
| 7.72                  | 0.628   | menor_sensibilidad |        |        |        |
| bar_p50_h90_tp10_sl10 | p50_h90 | TP 1.0 / SL 1.0    | 5      | 41.38% |        |
| 55.05%                | 95.62%  | 2.63%              | 4.38%  | 2.63%  | 0.00%  |
| 0.00%                 | 100.00% | 0.00%              |        | 95.62% |        |
| 2.63%                 | 0.912   | 0.053              |        | 42.94  |        |

|                       |         |                         |                    |       |        |
|-----------------------|---------|-------------------------|--------------------|-------|--------|
| 13.40                 |         | 0.629                   | menor_sensibilidad |       |        |
| bar_p50_h60_tp10_sl05 |         | p50_h60 TP 1.0 / SL 0.5 |                    | 5     | 42.30% |
| 54.68%                | 81.75%  | 2.84%                   | 18.25%             | 2.84% | 0.00%  |
| 0.00%                 | 100.00% |                         | 0.00%              |       | 81.75% |
| 2.84%                 |         | 0.726                   |                    | 0.043 | 25.38  |
| 9.41                  |         | 0.632                   | menor_sensibilidad |       |        |
| bar_p50_h60_tp15_sl05 |         | p50_h60 TP 1.5 / SL 0.5 |                    | 5     | 42.30% |
| 54.68%                | 44.06%  | 0.65%                   | 22.12%             | 2.92% | 33.81% |
| 2.61%                 | 66.19%  |                         | 2.61%              |       | 66.59% |
| 3.14%                 |         | 0.550                   |                    | 0.022 | 27.35  |
| 8.05                  |         | 0.640                   | menor_sensibilidad |       |        |
| bar_p60_h60_tp20_sl10 |         | p60_h60 TP 2.0 / SL 1.0 |                    | 5     | 34.73% |
| 45.10%                | 24.08%  | 6.22%                   | 4.38%              | 1.85% | 71.54% |
| 8.07%                 | 28.46%  |                         | 8.07%              |       | 84.67% |
| 2.84%                 |         | 0.438                   |                    | 0.109 | 35.35  |
| 6.96                  |         | 0.641                   | menor_sensibilidad |       |        |

Top 15 targets con mayor sensibilidad al régimen:

bar\_col target\_base tp\_sl\_config n\_regimes operable\_rate\_mean o  
perable\_rate\_range tp\_rate\_mean tp\_rate\_range sl\_rate\_mean sl\_rate\_range no\_event  
\_rate\_mean no\_event\_rate\_range resolved\_rate\_mean resolved\_rate\_range tp\_share\_ov  
er\_resolved\_mean tp\_share\_over\_resolved\_range expectancy\_units\_mean expectancy\_un  
its\_range event\_step\_mean\_mean event\_step\_mean\_range regime\_sensitivity\_score reg  
ime\_sensitivity\_profile

|                       |        |                         |                    |       |        |
|-----------------------|--------|-------------------------|--------------------|-------|--------|
| bar_p40_h30_tp25_sl10 |        | p40_h30 TP 2.5 / SL 1.0 |                    | 5     | 51.39% |
| 61.94%                | 22.69% | 8.98%                   | 9.01%              | 5.34% | 68.30% |
| 14.32%                | 31.70% |                         | 14.32%             |       | 71.78% |
| 3.66%                 |        | 0.477                   |                    | 0.171 | 16.62  |
| 2.25                  |        | 0.942                   | mayor_sensibilidad |       |        |
| bar_p40_h30_tp20_sl10 |        | p40_h30 TP 2.0 / SL 1.0 |                    | 5     | 51.39% |
| 61.94%                | 35.74% | 8.82%                   | 8.44%              | 4.86% | 55.82% |
| 13.68%                | 44.18% |                         | 13.68%             |       | 81.07% |
| 4.81%                 |        | 0.631                   |                    | 0.128 | 16.31  |
| 2.42                  |        | 0.941                   | mayor_sensibilidad |       |        |
| bar_p40_h30_tp20_sl05 |        | p40_h30 TP 2.0 / SL 0.5 |                    | 5     | 51.39% |
| 61.94%                | 29.58% | 5.83%                   | 27.03%             | 7.72% | 43.39% |
| 12.96%                | 56.61% |                         | 12.96%             |       | 52.30% |
| 3.36%                 |        | 0.456                   |                    | 0.092 | 13.26  |
| 1.96                  |        | 0.918                   | mayor_sensibilidad |       |        |
| bar_p40_h90_tp25_sl10 |        | p40_h90 TP 2.5 / SL 1.0 |                    | 5     | 48.61% |
| 64.33%                | 19.69% | 7.54%                   | 8.79%              | 3.44% | 71.52% |
| 10.52%                | 28.48% |                         | 10.52%             |       | 69.13% |
| 3.47%                 |        | 0.404                   |                    | 0.159 | 48.89  |
| 11.15                 |        | 0.893                   | mayor_sensibilidad |       |        |
| bar_p40_h30_tp15_sl05 |        | p40_h30 TP 1.5 / SL 0.5 |                    | 5     | 51.39% |
| 61.94%                | 48.14% | 2.86%                   | 25.03%             | 7.04% | 26.83% |
| 8.02%                 | 73.17% |                         | 8.02%              |       | 65.85% |
| 5.78%                 |        | 0.597                   |                    | 0.040 | 13.06  |
| 2.22                  |        | 0.856                   | mayor_sensibilidad |       |        |
| bar_p40_h90_tp20_sl10 |        | p40_h90 TP 2.0 / SL 1.0 |                    | 5     | 48.61% |
| 64.33%                | 33.06% | 5.54%                   | 8.26%              | 3.13% | 58.68% |
| 8.19%                 | 41.32% |                         | 8.19%              |       | 80.10% |
| 4.19%                 |        | 0.579                   |                    | 0.084 | 48.17  |
| 12.21                 |        | 0.854                   | mayor_sensibilidad |       |        |
| bar_p40_h30_tp15_sl10 |        | p40_h30 TP 1.5 / SL 1.0 |                    | 5     | 51.39% |
| 61.94%                | 57.66% | 4.94%                   | 7.41%              | 4.30% | 34.93% |
| 9.05%                 | 65.07% |                         | 9.05%              |       | 88.69% |
| 4.85%                 |        | 0.791                   |                    | 0.040 | 15.49  |
| 2.63                  |        | 0.851                   | mayor_sensibilidad |       |        |
| bar_p50_h30_tp25_sl10 |        | p50_h30 TP 2.5 / SL 1.0 |                    | 5     | 43.64% |

|                       |                         |       |                    |       |        |
|-----------------------|-------------------------|-------|--------------------|-------|--------|
| 52.60%                | 18.60%                  | 8.52% | 6.80%              | 4.37% | 74.60% |
| 12.85%                | 25.40%                  |       | 12.85%             |       | 73.47% |
| 6.18%                 | 0.397                   |       | 0.170              |       | 17.44  |
| 2.46                  | 0.845                   |       | mayor_sensibilidad |       |        |
| bar_p50_h30_tp20_sl10 | p50_h30 TP 2.0 / SL 1.0 |       | 5                  |       | 43.64% |
| 52.60%                | 31.34%                  | 8.58% | 6.36%              | 4.04% | 62.31% |
| 12.53%                | 37.69%                  |       | 12.53%             |       | 83.33% |
| 5.78%                 | 0.563                   |       | 0.132              |       | 17.30  |
| 2.90                  | 0.835                   |       | mayor_sensibilidad |       |        |
| bar_p50_h90_tp20_sl10 | p50_h90 TP 2.0 / SL 1.0 |       | 5                  |       | 41.38% |
| 55.05%                | 28.17%                  | 8.28% | 6.11%              | 3.28% | 65.72% |
| 11.46%                | 34.28%                  |       | 11.46%             |       | 82.36% |
| 4.43%                 | 0.502                   |       | 0.134              |       | 50.76  |
| 11.07                 | 0.825                   |       | mayor_sensibilidad |       |        |
| bar_p40_h30_tp10_sl05 | p40_h30 TP 1.0 / SL 0.5 |       | 5                  |       | 51.39% |
| 61.94%                | 78.95%                  | 6.67% | 21.05%             | 6.67% | 0.00%  |
| 0.00%                 | 100.00%                 |       | 0.00%              |       | 78.95% |
| 6.67%                 | 0.684                   |       | 0.100              |       | 11.81  |
| 2.49                  | 0.820                   |       | mayor_sensibilidad |       |        |
| bar_p40_h90_tp20_sl05 | p40_h90 TP 2.0 / SL 0.5 |       | 5                  |       | 48.61% |
| 64.33%                | 26.95%                  | 2.75% | 28.34%             | 5.05% | 44.71% |
| 6.24%                 | 55.29%                  |       | 6.24%              |       | 48.78% |
| 3.34%                 | 0.397                   |       | 0.039              |       | 37.84  |
| 12.41                 | 0.817                   |       | mayor_sensibilidad |       |        |
| bar_p50_h90_tp25_sl10 | p50_h90 TP 2.5 / SL 1.0 |       | 5                  |       | 41.38% |
| 55.05%                | 15.87%                  | 8.51% | 6.45%              | 3.53% | 77.68% |
| 11.98%                | 22.32%                  |       | 11.98%             |       | 71.13% |
| 2.30%                 | 0.332                   |       | 0.178              |       | 51.34  |
| 10.70                 | 0.814                   |       | mayor_sensibilidad |       |        |
| bar_p50_h30_tp20_sl05 | p50_h30 TP 2.0 / SL 0.5 |       | 5                  |       | 43.64% |
| 52.60%                | 26.47%                  | 5.44% | 23.14%             | 7.51% | 50.39% |
| 12.95%                | 49.61%                  |       | 12.95%             |       | 53.43% |
| 2.83%                 | 0.414                   |       | 0.071              |       | 14.10  |
| 2.49                  | 0.813                   |       | mayor_sensibilidad |       |        |
| bar_p40_h90_tp15_sl05 | p40_h90 TP 1.5 / SL 0.5 |       | 5                  |       | 48.61% |
| 64.33%                | 45.89%                  | 2.20% | 26.49%             | 4.81% | 27.62% |
| 2.61%                 | 72.38%                  |       | 2.61%              |       | 63.42% |
| 5.29%                 | 0.556                   |       | 0.057              |       | 37.43  |
| 13.11                 | 0.792                   |       | mayor_sensibilidad |       |        |

Conteo por perfil de sensibilidad al régimen:

regime\_sensitivity\_profile n\_targets regime\_sensitivity\_score\_mean operable\_rate  
\_range\_mean tp\_rate\_range\_mean sl\_rate\_range\_mean no\_event\_rate\_range\_mean tp\_sha  
re\_range\_mean expectancy\_range\_mean event\_step\_range\_mean

|                         |       |       |
|-------------------------|-------|-------|
| menor_sensibilidad      | 21    | 0.607 |
| 47.93%                  | 3.25% | 2.96% |
| 3.25%                   | 0.052 | 7.67  |
| sensibilidad_intermedia | 20    | 0.719 |
| 55.27%                  | 4.18% | 3.48% |
| 3.87%                   | 0.073 | 8.47  |
| mayor_sensibilidad      | 22    | 0.828 |
| 58.88%                  | 5.91% | 4.66% |
| 4.31%                   | 0.095 | 6.74  |

Archivos guardados:

Análisis BAR por régimen: c:\Users\heguu\OneDrive\Escritorio\neural\_  
profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_regime\_analysis.csv  
Resumen general por régimen: c:\Users\heguu\OneDrive\Escritorio\neural\_  
profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_regime\_by\_regime.csv  
Resumen por régimen y configuración: c:\Users\heguu\OneDrive\Escritorio\neural\_

profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_regime\_by\_config.csv  
 Resumen por régimen y target base: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_regime\_by\_target\_base.csv  
 Candidata TP 1.5 / SL 1.0 por régimen: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_regime\_candidate\_tp15\_sl10.csv  
 Sensibilidad por target: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_regime\_stability\_by\_target.csv  
 Sensibilidad por configuración: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_regime\_sensitivity\_by\_config.csv  
 Sensibilidad por percentil/horizonte: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_regime\_sensitivity\_by\_target\_base.csv  
 Tabla final por régimen: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_regime\_final\_table.csv  
 Reporte TXT: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_regime\_report.txt

Celda 4 finalizada correctamente.

## Observaciones preliminares — Análisis por régimen intradiario BAR

El análisis por régimen intradiario muestra que los 63 targets **BAR** fueron evaluados correctamente sobre los cinco regímenes definidos: **Overnight**, **Pre-market**, **Opening**, **Regular** y **Closing**. En total se generaron 315 combinaciones, correspondientes a los 63 targets por los 5 regímenes.

En términos generales, las métricas promedio son bastante consistentes entre **Overnight**, **Pre-market**, **Opening** y **Regular**. La tasa promedio de TP se mantiene cerca del 49%–50%, mientras que la tasa promedio de SL se ubica entre 11% y 14%. Esto indica que el comportamiento general de **BAR** no cambia de forma extrema entre los principales regímenes intradiarios.

El régimen **Opening** muestra una característica interesante: tiene la menor tasa promedio de SL, cercana al 10.90%, pero también el mayor **no\_event**, cercano al 40.43%. Esto sugiere que durante la apertura los targets tienden a tener menos pérdidas por SL, pero también más casos que no alcanzan ninguna barrera dentro del horizonte definido.

**Overnight** y **Regular** muestran un comportamiento más equilibrado. Ambos mantienen tasas de TP cercanas al 50%, tasas de SL moderadas y proporciones de eventos resueltos alrededor del 62%–64%. Esto los convierte en regímenes importantes para el análisis posterior, especialmente porque concentran una cantidad considerable de observaciones.

El **Pre-market** presenta una tasa de TP similar al resto de regímenes, pero un tiempo promedio hasta evento más alto. Esto sugiere que las operaciones en este régimen tienden a necesitar más barras para resolverse, especialmente en horizontes largos.

El caso de **Closing** debe interpretarse con mucha cautela. Aunque sus métricas promedio parecen razonables, la tasa de filas operables es extremadamente baja, cercana al 0.59%. Además, para horizontes **H60** y **H90** prácticamente no existen

ventanas futuras válidas sin cruzar el final del día. Por este motivo, **Closing** no debería compararse directamente con los demás regímenes y debería tratarse como un caso especial, principalmente limitado a **H30**.

Por configuración TP/SL, el patrón observado en los puntos anteriores se mantiene dentro de los regímenes. **TP 1.0 / SL 1.0** sigue siendo la configuración más estable y con mayor tasa de TP, pero debe interpretarse como benchmark porque está favorecida por la construcción del target **DIR**.

La configuración **TP 1.5 / SL 1.0** mantiene un perfil razonable en los principales regímenes. En promedio, su tasa de TP se mantiene alrededor del 51%–53%, con tasas de SL relativamente bajas, entre 4% y 7%, y una proporción de TP sobre eventos resueltos cercana al 89%–92%. Esto confirma que esta configuración sigue siendo una candidata operativa consistente.

Al revisar específicamente **TP 1.5 / SL 1.0**, se observa que las variantes **p50** mantienen un comportamiento bastante estable entre **Overnight**, **Pre-market**, **Opening** y **Regular**. Por ejemplo, en **p50\_h30**, la tasa de TP se ubica aproximadamente entre 52% y 58%, mientras que la tasa de SL permanece entre 4% y 8%. Esto refuerza la idea de que **p50** sigue siendo un punto intermedio razonable.

Las configuraciones más exigentes, como **TP 2.0 / SL 1.0** y **TP 2.5 / SL 1.0**, muestran mayor dependencia del régimen. Esto se observa en los scores de sensibilidad: **TP 2.0 / SL 1.0** tiene un **regime\_sensitivity\_score\_mean** de 0.780 y **TP 2.5 / SL 1.0** alcanza 0.794. Esto confirma que objetivos más lejanos son más sensibles al contexto intradiario.

Las configuraciones con **SL = 0.5** también muestran mayor sensibilidad. Aunque algunas resuelven rápido, tienen mayor exposición a movimientos adversos pequeños, por lo que pueden ser más sensibles al ruido específico de cada régimen.

El análisis por percentil y horizonte muestra que **p60\_h60** y **p50\_h60** son las combinaciones menos sensibles al régimen, con **regime\_sensitivity\_score\_mean** de 0.589 y 0.657 respectivamente. En cambio, **p40\_h30** aparece como la combinación más sensible, con un score promedio de 0.866. Esto sugiere que targets más activos y de horizonte corto pueden depender más del régimen intradiario.

Una observación metodológica importante es que el **regime\_sensitivity\_score** incluye la variación de **operable\_rate**. Por eso, parte de la sensibilidad está influenciada por la baja operabilidad del régimen **Closing**. Para conclusiones operativas finales, conviene analizar también las métricas excluyendo **Closing** o tratándolo por separado.

En síntesis, el análisis por régimen confirma que **BAR** mantiene una estructura razonable entre los principales regímenes intradiarios. Las diferencias existen, pero no cambian por completo la interpretación general obtenida en los puntos anteriores.

De forma preliminar:

Overnight → comportamiento equilibrado y representativo.  
 Pre-market → similar, pero con mayor tiempo promedio hasta evento.  
 Opening → menor SL, pero mayor no\_event.  
 Regular → régimen importante y relativamente estable.  
 Closing → tratar por separado; válido principalmente para H30.

En términos de configuración:

TP 1.0 / SL 1.0 → más estable, pero benchmark.  
 TP 1.5 / SL 1.0 → candidata operativa principal.  
 TP 2.0 / SL 1.0 → más exigente y más sensible al régimen.  
 TP 2.5 / SL 1.0 → demasiado dependiente del contexto.  
 SL 0.5 → más sensible al ruido intradiario.

La conclusión operativa se mantiene: TP 1.5 / SL 1.0 sigue siendo una de las configuraciones más razonables para continuar el análisis. No es la más estable en términos absolutos, pero ofrece un buen equilibrio entre tasa de TP, baja tasa de SL, proporción favorable de TP sobre eventos resueltos y comportamiento aceptable entre regímenes.

## 4.11. Análisis por contrato y contexto

En este punto se analiza el comportamiento de los targets **BAR** según contrato y contexto de mercado.

El objetivo es revisar si las métricas operativas cambian entre contratos individuales, familias de contrato, años y combinaciones con régimen intradiario.

Este análisis es importante porque, aunque un target pueda parecer razonable a nivel global, podría depender demasiado de un contrato específico o de un período histórico particular.

Se evaluarán los 63 targets **BAR**, considerando todos los percentiles, horizontes y configuraciones TP/SL.

La estructura sugerida es:

Celda 1 – Preparación de métricas BAR por contrato y contexto  
 Celda 2 – Tablas agregadas por contrato, familia de contrato y contexto  
 Celda 3 – Gráficos agrupados por contrato y contexto  
 Celda 4 – Guardado y tabla final para observaciones

### Celda 1 — Preparación de métricas BAR por contrato y contexto

Esta celda deja creado el dataframe principal:

`df_bar_context_analysis`

y también estas tablas auxiliares:

`df_bar_context_contract`  
`df_bar_context_contract_code`  
`df_bar_context_year`



```

df_bar_context_dataset_split
df_bar_context_year_contract
df_bar_context_contract_regime
df_bar_context_year_regime
df_bar_context_contract_code_regime

```

In [115...

```

# Celda 1 – Preparación de métricas BAR por contrato y contexto
# =====

print("4.11. Análisis por contrato y contexto - BAR")
print("=" * 80)

# -----
# Validar objetos necesarios
# -----

if "df_targets_bar" not in globals():
    raise NameError("No existe df_targets_bar. Primero debes construir los targe

if "df_bar_metadata" not in globals():
    raise NameError("No existe df_bar_metadata. Primero debes construir df_bar_m

# -----
# Copia de trabajo
# -----

df_bar_context_base = df_targets_bar.copy()

# -----
# Validar columnas necesarias
# -----

required_cols = ["contract", "regime_id"]

missing_cols = [
    col for col in required_cols
    if col not in df_bar_context_base.columns
]

if missing_cols:
    raise KeyError(
        "Faltan columnas necesarias para el análisis por contrato/contexto: "
        f"{missing_cols}"
    )

# -----
# Crear columnas temporales auxiliares
# -----

if isinstance(df_bar_context_base.index, pd.DatetimeIndex):
    date_series_tmp = pd.Series(
        df_bar_context_base.index,
        index=df_bar_context_base.index
    )

elif "date" in df_bar_context_base.columns:
    date_series_tmp = pd.to_datetime(df_bar_context_base["date"])

else:

```

```

        raise KeyError(
            "No se pudo crear la columna year. "
            "Se requiere DatetimeIndex o columna date."
        )

df_bar_context_base["year"] = date_series_tmp.dt.year

df_bar_context_base["dataset_split"] = np.where(
    df_bar_context_base["year"].between(2020, 2024),
    "development_2020_2024",
    np.where(
        df_bar_context_base["year"].between(2025, 2026),
        "final_test_2025_2026",
        "other"
    )
)

# -----
# Crear familia de contrato
# -----
# contract:      H20, M20, U20, Z20, ...
# contract_code: H, M, U, Z
# -----

df_bar_context_base["contract"] = (
    df_bar_context_base["contract"]
    .astype(str)
)

df_bar_context_base["contract_code"] = (
    df_bar_context_base["contract"]
    .str[0]
)

# -----
# Nombres de régimen
# -----

REGIME_NAME_MAP = {
    0: "Overnight",
    1: "Pre-market",
    2: "Opening",
    3: "Regular",
    4: "Closing",
}

df_bar_context_base["regime_name"] = (
    df_bar_context_base["regime_id"]
    .map(REGIME_NAME_MAP)
    .fillna("Unknown")
)

# -----
# Crear contextos combinados
# -----

df_bar_context_base["year_contract"] = (
    df_bar_context_base["year"].astype(str)
    + "_"
    + df_bar_context_base["contract"].astype(str)
)

```

```

)

df_bar_context_base["contract_regime"] = (
    df_bar_context_base["contract"].astype(str)
    + "_"
    + df_bar_context_base["regime_name"].astype(str)
)

df_bar_context_base["year_regime"] = (
    df_bar_context_base["year"].astype(str)
    + "_"
    + df_bar_context_base["regime_name"].astype(str)
)

df_bar_context_base["contract_code_regime"] = (
    df_bar_context_base["contract_code"].astype(str)
    + "_"
    + df_bar_context_base["regime_name"].astype(str)
)

# -----
# Función auxiliar para resumir un target BAR por contexto
# -----

def summarize_bar_target_for_context(df, bar_col, event_step_col, tp_mult, sl_mu
    """
    Resume un target BAR dentro de un contexto determinado.

    BAR:
        1 = TP primero
        -1 = SL primero
        0 = no_event
        NaN = no operable / sin ventana válida
    """

    bar_series = df[bar_col]
    step_series = df[event_step_col]

    valid_mask = bar_series.notna()
    tp_mask = bar_series == 1
    sl_mask = bar_series == -1
    no_event_mask = bar_series == 0
    resolved_mask = tp_mask | sl_mask

    total_rows = len(df)
    valid_rows = int(valid_mask.sum())
    nan_rows = int(bar_series.isna().sum())

    tp_count = int(tp_mask.sum())
    sl_count = int(sl_mask.sum())
    no_event_count = int(no_event_mask.sum())
    resolved_count = tp_count + sl_count

    operable_rate = valid_rows / total_rows if total_rows > 0 else np.nan

    if valid_rows > 0:
        tp_rate = tp_count / valid_rows
        sl_rate = sl_count / valid_rows
        no_event_rate = no_event_count / valid_rows
        resolved_rate = resolved_count / valid_rows

```

```

else:
    tp_rate = np.nan
    sl_rate = np.nan
    no_event_rate = np.nan
    resolved_rate = np.nan

if resolved_count > 0:
    tp_share_over_resolved = tp_count / resolved_count
    sl_share_over_resolved = sl_count / resolved_count
else:
    tp_share_over_resolved = np.nan
    sl_share_over_resolved = np.nan

expectancy_units = (
    tp_rate * tp_mult
    - sl_rate * sl_mult
    if valid_rows > 0
    else np.nan
)

event_steps = step_series.loc[resolved_mask].dropna()

if len(event_steps) > 0:
    event_step_mean = event_steps.mean()
    event_step_median = event_steps.median()
else:
    event_step_mean = np.nan
    event_step_median = np.nan

return {
    "total_rows": total_rows,
    "valid_rows": valid_rows,
    "nan_rows": nan_rows,

    "operable_rate": operable_rate,

    "tp_count": tp_count,
    "sl_count": sl_count,
    "no_event_count": no_event_count,
    "resolved_count": resolved_count,

    "tp_rate": tp_rate,
    "sl_rate": sl_rate,
    "no_event_rate": no_event_rate,
    "resolved_rate": resolved_rate,

    "tp_share_over_resolved": tp_share_over_resolved,
    "sl_share_over_resolved": sl_share_over_resolved,

    "expectancy_units": expectancy_units,

    "event_step_mean": event_step_mean,
    "event_step_median": event_step_median,
}

```

```

# -----
# Definición de contextos a analizar
# -----

```

```
context_definitions = [
```

```

{
    "context_type": "contract",
    "context_col": "contract",
    "description": "Contrato individual"
},
{
    "context_type": "contract_code",
    "context_col": "contract_code",
    "description": "Familia de contrato H/M/U/Z"
},
{
    "context_type": "year",
    "context_col": "year",
    "description": "Año calendario"
},
{
    "context_type": "dataset_split",
    "context_col": "dataset_split",
    "description": "Development vs final test"
},
{
    "context_type": "year_contract",
    "context_col": "year_contract",
    "description": "Año + contrato"
},
{
    "context_type": "contract_regime",
    "context_col": "contract_regime",
    "description": "Contrato + régimen intradiario"
},
{
    "context_type": "year_regime",
    "context_col": "year_regime",
    "description": "Año + régimen intradiario"
},
{
    "context_type": "contract_code_regime",
    "context_col": "contract_code_regime",
    "description": "Familia de contrato + régimen intradiario"
},
],

# -----
# Construir métricas por contrato/contexto para Los 63 targets BAR
# -----

context_rows = []

for _, meta in df_bar_metadata.iterrows():

    bar_col = meta["bar_col"]
    event_step_col = meta["event_step_col"]

    horizon = int(meta["horizon"])
    percentile = int(meta["percentile"])
    tp_mult = float(meta["tp_mult"])
    sl_mult = float(meta["sl_mult"])

    tp_sl_config = f"TP {tp_mult} / SL {sl_mult}"
    target_base = f"p{percentile}_h{horizon}"

```

```

if bar_col not in df_bar_context_base.columns:
    raise KeyError(f"No existe la columna BAR: {bar_col}")

if event_step_col not in df_bar_context_base.columns:
    raise KeyError(f"No existe la columna event_step: {event_step_col}")

for context_info in context_definitions:

    context_type = context_info["context_type"]
    context_col = context_info["context_col"]
    context_description = context_info["description"]

    context_values = (
        df_bar_context_base[context_col]
        .dropna()
        .unique()
    )

    for context_value in sorted(context_values):

        if context_value == "other":
            continue

        context_mask = df_bar_context_base[context_col] == context_value
        df_context = df_bar_context_base.loc[context_mask]

        summary = summarize_bar_target_for_context(
            df=df_context,
            bar_col=bar_col,
            event_step_col=event_step_col,
            tp_mult=tp_mult,
            sl_mult=sl_mult,
        )

        context_rows.append(
            {
                "bar_col": bar_col,
                "event_step_col": event_step_col,
                "dir_col": meta["dir_col"],

                "target_base": target_base,
                "horizon": horizon,
                "percentile": percentile,
                "tp_mult": tp_mult,
                "sl_mult": sl_mult,
                "tp_sl_config": tp_sl_config,

                "context_type": context_type,
                "context_col": context_col,
                "context_description": context_description,
                "context_value": str(context_value),

                **summary,
            }
        )

df_bar_context_analysis = pd.DataFrame(context_rows)

df_bar_context_analysis = (

```

```
df_bar_context_analysis
.sort_values(
    [
        "context_type",
        "context_value",
        "percentile",
        "horizon",
        "sl_mult",
        "tp_mult",
    ]
)
.reset_index(drop=True)
)

# -----
# Crear tablas auxiliares separadas por contexto
# -----

df_bar_context_contract = (
    df_bar_context_analysis[
        df_bar_context_analysis["context_type"] == "contract"
    ]
    .copy()
    .reset_index(drop=True)
)

df_bar_context_contract_code = (
    df_bar_context_analysis[
        df_bar_context_analysis["context_type"] == "contract_code"
    ]
    .copy()
    .reset_index(drop=True)
)

df_bar_context_year = (
    df_bar_context_analysis[
        df_bar_context_analysis["context_type"] == "year"
    ]
    .copy()
    .reset_index(drop=True)
)

df_bar_context_dataset_split = (
    df_bar_context_analysis[
        df_bar_context_analysis["context_type"] == "dataset_split"
    ]
    .copy()
    .reset_index(drop=True)
)

df_bar_context_year_contract = (
    df_bar_context_analysis[
        df_bar_context_analysis["context_type"] == "year_contract"
    ]
    .copy()
    .reset_index(drop=True)
)

df_bar_context_contract_regime = (
    df_bar_context_analysis[
```

```

        df_bar_context_analysis["context_type"] == "contract_regime"
    ]
    .copy()
    .reset_index(drop=True)
)

df_bar_context_year_regime = (
    df_bar_context_analysis[
        df_bar_context_analysis["context_type"] == "year_regime"
    ]
    .copy()
    .reset_index(drop=True)
)

df_bar_context_contract_code_regime = (
    df_bar_context_analysis[
        df_bar_context_analysis["context_type"] == "contract_code_regime"
    ]
    .copy()
    .reset_index(drop=True)
)

# -----
# Vista inicial
# -----

print(f"Targets BAR analizados: {df_bar_metadata['bar_col'].unique()}")
print(f"Filas generadas: {len(df_bar_context_analysis)}")

print("\nContextos analizados:")
print(
    df_bar_context_analysis[
        ["context_type", "context_description"]
    ]
    .drop_duplicates()
    .sort_values("context_type")
    .to_string(index=False)
)

print("\nCantidad de valores por contexto:")
print(
    df_bar_context_analysis
    .groupby("context_type", as_index=False)
    .agg(
        n_context_values=("context_value", "nunique"),
        n_rows=("bar_col", "count")
    )
    .sort_values("context_type")
    .to_string(index=False)
)

print("\nContratos detectados:")
print(
    df_bar_context_base[["contract", "contract_code"]]
    .drop_duplicates()
    .sort_values("contract")
    .to_string(index=False)
)

print("\nVista inicial de BAR por contrato/contexto:")

```



```
print(
    df_bar_context_analysis[
        [
            "bar_col",
            "context_type",
            "context_value",
            "horizon",
            "percentile",
            "tp_mult",
            "sl_mult",
            "total_rows",
            "valid_rows",
            "operable_rate",
            "tp_rate",
            "sl_rate",
            "no_event_rate",
            "resolved_rate",
            "tp_share_over_resolved",
            "expectancy_units",
            "event_step_mean",
        ]
    ].head(40)
    .to_string(
        index=False,
        formatters={
            "operable_rate": "{:.2%}".format,
            "tp_rate": "{:.2%}".format,
            "sl_rate": "{:.2%}".format,
            "no_event_rate": "{:.2%}".format,
            "resolved_rate": "{:.2%}".format,
            "tp_share_over_resolved": "{:.2%}".format,
            "expectancy_units": "{:.3f}".format,
            "event_step_mean": "{:.2f}".format,
        }
    )
)

print("\nPreparación de métricas BAR por contrato y contexto finalizada correcta")
```

## 4.11. Análisis por contrato y contexto - BAR

=====

Targets BAR analizados: 63

Filas generadas: 16128

Contextos analizados:

|                      |                                           |
|----------------------|-------------------------------------------|
| context_type         | context_description                       |
| contract             | Contrato individual                       |
| contract_code        | Familia de contrato H/M/U/Z               |
| contract_code_regime | Familia de contrato + régimen intradiario |
| contract_regime      | Contrato + régimen intradiario            |
| dataset_split        | Development vs final test                 |
| year                 | Año calendario                            |
| year_contract        | Año + contrato                            |
| year_regime          | Año + régimen intradiario                 |

Cantidad de valores por contexto:

| context_type         | n_context_values | n_rows |
|----------------------|------------------|--------|
| contract             | 26               | 1638   |
| contract_code        | 4                | 252    |
| contract_code_regime | 20               | 1260   |
| contract_regime      | 130              | 8190   |
| dataset_split        | 2                | 126    |
| year                 | 7                | 441    |
| year_contract        | 32               | 2016   |
| year_regime          | 35               | 2205   |

Contratos detectados:

| contract | contract_code |
|----------|---------------|
| H20      | H             |
| H21      | H             |
| H22      | H             |
| H23      | H             |
| H24      | H             |
| H25      | H             |
| H26      | H             |
| M20      | M             |
| M21      | M             |
| M22      | M             |
| M23      | M             |
| M24      | M             |
| M25      | M             |
| M26      | M             |
| U20      | U             |
| U21      | U             |
| U22      | U             |
| U23      | U             |
| U24      | U             |
| U25      | U             |
| Z20      | Z             |
| Z21      | Z             |
| Z22      | Z             |
| Z23      | Z             |
| Z24      | Z             |
| Z25      | Z             |

Vista inicial de BAR por contrato/contexto:

| bar_col       | context_type           | context_value    | horizon         | percentile | tp_mult | s             |
|---------------|------------------------|------------------|-----------------|------------|---------|---------------|
| l_mult        | total_rows             | valid_rows       | operable_rate   | tp_rate    | sl_rate | no_event_rate |
| resolved_rate | tp_share_over_resolved | expectancy_units | event_step_mean |            |         |               |

|                       |          |        |        |        |        |
|-----------------------|----------|--------|--------|--------|--------|
| bar_p40_h30_tp10_sl05 | contract | H20    | 30     | 40     | 1.0    |
| 0.5                   | 33168    | 11830  | 35.67% | 76.42% | 23.58% |
| 0.00%                 |          | 76.42% | 0.646  | 10.57  | 10     |
| bar_p40_h30_tp15_sl05 | contract | H20    | 30     | 40     | 1.5    |
| 0.5                   | 33168    | 11830  | 35.67% | 46.13% | 27.28% |
| 3.41%                 |          | 62.84% | 0.556  | 10.79  | 7      |
| bar_p40_h30_tp20_sl05 | contract | H20    | 30     | 40     | 2.0    |
| 0.5                   | 33168    | 11830  | 35.67% | 31.29% | 29.61% |
| 0.90%                 |          | 51.38% | 0.478  | 11.08  | 6      |
| bar_p40_h30_tp10_sl10 | contract | H20    | 30     | 40     | 1.0    |
| 1.0                   | 33168    | 11830  | 35.67% | 91.20% | 8.80%  |
| 0.00%                 |          | 91.20% | 0.824  | 12.37  | 10     |
| bar_p40_h30_tp15_sl10 | contract | H20    | 30     | 40     | 1.5    |
| 1.0                   | 33168    | 11830  | 35.67% | 57.36% | 10.19% |
| 7.56%                 |          | 84.91% | 0.758  | 13.13  | 6      |
| bar_p40_h30_tp20_sl10 | contract | H20    | 30     | 40     | 2.0    |
| 1.0                   | 33168    | 11830  | 35.67% | 39.44% | 11.37% |
| 0.81%                 |          | 77.62% | 0.675  | 13.90  | 5      |
| bar_p40_h30_tp25_sl10 | contract | H20    | 30     | 40     | 2.5    |
| 1.0                   | 33168    | 11830  | 35.67% | 27.62% | 12.57% |
| 0.19%                 |          | 68.72% | 0.565  | 14.75  | 4      |
| bar_p40_h60_tp10_sl05 | contract | H20    | 60     | 40     | 1.0    |
| 0.5                   | 33168    | 10961  | 33.05% | 74.88% | 25.12% |
| 0.00%                 |          | 74.88% | 0.623  | 20.03  | 10     |
| bar_p40_h60_tp15_sl05 | contract | H20    | 60     | 40     | 1.5    |
| 0.5                   | 33168    | 10961  | 33.05% | 43.23% | 29.35% |
| 2.58%                 |          | 59.56% | 0.502  | 19.86  | 7      |
| bar_p40_h60_tp20_sl05 | contract | H20    | 60     | 40     | 2.0    |
| 0.5                   | 33168    | 10961  | 33.05% | 28.88% | 31.71% |
| 0.59%                 |          | 47.66% | 0.419  | 20.77  | 6      |
| bar_p40_h60_tp10_sl10 | contract | H20    | 60     | 40     | 1.0    |
| 1.0                   | 33168    | 10961  | 33.05% | 91.45% | 8.55%  |
| 0.00%                 |          | 91.45% | 0.829  | 23.99  | 10     |
| bar_p40_h60_tp15_sl10 | contract | H20    | 60     | 40     | 1.5    |
| 1.0                   | 33168    | 10961  | 33.05% | 55.43% | 9.86%  |
| 5.30%                 |          | 84.90% | 0.733  | 25.04  | 6      |
| bar_p40_h60_tp20_sl10 | contract | H20    | 60     | 40     | 2.0    |
| 1.0                   | 33168    | 10961  | 33.05% | 37.47% | 11.41% |
| 8.88%                 |          | 76.65% | 0.635  | 27.21  | 4      |
| bar_p40_h60_tp25_sl10 | contract | H20    | 60     | 40     | 2.5    |
| 1.0                   | 33168    | 10961  | 33.05% | 25.03% | 12.75% |
| 7.78%                 |          | 66.26% | 0.498  | 29.10  | 3      |
| bar_p40_h90_tp10_sl05 | contract | H20    | 90     | 40     | 1.0    |
| 0.5                   | 33168    | 10172  | 30.67% | 73.75% | 26.25% |
| 0.00%                 |          | 73.75% | 0.606  | 30.19  | 10     |
| bar_p40_h90_tp15_sl05 | contract | H20    | 90     | 40     | 1.5    |
| 0.5                   | 33168    | 10172  | 30.67% | 42.42% | 30.41% |
| 2.83%                 |          | 58.25% | 0.484  | 30.49  | 7      |
| bar_p40_h90_tp20_sl05 | contract | H20    | 90     | 40     | 2.0    |
| 0.5                   | 33168    | 10172  | 30.67% | 27.07% | 33.26% |
| 0.33%                 |          | 44.88% | 0.375  | 31.37  | 6      |
| bar_p40_h90_tp10_sl10 | contract | H20    | 90     | 40     | 1.0    |
| 1.0                   | 33168    | 10172  | 30.67% | 90.75% | 9.25%  |
| 0.00%                 |          | 90.75% | 0.815  | 36.60  | 10     |
| bar_p40_h90_tp15_sl10 | contract | H20    | 90     | 40     | 1.5    |
| 1.0                   | 33168    | 10172  | 30.67% | 55.94% | 10.93% |
| 6.87%                 |          | 83.65% | 0.730  | 39.58  | 6      |
| bar_p40_h90_tp20_sl10 | contract | H20    | 90     | 40     | 2.0    |
| 1.0                   | 33168    | 10172  | 30.67% | 36.52% | 13.34% |
| 9.86%                 |          | 73.25% | 0.597  | 42.96  | 4      |

|                       |       |          |        |        |        |       |  |        |    |     |  |
|-----------------------|-------|----------|--------|--------|--------|-------|--|--------|----|-----|--|
| bar_p40_h90_tp25_sl10 |       | contract |        | H20    |        | 90    |  | 40     |    | 2.5 |  |
| 1.0                   | 33168 | 10172    | 30.67% | 24.13% | 14.82% |       |  | 61.06% | 3  |     |  |
| 8.94%                 |       | 61.95%   |        | 0.455  |        | 45.81 |  |        |    |     |  |
| bar_p50_h30_tp10_sl05 |       | contract |        | H20    |        | 30    |  | 50     |    | 1.0 |  |
| 0.5                   | 33168 | 9988     | 30.11% | 77.18% | 22.82% |       |  | 0.00%  | 10 |     |  |
| 0.00%                 |       | 77.18%   |        | 0.658  |        | 11.01 |  |        |    |     |  |
| bar_p50_h30_tp15_sl05 |       | contract |        | H20    |        | 30    |  | 50     |    | 1.5 |  |
| 0.5                   | 33168 | 9988     | 30.11% | 45.81% | 26.79% |       |  | 27.39% | 7  |     |  |
| 2.61%                 |       | 63.10%   |        | 0.553  |        | 11.37 |  |        |    |     |  |
| bar_p50_h30_tp20_sl05 |       | contract |        | H20    |        | 30    |  | 50     |    | 2.0 |  |
| 0.5                   | 33168 | 9988     | 30.11% | 29.98% | 29.04% |       |  | 40.98% | 5  |     |  |
| 9.02%                 |       | 50.79%   |        | 0.454  |        | 11.95 |  |        |    |     |  |
| bar_p50_h30_tp10_sl10 |       | contract |        | H20    |        | 30    |  | 50     |    | 1.0 |  |
| 1.0                   | 33168 | 9988     | 30.11% | 92.68% | 7.32%  |       |  | 0.00%  | 10 |     |  |
| 0.00%                 |       | 92.68%   |        | 0.854  |        | 12.97 |  |        |    |     |  |
| bar_p50_h30_tp15_sl10 |       | contract |        | H20    |        | 30    |  | 50     |    | 1.5 |  |
| 1.0                   | 33168 | 9988     | 30.11% | 56.98% | 8.37%  |       |  | 34.65% | 6  |     |  |
| 5.35%                 |       | 87.19%   |        | 0.771  |        | 13.91 |  |        |    |     |  |
| bar_p50_h30_tp20_sl10 |       | contract |        | H20    |        | 30    |  | 50     |    | 2.0 |  |
| 1.0                   | 33168 | 9988     | 30.11% | 37.66% | 9.68%  |       |  | 52.66% | 4  |     |  |
| 7.34%                 |       | 79.55%   |        | 0.656  |        | 15.17 |  |        |    |     |  |
| bar_p50_h30_tp25_sl10 |       | contract |        | H20    |        | 30    |  | 50     |    | 2.5 |  |
| 1.0                   | 33168 | 9988     | 30.11% | 25.27% | 10.47% |       |  | 64.26% | 3  |     |  |
| 5.74%                 |       | 70.70%   |        | 0.527  |        | 16.18 |  |        |    |     |  |
| bar_p50_h60_tp10_sl05 |       | contract |        | H20    |        | 60    |  | 50     |    | 1.0 |  |
| 0.5                   | 33168 | 9274     | 27.96% | 76.91% | 23.09% |       |  | 0.00%  | 10 |     |  |
| 0.00%                 |       | 76.91%   |        | 0.654  |        | 21.13 |  |        |    |     |  |
| bar_p50_h60_tp15_sl05 |       | contract |        | H20    |        | 60    |  | 50     |    | 1.5 |  |
| 0.5                   | 33168 | 9274     | 27.96% | 43.79% | 27.08% |       |  | 29.14% | 7  |     |  |
| 0.86%                 |       | 61.79%   |        | 0.521  |        | 21.58 |  |        |    |     |  |
| bar_p50_h60_tp20_sl05 |       | contract |        | H20    |        | 60    |  | 50     |    | 2.0 |  |
| 0.5                   | 33168 | 9274     | 27.96% | 27.90% | 29.93% |       |  | 42.17% | 5  |     |  |
| 7.83%                 |       | 48.24%   |        | 0.408  |        | 23.13 |  |        |    |     |  |
| bar_p50_h60_tp10_sl10 |       | contract |        | H20    |        | 60    |  | 50     |    | 1.0 |  |
| 1.0                   | 33168 | 9274     | 27.96% | 93.31% | 6.69%  |       |  | 0.00%  | 10 |     |  |
| 0.00%                 |       | 93.31%   |        | 0.866  |        | 25.43 |  |        |    |     |  |
| bar_p50_h60_tp15_sl10 |       | contract |        | H20    |        | 60    |  | 50     |    | 1.5 |  |
| 1.0                   | 33168 | 9274     | 27.96% | 55.29% | 7.98%  |       |  | 36.73% | 6  |     |  |
| 3.27%                 |       | 87.39%   |        | 0.750  |        | 27.36 |  |        |    |     |  |
| bar_p50_h60_tp20_sl10 |       | contract |        | H20    |        | 60    |  | 50     |    | 2.0 |  |
| 1.0                   | 33168 | 9274     | 27.96% | 34.95% | 9.56%  |       |  | 55.49% | 4  |     |  |
| 4.51%                 |       | 78.51%   |        | 0.603  |        | 30.20 |  |        |    |     |  |
| bar_p50_h60_tp25_sl10 |       | contract |        | H20    |        | 60    |  | 50     |    | 2.5 |  |
| 1.0                   | 33168 | 9274     | 27.96% | 21.34% | 10.68% |       |  | 67.99% | 3  |     |  |
| 2.01%                 |       | 66.66%   |        | 0.427  |        | 31.87 |  |        |    |     |  |
| bar_p50_h90_tp10_sl05 |       | contract |        | H20    |        | 90    |  | 50     |    | 1.0 |  |
| 0.5                   | 33168 | 8612     | 25.96% | 74.36% | 25.64% |       |  | 0.00%  | 10 |     |  |
| 0.00%                 |       | 74.36%   |        | 0.615  |        | 31.70 |  |        |    |     |  |
| bar_p50_h90_tp15_sl05 |       | contract |        | H20    |        | 90    |  | 50     |    | 1.5 |  |
| 0.5                   | 33168 | 8612     | 25.96% | 42.68% | 29.92% |       |  | 27.39% | 7  |     |  |
| 2.61%                 |       | 58.79%   |        | 0.491  |        | 33.15 |  |        |    |     |  |
| bar_p50_h90_tp20_sl05 |       | contract |        | H20    |        | 90    |  | 50     |    | 2.0 |  |
| 0.5                   | 33168 | 8612     | 25.96% | 25.94% | 33.30% |       |  | 40.76% | 5  |     |  |
| 9.24%                 |       | 43.79%   |        | 0.352  |        | 35.47 |  |        |    |     |  |
| bar_p50_h90_tp10_sl10 |       | contract |        | H20    |        | 90    |  | 50     |    | 1.0 |  |
| 1.0                   | 33168 | 8612     | 25.96% | 91.93% | 8.07%  |       |  | 0.00%  | 10 |     |  |
| 0.00%                 |       | 91.93%   |        | 0.839  |        | 38.66 |  |        |    |     |  |
| bar_p50_h90_tp15_sl10 |       | contract |        | H20    |        | 90    |  | 50     |    | 1.5 |  |
| 1.0                   | 33168 | 8612     | 25.96% | 55.39% | 10.02% |       |  | 34.59% | 6  |     |  |
| 5.41%                 |       | 84.68%   |        | 0.731  |        | 43.09 |  |        |    |     |  |

Preparación de métricas BAR por contrato y contexto finalizada correctamente.

## Celda 2 — Tablas agregadas por contrato, familia de contrato y contexto

Esta celda deja listas las tablas más importantes para graficar e interpretar después:

```
df_bar_context_by_context_type
df_bar_context_by_contract
df_bar_context_by_contract_code
df_bar_context_by_year
df_bar_context_by_dataset_split
df_bar_context_contract_by_config

df_bar_context_candidate_tp15_sl10_contract
df_bar_context_candidate_tp15_sl10_contract_code
df_bar_context_candidate_tp15_sl10_year

df_bar_context_sensitivity_by_target
df_bar_context_sensitivity_by_context_type
df_bar_context_sensitivity_by_config
df_bar_context_sensitivity_by_target_base
```

In [116...

```
# Celda 2 – Tablas agregadas por contrato, familia de contrato y contexto
# =====

print("Tablas agregadas por contrato y contexto - BAR")
print("=" * 80)

# -----
# Validar objetos necesarios
# -----

required_objects = [
    "df_bar_context_analysis",
    "df_bar_context_contract",
    "df_bar_context_contract_code",
    "df_bar_context_year",
    "df_bar_context_dataset_split",
    "df_bar_context_year_contract",
    "df_bar_context_contract_regime",
    "df_bar_context_year_regime",
    "df_bar_context_contract_code_regime",
]

missing_objects = [
    obj for obj in required_objects
    if obj not in globals()
]

if missing_objects:
    raise NameError(
        "Faltan objetos necesarios. "
        f"Debes correr la Celda 1: {missing_objects}"
    )

# -----
```

```

# Orden auxiliar de configuraciones TP/SL
# -----

config_order = [
    (1.0, 1.0),
    (1.5, 1.0),
    (2.0, 1.0),
    (2.5, 1.0),
    (1.0, 0.5),
    (1.5, 0.5),
    (2.0, 0.5),
]

# -----
# Funciones auxiliares
# -----

def value_range(series):
    """
    Rango simple: máximo - mínimo.
    """
    return series.max() - series.min()

def contract_sort_key(contract_value):
    """
    Orden cronológico aproximado de contratos MNQ:
    H = marzo, M = junio, U = septiembre, Z = diciembre.

    Ejemplo:
        H20 < M20 < U20 < Z20 < H21
    """

    contract_value = str(contract_value)

    month_order = {
        "H": 3,
        "M": 6,
        "U": 9,
        "Z": 12,
    }

    code = contract_value[0]
    year_suffix = contract_value[1:]

    try:
        year = 2000 + int(year_suffix)
    except ValueError:
        year = 9999

    month = month_order.get(code, 99)

    return year * 100 + month

def add_config_order(df):
    """
    Agrega orden auxiliar para las configuraciones TP/SL.
    """

```

```

df = df.copy()

df["config_order"] = df.apply(
    lambda row: config_order.index((row["tp_mult"], row["sl_mult"]))
    if (row["tp_mult"], row["sl_mult"]) in config_order
    else 999,
    axis=1
)

return df

def summarize_context_table(df, group_cols):
    """
    Resume métricas BAR para una tabla de contexto.
    """

    summary = (
        df
        .groupby(group_cols, as_index=False)
        .agg(
            n_targets=("bar_col", "count"),

            total_rows_mean=("total_rows", "mean"),
            valid_rows_mean=("valid_rows", "mean"),
            operable_rate_mean=("operable_rate", "mean"),

            tp_rate_mean=("tp_rate", "mean"),
            sl_rate_mean=("sl_rate", "mean"),
            no_event_rate_mean=("no_event_rate", "mean"),
            resolved_rate_mean=("resolved_rate", "mean"),

            tp_share_over_resolved_mean=("tp_share_over_resolved", "mean"),
            expectancy_units_mean=("expectancy_units", "mean"),

            event_step_mean=("event_step_mean", "mean"),
            event_step_median=("event_step_median", "mean"),
        )
        .reset_index(drop=True)
    )

    return summary

# -----
# Columnas identificadoras de target BAR
# -----

id_cols = [
    "bar_col",
    "target_base",
    "horizon",
    "percentile",
    "tp_mult",
    "sl_mult",
    "tp_sl_config",
]

# =====
# 1. Resumen general por tipo de contexto

```

```

# =====

df_bar_context_by_context_type = (
    df_bar_context_analysis
    .groupby(["context_type", "context_description"], as_index=False)
    .agg(
        n_rows=("bar_col", "count"),
        n_targets=("bar_col", "nunique"),
        n_context_values=("context_value", "nunique"),

        total_rows_mean=("total_rows", "mean"),
        valid_rows_mean=("valid_rows", "mean"),
        operable_rate_mean=("operable_rate", "mean"),

        tp_rate_mean=("tp_rate", "mean"),
        sl_rate_mean=("sl_rate", "mean"),
        no_event_rate_mean=("no_event_rate", "mean"),
        resolved_rate_mean=("resolved_rate", "mean"),

        tp_share_over_resolved_mean=("tp_share_over_resolved", "mean"),
        expectancy_units_mean=("expectancy_units", "mean"),

        event_step_mean=("event_step_mean", "mean"),
        event_step_median=("event_step_median", "mean"),
    )
    .sort_values("context_type")
    .reset_index(drop=True)
)

print("\nResumen general por tipo de contexto:")
print(
    df_bar_context_by_context_type.to_string(
        index=False,
        formatters={
            "total_rows_mean": "{:.0f}".format,
            "valid_rows_mean": "{:.0f}".format,
            "operable_rate_mean": "{:.2%}".format,
            "tp_rate_mean": "{:.2%}".format,
            "sl_rate_mean": "{:.2%}".format,
            "no_event_rate_mean": "{:.2%}".format,
            "resolved_rate_mean": "{:.2%}".format,
            "tp_share_over_resolved_mean": "{:.2%}".format,
            "expectancy_units_mean": "{:.3f}".format,
            "event_step_mean": "{:.2f}".format,
            "event_step_median": "{:.2f}".format,
        }
    )
)

# =====
# 2. Resumen por contrato individual
# =====

df_bar_context_by_contract = summarize_context_table(
    df=df_bar_context_contract,
    group_cols=["context_value"]
)

df_bar_context_by_contract = (
    df_bar_context_by_contract

```



```

        .rename(columns={"context_value": "contract"})
    )

df_bar_context_by_contract["contract_sort_key"] = (
    df_bar_context_by_contract["contract"]
    .apply(contract_sort_key)
)

df_bar_context_by_contract = (
    df_bar_context_by_contract
    .sort_values("contract_sort_key")
    .reset_index(drop=True)
)

print("\nResumen por contrato individual:")
print(
    df_bar_context_by_contract[
        [
            "contract",
            "n_targets",
            "total_rows_mean",
            "valid_rows_mean",
            "operable_rate_mean",
            "tp_rate_mean",
            "sl_rate_mean",
            "no_event_rate_mean",
            "resolved_rate_mean",
            "tp_share_over_resolved_mean",
            "expectancy_units_mean",
            "event_step_mean",
        ]
    ]
    .to_string(
        index=False,
        formatters={
            "total_rows_mean": "{:.0f}".format,
            "valid_rows_mean": "{:.0f}".format,
            "operable_rate_mean": "{:.2%}".format,
            "tp_rate_mean": "{:.2%}".format,
            "sl_rate_mean": "{:.2%}".format,
            "no_event_rate_mean": "{:.2%}".format,
            "resolved_rate_mean": "{:.2%}".format,
            "tp_share_over_resolved_mean": "{:.2%}".format,
            "expectancy_units_mean": "{:.3f}".format,
            "event_step_mean": "{:.2f}".format,
        }
    )
)

# =====
# 3. Resumen por familia de contrato H/M/U/Z
# =====

df_bar_context_by_contract_code = summarize_context_table(
    df=df_bar_context_contract_code,
    group_cols=["context_value"]
)

df_bar_context_by_contract_code = (
    df_bar_context_by_contract_code

```

```

        .rename(columns={"context_value": "contract_code"})
    )

    contract_code_order = {
        "H": 1,
        "M": 2,
        "U": 3,
        "Z": 4,
    }

    df_bar_context_by_contract_code["contract_code_order"] = (
        df_bar_context_by_contract_code["contract_code"]
        .map(contract_code_order)
        .fillna(99)
    )

    df_bar_context_by_contract_code = (
        df_bar_context_by_contract_code
        .sort_values("contract_code_order")
        .reset_index(drop=True)
    )

    print("\nResumen por familia de contrato:")
    print(
        df_bar_context_by_contract_code[
            [
                "contract_code",
                "n_targets",
                "total_rows_mean",
                "valid_rows_mean",
                "operable_rate_mean",
                "tp_rate_mean",
                "sl_rate_mean",
                "no_event_rate_mean",
                "resolved_rate_mean",
                "tp_share_over_resolved_mean",
                "expectancy_units_mean",
                "event_step_mean",
            ]
        ]
        .to_string(
            index=False,
            formatters={
                "total_rows_mean": "{:.0f}".format,
                "valid_rows_mean": "{:.0f}".format,
                "operable_rate_mean": "{:.2%}".format,
                "tp_rate_mean": "{:.2%}".format,
                "sl_rate_mean": "{:.2%}".format,
                "no_event_rate_mean": "{:.2%}".format,
                "resolved_rate_mean": "{:.2%}".format,
                "tp_share_over_resolved_mean": "{:.2%}".format,
                "expectancy_units_mean": "{:.3f}".format,
                "event_step_mean": "{:.2f}".format,
            }
        )
    )

# =====
# 4. Resumen por año
# =====

```

```

df_bar_context_by_year = summarize_context_table(
    df=df_bar_context_year,
    group_cols=["context_value"]
)

df_bar_context_by_year = (
    df_bar_context_by_year
    .rename(columns={"context_value": "year"})
)

df_bar_context_by_year["year"] = df_bar_context_by_year["year"].astype(int)

df_bar_context_by_year = (
    df_bar_context_by_year
    .sort_values("year")
    .reset_index(drop=True)
)

print("\nResumen por año:")
print(
    df_bar_context_by_year[
        [
            "year",
            "n_targets",
            "total_rows_mean",
            "valid_rows_mean",
            "operable_rate_mean",
            "tp_rate_mean",
            "sl_rate_mean",
            "no_event_rate_mean",
            "resolved_rate_mean",
            "tp_share_over_resolved_mean",
            "expectancy_units_mean",
            "event_step_mean",
        ]
    ]
    .to_string(
        index=False,
        formatters={
            "total_rows_mean": "{:.0f}".format,
            "valid_rows_mean": "{:.0f}".format,
            "operable_rate_mean": "{:.2%}".format,
            "tp_rate_mean": "{:.2%}".format,
            "sl_rate_mean": "{:.2%}".format,
            "no_event_rate_mean": "{:.2%}".format,
            "resolved_rate_mean": "{:.2%}".format,
            "tp_share_over_resolved_mean": "{:.2%}".format,
            "expectancy_units_mean": "{:.3f}".format,
            "event_step_mean": "{:.2f}".format,
        }
    )
)

# =====
# 5. Resumen por split metodológico
# =====

df_bar_context_by_dataset_split = summarize_context_table(
    df=df_bar_context_dataset_split,

```

```

    group_cols=["context_value"]
)

df_bar_context_by_dataset_split = (
    df_bar_context_by_dataset_split
    .rename(columns={"context_value": "dataset_split"})
)

print("\nResumen por split metodológico:")
print(
    df_bar_context_by_dataset_split[
        [
            "dataset_split",
            "n_targets",
            "total_rows_mean",
            "valid_rows_mean",
            "operable_rate_mean",
            "tp_rate_mean",
            "sl_rate_mean",
            "no_event_rate_mean",
            "resolved_rate_mean",
            "tp_share_over_resolved_mean",
            "expectancy_units_mean",
            "event_step_mean",
        ]
    ]
    .to_string(
        index=False,
        formatters={
            "total_rows_mean": "{:.0f}".format,
            "valid_rows_mean": "{:.0f}".format,
            "operable_rate_mean": "{:.2%}".format,
            "tp_rate_mean": "{:.2%}".format,
            "sl_rate_mean": "{:.2%}".format,
            "no_event_rate_mean": "{:.2%}".format,
            "resolved_rate_mean": "{:.2%}".format,
            "tp_share_over_resolved_mean": "{:.2%}".format,
            "expectancy_units_mean": "{:.3f}".format,
            "event_step_mean": "{:.2f}".format,
        }
    )
)

# =====
# 6. Resumen por contrato y configuración TP/SL
# =====

df_bar_context_contract_by_config = (
    df_bar_context_contract
    .groupby(
        [
            "context_value",
            "tp_mult",
            "sl_mult",
            "tp_sl_config",
        ],
        as_index=False
    )
    .agg(
        n_targets=("bar_col", "count"),

```

```

        operable_rate_mean=("operable_rate", "mean"),
        tp_rate_mean=("tp_rate", "mean"),
        sl_rate_mean=("sl_rate", "mean"),
        no_event_rate_mean=("no_event_rate", "mean"),
        resolved_rate_mean=("resolved_rate", "mean"),
        tp_share_over_resolved_mean=("tp_share_over_resolved", "mean"),
        expectancy_units_mean=("expectancy_units", "mean"),
        event_step_mean=("event_step_mean", "mean"),
    )
    .rename(columns={"context_value": "contract"})
)

df_bar_context_contract_by_config["contract_sort_key"] = (
    df_bar_context_contract_by_config["contract"]
    .apply(contract_sort_key)
)

df_bar_context_contract_by_config = add_config_order(
    df_bar_context_contract_by_config
)

df_bar_context_contract_by_config = (
    df_bar_context_contract_by_config
    .sort_values(["contract_sort_key", "config_order"])
    .reset_index(drop=True)
)

print("\nResumen por contrato y configuración TP/SL:")
print(
    df_bar_context_contract_by_config[
        [
            "contract",
            "tp_sl_config",
            "n_targets",
            "operable_rate_mean",
            "tp_rate_mean",
            "sl_rate_mean",
            "no_event_rate_mean",
            "resolved_rate_mean",
            "tp_share_over_resolved_mean",
            "expectancy_units_mean",
            "event_step_mean",
        ]
    ]
    .head(60)
    .to_string(
        index=False,
        formatters={
            "operable_rate_mean": "{:.2%}".format,
            "tp_rate_mean": "{:.2%}".format,
            "sl_rate_mean": "{:.2%}".format,
            "no_event_rate_mean": "{:.2%}".format,
            "resolved_rate_mean": "{:.2%}".format,
            "tp_share_over_resolved_mean": "{:.2%}".format,
            "expectancy_units_mean": "{:.3f}".format,
            "event_step_mean": "{:.2f}".format,
        }
    )
)

```

```

# =====
# 7. Detalle de candidata TP 1.5 / SL 1.0
# =====

df_bar_context_candidate_tp15_sl10_contract = (
    df_bar_context_contract[
        (df_bar_context_contract["tp_mult"] == 1.5)
        & (df_bar_context_contract["sl_mult"] == 1.0)
    ]
    .copy()
    .rename(columns={"context_value": "contract"})
)

df_bar_context_candidate_tp15_sl10_contract["contract_sort_key"] = (
    df_bar_context_candidate_tp15_sl10_contract["contract"]
    .apply(contract_sort_key)
)

df_bar_context_candidate_tp15_sl10_contract = (
    df_bar_context_candidate_tp15_sl10_contract
    .sort_values(["percentile", "horizon", "contract_sort_key"])
    .reset_index(drop=True)
)

print("\nDetalle por contrato para configuración candidata TP 1.5 / SL 1.0:")
print(
    df_bar_context_candidate_tp15_sl10_contract[
        [
            "bar_col",
            "target_base",
            "contract",
            "total_rows",
            "valid_rows",
            "operable_rate",
            "tp_rate",
            "sl_rate",
            "no_event_rate",
            "resolved_rate",
            "tp_share_over_resolved",
            "expectancy_units",
            "event_step_mean",
        ]
    ]
    .head(80)
    .to_string(
        index=False,
        formatters={
            "operable_rate": "{:.2%}".format,
            "tp_rate": "{:.2%}".format,
            "sl_rate": "{:.2%}".format,
            "no_event_rate": "{:.2%}".format,
            "resolved_rate": "{:.2%}".format,
            "tp_share_over_resolved": "{:.2%}".format,
            "expectancy_units": "{:.3f}".format,
            "event_step_mean": "{:.2f}".format,
        }
    )
)

df_bar_context_candidate_tp15_sl10_contract_code = (

```

```

df_bar_context_contract_code[
    (df_bar_context_contract_code["tp_mult"] == 1.5)
    & (df_bar_context_contract_code["sl_mult"] == 1.0)
]
    .copy()
    .rename(columns={"context_value": "contract_code"})
)

df_bar_context_candidate_tp15_sl10_year = (
    df_bar_context_year[
        (df_bar_context_year["tp_mult"] == 1.5)
        & (df_bar_context_year["sl_mult"] == 1.0)
    ]
    .copy()
    .rename(columns={"context_value": "year"})
)

df_bar_context_candidate_tp15_sl10_year["year"] = (
    df_bar_context_candidate_tp15_sl10_year["year"].astype(int)
)

df_bar_context_candidate_tp15_sl10_year = (
    df_bar_context_candidate_tp15_sl10_year
    .sort_values(["percentile", "horizon", "year"])
    .reset_index(drop=True)
)

# =====
# 8. Sensibilidad por contexto para cada target BAR
# =====

df_bar_context_sensitivity_by_target = (
    df_bar_context_analysis
    .groupby(["context_type"] + id_cols, as_index=False)
    .agg(
        n_context_values=("context_value", "nunique"),

        operable_rate_mean=("operable_rate", "mean"),
        operable_rate_range=("operable_rate", "value_range"),

        tp_rate_mean=("tp_rate", "mean"),
        tp_rate_range=("tp_rate", "value_range"),

        sl_rate_mean=("sl_rate", "mean"),
        sl_rate_range=("sl_rate", "value_range"),

        no_event_rate_mean=("no_event_rate", "mean"),
        no_event_rate_range=("no_event_rate", "value_range"),

        resolved_rate_mean=("resolved_rate", "mean"),
        resolved_rate_range=("resolved_rate", "value_range"),

        tp_share_over_resolved_mean=("tp_share_over_resolved", "mean"),
        tp_share_over_resolved_range=("tp_share_over_resolved", "value_range"),

        expectancy_units_mean=("expectancy_units", "mean"),
        expectancy_units_range=("expectancy_units", "value_range"),

        event_step_mean_mean=("event_step_mean", "mean"),
        event_step_mean_range=("event_step_mean", "value_range"),
    )
)

```

```

    )
    .reset_index(drop=True)
)

df_bar_context_sensitivity_by_target["context_sensitivity_score"] = (
    df_bar_context_sensitivity_by_target[
        [
            "operable_rate_range",
            "tp_rate_range",
            "sl_rate_range",
            "no_event_rate_range",
            "tp_share_over_resolved_range",
        ]
    ]
    .sum(axis=1)
)

df_bar_context_sensitivity_by_target = (
    df_bar_context_sensitivity_by_target
    .sort_values(
        [
            "context_type",
            "context_sensitivity_score",
            "percentile",
            "horizon",
            "sl_mult",
            "tp_mult",
        ],
        ascending=[True, False, True, True, True, True]
    )
    .reset_index(drop=True)
)

print("\nTop 20 targets con mayor sensibilidad por contrato individual:")
print(
    df_bar_context_sensitivity_by_target[
        df_bar_context_sensitivity_by_target["context_type"] == "contract"
    ][
        [
            "bar_col",
            "target_base",
            "tp_sl_config",
            "n_context_values",
            "operable_rate_range",
            "tp_rate_range",
            "sl_rate_range",
            "no_event_rate_range",
            "tp_share_over_resolved_range",
            "expectancy_units_range",
            "context_sensitivity_score",
        ]
    ]
    .head(20)
    .to_string(
        index=False,
        formatters={
            "operable_rate_range": "{:.2%}".format,
            "tp_rate_range": "{:.2%}".format,
            "sl_rate_range": "{:.2%}".format,
            "no_event_rate_range": "{:.2%}".format,

```



```

        "tp_share_over_resolved_range": "{:.2%}".format,
        "expectancy_units_range": "{:.3f}".format,
        "context_sensitivity_score": "{:.3f}".format,
    }
)
)

# =====
# 9. Sensibilidad agregada por tipo de contexto
# =====

df_bar_context_sensitivity_by_context_type = (
    df_bar_context_sensitivity_by_target
    .groupby("context_type", as_index=False)
    .agg(
        n_targets=("bar_col", "count"),
        n_context_values_mean=("n_context_values", "mean"),

        operable_rate_range_mean=("operable_rate_range", "mean"),
        tp_rate_range_mean=("tp_rate_range", "mean"),
        sl_rate_range_mean=("sl_rate_range", "mean"),
        no_event_rate_range_mean=("no_event_rate_range", "mean"),
        tp_share_range_mean=("tp_share_over_resolved_range", "mean"),
        expectancy_range_mean=("expectancy_units_range", "mean"),
        event_step_range_mean=("event_step_mean_range", "mean"),

        context_sensitivity_score_mean=("context_sensitivity_score", "mean"),
        context_sensitivity_score_max=("context_sensitivity_score", "max"),
    )
    .sort_values("context_sensitivity_score_mean", ascending=False)
    .reset_index(drop=True)
)

print("\nSensibilidad agregada por tipo de contexto:")
print(
    df_bar_context_sensitivity_by_context_type.to_string(
        index=False,
        formatters={
            "n_context_values_mean": "{:.1f}".format,
            "operable_rate_range_mean": "{:.2%}".format,
            "tp_rate_range_mean": "{:.2%}".format,
            "sl_rate_range_mean": "{:.2%}".format,
            "no_event_rate_range_mean": "{:.2%}".format,
            "tp_share_range_mean": "{:.2%}".format,
            "expectancy_range_mean": "{:.3f}".format,
            "event_step_range_mean": "{:.2f}".format,
            "context_sensitivity_score_mean": "{:.3f}".format,
            "context_sensitivity_score_max": "{:.3f}".format,
        }
    )
)

# =====
# 10. Sensibilidad agregada por configuración TP/SL y contexto
# =====

df_bar_context_sensitivity_by_config = (
    df_bar_context_sensitivity_by_target
    .groupby(
        [

```

```

        "context_type",
        "tp_mult",
        "sl_mult",
        "tp_sl_config",
    ],
    as_index=False
)
.agg(
    n_targets=("bar_col", "count"),

    operable_rate_range_mean=("operable_rate_range", "mean"),
    tp_rate_range_mean=("tp_rate_range", "mean"),
    sl_rate_range_mean=("sl_rate_range", "mean"),
    no_event_rate_range_mean=("no_event_rate_range", "mean"),
    tp_share_range_mean=("tp_share_over_resolved_range", "mean"),
    expectancy_range_mean=("expectancy_units_range", "mean"),

    context_sensitivity_score_mean=("context_sensitivity_score", "mean"),
    context_sensitivity_score_max=("context_sensitivity_score", "max"),
)
.reset_index(drop=True)
)

df_bar_context_sensitivity_by_config = add_config_order(
    df_bar_context_sensitivity_by_config
)

df_bar_context_sensitivity_by_config = (
    df_bar_context_sensitivity_by_config
    .sort_values(["context_type", "config_order"])
    .reset_index(drop=True)
)

print("\nSensibilidad por configuración TP/SL dentro de cada contexto:")
print(
    df_bar_context_sensitivity_by_config[
        [
            "context_type",
            "tp_sl_config",
            "n_targets",
            "operable_rate_range_mean",
            "tp_rate_range_mean",
            "sl_rate_range_mean",
            "no_event_rate_range_mean",
            "tp_share_range_mean",
            "expectancy_range_mean",
            "context_sensitivity_score_mean",
            "context_sensitivity_score_max",
        ]
    ]
    .to_string(
        index=False,
        formatters={
            "operable_rate_range_mean": "{:.2%}".format,
            "tp_rate_range_mean": "{:.2%}".format,
            "sl_rate_range_mean": "{:.2%}".format,
            "no_event_rate_range_mean": "{:.2%}".format,
            "tp_share_range_mean": "{:.2%}".format,
            "expectancy_range_mean": "{:.3f}".format,
            "context_sensitivity_score_mean": "{:.3f}".format,

```

```

        "context_sensitivity_score_max": "{:.3f}".format,
    }
)
)

# =====
# 11. Sensibilidad agregada por percentil/horizonte y contexto
# =====

df_bar_context_sensitivity_by_target_base = (
    df_bar_context_sensitivity_by_target
    .groupby(
        [
            "context_type",
            "percentile",
            "horizon",
            "target_base",
        ],
        as_index=False
    )
    .agg(
        n_configs=("bar_col", "count"),

        operable_rate_range_mean=("operable_rate_range", "mean"),
        tp_rate_range_mean=("tp_rate_range", "mean"),
        sl_rate_range_mean=("sl_rate_range", "mean"),
        no_event_rate_range_mean=("no_event_rate_range", "mean"),
        tp_share_range_mean=("tp_share_over_resolved_range", "mean"),
        expectancy_range_mean=("expectancy_units_range", "mean"),

        context_sensitivity_score_mean=("context_sensitivity_score", "mean"),
        context_sensitivity_score_max=("context_sensitivity_score", "max"),
    )
    .sort_values(["context_type", "percentile", "horizon"])
    .reset_index(drop=True)
)

print("\nSensibilidad por percentil/horizonte dentro de cada contexto:")
print(
    df_bar_context_sensitivity_by_target_base.to_string(
        index=False,
        formatters={
            "operable_rate_range_mean": "{:.2%}".format,
            "tp_rate_range_mean": "{:.2%}".format,
            "sl_rate_range_mean": "{:.2%}".format,
            "no_event_rate_range_mean": "{:.2%}".format,
            "tp_share_range_mean": "{:.2%}".format,
            "expectancy_range_mean": "{:.3f}".format,
            "context_sensitivity_score_mean": "{:.3f}".format,
            "context_sensitivity_score_max": "{:.3f}".format,
        },
    )
)

print("\nCelda 2 finalizada correctamente.")

```

## Tablas agregadas por contrato y contexto - BAR

| Resumen general por tipo de contexto: |                                           |                       |                    |                   |              |                    |
|---------------------------------------|-------------------------------------------|-----------------------|--------------------|-------------------|--------------|--------------------|
| context_type                          | context_description                       |                       | n_rows             | n_targets         |              |                    |
| n_context_values                      | total_rows_mean                           | valid_rows_mean       | operable_rate_mean | tp_rate_mean      | sl_rate_mean | no_event_rate_mean |
| resolved_rate_mean                    | tp_share_over_resolved_mean               | expectancy_units_mean | event_step_mean    | event_step_median |              |                    |
| contract                              | Contrato individual                       |                       | 1638               | 63                |              |                    |
| 26                                    | 39387                                     | 19443                 | 50.08%             | 48.79%            | 12.04%       |                    |
| 39.17%                                | 60.83%                                    |                       | 78.28%             | 0.590             |              |                    |
| 30.40                                 | 29.74                                     |                       |                    |                   |              |                    |
| contract_code                         | Familia de contrato H/M/U/Z               |                       | 252                | 63                |              |                    |
| 4                                     | 256016                                    | 126379                | 49.23%             | 49.38%            | 12.62%       |                    |
| 37.99%                                | 62.01%                                    |                       | 77.31%             | 0.599             |              |                    |
| 29.42                                 | 28.31                                     |                       |                    |                   |              |                    |
| contract_code_regime                  | Familia de contrato + régimen intradiario |                       | 1260               | 63                |              |                    |
| 20                                    | 51203                                     | 25276                 | 42.29%             | 49.20%            | 12.25%       |                    |
| 38.55%                                | 61.45%                                    |                       | 77.66%             | 0.596             |              |                    |
| 29.03                                 | 28.24                                     |                       |                    |                   |              |                    |
| contract_regime                       | Contrato + régimen intradiario            |                       | 8190               | 63                |              |                    |
| 130                                   | 7877                                      | 3889                  | 42.98%             | 48.44%            | 11.53%       |                    |
| 40.03%                                | 59.97%                                    |                       | 78.67%             | 0.585             |              |                    |
| 30.23                                 | 29.80                                     |                       |                    |                   |              |                    |
| dataset_split                         | Development vs final test                 |                       | 126                | 63                |              |                    |
| 2                                     | 512031                                    | 252757                | 54.66%             | 50.70%            | 14.69%       |                    |
| 34.61%                                | 65.39%                                    |                       | 75.80%             | 0.615             |              |                    |
| 28.00                                 | 26.40                                     |                       |                    |                   |              |                    |
| year                                  | Año calendario                            |                       | 441                | 63                |              |                    |
| 7                                     | 146295                                    | 72216                 | 51.86%             | 49.48%            | 12.99%       |                    |
| 37.53%                                | 62.47%                                    |                       | 77.29%             | 0.599             |              |                    |
| 29.47                                 | 28.49                                     |                       |                    |                   |              |                    |
| year_contract                         | Año + contrato                            |                       | 2016               | 63                |              |                    |
| 32                                    | 32002                                     | 15797                 | 47.87%             | 48.51%            | 11.23%       |                    |
| 40.26%                                | 59.74%                                    |                       | 79.34%             | 0.589             |              |                    |
| 30.99                                 | 30.49                                     |                       |                    |                   |              |                    |
| year_regime                           | Año + régimen intradiario                 |                       | 2205               | 63                |              |                    |
| 35                                    | 29259                                     | 14443                 | 44.64%             | 49.19%            | 12.55%       |                    |
| 38.25%                                | 61.75%                                    |                       | 77.52%             | 0.595             |              |                    |
| 29.19                                 | 28.56                                     |                       |                    |                   |              |                    |

## Resumen por contrato individual:

| contract | n_targets | total_rows_mean | valid_rows_mean | operable_rate_mean | tp_rate_mean | sl_rate_mean | no_event_rate_mean | resolved_rate_mean | tp_share_over_resolved_mean | expectancy_units_mean | event_step_mean |
|----------|-----------|-----------------|-----------------|--------------------|--------------|--------------|--------------------|--------------------|-----------------------------|-----------------------|-----------------|
| H20      | 63        | 33168           | 9358            | 28.21%             | 50.4         | 3%           | 17.06%             | 32.52%             | 67.48%                      | 0.600                 | 26.29           |
| M20      | 63        | 38696           | 16840           | 43.52%             | 46.7         | 6%           | 10.33%             | 42.91%             | 57.09%                      | 0.560                 | 31.00           |
| U20      | 63        | 42151           | 16752           | 39.74%             | 46.6         | 7%           | 9.59%              | 43.74%             | 56.26%                      | 0.561                 | 31.38           |
| Z20      | 63        | 36623           | 15298           | 41.77%             | 46.8         | 0%           | 9.45%              | 43.75%             | 56.25%                      | 0.563                 | 32.01           |
| H21      | 63        | 40769           | 17248           | 42.31%             | 50.2         | 7%           | 13.31%             | 36.42%             | 63.58%                      | 0.613                 | 28.63           |

|       |        |        |        |        |        |
|-------|--------|--------|--------|--------|--------|
| M21   | 63     | 42151  | 14081  | 33.41% | 44.8   |
| 0%    | 6.87%  | 48.33% | 51.67% |        | 82.50% |
| 0.537 | 34.42  |        |        |        |        |
| U21   | 63     | 41460  | 9779   | 23.59% | 41.6   |
| 1%    | 4.56%  | 53.83% | 46.17% |        | 84.46% |
| 0.487 | 37.19  |        |        |        |        |
| Z21   | 63     | 42842  | 18397  | 42.94% | 47.1   |
| 0%    | 9.78%  | 43.11% | 56.89% |        | 79.81% |
| 0.568 | 32.50  |        |        |        |        |
| H22   | 63     | 42151  | 29843  | 70.80% | 53.3   |
| 4%    | 19.58% | 27.08% | 72.92% |        | 72.24% |
| 0.644 | 25.64  |        |        |        |        |
| M22   | 63     | 42151  | 30863  | 73.22% | 52.8   |
| 3%    | 17.04% | 30.13% | 69.87% |        | 74.24% |
| 0.643 | 28.01  |        |        |        |        |
| U22   | 63     | 40769  | 23149  | 56.78% | 48.5   |
| 0%    | 10.28% | 41.22% | 58.78% |        | 80.32% |
| 0.592 | 32.34  |        |        |        |        |
| Z22   | 63     | 40769  | 23941  | 58.72% | 49.5   |
| 9%    | 11.90% | 38.51% | 61.49% |        | 78.52% |
| 0.606 | 30.21  |        |        |        |        |
| H23   | 63     | 41460  | 20511  | 49.47% | 47.1   |
| 2%    | 9.50%  | 43.38% | 56.62% |        | 79.68% |
| 0.568 | 32.06  |        |        |        |        |
| M23   | 63     | 38005  | 12404  | 32.64% | 41.8   |
| 4%    | 5.25%  | 52.91% | 47.09% |        | 82.84% |
| 0.489 | 35.41  |        |        |        |        |
| U23   | 63     | 41460  | 16949  | 40.88% | 44.2   |
| 1%    | 5.78%  | 50.01% | 49.99% |        | 84.49% |
| 0.531 | 35.89  |        |        |        |        |
| Z23   | 63     | 42842  | 17834  | 41.63% | 46.1   |
| 8%    | 7.01%  | 46.81% | 53.19% |        | 83.65% |
| 0.562 | 34.42  |        |        |        |        |
| H24   | 63     | 41460  | 16331  | 39.39% | 44.8   |
| 5%    | 7.01%  | 48.14% | 51.86% |        | 82.43% |
| 0.537 | 34.19  |        |        |        |        |
| M24   | 63     | 45606  | 18915  | 41.47% | 47.5   |
| 3%    | 8.17%  | 44.30% | 55.70% |        | 82.28% |
| 0.582 | 33.39  |        |        |        |        |
| U24   | 63     | 38696  | 23688  | 61.22% | 51.9   |
| 7%    | 13.56% | 34.48% | 65.52% |        | 77.81% |
| 0.644 | 29.04  |        |        |        |        |
| Z24   | 63     | 42842  | 21218  | 49.53% | 48.6   |
| 7%    | 9.09%  | 42.24% | 57.76% |        | 81.87% |
| 0.600 | 32.05  |        |        |        |        |
| H25   | 63     | 39387  | 26967  | 68.47% | 54.3   |
| 7%    | 18.86% | 26.77% | 73.23% |        | 73.26% |
| 0.666 | 25.27  |        |        |        |        |
| M25   | 63     | 34550  | 24453  | 70.78% | 53.7   |
| 5%    | 19.82% | 26.43% | 73.57% |        | 72.07% |
| 0.651 | 23.41  |        |        |        |        |
| U25   | 63     | 39387  | 16770  | 42.58% | 46.3   |
| 3%    | 10.16% | 43.51% | 56.49% |        | 78.43% |
| 0.553 | 31.53  |        |        |        |        |
| Z25   | 63     | 42151  | 26753  | 63.47% | 53.6   |
| 6%    | 18.16% | 28.19% | 71.81% |        | 73.56% |
| 0.658 | 25.18  |        |        |        |        |
| H26   | 63     | 39387  | 27081  | 68.76% | 54.1   |
| 3%    | 21.21% | 24.66% | 75.34% |        | 70.91% |
| 0.651 | 24.02  |        |        |        |        |

|       |        |        |        |        |        |
|-------|--------|--------|--------|--------|--------|
| M26   | 63     | 13129  | 10093  | 76.87% | 55.3   |
| 1%    | 19.68% | 25.01% | 74.99% |        | 72.93% |
| 0.679 | 24.84  |        |        |        |        |

## Resumen por familia de contrato:

| contract_code | n_targets | total_rows_mean | valid_rows_mean | operable_rate_mean | tp_rate_mean | sl_rate_mean | no_event_rate_mean | resolved_rate_mean | tp_share_over_resolved_mean | expectancy_units_mean | event_step_mean |
|---------------|-----------|-----------------|-----------------|--------------------|--------------|--------------|--------------------|--------------------|-----------------------------|-----------------------|-----------------|
| H             | 63        | 277782          | 147338          | 53.04%             |              |              |                    |                    |                             |                       |                 |
| 51.35%        | 16.06%    | 32.59%          | 67.41%          |                    | 7            |              |                    |                    |                             |                       |                 |
| 4.32%         |           | 0.621           | 27.03           |                    |              |              |                    |                    |                             |                       |                 |
| M             | 63        | 254288          | 127648          | 50.20%             |              |              |                    |                    |                             |                       |                 |
| 49.70%        | 13.32%    | 36.97%          | 63.03%          |                    | 7            |              |                    |                    |                             |                       |                 |
| 6.35%         |           | 0.601           | 28.63           |                    |              |              |                    |                    |                             |                       |                 |
| U             | 63        | 243923          | 107086          | 43.90%             |              |              |                    |                    |                             |                       |                 |
| 47.35%        | 9.64%     | 43.01%          | 56.99%          |                    | 8            |              |                    |                    |                             |                       |                 |
| 0.00%         |           | 0.574           | 31.95           |                    |              |              |                    |                    |                             |                       |                 |
| Z             | 63        | 248069          | 123442          | 49.76%             |              |              |                    |                    |                             |                       |                 |
| 49.13%        | 11.46%    | 39.41%          | 60.59%          |                    | 7            |              |                    |                    |                             |                       |                 |
| 8.56%         |           | 0.599           | 30.07           |                    |              |              |                    |                    |                             |                       |                 |

## Resumen por año:

| year   | n_targets | total_rows_mean | valid_rows_mean | operable_rate_mean | tp_rate_mean | sl_rate_mean | no_event_rate_mean | resolved_rate_mean | tp_share_over_resolved_mean | expectancy_units_mean | event_step_mean |
|--------|-----------|-----------------|-----------------|--------------------|--------------|--------------|--------------------|--------------------|-----------------------------|-----------------------|-----------------|
| 2020   | 63        | 155475          | 59018           | 37.96%             | 47.34%       |              |                    |                    |                             |                       |                 |
| 10.93% |           | 41.72%          | 58.28%          | 77.85%             |              |              |                    |                    |                             |                       |                 |
| 0.568  | 30.40     |                 |                 |                    |              |              |                    |                    |                             |                       |                 |
| 2021   | 63        | 168604          | 61043           | 36.21%             | 46.42%       |              |                    |                    |                             |                       |                 |
| 9.19%  |           | 44.39%          | 55.61%          | 79.86%             |              |              |                    |                    |                             |                       |                 |
| 0.558  | 32.23     |                 |                 |                    |              |              |                    |                    |                             |                       |                 |
| 2022   | 63        | 165840          | 108316          | 65.31%             | 51.40%       |              |                    |                    |                             |                       |                 |
| 15.09% |           | 33.51%          | 66.49%          | 75.63%             |              |              |                    |                    |                             |                       |                 |
| 0.626  | 28.52     |                 |                 |                    |              |              |                    |                    |                             |                       |                 |
| 2023   | 63        | 163767          | 65798           | 40.18%             | 45.15%       |              |                    |                    |                             |                       |                 |
| 7.17%  |           | 47.68%          | 52.32%          | 82.25%             |              |              |                    |                    |                             |                       |                 |
| 0.542  | 34.00     |                 |                 |                    |              |              |                    |                    |                             |                       |                 |
| 2024   | 63        | 165149          | 81015           | 49.06%             | 48.74%       |              |                    |                    |                             |                       |                 |
| 9.91%  |           | 41.35%          | 58.65%          | 80.56%             |              |              |                    |                    |                             |                       |                 |
| 0.598  | 31.57     |                 |                 |                    |              |              |                    |                    |                             |                       |                 |
| 2025   | 63        | 156857          | 94496           | 60.24%             | 52.48%       |              |                    |                    |                             |                       |                 |
| 17.28% |           | 30.25%          | 69.75%          | 73.76%             |              |              |                    |                    |                             |                       |                 |
| 0.638  | 25.62     |                 |                 |                    |              |              |                    |                    |                             |                       |                 |
| 2026   | 63        | 48370           | 35828           | 74.07%             | 54.81%       |              |                    |                    |                             |                       |                 |
| 21.38% |           | 23.81%          | 76.19%          | 71.15%             |              |              |                    |                    |                             |                       |                 |
| 0.663  | 23.98     |                 |                 |                    |              |              |                    |                    |                             |                       |                 |

## Resumen por split metodológico:

| dataset_split         | n_targets | total_rows_mean | valid_rows_mean | operable_rate_mean | tp_rate_mean | sl_rate_mean | no_event_rate_mean | resolved_rate_mean | tp_share_over_resolved_mean | expectancy_units_mean | event_step_mean |
|-----------------------|-----------|-----------------|-----------------|--------------------|--------------|--------------|--------------------|--------------------|-----------------------------|-----------------------|-----------------|
| development_2020_2024 | 63        | 818835          | 375190          | 45.8               |              |              |                    |                    |                             |                       |                 |
| 2%                    | 48.29%    | 10.97%          | 40.74%          | 59.26%             |              |              |                    |                    |                             |                       |                 |
| 78.61%                |           | 0.585           | 30.84           |                    |              |              |                    |                    |                             |                       |                 |
| final_test_2025_2026  | 63        | 205227          | 130324          | 63.5               |              |              |                    |                    |                             |                       |                 |
| 0%                    | 53.12%    | 18.40%          | 28.48%          | 71.52%             |              |              |                    |                    |                             |                       |                 |
| 73.00%                |           | 0.645           | 25.15           |                    |              |              |                    |                    |                             |                       |                 |

## Resumen por contrato y configuración TP/SL:

| contract | tp_sl_config | n_targets | operable_rate_mean | tp_rate_mean | sl_rate_mean |
|----------|--------------|-----------|--------------------|--------------|--------------|
|----------|--------------|-----------|--------------------|--------------|--------------|

| no_event_rate_mean  | resolved_rate_mean | tp_share_over_resolved_mean | expectancy_unit | s_mean | event_step_mean |
|---------------------|--------------------|-----------------------------|-----------------|--------|-----------------|
| H20 TP 1.0 / SL 1.0 | 9                  | 28.21%                      | 92.70%          | 7.30%  |                 |
| 0.00%               | 100.00%            | 92.70%                      |                 | 0.854  |                 |
| 25.75               |                    |                             |                 |        |                 |
| H20 TP 1.5 / SL 1.0 | 9                  | 28.21%                      | 55.39%          | 8.73%  |                 |
| 35.87%              | 64.13%             | 86.44%                      |                 | 0.744  |                 |
| 28.18               |                    |                             |                 |        |                 |
| H20 TP 2.0 / SL 1.0 | 9                  | 28.21%                      | 34.93%          | 10.31% |                 |
| 54.76%              | 45.24%             | 77.28%                      |                 | 0.595  |                 |
| 30.76               |                    |                             |                 |        |                 |
| H20 TP 2.5 / SL 1.0 | 9                  | 28.21%                      | 22.04%          | 11.19% |                 |
| 66.77%              | 33.23%             | 66.31%                      |                 | 0.439  |                 |
| 32.25               |                    |                             |                 |        |                 |
| H20 TP 1.0 / SL 0.5 | 9                  | 28.21%                      | 76.39%          | 23.61% |                 |
| 0.00%               | 100.00%            | 76.39%                      |                 | 0.646  |                 |
| 21.39               |                    |                             |                 |        |                 |
| H20 TP 1.5 / SL 0.5 | 9                  | 28.21%                      | 43.97%          | 27.80% |                 |
| 28.24%              | 71.76%             | 61.28%                      |                 | 0.521  |                 |
| 22.29               |                    |                             |                 |        |                 |
| H20 TP 2.0 / SL 0.5 | 9                  | 28.21%                      | 27.58%          | 30.44% |                 |
| 41.98%              | 58.02%             | 47.52%                      |                 | 0.399  |                 |
| 23.40               |                    |                             |                 |        |                 |
| M20 TP 1.0 / SL 1.0 | 9                  | 43.52%                      | 96.61%          | 3.39%  |                 |
| 0.00%               | 100.00%            | 96.61%                      |                 | 0.932  |                 |
| 30.83               |                    |                             |                 |        |                 |
| M20 TP 1.5 / SL 1.0 | 9                  | 43.52%                      | 47.33%          | 4.11%  |                 |
| 48.56%              | 51.44%             | 92.09%                      |                 | 0.669  |                 |
| 33.92               |                    |                             |                 |        |                 |
| M20 TP 2.0 / SL 1.0 | 9                  | 43.52%                      | 24.49%          | 4.68%  |                 |
| 70.83%              | 29.17%             | 84.11%                      |                 | 0.443  |                 |
| 35.01               |                    |                             |                 |        |                 |
| M20 TP 2.5 / SL 1.0 | 9                  | 43.52%                      | 13.23%          | 4.96%  |                 |
| 81.81%              | 18.19%             | 72.78%                      |                 | 0.281  |                 |
| 34.34               |                    |                             |                 |        |                 |
| M20 TP 1.0 / SL 0.5 | 9                  | 43.52%                      | 83.96%          | 16.04% |                 |
| 0.00%               | 100.00%            | 83.96%                      |                 | 0.759  |                 |
| 27.14               |                    |                             |                 |        |                 |
| M20 TP 1.5 / SL 0.5 | 9                  | 43.52%                      | 40.75%          | 19.01% |                 |
| 40.25%              | 59.75%             | 68.38%                      |                 | 0.516  |                 |
| 28.50               |                    |                             |                 |        |                 |
| M20 TP 2.0 / SL 0.5 | 9                  | 43.52%                      | 20.97%          | 20.10% |                 |
| 58.93%              | 41.07%             | 51.16%                      |                 | 0.319  |                 |
| 27.30               |                    |                             |                 |        |                 |
| U20 TP 1.0 / SL 1.0 | 9                  | 39.74%                      | 96.88%          | 3.12%  |                 |
| 0.00%               | 100.00%            | 96.88%                      |                 | 0.938  |                 |
| 31.34               |                    |                             |                 |        |                 |
| U20 TP 1.5 / SL 1.0 | 9                  | 39.74%                      | 46.33%          | 3.84%  |                 |
| 49.83%              | 50.17%             | 92.43%                      |                 | 0.657  |                 |
| 33.81               |                    |                             |                 |        |                 |
| U20 TP 2.0 / SL 1.0 | 9                  | 39.74%                      | 23.89%          | 4.41%  |                 |
| 71.70%              | 28.30%             | 84.57%                      |                 | 0.434  |                 |
| 34.99               |                    |                             |                 |        |                 |
| U20 TP 2.5 / SL 1.0 | 9                  | 39.74%                      | 13.18%          | 4.69%  |                 |
| 82.13%              | 17.87%             | 73.95%                      |                 | 0.283  |                 |
| 35.04               |                    |                             |                 |        |                 |
| U20 TP 1.0 / SL 0.5 | 9                  | 39.74%                      | 85.40%          | 14.60% |                 |
| 0.00%               | 100.00%            | 85.40%                      |                 | 0.781  |                 |
| 27.93               |                    |                             |                 |        |                 |
| U20 TP 1.5 / SL 0.5 | 9                  | 39.74%                      | 40.44%          | 17.63% |                 |

|                     |         |        |        |        |
|---------------------|---------|--------|--------|--------|
| 41.92%              | 58.08%  |        | 69.74% | 0.518  |
| 28.82               |         |        |        |        |
| U20 TP 2.0 / SL 0.5 | 9       | 39.74% | 20.59% | 18.83% |
| 60.59%              | 39.41%  | 52.26% |        | 0.318  |
| 27.73               |         |        |        |        |
| Z20 TP 1.0 / SL 1.0 | 9       | 41.77% | 97.29% | 2.71%  |
| 0.00%               | 100.00% | 97.29% |        | 0.946  |
| 30.72               |         |        |        |        |
| Z20 TP 1.5 / SL 1.0 | 9       | 41.77% | 47.43% | 3.37%  |
| 49.20%              | 50.80%  | 93.52% |        | 0.678  |
| 34.57               |         |        |        |        |
| Z20 TP 2.0 / SL 1.0 | 9       | 41.77% | 23.73% | 3.81%  |
| 72.46%              | 27.54%  | 86.54% |        | 0.436  |
| 36.47               |         |        |        |        |
| Z20 TP 2.5 / SL 1.0 | 9       | 41.77% | 12.03% | 4.04%  |
| 83.93%              | 16.07%  | 75.10% |        | 0.260  |
| 36.90               |         |        |        |        |
| Z20 TP 1.0 / SL 0.5 | 9       | 41.77% | 85.22% | 14.78% |
| 0.00%               | 100.00% | 85.22% |        | 0.778  |
| 27.13               |         |        |        |        |
| Z20 TP 1.5 / SL 0.5 | 9       | 41.77% | 41.34% | 18.18% |
| 40.48%              | 59.52%  | 69.67% |        | 0.529  |
| 29.49               |         |        |        |        |
| Z20 TP 2.0 / SL 0.5 | 9       | 41.77% | 20.56% | 19.27% |
| 60.17%              | 39.83%  | 51.69% |        | 0.315  |
| 28.77               |         |        |        |        |
| H21 TP 1.0 / SL 1.0 | 9       | 42.31% | 94.75% | 5.25%  |
| 0.00%               | 100.00% | 94.75% |        | 0.895  |
| 27.91               |         |        |        |        |
| H21 TP 1.5 / SL 1.0 | 9       | 42.31% | 53.78% | 6.24%  |
| 39.98%              | 60.02%  | 89.71% |        | 0.744  |
| 31.14               |         |        |        |        |
| H21 TP 2.0 / SL 1.0 | 9       | 42.31% | 31.67% | 6.98%  |
| 61.35%              | 38.65%  | 82.19% |        | 0.564  |
| 32.76               |         |        |        |        |
| H21 TP 2.5 / SL 1.0 | 9       | 42.31% | 19.16% | 7.39%  |
| 73.45%              | 26.55%  | 72.47% |        | 0.405  |
| 33.14               |         |        |        |        |
| H21 TP 1.0 / SL 0.5 | 9       | 42.31% | 80.54% | 19.46% |
| 0.00%               | 100.00% | 80.54% |        | 0.708  |
| 23.91               |         |        |        |        |
| H21 TP 1.5 / SL 0.5 | 9       | 42.31% | 45.38% | 23.17% |
| 31.45%              | 68.55%  | 66.31% |        | 0.565  |
| 25.84               |         |        |        |        |
| H21 TP 2.0 / SL 0.5 | 9       | 42.31% | 26.65% | 24.67% |
| 48.68%              | 51.32%  | 52.01% |        | 0.410  |
| 25.73               |         |        |        |        |
| M21 TP 1.0 / SL 1.0 | 9       | 33.41% | 98.15% | 1.85%  |
| 0.00%               | 100.00% | 98.15% |        | 0.963  |
| 33.28               |         |        |        |        |
| M21 TP 1.5 / SL 1.0 | 9       | 33.41% | 42.07% | 2.35%  |
| 55.59%              | 44.41%  | 94.81% |        | 0.607  |
| 36.86               |         |        |        |        |
| M21 TP 2.0 / SL 1.0 | 9       | 33.41% | 19.27% | 2.67%  |
| 78.06%              | 21.94%  | 88.10% |        | 0.359  |
| 38.26               |         |        |        |        |
| M21 TP 2.5 / SL 1.0 | 9       | 33.41% | 9.61%  | 2.78%  |
| 87.61%              | 12.39%  | 78.02% |        | 0.212  |
| 38.44               |         |        |        |        |
| M21 TP 1.0 / SL 0.5 | 9       | 33.41% | 89.18% | 10.82% |



|                     |         |        |        |        |
|---------------------|---------|--------|--------|--------|
| 0.00%               | 100.00% |        | 89.18% | 0.838  |
| 30.63               |         |        |        |        |
| M21 TP 1.5 / SL 0.5 | 9       | 33.41% | 38.02% | 13.43% |
| 48.55%              |         | 74.12% |        | 0.503  |
| 32.45               |         |        |        |        |
| M21 TP 2.0 / SL 0.5 | 9       | 33.41% | 17.29% | 14.18% |
| 68.53%              |         | 55.15% |        | 0.275  |
| 31.00               |         |        |        |        |
| U21 TP 1.0 / SL 1.0 | 9       | 23.59% | 98.76% | 1.24%  |
| 0.00%               |         | 98.76% |        | 0.975  |
| 35.83               |         |        |        |        |
| U21 TP 1.5 / SL 1.0 | 9       | 23.59% | 35.38% | 1.51%  |
| 63.10%              |         | 95.95% |        | 0.516  |
| 39.91               |         |        |        |        |
| U21 TP 2.0 / SL 1.0 | 9       | 23.59% | 13.63% | 1.60%  |
| 84.78%              |         | 89.74% |        | 0.257  |
| 40.97               |         |        |        |        |
| U21 TP 2.5 / SL 1.0 | 9       | 23.59% | 5.55%  | 1.62%  |
| 92.83%              |         | 77.88% |        | 0.123  |
| 39.62               |         |        |        |        |
| U21 TP 1.0 / SL 0.5 | 9       | 23.59% | 92.42% | 7.58%  |
| 0.00%               |         | 92.42% |        | 0.886  |
| 34.07               |         |        |        |        |
| U21 TP 1.5 / SL 0.5 | 9       | 23.59% | 32.92% | 9.06%  |
| 58.02%              |         | 78.71% |        | 0.448  |
| 36.14               |         |        |        |        |
| U21 TP 2.0 / SL 0.5 | 9       | 23.59% | 12.61% | 9.30%  |
| 78.09%              |         | 57.76% |        | 0.206  |
| 33.76               |         |        |        |        |
| Z21 TP 1.0 / SL 1.0 | 9       | 42.94% | 97.19% | 2.81%  |
| 0.00%               |         | 97.19% |        | 0.944  |
| 30.86               |         |        |        |        |
| Z21 TP 1.5 / SL 1.0 | 9       | 42.94% | 48.32% | 3.50%  |
| 48.18%              |         | 93.42% |        | 0.690  |
| 35.16               |         |        |        |        |
| Z21 TP 2.0 / SL 1.0 | 9       | 42.94% | 24.44% | 3.93%  |
| 71.63%              |         | 86.64% |        | 0.450  |
| 37.48               |         |        |        |        |
| Z21 TP 2.5 / SL 1.0 | 9       | 42.94% | 12.51% | 4.10%  |
| 83.40%              |         | 76.02% |        | 0.272  |
| 37.70               |         |        |        |        |
| Z21 TP 1.0 / SL 0.5 | 9       | 42.94% | 84.33% | 15.67% |
| 0.00%               |         | 84.33% |        | 0.765  |
| 27.19               |         |        |        |        |
| Z21 TP 1.5 / SL 0.5 | 9       | 42.94% | 41.82% | 18.78% |
| 39.40%              |         | 69.24% |        | 0.533  |
| 29.82               |         |        |        |        |
| Z21 TP 2.0 / SL 0.5 | 9       | 42.94% | 21.10% | 19.70% |
| 59.20%              |         | 51.82% |        | 0.324  |
| 29.29               |         |        |        |        |
| H22 TP 1.0 / SL 1.0 | 9       | 70.80% | 91.67% | 8.33%  |
| 0.00%               |         | 91.67% |        | 0.833  |
| 23.54               |         |        |        |        |
| H22 TP 1.5 / SL 1.0 | 9       | 70.80% | 61.84% | 10.00% |
| 28.15%              |         | 86.28% |        | 0.828  |
| 28.20               |         |        |        |        |
| H22 TP 2.0 / SL 1.0 | 9       | 70.80% | 40.32% | 11.29% |
| 48.39%              |         | 78.50% |        | 0.693  |
| 30.91               |         |        |        |        |
| H22 TP 2.5 / SL 1.0 | 9       | 70.80% | 25.98% | 11.92% |

62.10% 37.90% 68.98% 0.530  
31.92

Detalle por contrato para configuración candidata TP 1.5 / SL 1.0:

| bar_col               | target_base     | contract      | total_rows    | valid_rows             | operable_rate  |
|-----------------------|-----------------|---------------|---------------|------------------------|----------------|
| tp_rate               | sl_rate         | no_event_rate | resolved_rate | tp_share_over_resolved | expectancy_uni |
| ts                    | event_step_mean |               |               |                        |                |
| bar_p40_h30_tp15_sl10 | p40_h30         | H20           | 33168         | 11830                  | 35.67%         |
| 57.36%                | 10.19%          | 32.44%        | 67.56%        | 84.91%                 | 0.75           |
| 8                     | 13.13           |               |               |                        |                |
| bar_p40_h30_tp15_sl10 | p40_h30         | M20           | 38696         | 22026                  | 56.92%         |
| 53.25%                | 5.64%           | 41.11%        | 58.89%        | 90.42%                 | 0.74           |
| 2                     | 15.87           |               |               |                        |                |
| bar_p40_h30_tp15_sl10 | p40_h30         | U20           | 42151         | 21802                  | 51.72%         |
| 52.16%                | 4.97%           | 42.87%        | 57.13%        | 91.30%                 | 0.73           |
| 3                     | 15.91           |               |               |                        |                |
| bar_p40_h30_tp15_sl10 | p40_h30         | Z20           | 36623         | 19859                  | 54.23%         |
| 54.15%                | 4.86%           | 40.98%        | 59.02%        | 91.76%                 | 0.76           |
| 4                     | 16.39           |               |               |                        |                |
| bar_p40_h30_tp15_sl10 | p40_h30         | H21           | 40769         | 21753                  | 53.36%         |
| 57.33%                | 7.50%           | 35.16%        | 64.84%        | 88.43%                 | 0.78           |
| 5                     | 14.50           |               |               |                        |                |
| bar_p40_h30_tp15_sl10 | p40_h30         | M21           | 42151         | 18843                  | 44.70%         |
| 48.89%                | 3.04%           | 48.07%        | 51.93%        | 94.14%                 | 0.70           |
| 3                     | 17.49           |               |               |                        |                |
| bar_p40_h30_tp15_sl10 | p40_h30         | U21           | 41460         | 13194                  | 31.82%         |
| 43.50%                | 2.28%           | 54.21%        | 45.79%        | 95.02%                 | 0.63           |
| 0                     | 18.59           |               |               |                        |                |
| bar_p40_h30_tp15_sl10 | p40_h30         | Z21           | 42842         | 23317                  | 54.43%         |
| 55.14%                | 5.22%           | 39.64%        | 60.36%        | 91.35%                 | 0.77           |
| 5                     | 16.29           |               |               |                        |                |
| bar_p40_h30_tp15_sl10 | p40_h30         | H22           | 42151         | 34096                  | 80.89%         |
| 66.78%                | 13.21%          | 20.02%        | 79.98%        | 83.49%                 | 0.87           |
| 0                     | 12.63           |               |               |                        |                |
| bar_p40_h30_tp15_sl10 | p40_h30         | M22           | 42151         | 35098                  | 83.27%         |
| 67.02%                | 10.64%          | 22.33%        | 77.67%        | 86.29%                 | 0.89           |
| 9                     | 13.85           |               |               |                        |                |
| bar_p40_h30_tp15_sl10 | p40_h30         | U22           | 40769         | 28546                  | 70.02%         |
| 57.86%                | 5.31%           | 36.83%        | 63.17%        | 91.59%                 | 0.81           |
| 5                     | 16.29           |               |               |                        |                |
| bar_p40_h30_tp15_sl10 | p40_h30         | Z22           | 40769         | 29341                  | 71.97%         |
| 59.46%                | 6.66%           | 33.89%        | 66.11%        | 89.93%                 | 0.82           |
| 5                     | 15.52           |               |               |                        |                |
| bar_p40_h30_tp15_sl10 | p40_h30         | H23           | 41460         | 25805                  | 62.24%         |
| 53.73%                | 5.05%           | 41.23%        | 58.77%        | 91.42%                 | 0.75           |
| 5                     | 16.37           |               |               |                        |                |
| bar_p40_h30_tp15_sl10 | p40_h30         | M23           | 38005         | 17346                  | 45.64%         |
| 44.07%                | 2.26%           | 53.67%        | 46.33%        | 95.12%                 | 0.63           |
| 9                     | 18.02           |               |               |                        |                |
| bar_p40_h30_tp15_sl10 | p40_h30         | U23           | 41460         | 22646                  | 54.62%         |
| 47.91%                | 2.58%           | 49.51%        | 50.49%        | 94.88%                 | 0.69           |
| 3                     | 18.04           |               |               |                        |                |
| bar_p40_h30_tp15_sl10 | p40_h30         | Z23           | 42842         | 23269                  | 54.31%         |
| 50.77%                | 2.73%           | 46.50%        | 53.50%        | 94.89%                 | 0.73           |
| 4                     | 17.51           |               |               |                        |                |
| bar_p40_h30_tp15_sl10 | p40_h30         | H24           | 41460         | 21490                  | 51.83%         |
| 49.52%                | 3.01%           | 47.47%        | 52.53%        | 94.27%                 | 0.71           |
| 3                     | 17.60           |               |               |                        |                |
| bar_p40_h30_tp15_sl10 | p40_h30         | M24           | 45606         | 24464                  | 53.64%         |
| 52.03%                | 3.96%           | 44.01%        | 55.99%        | 92.93%                 | 0.74           |

|                       |         |        |        |        |        |  |
|-----------------------|---------|--------|--------|--------|--------|--|
| 1                     | 16.94   |        |        |        |        |  |
| bar_p40_h30_tp15_sl10 | p40_h30 | U24    | 38696  | 27668  | 71.50% |  |
| 62.73%                | 7.98%   | 29.29% | 70.71% | 88.71% | 0.86   |  |
| 1                     | 14.69   |        |        |        |        |  |
| bar_p40_h30_tp15_sl10 | p40_h30 | Z24    | 42842  | 26957  | 62.92% |  |
| 54.87%                | 4.46%   | 40.67% | 59.33% | 92.48% | 0.77   |  |
| 8                     | 15.99   |        |        |        |        |  |
| bar_p40_h30_tp15_sl10 | p40_h30 | H25    | 39387  | 31146  | 79.08% |  |
| 65.41%                | 11.77%  | 22.82% | 77.18% | 84.75% | 0.86   |  |
| 3                     | 12.56   |        |        |        |        |  |
| bar_p40_h30_tp15_sl10 | p40_h30 | M25    | 34550  | 28074  | 81.26% |  |
| 65.61%                | 13.08%  | 21.31% | 78.69% | 83.38% | 0.85   |  |
| 3                     | 12.17   |        |        |        |        |  |
| bar_p40_h30_tp15_sl10 | p40_h30 | U25    | 39387  | 21840  | 55.45% |  |
| 52.61%                | 5.38%   | 42.01% | 57.99% | 90.72% | 0.73   |  |
| 5                     | 16.27   |        |        |        |        |  |
| bar_p40_h30_tp15_sl10 | p40_h30 | Z25    | 42151  | 31195  | 74.01% |  |
| 63.00%                | 11.78%  | 25.22% | 74.78% | 84.24% | 0.82   |  |
| 7                     | 12.58   |        |        |        |        |  |
| bar_p40_h30_tp15_sl10 | p40_h30 | H26    | 39387  | 30922  | 78.51% |  |
| 65.66%                | 13.93%  | 20.41% | 79.59% | 82.50% | 0.84   |  |
| 6                     | 11.78   |        |        |        |        |  |
| bar_p40_h30_tp15_sl10 | p40_h30 | M26    | 13129  | 11209  | 85.38% |  |
| 68.46%                | 13.26%  | 18.28% | 81.72% | 83.78% | 0.89   |  |
| 4                     | 12.50   |        |        |        |        |  |
| bar_p40_h60_tp15_sl10 | p40_h60 | H20    | 33168  | 10961  | 33.05% |  |
| 55.43%                | 9.86%   | 34.70% | 65.30% | 84.90% | 0.73   |  |
| 3                     | 25.04   |        |        |        |        |  |
| bar_p40_h60_tp15_sl10 | p40_h60 | M20    | 38696  | 20727  | 53.56% |  |
| 50.80%                | 4.99%   | 44.21% | 55.79% | 91.06% | 0.71   |  |
| 2                     | 31.89   |        |        |        |        |  |
| bar_p40_h60_tp15_sl10 | p40_h60 | U20    | 42151  | 20494  | 48.62% |  |
| 50.21%                | 5.07%   | 44.72% | 55.28% | 90.83% | 0.70   |  |
| 2                     | 31.71   |        |        |        |        |  |
| bar_p40_h60_tp15_sl10 | p40_h60 | Z20    | 36623  | 18570  | 50.71% |  |
| 52.42%                | 4.63%   | 42.96% | 57.04% | 91.89% | 0.74   |  |
| 0                     | 32.08   |        |        |        |        |  |
| bar_p40_h60_tp15_sl10 | p40_h60 | H21    | 40769  | 20261  | 49.70% |  |
| 55.42%                | 8.04%   | 36.54% | 63.46% | 87.33% | 0.75   |  |
| 1                     | 28.31   |        |        |        |        |  |
| bar_p40_h60_tp15_sl10 | p40_h60 | M21    | 42151  | 17654  | 41.88% |  |
| 47.60%                | 3.38%   | 49.02% | 50.98% | 93.37% | 0.68   |  |
| 0                     | 34.56   |        |        |        |        |  |
| bar_p40_h60_tp15_sl10 | p40_h60 | U21    | 41460  | 12801  | 30.88% |  |
| 41.13%                | 1.73%   | 57.14% | 42.86% | 95.95% | 0.60   |  |
| 0                     | 37.38   |        |        |        |        |  |
| bar_p40_h60_tp15_sl10 | p40_h60 | Z21    | 42842  | 22580  | 52.71% |  |
| 52.22%                | 5.03%   | 42.75% | 57.25% | 91.21% | 0.73   |  |
| 3                     | 32.48   |        |        |        |        |  |
| bar_p40_h60_tp15_sl10 | p40_h60 | H22    | 42151  | 32852  | 77.94% |  |
| 65.54%                | 13.19%  | 21.27% | 78.73% | 83.25% | 0.85   |  |
| 1                     | 25.03   |        |        |        |        |  |
| bar_p40_h60_tp15_sl10 | p40_h60 | M22    | 42151  | 34326  | 81.44% |  |
| 65.74%                | 10.75%  | 23.50% | 76.50% | 85.94% | 0.87   |  |
| 9                     | 27.57   |        |        |        |        |  |
| bar_p40_h60_tp15_sl10 | p40_h60 | U22    | 40769  | 27651  | 67.82% |  |
| 56.20%                | 5.07%   | 38.73% | 61.27% | 91.72% | 0.79   |  |
| 2                     | 32.51   |        |        |        |        |  |
| bar_p40_h60_tp15_sl10 | p40_h60 | Z22    | 40769  | 27754  | 68.08% |  |
| 59.27%                | 6.29%   | 34.45% | 65.55% | 90.41% | 0.82   |  |

|                       |         |        |        |        |        |  |
|-----------------------|---------|--------|--------|--------|--------|--|
| 6                     | 30.88   |        |        |        |        |  |
| bar_p40_h60_tp15_sl10 | p40_h60 | H23    | 41460  | 24969  | 60.22% |  |
| 52.98%                | 4.68%   | 42.34% | 57.66% | 91.88% | 0.74   |  |
| 8                     | 32.84   |        |        |        |        |  |
| bar_p40_h60_tp15_sl10 | p40_h60 | M23    | 38005  | 16188  | 42.59% |  |
| 41.87%                | 1.80%   | 56.33% | 43.67% | 95.87% | 0.61   |  |
| 0                     | 36.61   |        |        |        |        |  |
| bar_p40_h60_tp15_sl10 | p40_h60 | U23    | 41460  | 21877  | 52.77% |  |
| 45.68%                | 1.90%   | 52.42% | 47.58% | 96.01% | 0.66   |  |
| 6                     | 36.46   |        |        |        |        |  |
| bar_p40_h60_tp15_sl10 | p40_h60 | Z23    | 42842  | 22229  | 51.89% |  |
| 49.79%                | 2.73%   | 47.49% | 52.51% | 94.81% | 0.72   |  |
| 0                     | 34.66   |        |        |        |        |  |
| bar_p40_h60_tp15_sl10 | p40_h60 | H24    | 41460  | 20282  | 48.92% |  |
| 49.60%                | 2.90%   | 47.50% | 52.50% | 94.47% | 0.71   |  |
| 5                     | 35.54   |        |        |        |        |  |
| bar_p40_h60_tp15_sl10 | p40_h60 | M24    | 45606  | 23239  | 50.96% |  |
| 52.55%                | 3.52%   | 43.94% | 56.06% | 93.73% | 0.75   |  |
| 3                     | 33.96   |        |        |        |        |  |
| bar_p40_h60_tp15_sl10 | p40_h60 | U24    | 38696  | 27240  | 70.39% |  |
| 62.87%                | 7.83%   | 29.30% | 70.70% | 88.92% | 0.86   |  |
| 5                     | 29.13   |        |        |        |        |  |
| bar_p40_h60_tp15_sl10 | p40_h60 | Z24    | 42842  | 25607  | 59.77% |  |
| 54.68%                | 4.28%   | 41.04% | 58.96% | 92.74% | 0.77   |  |
| 7                     | 32.15   |        |        |        |        |  |
| bar_p40_h60_tp15_sl10 | p40_h60 | H25    | 39387  | 29827  | 75.73% |  |
| 64.53%                | 12.41%  | 23.06% | 76.94% | 83.87% | 0.84   |  |
| 4                     | 24.88   |        |        |        |        |  |
| bar_p40_h60_tp15_sl10 | p40_h60 | M25    | 34550  | 26955  | 78.02% |  |
| 64.08%                | 13.66%  | 22.26% | 77.74% | 82.43% | 0.82   |  |
| 5                     | 23.89   |        |        |        |        |  |
| bar_p40_h60_tp15_sl10 | p40_h60 | U25    | 39387  | 20831  | 52.89% |  |
| 49.42%                | 4.86%   | 45.72% | 54.28% | 91.05% | 0.69   |  |
| 3                     | 32.02   |        |        |        |        |  |
| bar_p40_h60_tp15_sl10 | p40_h60 | Z25    | 42151  | 30319  | 71.93% |  |
| 61.32%                | 11.95%  | 26.73% | 73.27% | 83.69% | 0.80   |  |
| 0                     | 24.85   |        |        |        |        |  |
| bar_p40_h60_tp15_sl10 | p40_h60 | H26    | 39387  | 29887  | 75.88% |  |
| 64.61%                | 14.28%  | 21.11% | 78.89% | 81.90% | 0.82   |  |
| 6                     | 23.27   |        |        |        |        |  |
| bar_p40_h60_tp15_sl10 | p40_h60 | M26    | 13129  | 10943  | 83.35% |  |
| 68.84%                | 13.44%  | 17.72% | 82.28% | 83.66% | 0.89   |  |
| 8                     | 24.28   |        |        |        |        |  |
| bar_p40_h90_tp15_sl10 | p40_h90 | H20    | 33168  | 10172  | 30.67% |  |
| 55.94%                | 10.93%  | 33.13% | 66.87% | 83.65% | 0.73   |  |
| 0                     | 39.58   |        |        |        |        |  |
| bar_p40_h90_tp15_sl10 | p40_h90 | M20    | 38696  | 19130  | 49.44% |  |
| 50.96%                | 5.12%   | 43.92% | 56.08% | 90.87% | 0.71   |  |
| 3                     | 48.67   |        |        |        |        |  |
| bar_p40_h90_tp15_sl10 | p40_h90 | U20    | 42151  | 20020  | 47.50% |  |
| 47.56%                | 4.81%   | 47.63% | 52.37% | 90.82% | 0.66   |  |
| 5                     | 48.61   |        |        |        |        |  |
| bar_p40_h90_tp15_sl10 | p40_h90 | Z20    | 36623  | 17651  | 48.20% |  |
| 49.91%                | 5.10%   | 44.98% | 55.02% | 90.72% | 0.69   |  |
| 8                     | 49.00   |        |        |        |        |  |
| bar_p40_h90_tp15_sl10 | p40_h90 | H21    | 40769  | 19133  | 46.93% |  |
| 56.74%                | 8.71%   | 34.55% | 65.45% | 86.70% | 0.76   |  |
| 4                     | 44.23   |        |        |        |        |  |
| bar_p40_h90_tp15_sl10 | p40_h90 | M21    | 42151  | 17015  | 40.37% |  |
| 45.82%                | 3.29%   | 50.89% | 49.11% | 93.31% | 0.65   |  |

|                       |         |        |        |        |        |  |
|-----------------------|---------|--------|--------|--------|--------|--|
| 5                     | 53.20   |        |        |        |        |  |
| bar_p40_h90_tp15_sl10 | p40_h90 | U21    | 41460  | 12788  | 30.84% |  |
| 38.69%                | 2.49%   | 58.82% | 41.18% | 93.96% | 0.55   |  |
| 6                     | 58.31   |        |        |        |        |  |
| bar_p40_h90_tp15_sl10 | p40_h90 | Z21    | 42842  | 21291  | 49.70% |  |
| 51.17%                | 5.23%   | 43.60% | 56.40% | 90.72% | 0.71   |  |
| 5                     | 49.40   |        |        |        |        |  |
| bar_p40_h90_tp15_sl10 | p40_h90 | H22    | 42151  | 31285  | 74.22% |  |
| 64.41%                | 13.45%  | 22.14% | 77.86% | 82.73% | 0.83   |  |
| 2                     | 38.55   |        |        |        |        |  |
| bar_p40_h90_tp15_sl10 | p40_h90 | M22    | 42151  | 32576  | 77.28% |  |
| 66.73%                | 11.18%  | 22.09% | 77.91% | 85.65% | 0.88   |  |
| 9                     | 42.21   |        |        |        |        |  |
| bar_p40_h90_tp15_sl10 | p40_h90 | U22    | 40769  | 26242  | 64.37% |  |
| 55.77%                | 5.92%   | 38.31% | 61.69% | 90.41% | 0.77   |  |
| 7                     | 49.04   |        |        |        |        |  |
| bar_p40_h90_tp15_sl10 | p40_h90 | Z22    | 40769  | 26573  | 65.18% |  |
| 58.04%                | 6.65%   | 35.32% | 64.68% | 89.73% | 0.80   |  |
| 4                     | 46.71   |        |        |        |        |  |
| bar_p40_h90_tp15_sl10 | p40_h90 | H23    | 41460  | 24009  | 57.91% |  |
| 52.78%                | 4.61%   | 42.60% | 57.40% | 91.96% | 0.74   |  |
| 6                     | 49.89   |        |        |        |        |  |
| bar_p40_h90_tp15_sl10 | p40_h90 | M23    | 38005  | 15356  | 40.41% |  |
| 40.60%                | 2.68%   | 56.73% | 43.27% | 93.81% | 0.58   |  |
| 2                     | 56.81   |        |        |        |        |  |
| bar_p40_h90_tp15_sl10 | p40_h90 | U23    | 41460  | 20232  | 48.80% |  |
| 47.59%                | 2.40%   | 50.01% | 49.99% | 95.20% | 0.69   |  |
| 0                     | 55.84   |        |        |        |        |  |
| bar_p40_h90_tp15_sl10 | p40_h90 | Z23    | 42842  | 21413  | 49.98% |  |
| 49.91%                | 3.60%   | 46.50% | 53.50% | 93.28% | 0.71   |  |
| 3                     | 52.76   |        |        |        |        |  |
| bar_p40_h90_tp15_sl10 | p40_h90 | H24    | 41460  | 19381  | 46.75% |  |
| 50.34%                | 3.42%   | 46.25% | 53.75% | 93.65% | 0.72   |  |
| 1                     | 54.14   |        |        |        |        |  |
| bar_p40_h90_tp15_sl10 | p40_h90 | M24    | 45606  | 21920  | 48.06% |  |
| 53.47%                | 4.52%   | 42.02% | 57.98% | 92.21% | 0.75   |  |
| 7                     | 51.22   |        |        |        |        |  |
| bar_p40_h90_tp15_sl10 | p40_h90 | U24    | 38696  | 26054  | 67.33% |  |
| 62.01%                | 7.65%   | 30.34% | 69.66% | 89.01% | 0.85   |  |
| 4                     | 43.63   |        |        |        |        |  |
| bar_p40_h90_tp15_sl10 | p40_h90 | Z24    | 42842  | 24317  | 56.76% |  |
| 55.15%                | 4.77%   | 40.07% | 59.93% | 92.04% | 0.78   |  |
| 0                     | 48.72   |        |        |        |        |  |
| bar_p40_h90_tp15_sl10 | p40_h90 | H25    | 39387  | 28483  | 72.32% |  |
| 64.17%                | 14.02%  | 21.82% | 78.18% | 82.07% | 0.82   |  |
| 2                     | 37.75   |        |        |        |        |  |
| bar_p40_h90_tp15_sl10 | p40_h90 | M25    | 34550  | 25787  | 74.64% |  |
| 63.24%                | 13.88%  | 22.88% | 77.12% | 82.00% | 0.81   |  |
| 0                     | 36.09   |        |        |        |        |  |
| bar_p40_h90_tp15_sl10 | p40_h90 | U25    | 39387  | 19570  | 49.69% |  |
| 48.56%                | 5.28%   | 46.16% | 53.84% | 90.20% | 0.67   |  |
| 6                     | 48.71   |        |        |        |        |  |
| bar_p40_h90_tp15_sl10 | p40_h90 | Z25    | 42151  | 28652  | 67.97% |  |
| 62.33%                | 12.15%  | 25.52% | 74.48% | 83.69% | 0.81   |  |
| 4                     | 37.56   |        |        |        |        |  |
| bar_p40_h90_tp15_sl10 | p40_h90 | H26    | 39387  | 28438  | 72.20% |  |
| 63.37%                | 15.21%  | 21.42% | 78.58% | 80.65% | 0.79   |  |
| 9                     | 34.82   |        |        |        |        |  |
| bar_p40_h90_tp15_sl10 | p40_h90 | M26    | 13129  | 10482  | 79.84% |  |
| 71.74%                | 12.36%  | 15.89% | 84.11% | 85.30% | 0.95   |  |

|                       |         |        |        |        |        |  |
|-----------------------|---------|--------|--------|--------|--------|--|
| 2                     | 36.83   |        |        |        |        |  |
| bar_p50_h30_tp15_sl10 | p50_h30 | H20    | 33168  | 9988   | 30.11% |  |
| 56.98%                | 8.37%   | 34.65% | 65.35% | 87.19% | 0.77   |  |
| 1                     | 13.91   |        |        |        |        |  |
| bar_p50_h30_tp15_sl10 | p50_h30 | M20    | 38696  | 18151  | 46.91% |  |
| 48.01%                | 4.25%   | 47.73% | 52.27% | 91.86% | 0.67   |  |
| 8                     | 16.79   |        |        |        |        |  |

Top 20 targets con mayor sensibilidad por contrato individual:

| bar_col               | target_base   | tp_sl_config    | n_context_values    | operable_rate                |                        |                           |
|-----------------------|---------------|-----------------|---------------------|------------------------------|------------------------|---------------------------|
| _range                | tp_rate_range | sl_rate_range   | no_event_rate_range | tp_share_over_resolved_range | expectancy_units_range | context_sensitivity_score |
| bar_p60_h60_tp15_sl05 | p60_h60       | TP 1.5 / SL 0.5 |                     | 26                           |                        |                           |
| 54.14%                | 24.70%        | 25.11%          | 48.71%              |                              | 21.5                   |                           |
| 0%                    | 0.250         | 1.742           |                     |                              |                        |                           |
| bar_p50_h60_tp20_sl05 | p50_h60       | TP 2.0 / SL 0.5 |                     | 26                           |                        |                           |
| 54.31%                | 24.08%        | 29.08%          | 53.15%              |                              | 11.9                   |                           |
| 3%                    | 0.336         | 1.726           |                     |                              |                        |                           |
| bar_p40_h90_tp20_sl05 | p40_h90       | TP 2.0 / SL 0.5 |                     | 26                           |                        |                           |
| 49.17%                | 24.89%        | 30.69%          | 54.76%              |                              | 11.4                   |                           |
| 0%                    | 0.362         | 1.709           |                     |                              |                        |                           |
| bar_p50_h90_tp15_sl05 | p50_h90       | TP 1.5 / SL 0.5 |                     | 26                           |                        |                           |
| 52.11%                | 23.12%        | 26.45%          | 48.36%              |                              | 19.8                   |                           |
| 8%                    | 0.221         | 1.699           |                     |                              |                        |                           |
| bar_p60_h60_tp20_sl05 | p60_h60       | TP 2.0 / SL 0.5 |                     | 26                           |                        |                           |
| 54.14%                | 21.86%        | 27.93%          | 49.34%              |                              | 16.3                   |                           |
| 2%                    | 0.323         | 1.696           |                     |                              |                        |                           |
| bar_p40_h60_tp20_sl05 | p40_h60       | TP 2.0 / SL 0.5 |                     | 26                           |                        |                           |
| 52.47%                | 23.54%        | 30.00%          | 53.53%              |                              | 9.8                    |                           |
| 6%                    | 0.321         | 1.694           |                     |                              |                        |                           |
| bar_p50_h90_tp20_sl05 | p50_h90       | TP 2.0 / SL 0.5 |                     | 26                           |                        |                           |
| 52.11%                | 22.26%        | 30.19%          | 50.75%              |                              | 14.0                   |                           |
| 8%                    | 0.313         | 1.694           |                     |                              |                        |                           |
| bar_p50_h60_tp15_sl05 | p50_h60       | TP 1.5 / SL 0.5 |                     | 26                           |                        |                           |
| 54.31%                | 19.63%        | 26.79%          | 46.41%              |                              | 21.9                   |                           |
| 6%                    | 0.167         | 1.691           |                     |                              |                        |                           |
| bar_p60_h90_tp20_sl05 | p60_h90       | TP 2.0 / SL 0.5 |                     | 26                           |                        |                           |
| 53.30%                | 21.46%        | 27.81%          | 48.63%              |                              | 17.4                   |                           |
| 8%                    | 0.322         | 1.687           |                     |                              |                        |                           |
| bar_p40_h60_tp20_sl10 | p40_h60       | TP 2.0 / SL 1.0 |                     | 26                           |                        |                           |
| 52.47%                | 34.90%        | 14.29%          | 48.49%              |                              | 15.9                   |                           |
| 5%                    | 0.562         | 1.661           |                     |                              |                        |                           |
| bar_p60_h90_tp15_sl05 | p60_h90       | TP 1.5 / SL 0.5 |                     | 26                           |                        |                           |
| 53.30%                | 23.61%        | 24.84%          | 43.93%              |                              | 20.3                   |                           |
| 6%                    | 0.257         | 1.660           |                     |                              |                        |                           |
| bar_p40_h90_tp20_sl10 | p40_h90       | TP 2.0 / SL 1.0 |                     | 26                           |                        |                           |
| 49.17%                | 36.08%        | 15.06%          | 47.54%              |                              | 16.7                   |                           |
| 3%                    | 0.607         | 1.646           |                     |                              |                        |                           |
| bar_p40_h90_tp15_sl05 | p40_h90       | TP 1.5 / SL 0.5 |                     | 26                           |                        |                           |
| 49.17%                | 17.27%        | 27.74%          | 45.01%              |                              | 20.6                   |                           |
| 6%                    | 0.123         | 1.598           |                     |                              |                        |                           |
| bar_p60_h30_tp20_sl05 | p60_h30       | TP 2.0 / SL 0.5 |                     | 26                           |                        |                           |
| 54.76%                | 22.17%        | 24.01%          | 45.66%              |                              | 12.3                   |                           |
| 6%                    | 0.336         | 1.590           |                     |                              |                        |                           |
| bar_p50_h60_tp20_sl10 | p50_h60       | TP 2.0 / SL 1.0 |                     | 26                           |                        |                           |
| 54.31%                | 32.91%        | 11.90%          | 43.19%              |                              | 15.3                   |                           |
| 0%                    | 0.561         | 1.576           |                     |                              |                        |                           |
| bar_p50_h30_tp20_sl05 | p50_h30       | TP 2.0 / SL 0.5 |                     | 26                           |                        |                           |
| 55.94%                | 21.67%        | 24.31%          | 45.82%              |                              | 8.1                    |                           |
| 4%                    | 0.320         | 1.559           |                     |                              |                        |                           |

|                       |                         |        |        |
|-----------------------|-------------------------|--------|--------|
| bar_p40_h60_tp15_sl05 | p40_h60 TP 1.5 / SL 0.5 | 26     |        |
| 52.47%                | 13.83%                  | 26.93% | 40.76% |
| 7%                    | 0.106                   | 1.547  | 20.6   |
| bar_p40_h60_tp25_sl10 | p40_h60 TP 2.5 / SL 1.0 | 26     |        |
| 52.47%                | 27.62%                  | 15.93% | 42.78% |
| 5%                    | 0.544                   | 1.546  | 15.7   |
| bar_p60_h30_tp15_sl05 | p60_h30 TP 1.5 / SL 0.5 | 26     |        |
| 54.76%                | 19.03%                  | 21.89% | 40.00% |
| 3%                    | 0.191                   | 1.544  | 18.7   |
| bar_p50_h90_tp15_sl10 | p50_h90 TP 1.5 / SL 1.0 | 26     |        |
| 52.11%                | 35.29%                  | 10.85% | 42.65% |
| 4%                    | 0.456                   | 1.542  | 13.3   |

Sensibilidad agregada por tipo de contexto:

| context_type         | n_targets | n_context_values_mean | operable_rate_range_mean | tp_rate_range_mean | sl_rate_range_mean | no_event_rate_range_mean | tp_share_range_mean | expectancy_range_mean | event_step_range_mean | context_sensitivity_score_mean | context_sensitivity_score_max |
|----------------------|-----------|-----------------------|--------------------------|--------------------|--------------------|--------------------------|---------------------|-----------------------|-----------------------|--------------------------------|-------------------------------|
| contract_regime      | 63        | 130.0                 | 87.07%                   |                    |                    |                          |                     |                       |                       |                                |                               |
|                      |           |                       |                          | 36.55%             | 24.63%             | 42.13%                   | 45.70%              |                       |                       |                                |                               |
|                      |           |                       |                          | 0.652              | 28.26              |                          | 2.361               |                       |                       |                                |                               |
|                      |           |                       |                          | 3.303              |                    |                          |                     |                       |                       |                                |                               |
| year_regime          | 63        | 35.0                  | 86.40%                   |                    |                    |                          |                     |                       |                       |                                |                               |
|                      |           |                       |                          | 26.04%             | 20.02%             | 33.74%                   | 21.90%              |                       |                       |                                |                               |
|                      |           |                       |                          | 0.409              | 17.96              |                          | 1.881               |                       |                       |                                |                               |
|                      |           |                       |                          | 2.342              |                    |                          |                     |                       |                       |                                |                               |
| year_contract        | 63        | 32.0                  | 62.45%                   |                    |                    |                          |                     |                       |                       |                                |                               |
|                      |           |                       |                          | 24.11%             | 18.71%             | 31.95%                   | 22.63%              |                       |                       |                                |                               |
|                      |           |                       |                          | 0.374              | 17.04              |                          | 1.598               |                       |                       |                                |                               |
|                      |           |                       |                          | 2.179              |                    |                          |                     |                       |                       |                                |                               |
| contract             | 63        | 26.0                  | 53.31%                   |                    |                    |                          |                     |                       |                       |                                |                               |
|                      |           |                       |                          | 22.54%             | 16.81%             | 30.10%                   | 15.35%              |                       |                       |                                |                               |
|                      |           |                       |                          | 0.341              | 14.38              |                          | 1.381               |                       |                       |                                |                               |
|                      |           |                       |                          | 1.742              |                    |                          |                     |                       |                       |                                |                               |
| contract_code_regime | 63        | 20.0                  | 59.27%                   |                    |                    |                          |                     |                       |                       |                                |                               |
|                      |           |                       |                          | 12.42%             | 9.71%              | 15.67%                   | 11.46%              |                       |                       |                                |                               |
|                      |           |                       |                          | 0.201              | 12.58              |                          | 1.085               |                       |                       |                                |                               |
|                      |           |                       |                          | 1.389              |                    |                          |                     |                       |                       |                                |                               |
| year                 | 63        | 7.0                   | 37.87%                   |                    |                    |                          |                     |                       |                       |                                |                               |
|                      |           |                       |                          | 16.90%             | 14.21%             | 23.88%                   | 11.23%              |                       |                       |                                |                               |
|                      |           |                       |                          | 0.243              | 10.13              |                          | 1.041               |                       |                       |                                |                               |
|                      |           |                       |                          | 1.294              |                    |                          |                     |                       |                       |                                |                               |
| dataset_split        | 63        | 2.0                   | 17.68%                   |                    |                    |                          |                     |                       |                       |                                |                               |
|                      |           |                       |                          | 8.59%              | 7.44%              | 12.26%                   | 5.61%               |                       |                       |                                |                               |
|                      |           |                       |                          | 0.123              | 5.69               |                          | 0.516               |                       |                       |                                |                               |
|                      |           |                       |                          | 0.652              |                    |                          |                     |                       |                       |                                |                               |
| contract_code        | 63        | 4.0                   | 9.14%                    |                    |                    |                          |                     |                       |                       |                                |                               |
|                      |           |                       |                          | 7.26%              | 6.42%              | 10.42%                   | 5.69%               |                       |                       |                                |                               |
|                      |           |                       |                          | 0.102              | 4.96               |                          | 0.389               |                       |                       |                                |                               |
|                      |           |                       |                          | 0.535              |                    |                          |                     |                       |                       |                                |                               |

Sensibilidad por configuración TP/SL dentro de cada contexto:

| context_type | tp_sl_config    | n_targets | operable_rate_range_mean | tp_rate_range_mean | sl_rate_range_mean | no_event_rate_range_mean | tp_share_range_mean | expectancy_range_mean | context_sensitivity_score_mean | context_sensitivity_score_max |
|--------------|-----------------|-----------|--------------------------|--------------------|--------------------|--------------------------|---------------------|-----------------------|--------------------------------|-------------------------------|
| contract     | TP 1.0 / SL 1.0 | 9         | 53.31%                   |                    |                    |                          |                     |                       |                                |                               |
|              |                 |           |                          | 8.84%              | 8.84%              | 0.00%                    | 8.84%               |                       |                                |                               |
|              |                 | 0.798     | 0.845                    |                    |                    |                          |                     |                       |                                |                               |
| contract     | TP 1.5 / SL 1.0 | 9         | 53.31%                   |                    |                    |                          |                     |                       |                                |                               |
|              |                 |           |                          | 30.88%             | 10.33%             | 39.37%                   | 12.50%              |                       |                                |                               |

|        |                                      |        |  |        |
|--------|--------------------------------------|--------|--|--------|
| 0.382  |                                      | 1.464  |  | 1.542  |
|        | contract TP 2.0 / SL 1.0             | 9      |  | 53.31% |
| 30.96% | 11.98%                               | 41.93% |  | 15.99% |
| 0.517  |                                      | 1.542  |  | 1.661  |
|        | contract TP 2.5 / SL 1.0             | 9      |  | 53.31% |
| 24.16% | 13.03%                               | 36.61% |  | 17.41% |
| 0.491  |                                      | 1.445  |  | 1.546  |
|        | contract TP 1.0 / SL 0.5             | 9      |  | 53.31% |
| 21.00% | 21.00%                               | 0.00%  |  | 21.00% |
| 0.315  |                                      | 1.163  |  | 1.222  |
|        | contract TP 1.5 / SL 0.5             | 9      |  | 53.31% |
| 19.33% | 24.84%                               | 43.09% |  | 19.53% |
| 0.177  |                                      | 1.601  |  | 1.742  |
|        | contract TP 2.0 / SL 0.5             | 9      |  | 53.31% |
| 22.63% | 27.65%                               | 49.68% |  | 12.18% |
| 0.328  |                                      | 1.655  |  | 1.726  |
|        | contract_code TP 1.0 / SL 1.0        | 9      |  | 9.14%  |
| 3.76%  | 3.76%                                | 0.00%  |  | 3.76%  |
| 0.075  |                                      | 0.204  |  | 0.211  |
|        | contract_code TP 1.5 / SL 1.0        | 9      |  | 9.14%  |
| 8.81%  | 4.40%                                | 13.21% |  | 5.38%  |
| 0.088  |                                      | 0.409  |  | 0.446  |
|        | contract_code TP 2.0 / SL 1.0        | 9      |  | 9.14%  |
| 10.33% | 5.05%                                | 15.38% |  | 6.51%  |
| 0.156  |                                      | 0.464  |  | 0.490  |
|        | contract_code TP 2.5 / SL 1.0        | 9      |  | 9.14%  |
| 8.65%  | 5.45%                                | 14.10% |  | 6.46%  |
| 0.162  |                                      | 0.438  |  | 0.460  |
|        | contract_code TP 1.0 / SL 0.5        | 9      |  | 9.14%  |
| 7.65%  | 7.65%                                | 0.00%  |  | 7.65%  |
| 0.115  |                                      | 0.321  |  | 0.360  |
|        | contract_code TP 1.5 / SL 0.5        | 9      |  | 9.14%  |
| 4.62%  | 8.78%                                | 13.39% |  | 6.56%  |
| 0.029  |                                      | 0.425  |  | 0.521  |
|        | contract_code TP 2.0 / SL 0.5        | 9      |  | 9.14%  |
| 7.01%  | 9.85%                                | 16.86% |  | 3.49%  |
| 0.091  |                                      | 0.464  |  | 0.535  |
|        | contract_code_regime TP 1.0 / SL 1.0 | 9      |  | 59.27% |
| 5.88%  | 5.88%                                | 0.00%  |  | 5.88%  |
| 0.118  |                                      | 0.769  |  | 0.902  |
|        | contract_code_regime TP 1.5 / SL 1.0 | 9      |  | 59.27% |
| 13.89% | 6.76%                                | 19.39% |  | 8.97%  |
| 0.169  |                                      | 1.083  |  | 1.169  |
|        | contract_code_regime TP 2.0 / SL 1.0 | 9      |  | 59.27% |
| 17.28% | 7.75%                                | 24.16% |  | 12.54% |
| 0.284  |                                      | 1.210  |  | 1.324  |
|        | contract_code_regime TP 2.5 / SL 1.0 | 9      |  | 59.27% |
| 15.47% | 8.47%                                | 23.02% |  | 16.18% |
| 0.312  |                                      | 1.224  |  | 1.344  |
|        | contract_code_regime TP 1.0 / SL 0.5 | 9      |  | 59.27% |
| 11.76% | 11.76%                               | 0.00%  |  | 11.76% |
| 0.176  |                                      | 0.945  |  | 1.060  |
|        | contract_code_regime TP 1.5 / SL 0.5 | 9      |  | 59.27% |
| 10.26% | 12.98%                               | 18.93% |  | 12.26% |
| 0.140  |                                      | 1.137  |  | 1.236  |
|        | contract_code_regime TP 2.0 / SL 0.5 | 9      |  | 59.27% |
| 12.43% | 14.37%                               | 24.21% |  | 12.63% |
| 0.211  |                                      | 1.229  |  | 1.389  |
|        | contract_regime TP 1.0 / SL 1.0      | 9      |  | 87.07% |
| 12.62% | 12.62%                               | 0.00%  |  | 12.62% |



|        |                                 |        |  |        |
|--------|---------------------------------|--------|--|--------|
| 0.252  |                                 | 1.249  |  | 1.418  |
|        | contract_regime TP 1.5 / SL 1.0 | 9      |  | 87.07% |
| 45.65% | 14.72%                          | 54.78% |  | 24.73% |
| 0.657  |                                 | 2.270  |  | 2.467  |
|        | contract_regime TP 2.0 / SL 1.0 | 9      |  | 87.07% |
| 48.18% | 16.77%                          | 60.80% |  | 56.11% |
| 0.905  |                                 | 2.689  |  | 2.852  |
|        | contract_regime TP 2.5 / SL 1.0 | 9      |  | 87.07% |
| 40.54% | 18.42%                          | 53.75% |  | 85.05% |
| 0.995  |                                 | 2.848  |  | 3.185  |
|        | contract_regime TP 1.0 / SL 0.5 | 9      |  | 87.07% |
| 32.36% | 32.36%                          | 0.00%  |  | 32.36% |
| 0.485  |                                 | 1.841  |  | 2.004  |
|        | contract_regime TP 1.5 / SL 0.5 | 9      |  | 87.07% |
| 36.28% | 37.19%                          | 57.73% |  | 46.15% |
| 0.545  |                                 | 2.644  |  | 2.925  |
|        | contract_regime TP 2.0 / SL 0.5 | 9      |  | 87.07% |
| 40.21% | 40.34%                          | 67.86% |  | 62.86% |
| 0.727  |                                 | 2.984  |  | 3.303  |
|        | dataset_split TP 1.0 / SL 1.0   | 9      |  | 17.68% |
| 4.48%  | 4.48%                           | 0.00%  |  | 4.48%  |
| 0.090  |                                 | 0.311  |  | 0.328  |
|        | dataset_split TP 1.5 / SL 1.0   | 9      |  | 17.68% |
| 9.68%  | 5.20%                           | 14.88% |  | 5.72%  |
| 0.093  |                                 | 0.532  |  | 0.560  |
|        | dataset_split TP 2.0 / SL 1.0   | 9      |  | 17.68% |
| 12.52% | 5.98%                           | 18.50% |  | 6.03%  |
| 0.191  |                                 | 0.607  |  | 0.636  |
|        | dataset_split TP 2.5 / SL 1.0   | 9      |  | 17.68% |
| 11.24% | 6.50%                           | 17.74% |  | 4.85%  |
| 0.216  |                                 | 0.580  |  | 0.616  |
|        | dataset_split TP 1.0 / SL 0.5   | 9      |  | 17.68% |
| 8.70%  | 8.70%                           | 0.00%  |  | 8.70%  |
| 0.131  |                                 | 0.438  |  | 0.465  |
|        | dataset_split TP 1.5 / SL 0.5   | 9      |  | 17.68% |
| 4.96%  | 9.99%                           | 14.95% |  | 6.79%  |
| 0.028  |                                 | 0.544  |  | 0.622  |
|        | dataset_split TP 2.0 / SL 0.5   | 9      |  | 17.68% |
| 8.54%  | 11.22%                          | 19.76% |  | 2.71%  |
| 0.115  |                                 | 0.599  |  | 0.652  |
|        | year TP 1.0 / SL 1.0            | 9      |  | 37.87% |
| 7.78%  | 7.78%                           | 0.00%  |  | 7.78%  |
| 0.156  |                                 | 0.612  |  | 0.660  |
|        | year TP 1.5 / SL 1.0            | 9      |  | 37.87% |
| 21.27% | 9.12%                           | 30.39% |  | 9.82%  |
| 0.228  |                                 | 1.085  |  | 1.139  |
|        | year TP 2.0 / SL 1.0            | 9      |  | 37.87% |
| 24.07% | 10.57%                          | 34.64% |  | 11.08% |
| 0.376  |                                 | 1.182  |  | 1.278  |
|        | year TP 2.5 / SL 1.0            | 9      |  | 37.87% |
| 19.40% | 11.50%                          | 30.90% |  | 11.17% |
| 0.371  |                                 | 1.108  |  | 1.235  |
|        | year TP 1.0 / SL 0.5            | 9      |  | 37.87% |
| 17.53% | 17.53%                          | 0.00%  |  | 17.53% |
| 0.263  |                                 | 0.904  |  | 0.951  |
|        | year TP 1.5 / SL 0.5            | 9      |  | 37.87% |
| 11.64% | 20.28%                          | 31.88% |  | 14.18% |
| 0.086  |                                 | 1.159  |  | 1.287  |
|        | year TP 2.0 / SL 0.5            | 9      |  | 37.87% |
| 16.63% | 22.70%                          | 39.32% |  | 7.09%  |

|        |                               |        |  |        |
|--------|-------------------------------|--------|--|--------|
| 0.219  |                               | 1.236  |  | 1.294  |
|        | year_contract TP 1.0 / SL 1.0 | 9      |  | 62.45% |
| 9.96%  | 9.96%                         | 0.00%  |  | 9.96%  |
| 0.199  |                               | 0.923  |  | 1.011  |
|        | year_contract TP 1.5 / SL 1.0 | 9      |  | 62.45% |
| 32.32% | 11.71%                        | 41.21% |  | 14.97% |
| 0.404  |                               | 1.627  |  | 1.722  |
|        | year_contract TP 2.0 / SL 1.0 | 9      |  | 62.45% |
| 32.61% | 13.57%                        | 44.79% |  | 22.87% |
| 0.539  |                               | 1.763  |  | 1.903  |
|        | year_contract TP 2.5 / SL 1.0 | 9      |  | 62.45% |
| 25.88% | 14.72%                        | 39.79% |  | 35.71% |
| 0.527  |                               | 1.785  |  | 2.095  |
|        | year_contract TP 1.0 / SL 0.5 | 9      |  | 62.45% |
| 23.00% | 23.00%                        | 0.00%  |  | 23.00% |
| 0.345  |                               | 1.315  |  | 1.417  |
|        | year_contract TP 1.5 / SL 0.5 | 9      |  | 62.45% |
| 21.05% | 27.49%                        | 45.62% |  | 24.88% |
| 0.232  |                               | 1.815  |  | 2.007  |
|        | year_contract TP 2.0 / SL 0.5 | 9      |  | 62.45% |
| 23.92% | 30.48%                        | 52.25% |  | 27.00% |
| 0.375  |                               | 1.961  |  | 2.179  |
|        | year_regime TP 1.0 / SL 1.0   | 9      |  | 86.40% |
| 11.77% | 11.77%                        | 0.00%  |  | 11.77% |
| 0.235  |                               | 1.217  |  | 1.374  |
|        | year_regime TP 1.5 / SL 1.0   | 9      |  | 86.40% |
| 32.33% | 13.44%                        | 42.11% |  | 15.62% |
| 0.408  |                               | 1.899  |  | 2.024  |
|        | year_regime TP 2.0 / SL 1.0   | 9      |  | 86.40% |
| 36.14% | 15.19%                        | 50.15% |  | 22.09% |
| 0.604  |                               | 2.100  |  | 2.247  |
|        | year_regime TP 2.5 / SL 1.0   | 9      |  | 86.40% |
| 29.55% | 16.72%                        | 45.88% |  | 35.37% |
| 0.586  |                               | 2.139  |  | 2.279  |
|        | year_regime TP 1.0 / SL 0.5   | 9      |  | 86.40% |
| 24.97% | 24.97%                        | 0.00%  |  | 24.97% |
| 0.375  |                               | 1.613  |  | 1.716  |
|        | year_regime TP 1.5 / SL 0.5   | 9      |  | 86.40% |
| 21.07% | 27.63%                        | 43.59% |  | 22.09% |
| 0.237  |                               | 2.008  |  | 2.167  |
|        | year_regime TP 2.0 / SL 0.5   | 9      |  | 86.40% |
| 26.44% | 30.45%                        | 54.44% |  | 21.37% |
| 0.420  |                               | 2.191  |  | 2.342  |

Sensibilidad por percentil/horizonte dentro de cada contexto:

context\_type percentile horizon target\_base n\_configs operable\_rate\_range\_mean tp\_rate\_range\_mean sl\_rate\_range\_mean no\_event\_rate\_range\_mean tp\_share\_range\_mean expectancy\_range\_mean context\_sensitivity\_score\_mean context\_sensitivity\_score\_max

|        |          |    |        |         |        |
|--------|----------|----|--------|---------|--------|
|        | contract | 40 | 30     | p40_h30 | 7      |
| 53.55% | 21.17%   |    | 16.65% |         | 28.79% |
| 13.54% | 0.317    |    |        | 1.337   |        |
| 1.537  |          |    |        |         |        |
|        | contract | 40 | 60     | p40_h60 | 7      |
| 52.47% | 23.03%   |    | 19.04% |         | 32.14% |
| 15.70% | 0.341    |    |        | 1.424   |        |
| 1.694  |          |    |        |         |        |
|        | contract | 40 | 90     | p40_h90 | 7      |
| 49.17% | 24.70%   |    | 19.78% |         | 33.20% |
| 16.64% | 0.371    |    |        | 1.435   |        |

|        |               |    |        |         |        |
|--------|---------------|----|--------|---------|--------|
| 1.709  |               |    |        |         |        |
|        | contract      | 50 | 30     | p50_h30 | 7      |
| 55.94% | 21.02%        |    | 15.36% |         | 28.23% |
| 13.66% | 0.318         |    |        | 1.342   |        |
| 1.559  |               |    |        |         |        |
|        | contract      | 50 | 60     | p50_h60 | 7      |
| 54.31% | 23.65%        |    | 17.42% |         | 31.62% |
| 15.61% | 0.355         |    |        | 1.426   |        |
| 1.726  |               |    |        |         |        |
|        | contract      | 50 | 90     | p50_h90 | 7      |
| 52.11% | 23.72%        |    | 17.90% |         | 31.54% |
| 16.91% | 0.359         |    |        | 1.422   |        |
| 1.699  |               |    |        |         |        |
|        | contract      | 60 | 30     | p60_h30 | 7      |
| 54.76% | 20.46%        |    | 13.93% |         | 27.03% |
| 14.16% | 0.312         |    |        | 1.303   |        |
| 1.590  |               |    |        |         |        |
|        | contract      | 60 | 60     | p60_h60 | 7      |
| 54.14% | 22.84%        |    | 15.37% |         | 29.45% |
| 15.36% | 0.348         |    |        | 1.372   |        |
| 1.742  |               |    |        |         |        |
|        | contract      | 60 | 90     | p60_h90 | 7      |
| 53.30% | 22.30%        |    | 15.85% |         | 28.87% |
| 16.56% | 0.347         |    |        | 1.369   |        |
| 1.687  |               |    |        |         |        |
|        | contract_code | 40 | 30     | p40_h30 | 7      |
| 8.10%  | 6.67%         |    | 5.97%  |         | 9.62%  |
| 4.79%  | 0.095         |    |        | 0.352   |        |
| 0.444  |               |    |        |         |        |
|        | contract_code | 40 | 60     | p40_h60 | 7      |
| 7.19%  | 7.03%         |    | 6.70%  |         | 10.36% |
| 5.47%  | 0.098         |    |        | 0.368   |        |
| 0.471  |               |    |        |         |        |
|        | contract_code | 40 | 90     | p40_h90 | 7      |
| 6.72%  | 7.02%         |    | 7.19%  |         | 10.53% |
| 6.00%  | 0.094         |    |        | 0.375   |        |
| 0.467  |               |    |        |         |        |
|        | contract_code | 50 | 30     | p50_h30 | 7      |
| 9.77%  | 7.33%         |    | 5.92%  |         | 10.25% |
| 5.05%  | 0.106         |    |        | 0.383   |        |
| 0.469  |               |    |        |         |        |
|        | contract_code | 50 | 60     | p50_h60 | 7      |
| 9.25%  | 7.72%         |    | 6.74%  |         | 11.05% |
| 5.92%  | 0.109         |    |        | 0.407   |        |
| 0.491  |               |    |        |         |        |
|        | contract_code | 50 | 90     | p50_h90 | 7      |
| 8.83%  | 7.20%         |    | 6.91%  |         | 10.60% |
| 6.31%  | 0.098         |    |        | 0.398   |        |
| 0.469  |               |    |        |         |        |
|        | contract_code | 60 | 30     | p60_h30 | 7      |
| 11.20% | 7.60%         |    | 5.63%  |         | 10.39% |
| 5.23%  | 0.112         |    |        | 0.400   |        |
| 0.494  |               |    |        |         |        |
|        | contract_code | 60 | 60     | p60_h60 | 7      |
| 10.68% | 7.90%         |    | 6.35%  |         | 11.01% |
| 6.06%  | 0.113         |    |        | 0.420   |        |
| 0.535  |               |    |        |         |        |
|        | contract_code | 60 | 90     | p60_h90 | 7      |
| 10.52% | 6.87%         |    | 6.38%  |         | 9.99%  |
| 6.37%  | 0.095         |    |        | 0.401   |        |

|                      |        |        |         |        |  |
|----------------------|--------|--------|---------|--------|--|
| 0.504                |        |        |         |        |  |
| contract_code_regime | 40     | 30     | p40_h30 | 7      |  |
| 66.56%               | 12.57% | 11.52% |         | 15.47% |  |
| 12.31%               | 0.217  |        | 1.184   |        |  |
| 1.389                |        |        |         |        |  |
| contract_code_regime | 40     | 60     | p40_h60 | 7      |  |
| 68.30%               | 9.76%  | 8.88%  |         | 13.12% |  |
| 8.59%                | 0.155  |        | 1.087   |        |  |
| 1.227                |        |        |         |        |  |
| contract_code_regime | 40     | 90     | p40_h90 | 7      |  |
| 67.63%               | 11.62% | 9.49%  |         | 15.09% |  |
| 10.22%               | 0.187  |        | 1.140   |        |  |
| 1.344                |        |        |         |        |  |
| contract_code_regime | 50     | 30     | p50_h30 | 7      |  |
| 58.75%               | 12.99% | 11.78% |         | 16.13% |  |
| 14.23%               | 0.220  |        | 1.139   |        |  |
| 1.379                |        |        |         |        |  |
| contract_code_regime | 50     | 60     | p50_h60 | 7      |  |
| 59.80%               | 11.24% | 8.53%  |         | 14.27% |  |
| 9.63%                | 0.178  |        | 1.035   |        |  |
| 1.191                |        |        |         |        |  |
| contract_code_regime | 50     | 90     | p50_h90 | 7      |  |
| 59.91%               | 13.22% | 9.72%  |         | 17.03% |  |
| 11.56%               | 0.209  |        | 1.114   |        |  |
| 1.287                |        |        |         |        |  |
| contract_code_regime | 60     | 30     | p60_h30 | 7      |  |
| 50.50%               | 13.12% | 9.80%  |         | 16.40% |  |
| 12.47%               | 0.210  |        | 1.023   |        |  |
| 1.216                |        |        |         |        |  |
| contract_code_regime | 60     | 60     | p60_h60 | 7      |  |
| 50.93%               | 12.57% | 7.94%  |         | 15.12% |  |
| 10.32%               | 0.203  |        | 0.969   |        |  |
| 1.118                |        |        |         |        |  |
| contract_code_regime | 60     | 90     | p60_h90 | 7      |  |
| 51.05%               | 14.75% | 9.72%  |         | 18.42% |  |
| 13.81%               | 0.234  |        | 1.078   |        |  |
| 1.253                |        |        |         |        |  |
| contract_regime      | 40     | 30     | p40_h30 | 7      |  |
| 91.39%               | 40.57% | 28.87% |         | 41.80% |  |
| 43.27%               | 0.766  |        | 2.459   |        |  |
| 3.162                |        |        |         |        |  |
| contract_regime      | 40     | 60     | p40_h60 | 7      |  |
| 92.46%               | 32.00% | 26.31% |         | 39.44% |  |
| 32.33%               | 0.535  |        | 2.225   |        |  |
| 2.674                |        |        |         |        |  |
| contract_regime      | 40     | 90     | p40_h90 | 7      |  |
| 94.65%               | 33.46% | 26.12% |         | 43.00% |  |
| 43.58%               | 0.591  |        | 2.408   |        |  |
| 3.016                |        |        |         |        |  |
| contract_regime      | 50     | 30     | p50_h30 | 7      |  |
| 86.98%               | 45.12% | 26.65% |         | 42.44% |  |
| 45.94%               | 0.841  |        | 2.471   |        |  |
| 3.232                |        |        |         |        |  |
| contract_regime      | 50     | 60     | p50_h60 | 7      |  |
| 86.75%               | 31.67% | 24.44% |         | 40.69% |  |
| 39.02%               | 0.517  |        | 2.226   |        |  |
| 2.778                |        |        |         |        |  |
| contract_regime      | 50     | 90     | p50_h90 | 7      |  |
| 89.47%               | 34.34% | 24.23% |         | 43.97% |  |
| 49.29%               | 0.606  |        | 2.413   |        |  |

|        |                 |    |        |         |        |
|--------|-----------------|----|--------|---------|--------|
| 2.939  |                 |    |        |         |        |
|        | contract_regime | 60 | 30     | p60_h30 | 7      |
| 80.13% | 46.03%          |    | 22.14% |         | 45.25% |
| 52.13% | 0.896           |    |        | 2.457   |        |
| 3.303  |                 |    |        |         |        |
|        | contract_regime | 60 | 60     | p60_h60 | 7      |
| 80.35% | 31.95%          |    | 20.70% |         | 40.69% |
| 44.53% | 0.518           |    |        | 2.182   |        |
| 2.719  |                 |    |        |         |        |
|        | contract_regime | 60 | 90     | p60_h90 | 7      |
| 81.49% | 33.80%          |    | 22.22% |         | 41.91% |
| 61.20% | 0.603           |    |        | 2.406   |        |
| 3.145  |                 |    |        |         |        |
|        | dataset_split   | 40 | 30     | p40_h30 | 7      |
| 16.59% | 7.95%           |    | 7.15%  |         | 11.50% |
| 5.18%  | 0.113           |    |        | 0.484   |        |
| 0.589  |                 |    |        |         |        |
|        | dataset_split   | 40 | 60     | p40_h60 | 7      |
| 16.49% | 8.03%           |    | 8.10%  |         | 12.03% |
| 6.06%  | 0.111           |    |        | 0.507   |        |
| 0.616  |                 |    |        |         |        |
|        | dataset_split   | 40 | 90     | p40_h90 | 7      |
| 15.64% | 8.16%           |    | 8.39%  |         | 12.29% |
| 6.09%  | 0.113           |    |        | 0.506   |        |
| 0.612  |                 |    |        |         |        |
|        | dataset_split   | 50 | 30     | p50_h30 | 7      |
| 18.43% | 8.40%           |    | 6.81%  |         | 11.77% |
| 5.15%  | 0.122           |    |        | 0.506   |        |
| 0.600  |                 |    |        |         |        |
|        | dataset_split   | 50 | 60     | p50_h60 | 7      |
| 18.11% | 8.86%           |    | 7.82%  |         | 12.74% |
| 6.05%  | 0.126           |    |        | 0.536   |        |
| 0.636  |                 |    |        |         |        |
|        | dataset_split   | 50 | 90     | p50_h90 | 7      |
| 17.16% | 8.96%           |    | 7.95%  |         | 12.87% |
| 5.82%  | 0.127           |    |        | 0.528   |        |
| 0.623  |                 |    |        |         |        |
|        | dataset_split   | 60 | 30     | p60_h30 | 7      |
| 19.51% | 8.64%           |    | 6.42%  |         | 11.81% |
| 5.13%  | 0.128           |    |        | 0.515   |        |
| 0.611  |                 |    |        |         |        |
|        | dataset_split   | 60 | 60     | p60_h60 | 7      |
| 18.95% | 9.21%           |    | 7.14%  |         | 12.71% |
| 5.66%  | 0.135           |    |        | 0.537   |        |
| 0.652  |                 |    |        |         |        |
|        | dataset_split   | 60 | 90     | p60_h90 | 7      |
| 18.26% | 9.10%           |    | 7.15%  |         | 12.62% |
| 5.37%  | 0.132           |    |        | 0.525   |        |
| 0.639  |                 |    |        |         |        |
|        | year            | 40 | 30     | p40_h30 | 7      |
| 36.43% | 16.84%          |    | 14.34% |         | 23.95% |
| 10.49% | 0.243           |    |        | 1.021   |        |
| 1.221  |                 |    |        |         |        |
|        | year            | 40 | 60     | p40_h60 | 7      |
| 36.00% | 17.48%          |    | 16.13% |         | 25.43% |
| 11.96% | 0.249           |    |        | 1.070   |        |
| 1.278  |                 |    |        |         |        |
|        | year            | 40 | 90     | p40_h90 | 7      |
| 34.23% | 16.90%          |    | 16.51% |         | 24.91% |
| 12.16% | 0.240           |    |        | 1.047   |        |

|        |               |    |        |         |        |
|--------|---------------|----|--------|---------|--------|
| 1.230  |               |    |        |         |        |
|        | year          | 50 | 30     | p50_h30 | 7      |
| 39.37% | 17.15%        |    | 13.22% |         | 23.65% |
| 10.25% | 0.252         |    |        | 1.036   |        |
| 1.214  |               |    |        |         |        |
|        | year          | 50 | 60     | p50_h60 | 7      |
| 38.80% | 17.92%        |    | 14.98% |         | 25.27% |
| 11.94% | 0.255         |    |        | 1.089   |        |
| 1.294  |               |    |        |         |        |
|        | year          | 50 | 90     | p50_h90 | 7      |
| 37.14% | 16.75%        |    | 15.00% |         | 24.12% |
| 11.70% | 0.235         |    |        | 1.047   |        |
| 1.272  |               |    |        |         |        |
|        | year          | 60 | 30     | p60_h30 | 7      |
| 40.22% | 16.68%        |    | 11.89% |         | 22.51% |
| 10.05% | 0.248         |    |        | 1.013   |        |
| 1.234  |               |    |        |         |        |
|        | year          | 60 | 60     | p60_h60 | 7      |
| 40.07% | 16.93%        |    | 13.00% |         | 23.24% |
| 11.39% | 0.243         |    |        | 1.046   |        |
| 1.288  |               |    |        |         |        |
|        | year          | 60 | 90     | p60_h90 | 7      |
| 38.54% | 15.48%        |    | 12.82% |         | 21.82% |
| 11.17% | 0.219         |    |        | 0.998   |        |
| 1.249  |               |    |        |         |        |
|        | year_contract | 40 | 30     | p40_h30 | 7      |
| 63.21% | 24.28%        |    | 18.64% |         | 32.89% |
| 17.02% | 0.361         |    |        | 1.560   |        |
| 1.875  |               |    |        |         |        |
|        | year_contract | 40 | 60     | p40_h60 | 7      |
| 64.10% | 25.25%        |    | 20.22% |         | 33.77% |
| 18.83% | 0.376         |    |        | 1.622   |        |
| 1.872  |               |    |        |         |        |
|        | year_contract | 40 | 90     | p40_h90 | 7      |
| 61.51% | 25.43%        |    | 22.32% |         | 33.75% |
| 24.38% | 0.388         |    |        | 1.674   |        |
| 2.006  |               |    |        |         |        |
|        | year_contract | 50 | 30     | p50_h30 | 7      |
| 64.11% | 22.42%        |    | 16.61% |         | 29.95% |
| 18.49% | 0.340         |    |        | 1.516   |        |
| 1.782  |               |    |        |         |        |
|        | year_contract | 50 | 60     | p50_h60 | 7      |
| 64.73% | 24.63%        |    | 18.88% |         | 33.14% |
| 20.30% | 0.378         |    |        | 1.617   |        |
| 1.940  |               |    |        |         |        |
|        | year_contract | 50 | 90     | p50_h90 | 7      |
| 61.77% | 25.22%        |    | 21.39% |         | 32.85% |
| 28.13% | 0.396         |    |        | 1.694   |        |
| 2.179  |               |    |        |         |        |
|        | year_contract | 60 | 30     | p60_h30 | 7      |
| 61.91% | 21.45%        |    | 14.85% |         | 28.38% |
| 19.29% | 0.333         |    |        | 1.459   |        |
| 1.812  |               |    |        |         |        |
|        | year_contract | 60 | 60     | p60_h60 | 7      |
| 61.54% | 24.68%        |    | 17.09% |         | 31.77% |
| 23.11% | 0.400         |    |        | 1.582   |        |
| 2.043  |               |    |        |         |        |
|        | year_contract | 60 | 90     | p60_h90 | 7      |
| 59.14% | 23.62%        |    | 18.36% |         | 31.04% |
| 34.09% | 0.399         |    |        | 1.662   |        |

|        |             |    |        |         |        |
|--------|-------------|----|--------|---------|--------|
| 2.140  | year_regime | 40 | 30     | p40_h30 | 7      |
| 89.04% | 25.79%      |    | 22.42% |         | 34.16% |
| 21.76% | 0.408       |    |        | 1.932   |        |
| 2.229  | year_regime | 40 | 60     | p40_h60 | 7      |
| 92.40% | 24.13%      |    | 19.86% |         | 31.59% |
| 17.24% | 0.379       |    |        | 1.852   |        |
| 2.129  | year_regime | 40 | 90     | p40_h90 | 7      |
| 95.40% | 27.34%      |    | 21.67% |         | 35.17% |
| 19.20% | 0.428       |    |        | 1.988   |        |
| 2.279  | year_regime | 50 | 30     | p50_h30 | 7      |
| 83.32% | 25.04%      |    | 21.21% |         | 32.47% |
| 26.07% | 0.403       |    |        | 1.881   |        |
| 2.248  | year_regime | 50 | 60     | p50_h60 | 7      |
| 86.52% | 25.17%      |    | 19.20% |         | 33.30% |
| 19.04% | 0.393       |    |        | 1.832   |        |
| 2.187  | year_regime | 50 | 90     | p50_h90 | 7      |
| 91.45% | 27.66%      |    | 21.00% |         | 35.82% |
| 21.79% | 0.434       |    |        | 1.977   |        |
| 2.342  | year_regime | 60 | 30     | p60_h30 | 7      |
| 75.74% | 26.13%      |    | 18.91% |         | 32.85% |
| 25.52% | 0.425       |    |        | 1.792   |        |
| 2.137  | year_regime | 60 | 60     | p60_h60 | 7      |
| 80.71% | 25.51%      |    | 17.25% |         | 33.06% |
| 21.09% | 0.393       |    |        | 1.776   |        |
| 2.141  | year_regime | 60 | 90     | p60_h90 | 7      |
| 83.00% | 27.58%      |    | 18.69% |         | 35.23% |
| 25.36% | 0.422       |    |        | 1.899   |        |
| 2.275  |             |    |        |         |        |

Celda 2 finalizada correctamente.

## Celda 3 — Gráficos agrupados por contrato y contexto

Esta celda te permite revisar el comportamiento de BAR desde varios ángulos:

1. Contrato individual.
2. Familia de contrato H/M/U/Z.
3. Año.
4. Candidata TP 1.5 / SL 1.0 por contrato y año.
5. Sensibilidad por tipo de contexto.
6. Sensibilidad por configuración TP/SL.
7. Heatmaps de sensibilidad para contratos, años y contexto combinado.

In [117...

```
# Celda 3 – Gráficos agrupados por contrato y contexto
# =====

print("Gráficos agrupados por contrato y contexto - BAR")
print("=" * 80)
```

```

# -----
# Validar objetos necesarios
# -----

required_objects = [
    "df_bar_context_by_contract",
    "df_bar_context_by_contract_code",
    "df_bar_context_by_year",
    "df_bar_context_by_dataset_split",
    "df_bar_context_contract_by_config",
    "df_bar_context_candidate_tp15_sl10_contract",
    "df_bar_context_candidate_tp15_sl10_contract_code",
    "df_bar_context_candidate_tp15_sl10_year",
    "df_bar_context_sensitivity_by_context_type",
    "df_bar_context_sensitivity_by_config",
    "df_bar_context_sensitivity_by_target_base",
    "BAR_PERCENTILES",
    "BAR_HORIZONS",
]

missing_objects = [
    obj for obj in required_objects
    if obj not in globals()
]

if missing_objects:
    raise NameError(
        "Faltan objetos necesarios. "
        f"Debes correr las celdas anteriores: {missing_objects}"
    )

# -----
# Orden auxiliar de configuraciones TP/SL
# -----

config_order = [
    (1.0, 1.0),
    (1.5, 1.0),
    (2.0, 1.0),
    (2.5, 1.0),
    (1.0, 0.5),
    (1.5, 0.5),
    (2.0, 0.5),
]

config_labels = [
    f"TP {tp} / SL {sl}"
    for tp, sl in config_order
]

target_base_order = [
    f"p{p}_h{h}"
    for p in BAR_PERCENTILES
    for h in BAR_HORIZONS
]

contract_code_order = {
    "H": 1,
    "M": 2,
}

```



```

    "U": 3,
    "Z": 4,
}

# -----
# Funciones auxiliares
# -----

def contract_sort_key(contract_value):
    """
    Orden cronológico aproximado de contratos MNQ:
    H = marzo, M = junio, U = septiembre, Z = diciembre.
    """

    contract_value = str(contract_value)

    month_order = {
        "H": 3,
        "M": 6,
        "U": 9,
        "Z": 12,
    }

    code = contract_value[0]
    year_suffix = contract_value[1:]

    try:
        year = 2000 + int(year_suffix)
    except ValueError:
        year = 9999

    month = month_order.get(code, 99)

    return year * 100 + month

def add_contract_sort(df, contract_col="contract"):
    """
    Agrega columna de orden cronológico de contrato.
    """

    df = df.copy()

    df["contract_sort_key"] = (
        df[contract_col]
        .astype(str)
        .apply(contract_sort_key)
    )

    df = (
        df
        .sort_values("contract_sort_key")
        .reset_index(drop=True)
    )

    return df

def add_config_label_and_order(df):
    """

```

```

Agrega etiqueta y orden auxiliar para configuraciones TP/SL.
"""

df = df.copy()

df["config_label"] = df.apply(
    lambda row: f"TP {row['tp_mult']} / SL {row['sl_mult']}",
    axis=1
)

df["config_order"] = df.apply(
    lambda row: config_order.index((row["tp_mult"], row["sl_mult"]))
    if (row["tp_mult"], row["sl_mult"]) in config_order
    else 999,
    axis=1
)

return df

def plot_heatmap_with_values(
    pivot_data,
    title,
    value_format="{:.2f}",
    figsize=(15, 7),
    colorbar_label="Valor",
    vmin=None,
    vmax=None
):
    """
    Grafica un heatmap simple con valores anotados.
    """

    fig, ax = plt.subplots(figsize=figsize)

    data = np.ma.masked_invalid(pivot_data.values)

    im = ax.imshow(
        data,
        aspect="auto",
        vmin=vmin,
        vmax=vmax
    )

    ax.set_title(title)

    ax.set_yticks(np.arange(len(pivot_data.index)))
    ax.set_yticklabels(pivot_data.index)

    ax.set_xticks(np.arange(len(pivot_data.columns)))
    ax.set_xticklabels(pivot_data.columns, rotation=45, ha="right")

    for i in range(pivot_data.shape[0]):
        for j in range(pivot_data.shape[1]):
            value = pivot_data.iloc[i, j]

            if pd.notna(value):
                ax.text(
                    j,
                    i,

```

```

        value_format.format(value),
        ha="center",
        va="center",
        fontsize=8
    )

fig.colorbar(
    im,
    ax=ax,
    label=colorbar_label
)

plt.tight_layout()
plt.show()

# =====
# Figura 1 – Distribución promedio BAR por contrato individual
# =====

df_contract_plot = df_bar_context_by_contract.copy()
df_contract_plot = add_contract_sort(df_contract_plot, contract_col="contract")

x = np.arange(len(df_contract_plot))
bar_width = 0.25

fig, ax = plt.subplots(figsize=(18, 6))

ax.bar(
    x - bar_width,
    df_contract_plot["tp_rate_mean"],
    width=bar_width,
    label="TP primero"
)

ax.bar(
    x,
    df_contract_plot["sl_rate_mean"],
    width=bar_width,
    label="SL primero"
)

ax.bar(
    x + bar_width,
    df_contract_plot["no_event_rate_mean"],
    width=bar_width,
    label="no_event"
)

ax.set_title("Distribución promedio de BAR por contrato individual")
ax.set_ylabel("Proporción promedio")
ax.set_xticks(x)
ax.set_xticklabels(df_contract_plot["contract"], rotation=45, ha="right")
ax.legend()

ax.yaxis.set_major_formatter(
    plt.FuncFormatter(lambda y, _: f"{y:.0%}")
)

plt.tight_layout()

```

```

plt.show()

# =====
# Figura 2 – Operabilidad y eventos resueltos por contrato
# =====

fig, ax = plt.subplots(figsize=(18, 6))

ax.plot(
    df_contract_plot["contract"],
    df_contract_plot["operable_rate_mean"],
    marker="o",
    linewidth=1.5,
    label="Operable rate"
)

ax.plot(
    df_contract_plot["contract"],
    df_contract_plot["resolved_rate_mean"],
    marker="o",
    linewidth=1.5,
    label="Resolved rate"
)

ax.plot(
    df_contract_plot["contract"],
    df_contract_plot["no_event_rate_mean"],
    marker="o",
    linewidth=1.5,
    label="no_event rate"
)

ax.set_title("Operabilidad, eventos resueltos y no_event por contrato")
ax.set_ylabel("Proporción promedio")
ax.set_xticks(np.arange(len(df_contract_plot)))
ax.set_xticklabels(df_contract_plot["contract"], rotation=45, ha="right")
ax.legend()

ax.yaxis.set_major_formatter(
    plt.FuncFormatter(lambda y, _: f"{y:.0%}")
)

ax.grid(True, alpha=0.3)

plt.tight_layout()
plt.show()

# =====
# Figura 3 – Distribución promedio BAR por familia de contrato
# =====

df_contract_code_plot = df_bar_context_by_contract_code.copy()

df_contract_code_plot["contract_code_order"] = (
    df_contract_code_plot["contract_code"]
    .map(contract_code_order)
    .fillna(99)
)

df_contract_code_plot = (

```

```

df_contract_code_plot
.sort_values("contract_code_order")
.reset_index(drop=True)
)

x = np.arange(len(df_contract_code_plot))
bar_width = 0.25

fig, ax = plt.subplots(figsize=(10, 5))

ax.bar(
    x - bar_width,
    df_contract_code_plot["tp_rate_mean"],
    width=bar_width,
    label="TP primero"
)

ax.bar(
    x,
    df_contract_code_plot["sl_rate_mean"],
    width=bar_width,
    label="SL primero"
)

ax.bar(
    x + bar_width,
    df_contract_code_plot["no_event_rate_mean"],
    width=bar_width,
    label="no_event"
)

ax.set_title("Distribución promedio de BAR por familia de contrato")
ax.set_ylabel("Proporción promedio")
ax.set_xticks(x)
ax.set_xticklabels(df_contract_code_plot["contract_code"])
ax.legend()

ax.yaxis.set_major_formatter(
    plt.FuncFormatter(lambda y, _: f"{y:.0%}")
)

for i, row in df_contract_code_plot.iterrows():
    ax.text(
        i - bar_width,
        row["tp_rate_mean"],
        f"{row['tp_rate_mean']:.0%}",
        ha="center",
        va="bottom",
        fontsize=8
    )

    ax.text(
        i,
        row["sl_rate_mean"],
        f"{row['sl_rate_mean']:.0%}",
        ha="center",
        va="bottom",
        fontsize=8
    )

```

```

        ax.text(
            i + bar_width,
            row["no_event_rate_mean"],
            f"{row['no_event_rate_mean']:.0%}",
            ha="center",
            va="bottom",
            fontsize=8
        )

plt.tight_layout()
plt.show()

# =====
# Figura 4 – Distribución promedio BAR por año
# =====

df_year_plot = df_bar_context_by_year.copy()
df_year_plot = df_year_plot.sort_values("year").reset_index(drop=True)

x = np.arange(len(df_year_plot))
bar_width = 0.25

fig, ax = plt.subplots(figsize=(12, 5))

ax.bar(
    x - bar_width,
    df_year_plot["tp_rate_mean"],
    width=bar_width,
    label="TP primero"
)

ax.bar(
    x,
    df_year_plot["sl_rate_mean"],
    width=bar_width,
    label="SL primero"
)

ax.bar(
    x + bar_width,
    df_year_plot["no_event_rate_mean"],
    width=bar_width,
    label="no_event"
)

ax.set_title("Distribución promedio de BAR por año")
ax.set_ylabel("Proporción promedio")
ax.set_xticks(x)
ax.set_xticklabels(df_year_plot["year"])
ax.legend()

ax.yaxis.set_major_formatter(
    plt.FuncFormatter(lambda y, _: f"{y:.0%}")
)

for i, row in df_year_plot.iterrows():
    ax.text(
        i - bar_width,
        row["tp_rate_mean"],
        f"{row['tp_rate_mean']:.0%}",

```

```

        ha="center",
        va="bottom",
        fontsize=8
    )

    ax.text(
        i,
        row["sl_rate_mean"],
        f"{row['sl_rate_mean']:.0%}",
        ha="center",
        va="bottom",
        fontsize=8
    )

    ax.text(
        i + bar_width,
        row["no_event_rate_mean"],
        f"{row['no_event_rate_mean']:.0%}",
        ha="center",
        va="bottom",
        fontsize=8
    )

plt.tight_layout()
plt.show()

# =====
# Figura 5 – Candidata TP 1.5 / SL 1.0 por contrato
# Filas: percentil | Columnas: horizonte
# =====

df_candidate_contract_plot = df_bar_context_candidate_tp15_sl10_contract.copy()
df_candidate_contract_plot = add_contract_sort(df_candidate_contract_plot, contr

contracts_order = (
    df_candidate_contract_plot[["contract", "contract_sort_key"]]
    .drop_duplicates()
    .sort_values("contract_sort_key")
    ["contract"]
    .tolist()
)

fig, axes = plt.subplots(
    nrows=len(BAR_PERCENTILES),
    ncols=len(BAR_HORIZONS),
    figsize=(22, 12),
    sharex=True,
    sharey=True
)

for i, percentile in enumerate(BAR_PERCENTILES):
    for j, horizon in enumerate(BAR_HORIZONS):

        ax = axes[i, j]

        tmp = df_candidate_contract_plot[
            (df_candidate_contract_plot["percentile"] == percentile)
            & (df_candidate_contract_plot["horizon"] == horizon)
        ].copy()

```

```

tmp = (
    tmp
    .set_index("contract")
    .reindex(contracts_order)
    .reset_index()
)

ax.plot(
    tmp["contract"],
    tmp["tp_rate"],
    marker="o",
    linewidth=1.2,
    label="TP rate"
)

ax.plot(
    tmp["contract"],
    tmp["sl_rate"],
    marker="o",
    linewidth=1.2,
    label="SL rate"
)

ax.plot(
    tmp["contract"],
    tmp["no_event_rate"],
    marker="o",
    linewidth=1.2,
    label="no_event"
)

ax.set_title(f"p{percentile} | H{horizon}")

if j == 0:
    ax.set_ylabel("Proporción")

if i == len(BAR_PERCENTILES) - 1:
    ax.set_xticks(np.arange(len(contracts_order)))
    ax.set_xticklabels(contracts_order, rotation=45, ha="right")

ax.yaxis.set_major_formatter(
    plt.FuncFormatter(lambda y, _: f"{y:.0%}")
)

ax.grid(True, alpha=0.3)

handles, labels = axes[0, 0].get_legend_handles_labels()

fig.legend(
    handles,
    labels,
    loc="upper center",
    ncol=3,
    bbox_to_anchor=(0.5, 1.03)
)

fig.suptitle(
    "Candidata TP 1.5 / SL 1.0 por contrato – todos los percentiles y horizontes"
    fontsize=14,
    y=1.08

```



```

)

plt.tight_layout()
plt.show()

# =====
# Figura 6 – Candidata TP 1.5 / SL 1.0 por año
# =====

df_candidate_year_plot = df_bar_context_candidate_tp15_sl10_year.copy()
df_candidate_year_plot = df_candidate_year_plot.sort_values(["percentile", "horizon"])

fig, axes = plt.subplots(
    nrows=len(BAR_PERCENTILES),
    ncols=len(BAR_HORIZONS),
    figsize=(18, 12),
    sharex=True,
    sharey=True
)

for i, percentile in enumerate(BAR_PERCENTILES):
    for j, horizon in enumerate(BAR_HORIZONS):

        ax = axes[i, j]

        tmp = df_candidate_year_plot[
            (df_candidate_year_plot["percentile"] == percentile)
            & (df_candidate_year_plot["horizon"] == horizon)
        ].copy()

        tmp = tmp.sort_values("year")

        ax.plot(
            tmp["year"],
            tmp["tp_rate"],
            marker="o",
            linewidth=1.2,
            label="TP rate"
        )

        ax.plot(
            tmp["year"],
            tmp["sl_rate"],
            marker="o",
            linewidth=1.2,
            label="SL rate"
        )

        ax.plot(
            tmp["year"],
            tmp["no_event_rate"],
            marker="o",
            linewidth=1.2,
            label="no_event"
        )

        ax.set_title(f"p{percentile} | H{horizon}")

        if j == 0:
            ax.set_ylabel("Proporción")

```

```

        if i == len(BAR_PERCENTILES) - 1:
            ax.set_xlabel("Año")

        ax.yaxis.set_major_formatter(
            plt.FuncFormatter(lambda y, _: f"{y:.0%}")
        )

        ax.grid(True, alpha=0.3)

handles, labels = axes[0, 0].get_legend_handles_labels()

fig.legend(
    handles,
    labels,
    loc="upper center",
    ncol=3,
    bbox_to_anchor=(0.5, 1.03)
)

fig.suptitle(
    "Candidata TP 1.5 / SL 1.0 por año – todos los percentiles y horizontes",
    fontsize=14,
    y=1.08
)

plt.tight_layout()
plt.show()

# =====
# Figura 7 – Sensibilidad agregada por tipo de contexto
# =====

df_context_type_plot = df_bar_context_sensitivity_by_context_type.copy()

fig, ax = plt.subplots(figsize=(14, 6))

x = np.arange(len(df_context_type_plot))

ax.bar(
    x,
    df_context_type_plot["context_sensitivity_score_mean"],
    width=0.6
)

ax.set_title("Sensibilidad promedio por tipo de contexto")
ax.set_ylabel("context_sensitivity_score promedio")
ax.set_xticks(x)
ax.set_xticklabels(df_context_type_plot["context_type"], rotation=45, ha="right")

for i, row in df_context_type_plot.iterrows():
    ax.text(
        i,
        row["context_sensitivity_score_mean"],
        f"{row['context_sensitivity_score_mean']:.2f}",
        ha="center",
        va="bottom",
        fontsize=8
    )

```

```

plt.tight_layout()
plt.show()

# =====
# Figura 8 – Sensibilidad por configuración TP/SL dentro de cada contexto
# =====

df_sensitivity_config_plot = df_bar_context_sensitivity_by_config.copy()
df_sensitivity_config_plot = add_config_label_and_order(df_sensitivity_config_pl

context_types_order = (
    df_sensitivity_config_plot["context_type"]
    .drop_duplicates()
    .tolist()
)

fig, axes = plt.subplots(
    nrows=len(context_types_order),
    ncols=1,
    figsize=(15, 3.2 * len(context_types_order)),
    sharex=True
)

if len(context_types_order) == 1:
    axes = [axes]

for ax, context_type in zip(axes, context_types_order):

    tmp = df_sensitivity_config_plot[
        df_sensitivity_config_plot["context_type"] == context_type
    ].copy()

    tmp = (
        tmp
        .sort_values("config_order")
        .reset_index(drop=True)
    )

    x = np.arange(len(tmp))

    ax.bar(
        x,
        tmp["context_sensitivity_score_mean"],
        width=0.6
    )

    ax.set_title(f"Contexto: {context_type}")
    ax.set_ylabel("Score promedio")

    for i, row in tmp.iterrows():
        ax.text(
            i,
            row["context_sensitivity_score_mean"],
            f"{row['context_sensitivity_score_mean']:.2f}",
            ha="center",
            va="bottom",
            fontsize=8
        )

axes[-1].set_xticks(np.arange(len(config_labels)))

```

```

axes[-1].set_xticklabels(config_labels, rotation=45, ha="right")

fig.suptitle(
    "Sensibilidad por configuración TP/SL dentro de cada contexto",
    fontsize=14,
    y=1.01
)

plt.tight_layout()
plt.show()

# =====
# Figura 9 – Heatmaps de sensibilidad por target_base y configuración
# Para contextos principales
# =====

main_context_types = [
    "contract",
    "contract_code",
    "year",
    "contract_regime",
    "year_regime",
]

# Para este heatmap necesitamos una tabla que todavía conserve
# tp_mult y sl_mult. Por eso usamos df_bar_context_sensitivity_by_target,
# no df_bar_context_sensitivity_by_target_base.
df_sensitivity_target_plot = df_bar_context_sensitivity_by_target.copy()
df_sensitivity_target_plot = add_config_label_and_order(df_sensitivity_target_pl

for context_type in main_context_types:

    tmp = df_sensitivity_target_plot[
        df_sensitivity_target_plot["context_type"] == context_type
    ].copy()

    if tmp.empty:
        continue

    pivot_tmp = (
        tmp
        .pivot_table(
            index="target_base",
            columns="config_label",
            values="context_sensitivity_score",
            aggfunc="first"
        )
        .reindex(index=target_base_order)
        .reindex(columns=config_labels)
    )

    plot_heatmap_with_values(
        pivot_data=pivot_tmp,
        title=f"Sensibilidad por target y configuración – contexto: {context_type}",
        value_format="{:.2f}",
        figsize=(15, 7),
        colorbar_label="context_sensitivity_score"
    )

# =====
# Figura 10 – Heatmap por contrato y configuración TP/SL

```

```

# Promedio de Los 9 targets base para cada configuración
# =====

df_contract_config_plot = df_bar_context_contract_by_config.copy()
df_contract_config_plot = add_contract_sort(df_contract_config_plot, contract_co
df_contract_config_plot = add_config_label_and_order(df_contract_config_plot)

pivot_contract_tp = (
    df_contract_config_plot
    .pivot_table(
        index="contract",
        columns="config_label",
        values="tp_rate_mean",
        aggfunc="first"
    )
    .reindex(index=contracts_order)
    .reindex(columns=config_labels)
)

plot_heatmap_with_values(
    pivot_data=pivot_contract_tp,
    title="TP rate promedio por contrato y configuración TP/SL",
    value_format="{:.0%}",
    figsize=(15, 10),
    colorbar_label="TP rate",
    vmin=0,
    vmax=1
)

pivot_contract_no_event = (
    df_contract_config_plot
    .pivot_table(
        index="contract",
        columns="config_label",
        values="no_event_rate_mean",
        aggfunc="first"
    )
    .reindex(index=contracts_order)
    .reindex(columns=config_labels)
)

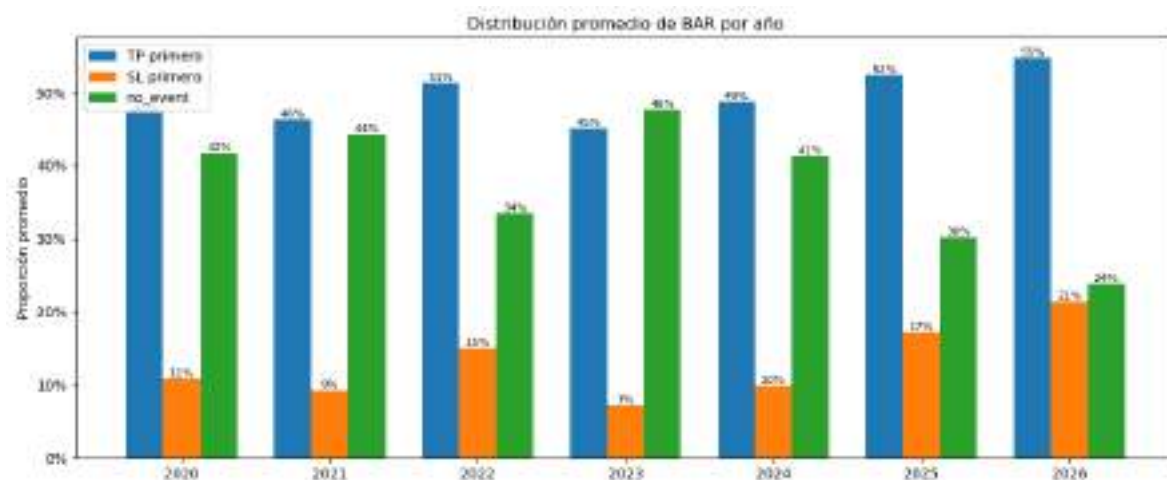
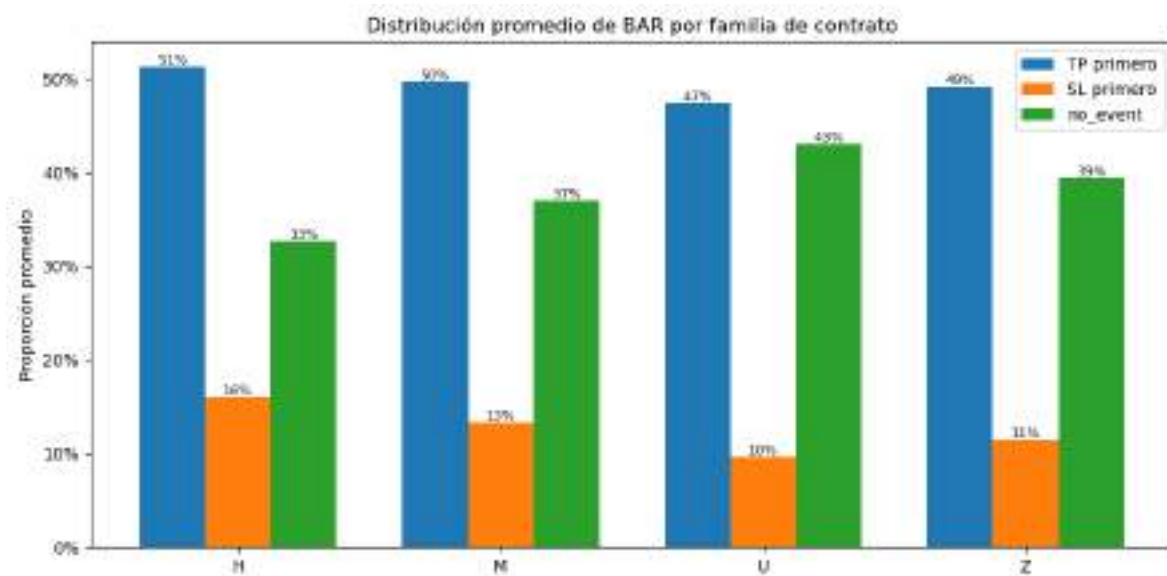
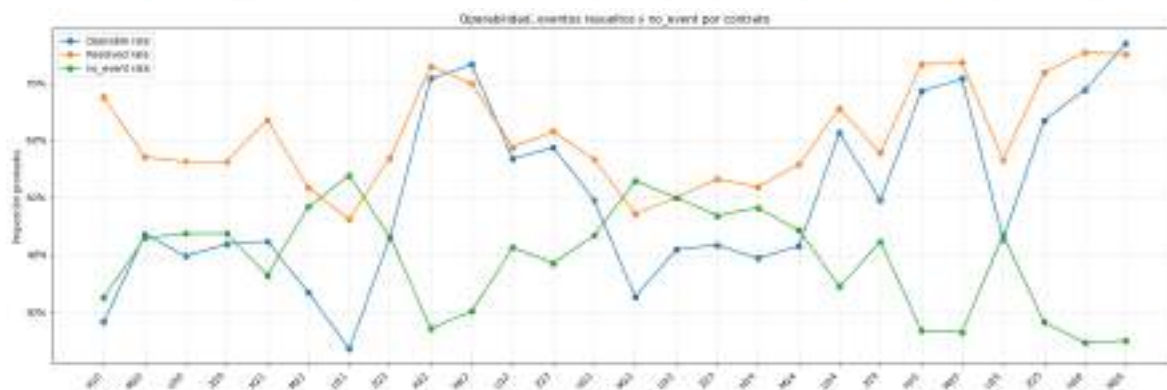
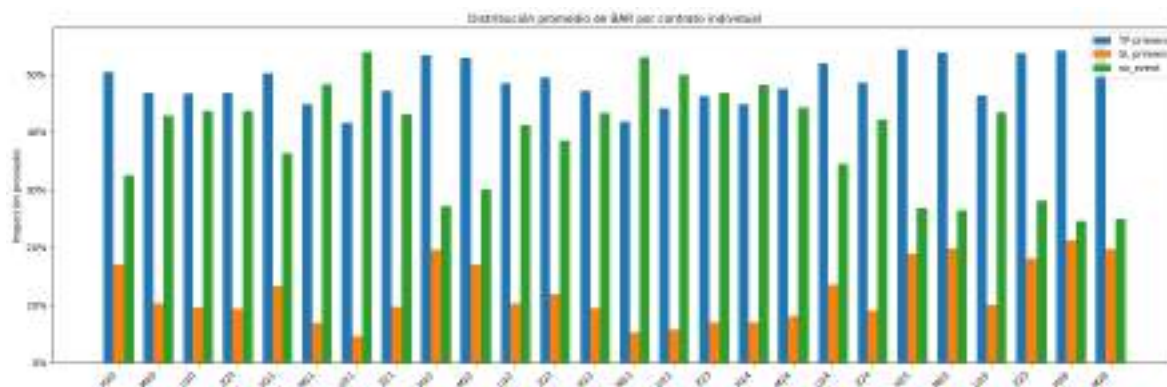
plot_heatmap_with_values(
    pivot_data=pivot_contract_no_event,
    title="no_event rate promedio por contrato y configuración TP/SL",
    value_format="{:.0%}",
    figsize=(15, 10),
    colorbar_label="no_event rate",
    vmin=0,
    vmax=1
)

print("\nCelda 3 finalizada correctamente.")

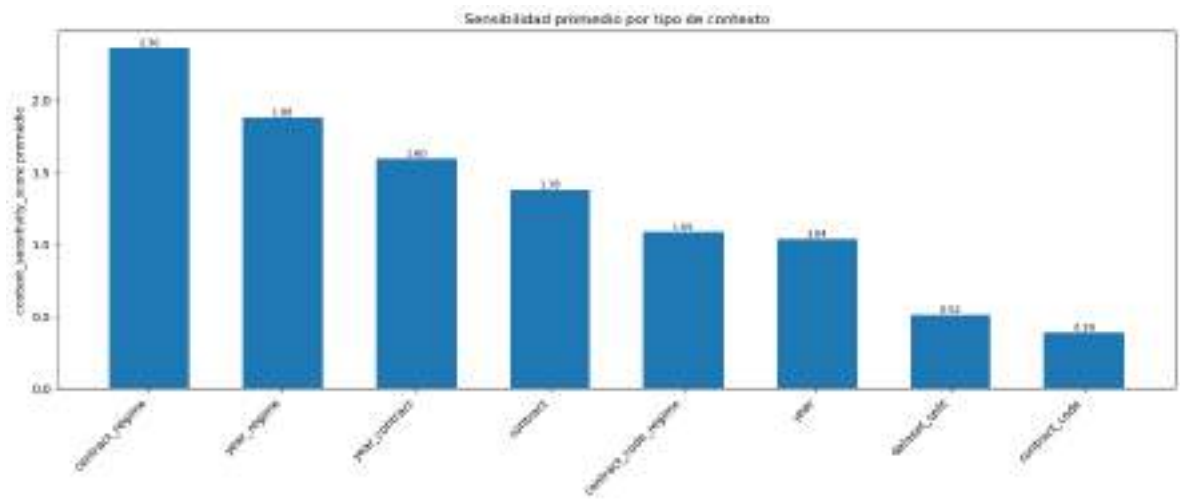
```

Gráficos agrupados por contrato y contexto - BAR

=====

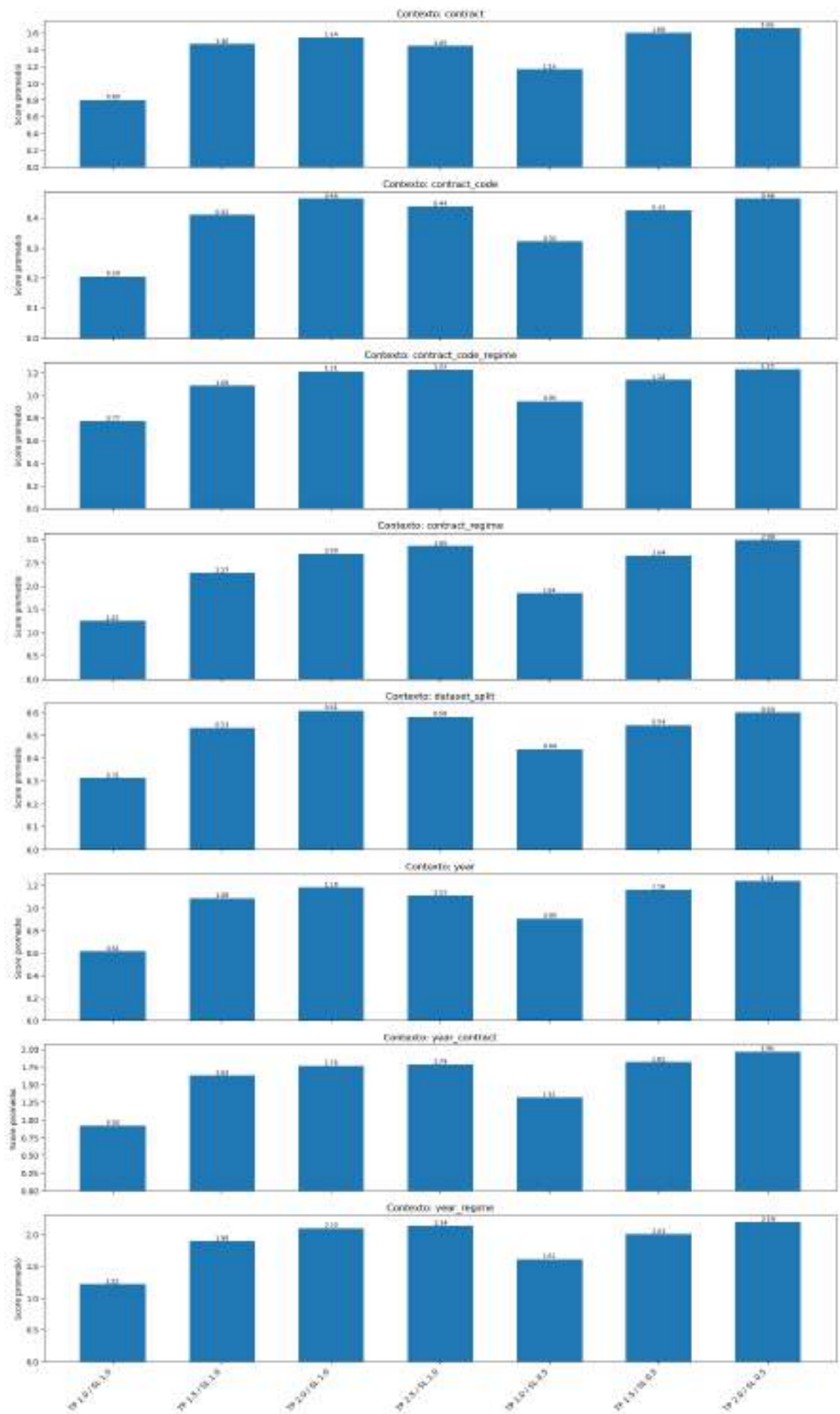


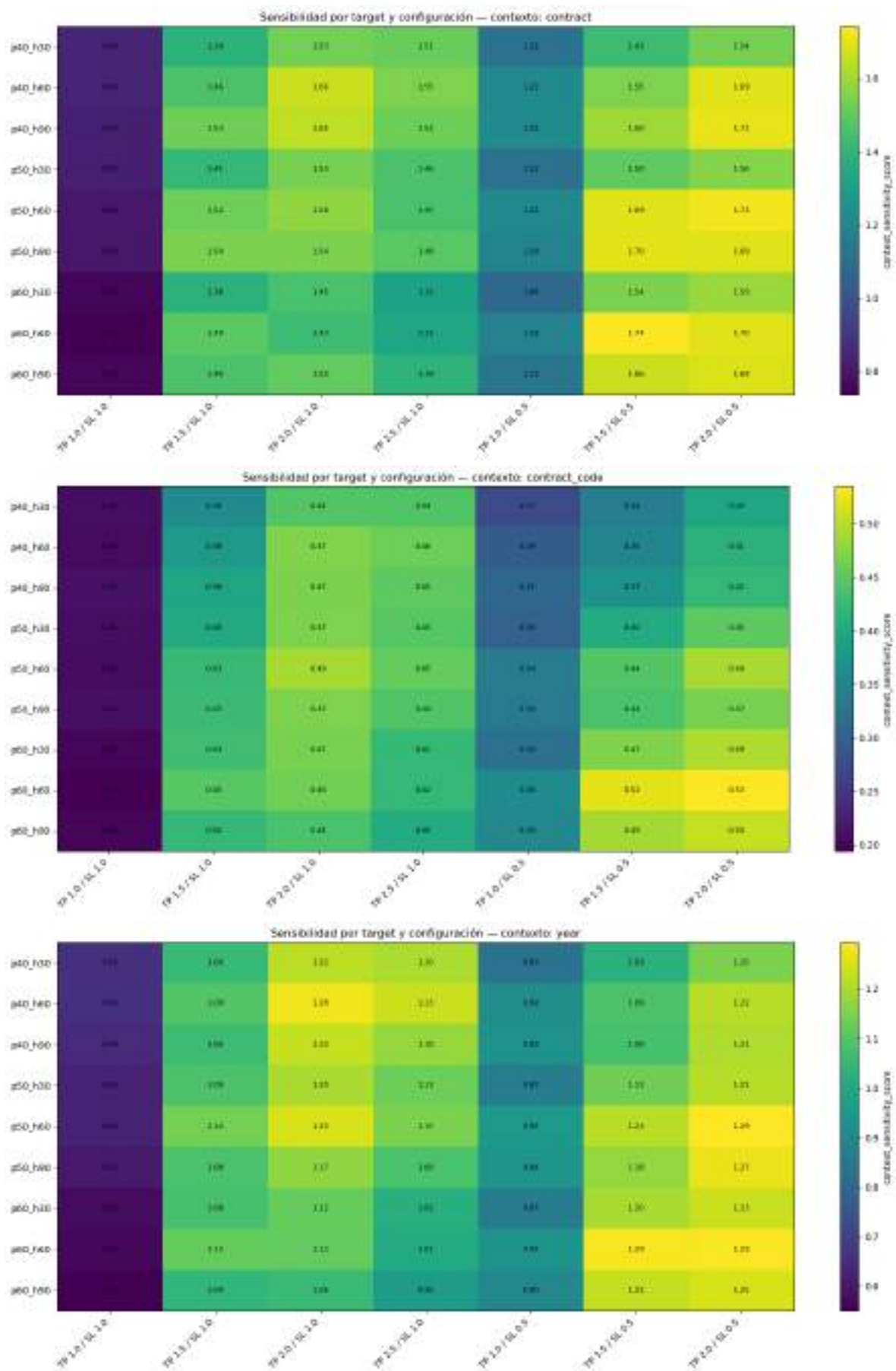


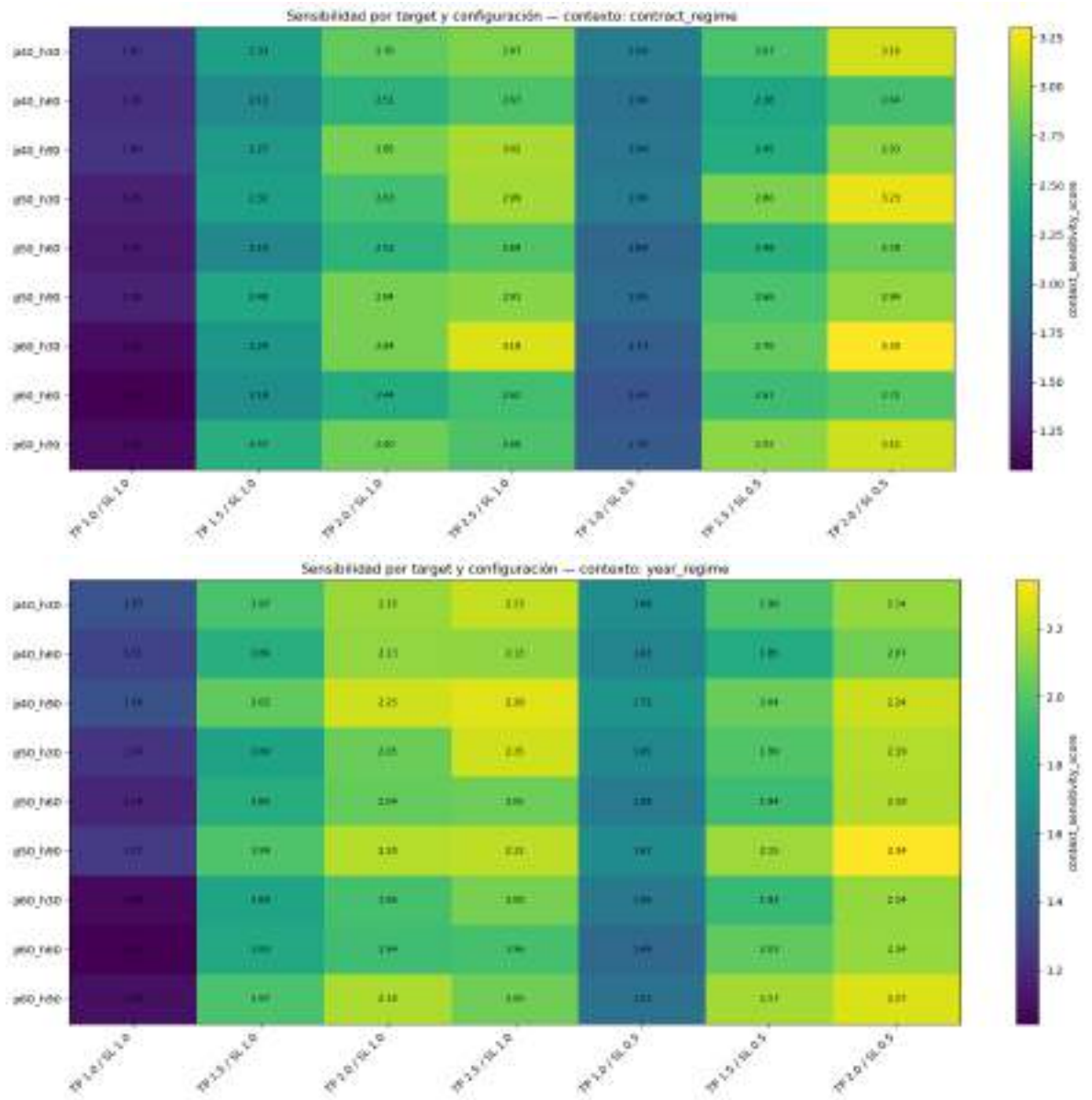


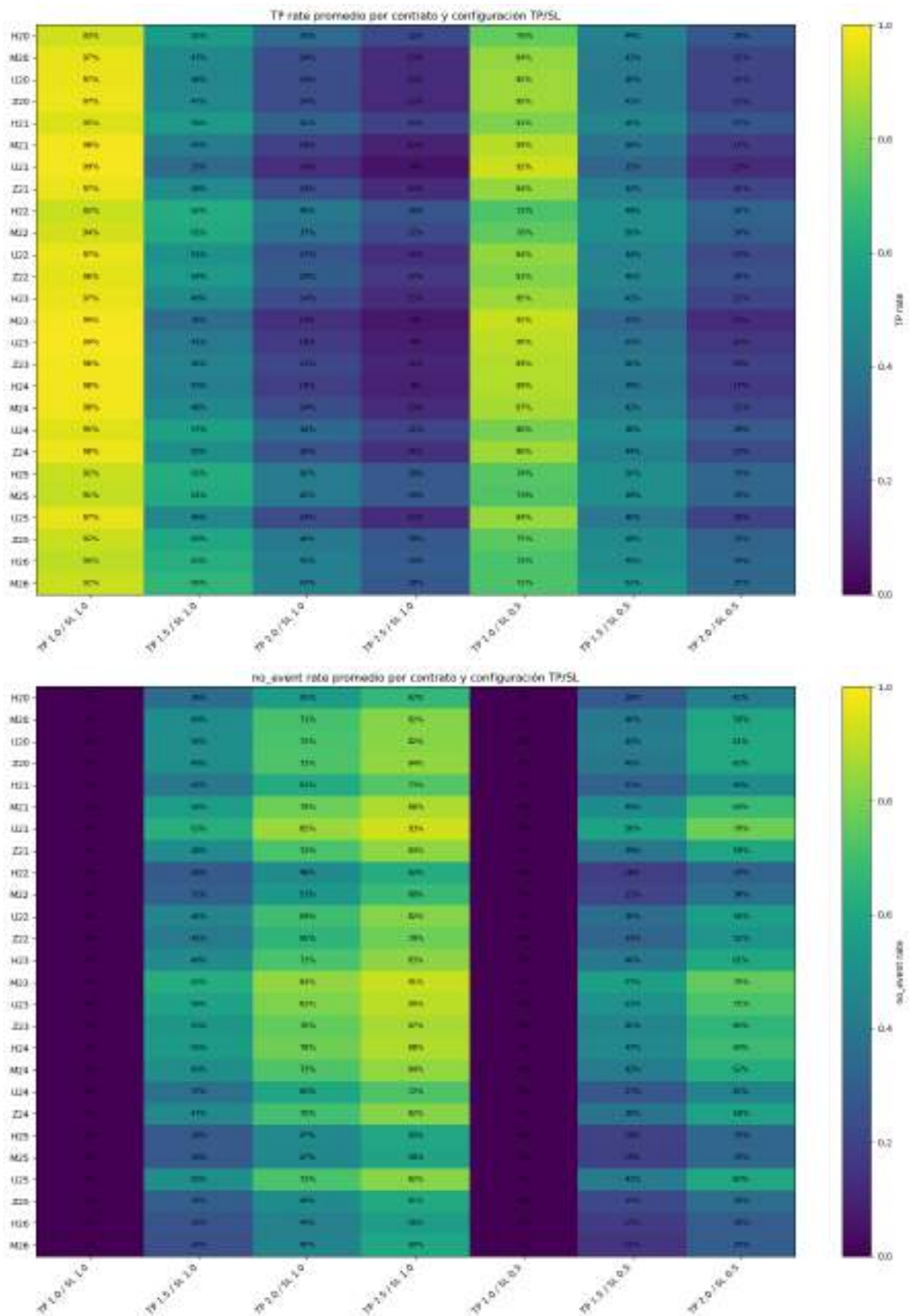


Sensibilidad por configuración TP/SL dentro de cada contexto









Celda 3 finalizada correctamente.

## Celda 4 — Guardado y tabla final para observaciones

In [118...

```
# Celda 4 - Guardado y tabla final para observaciones
# =====

print("Guardado y tabla final de análisis por contrato y contexto - BAR")
print("=" * 80)
```

```

# -----
# Validar objetos necesarios
# -----

required_objects = [
    "df_bar_context_analysis",
    "df_bar_context_contract",
    "df_bar_context_contract_code",
    "df_bar_context_year",
    "df_bar_context_dataset_split",
    "df_bar_context_year_contract",
    "df_bar_context_contract_regime",
    "df_bar_context_year_regime",
    "df_bar_context_contract_code_regime",

    "df_bar_context_by_context_type",
    "df_bar_context_by_contract",
    "df_bar_context_by_contract_code",
    "df_bar_context_by_year",
    "df_bar_context_by_dataset_split",
    "df_bar_context_contract_by_config",

    "df_bar_context_candidate_tp15_sl10_contract",
    "df_bar_context_candidate_tp15_sl10_contract_code",
    "df_bar_context_candidate_tp15_sl10_year",

    "df_bar_context_sensitivity_by_target",
    "df_bar_context_sensitivity_by_context_type",
    "df_bar_context_sensitivity_by_config",
    "df_bar_context_sensitivity_by_target_base",

    "MNQ_TARGETS_PATH",
]

missing_objects = [
    obj for obj in required_objects
    if obj not in globals()
]

if missing_objects:
    raise NameError(
        "Faltan objetos necesarios. "
        f"Debes correr las celdas anteriores: {missing_objects}"
    )

# -----
# Definir rutas de guardado
# -----

MNQ_TARGETS_BAR_CONTEXT_ANALYSIS_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_context_analysis.csv"
)

MNQ_TARGETS_BAR_CONTEXT_CONTRACT_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_context_contract.csv"
)

MNQ_TARGETS_BAR_CONTEXT_CONTRACT_CODE_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_context_contract_code.csv"
)

```



```
)

MNQ_TARGETS_BAR_CONTEXT_YEAR_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_context_year.csv"
)

MNQ_TARGETS_BAR_CONTEXT_DATASET_SPLIT_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_context_dataset_split.csv"
)

MNQ_TARGETS_BAR_CONTEXT_YEAR_CONTRACT_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_context_year_contract.csv"
)

MNQ_TARGETS_BAR_CONTEXT_CONTRACT_REGIME_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_context_contract_regime.csv"
)

MNQ_TARGETS_BAR_CONTEXT_YEAR_REGIME_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_context_year_regime.csv"
)

MNQ_TARGETS_BAR_CONTEXT_CONTRACT_CODE_REGIME_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_context_contract_code_regime.csv"
)

MNQ_TARGETS_BAR_CONTEXT_BY_CONTEXT_TYPE_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_context_by_context_type.csv"
)

MNQ_TARGETS_BAR_CONTEXT_BY_CONTRACT_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_context_by_contract.csv"
)

MNQ_TARGETS_BAR_CONTEXT_BY_CONTRACT_CODE_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_context_by_contract_code.csv"
)

MNQ_TARGETS_BAR_CONTEXT_BY_YEAR_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_context_by_year.csv"
)

MNQ_TARGETS_BAR_CONTEXT_BY_DATASET_SPLIT_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_context_by_dataset_split.csv"
)

MNQ_TARGETS_BAR_CONTEXT_CONTRACT_BY_CONFIG_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_context_contract_by_config.csv"
)

MNQ_TARGETS_BAR_CONTEXT_CANDIDATE_TP15_SL10_CONTRACT_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_context_candidate_tp15_sl10_contract.csv"
)

MNQ_TARGETS_BAR_CONTEXT_CANDIDATE_TP15_SL10_CONTRACT_CODE_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_context_candidate_tp15_sl10_contract_cod
)

MNQ_TARGETS_BAR_CONTEXT_CANDIDATE_TP15_SL10_YEAR_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_context_candidate_tp15_sl10_year.csv"
```

```

)

MNQ_TARGETS_BAR_CONTEXT_SENSITIVITY_BY_TARGET_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_context_sensitivity_by_target.csv"
)

MNQ_TARGETS_BAR_CONTEXT_SENSITIVITY_BY_CONTEXT_TYPE_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_context_sensitivity_by_context_type.csv"
)

MNQ_TARGETS_BAR_CONTEXT_SENSITIVITY_BY_CONFIG_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_context_sensitivity_by_config.csv"
)

MNQ_TARGETS_BAR_CONTEXT_SENSITIVITY_BY_TARGET_BASE_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_context_sensitivity_by_target_base.csv"
)

MNQ_TARGETS_BAR_CONTEXT_FINAL_TABLE_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_context_final_table.csv"
)

MNQ_TARGETS_BAR_CONTEXT_PROFILE_COUNTS_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_context_profile_counts.csv"
)

MNQ_TARGETS_BAR_CONTEXT_REPORT_TXT = (
    MNQ_TARGETS_PATH / "mnq_targets_bar_context_report.txt"
)

# -----
# Crear tabla final de sensibilidad por contexto
# -----

df_bar_context_final_table = df_bar_context_sensitivity_by_target.copy()

if "context_type" not in df_bar_context_final_table.columns:
    raise KeyError(
        "No existe la columna context_type en df_bar_context_sensitivity_by_targ
        "Revisa la salida de la Celda 2."
    )

# -----
# Clasificación relativa dentro de cada tipo de contexto
# -----
# Menor score = menor dependencia de ese contexto.
# Mayor score = mayor dependencia de ese contexto.
#
# Se evita groupby().apply() para no perder la columna context_type
# en algunas versiones de pandas.
# -----

classified_context_groups = []

for context_type, group in df_bar_context_final_table.groupby("context_type", dr

    group = group.copy()

    q33 = group["context_sensitivity_score"].quantile(0.33)
    q66 = group["context_sensitivity_score"].quantile(0.66)

```

```

group["context_sensitivity_profile"] = np.where(
    group["context_sensitivity_score"].isna(),
    "sin_clasificar",
    np.where(
        group["context_sensitivity_score"] <= q33,
        "menor_sensibilidad",
        np.where(
            group["context_sensitivity_score"] <= q66,
            "sensibilidad_intermedia",
            "mayor_sensibilidad"
        )
    )
)

classified_context_groups.append(group)

df_bar_context_final_table = (
    pd.concat(classified_context_groups, ignore_index=True)
    if classified_context_groups
    else pd.DataFrame()
)

# -----
# Validar columnas necesarias para ordenar
# -----

sort_cols = [
    "context_type",
    "context_sensitivity_score",
    "percentile",
    "horizon",
    "sl_mult",
    "tp_mult",
]

missing_sort_cols = [
    col for col in sort_cols
    if col not in df_bar_context_final_table.columns
]

if missing_sort_cols:
    raise KeyError(
        "Faltan columnas necesarias para ordenar df_bar_context_final_table: "
        f"{missing_sort_cols}\n\n"
        "Columnas disponibles:\n"
        f"{df_bar_context_final_table.columns.tolist()}"
    )

df_bar_context_final_table = (
    df_bar_context_final_table
    .sort_values(
        sort_cols,
        ascending=[True, True, True, True, True, True]
    )
    .reset_index(drop=True)
)

# -----
# Conteo por perfil de sensibilidad

```



```

# -----
df_bar_context_sensitivity_profile_counts = (
    df_bar_context_final_table
    .groupby(
        [
            "context_type",
            "context_sensitivity_profile",
        ],
        as_index=False
    )
    .agg(
        n_targets=("bar_col", "count"),

        context_sensitivity_score_mean=("context_sensitivity_score", "mean"),

        operable_rate_range_mean=("operable_rate_range", "mean"),
        tp_rate_range_mean=("tp_rate_range", "mean"),
        sl_rate_range_mean=("sl_rate_range", "mean"),
        no_event_rate_range_mean=("no_event_rate_range", "mean"),
        tp_share_range_mean=("tp_share_over_resolved_range", "mean"),

        expectancy_range_mean=("expectancy_units_range", "mean"),
        event_step_range_mean=("event_step_mean_range", "mean"),
    )
    .sort_values(["context_type", "context_sensitivity_score_mean"])
    .reset_index(drop=True)
)

# -----
# Guardar tablas principales
# -----

df_bar_context_analysis.to_csv(
    MNQ_TARGETS_BAR_CONTEXT_ANALYSIS_CSV,
    index=False
)

df_bar_context_contract.to_csv(
    MNQ_TARGETS_BAR_CONTEXT_CONTRACT_CSV,
    index=False
)

df_bar_context_contract_code.to_csv(
    MNQ_TARGETS_BAR_CONTEXT_CONTRACT_CODE_CSV,
    index=False
)

df_bar_context_year.to_csv(
    MNQ_TARGETS_BAR_CONTEXT_YEAR_CSV,
    index=False
)

df_bar_context_dataset_split.to_csv(
    MNQ_TARGETS_BAR_CONTEXT_DATASET_SPLIT_CSV,
    index=False
)

df_bar_context_year_contract.to_csv(
    MNQ_TARGETS_BAR_CONTEXT_YEAR_CONTRACT_CSV,

```

```
        index=False
    )

    df_bar_context_contract_regime.to_csv(
        MNQ_TARGETS_BAR_CONTEXT_CONTRACT_REGIME_CSV,
        index=False
    )

    df_bar_context_year_regime.to_csv(
        MNQ_TARGETS_BAR_CONTEXT_YEAR_REGIME_CSV,
        index=False
    )

    df_bar_context_contract_code_regime.to_csv(
        MNQ_TARGETS_BAR_CONTEXT_CONTRACT_CODE_REGIME_CSV,
        index=False
    )

    df_bar_context_by_context_type.to_csv(
        MNQ_TARGETS_BAR_CONTEXT_BY_CONTEXT_TYPE_CSV,
        index=False
    )

    df_bar_context_by_contract.to_csv(
        MNQ_TARGETS_BAR_CONTEXT_BY_CONTRACT_CSV,
        index=False
    )

    df_bar_context_by_contract_code.to_csv(
        MNQ_TARGETS_BAR_CONTEXT_BY_CONTRACT_CODE_CSV,
        index=False
    )

    df_bar_context_by_year.to_csv(
        MNQ_TARGETS_BAR_CONTEXT_BY_YEAR_CSV,
        index=False
    )

    df_bar_context_by_dataset_split.to_csv(
        MNQ_TARGETS_BAR_CONTEXT_BY_DATASET_SPLIT_CSV,
        index=False
    )

    df_bar_context_contract_by_config.to_csv(
        MNQ_TARGETS_BAR_CONTEXT_CONTRACT_BY_CONFIG_CSV,
        index=False
    )

    df_bar_context_candidate_tp15_sl10_contract.to_csv(
        MNQ_TARGETS_BAR_CONTEXT_CANDIDATE_TP15_SL10_CONTRACT_CSV,
        index=False
    )

    df_bar_context_candidate_tp15_sl10_contract_code.to_csv(
        MNQ_TARGETS_BAR_CONTEXT_CANDIDATE_TP15_SL10_CONTRACT_CODE_CSV,
        index=False
    )

    df_bar_context_candidate_tp15_sl10_year.to_csv(
        MNQ_TARGETS_BAR_CONTEXT_CANDIDATE_TP15_SL10_YEAR_CSV,
```

```
        index=False
    )

    df_bar_context_sensitivity_by_target.to_csv(
        MNQ_TARGETS_BAR_CONTEXT_SENSITIVITY_BY_TARGET_CSV,
        index=False
    )

    df_bar_context_sensitivity_by_context_type.to_csv(
        MNQ_TARGETS_BAR_CONTEXT_SENSITIVITY_BY_CONTEXT_TYPE_CSV,
        index=False
    )

    df_bar_context_sensitivity_by_config.to_csv(
        MNQ_TARGETS_BAR_CONTEXT_SENSITIVITY_BY_CONFIG_CSV,
        index=False
    )

    df_bar_context_sensitivity_by_target_base.to_csv(
        MNQ_TARGETS_BAR_CONTEXT_SENSITIVITY_BY_TARGET_BASE_CSV,
        index=False
    )

    df_bar_context_final_table.to_csv(
        MNQ_TARGETS_BAR_CONTEXT_FINAL_TABLE_CSV,
        index=False
    )

    df_bar_context_sensitivity_profile_counts.to_csv(
        MNQ_TARGETS_BAR_CONTEXT_PROFILE_COUNTS_CSV,
        index=False
    )

    # -----
    # Columnas para impresión
    # -----

    context_type_cols = [
        "context_type",
        "context_description",
        "n_rows",
        "n_targets",
        "n_context_values",
        "operable_rate_mean",
        "tp_rate_mean",
        "sl_rate_mean",
        "no_event_rate_mean",
        "resolved_rate_mean",
        "tp_share_over_resolved_mean",
        "expectancy_units_mean",
        "event_step_mean",
    ]

    contract_cols = [
        "contract",
        "n_targets",
        "total_rows_mean",
        "valid_rows_mean",
        "operable_rate_mean",
        "tp_rate_mean",
```

```
        "sl_rate_mean",
        "no_event_rate_mean",
        "resolved_rate_mean",
        "tp_share_over_resolved_mean",
        "expectancy_units_mean",
        "event_step_mean",
    ]

    contract_code_cols = [
        "contract_code",
        "n_targets",
        "total_rows_mean",
        "valid_rows_mean",
        "operable_rate_mean",
        "tp_rate_mean",
        "sl_rate_mean",
        "no_event_rate_mean",
        "resolved_rate_mean",
        "tp_share_over_resolved_mean",
        "expectancy_units_mean",
        "event_step_mean",
    ]

    year_cols = [
        "year",
        "n_targets",
        "total_rows_mean",
        "valid_rows_mean",
        "operable_rate_mean",
        "tp_rate_mean",
        "sl_rate_mean",
        "no_event_rate_mean",
        "resolved_rate_mean",
        "tp_share_over_resolved_mean",
        "expectancy_units_mean",
        "event_step_mean",
    ]

    dataset_split_cols = [
        "dataset_split",
        "n_targets",
        "total_rows_mean",
        "valid_rows_mean",
        "operable_rate_mean",
        "tp_rate_mean",
        "sl_rate_mean",
        "no_event_rate_mean",
        "resolved_rate_mean",
        "tp_share_over_resolved_mean",
        "expectancy_units_mean",
        "event_step_mean",
    ]

    candidate_contract_cols = [
        "bar_col",
        "target_base",
        "contract",
        "total_rows",
        "valid_rows",
        "operable_rate",
```

```
"tp_rate",
"sl_rate",
"no_event_rate",
"resolved_rate",
"tp_share_over_resolved",
"expectancy_units",
"event_step_mean",
]

candidate_year_cols = [
    "bar_col",
    "target_base",
    "year",
    "total_rows",
    "valid_rows",
    "operable_rate",
    "tp_rate",
    "sl_rate",
    "no_event_rate",
    "resolved_rate",
    "tp_share_over_resolved",
    "expectancy_units",
    "event_step_mean",
]

sensitivity_context_type_cols = [
    "context_type",
    "n_targets",
    "n_context_values_mean",
    "operable_rate_range_mean",
    "tp_rate_range_mean",
    "sl_rate_range_mean",
    "no_event_rate_range_mean",
    "tp_share_range_mean",
    "expectancy_range_mean",
    "event_step_range_mean",
    "context_sensitivity_score_mean",
    "context_sensitivity_score_max",
]

sensitivity_config_cols = [
    "context_type",
    "tp_sl_config",
    "n_targets",
    "operable_rate_range_mean",
    "tp_rate_range_mean",
    "sl_rate_range_mean",
    "no_event_rate_range_mean",
    "tp_share_range_mean",
    "expectancy_range_mean",
    "context_sensitivity_score_mean",
    "context_sensitivity_score_max",
]

sensitivity_target_base_cols = [
    "context_type",
    "target_base",
    "percentile",
    "horizon",
    "n_configs",
```

```

        "operable_rate_range_mean",
        "tp_rate_range_mean",
        "sl_rate_range_mean",
        "no_event_rate_range_mean",
        "tp_share_range_mean",
        "expectancy_range_mean",
        "context_sensitivity_score_mean",
        "context_sensitivity_score_max",
    ]

    final_table_cols = [
        "context_type",
        "bar_col",
        "target_base",
        "tp_sl_config",
        "n_context_values",
        "operable_rate_mean",
        "operable_rate_range",
        "tp_rate_mean",
        "tp_rate_range",
        "sl_rate_mean",
        "sl_rate_range",
        "no_event_rate_mean",
        "no_event_rate_range",
        "resolved_rate_mean",
        "resolved_rate_range",
        "tp_share_over_resolved_mean",
        "tp_share_over_resolved_range",
        "expectancy_units_mean",
        "expectancy_units_range",
        "event_step_mean_mean",
        "event_step_mean_range",
        "context_sensitivity_score",
        "context_sensitivity_profile",
    ]

    profile_count_cols = [
        "context_type",
        "context_sensitivity_profile",
        "n_targets",
        "context_sensitivity_score_mean",
        "operable_rate_range_mean",
        "tp_rate_range_mean",
        "sl_rate_range_mean",
        "no_event_rate_range_mean",
        "tp_share_range_mean",
        "expectancy_range_mean",
        "event_step_range_mean",
    ]

    # -----
    # Formateadores
    # -----

    percent_formatters = {
        "operable_rate_mean": "{:.2%}".format,
        "tp_rate_mean": "{:.2%}".format,
        "sl_rate_mean": "{:.2%}".format,
        "no_event_rate_mean": "{:.2%}".format,
        "resolved_rate_mean": "{:.2%}".format,
    }

```

```

"tp_share_over_resolved_mean": "{:.2%}".format,

"operable_rate": "{:.2%}".format,
"tp_rate": "{:.2%}".format,
"sl_rate": "{:.2%}".format,
"no_event_rate": "{:.2%}".format,
"resolved_rate": "{:.2%}".format,
"tp_share_over_resolved": "{:.2%}".format,

"operable_rate_range": "{:.2%}".format,
"tp_rate_range": "{:.2%}".format,
"sl_rate_range": "{:.2%}".format,
"no_event_rate_range": "{:.2%}".format,
"resolved_rate_range": "{:.2%}".format,
"tp_share_over_resolved_range": "{:.2%}".format,

"operable_rate_range_mean": "{:.2%}".format,
"tp_rate_range_mean": "{:.2%}".format,
"sl_rate_range_mean": "{:.2%}".format,
"no_event_rate_range_mean": "{:.2%}".format,
"tp_share_range_mean": "{:.2%}".format,
}

numeric_formatters = {
    "total_rows_mean": "{:.0f}".format,
    "valid_rows_mean": "{:.0f}".format,
    "expectancy_units_mean": "{:.3f}".format,
    "expectancy_units": "{:.3f}".format,
    "expectancy_units_range": "{:.3f}".format,
    "expectancy_range_mean": "{:.3f}".format,
    "event_step_mean": "{:.2f}".format,
    "event_step_mean_mean": "{:.2f}".format,
    "event_step_mean_range": "{:.2f}".format,
    "event_step_range_mean": "{:.2f}".format,
    "n_context_values_mean": "{:.1f}".format,
    "context_sensitivity_score": "{:.3f}".format,
    "context_sensitivity_score_mean": "{:.3f}".format,
    "context_sensitivity_score_max": "{:.3f}".format,
}

formatters = {
    **percent_formatters,
    **numeric_formatters,
}

# -----
# Impresiones principales
# -----

print("\nResumen general por tipo de contexto:")
print(
    df_bar_context_by_context_type[context_type_cols]
    .to_string(
        index=False,
        formatters=formatters
    )
)

print("\nResumen por contrato individual:")
print(

```

```
df_bar_context_by_contract[contract_cols]
.to_string(
    index=False,
    formatters=formatters
)
)

print("\nResumen por familia de contrato:")
print(
    df_bar_context_by_contract_code[contract_code_cols]
    .to_string(
        index=False,
        formatters=formatters
    )
)

print("\nResumen por año:")
print(
    df_bar_context_by_year[year_cols]
    .to_string(
        index=False,
        formatters=formatters
    )
)

print("\nResumen por split metodológico:")
print(
    df_bar_context_by_dataset_split[dataset_split_cols]
    .to_string(
        index=False,
        formatters=formatters
    )
)

print("\nDetalle por contrato para candidata TP 1.5 / SL 1.0:")
print(
    df_bar_context_candidate_tp15_sl10_contract[candidate_contract_cols]
    .head(120)
    .to_string(
        index=False,
        formatters=formatters
    )
)

print("\nDetalle por año para candidata TP 1.5 / SL 1.0:")
print(
    df_bar_context_candidate_tp15_sl10_year[candidate_year_cols]
    .to_string(
        index=False,
        formatters=formatters
    )
)

print("\nSensibilidad agregada por tipo de contexto:")
print(
    df_bar_context_sensitivity_by_context_type[sensitivity_context_type_cols]
    .to_string(
        index=False,
        formatters=formatters
    )
)
```



```

)

print("\nSensibilidad por configuración TP/SL dentro de cada contexto:")
print(
    df_bar_context_sensitivity_by_config[sensitivity_config_cols]
    .to_string(
        index=False,
        formatters=formatters
    )
)

print("\nSensibilidad por percentil/horizonte dentro de cada contexto:")
print(
    df_bar_context_sensitivity_by_target_base[sensitivity_target_base_cols]
    .to_string(
        index=False,
        formatters=formatters
    )
)

print("\nTabla final de sensibilidad por contrato/contexto:")
print(
    df_bar_context_final_table[final_table_cols]
    .to_string(
        index=False,
        formatters=formatters
    )
)

# -----
# Top targets por contextos principales
# -----

main_contexts_for_print = [
    "contract",
    "contract_code",
    "year",
    "year_contract",
    "contract_regime",
    "year_regime",
]

for context_type in main_contexts_for_print:

    tmp = df_bar_context_final_table[
        df_bar_context_final_table["context_type"] == context_type
    ].copy()

    if tmp.empty:
        continue

    print(f"\nTop 15 targets con menor sensibilidad – contexto: {context_type}")
    print(
        tmp[final_table_cols]
        .head(15)
        .to_string(
            index=False,
            formatters=formatters
        )
    )

```

```

print(f"\nTop 15 targets con mayor sensibilidad – contexto: {context_type}")
print(
    tmp[final_table_cols]
    .tail(15)
    .sort_values("context_sensitivity_score", ascending=False)
    .to_string(
        index=False,
        formatters=formatters
    )
)

# -----
# Conteo por perfil de sensibilidad
# -----

print("\nConteo por perfil de sensibilidad contextual:")
print(
    df_bar_context_sensitivity_profile_counts[profile_count_cols]
    .to_string(
        index=False,
        formatters=formatters
    )
)

# -----
# Guardar reporte TXT
# -----

with MNQ_TARGETS_BAR_CONTEXT_REPORT_TXT.open("w", encoding="utf-8") as f:
    f.write("4.11. Análisis por contrato y contexto - BAR\n")
    f.write("=" * 100 + "\n\n")

    f.write("Resumen general por tipo de contexto\n")
    f.write("-" * 100 + "\n")
    f.write(
        df_bar_context_by_context_type[context_type_cols]
        .to_string(
            index=False,
            formatters=formatters
        )
    )

    f.write("\n\n")
    f.write("Resumen por contrato individual\n")
    f.write("-" * 100 + "\n")
    f.write(
        df_bar_context_by_contract[contract_cols]
        .to_string(
            index=False,
            formatters=formatters
        )
    )

    f.write("\n\n")
    f.write("Resumen por familia de contrato\n")
    f.write("-" * 100 + "\n")
    f.write(
        df_bar_context_by_contract_code[contract_code_cols]
        .to_string(

```

```

        index=False,
        formatters=formatters
    )
)

f.write("\n\n")
f.write("Resumen por año\n")
f.write("-" * 100 + "\n")
f.write(
    df_bar_context_by_year[year_cols]
    .to_string(
        index=False,
        formatters=formatters
    )
)

f.write("\n\n")
f.write("Resumen por split metodológico\n")
f.write("-" * 100 + "\n")
f.write(
    df_bar_context_by_dataset_split[dataset_split_cols]
    .to_string(
        index=False,
        formatters=formatters
    )
)

f.write("\n\n")
f.write("Detalle por contrato para candidata TP 1.5 / SL 1.0\n")
f.write("-" * 100 + "\n")
f.write(
    df_bar_context_candidate_tp15_sl10_contract[candidate_contract_cols]
    .to_string(
        index=False,
        formatters=formatters
    )
)

f.write("\n\n")
f.write("Detalle por año para candidata TP 1.5 / SL 1.0\n")
f.write("-" * 100 + "\n")
f.write(
    df_bar_context_candidate_tp15_sl10_year[candidate_year_cols]
    .to_string(
        index=False,
        formatters=formatters
    )
)

f.write("\n\n")
f.write("Sensibilidad agregada por tipo de contexto\n")
f.write("-" * 100 + "\n")
f.write(
    df_bar_context_sensitivity_by_context_type[sensitivity_context_type_cols]
    .to_string(
        index=False,
        formatters=formatters
    )
)

```

```

f.write("\n\n")
f.write("Sensibilidad por configuración TP/SL dentro de cada contexto\n")
f.write("-" * 100 + "\n")
f.write(
    df_bar_context_sensitivity_by_config[sensitivity_config_cols]
    .to_string(
        index=False,
        formatters=formatters
    )
)

f.write("\n\n")
f.write("Sensibilidad por percentil/horizonte dentro de cada contexto\n")
f.write("-" * 100 + "\n")
f.write(
    df_bar_context_sensitivity_by_target_base[sensitivity_target_base_cols]
    .to_string(
        index=False,
        formatters=formatters
    )
)

f.write("\n\n")
f.write("Tabla final de sensibilidad por contrato/contexto\n")
f.write("-" * 100 + "\n")
f.write(
    df_bar_context_final_table[final_table_cols]
    .to_string(
        index=False,
        formatters=formatters
    )
)

f.write("\n\n")
f.write("Conteo por perfil de sensibilidad contextual\n")
f.write("-" * 100 + "\n")
f.write(
    df_bar_context_sensitivity_profile_counts[profile_count_cols]
    .to_string(
        index=False,
        formatters=formatters
    )
)

# -----
# Confirmación final
# -----

print("\nArchivos guardados:")
print(f"Análisis BAR por contexto: {MNQ_TARGETS_BAR_CONTEXT_ANAL}")
print(f"Contexto contrato individual: {MNQ_TARGETS_BAR_CONTEXT_CONT}")
print(f"Contexto familia de contrato: {MNQ_TARGETS_BAR_CONTEXT_CONT}")
print(f"Contexto año: {MNQ_TARGETS_BAR_CONTEXT_YEAR}")
print(f"Contexto dataset split: {MNQ_TARGETS_BAR_CONTEXT_DATA}")
print(f"Contexto año + contrato: {MNQ_TARGETS_BAR_CONTEXT_YEAR}")
print(f"Contexto contrato + régimen: {MNQ_TARGETS_BAR_CONTEXT_CONT}")
print(f"Contexto año + régimen: {MNQ_TARGETS_BAR_CONTEXT_YEAR}")
print(f"Contexto familia contrato + régimen: {MNQ_TARGETS_BAR_CONTEXT_CONT}")

print(f"Resumen por tipo de contexto: {MNQ_TARGETS_BAR_CONTEXT_BY_C")

```

```

print(f"Resumen por contrato:                                {MNQ_TARGETS_BAR_CONTEXT_BY_C
print(f"Resumen por familia de contrato:                    {MNQ_TARGETS_BAR_CONTEXT_BY_C
print(f"Resumen por año:                                    {MNQ_TARGETS_BAR_CONTEXT_BY_Y
print(f"Resumen por split metodológico:                     {MNQ_TARGETS_BAR_CONTEXT_BY_D
print(f"Resumen contrato por configuración:                 {MNQ_TARGETS_BAR_CONTEXT_CONT

print(f"Candidata TP 1.5 / SL 1.0 por contrato:            {MNQ_TARGETS_BAR_CONTEXT_CAND
print(f"Candidata TP 1.5 / SL 1.0 por familia:              {MNQ_TARGETS_BAR_CONTEXT_CAND
print(f"Candidata TP 1.5 / SL 1.0 por año:                   {MNQ_TARGETS_BAR_CONTEXT_CAND

print(f"Sensibilidad por target/contexto:                    {MNQ_TARGETS_BAR_CONTEXT_SENS
print(f"Sensibilidad por tipo de contexto:                   {MNQ_TARGETS_BAR_CONTEXT_SENS
print(f"Sensibilidad por configuración/contexto:             {MNQ_TARGETS_BAR_CONTEXT_SENS
print(f"Sensibilidad por percentil/horizonte:                {MNQ_TARGETS_BAR_CONTEXT_SENS

print(f"Tabla final contexto:                                {MNQ_TARGETS_BAR_CONTEXT_FINA
print(f"Conteo perfiles contexto:                            {MNQ_TARGETS_BAR_CONTEXT_PROF
print(f"Reporte TXT:                                          {MNQ_TARGETS_BAR_CONTEXT_REPO

print("\nCelda 4 finalizada correctamente.")

```

## Guardado y tabla final de análisis por contrato y contexto - BAR

=====

## Resumen general por tipo de contexto:

|                    | context_type                | context_description                       | n_rows          | n_targets          |
|--------------------|-----------------------------|-------------------------------------------|-----------------|--------------------|
| n_context_values   | operable_rate_mean          | tp_rate_mean                              | sl_rate_mean    | no_event_rate_mean |
| resolved_rate_mean | tp_share_over_resolved_mean | expectancy_units_mean                     | event_step_mean |                    |
|                    | contract                    | Contrato individual                       | 1638            | 63                 |
| 26                 | 50.08%                      | 48.79%                                    | 12.04%          | 39.17%             |
| 0.83%              |                             |                                           |                 | 6                  |
|                    | contract_code               | Familia de contrato H/M/U/Z               | 252             | 63                 |
| 4                  | 49.23%                      | 49.38%                                    | 12.62%          | 37.99%             |
| 01%                |                             |                                           |                 | 62.                |
|                    | contract_code_regime        | Familia de contrato + régimen intradiario | 1260            | 63                 |
| 20                 | 42.29%                      | 49.20%                                    | 12.25%          | 38.55%             |
| 1.45%              |                             |                                           |                 | 6                  |
|                    | contract_regime             | Contrato + régimen intradiario            | 8190            | 63                 |
| 130                | 42.98%                      | 48.44%                                    | 11.53%          | 40.03%             |
| 9.97%              |                             |                                           |                 | 5                  |
|                    | dataset_split               | Development vs final test                 | 126             | 63                 |
| 2                  | 54.66%                      | 50.70%                                    | 14.69%          | 34.61%             |
| 39%                |                             |                                           |                 | 65.                |
|                    | year                        | Año calendario                            | 441             | 63                 |
| 7                  | 51.86%                      | 49.48%                                    | 12.99%          | 37.53%             |
| 47%                |                             |                                           |                 | 62.                |
|                    | year_contract               | Año + contrato                            | 2016            | 63                 |
| 32                 | 47.87%                      | 48.51%                                    | 11.23%          | 40.26%             |
| 9.74%              |                             |                                           |                 | 5                  |
|                    | year_regime                 | Año + régimen intradiario                 | 2205            | 63                 |
| 35                 | 44.64%                      | 49.19%                                    | 12.55%          | 38.25%             |
| 1.75%              |                             |                                           |                 | 6                  |

## Resumen por contrato individual:

| contract        | n_targets    | total_rows_mean    | valid_rows_mean    | operable_rate_mean          | tp_rate_mean          |
|-----------------|--------------|--------------------|--------------------|-----------------------------|-----------------------|
| an              | sl_rate_mean | no_event_rate_mean | resolved_rate_mean | tp_share_over_resolved_mean | expectancy_units_mean |
| event_step_mean |              |                    |                    |                             |                       |
| H20             | 63           | 33168              | 9358               | 28.21%                      | 50.4                  |
| 3%              | 17.06%       | 32.52%             | 67.48%             |                             | 72.56%                |
| 0.600           | 26.29        |                    |                    |                             |                       |
| M20             | 63           | 38696              | 16840              | 43.52%                      | 46.7                  |
| 6%              | 10.33%       | 42.91%             | 57.09%             |                             | 78.44%                |
| 0.560           | 31.00        |                    |                    |                             |                       |
| U20             | 63           | 42151              | 16752              | 39.74%                      | 46.6                  |
| 7%              | 9.59%        | 43.74%             | 56.26%             |                             | 79.32%                |
| 0.561           | 31.38        |                    |                    |                             |                       |
| Z20             | 63           | 36623              | 15298              | 41.77%                      | 46.8                  |
| 0%              | 9.45%        | 43.75%             | 56.25%             |                             | 79.86%                |
| 0.563           | 32.01        |                    |                    |                             |                       |
| H21             | 63           | 40769              | 17248              | 42.31%                      | 50.2                  |
| 7%              | 13.31%       | 36.42%             | 63.58%             |                             | 76.85%                |
| 0.613           | 28.63        |                    |                    |                             |                       |
| M21             | 63           | 42151              | 14081              | 33.41%                      | 44.8                  |
| 0%              | 6.87%        | 48.33%             | 51.67%             |                             | 82.50%                |
| 0.537           | 34.42        |                    |                    |                             |                       |
| U21             | 63           | 41460              | 9779               | 23.59%                      | 41.6                  |
| 1%              | 4.56%        | 53.83%             | 46.17%             |                             | 84.46%                |
| 0.487           | 37.19        |                    |                    |                             |                       |
| Z21             | 63           | 42842              | 18397              | 42.94%                      | 47.1                  |
| 0%              | 9.78%        | 43.11%             | 56.89%             |                             | 79.81%                |

|       |        |       |        |        |        |        |
|-------|--------|-------|--------|--------|--------|--------|
| 0.568 |        | 32.50 |        |        |        |        |
| H22   |        | 63    | 42151  | 29843  | 70.80% | 53.3   |
| 4%    | 19.58% |       | 27.08% | 72.92% |        | 72.24% |
| 0.644 |        | 25.64 |        |        |        |        |
| M22   |        | 63    | 42151  | 30863  | 73.22% | 52.8   |
| 3%    | 17.04% |       | 30.13% | 69.87% |        | 74.24% |
| 0.643 |        | 28.01 |        |        |        |        |
| U22   |        | 63    | 40769  | 23149  | 56.78% | 48.5   |
| 0%    | 10.28% |       | 41.22% | 58.78% |        | 80.32% |
| 0.592 |        | 32.34 |        |        |        |        |
| Z22   |        | 63    | 40769  | 23941  | 58.72% | 49.5   |
| 9%    | 11.90% |       | 38.51% | 61.49% |        | 78.52% |
| 0.606 |        | 30.21 |        |        |        |        |
| H23   |        | 63    | 41460  | 20511  | 49.47% | 47.1   |
| 2%    | 9.50%  |       | 43.38% | 56.62% |        | 79.68% |
| 0.568 |        | 32.06 |        |        |        |        |
| M23   |        | 63    | 38005  | 12404  | 32.64% | 41.8   |
| 4%    | 5.25%  |       | 52.91% | 47.09% |        | 82.84% |
| 0.489 |        | 35.41 |        |        |        |        |
| U23   |        | 63    | 41460  | 16949  | 40.88% | 44.2   |
| 1%    | 5.78%  |       | 50.01% | 49.99% |        | 84.49% |
| 0.531 |        | 35.89 |        |        |        |        |
| Z23   |        | 63    | 42842  | 17834  | 41.63% | 46.1   |
| 8%    | 7.01%  |       | 46.81% | 53.19% |        | 83.65% |
| 0.562 |        | 34.42 |        |        |        |        |
| H24   |        | 63    | 41460  | 16331  | 39.39% | 44.8   |
| 5%    | 7.01%  |       | 48.14% | 51.86% |        | 82.43% |
| 0.537 |        | 34.19 |        |        |        |        |
| M24   |        | 63    | 45606  | 18915  | 41.47% | 47.5   |
| 3%    | 8.17%  |       | 44.30% | 55.70% |        | 82.28% |
| 0.582 |        | 33.39 |        |        |        |        |
| U24   |        | 63    | 38696  | 23688  | 61.22% | 51.9   |
| 7%    | 13.56% |       | 34.48% | 65.52% |        | 77.81% |
| 0.644 |        | 29.04 |        |        |        |        |
| Z24   |        | 63    | 42842  | 21218  | 49.53% | 48.6   |
| 7%    | 9.09%  |       | 42.24% | 57.76% |        | 81.87% |
| 0.600 |        | 32.05 |        |        |        |        |
| H25   |        | 63    | 39387  | 26967  | 68.47% | 54.3   |
| 7%    | 18.86% |       | 26.77% | 73.23% |        | 73.26% |
| 0.666 |        | 25.27 |        |        |        |        |
| M25   |        | 63    | 34550  | 24453  | 70.78% | 53.7   |
| 5%    | 19.82% |       | 26.43% | 73.57% |        | 72.07% |
| 0.651 |        | 23.41 |        |        |        |        |
| U25   |        | 63    | 39387  | 16770  | 42.58% | 46.3   |
| 3%    | 10.16% |       | 43.51% | 56.49% |        | 78.43% |
| 0.553 |        | 31.53 |        |        |        |        |
| Z25   |        | 63    | 42151  | 26753  | 63.47% | 53.6   |
| 6%    | 18.16% |       | 28.19% | 71.81% |        | 73.56% |
| 0.658 |        | 25.18 |        |        |        |        |
| H26   |        | 63    | 39387  | 27081  | 68.76% | 54.1   |
| 3%    | 21.21% |       | 24.66% | 75.34% |        | 70.91% |
| 0.651 |        | 24.02 |        |        |        |        |
| M26   |        | 63    | 13129  | 10093  | 76.87% | 55.3   |
| 1%    | 19.68% |       | 25.01% | 74.99% |        | 72.93% |
| 0.679 |        | 24.84 |        |        |        |        |

Resumen por familia de contrato:

contract\_code n\_targets total\_rows\_mean valid\_rows\_mean operable\_rate\_mean tp\_rate\_mean sl\_rate\_mean no\_event\_rate\_mean resolved\_rate\_mean tp\_share\_over\_resolved\_mean expectancy\_units\_mean event\_step\_mean

|        |        |       |        |        |        |   |
|--------|--------|-------|--------|--------|--------|---|
|        | H      | 63    | 277782 | 147338 | 53.04% |   |
| 51.35% | 16.06% |       | 32.59% | 67.41% |        | 7 |
| 4.32%  |        | 0.621 | 27.03  |        |        |   |
|        | M      | 63    | 254288 | 127648 | 50.20% |   |
| 49.70% | 13.32% |       | 36.97% | 63.03% |        | 7 |
| 6.35%  |        | 0.601 | 28.63  |        |        |   |
|        | U      | 63    | 243923 | 107086 | 43.90% |   |
| 47.35% | 9.64%  |       | 43.01% | 56.99% |        | 8 |
| 0.00%  |        | 0.574 | 31.95  |        |        |   |
|        | Z      | 63    | 248069 | 123442 | 49.76% |   |
| 49.13% | 11.46% |       | 39.41% | 60.59% |        | 7 |
| 8.56%  |        | 0.599 | 30.07  |        |        |   |

## Resumen por año:

| year | n_targets | total_rows_mean | valid_rows_mean | operable_rate_mean | tp_rate_mean | sl_rate_mean | no_event_rate_mean | resolved_rate_mean | tp_share_over_resolved_mean | expectancy_units_mean | event_step_mean |
|------|-----------|-----------------|-----------------|--------------------|--------------|--------------|--------------------|--------------------|-----------------------------|-----------------------|-----------------|
| 2020 | 63        | 155475          | 59018           | 37.96%             | 47.34%       | 10.93%       | 41.72%             | 58.28%             | 77.85%                      | 0.568                 | 30.40           |
| 2021 | 63        | 168604          | 61043           | 36.21%             | 46.42%       | 9.19%        | 44.39%             | 55.61%             | 79.86%                      | 0.558                 | 32.23           |
| 2022 | 63        | 165840          | 108316          | 65.31%             | 51.40%       | 15.09%       | 33.51%             | 66.49%             | 75.63%                      | 0.626                 | 28.52           |
| 2023 | 63        | 163767          | 65798           | 40.18%             | 45.15%       | 7.17%        | 47.68%             | 52.32%             | 82.25%                      | 0.542                 | 34.00           |
| 2024 | 63        | 165149          | 81015           | 49.06%             | 48.74%       | 9.91%        | 41.35%             | 58.65%             | 80.56%                      | 0.598                 | 31.57           |
| 2025 | 63        | 156857          | 94496           | 60.24%             | 52.48%       | 17.28%       | 30.25%             | 69.75%             | 73.76%                      | 0.638                 | 25.62           |
| 2026 | 63        | 48370           | 35828           | 74.07%             | 54.81%       | 21.38%       | 23.81%             | 76.19%             | 71.15%                      | 0.663                 | 23.98           |

## Resumen por split metodológico:

| dataset_split         | n_targets | total_rows_mean | valid_rows_mean | operable_rate_mean | tp_rate_mean | sl_rate_mean | no_event_rate_mean | resolved_rate_mean | tp_share_over_resolved_mean | expectancy_units_mean | event_step_mean |
|-----------------------|-----------|-----------------|-----------------|--------------------|--------------|--------------|--------------------|--------------------|-----------------------------|-----------------------|-----------------|
| development_2020_2024 | 63        | 818835          | 375190          | 45.8%              |              | 2%           | 48.29%             | 10.97%             | 40.74%                      | 59.26%                | 78.61%          |
| final_test_2025_2026  | 63        | 205227          | 130324          | 63.5%              |              | 0%           | 53.12%             | 18.40%             | 28.48%                      | 71.52%                | 73.00%          |

## Detalle por contrato para candidata TP 1.5 / SL 1.0:

| bar_col               | target_base | contract | total_rows | valid_rows | operable_rate | tp_rate | sl_rate | no_event_rate | resolved_rate | tp_share_over_resolved | expectancy_units | event_step_mean |
|-----------------------|-------------|----------|------------|------------|---------------|---------|---------|---------------|---------------|------------------------|------------------|-----------------|
| bar_p40_h30_tp15_sl10 | p40_h30     | H20      | 33168      | 11830      | 35.67%        | 57.36%  | 10.19%  | 32.44%        | 67.56%        | 84.91%                 | 0.75             | 13.13           |
| bar_p40_h30_tp15_sl10 | p40_h30     | M20      | 38696      | 22026      | 56.92%        | 53.25%  | 5.64%   | 41.11%        | 58.89%        | 90.42%                 | 0.74             | 15.87           |



|                       |         |        |        |        |        |
|-----------------------|---------|--------|--------|--------|--------|
| bar_p40_h30_tp15_sl10 | p40_h30 | U20    | 42151  | 21802  | 51.72% |
| 52.16%                | 4.97%   | 42.87% | 57.13% | 91.30% | 0.73   |
| 3                     | 15.91   |        |        |        |        |
| bar_p40_h30_tp15_sl10 | p40_h30 | Z20    | 36623  | 19859  | 54.23% |
| 54.15%                | 4.86%   | 40.98% | 59.02% | 91.76% | 0.76   |
| 4                     | 16.39   |        |        |        |        |
| bar_p40_h30_tp15_sl10 | p40_h30 | H21    | 40769  | 21753  | 53.36% |
| 57.33%                | 7.50%   | 35.16% | 64.84% | 88.43% | 0.78   |
| 5                     | 14.50   |        |        |        |        |
| bar_p40_h30_tp15_sl10 | p40_h30 | M21    | 42151  | 18843  | 44.70% |
| 48.89%                | 3.04%   | 48.07% | 51.93% | 94.14% | 0.70   |
| 3                     | 17.49   |        |        |        |        |
| bar_p40_h30_tp15_sl10 | p40_h30 | U21    | 41460  | 13194  | 31.82% |
| 43.50%                | 2.28%   | 54.21% | 45.79% | 95.02% | 0.63   |
| 0                     | 18.59   |        |        |        |        |
| bar_p40_h30_tp15_sl10 | p40_h30 | Z21    | 42842  | 23317  | 54.43% |
| 55.14%                | 5.22%   | 39.64% | 60.36% | 91.35% | 0.77   |
| 5                     | 16.29   |        |        |        |        |
| bar_p40_h30_tp15_sl10 | p40_h30 | H22    | 42151  | 34096  | 80.89% |
| 66.78%                | 13.21%  | 20.02% | 79.98% | 83.49% | 0.87   |
| 0                     | 12.63   |        |        |        |        |
| bar_p40_h30_tp15_sl10 | p40_h30 | M22    | 42151  | 35098  | 83.27% |
| 67.02%                | 10.64%  | 22.33% | 77.67% | 86.29% | 0.89   |
| 9                     | 13.85   |        |        |        |        |
| bar_p40_h30_tp15_sl10 | p40_h30 | U22    | 40769  | 28546  | 70.02% |
| 57.86%                | 5.31%   | 36.83% | 63.17% | 91.59% | 0.81   |
| 5                     | 16.29   |        |        |        |        |
| bar_p40_h30_tp15_sl10 | p40_h30 | Z22    | 40769  | 29341  | 71.97% |
| 59.46%                | 6.66%   | 33.89% | 66.11% | 89.93% | 0.82   |
| 5                     | 15.52   |        |        |        |        |
| bar_p40_h30_tp15_sl10 | p40_h30 | H23    | 41460  | 25805  | 62.24% |
| 53.73%                | 5.05%   | 41.23% | 58.77% | 91.42% | 0.75   |
| 5                     | 16.37   |        |        |        |        |
| bar_p40_h30_tp15_sl10 | p40_h30 | M23    | 38005  | 17346  | 45.64% |
| 44.07%                | 2.26%   | 53.67% | 46.33% | 95.12% | 0.63   |
| 9                     | 18.02   |        |        |        |        |
| bar_p40_h30_tp15_sl10 | p40_h30 | U23    | 41460  | 22646  | 54.62% |
| 47.91%                | 2.58%   | 49.51% | 50.49% | 94.88% | 0.69   |
| 3                     | 18.04   |        |        |        |        |
| bar_p40_h30_tp15_sl10 | p40_h30 | Z23    | 42842  | 23269  | 54.31% |
| 50.77%                | 2.73%   | 46.50% | 53.50% | 94.89% | 0.73   |
| 4                     | 17.51   |        |        |        |        |
| bar_p40_h30_tp15_sl10 | p40_h30 | H24    | 41460  | 21490  | 51.83% |
| 49.52%                | 3.01%   | 47.47% | 52.53% | 94.27% | 0.71   |
| 3                     | 17.60   |        |        |        |        |
| bar_p40_h30_tp15_sl10 | p40_h30 | M24    | 45606  | 24464  | 53.64% |
| 52.03%                | 3.96%   | 44.01% | 55.99% | 92.93% | 0.74   |
| 1                     | 16.94   |        |        |        |        |
| bar_p40_h30_tp15_sl10 | p40_h30 | U24    | 38696  | 27668  | 71.50% |
| 62.73%                | 7.98%   | 29.29% | 70.71% | 88.71% | 0.86   |
| 1                     | 14.69   |        |        |        |        |
| bar_p40_h30_tp15_sl10 | p40_h30 | Z24    | 42842  | 26957  | 62.92% |
| 54.87%                | 4.46%   | 40.67% | 59.33% | 92.48% | 0.77   |
| 8                     | 15.99   |        |        |        |        |
| bar_p40_h30_tp15_sl10 | p40_h30 | H25    | 39387  | 31146  | 79.08% |
| 65.41%                | 11.77%  | 22.82% | 77.18% | 84.75% | 0.86   |
| 3                     | 12.56   |        |        |        |        |
| bar_p40_h30_tp15_sl10 | p40_h30 | M25    | 34550  | 28074  | 81.26% |
| 65.61%                | 13.08%  | 21.31% | 78.69% | 83.38% | 0.85   |
| 3                     | 12.17   |        |        |        |        |

|                       |         |        |        |        |        |
|-----------------------|---------|--------|--------|--------|--------|
| bar_p40_h30_tp15_sl10 | p40_h30 | U25    | 39387  | 21840  | 55.45% |
| 52.61%                | 5.38%   | 42.01% | 57.99% | 90.72% | 0.73   |
| 5                     | 16.27   |        |        |        |        |
| bar_p40_h30_tp15_sl10 | p40_h30 | Z25    | 42151  | 31195  | 74.01% |
| 63.00%                | 11.78%  | 25.22% | 74.78% | 84.24% | 0.82   |
| 7                     | 12.58   |        |        |        |        |
| bar_p40_h30_tp15_sl10 | p40_h30 | H26    | 39387  | 30922  | 78.51% |
| 65.66%                | 13.93%  | 20.41% | 79.59% | 82.50% | 0.84   |
| 6                     | 11.78   |        |        |        |        |
| bar_p40_h30_tp15_sl10 | p40_h30 | M26    | 13129  | 11209  | 85.38% |
| 68.46%                | 13.26%  | 18.28% | 81.72% | 83.78% | 0.89   |
| 4                     | 12.50   |        |        |        |        |
| bar_p40_h60_tp15_sl10 | p40_h60 | H20    | 33168  | 10961  | 33.05% |
| 55.43%                | 9.86%   | 34.70% | 65.30% | 84.90% | 0.73   |
| 3                     | 25.04   |        |        |        |        |
| bar_p40_h60_tp15_sl10 | p40_h60 | M20    | 38696  | 20727  | 53.56% |
| 50.80%                | 4.99%   | 44.21% | 55.79% | 91.06% | 0.71   |
| 2                     | 31.89   |        |        |        |        |
| bar_p40_h60_tp15_sl10 | p40_h60 | U20    | 42151  | 20494  | 48.62% |
| 50.21%                | 5.07%   | 44.72% | 55.28% | 90.83% | 0.70   |
| 2                     | 31.71   |        |        |        |        |
| bar_p40_h60_tp15_sl10 | p40_h60 | Z20    | 36623  | 18570  | 50.71% |
| 52.42%                | 4.63%   | 42.96% | 57.04% | 91.89% | 0.74   |
| 0                     | 32.08   |        |        |        |        |
| bar_p40_h60_tp15_sl10 | p40_h60 | H21    | 40769  | 20261  | 49.70% |
| 55.42%                | 8.04%   | 36.54% | 63.46% | 87.33% | 0.75   |
| 1                     | 28.31   |        |        |        |        |
| bar_p40_h60_tp15_sl10 | p40_h60 | M21    | 42151  | 17654  | 41.88% |
| 47.60%                | 3.38%   | 49.02% | 50.98% | 93.37% | 0.68   |
| 0                     | 34.56   |        |        |        |        |
| bar_p40_h60_tp15_sl10 | p40_h60 | U21    | 41460  | 12801  | 30.88% |
| 41.13%                | 1.73%   | 57.14% | 42.86% | 95.95% | 0.60   |
| 0                     | 37.38   |        |        |        |        |
| bar_p40_h60_tp15_sl10 | p40_h60 | Z21    | 42842  | 22580  | 52.71% |
| 52.22%                | 5.03%   | 42.75% | 57.25% | 91.21% | 0.73   |
| 3                     | 32.48   |        |        |        |        |
| bar_p40_h60_tp15_sl10 | p40_h60 | H22    | 42151  | 32852  | 77.94% |
| 65.54%                | 13.19%  | 21.27% | 78.73% | 83.25% | 0.85   |
| 1                     | 25.03   |        |        |        |        |
| bar_p40_h60_tp15_sl10 | p40_h60 | M22    | 42151  | 34326  | 81.44% |
| 65.74%                | 10.75%  | 23.50% | 76.50% | 85.94% | 0.87   |
| 9                     | 27.57   |        |        |        |        |
| bar_p40_h60_tp15_sl10 | p40_h60 | U22    | 40769  | 27651  | 67.82% |
| 56.20%                | 5.07%   | 38.73% | 61.27% | 91.72% | 0.79   |
| 2                     | 32.51   |        |        |        |        |
| bar_p40_h60_tp15_sl10 | p40_h60 | Z22    | 40769  | 27754  | 68.08% |
| 59.27%                | 6.29%   | 34.45% | 65.55% | 90.41% | 0.82   |
| 6                     | 30.88   |        |        |        |        |
| bar_p40_h60_tp15_sl10 | p40_h60 | H23    | 41460  | 24969  | 60.22% |
| 52.98%                | 4.68%   | 42.34% | 57.66% | 91.88% | 0.74   |
| 8                     | 32.84   |        |        |        |        |
| bar_p40_h60_tp15_sl10 | p40_h60 | M23    | 38005  | 16188  | 42.59% |
| 41.87%                | 1.80%   | 56.33% | 43.67% | 95.87% | 0.61   |
| 0                     | 36.61   |        |        |        |        |
| bar_p40_h60_tp15_sl10 | p40_h60 | U23    | 41460  | 21877  | 52.77% |
| 45.68%                | 1.90%   | 52.42% | 47.58% | 96.01% | 0.66   |
| 6                     | 36.46   |        |        |        |        |
| bar_p40_h60_tp15_sl10 | p40_h60 | Z23    | 42842  | 22229  | 51.89% |
| 49.79%                | 2.73%   | 47.49% | 52.51% | 94.81% | 0.72   |
| 0                     | 34.66   |        |        |        |        |

|                       |         |        |        |        |        |
|-----------------------|---------|--------|--------|--------|--------|
| bar_p40_h60_tp15_sl10 | p40_h60 | H24    | 41460  | 20282  | 48.92% |
| 49.60%                | 2.90%   | 47.50% | 52.50% | 94.47% | 0.71   |
| 5                     | 35.54   |        |        |        |        |
| bar_p40_h60_tp15_sl10 | p40_h60 | M24    | 45606  | 23239  | 50.96% |
| 52.55%                | 3.52%   | 43.94% | 56.06% | 93.73% | 0.75   |
| 3                     | 33.96   |        |        |        |        |
| bar_p40_h60_tp15_sl10 | p40_h60 | U24    | 38696  | 27240  | 70.39% |
| 62.87%                | 7.83%   | 29.30% | 70.70% | 88.92% | 0.86   |
| 5                     | 29.13   |        |        |        |        |
| bar_p40_h60_tp15_sl10 | p40_h60 | Z24    | 42842  | 25607  | 59.77% |
| 54.68%                | 4.28%   | 41.04% | 58.96% | 92.74% | 0.77   |
| 7                     | 32.15   |        |        |        |        |
| bar_p40_h60_tp15_sl10 | p40_h60 | H25    | 39387  | 29827  | 75.73% |
| 64.53%                | 12.41%  | 23.06% | 76.94% | 83.87% | 0.84   |
| 4                     | 24.88   |        |        |        |        |
| bar_p40_h60_tp15_sl10 | p40_h60 | M25    | 34550  | 26955  | 78.02% |
| 64.08%                | 13.66%  | 22.26% | 77.74% | 82.43% | 0.82   |
| 5                     | 23.89   |        |        |        |        |
| bar_p40_h60_tp15_sl10 | p40_h60 | U25    | 39387  | 20831  | 52.89% |
| 49.42%                | 4.86%   | 45.72% | 54.28% | 91.05% | 0.69   |
| 3                     | 32.02   |        |        |        |        |
| bar_p40_h60_tp15_sl10 | p40_h60 | Z25    | 42151  | 30319  | 71.93% |
| 61.32%                | 11.95%  | 26.73% | 73.27% | 83.69% | 0.80   |
| 0                     | 24.85   |        |        |        |        |
| bar_p40_h60_tp15_sl10 | p40_h60 | H26    | 39387  | 29887  | 75.88% |
| 64.61%                | 14.28%  | 21.11% | 78.89% | 81.90% | 0.82   |
| 6                     | 23.27   |        |        |        |        |
| bar_p40_h60_tp15_sl10 | p40_h60 | M26    | 13129  | 10943  | 83.35% |
| 68.84%                | 13.44%  | 17.72% | 82.28% | 83.66% | 0.89   |
| 8                     | 24.28   |        |        |        |        |
| bar_p40_h90_tp15_sl10 | p40_h90 | H20    | 33168  | 10172  | 30.67% |
| 55.94%                | 10.93%  | 33.13% | 66.87% | 83.65% | 0.73   |
| 0                     | 39.58   |        |        |        |        |
| bar_p40_h90_tp15_sl10 | p40_h90 | M20    | 38696  | 19130  | 49.44% |
| 50.96%                | 5.12%   | 43.92% | 56.08% | 90.87% | 0.71   |
| 3                     | 48.67   |        |        |        |        |
| bar_p40_h90_tp15_sl10 | p40_h90 | U20    | 42151  | 20020  | 47.50% |
| 47.56%                | 4.81%   | 47.63% | 52.37% | 90.82% | 0.66   |
| 5                     | 48.61   |        |        |        |        |
| bar_p40_h90_tp15_sl10 | p40_h90 | Z20    | 36623  | 17651  | 48.20% |
| 49.91%                | 5.10%   | 44.98% | 55.02% | 90.72% | 0.69   |
| 8                     | 49.00   |        |        |        |        |
| bar_p40_h90_tp15_sl10 | p40_h90 | H21    | 40769  | 19133  | 46.93% |
| 56.74%                | 8.71%   | 34.55% | 65.45% | 86.70% | 0.76   |
| 4                     | 44.23   |        |        |        |        |
| bar_p40_h90_tp15_sl10 | p40_h90 | M21    | 42151  | 17015  | 40.37% |
| 45.82%                | 3.29%   | 50.89% | 49.11% | 93.31% | 0.65   |
| 5                     | 53.20   |        |        |        |        |
| bar_p40_h90_tp15_sl10 | p40_h90 | U21    | 41460  | 12788  | 30.84% |
| 38.69%                | 2.49%   | 58.82% | 41.18% | 93.96% | 0.55   |
| 6                     | 58.31   |        |        |        |        |
| bar_p40_h90_tp15_sl10 | p40_h90 | Z21    | 42842  | 21291  | 49.70% |
| 51.17%                | 5.23%   | 43.60% | 56.40% | 90.72% | 0.71   |
| 5                     | 49.40   |        |        |        |        |
| bar_p40_h90_tp15_sl10 | p40_h90 | H22    | 42151  | 31285  | 74.22% |
| 64.41%                | 13.45%  | 22.14% | 77.86% | 82.73% | 0.83   |
| 2                     | 38.55   |        |        |        |        |
| bar_p40_h90_tp15_sl10 | p40_h90 | M22    | 42151  | 32576  | 77.28% |
| 66.73%                | 11.18%  | 22.09% | 77.91% | 85.65% | 0.88   |
| 9                     | 42.21   |        |        |        |        |

|                       |         |        |        |        |        |
|-----------------------|---------|--------|--------|--------|--------|
| bar_p40_h90_tp15_sl10 | p40_h90 | U22    | 40769  | 26242  | 64.37% |
| 55.77%                | 5.92%   | 38.31% | 61.69% | 90.41% | 0.77   |
| 7                     | 49.04   |        |        |        |        |
| bar_p40_h90_tp15_sl10 | p40_h90 | Z22    | 40769  | 26573  | 65.18% |
| 58.04%                | 6.65%   | 35.32% | 64.68% | 89.73% | 0.80   |
| 4                     | 46.71   |        |        |        |        |
| bar_p40_h90_tp15_sl10 | p40_h90 | H23    | 41460  | 24009  | 57.91% |
| 52.78%                | 4.61%   | 42.60% | 57.40% | 91.96% | 0.74   |
| 6                     | 49.89   |        |        |        |        |
| bar_p40_h90_tp15_sl10 | p40_h90 | M23    | 38005  | 15356  | 40.41% |
| 40.60%                | 2.68%   | 56.73% | 43.27% | 93.81% | 0.58   |
| 2                     | 56.81   |        |        |        |        |
| bar_p40_h90_tp15_sl10 | p40_h90 | U23    | 41460  | 20232  | 48.80% |
| 47.59%                | 2.40%   | 50.01% | 49.99% | 95.20% | 0.69   |
| 0                     | 55.84   |        |        |        |        |
| bar_p40_h90_tp15_sl10 | p40_h90 | Z23    | 42842  | 21413  | 49.98% |
| 49.91%                | 3.60%   | 46.50% | 53.50% | 93.28% | 0.71   |
| 3                     | 52.76   |        |        |        |        |
| bar_p40_h90_tp15_sl10 | p40_h90 | H24    | 41460  | 19381  | 46.75% |
| 50.34%                | 3.42%   | 46.25% | 53.75% | 93.65% | 0.72   |
| 1                     | 54.14   |        |        |        |        |
| bar_p40_h90_tp15_sl10 | p40_h90 | M24    | 45606  | 21920  | 48.06% |
| 53.47%                | 4.52%   | 42.02% | 57.98% | 92.21% | 0.75   |
| 7                     | 51.22   |        |        |        |        |
| bar_p40_h90_tp15_sl10 | p40_h90 | U24    | 38696  | 26054  | 67.33% |
| 62.01%                | 7.65%   | 30.34% | 69.66% | 89.01% | 0.85   |
| 4                     | 43.63   |        |        |        |        |
| bar_p40_h90_tp15_sl10 | p40_h90 | Z24    | 42842  | 24317  | 56.76% |
| 55.15%                | 4.77%   | 40.07% | 59.93% | 92.04% | 0.78   |
| 0                     | 48.72   |        |        |        |        |
| bar_p40_h90_tp15_sl10 | p40_h90 | H25    | 39387  | 28483  | 72.32% |
| 64.17%                | 14.02%  | 21.82% | 78.18% | 82.07% | 0.82   |
| 2                     | 37.75   |        |        |        |        |
| bar_p40_h90_tp15_sl10 | p40_h90 | M25    | 34550  | 25787  | 74.64% |
| 63.24%                | 13.88%  | 22.88% | 77.12% | 82.00% | 0.81   |
| 0                     | 36.09   |        |        |        |        |
| bar_p40_h90_tp15_sl10 | p40_h90 | U25    | 39387  | 19570  | 49.69% |
| 48.56%                | 5.28%   | 46.16% | 53.84% | 90.20% | 0.67   |
| 6                     | 48.71   |        |        |        |        |
| bar_p40_h90_tp15_sl10 | p40_h90 | Z25    | 42151  | 28652  | 67.97% |
| 62.33%                | 12.15%  | 25.52% | 74.48% | 83.69% | 0.81   |
| 4                     | 37.56   |        |        |        |        |
| bar_p40_h90_tp15_sl10 | p40_h90 | H26    | 39387  | 28438  | 72.20% |
| 63.37%                | 15.21%  | 21.42% | 78.58% | 80.65% | 0.79   |
| 9                     | 34.82   |        |        |        |        |
| bar_p40_h90_tp15_sl10 | p40_h90 | M26    | 13129  | 10482  | 79.84% |
| 71.74%                | 12.36%  | 15.89% | 84.11% | 85.30% | 0.95   |
| 2                     | 36.83   |        |        |        |        |
| bar_p50_h30_tp15_sl10 | p50_h30 | H20    | 33168  | 9988   | 30.11% |
| 56.98%                | 8.37%   | 34.65% | 65.35% | 87.19% | 0.77   |
| 1                     | 13.91   |        |        |        |        |
| bar_p50_h30_tp15_sl10 | p50_h30 | M20    | 38696  | 18151  | 46.91% |
| 48.01%                | 4.25%   | 47.73% | 52.27% | 91.86% | 0.67   |
| 8                     | 16.79   |        |        |        |        |
| bar_p50_h30_tp15_sl10 | p50_h30 | U20    | 42151  | 17562  | 41.66% |
| 48.37%                | 3.87%   | 47.76% | 52.24% | 92.59% | 0.68   |
| 7                     | 16.70   |        |        |        |        |
| bar_p50_h30_tp15_sl10 | p50_h30 | Z20    | 36623  | 16339  | 44.61% |
| 50.50%                | 3.36%   | 46.14% | 53.86% | 93.76% | 0.72   |
| 4                     | 17.67   |        |        |        |        |

|                       |         |        |        |        |        |
|-----------------------|---------|--------|--------|--------|--------|
| bar_p50_h30_tp15_sl10 | p50_h30 | H21    | 40769  | 18255  | 44.78% |
| 54.95%                | 5.74%   | 39.30% | 60.70% | 90.54% | 0.76   |
| 7                     | 15.54   |        |        |        |        |
| bar_p50_h30_tp15_sl10 | p50_h30 | M21    | 42151  | 14837  | 35.20% |
| 43.22%                | 2.14%   | 54.65% | 45.35% | 95.29% | 0.62   |
| 7                     | 18.26   |        |        |        |        |
| bar_p50_h30_tp15_sl10 | p50_h30 | U21    | 41460  | 9873   | 23.81% |
| 38.88%                | 1.09%   | 60.02% | 39.98% | 97.26% | 0.57   |
| 2                     | 19.52   |        |        |        |        |
| bar_p50_h30_tp15_sl10 | p50_h30 | Z21    | 42842  | 19268  | 44.97% |
| 50.35%                | 3.76%   | 45.89% | 54.11% | 93.05% | 0.71   |
| 8                     | 17.41   |        |        |        |        |
| bar_p50_h30_tp15_sl10 | p50_h30 | H22    | 42151  | 31381  | 74.45% |
| 63.67%                | 9.96%   | 26.38% | 73.62% | 86.48% | 0.85   |
| 5                     | 13.94   |        |        |        |        |
| bar_p50_h30_tp15_sl10 | p50_h30 | M22    | 42151  | 32265  | 76.55% |
| 61.52%                | 7.70%   | 30.78% | 69.22% | 88.88% | 0.84   |
| 6                     | 15.14   |        |        |        |        |
| bar_p50_h30_tp15_sl10 | p50_h30 | U22    | 40769  | 24200  | 59.36% |
| 51.89%                | 3.66%   | 44.45% | 55.55% | 93.41% | 0.74   |
| 2                     | 17.54   |        |        |        |        |
| bar_p50_h30_tp15_sl10 | p50_h30 | Z22    | 40769  | 25349  | 62.18% |
| 54.22%                | 4.57%   | 41.21% | 58.79% | 92.22% | 0.76   |
| 8                     | 16.69   |        |        |        |        |
| bar_p50_h30_tp15_sl10 | p50_h30 | H23    | 41460  | 21198  | 51.13% |
| 48.52%                | 4.11%   | 47.37% | 52.63% | 92.19% | 0.68   |
| 7                     | 17.25   |        |        |        |        |
| bar_p50_h30_tp15_sl10 | p50_h30 | M23    | 38005  | 13159  | 34.62% |
| 37.63%                | 1.69%   | 60.67% | 39.33% | 95.69% | 0.54   |
| 8                     | 18.71   |        |        |        |        |
| bar_p50_h30_tp15_sl10 | p50_h30 | U23    | 41460  | 17773  | 42.87% |
| 41.73%                | 1.50%   | 56.77% | 43.23% | 96.52% | 0.61   |
| 1                     | 19.16   |        |        |        |        |
| bar_p50_h30_tp15_sl10 | p50_h30 | Z23    | 42842  | 18483  | 43.14% |
| 45.57%                | 1.72%   | 52.72% | 47.28% | 96.37% | 0.66   |
| 6                     | 18.38   |        |        |        |        |
| bar_p50_h30_tp15_sl10 | p50_h30 | H24    | 41460  | 17014  | 41.04% |
| 43.23%                | 2.36%   | 54.41% | 45.59% | 94.83% | 0.62   |
| 5                     | 18.32   |        |        |        |        |
| bar_p50_h30_tp15_sl10 | p50_h30 | M24    | 45606  | 19681  | 43.15% |
| 47.32%                | 2.73%   | 49.95% | 50.05% | 94.55% | 0.68   |
| 3                     | 17.88   |        |        |        |        |
| bar_p50_h30_tp15_sl10 | p50_h30 | U24    | 38696  | 24454  | 63.20% |
| 57.78%                | 5.60%   | 36.62% | 63.38% | 91.17% | 0.81   |
| 1                     | 15.77   |        |        |        |        |
| bar_p50_h30_tp15_sl10 | p50_h30 | Z24    | 42842  | 22348  | 52.16% |
| 49.73%                | 3.16%   | 47.10% | 52.90% | 94.02% | 0.71   |
| 4                     | 17.01   |        |        |        |        |
| bar_p50_h30_tp15_sl10 | p50_h30 | H25    | 39387  | 28256  | 71.74% |
| 62.61%                | 9.10%   | 28.30% | 71.70% | 87.31% | 0.84   |
| 8                     | 13.74   |        |        |        |        |
| bar_p50_h30_tp15_sl10 | p50_h30 | M25    | 34550  | 25657  | 74.26% |
| 62.12%                | 10.57%  | 27.30% | 72.70% | 85.45% | 0.82   |
| 6                     | 13.11   |        |        |        |        |
| bar_p50_h30_tp15_sl10 | p50_h30 | U25    | 39387  | 17869  | 45.37% |
| 47.76%                | 3.92%   | 48.31% | 51.69% | 92.41% | 0.67   |
| 7                     | 17.04   |        |        |        |        |
| bar_p50_h30_tp15_sl10 | p50_h30 | Z25    | 42151  | 27875  | 66.13% |
| 60.42%                | 9.59%   | 29.99% | 70.01% | 86.31% | 0.81   |
| 0                     | 13.69   |        |        |        |        |

|                       |         |        |        |        |        |
|-----------------------|---------|--------|--------|--------|--------|
| bar_p50_h30_tp15_sl10 | p50_h30 | H26    | 39387  | 28276  | 71.79% |
| 63.91%                | 11.21%  | 24.88% | 75.12% | 85.07% | 0.84   |
| 6                     | 12.99   |        |        |        |        |
| bar_p50_h30_tp15_sl10 | p50_h30 | M26    | 13129  | 10471  | 79.75% |
| 64.23%                | 9.90%   | 25.86% | 74.14% | 86.64% | 0.86   |
| 4                     | 13.60   |        |        |        |        |
| bar_p50_h60_tp15_sl10 | p50_h60 | H20    | 33168  | 9274   | 27.96% |
| 55.29%                | 7.98%   | 36.73% | 63.27% | 87.39% | 0.75   |
| 0                     | 27.36   |        |        |        |        |
| bar_p50_h60_tp15_sl10 | p50_h60 | M20    | 38696  | 16710  | 43.18% |
| 46.76%                | 3.45%   | 49.79% | 50.21% | 93.12% | 0.66   |
| 7                     | 33.89   |        |        |        |        |
| bar_p50_h60_tp15_sl10 | p50_h60 | U20    | 42151  | 16419  | 38.95% |
| 46.18%                | 3.54%   | 50.28% | 49.72% | 92.88% | 0.65   |
| 7                     | 33.55   |        |        |        |        |
| bar_p50_h60_tp15_sl10 | p50_h60 | Z20    | 36623  | 15135  | 41.33% |
| 46.65%                | 3.18%   | 50.18% | 49.82% | 93.62% | 0.66   |
| 8                     | 33.96   |        |        |        |        |
| bar_p50_h60_tp15_sl10 | p50_h60 | H21    | 40769  | 16990  | 41.67% |
| 53.06%                | 6.15%   | 40.79% | 59.21% | 89.61% | 0.73   |
| 4                     | 30.52   |        |        |        |        |
| bar_p50_h60_tp15_sl10 | p50_h60 | M21    | 42151  | 13764  | 32.65% |
| 41.80%                | 2.22%   | 55.99% | 44.01% | 94.97% | 0.60   |
| 5                     | 36.47   |        |        |        |        |
| bar_p50_h60_tp15_sl10 | p50_h60 | U21    | 41460  | 9604   | 23.16% |
| 33.25%                | 1.79%   | 64.96% | 35.04% | 94.89% | 0.48   |
| 1                     | 38.76   |        |        |        |        |
| bar_p50_h60_tp15_sl10 | p50_h60 | Z21    | 42842  | 18569  | 43.34% |
| 46.45%                | 2.99%   | 50.56% | 49.44% | 93.95% | 0.66   |
| 7                     | 34.74   |        |        |        |        |
| bar_p50_h60_tp15_sl10 | p50_h60 | H22    | 42151  | 30153  | 71.54% |
| 61.34%                | 9.90%   | 28.76% | 71.24% | 86.11% | 0.82   |
| 1                     | 27.82   |        |        |        |        |
| bar_p50_h60_tp15_sl10 | p50_h60 | M22    | 42151  | 31373  | 74.43% |
| 61.11%                | 7.68%   | 31.21% | 68.79% | 88.83% | 0.84   |
| 0                     | 30.44   |        |        |        |        |
| bar_p50_h60_tp15_sl10 | p50_h60 | U22    | 40769  | 23243  | 57.01% |
| 50.39%                | 2.87%   | 46.74% | 53.26% | 94.61% | 0.72   |
| 7                     | 35.12   |        |        |        |        |
| bar_p50_h60_tp15_sl10 | p50_h60 | Z22    | 40769  | 23960  | 58.77% |
| 53.34%                | 4.34%   | 42.32% | 57.68% | 92.47% | 0.75   |
| 7                     | 33.43   |        |        |        |        |
| bar_p50_h60_tp15_sl10 | p50_h60 | H23    | 41460  | 20460  | 49.35% |
| 47.53%                | 3.13%   | 49.34% | 50.66% | 93.82% | 0.68   |
| 2                     | 34.66   |        |        |        |        |
| bar_p50_h60_tp15_sl10 | p50_h60 | M23    | 38005  | 12153  | 31.98% |
| 34.35%                | 1.28%   | 64.36% | 35.64% | 96.40% | 0.50   |
| 2                     | 37.69   |        |        |        |        |
| bar_p50_h60_tp15_sl10 | p50_h60 | U23    | 41460  | 16760  | 40.42% |
| 39.29%                | 1.27%   | 59.44% | 40.56% | 96.87% | 0.57   |
| 7                     | 38.37   |        |        |        |        |
| bar_p50_h60_tp15_sl10 | p50_h60 | Z23    | 42842  | 17584  | 41.04% |
| 44.23%                | 1.61%   | 54.16% | 45.84% | 96.49% | 0.64   |
| 7                     | 36.32   |        |        |        |        |

Detalle por año para candidata TP 1.5 / SL 1.0:

| bar_col               | target_base   | year          | total_rows             | valid_rows       | operable_rate   | tp_rate |
|-----------------------|---------------|---------------|------------------------|------------------|-----------------|---------|
| sl_rate               | no_event_rate | resolved_rate | tp_share_over_resolved | expectancy_units | event_step_mean |         |
| bar_p40_h30_tp15_sl10 | p40_h30       | 2020          | 155475                 | 76643            | 49.30%          | 5       |

|                       |         |        |        |        |          |
|-----------------------|---------|--------|--------|--------|----------|
| 3.67%                 | 5.93%   | 40.40% | 59.60% | 90.05% | 0.746    |
| 15.53                 |         |        |        |        |          |
| bar_p40_h30_tp15_sl10 | p40_h30 | 2021   | 168604 | 79001  | 46.86% 5 |
| 2.15%                 | 4.82%   | 43.03% | 56.97% | 91.53% | 0.734    |
| 16.36                 |         |        |        |        |          |
| bar_p40_h30_tp15_sl10 | p40_h30 | 2022   | 165840 | 127656 | 76.98% 6 |
| 3.20%                 | 9.14%   | 27.66% | 72.34% | 87.37% | 0.857    |
| 14.34                 |         |        |        |        |          |
| bar_p40_h30_tp15_sl10 | p40_h30 | 2023   | 163767 | 86966  | 53.10% 4 |
| 9.35%                 | 3.28%   | 47.37% | 52.63% | 93.76% | 0.707    |
| 17.32                 |         |        |        |        |          |
| bar_p40_h30_tp15_sl10 | p40_h30 | 2024   | 165149 | 101242 | 61.30% 5 |
| 5.66%                 | 5.12%   | 39.21% | 60.79% | 91.57% | 0.784    |
| 16.02                 |         |        |        |        |          |
| bar_p40_h30_tp15_sl10 | p40_h30 | 2025   | 156857 | 111943 | 71.37% 6 |
| 1.99%                 | 10.75%  | 27.26% | 72.74% | 85.22% | 0.822    |
| 13.08                 |         |        |        |        |          |
| bar_p40_h30_tp15_sl10 | p40_h30 | 2026   | 48370  | 40285  | 83.29% 6 |
| 7.30%                 | 14.29%  | 18.42% | 81.58% | 82.49% | 0.867    |
| 11.79                 |         |        |        |        |          |
| bar_p40_h60_tp15_sl10 | p40_h60 | 2020   | 155475 | 71768  | 46.16% 5 |
| 1.60%                 | 5.65%   | 42.74% | 57.26% | 90.13% | 0.718    |
| 30.66                 |         |        |        |        |          |
| bar_p40_h60_tp15_sl10 | p40_h60 | 2021   | 168604 | 75199  | 44.60% 5 |
| 0.11%                 | 4.78%   | 45.11% | 54.89% | 91.29% | 0.704    |
| 32.48                 |         |        |        |        |          |
| bar_p40_h60_tp15_sl10 | p40_h60 | 2022   | 165840 | 123282 | 74.34% 6 |
| 2.04%                 | 9.06%   | 28.90% | 71.10% | 87.25% | 0.840    |
| 28.49                 |         |        |        |        |          |
| bar_p40_h60_tp15_sl10 | p40_h60 | 2023   | 163767 | 82842  | 50.59% 4 |
| 7.95%                 | 2.96%   | 49.09% | 50.91% | 94.18% | 0.690    |
| 34.72                 |         |        |        |        |          |
| bar_p40_h60_tp15_sl10 | p40_h60 | 2024   | 165149 | 97230  | 58.87% 5 |
| 5.78%                 | 4.89%   | 39.33% | 60.67% | 91.94% | 0.788    |
| 32.05                 |         |        |        |        |          |
| bar_p40_h60_tp15_sl10 | p40_h60 | 2025   | 156857 | 107716 | 68.67% 6 |
| 0.24%                 | 11.05%  | 28.71% | 71.29% | 84.50% | 0.793    |
| 25.71                 |         |        |        |        |          |
| bar_p40_h60_tp15_sl10 | p40_h60 | 2026   | 48370  | 38987  | 80.60% 6 |
| 6.90%                 | 14.65%  | 18.45% | 81.55% | 82.04% | 0.857    |
| 23.27                 |         |        |        |        |          |
| bar_p40_h90_tp15_sl10 | p40_h90 | 2020   | 155475 | 67923  | 43.69% 5 |
| 0.37%                 | 5.87%   | 43.76% | 56.24% | 89.57% | 0.697    |
| 47.09                 |         |        |        |        |          |
| bar_p40_h90_tp15_sl10 | p40_h90 | 2021   | 168604 | 72130  | 42.78% 4 |
| 9.13%                 | 5.11%   | 45.76% | 54.24% | 90.58% | 0.686    |
| 50.15                 |         |        |        |        |          |
| bar_p40_h90_tp15_sl10 | p40_h90 | 2022   | 165840 | 117217 | 70.68% 6 |
| 1.73%                 | 9.53%   | 28.74% | 71.26% | 86.63% | 0.831    |
| 43.36                 |         |        |        |        |          |
| bar_p40_h90_tp15_sl10 | p40_h90 | 2023   | 163767 | 78756  | 48.09% 4 |
| 8.16%                 | 3.45%   | 48.39% | 51.61% | 93.32% | 0.688    |
| 53.19                 |         |        |        |        |          |
| bar_p40_h90_tp15_sl10 | p40_h90 | 2024   | 165149 | 92460  | 55.99% 5 |
| 5.96%                 | 5.32%   | 38.72% | 61.28% | 91.32% | 0.786    |
| 48.44                 |         |        |        |        |          |
| bar_p40_h90_tp15_sl10 | p40_h90 | 2025   | 156857 | 102236 | 65.18% 6 |
| 0.06%                 | 11.74%  | 28.20% | 71.80% | 83.65% | 0.783    |
| 38.88                 |         |        |        |        |          |
| bar_p40_h90_tp15_sl10 | p40_h90 | 2026   | 48370  | 37248  | 77.01% 6 |

|                       |         |        |        |        |          |
|-----------------------|---------|--------|--------|--------|----------|
| 6.81%                 | 15.00%  | 18.19% | 81.81% | 81.67% | 0.852    |
| 34.99                 |         |        |        |        |          |
| bar_p50_h30_tp15_sl10 | p50_h30 | 2020   | 155475 | 62849  | 40.42% 5 |
| 0.19%                 | 4.56%   | 45.25% | 54.75% | 91.67% | 0.707    |
| 16.44                 |         |        |        |        |          |
| bar_p50_h30_tp15_sl10 | p50_h30 | 2021   | 168604 | 63758  | 37.82% 4 |
| 7.92%                 | 3.49%   | 48.59% | 51.41% | 93.21% | 0.684    |
| 17.25                 |         |        |        |        |          |
| bar_p50_h30_tp15_sl10 | p50_h30 | 2022   | 165840 | 113734 | 68.58% 5 |
| 8.41%                 | 6.73%   | 34.85% | 65.15% | 89.67% | 0.809    |
| 15.54                 |         |        |        |        |          |
| bar_p50_h30_tp15_sl10 | p50_h30 | 2023   | 163767 | 68783  | 42.00% 4 |
| 3.88%                 | 2.40%   | 53.72% | 46.28% | 94.82% | 0.634    |
| 18.18                 |         |        |        |        |          |
| bar_p50_h30_tp15_sl10 | p50_h30 | 2024   | 165149 | 84401  | 51.11% 5 |
| 0.60%                 | 3.66%   | 45.74% | 54.26% | 93.25% | 0.722    |
| 16.96                 |         |        |        |        |          |
| bar_p50_h30_tp15_sl10 | p50_h30 | 2025   | 156857 | 99124  | 63.19% 5 |
| 8.95%                 | 8.63%   | 32.43% | 67.57% | 87.23% | 0.798    |
| 14.02                 |         |        |        |        |          |
| bar_p50_h30_tp15_sl10 | p50_h30 | 2026   | 48370  | 37333  | 77.18% 6 |
| 4.82%                 | 11.26%  | 23.91% | 76.09% | 85.20% | 0.860    |
| 13.02                 |         |        |        |        |          |
| bar_p50_h60_tp15_sl10 | p50_h60 | 2020   | 155475 | 58281  | 37.49% 4 |
| 7.89%                 | 4.11%   | 48.00% | 52.00% | 92.10% | 0.677    |
| 32.52                 |         |        |        |        |          |
| bar_p50_h60_tp15_sl10 | p50_h60 | 2021   | 168604 | 60450  | 35.85% 4 |
| 4.98%                 | 3.42%   | 51.60% | 48.40% | 92.93% | 0.640    |
| 34.34                 |         |        |        |        |          |
| bar_p50_h60_tp15_sl10 | p50_h60 | 2022   | 165840 | 109319 | 65.92% 5 |
| 7.21%                 | 6.50%   | 36.29% | 63.71% | 89.80% | 0.793    |
| 31.08                 |         |        |        |        |          |
| bar_p50_h60_tp15_sl10 | p50_h60 | 2023   | 163767 | 64902  | 39.63% 4 |
| 2.07%                 | 1.99%   | 55.94% | 44.06% | 95.49% | 0.611    |
| 36.28                 |         |        |        |        |          |
| bar_p50_h60_tp15_sl10 | p50_h60 | 2024   | 165149 | 81142  | 49.13% 5 |
| 0.47%                 | 3.15%   | 46.38% | 53.62% | 94.13% | 0.726    |
| 34.03                 |         |        |        |        |          |
| bar_p50_h60_tp15_sl10 | p50_h60 | 2025   | 156857 | 94814  | 60.45% 5 |
| 7.38%                 | 8.91%   | 33.70% | 66.30% | 86.56% | 0.772    |
| 27.55                 |         |        |        |        |          |
| bar_p50_h60_tp15_sl10 | p50_h60 | 2026   | 48370  | 36112  | 74.66% 6 |
| 5.04%                 | 11.38%  | 23.58% | 76.42% | 85.11% | 0.862    |
| 26.12                 |         |        |        |        |          |
| bar_p50_h90_tp15_sl10 | p50_h90 | 2020   | 155475 | 55089  | 35.43% 4 |
| 7.40%                 | 4.63%   | 47.98% | 52.02% | 91.11% | 0.665    |
| 50.08                 |         |        |        |        |          |
| bar_p50_h90_tp15_sl10 | p50_h90 | 2021   | 168604 | 58257  | 34.55% 4 |
| 5.10%                 | 3.51%   | 51.39% | 48.61% | 92.79% | 0.642    |
| 53.30                 |         |        |        |        |          |
| bar_p50_h90_tp15_sl10 | p50_h90 | 2022   | 165840 | 104127 | 62.79% 5 |
| 7.35%                 | 6.75%   | 35.91% | 64.09% | 89.47% | 0.793    |
| 47.31                 |         |        |        |        |          |
| bar_p50_h90_tp15_sl10 | p50_h90 | 2023   | 163767 | 62514  | 38.17% 4 |
| 2.64%                 | 2.56%   | 54.80% | 45.20% | 94.33% | 0.614    |
| 56.10                 |         |        |        |        |          |
| bar_p50_h90_tp15_sl10 | p50_h90 | 2024   | 165149 | 77739  | 47.07% 5 |
| 0.32%                 | 3.86%   | 45.82% | 54.18% | 92.88% | 0.716    |
| 51.79                 |         |        |        |        |          |
| bar_p50_h90_tp15_sl10 | p50_h90 | 2025   | 156857 | 90201  | 57.51% 5 |



|                       |        |         |        |        |        |          |
|-----------------------|--------|---------|--------|--------|--------|----------|
| 8.35%                 | 9.27%  | 32.38%  | 67.62% |        | 86.29% | 0.782    |
| 41.89                 |        |         |        |        |        |          |
| bar_p50_h90_tp15_sl10 |        | p50_h90 | 2026   | 48370  | 34676  | 71.69% 6 |
| 4.30%                 | 11.65% | 24.05%  | 75.95% |        | 84.66% | 0.848    |
| 39.13                 |        |         |        |        |        |          |
| bar_p60_h30_tp15_sl10 |        | p60_h30 | 2020   | 155475 | 49927  | 32.11% 4 |
| 6.06%                 | 3.46%  | 50.48%  | 49.52% |        | 93.02% | 0.656    |
| 17.24                 |        |         |        |        |        |          |
| bar_p60_h30_tp15_sl10 |        | p60_h30 | 2021   | 168604 | 49673  | 29.46% 4 |
| 3.17%                 | 2.55%  | 54.28%  | 45.72% |        | 94.42% | 0.622    |
| 18.13                 |        |         |        |        |        |          |
| bar_p60_h30_tp15_sl10 |        | p60_h30 | 2022   | 165840 | 97353  | 58.70% 5 |
| 3.05%                 | 4.79%  | 42.17%  | 57.83% |        | 91.72% | 0.748    |
| 16.65                 |        |         |        |        |        |          |
| bar_p60_h30_tp15_sl10 |        | p60_h30 | 2023   | 163767 | 51794  | 31.63% 3 |
| 8.82%                 | 1.67%  | 59.51%  | 40.49% |        | 95.88% | 0.566    |
| 18.94                 |        |         |        |        |        |          |
| bar_p60_h30_tp15_sl10 |        | p60_h30 | 2024   | 165149 | 67057  | 40.60% 4 |
| 6.17%                 | 2.45%  | 51.38%  | 48.62% |        | 94.96% | 0.668    |
| 17.87                 |        |         |        |        |        |          |
| bar_p60_h30_tp15_sl10 |        | p60_h30 | 2025   | 156857 | 85488  | 54.50% 5 |
| 5.59%                 | 6.95%  | 37.46%  | 62.54% |        | 88.89% | 0.764    |
| 14.98                 |        |         |        |        |        |          |
| bar_p60_h30_tp15_sl10 |        | p60_h30 | 2026   | 48370  | 33703  | 69.68% 6 |
| 1.60%                 | 8.57%  | 29.83%  | 70.17% |        | 87.79% | 0.838    |
| 14.34                 |        |         |        |        |        |          |
| bar_p60_h60_tp15_sl10 |        | p60_h60 | 2020   | 155475 | 45856  | 29.49% 4 |
| 4.33%                 | 3.28%  | 52.39%  | 47.61% |        | 93.10% | 0.632    |
| 34.14                 |        |         |        |        |        |          |
| bar_p60_h60_tp15_sl10 |        | p60_h60 | 2021   | 168604 | 46518  | 27.59% 4 |
| 0.78%                 | 2.43%  | 56.79%  | 43.21% |        | 94.38% | 0.587    |
| 36.06                 |        |         |        |        |        |          |
| bar_p60_h60_tp15_sl10 |        | p60_h60 | 2022   | 165840 | 93262  | 56.24% 5 |
| 2.21%                 | 4.50%  | 43.29%  | 56.71% |        | 92.06% | 0.738    |
| 33.42                 |        |         |        |        |        |          |
| bar_p60_h60_tp15_sl10 |        | p60_h60 | 2023   | 163767 | 48975  | 29.91% 3 |
| 6.31%                 | 1.53%  | 62.16%  | 37.84% |        | 95.96% | 0.529    |
| 37.51                 |        |         |        |        |        |          |
| bar_p60_h60_tp15_sl10 |        | p60_h60 | 2024   | 165149 | 65160  | 39.46% 4 |
| 4.87%                 | 2.05%  | 53.08%  | 46.92% |        | 95.63% | 0.653    |
| 35.62                 |        |         |        |        |        |          |
| bar_p60_h60_tp15_sl10 |        | p60_h60 | 2025   | 156857 | 81300  | 51.83% 5 |
| 4.88%                 | 6.76%  | 38.36%  | 61.64% |        | 89.04% | 0.756    |
| 29.38                 |        |         |        |        |        |          |
| bar_p60_h60_tp15_sl10 |        | p60_h60 | 2026   | 48370  | 32727  | 67.66% 6 |
| 1.51%                 | 8.32%  | 30.16%  | 69.84% |        | 88.08% | 0.839    |
| 28.71                 |        |         |        |        |        |          |
| bar_p60_h90_tp15_sl10 |        | p60_h90 | 2020   | 155475 | 42830  | 27.55% 4 |
| 3.69%                 | 3.58%  | 52.73%  | 47.27% |        | 92.42% | 0.619    |
| 52.56                 |        |         |        |        |        |          |
| bar_p60_h90_tp15_sl10 |        | p60_h90 | 2021   | 168604 | 44404  | 26.34% 4 |
| 1.70%                 | 2.44%  | 55.87%  | 44.13% |        | 94.48% | 0.601    |
| 55.92                 |        |         |        |        |        |          |
| bar_p60_h90_tp15_sl10 |        | p60_h90 | 2022   | 165840 | 88891  | 53.60% 5 |
| 2.88%                 | 4.78%  | 42.34%  | 57.66% |        | 91.71% | 0.745    |
| 51.27                 |        |         |        |        |        |          |
| bar_p60_h90_tp15_sl10 |        | p60_h90 | 2023   | 163767 | 46650  | 28.49% 3 |
| 7.81%                 | 2.06%  | 60.13%  | 39.87% |        | 94.84% | 0.547    |
| 58.10                 |        |         |        |        |        |          |
| bar_p60_h90_tp15_sl10 |        | p60_h90 | 2024   | 165149 | 62700  | 37.97% 4 |

|                       |         |        |        |        |          |
|-----------------------|---------|--------|--------|--------|----------|
| 4.45%                 | 2.75%   | 52.80% | 47.20% | 94.18% | 0.639    |
| 54.23                 |         |        |        |        |          |
| bar_p60_h90_tp15_sl10 | p60_h90 | 2025   | 156857 | 77645  | 49.50% 5 |
| 6.43%                 | 7.08%   | 36.48% | 63.52% | 88.85% | 0.776    |
| 44.95                 |         |        |        |        |          |
| bar_p60_h90_tp15_sl10 | p60_h90 | 2026   | 48370  | 31381  | 64.88% 6 |
| 0.15%                 | 8.84%   | 31.01% | 68.99% | 87.19% | 0.814    |
| 43.91                 |         |        |        |        |          |

Sensibilidad agregada por tipo de contexto:

| context_type         | n_targets | n_context_values_mean | operable_rate_range_mean | tp_rate_range_mean | sl_rate_range_mean | no_event_rate_range_mean | tp_share_range_mean | expectancy_range_mean | event_step_range_mean | context_sensitivity_score_mean | context_sensitivity_score_max |
|----------------------|-----------|-----------------------|--------------------------|--------------------|--------------------|--------------------------|---------------------|-----------------------|-----------------------|--------------------------------|-------------------------------|
| contract_regime      | 63        |                       | 130.0                    |                    |                    |                          | 87.07%              |                       |                       |                                |                               |
| 36.55%               |           | 24.63%                | 42.13%                   |                    |                    | 45.70%                   |                     |                       |                       |                                |                               |
| 0.652                |           | 28.26                 |                          |                    | 2.361              |                          |                     |                       |                       |                                |                               |
| 3.303                |           |                       |                          |                    |                    |                          |                     |                       |                       |                                |                               |
| year_regime          | 63        |                       | 35.0                     |                    |                    |                          | 86.40%              |                       |                       |                                |                               |
| 26.04%               |           | 20.02%                | 33.74%                   |                    |                    | 21.90%                   |                     |                       |                       |                                |                               |
| 0.409                |           | 17.96                 |                          |                    | 1.881              |                          |                     |                       |                       |                                |                               |
| 2.342                |           |                       |                          |                    |                    |                          |                     |                       |                       |                                |                               |
| year_contract        | 63        |                       | 32.0                     |                    |                    |                          | 62.45%              |                       |                       |                                |                               |
| 24.11%               |           | 18.71%                | 31.95%                   |                    |                    | 22.63%                   |                     |                       |                       |                                |                               |
| 0.374                |           | 17.04                 |                          |                    | 1.598              |                          |                     |                       |                       |                                |                               |
| 2.179                |           |                       |                          |                    |                    |                          |                     |                       |                       |                                |                               |
| contract             | 63        |                       | 26.0                     |                    |                    |                          | 53.31%              |                       |                       |                                |                               |
| 22.54%               |           | 16.81%                | 30.10%                   |                    |                    | 15.35%                   |                     |                       |                       |                                |                               |
| 0.341                |           | 14.38                 |                          |                    | 1.381              |                          |                     |                       |                       |                                |                               |
| 1.742                |           |                       |                          |                    |                    |                          |                     |                       |                       |                                |                               |
| contract_code_regime | 63        |                       | 20.0                     |                    |                    |                          | 59.27%              |                       |                       |                                |                               |
| 12.42%               |           | 9.71%                 | 15.67%                   |                    |                    | 11.46%                   |                     |                       |                       |                                |                               |
| 0.201                |           | 12.58                 |                          |                    | 1.085              |                          |                     |                       |                       |                                |                               |
| 1.389                |           |                       |                          |                    |                    |                          |                     |                       |                       |                                |                               |
| year                 | 63        |                       | 7.0                      |                    |                    |                          | 37.87%              |                       |                       |                                |                               |
| 16.90%               |           | 14.21%                | 23.88%                   |                    |                    | 11.23%                   |                     |                       |                       |                                |                               |
| 0.243                |           | 10.13                 |                          |                    | 1.041              |                          |                     |                       |                       |                                |                               |
| 1.294                |           |                       |                          |                    |                    |                          |                     |                       |                       |                                |                               |
| dataset_split        | 63        |                       | 2.0                      |                    |                    |                          | 17.68%              |                       |                       |                                |                               |
| 8.59%                |           | 7.44%                 | 12.26%                   |                    |                    | 5.61%                    |                     |                       |                       |                                |                               |
| 0.123                |           | 5.69                  |                          |                    | 0.516              |                          |                     |                       |                       |                                |                               |
| 0.652                |           |                       |                          |                    |                    |                          |                     |                       |                       |                                |                               |
| contract_code        | 63        |                       | 4.0                      |                    |                    |                          | 9.14%               |                       |                       |                                |                               |
| 7.26%                |           | 6.42%                 | 10.42%                   |                    |                    | 5.69%                    |                     |                       |                       |                                |                               |
| 0.102                |           | 4.96                  |                          |                    | 0.389              |                          |                     |                       |                       |                                |                               |
| 0.535                |           |                       |                          |                    |                    |                          |                     |                       |                       |                                |                               |

Sensibilidad por configuración TP/SL dentro de cada contexto:

| context_type | tp_sl_config    | n_targets | operable_rate_range_mean | tp_rate_range_mean | sl_rate_range_mean | no_event_rate_range_mean | tp_share_range_mean | expectancy_range_mean | context_sensitivity_score_mean | context_sensitivity_score_max |
|--------------|-----------------|-----------|--------------------------|--------------------|--------------------|--------------------------|---------------------|-----------------------|--------------------------------|-------------------------------|
| contract     | TP 1.0 / SL 1.0 | 9         |                          | 53.31%             |                    |                          |                     |                       |                                |                               |
| 8.84%        |                 |           | 0.00%                    | 8.84%              |                    |                          |                     |                       |                                |                               |
| 0.177        |                 | 0.798     |                          | 0.845              |                    |                          |                     |                       |                                |                               |
| contract     | TP 1.5 / SL 1.0 | 9         |                          | 53.31%             |                    |                          |                     |                       |                                |                               |
| 30.88%       |                 | 10.33%    | 39.37%                   | 12.50%             |                    |                          |                     |                       |                                |                               |
| 0.382        |                 | 1.464     |                          | 1.542              |                    |                          |                     |                       |                                |                               |
| contract     | TP 2.0 / SL 1.0 | 9         |                          | 53.31%             |                    |                          |                     |                       |                                |                               |
| 30.96%       |                 | 11.98%    | 41.93%                   | 15.99%             |                    |                          |                     |                       |                                |                               |
| 0.517        |                 | 1.542     |                          | 1.661              |                    |                          |                     |                       |                                |                               |

|        |                                      |        |        |
|--------|--------------------------------------|--------|--------|
|        | contract TP 2.5 / SL 1.0             | 9      | 53.31% |
| 24.16% | 13.03%                               | 36.61% | 17.41% |
| 0.491  | 1.445                                |        | 1.546  |
|        | contract TP 1.0 / SL 0.5             | 9      | 53.31% |
| 21.00% | 21.00%                               | 0.00%  | 21.00% |
| 0.315  | 1.163                                |        | 1.222  |
|        | contract TP 1.5 / SL 0.5             | 9      | 53.31% |
| 19.33% | 24.84%                               | 43.09% | 19.53% |
| 0.177  | 1.601                                |        | 1.742  |
|        | contract TP 2.0 / SL 0.5             | 9      | 53.31% |
| 22.63% | 27.65%                               | 49.68% | 12.18% |
| 0.328  | 1.655                                |        | 1.726  |
|        | contract_code TP 1.0 / SL 1.0        | 9      | 9.14%  |
| 3.76%  | 3.76%                                | 0.00%  | 3.76%  |
| 0.075  | 0.204                                |        | 0.211  |
|        | contract_code TP 1.5 / SL 1.0        | 9      | 9.14%  |
| 8.81%  | 4.40%                                | 13.21% | 5.38%  |
| 0.088  | 0.409                                |        | 0.446  |
|        | contract_code TP 2.0 / SL 1.0        | 9      | 9.14%  |
| 10.33% | 5.05%                                | 15.38% | 6.51%  |
| 0.156  | 0.464                                |        | 0.490  |
|        | contract_code TP 2.5 / SL 1.0        | 9      | 9.14%  |
| 8.65%  | 5.45%                                | 14.10% | 6.46%  |
| 0.162  | 0.438                                |        | 0.460  |
|        | contract_code TP 1.0 / SL 0.5        | 9      | 9.14%  |
| 7.65%  | 7.65%                                | 0.00%  | 7.65%  |
| 0.115  | 0.321                                |        | 0.360  |
|        | contract_code TP 1.5 / SL 0.5        | 9      | 9.14%  |
| 4.62%  | 8.78%                                | 13.39% | 6.56%  |
| 0.029  | 0.425                                |        | 0.521  |
|        | contract_code TP 2.0 / SL 0.5        | 9      | 9.14%  |
| 7.01%  | 9.85%                                | 16.86% | 3.49%  |
| 0.091  | 0.464                                |        | 0.535  |
|        | contract_code_regime TP 1.0 / SL 1.0 | 9      | 59.27% |
| 5.88%  | 5.88%                                | 0.00%  | 5.88%  |
| 0.118  | 0.769                                |        | 0.902  |
|        | contract_code_regime TP 1.5 / SL 1.0 | 9      | 59.27% |
| 13.89% | 6.76%                                | 19.39% | 8.97%  |
| 0.169  | 1.083                                |        | 1.169  |
|        | contract_code_regime TP 2.0 / SL 1.0 | 9      | 59.27% |
| 17.28% | 7.75%                                | 24.16% | 12.54% |
| 0.284  | 1.210                                |        | 1.324  |
|        | contract_code_regime TP 2.5 / SL 1.0 | 9      | 59.27% |
| 15.47% | 8.47%                                | 23.02% | 16.18% |
| 0.312  | 1.224                                |        | 1.344  |
|        | contract_code_regime TP 1.0 / SL 0.5 | 9      | 59.27% |
| 11.76% | 11.76%                               | 0.00%  | 11.76% |
| 0.176  | 0.945                                |        | 1.060  |
|        | contract_code_regime TP 1.5 / SL 0.5 | 9      | 59.27% |
| 10.26% | 12.98%                               | 18.93% | 12.26% |
| 0.140  | 1.137                                |        | 1.236  |
|        | contract_code_regime TP 2.0 / SL 0.5 | 9      | 59.27% |
| 12.43% | 14.37%                               | 24.21% | 12.63% |
| 0.211  | 1.229                                |        | 1.389  |
|        | contract_regime TP 1.0 / SL 1.0      | 9      | 87.07% |
| 12.62% | 12.62%                               | 0.00%  | 12.62% |
| 0.252  | 1.249                                |        | 1.418  |
|        | contract_regime TP 1.5 / SL 1.0      | 9      | 87.07% |
| 45.65% | 14.72%                               | 54.78% | 24.73% |
| 0.657  | 2.270                                |        | 2.467  |

|                                 |        |        |
|---------------------------------|--------|--------|
| contract_regime TP 2.0 / SL 1.0 | 9      | 87.07% |
| 48.18%                          | 16.77% | 60.80% |
| 0.905                           | 2.689  | 56.11% |
| contract_regime TP 2.5 / SL 1.0 | 9      | 87.07% |
| 40.54%                          | 18.42% | 53.75% |
| 0.995                           | 2.848  | 85.05% |
| contract_regime TP 1.0 / SL 0.5 | 9      | 87.07% |
| 32.36%                          | 32.36% | 0.00%  |
| 0.485                           | 1.841  | 32.36% |
| contract_regime TP 1.5 / SL 0.5 | 9      | 87.07% |
| 36.28%                          | 37.19% | 57.73% |
| 0.545                           | 2.644  | 46.15% |
| contract_regime TP 2.0 / SL 0.5 | 9      | 87.07% |
| 40.21%                          | 40.34% | 67.86% |
| 0.727                           | 2.984  | 62.86% |
| dataset_split TP 1.0 / SL 1.0   | 9      | 17.68% |
| 4.48%                           | 4.48%  | 0.00%  |
| 0.090                           | 0.311  | 4.48%  |
| dataset_split TP 1.5 / SL 1.0   | 9      | 17.68% |
| 9.68%                           | 5.20%  | 14.88% |
| 0.093                           | 0.532  | 5.72%  |
| dataset_split TP 2.0 / SL 1.0   | 9      | 17.68% |
| 12.52%                          | 5.98%  | 18.50% |
| 0.191                           | 0.607  | 6.03%  |
| dataset_split TP 2.5 / SL 1.0   | 9      | 17.68% |
| 11.24%                          | 6.50%  | 17.74% |
| 0.216                           | 0.580  | 4.85%  |
| dataset_split TP 1.0 / SL 0.5   | 9      | 17.68% |
| 8.70%                           | 8.70%  | 0.00%  |
| 0.131                           | 0.438  | 8.70%  |
| dataset_split TP 1.5 / SL 0.5   | 9      | 17.68% |
| 4.96%                           | 9.99%  | 14.95% |
| 0.028                           | 0.544  | 6.79%  |
| dataset_split TP 2.0 / SL 0.5   | 9      | 17.68% |
| 8.54%                           | 11.22% | 19.76% |
| 0.115                           | 0.599  | 2.71%  |
| year TP 1.0 / SL 1.0            | 9      | 37.87% |
| 7.78%                           | 7.78%  | 0.00%  |
| 0.156                           | 0.612  | 7.78%  |
| year TP 1.5 / SL 1.0            | 9      | 37.87% |
| 21.27%                          | 9.12%  | 30.39% |
| 0.228                           | 1.085  | 9.82%  |
| year TP 2.0 / SL 1.0            | 9      | 37.87% |
| 24.07%                          | 10.57% | 34.64% |
| 0.376                           | 1.182  | 11.08% |
| year TP 2.5 / SL 1.0            | 9      | 37.87% |
| 19.40%                          | 11.50% | 30.90% |
| 0.371                           | 1.108  | 11.17% |
| year TP 1.0 / SL 0.5            | 9      | 37.87% |
| 17.53%                          | 17.53% | 0.00%  |
| 0.263                           | 0.904  | 17.53% |
| year TP 1.5 / SL 0.5            | 9      | 37.87% |
| 11.64%                          | 20.28% | 31.88% |
| 0.086                           | 1.159  | 14.18% |
| year TP 2.0 / SL 0.5            | 9      | 37.87% |
| 16.63%                          | 22.70% | 39.32% |
| 0.219                           | 1.236  | 7.09%  |
| year_contract TP 1.0 / SL 1.0   | 9      | 62.45% |
| 9.96%                           | 9.96%  | 0.00%  |
| 0.199                           | 0.923  | 9.96%  |
|                                 |        | 1.011  |

|        |                               |        |        |
|--------|-------------------------------|--------|--------|
|        | year_contract TP 1.5 / SL 1.0 | 9      | 62.45% |
| 32.32% | 11.71%                        | 41.21% | 14.97% |
| 0.404  | 1.627                         |        | 1.722  |
|        | year_contract TP 2.0 / SL 1.0 | 9      | 62.45% |
| 32.61% | 13.57%                        | 44.79% | 22.87% |
| 0.539  | 1.763                         |        | 1.903  |
|        | year_contract TP 2.5 / SL 1.0 | 9      | 62.45% |
| 25.88% | 14.72%                        | 39.79% | 35.71% |
| 0.527  | 1.785                         |        | 2.095  |
|        | year_contract TP 1.0 / SL 0.5 | 9      | 62.45% |
| 23.00% | 23.00%                        | 0.00%  | 23.00% |
| 0.345  | 1.315                         |        | 1.417  |
|        | year_contract TP 1.5 / SL 0.5 | 9      | 62.45% |
| 21.05% | 27.49%                        | 45.62% | 24.88% |
| 0.232  | 1.815                         |        | 2.007  |
|        | year_contract TP 2.0 / SL 0.5 | 9      | 62.45% |
| 23.92% | 30.48%                        | 52.25% | 27.00% |
| 0.375  | 1.961                         |        | 2.179  |
|        | year_regime TP 1.0 / SL 1.0   | 9      | 86.40% |
| 11.77% | 11.77%                        | 0.00%  | 11.77% |
| 0.235  | 1.217                         |        | 1.374  |
|        | year_regime TP 1.5 / SL 1.0   | 9      | 86.40% |
| 32.33% | 13.44%                        | 42.11% | 15.62% |
| 0.408  | 1.899                         |        | 2.024  |
|        | year_regime TP 2.0 / SL 1.0   | 9      | 86.40% |
| 36.14% | 15.19%                        | 50.15% | 22.09% |
| 0.604  | 2.100                         |        | 2.247  |
|        | year_regime TP 2.5 / SL 1.0   | 9      | 86.40% |
| 29.55% | 16.72%                        | 45.88% | 35.37% |
| 0.586  | 2.139                         |        | 2.279  |
|        | year_regime TP 1.0 / SL 0.5   | 9      | 86.40% |
| 24.97% | 24.97%                        | 0.00%  | 24.97% |
| 0.375  | 1.613                         |        | 1.716  |
|        | year_regime TP 1.5 / SL 0.5   | 9      | 86.40% |
| 21.07% | 27.63%                        | 43.59% | 22.09% |
| 0.237  | 2.008                         |        | 2.167  |
|        | year_regime TP 2.0 / SL 0.5   | 9      | 86.40% |
| 26.44% | 30.45%                        | 54.44% | 21.37% |
| 0.420  | 2.191                         |        | 2.342  |

Sensibilidad por percentil/horizonte dentro de cada contexto:

context\_type target\_base percentile horizon n\_configs operable\_rate\_range\_mean tp\_rate\_range\_mean sl\_rate\_range\_mean no\_event\_rate\_range\_mean tp\_share\_range\_mean expectancy\_range\_mean context\_sensitivity\_score\_mean context\_sensitivity\_score\_max

|        |          |         |    |       |        |
|--------|----------|---------|----|-------|--------|
|        | contract | p40_h30 | 40 | 30    | 7      |
| 53.55% | 21.17%   | 16.65%  |    |       | 28.79% |
| 13.54% | 0.317    |         |    | 1.337 |        |
| 1.537  |          |         |    |       |        |
|        | contract | p40_h60 | 40 | 60    | 7      |
| 52.47% | 23.03%   | 19.04%  |    |       | 32.14% |
| 15.70% | 0.341    |         |    | 1.424 |        |
| 1.694  |          |         |    |       |        |
|        | contract | p40_h90 | 40 | 90    | 7      |
| 49.17% | 24.70%   | 19.78%  |    |       | 33.20% |
| 16.64% | 0.371    |         |    | 1.435 |        |
| 1.709  |          |         |    |       |        |
|        | contract | p50_h30 | 50 | 30    | 7      |
| 55.94% | 21.02%   | 15.36%  |    |       | 28.23% |
| 13.66% | 0.318    |         |    | 1.342 |        |

|        |                      |         |        |       |        |
|--------|----------------------|---------|--------|-------|--------|
| 1.559  |                      |         |        |       |        |
|        | contract             | p50_h60 | 50     | 60    | 7      |
| 54.31% |                      | 23.65%  | 17.42% |       | 31.62% |
| 15.61% |                      | 0.355   |        | 1.426 |        |
| 1.726  |                      |         |        |       |        |
|        | contract             | p50_h90 | 50     | 90    | 7      |
| 52.11% |                      | 23.72%  | 17.90% |       | 31.54% |
| 16.91% |                      | 0.359   |        | 1.422 |        |
| 1.699  |                      |         |        |       |        |
|        | contract             | p60_h30 | 60     | 30    | 7      |
| 54.76% |                      | 20.46%  | 13.93% |       | 27.03% |
| 14.16% |                      | 0.312   |        | 1.303 |        |
| 1.590  |                      |         |        |       |        |
|        | contract             | p60_h60 | 60     | 60    | 7      |
| 54.14% |                      | 22.84%  | 15.37% |       | 29.45% |
| 15.36% |                      | 0.348   |        | 1.372 |        |
| 1.742  |                      |         |        |       |        |
|        | contract             | p60_h90 | 60     | 90    | 7      |
| 53.30% |                      | 22.30%  | 15.85% |       | 28.87% |
| 16.56% |                      | 0.347   |        | 1.369 |        |
| 1.687  |                      |         |        |       |        |
|        | contract_code        | p40_h30 | 40     | 30    | 7      |
| 8.10%  |                      | 6.67%   | 5.97%  |       | 9.62%  |
| 4.79%  |                      | 0.095   |        | 0.352 |        |
| 0.444  |                      |         |        |       |        |
|        | contract_code        | p40_h60 | 40     | 60    | 7      |
| 7.19%  |                      | 7.03%   | 6.70%  |       | 10.36% |
| 5.47%  |                      | 0.098   |        | 0.368 |        |
| 0.471  |                      |         |        |       |        |
|        | contract_code        | p40_h90 | 40     | 90    | 7      |
| 6.72%  |                      | 7.02%   | 7.19%  |       | 10.53% |
| 6.00%  |                      | 0.094   |        | 0.375 |        |
| 0.467  |                      |         |        |       |        |
|        | contract_code        | p50_h30 | 50     | 30    | 7      |
| 9.77%  |                      | 7.33%   | 5.92%  |       | 10.25% |
| 5.05%  |                      | 0.106   |        | 0.383 |        |
| 0.469  |                      |         |        |       |        |
|        | contract_code        | p50_h60 | 50     | 60    | 7      |
| 9.25%  |                      | 7.72%   | 6.74%  |       | 11.05% |
| 5.92%  |                      | 0.109   |        | 0.407 |        |
| 0.491  |                      |         |        |       |        |
|        | contract_code        | p50_h90 | 50     | 90    | 7      |
| 8.83%  |                      | 7.20%   | 6.91%  |       | 10.60% |
| 6.31%  |                      | 0.098   |        | 0.398 |        |
| 0.469  |                      |         |        |       |        |
|        | contract_code        | p60_h30 | 60     | 30    | 7      |
| 11.20% |                      | 7.60%   | 5.63%  |       | 10.39% |
| 5.23%  |                      | 0.112   |        | 0.400 |        |
| 0.494  |                      |         |        |       |        |
|        | contract_code        | p60_h60 | 60     | 60    | 7      |
| 10.68% |                      | 7.90%   | 6.35%  |       | 11.01% |
| 6.06%  |                      | 0.113   |        | 0.420 |        |
| 0.535  |                      |         |        |       |        |
|        | contract_code        | p60_h90 | 60     | 90    | 7      |
| 10.52% |                      | 6.87%   | 6.38%  |       | 9.99%  |
| 6.37%  |                      | 0.095   |        | 0.401 |        |
| 0.504  |                      |         |        |       |        |
|        | contract_code_regime | p40_h30 | 40     | 30    | 7      |
| 66.56% |                      | 12.57%  | 11.52% |       | 15.47% |
| 12.31% |                      | 0.217   |        | 1.184 |        |

|                      |         |        |       |        |
|----------------------|---------|--------|-------|--------|
| 1.389                |         |        |       |        |
| contract_code_regime | p40_h60 | 40     | 60    | 7      |
| 68.30%               | 9.76%   | 8.88%  |       | 13.12% |
| 8.59%                | 0.155   |        | 1.087 |        |
| 1.227                |         |        |       |        |
| contract_code_regime | p40_h90 | 40     | 90    | 7      |
| 67.63%               | 11.62%  | 9.49%  |       | 15.09% |
| 10.22%               | 0.187   |        | 1.140 |        |
| 1.344                |         |        |       |        |
| contract_code_regime | p50_h30 | 50     | 30    | 7      |
| 58.75%               | 12.99%  | 11.78% |       | 16.13% |
| 14.23%               | 0.220   |        | 1.139 |        |
| 1.379                |         |        |       |        |
| contract_code_regime | p50_h60 | 50     | 60    | 7      |
| 59.80%               | 11.24%  | 8.53%  |       | 14.27% |
| 9.63%                | 0.178   |        | 1.035 |        |
| 1.191                |         |        |       |        |
| contract_code_regime | p50_h90 | 50     | 90    | 7      |
| 59.91%               | 13.22%  | 9.72%  |       | 17.03% |
| 11.56%               | 0.209   |        | 1.114 |        |
| 1.287                |         |        |       |        |
| contract_code_regime | p60_h30 | 60     | 30    | 7      |
| 50.50%               | 13.12%  | 9.80%  |       | 16.40% |
| 12.47%               | 0.210   |        | 1.023 |        |
| 1.216                |         |        |       |        |
| contract_code_regime | p60_h60 | 60     | 60    | 7      |
| 50.93%               | 12.57%  | 7.94%  |       | 15.12% |
| 10.32%               | 0.203   |        | 0.969 |        |
| 1.118                |         |        |       |        |
| contract_code_regime | p60_h90 | 60     | 90    | 7      |
| 51.05%               | 14.75%  | 9.72%  |       | 18.42% |
| 13.81%               | 0.234   |        | 1.078 |        |
| 1.253                |         |        |       |        |
| contract_regime      | p40_h30 | 40     | 30    | 7      |
| 91.39%               | 40.57%  | 28.87% |       | 41.80% |
| 43.27%               | 0.766   |        | 2.459 |        |
| 3.162                |         |        |       |        |
| contract_regime      | p40_h60 | 40     | 60    | 7      |
| 92.46%               | 32.00%  | 26.31% |       | 39.44% |
| 32.33%               | 0.535   |        | 2.225 |        |
| 2.674                |         |        |       |        |
| contract_regime      | p40_h90 | 40     | 90    | 7      |
| 94.65%               | 33.46%  | 26.12% |       | 43.00% |
| 43.58%               | 0.591   |        | 2.408 |        |
| 3.016                |         |        |       |        |
| contract_regime      | p50_h30 | 50     | 30    | 7      |
| 86.98%               | 45.12%  | 26.65% |       | 42.44% |
| 45.94%               | 0.841   |        | 2.471 |        |
| 3.232                |         |        |       |        |
| contract_regime      | p50_h60 | 50     | 60    | 7      |
| 86.75%               | 31.67%  | 24.44% |       | 40.69% |
| 39.02%               | 0.517   |        | 2.226 |        |
| 2.778                |         |        |       |        |
| contract_regime      | p50_h90 | 50     | 90    | 7      |
| 89.47%               | 34.34%  | 24.23% |       | 43.97% |
| 49.29%               | 0.606   |        | 2.413 |        |
| 2.939                |         |        |       |        |
| contract_regime      | p60_h30 | 60     | 30    | 7      |
| 80.13%               | 46.03%  | 22.14% |       | 45.25% |
| 52.13%               | 0.896   |        | 2.457 |        |

3.303

|        |                 |         |        |       |        |
|--------|-----------------|---------|--------|-------|--------|
|        | contract_regime | p60_h60 | 60     | 60    | 7      |
| 80.35% | 31.95%          |         | 20.70% |       | 40.69% |
| 44.53% | 0.518           |         |        | 2.182 |        |
| 2.719  |                 |         |        |       |        |
|        | contract_regime | p60_h90 | 60     | 90    | 7      |
| 81.49% | 33.80%          |         | 22.22% |       | 41.91% |
| 61.20% | 0.603           |         |        | 2.406 |        |
| 3.145  |                 |         |        |       |        |
|        | dataset_split   | p40_h30 | 40     | 30    | 7      |
| 16.59% | 7.95%           |         | 7.15%  |       | 11.50% |
| 5.18%  | 0.113           |         |        | 0.484 |        |
| 0.589  |                 |         |        |       |        |
|        | dataset_split   | p40_h60 | 40     | 60    | 7      |
| 16.49% | 8.03%           |         | 8.10%  |       | 12.03% |
| 6.06%  | 0.111           |         |        | 0.507 |        |
| 0.616  |                 |         |        |       |        |
|        | dataset_split   | p40_h90 | 40     | 90    | 7      |
| 15.64% | 8.16%           |         | 8.39%  |       | 12.29% |
| 6.09%  | 0.113           |         |        | 0.506 |        |
| 0.612  |                 |         |        |       |        |
|        | dataset_split   | p50_h30 | 50     | 30    | 7      |
| 18.43% | 8.40%           |         | 6.81%  |       | 11.77% |
| 5.15%  | 0.122           |         |        | 0.506 |        |
| 0.600  |                 |         |        |       |        |
|        | dataset_split   | p50_h60 | 50     | 60    | 7      |
| 18.11% | 8.86%           |         | 7.82%  |       | 12.74% |
| 6.05%  | 0.126           |         |        | 0.536 |        |
| 0.636  |                 |         |        |       |        |
|        | dataset_split   | p50_h90 | 50     | 90    | 7      |
| 17.16% | 8.96%           |         | 7.95%  |       | 12.87% |
| 5.82%  | 0.127           |         |        | 0.528 |        |
| 0.623  |                 |         |        |       |        |
|        | dataset_split   | p60_h30 | 60     | 30    | 7      |
| 19.51% | 8.64%           |         | 6.42%  |       | 11.81% |
| 5.13%  | 0.128           |         |        | 0.515 |        |
| 0.611  |                 |         |        |       |        |
|        | dataset_split   | p60_h60 | 60     | 60    | 7      |
| 18.95% | 9.21%           |         | 7.14%  |       | 12.71% |
| 5.66%  | 0.135           |         |        | 0.537 |        |
| 0.652  |                 |         |        |       |        |
|        | dataset_split   | p60_h90 | 60     | 90    | 7      |
| 18.26% | 9.10%           |         | 7.15%  |       | 12.62% |
| 5.37%  | 0.132           |         |        | 0.525 |        |
| 0.639  |                 |         |        |       |        |
|        | year            | p40_h30 | 40     | 30    | 7      |
| 36.43% | 16.84%          |         | 14.34% |       | 23.95% |
| 10.49% | 0.243           |         |        | 1.021 |        |
| 1.221  |                 |         |        |       |        |
|        | year            | p40_h60 | 40     | 60    | 7      |
| 36.00% | 17.48%          |         | 16.13% |       | 25.43% |
| 11.96% | 0.249           |         |        | 1.070 |        |
| 1.278  |                 |         |        |       |        |
|        | year            | p40_h90 | 40     | 90    | 7      |
| 34.23% | 16.90%          |         | 16.51% |       | 24.91% |
| 12.16% | 0.240           |         |        | 1.047 |        |
| 1.230  |                 |         |        |       |        |
|        | year            | p50_h30 | 50     | 30    | 7      |
| 39.37% | 17.15%          |         | 13.22% |       | 23.65% |
| 10.25% | 0.252           |         |        | 1.036 |        |



|        |               |         |        |       |        |
|--------|---------------|---------|--------|-------|--------|
| 1.214  |               |         |        |       |        |
|        | year          | p50_h60 | 50     | 60    | 7      |
| 38.80% | 17.92%        |         | 14.98% |       | 25.27% |
| 11.94% |               | 0.255   |        | 1.089 |        |
| 1.294  |               |         |        |       |        |
|        | year          | p50_h90 | 50     | 90    | 7      |
| 37.14% | 16.75%        |         | 15.00% |       | 24.12% |
| 11.70% |               | 0.235   |        | 1.047 |        |
| 1.272  |               |         |        |       |        |
|        | year          | p60_h30 | 60     | 30    | 7      |
| 40.22% | 16.68%        |         | 11.89% |       | 22.51% |
| 10.05% |               | 0.248   |        | 1.013 |        |
| 1.234  |               |         |        |       |        |
|        | year          | p60_h60 | 60     | 60    | 7      |
| 40.07% | 16.93%        |         | 13.00% |       | 23.24% |
| 11.39% |               | 0.243   |        | 1.046 |        |
| 1.288  |               |         |        |       |        |
|        | year          | p60_h90 | 60     | 90    | 7      |
| 38.54% | 15.48%        |         | 12.82% |       | 21.82% |
| 11.17% |               | 0.219   |        | 0.998 |        |
| 1.249  |               |         |        |       |        |
|        | year_contract | p40_h30 | 40     | 30    | 7      |
| 63.21% | 24.28%        |         | 18.64% |       | 32.89% |
| 17.02% |               | 0.361   |        | 1.560 |        |
| 1.875  |               |         |        |       |        |
|        | year_contract | p40_h60 | 40     | 60    | 7      |
| 64.10% | 25.25%        |         | 20.22% |       | 33.77% |
| 18.83% |               | 0.376   |        | 1.622 |        |
| 1.872  |               |         |        |       |        |
|        | year_contract | p40_h90 | 40     | 90    | 7      |
| 61.51% | 25.43%        |         | 22.32% |       | 33.75% |
| 24.38% |               | 0.388   |        | 1.674 |        |
| 2.006  |               |         |        |       |        |
|        | year_contract | p50_h30 | 50     | 30    | 7      |
| 64.11% | 22.42%        |         | 16.61% |       | 29.95% |
| 18.49% |               | 0.340   |        | 1.516 |        |
| 1.782  |               |         |        |       |        |
|        | year_contract | p50_h60 | 50     | 60    | 7      |
| 64.73% | 24.63%        |         | 18.88% |       | 33.14% |
| 20.30% |               | 0.378   |        | 1.617 |        |
| 1.940  |               |         |        |       |        |
|        | year_contract | p50_h90 | 50     | 90    | 7      |
| 61.77% | 25.22%        |         | 21.39% |       | 32.85% |
| 28.13% |               | 0.396   |        | 1.694 |        |
| 2.179  |               |         |        |       |        |
|        | year_contract | p60_h30 | 60     | 30    | 7      |
| 61.91% | 21.45%        |         | 14.85% |       | 28.38% |
| 19.29% |               | 0.333   |        | 1.459 |        |
| 1.812  |               |         |        |       |        |
|        | year_contract | p60_h60 | 60     | 60    | 7      |
| 61.54% | 24.68%        |         | 17.09% |       | 31.77% |
| 23.11% |               | 0.400   |        | 1.582 |        |
| 2.043  |               |         |        |       |        |
|        | year_contract | p60_h90 | 60     | 90    | 7      |
| 59.14% | 23.62%        |         | 18.36% |       | 31.04% |
| 34.09% |               | 0.399   |        | 1.662 |        |
| 2.140  |               |         |        |       |        |
|        | year_regime   | p40_h30 | 40     | 30    | 7      |
| 89.04% | 25.79%        |         | 22.42% |       | 34.16% |
| 21.76% |               | 0.408   |        | 1.932 |        |

|        |             |         |        |       |        |
|--------|-------------|---------|--------|-------|--------|
| 2.229  | year_regime | p40_h60 | 40     | 60    | 7      |
| 92.40% | 24.13%      |         | 19.86% |       | 31.59% |
| 17.24% | 0.379       |         |        | 1.852 |        |
| 2.129  | year_regime | p40_h90 | 40     | 90    | 7      |
| 95.40% | 27.34%      |         | 21.67% |       | 35.17% |
| 19.20% | 0.428       |         |        | 1.988 |        |
| 2.279  | year_regime | p50_h30 | 50     | 30    | 7      |
| 83.32% | 25.04%      |         | 21.21% |       | 32.47% |
| 26.07% | 0.403       |         |        | 1.881 |        |
| 2.248  | year_regime | p50_h60 | 50     | 60    | 7      |
| 86.52% | 25.17%      |         | 19.20% |       | 33.30% |
| 19.04% | 0.393       |         |        | 1.832 |        |
| 2.187  | year_regime | p50_h90 | 50     | 90    | 7      |
| 91.45% | 27.66%      |         | 21.00% |       | 35.82% |
| 21.79% | 0.434       |         |        | 1.977 |        |
| 2.342  | year_regime | p60_h30 | 60     | 30    | 7      |
| 75.74% | 26.13%      |         | 18.91% |       | 32.85% |
| 25.52% | 0.425       |         |        | 1.792 |        |
| 2.137  | year_regime | p60_h60 | 60     | 60    | 7      |
| 80.71% | 25.51%      |         | 17.25% |       | 33.06% |
| 21.09% | 0.393       |         |        | 1.776 |        |
| 2.141  | year_regime | p60_h90 | 60     | 90    | 7      |
| 83.00% | 27.58%      |         | 18.69% |       | 35.23% |
| 25.36% | 0.422       |         |        | 1.899 |        |
| 2.275  |             |         |        |       |        |

Tabla final de sensibilidad por contrato/contexto:

| context_type        | bar_col                     | target_base                  | tp_sl_config                | n_context              |
|---------------------|-----------------------------|------------------------------|-----------------------------|------------------------|
| _values             | operable_rate_mean          | operable_rate_range          | tp_rate_mean                | tp_rate_range          |
| sl_rate_mean        | sl_rate_range               | no_event_rate_mean           | no_event_rate_range         | resolved_rate_mean     |
| resolved_rate_range | tp_share_over_resolved_mean | tp_share_over_resolved_range | expectancy_units_mean       | expectancy_units_range |
| event_step_mean     | event_step_mean_range       | context_sensitivity_score    | context_sensitivity_profile |                        |
| contract            | bar_p60_h60_tp10_sl10       | p60_h60                      | TP 1.0 / SL 1.0             |                        |
| 26                  | 41.23%                      | 54.14%                       | 97.06%                      | 6.50%                  |
| 6.50%               | 0.00%                       | 0.00%                        | 100.00%                     | 0.0                    |
| 0%                  | 97.06%                      |                              | 6.50%                       | 0.941                  |
| 0.130               | 30.92                       | 12.65                        |                             | 0.736                  |
| menor_sensibilidad  | contract                    | bar_p60_h90_tp10_sl10        | p60_h90                     | TP 1.0 / SL 1.0        |
| 26                  | 39.34%                      | 53.30%                       | 96.83%                      | 7.37%                  |
| 7.37%               | 0.00%                       | 0.00%                        | 100.00%                     | 0.0                    |
| 0%                  | 96.83%                      |                              | 7.37%                       | 0.937                  |
| 0.147               | 47.13                       | 21.18                        |                             | 0.754                  |
| menor_sensibilidad  | contract                    | bar_p60_h30_tp10_sl10        | p60_h30                     | TP 1.0 / SL 1.0        |
| 26                  | 43.30%                      | 54.76%                       | 96.85%                      | 6.93%                  |
| 6.93%               | 0.00%                       | 0.00%                        | 100.00%                     | 0.0                    |
| 0%                  | 96.85%                      |                              | 6.93%                       | 0.937                  |
| 0.139               | 15.57                       | 6.25                         |                             | 0.755                  |
| menor_sensibilidad  | contract                    | bar_p50_h90_tp10_sl10        | p50_h90                     | TP 1.0 / SL 1.0        |

|                    |                                |        |                         |        |        |
|--------------------|--------------------------------|--------|-------------------------|--------|--------|
| 26                 | 47.85%                         | 52.11% | 95.55%                  | 9.30%  | 4.45%  |
| 9.30%              | 0.00%                          | 0.00%  | 100.00%                 |        | 0.0    |
| 0%                 | 95.55%                         |        | 9.30%                   |        | 0.911  |
| 0.186              | 43.81                          | 21.76  |                         | 0.800  |        |
| menor_sensibilidad |                                |        |                         |        |        |
|                    | contract bar_p50_h60_tp10_sl10 |        | p50_h60 TP 1.0 / SL 1.0 |        |        |
| 26                 | 50.05%                         | 54.31% | 95.89%                  | 8.69%  | 4.11%  |
| 8.69%              | 0.00%                          | 0.00%  | 100.00%                 |        | 0.0    |
| 0%                 | 95.89%                         |        | 8.69%                   |        | 0.918  |
| 0.174              | 28.92                          | 13.24  |                         | 0.804  |        |
| menor_sensibilidad |                                |        |                         |        |        |
|                    | contract bar_p50_h30_tp10_sl10 |        | p50_h30 TP 1.0 / SL 1.0 |        |        |
| 26                 | 52.50%                         | 55.94% | 95.67%                  | 8.80%  | 4.33%  |
| 8.80%              | 0.00%                          | 0.00%  | 100.00%                 |        | 0.0    |
| 0%                 | 95.67%                         |        | 8.80%                   |        | 0.913  |
| 0.176              | 14.55                          | 6.49   |                         | 0.823  |        |
| menor_sensibilidad |                                |        |                         |        |        |
|                    | contract bar_p40_h90_tp10_sl10 |        | p40_h90 TP 1.0 / SL 1.0 |        |        |
| 26                 | 56.05%                         | 49.17% | 93.97%                  | 11.15% | 6.03%  |
| 11.15%             | 0.00%                          | 0.00%  | 100.00%                 |        | 0.     |
| 00%                | 93.97%                         |        | 11.15%                  |        | 0.87   |
| 9                  | 0.223                          | 40.35  |                         | 22.30  |        |
| 0.826              | menor_sensibilidad             |        |                         |        |        |
|                    | contract bar_p40_h60_tp10_sl10 |        | p40_h60 TP 1.0 / SL 1.0 |        |        |
| 26                 | 58.91%                         | 52.47% | 94.36%                  | 10.55% | 5.64%  |
| 10.55%             | 0.00%                          | 0.00%  | 100.00%                 |        | 0.     |
| 00%                | 94.36%                         |        | 10.55%                  |        | 0.88   |
| 7                  | 0.211                          | 26.70  |                         | 13.69  |        |
| 0.841              | menor_sensibilidad             |        |                         |        |        |
|                    | contract bar_p40_h30_tp10_sl10 |        | p40_h30 TP 1.0 / SL 1.0 |        |        |
| 26                 | 61.51%                         | 53.55% | 94.18%                  | 10.30% | 5.82%  |
| 10.30%             | 0.00%                          | 0.00%  | 100.00%                 |        | 0.     |
| 00%                | 94.18%                         |        | 10.30%                  |        | 0.88   |
| 4                  | 0.206                          | 13.45  |                         | 6.71   |        |
| 0.845              | menor_sensibilidad             |        |                         |        |        |
|                    | contract bar_p60_h30_tp10_sl05 |        | p60_h30 TP 1.0 / SL 0.5 |        |        |
| 26                 | 43.30%                         | 54.76% | 86.03%                  | 17.75% | 13.97% |
| 17.75%             | 0.00%                          | 0.00%  | 100.00%                 |        | 0.     |
| 00%                | 86.03%                         |        | 17.75%                  |        | 0.79   |
| 1                  | 0.266                          | 14.05  |                         | 7.73   |        |
| 1.080              | menor_sensibilidad             |        |                         |        |        |
|                    | contract bar_p50_h30_tp10_sl05 |        | p50_h30 TP 1.0 / SL 0.5 |        |        |
| 26                 | 52.50%                         | 55.94% | 82.92%                  | 18.53% | 17.08% |
| 18.53%             | 0.00%                          | 0.00%  | 100.00%                 |        | 0.     |
| 00%                | 82.92%                         |        | 18.53%                  |        | 0.74   |
| 4                  | 0.278                          | 12.78  |                         | 7.67   |        |
| 1.115              | menor_sensibilidad             |        |                         |        |        |
|                    | contract bar_p40_h30_tp10_sl05 |        | p40_h30 TP 1.0 / SL 0.5 |        |        |
| 26                 | 61.51%                         | 53.55% | 79.74%                  | 19.41% | 20.26% |
| 19.41%             | 0.00%                          | 0.00%  | 100.00%                 |        | 0.     |
| 00%                | 79.74%                         |        | 19.41%                  |        | 0.69   |
| 6                  | 0.291                          | 11.48  |                         | 7.67   |        |
| 1.118              | menor_sensibilidad             |        |                         |        |        |
|                    | contract bar_p60_h90_tp10_sl05 |        | p60_h90 TP 1.0 / SL 0.5 |        |        |
| 26                 | 39.34%                         | 53.30% | 84.64%                  | 19.93% | 15.36% |
| 19.93%             | 0.00%                          | 0.00%  | 100.00%                 |        | 0.     |
| 00%                | 84.64%                         |        | 19.93%                  |        | 0.77   |
| 0                  | 0.299                          | 41.97  |                         | 26.11  |        |
| 1.131              | menor_sensibilidad             |        |                         |        |        |
|                    | contract bar_p60_h60_tp10_sl05 |        | p60_h60 TP 1.0 / SL 0.5 |        |        |

|        |                                |        |                         |        |        |
|--------|--------------------------------|--------|-------------------------|--------|--------|
| 26     | 41.23%                         | 54.14% | 85.70%                  | 21.31% | 14.30% |
| 21.31% | 0.00%                          | 0.00%  | 100.00%                 |        | 0.     |
| 00%    | 85.70%                         |        | 21.31%                  |        | 0.78   |
| 5      | 0.320                          | 27.76  |                         | 16.44  |        |
| 1.181  | menor_sensibilidad             |        |                         |        |        |
|        | contract bar_p50_h90_tp10_sl05 |        | p50_h90 TP 1.0 / SL 0.5 |        |        |
| 26     | 47.85%                         | 52.11% | 81.41%                  | 22.29% | 18.59% |
| 22.29% | 0.00%                          | 0.00%  | 100.00%                 |        | 0.     |
| 00%    | 81.41%                         |        | 22.29%                  |        | 0.72   |
| 1      | 0.334                          | 37.97  |                         | 26.81  |        |
| 1.190  | menor_sensibilidad             |        |                         |        |        |
|        | contract bar_p40_h60_tp10_sl05 |        | p40_h60 TP 1.0 / SL 0.5 |        |        |
| 26     | 58.91%                         | 52.47% | 78.83%                  | 23.04% | 21.17% |
| 23.04% | 0.00%                          | 0.00%  | 100.00%                 |        | 0.     |
| 00%    | 78.83%                         |        | 23.04%                  |        | 0.68   |
| 3      | 0.346                          | 22.59  |                         | 16.40  |        |
| 1.216  | menor_sensibilidad             |        |                         |        |        |
|        | contract bar_p50_h60_tp10_sl05 |        | p50_h60 TP 1.0 / SL 0.5 |        |        |
| 26     | 50.05%                         | 54.31% | 82.52%                  | 22.43% | 17.48% |
| 22.43% | 0.00%                          | 0.00%  | 100.00%                 |        | 0.     |
| 00%    | 82.52%                         |        | 22.43%                  |        | 0.73   |
| 8      | 0.336                          | 25.29  |                         | 16.57  |        |
| 1.216  | menor_sensibilidad             |        |                         |        |        |
|        | contract bar_p40_h90_tp10_sl05 |        | p40_h90 TP 1.0 / SL 0.5 |        |        |
| 26     | 56.05%                         | 49.17% | 78.03%                  | 24.33% | 21.97% |
| 24.33% | 0.00%                          | 0.00%  | 100.00%                 |        | 0.     |
| 00%    | 78.03%                         |        | 24.33%                  |        | 0.67   |
| 0      | 0.365                          | 33.93  |                         | 26.88  |        |
| 1.222  | menor_sensibilidad             |        |                         |        |        |
|        | contract bar_p60_h30_tp25_sl10 |        | p60_h30 TP 2.5 / SL 1.0 |        |        |
| 26     | 43.30%                         | 54.76% | 13.69%                  | 20.65% | 4.57%  |
| 9.63%  | 81.74%                         | 30.28% |                         | 18.26% | 30.2   |
| 8%     | 77.10%                         |        |                         | 16.66% | 0.296  |
| 0.425  | 18.17                          | 6.69   |                         | 1.320  |        |
|        | menor_sensibilidad             |        |                         |        |        |
|        | contract bar_p60_h60_tp25_sl10 |        | p60_h60 TP 2.5 / SL 1.0 |        |        |
| 26     | 41.23%                         | 54.14% | 12.71%                  | 21.06% | 4.37%  |
| 9.81%  | 82.91%                         | 29.42% |                         | 17.09% | 29.4   |
| 2%     | 76.34%                         |        |                         | 18.26% | 0.274  |
| 0.443  | 35.51                          | 12.47  |                         | 1.327  |        |
|        | menor_sensibilidad             |        |                         |        |        |
|        | contract bar_p60_h30_tp15_sl10 |        | p60_h30 TP 1.5 / SL 1.0 |        |        |
| 26     | 43.30%                         | 54.76% | 47.47%                  | 28.16% | 3.83%  |
| 8.13%  | 48.70%                         | 35.76% |                         | 51.30% | 35.7   |
| 6%     | 93.21%                         |        |                         | 11.24% | 0.674  |
| 0.348  | 17.34                          | 6.37   |                         | 1.381  |        |
|        | menor_sensibilidad             |        |                         |        |        |
|        | contract bar_p40_h30_tp15_sl10 |        | p40_h30 TP 1.5 / SL 1.0 |        |        |
| 26     | 61.51%                         | 53.55% | 56.51%                  | 24.96% | 6.95%  |
| 11.67% | 36.54%                         | 35.93% |                         | 63.46% | 35.    |
| 93%    | 89.72%                         |        |                         | 12.62% | 0.77   |
| 8      | 0.269                          | 15.36  |                         | 6.81   |        |
| 1.387  | sensibilidad_intermedia        |        |                         |        |        |
|        | contract bar_p60_h90_tp25_sl10 |        | p60_h90 TP 2.5 / SL 1.0 |        |        |
| 26     | 39.34%                         | 53.30% | 12.80%                  | 22.34% | 4.78%  |
| 11.43% | 82.41%                         | 31.97% |                         | 17.59% | 31.    |
| 97%    | 74.32%                         |        |                         | 19.80% | 0.27   |
| 2      | 0.462                          | 56.03  |                         | 24.50  |        |
| 1.388  | sensibilidad_intermedia        |        |                         |        |        |
|        | contract bar_p50_h30_tp15_sl10 |        | p50_h30 TP 1.5 / SL 1.0 |        |        |

|        |                                |        |                         |        |        |
|--------|--------------------------------|--------|-------------------------|--------|--------|
| 26     | 52.50%                         | 55.94% | 52.12%                  | 26.60% | 5.22%  |
| 10.12% | 42.66%                         | 35.79% |                         | 57.34% | 35.    |
| 79%    | 91.58%                         |        | 12.19%                  |        | 0.73   |
| 0      | 0.317                          | 16.38  |                         | 6.54   |        |
| 1.407  | sensibilidad_intermedia        |        |                         |        |        |
|        | contract bar_p40_h30_tp15_sl05 |        | p40_h30 TP 1.5 / SL 0.5 |        |        |
| 26     | 61.51%                         | 53.55% | 47.50%                  | 15.13% | 24.16% |
| 22.18% | 28.34%                         | 36.68% |                         | 71.66% | 36.    |
| 68%    | 66.94%                         |        | 15.41%                  |        | 0.59   |
| 2      | 0.119                          | 12.97  |                         | 7.14   |        |
| 1.429  | sensibilidad_intermedia        |        |                         |        |        |
|        | contract bar_p60_h60_tp20_sl10 |        | p60_h60 TP 2.0 / SL 1.0 |        |        |
| 26     | 41.23%                         | 54.14% | 23.34%                  | 28.64% | 4.14%  |
| 9.19%  | 72.52%                         | 37.42% |                         | 27.48% | 37.4   |
| 2%     | 86.50%                         |        | 13.61%                  |        | 0.425  |
| 0.485  | 35.73                          | 12.03  |                         | 1.430  | se     |
|        | nsibilidad_intermedia          |        |                         |        |        |
|        | contract bar_p50_h60_tp25_sl10 |        | p50_h60 TP 2.5 / SL 1.0 |        |        |
| 26     | 50.05%                         | 54.31% | 15.98%                  | 24.20% | 6.08%  |
| 12.97% | 77.94%                         | 36.88% |                         | 22.06% | 36.    |
| 88%    | 74.53%                         |        | 16.80%                  |        | 0.33   |
| 9      | 0.486                          | 34.46  |                         | 11.69  |        |
| 1.452  | sensibilidad_intermedia        |        |                         |        |        |
|        | contract bar_p60_h30_tp20_sl10 |        | p60_h30 TP 2.0 / SL 1.0 |        |        |
| 26     | 43.30%                         | 54.76% | 24.88%                  | 28.52% | 4.34%  |
| 9.19%  | 70.78%                         | 37.51% |                         | 29.22% | 37.5   |
| 1%     | 86.78%                         |        | 15.46%                  |        | 0.454  |
| 0.480  | 18.12                          | 6.20   |                         | 1.454  | se     |
|        | nsibilidad_intermedia          |        |                         |        |        |
|        | contract bar_p50_h30_tp25_sl10 |        | p50_h30 TP 2.5 / SL 1.0 |        |        |
| 26     | 52.50%                         | 55.94% | 17.47%                  | 24.46% | 6.30%  |
| 12.57% | 76.23%                         | 37.04% |                         | 23.77% | 37.    |
| 04%    | 75.42%                         |        | 15.93%                  |        | 0.37   |
| 4      | 0.499                          | 17.59  |                         | 6.38   |        |
| 1.459  | sensibilidad_intermedia        |        |                         |        |        |
|        | contract bar_p60_h90_tp15_sl10 |        | p60_h90 TP 1.5 / SL 1.0 |        |        |
| 26     | 39.34%                         | 53.30% | 46.41%                  | 32.72% | 3.98%  |
| 9.00%  | 49.61%                         | 38.78% |                         | 50.39% | 38.7   |
| 8%     | 92.76%                         |        | 12.18%                  |        | 0.656  |
| 0.442  | 53.10                          | 22.85  |                         | 1.460  | se     |
|        | nsibilidad_intermedia          |        |                         |        |        |
|        | contract bar_p40_h60_tp15_sl10 |        | p40_h60 TP 1.5 / SL 1.0 |        |        |
| 26     | 58.91%                         | 52.47% | 55.18%                  | 27.71% | 6.86%  |
| 12.55% | 37.96%                         | 39.42% |                         | 62.04% | 39.    |
| 42%    | 89.73%                         |        | 14.11%                  |        | 0.75   |
| 9      | 0.299                          | 30.54  |                         | 14.11  |        |
| 1.463  | sensibilidad_intermedia        |        |                         |        |        |
|        | contract bar_p50_h90_tp25_sl10 |        | p50_h90 TP 2.5 / SL 1.0 |        |        |
| 26     | 47.85%                         | 52.11% | 16.25%                  | 23.91% | 6.58%  |
| 13.63% | 77.16%                         | 37.54% |                         | 22.84% | 37.    |
| 54%    | 72.72%                         |        | 21.28%                  |        | 0.34   |
| 0      | 0.493                          | 53.52  |                         | 22.21  |        |
| 1.485  | sensibilidad_intermedia        |        |                         |        |        |
|        | contract bar_p60_h60_tp15_sl10 |        | p60_h60 TP 1.5 / SL 1.0 |        |        |
| 26     | 41.23%                         | 54.14% | 45.93%                  | 35.82% | 3.62%  |
| 7.71%  | 50.45%                         | 41.29% |                         | 49.55% | 41.2   |
| 9%     | 93.39%                         |        | 10.05%                  |        | 0.653  |
| 0.484  | 34.49                          | 13.58  |                         | 1.490  | se     |
|        | nsibilidad_intermedia          |        |                         |        |        |
|        | contract bar_p50_h30_tp15_sl05 |        | p50_h30 TP 1.5 / SL 0.5 |        |        |

|        |                                |        |                         |        |        |
|--------|--------------------------------|--------|-------------------------|--------|--------|
| 26     | 52.50%                         | 55.94% | 44.76%                  | 17.61% | 20.47% |
| 21.62% | 34.77%                         | 37.93% |                         | 65.23% | 37.    |
| 93%    | 69.49%                         |        | 16.63%                  |        | 0.56   |
| 9      | 0.163                          | 14.05  |                         | 7.12   |        |
| 1.497  | sensibilidad_intermedia        |        |                         |        |        |
|        | contract bar_p60_h90_tp20_sl10 |        | p60_h90 TP 2.0 / SL 1.0 |        |        |
| 26     | 39.34%                         | 53.30% | 23.67%                  | 28.68% | 4.52%  |
| 10.58% | 71.81%                         | 38.76% |                         | 28.19% | 38.    |
| 76%    | 85.41%                         |        | 18.82%                  |        | 0.42   |
| 8      | 0.502                          | 55.53  |                         | 20.99  |        |
| 1.501  | sensibilidad_intermedia        |        |                         |        |        |
|        | contract bar_p40_h30_tp25_sl10 |        | p40_h30 TP 2.5 / SL 1.0 |        |        |
| 26     | 61.51%                         | 53.55% | 21.65%                  | 27.04% | 8.42%  |
| 14.66% | 69.93%                         | 41.42% |                         | 30.07% | 41.    |
| 42%    | 73.63%                         |        | 14.52%                  |        | 0.45   |
| 7      | 0.541                          | 16.79  |                         | 6.33   |        |
| 1.512  | sensibilidad_intermedia        |        |                         |        |        |
|        | contract bar_p40_h90_tp25_sl10 |        | p40_h90 TP 2.5 / SL 1.0 |        |        |
| 26     | 56.05%                         | 49.17% | 20.12%                  | 26.12% | 8.90%  |
| 16.69% | 70.97%                         | 42.14% |                         | 29.03% | 42.    |
| 14%    | 70.83%                         |        | 17.67%                  |        | 0.41   |
| 4      | 0.522                          | 51.11  |                         | 20.28  |        |
| 1.518  | sensibilidad_intermedia        |        |                         |        |        |
|        | contract bar_p50_h60_tp15_sl10 |        | p50_h60 TP 1.5 / SL 1.0 |        |        |
| 26     | 50.05%                         | 54.31% | 50.58%                  | 33.61% | 5.01%  |
| 10.12% | 44.41%                         | 41.73% |                         | 55.59% | 41.    |
| 73%    | 91.77%                         |        | 12.18%                  |        | 0.70   |
| 9      | 0.423                          | 32.64  |                         | 12.91  |        |
| 1.520  | sensibilidad_intermedia        |        |                         |        |        |
|        | contract bar_p40_h90_tp15_sl10 |        | p40_h90 TP 1.5 / SL 1.0 |        |        |
| 26     | 56.05%                         | 49.17% | 54.88%                  | 33.05% | 7.29%  |
| 12.81% | 37.83%                         | 42.93% |                         | 62.17% | 42.    |
| 93%    | 89.01%                         |        | 14.56%                  |        | 0.75   |
| 0      | 0.397                          | 46.63  |                         | 23.49  |        |
| 1.525  | sensibilidad_intermedia        |        |                         |        |        |
|        | contract bar_p40_h30_tp20_sl10 |        | p40_h30 TP 2.0 / SL 1.0 |        |        |
| 26     | 61.51%                         | 53.55% | 34.50%                  | 29.61% | 7.88%  |
| 13.51% | 57.62%                         | 41.96% |                         | 42.38% | 41.    |
| 96%    | 82.78%                         |        | 14.47%                  |        | 0.61   |
| 1      | 0.470                          | 16.40  |                         | 6.43   |        |
| 1.531  | sensibilidad_intermedia        |        |                         |        |        |
|        | contract bar_p50_h30_tp20_sl10 |        | p50_h30 TP 2.0 / SL 1.0 |        |        |
| 26     | 52.50%                         | 55.94% | 29.54%                  | 29.50% | 5.93%  |
| 11.54% | 64.53%                         | 41.03% |                         | 35.47% | 41.    |
| 03%    | 84.82%                         |        | 15.38%                  |        | 0.53   |
| 2      | 0.475                          | 17.34  |                         | 7.02   |        |
| 1.534  | sensibilidad_intermedia        |        |                         |        |        |
|        | contract bar_p40_h30_tp20_sl05 |        | p40_h30 TP 2.0 / SL 0.5 |        |        |
| 26     | 61.51%                         | 53.55% | 28.83%                  | 21.73% | 25.93% |
| 24.85% | 45.24%                         | 45.51% |                         | 54.76% | 45.    |
| 51%    | 53.11%                         |        | 8.08%                   |        | 0.44   |
| 7      | 0.320                          | 13.23  |                         | 5.79   |        |
| 1.537  | mayor_sensibilidad             |        |                         |        |        |
|        | contract bar_p50_h90_tp20_sl10 |        | p50_h90 TP 2.0 / SL 1.0 |        |        |
| 26     | 47.85%                         | 52.11% | 28.07%                  | 29.85% | 6.22%  |
| 12.59% | 65.71%                         | 41.47% |                         | 34.29% | 41.    |
| 47%    | 83.32%                         |        | 18.20%                  |        | 0.49   |
| 9      | 0.510                          | 53.00  |                         | 23.50  |        |
| 1.542  | mayor_sensibilidad             |        |                         |        |        |
|        | contract bar_p50_h90_tp15_sl10 |        | p50_h90 TP 1.5 / SL 1.0 |        |        |

|        |                                |        |                         |        |        |
|--------|--------------------------------|--------|-------------------------|--------|--------|
| 26     | 47.85%                         | 52.11% | 50.73%                  | 35.29% | 5.42%  |
| 10.85% | 43.85%                         | 42.65% |                         | 56.15% | 42.    |
| 65%    | 91.05%                         |        | 13.34%                  |        | 0.70   |
| 7      | 0.456                          | 50.06  |                         | 23.09  |        |
| 1.542  | mayor_sensibilidad             |        |                         |        |        |
|        | contract bar_p60_h30_tp15_sl05 |        | p60_h30 TP 1.5 / SL 0.5 |        |        |
| 26     | 43.30%                         | 54.76% | 41.64%                  | 19.03% | 16.90% |
| 21.89% | 41.46%                         | 40.00% |                         | 58.54% | 40.    |
| 00%    | 72.26%                         |        | 18.73%                  |        | 0.54   |
| 0      | 0.191                          | 15.15  |                         | 6.86   |        |
| 1.544  | mayor_sensibilidad             |        |                         |        |        |
|        | contract bar_p40_h60_tp25_sl10 |        | p40_h60 TP 2.5 / SL 1.0 |        |        |
| 26     | 58.91%                         | 52.47% | 20.00%                  | 27.62% | 8.30%  |
| 15.93% | 71.69%                         | 42.78% |                         | 28.31% | 42.    |
| 78%    | 72.63%                         |        | 15.75%                  |        | 0.41   |
| 7      | 0.544                          | 33.15  |                         | 12.48  |        |
| 1.546  | mayor_sensibilidad             |        |                         |        |        |
|        | contract bar_p40_h60_tp15_sl05 |        | p40_h60 TP 1.5 / SL 0.5 |        |        |
| 26     | 58.91%                         | 52.47% | 45.81%                  | 13.83% | 25.36% |
| 26.93% | 28.83%                         | 40.76% |                         | 71.17% | 40.    |
| 76%    | 65.23%                         |        | 20.67%                  |        | 0.56   |
| 0      | 0.106                          | 25.45  |                         | 15.25  |        |
| 1.547  | mayor_sensibilidad             |        |                         |        |        |
|        | contract bar_p50_h30_tp20_sl05 |        | p50_h30 TP 2.0 / SL 0.5 |        |        |
| 26     | 52.50%                         | 55.94% | 25.13%                  | 21.67% | 21.86% |
| 24.31% | 53.00%                         | 45.82% |                         | 47.00% | 45.    |
| 82%    | 54.15%                         |        | 8.14%                   |        | 0.39   |
| 3      | 0.320                          | 14.02  |                         | 5.57   |        |
| 1.559  | mayor_sensibilidad             |        |                         |        |        |
|        | contract bar_p50_h60_tp20_sl10 |        | p50_h60 TP 2.0 / SL 1.0 |        |        |
| 26     | 50.05%                         | 54.31% | 27.79%                  | 32.91% | 5.71%  |
| 11.90% | 66.50%                         | 43.19% |                         | 33.50% | 43.    |
| 19%    | 84.68%                         |        | 15.30%                  |        | 0.49   |
| 9      | 0.561                          | 34.40  |                         | 12.97  |        |
| 1.576  | mayor_sensibilidad             |        |                         |        |        |
|        | contract bar_p60_h30_tp20_sl05 |        | p60_h30 TP 2.0 / SL 0.5 |        |        |
| 26     | 43.30%                         | 54.76% | 21.63%                  | 22.17% | 17.93% |
| 24.01% | 60.45%                         | 45.66% |                         | 39.55% | 45.    |
| 66%    | 55.64%                         |        | 12.36%                  |        | 0.34   |
| 3      | 0.336                          | 14.80  |                         | 5.19   |        |
| 1.590  | mayor_sensibilidad             |        |                         |        |        |
|        | contract bar_p40_h90_tp15_sl05 |        | p40_h90 TP 1.5 / SL 0.5 |        |        |
| 26     | 56.05%                         | 49.17% | 44.88%                  | 17.27% | 26.22% |
| 27.74% | 28.91%                         | 45.01% |                         | 71.09% | 45.    |
| 01%    | 64.00%                         |        | 20.66%                  |        | 0.54   |
| 2      | 0.123                          | 38.44  |                         | 25.36  |        |
| 1.598  | mayor_sensibilidad             |        |                         |        |        |
|        | contract bar_p40_h90_tp20_sl10 |        | p40_h90 TP 2.0 / SL 1.0 |        |        |
| 26     | 56.05%                         | 49.17% | 32.85%                  | 36.08% | 8.32%  |
| 15.06% | 58.83%                         | 47.54% |                         | 41.17% | 47.    |
| 54%    | 81.17%                         |        | 16.73%                  |        | 0.57   |
| 4      | 0.607                          | 50.09  |                         | 22.17  |        |
| 1.646  | mayor_sensibilidad             |        |                         |        |        |
|        | contract bar_p60_h90_tp15_sl05 |        | p60_h90 TP 1.5 / SL 0.5 |        |        |
| 26     | 39.34%                         | 53.30% | 40.02%                  | 23.61% | 18.50% |
| 24.84% | 41.48%                         | 43.93% |                         | 58.52% | 43.    |
| 93%    | 69.62%                         |        | 20.36%                  |        | 0.50   |
| 8      | 0.257                          | 45.51  |                         | 23.86  |        |
| 1.660  | mayor_sensibilidad             |        |                         |        |        |
|        | contract bar_p40_h60_tp20_sl10 |        | p40_h60 TP 2.0 / SL 1.0 |        |        |

|        |                                     |        |                         |        |        |
|--------|-------------------------------------|--------|-------------------------|--------|--------|
| 26     | 58.91%                              | 52.47% | 32.83%                  | 34.90% | 7.79%  |
| 14.29% | 59.38%                              | 48.49% |                         | 40.62% | 48.    |
| 49%    | 82.48%                              |        | 15.95%                  |        | 0.57   |
| 9      | 0.562                               | 32.64  |                         | 13.98  |        |
| 1.661  | mayor_sensibilidad                  |        |                         |        |        |
|        | contract_bar_p60_h90_tp20_sl05      |        | p60_h90 TP 2.0 / SL 0.5 |        |        |
| 26     | 39.34%                              | 53.30% | 20.15%                  | 21.46% | 19.63% |
| 27.81% | 60.22%                              | 48.63% |                         | 39.78% | 48.    |
| 63%    | 51.66%                              |        | 17.48%                  |        | 0.30   |
| 5      | 0.322                               | 44.07  |                         | 18.29  |        |
| 1.687  | mayor_sensibilidad                  |        |                         |        |        |
|        | contract_bar_p50_h60_tp15_sl05      |        | p50_h60 TP 1.5 / SL 0.5 |        |        |
| 26     | 50.05%                              | 54.31% | 42.95%                  | 19.63% | 21.07% |
| 26.79% | 35.98%                              | 46.41% |                         | 64.02% | 46.    |
| 41%    | 68.30%                              |        | 21.96%                  |        | 0.53   |
| 9      | 0.167                               | 27.73  |                         | 15.19  |        |
| 1.691  | mayor_sensibilidad                  |        |                         |        |        |
|        | contract_bar_p50_h90_tp20_sl05      |        | p50_h90 TP 2.0 / SL 0.5 |        |        |
| 26     | 47.85%                              | 52.11% | 23.29%                  | 22.26% | 23.76% |
| 30.19% | 52.95%                              | 50.75% |                         | 47.05% | 50.    |
| 75%    | 50.33%                              |        | 14.08%                  |        | 0.34   |
| 7      | 0.313                               | 41.67  |                         | 20.07  |        |
| 1.694  | mayor_sensibilidad                  |        |                         |        |        |
|        | contract_bar_p40_h60_tp20_sl05      |        | p40_h60 TP 2.0 / SL 0.5 |        |        |
| 26     | 58.91%                              | 52.47% | 26.93%                  | 23.54% | 27.13% |
| 30.00% | 45.94%                              | 53.53% |                         | 54.06% | 53.    |
| 53%    | 50.43%                              |        | 9.86%                   |        | 0.40   |
| 3      | 0.321                               | 25.69  |                         | 11.85  |        |
| 1.694  | mayor_sensibilidad                  |        |                         |        |        |
|        | contract_bar_p60_h60_tp20_sl05      |        | p60_h60 TP 2.0 / SL 0.5 |        |        |
| 26     | 41.23%                              | 54.14% | 19.98%                  | 21.86% | 18.34% |
| 27.93% | 61.69%                              | 49.34% |                         | 38.31% | 49.    |
| 34%    | 53.60%                              |        | 16.32%                  |        | 0.30   |
| 8      | 0.323                               | 28.64  |                         | 11.47  |        |
| 1.696  | mayor_sensibilidad                  |        |                         |        |        |
|        | contract_bar_p50_h90_tp15_sl05      |        | p50_h90 TP 1.5 / SL 0.5 |        |        |
| 26     | 47.85%                              | 52.11% | 42.58%                  | 23.12% | 22.31% |
| 26.45% | 35.12%                              | 48.36% |                         | 64.88% | 48.    |
| 36%    | 66.69%                              |        | 19.88%                  |        | 0.52   |
| 7      | 0.221                               | 42.10  |                         | 25.53  |        |
| 1.699  | mayor_sensibilidad                  |        |                         |        |        |
|        | contract_bar_p40_h90_tp20_sl05      |        | p40_h90 TP 2.0 / SL 0.5 |        |        |
| 26     | 56.05%                              | 49.17% | 26.57%                  | 24.89% | 28.01% |
| 30.69% | 45.41%                              | 54.76% |                         | 54.59% | 54.    |
| 76%    | 49.28%                              |        | 11.40%                  |        | 0.39   |
| 1      | 0.362                               | 39.01  |                         | 20.13  |        |
| 1.709  | mayor_sensibilidad                  |        |                         |        |        |
|        | contract_bar_p50_h60_tp20_sl05      |        | p50_h60 TP 2.0 / SL 0.5 |        |        |
| 26     | 50.05%                              | 54.31% | 23.32%                  | 24.08% | 22.42% |
| 29.08% | 54.26%                              | 53.15% |                         | 45.74% | 53.    |
| 15%    | 51.91%                              |        | 11.93%                  |        | 0.35   |
| 4      | 0.336                               | 27.31  |                         | 11.56  |        |
| 1.726  | mayor_sensibilidad                  |        |                         |        |        |
|        | contract_bar_p60_h60_tp15_sl05      |        | p60_h60 TP 1.5 / SL 0.5 |        |        |
| 26     | 41.23%                              | 54.14% | 39.84%                  | 24.70% | 17.30% |
| 25.11% | 42.86%                              | 48.71% |                         | 57.14% | 48.    |
| 71%    | 71.28%                              |        | 21.50%                  |        | 0.51   |
| 1      | 0.250                               | 29.82  |                         | 15.18  |        |
| 1.742  | mayor_sensibilidad                  |        |                         |        |        |
|        | contract_code_bar_p60_h60_tp10_sl10 |        | p60_h60 TP 1.0 / SL 1.0 |        |        |



|                    |               |                       |         |                 |        |
|--------------------|---------------|-----------------------|---------|-----------------|--------|
| 4                  | 40.24%        | 10.68%                | 96.78%  | 2.92%           | 3.22%  |
| 2.92%              | 0.00%         | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 96.78%        |                       | 2.92%   |                 | 0.936  |
| 0.058              | 30.28         | 4.80                  |         | 0.195           |        |
| menor_sensibilidad |               |                       |         |                 |        |
|                    | contract_code | bar_p60_h90_tp10_sl10 | p60_h90 | TP 1.0 / SL 1.0 |        |
| 4                  | 38.35%        | 10.52%                | 96.54%  | 3.14%           | 3.46%  |
| 3.14%              | 0.00%         | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 96.54%        |                       | 3.14%   |                 | 0.931  |
| 0.063              | 46.05         | 6.92                  |         | 0.199           |        |
| menor_sensibilidad |               |                       |         |                 |        |
|                    | contract_code | bar_p60_h30_tp10_sl10 | p60_h30 | TP 1.0 / SL 1.0 |        |
| 4                  | 42.30%        | 11.20%                | 96.56%  | 2.91%           | 3.44%  |
| 2.91%              | 0.00%         | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 96.56%        |                       | 2.91%   |                 | 0.931  |
| 0.058              | 15.26         | 2.38                  |         | 0.199           |        |
| menor_sensibilidad |               |                       |         |                 |        |
|                    | contract_code | bar_p40_h60_tp10_sl10 | p40_h60 | TP 1.0 / SL 1.0 |        |
| 4                  | 58.20%        | 7.19%                 | 94.14%  | 4.41%           | 5.86%  |
| 4.41%              | 0.00%         | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 94.14%        |                       | 4.41%   |                 | 0.883  |
| 0.088              | 26.34         | 4.82                  |         | 0.204           |        |
| menor_sensibilidad |               |                       |         |                 |        |
|                    | contract_code | bar_p50_h60_tp10_sl10 | p50_h60 | TP 1.0 / SL 1.0 |        |
| 4                  | 49.18%        | 9.25%                 | 95.61%  | 3.77%           | 4.39%  |
| 3.77%              | 0.00%         | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 95.61%        |                       | 3.77%   |                 | 0.912  |
| 0.075              | 28.42         | 4.87                  |         | 0.206           |        |
| menor_sensibilidad |               |                       |         |                 |        |
|                    | contract_code | bar_p40_h30_tp10_sl10 | p40_h30 | TP 1.0 / SL 1.0 |        |
| 4                  | 60.80%        | 8.10%                 | 93.99%  | 4.19%           | 6.01%  |
| 4.19%              | 0.00%         | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 93.99%        |                       | 4.19%   |                 | 0.880  |
| 0.084              | 13.28         | 2.32                  |         | 0.207           |        |
| menor_sensibilidad |               |                       |         |                 |        |
|                    | contract_code | bar_p50_h30_tp10_sl10 | p50_h30 | TP 1.0 / SL 1.0 |        |
| 4                  | 51.61%        | 9.77%                 | 95.41%  | 3.67%           | 4.59%  |
| 3.67%              | 0.00%         | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 95.41%        |                       | 3.67%   |                 | 0.908  |
| 0.073              | 14.31         | 2.38                  |         | 0.208           |        |
| menor_sensibilidad |               |                       |         |                 |        |
|                    | contract_code | bar_p50_h90_tp10_sl10 | p50_h90 | TP 1.0 / SL 1.0 |        |
| 4                  | 46.99%        | 8.83%                 | 95.28%  | 4.05%           | 4.72%  |
| 4.05%              | 0.00%         | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 95.28%        |                       | 4.05%   |                 | 0.906  |
| 0.081              | 42.99         | 7.24                  |         | 0.210           |        |
| menor_sensibilidad |               |                       |         |                 |        |
|                    | contract_code | bar_p40_h90_tp10_sl10 | p40_h90 | TP 1.0 / SL 1.0 |        |
| 4                  | 55.37%        | 6.72%                 | 93.72%  | 4.81%           | 6.28%  |
| 4.81%              | 0.00%         | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 93.72%        |                       | 4.81%   |                 | 0.874  |
| 0.096              | 39.75         | 7.29                  |         | 0.211           |        |
| menor_sensibilidad |               |                       |         |                 |        |
|                    | contract_code | bar_p40_h30_tp10_sl05 | p40_h30 | TP 1.0 / SL 0.5 |        |
| 4                  | 60.80%        | 8.10%                 | 79.20%  | 6.40%           | 20.80% |
| 6.40%              | 0.00%         | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 79.20%        |                       | 6.40%   |                 | 0.688  |
| 0.096              | 11.26         | 2.46                  |         | 0.273           |        |
| menor_sensibilidad |               |                       |         |                 |        |
|                    | contract_code | bar_p40_h60_tp10_sl05 | p40_h60 | TP 1.0 / SL 0.5 |        |

|                    |               |                       |         |                 |        |
|--------------------|---------------|-----------------------|---------|-----------------|--------|
| 4                  | 58.20%        | 7.19%                 | 78.18%  | 7.42%           | 21.82% |
| 7.42%              | 0.00%         | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 78.18%        |                       | 7.42%   |                 | 0.673  |
| 0.111              | 22.12         | 5.19                  |         | 0.294           |        |
| menor_sensibilidad |               |                       |         |                 |        |
|                    | contract_code | bar_p50_h30_tp10_sl05 | p50_h30 | TP 1.0 / SL 0.5 |        |
| 4                  | 51.61%        | 9.77%                 | 82.21%  | 6.81%           | 17.79% |
| 6.81%              | 0.00%         | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 82.21%        |                       | 6.81%   |                 | 0.733  |
| 0.102              | 12.48         | 2.69                  |         | 0.302           |        |
| menor_sensibilidad |               |                       |         |                 |        |
|                    | contract_code | bar_p40_h90_tp10_sl05 | p40_h90 | TP 1.0 / SL 0.5 |        |
| 4                  | 55.37%        | 6.72%                 | 77.36%  | 8.09%           | 22.64% |
| 8.09%              | 0.00%         | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 77.36%        |                       | 8.09%   |                 | 0.660  |
| 0.121              | 33.18         | 7.93                  |         | 0.310           |        |
| menor_sensibilidad |               |                       |         |                 |        |
|                    | contract_code | bar_p60_h30_tp10_sl05 | p60_h30 | TP 1.0 / SL 0.5 |        |
| 4                  | 42.30%        | 11.20%                | 85.15%  | 7.00%           | 14.85% |
| 7.00%              | 0.00%         | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 85.15%        |                       | 7.00%   |                 | 0.777  |
| 0.105              | 13.66         | 2.86                  |         | 0.322           |        |
| menor_sensibilidad |               |                       |         |                 |        |
|                    | contract_code | bar_p50_h90_tp10_sl05 | p50_h90 | TP 1.0 / SL 0.5 |        |
| 4                  | 46.99%        | 8.83%                 | 80.57%  | 8.25%           | 19.43% |
| 8.25%              | 0.00%         | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 80.57%        |                       | 8.25%   |                 | 0.709  |
| 0.124              | 36.92         | 8.52                  |         | 0.336           |        |
| menor_sensibilidad |               |                       |         |                 |        |
|                    | contract_code | bar_p50_h60_tp10_sl05 | p50_h60 | TP 1.0 / SL 0.5 |        |
| 4                  | 49.18%        | 9.25%                 | 81.66%  | 8.17%           | 18.34% |
| 8.17%              | 0.00%         | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 81.66%        |                       | 8.17%   |                 | 0.725  |
| 0.122              | 24.64         | 5.69                  |         | 0.337           |        |
| menor_sensibilidad |               |                       |         |                 |        |
|                    | contract_code | bar_p40_h30_tp15_sl05 | p40_h30 | TP 1.5 / SL 0.5 |        |
| 4                  | 60.80%        | 8.10%                 | 48.03%  | 3.40%           | 24.83% |
| 7.10%              | 27.15%        | 10.51%                | 72.85%  |                 | 10.5   |
| 1%                 | 66.01%        |                       | 4.84%   |                 | 0.596  |
| 0.016              | 12.60         | 2.53                  |         | 0.340           |        |
| menor_sensibilidad |               |                       |         |                 |        |
|                    | contract_code | bar_p40_h60_tp15_sl05 | p40_h60 | TP 1.5 / SL 0.5 |        |
| 4                  | 58.20%        | 7.19%                 | 46.36%  | 2.90%           | 26.16% |
| 8.15%              | 27.48%        | 11.05%                | 72.52%  |                 | 11.0   |
| 5%                 | 64.04%        |                       | 5.74%   |                 | 0.565  |
| 0.010              | 24.64         | 5.40                  |         | 0.350           |        |
| menor_sensibilidad |               |                       |         |                 |        |
|                    | contract_code | bar_p60_h90_tp10_sl05 | p60_h90 | TP 1.0 / SL 0.5 |        |
| 4                  | 38.35%        | 10.52%                | 83.64%  | 8.30%           | 16.36% |
| 8.30%              | 0.00%         | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 83.64%        |                       | 8.30%   |                 | 0.755  |
| 0.125              | 40.61         | 8.88                  |         | 0.354           |        |
| menor_sensibilidad |               |                       |         |                 |        |
|                    | contract_code | bar_p40_h30_tp15_sl10 | p40_h30 | TP 1.5 / SL 1.0 |        |
| 4                  | 60.80%        | 8.10%                 | 57.30%  | 6.52%           | 7.19%  |
| 4.69%              | 35.51%        | 11.21%                | 64.49%  |                 | 11.2   |
| 1%                 | 88.97%        |                       | 5.30%   |                 | 0.788  |
| 0.051              | 14.98         | 2.66                  |         | 0.358           |        |
| menor_sensibilidad |               |                       |         |                 |        |
|                    | contract_code | bar_p60_h60_tp10_sl05 | p60_h60 | TP 1.0 / SL 0.5 |        |

|                       |               |                       |         |                 |        |
|-----------------------|---------------|-----------------------|---------|-----------------|--------|
| 4                     | 40.24%        | 10.68%                | 84.63%  | 8.44%           | 15.37% |
| 8.44%                 | 0.00%         | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                    | 84.63%        |                       | 8.44%   |                 | 0.769  |
| 0.127                 | 26.93         | 6.08                  |         | 0.360           |        |
| menor_sensibilidad    |               |                       |         |                 |        |
|                       | contract_code | bar_p40_h90_tp15_sl05 | p40_h90 | TP 1.5 / SL 0.5 |        |
| 4                     | 55.37%        | 6.72%                 | 45.41%  | 3.38%           | 27.04% |
| 8.77%                 | 27.55%        | 12.15%                |         | 72.45%          | 12.1   |
| 5%                    | 62.81%        |                       | 5.80%   |                 | 0.546  |
| 0.015                 | 37.06         | 8.01                  |         | 0.368           | se     |
| nsibilidad_intermedia |               |                       |         |                 |        |
|                       | contract_code | bar_p40_h60_tp15_sl10 | p40_h60 | TP 1.5 / SL 1.0 |        |
| 4                     | 58.20%        | 7.19%                 | 56.07%  | 7.22%           | 7.14%  |
| 5.17%                 | 36.79%        | 12.39%                |         | 63.21%          | 12.3   |
| 9%                    | 88.86%        |                       | 5.95%   |                 | 0.770  |
| 0.057                 | 29.72         | 5.43                  |         | 0.379           | se     |
| nsibilidad_intermedia |               |                       |         |                 |        |
|                       | contract_code | bar_p40_h90_tp15_sl10 | p40_h90 | TP 1.5 / SL 1.0 |        |
| 4                     | 55.37%        | 6.72%                 | 55.74%  | 7.75%           | 7.58%  |
| 5.53%                 | 36.69%        | 13.28%                |         | 63.31%          | 13.2   |
| 8%                    | 88.20%        |                       | 6.20%   |                 | 0.760  |
| 0.061                 | 45.25         | 8.02                  |         | 0.395           | se     |
| nsibilidad_intermedia |               |                       |         |                 |        |
|                       | contract_code | bar_p40_h30_tp20_sl05 | p40_h30 | TP 2.0 / SL 0.5 |        |
| 4                     | 60.80%        | 8.10%                 | 29.46%  | 6.71%           | 26.66% |
| 8.09%                 | 43.88%        | 14.81%                |         | 56.12%          | 14.8   |
| 1%                    | 52.56%        |                       | 1.89%   |                 | 0.456  |
| 0.094                 | 12.84         | 2.12                  |         | 0.396           | se     |
| nsibilidad_intermedia |               |                       |         |                 |        |
|                       | contract_code | bar_p50_h30_tp15_sl10 | p50_h30 | TP 1.5 / SL 1.0 |        |
| 4                     | 51.61%        | 9.77%                 | 53.25%  | 8.52%           | 5.53%  |
| 4.20%                 | 41.22%        | 12.72%                |         | 58.78%          | 12.7   |
| 2%                    | 90.74%        |                       | 5.09%   |                 | 0.743  |
| 0.086                 | 15.92         | 2.64                  |         | 0.403           | se     |
| nsibilidad_intermedia |               |                       |         |                 |        |
|                       | contract_code | bar_p60_h90_tp25_sl10 | p60_h90 | TP 2.5 / SL 1.0 |        |
| 4                     | 38.35%        | 10.52%                | 14.04%  | 6.94%           | 5.18%  |
| 4.42%                 | 80.79%        | 11.36%                |         | 19.21%          | 11.3   |
| 6%                    | 73.68%        |                       | 7.09%   |                 | 0.299  |
| 0.129                 | 53.86         | 7.31                  |         | 0.403           | se     |
| nsibilidad_intermedia |               |                       |         |                 |        |
|                       | contract_code | bar_p50_h30_tp15_sl05 | p50_h30 | TP 1.5 / SL 0.5 |        |
| 4                     | 51.61%        | 9.77%                 | 45.54%  | 4.76%           | 21.34% |
| 7.75%                 | 33.11%        | 12.52%                |         | 66.89%          | 12.5   |
| 2%                    | 68.23%        |                       | 5.59%   |                 | 0.576  |
| 0.033                 | 13.58         | 2.67                  |         | 0.404           | se     |
| nsibilidad_intermedia |               |                       |         |                 |        |
|                       | contract_code | bar_p40_h60_tp20_sl05 | p40_h60 | TP 2.0 / SL 0.5 |        |
| 4                     | 58.20%        | 7.19%                 | 27.62%  | 6.83%           | 28.01% |
| 9.24%                 | 44.37%        | 16.07%                |         | 55.63%          | 16.0   |
| 7%                    | 49.73%        |                       | 2.05%   |                 | 0.412  |
| 0.090                 | 24.88         | 4.39                  |         | 0.414           | se     |
| nsibilidad_intermedia |               |                       |         |                 |        |
|                       | contract_code | bar_p60_h90_tp15_sl10 | p60_h90 | TP 1.5 / SL 1.0 |        |
| 4                     | 38.35%        | 10.52%                | 48.29%  | 9.47%           | 4.31%  |
| 3.77%                 | 47.40%        | 13.23%                |         | 52.60%          | 13.2   |
| 3%                    | 91.98%        |                       | 5.06%   |                 | 0.681  |
| 0.104                 | 51.20         | 6.86                  |         | 0.420           | se     |
| nsibilidad_intermedia |               |                       |         |                 |        |
|                       | contract_code | bar_p60_h60_tp25_sl10 | p60_h60 | TP 2.5 / SL 1.0 |        |

|                       |                                     |        |                         |        |        |
|-----------------------|-------------------------------------|--------|-------------------------|--------|--------|
| 4                     | 40.24%                              | 10.68% | 13.82%                  | 8.05%  | 4.74%  |
| 4.34%                 | 81.44%                              | 12.40% |                         | 18.56% | 12.4   |
| 0%                    | 75.14%                              |        |                         | 6.60%  | 0.298  |
| 0.158                 | 34.56                               | 4.52   |                         | 0.421  | se     |
| nsibilidad_intermedia |                                     |        |                         |        |        |
|                       | contract_code bar_p40_h90_tp20_sl05 |        | p40_h90 TP 2.0 / SL 0.5 |        |        |
| 4                     | 55.37%                              | 6.72%  | 27.25%                  | 6.36%  | 28.89% |
| 9.86%                 | 43.86%                              | 16.22% |                         | 56.14% | 16.2   |
| 2%                    | 48.65%                              |        |                         | 2.94%  | 0.401  |
| 0.078                 | 37.55                               | 6.83   |                         | 0.421  | se     |
| nsibilidad_intermedia |                                     |        |                         |        |        |
|                       | contract_code bar_p60_h30_tp25_sl10 |        | p60_h30 TP 2.5 / SL 1.0 |        |        |
| 4                     | 42.30%                              | 11.20% | 14.64%                  | 8.10%  | 4.99%  |
| 4.24%                 | 80.37%                              | 12.34% |                         | 19.63% | 12.3   |
| 4%                    | 75.16%                              |        |                         | 6.29%  | 0.316  |
| 0.160                 | 17.56                               | 2.14   |                         | 0.422  | se     |
| nsibilidad_intermedia |                                     |        |                         |        |        |
|                       | contract_code bar_p50_h60_tp15_sl10 |        | p50_h60 TP 1.5 / SL 1.0 |        |        |
| 4                     | 49.18%                              | 9.25%  | 51.85%                  | 9.38%  | 5.35%  |
| 4.48%                 | 42.80%                              | 13.86% |                         | 57.20% | 13.8   |
| 6%                    | 90.83%                              |        |                         | 5.56%  | 0.724  |
| 0.096                 | 31.66                               | 5.20   |                         | 0.425  | se     |
| nsibilidad_intermedia |                                     |        |                         |        |        |
|                       | contract_code bar_p50_h90_tp15_sl10 |        | p50_h90 TP 1.5 / SL 1.0 |        |        |
| 4                     | 46.99%                              | 8.83%  | 52.01%                  | 9.19%  | 5.73%  |
| 4.81%                 | 42.26%                              | 14.00% |                         | 57.74% | 14.0   |
| 0%                    | 90.27%                              |        |                         | 5.93%  | 0.723  |
| 0.090                 | 48.37                               | 7.65   |                         | 0.428  | se     |
| nsibilidad_intermedia |                                     |        |                         |        |        |
|                       | contract_code bar_p60_h30_tp15_sl10 |        | p60_h30 TP 1.5 / SL 1.0 |        |        |
| 4                     | 42.30%                              | 11.20% | 49.01%                  | 10.08% | 4.18%  |
| 3.50%                 | 46.81%                              | 13.57% |                         | 53.19% | 13.5   |
| 7%                    | 92.29%                              |        |                         | 4.64%  | 0.693  |
| 0.116                 | 16.82                               | 2.42   |                         | 0.430  | se     |
| nsibilidad_intermedia |                                     |        |                         |        |        |
|                       | contract_code bar_p60_h90_tp20_sl10 |        | p60_h90 TP 2.0 / SL 1.0 |        |        |
| 4                     | 38.35%                              | 10.52% | 25.41%                  | 9.29%  | 4.90%  |
| 4.13%                 | 69.69%                              | 13.42% |                         | 30.31% | 13.4   |
| 2%                    | 84.24%                              |        |                         | 6.42%  | 0.459  |
| 0.145                 | 53.55                               | 7.12   |                         | 0.438  | se     |
| nsibilidad_intermedia |                                     |        |                         |        |        |
|                       | contract_code bar_p50_h90_tp15_sl05 |        | p50_h90 TP 1.5 / SL 0.5 |        |        |
| 4                     | 46.99%                              | 8.83%  | 43.41%                  | 4.75%  | 23.31% |
| 9.41%                 | 33.28%                              | 14.16% |                         | 66.72% | 14.1   |
| 6%                    | 65.25%                              |        |                         | 6.65%  | 0.535  |
| 0.031                 | 40.29                               | 8.29   |                         | 0.438  | se     |
| nsibilidad_intermedia |                                     |        |                         |        |        |
|                       | contract_code bar_p50_h90_tp25_sl10 |        | p50_h90 TP 2.5 / SL 1.0 |        |        |
| 4                     | 46.99%                              | 8.83%  | 17.31%                  | 8.26%  | 6.93%  |
| 5.75%                 | 75.76%                              | 14.02% |                         | 24.24% | 14.0   |
| 2%                    | 72.02%                              |        |                         | 7.37%  | 0.363  |
| 0.149                 | 51.62                               | 7.21   |                         | 0.442  | se     |
| nsibilidad_intermedia |                                     |        |                         |        |        |
|                       | contract_code bar_p40_h30_tp25_sl10 |        | p40_h30 TP 2.5 / SL 1.0 |        |        |
| 4                     | 60.80%                              | 8.10%  | 22.26%                  | 9.66%  | 8.72%  |
| 5.92%                 | 69.01%                              | 15.58% |                         | 30.99% | 15.5   |
| 8%                    | 72.21%                              |        |                         | 5.04%  | 0.469  |
| 0.182                 | 16.20                               | 2.26   |                         | 0.443  | se     |
| nsibilidad_intermedia |                                     |        |                         |        |        |
|                       | contract_code bar_p50_h60_tp15_sl05 |        | p50_h60 TP 1.5 / SL 0.5 |        |        |

|                       |               |                       |         |                 |        |
|-----------------------|---------------|-----------------------|---------|-----------------|--------|
| 4                     | 49.18%        | 9.25%                 | 43.81%  | 4.64%           | 22.13% |
| 9.37%                 | 34.06%        | 14.01%                |         | 65.94%          | 14.0   |
| 1%                    | 66.64%        |                       |         | 7.07%           | 0.547  |
| 0.023                 | 26.67         | 5.60                  |         | 0.443           | se     |
| nsibilidad_intermedia |               |                       |         |                 |        |
|                       | contract_code | bar_p40_h30_tp20_sl10 | p40_h30 | TP 2.0 / SL 1.0 |        |
| 4                     | 60.80%        | 8.10%                 | 35.30%  | 9.81%           | 8.16%  |
| 5.40%                 | 56.54%        | 15.21%                |         | 43.46%          | 15.2   |
| 1%                    | 81.49%        |                       |         | 5.86%           | 0.624  |
| 0.142                 | 15.87         | 2.64                  |         | 0.444           |        |
| mayor_sensibilidad    |               |                       |         |                 |        |
|                       | contract_code | bar_p60_h60_tp15_sl10 | p60_h60 | TP 1.5 / SL 1.0 |        |
| 4                     | 40.24%        | 10.68%                | 47.71%  | 11.12%          | 3.96%  |
| 3.51%                 | 48.34%        | 14.62%                |         | 51.66%          | 14.6   |
| 2%                    | 92.52%        |                       |         | 4.65%           | 0.676  |
| 0.132                 | 33.38         | 4.67                  |         | 0.446           |        |
| mayor_sensibilidad    |               |                       |         |                 |        |
|                       | contract_code | bar_p50_h30_tp25_sl10 | p50_h30 | TP 2.5 / SL 1.0 |        |
| 4                     | 51.61%        | 9.77%                 | 18.25%  | 9.30%           | 6.68%  |
| 5.26%                 | 75.07%        | 14.56%                |         | 24.93%          | 14.5   |
| 6%                    | 73.66%        |                       |         | 5.69%           | 0.389  |
| 0.180                 | 16.93         | 1.96                  |         | 0.446           |        |
| mayor_sensibilidad    |               |                       |         |                 |        |
|                       | contract_code | bar_p50_h30_tp20_sl05 | p50_h30 | TP 2.0 / SL 0.5 |        |
| 4                     | 51.61%        | 9.77%                 | 26.00%  | 7.54%           | 22.82% |
| 8.87%                 | 51.18%        | 16.42%                |         | 48.82%          | 16.4   |
| 2%                    | 53.37%        |                       |         | 2.43%           | 0.406  |
| 0.107                 | 13.57         | 2.04                  |         | 0.450           |        |
| mayor_sensibilidad    |               |                       |         |                 |        |
|                       | contract_code | bar_p40_h90_tp25_sl10 | p40_h90 | TP 2.5 / SL 1.0 |        |
| 4                     | 55.37%        | 6.72%                 | 20.90%  | 8.75%           | 9.21%  |
| 6.91%                 | 69.89%        | 15.66%                |         | 30.11%          | 15.6   |
| 6%                    | 69.93%        |                       |         | 7.00%           | 0.431  |
| 0.150                 | 49.11         | 7.79                  |         | 0.450           |        |
| mayor_sensibilidad    |               |                       |         |                 |        |
|                       | contract_code | bar_p50_h60_tp25_sl10 | p50_h60 | TP 2.5 / SL 1.0 |        |
| 4                     | 49.18%        | 9.25%                 | 16.96%  | 9.06%           | 6.45%  |
| 5.60%                 | 76.59%        | 14.66%                |         | 23.41%          | 14.6   |
| 6%                    | 73.06%        |                       |         | 6.86%           | 0.359  |
| 0.170                 | 33.35         | 3.99                  |         | 0.454           |        |
| mayor_sensibilidad    |               |                       |         |                 |        |
|                       | contract_code | bar_p40_h60_tp25_sl10 | p40_h60 | TP 2.5 / SL 1.0 |        |
| 4                     | 58.20%        | 7.19%                 | 20.73%  | 9.72%           | 8.62%  |
| 6.59%                 | 70.64%        | 16.32%                |         | 29.36%          | 16.3   |
| 2%                    | 71.11%        |                       |         | 6.18%           | 0.432  |
| 0.177                 | 31.94         | 4.40                  |         | 0.460           |        |
| mayor_sensibilidad    |               |                       |         |                 |        |
|                       | contract_code | bar_p60_h60_tp20_sl10 | p60_h60 | TP 2.0 / SL 1.0 |        |
| 4                     | 40.24%        | 10.68%                | 24.94%  | 10.53%          | 4.50%  |
| 4.09%                 | 70.55%        | 14.62%                |         | 29.45%          | 14.6   |
| 2%                    | 85.17%        |                       |         | 6.37%           | 0.454  |
| 0.170                 | 34.58         | 3.91                  |         | 0.463           |        |
| mayor_sensibilidad    |               |                       |         |                 |        |
|                       | contract_code | bar_p60_h30_tp20_sl10 | p60_h30 | TP 2.0 / SL 1.0 |        |
| 4                     | 42.30%        | 11.20%                | 26.22%  | 10.85%          | 4.74%  |
| 3.94%                 | 69.05%        | 14.80%                |         | 30.95%          | 14.8   |
| 0%                    | 85.09%        |                       |         | 5.78%           | 0.477  |
| 0.178                 | 17.49         | 2.03                  |         | 0.466           |        |
| mayor_sensibilidad    |               |                       |         |                 |        |
|                       | contract_code | bar_p50_h90_tp20_sl05 | p50_h90 | TP 2.0 / SL 0.5 |        |

|        |                    |                       |         |                 |        |
|--------|--------------------|-----------------------|---------|-----------------|--------|
| 4      | 46.99%             | 8.83%                 | 24.27%  | 6.12%           | 24.81% |
| 10.63% | 50.92%             | 16.74%                | 49.08%  | 16.             |        |
| 74%    | 49.66%             |                       | 4.32%   | 0.36            |        |
| 1      | 0.069              | 39.91                 | 7.00    |                 |        |
| 0.466  | mayor_sensibilidad |                       |         |                 |        |
|        | contract_code      | bar_p40_h90_tp20_sl10 | p40_h90 | TP 2.0 / SL 1.0 |        |
| 4      | 55.37%             | 6.72%                 | 33.77%  | 10.03%          | 8.62%  |
| 6.39%  | 57.60%             | 16.42%                | 42.40%  | 16.4            |        |
| 2%     | 80.04%             |                       | 7.14%   | 0.589           |        |
| 0.137  | 48.17              | 7.97                  | 0.467   |                 |        |
|        | mayor_sensibilidad |                       |         |                 |        |
|        | contract_code      | bar_p50_h90_tp20_sl10 | p50_h90 | TP 2.0 / SL 1.0 |        |
| 4      | 46.99%             | 8.83%                 | 29.38%  | 9.75%           | 6.54%  |
| 5.49%  | 64.08%             | 15.24%                | 35.92%  | 15.2            |        |
| 4%     | 82.24%             |                       | 7.57%   | 0.522           |        |
| 0.140  | 50.86              | 7.53                  | 0.469   |                 |        |
|        | mayor_sensibilidad |                       |         |                 |        |
|        | contract_code      | bar_p50_h30_tp20_sl10 | p50_h30 | TP 2.0 / SL 1.0 |        |
| 4      | 51.61%             | 9.77%                 | 30.63%  | 10.67%          | 6.29%  |
| 4.85%  | 63.08%             | 15.53%                | 36.92%  | 15.5            |        |
| 3%     | 83.32%             |                       | 6.09%   | 0.550           |        |
| 0.165  | 16.74              | 2.45                  | 0.469   |                 |        |
|        | mayor_sensibilidad |                       |         |                 |        |
|        | contract_code      | bar_p60_h30_tp15_sl05 | p60_h30 | TP 1.5 / SL 0.5 |        |
| 4      | 42.30%             | 11.20%                | 42.76%  | 6.16%           | 17.99% |
| 8.40%  | 39.26%             | 14.56%                | 60.74%  | 14.5            |        |
| 6%     | 70.61%             |                       | 6.70%   | 0.551           |        |
| 0.050  | 14.57              | 2.64                  | 0.470   |                 |        |
|        | mayor_sensibilidad |                       |         |                 |        |
|        | contract_code      | bar_p40_h60_tp20_sl10 | p40_h60 | TP 2.0 / SL 1.0 |        |
| 4      | 58.20%             | 7.19%                 | 33.73%  | 10.73%          | 8.11%  |
| 5.94%  | 58.16%             | 16.67%                | 41.84%  | 16.6            |        |
| 7%     | 80.98%             |                       | 6.53%   | 0.594           |        |
| 0.155  | 31.49              | 5.20                  | 0.471   |                 |        |
|        | mayor_sensibilidad |                       |         |                 |        |
|        | contract_code      | bar_p60_h90_tp15_sl05 | p60_h90 | TP 1.5 / SL 0.5 |        |
| 4      | 38.35%             | 10.52%                | 41.34%  | 4.86%           | 19.65% |
| 10.06% | 39.01%             | 14.92%                | 60.99%  | 14.             |        |
| 92%    | 68.06%             |                       | 8.59%   | 0.52            |        |
| 2      | 0.035              | 43.38                 | 8.37    |                 |        |
| 0.489  | mayor_sensibilidad |                       |         |                 |        |
|        | contract_code      | bar_p50_h60_tp20_sl10 | p50_h60 | TP 2.0 / SL 1.0 |        |
| 4      | 49.18%             | 9.25%                 | 29.03%  | 11.27%          | 6.07%  |
| 5.21%  | 64.90%             | 16.48%                | 35.10%  | 16.4            |        |
| 8%     | 83.14%             |                       | 6.83%   | 0.520           |        |
| 0.173  | 33.15              | 4.54                  | 0.490   |                 |        |
|        | mayor_sensibilidad |                       |         |                 |        |
|        | contract_code      | bar_p50_h60_tp20_sl05 | p50_h60 | TP 2.0 / SL 0.5 |        |
| 4      | 49.18%             | 9.25%                 | 24.26%  | 7.77%           | 23.55% |
| 10.57% | 52.19%             | 18.34%                | 47.81%  | 18.             |        |
| 34%    | 50.91%             |                       | 3.18%   | 0.36            |        |
| 7      | 0.102              | 26.32                 | 4.05    |                 |        |
| 0.491  | mayor_sensibilidad |                       |         |                 |        |
|        | contract_code      | bar_p60_h30_tp20_sl05 | p60_h30 | TP 2.0 / SL 0.5 |        |
| 4      | 42.30%             | 11.20%                | 22.70%  | 8.09%           | 19.10% |
| 9.38%  | 58.20%             | 17.47%                | 41.80%  | 17.4            |        |
| 7%     | 54.50%             |                       | 3.27%   | 0.358           |        |
| 0.115  | 14.27              | 1.78                  | 0.494   |                 |        |
|        | mayor_sensibilidad |                       |         |                 |        |
|        | contract_code      | bar_p60_h90_tp20_sl05 | p60_h90 | TP 2.0 / SL 0.5 |        |

|        |                      |                       |         |                 |        |
|--------|----------------------|-----------------------|---------|-----------------|--------|
| 4      | 38.35%               | 10.52%                | 21.51%  | 6.10%           | 20.85% |
| 10.87% | 57.64%               | 16.97%                | 42.36%  | 16.             |        |
| 97%    | 51.11%               |                       | 5.99%   | 0.32            |        |
| 6      | 0.068                | 42.34                 | 7.25    |                 |        |
| 0.504  | mayor_sensibilidad   |                       |         |                 |        |
|        | contract_code        | bar_p60_h60_tp15_sl05 | p60_h60 | TP 1.5 / SL 0.5 |        |
| 4      | 40.24%               | 10.68%                | 41.10%  | 6.69%           | 18.57% |
| 9.99%  | 40.32%               | 16.68%                | 59.68%  | 16.6            |        |
| 8%     | 69.18%               |                       | 8.07%   | 0.524           |        |
| 0.050  | 28.50                | 5.52                  | 0.521   |                 |        |
|        | mayor_sensibilidad   |                       |         |                 |        |
|        | contract_code        | bar_p60_h60_tp20_sl05 | p60_h60 | TP 2.0 / SL 0.5 |        |
| 4      | 40.24%               | 10.68%                | 21.23%  | 7.56%           | 19.67% |
| 11.17% | 59.10%               | 18.73%                | 40.90%  | 18.             |        |
| 73%    | 52.25%               |                       | 5.33%   | 0.32            |        |
| 6      | 0.095                | 27.55                 | 3.93    |                 |        |
| 0.535  | mayor_sensibilidad   |                       |         |                 |        |
|        | contract_code_regime | bar_p60_h60_tp10_sl10 | p60_h60 | TP 1.0 / SL 1.0 |        |
| 20     | 34.60%               | 50.93%                | 97.05%  | 3.97%           | 2.95%  |
| 3.97%  | 0.00%                | 0.00%                 | 100.00% | 0.0             |        |
| 0%     | 97.05%               |                       | 3.97%   | 0.941           |        |
| 0.079  | 31.14                | 13.74                 | 0.628   |                 |        |
|        | menor_sensibilidad   |                       |         |                 |        |
|        | contract_code_regime | bar_p60_h30_tp10_sl10 | p60_h30 | TP 1.0 / SL 1.0 |        |
| 20     | 35.62%               | 50.50%                | 96.72%  | 5.67%           | 3.28%  |
| 5.67%  | 0.00%                | 0.00%                 | 100.00% | 0.0             |        |
| 0%     | 96.72%               |                       | 5.67%   | 0.934           |        |
| 0.113  | 15.94                | 5.13                  | 0.675   |                 |        |
|        | menor_sensibilidad   |                       |         |                 |        |
|        | contract_code_regime | bar_p60_h90_tp10_sl10 | p60_h90 | TP 1.0 / SL 1.0 |        |
| 20     | 33.77%               | 51.05%                | 96.99%  | 5.62%           | 3.01%  |
| 5.62%  | 0.00%                | 0.00%                 | 100.00% | 0.0             |        |
| 0%     | 96.99%               |                       | 5.62%   | 0.940           |        |
| 0.112  | 46.22                | 20.28                 | 0.679   |                 |        |
|        | menor_sensibilidad   |                       |         |                 |        |
|        | contract_code_regime | bar_p50_h60_tp10_sl10 | p50_h60 | TP 1.0 / SL 1.0 |        |
| 20     | 42.19%               | 59.80%                | 95.88%  | 4.79%           | 4.12%  |
| 4.79%  | 0.00%                | 0.00%                 | 100.00% | 0.0             |        |
| 0%     | 95.88%               |                       | 4.79%   | 0.918           |        |
| 0.096  | 29.28                | 14.14                 | 0.742   |                 |        |
|        | menor_sensibilidad   |                       |         |                 |        |
|        | contract_code_regime | bar_p50_h90_tp10_sl10 | p50_h90 | TP 1.0 / SL 1.0 |        |
| 20     | 41.27%               | 59.91%                | 95.78%  | 6.38%           | 4.22%  |
| 6.38%  | 0.00%                | 0.00%                 | 100.00% | 0.0             |        |
| 0%     | 95.78%               |                       | 6.38%   | 0.916           |        |
| 0.128  | 43.22                | 20.26                 | 0.791   |                 |        |
|        | menor_sensibilidad   |                       |         |                 |        |
|        | contract_code_regime | bar_p50_h30_tp10_sl10 | p50_h30 | TP 1.0 / SL 1.0 |        |
| 20     | 43.52%               | 58.75%                | 95.49%  | 6.86%           | 4.51%  |
| 6.86%  | 0.00%                | 0.00%                 | 100.00% | 0.0             |        |
| 0%     | 95.49%               |                       | 6.86%   | 0.910           |        |
| 0.137  | 15.01                | 5.30                  | 0.793   |                 |        |
|        | menor_sensibilidad   |                       |         |                 |        |
|        | contract_code_regime | bar_p60_h60_tp10_sl05 | p60_h60 | TP 1.0 / SL 0.5 |        |
| 20     | 34.60%               | 50.93%                | 85.07%  | 10.47%          | 14.93% |
| 10.47% | 0.00%                | 0.00%                 | 100.00% | 0.              |        |
| 00%    | 85.07%               |                       | 10.47%  | 0.77            |        |
| 6      | 0.157                | 27.89                 | 14.76   |                 |        |
| 0.823  | menor_sensibilidad   |                       |         |                 |        |
|        | contract_code_regime | bar_p40_h60_tp10_sl10 | p40_h60 | TP 1.0 / SL 1.0 |        |

|                                            |                    |        |                         |        |        |
|--------------------------------------------|--------------------|--------|-------------------------|--------|--------|
| 20                                         | 49.83%             | 68.30% | 94.46%                  | 5.17%  | 5.54%  |
| 5.17%                                      | 0.00%              | 0.00%  | 100.00%                 |        | 0.0    |
| 0%                                         | 94.46%             |        | 5.17%                   |        | 0.889  |
| 0.103                                      | 27.28              | 14.67  |                         | 0.838  |        |
| menor_sensibilidad                         |                    |        |                         |        |        |
| contract_code_regime bar_p60_h90_tp10_sl05 |                    |        | p60_h90 TP 1.0 / SL 0.5 |        |        |
| 20                                         | 33.77%             | 51.05% | 84.53%                  | 11.44% | 15.47% |
| 11.44%                                     | 0.00%              | 0.00%  | 100.00%                 |        | 0.     |
| 00%                                        | 84.53%             |        | 11.44%                  |        | 0.76   |
| 8                                          | 0.172              | 41.07  |                         | 22.30  |        |
| 0.854                                      | menor_sensibilidad |        |                         |        |        |
| contract_code_regime bar_p40_h90_tp10_sl10 |                    |        | p40_h90 TP 1.0 / SL 1.0 |        |        |
| 20                                         | 48.54%             | 67.63% | 94.20%                  | 6.61%  | 5.80%  |
| 6.61%                                      | 0.00%              | 0.00%  | 100.00%                 |        | 0.0    |
| 0%                                         | 94.20%             |        | 6.61%                   |        | 0.884  |
| 0.132                                      | 40.07              | 21.61  |                         | 0.874  |        |
| menor_sensibilidad                         |                    |        |                         |        |        |
| contract_code_regime bar_p60_h30_tp10_sl05 |                    |        | p60_h30 TP 1.0 / SL 0.5 |        |        |
| 20                                         | 35.62%             | 50.50% | 85.26%                  | 12.38% | 14.74% |
| 12.38%                                     | 0.00%              | 0.00%  | 100.00%                 |        | 0.     |
| 00%                                        | 85.26%             |        | 12.38%                  |        | 0.77   |
| 9                                          | 0.186              | 14.35  |                         | 5.42   |        |
| 0.876                                      | menor_sensibilidad |        |                         |        |        |
| contract_code_regime bar_p40_h30_tp10_sl10 |                    |        | p40_h30 TP 1.0 / SL 1.0 |        |        |
| 20                                         | 51.30%             | 66.56% | 93.91%                  | 7.88%  | 6.09%  |
| 7.88%                                      | 0.00%              | 0.00%  | 100.00%                 |        | 0.0    |
| 0%                                         | 93.91%             |        | 7.88%                   |        | 0.878  |
| 0.158                                      | 13.92              | 5.93   |                         | 0.902  |        |
| menor_sensibilidad                         |                    |        |                         |        |        |
| contract_code_regime bar_p50_h60_tp10_sl05 |                    |        | p50_h60 TP 1.0 / SL 0.5 |        |        |
| 20                                         | 42.19%             | 59.80% | 82.08%                  | 10.46% | 17.92% |
| 10.46%                                     | 0.00%              | 0.00%  | 100.00%                 |        | 0.     |
| 00%                                        | 82.08%             |        | 10.46%                  |        | 0.73   |
| 1                                          | 0.157              | 25.60  |                         | 15.00  |        |
| 0.912                                      | menor_sensibilidad |        |                         |        |        |
| contract_code_regime bar_p50_h90_tp10_sl05 |                    |        | p50_h90 TP 1.0 / SL 0.5 |        |        |
| 20                                         | 41.27%             | 59.91% | 81.36%                  | 11.15% | 18.64% |
| 11.15%                                     | 0.00%              | 0.00%  | 100.00%                 |        | 0.     |
| 00%                                        | 81.36%             |        | 11.15%                  |        | 0.72   |
| 0                                          | 0.167              | 37.43  |                         | 21.73  |        |
| 0.934                                      | menor_sensibilidad |        |                         |        |        |
| contract_code_regime bar_p60_h60_tp15_sl10 |                    |        | p60_h60 TP 1.5 / SL 1.0 |        |        |
| 20                                         | 34.60%             | 50.93% | 47.08%                  | 16.07% | 3.67%  |
| 4.58%                                      | 49.26%             | 20.07% | 50.74%                  |        | 20.0   |
| 7%                                         | 92.98%             |        | 6.50%                   |        | 0.670  |
| 0.201                                      | 34.47              | 12.50  |                         | 0.982  |        |
| menor_sensibilidad                         |                    |        |                         |        |        |
| contract_code_regime bar_p40_h90_tp10_sl05 |                    |        | p40_h90 TP 1.0 / SL 0.5 |        |        |
| 20                                         | 48.54%             | 67.63% | 78.18%                  | 10.88% | 21.82% |
| 10.88%                                     | 0.00%              | 0.00%  | 100.00%                 |        | 0.     |
| 00%                                        | 78.18%             |        | 10.88%                  |        | 0.67   |
| 3                                          | 0.163              | 33.81  |                         | 22.47  |        |
| 1.003                                      | menor_sensibilidad |        |                         |        |        |
| contract_code_regime bar_p40_h60_tp10_sl05 |                    |        | p40_h60 TP 1.0 / SL 0.5 |        |        |
| 20                                         | 49.83%             | 68.30% | 78.56%                  | 11.02% | 21.44% |
| 11.02%                                     | 0.00%              | 0.00%  | 100.00%                 |        | 0.     |
| 00%                                        | 78.56%             |        | 11.02%                  |        | 0.67   |
| 8                                          | 0.165              | 23.14  |                         | 15.05  |        |
| 1.014                                      | menor_sensibilidad |        |                         |        |        |
| contract_code_regime bar_p60_h30_tp15_sl10 |                    |        | p60_h30 TP 1.5 / SL 1.0 |        |        |



|                                            |                         |        |                         |         |        |
|--------------------------------------------|-------------------------|--------|-------------------------|---------|--------|
| 20                                         | 35.62%                  | 50.50% | 49.00%                  | 15.61%  | 3.97%  |
| 6.56%                                      | 47.03%                  | 20.25% |                         | 52.97%  | 20.2   |
| 5%                                         | 92.70%                  |        |                         | 9.23%   | 0.695  |
| 0.195                                      | 17.60                   | 4.80   |                         | 1.021   |        |
| menor_sensibilidad                         |                         |        |                         |         |        |
| contract_code_regime bar_p50_h60_tp15_sl10 |                         |        | p50_h60 TP 1.5 / SL 1.0 |         |        |
| 20                                         | 42.19%                  | 59.80% | 51.68%                  | 12.60%  | 5.08%  |
| 5.66%                                      | 43.23%                  | 17.09% |                         | 56.77%  | 17.0   |
| 9%                                         | 91.24%                  |        |                         | 7.45%   | 0.724  |
| 0.144                                      | 32.67                   | 13.48  |                         | 1.026   |        |
| menor_sensibilidad                         |                         |        |                         |         |        |
| contract_code_regime bar_p50_h30_tp10_sl05 |                         |        | p50_h30 TP 1.0 / SL 0.5 |         |        |
| 20                                         | 43.52%                  | 58.75% | 82.32%                  | 14.89%  | 17.68% |
| 14.89%                                     | 0.00%                   | 0.00%  |                         | 100.00% | 0.     |
| 00%                                        | 82.32%                  |        |                         | 14.89%  | 0.73   |
| 5                                          | 0.223                   | 13.18  |                         | 5.92    |        |
| 1.034                                      | menor_sensibilidad      |        |                         |         |        |
| contract_code_regime bar_p60_h60_tp15_sl05 |                         |        | p60_h60 TP 1.5 / SL 0.5 |         |        |
| 20                                         | 34.60%                  | 50.93% | 40.75%                  | 11.25%  | 18.17% |
| 11.84%                                     | 41.08%                  | 20.09% |                         | 58.92%  | 20.    |
| 09%                                        | 69.46%                  |        |                         | 10.74%  | 0.52   |
| 0                                          | 0.129                   | 29.60  |                         | 12.92   |        |
| 1.049                                      | menor_sensibilidad      |        |                         |         |        |
| contract_code_regime bar_p40_h30_tp10_sl05 |                         |        | p40_h30 TP 1.0 / SL 0.5 |         |        |
| 20                                         | 51.30%                  | 66.56% | 79.17%                  | 13.14%  | 20.83% |
| 13.14%                                     | 0.00%                   | 0.00%  |                         | 100.00% | 0.     |
| 00%                                        | 79.17%                  |        |                         | 13.14%  | 0.68   |
| 8                                          | 0.197                   | 11.91  |                         | 6.55    |        |
| 1.060                                      | sensibilidad_intermedia |        |                         |         |        |
| contract_code_regime bar_p50_h60_tp15_sl05 |                         |        | p50_h60 TP 1.5 / SL 0.5 |         |        |
| 20                                         | 42.19%                  | 59.80% | 43.87%                  | 7.14%   | 21.75% |
| 11.66%                                     | 34.38%                  | 18.62% |                         | 65.62%  | 18.    |
| 62%                                        | 67.07%                  |        |                         | 9.68%   | 0.54   |
| 9                                          | 0.055                   | 27.73  |                         | 13.99   |        |
| 1.069                                      | sensibilidad_intermedia |        |                         |         |        |
| contract_code_regime bar_p40_h60_tp15_sl10 |                         |        | p40_h60 TP 1.5 / SL 1.0 |         |        |
| 20                                         | 49.83%                  | 68.30% | 56.17%                  | 9.90%   | 6.83%  |
| 6.22%                                      | 37.00%                  | 15.20% |                         | 63.00%  | 15.2   |
| 0%                                         | 89.32%                  |        |                         | 7.59%   | 0.774  |
| 0.104                                      | 30.71                   | 13.97  |                         | 1.072   | se     |
| nsibilidad_intermedia                      |                         |        |                         |         |        |
| contract_code_regime bar_p50_h30_tp15_sl10 |                         |        | p50_h30 TP 1.5 / SL 1.0 |         |        |
| 20                                         | 43.52%                  | 58.75% | 53.76%                  | 11.72%  | 5.47%  |
| 7.82%                                      | 40.77%                  | 19.05% |                         | 59.23%  | 19.0   |
| 5%                                         | 90.99%                  |        |                         | 10.27%  | 0.752  |
| 0.139                                      | 16.67                   | 5.48   |                         | 1.076   | se     |
| nsibilidad_intermedia                      |                         |        |                         |         |        |
| contract_code_regime bar_p60_h60_tp25_sl10 |                         |        | p60_h60 TP 2.5 / SL 1.0 |         |        |
| 20                                         | 34.60%                  | 50.93% | 12.38%                  | 14.91%  | 4.37%  |
| 5.68%                                      | 83.25%                  | 19.81% |                         | 16.75%  | 19.8   |
| 1%                                         | 74.34%                  |        |                         | 16.59%  | 0.266  |
| 0.324                                      | 35.67                   | 13.09  |                         | 1.079   | se     |
| nsibilidad_intermedia                      |                         |        |                         |         |        |
| contract_code_regime bar_p40_h90_tp15_sl10 |                         |        | p40_h90 TP 1.5 / SL 1.0 |         |        |
| 20                                         | 48.54%                  | 67.63% | 55.68%                  | 10.12%  | 7.07%  |
| 7.36%                                      | 37.25%                  | 16.11% |                         | 62.75%  | 16.1   |
| 1%                                         | 88.94%                  |        |                         | 8.87%   | 0.764  |
| 0.124                                      | 45.75                   | 22.09  |                         | 1.101   | se     |
| nsibilidad_intermedia                      |                         |        |                         |         |        |
| contract_code_regime bar_p60_h60_tp20_sl10 |                         |        | p60_h60 TP 2.0 / SL 1.0 |         |        |

|                                                                    |                         |        |        |        |        |
|--------------------------------------------------------------------|-------------------------|--------|--------|--------|--------|
| 20                                                                 | 34.60%                  | 50.93% | 23.54% | 18.18% | 4.18%  |
| 5.48%                                                              | 72.28%                  | 22.71% | 27.72% |        | 22.7   |
| 1%                                                                 | 85.41%                  |        | 12.95% |        | 0.429  |
| 0.318                                                              | 35.77                   | 11.03  |        | 1.103  | se     |
| nsibilidad_intermedia                                              |                         |        |        |        |        |
| contract_code_regime bar_p40_h60_tp15_sl05 p40_h60 TP 1.5 / SL 0.5 |                         |        |        |        |        |
| 20                                                                 | 49.83%                  | 68.30% | 46.64% | 5.22%  | 25.75% |
| 11.66%                                                             | 27.61%                  | 15.46% | 72.39% |        | 15.    |
| 46%                                                                | 64.57%                  |        | 9.85%  |        | 0.57   |
| 1                                                                  | 0.059                   | 25.68  |        | 13.82  |        |
| 1.105                                                              | sensibilidad_intermedia |        |        |        |        |
| contract_code_regime bar_p60_h30_tp25_sl10 p60_h30 TP 2.5 / SL 1.0 |                         |        |        |        |        |
| 20                                                                 | 35.62%                  | 50.50% | 14.31% | 14.24% | 4.74%  |
| 7.82%                                                              | 80.95%                  | 21.93% | 19.05% |        | 21.9   |
| 3%                                                                 | 75.83%                  |        | 16.11% |        | 0.310  |
| 0.279                                                              | 18.34                   | 4.29   |        | 1.106  | se     |
| nsibilidad_intermedia                                              |                         |        |        |        |        |
| contract_code_regime bar_p40_h90_tp15_sl05 p40_h90 TP 1.5 / SL 0.5 |                         |        |        |        |        |
| 20                                                                 | 48.54%                  | 67.63% | 45.77% | 7.26%  | 26.21% |
| 11.46%                                                             | 28.02%                  | 15.15% | 71.98% |        | 15.    |
| 15%                                                                | 63.75%                  |        | 9.80%  |        | 0.55   |
| 6                                                                  | 0.084                   | 37.85  |        | 21.90  |        |
| 1.113                                                              | sensibilidad_intermedia |        |        |        |        |
| contract_code_regime bar_p60_h60_tp20_sl05 p60_h60 TP 2.0 / SL 0.5 |                         |        |        |        |        |
| 20                                                                 | 34.60%                  | 50.93% | 20.22% | 13.11% | 19.24% |
| 13.56%                                                             | 60.55%                  | 23.15% | 39.45% |        | 23.    |
| 15%                                                                | 51.46%                  |        | 11.02% |        | 0.30   |
| 8                                                                  | 0.212                   | 28.49  |        | 10.21  |        |
| 1.118                                                              | sensibilidad_intermedia |        |        |        |        |
| contract_code_regime bar_p60_h30_tp20_sl10 p60_h30 TP 2.0 / SL 1.0 |                         |        |        |        |        |
| 20                                                                 | 35.62%                  | 50.50% | 25.98% | 17.57% | 4.47%  |
| 7.11%                                                              | 69.55%                  | 24.65% | 30.45% |        | 24.6   |
| 5%                                                                 | 85.87%                  |        | 12.79% |        | 0.475  |
| 0.281                                                              | 18.30                   | 5.74   |        | 1.126  | se     |
| nsibilidad_intermedia                                              |                         |        |        |        |        |
| contract_code_regime bar_p50_h90_tp15_sl05 p50_h90 TP 1.5 / SL 0.5 |                         |        |        |        |        |
| 20                                                                 | 41.27%                  | 59.91% | 43.33% | 10.19% | 22.50% |
| 12.35%                                                             | 34.17%                  | 19.95% | 65.83% |        | 19.    |
| 95%                                                                | 66.05%                  |        | 11.00% |        | 0.53   |
| 8                                                                  | 0.123                   | 41.03  |        | 21.16  |        |
| 1.134                                                              | sensibilidad_intermedia |        |        |        |        |
| contract_code_regime bar_p50_h60_tp20_sl05 p50_h60 TP 2.0 / SL 0.5 |                         |        |        |        |        |
| 20                                                                 | 42.19%                  | 59.80% | 23.70% | 11.68% | 23.14% |
| 12.66%                                                             | 53.16%                  | 20.96% | 46.84% |        | 20.    |
| 96%                                                                | 50.72%                  |        | 8.50%  |        | 0.35   |
| 8                                                                  | 0.188                   | 27.34  |        | 11.08  |        |
| 1.136                                                              | sensibilidad_intermedia |        |        |        |        |
| contract_code_regime bar_p60_h30_tp15_sl05 p60_h30 TP 1.5 / SL 0.5 |                         |        |        |        |        |
| 20                                                                 | 35.62%                  | 50.50% | 42.69% | 13.87% | 18.00% |
| 13.97%                                                             | 39.31%                  | 20.52% | 60.69% |        | 20.    |
| 52%                                                                | 70.59%                  |        | 15.04% |        | 0.55   |
| 0                                                                  | 0.204                   | 15.34  |        | 5.26   |        |
| 1.139                                                              | sensibilidad_intermedia |        |        |        |        |
| contract_code_regime bar_p40_h30_tp15_sl10 p40_h30 TP 1.5 / SL 1.0 |                         |        |        |        |        |
| 20                                                                 | 51.30%                  | 66.56% | 57.43% | 11.18% | 7.26%  |
| 8.64%                                                              | 35.31%                  | 17.49% | 64.69% |        | 17.4   |
| 9%                                                                 | 88.97%                  |        | 10.36% |        | 0.789  |
| 0.145                                                              | 15.64                   | 5.22   |        | 1.142  | se     |
| nsibilidad_intermedia                                              |                         |        |        |        |        |
| contract code regime bar p60 h90 tp15 sl05 p60 h90 TP 1.5 / SL 0.5 |                         |        |        |        |        |

|                                                                    |                         |        |        |        |        |
|--------------------------------------------------------------------|-------------------------|--------|--------|--------|--------|
| 20                                                                 | 33.77%                  | 51.05% | 40.63% | 13.25% | 18.74% |
| 13.38%                                                             | 40.63%                  | 24.74% |        | 59.37% | 24.    |
| 74%                                                                | 68.76%                  |        |        | 13.07% | 0.51   |
| 6                                                                  | 0.170                   | 44.03  |        | 19.71  |        |
| 1.155                                                              | sensibilidad_intermedia |        |        |        |        |
| contract_code_regime bar_p50_h90_tp15_sl10 p50_h90 TP 1.5 / SL 1.0 |                         |        |        |        |        |
| 20                                                                 | 41.27%                  | 59.91% | 51.41% | 16.18% | 5.17%  |
| 7.12%                                                              | 43.42%                  | 22.49% |        | 56.58% | 22.4   |
| 9%                                                                 | 91.10%                  |        |        | 9.86%  | 0.720  |
| 0.199                                                              | 48.91                   | 20.98  |        | 1.156  | se     |
| nsibilidad_intermedia                                              |                         |        |        |        |        |
| contract_code_regime bar_p50_h60_tp20_sl10 p50_h60 TP 2.0 / SL 1.0 |                         |        |        |        |        |
| 20                                                                 | 42.19%                  | 59.80% | 28.17% | 16.96% | 5.78%  |
| 6.78%                                                              | 66.05%                  | 22.40% |        | 33.95% | 22.4   |
| 0%                                                                 | 83.44%                  |        |        | 10.76% | 0.506  |
| 0.285                                                              | 34.30                   | 11.24  |        | 1.167  | se     |
| nsibilidad_intermedia                                              |                         |        |        |        |        |
| contract_code_regime bar_p60_h90_tp15_sl10 p60_h90 TP 1.5 / SL 1.0 |                         |        |        |        |        |
| 20                                                                 | 33.77%                  | 51.05% | 46.94% | 21.61% | 3.77%  |
| 6.89%                                                              | 49.30%                  | 26.77% |        | 50.70% | 26.7   |
| 7%                                                                 | 92.82%                  |        |        | 10.59% | 0.666  |
| 0.272                                                              | 51.67                   | 18.88  |        | 1.169  | se     |
| nsibilidad_intermedia                                              |                         |        |        |        |        |
| contract_code_regime bar_p40_h60_tp20_sl05 p40_h60 TP 2.0 / SL 0.5 |                         |        |        |        |        |
| 20                                                                 | 49.83%                  | 68.30% | 27.43% | 8.99%  | 27.62% |
| 12.60%                                                             | 44.95%                  | 20.73% |        | 55.05% | 20.    |
| 73%                                                                | 49.93%                  |        |        | 6.44%  | 0.41   |
| 0                                                                  | 0.149                   | 25.96  |        | 12.12  |        |
| 1.171                                                              | mayor_sensibilidad      |        |        |        |        |
| contract_code_regime bar_p40_h60_tp20_sl10 p40_h60 TP 2.0 / SL 1.0 |                         |        |        |        |        |
| 20                                                                 | 49.83%                  | 68.30% | 33.32% | 13.85% | 7.81%  |
| 7.18%                                                              | 58.88%                  | 19.84% |        | 41.12% | 19.8   |
| 4%                                                                 | 81.39%                  |        |        | 8.77%  | 0.588  |
| 0.219                                                              | 32.65                   | 12.87  |        | 1.179  |        |
| mayor_sensibilidad                                                 |                         |        |        |        |        |
| contract_code_regime bar_p50_h60_tp25_sl10 p50_h60 TP 2.5 / SL 1.0 |                         |        |        |        |        |
| 20                                                                 | 42.19%                  | 59.80% | 15.76% | 15.04% | 6.14%  |
| 7.70%                                                              | 78.09%                  | 20.81% |        | 21.91% | 20.8   |
| 1%                                                                 | 72.48%                  |        |        | 15.75% | 0.333  |
| 0.318                                                              | 34.50                   | 10.88  |        | 1.191  |        |
| mayor_sensibilidad                                                 |                         |        |        |        |        |
| contract_code_regime bar_p60_h90_tp20_sl05 p60_h90 TP 2.0 / SL 0.5 |                         |        |        |        |        |
| 20                                                                 | 33.77%                  | 51.05% | 20.03% | 14.32% | 19.84% |
| 14.67%                                                             | 60.13%                  | 27.91% |        | 39.87% | 27.    |
| 91%                                                                | 50.46%                  |        |        | 12.71% | 0.30   |
| 1                                                                  | 0.218                   | 42.53  |        | 15.85  |        |
| 1.207                                                              | mayor_sensibilidad      |        |        |        |        |
| contract_code_regime bar_p60_h30_tp20_sl05 p60_h30 TP 2.0 / SL 0.5 |                         |        |        |        |        |
| 20                                                                 | 35.62%                  | 50.50% | 22.43% | 12.47% | 19.10% |
| 15.11%                                                             | 58.47%                  | 27.42% |        | 41.53% | 27.    |
| 42%                                                                | 54.26%                  |        |        | 16.08% | 0.35   |
| 3                                                                  | 0.212                   | 14.95  |        | 4.07   |        |
| 1.216                                                              | mayor_sensibilidad      |        |        |        |        |
| contract_code_regime bar_p50_h30_tp20_sl10 p50_h30 TP 2.0 / SL 1.0 |                         |        |        |        |        |
| 20                                                                 | 43.52%                  | 58.75% | 30.95% | 15.86% | 6.16%  |
| 9.01%                                                              | 62.89%                  | 24.17% |        | 37.11% | 24.1   |
| 7%                                                                 | 83.97%                  |        |        | 14.33% | 0.557  |
| 0.235                                                              | 17.52                   | 5.21   |        | 1.221  |        |
| mayor_sensibilidad                                                 |                         |        |        |        |        |
| contract code regime bar p50 h90 tp20 sl05 p50 h90 TP 2.0 / SL 0.5 |                         |        |        |        |        |

|                      |                       |        |                         |        |        |
|----------------------|-----------------------|--------|-------------------------|--------|--------|
| 20                   | 41.27%                | 59.91% | 23.31%                  | 12.16% | 23.92% |
| 13.74%               | 52.77%                | 25.22% |                         | 47.23% | 25.    |
| 22%                  | 49.56%                |        | 11.21%                  |        | 0.34   |
| 7                    | 0.181                 | 40.45  |                         | 17.28  |        |
| 1.222                | mayor_sensibilidad    |        |                         |        |        |
| contract_code_regime | bar_p40_h90_tp20_sl05 |        | p40_h90 TP 2.0 / SL 0.5 |        |        |
| 20                   | 48.54%                | 67.63% | 26.74%                  | 11.74% | 28.02% |
| 12.58%               | 45.24%                | 22.79% |                         | 54.76% | 22.    |
| 79%                  | 48.96%                |        | 7.69%                   |        | 0.39   |
| 5                    | 0.197                 | 38.29  |                         | 18.14  |        |
| 1.224                | mayor_sensibilidad    |        |                         |        |        |
| contract_code_regime | bar_p60_h90_tp20_sl10 |        | p60_h90 TP 2.0 / SL 1.0 |        |        |
| 20                   | 33.77%                | 51.05% | 23.32%                  | 21.04% | 4.28%  |
| 7.66%                | 72.40%                | 26.96% |                         | 27.60% | 26.9   |
| 6%                   | 85.13%                |        | 15.96%                  |        | 0.423  |
| 0.361                | 54.00                 | 15.46  |                         |        | 1.227  |
| mayor_sensibilidad   |                       |        |                         |        |        |
| contract_code_regime | bar_p40_h60_tp25_sl10 |        | p40_h60 TP 2.5 / SL 1.0 |        |        |
| 20                   | 49.83%                | 68.30% | 19.94%                  | 14.15% | 8.31%  |
| 8.33%                | 71.75%                | 20.63% |                         | 28.25% | 20.6   |
| 3%                   | 71.05%                |        | 11.31%                  |        | 0.415  |
| 0.289                | 33.12                 | 10.91  |                         |        | 1.227  |
| mayor_sensibilidad   |                       |        |                         |        |        |
| contract_code_regime | bar_p50_h30_tp25_sl10 |        | p50_h30 TP 2.5 / SL 1.0 |        |        |
| 20                   | 43.52%                | 58.75% | 18.23%                  | 16.32% | 6.59%  |
| 9.40%                | 75.17%                | 25.72% |                         | 24.83% | 25.7   |
| 2%                   | 74.16%                |        | 12.95%                  |        | 0.390  |
| 0.314                | 17.71                 | 5.85   |                         |        | 1.231  |
| mayor_sensibilidad   |                       |        |                         |        |        |
| contract_code_regime | bar_p40_h30_tp15_sl05 |        | p40_h30 TP 1.5 / SL 0.5 |        |        |
| 20                   | 51.30%                | 66.56% | 48.02%                  | 11.94% | 24.79% |
| 14.42%               | 27.19%                | 16.95% |                         | 72.81% | 16.    |
| 95%                  | 66.11%                |        | 13.54%                  |        | 0.59   |
| 6                    | 0.201                 | 13.21  |                         | 5.79   |        |
| 1.234                | mayor_sensibilidad    |        |                         |        |        |
| contract_code_regime | bar_p50_h30_tp15_sl05 |        | p50_h30 TP 1.5 / SL 0.5 |        |        |
| 20                   | 43.52%                | 58.75% | 45.81%                  | 12.23% | 21.31% |
| 16.11%               | 32.88%                | 18.86% |                         | 67.12% | 18.    |
| 86%                  | 68.45%                |        | 17.61%                  |        | 0.58   |
| 1                    | 0.235                 | 14.28  |                         | 5.73   |        |
| 1.236                | mayor_sensibilidad    |        |                         |        |        |
| contract_code_regime | bar_p60_h90_tp25_sl10 |        | p60_h90 TP 2.5 / SL 1.0 |        |        |
| 20                   | 33.77%                | 51.05% | 12.10%                  | 15.97% | 4.49%  |
| 8.34%                | 83.41%                | 22.57% |                         | 16.59% | 22.5   |
| 7%                   | 73.61%                |        | 27.32%                  |        | 0.258  |
| 0.333                | 54.21                 | 19.06  |                         |        | 1.253  |
| mayor_sensibilidad   |                       |        |                         |        |        |
| contract_code_regime | bar_p40_h30_tp20_sl10 |        | p40_h30 TP 2.0 / SL 1.0 |        |        |
| 20                   | 51.30%                | 66.56% | 35.40%                  | 14.60% | 8.25%  |
| 9.76%                | 56.35%                | 23.96% |                         | 43.65% | 23.9   |
| 6%                   | 81.53%                |        | 11.61%                  |        | 0.625  |
| 0.262                | 16.50                 | 4.54   |                         |        | 1.265  |
| mayor_sensibilidad   |                       |        |                         |        |        |
| contract_code_regime | bar_p50_h90_tp20_sl10 |        | p50_h90 TP 2.0 / SL 1.0 |        |        |
| 20                   | 41.27%                | 59.91% | 27.78%                  | 19.47% | 5.89%  |
| 8.39%                | 66.33%                | 26.72% |                         | 33.67% | 26.7   |
| 2%                   | 83.15%                |        | 13.27%                  |        | 0.497  |
| 0.317                | 51.46                 | 17.87  |                         |        | 1.278  |
| mayor_sensibilidad   |                       |        |                         |        |        |
| contract_code_regime | bar_p50_h90_tp25_sl10 |        | p50_h90 TP 2.5 / SL 1.0 |        |        |

|                      |                       |        |                         |         |        |
|----------------------|-----------------------|--------|-------------------------|---------|--------|
| 20                   | 41.27%                | 59.91% | 15.54%                  | 16.98%  | 6.22%  |
| 8.91%                | 78.24%                | 24.85% |                         | 21.76%  | 24.8   |
| 5%                   | 72.08%                |        | 18.05%                  |         | 0.326  |
| 0.346                | 52.00                 | 15.20  |                         | 1.287   |        |
| mayor_sensibilidad   |                       |        |                         |         |        |
| contract_code_regime | bar_p40_h30_tp25_sl10 |        | p40_h30 TP 2.5 / SL 1.0 |         |        |
| 20                   | 51.30%                | 66.56% | 22.36%                  | 14.89%  | 8.81%  |
| 10.82%               | 68.83%                | 25.27% |                         | 31.17%  | 25.    |
| 27%                  | 72.29%                |        | 12.28%                  |         | 0.47   |
| 1                    | 0.270                 | 16.84  |                         | 4.49    |        |
| 1.298                | mayor_sensibilidad    |        |                         |         |        |
| contract_code_regime | bar_p40_h90_tp20_sl10 |        | p40_h90 TP 2.0 / SL 1.0 |         |        |
| 20                   | 48.54%                | 67.63% | 32.74%                  | 18.00%  | 8.05%  |
| 8.36%                | 59.20%                | 25.98% |                         | 40.80%  | 25.9   |
| 8%                   | 80.74%                |        | 12.39%                  |         | 0.574  |
| 0.280                | 48.82                 | 19.58  |                         | 1.324   |        |
| mayor_sensibilidad   |                       |        |                         |         |        |
| contract_code_regime | bar_p40_h90_tp25_sl10 |        | p40_h90 TP 2.5 / SL 1.0 |         |        |
| 20                   | 48.54%                | 67.63% | 19.42%                  | 16.75%  | 8.57%  |
| 9.19%                | 72.01%                | 25.61% |                         | 27.99%  | 25.6   |
| 1%                   | 69.99%                |        | 15.27%                  |         | 0.400  |
| 0.330                | 49.64                 | 17.68  |                         | 1.344   |        |
| mayor_sensibilidad   |                       |        |                         |         |        |
| contract_code_regime | bar_p50_h30_tp20_sl05 |        | p50_h30 TP 2.0 / SL 0.5 |         |        |
| 20                   | 43.52%                | 58.75% | 26.20%                  | 13.05%  | 22.73% |
| 18.39%               | 51.07%                | 25.07% |                         | 48.93%  | 25.    |
| 07%                  | 53.79%                |        | 22.67%                  |         | 0.41   |
| 0                    | 0.255                 | 14.28  |                         | 5.64    |        |
| 1.379                | mayor_sensibilidad    |        |                         |         |        |
| contract_code_regime | bar_p40_h30_tp20_sl05 |        | p40_h30 TP 2.0 / SL 0.5 |         |        |
| 20                   | 51.30%                | 66.56% | 29.32%                  | 14.35%  | 26.73% |
| 15.99%               | 43.94%                | 24.63% |                         | 56.06%  | 24.    |
| 63%                  | 52.44%                |        | 17.34%                  |         | 0.45   |
| 3                    | 0.286                 | 13.42  |                         | 5.23    |        |
| 1.389                | mayor_sensibilidad    |        |                         |         |        |
| contract_regime      | bar_p60_h60_tp10_sl10 |        | p60_h60 TP 1.0 / SL 1.0 |         |        |
| 130                  | 35.39%                | 80.35% | 97.36%                  | 8.45%   | 2.6    |
| 4%                   | 8.45%                 | 0.00%  | 0.00%                   | 100.00% |        |
| 0.00%                | 97.36%                |        | 8.45%                   |         | 0.     |
| 947                  | 0.169                 | 31.89  |                         | 22.02   |        |
| 1.057                | menor_sensibilidad    |        |                         |         |        |
| contract_regime      | bar_p60_h90_tp10_sl10 |        | p60_h90 TP 1.0 / SL 1.0 |         |        |
| 130                  | 34.62%                | 81.49% | 97.26%                  | 10.30%  | 2.7    |
| 4%                   | 10.30%                | 0.00%  | 0.00%                   | 100.00% |        |
| 0.00%                | 97.26%                |        | 10.30%                  |         | 0.     |
| 945                  | 0.206                 | 47.49  |                         | 35.66   |        |
| 1.124                | menor_sensibilidad    |        |                         |         |        |
| contract_regime      | bar_p60_h30_tp10_sl10 |        | p60_h30 TP 1.0 / SL 1.0 |         |        |
| 130                  | 36.38%                | 80.13% | 97.06%                  | 10.88%  | 2.9    |
| 4%                   | 10.88%                | 0.00%  | 0.00%                   | 100.00% |        |
| 0.00%                | 97.06%                |        | 10.88%                  |         | 0.     |
| 941                  | 0.218                 | 16.31  |                         | 12.16   |        |
| 1.128                | menor_sensibilidad    |        |                         |         |        |
| contract_regime      | bar_p50_h60_tp10_sl10 |        | p50_h60 TP 1.0 / SL 1.0 |         |        |
| 130                  | 42.89%                | 86.75% | 96.20%                  | 11.62%  | 3.8    |
| 0%                   | 11.62%                | 0.00%  | 0.00%                   | 100.00% |        |
| 0.00%                | 96.20%                |        | 11.62%                  |         | 0.     |
| 924                  | 0.232                 | 29.89  |                         | 23.09   |        |
| 1.216                | menor_sensibilidad    |        |                         |         |        |
| contract_regime      | bar_p50_h30_tp10_sl10 |        | p50_h30 TP 1.0 / SL 1.0 |         |        |

|       |                                       |        |                         |         |      |
|-------|---------------------------------------|--------|-------------------------|---------|------|
| 130   | 44.19%                                | 86.98% | 95.81%                  | 13.02%  | 4.1  |
| 9%    | 13.02%                                | 0.00%  | 0.00%                   | 100.00% |      |
| 0.00% |                                       | 95.81% |                         | 13.02%  | 0.   |
| 916   | 0.260                                 | 15.29  |                         | 12.75   |      |
| 1.260 | menor_sensibilidad                    |        |                         |         |      |
|       | contract_regime bar_p50_h90_tp10_sl10 |        | p50_h90 TP 1.0 / SL 1.0 |         |      |
| 130   | 42.01%                                | 89.47% | 96.09%                  | 12.86%  | 3.9  |
| 1%    | 12.86%                                | 0.00%  | 0.00%                   | 100.00% |      |
| 0.00% |                                       | 96.09% |                         | 12.86%  | 0.   |
| 922   | 0.257                                 | 44.21  |                         | 34.76   |      |
| 1.281 | menor_sensibilidad                    |        |                         |         |      |
|       | contract_regime bar_p40_h60_tp10_sl10 |        | p40_h60 TP 1.0 / SL 1.0 |         |      |
| 130   | 50.40%                                | 92.46% | 94.72%                  | 14.58%  | 5.2  |
| 8%    | 14.58%                                | 0.00%  | 0.00%                   | 100.00% |      |
| 0.00% |                                       | 94.72% |                         | 14.58%  | 0.   |
| 894   | 0.292                                 | 27.72  |                         | 23.38   |      |
| 1.362 | menor_sensibilidad                    |        |                         |         |      |
|       | contract_regime bar_p40_h90_tp10_sl10 |        | p40_h90 TP 1.0 / SL 1.0 |         |      |
| 130   | 49.12%                                | 94.65% | 94.51%                  | 15.07%  | 5.4  |
| 9%    | 15.07%                                | 0.00%  | 0.00%                   | 100.00% |      |
| 0.00% |                                       | 94.51% |                         | 15.07%  | 0.   |
| 890   | 0.301                                 | 40.80  |                         | 36.05   |      |
| 1.399 | menor_sensibilidad                    |        |                         |         |      |
|       | contract_regime bar_p40_h30_tp10_sl10 |        | p40_h30 TP 1.0 / SL 1.0 |         |      |
| 130   | 51.84%                                | 91.39% | 94.18%                  | 16.79%  | 5.8  |
| 2%    | 16.79%                                | 0.00%  | 0.00%                   | 100.00% |      |
| 0.00% |                                       | 94.18% |                         | 16.79%  | 0.   |
| 884   | 0.336                                 | 14.13  |                         | 12.10   |      |
| 1.418 | menor_sensibilidad                    |        |                         |         |      |
|       | contract_regime bar_p60_h60_tp10_sl05 |        | p60_h60 TP 1.0 / SL 0.5 |         |      |
| 130   | 35.39%                                | 80.35% | 86.31%                  | 28.35%  | 13.6 |
| 9%    | 28.35%                                | 0.00%  | 0.00%                   | 100.00% |      |
| 0.00% |                                       | 86.31% |                         | 28.35%  | 0.   |
| 795   | 0.425                                 | 28.86  |                         | 26.14   |      |
| 1.654 | menor_sensibilidad                    |        |                         |         |      |
|       | contract_regime bar_p60_h90_tp10_sl05 |        | p60_h90 TP 1.0 / SL 0.5 |         |      |
| 130   | 34.62%                                | 81.49% | 85.76%                  | 29.54%  | 14.2 |
| 4%    | 29.54%                                | 0.00%  | 0.00%                   | 100.00% |      |
| 0.00% |                                       | 85.76% |                         | 29.54%  | 0.   |
| 786   | 0.443                                 | 42.68  |                         | 41.42   |      |
| 1.701 | menor_sensibilidad                    |        |                         |         |      |
|       | contract_regime bar_p60_h30_tp10_sl05 |        | p60_h30 TP 1.0 / SL 0.5 |         |      |
| 130   | 36.38%                                | 80.13% | 86.44%                  | 31.09%  | 13.5 |
| 6%    | 31.09%                                | 0.00%  | 0.00%                   | 100.00% |      |
| 0.00% |                                       | 86.44% |                         | 31.09%  | 0.   |
| 797   | 0.466                                 | 14.83  |                         | 14.56   |      |
| 1.734 | menor_sensibilidad                    |        |                         |         |      |
|       | contract_regime bar_p50_h60_tp10_sl05 |        | p50_h60 TP 1.0 / SL 0.5 |         |      |
| 130   | 42.89%                                | 86.75% | 83.04%                  | 31.02%  | 16.9 |
| 6%    | 31.02%                                | 0.00%  | 0.00%                   | 100.00% |      |
| 0.00% |                                       | 83.04% |                         | 31.02%  | 0.   |
| 746   | 0.465                                 | 26.36  |                         | 26.24   |      |
| 1.798 | menor_sensibilidad                    |        |                         |         |      |
|       | contract_regime bar_p50_h90_tp10_sl05 |        | p50_h90 TP 1.0 / SL 0.5 |         |      |
| 130   | 42.01%                                | 89.47% | 82.45%                  | 31.87%  | 17.5 |
| 5%    | 31.87%                                | 0.00%  | 0.00%                   | 100.00% |      |
| 0.00% |                                       | 82.45% |                         | 31.87%  | 0.   |
| 737   | 0.478                                 | 38.68  |                         | 39.90   |      |
| 1.851 | menor_sensibilidad                    |        |                         |         |      |
|       | contract_regime bar_p40_h60_tp10_sl05 |        | p40_h60 TP 1.0 / SL 0.5 |         |      |

|        |                                                               |        |        |         |      |
|--------|---------------------------------------------------------------|--------|--------|---------|------|
| 130    | 50.40%                                                        | 92.46% | 79.30% | 32.37%  | 20.7 |
| 0%     | 32.37%                                                        | 0.00%  | 0.00%  | 100.00% |      |
| 0.00%  |                                                               | 79.30% |        | 32.37%  | 0.   |
| 689    | 0.486                                                         | 23.69  |        | 25.51   |      |
| 1.896  | menor_sensibilidad                                            |        |        |         |      |
|        | contract_regime bar_p40_h90_tp10_sl05 p40_h90 TP 1.0 / SL 0.5 |        |        |         |      |
| 130    | 49.12%                                                        | 94.65% | 79.01% | 33.18%  | 20.9 |
| 9%     | 33.18%                                                        | 0.00%  | 0.00%  | 100.00% |      |
| 0.00%  |                                                               | 79.01% |        | 33.18%  | 0.   |
| 685    | 0.498                                                         | 34.70  |        | 40.18   |      |
| 1.942  | menor_sensibilidad                                            |        |        |         |      |
|        | contract_regime bar_p50_h30_tp10_sl05 p50_h30 TP 1.0 / SL 0.5 |        |        |         |      |
| 130    | 44.19%                                                        | 86.98% | 83.22% | 37.43%  | 16.7 |
| 8%     | 37.43%                                                        | 0.00%  | 0.00%  | 100.00% |      |
| 0.00%  |                                                               | 83.22% |        | 37.43%  | 0.   |
| 748    | 0.561                                                         | 13.55  |        | 15.01   |      |
| 1.993  | menor_sensibilidad                                            |        |        |         |      |
|        | contract_regime bar_p40_h30_tp10_sl05 p40_h30 TP 1.0 / SL 0.5 |        |        |         |      |
| 130    | 51.84%                                                        | 91.39% | 79.82% | 36.35%  | 20.1 |
| 8%     | 36.35%                                                        | 0.00%  | 0.00%  | 100.00% |      |
| 0.00%  |                                                               | 79.82% |        | 36.35%  | 0.   |
| 697    | 0.545                                                         | 12.18  |        | 12.60   |      |
| 2.004  | menor_sensibilidad                                            |        |        |         |      |
|        | contract_regime bar_p50_h60_tp15_sl10 p50_h60 TP 1.5 / SL 1.0 |        |        |         |      |
| 130    | 42.89%                                                        | 86.75% | 50.07% | 39.44%  | 4.6  |
| 8%     | 13.77%                                                        | 45.25% | 51.12% | 54.75%  |      |
| 51.12% |                                                               | 92.34% |        | 18.56%  |      |
| 0.704  | 0.495                                                         | 33.87  |        | 23.49   |      |
| 2.096  | menor_sensibilidad                                            |        |        |         |      |
|        | contract_regime bar_p40_h60_tp15_sl10 p40_h60 TP 1.5 / SL 1.0 |        |        |         |      |
| 130    | 50.40%                                                        | 92.46% | 55.09% | 35.64%  | 6.4  |
| 8%     | 17.03%                                                        | 38.44% | 46.00% | 61.56%  |      |
| 46.00% |                                                               | 90.32% |        | 20.43%  |      |
| 0.762  | 0.490                                                         | 31.68  |        | 23.88   |      |
| 2.116  | menor_sensibilidad                                            |        |        |         |      |
|        | contract_regime bar_p60_h60_tp15_sl10 p60_h60 TP 1.5 / SL 1.0 |        |        |         |      |
| 130    | 35.39%                                                        | 80.35% | 44.72% | 49.41%  | 3.2  |
| 9%     | 10.96%                                                        | 52.00% | 60.20% | 48.00%  |      |
| 60.20% |                                                               | 93.96% |        | 17.05%  |      |
| 0.638  | 0.661                                                         | 35.89  |        | 26.51   |      |
| 2.180  | menor_sensibilidad                                            |        |        |         |      |
|        | contract_regime bar_p60_h30_tp15_sl10 p60_h30 TP 1.5 / SL 1.0 |        |        |         |      |
| 130    | 36.38%                                                        | 80.13% | 46.95% | 52.27%  | 3.5  |
| 3%     | 12.56%                                                        | 49.52% | 59.35% | 50.48%  |      |
| 59.35% |                                                               | 93.77% |        | 20.00%  |      |
| 0.669  | 0.727                                                         | 18.22  |        | 16.10   |      |
| 2.243  | sensibilidad_intermedia                                       |        |        |         |      |
|        | contract_regime bar_p40_h90_tp15_sl10 p40_h90 TP 1.5 / SL 1.0 |        |        |         |      |
| 130    | 49.12%                                                        | 94.65% | 54.49% | 41.85%  | 6.6  |
| 9%     | 17.37%                                                        | 38.82% | 50.75% | 61.18%  |      |
| 50.75% |                                                               | 89.94% |        | 22.10%  |      |
| 0.750  | 0.565                                                         | 47.41  |        | 42.97   |      |
| 2.267  | sensibilidad_intermedia                                       |        |        |         |      |
|        | contract_regime bar_p50_h30_tp15_sl10 p50_h30 TP 1.5 / SL 1.0 |        |        |         |      |
| 130    | 44.19%                                                        | 86.98% | 52.46% | 52.26%  | 5.0  |
| 7%     | 14.58%                                                        | 42.47% | 53.52% | 57.53%  |      |
| 53.52% |                                                               | 91.99% |        | 25.00%  |      |
| 0.736  | 0.784                                                         | 17.24  |        | 13.93   |      |
| 2.323  | sensibilidad_intermedia                                       |        |        |         |      |
|        | contract_regime bar_p40_h30_tp15_sl10 p40_h30 TP 1.5 / SL 1.0 |        |        |         |      |

|        |                                       |        |                         |        |      |
|--------|---------------------------------------|--------|-------------------------|--------|------|
| 130    | 51.84%                                | 91.39% | 56.59%                  | 43.00% | 6.9  |
| 2%     | 19.25%                                | 36.49% | 52.94%                  | 63.51% |      |
| 52.94% |                                       | 89.86% |                         | 26.32% |      |
| 0.780  | 0.704                                 |        | 16.11                   | 13.53  |      |
| 2.329  | sensibilidad_intermedia               |        |                         |        |      |
|        | contract_regime bar_p40_h60_tp15_sl05 |        | p40_h60 TP 1.5 / SL 0.5 |        |      |
| 130    | 50.40%                                | 92.46% | 45.95%                  | 25.58% | 24.8 |
| 2%     | 37.06%                                | 29.23% | 45.76%                  | 70.77% |      |
| 45.76% |                                       | 65.94% |                         | 37.33% |      |
| 0.565  | 0.436                                 |        | 26.63                   | 24.67  |      |
| 2.382  | sensibilidad_intermedia               |        |                         |        |      |
|        | contract_regime bar_p50_h90_tp15_sl10 |        | p50_h90 TP 1.5 / SL 1.0 |        |      |
| 130    | 42.01%                                | 89.47% | 49.58%                  | 46.84% | 4.8  |
| 2%     | 14.93%                                | 45.60% | 59.53%                  | 54.40% |      |
| 59.53% |                                       | 91.97% |                         | 29.71% |      |
| 0.695  | 0.685                                 |        | 51.05                   | 44.15  |      |
| 2.405  | sensibilidad_intermedia               |        |                         |        |      |
|        | contract_regime bar_p60_h60_tp20_sl10 |        | p60_h60 TP 2.0 / SL 1.0 |        |      |
| 130    | 35.39%                                | 80.35% | 21.39%                  | 40.14% | 3.7  |
| 6%     | 12.99%                                | 74.85% | 52.67%                  | 25.15% |      |
| 52.67% |                                       | 86.96% |                         | 57.41% |      |
| 0.390  | 0.714                                 |        | 37.69                   | 28.64  |      |
| 2.435  | sensibilidad_intermedia               |        |                         |        |      |
|        | contract_regime bar_p40_h90_tp15_sl05 |        | p40_h90 TP 1.5 / SL 0.5 |        |      |
| 130    | 49.12%                                | 94.65% | 45.02%                  | 25.31% | 25.1 |
| 9%     | 35.84%                                | 29.79% | 55.43%                  | 70.21% |      |
| 55.43% |                                       | 65.27% |                         | 33.62% |      |
| 0.549  | 0.381                                 |        | 39.52                   | 47.30  |      |
| 2.449  | sensibilidad_intermedia               |        |                         |        |      |
|        | contract_regime bar_p60_h90_tp15_sl10 |        | p60_h90 TP 1.5 / SL 1.0 |        |      |
| 130    | 34.62%                                | 81.49% | 44.37%                  | 50.13% | 3.4  |
| 6%     | 12.02%                                | 52.18% | 59.62%                  | 47.82% |      |
| 59.62% |                                       | 93.50% |                         | 43.43% |      |
| 0.631  | 0.803                                 |        | 54.16                   | 41.31  |      |
| 2.467  | sensibilidad_intermedia               |        |                         |        |      |
|        | contract_regime bar_p50_h60_tp15_sl05 |        | p50_h60 TP 1.5 / SL 0.5 |        |      |
| 130    | 42.89%                                | 86.75% | 42.75%                  | 27.55% | 20.5 |
| 2%     | 38.65%                                | 36.73% | 53.62%                  | 63.27% |      |
| 53.62% |                                       | 69.06% |                         | 41.58% |      |
| 0.539  | 0.395                                 |        | 29.00                   | 26.36  |      |
| 2.482  | sensibilidad_intermedia               |        |                         |        |      |
|        | contract_regime bar_p40_h60_tp20_sl10 |        | p40_h60 TP 2.0 / SL 1.0 |        |      |
| 130    | 50.40%                                | 92.46% | 32.09%                  | 44.89% | 7.4  |
| 0%     | 19.55%                                | 60.51% | 62.04%                  | 39.49% |      |
| 62.04% |                                       | 83.25% |                         | 31.82% |      |
| 0.568  | 0.728                                 |        | 34.13                   | 28.65  |      |
| 2.508  | sensibilidad_intermedia               |        |                         |        |      |
|        | contract_regime bar_p50_h60_tp20_sl10 |        | p50_h60 TP 2.0 / SL 1.0 |        |      |
| 130    | 42.89%                                | 86.75% | 26.46%                  | 45.19% | 5.3  |
| 3%     | 16.68%                                | 68.21% | 61.78%                  | 31.79% |      |
| 61.78% |                                       | 85.30% |                         | 41.49% |      |
| 0.476  | 0.738                                 |        | 36.03                   | 25.65  |      |
| 2.519  | sensibilidad_intermedia               |        |                         |        |      |
|        | contract_regime bar_p60_h60_tp15_sl05 |        | p60_h60 TP 1.5 / SL 0.5 |        |      |
| 130    | 35.39%                                | 80.35% | 39.03%                  | 38.27% | 16.6 |
| 8%     | 34.60%                                | 44.30% | 66.83%                  | 55.70% |      |
| 66.83% |                                       | 72.02% |                         | 40.68% |      |
| 0.502  | 0.477                                 |        | 31.26                   | 32.52  |      |
| 2.607  | sensibilidad_intermedia               |        |                         |        |      |
|        | contract_regime bar_p60_h60_tp25_sl10 |        | p60_h60 TP 2.5 / SL 1.0 |        |      |



|        |                                       |        |                         |         |      |
|--------|---------------------------------------|--------|-------------------------|---------|------|
| 130    | 35.39%                                | 80.35% | 10.99%                  | 27.37%  | 3.9  |
| 5%     | 14.24%                                | 85.06% | 40.27%                  | 14.94%  |      |
| 40.27% |                                       | 73.88% |                         | 100.00% |      |
| 0.235  | 0.679                                 |        | 36.80                   | 44.30   |      |
| 2.622  | sensibilidad_intermedia               |        |                         |         |      |
|        | contract_regime bar_p50_h30_tp20_sl10 |        | p50_h30 TP 2.0 / SL 1.0 |         |      |
| 130    | 44.19%                                | 86.98% | 29.66%                  | 52.80%  | 5.7  |
| 1%     | 15.95%                                | 64.63% | 63.32%                  | 35.37%  |      |
| 63.32% |                                       | 85.81% |                         | 44.44%  |      |
| 0.536  | 1.056                                 |        | 18.29                   | 15.50   |      |
| 2.635  | sensibilidad_intermedia               |        |                         |         |      |
|        | contract_regime bar_p40_h60_tp20_sl05 |        | p40_h60 TP 2.0 / SL 0.5 |         |      |
| 130    | 50.40%                                | 92.46% | 26.49%                  | 32.67%  | 26.5 |
| 9%     | 42.03%                                | 46.92% | 62.95%                  | 53.08%  |      |
| 62.95% |                                       | 50.64% |                         | 33.97%  |      |
| 0.397  | 0.557                                 |        | 26.92                   | 23.59   |      |
| 2.641  | sensibilidad_intermedia               |        |                         |         |      |
|        | contract_regime bar_p50_h90_tp15_sl05 |        | p50_h90 TP 1.5 / SL 0.5 |         |      |
| 130    | 42.01%                                | 89.47% | 42.10%                  | 35.40%  | 21.1 |
| 9%     | 34.68%                                | 36.71% | 64.90%                  | 63.29%  |      |
| 64.90% |                                       | 68.04% |                         | 41.86%  |      |
| 0.525  | 0.521                                 |        | 43.31                   | 49.69   |      |
| 2.663  | sensibilidad_intermedia               |        |                         |         |      |
|        | contract_regime bar_p40_h30_tp15_sl05 |        | p40_h30 TP 1.5 / SL 0.5 |         |      |
| 130    | 51.84%                                | 91.39% | 47.46%                  | 38.50%  | 23.9 |
| 9%     | 42.12%                                | 28.55% | 48.09%                  | 71.45%  |      |
| 48.09% |                                       | 67.24% |                         | 46.47%  |      |
| 0.592  | 0.615                                 |        | 13.69                   | 13.65   |      |
| 2.666  | sensibilidad_intermedia               |        |                         |         |      |
|        | contract_regime bar_p40_h60_tp25_sl10 |        | p40_h60 TP 2.5 / SL 1.0 |         |      |
| 130    | 50.40%                                | 92.46% | 18.87%                  | 38.26%  | 7.8  |
| 8%     | 21.52%                                | 73.24% | 59.33%                  | 26.76%  |      |
| 59.33% |                                       | 72.67% |                         | 55.80%  |      |
| 0.393  | 0.758                                 |        | 34.81                   | 25.74   |      |
| 2.674  | sensibilidad_intermedia               |        |                         |         |      |
|        | contract_regime bar_p60_h90_tp25_sl10 |        | p60_h90 TP 2.5 / SL 1.0 |         |      |
| 130    | 34.62%                                | 81.49% | 10.65%                  | 28.12%  | 4.1  |
| 1%     | 15.84%                                | 85.24% | 42.17%                  | 14.76%  |      |
| 42.17% |                                       | 71.71% |                         | 100.00% |      |
| 0.225  | 0.705                                 |        | 56.67                   | 43.99   |      |
| 2.676  | sensibilidad_intermedia               |        |                         |         |      |
|        | contract_regime bar_p50_h60_tp25_sl10 |        | p50_h60 TP 2.5 / SL 1.0 |         |      |
| 130    | 42.89%                                | 86.75% | 14.45%                  | 31.99%  | 5.6  |
| 8%     | 18.45%                                | 79.87% | 49.73%                  | 20.13%  |      |
| 49.73% |                                       | 73.58% |                         | 82.20%  |      |
| 0.304  | 0.736                                 |        | 36.27                   | 30.15   |      |
| 2.691  | mayor_sensibilidad                    |        |                         |         |      |
|        | contract_regime bar_p60_h60_tp20_sl05 |        | p60_h60 TP 2.0 / SL 0.5 |         |      |
| 130    | 35.39%                                | 80.35% | 18.48%                  | 31.66%  | 17.6 |
| 5%     | 35.33%                                | 63.87% | 64.86%                  | 36.13%  |      |
| 64.86% |                                       | 52.40% |                         | 59.74%  |      |
| 0.281  | 0.504                                 |        | 29.93                   | 31.39   |      |
| 2.719  | mayor_sensibilidad                    |        |                         |         |      |
|        | contract_regime bar_p60_h30_tp15_sl05 |        | p60_h30 TP 1.5 / SL 0.5 |         |      |
| 130    | 36.38%                                | 80.13% | 41.24%                  | 43.36%  | 16.5 |
| 2%     | 34.07%                                | 42.24% | 62.41%                  | 57.76%  |      |
| 62.41% |                                       | 72.93% |                         | 56.25%  |      |
| 0.536  | 0.716                                 |        | 16.13                   | 16.65   |      |
| 2.762  | mayor_sensibilidad                    |        |                         |         |      |
|        | contract_regime bar_p40_h30_tp20_sl10 |        | p40_h30 TP 2.0 / SL 1.0 |         |      |

|        |                                       |        |                         |        |      |
|--------|---------------------------------------|--------|-------------------------|--------|------|
| 130    | 51.84%                                | 91.39% | 34.45%                  | 54.50% | 7.8  |
| 4%     | 20.21%                                | 57.71% | 64.09%                  | 42.29% |      |
| 64.09% |                                       | 83.18% |                         | 46.15% |      |
| 0.611  | 1.090                                 |        | 17.14                   | 15.75  |      |
| 2.763  | mayor_sensibilidad                    |        |                         |        |      |
|        | contract_regime bar_p50_h60_tp20_sl05 |        | p50_h60 TP 2.0 / SL 0.5 |        |      |
| 130    | 42.89%                                | 86.75% | 22.36%                  | 34.89% | 21.8 |
| 3%     | 40.86%                                | 55.81% | 68.55%                  | 44.19% |      |
| 68.55% |                                       | 51.43% |                         | 46.70% |      |
| 0.338  | 0.555                                 |        | 28.47                   | 21.59  |      |
| 2.778  | mayor_sensibilidad                    |        |                         |        |      |
|        | contract_regime bar_p60_h90_tp20_sl10 |        | p60_h90 TP 2.0 / SL 1.0 |        |      |
| 130    | 34.62%                                | 81.49% | 21.01%                  | 42.91% | 3.9  |
| 2%     | 14.17%                                | 75.07% | 56.36%                  | 24.93% |      |
| 56.36% |                                       | 85.46% |                         | 85.56% |      |
| 0.381  | 0.823                                 |        | 57.00                   | 51.35  |      |
| 2.805  | mayor_sensibilidad                    |        |                         |        |      |
|        | contract_regime bar_p60_h30_tp20_sl10 |        | p60_h30 TP 2.0 / SL 1.0 |        |      |
| 130    | 36.38%                                | 80.13% | 24.56%                  | 61.54% | 3.9  |
| 7%     | 14.29%                                | 71.47% | 61.54%                  | 28.53% |      |
| 61.54% |                                       | 87.63% |                         | 66.67% |      |
| 0.451  | 1.231                                 |        | 19.11                   | 17.42  |      |
| 2.842  | mayor_sensibilidad                    |        |                         |        |      |
|        | contract_regime bar_p50_h90_tp20_sl10 |        | p50_h90 TP 2.0 / SL 1.0 |        |      |
| 130    | 42.01%                                | 89.47% | 25.98%                  | 44.46% | 5.5  |
| 2%     | 17.25%                                | 68.50% | 60.73%                  | 31.50% |      |
| 60.73% |                                       | 84.15% |                         | 72.54% |      |
| 0.464  | 0.852                                 |        | 54.47                   | 40.70  |      |
| 2.845  | mayor_sensibilidad                    |        |                         |        |      |
|        | contract_regime bar_p40_h90_tp20_sl10 |        | p40_h90 TP 2.0 / SL 1.0 |        |      |
| 130    | 49.12%                                | 94.65% | 31.41%                  | 47.17% | 7.6  |
| 7%     | 19.85%                                | 60.93% | 64.62%                  | 39.07% |      |
| 64.62% |                                       | 82.24% |                         | 58.91% |      |
| 0.551  | 0.915                                 |        | 51.35                   | 40.64  |      |
| 2.852  | mayor_sensibilidad                    |        |                         |        |      |
|        | contract_regime bar_p50_h30_tp15_sl05 |        | p50_h30 TP 1.5 / SL 0.5 |        |      |
| 130    | 44.19%                                | 86.98% | 44.98%                  | 50.38% | 20.2 |
| 1%     | 42.76%                                | 34.81% | 55.92%                  | 65.19% |      |
| 55.92% |                                       | 70.19% |                         | 50.27% |      |
| 0.574  | 0.737                                 |        | 14.91                   | 16.45  |      |
| 2.863  | mayor_sensibilidad                    |        |                         |        |      |
|        | contract_regime bar_p40_h30_tp25_sl10 |        | p40_h30 TP 2.5 / SL 1.0 |        |      |
| 130    | 51.84%                                | 91.39% | 21.67%                  | 50.00% | 8.3  |
| 7%     | 21.60%                                | 69.96% | 64.13%                  | 30.04% |      |
| 64.13% |                                       | 74.02% |                         | 60.00% |      |
| 0.458  | 1.250                                 |        | 17.56                   | 17.07  |      |
| 2.871  | mayor_sensibilidad                    |        |                         |        |      |
|        | contract_regime bar_p50_h90_tp25_sl10 |        | p50_h90 TP 2.5 / SL 1.0 |        |      |
| 130    | 42.01%                                | 89.47% | 14.17%                  | 33.47% | 5.8  |
| 2%     | 18.92%                                | 80.01% | 51.74%                  | 19.99% |      |
| 51.74% |                                       | 71.60% |                         | 97.20% |      |
| 0.296  | 0.826                                 |        | 54.78                   | 46.33  |      |
| 2.908  | mayor_sensibilidad                    |        |                         |        |      |
|        | contract_regime bar_p60_h90_tp15_sl05 |        | p60_h90 TP 1.5 / SL 0.5 |        |      |
| 130    | 34.62%                                | 81.49% | 38.71%                  | 42.16% | 17.2 |
| 9%     | 34.92%                                | 44.00% | 66.58%                  | 56.00% |      |
| 66.58% |                                       | 70.86% |                         | 67.33% |      |
| 0.494  | 0.631                                 |        | 46.76                   | 45.04  |      |
| 2.925  | mayor_sensibilidad                    |        |                         |        |      |
|        | contract_regime bar_p40_h90_tp20_sl05 |        | p40_h90 TP 2.0 / SL 0.5 |        |      |

|        |                                       |        |                         |         |       |
|--------|---------------------------------------|--------|-------------------------|---------|-------|
| 130    | 49.12%                                | 94.65% | 25.73%                  | 34.65%  | 26.9  |
| 2%     | 39.60%                                | 47.35% | 72.45%                  | 52.65%  |       |
| 72.45% |                                       | 49.53% |                         | 51.87%  |       |
| 0.380  | 0.640                                 |        | 40.03                   | 40.10   |       |
| 2.932  | mayor_sensibilidad                    |        |                         |         |       |
|        | contract_regime bar_p50_h90_tp20_sl05 |        | p50_h90 TP 2.0 / SL 0.5 |         |       |
| 130    | 42.01%                                | 89.47% | 21.89%                  | 35.44%  | 22.5  |
| 4%     | 39.12%                                | 55.57% | 70.87%                  | 44.43%  |       |
| 70.87% |                                       | 49.90% |                         | 58.98%  |       |
| 0.325  | 0.620                                 |        | 42.55                   | 42.25   |       |
| 2.939  | mayor_sensibilidad                    |        |                         |         |       |
|        | contract_regime bar_p50_h30_tp25_sl10 |        | p50_h30 TP 2.5 / SL 1.0 |         |       |
| 130    | 44.19%                                | 86.98% | 17.39%                  | 57.14%  | 6.1   |
| 0%     | 17.96%                                | 76.51% | 57.14%                  | 23.49%  |       |
| 57.14% |                                       | 76.16% |                         | 80.00%  |       |
| 0.374  | 1.482                                 |        | 18.60                   | 17.20   |       |
| 2.992  | mayor_sensibilidad                    |        |                         |         |       |
|        | contract_regime bar_p40_h90_tp25_sl10 |        | p40_h90 TP 2.5 / SL 1.0 |         |       |
| 130    | 49.12%                                | 94.65% | 18.31%                  | 36.99%  | 8.1   |
| 6%     | 21.95%                                | 73.53% | 57.73%                  | 26.47%  |       |
| 57.73% |                                       | 70.79% |                         | 90.29%  |       |
| 0.376  | 0.834                                 |        | 52.42                   | 44.60   |       |
| 3.016  | mayor_sensibilidad                    |        |                         |         |       |
|        | contract_regime bar_p60_h90_tp20_sl05 |        | p60_h90 TP 2.0 / SL 0.5 |         |       |
| 130    | 34.62%                                | 81.49% | 18.15%                  | 33.44%  | 18.2  |
| 9%     | 38.72%                                | 63.57% | 68.62%                  | 36.43%  |       |
| 68.62% |                                       | 50.23% |                         | 92.24%  |       |
| 0.271  | 0.610                                 |        | 44.69                   | 48.72   |       |
| 3.145  | mayor_sensibilidad                    |        |                         |         |       |
|        | contract_regime bar_p40_h30_tp20_sl05 |        | p40_h30 TP 2.0 / SL 0.5 |         |       |
| 130    | 51.84%                                | 91.39% | 28.54%                  | 44.84%  | 25.8  |
| 8%     | 45.76%                                | 45.59% | 63.36%                  | 54.41%  |       |
| 63.36% |                                       | 52.93% |                         | 70.83%  |       |
| 0.441  | 0.822                                 |        | 13.90                   | 11.62   |       |
| 3.162  | mayor_sensibilidad                    |        |                         |         |       |
|        | contract_regime bar_p60_h30_tp25_sl10 |        | p60_h30 TP 2.5 / SL 1.0 |         |       |
| 130    | 36.38%                                | 80.13% | 13.28%                  | 61.54%  | 4.2   |
| 4%     | 15.28%                                | 82.47% | 61.54%                  | 17.53%  |       |
| 61.54% |                                       | 76.88% |                         | 100.00% |       |
| 0.290  | 1.681                                 |        | 19.32                   | 22.00   |       |
| 3.185  | mayor_sensibilidad                    |        |                         |         |       |
|        | contract_regime bar_p50_h30_tp20_sl05 |        | p50_h30 TP 2.0 / SL 0.5 |         |       |
| 130    | 44.19%                                | 86.98% | 25.24%                  | 52.80%  | 21.5  |
| 4%     | 44.84%                                | 53.22% | 67.20%                  | 46.78%  |       |
| 67.20% |                                       | 54.83% |                         | 71.43%  |       |
| 0.397  | 1.006                                 |        | 14.94                   | 16.37   |       |
| 3.232  | mayor_sensibilidad                    |        |                         |         |       |
|        | contract_regime bar_p60_h30_tp20_sl05 |        | p60_h30 TP 2.0 / SL 0.5 |         |       |
| 130    | 36.38%                                | 80.13% | 21.39%                  | 61.54%  | 17.4  |
| 8%     | 36.79%                                | 61.13% | 71.88%                  | 38.87%  |       |
| 71.88% |                                       | 55.85% |                         | 80.00%  |       |
| 0.340  | 1.231                                 |        | 15.71                   | 17.50   |       |
| 3.303  | mayor_sensibilidad                    |        |                         |         |       |
|        | dataset_split bar_p60_h90_tp10_sl10   |        | p60_h90 TP 1.0 / SL 1.0 |         |       |
| 2      | 43.99%                                | 18.26% | 95.58%                  | 3.50%   | 4.42% |
| 3.50%  | 0.00%                                 | 0.00%  |                         | 100.00% | 0.0   |
| 0%     |                                       | 95.58% |                         | 3.50%   | 0.912 |
| 0.070  | 43.68                                 |        | 8.92                    | 0.288   |       |
|        | menor_sensibilidad                    |        |                         |         |       |
|        | dataset_split bar_p60_h60_tp10_sl10   |        | p60_h60 TP 1.0 / SL 1.0 |         |       |

|                    |               |                       |         |                 |        |
|--------------------|---------------|-----------------------|---------|-----------------|--------|
| 2                  | 46.09%        | 18.95%                | 95.82%  | 3.55%           | 4.18%  |
| 3.55%              | 0.00%         | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 95.82%        |                       | 3.55%   |                 | 0.916  |
| 0.071              | 28.78         | 5.59                  |         | 0.296           |        |
| menor_sensibilidad |               |                       |         |                 |        |
|                    | dataset_split | bar_p60_h30_tp10_sl10 | p60_h30 | TP 1.0 / SL 1.0 |        |
| 2                  | 48.32%        | 19.51%                | 95.61%  | 3.56%           | 4.39%  |
| 3.56%              | 0.00%         | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 95.61%        |                       | 3.56%   |                 | 0.912  |
| 0.071              | 14.53         | 2.71                  |         | 0.302           |        |
| menor_sensibilidad |               |                       |         |                 |        |
|                    | dataset_split | bar_p50_h90_tp10_sl10 | p50_h90 | TP 1.0 / SL 1.0 |        |
| 2                  | 52.27%        | 17.16%                | 94.00%  | 4.63%           | 6.00%  |
| 4.63%              | 0.00%         | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 94.00%        |                       | 4.63%   |                 | 0.880  |
| 0.093              | 40.50         | 9.09                  |         | 0.311           |        |
| menor_sensibilidad |               |                       |         |                 |        |
|                    | dataset_split | bar_p50_h30_tp10_sl10 | p50_h30 | TP 1.0 / SL 1.0 |        |
| 2                  | 57.27%        | 18.43%                | 94.21%  | 4.34%           | 5.79%  |
| 4.34%              | 0.00%         | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 94.21%        |                       | 4.34%   |                 | 0.884  |
| 0.087              | 13.53         | 2.81                  |         | 0.314           |        |
| menor_sensibilidad |               |                       |         |                 |        |
|                    | dataset_split | bar_p40_h30_tp10_sl10 | p40_h30 | TP 1.0 / SL 1.0 |        |
| 2                  | 65.88%        | 16.59%                | 92.53%  | 5.12%           | 7.47%  |
| 5.12%              | 0.00%         | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 92.53%        |                       | 5.12%   |                 | 0.851  |
| 0.102              | 12.47         | 2.87                  |         | 0.320           |        |
| menor_sensibilidad |               |                       |         |                 |        |
|                    | dataset_split | bar_p50_h60_tp10_sl10 | p50_h60 | TP 1.0 / SL 1.0 |        |
| 2                  | 54.74%        | 18.11%                | 94.33%  | 4.67%           | 5.67%  |
| 4.67%              | 0.00%         | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 94.33%        |                       | 4.67%   |                 | 0.887  |
| 0.093              | 26.85         | 5.70                  |         | 0.321           |        |
| menor_sensibilidad |               |                       |         |                 |        |
|                    | dataset_split | bar_p40_h90_tp10_sl10 | p40_h90 | TP 1.0 / SL 1.0 |        |
| 2                  | 60.15%        | 15.64%                | 92.16%  | 5.52%           | 7.84%  |
| 5.52%              | 0.00%         | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 92.16%        |                       | 5.52%   |                 | 0.843  |
| 0.110              | 37.23         | 9.01                  |         | 0.322           |        |
| menor_sensibilidad |               |                       |         |                 |        |
|                    | dataset_split | bar_p40_h60_tp10_sl10 | p40_h60 | TP 1.0 / SL 1.0 |        |
| 2                  | 63.24%        | 16.49%                | 92.62%  | 5.44%           | 7.38%  |
| 5.44%              | 0.00%         | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 92.62%        |                       | 5.44%   |                 | 0.852  |
| 0.109              | 24.75         | 5.67                  |         | 0.328           |        |
| menor_sensibilidad |               |                       |         |                 |        |
|                    | dataset_split | bar_p40_h30_tp10_sl05 | p40_h30 | TP 1.0 / SL 0.5 |        |
| 2                  | 65.88%        | 16.59%                | 77.06%  | 7.50%           | 22.94% |
| 7.50%              | 0.00%         | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 77.06%        |                       | 7.50%   |                 | 0.656  |
| 0.113              | 10.41         | 3.02                  |         | 0.391           |        |
| menor_sensibilidad |               |                       |         |                 |        |
|                    | dataset_split | bar_p50_h30_tp10_sl05 | p50_h30 | TP 1.0 / SL 0.5 |        |
| 2                  | 57.27%        | 18.43%                | 80.06%  | 7.69%           | 19.94% |
| 7.69%              | 0.00%         | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 80.06%        |                       | 7.69%   |                 | 0.701  |
| 0.115              | 11.60         | 3.14                  |         | 0.415           |        |
| menor_sensibilidad |               |                       |         |                 |        |
|                    | dataset_split | bar_p60_h30_tp10_sl05 | p60_h30 | TP 1.0 / SL 0.5 |        |

|                    |                    |                       |         |                 |        |
|--------------------|--------------------|-----------------------|---------|-----------------|--------|
| 2                  | 48.32%             | 19.51%                | 83.02%  | 7.79%           | 16.98% |
| 7.79%              | 0.00%              | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 83.02%             |                       | 7.79%   |                 | 0.745  |
| 0.117              | 12.79              | 3.19                  |         | 0.429           |        |
| menor_sensibilidad |                    |                       |         |                 |        |
|                    | dataset_split      | bar_p40_h60_tp10_sl05 | p40_h60 | TP 1.0 / SL 0.5 |        |
| 2                  | 63.24%             | 16.49%                | 75.68%  | 8.90%           | 24.32% |
| 8.90%              | 0.00%              | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 75.68%             |                       | 8.90%   |                 | 0.635  |
| 0.133              | 20.39              | 6.12                  |         | 0.432           |        |
| menor_sensibilidad |                    |                       |         |                 |        |
|                    | dataset_split      | bar_p40_h90_tp10_sl05 | p40_h90 | TP 1.0 / SL 0.5 |        |
| 2                  | 60.15%             | 15.64%                | 74.70%  | 9.41%           | 25.30% |
| 9.41%              | 0.00%              | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 74.70%             |                       | 9.41%   |                 | 0.621  |
| 0.141              | 30.46              | 9.69                  |         | 0.439           |        |
| menor_sensibilidad |                    |                       |         |                 |        |
|                    | dataset_split      | bar_p50_h60_tp10_sl05 | p50_h60 | TP 1.0 / SL 0.5 |        |
| 2                  | 54.74%             | 18.11%                | 79.13%  | 9.11%           | 20.87% |
| 9.11%              | 0.00%              | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 79.13%             |                       | 9.11%   |                 | 0.687  |
| 0.137              | 22.84              | 6.49                  |         | 0.454           |        |
| menor_sensibilidad |                    |                       |         |                 |        |
|                    | dataset_split      | bar_p50_h90_tp10_sl05 | p50_h90 | TP 1.0 / SL 0.5 |        |
| 2                  | 52.27%             | 17.16%                | 77.94%  | 9.50%           | 22.06% |
| 9.50%              | 0.00%              | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 77.94%             |                       | 9.50%   |                 | 0.669  |
| 0.142              | 34.11              | 10.23                 |         | 0.457           |        |
| menor_sensibilidad |                    |                       |         |                 |        |
|                    | dataset_split      | bar_p60_h90_tp10_sl05 | p60_h90 | TP 1.0 / SL 0.5 |        |
| 2                  | 43.99%             | 18.26%                | 81.13%  | 9.21%           | 18.87% |
| 9.21%              | 0.00%              | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 81.13%             |                       | 9.21%   |                 | 0.717  |
| 0.138              | 37.77              | 10.60                 |         | 0.459           |        |
| menor_sensibilidad |                    |                       |         |                 |        |
|                    | dataset_split      | bar_p60_h60_tp10_sl05 | p60_h60 | TP 1.0 / SL 0.5 |        |
| 2                  | 46.09%             | 18.95%                | 82.13%  | 9.19%           | 17.87% |
| 9.19%              | 0.00%              | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 82.13%             |                       | 9.19%   |                 | 0.732  |
| 0.138              | 25.09              | 6.79                  |         | 0.465           |        |
| menor_sensibilidad |                    |                       |         |                 |        |
|                    | dataset_split      | bar_p40_h30_tp15_sl05 | p40_h30 | TP 1.5 / SL 0.5 |        |
| 2                  | 65.88%             | 16.59%                | 49.22%  | 4.21%           | 27.23% |
| 8.44%              | 23.56%             | 12.65%                | 76.44%  |                 | 12.6   |
| 5%                 | 64.60%             |                       | 5.19%   |                 | 0.602  |
| 0.021              | 11.83              | 2.83                  |         | 0.471           |        |
| menor_sensibilidad |                    |                       |         |                 |        |
|                    | dataset_split      | bar_p40_h60_tp15_sl05 | p40_h60 | TP 1.5 / SL 0.5 |        |
| 2                  | 63.24%             | 16.49%                | 47.21%  | 2.97%           | 28.95% |
| 9.91%              | 23.84%             | 12.88%                | 76.16%  |                 | 12.8   |
| 8%                 | 62.27%             |                       | 6.63%   |                 | 0.563  |
| 0.005              | 23.02              | 6.03                  |         | 0.489           |        |
| menor_sensibilidad |                    |                       |         |                 |        |
|                    | dataset_split      | bar_p40_h90_tp15_sl05 | p40_h90 | TP 1.5 / SL 0.5 |        |
| 2                  | 60.15%             | 15.64%                | 46.24%  | 2.81%           | 29.96% |
| 10.38%             | 23.80%             | 13.19%                | 76.20%  |                 | 13.    |
| 19%                | 60.98%             |                       | 6.86%   |                 | 0.54   |
| 4                  | 0.010              | 34.47                 |         | 9.76            |        |
| 0.489              | menor_sensibilidad |                       |         |                 |        |
|                    | dataset_split      | bar_p40_h30_tp15_sl10 | p40_h30 | TP 1.5 / SL 1.0 |        |

|                       |                                     |        |                         |        |        |
|-----------------------|-------------------------------------|--------|-------------------------|--------|--------|
| 2                     | 65.88%                              | 16.59% | 59.51%                  | 7.77%  | 8.82%  |
| 5.73%                 | 31.67%                              | 13.50% | 68.33%                  |        | 13.5   |
| 0%                    | 87.38%                              |        | 5.90%                   |        | 0.804  |
| 0.059                 | 14.19                               | 2.96   |                         | 0.495  | se     |
| nsibilidad_intermedia |                                     |        |                         |        |        |
|                       | dataset_split bar_p40_h60_tp15_sl10 |        | p40_h60 TP 1.5 / SL 1.0 |        |        |
| 2                     | 63.24%                              | 16.49% | 58.22%                  | 7.57%  | 8.89%  |
| 6.22%                 | 32.88%                              | 13.79% | 67.12%                  |        | 13.7   |
| 9%                    | 87.09%                              |        | 6.62%                   |        | 0.784  |
| 0.051                 | 28.08                               | 6.18   |                         | 0.507  | se     |
| nsibilidad_intermedia |                                     |        |                         |        |        |
|                       | dataset_split bar_p40_h90_tp15_sl10 |        | p40_h90 TP 1.5 / SL 1.0 |        |        |
| 2                     | 60.15%                              | 15.64% | 57.97%                  | 7.79%  | 9.40%  |
| 6.43%                 | 32.64%                              | 14.22% | 67.36%                  |        | 14.2   |
| 2%                    | 86.40%                              |        | 6.68%                   |        | 0.776  |
| 0.053                 | 42.67                               | 9.86   |                         | 0.508  | se     |
| nsibilidad_intermedia |                                     |        |                         |        |        |
|                       | dataset_split bar_p50_h30_tp15_sl10 |        | p50_h30 TP 1.5 / SL 1.0 |        |        |
| 2                     | 57.27%                              | 18.43% | 55.87%                  | 9.37%  | 6.90%  |
| 4.91%                 | 37.23%                              | 14.28% | 62.77%                  |        | 14.2   |
| 8%                    | 89.32%                              |        | 5.39%                   |        | 0.769  |
| 0.092                 | 15.17                               | 2.89   |                         | 0.524  | se     |
| nsibilidad_intermedia |                                     |        |                         |        |        |
|                       | dataset_split bar_p50_h30_tp15_sl05 |        | p50_h30 TP 1.5 / SL 0.5 |        |        |
| 2                     | 57.27%                              | 18.43% | 47.03%                  | 5.34%  | 23.80% |
| 8.80%                 | 29.17%                              | 14.14% | 70.83%                  |        | 14.1   |
| 4%                    | 66.69%                              |        | 5.77%                   |        | 0.586  |
| 0.036                 | 12.82                               | 2.89   |                         | 0.525  | se     |
| nsibilidad_intermedia |                                     |        |                         |        |        |
|                       | dataset_split bar_p60_h90_tp25_sl10 |        | p60_h90 TP 2.5 / SL 1.0 |        |        |
| 2                     | 43.99%                              | 18.26% | 16.66%                  | 10.10% | 6.58%  |
| 5.20%                 | 76.76%                              | 15.30% | 23.24%                  |        | 15.3   |
| 0%                    | 72.38%                              |        | 4.19%                   |        | 0.351  |
| 0.201                 | 52.54                               | 6.73   |                         | 0.531  | se     |
| nsibilidad_intermedia |                                     |        |                         |        |        |
|                       | dataset_split bar_p40_h30_tp20_sl05 |        | p40_h30 TP 2.0 / SL 0.5 |        |        |
| 2                     | 65.88%                              | 16.59% | 31.71%                  | 7.94%  | 29.40% |
| 9.61%                 | 38.89%                              | 17.55% | 61.11%                  |        | 17.5   |
| 5%                    | 52.04%                              |        | 1.96%                   |        | 0.487  |
| 0.111                 | 12.23                               | 2.36   |                         | 0.537  | se     |
| nsibilidad_intermedia |                                     |        |                         |        |        |
|                       | dataset_split bar_p60_h30_tp15_sl10 |        | p60_h30 TP 1.5 / SL 1.0 |        |        |
| 2                     | 48.32%                              | 19.51% | 51.94%                  | 10.70% | 5.31%  |
| 4.19%                 | 42.74%                              | 14.89% | 57.26%                  |        | 14.8   |
| 9%                    | 91.05%                              |        | 4.99%                   |        | 0.726  |
| 0.119                 | 16.15                               | 2.73   |                         | 0.543  | se     |
| nsibilidad_intermedia |                                     |        |                         |        |        |
|                       | dataset_split bar_p50_h90_tp15_sl10 |        | p50_h90 TP 1.5 / SL 1.0 |        |        |
| 2                     | 52.27%                              | 17.16% | 54.86%                  | 10.28% | 7.23%  |
| 5.40%                 | 37.91%                              | 15.68% | 62.09%                  |        | 15.6   |
| 8%                    | 88.72%                              |        | 5.85%                   |        | 0.751  |
| 0.100                 | 45.95                               | 9.79   |                         | 0.544  | se     |
| nsibilidad_intermedia |                                     |        |                         |        |        |
|                       | dataset_split bar_p50_h60_tp15_sl10 |        | p50_h60 TP 1.5 / SL 1.0 |        |        |
| 2                     | 54.74%                              | 18.11% | 54.60%                  | 9.80%  | 6.86%  |
| 5.47%                 | 38.55%                              | 15.28% | 61.45%                  |        | 15.2   |
| 8%                    | 89.23%                              |        | 6.23%                   |        | 0.750  |
| 0.092                 | 30.13                               | 6.03   |                         | 0.549  | se     |
| nsibilidad_intermedia |                                     |        |                         |        |        |
|                       | dataset_split bar_p60_h30_tp25_sl10 |        | p60_h30 TP 2.5 / SL 1.0 |        |        |

|                       |                                     |        |                         |        |        |
|-----------------------|-------------------------------------|--------|-------------------------|--------|--------|
| 2                     | 48.32%                              | 19.51% | 17.40%                  | 10.37% | 6.37%  |
| 5.14%                 | 76.22%                              | 15.51% |                         | 23.78% | 15.5   |
| 1%                    | 73.96%                              |        |                         | 4.65%  | 0.371  |
| 0.208                 | 17.16                               | 2.13   |                         | 0.552  | se     |
| nsibilidad_intermedia |                                     |        |                         |        |        |
|                       | dataset_split bar_p60_h60_tp25_sl10 |        | p60_h60 TP 2.5 / SL 1.0 |        |        |
| 2                     | 46.09%                              | 18.95% | 16.56%                  | 10.37% | 6.15%  |
| 5.26%                 | 77.29%                              | 15.63% |                         | 22.71% | 15.6   |
| 3%                    | 73.81%                              |        |                         | 5.13%  | 0.353  |
| 0.207                 | 33.84                               | 3.98   |                         | 0.553  | se     |
| nsibilidad_intermedia |                                     |        |                         |        |        |
|                       | dataset_split bar_p50_h90_tp15_sl05 |        | p50_h90 TP 1.5 / SL 0.5 |        |        |
| 2                     | 52.27%                              | 17.16% | 44.75%                  | 4.77%  | 26.32% |
| 10.84%                | 28.93%                              | 15.61% |                         | 71.07% | 15.    |
| 61%                   | 63.36%                              |        |                         | 7.21%  | 0.54   |
| 0                     | 0.017                               | 37.76  |                         | 9.99   |        |
| 0.556                 | sensibilidad_intermedia             |        |                         |        |        |
|                       | dataset_split bar_p60_h90_tp15_sl10 |        | p60_h90 TP 1.5 / SL 1.0 |        |        |
| 2                     | 43.99%                              | 18.26% | 51.48%                  | 12.06% | 5.47%  |
| 4.24%                 | 43.06%                              | 16.30% |                         | 56.94% | 16.3   |
| 0%                    | 90.74%                              |        |                         | 4.80%  | 0.717  |
| 0.138                 | 49.14                               | 9.02   |                         | 0.557  | se     |
| nsibilidad_intermedia |                                     |        |                         |        |        |
|                       | dataset_split bar_p50_h60_tp15_sl05 |        | p50_h60 TP 1.5 / SL 0.5 |        |        |
| 2                     | 54.74%                              | 18.11% | 45.13%                  | 4.68%  | 25.06% |
| 10.56%                | 29.81%                              | 15.24% |                         | 70.19% | 15.    |
| 24%                   | 64.70%                              |        |                         | 7.38%  | 0.55   |
| 2                     | 0.017                               | 25.06  |                         | 6.20   |        |
| 0.560                 | sensibilidad_intermedia             |        |                         |        |        |
|                       | dataset_split bar_p60_h60_tp15_sl10 |        | p60_h60 TP 1.5 / SL 1.0 |        |        |
| 2                     | 46.09%                              | 18.95% | 50.91%                  | 11.75% | 5.09%  |
| 4.23%                 | 44.00%                              | 15.98% |                         | 56.00% | 15.9   |
| 8%                    | 91.27%                              |        |                         | 5.07%  | 0.713  |
| 0.134                 | 32.03                               | 5.72   |                         | 0.560  | se     |
| nsibilidad_intermedia |                                     |        |                         |        |        |
|                       | dataset_split bar_p40_h90_tp20_sl05 |        | p40_h90 TP 2.0 / SL 0.5 |        |        |
| 2                     | 60.15%                              | 15.64% | 29.33%                  | 7.38%  | 32.15% |
| 11.57%                | 38.52%                              | 18.95% |                         | 61.48% | 18.    |
| 95%                   | 47.92%                              |        |                         | 2.77%  | 0.42   |
| 6                     | 0.090                               | 35.49  |                         | 8.26   |        |
| 0.563                 | sensibilidad_intermedia             |        |                         |        |        |
|                       | dataset_split bar_p40_h60_tp20_sl05 |        | p40_h60 TP 2.0 / SL 0.5 |        |        |
| 2                     | 63.24%                              | 16.49% | 29.72%                  | 7.38%  | 31.16% |
| 11.23%                | 39.12%                              | 18.61% |                         | 60.88% | 18.    |
| 61%                   | 49.03%                              |        |                         | 2.87%  | 0.43   |
| 9                     | 0.091                               | 23.60  |                         | 5.02   |        |
| 0.566                 | sensibilidad_intermedia             |        |                         |        |        |
|                       | dataset_split bar_p60_h30_tp15_sl05 |        | p60_h30 TP 1.5 / SL 0.5 |        |        |
| 2                     | 48.32%                              | 19.51% | 44.53%                  | 6.48%  | 20.49% |
| 9.15%                 | 34.98%                              | 15.63% |                         | 65.02% | 15.6   |
| 3%                    | 68.88%                              |        |                         | 6.59%  | 0.566  |
| 0.052                 | 13.84                               | 2.88   |                         | 0.574  | se     |
| nsibilidad_intermedia |                                     |        |                         |        |        |
|                       | dataset_split bar_p50_h30_tp25_sl10 |        | p50_h30 TP 2.5 / SL 1.0 |        |        |
| 2                     | 57.27%                              | 18.43% | 21.34%                  | 11.17% | 8.41%  |
| 6.22%                 | 70.25%                              | 17.39% |                         | 29.75% | 17.3   |
| 9%                    | 72.43%                              |        |                         | 4.80%  | 0.449  |
| 0.217                 | 16.43                               | 2.37   |                         | 0.580  | se     |
| nsibilidad_intermedia |                                     |        |                         |        |        |
|                       | dataset_split bar_p50_h30_tp20_sl05 |        | p50_h30 TP 2.0 / SL 0.5 |        |        |

|        |                    |                       |         |                 |        |
|--------|--------------------|-----------------------|---------|-----------------|--------|
| 2      | 57.27%             | 18.43%                | 28.41%  | 8.68%           | 25.63% |
| 10.05% | 45.96%             | 18.73%                | 54.04%  |                 | 18.    |
| 73%    | 52.77%             |                       | 2.23%   |                 | 0.44   |
| 0      | 0.123              | 13.01                 |         | 2.25            |        |
| 0.581  | mayor_sensibilidad |                       |         |                 |        |
|        | dataset_split      | bar_p50_h90_tp25_sl10 | p50_h90 | TP 2.5 / SL 1.0 |        |
| 2      | 52.27%             | 17.16%                | 20.44%  | 11.55%          | 8.79%  |
| 6.73%  | 70.77%             | 18.28%                |         | 29.23%          | 18.2   |
| 8%     | 70.65%             |                       | 4.66%   |                 | 0.423  |
| 0.222  | 50.07              | 7.19                  |         |                 | 0.584  |
|        | mayor_sensibilidad |                       |         |                 |        |
|        | dataset_split      | bar_p40_h30_tp20_sl10 | p40_h30 | TP 2.0 / SL 1.0 |        |
| 2      | 65.88%             | 16.59%                | 38.57%  | 11.48%          | 10.01% |
| 6.49%  | 51.43%             | 17.98%                |         | 48.57%          | 17.9   |
| 8%     | 79.95%             |                       | 5.95%   |                 | 0.671  |
| 0.165  | 15.14              | 2.88                  |         |                 | 0.585  |
|        | mayor_sensibilidad |                       |         |                 |        |
|        | dataset_split      | bar_p40_h30_tp25_sl10 | p40_h30 | TP 2.5 / SL 1.0 |        |
| 2      | 65.88%             | 16.59%                | 25.57%  | 11.66%          | 10.75% |
| 7.15%  | 63.67%             | 18.81%                |         | 36.33%          | 18.8   |
| 1%     | 71.01%             |                       | 4.67%   |                 | 0.532  |
| 0.220  | 15.59              | 2.58                  |         |                 | 0.589  |
|        | mayor_sensibilidad |                       |         |                 |        |
|        | dataset_split      | bar_p60_h90_tp20_sl10 | p60_h90 | TP 2.0 / SL 1.0 |        |
| 2      | 43.99%             | 18.26%                | 28.83%  | 13.11%          | 6.21%  |
| 4.86%  | 64.97%             | 17.96%                |         | 35.03%          | 17.9   |
| 6%     | 82.93%             |                       | 5.11%   |                 | 0.514  |
| 0.214  | 51.95              | 7.57                  |         |                 | 0.593  |
|        | mayor_sensibilidad |                       |         |                 |        |
|        | dataset_split      | bar_p60_h30_tp20_sl10 | p60_h30 | TP 2.0 / SL 1.0 |        |
| 2      | 48.32%             | 19.51%                | 29.57%  | 12.37%          | 6.03%  |
| 4.82%  | 64.40%             | 17.18%                |         | 35.60%          | 17.1   |
| 8%     | 83.75%             |                       | 5.68%   |                 | 0.531  |
| 0.199  | 16.96              | 2.47                  |         |                 | 0.596  |
|        | mayor_sensibilidad |                       |         |                 |        |
|        | dataset_split      | bar_p50_h30_tp20_sl10 | p50_h30 | TP 2.0 / SL 1.0 |        |
| 2      | 57.27%             | 18.43%                | 34.03%  | 12.19%          | 7.87%  |
| 5.69%  | 58.10%             | 17.88%                |         | 41.90%          | 17.8   |
| 8%     | 81.84%             |                       | 5.83%   |                 | 0.602  |
| 0.187  | 16.09              | 2.71                  |         |                 | 0.600  |
|        | mayor_sensibilidad |                       |         |                 |        |
|        | dataset_split      | bar_p50_h60_tp25_sl10 | p50_h60 | TP 2.5 / SL 1.0 |        |
| 2      | 54.74%             | 18.11%                | 20.16%  | 11.76%          | 8.33%  |
| 6.84%  | 71.50%             | 18.60%                |         | 28.50%          | 18.6   |
| 0%     | 71.65%             |                       | 5.50%   |                 | 0.421  |
| 0.226  | 32.47              | 4.36                  |         |                 | 0.608  |
|        | mayor_sensibilidad |                       |         |                 |        |
|        | dataset_split      | bar_p40_h90_tp25_sl10 | p40_h90 | TP 2.5 / SL 1.0 |        |
| 2      | 60.15%             | 15.64%                | 24.25%  | 12.03%          | 11.49% |
| 8.10%  | 64.26%             | 20.13%                |         | 35.74%          | 20.1   |
| 3%     | 68.54%             |                       | 4.94%   |                 | 0.491  |
| 0.220  | 47.26              | 8.09                  |         |                 | 0.608  |
|        | mayor_sensibilidad |                       |         |                 |        |
|        | dataset_split      | bar_p60_h60_tp20_sl10 | p60_h60 | TP 2.0 / SL 1.0 |        |
| 2      | 46.09%             | 18.95%                | 28.45%  | 13.23%          | 5.81%  |
| 4.88%  | 65.73%             | 18.11%                |         | 34.27%          | 18.1   |
| 1%     | 83.78%             |                       | 5.68%   |                 | 0.511  |
| 0.216  | 33.61              | 4.84                  |         |                 | 0.609  |
|        | mayor_sensibilidad |                       |         |                 |        |
|        | dataset_split      | bar_p60_h90_tp15_sl05 | p60_h90 | TP 1.5 / SL 0.5 |        |



|        |                                     |                         |        |        |        |
|--------|-------------------------------------|-------------------------|--------|--------|--------|
| 2      | 43.99%                              | 18.26%                  | 43.06% | 6.65%  | 22.63% |
| 10.90% | 34.31%                              | 17.54%                  | 65.69% | 17.    |        |
| 54%    | 66.06%                              |                         | 7.52%  | 0.53   |        |
| 3      | 0.045                               | 41.08                   | 9.55   |        |        |
| 0.609  | mayor_sensibilidad                  |                         |        |        |        |
|        | dataset_split bar_p60_h30_tp20_sl05 | p60_h30 TP 2.0 / SL 0.5 |        |        |        |
| 2      | 48.32%                              | 19.51%                  | 25.20% | 9.23%  | 21.91% |
| 10.25% | 52.89%                              | 19.49%                  | 47.11% | 19.    |        |
| 49%    | 53.76%                              |                         | 2.64%  | 0.39   |        |
| 4      | 0.133                               | 13.79                   | 2.09   |        |        |
| 0.611  | mayor_sensibilidad                  |                         |        |        |        |
|        | dataset_split bar_p40_h90_tp20_sl10 | p40_h90 TP 2.0 / SL 1.0 |        |        |        |
| 2      | 60.15%                              | 15.64%                  | 37.20% | 12.20% | 10.70% |
| 7.33%  | 52.11%                              | 19.53%                  | 47.89% | 19.5   |        |
| 3%     | 78.33%                              |                         | 6.47%  | 0.637  |        |
| 0.171  | 45.88                               | 9.55                    |        | 0.612  |        |
|        | mayor_sensibilidad                  |                         |        |        |        |
|        | dataset_split bar_p40_h60_tp20_sl10 | p40_h60 TP 2.0 / SL 1.0 |        |        |        |
| 2      | 63.24%                              | 16.49%                  | 37.08% | 11.80% | 10.13% |
| 7.17%  | 52.79%                              | 18.97%                  | 47.21% | 18.9   |        |
| 7%     | 79.24%                              |                         | 6.84%  | 0.640  |        |
| 0.164  | 30.02                               | 5.98                    |        | 0.613  |        |
|        | mayor_sensibilidad                  |                         |        |        |        |
|        | dataset_split bar_p40_h60_tp25_sl10 | p40_h60 TP 2.5 / SL 1.0 |        |        |        |
| 2      | 63.24%                              | 16.49%                  | 24.14% | 12.14% | 10.83% |
| 7.83%  | 65.03%                              | 19.98%                  | 34.97% | 19.9   |        |
| 8%     | 69.76%                              |                         | 5.12%  | 0.495  |        |
| 0.225  | 30.81                               | 5.15                    |        | 0.616  |        |
|        | mayor_sensibilidad                  |                         |        |        |        |
|        | dataset_split bar_p50_h90_tp20_sl05 | p50_h90 TP 2.0 / SL 0.5 |        |        |        |
| 2      | 52.27%                              | 17.16%                  | 26.62% | 8.70%  | 28.22% |
| 12.31% | 45.16%                              | 21.01%                  | 54.84% | 21.    |        |
| 01%    | 48.81%                              |                         | 2.83%  | 0.39   |        |
| 1      | 0.113                               | 38.13                   | 7.38   |        |        |
| 0.620  | mayor_sensibilidad                  |                         |        |        |        |
|        | dataset_split bar_p60_h60_tp15_sl05 | p60_h60 TP 1.5 / SL 0.5 |        |        |        |
| 2      | 46.09%                              | 18.95%                  | 42.95% | 6.72%  | 21.54% |
| 10.91% | 35.50%                              | 17.64%                  | 64.50% | 17.    |        |
| 64%    | 67.14%                              |                         | 7.94%  | 0.53   |        |
| 7      | 0.046                               | 27.00                   | 6.09   |        |        |
| 0.622  | mayor_sensibilidad                  |                         |        |        |        |
|        | dataset_split bar_p50_h60_tp20_sl05 | p50_h60 TP 2.0 / SL 0.5 |        |        |        |
| 2      | 54.74%                              | 18.11%                  | 26.72% | 8.89%  | 26.82% |
| 11.78% | 46.46%                              | 20.67%                  | 53.54% | 20.    |        |
| 67%    | 50.17%                              |                         | 2.76%  | 0.40   |        |
| 0      | 0.119                               | 25.20                   | 4.68   |        |        |
| 0.622  | mayor_sensibilidad                  |                         |        |        |        |
|        | dataset_split bar_p50_h90_tp20_sl10 | p50_h90 TP 2.0 / SL 1.0 |        |        |        |
| 2      | 52.27%                              | 17.16%                  | 32.98% | 13.25% | 8.28%  |
| 6.27%  | 58.74%                              | 19.53%                  | 41.26% | 19.5   |        |
| 3%     | 80.64%                              |                         | 6.04%  | 0.577  |        |
| 0.202  | 48.91                               | 8.82                    |        | 0.623  |        |
|        | mayor_sensibilidad                  |                         |        |        |        |
|        | dataset_split bar_p50_h60_tp20_sl10 | p50_h60 TP 2.0 / SL 1.0 |        |        |        |
| 2      | 54.74%                              | 18.11%                  | 32.64% | 13.10% | 7.81%  |
| 6.30%  | 59.55%                              | 19.40%                  | 40.45% | 19.4   |        |
| 0%     | 81.50%                              |                         | 6.70%  | 0.575  |        |
| 0.199  | 31.92                               | 5.56                    |        | 0.636  |        |
|        | mayor_sensibilidad                  |                         |        |        |        |
|        | dataset_split bar_p60_h90_tp20_sl05 | p60_h90 TP 2.0 / SL 0.5 |        |        |        |

|        |                    |                       |         |                 |        |
|--------|--------------------|-----------------------|---------|-----------------|--------|
| 2      | 43.99%             | 18.26%                | 23.85%  | 9.04%           | 24.15% |
| 12.16% | 52.00%             | 21.20%                | 48.00%  | 21.             |        |
| 20%    | 50.04%             |                       | 3.26%   | 0.35            |        |
| 6      | 0.120              | 40.78                 | 6.45    |                 |        |
| 0.639  | mayor_sensibilidad |                       |         |                 |        |
|        | dataset_split      | bar_p60_h60_tp20_sl05 | p60_h60 | TP 2.0 / SL 0.5 |        |
| 2      | 46.09%             | 18.95%                | 23.77%  | 9.62%           | 22.94% |
| 11.97% | 53.29%             | 21.59%                | 46.71%  | 21.             |        |
| 59%    | 51.25%             |                       | 3.10%   | 0.36            |        |
| 1      | 0.133              | 26.62                 | 4.09    |                 |        |
| 0.652  | mayor_sensibilidad |                       |         |                 |        |
|        | year               | bar_p60_h90_tp10_sl10 | p60_h90 | TP 1.0 / SL 1.0 |        |
| 7      | 41.19%             | 38.54%                | 96.39%  | 5.52%           | 3.61%  |
| 5.52%  | 0.00%              | 0.00%                 | 100.00% | 0.0             |        |
| 0%     | 96.39%             |                       | 5.52%   | 0.928           |        |
| 0.110  | 45.86              | 16.67                 | 0.551   |                 |        |
|        | menor_sensibilidad |                       |         |                 |        |
|        | year               | bar_p60_h60_tp10_sl10 | p60_h60 | TP 1.0 / SL 1.0 |        |
| 7      | 43.17%             | 40.07%                | 96.65%  | 5.57%           | 3.35%  |
| 5.57%  | 0.00%              | 0.00%                 | 100.00% | 0.0             |        |
| 0%     | 96.65%             |                       | 5.57%   | 0.933           |        |
| 0.111  | 30.15              | 10.31                 | 0.568   |                 |        |
|        | menor_sensibilidad |                       |         |                 |        |
|        | year               | bar_p60_h30_tp10_sl10 | p60_h30 | TP 1.0 / SL 1.0 |        |
| 7      | 45.24%             | 40.22%                | 96.42%  | 5.76%           | 3.58%  |
| 5.76%  | 0.00%              | 0.00%                 | 100.00% | 0.0             |        |
| 0%     | 96.42%             |                       | 5.76%   | 0.928           |        |
| 0.115  | 15.18              | 5.08                  | 0.575   |                 |        |
|        | menor_sensibilidad |                       |         |                 |        |
|        | year               | bar_p50_h90_tp10_sl10 | p50_h90 | TP 1.0 / SL 1.0 |        |
| 7      | 49.60%             | 37.14%                | 95.04%  | 7.81%           | 4.96%  |
| 7.81%  | 0.00%              | 0.00%                 | 100.00% | 0.0             |        |
| 0%     | 95.04%             |                       | 7.81%   | 0.901           |        |
| 0.156  | 42.64              | 17.44                 | 0.606   |                 |        |
|        | menor_sensibilidad |                       |         |                 |        |
|        | year               | bar_p50_h30_tp10_sl10 | p50_h30 | TP 1.0 / SL 1.0 |        |
| 7      | 54.33%             | 39.37%                | 95.15%  | 7.74%           | 4.85%  |
| 7.74%  | 0.00%              | 0.00%                 | 100.00% | 0.0             |        |
| 0%     | 95.15%             |                       | 7.74%   | 0.903           |        |
| 0.155  | 14.18              | 5.40                  | 0.626   |                 |        |
|        | menor_sensibilidad |                       |         |                 |        |
|        | year               | bar_p50_h60_tp10_sl10 | p50_h60 | TP 1.0 / SL 1.0 |        |
| 7      | 51.87%             | 38.80%                | 95.36%  | 7.94%           | 4.64%  |
| 7.94%  | 0.00%              | 0.00%                 | 100.00% | 0.0             |        |
| 0%     | 95.36%             |                       | 7.94%   | 0.907           |        |
| 0.159  | 28.18              | 11.00                 | 0.626   |                 |        |
|        | menor_sensibilidad |                       |         |                 |        |
|        | year               | bar_p40_h90_tp10_sl10 | p40_h90 | TP 1.0 / SL 1.0 |        |
| 7      | 57.63%             | 34.23%                | 93.36%  | 9.94%           | 6.64%  |
| 9.94%  | 0.00%              | 0.00%                 | 100.00% | 0.0             |        |
| 0%     | 93.36%             |                       | 9.94%   | 0.867           |        |
| 0.199  | 39.25              | 17.86                 | 0.640   |                 |        |
|        | menor_sensibilidad |                       |         |                 |        |
|        | year               | bar_p40_h30_tp10_sl10 | p40_h30 | TP 1.0 / SL 1.0 |        |
| 7      | 63.17%             | 36.43%                | 93.59%  | 9.72%           | 6.41%  |
| 9.72%  | 0.00%              | 0.00%                 | 100.00% | 0.0             |        |
| 0%     | 93.59%             |                       | 9.72%   | 0.872           |        |
| 0.194  | 13.10              | 5.61                  | 0.656   |                 |        |
|        | menor_sensibilidad |                       |         |                 |        |
|        | year               | bar_p40_h60_tp10_sl10 | p40_h60 | TP 1.0 / SL 1.0 |        |

|                    |                            |        |                         |        |        |
|--------------------|----------------------------|--------|-------------------------|--------|--------|
| 7                  | 60.55%                     | 36.00% | 93.76%                  | 9.99%  | 6.24%  |
| 9.99%              | 0.00%                      | 0.00%  | 100.00%                 |        | 0.0    |
| 0%                 | 93.76%                     |        | 9.99%                   |        | 0.875  |
| 0.200              | 26.00                      | 11.42  |                         | 0.660  |        |
| menor_sensibilidad |                            |        |                         |        |        |
|                    | year bar_p40_h30_tp10_sl05 |        | p40_h30 TP 1.0 / SL 0.5 |        |        |
| 7                  | 63.17%                     | 36.43% | 78.74%                  | 15.61% | 21.26% |
| 15.61%             | 0.00%                      | 0.00%  | 100.00%                 |        | 0.     |
| 00%                | 78.74%                     |        | 15.61%                  |        | 0.68   |
| 1                  | 0.234                      | 11.09  |                         | 6.06   |        |
| 0.833              | menor_sensibilidad         |        |                         |        |        |
|                    | year bar_p60_h30_tp10_sl05 |        | p60_h30 TP 1.0 / SL 0.5 |        |        |
| 7                  | 45.24%                     | 40.22% | 84.90%                  | 15.47% | 15.10% |
| 15.47%             | 0.00%                      | 0.00%  | 100.00%                 |        | 0.     |
| 00%                | 84.90%                     |        | 15.47%                  |        | 0.77   |
| 3                  | 0.232                      | 13.57  |                         | 6.25   |        |
| 0.866              | menor_sensibilidad         |        |                         |        |        |
|                    | year bar_p50_h30_tp10_sl05 |        | p50_h30 TP 1.0 / SL 0.5 |        |        |
| 7                  | 54.33%                     | 39.37% | 81.86%                  | 15.77% | 18.14% |
| 15.77%             | 0.00%                      | 0.00%  | 100.00%                 |        | 0.     |
| 00%                | 81.86%                     |        | 15.77%                  |        | 0.72   |
| 8                  | 0.237                      | 12.35  |                         | 6.25   |        |
| 0.867              | menor_sensibilidad         |        |                         |        |        |
|                    | year bar_p60_h90_tp10_sl05 |        | p60_h90 TP 1.0 / SL 0.5 |        |        |
| 7                  | 41.19%                     | 38.54% | 83.43%                  | 17.15% | 16.57% |
| 17.15%             | 0.00%                      | 0.00%  | 100.00%                 |        | 0.     |
| 00%                | 83.43%                     |        | 17.15%                  |        | 0.75   |
| 1                  | 0.257                      | 40.43  |                         | 20.64  |        |
| 0.900              | menor_sensibilidad         |        |                         |        |        |
|                    | year bar_p40_h60_tp10_sl05 |        | p40_h60 TP 1.0 / SL 0.5 |        |        |
| 7                  | 60.55%                     | 36.00% | 77.68%                  | 18.64% | 22.32% |
| 18.64%             | 0.00%                      | 0.00%  | 100.00%                 |        | 0.     |
| 00%                | 77.68%                     |        | 18.64%                  |        | 0.66   |
| 5                  | 0.280                      | 21.80  |                         | 12.64  |        |
| 0.919              | menor_sensibilidad         |        |                         |        |        |
|                    | year bar_p40_h90_tp10_sl05 |        | p40_h90 TP 1.0 / SL 0.5 |        |        |
| 7                  | 57.63%                     | 34.23% | 76.84%                  | 19.57% | 23.16% |
| 19.57%             | 0.00%                      | 0.00%  | 100.00%                 |        | 0.     |
| 00%                | 76.84%                     |        | 19.57%                  |        | 0.65   |
| 3                  | 0.294                      | 32.70  |                         | 20.17  |        |
| 0.929              | menor_sensibilidad         |        |                         |        |        |
|                    | year bar_p60_h60_tp10_sl05 |        | p60_h60 TP 1.0 / SL 0.5 |        |        |
| 7                  | 43.17%                     | 40.07% | 84.38%                  | 17.85% | 15.62% |
| 17.85%             | 0.00%                      | 0.00%  | 100.00%                 |        | 0.     |
| 00%                | 84.38%                     |        | 17.85%                  |        | 0.76   |
| 6                  | 0.268                      | 26.78  |                         | 13.27  |        |
| 0.936              | menor_sensibilidad         |        |                         |        |        |
|                    | year bar_p50_h90_tp10_sl05 |        | p50_h90 TP 1.0 / SL 0.5 |        |        |
| 7                  | 49.60%                     | 37.14% | 80.19%                  | 18.91% | 19.81% |
| 18.91%             | 0.00%                      | 0.00%  | 100.00%                 |        | 0.     |
| 00%                | 80.19%                     |        | 18.91%                  |        | 0.70   |
| 3                  | 0.284                      | 36.58  |                         | 20.72  |        |
| 0.939              | menor_sensibilidad         |        |                         |        |        |
|                    | year bar_p50_h60_tp10_sl05 |        | p50_h60 TP 1.0 / SL 0.5 |        |        |
| 7                  | 51.87%                     | 38.80% | 81.25%                  | 18.76% | 18.75% |
| 18.76%             | 0.00%                      | 0.00%  | 100.00%                 |        | 0.     |
| 00%                | 81.25%                     |        | 18.76%                  |        | 0.71   |
| 9                  | 0.281                      | 24.39  |                         | 13.24  |        |
| 0.951              | menor_sensibilidad         |        |                         |        |        |
|                    | year bar_p60_h90_tp25_sl10 |        | p60_h90 TP 2.5 / SL 1.0 |        |        |

|                       |                            |        |                         |        |        |
|-----------------------|----------------------------|--------|-------------------------|--------|--------|
| 7                     | 41.19%                     | 38.54% | 14.05%                  | 14.42% | 5.43%  |
| 8.36%                 | 80.52%                     | 22.78% |                         | 19.48% | 22.7   |
| 8%                    | 73.29%                     |        |                         | 12.15% | 0.297  |
| 0.288                 | 54.56                      | 10.69  |                         |        | 0.962  |
| menor_sensibilidad    |                            |        |                         |        |        |
|                       | year bar_p60_h60_tp25_sl10 |        | p60_h60 TP 2.5 / SL 1.0 |        |        |
| 7                     | 43.17%                     | 40.07% | 13.95%                  | 15.35% | 5.01%  |
| 8.62%                 | 81.04%                     | 23.97% |                         | 18.96% | 23.9   |
| 7%                    | 75.12%                     |        |                         | 12.90% | 0.299  |
| 0.298                 | 34.96                      | 6.08   |                         |        | 1.009  |
| menor_sensibilidad    |                            |        |                         |        |        |
|                       | year bar_p60_h30_tp25_sl10 |        | p60_h30 TP 2.5 / SL 1.0 |        |        |
| 7                     | 45.24%                     | 40.22% | 14.97%                  | 17.74% | 5.19%  |
| 8.44%                 | 79.84%                     | 26.18% |                         | 20.16% | 26.1   |
| 8%                    | 75.58%                     |        |                         | 9.72%  | 0.322  |
| 0.359                 | 17.76                      | 2.80   |                         |        | 1.023  |
| menor_sensibilidad    |                            |        |                         |        |        |
|                       | year bar_p40_h30_tp15_sl05 |        | p40_h30 TP 1.5 / SL 0.5 |        |        |
| 7                     | 63.17%                     | 36.43% | 48.11%                  | 9.91%  | 25.29% |
| 17.73%                | 26.59%                     | 27.64% |                         | 73.41% | 27.    |
| 64%                   | 66.01%                     |        |                         | 10.95% | 0.59   |
| 5                     | 0.066                      | 12.53  |                         | 5.46   |        |
| 1.027                 | sensibilidad_intermedia    |        |                         |        |        |
|                       | year bar_p60_h90_tp15_sl10 |        | p60_h90 TP 1.5 / SL 1.0 |        |        |
| 7                     | 41.19%                     | 38.54% | 48.16%                  | 22.35% | 4.50%  |
| 6.78%                 | 47.34%                     | 29.12% |                         | 52.66% | 29.1   |
| 2%                    | 91.95%                     |        |                         | 7.65%  | 0.677  |
| 0.267                 | 51.56                      | 14.19  |                         |        | 1.044  |
| nsibilidad_intermedia |                            |        |                         |        |        |
|                       | year bar_p40_h30_tp15_sl10 |        | p40_h30 TP 1.5 / SL 1.0 |        |        |
| 7                     | 63.17%                     | 36.43% | 57.61%                  | 17.95% | 7.62%  |
| 11.00%                | 34.77%                     | 28.95% |                         | 65.23% | 28.    |
| 95%                   | 88.86%                     |        |                         | 11.27% | 0.78   |
| 8                     | 0.159                      | 14.92  |                         | 5.53   |        |
| 1.056                 | sensibilidad_intermedia    |        |                         |        |        |
|                       | year bar_p60_h90_tp20_sl10 |        | p60_h90 TP 2.0 / SL 1.0 |        |        |
| 7                     | 41.19%                     | 38.54% | 25.40%                  | 20.93% | 5.12%  |
| 7.87%                 | 69.47%                     | 28.80% |                         | 30.53% | 28.8   |
| 0%                    | 84.21%                     |        |                         | 9.50%  | 0.457  |
| 0.340                 | 54.07                      | 11.64  |                         |        | 1.057  |
| nsibilidad_intermedia |                            |        |                         |        |        |
|                       | year bar_p40_h90_tp15_sl10 |        | p40_h90 TP 1.5 / SL 1.0 |        |        |
| 7                     | 57.63%                     | 34.23% | 56.03%                  | 18.65% | 8.00%  |
| 11.55%                | 35.97%                     | 30.20% |                         | 64.03% | 30.    |
| 20%                   | 88.10%                     |        |                         | 11.65% | 0.76   |
| 0                     | 0.166                      | 45.16  |                         | 18.20  |        |
| 1.063                 | sensibilidad_intermedia    |        |                         |        |        |
|                       | year bar_p60_h30_tp15_sl10 |        | p60_h30 TP 1.5 / SL 1.0 |        |        |
| 7                     | 45.24%                     | 40.22% | 49.21%                  | 22.78% | 4.35%  |
| 6.90%                 | 46.44%                     | 29.68% |                         | 53.56% | 29.6   |
| 8%                    | 92.38%                     |        |                         | 8.08%  | 0.695  |
| 0.273                 | 16.88                      | 4.60   |                         |        | 1.077  |
| nsibilidad_intermedia |                            |        |                         |        |        |
|                       | year bar_p40_h90_tp15_sl05 |        | p40_h90 TP 1.5 / SL 0.5 |        |        |
| 7                     | 57.63%                     | 34.23% | 45.42%                  | 8.14%  | 27.55% |
| 21.74%                | 27.03%                     | 29.43% |                         | 72.97% | 29.    |
| 43%                   | 62.87%                     |        |                         | 14.35% | 0.54   |
| 3                     | 0.079                      | 36.95  |                         | 19.11  |        |
| 1.079                 | sensibilidad_intermedia    |        |                         |        |        |
|                       | year bar_p50_h90_tp25_sl10 |        | p50_h90 TP 2.5 / SL 1.0 |        |        |

|        |                            |         |                 |        |        |
|--------|----------------------------|---------|-----------------|--------|--------|
| 7      | 49.60%                     | 37.14%  | 17.62%          | 18.60% | 7.34%  |
| 11.53% | 75.04%                     | 30.13%  | 24.96%          | 30.    |        |
| 13%    | 71.87%                     |         | 10.88%          | 0.36   |        |
| 7      | 0.350                      | 52.17   | 11.70           |        |        |
| 1.083  | sensibilidad_intermedia    |         |                 |        |        |
|        | year bar_p50_h90_tp15_sl10 | p50_h90 | TP 1.5 / SL 1.0 |        |        |
| 7      | 49.60%                     | 37.14%  | 52.21%          | 21.66% | 6.03%  |
| 9.09%  | 41.76%                     | 30.75%  | 58.24%          | 30.7   |        |
| 5%     | 90.22%                     |         | 9.67%           | 0.723  |        |
| 0.234  | 48.51                      | 16.97   | 1.083           | se     |        |
|        | nsibilidad_intermedia      |         |                 |        |        |
|        | year bar_p40_h60_tp15_sl05 | p40_h60 | TP 1.5 / SL 0.5 |        |        |
| 7      | 60.55%                     | 36.00%  | 46.36%          | 8.49%  | 26.67% |
| 20.93% | 26.97%                     | 29.43%  | 73.03%          | 29.    |        |
| 43%    | 64.09%                     |         | 13.69%          | 0.56   |        |
| 2      | 0.052                      | 24.54   | 11.76           |        |        |
| 1.085  | sensibilidad_intermedia    |         |                 |        |        |
|        | year bar_p50_h30_tp15_sl10 | p50_h30 | TP 1.5 / SL 1.0 |        |        |
| 7      | 54.33%                     | 39.37%  | 53.54%          | 20.94% | 5.82%  |
| 8.86%  | 40.64%                     | 29.81%  | 59.36%          | 29.8   |        |
| 1%     | 90.72%                     |         | 9.62%           | 0.745  |        |
| 0.226  | 15.92                      | 5.15    | 1.086           | se     |        |
|        | nsibilidad_intermedia      |         |                 |        |        |
|        | year bar_p40_h60_tp15_sl10 | p40_h60 | TP 1.5 / SL 1.0 |        |        |
| 7      | 60.55%                     | 36.00%  | 56.37%          | 18.95% | 7.58%  |
| 11.68% | 36.05%                     | 30.63%  | 63.95%          | 30.    |        |
| 63%    | 88.76%                     |         | 12.14%          | 0.77   |        |
| 0      | 0.167                      | 29.63   | 11.45           |        |        |
| 1.094  | sensibilidad_intermedia    |         |                 |        |        |
|        | year bar_p60_h60_tp20_sl10 | p60_h60 | TP 2.0 / SL 1.0 |        |        |
| 7      | 43.17%                     | 40.07%  | 25.06%          | 22.61% | 4.74%  |
| 8.05%  | 70.21%                     | 30.66%  | 29.79%          | 30.6   |        |
| 6%     | 85.29%                     |         | 10.14%          | 0.454  |        |
| 0.372  | 34.90                      | 6.62    | 1.115           | se     |        |
|        | nsibilidad_intermedia      |         |                 |        |        |
|        | year bar_p60_h60_tp15_sl10 | p60_h60 | TP 1.5 / SL 1.0 |        |        |
| 7      | 43.17%                     | 40.07%  | 47.84%          | 25.21% | 4.12%  |
| 6.80%  | 48.03%                     | 32.00%  | 51.97%          | 32.0   |        |
| 0%     | 92.61%                     |         | 7.88%           | 0.676  |        |
| 0.310  | 33.55                      | 8.80    | 1.120           | se     |        |
|        | nsibilidad_intermedia      |         |                 |        |        |
|        | year bar_p60_h30_tp20_sl10 | p60_h30 | TP 2.0 / SL 1.0 |        |        |
| 7      | 45.24%                     | 40.22%  | 26.61%          | 23.16% | 4.91%  |
| 7.86%  | 68.48%                     | 31.02%  | 31.52%          | 31.0   |        |
| 2%     | 85.51%                     |         | 9.73%           | 0.483  |        |
| 0.385  | 17.65                      | 3.71    | 1.120           | se     |        |
|        | nsibilidad_intermedia      |         |                 |        |        |
|        | year bar_p50_h30_tp25_sl10 | p50_h30 | TP 2.5 / SL 1.0 |        |        |
| 7      | 54.33%                     | 39.37%  | 18.83%          | 20.62% | 7.05%  |
| 11.13% | 74.12%                     | 31.75%  | 25.88%          | 31.    |        |
| 75%    | 74.09%                     |         | 10.13%          | 0.40   |        |
| 0      | 0.404                      | 17.10   | 3.63            |        |        |
| 1.130  | sensibilidad_intermedia    |         |                 |        |        |
|        | year bar_p50_h30_tp15_sl05 | p50_h30 | TP 1.5 / SL 0.5 |        |        |
| 7      | 54.33%                     | 39.37%  | 45.59%          | 12.29% | 21.70% |
| 18.39% | 32.70%                     | 30.69%  | 67.30%          | 30.    |        |
| 69%    | 68.36%                     |         | 12.52%          | 0.57   |        |
| 5      | 0.092                      | 13.57   | 5.41            |        |        |
| 1.133  | sensibilidad_intermedia    |         |                 |        |        |
|        | year bar_p50_h60_tp15_sl10 | p50_h60 | TP 1.5 / SL 1.0 |        |        |

|                       |                            |        |                         |        |        |
|-----------------------|----------------------------|--------|-------------------------|--------|--------|
| 7                     | 51.87%                     | 38.80% | 52.15%                  | 22.97% | 5.64%  |
| 9.39%                 | 42.21%                     | 32.36% |                         | 57.79% | 32.3   |
| 6%                    | 90.87%                     |        | 10.37%                  |        | 0.726  |
| 0.251                 | 31.70                      | 10.16  |                         | 1.139  | se     |
| nsibilidad_intermedia |                            |        |                         |        |        |
|                       | year bar_p40_h30_tp20_sl05 |        | p40_h30 TP 2.0 / SL 0.5 |        |        |
| 7                     | 63.17%                     | 36.43% | 29.86%                  | 16.86% | 27.22% |
| 20.19%                | 42.92%                     | 37.05% |                         | 57.08% | 37.    |
| 05%                   | 52.63%                     |        | 4.24%                   |        | 0.46   |
| 1                     | 0.236                      | 12.84  |                         | 4.19   |        |
| 1.148                 | sensibilidad_intermedia    |        |                         |        |        |
|                       | year bar_p50_h60_tp25_sl10 |        | p50_h60 TP 2.5 / SL 1.0 |        |        |
| 7                     | 51.87%                     | 38.80% | 17.35%                  | 19.69% | 6.89%  |
| 12.07%                | 75.76%                     | 31.77% |                         | 24.24% | 31.    |
| 77%                   | 73.24%                     |        | 12.49%                  |        | 0.36   |
| 5                     | 0.372                      | 33.67  |                         | 6.43   |        |
| 1.148                 | sensibilidad_intermedia    |        |                         |        |        |
|                       | year bar_p50_h90_tp20_sl10 |        | p50_h90 TP 2.0 / SL 1.0 |        |        |
| 7                     | 49.60%                     | 37.14% | 29.68%                  | 23.68% | 6.92%  |
| 10.78%                | 63.39%                     | 34.46% |                         | 36.61% | 34.    |
| 46%                   | 82.25%                     |        | 11.01%                  |        | 0.52   |
| 4                     | 0.366                      | 51.36  |                         | 13.39  |        |
| 1.171                 | mayor_sensibilidad         |        |                         |        |        |
|                       | year bar_p50_h90_tp15_sl05 |        | p50_h90 TP 1.5 / SL 0.5 |        |        |
| 7                     | 49.60%                     | 37.14% | 43.41%                  | 10.81% | 23.75% |
| 21.84%                | 32.85%                     | 32.65% |                         | 67.15% | 32.    |
| 65%                   | 65.42%                     |        | 15.18%                  |        | 0.53   |
| 2                     | 0.066                      | 40.44  |                         | 18.64  |        |
| 1.176                 | mayor_sensibilidad         |        |                         |        |        |
|                       | year bar_p40_h90_tp25_sl10 |        | p40_h90 TP 2.5 / SL 1.0 |        |        |
| 7                     | 57.63%                     | 34.23% | 21.48%                  | 21.51% | 9.81%  |
| 14.91%                | 68.71%                     | 36.42% |                         | 31.29% | 36.    |
| 42%                   | 69.90%                     |        | 10.99%                  |        | 0.43   |
| 9                     | 0.389                      | 49.57  |                         | 12.99  |        |
| 1.181                 | mayor_sensibilidad         |        |                         |        |        |
|                       | year bar_p50_h30_tp20_sl10 |        | p50_h30 TP 2.0 / SL 1.0 |        |        |
| 7                     | 54.33%                     | 39.37% | 31.21%                  | 24.81% | 6.61%  |
| 10.15%                | 62.17%                     | 34.96% |                         | 37.83% | 34.    |
| 96%                   | 83.58%                     |        | 10.64%                  |        | 0.55   |
| 8                     | 0.395                      | 16.84  |                         | 4.45   |        |
| 1.199                 | mayor_sensibilidad         |        |                         |        |        |
|                       | year bar_p60_h30_tp15_sl05 |        | p60_h30 TP 1.5 / SL 0.5 |        |        |
| 7                     | 45.24%                     | 40.22% | 42.75%                  | 14.37% | 18.28% |
| 18.46%                | 38.97%                     | 32.83% |                         | 61.03% | 32.    |
| 83%                   | 70.82%                     |        | 14.04%                  |        | 0.55   |
| 0                     | 0.123                      | 14.63  |                         | 5.23   |        |
| 1.199                 | mayor_sensibilidad         |        |                         |        |        |
|                       | year bar_p40_h30_tp25_sl10 |        | p40_h30 TP 2.5 / SL 1.0 |        |        |
| 7                     | 63.17%                     | 36.43% | 23.08%                  | 23.48% | 9.26%  |
| 13.61%                | 67.66%                     | 37.08% |                         | 32.34% | 37.    |
| 08%                   | 72.54%                     |        | 9.80%                   |        | 0.48   |
| 5                     | 0.451                      | 16.31  |                         | 4.19   |        |
| 1.204                 | mayor_sensibilidad         |        |                         |        |        |
|                       | year bar_p40_h90_tp20_sl05 |        | p40_h90 TP 2.0 / SL 0.5 |        |        |
| 7                     | 57.63%                     | 34.23% | 27.55%                  | 15.51% | 29.52% |
| 24.41%                | 42.93%                     | 39.91% |                         | 57.07% | 39.    |
| 91%                   | 48.75%                     |        | 6.64%                   |        | 0.40   |
| 3                     | 0.188                      | 37.67  |                         | 14.71  |        |
| 1.207                 | mayor_sensibilidad         |        |                         |        |        |
|                       | year bar_p50_h30_tp20_sl05 |        | p50_h30 TP 2.0 / SL 0.5 |        |        |

|        |                            |                         |        |        |        |
|--------|----------------------------|-------------------------|--------|--------|--------|
| 7      | 54.33%                     | 39.37%                  | 26.34% | 17.84% | 23.25% |
| 20.54% | 50.41%                     | 38.38%                  | 49.59% | 38.    |        |
| 38%    | 53.55%                     |                         | 5.33%  | 0.41   |        |
| 0      | 0.254                      | 13.61                   | 3.87   |        |        |
| 1.214  | mayor_sensibilidad         |                         |        |        |        |
|        | year bar_p50_h60_tp15_sl05 | p50_h60 TP 1.5 / SL 0.5 |        |        |        |
| 7      | 51.87%                     | 38.80%                  | 43.82% | 12.09% | 22.54% |
| 21.69% | 33.64%                     | 33.78%                  | 66.36% | 33.    |        |
| 78%    | 66.85%                     |                         | 15.26% | 0.54   |        |
| 5      | 0.073                      | 26.69                   | 11.41  |        |        |
| 1.216  | mayor_sensibilidad         |                         |        |        |        |
|        | year bar_p40_h60_tp20_sl05 | p40_h60 TP 2.0 / SL 0.5 |        |        |        |
| 7      | 60.55%                     | 36.00%                  | 27.97% | 16.96% | 28.59% |
| 23.38% | 43.43%                     | 40.34%                  | 56.57% | 40.    |        |
| 34%    | 49.88%                     |                         | 5.10%  | 0.41   |        |
| 7      | 0.222                      | 24.87                   | 8.70   |        |        |
| 1.218  | mayor_sensibilidad         |                         |        |        |        |
|        | year bar_p40_h30_tp20_sl10 | p40_h30 TP 2.0 / SL 1.0 |        |        |        |
| 7      | 63.17%                     | 36.43%                  | 35.99% | 24.36% | 8.65%  |
| 12.54% | 55.36%                     | 36.90%                  | 44.64% | 36.    |        |
| 90%    | 81.65%                     |                         | 11.86% | 0.63   |        |
| 3      | 0.362                      | 15.91                   | 5.07   |        |        |
| 1.221  | mayor_sensibilidad         |                         |        |        |        |
|        | year bar_p60_h90_tp15_sl05 | p60_h90 TP 1.5 / SL 0.5 |        |        |        |
| 7      | 41.19%                     | 38.54%                  | 41.12% | 13.17% | 19.98% |
| 20.83% | 38.90%                     | 34.00%                  | 61.10% | 34.    |        |
| 00%    | 68.21%                     |                         | 15.91% | 0.51   |        |
| 7      | 0.093                      | 43.77                   | 16.58  |        |        |
| 1.225  | mayor_sensibilidad         |                         |        |        |        |
|        | year bar_p40_h90_tp20_sl10 | p40_h90 TP 2.0 / SL 1.0 |        |        |        |
| 7      | 57.63%                     | 34.23%                  | 34.36% | 24.96% | 9.15%  |
| 13.43% | 56.50%                     | 38.39%                  | 43.50% | 38.    |        |
| 39%    | 80.10%                     |                         | 12.01% | 0.59   |        |
| 6      | 0.365                      | 48.49                   | 15.94  |        |        |
| 1.230  | mayor_sensibilidad         |                         |        |        |        |
|        | year bar_p60_h30_tp20_sl05 | p60_h30 TP 2.0 / SL 0.5 |        |        |        |
| 7      | 45.24%                     | 40.22%                  | 22.93% | 17.46% | 19.45% |
| 20.36% | 57.62%                     | 37.83%                  | 42.38% | 37.    |        |
| 83%    | 54.79%                     |                         | 7.54%  | 0.36   |        |
| 1      | 0.247                      | 14.38                   | 3.54   |        |        |
| 1.234  | mayor_sensibilidad         |                         |        |        |        |
|        | year bar_p40_h60_tp25_sl10 | p40_h60 TP 2.5 / SL 1.0 |        |        |        |
| 7      | 60.55%                     | 36.00%                  | 21.43% | 23.18% | 9.22%  |
| 14.84% | 69.35%                     | 38.02%                  | 30.65% | 38.    |        |
| 02%    | 71.31%                     |                         | 11.45% | 0.44   |        |
| 4      | 0.431                      | 32.19                   | 8.02   |        |        |
| 1.235  | mayor_sensibilidad         |                         |        |        |        |
|        | year bar_p50_h60_tp20_sl10 | p50_h60 TP 2.0 / SL 1.0 |        |        |        |
| 7      | 51.87%                     | 38.80%                  | 29.55% | 25.94% | 6.44%  |
| 11.04% | 64.01%                     | 36.98%                  | 35.99% | 36.    |        |
| 98%    | 83.43%                     |                         | 12.11% | 0.52   |        |
| 7      | 0.408                      | 33.44                   | 8.40   |        |        |
| 1.249  | mayor_sensibilidad         |                         |        |        |        |
|        | year bar_p60_h90_tp20_sl05 | p60_h90 TP 2.0 / SL 0.5 |        |        |        |
| 7      | 41.19%                     | 38.54%                  | 21.42% | 14.81% | 21.28% |
| 23.21% | 57.30%                     | 38.02%                  | 42.70% | 38.    |        |
| 02%    | 50.99%                     |                         | 10.33% | 0.32   |        |
| 2      | 0.180                      | 42.73                   | 11.16  |        |        |
| 1.249  | mayor_sensibilidad         |                         |        |        |        |
|        | year bar_p50_h90_tp20_sl05 | p50_h90 TP 2.0 / SL 0.5 |        |        |        |

|        |                                     |                         |        |         |        |
|--------|-------------------------------------|-------------------------|--------|---------|--------|
| 7      | 49.60%                              | 37.14%                  | 24.40% | 15.77%  | 25.40% |
| 25.06% | 50.20%                              | 40.83%                  |        | 49.80%  | 40.    |
| 83%    | 49.64%                              |                         |        | 8.42%   | 0.36   |
| 1      | 0.190                               | 40.24                   |        | 12.44   |        |
| 1.272  | mayor_sensibilidad                  |                         |        |         |        |
|        | year_bar_p40_h60_tp20_sl10          | p40_h60 TP 2.0 / SL 1.0 |        |         |        |
| 7      | 60.55%                              | 36.00%                  | 34.42% | 26.14%  | 8.61%  |
| 13.42% | 56.97%                              | 39.56%                  |        | 43.03%  | 39.    |
| 56%    | 81.20%                              |                         |        | 12.71%  | 0.60   |
| 2      | 0.389                               | 31.59                   |        | 9.94    |        |
| 1.278  | mayor_sensibilidad                  |                         |        |         |        |
|        | year_bar_p60_h60_tp15_sl05          | p60_h60 TP 1.5 / SL 0.5 |        |         |        |
| 7      | 43.17%                              | 40.07%                  | 41.08% | 15.51%  | 18.86% |
| 20.93% | 40.07%                              | 36.44%                  |        | 59.93%  | 36.    |
| 44%    | 69.55%                              |                         |        | 15.76%  | 0.52   |
| 2      | 0.128                               | 28.66                   |        | 10.67   |        |
| 1.287  | mayor_sensibilidad                  |                         |        |         |        |
|        | year_bar_p60_h60_tp20_sl05          | p60_h60 TP 2.0 / SL 0.5 |        |         |        |
| 7      | 43.17%                              | 40.07%                  | 21.26% | 16.40%  | 20.07% |
| 23.18% | 58.67%                              | 39.58%                  |        | 41.33%  | 39.    |
| 58%    | 52.33%                              |                         |        | 9.61%   | 0.32   |
| 5      | 0.212                               | 27.75                   |        | 6.92    |        |
| 1.288  | mayor_sensibilidad                  |                         |        |         |        |
|        | year_bar_p50_h60_tp20_sl05          | p50_h60 TP 2.0 / SL 0.5 |        |         |        |
| 7      | 51.87%                              | 38.80%                  | 24.57% | 18.04%  | 24.05% |
| 23.94% | 51.39%                              | 41.98%                  |        | 48.61%  | 41.    |
| 98%    | 51.14%                              |                         |        | 6.64%   | 0.37   |
| 1      | 0.241                               | 26.45                   |        | 7.93    |        |
| 1.294  | mayor_sensibilidad                  |                         |        |         |        |
|        | year_contract_bar_p60_h90_tp10_sl10 | p60_h90 TP 1.0 / SL 1.0 |        |         |        |
| 32     | 37.28%                              | 59.14%                  | 97.13% | 7.95%   | 2.87%  |
| 7.95%  | 0.00%                               | 0.00%                   |        | 100.00% | 0.0    |
| 0%     | 97.13%                              |                         |        | 7.95%   | 0.943  |
| 0.159  | 47.76                               | 21.72                   |        |         | 0.830  |
|        | menor_sensibilidad                  |                         |        |         |        |
|        | year_contract_bar_p60_h60_tp10_sl10 | p60_h60 TP 1.0 / SL 1.0 |        |         |        |
| 32     | 39.02%                              | 61.54%                  | 97.51% | 7.31%   | 2.49%  |
| 7.31%  | 0.00%                               | 0.00%                   |        | 100.00% | 0.0    |
| 0%     | 97.51%                              |                         |        | 7.31%   | 0.950  |
| 0.146  | 31.20                               | 13.01                   |        |         | 0.835  |
|        | menor_sensibilidad                  |                         |        |         |        |
|        | year_contract_bar_p60_h30_tp10_sl10 | p60_h30 TP 1.0 / SL 1.0 |        |         |        |
| 32     | 41.17%                              | 61.91%                  | 97.16% | 7.53%   | 2.84%  |
| 7.53%  | 0.00%                               | 0.00%                   |        | 100.00% | 0.0    |
| 0%     | 97.16%                              |                         |        | 7.53%   | 0.943  |
| 0.151  | 15.82                               | 6.51                    |        |         | 0.845  |
|        | menor_sensibilidad                  |                         |        |         |        |
|        | year_contract_bar_p50_h30_tp10_sl10 | p50_h30 TP 1.0 / SL 1.0 |        |         |        |
| 32     | 50.21%                              | 64.11%                  | 96.01% | 9.74%   | 3.99%  |
| 9.74%  | 0.00%                               | 0.00%                   |        | 100.00% | 0.0    |
| 0%     | 96.01%                              |                         |        | 9.74%   | 0.920  |
| 0.195  | 14.84                               | 6.82                    |        |         | 0.933  |
|        | menor_sensibilidad                  |                         |        |         |        |
|        | year_contract_bar_p50_h90_tp10_sl10 | p50_h90 TP 1.0 / SL 1.0 |        |         |        |
| 32     | 45.59%                              | 61.77%                  | 96.02% | 10.60%  | 3.98%  |
| 10.60% | 0.00%                               | 0.00%                   |        | 100.00% | 0.     |
| 00%    | 96.02%                              |                         |        | 10.60%  | 0.92   |
| 0      | 0.212                               | 44.61                   |        | 22.71   |        |
| 0.936  | menor_sensibilidad                  |                         |        |         |        |
|        | year_contract_bar_p50_h60_tp10_sl10 | p50_h60 TP 1.0 / SL 1.0 |        |         |        |



|        |                    |                       |                         |        |        |
|--------|--------------------|-----------------------|-------------------------|--------|--------|
| 32     | 47.79%             | 64.73%                | 96.44%                  | 10.09% | 3.56%  |
| 10.09% | 0.00%              | 0.00%                 | 100.00%                 |        | 0.     |
| 00%    | 96.44%             |                       | 10.09%                  |        | 0.92   |
| 9      | 0.202              | 29.42                 | 13.85                   |        |        |
| 0.950  | menor_sensibilidad |                       |                         |        |        |
|        | year_contract      | bar_p40_h30_tp10_sl10 | p40_h30 TP 1.0 / SL 1.0 |        |        |
| 32     | 59.22%             | 63.21%                | 94.56%                  | 11.11% | 5.44%  |
| 11.11% | 0.00%              | 0.00%                 | 100.00%                 |        | 0.     |
| 00%    | 94.56%             |                       | 11.11%                  |        | 0.89   |
| 1      | 0.222              | 13.75                 | 7.10                    |        |        |
| 0.965  | menor_sensibilidad |                       |                         |        |        |
|        | year_contract      | bar_p40_h60_tp10_sl10 | p40_h60 TP 1.0 / SL 1.0 |        |        |
| 32     | 56.71%             | 64.10%                | 94.91%                  | 12.12% | 5.09%  |
| 12.12% | 0.00%              | 0.00%                 | 100.00%                 |        | 0.     |
| 00%    | 94.91%             |                       | 12.12%                  |        | 0.89   |
| 8      | 0.242              | 27.31                 | 14.48                   |        |        |
| 1.005  | menor_sensibilidad |                       |                         |        |        |
|        | year_contract      | bar_p40_h90_tp10_sl10 | p40_h90 TP 1.0 / SL 1.0 |        |        |
| 32     | 53.87%             | 61.51%                | 94.63%                  | 13.20% | 5.37%  |
| 13.20% | 0.00%              | 0.00%                 | 100.00%                 |        | 0.     |
| 00%    | 94.63%             |                       | 13.20%                  |        | 0.89   |
| 3      | 0.264              | 41.40                 | 25.38                   |        |        |
| 1.011  | menor_sensibilidad |                       |                         |        |        |
|        | year_contract      | bar_p60_h30_tp10_sl05 | p60_h30 TP 1.0 / SL 0.5 |        |        |
| 32     | 41.17%             | 61.91%                | 86.67%                  | 18.48% | 13.33% |
| 18.48% | 0.00%              | 0.00%                 | 100.00%                 |        | 0.     |
| 00%    | 86.67%             |                       | 18.48%                  |        | 0.80   |
| 0      | 0.277              | 14.35                 | 8.05                    |        |        |
| 1.173  | menor_sensibilidad |                       |                         |        |        |
|        | year_contract      | bar_p50_h30_tp10_sl05 | p50_h30 TP 1.0 / SL 0.5 |        |        |
| 32     | 50.21%             | 64.11%                | 83.63%                  | 19.44% | 16.37% |
| 19.44% | 0.00%              | 0.00%                 | 100.00%                 |        | 0.     |
| 00%    | 83.63%             |                       | 19.44%                  |        | 0.75   |
| 4      | 0.292              | 13.11                 | 8.24                    |        |        |
| 1.224  | menor_sensibilidad |                       |                         |        |        |
|        | year_contract      | bar_p40_h30_tp10_sl05 | p40_h30 TP 1.0 / SL 0.5 |        |        |
| 32     | 59.22%             | 63.21%                | 80.61%                  | 22.33% | 19.39% |
| 22.33% | 0.00%              | 0.00%                 | 100.00%                 |        | 0.     |
| 00%    | 80.61%             |                       | 22.33%                  |        | 0.70   |
| 9      | 0.335              | 11.84                 | 8.38                    |        |        |
| 1.302  | menor_sensibilidad |                       |                         |        |        |
|        | year_contract      | bar_p60_h60_tp10_sl05 | p60_h60 TP 1.0 / SL 0.5 |        |        |
| 32     | 39.02%             | 61.54%                | 86.75%                  | 23.21% | 13.25% |
| 23.21% | 0.00%              | 0.00%                 | 100.00%                 |        | 0.     |
| 00%    | 86.75%             |                       | 23.21%                  |        | 0.80   |
| 1      | 0.348              | 28.21                 | 16.95                   |        |        |
| 1.312  | menor_sensibilidad |                       |                         |        |        |
|        | year_contract      | bar_p60_h90_tp10_sl05 | p60_h90 TP 1.0 / SL 0.5 |        |        |
| 32     | 37.28%             | 59.14%                | 85.95%                  | 24.06% | 14.05% |
| 24.06% | 0.00%              | 0.00%                 | 100.00%                 |        | 0.     |
| 00%    | 85.95%             |                       | 24.06%                  |        | 0.78   |
| 9      | 0.361              | 42.95                 | 26.90                   |        |        |
| 1.313  | menor_sensibilidad |                       |                         |        |        |
|        | year_contract      | bar_p40_h60_tp10_sl05 | p40_h60 TP 1.0 / SL 0.5 |        |        |
| 32     | 56.71%             | 64.10%                | 79.70%                  | 23.58% | 20.30% |
| 23.58% | 0.00%              | 0.00%                 | 100.00%                 |        | 0.     |
| 00%    | 79.70%             |                       | 23.58%                  |        | 0.69   |
| 5      | 0.354              | 23.29                 | 17.04                   |        |        |
| 1.348  | menor_sensibilidad |                       |                         |        |        |
|        | year_contract      | bar_p50_h60_tp10_sl05 | p50_h60 TP 1.0 / SL 0.5 |        |        |

|        |                                     |        |                         |        |        |
|--------|-------------------------------------|--------|-------------------------|--------|--------|
| 32     | 47.79%                              | 64.73% | 83.41%                  | 23.52% | 16.59% |
| 23.52% | 0.00%                               | 0.00%  | 100.00%                 |        | 0.     |
| 00%    | 83.41%                              |        | 23.52%                  |        | 0.75   |
| 1      | 0.353                               | 25.87  |                         | 17.25  |        |
| 1.353  | menor_sensibilidad                  |        |                         |        |        |
|        | year_contract bar_p40_h90_tp10_sl05 |        | p40_h90 TP 1.0 / SL 0.5 |        |        |
| 32     | 53.87%                              | 61.51% | 79.14%                  | 25.76% | 20.86% |
| 25.76% | 0.00%                               | 0.00%  | 100.00%                 |        | 0.     |
| 00%    | 79.14%                              |        | 25.76%                  |        | 0.68   |
| 7      | 0.386                               | 35.15  |                         | 30.93  |        |
| 1.388  | menor_sensibilidad                  |        |                         |        |        |
|        | year_contract bar_p50_h90_tp10_sl05 |        | p50_h90 TP 1.0 / SL 0.5 |        |        |
| 32     | 45.59%                              | 61.77% | 82.73%                  | 26.65% | 17.27% |
| 26.65% | 0.00%                               | 0.00%  | 100.00%                 |        | 0.     |
| 00%    | 82.73%                              |        | 26.65%                  |        | 0.74   |
| 1      | 0.400                               | 39.05  |                         | 29.41  |        |
| 1.417  | menor_sensibilidad                  |        |                         |        |        |
|        | year_contract bar_p60_h30_tp15_sl10 |        | p60_h30 TP 1.5 / SL 1.0 |        |        |
| 32     | 41.17%                              | 61.91% | 46.58%                  | 29.09% | 3.44%  |
| 8.95%  | 49.97%                              | 37.06% |                         | 50.03% | 37.0   |
| 6%     | 93.81%                              |        | 13.14%                  |        | 0.664  |
| 0.358  | 17.55                               | 6.37   |                         | 1.502  |        |
|        | menor_sensibilidad                  |        |                         |        |        |
|        | year_contract bar_p60_h90_tp15_sl10 |        | p60_h90 TP 1.5 / SL 1.0 |        |        |
| 32     | 37.28%                              | 59.14% | 45.98%                  | 32.77% | 3.61%  |
| 9.73%  | 50.42%                              | 41.62% |                         | 49.58% | 41.6   |
| 2%     | 93.32%                              |        | 13.98%                  |        | 0.654  |
| 0.442  | 54.06                               | 29.44  |                         | 1.572  |        |
|        | menor_sensibilidad                  |        |                         |        |        |
|        | year_contract bar_p50_h30_tp15_sl10 |        | p50_h30 TP 1.5 / SL 1.0 |        |        |
| 32     | 50.21%                              | 64.11% | 51.01%                  | 29.32% | 4.82%  |
| 11.51% | 44.17%                              | 37.82% |                         | 55.83% | 37.    |
| 82%    | 92.08%                              |        | 14.67%                  |        | 0.71   |
| 7      | 0.363                               | 16.59  |                         | 6.71   |        |
| 1.574  | menor_sensibilidad                  |        |                         |        |        |
|        | year_contract bar_p60_h30_tp25_sl10 |        | p60_h30 TP 2.5 / SL 1.0 |        |        |
| 32     | 41.17%                              | 61.91% | 13.08%                  | 23.74% | 4.08%  |
| 10.76% | 82.84%                              | 33.86% |                         | 17.16% | 33.    |
| 86%    | 78.58%                              |        | 29.56%                  |        | 0.28   |
| 6      | 0.492                               | 18.38  |                         | 9.65   |        |
| 1.598  | sensibilidad_intermedia             |        |                         |        |        |
|        | year_contract bar_p60_h30_tp20_sl10 |        | p60_h30 TP 2.0 / SL 1.0 |        |        |
| 32     | 41.17%                              | 61.91% | 23.96%                  | 29.39% | 3.87%  |
| 10.07% | 72.17%                              | 38.80% |                         | 27.83% | 38.    |
| 80%    | 87.92%                              |        | 21.05%                  |        | 0.44   |
| 0      | 0.494                               | 18.32  |                         | 7.81   |        |
| 1.612  | sensibilidad_intermedia             |        |                         |        |        |
|        | year_contract bar_p60_h60_tp15_sl10 |        | p60_h60 TP 1.5 / SL 1.0 |        |        |
| 32     | 39.02%                              | 61.54% | 45.65%                  | 35.82% | 3.11%  |
| 8.92%  | 51.24%                              | 42.40% |                         | 48.76% | 42.4   |
| 0%     | 94.34%                              |        | 12.71%                  |        | 0.654  |
| 0.484  | 35.01                               | 15.71  |                         | 1.614  | se     |
|        | nsibilidad_intermedia               |        |                         |        |        |
|        | year_contract bar_p40_h30_tp15_sl10 |        | p40_h30 TP 1.5 / SL 1.0 |        |        |
| 32     | 59.22%                              | 63.21% | 55.21%                  | 30.35% | 6.50%  |
| 12.68% | 38.29%                              | 41.99% |                         | 61.71% | 41.    |
| 99%    | 90.19%                              |        | 13.92%                  |        | 0.76   |
| 3      | 0.347                               | 15.62  |                         | 7.17   |        |
| 1.621  | sensibilidad_intermedia             |        |                         |        |        |
|        | year_contract bar_p60_h60_tp25_sl10 |        | p60_h60 TP 2.5 / SL 1.0 |        |        |

|        |                                     |        |                         |        |        |
|--------|-------------------------------------|--------|-------------------------|--------|--------|
| 32     | 39.02%                              | 61.54% | 12.44%                  | 23.41% | 3.77%  |
| 11.33% | 83.79%                              | 34.07% |                         | 16.21% | 34.    |
| 07%    | 79.54%                              |        |                         | 32.71% | 0.27   |
| 3      | 0.498                               | 36.18  |                         | 24.55  |        |
| 1.631  | sensibilidad_intermedia             |        |                         |        |        |
|        | year_contract bar_p50_h30_tp15_sl05 |        | p50_h30 TP 1.5 / SL 0.5 |        |        |
| 32     | 50.21%                              | 64.11% | 43.93%                  | 19.80% | 19.65% |
| 22.72% | 36.41%                              | 42.17% |                         | 63.59% | 42.    |
| 17%    | 70.07%                              |        |                         | 17.10% | 0.56   |
| 1      | 0.188                               | 14.29  |                         | 7.30   |        |
| 1.659  | sensibilidad_intermedia             |        |                         |        |        |
|        | year_contract bar_p50_h60_tp15_sl10 |        | p50_h60 TP 1.5 / SL 1.0 |        |        |
| 32     | 47.79%                              | 64.73% | 49.64%                  | 33.61% | 4.38%  |
| 11.64% | 45.98%                              | 41.73% |                         | 54.02% | 41.    |
| 73%    | 92.77%                              |        |                         | 14.92% | 0.70   |
| 1      | 0.423                               | 33.02  |                         | 15.01  |        |
| 1.666  | sensibilidad_intermedia             |        |                         |        |        |
|        | year_contract bar_p60_h30_tp15_sl05 |        | p60_h30 TP 1.5 / SL 0.5 |        |        |
| 32     | 41.17%                              | 61.91% | 40.98%                  | 19.56% | 16.08% |
| 22.91% | 42.94%                              | 41.55% |                         | 57.06% | 41.    |
| 55%    | 73.01%                              |        |                         | 20.93% | 0.53   |
| 4      | 0.199                               | 15.39  |                         | 6.96   |        |
| 1.669  | sensibilidad_intermedia             |        |                         |        |        |
|        | year_contract bar_p40_h60_tp15_sl10 |        | p40_h60 TP 1.5 / SL 1.0 |        |        |
| 32     | 56.71%                              | 64.10% | 53.70%                  | 31.58% | 6.25%  |
| 13.57% | 40.05%                              | 42.68% |                         | 59.95% | 42.    |
| 68%    | 90.44%                              |        |                         | 15.56% | 0.74   |
| 3      | 0.363                               | 30.90  |                         | 14.51  |        |
| 1.675  | sensibilidad_intermedia             |        |                         |        |        |
|        | year_contract bar_p50_h90_tp15_sl10 |        | p50_h90 TP 1.5 / SL 1.0 |        |        |
| 32     | 45.59%                              | 61.77% | 50.24%                  | 35.29% | 4.84%  |
| 12.71% | 44.92%                              | 42.65% |                         | 55.08% | 42.    |
| 65%    | 91.93%                              |        |                         | 16.84% | 0.70   |
| 5      | 0.456                               | 51.04  |                         | 26.49  |        |
| 1.693  | sensibilidad_intermedia             |        |                         |        |        |
|        | year_contract bar_p40_h30_tp15_sl05 |        | p40_h30 TP 1.5 / SL 0.5 |        |        |
| 32     | 59.22%                              | 63.21% | 46.58%                  | 19.48% | 23.15% |
| 25.58% | 30.26%                              | 44.65% |                         | 69.74% | 44.    |
| 65%    | 67.63%                              |        |                         | 17.19% | 0.58   |
| 3      | 0.174                               | 13.23  |                         | 7.42   |        |
| 1.701  | sensibilidad_intermedia             |        |                         |        |        |
|        | year_contract bar_p60_h60_tp20_sl10 |        | p60_h60 TP 2.0 / SL 1.0 |        |        |
| 32     | 39.02%                              | 61.54% | 22.77%                  | 33.09% | 3.58%  |
| 10.61% | 73.65%                              | 43.65% |                         | 26.35% | 43.    |
| 65%    | 88.34%                              |        |                         | 21.50% | 0.42   |
| 0      | 0.556                               | 36.52  |                         | 19.84  |        |
| 1.704  | sensibilidad_intermedia             |        |                         |        |        |
|        | year_contract bar_p40_h30_tp25_sl10 |        | p40_h30 TP 2.5 / SL 1.0 |        |        |
| 32     | 59.22%                              | 63.21% | 20.59%                  | 28.19% | 7.82%  |
| 16.04% | 71.59%                              | 43.81% |                         | 28.41% | 43.    |
| 81%    | 74.30%                              |        |                         | 19.43% | 0.43   |
| 7      | 0.549                               | 16.97  |                         | 6.33   |        |
| 1.707  | sensibilidad_intermedia             |        |                         |        |        |
|        | year_contract bar_p60_h90_tp20_sl10 |        | p60_h90 TP 2.0 / SL 1.0 |        |        |
| 32     | 37.28%                              | 59.14% | 22.88%                  | 30.60% | 4.16%  |
| 11.24% | 72.97%                              | 41.79% |                         | 27.03% | 41.    |
| 79%    | 86.25%                              |        |                         | 28.57% | 0.41   |
| 6      | 0.532                               | 56.09  |                         | 31.74  |        |
| 1.713  | sensibilidad_intermedia             |        |                         |        |        |
|        | year_contract bar_p50_h30_tp25_sl10 |        | p50_h30 TP 2.5 / SL 1.0 |        |        |

|        |                                     |                         |        |        |        |
|--------|-------------------------------------|-------------------------|--------|--------|--------|
| 32     | 50.21%                              | 64.11%                  | 16.61% | 25.42% | 5.75%  |
| 14.18% | 77.64%                              | 38.70%                  | 22.36% | 38.    |        |
| 70%    | 76.42%                              |                         | 29.28% | 0.35   |        |
| 8      | 0.503                               | 17.74                   | 7.02   |        |        |
| 1.717  | sensibilidad_intermedia             |                         |        |        |        |
|        | year_contract bar_p50_h30_tp20_sl10 | p50_h30 TP 2.0 / SL 1.0 |        |        |        |
| 32     | 50.21%                              | 64.11%                  | 28.49% | 30.73% | 5.43%  |
| 13.09% | 66.08%                              | 42.93%                  | 33.92% | 42.    |        |
| 93%    | 85.62%                              |                         | 21.21% | 0.51   |        |
| 6      | 0.493                               | 17.53                   | 7.02   |        |        |
| 1.721  | sensibilidad_intermedia             |                         |        |        |        |
|        | year_contract bar_p40_h90_tp15_sl10 | p40_h90 TP 1.5 / SL 1.0 |        |        |        |
| 32     | 53.87%                              | 61.51%                  | 53.91% | 33.05% | 6.50%  |
| 15.68% | 39.59%                              | 42.93%                  | 60.41% | 42.    |        |
| 93%    | 90.11%                              |                         | 19.02% | 0.74   |        |
| 4      | 0.397                               | 47.49                   | 25.09  |        |        |
| 1.722  | sensibilidad_intermedia             |                         |        |        |        |
|        | year_contract bar_p40_h30_tp20_sl10 | p40_h30 TP 2.0 / SL 1.0 |        |        |        |
| 32     | 59.22%                              | 63.21%                  | 33.25% | 33.27% | 7.35%  |
| 14.76% | 59.40%                              | 46.63%                  | 40.60% | 46.    |        |
| 63%    | 83.38%                              |                         | 17.28% | 0.59   |        |
| 2      | 0.532                               | 16.63                   | 6.68   |        |        |
| 1.751  | sensibilidad_intermedia             |                         |        |        |        |
|        | year_contract bar_p40_h90_tp15_sl05 | p40_h90 TP 1.5 / SL 0.5 |        |        |        |
| 32     | 53.87%                              | 61.51%                  | 44.52% | 17.27% | 24.90% |
| 30.04% | 30.59%                              | 45.01%                  | 69.41% | 45.    |        |
| 01%    | 65.20%                              |                         | 23.91% | 0.54   |        |
| 3      | 0.164                               | 39.43                   | 26.12  |        |        |
| 1.777  | sensibilidad_intermedia             |                         |        |        |        |
|        | year_contract bar_p50_h30_tp20_sl05 | p50_h30 TP 2.0 / SL 0.5 |        |        |        |
| 32     | 50.21%                              | 64.11%                  | 24.28% | 22.48% | 20.91% |
| 25.56% | 54.81%                              | 48.04%                  | 45.19% | 48.    |        |
| 04%    | 54.54%                              |                         | 17.99% | 0.38   |        |
| 1      | 0.346                               | 14.21                   | 5.57   |        |        |
| 1.782  | sensibilidad_intermedia             |                         |        |        |        |
|        | year_contract bar_p50_h60_tp25_sl10 | p50_h60 TP 2.5 / SL 1.0 |        |        |        |
| 32     | 47.79%                              | 64.73%                  | 15.34% | 27.28% | 5.35%  |
| 14.97% | 79.31%                              | 42.24%                  | 20.69% | 42.    |        |
| 24%    | 77.14%                              |                         | 29.02% | 0.33   |        |
| 0      | 0.532                               | 35.28                   | 18.64  |        |        |
| 1.782  | mayor_sensibilidad                  |                         |        |        |        |
|        | year_contract bar_p40_h60_tp15_sl05 | p40_h60 TP 1.5 / SL 0.5 |        |        |        |
| 32     | 56.71%                              | 64.10%                  | 44.79% | 21.68% | 24.36% |
| 27.95% | 30.85%                              | 45.29%                  | 69.15% | 45.    |        |
| 29%    | 65.68%                              |                         | 21.11% | 0.55   |        |
| 0      | 0.221                               | 25.85                   | 15.70  |        |        |
| 1.801  | mayor_sensibilidad                  |                         |        |        |        |
|        | year_contract bar_p50_h90_tp20_sl10 | p50_h90 TP 2.0 / SL 1.0 |        |        |        |
| 32     | 45.59%                              | 61.77%                  | 27.34% | 31.74% | 5.58%  |
| 14.77% | 67.08%                              | 45.89%                  | 32.92% | 45.    |        |
| 89%    | 84.66%                              |                         | 26.72% | 0.49   |        |
| 1      | 0.514                               | 53.97                   | 30.27  |        |        |
| 1.809  | mayor_sensibilidad                  |                         |        |        |        |
|        | year_contract bar_p50_h60_tp15_sl05 | p50_h60 TP 1.5 / SL 0.5 |        |        |        |
| 32     | 47.79%                              | 64.73%                  | 42.56% | 19.63% | 20.00% |
| 27.73% | 37.45%                              | 46.41%                  | 62.55% | 46.    |        |
| 41%    | 69.35%                              |                         | 22.55% | 0.53   |        |
| 8      | 0.177                               | 28.22                   | 15.51  |        |        |
| 1.811  | mayor_sensibilidad                  |                         |        |        |        |
|        | year_contract bar_p60_h30_tp20_sl05 | p60_h30 TP 2.0 / SL 0.5 |        |        |        |

|        |                    |                       |                         |        |        |
|--------|--------------------|-----------------------|-------------------------|--------|--------|
| 32     | 41.17%             | 61.91%                | 20.89%                  | 22.33% | 16.97% |
| 25.26% | 62.14%             | 47.37%                |                         | 37.86% | 47.    |
| 37%    | 56.25%             |                       | 24.35%                  |        | 0.33   |
| 3      | 0.360              | 14.96                 |                         | 5.36   |        |
| 1.812  | mayor_sensibilidad |                       |                         |        |        |
|        | year_contract      | bar_p40_h60_tp25_sl10 | p40_h60 TP 2.5 / SL 1.0 |        |        |
| 32     | 56.71%             | 64.10%                | 19.10%                  | 29.37% | 7.55%  |
| 17.46% | 73.35%             | 46.38%                |                         | 26.65% | 46.    |
| 38%    | 74.00%             |                       | 23.95%                  |        | 0.40   |
| 2      | 0.568              | 33.87                 |                         | 15.66  |        |
| 1.813  | mayor_sensibilidad |                       |                         |        |        |
|        | year_contract      | bar_p50_h90_tp25_sl10 | p50_h90 TP 2.5 / SL 1.0 |        |        |
| 32     | 45.59%             | 61.77%                | 15.57%                  | 25.05% | 5.96%  |
| 15.75% | 78.48%             | 40.18%                |                         | 21.52% | 40.    |
| 18%    | 74.46%             |                       | 38.75%                  |        | 0.33   |
| 0      | 0.493              | 54.18                 |                         | 30.83  |        |
| 1.815  | mayor_sensibilidad |                       |                         |        |        |
|        | year_contract      | bar_p50_h60_tp20_sl10 | p50_h60 TP 2.0 / SL 1.0 |        |        |
| 32     | 47.79%             | 64.73%                | 26.91%                  | 33.73% | 5.03%  |
| 13.70% | 68.06%             | 47.42%                |                         | 31.94% | 47.    |
| 42%    | 86.35%             |                       | 21.94%                  |        | 0.48   |
| 8      | 0.561              | 35.17                 |                         | 18.31  |        |
| 1.815  | mayor_sensibilidad |                       |                         |        |        |
|        | year_contract      | bar_p40_h60_tp20_sl10 | p40_h60 TP 2.0 / SL 1.0 |        |        |
| 32     | 56.71%             | 64.10%                | 31.81%                  | 34.90% | 7.07%  |
| 15.68% | 61.12%             | 48.49%                |                         | 38.88% | 48.    |
| 49%    | 83.72%             |                       | 20.64%                  |        | 0.56   |
| 5      | 0.562              | 33.23                 |                         | 16.02  |        |
| 1.838  | mayor_sensibilidad |                       |                         |        |        |
|        | year_contract      | bar_p40_h60_tp20_sl05 | p40_h60 TP 2.0 / SL 0.5 |        |        |
| 32     | 56.71%             | 64.10%                | 26.34%                  | 23.54% | 25.95% |
| 31.20% | 47.71%             | 53.53%                |                         | 52.29% | 53.    |
| 53%    | 51.12%             |                       | 14.85%                  |        | 0.39   |
| 7      | 0.321              | 26.20                 |                         | 12.49  |        |
| 1.872  | mayor_sensibilidad |                       |                         |        |        |
|        | year_contract      | bar_p40_h30_tp20_sl05 | p40_h30 TP 2.0 / SL 0.5 |        |        |
| 32     | 59.22%             | 63.21%                | 27.85%                  | 25.21% | 24.84% |
| 27.96% | 47.32%             | 53.16%                |                         | 52.68% | 53.    |
| 16%    | 53.46%             |                       | 17.92%                  |        | 0.43   |
| 3      | 0.367              | 13.44                 |                         | 5.95   |        |
| 1.875  | mayor_sensibilidad |                       |                         |        |        |
|        | year_contract      | bar_p40_h90_tp20_sl10 | p40_h90 TP 2.0 / SL 1.0 |        |        |
| 32     | 53.87%             | 61.51%                | 32.04%                  | 36.08% | 7.43%  |
| 18.22% | 60.53%             | 47.54%                |                         | 39.47% | 47.    |
| 54%    | 82.85%             |                       | 26.91%                  |        | 0.56   |
| 6      | 0.607              | 51.27                 |                         | 28.93  |        |
| 1.903  | mayor_sensibilidad |                       |                         |        |        |
|        | year_contract      | bar_p40_h90_tp25_sl10 | p40_h90 TP 2.5 / SL 1.0 |        |        |
| 32     | 53.87%             | 61.51%                | 19.28%                  | 27.75% | 7.97%  |
| 19.95% | 72.75%             | 46.05%                |                         | 27.25% | 46.    |
| 05%    | 72.83%             |                       | 35.85%                  |        | 0.40   |
| 2      | 0.536              | 52.16                 |                         | 27.41  |        |
| 1.911  | mayor_sensibilidad |                       |                         |        |        |
|        | year_contract      | bar_p60_h60_tp15_sl05 | p60_h60 TP 1.5 / SL 0.5 |        |        |
| 32     | 39.02%             | 61.54%                | 40.10%                  | 24.76% | 15.96% |
| 27.64% | 43.93%             | 49.03%                |                         | 56.07% | 49.    |
| 03%    | 73.19%             |                       | 30.65%                  |        | 0.52   |
| 2      | 0.310              | 30.49                 |                         | 15.74  |        |
| 1.936  | mayor_sensibilidad |                       |                         |        |        |
|        | year_contract      | bar_p50_h60_tp20_sl05 | p50_h60 TP 2.0 / SL 0.5 |        |        |

|        |                    |                       |                         |         |        |
|--------|--------------------|-----------------------|-------------------------|---------|--------|
| 32     | 47.79%             | 64.73%                | 22.85%                  | 24.56%  | 21.21% |
| 30.48% | 55.94%             | 54.20%                |                         | 44.06%  | 54.    |
| 20%    | 52.87%             |                       | 20.07%                  |         | 0.35   |
| 1      | 0.399              | 27.86                 |                         | 12.09   |        |
| 1.940  | mayor_sensibilidad |                       |                         |         |        |
|        | year_contract      | bar_p60_h90_tp15_sl05 | p60_h90 TP 1.5 / SL 0.5 |         |        |
| 32     | 37.28%             | 59.14%                | 40.26%                  | 24.18%  | 16.92% |
| 30.24% | 42.82%             | 48.07%                |                         | 57.18%  | 48.    |
| 07%    | 71.85%             |                       | 35.66%                  |         | 0.51   |
| 9      | 0.380              | 46.87                 |                         | 29.88   |        |
| 1.973  | mayor_sensibilidad |                       |                         |         |        |
|        | year_contract      | bar_p40_h90_tp20_sl05 | p40_h90 TP 2.0 / SL 0.5 |         |        |
| 32     | 53.87%             | 61.51%                | 26.21%                  | 24.89%  | 26.51% |
| 33.40% | 47.28%             | 54.76%                |                         | 52.72%  | 54.    |
| 76%    | 50.60%             |                       | 26.03%                  |         | 0.39   |
| 2      | 0.362              | 40.04                 |                         | 22.38   |        |
| 2.006  | mayor_sensibilidad |                       |                         |         |        |
|        | year_contract      | bar_p50_h90_tp15_sl05 | p50_h90 TP 1.5 / SL 0.5 |         |        |
| 32     | 45.59%             | 61.77%                | 42.75%                  | 23.12%  | 20.62% |
| 32.63% | 36.63%             | 48.36%                |                         | 63.37%  | 48.    |
| 36%    | 68.82%             |                       | 34.81%                  |         | 0.53   |
| 8      | 0.275              | 43.34                 |                         | 26.21   |        |
| 2.007  | mayor_sensibilidad |                       |                         |         |        |
|        | year_contract      | bar_p60_h60_tp20_sl05 | p60_h60 TP 2.0 / SL 0.5 |         |        |
| 32     | 39.02%             | 61.54%                | 19.83%                  | 25.19%  | 16.92% |
| 30.63% | 63.25%             | 53.26%                |                         | 36.75%  | 53.    |
| 26%    | 55.62%             |                       | 33.67%                  |         | 0.31   |
| 2      | 0.455              | 29.27                 |                         | 16.27   |        |
| 2.043  | mayor_sensibilidad |                       |                         |         |        |
|        | year_contract      | bar_p60_h90_tp25_sl10 | p60_h90 TP 2.5 / SL 1.0 |         |        |
| 32     | 37.28%             | 59.14%                | 12.21%                  | 22.73%  | 4.51%  |
| 12.02% | 83.27%             | 32.78%                |                         | 16.73%  | 32.    |
| 78%    | 74.43%             |                       | 82.86%                  |         | 0.26   |
| 0      | 0.571              | 56.82                 |                         | 44.92   |        |
| 2.095  | mayor_sensibilidad |                       |                         |         |        |
|        | year_contract      | bar_p60_h90_tp20_sl05 | p60_h90 TP 2.0 / SL 0.5 |         |        |
| 32     | 37.28%             | 59.14%                | 19.82%                  | 23.02%  | 17.98% |
| 33.26% | 62.21%             | 53.01%                |                         | 37.79%  | 53.    |
| 01%    | 53.91%             |                       | 45.58%                  |         | 0.30   |
| 6      | 0.346              | 44.99                 |                         | 26.91   |        |
| 2.140  | mayor_sensibilidad |                       |                         |         |        |
|        | year_contract      | bar_p50_h90_tp20_sl05 | p50_h90 TP 2.0 / SL 0.5 |         |        |
| 32     | 45.59%             | 61.77%                | 23.06%                  | 24.11%  | 21.97% |
| 36.60% | 54.97%             | 52.88%                |                         | 45.03%  | 52.    |
| 88%    | 52.54%             |                       | 42.51%                  |         | 0.35   |
| 1      | 0.422              | 42.73                 |                         | 22.82   |        |
| 2.179  | mayor_sensibilidad |                       |                         |         |        |
|        | year_regime        | bar_p60_h60_tp10_sl10 | p60_h60 TP 1.0 / SL 1.0 |         |        |
| 35     | 37.25%             | 80.71%                | 96.94%                  | 7.89%   | 3.06%  |
| 7.89%  | 0.00%              | 0.00%                 |                         | 100.00% | 0.0    |
| 0%     | 96.94%             |                       | 7.89%                   |         | 0.939  |
| 0.158  | 31.10              | 18.39                 |                         | 1.044   |        |
|        | menor_sensibilidad |                       |                         |         |        |
|        | year_regime        | bar_p60_h30_tp10_sl10 | p60_h30 TP 1.0 / SL 1.0 |         |        |
| 35     | 38.10%             | 75.74%                | 96.63%                  | 10.61%  | 3.37%  |
| 10.61% | 0.00%              | 0.00%                 |                         | 100.00% | 0.     |
| 00%    | 96.63%             |                       | 10.61%                  |         | 0.93   |
| 3      | 0.212              | 15.92                 |                         | 8.60    |        |
| 1.076  | menor_sensibilidad |                       |                         |         |        |
|        | year_regime        | bar_p60_h90_tp10_sl10 | p60_h90 TP 1.0 / SL 1.0 |         |        |

|                    |                    |                       |         |                 |        |
|--------------------|--------------------|-----------------------|---------|-----------------|--------|
| 35                 | 36.43%             | 83.00%                | 96.81%  | 8.99%           | 3.19%  |
| 8.99%              | 0.00%              | 0.00%                 | 100.00% |                 | 0.0    |
| 0%                 | 96.81%             |                       | 8.99%   |                 | 0.936  |
| 0.180              | 46.18              | 29.14                 |         | 1.100           |        |
| menor_sensibilidad |                    |                       |         |                 |        |
|                    | year_regime        | bar_p50_h60_tp10_sl10 | p50_h60 | TP 1.0 / SL 1.0 |        |
| 35                 | 44.61%             | 86.52%                | 95.66%  | 10.52%          | 4.34%  |
| 10.52%             | 0.00%              | 0.00%                 | 100.00% |                 | 0.     |
| 00%                | 95.66%             |                       | 10.52%  |                 | 0.91   |
| 3                  | 0.210              | 29.12                 |         | 19.56           |        |
| 1.181              | menor_sensibilidad |                       |         |                 |        |
|                    | year_regime        | bar_p50_h30_tp10_sl10 | p50_h30 | TP 1.0 / SL 1.0 |        |
| 35                 | 45.81%             | 83.32%                | 95.25%  | 13.47%          | 4.75%  |
| 13.47%             | 0.00%              | 0.00%                 | 100.00% |                 | 0.     |
| 00%                | 95.25%             |                       | 13.47%  |                 | 0.90   |
| 5                  | 0.269              | 14.92                 |         | 9.18            |        |
| 1.237              | menor_sensibilidad |                       |         |                 |        |
|                    | year_regime        | bar_p50_h90_tp10_sl10 | p50_h90 | TP 1.0 / SL 1.0 |        |
| 35                 | 43.74%             | 91.45%                | 95.56%  | 11.90%          | 4.44%  |
| 11.90%             | 0.00%              | 0.00%                 | 100.00% |                 | 0.     |
| 00%                | 95.56%             |                       | 11.90%  |                 | 0.91   |
| 1                  | 0.238              | 43.01                 |         | 29.44           |        |
| 1.272              | menor_sensibilidad |                       |         |                 |        |
|                    | year_regime        | bar_p40_h60_tp10_sl10 | p40_h60 | TP 1.0 / SL 1.0 |        |
| 35                 | 51.92%             | 92.40%                | 94.10%  | 12.71%          | 5.90%  |
| 12.71%             | 0.00%              | 0.00%                 | 100.00% |                 | 0.     |
| 00%                | 94.10%             |                       | 12.71%  |                 | 0.88   |
| 2                  | 0.254              | 26.99                 |         | 20.75           |        |
| 1.305              | menor_sensibilidad |                       |         |                 |        |
|                    | year_regime        | bar_p40_h90_tp10_sl10 | p40_h90 | TP 1.0 / SL 1.0 |        |
| 35                 | 50.64%             | 95.40%                | 93.87%  | 13.69%          | 6.13%  |
| 13.69%             | 0.00%              | 0.00%                 | 100.00% |                 | 0.     |
| 00%                | 93.87%             |                       | 13.69%  |                 | 0.87   |
| 7                  | 0.274              | 39.67                 |         | 31.37           |        |
| 1.365              | menor_sensibilidad |                       |         |                 |        |
|                    | year_regime        | bar_p40_h30_tp10_sl10 | p40_h30 | TP 1.0 / SL 1.0 |        |
| 35                 | 53.29%             | 89.04%                | 93.52%  | 16.12%          | 6.48%  |
| 16.12%             | 0.00%              | 0.00%                 | 100.00% |                 | 0.     |
| 00%                | 93.52%             |                       | 16.12%  |                 | 0.87   |
| 0                  | 0.322              | 13.77                 |         | 9.74            |        |
| 1.374              | menor_sensibilidad |                       |         |                 |        |
|                    | year_regime        | bar_p60_h60_tp10_sl05 | p60_h60 | TP 1.0 / SL 0.5 |        |
| 35                 | 37.25%             | 80.71%                | 84.99%  | 22.82%          | 15.01% |
| 22.82%             | 0.00%              | 0.00%                 | 100.00% |                 | 0.     |
| 00%                | 84.99%             |                       | 22.82%  |                 | 0.77   |
| 5                  | 0.342              | 27.87                 |         | 21.13           |        |
| 1.492              | menor_sensibilidad |                       |         |                 |        |
|                    | year_regime        | bar_p60_h90_tp10_sl05 | p60_h90 | TP 1.0 / SL 0.5 |        |
| 35                 | 36.43%             | 83.00%                | 84.43%  | 23.45%          | 15.57% |
| 23.45%             | 0.00%              | 0.00%                 | 100.00% |                 | 0.     |
| 00%                | 84.43%             |                       | 23.45%  |                 | 0.76   |
| 6                  | 0.352              | 41.07                 |         | 32.69           |        |
| 1.534              | menor_sensibilidad |                       |         |                 |        |
|                    | year_regime        | bar_p60_h30_tp10_sl05 | p60_h30 | TP 1.0 / SL 0.5 |        |
| 35                 | 38.10%             | 75.74%                | 85.02%  | 26.82%          | 14.98% |
| 26.82%             | 0.00%              | 0.00%                 | 100.00% |                 | 0.     |
| 00%                | 85.02%             |                       | 26.82%  |                 | 0.77   |
| 5                  | 0.402              | 14.31                 |         | 9.50            |        |
| 1.562              | menor_sensibilidad |                       |         |                 |        |
|                    | year_regime        | bar_p50_h60_tp10_sl05 | p50_h60 | TP 1.0 / SL 0.5 |        |

|        |                                   |        |                         |        |        |
|--------|-----------------------------------|--------|-------------------------|--------|--------|
| 35     | 44.61%                            | 86.52% | 81.76%                  | 23.73% | 18.24% |
| 23.73% | 0.00%                             | 0.00%  | 100.00%                 |        | 0.     |
| 00%    | 81.76%                            |        | 23.73%                  |        | 0.72   |
| 6      | 0.356                             | 25.45  |                         | 21.82  |        |
| 1.577  | menor_sensibilidad                |        |                         |        |        |
|        | year_regime bar_p40_h60_tp10_sl05 |        | p40_h60 TP 1.0 / SL 0.5 |        |        |
| 35     | 51.92%                            | 92.40% | 78.12%                  | 23.54% | 21.88% |
| 23.54% | 0.00%                             | 0.00%  | 100.00%                 |        | 0.     |
| 00%    | 78.12%                            |        | 23.54%                  |        | 0.67   |
| 2      | 0.353                             | 22.88  |                         | 22.23  |        |
| 1.630  | menor_sensibilidad                |        |                         |        |        |
|        | year_regime bar_p50_h30_tp10_sl05 |        | p50_h30 TP 1.0 / SL 0.5 |        |        |
| 35     | 45.81%                            | 83.32% | 81.86%                  | 27.25% | 18.14% |
| 27.25% | 0.00%                             | 0.00%  | 100.00%                 |        | 0.     |
| 00%    | 81.86%                            |        | 27.25%                  |        | 0.72   |
| 8      | 0.409                             | 13.07  |                         | 10.78  |        |
| 1.651  | menor_sensibilidad                |        |                         |        |        |
|        | year_regime bar_p50_h90_tp10_sl05 |        | p50_h90 TP 1.0 / SL 0.5 |        |        |
| 35     | 43.74%                            | 91.45% | 81.13%                  | 25.33% | 18.87% |
| 25.33% | 0.00%                             | 0.00%  | 100.00%                 |        | 0.     |
| 00%    | 81.13%                            |        | 25.33%                  |        | 0.71   |
| 7      | 0.380                             | 37.25  |                         | 32.62  |        |
| 1.675  | menor_sensibilidad                |        |                         |        |        |
|        | year_regime bar_p40_h30_tp10_sl05 |        | p40_h30 TP 1.0 / SL 0.5 |        |        |
| 35     | 53.29%                            | 89.04% | 78.73%                  | 26.37% | 21.27% |
| 26.37% | 0.00%                             | 0.00%  | 100.00%                 |        | 0.     |
| 00%    | 78.73%                            |        | 26.37%                  |        | 0.68   |
| 1      | 0.396                             | 11.77  |                         | 10.96  |        |
| 1.681  | menor_sensibilidad                |        |                         |        |        |
|        | year_regime bar_p40_h90_tp10_sl05 |        | p40_h90 TP 1.0 / SL 0.5 |        |        |
| 35     | 50.64%                            | 95.40% | 77.75%                  | 25.40% | 22.25% |
| 25.40% | 0.00%                             | 0.00%  | 100.00%                 |        | 0.     |
| 00%    | 77.75%                            |        | 25.40%                  |        | 0.66   |
| 6      | 0.381                             | 33.44  |                         | 34.02  |        |
| 1.716  | menor_sensibilidad                |        |                         |        |        |
|        | year_regime bar_p50_h30_tp15_sl10 |        | p50_h30 TP 1.5 / SL 1.0 |        |        |
| 35     | 45.81%                            | 83.32% | 53.90%                  | 27.93% | 5.71%  |
| 13.91% | 40.38%                            | 38.38% |                         | 59.62% | 38.    |
| 38%    | 90.99%                            |        |                         | 16.28% | 0.75   |
| 1      | 0.337                             | 16.77  |                         | 9.83   |        |
| 1.798  | menor_sensibilidad                |        |                         |        |        |
|        | year_regime bar_p60_h30_tp15_sl10 |        | p60_h30 TP 1.5 / SL 1.0 |        |        |
| 35     | 38.10%                            | 75.74% | 48.70%                  | 34.35% | 4.05%  |
| 11.71% | 47.25%                            | 42.69% |                         | 52.75% | 42.    |
| 69%    | 92.94%                            |        |                         | 15.84% | 0.69   |
| 0      | 0.450                             | 17.75  |                         | 9.44   |        |
| 1.803  | menor_sensibilidad                |        |                         |        |        |
|        | year_regime bar_p60_h60_tp15_sl10 |        | p60_h60 TP 1.5 / SL 1.0 |        |        |
| 35     | 37.25%                            | 80.71% | 46.95%                  | 36.45% | 3.81%  |
| 9.69%  | 49.25%                            | 43.66% |                         | 50.75% | 43.6   |
| 6%     | 93.06%                            |        |                         | 12.01% | 0.666  |
| 0.479  | 34.75                             | 15.48  |                         | 1.825  |        |
|        | menor_sensibilidad                |        |                         |        |        |
|        | year_regime bar_p40_h60_tp15_sl05 |        | p40_h60 TP 1.5 / SL 0.5 |        |        |
| 35     | 51.92%                            | 92.40% | 46.57%                  | 14.75% | 26.18% |
| 25.13% | 27.24%                            | 34.26% |                         | 72.76% | 34.    |
| 26%    | 64.73%                            |        |                         | 18.34% | 0.56   |
| 8      | 0.160                             | 25.68  |                         | 19.31  |        |
| 1.849  | sensibilidad_intermedia           |        |                         |        |        |
|        | year_regime bar_p50_h60_tp15_sl10 |        | p50_h60 TP 1.5 / SL 1.0 |        |        |



|        |                                   |                         |        |        |        |
|--------|-----------------------------------|-------------------------|--------|--------|--------|
| 35     | 44.61%                            | 86.52%                  | 51.83% | 31.05% | 5.32%  |
| 12.37% | 42.85%                            | 41.48%                  | 57.15% | 41.    |        |
| 48%    | 91.38%                            |                         | 13.99% | 0.72   |        |
| 4      | 0.368                             | 32.84                   | 17.61  |        |        |
| 1.854  | sensibilidad_intermedia           |                         |        |        |        |
|        | year_regime bar_p40_h60_tp15_sl10 | p40_h60 TP 1.5 / SL 1.0 |        |        |        |
| 35     | 51.92%                            | 92.40%                  | 56.41% | 26.35% | 7.23%  |
| 14.49% | 36.36%                            | 36.83%                  | 63.64% | 36.    |        |
| 83%    | 89.32%                            |                         | 15.51% | 0.77   |        |
| 4      | 0.320                             | 30.73                   | 18.84  |        |        |
| 1.856  | sensibilidad_intermedia           |                         |        |        |        |
|        | year_regime bar_p60_h30_tp15_sl05 | p60_h30 TP 1.5 / SL 0.5 |        |        |        |
| 35     | 38.10%                            | 75.74%                  | 42.22% | 22.76% | 18.22% |
| 26.64% | 39.56%                            | 44.59%                  | 60.44% | 44.    |        |
| 59%    | 70.72%                            |                         | 22.45% | 0.54   |        |
| 2      | 0.302                             | 15.47                   | 9.55   |        |        |
| 1.922  | sensibilidad_intermedia           |                         |        |        |        |
|        | year_regime bar_p50_h60_tp15_sl05 | p50_h60 TP 1.5 / SL 0.5 |        |        |        |
| 35     | 44.61%                            | 86.52%                  | 43.76% | 17.23% | 22.01% |
| 27.11% | 34.23%                            | 41.81%                  | 65.77% | 41.    |        |
| 81%    | 67.51%                            |                         | 21.04% | 0.54   |        |
| 6      | 0.185                             | 27.88                   | 18.68  |        |        |
| 1.937  | sensibilidad_intermedia           |                         |        |        |        |
|        | year_regime bar_p60_h60_tp20_sl10 | p60_h60 TP 2.0 / SL 1.0 |        |        |        |
| 35     | 37.25%                            | 80.71%                  | 23.51% | 34.65% | 4.40%  |
| 11.34% | 72.09%                            | 45.85%                  | 27.91% | 45.    |        |
| 85%    | 85.16%                            |                         | 21.79% | 0.42   |        |
| 6      | 0.587                             | 36.24                   | 14.61  |        |        |
| 1.944  | sensibilidad_intermedia           |                         |        |        |        |
|        | year_regime bar_p60_h60_tp25_sl10 | p60_h60 TP 2.5 / SL 1.0 |        |        |        |
| 35     | 37.25%                            | 80.71%                  | 12.52% | 25.58% | 4.62%  |
| 12.42% | 82.86%                            | 38.00%                  | 17.14% | 38.    |        |
| 00%    | 72.88%                            |                         | 39.66% | 0.26   |        |
| 7      | 0.515                             | 36.07                   | 13.06  |        |        |
| 1.964  | sensibilidad_intermedia           |                         |        |        |        |
|        | year_regime bar_p60_h30_tp20_sl10 | p60_h30 TP 2.0 / SL 1.0 |        |        |        |
| 35     | 38.10%                            | 75.74%                  | 25.96% | 34.99% | 4.54%  |
| 13.02% | 69.50%                            | 47.70%                  | 30.50% | 47.    |        |
| 70%    | 86.38%                            |                         | 25.00% | 0.47   |        |
| 4      | 0.584                             | 18.49                   | 7.22   |        |        |
| 1.964  | sensibilidad_intermedia           |                         |        |        |        |
|        | year_regime bar_p40_h30_tp15_sl10 | p40_h30 TP 1.5 / SL 1.0 |        |        |        |
| 35     | 53.29%                            | 89.04%                  | 57.66% | 28.54% | 7.66%  |
| 17.93% | 34.68%                            | 42.12%                  | 65.32% | 42.    |        |
| 12%    | 88.89%                            |                         | 19.32% | 0.78   |        |
| 8      | 0.327                             | 15.67                   | 8.95   |        |        |
| 1.969  | sensibilidad_intermedia           |                         |        |        |        |
|        | year_regime bar_p60_h90_tp15_sl10 | p60_h90 TP 1.5 / SL 1.0 |        |        |        |
| 35     | 36.43%                            | 83.00%                  | 46.55% | 40.04% | 4.00%  |
| 11.01% | 49.45%                            | 49.37%                  | 50.55% | 49.    |        |
| 37%    | 92.65%                            |                         | 13.95% | 0.65   |        |
| 8      | 0.538                             | 52.30                   | 27.58  |        |        |
| 1.974  | sensibilidad_intermedia           |                         |        |        |        |
|        | year_regime bar_p40_h30_tp15_sl05 | p40_h30 TP 1.5 / SL 0.5 |        |        |        |
| 35     | 53.29%                            | 89.04%                  | 47.99% | 18.54% | 25.20% |
| 28.07% | 26.81%                            | 42.12%                  | 73.19% | 42.    |        |
| 12%    | 66.15%                            |                         | 20.52% | 0.59   |        |
| 4      | 0.188                             | 13.22                   | 9.17   |        |        |
| 1.983  | sensibilidad_intermedia           |                         |        |        |        |
|        | year_regime bar_p50_h90_tp15_sl10 | p50_h90 TP 1.5 / SL 1.0 |        |        |        |

|        |                                   |        |                         |        |        |
|--------|-----------------------------------|--------|-------------------------|--------|--------|
| 35     | 43.74%                            | 91.45% | 51.35%                  | 34.06% | 5.43%  |
| 13.93% | 43.22%                            | 43.06% |                         | 56.78% | 43.    |
| 06%    | 91.07%                            |        | 16.21%                  |        | 0.71   |
| 6      | 0.454                             | 49.29  |                         | 29.51  |        |
| 1.987  | sensibilidad_intermedia           |        |                         |        |        |
|        | year_regime bar_p50_h30_tp15_sl05 |        | p50_h30 TP 1.5 / SL 0.5 |        |        |
| 35     | 45.81%                            | 83.32% | 45.66%                  | 16.71% | 21.71% |
| 30.60% | 32.63%                            | 38.99% |                         | 67.37% | 38.    |
| 99%    | 68.59%                            |        | 29.39%                  |        | 0.57   |
| 6      | 0.176                             | 14.36  |                         | 9.97   |        |
| 1.990  | sensibilidad_intermedia           |        |                         |        |        |
|        | year_regime bar_p40_h90_tp15_sl10 |        | p40_h90 TP 1.5 / SL 1.0 |        |        |
| 35     | 50.64%                            | 95.40% | 55.84%                  | 32.20% | 7.44%  |
| 15.92% | 36.71%                            | 41.37% |                         | 63.29% | 41.    |
| 37%    | 88.93%                            |        | 17.51%                  |        | 0.76   |
| 3      | 0.398                             | 45.86  |                         | 31.04  |        |
| 2.024  | sensibilidad_intermedia           |        |                         |        |        |
|        | year_regime bar_p60_h60_tp15_sl05 |        | p60_h60 TP 1.5 / SL 0.5 |        |        |
| 35     | 37.25%                            | 80.71% | 40.54%                  | 25.43% | 18.23% |
| 26.84% | 41.23%                            | 48.92% |                         | 58.77% | 48.    |
| 92%    | 70.15%                            |        | 20.67%                  |        | 0.51   |
| 7      | 0.283                             | 29.91  |                         | 18.25  |        |
| 2.026  | sensibilidad_intermedia           |        |                         |        |        |
|        | year_regime bar_p50_h60_tp20_sl10 |        | p50_h60 TP 2.0 / SL 1.0 |        |        |
| 35     | 44.61%                            | 86.52% | 28.56%                  | 35.99% | 6.10%  |
| 14.52% | 65.34%                            | 49.41% |                         | 34.66% | 49.    |
| 41%    | 83.71%                            |        | 17.13%                  |        | 0.51   |
| 0      | 0.599                             | 34.72  |                         | 15.99  |        |
| 2.036  | sensibilidad_intermedia           |        |                         |        |        |
|        | year_regime bar_p40_h90_tp15_sl05 |        | p40_h90 TP 1.5 / SL 0.5 |        |        |
| 35     | 50.64%                            | 95.40% | 45.70%                  | 19.64% | 26.61% |
| 27.28% | 27.69%                            | 41.52% |                         | 72.31% | 41.    |
| 52%    | 64.01%                            |        | 20.58%                  |        | 0.55   |
| 2      | 0.234                             | 37.95  |                         | 32.53  |        |
| 2.044  | sensibilidad_intermedia           |        |                         |        |        |
|        | year_regime bar_p50_h30_tp20_sl10 |        | p50_h30 TP 2.0 / SL 1.0 |        |        |
| 35     | 45.81%                            | 83.32% | 31.20%                  | 34.32% | 6.44%  |
| 14.93% | 62.36%                            | 48.84% |                         | 37.64% | 48.    |
| 84%    | 83.93%                            |        | 23.61%                  |        | 0.56   |
| 0      | 0.586                             | 17.70  |                         | 7.59   |        |
| 2.050  | sensibilidad_intermedia           |        |                         |        |        |
|        | year_regime bar_p50_h60_tp25_sl10 |        | p50_h60 TP 2.5 / SL 1.0 |        |        |
| 35     | 44.61%                            | 86.52% | 16.14%                  | 29.65% | 6.53%  |
| 16.06% | 77.33%                            | 45.65% |                         | 22.67% | 45.    |
| 65%    | 72.08%                            |        | 27.44%                  |        | 0.33   |
| 8      | 0.587                             | 34.86  |                         | 14.57  |        |
| 2.053  | sensibilidad_intermedia           |        |                         |        |        |
|        | year_regime bar_p60_h90_tp25_sl10 |        | p60_h90 TP 2.5 / SL 1.0 |        |        |
| 35     | 36.43%                            | 83.00% | 12.12%                  | 24.58% | 4.79%  |
| 14.24% | 83.08%                            | 38.79% |                         | 16.92% | 38.    |
| 79%    | 71.60%                            |        | 45.26%                  |        | 0.25   |
| 5      | 0.486                             | 55.11  |                         | 17.66  |        |
| 2.059  | sensibilidad_intermedia           |        |                         |        |        |
|        | year_regime bar_p40_h60_tp20_sl05 |        | p40_h60 TP 2.0 / SL 0.5 |        |        |
| 35     | 51.92%                            | 92.40% | 27.74%                  | 24.38% | 28.10% |
| 28.20% | 44.16%                            | 49.78% |                         | 55.84% | 49.    |
| 78%    | 50.07%                            |        | 11.98%                  |        | 0.41   |
| 4      | 0.374                             | 26.02  |                         | 16.18  |        |
| 2.067  | mayor_sensibilidad                |        |                         |        |        |
|        | year_regime bar_p60_h30_tp25_sl10 |        | p60_h30 TP 2.5 / SL 1.0 |        |        |

|        |                                   |        |                         |        |        |
|--------|-----------------------------------|--------|-------------------------|--------|--------|
| 35     | 38.10%                            | 75.74% | 14.54%                  | 26.81% | 4.85%  |
| 14.39% | 80.61%                            | 40.73% |                         | 19.39% | 40.    |
| 73%    | 76.31%                            |        | 50.00%                  |        | 0.31   |
| 5      | 0.553                             | 18.59  |                         | 7.28   |        |
| 2.077  | mayor_sensibilidad                |        |                         |        |        |
|        | year_regime bar_p40_h60_tp25_sl10 |        | p40_h60 TP 2.5 / SL 1.0 |        |        |
| 35     | 51.92%                            | 92.40% | 20.63%                  | 31.19% | 8.85%  |
| 18.35% | 70.52%                            | 49.24% |                         | 29.48% | 49.    |
| 24%    | 71.14%                            |        | 21.66%                  |        | 0.42   |
| 7      | 0.617                             | 33.47  |                         | 16.16  |        |
| 2.129  | mayor_sensibilidad                |        |                         |        |        |
|        | year_regime bar_p40_h60_tp20_sl10 |        | p40_h60 TP 2.0 / SL 1.0 |        |        |
| 35     | 51.92%                            | 92.40% | 33.98%                  | 36.02% | 8.25%  |
| 16.57% | 57.77%                            | 51.02% |                         | 42.23% | 51.    |
| 02%    | 81.80%                            |        | 16.91%                  |        | 0.59   |
| 7      | 0.574                             | 32.89  |                         | 16.95  |        |
| 2.129  | mayor_sensibilidad                |        |                         |        |        |
|        | year_regime bar_p60_h30_tp20_sl05 |        | p60_h30 TP 2.0 / SL 0.5 |        |        |
| 35     | 38.10%                            | 75.74% | 22.40%                  | 26.55% | 19.33% |
| 29.22% | 58.28%                            | 54.26% |                         | 41.72% | 54.    |
| 26%    | 53.90%                            |        | 27.89%                  |        | 0.35   |
| 1      | 0.471                             | 15.11  |                         | 9.85   |        |
| 2.137  | mayor_sensibilidad                |        |                         |        |        |
|        | year_regime bar_p40_h30_tp20_sl05 |        | p40_h30 TP 2.0 / SL 0.5 |        |        |
| 35     | 53.29%                            | 89.04% | 29.52%                  | 24.73% | 27.21% |
| 30.08% | 43.27%                            | 51.54% |                         | 56.73% | 51.    |
| 54%    | 52.32%                            |        | 18.41%                  |        | 0.45   |
| 4      | 0.397                             | 13.45  |                         | 7.19   |        |
| 2.138  | mayor_sensibilidad                |        |                         |        |        |
|        | year_regime bar_p60_h60_tp20_sl05 |        | p60_h60 TP 2.0 / SL 0.5 |        |        |
| 35     | 37.25%                            | 80.71% | 20.13%                  | 25.76% | 19.40% |
| 29.78% | 60.47%                            | 55.02% |                         | 39.53% | 55.    |
| 02%    | 51.31%                            |        | 22.81%                  |        | 0.30   |
| 6      | 0.384                             | 28.75  |                         | 13.43  |        |
| 2.141  | mayor_sensibilidad                |        |                         |        |        |
|        | year_regime bar_p40_h30_tp20_sl10 |        | p40_h30 TP 2.0 / SL 1.0 |        |        |
| 35     | 53.29%                            | 89.04% | 35.88%                  | 33.98% | 8.67%  |
| 18.54% | 55.45%                            | 51.24% |                         | 44.55% | 51.    |
| 24%    | 81.60%                            |        | 21.84%                  |        | 0.63   |
| 1      | 0.588                             | 16.63  |                         | 8.21   |        |
| 2.146  | mayor_sensibilidad                |        |                         |        |        |
|        | year_regime bar_p50_h90_tp15_sl05 |        | p50_h90 TP 1.5 / SL 0.5 |        |        |
| 35     | 43.74%                            | 91.45% | 43.15%                  | 23.96% | 22.74% |
| 28.78% | 34.11%                            | 48.39% |                         | 65.89% | 48.    |
| 39%    | 66.49%                            |        | 22.61%                  |        | 0.53   |
| 4      | 0.256                             | 41.42  |                         | 30.61  |        |
| 2.152  | mayor_sensibilidad                |        |                         |        |        |
|        | year_regime bar_p60_h90_tp15_sl05 |        | p60_h90 TP 1.5 / SL 0.5 |        |        |
| 35     | 36.43%                            | 83.00% | 40.16%                  | 30.58% | 18.91% |
| 28.22% | 40.92%                            | 51.73% |                         | 59.08% | 51.    |
| 73%    | 69.05%                            |        | 23.19%                  |        | 0.50   |
| 8      | 0.353                             | 44.65  |                         | 28.38  |        |
| 2.167  | mayor_sensibilidad                |        |                         |        |        |
|        | year_regime bar_p60_h90_tp20_sl10 |        | p60_h90 TP 2.0 / SL 1.0 |        |        |
| 35     | 36.43%                            | 83.00% | 23.13%                  | 37.79% | 4.57%  |
| 13.02% | 72.31%                            | 50.01% |                         | 27.69% | 50.    |
| 01%    | 84.32%                            |        | 34.46%                  |        | 0.41   |
| 7      | 0.634                             | 54.70  |                         | 19.75  |        |
| 2.183  | mayor_sensibilidad                |        |                         |        |        |
|        | year_regime bar_p50_h60_tp20_sl05 |        | p50_h60 TP 2.0 / SL 0.5 |        |        |

|        |                                   |        |                         |        |        |
|--------|-----------------------------------|--------|-------------------------|--------|--------|
| 35     | 44.61%                            | 86.52% | 23.91%                  | 27.99% | 23.47% |
| 30.05% | 52.63%                            | 54.77% |                         | 47.37% | 54.    |
| 77%    | 50.82%                            |        | 19.40%                  |        | 0.36   |
| 1      | 0.449                             | 27.49  |                         | 15.24  |        |
| 2.187  | mayor_sensibilidad                |        |                         |        |        |
|        | year_regime bar_p50_h30_tp20_sl05 |        | p50_h30 TP 2.0 / SL 0.5 |        |        |
| 35     | 45.81%                            | 83.32% | 26.25%                  | 24.18% | 23.21% |
| 31.68% | 50.55%                            | 52.98% |                         | 49.45% | 52.    |
| 98%    | 53.60%                            |        | 27.16%                  |        | 0.40   |
| 9      | 0.414                             | 14.38  |                         | 7.80   |        |
| 2.193  | mayor_sensibilidad                |        |                         |        |        |
|        | year_regime bar_p50_h90_tp20_sl10 |        | p50_h90 TP 2.0 / SL 1.0 |        |        |
| 35     | 43.74%                            | 91.45% | 27.96%                  | 38.98% | 6.25%  |
| 16.42% | 65.79%                            | 54.23% |                         | 34.21% | 54.    |
| 23%    | 82.79%                            |        | 18.76%                  |        | 0.49   |
| 7      | 0.645                             | 52.22  |                         | 23.82  |        |
| 2.198  | mayor_sensibilidad                |        |                         |        |        |
|        | year_regime bar_p50_h90_tp25_sl10 |        | p50_h90 TP 2.5 / SL 1.0 |        |        |
| 35     | 43.74%                            | 91.45% | 15.81%                  | 30.78% | 6.60%  |
| 17.88% | 77.59%                            | 47.48% |                         | 22.41% | 47.    |
| 48%    | 70.87%                            |        | 33.93%                  |        | 0.32   |
| 9      | 0.603                             | 52.69  |                         | 20.51  |        |
| 2.215  | mayor_sensibilidad                |        |                         |        |        |
|        | year_regime bar_p40_h30_tp25_sl10 |        | p40_h30 TP 2.5 / SL 1.0 |        |        |
| 35     | 53.29%                            | 89.04% | 22.93%                  | 32.26% | 9.26%  |
| 19.85% | 67.81%                            | 52.07% |                         | 32.19% | 52.    |
| 07%    | 72.32%                            |        | 29.73%                  |        | 0.48   |
| 1      | 0.635                             | 17.01  |                         | 7.40   |        |
| 2.229  | mayor_sensibilidad                |        |                         |        |        |
|        | year_regime bar_p40_h90_tp20_sl05 |        | p40_h90 TP 2.0 / SL 0.5 |        |        |
| 35     | 50.64%                            | 95.40% | 27.00%                  | 28.17% | 28.52% |
| 30.35% | 44.48%                            | 57.33% |                         | 55.52% | 57.    |
| 33%    | 49.06%                            |        | 12.66%                  |        | 0.39   |
| 7      | 0.418                             | 38.57  |                         | 25.99  |        |
| 2.239  | mayor_sensibilidad                |        |                         |        |        |
|        | year_regime bar_p40_h90_tp20_sl10 |        | p40_h90 TP 2.0 / SL 1.0 |        |        |
| 35     | 50.64%                            | 95.40% | 33.27%                  | 38.56% | 8.53%  |
| 18.39% | 58.20%                            | 53.09% |                         | 41.80% | 53.    |
| 09%    | 80.83%                            |        | 19.26%                  |        | 0.58   |
| 0      | 0.641                             | 49.43  |                         | 27.82  |        |
| 2.247  | mayor_sensibilidad                |        |                         |        |        |
|        | year_regime bar_p50_h30_tp25_sl10 |        | p50_h30 TP 2.5 / SL 1.0 |        |        |
| 35     | 45.81%                            | 83.32% | 18.59%                  | 31.42% | 6.89%  |
| 16.66% | 74.52%                            | 48.08% |                         | 25.48% | 48.    |
| 08%    | 73.89%                            |        | 45.33%                  |        | 0.39   |
| 6      | 0.631                             | 17.88  |                         | 9.57   |        |
| 2.248  | mayor_sensibilidad                |        |                         |        |        |
|        | year_regime bar_p60_h90_tp20_sl05 |        | p60_h90 TP 2.0 / SL 0.5 |        |        |
| 35     | 36.43%                            | 83.00% | 19.77%                  | 27.61% | 20.10% |
| 31.93% | 60.13%                            | 56.72% |                         | 39.87% | 56.    |
| 72%    | 49.58%                            |        | 28.23%                  |        | 0.29   |
| 5      | 0.412                             | 42.79  |                         | 23.11  |        |
| 2.275  | mayor_sensibilidad                |        |                         |        |        |
|        | year_regime bar_p40_h90_tp25_sl10 |        | p40_h90 TP 2.5 / SL 1.0 |        |        |
| 35     | 50.64%                            | 95.40% | 19.98%                  | 33.71% | 9.13%  |
| 20.64% | 70.89%                            | 52.87% |                         | 29.11% | 52.    |
| 87%    | 69.30%                            |        | 25.29%                  |        | 0.40   |
| 8      | 0.651                             | 50.17  |                         | 23.57  |        |
| 2.279  | mayor_sensibilidad                |        |                         |        |        |
|        | year_regime bar_p50_h90_tp20_sl05 |        | p50_h90 TP 2.0 / SL 0.5 |        |        |

|        |                    |        |        |        |        |
|--------|--------------------|--------|--------|--------|--------|
| 35     | 43.74%             | 91.45% | 23.31% | 28.63% | 24.31% |
| 32.78% | 52.37%             | 57.59% |        | 47.63% | 57.    |
| 59%    | 49.26%             |        | 23.78% |        | 0.34   |
| 5      | 0.462              | 40.83  |        | 24.35  |        |
| 2.342  | mayor_sensibilidad |        |        |        |        |

Top 15 targets con menor sensibilidad – contexto: contract

| context_type           | bar_col                | target_base          | tp_sl_config          | n_context_values            |                              |
|------------------------|------------------------|----------------------|-----------------------|-----------------------------|------------------------------|
| operable_rate_mean     | operable_rate_range    | tp_rate_mean         | tp_rate_range         | sl_rate_mean                | sl_rate_range                |
| no_event_rate_mean     | no_event_rate_range    | resolved_rate_mean   | resolved_rate_range   | tp_share_over_resolved_mean | tp_share_over_resolved_range |
| expectancy_unit_s_mean | expectancy_units_range | event_step_mean_mean | event_step_mean_range | context_sensitivity_score   | context_sensitivity_profile  |
| contract               | bar_p60_h60_tp10_sl10  | p60_h60              | TP 1.0 / SL 1.0       | 26                          |                              |
| 41.23%                 | 54.14%                 | 97.06%               | 6.50%                 | 2.94%                       | 6.50%                        |
| 0.00%                  | 0.00%                  | 100.00%              |                       | 0.00%                       |                              |
| 97.06%                 |                        | 6.50%                |                       | 0.941                       | 0.130                        |
| 30.92                  | 12.65                  |                      | 0.736                 |                             | menor_sensibilidad           |
| contract               | bar_p60_h90_tp10_sl10  | p60_h90              | TP 1.0 / SL 1.0       | 26                          |                              |
| 39.34%                 | 53.30%                 | 96.83%               | 7.37%                 | 3.17%                       | 7.37%                        |
| 0.00%                  | 0.00%                  | 100.00%              |                       | 0.00%                       |                              |
| 96.83%                 |                        | 7.37%                |                       | 0.937                       | 0.147                        |
| 47.13                  | 21.18                  |                      | 0.754                 |                             | menor_sensibilidad           |
| contract               | bar_p60_h30_tp10_sl10  | p60_h30              | TP 1.0 / SL 1.0       | 26                          |                              |
| 43.30%                 | 54.76%                 | 96.85%               | 6.93%                 | 3.15%                       | 6.93%                        |
| 0.00%                  | 0.00%                  | 100.00%              |                       | 0.00%                       |                              |
| 96.85%                 |                        | 6.93%                |                       | 0.937                       | 0.139                        |
| 15.57                  | 6.25                   |                      | 0.755                 |                             | menor_sensibilidad           |
| contract               | bar_p50_h90_tp10_sl10  | p50_h90              | TP 1.0 / SL 1.0       | 26                          |                              |
| 47.85%                 | 52.11%                 | 95.55%               | 9.30%                 | 4.45%                       | 9.30%                        |
| 0.00%                  | 0.00%                  | 100.00%              |                       | 0.00%                       |                              |
| 95.55%                 |                        | 9.30%                |                       | 0.911                       | 0.186                        |
| 43.81                  | 21.76                  |                      | 0.800                 |                             | menor_sensibilidad           |
| contract               | bar_p50_h60_tp10_sl10  | p50_h60              | TP 1.0 / SL 1.0       | 26                          |                              |
| 50.05%                 | 54.31%                 | 95.89%               | 8.69%                 | 4.11%                       | 8.69%                        |
| 0.00%                  | 0.00%                  | 100.00%              |                       | 0.00%                       |                              |
| 95.89%                 |                        | 8.69%                |                       | 0.918                       | 0.174                        |
| 28.92                  | 13.24                  |                      | 0.804                 |                             | menor_sensibilidad           |
| contract               | bar_p50_h30_tp10_sl10  | p50_h30              | TP 1.0 / SL 1.0       | 26                          |                              |
| 52.50%                 | 55.94%                 | 95.67%               | 8.80%                 | 4.33%                       | 8.80%                        |
| 0.00%                  | 0.00%                  | 100.00%              |                       | 0.00%                       |                              |
| 95.67%                 |                        | 8.80%                |                       | 0.913                       | 0.176                        |
| 14.55                  | 6.49                   |                      | 0.823                 |                             | menor_sensibilidad           |
| contract               | bar_p40_h90_tp10_sl10  | p40_h90              | TP 1.0 / SL 1.0       | 26                          |                              |
| 56.05%                 | 49.17%                 | 93.97%               | 11.15%                | 6.03%                       | 11.15%                       |
| 0.00%                  | 0.00%                  | 100.00%              |                       | 0.00%                       |                              |
| 93.97%                 |                        | 11.15%               |                       | 0.879                       | 0.223                        |
| 40.35                  | 22.30                  |                      | 0.826                 |                             | menor_sensibilidad           |
| contract               | bar_p40_h60_tp10_sl10  | p40_h60              | TP 1.0 / SL 1.0       | 26                          |                              |
| 58.91%                 | 52.47%                 | 94.36%               | 10.55%                | 5.64%                       | 10.55%                       |
| 0.00%                  | 0.00%                  | 100.00%              |                       | 0.00%                       |                              |
| 94.36%                 |                        | 10.55%               |                       | 0.887                       | 0.211                        |
| 26.70                  | 13.69                  |                      | 0.841                 |                             | menor_sensibilidad           |
| contract               | bar_p40_h30_tp10_sl10  | p40_h30              | TP 1.0 / SL 1.0       | 26                          |                              |
| 61.51%                 | 53.55%                 | 94.18%               | 10.30%                | 5.82%                       | 10.30%                       |
| 0.00%                  | 0.00%                  | 100.00%              |                       | 0.00%                       |                              |
| 94.18%                 |                        | 10.30%               |                       | 0.884                       | 0.206                        |
| 13.45                  | 6.71                   |                      | 0.845                 |                             | menor_sensibilidad           |
| contract               | bar_p60_h30_tp10_sl05  | p60_h30              | TP 1.0 / SL 0.5       | 26                          |                              |
| 43.30%                 | 54.76%                 | 86.03%               | 17.75%                | 13.97%                      | 17.75%                       |

|                                |        |                         |        |                    |
|--------------------------------|--------|-------------------------|--------|--------------------|
| 0.00%                          | 0.00%  | 100.00%                 | 0.00%  |                    |
| 86.03%                         |        | 17.75%                  | 0.791  | 0.266              |
| 14.05                          | 7.73   |                         | 1.080  | menor_sensibilidad |
| contract bar_p50_h30_tp10_sl05 |        | p50_h30 TP 1.0 / SL 0.5 |        | 26                 |
| 52.50%                         | 55.94% | 82.92%                  | 18.53% | 17.08%             |
| 0.00%                          | 0.00%  | 100.00%                 | 0.00%  |                    |
| 82.92%                         |        | 18.53%                  | 0.744  | 0.278              |
| 12.78                          | 7.67   |                         | 1.115  | menor_sensibilidad |
| contract bar_p40_h30_tp10_sl05 |        | p40_h30 TP 1.0 / SL 0.5 |        | 26                 |
| 61.51%                         | 53.55% | 79.74%                  | 19.41% | 20.26%             |
| 0.00%                          | 0.00%  | 100.00%                 | 0.00%  |                    |
| 79.74%                         |        | 19.41%                  | 0.696  | 0.291              |
| 11.48                          | 7.67   |                         | 1.118  | menor_sensibilidad |
| contract bar_p60_h90_tp10_sl05 |        | p60_h90 TP 1.0 / SL 0.5 |        | 26                 |
| 39.34%                         | 53.30% | 84.64%                  | 19.93% | 15.36%             |
| 0.00%                          | 0.00%  | 100.00%                 | 0.00%  |                    |
| 84.64%                         |        | 19.93%                  | 0.770  | 0.299              |
| 41.97                          | 26.11  |                         | 1.131  | menor_sensibilidad |
| contract bar_p60_h60_tp10_sl05 |        | p60_h60 TP 1.0 / SL 0.5 |        | 26                 |
| 41.23%                         | 54.14% | 85.70%                  | 21.31% | 14.30%             |
| 0.00%                          | 0.00%  | 100.00%                 | 0.00%  |                    |
| 85.70%                         |        | 21.31%                  | 0.785  | 0.320              |
| 27.76                          | 16.44  |                         | 1.181  | menor_sensibilidad |
| contract bar_p50_h90_tp10_sl05 |        | p50_h90 TP 1.0 / SL 0.5 |        | 26                 |
| 47.85%                         | 52.11% | 81.41%                  | 22.29% | 18.59%             |
| 0.00%                          | 0.00%  | 100.00%                 | 0.00%  |                    |
| 81.41%                         |        | 22.29%                  | 0.721  | 0.334              |
| 37.97                          | 26.81  |                         | 1.190  | menor_sensibilidad |

Top 15 targets con mayor sensibilidad – contexto: contract

|                                |                        |                         |                       |                             |                              |
|--------------------------------|------------------------|-------------------------|-----------------------|-----------------------------|------------------------------|
| context_type                   | bar_col                | target_base             | tp_sl_config          | n_context_values            |                              |
| operable_rate_mean             | operable_rate_range    | tp_rate_mean            | tp_rate_range         | sl_rate_mean                | sl_rate_range                |
| no_event_rate_mean             | no_event_rate_range    | resolved_rate_mean      | resolved_rate_range   | tp_share_over_resolved_mean | tp_share_over_resolved_range |
| expectancy_unit_s_mean         | expectancy_units_range | event_step_mean_mean    | event_step_mean_range | context_sensitivity_score   | context_sensitivity_profile  |
| contract bar_p60_h60_tp15_sl05 |                        | p60_h60 TP 1.5 / SL 0.5 |                       |                             | 26                           |
| 41.23%                         | 54.14%                 | 39.84%                  | 24.70%                | 17.30%                      | 25.11%                       |
| 42.86%                         | 48.71%                 |                         | 57.14%                | 48.71%                      |                              |
| 71.28%                         |                        | 21.50%                  | 0.511                 |                             | 0.250                        |
| 29.82                          | 15.18                  |                         | 1.742                 |                             | mayor_sensibilidad           |
| contract bar_p50_h60_tp20_sl05 |                        | p50_h60 TP 2.0 / SL 0.5 |                       |                             | 26                           |
| 50.05%                         | 54.31%                 | 23.32%                  | 24.08%                | 22.42%                      | 29.08%                       |
| 54.26%                         | 53.15%                 |                         | 45.74%                | 53.15%                      |                              |
| 51.91%                         |                        | 11.93%                  | 0.354                 |                             | 0.336                        |
| 27.31                          | 11.56                  |                         | 1.726                 |                             | mayor_sensibilidad           |
| contract bar_p40_h90_tp20_sl05 |                        | p40_h90 TP 2.0 / SL 0.5 |                       |                             | 26                           |
| 56.05%                         | 49.17%                 | 26.57%                  | 24.89%                | 28.01%                      | 30.69%                       |
| 45.41%                         | 54.76%                 |                         | 54.59%                | 54.76%                      |                              |
| 49.28%                         |                        | 11.40%                  | 0.391                 |                             | 0.362                        |
| 39.01                          | 20.13                  |                         | 1.709                 |                             | mayor_sensibilidad           |
| contract bar_p50_h90_tp15_sl05 |                        | p50_h90 TP 1.5 / SL 0.5 |                       |                             | 26                           |
| 47.85%                         | 52.11%                 | 42.58%                  | 23.12%                | 22.31%                      | 26.45%                       |
| 35.12%                         | 48.36%                 |                         | 64.88%                | 48.36%                      |                              |
| 66.69%                         |                        | 19.88%                  | 0.527                 |                             | 0.221                        |
| 42.10                          | 25.53                  |                         | 1.699                 |                             | mayor_sensibilidad           |
| contract bar_p60_h60_tp20_sl05 |                        | p60_h60 TP 2.0 / SL 0.5 |                       |                             | 26                           |
| 41.23%                         | 54.14%                 | 19.98%                  | 21.86%                | 18.34%                      | 27.93%                       |
| 61.69%                         | 49.34%                 |                         | 38.31%                | 49.34%                      |                              |
| 53.60%                         |                        | 16.32%                  | 0.308                 |                             | 0.323                        |

|                                |                         |        |                    |
|--------------------------------|-------------------------|--------|--------------------|
| 28.64                          | 11.47                   | 1.696  | mayor_sensibilidad |
| contract bar_p40_h60_tp20_sl05 | p40_h60 TP 2.0 / SL 0.5 | 26     |                    |
| 58.91%                         | 52.47%                  | 26.93% | 23.54%             |
| 45.94%                         | 53.53%                  | 54.06% | 53.53%             |
| 50.43%                         | 9.86%                   | 0.403  | 0.321              |
| 25.69                          | 11.85                   | 1.694  | mayor_sensibilidad |
| contract bar_p50_h90_tp20_sl05 | p50_h90 TP 2.0 / SL 0.5 | 26     |                    |
| 47.85%                         | 52.11%                  | 23.29% | 22.26%             |
| 52.95%                         | 50.75%                  | 47.05% | 50.75%             |
| 50.33%                         | 14.08%                  | 0.347  | 0.313              |
| 41.67                          | 20.07                   | 1.694  | mayor_sensibilidad |
| contract bar_p50_h60_tp15_sl05 | p50_h60 TP 1.5 / SL 0.5 | 26     |                    |
| 50.05%                         | 54.31%                  | 42.95% | 19.63%             |
| 35.98%                         | 46.41%                  | 64.02% | 46.41%             |
| 68.30%                         | 21.96%                  | 0.539  | 0.167              |
| 27.73                          | 15.19                   | 1.691  | mayor_sensibilidad |
| contract bar_p60_h90_tp20_sl05 | p60_h90 TP 2.0 / SL 0.5 | 26     |                    |
| 39.34%                         | 53.30%                  | 20.15% | 21.46%             |
| 60.22%                         | 48.63%                  | 39.78% | 48.63%             |
| 51.66%                         | 17.48%                  | 0.305  | 0.322              |
| 44.07                          | 18.29                   | 1.687  | mayor_sensibilidad |
| contract bar_p40_h60_tp20_sl10 | p40_h60 TP 2.0 / SL 1.0 | 26     |                    |
| 58.91%                         | 52.47%                  | 32.83% | 34.90%             |
| 59.38%                         | 48.49%                  | 40.62% | 48.49%             |
| 82.48%                         | 15.95%                  | 0.579  | 0.562              |
| 32.64                          | 13.98                   | 1.661  | mayor_sensibilidad |
| contract bar_p60_h90_tp15_sl05 | p60_h90 TP 1.5 / SL 0.5 | 26     |                    |
| 39.34%                         | 53.30%                  | 40.02% | 23.61%             |
| 41.48%                         | 43.93%                  | 58.52% | 43.93%             |
| 69.62%                         | 20.36%                  | 0.508  | 0.257              |
| 45.51                          | 23.86                   | 1.660  | mayor_sensibilidad |
| contract bar_p40_h90_tp20_sl10 | p40_h90 TP 2.0 / SL 1.0 | 26     |                    |
| 56.05%                         | 49.17%                  | 32.85% | 36.08%             |
| 58.83%                         | 47.54%                  | 41.17% | 47.54%             |
| 81.17%                         | 16.73%                  | 0.574  | 0.607              |
| 50.09                          | 22.17                   | 1.646  | mayor_sensibilidad |
| contract bar_p40_h90_tp15_sl05 | p40_h90 TP 1.5 / SL 0.5 | 26     |                    |
| 56.05%                         | 49.17%                  | 44.88% | 17.27%             |
| 28.91%                         | 45.01%                  | 71.09% | 45.01%             |
| 64.00%                         | 20.66%                  | 0.542  | 0.123              |
| 38.44                          | 25.36                   | 1.598  | mayor_sensibilidad |
| contract bar_p60_h30_tp20_sl05 | p60_h30 TP 2.0 / SL 0.5 | 26     |                    |
| 43.30%                         | 54.76%                  | 21.63% | 22.17%             |
| 60.45%                         | 45.66%                  | 39.55% | 45.66%             |
| 55.64%                         | 12.36%                  | 0.343  | 0.336              |
| 14.80                          | 5.19                    | 1.590  | mayor_sensibilidad |
| contract bar_p50_h60_tp20_sl10 | p50_h60 TP 2.0 / SL 1.0 | 26     |                    |
| 50.05%                         | 54.31%                  | 27.79% | 32.91%             |
| 66.50%                         | 43.19%                  | 33.50% | 43.19%             |
| 84.68%                         | 15.30%                  | 0.499  | 0.561              |
| 34.40                          | 12.97                   | 1.576  | mayor_sensibilidad |

Top 15 targets con menor sensibilidad – contexto: contract\_code

| context_type                | bar_col                      | target_base                 | tp_sl_config           | n_context_values    |
|-----------------------------|------------------------------|-----------------------------|------------------------|---------------------|
| operable_rate_mean          | operable_rate_range          | tp_rate_mean                | tp_rate_range          | sl_rate_mean        |
| sl_rate_range               | no_event_rate_mean           | no_event_rate_range         | resolved_rate_mean     | resolved_rate_range |
| tp_share_over_resolved_mean | tp_share_over_resolved_range | expectancy_unit_s_mean      | expectancy_units_range | event_step_mean     |
| event_step_mean_range       | context_sensitivity_score    | context_sensitivity_profile |                        |                     |
| contract_code               | bar_p60_h60_tp10_sl10        | p60_h60                     | TP 1.0 / SL 1.0        | 4                   |

|                                     |        |         |                 |                    |       |
|-------------------------------------|--------|---------|-----------------|--------------------|-------|
| 40.24%                              | 10.68% | 96.78%  | 2.92%           | 3.22%              | 2.92% |
| 0.00%                               | 0.00%  | 100.00% |                 | 0.00%              |       |
| 96.78%                              |        | 2.92%   | 0.936           |                    | 0.058 |
| 30.28                               | 4.80   |         | 0.195           | menor_sensibilidad |       |
| contract_code bar_p60_h90_tp10_sl10 |        | p60_h90 | TP 1.0 / SL 1.0 |                    | 4     |
| 38.35%                              | 10.52% | 96.54%  | 3.14%           | 3.46%              | 3.14% |
| 0.00%                               | 0.00%  | 100.00% |                 | 0.00%              |       |
| 96.54%                              |        | 3.14%   | 0.931           |                    | 0.063 |
| 46.05                               | 6.92   |         | 0.199           | menor_sensibilidad |       |
| contract_code bar_p60_h30_tp10_sl10 |        | p60_h30 | TP 1.0 / SL 1.0 |                    | 4     |
| 42.30%                              | 11.20% | 96.56%  | 2.91%           | 3.44%              | 2.91% |
| 0.00%                               | 0.00%  | 100.00% |                 | 0.00%              |       |
| 96.56%                              |        | 2.91%   | 0.931           |                    | 0.058 |
| 15.26                               | 2.38   |         | 0.199           | menor_sensibilidad |       |
| contract_code bar_p40_h60_tp10_sl10 |        | p40_h60 | TP 1.0 / SL 1.0 |                    | 4     |
| 58.20%                              | 7.19%  | 94.14%  | 4.41%           | 5.86%              | 4.41% |
| 0.00%                               | 0.00%  | 100.00% |                 | 0.00%              |       |
| 94.14%                              |        | 4.41%   | 0.883           |                    | 0.088 |
| 26.34                               | 4.82   |         | 0.204           | menor_sensibilidad |       |
| contract_code bar_p50_h60_tp10_sl10 |        | p50_h60 | TP 1.0 / SL 1.0 |                    | 4     |
| 49.18%                              | 9.25%  | 95.61%  | 3.77%           | 4.39%              | 3.77% |
| 0.00%                               | 0.00%  | 100.00% |                 | 0.00%              |       |
| 95.61%                              |        | 3.77%   | 0.912           |                    | 0.075 |
| 28.42                               | 4.87   |         | 0.206           | menor_sensibilidad |       |
| contract_code bar_p40_h30_tp10_sl10 |        | p40_h30 | TP 1.0 / SL 1.0 |                    | 4     |
| 60.80%                              | 8.10%  | 93.99%  | 4.19%           | 6.01%              | 4.19% |
| 0.00%                               | 0.00%  | 100.00% |                 | 0.00%              |       |
| 93.99%                              |        | 4.19%   | 0.880           |                    | 0.084 |
| 13.28                               | 2.32   |         | 0.207           | menor_sensibilidad |       |
| contract_code bar_p50_h30_tp10_sl10 |        | p50_h30 | TP 1.0 / SL 1.0 |                    | 4     |
| 51.61%                              | 9.77%  | 95.41%  | 3.67%           | 4.59%              | 3.67% |
| 0.00%                               | 0.00%  | 100.00% |                 | 0.00%              |       |
| 95.41%                              |        | 3.67%   | 0.908           |                    | 0.073 |
| 14.31                               | 2.38   |         | 0.208           | menor_sensibilidad |       |
| contract_code bar_p50_h90_tp10_sl10 |        | p50_h90 | TP 1.0 / SL 1.0 |                    | 4     |
| 46.99%                              | 8.83%  | 95.28%  | 4.05%           | 4.72%              | 4.05% |
| 0.00%                               | 0.00%  | 100.00% |                 | 0.00%              |       |
| 95.28%                              |        | 4.05%   | 0.906           |                    | 0.081 |
| 42.99                               | 7.24   |         | 0.210           | menor_sensibilidad |       |
| contract_code bar_p40_h90_tp10_sl10 |        | p40_h90 | TP 1.0 / SL 1.0 |                    | 4     |
| 55.37%                              | 6.72%  | 93.72%  | 4.81%           | 6.28%              | 4.81% |
| 0.00%                               | 0.00%  | 100.00% |                 | 0.00%              |       |
| 93.72%                              |        | 4.81%   | 0.874           |                    | 0.096 |
| 39.75                               | 7.29   |         | 0.211           | menor_sensibilidad |       |
| contract_code bar_p40_h30_tp10_sl05 |        | p40_h30 | TP 1.0 / SL 0.5 |                    | 4     |
| 60.80%                              | 8.10%  | 79.20%  | 6.40%           | 20.80%             | 6.40% |
| 0.00%                               | 0.00%  | 100.00% |                 | 0.00%              |       |
| 79.20%                              |        | 6.40%   | 0.688           |                    | 0.096 |
| 11.26                               | 2.46   |         | 0.273           | menor_sensibilidad |       |
| contract_code bar_p40_h60_tp10_sl05 |        | p40_h60 | TP 1.0 / SL 0.5 |                    | 4     |
| 58.20%                              | 7.19%  | 78.18%  | 7.42%           | 21.82%             | 7.42% |
| 0.00%                               | 0.00%  | 100.00% |                 | 0.00%              |       |
| 78.18%                              |        | 7.42%   | 0.673           |                    | 0.111 |
| 22.12                               | 5.19   |         | 0.294           | menor_sensibilidad |       |
| contract_code bar_p50_h30_tp10_sl05 |        | p50_h30 | TP 1.0 / SL 0.5 |                    | 4     |
| 51.61%                              | 9.77%  | 82.21%  | 6.81%           | 17.79%             | 6.81% |
| 0.00%                               | 0.00%  | 100.00% |                 | 0.00%              |       |
| 82.21%                              |        | 6.81%   | 0.733           |                    | 0.102 |
| 12.48                               | 2.69   |         | 0.302           | menor_sensibilidad |       |
| contract_code bar_p40_h90_tp10_sl05 |        | p40_h90 | TP 1.0 / SL 0.5 |                    | 4     |



|               |                       |         |                 |                    |       |
|---------------|-----------------------|---------|-----------------|--------------------|-------|
| 55.37%        | 6.72%                 | 77.36%  | 8.09%           | 22.64%             | 8.09% |
| 0.00%         | 0.00%                 | 100.00% |                 | 0.00%              |       |
| 77.36%        |                       | 8.09%   | 0.660           |                    | 0.121 |
| 33.18         | 7.93                  |         | 0.310           | menor_sensibilidad |       |
| contract_code | bar_p60_h30_tp10_sl05 | p60_h30 | TP 1.0 / SL 0.5 |                    | 4     |
| 42.30%        | 11.20%                | 85.15%  | 7.00%           | 14.85%             | 7.00% |
| 0.00%         | 0.00%                 | 100.00% |                 | 0.00%              |       |
| 85.15%        |                       | 7.00%   | 0.777           |                    | 0.105 |
| 13.66         | 2.86                  |         | 0.322           | menor_sensibilidad |       |
| contract_code | bar_p50_h90_tp10_sl05 | p50_h90 | TP 1.0 / SL 0.5 |                    | 4     |
| 46.99%        | 8.83%                 | 80.57%  | 8.25%           | 19.43%             | 8.25% |
| 0.00%         | 0.00%                 | 100.00% |                 | 0.00%              |       |
| 80.57%        |                       | 8.25%   | 0.709           |                    | 0.124 |
| 36.92         | 8.52                  |         | 0.336           | menor_sensibilidad |       |

Top 15 targets con mayor sensibilidad – contexto: contract\_code

| context_type           | bar_col                | target_base          | tp_sl_config          | n_context_values            |                              |
|------------------------|------------------------|----------------------|-----------------------|-----------------------------|------------------------------|
| operable_rate_mean     | operable_rate_range    | tp_rate_mean         | tp_rate_range         | sl_rate_mean                | sl_rate_range                |
| no_event_rate_mean     | no_event_rate_range    | resolved_rate_mean   | resolved_rate_range   | tp_share_over_resolved_mean | tp_share_over_resolved_range |
| expectancy_unit_s_mean | expectancy_units_range | event_step_mean_mean | event_step_mean_range | context_sensitivity_score   | context_sensitivity_profile  |
| contract_code          | bar_p60_h60_tp20_sl05  | p60_h60              | TP 2.0 / SL 0.5       |                             | 4                            |
| 40.24%                 | 10.68%                 | 21.23%               | 7.56%                 | 19.67%                      | 11.17%                       |
| 59.10%                 | 18.73%                 | 40.90%               |                       | 18.73%                      |                              |
| 52.25%                 |                        | 5.33%                | 0.326                 |                             | 0.095                        |
| 27.55                  | 3.93                   |                      | 0.535                 | mayor_sensibilidad          |                              |
| contract_code          | bar_p60_h60_tp15_sl05  | p60_h60              | TP 1.5 / SL 0.5       |                             | 4                            |
| 40.24%                 | 10.68%                 | 41.10%               | 6.69%                 | 18.57%                      | 9.99%                        |
| 40.32%                 | 16.68%                 | 59.68%               |                       | 16.68%                      |                              |
| 69.18%                 |                        | 8.07%                | 0.524                 |                             | 0.050                        |
| 28.50                  | 5.52                   |                      | 0.521                 | mayor_sensibilidad          |                              |
| contract_code          | bar_p60_h90_tp20_sl05  | p60_h90              | TP 2.0 / SL 0.5       |                             | 4                            |
| 38.35%                 | 10.52%                 | 21.51%               | 6.10%                 | 20.85%                      | 10.87%                       |
| 57.64%                 | 16.97%                 | 42.36%               |                       | 16.97%                      |                              |
| 51.11%                 |                        | 5.99%                | 0.326                 |                             | 0.068                        |
| 42.34                  | 7.25                   |                      | 0.504                 | mayor_sensibilidad          |                              |
| contract_code          | bar_p60_h30_tp20_sl05  | p60_h30              | TP 2.0 / SL 0.5       |                             | 4                            |
| 42.30%                 | 11.20%                 | 22.70%               | 8.09%                 | 19.10%                      | 9.38%                        |
| 58.20%                 | 17.47%                 | 41.80%               |                       | 17.47%                      |                              |
| 54.50%                 |                        | 3.27%                | 0.358                 |                             | 0.115                        |
| 14.27                  | 1.78                   |                      | 0.494                 | mayor_sensibilidad          |                              |
| contract_code          | bar_p50_h60_tp20_sl05  | p50_h60              | TP 2.0 / SL 0.5       |                             | 4                            |
| 49.18%                 | 9.25%                  | 24.26%               | 7.77%                 | 23.55%                      | 10.57%                       |
| 52.19%                 | 18.34%                 | 47.81%               |                       | 18.34%                      |                              |
| 50.91%                 |                        | 3.18%                | 0.367                 |                             | 0.102                        |
| 26.32                  | 4.05                   |                      | 0.491                 | mayor_sensibilidad          |                              |
| contract_code          | bar_p50_h60_tp20_sl10  | p50_h60              | TP 2.0 / SL 1.0       |                             | 4                            |
| 49.18%                 | 9.25%                  | 29.03%               | 11.27%                | 6.07%                       | 5.21%                        |
| 64.90%                 | 16.48%                 | 35.10%               |                       | 16.48%                      |                              |
| 83.14%                 |                        | 6.83%                | 0.520                 |                             | 0.173                        |
| 33.15                  | 4.54                   |                      | 0.490                 | mayor_sensibilidad          |                              |
| contract_code          | bar_p60_h90_tp15_sl05  | p60_h90              | TP 1.5 / SL 0.5       |                             | 4                            |
| 38.35%                 | 10.52%                 | 41.34%               | 4.86%                 | 19.65%                      | 10.06%                       |
| 39.01%                 | 14.92%                 | 60.99%               |                       | 14.92%                      |                              |
| 68.06%                 |                        | 8.59%                | 0.522                 |                             | 0.035                        |
| 43.38                  | 8.37                   |                      | 0.489                 | mayor_sensibilidad          |                              |
| contract_code          | bar_p40_h60_tp20_sl10  | p40_h60              | TP 2.0 / SL 1.0       |                             | 4                            |
| 58.20%                 | 7.19%                  | 33.73%               | 10.73%                | 8.11%                       | 5.94%                        |
| 58.16%                 | 16.67%                 | 41.84%               |                       | 16.67%                      |                              |

|               |                        |                         |        |                    |
|---------------|------------------------|-------------------------|--------|--------------------|
| 80.98%        |                        | 6.53%                   | 0.594  | 0.155              |
| 31.49         | 5.20                   |                         | 0.471  | mayor_sensibilidad |
| contract_code | bar_p60_h30_tp15_sl105 | p60_h30 TP 1.5 / SL 0.5 |        | 4                  |
| 42.30%        | 11.20%                 | 42.76%                  | 6.16%  | 17.99%             |
| 39.26%        | 14.56%                 | 60.74%                  |        | 14.56%             |
| 70.61%        |                        | 6.70%                   | 0.551  | 0.050              |
| 14.57         | 2.64                   |                         | 0.470  | mayor_sensibilidad |
| contract_code | bar_p50_h30_tp20_sl110 | p50_h30 TP 2.0 / SL 1.0 |        | 4                  |
| 51.61%        | 9.77%                  | 30.63%                  | 10.67% | 6.29%              |
| 63.08%        | 15.53%                 | 36.92%                  |        | 15.53%             |
| 83.32%        |                        | 6.09%                   | 0.550  | 0.165              |
| 16.74         | 2.45                   |                         | 0.469  | mayor_sensibilidad |
| contract_code | bar_p50_h90_tp20_sl110 | p50_h90 TP 2.0 / SL 1.0 |        | 4                  |
| 46.99%        | 8.83%                  | 29.38%                  | 9.75%  | 6.54%              |
| 64.08%        | 15.24%                 | 35.92%                  |        | 15.24%             |
| 82.24%        |                        | 7.57%                   | 0.522  | 0.140              |
| 50.86         | 7.53                   |                         | 0.469  | mayor_sensibilidad |
| contract_code | bar_p40_h90_tp20_sl110 | p40_h90 TP 2.0 / SL 1.0 |        | 4                  |
| 55.37%        | 6.72%                  | 33.77%                  | 10.03% | 8.62%              |
| 57.60%        | 16.42%                 | 42.40%                  |        | 16.42%             |
| 80.04%        |                        | 7.14%                   | 0.589  | 0.137              |
| 48.17         | 7.97                   |                         | 0.467  | mayor_sensibilidad |
| contract_code | bar_p50_h90_tp20_sl105 | p50_h90 TP 2.0 / SL 0.5 |        | 4                  |
| 46.99%        | 8.83%                  | 24.27%                  | 6.12%  | 24.81%             |
| 50.92%        | 16.74%                 | 49.08%                  |        | 16.74%             |
| 49.66%        |                        | 4.32%                   | 0.361  | 0.069              |
| 39.91         | 7.00                   |                         | 0.466  | mayor_sensibilidad |
| contract_code | bar_p60_h30_tp20_sl110 | p60_h30 TP 2.0 / SL 1.0 |        | 4                  |
| 42.30%        | 11.20%                 | 26.22%                  | 10.85% | 4.74%              |
| 69.05%        | 14.80%                 | 30.95%                  |        | 14.80%             |
| 85.09%        |                        | 5.78%                   | 0.477  | 0.178              |
| 17.49         | 2.03                   |                         | 0.466  | mayor_sensibilidad |
| contract_code | bar_p60_h60_tp20_sl110 | p60_h60 TP 2.0 / SL 1.0 |        | 4                  |
| 40.24%        | 10.68%                 | 24.94%                  | 10.53% | 4.50%              |
| 70.55%        | 14.62%                 | 29.45%                  |        | 14.62%             |
| 85.17%        |                        | 6.37%                   | 0.454  | 0.170              |
| 34.58         | 3.91                   |                         | 0.463  | mayor_sensibilidad |

Top 15 targets con menor sensibilidad – contexto: year

| context_type                | bar_col                      | target_base               | tp_sl_config                | n_context_values       |
|-----------------------------|------------------------------|---------------------------|-----------------------------|------------------------|
| operable_rate_mean          | operable_rate_range          | tp_rate_mean              | tp_rate_range               | sl_rate_mean           |
| sl_rate_range               | no_event_rate_mean           | no_event_rate_range       | resolved_rate_mean          | resolved_rate_range    |
| tp_share_over_resolved_mean | tp_share_over_resolved_range | expectancy_unit           | s_mean                      | expectancy_units_range |
| event_step_mean_mean        | event_step_mean_range        | context_sensitivity_score | context_sensitivity_profile |                        |
| year                        | bar_p60_h90_tp10_sl110       | p60_h90 TP 1.0 / SL 1.0   |                             | 7                      |
| 41.19%                      | 38.54%                       | 96.39%                    | 5.52%                       | 3.61%                  |
| 0.00%                       | 0.00%                        | 100.00%                   |                             | 0.00%                  |
| 96.39%                      |                              | 5.52%                     | 0.928                       | 0.110                  |
| 45.86                       | 16.67                        |                           | 0.551                       | menor_sensibilidad     |
| year                        | bar_p60_h60_tp10_sl110       | p60_h60 TP 1.0 / SL 1.0   |                             | 7                      |
| 43.17%                      | 40.07%                       | 96.65%                    | 5.57%                       | 3.35%                  |
| 0.00%                       | 0.00%                        | 100.00%                   |                             | 0.00%                  |
| 96.65%                      |                              | 5.57%                     | 0.933                       | 0.111                  |
| 30.15                       | 10.31                        |                           | 0.568                       | menor_sensibilidad     |
| year                        | bar_p60_h30_tp10_sl110       | p60_h30 TP 1.0 / SL 1.0   |                             | 7                      |
| 45.24%                      | 40.22%                       | 96.42%                    | 5.76%                       | 3.58%                  |
| 0.00%                       | 0.00%                        | 100.00%                   |                             | 0.00%                  |
| 96.42%                      |                              | 5.76%                     | 0.928                       | 0.115                  |
| 15.18                       | 5.08                         |                           | 0.575                       | menor_sensibilidad     |

|        |      |                       |         |                 |                    |
|--------|------|-----------------------|---------|-----------------|--------------------|
|        | year | bar_p50_h90_tp10_sl10 | p50_h90 | TP 1.0 / SL 1.0 | 7                  |
| 49.60% |      | 37.14%                | 95.04%  | 7.81%           | 4.96%              |
| 0.00%  |      | 0.00%                 | 100.00% |                 | 0.00%              |
| 95.04% |      | 7.81%                 |         | 0.901           | 0.156              |
| 42.64  |      | 17.44                 |         | 0.606           | menor_sensibilidad |
|        | year | bar_p50_h30_tp10_sl10 | p50_h30 | TP 1.0 / SL 1.0 | 7                  |
| 54.33% |      | 39.37%                | 95.15%  | 7.74%           | 4.85%              |
| 0.00%  |      | 0.00%                 | 100.00% |                 | 0.00%              |
| 95.15% |      | 7.74%                 |         | 0.903           | 0.155              |
| 14.18  |      | 5.40                  |         | 0.626           | menor_sensibilidad |
|        | year | bar_p50_h60_tp10_sl10 | p50_h60 | TP 1.0 / SL 1.0 | 7                  |
| 51.87% |      | 38.80%                | 95.36%  | 7.94%           | 4.64%              |
| 0.00%  |      | 0.00%                 | 100.00% |                 | 0.00%              |
| 95.36% |      | 7.94%                 |         | 0.907           | 0.159              |
| 28.18  |      | 11.00                 |         | 0.626           | menor_sensibilidad |
|        | year | bar_p40_h90_tp10_sl10 | p40_h90 | TP 1.0 / SL 1.0 | 7                  |
| 57.63% |      | 34.23%                | 93.36%  | 9.94%           | 6.64%              |
| 0.00%  |      | 0.00%                 | 100.00% |                 | 0.00%              |
| 93.36% |      | 9.94%                 |         | 0.867           | 0.199              |
| 39.25  |      | 17.86                 |         | 0.640           | menor_sensibilidad |
|        | year | bar_p40_h30_tp10_sl10 | p40_h30 | TP 1.0 / SL 1.0 | 7                  |
| 63.17% |      | 36.43%                | 93.59%  | 9.72%           | 6.41%              |
| 0.00%  |      | 0.00%                 | 100.00% |                 | 0.00%              |
| 93.59% |      | 9.72%                 |         | 0.872           | 0.194              |
| 13.10  |      | 5.61                  |         | 0.656           | menor_sensibilidad |
|        | year | bar_p40_h60_tp10_sl10 | p40_h60 | TP 1.0 / SL 1.0 | 7                  |
| 60.55% |      | 36.00%                | 93.76%  | 9.99%           | 6.24%              |
| 0.00%  |      | 0.00%                 | 100.00% |                 | 0.00%              |
| 93.76% |      | 9.99%                 |         | 0.875           | 0.200              |
| 26.00  |      | 11.42                 |         | 0.660           | menor_sensibilidad |
|        | year | bar_p40_h30_tp10_sl05 | p40_h30 | TP 1.0 / SL 0.5 | 7                  |
| 63.17% |      | 36.43%                | 78.74%  | 15.61%          | 21.26%             |
| 0.00%  |      | 0.00%                 | 100.00% |                 | 0.00%              |
| 78.74% |      | 15.61%                |         | 0.681           | 0.234              |
| 11.09  |      | 6.06                  |         | 0.833           | menor_sensibilidad |
|        | year | bar_p60_h30_tp10_sl05 | p60_h30 | TP 1.0 / SL 0.5 | 7                  |
| 45.24% |      | 40.22%                | 84.90%  | 15.47%          | 15.10%             |
| 0.00%  |      | 0.00%                 | 100.00% |                 | 0.00%              |
| 84.90% |      | 15.47%                |         | 0.773           | 0.232              |
| 13.57  |      | 6.25                  |         | 0.866           | menor_sensibilidad |
|        | year | bar_p50_h30_tp10_sl05 | p50_h30 | TP 1.0 / SL 0.5 | 7                  |
| 54.33% |      | 39.37%                | 81.86%  | 15.77%          | 18.14%             |
| 0.00%  |      | 0.00%                 | 100.00% |                 | 0.00%              |
| 81.86% |      | 15.77%                |         | 0.728           | 0.237              |
| 12.35  |      | 6.25                  |         | 0.867           | menor_sensibilidad |
|        | year | bar_p60_h90_tp10_sl05 | p60_h90 | TP 1.0 / SL 0.5 | 7                  |
| 41.19% |      | 38.54%                | 83.43%  | 17.15%          | 16.57%             |
| 0.00%  |      | 0.00%                 | 100.00% |                 | 0.00%              |
| 83.43% |      | 17.15%                |         | 0.751           | 0.257              |
| 40.43  |      | 20.64                 |         | 0.900           | menor_sensibilidad |
|        | year | bar_p40_h60_tp10_sl05 | p40_h60 | TP 1.0 / SL 0.5 | 7                  |
| 60.55% |      | 36.00%                | 77.68%  | 18.64%          | 22.32%             |
| 0.00%  |      | 0.00%                 | 100.00% |                 | 0.00%              |
| 77.68% |      | 18.64%                |         | 0.665           | 0.280              |
| 21.80  |      | 12.64                 |         | 0.919           | menor_sensibilidad |
|        | year | bar_p40_h90_tp10_sl05 | p40_h90 | TP 1.0 / SL 0.5 | 7                  |
| 57.63% |      | 34.23%                | 76.84%  | 19.57%          | 23.16%             |
| 0.00%  |      | 0.00%                 | 100.00% |                 | 0.00%              |
| 76.84% |      | 19.57%                |         | 0.653           | 0.294              |
| 32.70  |      | 20.17                 |         | 0.929           | menor_sensibilidad |

Top 15 targets con mayor sensibilidad – contexto: year

| context_type                | bar_col                      | target_base                 | tp_sl_config           | n_context_values     |
|-----------------------------|------------------------------|-----------------------------|------------------------|----------------------|
| operable_rate_mean          | operable_rate_range          | tp_rate_mean                | tp_rate_range          | sl_rate_mean         |
| sl_rate_range               | no_event_rate_mean           | no_event_rate_range         | resolved_rate_mean     | resolved_rate_range  |
| tp_share_over_resolved_mean | tp_share_over_resolved_range | expectancy_unit_s_mean      | expectancy_units_range | event_step_mean_mean |
| event_step_mean_range       | context_sensitivity_score    | context_sensitivity_profile |                        |                      |
| year                        | bar_p50_h60_tp20_sl05        | p50_h60                     | TP 2.0 / SL 0.5        | 7                    |
| 51.87%                      | 38.80%                       | 24.57%                      | 18.04%                 | 24.05%               |
| 51.39%                      | 41.98%                       | 48.61%                      | 41.98%                 |                      |
| 51.14%                      | 6.64%                        | 0.371                       | 0.241                  |                      |
| 26.45                       | 7.93                         | 1.294                       | mayor_sensibilidad     |                      |
| year                        | bar_p60_h60_tp20_sl05        | p60_h60                     | TP 2.0 / SL 0.5        | 7                    |
| 43.17%                      | 40.07%                       | 21.26%                      | 16.40%                 | 20.07%               |
| 58.67%                      | 39.58%                       | 41.33%                      | 39.58%                 |                      |
| 52.33%                      | 9.61%                        | 0.325                       | 0.212                  |                      |
| 27.75                       | 6.92                         | 1.288                       | mayor_sensibilidad     |                      |
| year                        | bar_p60_h60_tp15_sl05        | p60_h60                     | TP 1.5 / SL 0.5        | 7                    |
| 43.17%                      | 40.07%                       | 41.08%                      | 15.51%                 | 18.86%               |
| 40.07%                      | 36.44%                       | 59.93%                      | 36.44%                 |                      |
| 69.55%                      | 15.76%                       | 0.522                       | 0.128                  |                      |
| 28.66                       | 10.67                        | 1.287                       | mayor_sensibilidad     |                      |
| year                        | bar_p40_h60_tp20_sl10        | p40_h60                     | TP 2.0 / SL 1.0        | 7                    |
| 60.55%                      | 36.00%                       | 34.42%                      | 26.14%                 | 8.61%                |
| 56.97%                      | 39.56%                       | 43.03%                      | 39.56%                 |                      |
| 81.20%                      | 12.71%                       | 0.602                       | 0.389                  |                      |
| 31.59                       | 9.94                         | 1.278                       | mayor_sensibilidad     |                      |
| year                        | bar_p50_h90_tp20_sl05        | p50_h90                     | TP 2.0 / SL 0.5        | 7                    |
| 49.60%                      | 37.14%                       | 24.40%                      | 15.77%                 | 25.40%               |
| 50.20%                      | 40.83%                       | 49.80%                      | 40.83%                 |                      |
| 49.64%                      | 8.42%                        | 0.361                       | 0.190                  |                      |
| 40.24                       | 12.44                        | 1.272                       | mayor_sensibilidad     |                      |
| year                        | bar_p60_h90_tp20_sl05        | p60_h90                     | TP 2.0 / SL 0.5        | 7                    |
| 41.19%                      | 38.54%                       | 21.42%                      | 14.81%                 | 21.28%               |
| 57.30%                      | 38.02%                       | 42.70%                      | 38.02%                 |                      |
| 50.99%                      | 10.33%                       | 0.322                       | 0.180                  |                      |
| 42.73                       | 11.16                        | 1.249                       | mayor_sensibilidad     |                      |
| year                        | bar_p50_h60_tp20_sl10        | p50_h60                     | TP 2.0 / SL 1.0        | 7                    |
| 51.87%                      | 38.80%                       | 29.55%                      | 25.94%                 | 6.44%                |
| 64.01%                      | 36.98%                       | 35.99%                      | 36.98%                 |                      |
| 83.43%                      | 12.11%                       | 0.527                       | 0.408                  |                      |
| 33.44                       | 8.40                         | 1.249                       | mayor_sensibilidad     |                      |
| year                        | bar_p40_h60_tp25_sl10        | p40_h60                     | TP 2.5 / SL 1.0        | 7                    |
| 60.55%                      | 36.00%                       | 21.43%                      | 23.18%                 | 9.22%                |
| 69.35%                      | 38.02%                       | 30.65%                      | 38.02%                 |                      |
| 71.31%                      | 11.45%                       | 0.444                       | 0.431                  |                      |
| 32.19                       | 8.02                         | 1.235                       | mayor_sensibilidad     |                      |
| year                        | bar_p60_h30_tp20_sl05        | p60_h30                     | TP 2.0 / SL 0.5        | 7                    |
| 45.24%                      | 40.22%                       | 22.93%                      | 17.46%                 | 19.45%               |
| 57.62%                      | 37.83%                       | 42.38%                      | 37.83%                 |                      |
| 54.79%                      | 7.54%                        | 0.361                       | 0.247                  |                      |
| 14.38                       | 3.54                         | 1.234                       | mayor_sensibilidad     |                      |
| year                        | bar_p40_h90_tp20_sl10        | p40_h90                     | TP 2.0 / SL 1.0        | 7                    |
| 57.63%                      | 34.23%                       | 34.36%                      | 24.96%                 | 9.15%                |
| 56.50%                      | 38.39%                       | 43.50%                      | 38.39%                 |                      |
| 80.10%                      | 12.01%                       | 0.596                       | 0.365                  |                      |
| 48.49                       | 15.94                        | 1.230                       | mayor_sensibilidad     |                      |
| year                        | bar_p60_h90_tp15_sl05        | p60_h90                     | TP 1.5 / SL 0.5        | 7                    |
| 41.19%                      | 38.54%                       | 41.12%                      | 13.17%                 | 19.98%               |
|                             |                              |                             | 20.83%                 |                      |

|        |                            |                         |        |                    |
|--------|----------------------------|-------------------------|--------|--------------------|
| 38.90% | 34.00%                     | 61.10%                  | 34.00% |                    |
| 68.21% |                            | 15.91%                  | 0.517  | 0.093              |
| 43.77  | 16.58                      |                         | 1.225  | mayor_sensibilidad |
|        | year bar_p40_h30_tp20_sl10 | p40_h30 TP 2.0 / SL 1.0 |        | 7                  |
| 63.17% | 36.43%                     | 35.99%                  | 24.36% | 8.65%              |
| 55.36% | 36.90%                     | 44.64%                  |        | 36.90%             |
| 81.65% |                            | 11.86%                  | 0.633  | 0.362              |
| 15.91  | 5.07                       |                         | 1.221  | mayor_sensibilidad |
|        | year bar_p40_h60_tp20_sl05 | p40_h60 TP 2.0 / SL 0.5 |        | 7                  |
| 60.55% | 36.00%                     | 27.97%                  | 16.96% | 28.59%             |
| 43.43% | 40.34%                     | 56.57%                  |        | 40.34%             |
| 49.88% |                            | 5.10%                   | 0.417  | 0.222              |
| 24.87  | 8.70                       |                         | 1.218  | mayor_sensibilidad |
|        | year bar_p50_h60_tp15_sl05 | p50_h60 TP 1.5 / SL 0.5 |        | 7                  |
| 51.87% | 38.80%                     | 43.82%                  | 12.09% | 22.54%             |
| 33.64% | 33.78%                     | 66.36%                  |        | 33.78%             |
| 66.85% |                            | 15.26%                  | 0.545  | 0.073              |
| 26.69  | 11.41                      |                         | 1.216  | mayor_sensibilidad |
|        | year bar_p50_h30_tp20_sl05 | p50_h30 TP 2.0 / SL 0.5 |        | 7                  |
| 54.33% | 39.37%                     | 26.34%                  | 17.84% | 23.25%             |
| 50.41% | 38.38%                     | 49.59%                  |        | 38.38%             |
| 53.55% |                            | 5.33%                   | 0.410  | 0.254              |
| 13.61  | 3.87                       |                         | 1.214  | mayor_sensibilidad |

Top 15 targets con menor sensibilidad – contexto: year\_contract

| context_type                | bar_col                      | target_base               | tp_sl_config                | n_context_values       |
|-----------------------------|------------------------------|---------------------------|-----------------------------|------------------------|
| operable_rate_mean          | operable_rate_range          | tp_rate_mean              | tp_rate_range               | sl_rate_mean           |
| sl_rate_range               | no_event_rate_mean           | no_event_rate_range       | resolved_rate_mean          | resolved_rate_range    |
| tp_share_over_resolved_mean | tp_share_over_resolved_range | expectancy_unit           | s_mean                      | expectancy_units_range |
| event_step_mean_mean        | event_step_mean_range        | context_sensitivity_score | context_sensitivity_profile |                        |
| year_contract               | bar_p60_h90_tp10_sl10        | p60_h90 TP 1.0 / SL 1.0   |                             | 32                     |
| 37.28%                      | 59.14%                       | 97.13%                    | 7.95%                       | 2.87%                  |
| 0.00%                       | 0.00%                        | 100.00%                   |                             | 0.00%                  |
| 97.13%                      |                              | 7.95%                     | 0.943                       | 0.159                  |
| 47.76                       | 21.72                        |                           | 0.830                       | menor_sensibilidad     |
| year_contract               | bar_p60_h60_tp10_sl10        | p60_h60 TP 1.0 / SL 1.0   |                             | 32                     |
| 39.02%                      | 61.54%                       | 97.51%                    | 7.31%                       | 2.49%                  |
| 0.00%                       | 0.00%                        | 100.00%                   |                             | 0.00%                  |
| 97.51%                      |                              | 7.31%                     | 0.950                       | 0.146                  |
| 31.20                       | 13.01                        |                           | 0.835                       | menor_sensibilidad     |
| year_contract               | bar_p60_h30_tp10_sl10        | p60_h30 TP 1.0 / SL 1.0   |                             | 32                     |
| 41.17%                      | 61.91%                       | 97.16%                    | 7.53%                       | 2.84%                  |
| 0.00%                       | 0.00%                        | 100.00%                   |                             | 0.00%                  |
| 97.16%                      |                              | 7.53%                     | 0.943                       | 0.151                  |
| 15.82                       | 6.51                         |                           | 0.845                       | menor_sensibilidad     |
| year_contract               | bar_p50_h30_tp10_sl10        | p50_h30 TP 1.0 / SL 1.0   |                             | 32                     |
| 50.21%                      | 64.11%                       | 96.01%                    | 9.74%                       | 3.99%                  |
| 0.00%                       | 0.00%                        | 100.00%                   |                             | 0.00%                  |
| 96.01%                      |                              | 9.74%                     | 0.920                       | 0.195                  |
| 14.84                       | 6.82                         |                           | 0.933                       | menor_sensibilidad     |
| year_contract               | bar_p50_h90_tp10_sl10        | p50_h90 TP 1.0 / SL 1.0   |                             | 32                     |
| 45.59%                      | 61.77%                       | 96.02%                    | 10.60%                      | 3.98%                  |
| 0.00%                       | 0.00%                        | 100.00%                   |                             | 0.00%                  |
| 96.02%                      |                              | 10.60%                    | 0.920                       | 0.212                  |
| 44.61                       | 22.71                        |                           | 0.936                       | menor_sensibilidad     |
| year_contract               | bar_p50_h60_tp10_sl10        | p50_h60 TP 1.0 / SL 1.0   |                             | 32                     |
| 47.79%                      | 64.73%                       | 96.44%                    | 10.09%                      | 3.56%                  |
| 0.00%                       | 0.00%                        | 100.00%                   |                             | 0.00%                  |
| 96.44%                      |                              | 10.09%                    | 0.929                       | 0.202                  |

|               |                       |                         |                    |
|---------------|-----------------------|-------------------------|--------------------|
| 29.42         | 13.85                 | 0.950                   | menor_sensibilidad |
| year_contract | bar_p40_h30_tp10_sl10 | p40_h30 TP 1.0 / SL 1.0 | 32                 |
| 59.22%        | 63.21%                | 94.56%                  | 11.11%             |
| 0.00%         | 0.00%                 | 100.00%                 | 0.00%              |
| 94.56%        | 11.11%                | 0.891                   | 0.222              |
| 13.75         | 7.10                  | 0.965                   | menor_sensibilidad |
| year_contract | bar_p40_h60_tp10_sl10 | p40_h60 TP 1.0 / SL 1.0 | 32                 |
| 56.71%        | 64.10%                | 94.91%                  | 12.12%             |
| 0.00%         | 0.00%                 | 100.00%                 | 0.00%              |
| 94.91%        | 12.12%                | 0.898                   | 0.242              |
| 27.31         | 14.48                 | 1.005                   | menor_sensibilidad |
| year_contract | bar_p40_h90_tp10_sl10 | p40_h90 TP 1.0 / SL 1.0 | 32                 |
| 53.87%        | 61.51%                | 94.63%                  | 13.20%             |
| 0.00%         | 0.00%                 | 100.00%                 | 0.00%              |
| 94.63%        | 13.20%                | 0.893                   | 0.264              |
| 41.40         | 25.38                 | 1.011                   | menor_sensibilidad |
| year_contract | bar_p60_h30_tp10_sl05 | p60_h30 TP 1.0 / SL 0.5 | 32                 |
| 41.17%        | 61.91%                | 86.67%                  | 18.48%             |
| 0.00%         | 0.00%                 | 100.00%                 | 0.00%              |
| 86.67%        | 18.48%                | 0.800                   | 0.277              |
| 14.35         | 8.05                  | 1.173                   | menor_sensibilidad |
| year_contract | bar_p50_h30_tp10_sl05 | p50_h30 TP 1.0 / SL 0.5 | 32                 |
| 50.21%        | 64.11%                | 83.63%                  | 19.44%             |
| 0.00%         | 0.00%                 | 100.00%                 | 0.00%              |
| 83.63%        | 19.44%                | 0.754                   | 0.292              |
| 13.11         | 8.24                  | 1.224                   | menor_sensibilidad |
| year_contract | bar_p40_h30_tp10_sl05 | p40_h30 TP 1.0 / SL 0.5 | 32                 |
| 59.22%        | 63.21%                | 80.61%                  | 22.33%             |
| 0.00%         | 0.00%                 | 100.00%                 | 0.00%              |
| 80.61%        | 22.33%                | 0.709                   | 0.335              |
| 11.84         | 8.38                  | 1.302                   | menor_sensibilidad |
| year_contract | bar_p60_h60_tp10_sl05 | p60_h60 TP 1.0 / SL 0.5 | 32                 |
| 39.02%        | 61.54%                | 86.75%                  | 23.21%             |
| 0.00%         | 0.00%                 | 100.00%                 | 0.00%              |
| 86.75%        | 23.21%                | 0.801                   | 0.348              |
| 28.21         | 16.95                 | 1.312                   | menor_sensibilidad |
| year_contract | bar_p60_h90_tp10_sl05 | p60_h90 TP 1.0 / SL 0.5 | 32                 |
| 37.28%        | 59.14%                | 85.95%                  | 24.06%             |
| 0.00%         | 0.00%                 | 100.00%                 | 0.00%              |
| 85.95%        | 24.06%                | 0.789                   | 0.361              |
| 42.95         | 26.90                 | 1.313                   | menor_sensibilidad |
| year_contract | bar_p40_h60_tp10_sl05 | p40_h60 TP 1.0 / SL 0.5 | 32                 |
| 56.71%        | 64.10%                | 79.70%                  | 23.58%             |
| 0.00%         | 0.00%                 | 100.00%                 | 0.00%              |
| 79.70%        | 23.58%                | 0.695                   | 0.354              |
| 23.29         | 17.04                 | 1.348                   | menor_sensibilidad |

Top 15 targets con mayor sensibilidad – contexto: year\_contract

| context_type                | bar_col                      | target_base               | tp_sl_config                | n_context_values       |
|-----------------------------|------------------------------|---------------------------|-----------------------------|------------------------|
| operable_rate_mean          | operable_rate_range          | tp_rate_mean              | tp_rate_range               | sl_rate_mean           |
| sl_rate_range               | no_event_rate_mean           | no_event_rate_range       | resolved_rate_mean          | resolved_rate_range    |
| tp_share_over_resolved_mean | tp_share_over_resolved_range | expectancy_unit           | s_mean                      | expectancy_units_range |
| event_step_mean_mean        | event_step_mean_range        | context_sensitivity_score | context_sensitivity_profile |                        |
| year_contract               | bar_p50_h90_tp20_sl05        | p50_h90 TP 2.0 / SL 0.5   | 32                          |                        |
| 45.59%                      | 61.77%                       | 23.06%                    | 24.11%                      | 21.97%                 |
| 54.97%                      | 52.88%                       | 45.03%                    | 52.88%                      |                        |
| 52.54%                      | 42.51%                       | 0.351                     | 0.422                       |                        |
| 42.73                       | 22.82                        | 2.179                     | mayor_sensibilidad          |                        |
| year_contract               | bar_p60_h90_tp20_sl05        | p60_h90 TP 2.0 / SL 0.5   | 32                          |                        |

|               |                       |         |                 |                    |        |
|---------------|-----------------------|---------|-----------------|--------------------|--------|
| 37.28%        | 59.14%                | 19.82%  | 23.02%          | 17.98%             | 33.26% |
| 62.21%        | 53.01%                |         | 37.79%          | 53.01%             |        |
| 53.91%        |                       | 45.58%  |                 | 0.306              | 0.346  |
| 44.99         | 26.91                 |         | 2.140           | mayor_sensibilidad |        |
| year_contract | bar_p60_h90_tp25_sl10 | p60_h90 | TP 2.5 / SL 1.0 |                    | 32     |
| 37.28%        | 59.14%                | 12.21%  | 22.73%          | 4.51%              | 12.02% |
| 83.27%        | 32.78%                |         | 16.73%          | 32.78%             |        |
| 74.43%        |                       | 82.86%  |                 | 0.260              | 0.571  |
| 56.82         | 44.92                 |         | 2.095           | mayor_sensibilidad |        |
| year_contract | bar_p60_h60_tp20_sl05 | p60_h60 | TP 2.0 / SL 0.5 |                    | 32     |
| 39.02%        | 61.54%                | 19.83%  | 25.19%          | 16.92%             | 30.63% |
| 63.25%        | 53.26%                |         | 36.75%          | 53.26%             |        |
| 55.62%        |                       | 33.67%  |                 | 0.312              | 0.455  |
| 29.27         | 16.27                 |         | 2.043           | mayor_sensibilidad |        |
| year_contract | bar_p50_h90_tp15_sl05 | p50_h90 | TP 1.5 / SL 0.5 |                    | 32     |
| 45.59%        | 61.77%                | 42.75%  | 23.12%          | 20.62%             | 32.63% |
| 36.63%        | 48.36%                |         | 63.37%          | 48.36%             |        |
| 68.82%        |                       | 34.81%  |                 | 0.538              | 0.275  |
| 43.34         | 26.21                 |         | 2.007           | mayor_sensibilidad |        |
| year_contract | bar_p40_h90_tp20_sl05 | p40_h90 | TP 2.0 / SL 0.5 |                    | 32     |
| 53.87%        | 61.51%                | 26.21%  | 24.89%          | 26.51%             | 33.40% |
| 47.28%        | 54.76%                |         | 52.72%          | 54.76%             |        |
| 50.60%        |                       | 26.03%  |                 | 0.392              | 0.362  |
| 40.04         | 22.38                 |         | 2.006           | mayor_sensibilidad |        |
| year_contract | bar_p60_h90_tp15_sl05 | p60_h90 | TP 1.5 / SL 0.5 |                    | 32     |
| 37.28%        | 59.14%                | 40.26%  | 24.18%          | 16.92%             | 30.24% |
| 42.82%        | 48.07%                |         | 57.18%          | 48.07%             |        |
| 71.85%        |                       | 35.66%  |                 | 0.519              | 0.380  |
| 46.87         | 29.88                 |         | 1.973           | mayor_sensibilidad |        |
| year_contract | bar_p50_h60_tp20_sl05 | p50_h60 | TP 2.0 / SL 0.5 |                    | 32     |
| 47.79%        | 64.73%                | 22.85%  | 24.56%          | 21.21%             | 30.48% |
| 55.94%        | 54.20%                |         | 44.06%          | 54.20%             |        |
| 52.87%        |                       | 20.07%  |                 | 0.351              | 0.399  |
| 27.86         | 12.09                 |         | 1.940           | mayor_sensibilidad |        |
| year_contract | bar_p60_h60_tp15_sl05 | p60_h60 | TP 1.5 / SL 0.5 |                    | 32     |
| 39.02%        | 61.54%                | 40.10%  | 24.76%          | 15.96%             | 27.64% |
| 43.93%        | 49.03%                |         | 56.07%          | 49.03%             |        |
| 73.19%        |                       | 30.65%  |                 | 0.522              | 0.310  |
| 30.49         | 15.74                 |         | 1.936           | mayor_sensibilidad |        |
| year_contract | bar_p40_h90_tp25_sl10 | p40_h90 | TP 2.5 / SL 1.0 |                    | 32     |
| 53.87%        | 61.51%                | 19.28%  | 27.75%          | 7.97%              | 19.95% |
| 72.75%        | 46.05%                |         | 27.25%          | 46.05%             |        |
| 72.83%        |                       | 35.85%  |                 | 0.402              | 0.536  |
| 52.16         | 27.41                 |         | 1.911           | mayor_sensibilidad |        |
| year_contract | bar_p40_h90_tp20_sl10 | p40_h90 | TP 2.0 / SL 1.0 |                    | 32     |
| 53.87%        | 61.51%                | 32.04%  | 36.08%          | 7.43%              | 18.22% |
| 60.53%        | 47.54%                |         | 39.47%          | 47.54%             |        |
| 82.85%        |                       | 26.91%  |                 | 0.566              | 0.607  |
| 51.27         | 28.93                 |         | 1.903           | mayor_sensibilidad |        |
| year_contract | bar_p40_h30_tp20_sl05 | p40_h30 | TP 2.0 / SL 0.5 |                    | 32     |
| 59.22%        | 63.21%                | 27.85%  | 25.21%          | 24.84%             | 27.96% |
| 47.32%        | 53.16%                |         | 52.68%          | 53.16%             |        |
| 53.46%        |                       | 17.92%  |                 | 0.433              | 0.367  |
| 13.44         | 5.95                  |         | 1.875           | mayor_sensibilidad |        |
| year_contract | bar_p40_h60_tp20_sl05 | p40_h60 | TP 2.0 / SL 0.5 |                    | 32     |
| 56.71%        | 64.10%                | 26.34%  | 23.54%          | 25.95%             | 31.20% |
| 47.71%        | 53.53%                |         | 52.29%          | 53.53%             |        |
| 51.12%        |                       | 14.85%  |                 | 0.397              | 0.321  |
| 26.20         | 12.49                 |         | 1.872           | mayor_sensibilidad |        |
| year_contract | bar_p40_h60_tp20_sl10 | p40_h60 | TP 2.0 / SL 1.0 |                    | 32     |

|               |                       |                         |                    |       |        |
|---------------|-----------------------|-------------------------|--------------------|-------|--------|
| 56.71%        | 64.10%                | 31.81%                  | 34.90%             | 7.07% | 15.68% |
| 61.12%        | 48.49%                | 38.88%                  | 48.49%             |       |        |
| 83.72%        | 20.64%                | 0.565                   |                    | 0.562 |        |
| 33.23         | 16.02                 | 1.838                   | mayor_sensibilidad |       |        |
| year_contract | bar_p50_h60_tp20_sl10 | p50_h60 TP 2.0 / SL 1.0 |                    | 32    |        |
| 47.79%        | 64.73%                | 26.91%                  | 33.73%             | 5.03% | 13.70% |
| 68.06%        | 47.42%                | 31.94%                  | 47.42%             |       |        |
| 86.35%        | 21.94%                | 0.488                   |                    | 0.561 |        |
| 35.17         | 18.31                 | 1.815                   | mayor_sensibilidad |       |        |

Top 15 targets con menor sensibilidad – contexto: contract\_regime

| context_type                | bar_col               | target_base             | tp_sl_config         | n_context_valu              | es                           |
|-----------------------------|-----------------------|-------------------------|----------------------|-----------------------------|------------------------------|
| operable_rate_mean          | operable_rate_range   | tp_rate_mean            | tp_rate_range        | sl_rate_mean                | sl_rate_range                |
| no_event_rate_mean          | no_event_rate_range   | resolved_rate_mean      | resolved_rate_range  | tp_share_over_resolved_mean | tp_share_over_resolved_range |
| expectancy_un               | its_mean              | expectancy_units_range  | event_step_mean_mean | event_step_mean_range       | context_sensitivity_score    |
| context_sensitivity_profile |                       |                         |                      |                             |                              |
| contract_regime             | bar_p60_h60_tp10_sl10 | p60_h60 TP 1.0 / SL 1.0 |                      | 1                           |                              |
| 30                          | 35.39%                | 80.35%                  | 97.36%               | 8.45%                       | 2.64%                        |
| 8.45%                       | 0.00%                 | 0.00%                   | 100.00%              | 0.0                         |                              |
| 0%                          | 97.36%                |                         | 8.45%                | 0.947                       |                              |
| 0.169                       | 31.89                 | 22.02                   |                      | 1.057                       |                              |
| menor_sensibilidad          |                       |                         |                      |                             |                              |
| contract_regime             | bar_p60_h90_tp10_sl10 | p60_h90 TP 1.0 / SL 1.0 |                      | 1                           |                              |
| 30                          | 34.62%                | 81.49%                  | 97.26%               | 10.30%                      | 2.74%                        |
| 10.30%                      | 0.00%                 | 0.00%                   | 100.00%              | 0.                          |                              |
| 00%                         | 97.26%                |                         | 10.30%               | 0.94                        |                              |
| 5                           | 0.206                 | 47.49                   |                      | 35.66                       |                              |
| 1.124                       | menor_sensibilidad    |                         |                      |                             |                              |
| contract_regime             | bar_p60_h30_tp10_sl10 | p60_h30 TP 1.0 / SL 1.0 |                      | 1                           |                              |
| 30                          | 36.38%                | 80.13%                  | 97.06%               | 10.88%                      | 2.94%                        |
| 10.88%                      | 0.00%                 | 0.00%                   | 100.00%              | 0.                          |                              |
| 00%                         | 97.06%                |                         | 10.88%               | 0.94                        |                              |
| 1                           | 0.218                 | 16.31                   |                      | 12.16                       |                              |
| 1.128                       | menor_sensibilidad    |                         |                      |                             |                              |
| contract_regime             | bar_p50_h60_tp10_sl10 | p50_h60 TP 1.0 / SL 1.0 |                      | 1                           |                              |
| 30                          | 42.89%                | 86.75%                  | 96.20%               | 11.62%                      | 3.80%                        |
| 11.62%                      | 0.00%                 | 0.00%                   | 100.00%              | 0.                          |                              |
| 00%                         | 96.20%                |                         | 11.62%               | 0.92                        |                              |
| 4                           | 0.232                 | 29.89                   |                      | 23.09                       |                              |
| 1.216                       | menor_sensibilidad    |                         |                      |                             |                              |
| contract_regime             | bar_p50_h30_tp10_sl10 | p50_h30 TP 1.0 / SL 1.0 |                      | 1                           |                              |
| 30                          | 44.19%                | 86.98%                  | 95.81%               | 13.02%                      | 4.19%                        |
| 13.02%                      | 0.00%                 | 0.00%                   | 100.00%              | 0.                          |                              |
| 00%                         | 95.81%                |                         | 13.02%               | 0.91                        |                              |
| 6                           | 0.260                 | 15.29                   |                      | 12.75                       |                              |
| 1.260                       | menor_sensibilidad    |                         |                      |                             |                              |
| contract_regime             | bar_p50_h90_tp10_sl10 | p50_h90 TP 1.0 / SL 1.0 |                      | 1                           |                              |
| 30                          | 42.01%                | 89.47%                  | 96.09%               | 12.86%                      | 3.91%                        |
| 12.86%                      | 0.00%                 | 0.00%                   | 100.00%              | 0.                          |                              |
| 00%                         | 96.09%                |                         | 12.86%               | 0.92                        |                              |
| 2                           | 0.257                 | 44.21                   |                      | 34.76                       |                              |
| 1.281                       | menor_sensibilidad    |                         |                      |                             |                              |
| contract_regime             | bar_p40_h60_tp10_sl10 | p40_h60 TP 1.0 / SL 1.0 |                      | 1                           |                              |
| 30                          | 50.40%                | 92.46%                  | 94.72%               | 14.58%                      | 5.28%                        |
| 14.58%                      | 0.00%                 | 0.00%                   | 100.00%              | 0.                          |                              |
| 00%                         | 94.72%                |                         | 14.58%               | 0.89                        |                              |
| 4                           | 0.292                 | 27.72                   |                      | 23.38                       |                              |
| 1.362                       | menor_sensibilidad    |                         |                      |                             |                              |
| contract_regime             | bar_p40_h90_tp10_sl10 | p40_h90 TP 1.0 / SL 1.0 |                      | 1                           |                              |



|                 |                       |                         |         |        |        |
|-----------------|-----------------------|-------------------------|---------|--------|--------|
| 30              | 49.12%                | 94.65%                  | 94.51%  | 15.07% | 5.49%  |
| 15.07%          | 0.00%                 | 0.00%                   | 100.00% | 0.     |        |
| 00%             | 94.51%                |                         | 15.07%  | 0.89   |        |
| 0               | 0.301                 | 40.80                   |         | 36.05  |        |
| 1.399           | menor_sensibilidad    |                         |         |        |        |
| contract_regime | bar_p40_h30_tp10_sl10 | p40_h30 TP 1.0 / SL 1.0 |         |        | 1      |
| 30              | 51.84%                | 91.39%                  | 94.18%  | 16.79% | 5.82%  |
| 16.79%          | 0.00%                 | 0.00%                   | 100.00% | 0.     |        |
| 00%             | 94.18%                |                         | 16.79%  | 0.88   |        |
| 4               | 0.336                 | 14.13                   |         | 12.10  |        |
| 1.418           | menor_sensibilidad    |                         |         |        |        |
| contract_regime | bar_p60_h60_tp10_sl05 | p60_h60 TP 1.0 / SL 0.5 |         |        | 1      |
| 30              | 35.39%                | 80.35%                  | 86.31%  | 28.35% | 13.69% |
| 28.35%          | 0.00%                 | 0.00%                   | 100.00% | 0.     |        |
| 00%             | 86.31%                |                         | 28.35%  | 0.79   |        |
| 5               | 0.425                 | 28.86                   |         | 26.14  |        |
| 1.654           | menor_sensibilidad    |                         |         |        |        |
| contract_regime | bar_p60_h90_tp10_sl05 | p60_h90 TP 1.0 / SL 0.5 |         |        | 1      |
| 30              | 34.62%                | 81.49%                  | 85.76%  | 29.54% | 14.24% |
| 29.54%          | 0.00%                 | 0.00%                   | 100.00% | 0.     |        |
| 00%             | 85.76%                |                         | 29.54%  | 0.78   |        |
| 6               | 0.443                 | 42.68                   |         | 41.42  |        |
| 1.701           | menor_sensibilidad    |                         |         |        |        |
| contract_regime | bar_p60_h30_tp10_sl05 | p60_h30 TP 1.0 / SL 0.5 |         |        | 1      |
| 30              | 36.38%                | 80.13%                  | 86.44%  | 31.09% | 13.56% |
| 31.09%          | 0.00%                 | 0.00%                   | 100.00% | 0.     |        |
| 00%             | 86.44%                |                         | 31.09%  | 0.79   |        |
| 7               | 0.466                 | 14.83                   |         | 14.56  |        |
| 1.734           | menor_sensibilidad    |                         |         |        |        |
| contract_regime | bar_p50_h60_tp10_sl05 | p50_h60 TP 1.0 / SL 0.5 |         |        | 1      |
| 30              | 42.89%                | 86.75%                  | 83.04%  | 31.02% | 16.96% |
| 31.02%          | 0.00%                 | 0.00%                   | 100.00% | 0.     |        |
| 00%             | 83.04%                |                         | 31.02%  | 0.74   |        |
| 6               | 0.465                 | 26.36                   |         | 26.24  |        |
| 1.798           | menor_sensibilidad    |                         |         |        |        |
| contract_regime | bar_p50_h90_tp10_sl05 | p50_h90 TP 1.0 / SL 0.5 |         |        | 1      |
| 30              | 42.01%                | 89.47%                  | 82.45%  | 31.87% | 17.55% |
| 31.87%          | 0.00%                 | 0.00%                   | 100.00% | 0.     |        |
| 00%             | 82.45%                |                         | 31.87%  | 0.73   |        |
| 7               | 0.478                 | 38.68                   |         | 39.90  |        |
| 1.851           | menor_sensibilidad    |                         |         |        |        |
| contract_regime | bar_p40_h60_tp10_sl05 | p40_h60 TP 1.0 / SL 0.5 |         |        | 1      |
| 30              | 50.40%                | 92.46%                  | 79.30%  | 32.37% | 20.70% |
| 32.37%          | 0.00%                 | 0.00%                   | 100.00% | 0.     |        |
| 00%             | 79.30%                |                         | 32.37%  | 0.68   |        |
| 9               | 0.486                 | 23.69                   |         | 25.51  |        |
| 1.896           | menor_sensibilidad    |                         |         |        |        |

Top 15 targets con mayor sensibilidad – contexto: contract\_regime

| context_type                | bar_col                      | target_base               | tp_sl_config                | n_context_valu         |        |
|-----------------------------|------------------------------|---------------------------|-----------------------------|------------------------|--------|
| es operable_rate_mean       | operable_rate_range          | tp_rate_mean              | tp_rate_range               | sl_rate_mean           |        |
| sl_rate_range               | no_event_rate_mean           | no_event_rate_range       | resolved_rate_mean          | resolved_rate_range    |        |
| tp_share_over_resolved_mean | tp_share_over_resolved_range | expectancy_un             | its_mean                    | expectancy_units_range |        |
| event_step_mean_mean        | event_step_mean_range        | context_sensitivity_score | context_sensitivity_profile |                        |        |
| contract_regime             | bar_p60_h30_tp20_sl05        | p60_h30 TP 2.0 / SL 0.5   |                             |                        | 1      |
| 30                          | 36.38%                       | 80.13%                    | 21.39%                      | 61.54%                 | 17.48% |
| 36.79%                      | 61.13%                       | 71.88%                    | 38.87%                      | 71.                    |        |
| 88%                         | 55.85%                       |                           | 80.00%                      | 0.34                   |        |
| 0                           | 1.231                        | 15.71                     |                             | 17.50                  |        |

|                 |                       |                         |        |         |  |        |
|-----------------|-----------------------|-------------------------|--------|---------|--|--------|
| 3.303           | mayor_sensibilidad    |                         |        |         |  |        |
| contract_regime | bar_p50_h30_tp20_sl05 | p50_h30 TP 2.0 / SL 0.5 |        |         |  | 1      |
| 30              | 44.19%                | 86.98%                  | 25.24% | 52.80%  |  | 21.54% |
| 44.84%          | 53.22%                | 67.20%                  |        | 46.78%  |  | 67.    |
| 20%             | 54.83%                |                         |        | 71.43%  |  | 0.39   |
| 7               | 1.006                 | 14.94                   |        | 16.37   |  |        |
| 3.232           | mayor_sensibilidad    |                         |        |         |  |        |
| contract_regime | bar_p60_h30_tp25_sl10 | p60_h30 TP 2.5 / SL 1.0 |        |         |  | 1      |
| 30              | 36.38%                | 80.13%                  | 13.28% | 61.54%  |  | 4.24%  |
| 15.28%          | 82.47%                | 61.54%                  |        | 17.53%  |  | 61.    |
| 54%             | 76.88%                |                         |        | 100.00% |  | 0.29   |
| 0               | 1.681                 | 19.32                   |        | 22.00   |  |        |
| 3.185           | mayor_sensibilidad    |                         |        |         |  |        |
| contract_regime | bar_p40_h30_tp20_sl05 | p40_h30 TP 2.0 / SL 0.5 |        |         |  | 1      |
| 30              | 51.84%                | 91.39%                  | 28.54% | 44.84%  |  | 25.88% |
| 45.76%          | 45.59%                | 63.36%                  |        | 54.41%  |  | 63.    |
| 36%             | 52.93%                |                         |        | 70.83%  |  | 0.44   |
| 1               | 0.822                 | 13.90                   |        | 11.62   |  |        |
| 3.162           | mayor_sensibilidad    |                         |        |         |  |        |
| contract_regime | bar_p60_h90_tp20_sl05 | p60_h90 TP 2.0 / SL 0.5 |        |         |  | 1      |
| 30              | 34.62%                | 81.49%                  | 18.15% | 33.44%  |  | 18.29% |
| 38.72%          | 63.57%                | 68.62%                  |        | 36.43%  |  | 68.    |
| 62%             | 50.23%                |                         |        | 92.24%  |  | 0.27   |
| 1               | 0.610                 | 44.69                   |        | 48.72   |  |        |
| 3.145           | mayor_sensibilidad    |                         |        |         |  |        |
| contract_regime | bar_p40_h90_tp25_sl10 | p40_h90 TP 2.5 / SL 1.0 |        |         |  | 1      |
| 30              | 49.12%                | 94.65%                  | 18.31% | 36.99%  |  | 8.16%  |
| 21.95%          | 73.53%                | 57.73%                  |        | 26.47%  |  | 57.    |
| 73%             | 70.79%                |                         |        | 90.29%  |  | 0.37   |
| 6               | 0.834                 | 52.42                   |        | 44.60   |  |        |
| 3.016           | mayor_sensibilidad    |                         |        |         |  |        |
| contract_regime | bar_p50_h30_tp25_sl10 | p50_h30 TP 2.5 / SL 1.0 |        |         |  | 1      |
| 30              | 44.19%                | 86.98%                  | 17.39% | 57.14%  |  | 6.10%  |
| 17.96%          | 76.51%                | 57.14%                  |        | 23.49%  |  | 57.    |
| 14%             | 76.16%                |                         |        | 80.00%  |  | 0.37   |
| 4               | 1.482                 | 18.60                   |        | 17.20   |  |        |
| 2.992           | mayor_sensibilidad    |                         |        |         |  |        |
| contract_regime | bar_p50_h90_tp20_sl05 | p50_h90 TP 2.0 / SL 0.5 |        |         |  | 1      |
| 30              | 42.01%                | 89.47%                  | 21.89% | 35.44%  |  | 22.54% |
| 39.12%          | 55.57%                | 70.87%                  |        | 44.43%  |  | 70.    |
| 87%             | 49.90%                |                         |        | 58.98%  |  | 0.32   |
| 5               | 0.620                 | 42.55                   |        | 42.25   |  |        |
| 2.939           | mayor_sensibilidad    |                         |        |         |  |        |
| contract_regime | bar_p40_h90_tp20_sl05 | p40_h90 TP 2.0 / SL 0.5 |        |         |  | 1      |
| 30              | 49.12%                | 94.65%                  | 25.73% | 34.65%  |  | 26.92% |
| 39.60%          | 47.35%                | 72.45%                  |        | 52.65%  |  | 72.    |
| 45%             | 49.53%                |                         |        | 51.87%  |  | 0.38   |
| 0               | 0.640                 | 40.03                   |        | 40.10   |  |        |
| 2.932           | mayor_sensibilidad    |                         |        |         |  |        |
| contract_regime | bar_p60_h90_tp15_sl05 | p60_h90 TP 1.5 / SL 0.5 |        |         |  | 1      |
| 30              | 34.62%                | 81.49%                  | 38.71% | 42.16%  |  | 17.29% |
| 34.92%          | 44.00%                | 66.58%                  |        | 56.00%  |  | 66.    |
| 58%             | 70.86%                |                         |        | 67.33%  |  | 0.49   |
| 4               | 0.631                 | 46.76                   |        | 45.04   |  |        |
| 2.925           | mayor_sensibilidad    |                         |        |         |  |        |
| contract_regime | bar_p50_h90_tp25_sl10 | p50_h90 TP 2.5 / SL 1.0 |        |         |  | 1      |
| 30              | 42.01%                | 89.47%                  | 14.17% | 33.47%  |  | 5.82%  |
| 18.92%          | 80.01%                | 51.74%                  |        | 19.99%  |  | 51.    |
| 74%             | 71.60%                |                         |        | 97.20%  |  | 0.29   |
| 6               | 0.826                 | 54.78                   |        | 46.33   |  |        |

|                 |                       |         |                 |        |        |
|-----------------|-----------------------|---------|-----------------|--------|--------|
| 2.908           | mayor_sensibilidad    |         |                 |        |        |
| contract_regime | bar_p40_h30_tp25_sl10 | p40_h30 | TP 2.5 / SL 1.0 |        | 1      |
| 30              | 51.84%                | 91.39%  | 21.67%          | 50.00% | 8.37%  |
| 21.60%          | 69.96%                | 64.13%  |                 | 30.04% | 64.    |
| 13%             | 74.02%                |         | 60.00%          |        | 0.45   |
| 8               | 1.250                 | 17.56   |                 | 17.07  |        |
| 2.871           | mayor_sensibilidad    |         |                 |        |        |
| contract_regime | bar_p50_h30_tp15_sl05 | p50_h30 | TP 1.5 / SL 0.5 |        | 1      |
| 30              | 44.19%                | 86.98%  | 44.98%          | 50.38% | 20.21% |
| 42.76%          | 34.81%                | 55.92%  |                 | 65.19% | 55.    |
| 92%             | 70.19%                |         | 50.27%          |        | 0.57   |
| 4               | 0.737                 | 14.91   |                 | 16.45  |        |
| 2.863           | mayor_sensibilidad    |         |                 |        |        |
| contract_regime | bar_p40_h90_tp20_sl10 | p40_h90 | TP 2.0 / SL 1.0 |        | 1      |
| 30              | 49.12%                | 94.65%  | 31.41%          | 47.17% | 7.67%  |
| 19.85%          | 60.93%                | 64.62%  |                 | 39.07% | 64.    |
| 62%             | 82.24%                |         | 58.91%          |        | 0.55   |
| 1               | 0.915                 | 51.35   |                 | 40.64  |        |
| 2.852           | mayor_sensibilidad    |         |                 |        |        |
| contract_regime | bar_p50_h90_tp20_sl10 | p50_h90 | TP 2.0 / SL 1.0 |        | 1      |
| 30              | 42.01%                | 89.47%  | 25.98%          | 44.46% | 5.52%  |
| 17.25%          | 68.50%                | 60.73%  |                 | 31.50% | 60.    |
| 73%             | 84.15%                |         | 72.54%          |        | 0.46   |
| 4               | 0.852                 | 54.47   |                 | 40.70  |        |
| 2.845           | mayor_sensibilidad    |         |                 |        |        |

Top 15 targets con menor sensibilidad – contexto: year\_regime

|                    |                        |                      |                       |                             |                              |
|--------------------|------------------------|----------------------|-----------------------|-----------------------------|------------------------------|
| context_type       | bar_col                | target_base          | tp_sl_config          | n_context_values            |                              |
| operable_rate_mean | operable_rate_range    | tp_rate_mean         | tp_rate_range         | sl_rate_mean                | sl_rate_range                |
| no_event_rate_mean | no_event_rate_range    | resolved_rate_mean   | resolved_rate_range   | tp_share_over_resolved_mean | tp_share_over_resolved_range |
| expectancy_unit    | expectancy_units_range | event_step_mean_mean | event_step_mean_range | context_sensitivity_score   | context_sensitivity_profile  |
| year_regime        | bar_p60_h60_tp10_sl10  | p60_h60              | TP 1.0 / SL 1.0       |                             | 35                           |
| 37.25%             | 80.71%                 | 96.94%               | 7.89%                 | 3.06%                       | 7.89%                        |
| 0.00%              | 0.00%                  | 100.00%              |                       | 0.00%                       |                              |
| 96.94%             | 7.89%                  |                      | 0.939                 |                             | 0.158                        |
| 31.10              | 18.39                  |                      | 1.044                 |                             | menor_sensibilidad           |
| year_regime        | bar_p60_h30_tp10_sl10  | p60_h30              | TP 1.0 / SL 1.0       |                             | 35                           |
| 38.10%             | 75.74%                 | 96.63%               | 10.61%                | 3.37%                       | 10.61%                       |
| 0.00%              | 0.00%                  | 100.00%              |                       | 0.00%                       |                              |
| 96.63%             | 10.61%                 |                      | 0.933                 |                             | 0.212                        |
| 15.92              | 8.60                   |                      | 1.076                 |                             | menor_sensibilidad           |
| year_regime        | bar_p60_h90_tp10_sl10  | p60_h90              | TP 1.0 / SL 1.0       |                             | 35                           |
| 36.43%             | 83.00%                 | 96.81%               | 8.99%                 | 3.19%                       | 8.99%                        |
| 0.00%              | 0.00%                  | 100.00%              |                       | 0.00%                       |                              |
| 96.81%             | 8.99%                  |                      | 0.936                 |                             | 0.180                        |
| 46.18              | 29.14                  |                      | 1.100                 |                             | menor_sensibilidad           |
| year_regime        | bar_p50_h60_tp10_sl10  | p50_h60              | TP 1.0 / SL 1.0       |                             | 35                           |
| 44.61%             | 86.52%                 | 95.66%               | 10.52%                | 4.34%                       | 10.52%                       |
| 0.00%              | 0.00%                  | 100.00%              |                       | 0.00%                       |                              |
| 95.66%             | 10.52%                 |                      | 0.913                 |                             | 0.210                        |
| 29.12              | 19.56                  |                      | 1.181                 |                             | menor_sensibilidad           |
| year_regime        | bar_p50_h30_tp10_sl10  | p50_h30              | TP 1.0 / SL 1.0       |                             | 35                           |
| 45.81%             | 83.32%                 | 95.25%               | 13.47%                | 4.75%                       | 13.47%                       |
| 0.00%              | 0.00%                  | 100.00%              |                       | 0.00%                       |                              |
| 95.25%             | 13.47%                 |                      | 0.905                 |                             | 0.269                        |
| 14.92              | 9.18                   |                      | 1.237                 |                             | menor_sensibilidad           |
| year_regime        | bar_p50_h90_tp10_sl10  | p50_h90              | TP 1.0 / SL 1.0       |                             | 35                           |
| 43.74%             | 91.45%                 | 95.56%               | 11.90%                | 4.44%                       | 11.90%                       |

|             |                       |         |                 |                    |
|-------------|-----------------------|---------|-----------------|--------------------|
| 0.00%       | 0.00%                 | 100.00% | 0.00%           |                    |
| 95.56%      |                       | 11.90%  | 0.911           | 0.238              |
| 43.01       | 29.44                 |         | 1.272           | menor_sensibilidad |
| year_regime | bar_p40_h60_tp10_sl10 | p40_h60 | TP 1.0 / SL 1.0 | 35                 |
| 51.92%      | 92.40%                | 94.10%  | 12.71%          | 5.90% 12.71%       |
| 0.00%       | 0.00%                 | 100.00% | 0.00%           |                    |
| 94.10%      |                       | 12.71%  | 0.882           | 0.254              |
| 26.99       | 20.75                 |         | 1.305           | menor_sensibilidad |
| year_regime | bar_p40_h90_tp10_sl10 | p40_h90 | TP 1.0 / SL 1.0 | 35                 |
| 50.64%      | 95.40%                | 93.87%  | 13.69%          | 6.13% 13.69%       |
| 0.00%       | 0.00%                 | 100.00% | 0.00%           |                    |
| 93.87%      |                       | 13.69%  | 0.877           | 0.274              |
| 39.67       | 31.37                 |         | 1.365           | menor_sensibilidad |
| year_regime | bar_p40_h30_tp10_sl10 | p40_h30 | TP 1.0 / SL 1.0 | 35                 |
| 53.29%      | 89.04%                | 93.52%  | 16.12%          | 6.48% 16.12%       |
| 0.00%       | 0.00%                 | 100.00% | 0.00%           |                    |
| 93.52%      |                       | 16.12%  | 0.870           | 0.322              |
| 13.77       | 9.74                  |         | 1.374           | menor_sensibilidad |
| year_regime | bar_p60_h60_tp10_sl05 | p60_h60 | TP 1.0 / SL 0.5 | 35                 |
| 37.25%      | 80.71%                | 84.99%  | 22.82%          | 15.01% 22.82%      |
| 0.00%       | 0.00%                 | 100.00% | 0.00%           |                    |
| 84.99%      |                       | 22.82%  | 0.775           | 0.342              |
| 27.87       | 21.13                 |         | 1.492           | menor_sensibilidad |
| year_regime | bar_p60_h90_tp10_sl05 | p60_h90 | TP 1.0 / SL 0.5 | 35                 |
| 36.43%      | 83.00%                | 84.43%  | 23.45%          | 15.57% 23.45%      |
| 0.00%       | 0.00%                 | 100.00% | 0.00%           |                    |
| 84.43%      |                       | 23.45%  | 0.766           | 0.352              |
| 41.07       | 32.69                 |         | 1.534           | menor_sensibilidad |
| year_regime | bar_p60_h30_tp10_sl05 | p60_h30 | TP 1.0 / SL 0.5 | 35                 |
| 38.10%      | 75.74%                | 85.02%  | 26.82%          | 14.98% 26.82%      |
| 0.00%       | 0.00%                 | 100.00% | 0.00%           |                    |
| 85.02%      |                       | 26.82%  | 0.775           | 0.402              |
| 14.31       | 9.50                  |         | 1.562           | menor_sensibilidad |
| year_regime | bar_p50_h60_tp10_sl05 | p50_h60 | TP 1.0 / SL 0.5 | 35                 |
| 44.61%      | 86.52%                | 81.76%  | 23.73%          | 18.24% 23.73%      |
| 0.00%       | 0.00%                 | 100.00% | 0.00%           |                    |
| 81.76%      |                       | 23.73%  | 0.726           | 0.356              |
| 25.45       | 21.82                 |         | 1.577           | menor_sensibilidad |
| year_regime | bar_p40_h60_tp10_sl05 | p40_h60 | TP 1.0 / SL 0.5 | 35                 |
| 51.92%      | 92.40%                | 78.12%  | 23.54%          | 21.88% 23.54%      |
| 0.00%       | 0.00%                 | 100.00% | 0.00%           |                    |
| 78.12%      |                       | 23.54%  | 0.672           | 0.353              |
| 22.88       | 22.23                 |         | 1.630           | menor_sensibilidad |
| year_regime | bar_p50_h30_tp10_sl05 | p50_h30 | TP 1.0 / SL 0.5 | 35                 |
| 45.81%      | 83.32%                | 81.86%  | 27.25%          | 18.14% 27.25%      |
| 0.00%       | 0.00%                 | 100.00% | 0.00%           |                    |
| 81.86%      |                       | 27.25%  | 0.728           | 0.409              |
| 13.07       | 10.78                 |         | 1.651           | menor_sensibilidad |

Top 15 targets con mayor sensibilidad – contexto: year\_regime

|                             |                              |                           |                             |                        |
|-----------------------------|------------------------------|---------------------------|-----------------------------|------------------------|
| context_type                | bar_col                      | target_base               | tp_sl_config                | n_context_values       |
| operable_rate_mean          | operable_rate_range          | tp_rate_mean              | tp_rate_range               | sl_rate_mean           |
| sl_rate_range               | no_event_rate_mean           | no_event_rate_range       | resolved_rate_mean          | resolved_rate_range    |
| tp_share_over_resolved_mean | tp_share_over_resolved_range | expectancy_unit           | s_mean                      | expectancy_units_range |
| event_step_mean_mean        | event_step_mean_range        | context_sensitivity_score | context_sensitivity_profile |                        |
| year_regime                 | bar_p50_h90_tp20_sl05        | p50_h90                   | TP 2.0 / SL 0.5             | 35                     |
| 43.74%                      | 91.45%                       | 23.31%                    | 28.63%                      | 24.31% 32.78%          |
| 52.37%                      | 57.59%                       | 47.63%                    |                             | 57.59%                 |
| 49.26%                      |                              | 23.78%                    | 0.345                       | 0.462                  |

|             |                       |                         |                    |
|-------------|-----------------------|-------------------------|--------------------|
| 40.83       | 24.35                 | 2.342                   | mayor_sensibilidad |
| year_regime | bar_p40_h90_tp25_sl10 | p40_h90 TP 2.5 / SL 1.0 | 35                 |
| 50.64%      | 95.40%                | 19.98%                  | 33.71%             |
| 70.89%      | 52.87%                | 29.11%                  | 52.87%             |
| 69.30%      | 25.29%                | 0.408                   | 0.651              |
| 50.17       | 23.57                 | 2.279                   | mayor_sensibilidad |
| year_regime | bar_p60_h90_tp20_sl05 | p60_h90 TP 2.0 / SL 0.5 | 35                 |
| 36.43%      | 83.00%                | 19.77%                  | 27.61%             |
| 60.13%      | 56.72%                | 39.87%                  | 56.72%             |
| 49.58%      | 28.23%                | 0.295                   | 0.412              |
| 42.79       | 23.11                 | 2.275                   | mayor_sensibilidad |
| year_regime | bar_p50_h30_tp25_sl10 | p50_h30 TP 2.5 / SL 1.0 | 35                 |
| 45.81%      | 83.32%                | 18.59%                  | 31.42%             |
| 74.52%      | 48.08%                | 25.48%                  | 48.08%             |
| 73.89%      | 45.33%                | 0.396                   | 0.631              |
| 17.88       | 9.57                  | 2.248                   | mayor_sensibilidad |
| year_regime | bar_p40_h90_tp20_sl10 | p40_h90 TP 2.0 / SL 1.0 | 35                 |
| 50.64%      | 95.40%                | 33.27%                  | 38.56%             |
| 58.20%      | 53.09%                | 41.80%                  | 53.09%             |
| 80.83%      | 19.26%                | 0.580                   | 0.641              |
| 49.43       | 27.82                 | 2.247                   | mayor_sensibilidad |
| year_regime | bar_p40_h90_tp20_sl05 | p40_h90 TP 2.0 / SL 0.5 | 35                 |
| 50.64%      | 95.40%                | 27.00%                  | 28.17%             |
| 44.48%      | 57.33%                | 55.52%                  | 57.33%             |
| 49.06%      | 12.66%                | 0.397                   | 0.418              |
| 38.57       | 25.99                 | 2.239                   | mayor_sensibilidad |
| year_regime | bar_p40_h30_tp25_sl10 | p40_h30 TP 2.5 / SL 1.0 | 35                 |
| 53.29%      | 89.04%                | 22.93%                  | 32.26%             |
| 67.81%      | 52.07%                | 32.19%                  | 52.07%             |
| 72.32%      | 29.73%                | 0.481                   | 0.635              |
| 17.01       | 7.40                  | 2.229                   | mayor_sensibilidad |
| year_regime | bar_p50_h90_tp25_sl10 | p50_h90 TP 2.5 / SL 1.0 | 35                 |
| 43.74%      | 91.45%                | 15.81%                  | 30.78%             |
| 77.59%      | 47.48%                | 22.41%                  | 47.48%             |
| 70.87%      | 33.93%                | 0.329                   | 0.603              |
| 52.69       | 20.51                 | 2.215                   | mayor_sensibilidad |
| year_regime | bar_p50_h90_tp20_sl10 | p50_h90 TP 2.0 / SL 1.0 | 35                 |
| 43.74%      | 91.45%                | 27.96%                  | 38.98%             |
| 65.79%      | 54.23%                | 34.21%                  | 54.23%             |
| 82.79%      | 18.76%                | 0.497                   | 0.645              |
| 52.22       | 23.82                 | 2.198                   | mayor_sensibilidad |
| year_regime | bar_p50_h30_tp20_sl05 | p50_h30 TP 2.0 / SL 0.5 | 35                 |
| 45.81%      | 83.32%                | 26.25%                  | 24.18%             |
| 50.55%      | 52.98%                | 49.45%                  | 52.98%             |
| 53.60%      | 27.16%                | 0.409                   | 0.414              |
| 14.38       | 7.80                  | 2.193                   | mayor_sensibilidad |
| year_regime | bar_p50_h60_tp20_sl05 | p50_h60 TP 2.0 / SL 0.5 | 35                 |
| 44.61%      | 86.52%                | 23.91%                  | 27.99%             |
| 52.63%      | 54.77%                | 47.37%                  | 54.77%             |
| 50.82%      | 19.40%                | 0.361                   | 0.449              |
| 27.49       | 15.24                 | 2.187                   | mayor_sensibilidad |
| year_regime | bar_p60_h90_tp20_sl10 | p60_h90 TP 2.0 / SL 1.0 | 35                 |
| 36.43%      | 83.00%                | 23.13%                  | 37.79%             |
| 72.31%      | 50.01%                | 27.69%                  | 50.01%             |
| 84.32%      | 34.46%                | 0.417                   | 0.634              |
| 54.70       | 19.75                 | 2.183                   | mayor_sensibilidad |
| year_regime | bar_p60_h90_tp15_sl05 | p60_h90 TP 1.5 / SL 0.5 | 35                 |
| 36.43%      | 83.00%                | 40.16%                  | 30.58%             |
| 40.92%      | 51.73%                | 59.08%                  | 51.73%             |
| 69.05%      | 23.19%                | 0.508                   | 0.353              |

|             |                       |                         |                    |
|-------------|-----------------------|-------------------------|--------------------|
| 44.65       | 28.38                 | 2.167                   | mayor_sensibilidad |
| year_regime | bar_p50_h90_tp15_sl05 | p50_h90 TP 1.5 / SL 0.5 | 35                 |
| 43.74%      | 91.45%                | 43.15%                  | 23.96%             |
| 34.11%      | 48.39%                | 65.89%                  | 48.39%             |
| 66.49%      | 22.61%                | 0.534                   | 0.256              |
| 41.42       | 30.61                 | 2.152                   | mayor_sensibilidad |
| year_regime | bar_p40_h30_tp20_sl10 | p40_h30 TP 2.0 / SL 1.0 | 35                 |
| 53.29%      | 89.04%                | 35.88%                  | 33.98%             |
| 55.45%      | 51.24%                | 44.55%                  | 51.24%             |
| 81.60%      | 21.84%                | 0.631                   | 0.588              |
| 16.63       | 8.21                  | 2.146                   | mayor_sensibilidad |

Conteo por perfil de sensibilidad contextual:

| context_type             | context_sensitivity_profile | n_targets             | context_sensitivity_score_mean |
|--------------------------|-----------------------------|-----------------------|--------------------------------|
| operable_rate_range_mean | tp_rate_range_mean          | sl_rate_range_mean    | no_event_rate_range_mean       |
| tp_share_range_mean      | expectancy_range_mean       | event_step_range_mean |                                |
| contract                 | menor_sensibilidad          | 21                    |                                |
| 1.032                    | 53.48%                      | 16.12%                | 14.10%                         |
| 4.55%                    | 14.99%                      | 0.269                 | 14.38                          |
| contract                 | sensibilidad_intermedia     | 20                    |                                |
| 1.472                    | 53.62%                      | 27.01%                | 12.69%                         |
| 38.71%                   | 15.18%                      | 0.416                 | 12.93                          |
| contract                 | mayor_sensibilidad          | 22                    |                                |
| 1.631                    | 52.85%                      | 24.61%                | 23.15%                         |
| 46.66%                   | 15.85%                      | 0.341                 | 15.70                          |
| contract_code            | menor_sensibilidad          | 21                    |                                |
| 0.275                    | 8.95%                       | 5.50%                 | 5.84%                          |
| 1.56%                    | 5.65%                       | 0.085                 | 4.95                           |
| contract_code            | sensibilidad_intermedia     | 20                    |                                |
| 0.417                    | 8.99%                       | 7.47%                 | 6.14%                          |
| 13.61%                   | 5.46%                       | 0.095                 | 5.33                           |
| contract_code            | mayor_sensibilidad          | 22                    |                                |
| 0.473                    | 9.45%                       | 8.75%                 | 7.23%                          |
| 15.98%                   | 5.93%                       | 0.125                 | 4.62                           |
| contract_code_regime     | menor_sensibilidad          | 21                    |                                |
| 0.879                    | 57.73%                      | 9.58%                 | 8.30%                          |
| 3.69%                    | 8.55%                       | 0.149                 | 13.69                          |
| contract_code_regime     | sensibilidad_intermedia     | 20                    |                                |
| 1.116                    | 58.47%                      | 12.87%                | 9.54%                          |
| 19.36%                   | 11.39%                      | 0.188                 | 12.78                          |
| contract_code_regime     | mayor_sensibilidad          | 22                    |                                |
| 1.255                    | 61.46%                      | 14.73%                | 11.21%                         |
| 23.76%                   | 14.30%                      | 0.264                 | 11.34                          |
| contract_regime          | menor_sensibilidad          | 21                    |                                |
| 1.629                    | 87.00%                      | 25.20%                | 21.26%                         |
| 7.49%                    | 21.94%                      | 0.395                 | 25.11                          |
| contract_regime          | sensibilidad_intermedia     | 20                    |                                |
| 2.500                    | 87.62%                      | 39.32%                | 23.62%                         |
| 55.74%                   | 43.65%                      | 0.651                 | 30.11                          |
| contract_regime          | mayor_sensibilidad          | 22                    |                                |
| 2.933                    | 86.65%                      | 44.86%                | 28.76%                         |
| 62.82%                   | 70.24%                      | 0.900                 | 29.59                          |
| dataset_split            | menor_sensibilidad          | 21                    |                                |
| 0.390                    | 17.48%                      | 6.13%                 | 7.02%                          |
| 1.84%                    | 6.54%                       | 0.096                 | 6.20                           |
| dataset_split            | sensibilidad_intermedia     | 20                    |                                |
| 0.544                    | 17.81%                      | 8.65%                 | 7.02%                          |
| 15.67%                   | 5.24%                       | 0.104                 | 5.40                           |
| dataset_split            | mayor_sensibilidad          | 22                    |                                |
| 0.610                    | 17.76%                      | 10.88%                | 8.22%                          |

|        |               |                         |        |
|--------|---------------|-------------------------|--------|
| 19.10% | 5.07%         | 0.166                   | 5.47   |
|        | year          | menor_sensibilidad      | 21     |
| 0.792  | 38.11%        | 13.11%                  | 12.05% |
| 3.47%  | 12.50%        | 0.224                   | 11.41  |
|        | year          | sensibilidad_intermedia | 20     |
| 1.094  | 37.87%        | 18.64%                  | 11.98% |
| 30.59% | 10.35%        | 0.240                   | 9.24   |
|        | year          | mayor_sensibilidad      | 22     |
| 1.229  | 37.62%        | 18.95%                  | 18.30% |
| 37.24% | 10.83%        | 0.262                   | 9.74   |
|        | year_contract | menor_sensibilidad      | 21     |
| 1.180  | 62.34%        | 18.47%                  | 15.57% |
| 5.55%  | 16.12%        | 0.289                   | 16.06  |
|        | year_contract | sensibilidad_intermedia | 20     |
| 1.687  | 62.62%        | 27.81%                  | 15.70% |
| 42.00% | 20.54%        | 0.409                   | 13.90  |
|        | year_contract | mayor_sensibilidad      | 22     |
| 1.917  | 62.39%        | 26.13%                  | 24.43% |
| 48.01% | 30.74%        | 0.424                   | 20.83  |
|        | year_regime   | menor_sensibilidad      | 21     |
| 1.471  | 85.47%        | 20.45%                  | 17.43% |
| 5.94%  | 17.85%        | 0.322                   | 19.37  |
|        | year_regime   | sensibilidad_intermedia | 20     |
| 1.974  | 85.87%        | 27.35%                  | 18.69% |
| 43.03% | 22.47%        | 0.394                   | 17.09  |
|        | year_regime   | mayor_sensibilidad      | 22     |
| 2.187  | 87.76%        | 30.18%                  | 23.72% |
| 51.83% | 25.24%        | 0.507                   | 17.42  |

## Archivos guardados:

Análisis BAR por contexto: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_context\_analysis.csv  
 Contexto contrato individual: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_context\_contract.csv  
 Contexto familia de contrato: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_context\_contract\_code.csv  
 Contexto año: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_context\_year.csv  
 Contexto dataset split: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_context\_dataset\_split.csv  
 Contexto año + contrato: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_context\_year\_contract.csv  
 Contexto contrato + régimen: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_context\_contract\_regime.csv  
 Contexto año + régimen: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_context\_year\_regime.csv  
 Contexto familia contrato + régimen: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_context\_contract\_code\_regime.csv  
 Resumen por tipo de contexto: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_context\_by\_context\_type.csv  
 Resumen por contrato: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_context\_by\_contract.csv  
 Resumen por familia de contrato: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_context\_by\_contract\_code.csv  
 Resumen por año: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_context\_by\_year.csv  
 Resumen por split metodológico: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_context\_by\_dataset\_split.csv  
 Resumen contrato por configuración: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_bar\_context\_by\_contract\_config.csv

```

ral_profit_local\data\04_mnq_targets\mnq_targets_bar_context_contract_by_config.c
sv
Candidata TP 1.5 / SL 1.0 por contrato:      c:\Users\heguu\OneDrive\Escritorio\neu
ral_profit_local\data\04_mnq_targets\mnq_targets_bar_context_candidate_tp15_sl10_
contract.csv
Candidata TP 1.5 / SL 1.0 por familia:        c:\Users\heguu\OneDrive\Escritorio\neu
ral_profit_local\data\04_mnq_targets\mnq_targets_bar_context_candidate_tp15_sl10_
contract_code.csv
Candidata TP 1.5 / SL 1.0 por año:             c:\Users\heguu\OneDrive\Escritorio\neu
ral_profit_local\data\04_mnq_targets\mnq_targets_bar_context_candidate_tp15_sl10_
year.csv
Sensibilidad por target/contexto:              c:\Users\heguu\OneDrive\Escritorio\neu
ral_profit_local\data\04_mnq_targets\mnq_targets_bar_context_sensitivity_by_targe
t.csv
Sensibilidad por tipo de contexto:             c:\Users\heguu\OneDrive\Escritorio\neu
ral_profit_local\data\04_mnq_targets\mnq_targets_bar_context_sensitivity_by_conte
xt_type.csv
Sensibilidad por configuración/contexto:       c:\Users\heguu\OneDrive\Escritorio\neu
ral_profit_local\data\04_mnq_targets\mnq_targets_bar_context_sensitivity_by_confi
g.csv
Sensibilidad por percentil/horizonte:         c:\Users\heguu\OneDrive\Escritorio\neu
ral_profit_local\data\04_mnq_targets\mnq_targets_bar_context_sensitivity_by_targe
t_base.csv
Tabla final contexto:                         c:\Users\heguu\OneDrive\Escritorio\neu
ral_profit_local\data\04_mnq_targets\mnq_targets_bar_context_final_table.csv
Conteo perfiles contexto:                     c:\Users\heguu\OneDrive\Escritorio\neu
ral_profit_local\data\04_mnq_targets\mnq_targets_bar_context_profile_counts.csv
Reporte TXT:                                 c:\Users\heguu\OneDrive\Escritorio\neu
ral_profit_local\data\04_mnq_targets\mnq_targets_bar_context_report.txt

```

Celda 4 finalizada correctamente.

## Observaciones principales — 4.11. Análisis por contrato y contexto

El análisis se construyó correctamente sobre los **63 targets BAR**, generando **16.128 filas** al combinar targets con distintos contextos: contrato individual, familia de contrato, año, split metodológico, año+contrato, contrato+régimen, año+régimen y familia de contrato+régimen. Se detectaron **26 contratos individuales, 4 familias de contrato, 7 años, 2 splits metodológicos** y combinaciones contextuales más detalladas.

A nivel general, las métricas promedio de **BAR** son relativamente coherentes entre contextos: el **tp\_rate\_mean** se mantiene cerca del 48%–51%, el **sl\_rate\_mean** entre 11%–15%, y el **no\_event\_rate\_mean** entre 34%–40%. Sin embargo, esta estabilidad promedio esconde diferencias importantes cuando se analiza por contrato individual y por año.

Por contrato individual, se observa una heterogeneidad fuerte. Algunos contratos tienen alta operabilidad y alta resolución de eventos, como **H22**, **M22**, **H25**, **M25**, **Z25**, **H26** y **M26**. En cambio, otros contratos como **U21**, **M23**, **U23** y parte de 2023 muestran menor operabilidad, mayor **no\_event** y menor velocidad de resolución. Esto indica que no todos los contratos ofrecen el mismo tipo de entorno operativo para **BAR**.



Por familia de contrato, la diferencia es menor. La familia **H** muestra mayor operabilidad promedio, mayor **tp\_rate**, pero también mayor **sl\_rate**. La familia **U** aparece como la más conservadora: menor operabilidad, menor **tp\_rate**, menor **sl\_rate** y mayor **no\_event**. Sin embargo, la sensibilidad por **contract\_code** es la más baja de todos los contextos, con un **context\_sensitivity\_score\_mean** de apenas **0.389**. Esto sugiere que la letra del contrato por sí sola no explica gran parte del comportamiento de **BAR**.

El análisis por año es mucho más relevante. Los años **2020 y 2021** muestran menor operabilidad y mayor **no\_event**. El año **2022** aparece como un año claramente más activo, con mayor operabilidad, mayor **tp\_rate**, mayor **sl\_rate** y menor **no\_event**. El año **2023** se comporta como un período más lento o menos resolutivo, con el mayor **no\_event** y bajo **sl\_rate**. Los años **2025 y 2026** muestran un entorno mucho más activo: suben la operabilidad, el **tp\_rate**, el **sl\_rate** y baja fuertemente el **no\_event**.

La comparación por split metodológico confirma lo anterior. En **development\_2020\_2024**, la operabilidad promedio es **45.82%**, el **tp\_rate\_mean** **48.29%**, el **sl\_rate\_mean** **10.97%** y el **no\_event\_rate\_mean** **40.74%**. En cambio, en **final\_test\_2025\_2026**, la operabilidad sube a **63.50%**, el **tp\_rate\_mean** a **53.12%**, el **sl\_rate\_mean** a **18.40%** y el **no\_event\_rate\_mean** baja a **28.48%**. Esto confirma que 2025–2026 no es simplemente “más rentable”, sino un entorno más activo y también más riesgoso.

La candidata **TP 1.5 / SL 1.0** sigue teniendo buen comportamiento general, pero claramente no es estacionaria. En años más activos como 2022, 2025 y 2026 aumenta la tasa de TP y baja el **no\_event**, pero también aumenta la tasa de SL. En años más lentos como 2021 y 2023, el **no\_event** sube y las operaciones tardan más en resolverse. Esto refuerza que **TP 1.5 / SL 1.0** sigue siendo una candidata razonable, pero debe evaluarse con variables de contexto.

El ranking de sensibilidad contextual es muy claro. El contexto más sensible es **contract\_regime**, con un score promedio de **2.361**. Luego aparecen **year\_regime** con **1.881**, **year\_contract** con **1.598** y **contract** con **1.381**. Los contextos menos sensibles son **dataset\_split** con **0.516** y **contract\_code** con **0.389**. Esto indica que el comportamiento de **BAR** cambia mucho más cuando se combina contrato o año con régimen intradiario, y mucho menos cuando solo se mira la familia de contrato.

También se confirma el patrón visto antes: **TP 1.0 / SL 1.0** es la configuración más estable, pero sigue siendo un benchmark favorecido por la construcción del target. Las configuraciones más exigentes, como **TP 2.0 / SL 1.0**, **TP 2.5 / SL 1.0** y las variantes con **SL 0.5**, son más sensibles al contexto. Esto significa que cuanto más exigente o más ajustada sea la barrera, más depende el resultado del entorno de mercado.

Un punto metodológico importante: el **context\_sensitivity\_score** incluye la variación de **operable\_rate**. Por eso, los contextos combinados con régimen, especialmente **contract\_regime** y **year\_regime**, pueden verse muy sensibles en

parte porque incluyen regímenes con baja cantidad de ventanas válidas, como **Closing**. Conviene interpretar esos resultados con cautela y, si seguimos profundizando, hacer una versión excluyendo **Closing** o tratándolo por separado.

En síntesis:

|                             |                                                                            |
|-----------------------------|----------------------------------------------------------------------------|
| Contrato individual         | → sí importa, pero principalmente por el período histórico que representa. |
| Familia H/M/U/Z             | → aporta información, pero no parece ser el factor dominante.              |
| Año                         | → muy importante; 2022, 2025 y 2026 son claramente más activos.            |
| Split 2020-2024 / 2025-2026 | → confirma cambio fuerte fuera de muestra.                                 |
| Contrato + régimen          | → máxima sensibilidad contextual.                                          |
| Año + régimen               | → también muy sensible.                                                    |

La conclusión operativa es que **BAR** no debe modelarse como un target completamente estacionario. El modelo probablemente necesitará capturar información de **año, contrato, régimen intradiario, volatilidad reciente y contexto temporal** para generalizar mejor.

Para continuar, mantendría como candidata principal:

```
bar_p50_h30_tp15_sl10
bar_p50_h60_tp15_sl10
bar_p50_h90_tp15_sl10
```

y usaría como comparación selectiva:

```
bar_p60_h30_tp15_sl10
bar_p60_h60_tp15_sl10
bar_p60_h90_tp15_sl10
```

Pero con una advertencia importante: la selección final no debería basarse solo en métricas globales. Debe considerar estabilidad temporal, sensibilidad por contrato, comportamiento por régimen y desempeño fuera de muestra.

## 4.12. Observaciones e interpretación

El análisis del target **BAR** permitió evaluar el comportamiento operativo de las señales generadas a partir de **DIR**, incorporando explícitamente la dinámica entre **Take Profit**, **Stop Loss** y ausencia de evento dentro de cada horizonte temporal.

A diferencia de **DIR**, que solo identifica si existe una dirección significativa dentro de la ventana futura, **BAR** introduce una lógica más cercana a la ejecución operativa. En este caso, no solo importa si el precio se mueve a favor o en contra, sino también qué barrera se alcanza primero. Por este motivo, **BAR** representa un target más exigente y más alineado con la lógica real de una operación.

La construcción de **BAR** se realizó correctamente para todas las combinaciones definidas de percentil, horizonte y configuración TP/SL. Se evaluaron 63 variantes, derivadas de 9 targets base **DIR** y 7 configuraciones de barreras. La validación confirmó que los targets respetan la estructura esperada: solo existen valores válidos cuando **DIR**

genera una operación, los eventos se clasifican correctamente como TP , SL o no\_event , y no se cruzan días en la construcción de las ventanas futuras.

Una primera conclusión importante es que las configuraciones TP/SL no deben interpretarse de forma aislada. Cada configuración cambia de manera significativa la frecuencia de eventos, la proporción de operaciones resueltas, el tiempo hasta evento y la sensibilidad al contexto. Por lo tanto, la selección de una configuración operativa no puede basarse solamente en la tasa de TP global.

La configuración TP 1.0 / SL 1.0 mostró el comportamiento más estable y la mayor tasa de resolución. Sin embargo, debe interpretarse principalmente como un benchmark metodológico, ya que está muy favorecida por la propia construcción del target DIR . Al utilizar un TP igual al umbral que define la señal direccional, muchas operaciones tienden naturalmente a alcanzar el TP. Por este motivo, aunque es útil como referencia, no parece ser la configuración más realista para evaluar una estrategia operativa más exigente.

La configuración TP 1.5 / SL 1.0 aparece como la alternativa más equilibrada. Mantiene una tasa de TP razonable, una tasa de SL relativamente baja y una proporción favorable de TP sobre eventos resueltos. Además, no es tan fácil como TP 1.0 / SL 1.0 , pero tampoco tan restrictiva como TP 2.0 / SL 1.0 o TP 2.5 / SL 1.0 . Por este motivo, se mantiene como la principal candidata para continuar el análisis.

Las configuraciones TP 2.0 / SL 1.0 y TP 2.5 / SL 1.0 son más exigentes y generan un aumento importante de no\_event . Esto indica que, aunque pueden representar objetivos operativos más ambiciosos, muchas operaciones no alcanzan ninguna barrera dentro del horizonte evaluado. En consecuencia, estas configuraciones son más difíciles de modelar y más dependientes del contexto de mercado.

Las configuraciones con SL 0.5 permiten evaluar escenarios de stop más ajustado. Sin embargo, muestran mayor sensibilidad al ruido y al contexto. Un stop más cercano puede mejorar el control de pérdida por operación, pero también aumenta la probabilidad de salida prematura por movimientos adversos pequeños. Por eso, estas configuraciones deben interpretarse como escenarios comparativos más agresivos, no necesariamente como candidatas principales.

El análisis del tiempo hasta evento mostró que los eventos tienden a resolverse más tarde a medida que aumenta el horizonte. Esto es esperable, ya que los horizontes de 60 y 90 minutos ofrecen más espacio temporal para alcanzar una barrera. También se observó que los SL , cuando ocurren, suelen presentarse antes que los TP , especialmente en configuraciones donde el TP está más lejos que el SL. Esto no significa que las operaciones ganadoras sean lentas en todos los casos, sino que los eventos perdedores tienden a materializarse con mayor rapidez cuando la operación se mueve en contra.

El análisis de estabilidad temporal mostró que BAR no es estacionario. Los años 2022, 2025 y 2026 presentan mayor actividad, mayor operabilidad y mayor tasa de eventos resueltos. En cambio, años como 2021 y 2023 muestran más casos de no\_event y

menor resolución operativa. Esto confirma que el comportamiento del target depende fuertemente del contexto histórico del mercado.

La comparación entre `development_2020_2024` y `final_test_2025_2026` es especialmente relevante. En el período final aumentan la operabilidad, la tasa de TP y la tasa de SL, mientras disminuye el `no_event`. Esto indica que 2025–2026 no representa simplemente un entorno más favorable, sino un mercado más activo y también más riesgoso. Por lo tanto, la mejora en TP debe interpretarse junto con el aumento del riesgo de SL.

El análisis por régimen intradiario mostró que `BAR` mantiene una estructura razonablemente consistente entre `Overnight`, `Pre-market`, `Opening` y `Regular`, aunque existen diferencias. `Opening` tiende a mostrar menor tasa de SL, pero mayor proporción de `no_event`. `Pre-market` presenta tiempos promedio hasta evento más altos. `Regular` y `Overnight` muestran un comportamiento más equilibrado y representativo. El régimen `Closing` debe tratarse por separado, ya que para horizontes largos tiene muy pocas ventanas futuras válidas sin cruzar el cierre del día.

El análisis por contrato y contexto confirmó que el comportamiento de `BAR` cambia de forma importante según el período histórico. La familia de contrato `H/M/U/Z` por sí sola no parece explicar gran parte de la variación. En cambio, el año, el contrato individual y, sobre todo, las combinaciones de contrato con régimen o año con régimen muestran una sensibilidad mucho mayor. Esto sugiere que el modelo no debería depender únicamente de la letra del contrato, sino incorporar variables de contexto más informativas.

Una conclusión central es que `BAR` no debe considerarse un target completamente estable en el tiempo. Su comportamiento cambia con el régimen intradiario, el año, el contrato y el nivel general de actividad del mercado. Esto no invalida el target, pero sí implica que cualquier modelo entrenado sobre `BAR` deberá incorporar información contextual para generalizar mejor.

Desde una perspectiva de modelado, los resultados sugieren que no conviene entrenar un modelo ignorando el contexto temporal. Variables como régimen intradiario, volatilidad reciente, rango reciente, año, contrato, volumen y comportamiento previo del precio probablemente serán importantes para explicar cuándo una señal tiene mayor probabilidad de alcanzar TP, SL o no resolver dentro del horizonte.

En términos operativos, la familia de targets más razonable para continuar es:

```
bar_p50_h30_tp15_sl10  
bar_p50_h60_tp15_sl10  
bar_p50_h90_tp15_sl10
```

Estas variantes representan un punto intermedio entre frecuencia operativa, exigencia del TP, control del SL y estabilidad relativa. Como comparación más selectiva, también resulta conveniente conservar:

```
bar_p60_h30_tp15_sl10  
bar_p60_h60_tp15_sl10
```

bar\_p60\_h90\_tp15\_sl10

Estas variantes son más restrictivas y pueden ayudar a evaluar si señales más selectivas generan targets más limpios o más difíciles de modelar.

En síntesis, **BAR** aporta una visión operativa mucho más realista que **DIR**, porque incorpora explícitamente la dinámica entre beneficio, pérdida y tiempo. Sin embargo, también introduce mayor complejidad, mayor dependencia del contexto y mayor sensibilidad a la configuración TP/SL. Por este motivo, **BAR** debe interpretarse como un target operativo valioso, pero no como una solución definitiva por sí sola.

La selección final del target no debería basarse únicamente en métricas globales. Debe considerar la distribución de clases, la frecuencia de eventos, el tiempo hasta evento, la estabilidad temporal, el comportamiento por régimen, la sensibilidad por contrato y el desempeño fuera de muestra. Bajo estos criterios, **TP 1.5 / SL 1.0** se mantiene como la configuración más razonable para continuar hacia el análisis del target operativo combinado.

## 5. Preselección de targets DIR y BAR

Luego de construir y analizar estadísticamente los targets **DIR** y **BAR**, el siguiente paso consiste en reducir el universo de targets candidatos antes de avanzar hacia etapas posteriores del proyecto.

Hasta este punto se evaluaron múltiples variantes de targets direccionales y operativos, considerando distintos percentiles, horizontes temporales y configuraciones de barreras **TP/SL**. Este análisis permitió identificar fortalezas, limitaciones y comportamientos específicos de cada familia de targets.

Sin embargo, no resulta conveniente llevar todas las variantes estudiadas a una etapa posterior. Mantener un número demasiado amplio de targets aumentaría innecesariamente la complejidad del análisis, dificultaría la comparación entre alternativas y podría favorecer decisiones basadas en resultados aislados o poco robustos.

Por este motivo, en este punto se realiza una preselección metodológica de targets **DIR** y **BAR**, priorizando aquellas variantes que presentan mejor equilibrio entre interpretación operativa, frecuencia de observaciones, estabilidad relativa y coherencia con la lógica del proyecto.

La preselección no busca todavía identificar el target definitivo para modelado ni evaluar la existencia de señal predictiva. Ese análisis deberá realizarse posteriormente, utilizando únicamente información disponible en el instante **t** y respetando una separación temporal adecuada entre entrenamiento, validación y prueba final.

El objetivo de este punto es definir un universo reducido, coherente y justificable de targets candidatos que pueda ser utilizado en las siguientes etapas del estudio.

Un aspecto metodológico central es que los targets **BAR** no deben interpretarse de forma aislada. Cada target **BAR** está condicionado por un target **DIR** base, ya que **DIR** define primero si existe una operación **LONG** o **SHORT**, y luego **BAR** evalúa el resultado operativo de esa operación mediante **TP**, **SL** o **no\_event**.

Por lo tanto, la preselección se realizará por familias operativas **DIR → BAR**, conservando siempre el target **DIR** correspondiente a cada target **BAR** seleccionado.

## 5.1. Criterios de preselección

La preselección de targets se realiza utilizando criterios estadísticos, operativos y metodológicos derivados de los análisis previos.

En primer lugar, se priorizan targets que hayan sido contruidos de forma consistente. Esto implica que respeten la lógica temporal definida, que no crucen días de trading, que presenten valores válidos esperados y que mantengan una interpretación clara de sus clases.

En segundo lugar, se busca conservar targets con una frecuencia operativa razonable. En el caso de **DIR**, esto significa evitar variantes excesivamente activas que puedan introducir demasiado ruido, así como variantes demasiado restrictivas que reduzcan en exceso la cantidad de observaciones útiles. Bajo este criterio, los targets basados en **p50** representan un punto intermedio adecuado.

En tercer lugar, se considera el balance entre clases. Para **DIR**, resulta importante que las clases **LONG** y **SHORT** no estén fuertemente desbalanceadas. Para **BAR**, se evalúa la relación entre **TP**, **SL** y **no\_event**, evitando configuraciones demasiado triviales o excesivamente difíciles de resolver.

En cuarto lugar, se considera la estabilidad temporal. Los análisis previos mostraron que tanto **DIR** como **BAR** son sensibles al contexto histórico del mercado. Por lo tanto, se priorizan variantes que, aun mostrando variabilidad temporal, mantengan una estructura interpretable y no dependan de forma extrema de un único período, contrato o régimen.

En quinto lugar, se considera la coherencia operativa. En **BAR**, la configuración **TP 1.0 / SL 1.0** mostró buenos resultados, pero se interpreta principalmente como benchmark, ya que está favorecida por la construcción del target **DIR**. En cambio, la configuración **TP 1.5 / SL 1.0** representa una alternativa más exigente y operativamente más razonable, manteniendo al mismo tiempo una tasa de resolución aceptable y una baja frecuencia relativa de **SL**.

En sexto lugar, se evita ampliar innecesariamente el espacio de búsqueda. Aunque las variantes **p40** y **p60** aportan información útil como escenarios más activos o más selectivos, no resulta necesario conservar todas sus combinaciones en esta etapa. La preselección debe mantener diversidad suficiente sin perder foco metodológico.

Finalmente, se aplica una regla central para relacionar **DIR** y **BAR**:

Todo target BAR preseleccionado debe conservar su target DIR base correspondiente.

Esto se debe a que BAR depende directamente de DIR. El target DIR define la existencia y dirección de la operación, mientras que BAR evalúa el resultado operativo posterior. Por lo tanto, no sería metodológicamente correcto seleccionar un target BAR sin conservar también el DIR que le da origen.

Bajo estos criterios, la preselección se realizará por familias operativas, manteniendo pares coherentes de targets DIR → BAR.

## 5.2. Targets DIR preseleccionados

A partir del análisis realizado sobre los targets DIR, se decide reducir el universo inicial de variantes direccionales a un conjunto más pequeño y metodológicamente consistente.

Los targets DIR evaluados originalmente combinaban tres percentiles ( p40 , p50 , p60 ) y tres horizontes temporales ( H30 , H60 , H90 ). Esto permitió comparar targets más activos, intermedios y selectivos.

El análisis mostró que las variantes p40 generan una mayor frecuencia operativa, pero pueden incorporar más ruido al considerar movimientos menos exigentes. Por otro lado, las variantes p60 son más selectivas, pero reducen la cantidad de observaciones útiles. Las variantes p50 se ubican en un punto intermedio, manteniendo una frecuencia operativa razonable y un balance adecuado entre señales LONG y SHORT.

Por este motivo, los targets DIR principales seleccionados son:

```
dir_p50_h30  
dir_p50_h60  
dir_p50_h90
```

Estas tres variantes permiten conservar los tres horizontes temporales relevantes del estudio: corto, medio y largo. Además, al estar basadas en p50, representan un equilibrio entre frecuencia operativa, ruido y selectividad.

La inclusión de los tres horizontes es importante porque, aunque la frecuencia global de señales no cambia demasiado entre H30, H60 y H90, cada horizonte representa una lógica operativa diferente. H30 captura movimientos más rápidos, H60 funciona como un horizonte intermedio y H90 permite evaluar movimientos más amplios dentro de la sesión.

Además de los targets principales, se conservan dos variantes comparativas:

```
dir_p40_h60  
dir_p60_h60
```

La variante dir\_p40\_h60 representa un escenario más activo, con menor exigencia de movimiento significativo. Su función es servir como referencia para evaluar qué ocurre cuando se permite una mayor frecuencia de señales.

La variante `dir_p60_h60` representa un escenario más selectivo, con mayor exigencia de movimiento significativo. Su función es servir como referencia para evaluar si una señal más restrictiva ofrece un target más limpio o más estable.

Se utiliza `H60` como horizonte comparativo porque representa un punto medio entre `H30` y `H90`. De esta forma, se evita ampliar innecesariamente el espacio de búsqueda, pero se mantiene una comparación útil entre una versión más activa, una versión intermedia y una versión más selectiva.

Por lo tanto, los targets `DIR` preseleccionados son:

Targets `DIR` principales:

`dir_p50_h30`

`dir_p50_h60`

`dir_p50_h90`

Targets `DIR` comparativos:

`dir_p40_h60`

`dir_p60_h60`

Esta preselección reduce el universo de targets `DIR` de 9 variantes a 5 variantes, conservando diversidad suficiente para el análisis posterior sin perder foco metodológico.

Finalmente, es importante destacar que esta selección no se realiza de forma aislada. Los targets `DIR` seleccionados también deberán servir como base para los targets `BAR` preseleccionados. Esto asegura coherencia entre la definición direccional de la operación y su posterior evaluación operativa mediante barreras de `TP`, `SL` y `no_event`.

## 5.3. Targets `BAR` preseleccionados

A partir del análisis realizado sobre los targets `BAR`, se decide reducir el universo inicial de 63 variantes a un conjunto más pequeño de targets operativos candidatos.

Los targets `BAR` evaluados originalmente combinaban distintos percentiles, horizontes temporales y configuraciones de barreras `TP/SL`. Esta evaluación permitió comparar escenarios más simples, más exigentes y más sensibles al ruido.

Una conclusión importante del análisis es que no todas las configuraciones `TP/SL` tienen la misma utilidad metodológica. La configuración `TP 1.0 / SL 1.0` mostró resultados muy estables y alta tasa de resolución, pero debe interpretarse principalmente como benchmark, ya que está favorecida por la construcción del target `DIR`. Al usar un `TP` igual al umbral que define la señal direccional, el target tiende naturalmente a resolver muchas operaciones como `TP`.

Por otro lado, las configuraciones `TP 2.0 / SL 1.0` y `TP 2.5 / SL 1.0` son más exigentes, pero generan una proporción elevada de `no_event`. Esto significa que muchas operaciones no alcanzan ni `TP` ni `SL` dentro del horizonte evaluado, lo que puede dificultar el aprendizaje posterior y aumentar la dependencia del contexto.



Las configuraciones con **SL 0.5** también fueron útiles como escenarios comparativos, pero mostraron mayor sensibilidad al ruido y al contexto. Un **SL** demasiado ajustado puede provocar salidas prematuras por movimientos adversos pequeños, especialmente en datos intradiarios.

Bajo estos criterios, la configuración más razonable para continuar es:

**TP 1.5 / SL 1.0**

Esta configuración representa un punto intermedio entre exigencia operativa y frecuencia de resolución. No es tan simple como **TP 1.0 / SL 1.0**, pero tampoco tan restrictiva como **TP 2.0 / SL 1.0** o **TP 2.5 / SL 1.0**. Además, mantiene una tasa de **TP** razonable, una frecuencia de **SL** relativamente baja y una interpretación operativa clara.

Dado que todo target **BAR** está condicionado por un target **DIR** base, la preselección de **BAR** debe respetar las familias direccionales definidas previamente. Por lo tanto, los targets **BAR** principales corresponden a las variantes **p50** en los tres horizontes temporales:

bar\_p50\_h30\_tp15\_sl10  
bar\_p50\_h60\_tp15\_sl10  
bar\_p50\_h90\_tp15\_sl10

Estas variantes permiten evaluar el resultado operativo de las señales direccionales principales:

dir\_p50\_h30 → bar\_p50\_h30\_tp15\_sl10  
dir\_p50\_h60 → bar\_p50\_h60\_tp15\_sl10  
dir\_p50\_h90 → bar\_p50\_h90\_tp15\_sl10

Además, se conservan dos targets **BAR** comparativos asociados al horizonte intermedio **H60**:

bar\_p40\_h60\_tp15\_sl10  
bar\_p60\_h60\_tp15\_sl10

La variante **bar\_p40\_h60\_tp15\_sl10** permite analizar un escenario más activo, basado en un target direccional menos restrictivo. Su función es observar si una mayor frecuencia de señales puede traducirse en resultados operativos útiles o si introduce demasiado ruido.

La variante **bar\_p60\_h60\_tp15\_sl10** permite analizar un escenario más selectivo, basado en un target direccional más exigente. Su función es evaluar si una señal más restrictiva genera mejores condiciones operativas o si reduce demasiado la cantidad de oportunidades.

Por lo tanto, los targets **BAR** preseleccionados son:

Targets **BAR** principales:

bar\_p50\_h30\_tp15\_sl10  
bar\_p50\_h60\_tp15\_sl10  
bar\_p50\_h90\_tp15\_sl10

Targets **BAR** comparativos:

```
bar_p40_h60_tp15_sl10
bar_p60_h60_tp15_sl10
```

Esta preselección reduce el universo de targets **BAR** de 63 variantes a 5 variantes, manteniendo una relación directa con los targets **DIR** seleccionados.

La lógica final de selección queda organizada por familias operativas:

```
dir_p50_h30 → bar_p50_h30_tp15_sl10
dir_p50_h60 → bar_p50_h60_tp15_sl10
dir_p50_h90 → bar_p50_h90_tp15_sl10
dir_p40_h60 → bar_p40_h60_tp15_sl10
dir_p60_h60 → bar_p60_h60_tp15_sl10
```

De esta forma, cada target **BAR** conserva su **DIR** base correspondiente, asegurando coherencia entre la definición de la operación y su evaluación operativa posterior.

## 5.4. Universo reducido de targets candidatos

A partir de los criterios definidos y de la preselección realizada en los puntos anteriores, el universo inicial de targets se reduce de forma significativa.

Originalmente se evaluaron 9 variantes de **DIR** y 63 variantes de **BAR**, lo que daba un total de 72 targets analizados. Aunque este conjunto amplio fue útil para estudiar el comportamiento estadístico y operativo de distintas alternativas, no resulta conveniente trasladarlo completo a las siguientes etapas del proyecto.

La preselección reduce este universo a 5 familias operativas, donde cada familia conserva la relación metodológica entre un target direccional **DIR** y su correspondiente target operativo **BAR**.

El universo reducido queda definido de la siguiente manera:

Familias principales:

```
dir_p50_h30 → bar_p50_h30_tp15_sl10
dir_p50_h60 → bar_p50_h60_tp15_sl10
dir_p50_h90 → bar_p50_h90_tp15_sl10
```

Familias comparativas:

```
dir_p40_h60 → bar_p40_h60_tp15_sl10
dir_p60_h60 → bar_p60_h60_tp15_sl10
```

En total, se conservan 10 targets candidatos:

Targets **DIR**:

```
dir_p50_h30
dir_p50_h60
dir_p50_h90
dir_p40_h60
dir_p60_h60
```

Targets **BAR**:

```

bar_p50_h30_tp15_sl10
bar_p50_h60_tp15_sl10
bar_p50_h90_tp15_sl10
bar_p40_h60_tp15_sl10
bar_p60_h60_tp15_sl10

```

Esta selección mantiene tres familias principales basadas en **p50**, correspondientes a los horizontes **H30**, **H60** y **H90**. Estas variantes representan el núcleo central del análisis, ya que ofrecen un equilibrio razonable entre frecuencia operativa, balance de clases, interpretación y selectividad.

Además, se conservan dos familias comparativas en el horizonte **H60**: una más activa (**p40**) y una más selectiva (**p60**). Estas variantes permiten evaluar la sensibilidad de los resultados frente al nivel de exigencia del threshold, sin ampliar innecesariamente el espacio de búsqueda.

La elección de **H60** como horizonte comparativo responde a que representa un punto intermedio entre **H30** y **H90**. Esto permite comparar targets más activos y más selectivos bajo una ventana temporal equilibrada.

El universo reducido conserva diversidad suficiente para estudiar distintos enfoques:

|             |                                                 |
|-------------|-------------------------------------------------|
| DIR p50     | → dirección significativa equilibrada           |
| BAR p50     | → resultado operativo equilibrado               |
| DIR p40_h60 | → dirección más activa                          |
| BAR p40_h60 | → resultado operativo de señales más frecuentes |
| DIR p60_h60 | → dirección más selectiva                       |
| BAR p60_h60 | → resultado operativo de señales más exigentes  |

Esta reducción permite mantener el análisis enfocado y metodológicamente consistente. Al mismo tiempo, evita que las siguientes etapas se vean afectadas por un número excesivo de variantes, muchas de las cuales cumplen funciones similares o presentan limitaciones ya identificadas.

Por lo tanto, las siguientes etapas del estudio deberán concentrarse en este universo reducido de 5 familias operativas y 10 targets candidatos, manteniendo siempre la relación:

DIR define la dirección de la operación.  
 BAR evalúa el resultado operativo de esa operación.

La selección definitiva del target no se realiza todavía en este punto. Esta preselección simplemente define un conjunto razonable y justificable de candidatos sobre los cuales podrá profundizarse en etapas posteriores.

## 5.5. Consideraciones para la evaluación predictiva posterior

La preselección realizada en este punto no implica todavía que los targets seleccionados tengan señal predictiva suficiente para entrenar modelos rentables o robustos.

Hasta esta etapa, el análisis se concentró en la construcción, validación, distribución, estabilidad e interpretación operativa de los targets **DIR** y **BAR**. Es decir, se evaluó si

los targets están correctamente definidos, si tienen una frecuencia razonable, si presentan una estructura interpretable y si su comportamiento es coherente con la lógica del proyecto.

Sin embargo, estos análisis no responden todavía una pregunta distinta y más exigente:

¿Existe información disponible en el instante  $t$  que permita predecir estos targets de manera útil?

Responder esa pregunta requiere una etapa posterior específica, orientada a evaluar señal predictiva. Esa evaluación deberá realizarse utilizando únicamente variables observables en el momento de entrada, sin incorporar información futura ni resultados derivados de la ventana que se busca predecir.

Por este motivo, no se entrenan modelos en este punto. Tampoco se comparan métricas predictivas como `accuracy`, `f1_score`, `precision`, `recall` o métricas de performance de estrategia. Incluir ese análisis en esta etapa mezclaría dos objetivos distintos: la definición metodológica de targets y la evaluación predictiva de modelos.

La evaluación predictiva posterior deberá respetar algunos principios básicos:

1. Usar únicamente información disponible en  $t$ .
2. Mantener separación temporal estricta entre entrenamiento, validación y prueba.
3. Evitar splits aleatorios.
4. No utilizar 2025–2026 para seleccionar o ajustar targets.
5. Comparar los modelos contra benchmarks simples.
6. Evaluar estabilidad fuera de muestra.
7. Analizar resultados por año, régimen y contrato.

En particular, será importante verificar si los targets preseleccionados mantienen algún grado de predictibilidad en distintos contextos de mercado. Los análisis previos mostraron que tanto `DIR` como `BAR` son sensibles al año, al contrato y al régimen intradiario. Por lo tanto, la evaluación predictiva no deberá limitarse a una métrica global, sino que deberá considerar también la robustez temporal y contextual.

Para `DIR`, la evaluación futura deberá analizar si es posible anticipar de manera útil la dirección significativa del movimiento, distinguiendo entre señales `LONG`, `SHORT` y `NO_SIGNAL`.

Para `BAR`, la evaluación futura deberá analizar si es posible anticipar el resultado operativo de una señal ya definida por `DIR`, distinguiendo entre `TP`, `SL` y `no_event`. También podrá evaluarse una versión binaria más simple, como `TP` frente a `no_TP`, si resulta útil para una primera aproximación.

En consecuencia, el rol de esta sección no es demostrar que los targets son predictivos, sino dejar establecido que la predictibilidad deberá evaluarse en una etapa posterior, una vez definido el universo reducido de candidatos.

De esta manera, se mantiene una separación clara entre dos etapas del proyecto:

Stage actual:  
Preselección metodológica de targets.

Stage posterior:

Evaluación de señal predictiva y entrenamiento de modelos.

Esta separación permite evitar decisiones prematuras y reduce el riesgo de seleccionar targets únicamente por su comportamiento descriptivo. Un target puede estar bien construido y ser operativamente razonable, pero aun así no ser fácilmente predecible. Por eso, la selección final deberá depender de una evaluación posterior con modelos, validación temporal y análisis fuera de muestra.

## 5.6. Conclusiones y targets que pasan a la siguiente etapa

El proceso de preselección permitió reducir el universo inicial de targets **DIR** y **BAR** a un conjunto más acotado, coherente y metodológicamente justificable.

El objetivo de esta reducción no fue seleccionar todavía el target definitivo del proyecto, sino definir un grupo razonable de candidatos sobre los cuales continuar el análisis en etapas posteriores. Esta decisión permite evitar una exploración excesivamente amplia y facilita una comparación más ordenada entre alternativas.

La preselección se basó en los resultados obtenidos en los análisis anteriores, considerando principalmente la consistencia de construcción, la frecuencia operativa, el balance entre clases, la estabilidad temporal, la interpretación operativa y la relación metodológica entre **DIR** y **BAR**.

Una conclusión central es que los targets **DIR** basados en **p50** representan el núcleo principal de la preselección. Estas variantes ofrecen un equilibrio razonable entre frecuencia de señales y selectividad, evitando tanto el exceso de actividad de **p40** como la mayor restricción de **p60**.

Por este motivo, se conservan como familias principales:

```
dir_p50_h30 → bar_p50_h30_tp15_sl10  
dir_p50_h60 → bar_p50_h60_tp15_sl10  
dir_p50_h90 → bar_p50_h90_tp15_sl10
```

Estas tres familias permiten mantener los horizontes temporales principales del estudio: corto, medio y largo. De esta forma, se conserva la posibilidad de comparar si una lógica operativa de 30, 60 o 90 minutos resulta más adecuada en etapas posteriores.

Además, se conservan dos familias comparativas en el horizonte intermedio **H60**:

```
dir_p40_h60 → bar_p40_h60_tp15_sl10  
dir_p60_h60 → bar_p60_h60_tp15_sl10
```

La primera permite representar un escenario más activo, con mayor frecuencia de señales. La segunda permite representar un escenario más selectivo, con menor frecuencia de señales pero mayor exigencia de movimiento significativo. Ambas cumplen una función comparativa, no necesariamente principal.

En el caso de `BAR`, se selecciona la configuración `TP 1.5 / SL 1.0` como alternativa operativa principal. Esta configuración mostró un equilibrio más razonable que `TP 1.0 / SL 1.0`, que queda interpretada como benchmark, y que las configuraciones más exigentes como `TP 2.0 / SL 1.0` o `TP 2.5 / SL 1.0`, que generan una proporción elevada de `no_event`.

El universo reducido que pasa a la siguiente etapa queda compuesto por los siguientes targets:

Targets DIR:

```
dir_p50_h30
dir_p50_h60
dir_p50_h90
dir_p40_h60
dir_p60_h60
```

Targets BAR:

```
bar_p50_h30_tp15_sl10
bar_p50_h60_tp15_sl10
bar_p50_h90_tp15_sl10
bar_p40_h60_tp15_sl10
bar_p60_h60_tp15_sl10
```

En total, se conservan 10 targets organizados en 5 familias operativas `DIR → BAR`.

Esta organización es importante porque `BAR` no es independiente de `DIR`. Cada target `BAR` depende del target `DIR` que define previamente la dirección de la operación. Por lo tanto, los targets seleccionados deben interpretarse como pares metodológicos y no como columnas aisladas.

La etapa posterior deberá evaluar si estos targets presentan señal predictiva real utilizando únicamente información disponible en el instante `t`, con separación temporal estricta y sin utilizar el período 2025–2026 para ajustar decisiones metodológicas.

En síntesis, la preselección deja definido un conjunto reducido, interpretable y operativo de targets candidatos. Este conjunto mantiene diversidad suficiente para comparar horizontes, niveles de selectividad y lógica direccional versus lógica operativa, pero evita continuar con un espacio de búsqueda demasiado amplio.

La selección definitiva del target deberá realizarse más adelante, luego de evaluar la predictibilidad, la robustez temporal y el desempeño fuera de muestra de estas familias candidatas.

## 6. OPC — Target operativo combinado

Luego de analizar los targets `DIR` y `BAR`, el siguiente paso consiste en construir y estudiar un target operativo combinado, denominado `OPC`.

El target **DIR** permitió identificar si dentro de una ventana futura existía una dirección significativa del movimiento, clasificando cada observación como **LONG**, **SHORT** o **NO\_SIGNAL**.

El target **BAR**, por su parte, permitió evaluar el resultado operativo de una señal direccional, indicando si una operación alcanzaba primero el **Take Profit**, el **Stop Loss** o si no resolvía dentro del horizonte temporal definido.

El target **OPC** combina ambas dimensiones en una sola variable categórica:

Dirección de la operación + resultado operativo de esa dirección  
En términos simples:

$$\text{OPC} = \text{DIR} + \text{BAR}$$

Esta expresión no debe interpretarse como una suma matemática, sino como una combinación lógica entre:

**DIR** = dirección significativa del movimiento futuro

**BAR** = resultado operativo asociado a esa dirección

De esta manera, **OPC** busca representar una categoría operativa más completa. No solo indica si el mercado presentó una dirección significativa, sino también si esa dirección habría dado lugar a una operación ganadora o perdedora bajo una regla de barreras determinada.

En este estudio, **OPC** se construirá únicamente sobre el universo reducido de targets candidatos definido en el punto anterior. Es decir, no se volverá a abrir el análisis sobre todas las variantes originales de **DIR** y **BAR**, sino sobre las familias operativas preseleccionadas:

**dir\_p50\_h30** → **bar\_p50\_h30\_tp15\_sl10**

**dir\_p50\_h60** → **bar\_p50\_h60\_tp15\_sl10**

**dir\_p50\_h90** → **bar\_p50\_h90\_tp15\_sl10**

**dir\_p40\_h60** → **bar\_p40\_h60\_tp15\_sl10**

**dir\_p60\_h60** → **bar\_p60\_h60\_tp15\_sl10**

Esto permite mantener coherencia metodológica y evitar una expansión innecesaria del espacio de búsqueda.

El análisis de **OPC** no tiene como objetivo entrenar modelos en esta etapa. Al igual que con **DIR** y **BAR**, primero se busca construir, validar y estudiar estadísticamente el target antes de evaluar su posible señal predictiva en una etapa posterior.

## 6.1. Objetivo del target OPC

El objetivo del target **OPC** es integrar en una sola etiqueta la dirección operativa de una observación y el resultado operativo asociado a esa dirección.

Mientras que **DIR** responde a la pregunta:

¿Existe una dirección significativa del movimiento futuro?

y **BAR** responde a la pregunta:

¿Qué ocurre primero: TP, SL o no\_event?

el target **OPC** busca responder una pregunta más completa:

¿Qué tipo de evento operativo representa esta observación?

Esto permite clasificar cada observación en categorías como:

LONG ganador  
 LONG perdedor  
 SHORT ganador  
 SHORT perdedor  
 NO\_TRADE

La ventaja de **OPC** es que resume en una sola variable la lógica completa de una posible operación. En lugar de estudiar por separado la dirección y el resultado, **OPC** permite analizar directamente el tipo de evento operativo que se habría generado.

Esto puede resultar útil para etapas posteriores del proyecto, ya que un modelo podría intentar aprender no solo si existe una dirección significativa, sino también si esa dirección tiene mayor probabilidad de convertirse en una operación favorable.

Sin embargo, **OPC** también introduce mayor complejidad. Al combinar dirección y resultado, el número de clases aumenta y algunas categorías pueden quedar desbalanceadas. Por este motivo, será necesario analizar cuidadosamente su distribución, estabilidad temporal y comportamiento por régimen antes de considerarlo como candidato para modelado.

## 6.2. Lógica de construcción de OPC

La construcción de **OPC** parte de la relación entre un target **DIR** y su correspondiente target **BAR**.

Primero, **DIR** define si existe una dirección operativa válida:

DIR = 1 → dirección LONG  
 DIR = -1 → dirección SHORT  
 DIR = 0 → sin dirección operativa clara

Luego, para las observaciones con dirección válida, **BAR** evalúa el resultado operativo de esa dirección:

BAR = 1 → Take Profit alcanzado primero  
 BAR = -1 → Stop Loss alcanzado primero  
 BAR = 0 → sin resolución operativa dentro del horizonte

La combinación lógica entre **DIR** y **BAR** genera las clases de **OPC**.

Las combinaciones conceptuales son:

| DIR | BAR | Etiqueta lógica | Interpretación                          |
|-----|-----|-----------------|-----------------------------------------|
| 1   | 1   | LONG_TP         | LONG con Take Profit alcanzado primero  |
| 1   | -1  | LONG_SL         | LONG con Stop Loss alcanzado primero    |
| -1  | 1   | SHORT_TP        | SHORT con Take Profit alcanzado primero |



| DIR        | BAR        | Etiqueta lógica | Interpretación                                |
|------------|------------|-----------------|-----------------------------------------------|
| -1         | -1         | SHORT_SL        | SHORT con Stop Loss alcanzado primero         |
| 0          | cualquiera | NO_TRADE        | Sin operación válida                          |
| cualquiera | 0          | NO_TRADE        | Sin resolución operativa dentro del horizonte |

En esta definición, las observaciones sin dirección operativa clara y las observaciones que no resuelven dentro del horizonte se agrupan bajo la clase **NO\_TRADE**.

Esta decisión simplifica la interpretación operativa del target. Desde esta perspectiva, una observación solo se considera operativa si tiene una dirección válida y además alcanza una barrera de resultado dentro del horizonte definido.

Por lo tanto, **OPC** se enfocará en cinco clases principales:

LONG\_TP  
LONG\_SL  
SHORT\_TP  
SHORT\_SL  
NO\_TRADE

Para evitar ambigüedades, no se utilizarán etiquetas como **11**, **10**, **-11** o **-10**. En este estudio, **BAR = -1** representa Stop Loss alcanzado primero, mientras que **BAR = 0** representa ausencia de resolución operativa. Por lo tanto, una etiqueta como **10** podría confundirse con **DIR = 1** y **BAR = 0**, cuando en realidad se quiere distinguir claramente entre operación perdedora y operación sin resolución.

Por este motivo, se utilizarán etiquetas nominales explícitas.

## 6.3. Codificación de clases

Aunque conceptualmente **OPC** se interpreta mediante etiquetas nominales, para facilitar su análisis y almacenamiento también se podrá utilizar una codificación numérica.

La codificación propuesta es:

| Clase OPC | Etiqueta lógica | Interpretación                      |
|-----------|-----------------|-------------------------------------|
| 0         | NO_TRADE        | No trade / sin resolución operativa |
| 1         | LONG_TP         | LONG ganador                        |
| 2         | LONG_SL         | LONG perdedor                       |
| 3         | SHORT_TP        | SHORT ganador                       |
| 4         | SHORT_SL        | SHORT perdedor                      |

Esta codificación debe interpretarse únicamente como una representación categórica.

La clase **4** no es mayor que la clase **1**, ni la clase **2** representa una magnitud superior a la clase **0**. Cada número identifica un tipo distinto de evento operativo.

Por lo tanto, `OPC` no debe tratarse como una variable ordinal ni como una variable continua. En etapas posteriores, si se utiliza como target de Machine Learning, deberá tratarse como un problema de clasificación multiclase nominal.

Una consideración importante es que esta versión principal de `OPC` agrupa los casos `BAR = 0` dentro de `NO_TRADE`. Esto permite mantener un target más compacto y operativamente más claro. No obstante, si en etapas posteriores se desea estudiar con mayor detalle las operaciones que tuvieron dirección pero no alcanzaron ninguna barrera, podría construirse una versión auxiliar más granular que distinga entre `LONG_NO_EVENT` y `SHORT_NO_EVENT`.

Para este análisis principal, se mantiene la versión compacta de cinco clases.

## 6.4. Construcción práctica de OPC

Esta celda deja creados estos objetos principales:

```
df_targets_opc
df_opc_metadata
df_opc_summary
df_opc_selected_families
```

Y los targets OPC contruidos son:

```
opc_p50_h30_tp15_sl10
opc_p50_h60_tp15_sl10
opc_p50_h90_tp15_sl10
opc_p40_h60_tp15_sl10
opc_p60_h60_tp15_sl10
```

```
In [26]: # 6.4. Construcción práctica de OPC
# =====

import numpy as np
import pandas as pd

print("Construcción práctica de targets OPC")
print("=" * 80)

# -----
# 1) Cargar objetos si no están en memoria
# -----

# Intentar cargar DIR si no está en memoria
if "df_targets_dir" not in globals():
    if "MNQ_TARGETS_DIR_PARQUET" in globals():
        if MNQ_TARGETS_DIR_PARQUET.exists():
            df_targets_dir = pd.read_parquet(MNQ_TARGETS_DIR_PARQUET)
            print(f"df_targets_dir cargado desde: {MNQ_TARGETS_DIR_PARQUET}")
        else:
            print("MNQ_TARGETS_DIR_PARQUET existe como variable, pero el archivo")
    else:
        print("df_targets_dir no está en memoria y no se encontró MNQ_TARGETS_DI

# Intentar cargar BAR si no está en memoria
```

```

if "df_targets_bar" not in globals():
    if "MNQ_TARGETS_BAR_PARQUET" in globals():
        if MNQ_TARGETS_BAR_PARQUET.exists():
            df_targets_bar = pd.read_parquet(MNQ_TARGETS_BAR_PARQUET)
            print(f"df_targets_bar cargado desde: {MNQ_TARGETS_BAR_PARQUET}")
        else:
            print("MNQ_TARGETS_BAR_PARQUET existe como variable, pero el archivo")
    else:
        print("df_targets_bar no está en memoria y no se encontró MNQ_TARGETS_BA

# -----
# 2) Validar objetos base
# -----

required_objects = [
    "df_targets_bar",
]

missing_objects = [
    obj for obj in required_objects
    if obj not in globals()
]

if missing_objects:
    raise NameError(
        "Faltan objetos necesarios para construir OPC: "
        f"{missing_objects}"
    )

# -----
# 3) Definir familias operativas preseleccionadas
# -----

OPC_SELECTED_FAMILIES = [
    {
        "family_name": "p50_h30",
        "target_role": "principal",
        "dir_col": "dir_p50_h30",
        "bar_col": "bar_p50_h30_tp15_sl10",
        "percentile": 50,
        "horizon": 30,
        "tp_mult": 1.5,
        "sl_mult": 1.0,
    },
    {
        "family_name": "p50_h60",
        "target_role": "principal",
        "dir_col": "dir_p50_h60",
        "bar_col": "bar_p50_h60_tp15_sl10",
        "percentile": 50,
        "horizon": 60,
        "tp_mult": 1.5,
        "sl_mult": 1.0,
    },
    {
        "family_name": "p50_h90",
        "target_role": "principal",
        "dir_col": "dir_p50_h90",
        "bar_col": "bar_p50_h90_tp15_sl10",
        "percentile": 50,
    }
]

```

```

        "horizon": 90,
        "tp_mult": 1.5,
        "sl_mult": 1.0,
    },
    {
        "family_name": "p40_h60",
        "target_role": "comparativo_activo",
        "dir_col": "dir_p40_h60",
        "bar_col": "bar_p40_h60_tp15_sl10",
        "percentile": 40,
        "horizon": 60,
        "tp_mult": 1.5,
        "sl_mult": 1.0,
    },
    {
        "family_name": "p60_h60",
        "target_role": "comparativo_selectivo",
        "dir_col": "dir_p60_h60",
        "bar_col": "bar_p60_h60_tp15_sl10",
        "percentile": 60,
        "horizon": 60,
        "tp_mult": 1.5,
        "sl_mult": 1.0,
    },
]

df_opc_selected_families = pd.DataFrame(OPC_SELECTED_FAMILIES)

print("\nFamilias operativas seleccionadas para OPC:")
print(
    df_opc_selected_families[
        [
            "family_name",
            "target_role",
            "dir_col",
            "bar_col",
            "percentile",
            "horizon",
            "tp_mult",
            "sl_mult",
        ]
    ].to_string(index=False)
)

# -----
# 4) Funciones auxiliares para ubicar columnas
# -----

def get_column_from_available_sources(col_name):
    """
    Busca una columna en los dataframes disponibles.
    Prioridad:
    1. df_targets_bar
    2. df_targets_dir

    Devuelve la serie alineada al índice de df_targets_bar.
    """

    if col_name in df_targets_bar.columns:
        return df_targets_bar[col_name].copy()

```

```

if "df_targets_dir" in globals():
    if col_name in df_targets_dir.columns:
        return df_targets_dir[col_name].reindex(df_targets_bar.index).copy()

    raise KeyError(
        f"No se encontró la columna requerida: {col_name}"
    )

def build_opc_from_dir_bar(dir_series, bar_series):
    """
    Construye OPC a partir de DIR y BAR.

    Reglas:
    - DIR NaN                -> OPC NaN
    - DIR = 0                 -> NO_TRADE
    - DIR = 1 y BAR = 1       -> LONG_TP
    - DIR = 1 y BAR = -1      -> LONG_SL
    - DIR = -1 y BAR = 1      -> SHORT_TP
    - DIR = -1 y BAR = -1     -> SHORT_SL
    - DIR +/-1 y BAR = 0      -> NO_TRADE
    - DIR +/-1 y BAR NaN      -> NaN, caso no esperado
    """

    dir_series = dir_series.copy()
    bar_series = bar_series.copy()

    opc_label = pd.Series(
        data=np.nan,
        index=dir_series.index,
        dtype="object"
    )

    opc_class = pd.Series(
        data=np.nan,
        index=dir_series.index,
        dtype="float"
    )

    # NO_TRADE por ausencia de dirección operativa
    mask_no_signal = dir_series.eq(0)

    # NO_TRADE por operación sin resolución dentro del horizonte
    mask_no_event = dir_series.isin([1, -1]) & bar_series.eq(0)

    # Eventos operativos resueltos
    mask_long_tp = dir_series.eq(1) & bar_series.eq(1)
    mask_long_sl = dir_series.eq(1) & bar_series.eq(-1)
    mask_short_tp = dir_series.eq(-1) & bar_series.eq(1)
    mask_short_sl = dir_series.eq(-1) & bar_series.eq(-1)

    opc_label.loc[mask_no_signal | mask_no_event] = "NO_TRADE"
    opc_label.loc[mask_long_tp] = "LONG_TP"
    opc_label.loc[mask_long_sl] = "LONG_SL"
    opc_label.loc[mask_short_tp] = "SHORT_TP"
    opc_label.loc[mask_short_sl] = "SHORT_SL"

    class_map = {
        "NO_TRADE": 0,

```

```

        "LONG_TP": 1,
        "LONG_SL": 2,
        "SHORT_TP": 3,
        "SHORT_SL": 4,
    }

    opc_class = opc_label.map(class_map).astype("float")

    return opc_label, opc_class

# -----
# 5) Validar columnas requeridas
# -----

missing_required_columns = []

for family in OPC_SELECTED_FAMILIES:

    dir_col = family["dir_col"]
    bar_col = family["bar_col"]

    try:
        _ = get_column_from_available_sources(dir_col)
    except KeyError:
        missing_required_columns.append(dir_col)

    try:
        _ = get_column_from_available_sources(bar_col)
    except KeyError:
        missing_required_columns.append(bar_col)

if missing_required_columns:
    raise KeyError(
        "Faltan columnas necesarias para construir OPC:\n"
        f"{missing_required_columns}"
    )

print("\nTodas las columnas DIR y BAR requeridas fueron encontradas.")

# -----
# 6) Crear dataframe base de OPC
# -----

# Columnas de contexto que conviene conservar si existen
context_columns_candidates = [
    "date",
    "minute_of_day",
    "regime_id",
    "regime_name",
    "contract",
    "contract_code",
    "year",
    "dataset_split",
    "open",
    "high",
    "low",
    "close",
    "volume",
]

```

```

context_columns = [
    col for col in context_columns_candidates
    if col in df_targets_bar.columns
]

df_targets_opc = df_targets_bar[context_columns].copy()

print(f"\nColumnas de contexto conservadas: {context_columns}")

# -----
# 7) Construir targets OPC
# -----

opc_metadata_records = []
opc_summary_records = []

for family in OPC_SELECTED_FAMILIES:

    family_name = family["family_name"]
    target_role = family["target_role"]
    dir_col = family["dir_col"]
    bar_col = family["bar_col"]
    percentile = family["percentile"]
    horizon = family["horizon"]
    tp_mult = family["tp_mult"]
    sl_mult = family["sl_mult"]

    # Nombre del target OPC
    opc_col = bar_col.replace("bar_", "opc_")
    opc_label_col = f"{opc_col}_label"

    # Recuperar series
    dir_series = get_column_from_available_sources(dir_col)
    bar_series = get_column_from_available_sources(bar_col)

    # Construir OPC
    opc_label, opc_class = build_opc_from_dir_bar(
        dir_series=dir_series,
        bar_series=bar_series
    )

    # Guardar columnas originales necesarias para trazabilidad
    df_targets_opc[dir_col] = dir_series
    df_targets_opc[bar_col] = bar_series

    # Guardar columnas OPC
    df_targets_opc[opc_label_col] = opc_label
    df_targets_opc[opc_col] = opc_class

    # Casos no esperados:
    # DIR operativo pero BAR NaN
    unexpected_bar_nan = (
        dir_series.isin([1, -1])
        & bar_series.isna()
    ).sum()

    # Casos NaN esperados:
    # DIR NaN por falta de ventana futura
    dir_nan_count = dir_series.isna().sum()

```

```
opc_nan_count = opc_class.isna().sum()

# Distribución de clases
value_counts = (
    opc_label
    .value_counts(dropna=False)
    .rename_axis("opc_label")
    .reset_index(name="count")
)

total_rows = len(opc_label)
valid_rows = opc_label.notna().sum()
nan_rows = opc_label.isna().sum()

class_counts = {
    "NO_TRADE": int((opc_label == "NO_TRADE").sum()),
    "LONG_TP": int((opc_label == "LONG_TP").sum()),
    "LONG_SL": int((opc_label == "LONG_SL").sum()),
    "SHORT_TP": int((opc_label == "SHORT_TP").sum()),
    "SHORT_SL": int((opc_label == "SHORT_SL").sum()),
}

class_ratios = {
    key: value / valid_rows if valid_rows > 0 else np.nan
    for key, value in class_counts.items()
}

resolved_trade_count = (
    class_counts["LONG_TP"]
    + class_counts["LONG_SL"]
    + class_counts["SHORT_TP"]
    + class_counts["SHORT_SL"]
)

winner_count = (
    class_counts["LONG_TP"]
    + class_counts["SHORT_TP"]
)

loser_count = (
    class_counts["LONG_SL"]
    + class_counts["SHORT_SL"]
)

long_count = (
    class_counts["LONG_TP"]
    + class_counts["LONG_SL"]
)

short_count = (
    class_counts["SHORT_TP"]
    + class_counts["SHORT_SL"]
)

opc_summary_records.append(
    {
        "opc_col": opc_col,
        "opc_label_col": opc_label_col,
        "family_name": family_name,
        "target_role": target_role,
```



```

        "dir_col": dir_col,
        "bar_col": bar_col,
        "percentile": percentile,
        "horizon": horizon,
        "tp_mult": tp_mult,
        "sl_mult": sl_mult,

        "total_rows": total_rows,
        "valid_rows": valid_rows,
        "nan_rows": nan_rows,
        "nan_ratio": nan_rows / total_rows if total_rows > 0 else np.nan,

        "no_trade_count": class_counts["NO_TRADE"],
        "long_tp_count": class_counts["LONG_TP"],
        "long_sl_count": class_counts["LONG_SL"],
        "short_tp_count": class_counts["SHORT_TP"],
        "short_sl_count": class_counts["SHORT_SL"],

        "no_trade_ratio": class_ratios["NO_TRADE"],
        "long_tp_ratio": class_ratios["LONG_TP"],
        "long_sl_ratio": class_ratios["LONG_SL"],
        "short_tp_ratio": class_ratios["SHORT_TP"],
        "short_sl_ratio": class_ratios["SHORT_SL"],

        "resolved_trade_count": resolved_trade_count,
        "resolved_trade_ratio": resolved_trade_count / valid_rows if valid_r

        "winner_count": winner_count,
        "loser_count": loser_count,
        "winner_ratio": winner_count / valid_rows if valid_rows > 0 else np.
        "loser_ratio": loser_count / valid_rows if valid_rows > 0 else np.na

        "winner_share_over_resolved": winner_count / resolved_trade_count if
        "loser_share_over_resolved": loser_count / resolved_trade_count if r

        "long_count": long_count,
        "short_count": short_count,
        "long_ratio": long_count / valid_rows if valid_rows > 0 else np.nan,
        "short_ratio": short_count / valid_rows if valid_rows > 0 else np.na
        "long_share_over_resolved": long_count / resolved_trade_count if res
        "short_share_over_resolved": short_count / resolved_trade_count if r

        "dir_nan_count": dir_nan_count,
        "opc_nan_count": opc_nan_count,
        "unexpected_bar_nan_count": int(unexpected_bar_nan),
    }
)

opc_metadata_records.append(
    {
        "family_name": family_name,
        "target_role": target_role,
        "dir_col": dir_col,
        "bar_col": bar_col,
        "opc_col": opc_col,
        "opc_label_col": opc_label_col,
        "percentile": percentile,
        "horizon": horizon,
        "tp_mult": tp_mult,
        "sl_mult": sl_mult,
    }
)

```

```

        "class_0": "NO_TRADE",
        "class_1": "LONG_TP",
        "class_2": "LONG_SL",
        "class_3": "SHORT_TP",
        "class_4": "SHORT_SL",
        "description": "OPC combina DIR y BAR en una clase operativa nominal
    }
)

df_opc_metadata = pd.DataFrame(opc_metadata_records)
df_opc_summary = pd.DataFrame(opc_summary_records)

# -----
# 8) Ordenar columnas OPC
# -----

base_cols = context_columns.copy()

selected_target_cols = []

for family in OPC_SELECTED_FAMILIES:

    dir_col = family["dir_col"]
    bar_col = family["bar_col"]
    opc_col = bar_col.replace("bar_", "opc_")
    opc_label_col = f"{opc_col}_label"

    selected_target_cols.extend(
        [
            dir_col,
            bar_col,
            opc_label_col,
            opc_col,
        ]
    )

# Evitar duplicados manteniendo orden
ordered_cols = []
for col in base_cols + selected_target_cols:
    if col in df_targets_opc.columns and col not in ordered_cols:
        ordered_cols.append(col)

df_targets_opc = df_targets_opc[ordered_cols].copy()

# -----
# 9) Mostrar resumen
# -----

percent_cols = [
    "nan_ratio",
    "no_trade_ratio",
    "long_tp_ratio",
    "long_sl_ratio",
    "short_tp_ratio",
    "short_sl_ratio",
    "resolved_trade_ratio",
    "winner_ratio",
    "loser_ratio",
    "winner_share_over_resolved",
    "long_share_over_resolved",

```

```

        "short_share_over_resolved",
    ]

    formatters = {
        col: "{:.2%}".format
        for col in percent_cols
    }

    print("\nResumen de targets OPC contruidos:")
    print(
        df_opc_summary[
            [
                "opc_col",
                "target_role",
                "valid_rows",
                "nan_rows",
                "nan_ratio",
                "no_trade_ratio",
                "long_tp_ratio",
                "long_sl_ratio",
                "short_tp_ratio",
                "short_sl_ratio",
                "resolved_trade_ratio",
                "winner_share_over_resolved",
                "long_share_over_resolved",
                "short_share_over_resolved",
                "unexpected_bar_nan_count",
            ]
        ].to_string(
            index=False,
            formatters=formatters
        )
    )

    print("\nMetadata OPC:")
    print(
        df_opc_metadata[
            [
                "family_name",
                "target_role",
                "dir_col",
                "bar_col",
                "opc_col",
                "opc_label_col",
                "percentile",
                "horizon",
                "tp_mult",
                "sl_mult",
            ]
        ].to_string(index=False)
    )

    print("\nVista previa de df_targets_opc:")
    print(df_targets_opc.head().to_string())

    # -----
    # 10) Guardado de resultados
    # -----

    if "MNQ_TARGETS_PATH" in globals():

```

```
MNQ_TARGETS_PATH.mkdir(parents=True, exist_ok=True)

MNQ_TARGETS_OPC_PARQUET = (
    MNQ_TARGETS_PATH / "mnq_targets_opc.parquet"
)

MNQ_TARGETS_OPC_METADATA_PARQUET = (
    MNQ_TARGETS_PATH / "mnq_targets_opc_metadata.parquet"
)

MNQ_TARGETS_OPC_METADATA_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_opc_metadata.csv"
)

MNQ_TARGETS_OPC_SUMMARY_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_opc_summary.csv"
)

MNQ_TARGETS_OPC_SELECTED_FAMILIES_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_opc_selected_families.csv"
)

df_targets_opc.to_parquet(MNQ_TARGETS_OPC_PARQUET)
df_opc_metadata.to_parquet(MNQ_TARGETS_OPC_METADATA_PARQUET)
df_opc_metadata.to_csv(MNQ_TARGETS_OPC_METADATA_CSV, index=False)
df_opc_summary.to_csv(MNQ_TARGETS_OPC_SUMMARY_CSV, index=False)
df_opc_selected_families.to_csv(
    MNQ_TARGETS_OPC_SELECTED_FAMILIES_CSV,
    index=False
)

print("\nArchivos guardados:")
print(f"Targets OPC: {MNQ_TARGETS_OPC_PARQUET}")
print(f"Metadata OPC parquet: {MNQ_TARGETS_OPC_METADATA_PARQUET}")
print(f"Metadata OPC csv: {MNQ_TARGETS_OPC_METADATA_CSV}")
print(f"Resumen OPC: {MNQ_TARGETS_OPC_SUMMARY_CSV}")
print(f"Familias OPC seleccionadas: {MNQ_TARGETS_OPC_SELECTED_FAMILIES_CSV}")

else:
    print("\nMNQ_TARGETS_PATH no está definido. No se guardaron archivos.")

print("\nConstrucción práctica de OPC finalizada correctamente.")
```

## Construcción práctica de targets OPC

=====

Familias operativas seleccionadas para OPC:

| family_name             | target_role           | dir_col     | bar_col               | percentile |
|-------------------------|-----------------------|-------------|-----------------------|------------|
| horizon tp_mult sl_mult |                       |             |                       |            |
| p50_h30                 | principal             | dir_p50_h30 | bar_p50_h30_tp15_sl10 | 50         |
| 30 1.5 1.0              |                       |             |                       |            |
| p50_h60                 | principal             | dir_p50_h60 | bar_p50_h60_tp15_sl10 | 50         |
| 60 1.5 1.0              |                       |             |                       |            |
| p50_h90                 | principal             | dir_p50_h90 | bar_p50_h90_tp15_sl10 | 50         |
| 90 1.5 1.0              |                       |             |                       |            |
| p40_h60                 | comparativo_activo    | dir_p40_h60 | bar_p40_h60_tp15_sl10 | 40         |
| 60 1.5 1.0              |                       |             |                       |            |
| p60_h60                 | comparativo_selectivo | dir_p60_h60 | bar_p60_h60_tp15_sl10 | 60         |
| 60 1.5 1.0              |                       |             |                       |            |

Todas las columnas DIR y BAR requeridas fueron encontradas.

Columnas de contexto conservadas: ['date', 'minute\_of\_day', 'regime\_id', 'contract', 'year', 'open', 'high', 'low', 'close', 'volume']

Resumen de targets OPC contruidos:

| opc_col               | target_role           | valid_rows | nan_rows | nan_ratio | no_trade_ratio | long_tp_ratio | long_sl_ratio | short_tp_ratio | short_sl_ratio | resolved_trade_ratio | winner_share_over_resolved | long_share_over_resolved | short_share_over_resolved | unexpected_bar_nan_count |
|-----------------------|-----------------------|------------|----------|-----------|----------------|---------------|---------------|----------------|----------------|----------------------|----------------------------|--------------------------|---------------------------|--------------------------|
| opc_p50_h30_tp15_sl10 | principal             | 979602     | 44460    | 4.34%     |                |               |               |                |                |                      |                            |                          |                           |                          |
| 67.92%                |                       |            |          |           | 13.54%         | 1.54%         | 15.46%        | 1.55%          |                |                      | 3                          |                          |                           |                          |
| 2.08%                 |                       |            |          |           |                | 90.38%        |               | 47.00%         |                |                      | 53.0                       |                          |                           |                          |
| 0%                    |                       |            |          |           |                | 0             |               |                |                |                      |                            |                          |                           |                          |
| opc_p50_h60_tp15_sl10 | principal             | 935142     | 88920    | 8.68%     |                |               |               |                |                |                      |                            |                          |                           |                          |
| 68.80%                |                       |            |          |           | 12.87%         | 1.52%         | 15.34%        | 1.48%          |                |                      | 3                          |                          |                           |                          |
| 1.20%                 |                       |            |          |           |                | 90.41%        |               | 46.12%         |                |                      | 53.8                       |                          |                           |                          |
| 8%                    |                       |            |          |           |                | 0             |               |                |                |                      |                            |                          |                           |                          |
| opc_p50_h90_tp15_sl10 | principal             | 890682     | 133380   | 13.02%    |                |               |               |                |                |                      |                            |                          |                           |                          |
| 68.40%                |                       |            |          |           | 12.84%         | 1.63%         | 15.55%        | 1.58%          |                |                      | 3                          |                          |                           |                          |
| 1.60%                 |                       |            |          |           |                | 89.83%        |               | 45.79%         |                |                      | 54.2                       |                          |                           |                          |
| 1%                    |                       |            |          |           |                | 0             |               |                |                |                      |                            |                          |                           |                          |
| opc_p40_h60_tp15_sl10 | comparativo_activo    | 935142     | 88920    | 8.68%     |                |               |               |                |                |                      |                            |                          |                           |                          |
| 59.39%                |                       |            |          |           | 16.85%         | 2.34%         | 19.09%        | 2.33%          |                |                      | 4                          |                          |                           |                          |
| 0.61%                 |                       |            |          |           |                | 88.51%        |               | 47.26%         |                |                      | 52.7                       |                          |                           |                          |
| 4%                    |                       |            |          |           |                | 0             |               |                |                |                      |                            |                          |                           |                          |
| opc_p60_h60_tp15_sl10 | comparativo_selectivo | 935142     | 88920    | 8.68%     |                |               |               |                |                |                      |                            |                          |                           |                          |
| 76.81%                |                       |            |          |           | 9.47%          | 0.94%         | 11.89%        | 0.89%          |                |                      | 2                          |                          |                           |                          |
| 3.19%                 |                       |            |          |           |                | 92.10%        |               | 44.87%         |                |                      | 55.1                       |                          |                           |                          |
| 3%                    |                       |            |          |           |                | 0             |               |                |                |                      |                            |                          |                           |                          |

Metadata OPC:

| family_name | target_role           | dir_col     | bar_col               | opc_col               | opc_label_col               | percentile | horizon | tp_mult | sl_mult |
|-------------|-----------------------|-------------|-----------------------|-----------------------|-----------------------------|------------|---------|---------|---------|
| p50_h30     | principal             | dir_p50_h30 | bar_p50_h30_tp15_sl10 | opc_p50_h30_tp15_sl10 | opc_p50_h30_tp15_sl10_label | 50         | 30      | 1.5     | 1.0     |
| p50_h60     | principal             | dir_p50_h60 | bar_p50_h60_tp15_sl10 | opc_p50_h60_tp15_sl10 | opc_p50_h60_tp15_sl10_label | 50         | 60      | 1.5     | 1.0     |
| p50_h90     | principal             | dir_p50_h90 | bar_p50_h90_tp15_sl10 | opc_p50_h90_tp15_sl10 | opc_p50_h90_tp15_sl10_label | 50         | 90      | 1.5     | 1.0     |
| p40_h60     | comparativo_activo    | dir_p40_h60 | bar_p40_h60_tp15_sl10 | opc_p40_h60_tp15_sl10 | opc_p40_h60_tp15_sl10_label | 40         | 60      | 1.5     | 1.0     |
| p60_h60     | comparativo_selectivo | dir_p60_h60 | bar_p60_h60_tp15_sl10 | opc_p60_h60_tp15_sl10 | opc_p60_h60_tp15_sl10_label | 60         | 60      | 1.5     | 1.0     |

p15\_sl10 opc\_p60\_h60\_tp15\_sl10\_label 60 60 1.5 1.0

Vista previa de df\_targets\_opc:

| open     | high    | low     | close   | volume | dir_p50_h30 | bar_p50_h30_tp15_sl10 | opc_p50_h30_tp15_sl10_label | opc_p50_h30_tp15_sl10 | dir_p50_h60 | bar_p50_h60_tp15_sl10 | opc_p50_h60_tp15_sl10_label | opc_p50_h60_tp15_sl10 | dir_p50_h90 | bar_p50_h90_tp15_sl10 | opc_p50_h90_tp15_sl10_label | opc_p50_h90_tp15_sl10 | dir_p40_h60 | bar_p40_h60_tp15_sl10 | opc_p40_h60_tp15_sl10_label | opc_p40_h60_tp15_sl10 | dir_p60_h60 | bar_p60_h60_tp15_sl10 | opc_p60_h60_tp15_sl10_label | opc_p60_h60_tp15_sl10 | datetime                  | date       | minute_of_day | regime_id | contract | year |   |
|----------|---------|---------|---------|--------|-------------|-----------------------|-----------------------------|-----------------------|-------------|-----------------------|-----------------------------|-----------------------|-------------|-----------------------|-----------------------------|-----------------------|-------------|-----------------------|-----------------------------|-----------------------|-------------|-----------------------|-----------------------------|-----------------------|---------------------------|------------|---------------|-----------|----------|------|---|
| 813.25   | 8813.25 | 8812.50 | 8813.25 | 13     | 270         | 0                     | H20                         | 2020                  | 8           | <NA>                  |                             |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       | 2020-01-02 04:30:00-05:00 | 2020-01-02 | 270           | 0         | H20      | 2020 | 8 |
| NO_TRADE |         |         | 0.0     | 0      |             |                       | <NA>                        |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |                           |            |               |           |          |      |   |
| NO_TRADE |         |         | 0.0     | 0      |             |                       | <NA>                        |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |                           |            |               |           |          |      |   |
| NO_TRADE |         |         | 0.0     | 0      |             |                       | <NA>                        |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |                           |            |               |           |          |      |   |
| NO_TRADE |         |         | 0.0     | 0      |             |                       | <NA>                        |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |                           |            |               |           |          |      |   |
| NO_TRADE |         |         | 0.0     | 0      |             |                       | <NA>                        |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |                           |            |               |           |          |      |   |
| 812.50   | 8812.50 | 8811.00 | 8812.50 | 121    | 271         | 0                     | H20                         | 2020                  | 8           | <NA>                  |                             |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       | 2020-01-02 04:31:00-05:00 | 2020-01-02 | 271           | 0         | H20      | 2020 | 8 |
| NO_TRADE |         |         | 0.0     | 0      |             |                       | <NA>                        |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |                           |            |               |           |          |      |   |
| NO_TRADE |         |         | 0.0     | 0      |             |                       | <NA>                        |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |                           |            |               |           |          |      |   |
| NO_TRADE |         |         | 0.0     | 0      |             |                       | <NA>                        |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |                           |            |               |           |          |      |   |
| NO_TRADE |         |         | 0.0     | 0      |             |                       | <NA>                        |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |                           |            |               |           |          |      |   |
| NO_TRADE |         |         | 0.0     | 0      |             |                       | <NA>                        |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |                           |            |               |           |          |      |   |
| 812.50   | 8813.25 | 8811.75 | 8811.75 | 53     | 272         | 0                     | H20                         | 2020                  | 8           | <NA>                  |                             |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       | 2020-01-02 04:32:00-05:00 | 2020-01-02 | 272           | 0         | H20      | 2020 | 8 |
| NO_TRADE |         |         | 0.0     | 0      |             |                       | <NA>                        |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |                           |            |               |           |          |      |   |
| NO_TRADE |         |         | 0.0     | 0      |             |                       | <NA>                        |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |                           |            |               |           |          |      |   |
| NO_TRADE |         |         | 0.0     | 0      |             |                       | <NA>                        |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |                           |            |               |           |          |      |   |
| NO_TRADE |         |         | 0.0     | 0      |             |                       | <NA>                        |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |                           |            |               |           |          |      |   |
| NO_TRADE |         |         | 0.0     | 0      |             |                       | <NA>                        |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |                           |            |               |           |          |      |   |
| 812.00   | 8812.25 | 8810.50 | 8810.50 | 37     | 273         | 0                     | H20                         | 2020                  | 8           | <NA>                  |                             |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       | 2020-01-02 04:33:00-05:00 | 2020-01-02 | 273           | 0         | H20      | 2020 | 8 |
| NO_TRADE |         |         | 0.0     | 0      |             |                       | <NA>                        |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |                           |            |               |           |          |      |   |
| NO_TRADE |         |         | 0.0     | 0      |             |                       | <NA>                        |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |                           |            |               |           |          |      |   |
| NO_TRADE |         |         | 0.0     | 0      |             |                       | <NA>                        |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |                           |            |               |           |          |      |   |
| NO_TRADE |         |         | 0.0     | 0      |             |                       | <NA>                        |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |                           |            |               |           |          |      |   |
| NO_TRADE |         |         | 0.0     | 0      |             |                       | <NA>                        |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |                           |            |               |           |          |      |   |
| 810.75   | 8812.25 | 8810.50 | 8812.00 | 36     | 274         | 0                     | H20                         | 2020                  | 8           | <NA>                  |                             |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       | 2020-01-02 04:34:00-05:00 | 2020-01-02 | 274           | 0         | H20      | 2020 | 8 |
| NO_TRADE |         |         | 0.0     | 0      |             |                       | <NA>                        |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |                           |            |               |           |          |      |   |
| NO_TRADE |         |         | 0.0     | 0      |             |                       | <NA>                        |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |                           |            |               |           |          |      |   |
| NO_TRADE |         |         | 0.0     | 0      |             |                       | <NA>                        |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |                           |            |               |           |          |      |   |
| NO_TRADE |         |         | 0.0     | 0      |             |                       | <NA>                        |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |                           |            |               |           |          |      |   |
| NO_TRADE |         |         | 0.0     | 0      |             |                       | <NA>                        |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |             |                       |                             |                       |                           |            |               |           |          |      |   |

Archivos guardados:

Targets OPC: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_opc.parquet  
 Metadata OPC parquet: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_opc\_metadata.parquet  
 Metadata OPC csv: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_opc\_metadata.csv  
 Resumen OPC: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_opc\_summary.csv  
 Familias OPC seleccionadas: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_opc\_selected\_families.csv

Construcción práctica de OPC finalizada correctamente.

## Observaciones rápidas:

La construcción de `OPC` fue consistente: las 5 familias operativas fueron generadas correctamente y no aparecen casos inesperados de `BAR NaN` cuando `DIR` era operable.

Los `NaN` aumentan con el horizonte, como era esperable: `H30` tiene 4.34%, `H60` 8.68% y `H90` 13.02%. Esto confirma que se está respetando la regla de no cruzar días.

En los targets principales `p50`, la clase dominante es `NO_TRADE`, alrededor de 68%. Esto era esperable porque `NO_TRADE` agrupa tanto ausencia de dirección operativa como ausencia de resolución por barreras.

La proporción de trades resueltos en `p50` queda bastante estable entre horizontes: aproximadamente 31%–32%. Esto indica que `H30`, `H60` y `H90` tienen una estructura operativa similar bajo `TP 1.5 / SL 1.0`.

Entre los trades resueltos, el porcentaje ganador es alto: alrededor de 90%. Esto refleja lo que ya vimos en `BAR`: la configuración `TP 1.5 / SL 1.0` mantiene una proporción favorable de TP sobre operaciones resueltas.

Hay una leve inclinación hacia operaciones `SHORT`: en los targets `p50`, los `SHORT` representan aproximadamente 53%–54% de los trades resueltos, mientras que los `LONG` quedan cerca de 46%–47%.

El target comparativo activo `opc_p40_h60_tp15_sl10` reduce bastante `NO_TRADE` a 59.39% y aumenta los trades resueltos a 40.61%. Es más operativo, pero probablemente más ruidoso.

El target comparativo selectivo `opc_p60_h60_tp15_sl10` aumenta `NO_TRADE` a 76.81% y reduce los trades resueltos a 23.19%. Es más exigente, pero con menos muestras útiles.

`p60_h60` tiene el mayor `winner_share_over_resolved`, con 92.10%. Esto sugiere que la versión más selectiva genera menos operaciones, pero más limpias entre las que resuelven.

En síntesis: `OPC` quedó bien construido. Los targets principales `p50` son equilibrados y comparables entre horizontes. `p40_h60` sirve como versión activa y `p60_h60` como versión selectiva.

## 6.5. Validación de consistencia

Esta celda deja creados estos objetos:

```
df_opc_column_validation
df_opc_value_validation
df_opc_mapping_validation
df_opc_nan_validation
```

```
df_opc_class_balance_validation
df_opc_validation_summary
```

Y valida principalmente:

1. Existencia de columnas DIR, BAR y OPC.
2. Valores permitidos en etiquetas OPC.
3. Valores permitidos en clases numéricas.
4. Consistencia etiqueta ↔ clase.
5. Consistencia DIR + BAR → OPC.
6. Manejo correcto de NaN.
7. Reglas NO\_TRADE.
8. Ausencia de casos inesperados.

```
In [27]: # 6.5. Validación de consistencia de OPC
# =====

import numpy as np
import pandas as pd

print("Validación de consistencia de targets OPC")
print("=" * 80)

# -----
# 1) Validar objetos necesarios
# -----

required_objects = [
    "df_targets_opc",
    "df_opc_metadata",
    "df_opc_selected_families",
]

missing_objects = [
    obj for obj in required_objects
    if obj not in globals()
]

if missing_objects:
    raise NameError(
        "Faltan objetos necesarios para validar OPC: "
        f"{missing_objects}"
    )

print("Objetos base encontrados correctamente.")

# -----
# 2) Definir mapa de clases OPC
# -----

OPC_CLASS_MAP = {
    "NO_TRADE": 0,
    "LONG_TP": 1,
    "LONG_SL": 2,
    "SHORT_TP": 3,
    "SHORT_SL": 4,
}

OPC_ALLOWED_LABELS = set(OPC_CLASS_MAP.keys())
OPC_ALLOWED_CLASSES = set(OPC_CLASS_MAP.values())
```



```

print("\nMapa de clases OPC:")
for label, value in OPC_CLASS_MAP.items():
    print(f"{value}: {label}")

# -----
# 3) Función para construir etiqueta OPC esperada
# -----

def expected_opc_label_from_dir_bar(dir_series, bar_series):
    """
    Reconstruye la etiqueta OPC esperada a partir de DIR y BAR.

    Reglas:
    - DIR NaN                -> NaN
    - DIR = 0                 -> NO_TRADE
    - DIR = 1 y BAR = 1       -> LONG_TP
    - DIR = 1 y BAR = -1      -> LONG_SL
    - DIR = -1 y BAR = 1      -> SHORT_TP
    - DIR = -1 y BAR = -1     -> SHORT_SL
    - DIR +/-1 y BAR = 0      -> NO_TRADE
    - DIR +/-1 y BAR NaN      -> NaN, caso no esperado
    """

    expected_label = pd.Series(
        data=np.nan,
        index=dir_series.index,
        dtype="object"
    )

    mask_no_signal = dir_series.eq(0)
    mask_no_event = dir_series.isin([1, -1]) & bar_series.eq(0)

    mask_long_tp = dir_series.eq(1) & bar_series.eq(1)
    mask_long_sl = dir_series.eq(1) & bar_series.eq(-1)

    mask_short_tp = dir_series.eq(-1) & bar_series.eq(1)
    mask_short_sl = dir_series.eq(-1) & bar_series.eq(-1)

    expected_label.loc[mask_no_signal | mask_no_event] = "NO_TRADE"
    expected_label.loc[mask_long_tp] = "LONG_TP"
    expected_label.loc[mask_long_sl] = "LONG_SL"
    expected_label.loc[mask_short_tp] = "SHORT_TP"
    expected_label.loc[mask_short_sl] = "SHORT_SL"

    return expected_label

# -----
# 4) Validar existencia de columnas
# -----

required_columns_records = []

for _, row in df_opc_metadata.iterrows():

    dir_col = row["dir_col"]
    bar_col = row["bar_col"]
    opc_label_col = row["opc_label_col"]
    opc_col = row["opc_col"]

```

```

required_columns_records.append(
    {
        "family_name": row["family_name"],
        "dir_col": dir_col,
        "dir_col_exists": dir_col in df_targets_opc.columns,
        "bar_col": bar_col,
        "bar_col_exists": bar_col in df_targets_opc.columns,
        "opc_label_col": opc_label_col,
        "opc_label_col_exists": opc_label_col in df_targets_opc.columns,
        "opc_col": opc_col,
        "opc_col_exists": opc_col in df_targets_opc.columns,
    }
)

df_opc_column_validation = pd.DataFrame(required_columns_records)

print("\nValidación de columnas requeridas:")
print(df_opc_column_validation.to_string(index=False))

missing_cols = []

for col in [
    "dir_col",
    "bar_col",
    "opc_label_col",
    "opc_col",
]:
    missing_tmp = df_opc_column_validation.loc[
        ~df_opc_column_validation[f"{col}_exists"],
        col
    ].tolist()

    missing_cols.extend(missing_tmp)

if missing_cols:
    raise KeyError(
        "Faltan columnas necesarias para validar OPC:\n"
        f"{missing_cols}"
    )

print("\nTodas las columnas requeridas existen en df_targets_opc.")

# -----
# 5) Validaciones por target OPC
# -----

value_validation_records = []
mapping_validation_records = []
nan_validation_records = []
class_balance_records = []

for _, row in df_opc_metadata.iterrows():

    family_name = row["family_name"]
    target_role = row["target_role"]

    dir_col = row["dir_col"]
    bar_col = row["bar_col"]
    opc_label_col = row["opc_label_col"]

```

```

opc_col = row["opc_col"]

dir_series = df_targets_opc[dir_col]
bar_series = df_targets_opc[bar_col]
opc_label = df_targets_opc[opc_label_col]
opc_class = df_targets_opc[opc_col]

total_rows = len(df_targets_opc)

# -----
# Valores permitidos en labels
# -----

observed_labels = set(
    opc_label.dropna().unique().tolist()
)

invalid_labels = sorted(
    list(observed_labels - OPC_ALLOWED_LABELS)
)

# -----
# Valores permitidos en clases numéricas
# -----

observed_classes = set(
    opc_class.dropna().astype(int).unique().tolist()
)

invalid_classes = sorted(
    list(observed_classes - OPC_ALLOWED_CLASSES)
)

non_integer_class_count = (
    opc_class
    .dropna()
    .apply(lambda x: float(x).is_integer())
    .eq(False)
    .sum()
)

value_validation_records.append(
    {
        "family_name": family_name,
        "target_role": target_role,
        "opc_col": opc_col,
        "opc_label_col": opc_label_col,
        "total_rows": total_rows,
        "observed_labels": sorted(list(observed_labels)),
        "invalid_labels": invalid_labels,
        "n_invalid_labels": len(invalid_labels),
        "observed_classes": sorted(list(observed_classes)),
        "invalid_classes": invalid_classes,
        "n_invalid_classes": len(invalid_classes),
        "non_integer_class_count": int(non_integer_class_count),
    }
)

# -----
# Consistencia label ↔ clase numérica

```

```

# -----

expected_class_from_label = opc_label.map(OPC_CLASS_MAP).astype("float")

label_class_mismatch_mask = (
    expected_class_from_label.notna()
    & opc_class.notna()
    & expected_class_from_label.ne(opc_class)
)

label_class_nan_mismatch_mask = (
    expected_class_from_label.isna()
    ^ opc_class.isna()
)

label_class_mismatch_count = int(label_class_mismatch_mask.sum())
label_class_nan_mismatch_count = int(label_class_nan_mismatch_mask.sum())

# -----
# Consistencia DIR + BAR → OPC
# -----

expected_label = expected_opc_label_from_dir_bar(
    dir_series=dir_series,
    bar_series=bar_series
)

expected_class = expected_label.map(OPC_CLASS_MAP).astype("float")

mapping_label_mismatch_mask = (
    expected_label.notna()
    & opc_label.notna()
    & expected_label.ne(opc_label)
)

mapping_label_nan_mismatch_mask = (
    expected_label.isna()
    ^ opc_label.isna()
)

mapping_class_mismatch_mask = (
    expected_class.notna()
    & opc_class.notna()
    & expected_class.ne(opc_class)
)

mapping_class_nan_mismatch_mask = (
    expected_class.isna()
    ^ opc_class.isna()
)

mapping_validation_records.append(
    {
        "family_name": family_name,
        "target_role": target_role,
        "dir_col": dir_col,
        "bar_col": bar_col,
        "opc_col": opc_col,
        "opc_label_col": opc_label_col,
    }
)

```

```

        "label_class_mismatch_count": label_class_mismatch_count,
        "label_class_nan_mismatch_count": label_class_nan_mismatch_count,

        "mapping_label_mismatch_count": int(mapping_label_mismatch_mask.sum(
        "mapping_label_nan_mismatch_count": int(mapping_label_nan_mismatch_m

        "mapping_class_mismatch_count": int(mapping_class_mismatch_mask.sum(
        "mapping_class_nan_mismatch_count": int(mapping_class_nan_mismatch_m

    }
)

# -----
# Validación de NaN y casos no esperados
# -----

dir_nan_count = int(dir_series.isna().sum())
bar_nan_count = int(bar_series.isna().sum())
opc_label_nan_count = int(opc_label.isna().sum())
opc_class_nan_count = int(opc_class.isna().sum())

# Caso no esperado:
# DIR operativo pero BAR NaN
unexpected_bar_nan_count = int(
    (
        dir_series.isin([1, -1])
        & bar_series.isna()
    ).sum()
)

# DIR NaN debería producir OPC NaN
dir_nan_opc_not_nan_count = int(
    (
        dir_series.isna()
        & opc_label.notna()
    ).sum()
)

# DIR = 0 debería producir NO_TRADE
dir_zero_not_no_trade_count = int(
    (
        dir_series.eq(0)
        & opc_label.ne("NO_TRADE")
    ).sum()
)

# DIR operativo + BAR = 0 debería producir NO_TRADE
bar_zero_not_no_trade_count = int(
    (
        dir_series.isin([1, -1])
        & bar_series.eq(0)
        & opc_label.ne("NO_TRADE")
    ).sum()
)

nan_validation_records.append(
    {
        "family_name": family_name,
        "target_role": target_role,
        "opc_col": opc_col,

```

```

        "total_rows": total_rows,

        "dir_nan_count": dir_nan_count,
        "bar_nan_count": bar_nan_count,
        "opc_label_nan_count": opc_label_nan_count,
        "opc_class_nan_count": opc_class_nan_count,

        "dir_nan_ratio": dir_nan_count / total_rows if total_rows > 0 else n
        "opc_nan_ratio": opc_class_nan_count / total_rows if total_rows > 0

        "unexpected_bar_nan_count": unexpected_bar_nan_count,
        "dir_nan_opc_not_nan_count": dir_nan_opc_not_nan_count,
        "dir_zero_not_no_trade_count": dir_zero_not_no_trade_count,
        "bar_zero_not_no_trade_count": bar_zero_not_no_trade_count,
    }
)

# -----
# Distribución de clases para control
# -----

class_counts = (
    opc_label
    .value_counts(dropna=False)
    .rename_axis("opc_label")
    .reset_index(name="count")
)

valid_rows = opc_label.notna().sum()

for _, class_row in class_counts.iterrows():

    class_label = class_row["opc_label"]
    class_count = int(class_row["count"])

    class_balance_records.append(
        {
            "family_name": family_name,
            "target_role": target_role,
            "opc_col": opc_col,
            "opc_label": class_label,
            "count": class_count,
            "ratio_over_valid": (
                class_count / valid_rows
                if valid_rows > 0 and pd.notna(class_label)
                else np.nan
            ),
            "ratio_over_total": class_count / total_rows if total_rows > 0 e
        }
    )

# -----
# 6) Crear dataframes de validación
# -----

df_opc_value_validation = pd.DataFrame(value_validation_records)
df_opc_mapping_validation = pd.DataFrame(mapping_validation_records)
df_opc_nan_validation = pd.DataFrame(nan_validation_records)
df_opc_class_balance_validation = pd.DataFrame(class_balance_records)

```

```

# -----
# 7) Resumen general de validación
# -----

df_opc_validation_summary = pd.DataFrame(
    [
        {
            "validation_check": "columnas_requeridas",
            "status": "OK" if not missing_cols else "ERROR",
            "error_count": len(missing_cols),
        },
        {
            "validation_check": "labels_permitidos",
            "status": (
                "OK"
                if df_opc_value_validation["n_invalid_labels"].sum() == 0
                else "ERROR"
            ),
            "error_count": int(df_opc_value_validation["n_invalid_labels"].sum()),
        },
        {
            "validation_check": "clases_permitidas",
            "status": (
                "OK"
                if df_opc_value_validation["n_invalid_classes"].sum() == 0
                else "ERROR"
            ),
            "error_count": int(df_opc_value_validation["n_invalid_classes"].sum()),
        },
        {
            "validation_check": "clases_enteras",
            "status": (
                "OK"
                if df_opc_value_validation["non_integer_class_count"].sum() == 0
                else "ERROR"
            ),
            "error_count": int(df_opc_value_validation["non_integer_class_count"].sum()),
        },
        {
            "validation_check": "label_vs_clase",
            "status": (
                "OK"
                if (
                    df_opc_mapping_validation["label_class_mismatch_count"].sum()
                    and df_opc_mapping_validation["label_class_nan_mismatch_count"].sum()
                )
                else "ERROR"
            ),
            "error_count": int(
                df_opc_mapping_validation["label_class_mismatch_count"].sum()
                + df_opc_mapping_validation["label_class_nan_mismatch_count"].sum()
            ),
        },
        {
            "validation_check": "dir_bar_vs_opc",
            "status": (
                "OK"
                if (
                    df_opc_mapping_validation["mapping_label_mismatch_count"].sum()
                    and df_opc_mapping_validation["mapping_label_nan_mismatch_count"].sum()
                )
            ),
        },
    ]
)

```

```

        and df_opc_mapping_validation["mapping_class_mismatch_count"]
        and df_opc_mapping_validation["mapping_class_nan_mismatch_co
    )
    else "ERROR"
),
"error_count": int(
    df_opc_mapping_validation["mapping_label_mismatch_count"].sum()
    + df_opc_mapping_validation["mapping_label_nan_mismatch_count"].
    + df_opc_mapping_validation["mapping_class_mismatch_count"].sum(
    + df_opc_mapping_validation["mapping_class_nan_mismatch_count"].
),
},
{
    "validation_check": "dir_operativo_bar_nan",
    "status": (
        "OK"
        if df_opc_nan_validation["unexpected_bar_nan_count"].sum() == 0
        else "ERROR"
    ),
    "error_count": int(df_opc_nan_validation["unexpected_bar_nan_count"]
},
{
    "validation_check": "dir_nan_opc_nan",
    "status": (
        "OK"
        if df_opc_nan_validation["dir_nan_opc_not_nan_count"].sum() == 0
        else "ERROR"
    ),
    "error_count": int(df_opc_nan_validation["dir_nan_opc_not_nan_count"]
},
{
    "validation_check": "dir_0_no_trade",
    "status": (
        "OK"
        if df_opc_nan_validation["dir_zero_not_no_trade_count"].sum() ==
        else "ERROR"
    ),
    "error_count": int(df_opc_nan_validation["dir_zero_not_no_trade_coun
},
{
    "validation_check": "bar_0_no_trade",
    "status": (
        "OK"
        if df_opc_nan_validation["bar_zero_not_no_trade_count"].sum() ==
        else "ERROR"
    ),
    "error_count": int(df_opc_nan_validation["bar_zero_not_no_trade_coun
},
    ]
)

# -----
# 8) Imprimir resultados
# -----

print("\nResumen general de validación OPC:")
print(df_opc_validation_summary.to_string(index=False))

print("\nValidación de valores permitidos:")
print(

```



```

df_opc_value_validation[
    [
        "family_name",
        "target_role",
        "opc_col",
        "observed_labels",
        "invalid_labels",
        "observed_classes",
        "invalid_classes",
        "non_integer_class_count",
    ]
].to_string(index=False)
)

print("\nValidación de consistencia DIR + BAR -> OPC:")
print(
    df_opc_mapping_validation.to_string(index=False)
)

print("\nValidación de NaN y reglas NO_TRADE:")
print(
    df_opc_nan_validation.to_string(
        index=False,
        formatters={
            "dir_nan_ratio": "{:.2%}".format,
            "opc_nan_ratio": "{:.2%}".format,
        }
    )
)

print("\nDistribución de clases usada como control:")
print(
    df_opc_class_balance_validation.to_string(
        index=False,
        formatters={
            "ratio_over_valid": "{:.2%}".format,
            "ratio_over_total": "{:.2%}".format,
        }
    )
)

# -----
# 9) Guardado de validaciones
# -----

if "MNQ_TARGETS_PATH" in globals():

    MNQ_TARGETS_OPC_COLUMN_VALIDATION_CSV = (
        MNQ_TARGETS_PATH / "mnq_targets_opc_column_validation.csv"
    )

    MNQ_TARGETS_OPC_VALUE_VALIDATION_CSV = (
        MNQ_TARGETS_PATH / "mnq_targets_opc_value_validation.csv"
    )

    MNQ_TARGETS_OPC_MAPPING_VALIDATION_CSV = (
        MNQ_TARGETS_PATH / "mnq_targets_opc_mapping_validation.csv"
    )

    MNQ_TARGETS_OPC_NAN_VALIDATION_CSV = (

```

```

        MNQ_TARGETS_PATH / "mnq_targets_opc_nan_validation.csv"
    )

    MNQ_TARGETS_OPC_CLASS_BALANCE_VALIDATION_CSV = (
        MNQ_TARGETS_PATH / "mnq_targets_opc_class_balance_validation.csv"
    )

    MNQ_TARGETS_OPC_VALIDATION_SUMMARY_CSV = (
        MNQ_TARGETS_PATH / "mnq_targets_opc_validation_summary.csv"
    )

    df_opc_column_validation.to_csv(
        MNQ_TARGETS_OPC_COLUMN_VALIDATION_CSV,
        index=False
    )

    df_opc_value_validation.to_csv(
        MNQ_TARGETS_OPC_VALUE_VALIDATION_CSV,
        index=False
    )

    df_opc_mapping_validation.to_csv(
        MNQ_TARGETS_OPC_MAPPING_VALIDATION_CSV,
        index=False
    )

    df_opc_nan_validation.to_csv(
        MNQ_TARGETS_OPC_NAN_VALIDATION_CSV,
        index=False
    )

    df_opc_class_balance_validation.to_csv(
        MNQ_TARGETS_OPC_CLASS_BALANCE_VALIDATION_CSV,
        index=False
    )

    df_opc_validation_summary.to_csv(
        MNQ_TARGETS_OPC_VALIDATION_SUMMARY_CSV,
        index=False
    )

    print("\nArchivos de validación guardados:")
    print(f"Columnas OPC:           {MNQ_TARGETS_OPC_COLUMN_VALIDATION_CSV}")
    print(f"Valores OPC:              {MNQ_TARGETS_OPC_VALUE_VALIDATION_CSV}")
    print(f"Mapping DIR/BAR/OPC:      {MNQ_TARGETS_OPC_MAPPING_VALIDATION_CSV}")
    print(f"NaN OPC:                  {MNQ_TARGETS_OPC_NAN_VALIDATION_CSV}")
    print(f"Balance de clases OPC:    {MNQ_TARGETS_OPC_CLASS_BALANCE_VALIDATION_CSV}")
    print(f"Resumen validación OPC:   {MNQ_TARGETS_OPC_VALIDATION_SUMMARY_CSV}")

else:
    print("\nMNQ_TARGETS_PATH no está definido. No se guardaron archivos.")

# -----
# 10) Resultado final
# -----

n_errors = df_opc_validation_summary["error_count"].sum()

if n_errors == 0:
    print("\nValidación OPC finalizada correctamente: no se detectaron errores.")

```

```
else:  
    print("\nValidación OPC finalizada con errores.")  
    print(f"Cantidad total de errores detectados: {n_errors}")
```

## Validación de consistencia de targets OPC

=====

Objetos base encontrados correctamente.

## Mapa de clases OPC:

0: NO\_TRADE  
 1: LONG\_TP  
 2: LONG\_SL  
 3: SHORT\_TP  
 4: SHORT\_SL

## Validación de columnas requeridas:

| family_name              | dir_col              | dir_col_exists | bar_col               | bar_col_exists        |   |
|--------------------------|----------------------|----------------|-----------------------|-----------------------|---|
| opc_label_col            | opc_label_col_exists |                | opc_col               | opc_col_exists        |   |
| p50_h30                  | dir_p50_h30          | True           | bar_p50_h30_tp15_sl10 | True                  |   |
| _p50_h30_tp15_sl10_label |                      |                | True                  | opc_p50_h30_tp15_sl10 | T |
| rue                      |                      |                |                       |                       |   |
| p50_h60                  | dir_p50_h60          | True           | bar_p50_h60_tp15_sl10 | True                  |   |
| _p50_h60_tp15_sl10_label |                      |                | True                  | opc_p50_h60_tp15_sl10 | T |
| rue                      |                      |                |                       |                       |   |
| p50_h90                  | dir_p50_h90          | True           | bar_p50_h90_tp15_sl10 | True                  |   |
| _p50_h90_tp15_sl10_label |                      |                | True                  | opc_p50_h90_tp15_sl10 | T |
| rue                      |                      |                |                       |                       |   |
| p40_h60                  | dir_p40_h60          | True           | bar_p40_h60_tp15_sl10 | True                  |   |
| _p40_h60_tp15_sl10_label |                      |                | True                  | opc_p40_h60_tp15_sl10 | T |
| rue                      |                      |                |                       |                       |   |
| p60_h60                  | dir_p60_h60          | True           | bar_p60_h60_tp15_sl10 | True                  |   |
| _p60_h60_tp15_sl10_label |                      |                | True                  | opc_p60_h60_tp15_sl10 | T |
| rue                      |                      |                |                       |                       |   |

Todas las columnas requeridas existen en df\_targets\_opc.

## Resumen general de validación OPC:

| validation_check      | status | error_count |
|-----------------------|--------|-------------|
| columnas_requeridas   | OK     | 0           |
| labels_permitidos     | OK     | 0           |
| clases_permitidas     | OK     | 0           |
| clases_enteras        | OK     | 0           |
| label_vs_clase        | OK     | 0           |
| dir_bar_vs_opc        | OK     | 0           |
| dir_operativo_bar_nan | OK     | 0           |
| dir_nan_opc_nan       | OK     | 0           |
| dir_0_no_trade        | OK     | 0           |
| bar_0_no_trade        | OK     | 0           |

## Validación de valores permitidos:

| family_name | target_role        | opc_col               | observed_labels                                  | invalid_labels  | observed_classes | invalid_classes | non_integer_classes_count |
|-------------|--------------------|-----------------------|--------------------------------------------------|-----------------|------------------|-----------------|---------------------------|
| p50_h30     | principal          | opc_p50_h30_tp15_sl10 | [LONG_SL, LONG_TP, NO_TRADE, SHORT_SL, SHORT_TP] | [0, 1, 2, 3, 4] | [0, 1, 2, 3, 4]  | [0, 1, 2, 3, 4] | [0, 1, 2, 3, 4]           |
| p50_h60     | principal          | opc_p50_h60_tp15_sl10 | [LONG_SL, LONG_TP, NO_TRADE, SHORT_SL, SHORT_TP] | [0, 1, 2, 3, 4] | [0, 1, 2, 3, 4]  | [0, 1, 2, 3, 4] | [0, 1, 2, 3, 4]           |
| p50_h90     | principal          | opc_p50_h90_tp15_sl10 | [LONG_SL, LONG_TP, NO_TRADE, SHORT_SL, SHORT_TP] | [0, 1, 2, 3, 4] | [0, 1, 2, 3, 4]  | [0, 1, 2, 3, 4] | [0, 1, 2, 3, 4]           |
| p40_h60     | comparativo_activo | opc_p40_h60_tp15_sl10 | [LONG_SL, LONG_TP, NO_TRADE, SHORT_SL, SHORT_TP] | [0, 1, 2, 3, 4] | [0, 1, 2, 3, 4]  | [0, 1, 2, 3, 4] | [0, 1, 2, 3, 4]           |

```

0
    p60_h60 comparativo_selectivo opc_p60_h60_tp15_sl10 [LONG_SL, LONG_TP, NO_TRA
DE, SHORT_SL, SHORT_TP]          [] [0, 1, 2, 3, 4]          []
0

```

Validación de consistencia DIR + BAR -> OPC:

```

family_name      target_role      dir_col      bar_col
opc_col          opc_label_col  label_class_mismatch_count  label_class_nan_
mismatch_count  mapping_label_mismatch_count  mapping_label_nan_mismatch_count  m
apping_class_mismatch_count  mapping_class_nan_mismatch_count
    p50_h30          principal dir_p50_h30 bar_p50_h30_tp15_sl10 opc_p50_h30_t
p15_sl10 opc_p50_h30_tp15_sl10_label          0
0          0          0
0          0
    p50_h60          principal dir_p50_h60 bar_p50_h60_tp15_sl10 opc_p50_h60_t
p15_sl10 opc_p50_h60_tp15_sl10_label          0
0          0          0
0          0
    p50_h90          principal dir_p50_h90 bar_p50_h90_tp15_sl10 opc_p50_h90_t
p15_sl10 opc_p50_h90_tp15_sl10_label          0
0          0          0
0          0
    p40_h60    comparativo_activo dir_p40_h60 bar_p40_h60_tp15_sl10 opc_p40_h60_t
p15_sl10 opc_p40_h60_tp15_sl10_label          0
0          0          0
0          0
    p60_h60 comparativo_selectivo dir_p60_h60 bar_p60_h60_tp15_sl10 opc_p60_h60_t
p15_sl10 opc_p60_h60_tp15_sl10_label          0
0          0          0
0          0

```

Validación de NaN y reglas NO\_TRADE:

```

family_name      target_role      opc_col  total_rows  dir_nan_coun
t  bar_nan_count  opc_label_nan_count  opc_class_nan_count  dir_nan_ratio  opc_nan_
ratio  unexpected_bar_nan_count  dir_nan_opc_not_nan_count  dir_zero_not_no_trade
_count  bar_zero_not_no_trade_count
    p50_h30          principal opc_p50_h30_tp15_sl10      1024062      4446
0      494080          44460      44460      4.34%
4.34%          0          0
0          0
    p50_h60          principal opc_p50_h60_tp15_sl10      1024062      8892
0      519042          88920      88920      8.68%
8.68%          0          0
0          0
    p50_h90          principal opc_p50_h90_tp15_sl10      1024062      13338
0      541459          133380      133380      13.02%      1
3.02%          0          0
0          0
    p40_h60    comparativo_activo opc_p40_h60_tp15_sl10      1024062      8892
0      427038          88920      88920      8.68%
8.68%          0          0
0          0
    p60_h60 comparativo_selectivo opc_p60_h60_tp15_sl10      1024062      8892
0      610264          88920      88920      8.68%
8.68%          0          0
0          0

```

Distribución de clases usada como control:

```

family_name      target_role      opc_col  opc_label  count  ratio_ov
er_valid  ratio_over_total

```

|         |                       |                       |          |        |
|---------|-----------------------|-----------------------|----------|--------|
| p50_h30 | principal             | opc_p50_h30_tp15_sl10 | NO_TRADE | 665304 |
| 67.92%  | 64.97%                |                       |          |        |
| p50_h30 | principal             | opc_p50_h30_tp15_sl10 | SHORT_TP | 151418 |
| 15.46%  | 14.79%                |                       |          |        |
| p50_h30 | principal             | opc_p50_h30_tp15_sl10 | LONG_TP  | 132636 |
| 13.54%  | 12.95%                |                       |          |        |
| p50_h30 | principal             | opc_p50_h30_tp15_sl10 | NaN      | 44460  |
| NaN     | 4.34%                 |                       |          |        |
| p50_h30 | principal             | opc_p50_h30_tp15_sl10 | SHORT_SL | 15159  |
| 1.55%   | 1.48%                 |                       |          |        |
| p50_h30 | principal             | opc_p50_h30_tp15_sl10 | LONG_SL  | 15085  |
| 1.54%   | 1.47%                 |                       |          |        |
| p50_h60 | principal             | opc_p50_h60_tp15_sl10 | NO_TRADE | 643374 |
| 68.80%  | 62.83%                |                       |          |        |
| p50_h60 | principal             | opc_p50_h60_tp15_sl10 | SHORT_TP | 143412 |
| 15.34%  | 14.00%                |                       |          |        |
| p50_h60 | principal             | opc_p50_h60_tp15_sl10 | LONG_TP  | 120384 |
| 12.87%  | 11.76%                |                       |          |        |
| p50_h60 | principal             | opc_p50_h60_tp15_sl10 | NaN      | 88920  |
| NaN     | 8.68%                 |                       |          |        |
| p50_h60 | principal             | opc_p50_h60_tp15_sl10 | LONG_SL  | 14172  |
| 1.52%   | 1.38%                 |                       |          |        |
| p50_h60 | principal             | opc_p50_h60_tp15_sl10 | SHORT_SL | 13800  |
| 1.48%   | 1.35%                 |                       |          |        |
| p50_h90 | principal             | opc_p50_h90_tp15_sl10 | NO_TRADE | 609262 |
| 68.40%  | 59.49%                |                       |          |        |
| p50_h90 | principal             | opc_p50_h90_tp15_sl10 | SHORT_TP | 138461 |
| 15.55%  | 13.52%                |                       |          |        |
| p50_h90 | principal             | opc_p50_h90_tp15_sl10 | NaN      | 133380 |
| NaN     | 13.02%                |                       |          |        |
| p50_h90 | principal             | opc_p50_h90_tp15_sl10 | LONG_TP  | 114341 |
| 12.84%  | 11.17%                |                       |          |        |
| p50_h90 | principal             | opc_p50_h90_tp15_sl10 | LONG_SL  | 14518  |
| 1.63%   | 1.42%                 |                       |          |        |
| p50_h90 | principal             | opc_p50_h90_tp15_sl10 | SHORT_SL | 14100  |
| 1.58%   | 1.38%                 |                       |          |        |
| p40_h60 | comparativo_activo    | opc_p40_h60_tp15_sl10 | NO_TRADE | 555377 |
| 59.39%  | 54.23%                |                       |          |        |
| p40_h60 | comparativo_activo    | opc_p40_h60_tp15_sl10 | SHORT_TP | 178523 |
| 19.09%  | 17.43%                |                       |          |        |
| p40_h60 | comparativo_activo    | opc_p40_h60_tp15_sl10 | LONG_TP  | 157597 |
| 16.85%  | 15.39%                |                       |          |        |
| p40_h60 | comparativo_activo    | opc_p40_h60_tp15_sl10 | NaN      | 88920  |
| NaN     | 8.68%                 |                       |          |        |
| p40_h60 | comparativo_activo    | opc_p40_h60_tp15_sl10 | LONG_SL  | 21891  |
| 2.34%   | 2.14%                 |                       |          |        |
| p40_h60 | comparativo_activo    | opc_p40_h60_tp15_sl10 | SHORT_SL | 21754  |
| 2.33%   | 2.12%                 |                       |          |        |
| p60_h60 | comparativo_selectivo | opc_p60_h60_tp15_sl10 | NO_TRADE | 718245 |
| 76.81%  | 70.14%                |                       |          |        |
| p60_h60 | comparativo_selectivo | opc_p60_h60_tp15_sl10 | SHORT_TP | 111200 |
| 11.89%  | 10.86%                |                       |          |        |
| p60_h60 | comparativo_selectivo | opc_p60_h60_tp15_sl10 | NaN      | 88920  |
| NaN     | 8.68%                 |                       |          |        |
| p60_h60 | comparativo_selectivo | opc_p60_h60_tp15_sl10 | LONG_TP  | 88560  |
| 9.47%   | 8.65%                 |                       |          |        |
| p60_h60 | comparativo_selectivo | opc_p60_h60_tp15_sl10 | LONG_SL  | 8768   |
| 0.94%   | 0.86%                 |                       |          |        |
| p60_h60 | comparativo_selectivo | opc_p60_h60_tp15_sl10 | SHORT_SL | 8369   |
| 0.89%   | 0.82%                 |                       |          |        |

Archivos de validación guardados:

Columnas OPC: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_opc\_column\_validation.csv  
 Valores OPC: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_opc\_value\_validation.csv  
 Mapping DIR/BAR/OPC: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_opc\_mapping\_validation.csv  
 NaN OPC: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_opc\_nan\_validation.csv  
 Balance de clases OPC: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_opc\_class\_balance\_validation.csv  
 Resumen validación OPC: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_opc\_validation\_summary.csv

Validación OPC finalizada correctamente: no se detectaron errores.

Observaciones rápidas:

- La validación cerró perfecta: **0 errores** en todos los checks.
- Todas las columnas DIR , BAR , OPC\_label y OPC existen correctamente.
- Las clases son válidas: 0, 1, 2, 3, 4 .
- No hay inconsistencias entre etiqueta y clase numérica.
- La relación DIR + BAR → OPC está correctamente mapeada.
- Los NaN de OPC coinciden con los NaN de DIR , como corresponde.
- No hay casos problemáticos de DIR operativo con BAR NaN .
- DIR = 0 se convierte correctamente en NO\_TRADE .
- BAR = 0 también se convierte correctamente en NO\_TRADE .
- La clase dominante sigue siendo NO\_TRADE , especialmente en p60\_h60 .
- p40\_h60 es la versión más activa.
- p60\_h60 es la versión más selectiva.
- Podemos dar por cerrado el punto 6.5 y avanzar al análisis de distribución global.

## 6.6. Distribución global de clases

Esta celda deja creados estos objetos principales:

```
df_opc_global_distribution
df_opc_class_distribution_long
df_opc_distribution_by_role
df_opc_distribution_by_family
df_opc_most_active_ranking
df_opc_most_selective_ranking
df_opc_winner_share_ranking
df_opc_class_distribution_pivot
```

Con esto vamos a poder ver claramente:

- qué target OPC tiene más operaciones resueltas,
- cuál es más selectivo,
- qué proporción ocupa NO\_TRADE,
- qué peso tienen LONG\_TP y SHORT\_TP,
- qué tan desbalanceadas quedan las clases,

- y si p40/p50/p60 cumplen el rol esperado.

```
In [28]: # 6.6. Distribución global de clases OPC
# =====

import numpy as np
import pandas as pd

print("Distribución global de clases OPC")
print("=" * 80)

# -----
# 1) Validar objetos necesarios
# -----

required_objects = [
    "df_targets_opc",
    "df_opc_metadata",
]

missing_objects = [
    obj for obj in required_objects
    if obj not in globals()
]

if missing_objects:
    raise NameError(
        "Faltan objetos necesarios para analizar distribución OPC: "
        f"{missing_objects}"
    )

print("Objetos base encontrados correctamente.")

# -----
# 2) Definir clases OPC
# -----

OPC_CLASS_ORDER = [
    "NO_TRADE",
    "LONG_TP",
    "LONG_SL",
    "SHORT_TP",
    "SHORT_SL",
]

OPC_CLASS_MAP = {
    "NO_TRADE": 0,
    "LONG_TP": 1,
    "LONG_SL": 2,
    "SHORT_TP": 3,
    "SHORT_SL": 4,
}

print("\nClases OPC analizadas:")
for label in OPC_CLASS_ORDER:
    print(f"{OPC_CLASS_MAP[label]}: {label}")

# -----
# 3) Calcular distribución global por target OPC
```



```

# -----

global_distribution_records = []
class_distribution_records = []

for _, row in df_opc_metadata.iterrows():

    family_name = row["family_name"]
    target_role = row["target_role"]

    dir_col = row["dir_col"]
    bar_col = row["bar_col"]
    opc_col = row["opc_col"]
    opc_label_col = row["opc_label_col"]

    percentile = row["percentile"]
    horizon = row["horizon"]
    tp_mult = row["tp_mult"]
    sl_mult = row["sl_mult"]

    if opc_label_col not in df_targets_opc.columns:
        raise KeyError(f"No existe la columna {opc_label_col} en df_targets_opc.")

    opc_label = df_targets_opc[opc_label_col]

    total_rows = len(opc_label)
    valid_rows = int(opc_label.notna().sum())
    nan_rows = int(opc_label.isna().sum())

    class_counts = {
        label: int((opc_label == label).sum())
        for label in OPC_CLASS_ORDER
    }

    class_ratios = {
        label: class_counts[label] / valid_rows if valid_rows > 0 else np.nan
        for label in OPC_CLASS_ORDER
    }

    # Métricas agregadas
    no_trade_count = class_counts["NO_TRADE"]

    long_tp_count = class_counts["LONG_TP"]
    long_sl_count = class_counts["LONG_SL"]
    short_tp_count = class_counts["SHORT_TP"]
    short_sl_count = class_counts["SHORT_SL"]

    resolved_trade_count = (
        long_tp_count
        + long_sl_count
        + short_tp_count
        + short_sl_count
    )

    winner_count = long_tp_count + short_tp_count
    loser_count = long_sl_count + short_sl_count

    long_count = long_tp_count + long_sl_count
    short_count = short_tp_count + short_sl_count

```

```

# Ratios sobre filas válidas
no_trade_ratio = no_trade_count / valid_rows if valid_rows > 0 else np.nan
resolved_trade_ratio = resolved_trade_count / valid_rows if valid_rows > 0 else np.nan

winner_ratio = winner_count / valid_rows if valid_rows > 0 else np.nan
loser_ratio = loser_count / valid_rows if valid_rows > 0 else np.nan

long_ratio = long_count / valid_rows if valid_rows > 0 else np.nan
short_ratio = short_count / valid_rows if valid_rows > 0 else np.nan

# Ratios condicionados a trades resueltos
winner_share_over_resolved = (
    winner_count / resolved_trade_count
    if resolved_trade_count > 0
    else np.nan
)

loser_share_over_resolved = (
    loser_count / resolved_trade_count
    if resolved_trade_count > 0
    else np.nan
)

long_share_over_resolved = (
    long_count / resolved_trade_count
    if resolved_trade_count > 0
    else np.nan
)

short_share_over_resolved = (
    short_count / resolved_trade_count
    if resolved_trade_count > 0
    else np.nan
)

long_tp_share_over_long = (
    long_tp_count / long_count
    if long_count > 0
    else np.nan
)

short_tp_share_over_short = (
    short_tp_count / short_count
    if short_count > 0
    else np.nan
)

# Clase dominante
valid_class_counts = {
    label: class_counts[label]
    for label in OPC_CLASS_ORDER
}

majority_class = max(
    valid_class_counts,
    key=valid_class_counts.get
)

majority_class_count = valid_class_counts[majority_class]

```

```

majority_class_ratio = (
    majority_class_count / valid_rows
    if valid_rows > 0
    else np.nan
)

# Guardar resumen por target
global_distribution_records.append(
    {
        "opc_col": opc_col,
        "opc_label_col": opc_label_col,
        "family_name": family_name,
        "target_role": target_role,
        "dir_col": dir_col,
        "bar_col": bar_col,
        "percentile": percentile,
        "horizon": horizon,
        "tp_mult": tp_mult,
        "sl_mult": sl_mult,

        "total_rows": total_rows,
        "valid_rows": valid_rows,
        "nan_rows": nan_rows,
        "nan_ratio": nan_rows / total_rows if total_rows > 0 else np.nan,

        "no_trade_count": no_trade_count,
        "long_tp_count": long_tp_count,
        "long_sl_count": long_sl_count,
        "short_tp_count": short_tp_count,
        "short_sl_count": short_sl_count,

        "no_trade_ratio": no_trade_ratio,
        "long_tp_ratio": class_ratios["LONG_TP"],
        "long_sl_ratio": class_ratios["LONG_SL"],
        "short_tp_ratio": class_ratios["SHORT_TP"],
        "short_sl_ratio": class_ratios["SHORT_SL"],

        "resolved_trade_count": resolved_trade_count,
        "resolved_trade_ratio": resolved_trade_ratio,

        "winner_count": winner_count,
        "loser_count": loser_count,
        "winner_ratio": winner_ratio,
        "loser_ratio": loser_ratio,

        "winner_share_over_resolved": winner_share_over_resolved,
        "loser_share_over_resolved": loser_share_over_resolved,

        "long_count": long_count,
        "short_count": short_count,
        "long_ratio": long_ratio,
        "short_ratio": short_ratio,

        "long_share_over_resolved": long_share_over_resolved,
        "short_share_over_resolved": short_share_over_resolved,

        "long_tp_share_over_long": long_tp_share_over_long,
        "short_tp_share_over_short": short_tp_share_over_short,

        "majority_class": majority_class,
    }
)

```

```

        "majority_class_count": majority_class_count,
        "majority_class_ratio": majority_class_ratio,
    }
)

# Guardar distribución Larga por clase
for label in OPC_CLASS_ORDER:

    class_count = class_counts[label]

    class_distribution_records.append(
        {
            "opc_col": opc_col,
            "family_name": family_name,
            "target_role": target_role,
            "percentile": percentile,
            "horizon": horizon,
            "tp_mult": tp_mult,
            "sl_mult": sl_mult,

            "opc_label": label,
            "opc_class": OPC_CLASS_MAP[label],
            "count": class_count,
            "ratio_over_valid": class_count / valid_rows if valid_rows > 0 else 0,
            "ratio_over_total": class_count / total_rows if total_rows > 0 else 0
        }
    )

# Agregar NaN como control, no como clase OPC
class_distribution_records.append(
    {
        "opc_col": opc_col,
        "family_name": family_name,
        "target_role": target_role,
        "percentile": percentile,
        "horizon": horizon,
        "tp_mult": tp_mult,
        "sl_mult": sl_mult,

        "opc_label": "NaN",
        "opc_class": np.nan,
        "count": nan_rows,
        "ratio_over_valid": np.nan,
        "ratio_over_total": nan_rows / total_rows if total_rows > 0 else np.nan
    }
)

df_opc_global_distribution = pd.DataFrame(global_distribution_records)
df_opc_class_distribution_long = pd.DataFrame(class_distribution_records)

# -----
# 4) Tablas agregadas
# -----

df_opc_distribution_by_role = (
    df_opc_global_distribution
    .groupby("target_role", as_index=False)
    .agg(
        n_targets=("opc_col", "count"),

```

```

valid_rows_mean=("valid_rows", "mean"),
nan_ratio_mean=("nan_ratio", "mean"),

no_trade_ratio_mean=("no_trade_ratio", "mean"),
resolved_trade_ratio_mean=("resolved_trade_ratio", "mean"),

long_tp_ratio_mean=("long_tp_ratio", "mean"),
long_sl_ratio_mean=("long_sl_ratio", "mean"),
short_tp_ratio_mean=("short_tp_ratio", "mean"),
short_sl_ratio_mean=("short_sl_ratio", "mean"),

winner_ratio_mean=("winner_ratio", "mean"),
loser_ratio_mean=("loser_ratio", "mean"),
winner_share_over_resolved_mean=("winner_share_over_resolved", "mean"),

long_share_over_resolved_mean=("long_share_over_resolved", "mean"),
short_share_over_resolved_mean=("short_share_over_resolved", "mean"),

majority_class_ratio_mean=("majority_class_ratio", "mean"),
)
.sort_values("resolved_trade_ratio_mean", ascending=False)
.reset_index(drop=True)
)

df_opc_distribution_by_family = (
    df_opc_global_distribution
    .sort_values(
        ["target_role", "percentile", "horizon"]
    )
    .reset_index(drop=True)
)

# Ranking de targets más operativos
df_opc_most_active_ranking = (
    df_opc_global_distribution
    .sort_values(
        "resolved_trade_ratio",
        ascending=False
    )
    .reset_index(drop=True)
)

# Ranking de targets más selectivos
df_opc_most_selective_ranking = (
    df_opc_global_distribution
    .sort_values(
        "no_trade_ratio",
        ascending=False
    )
    .reset_index(drop=True)
)

# Ranking por proporción ganadora entre trades resueltos
df_opc_winner_share_ranking = (
    df_opc_global_distribution
    .sort_values(
        "winner_share_over_resolved",
        ascending=False
    )
    .reset_index(drop=True)
)

```

```

)

# Tabla pivot para lectura rápida de distribución por clase
df_opc_class_distribution_pivot = (
    df_opc_class_distribution_long[
        df_opc_class_distribution_long["opc_label"] != "NaN"
    ]
    .pivot_table(
        index="opc_col",
        columns="opc_label",
        values="ratio_over_valid",
        aggfunc="first"
    )
    .reset_index()
)

# Asegurar orden de columnas
pivot_cols = ["opc_col"] + OPC_CLASS_ORDER
df_opc_class_distribution_pivot = df_opc_class_distribution_pivot[
    [col for col in pivot_cols if col in df_opc_class_distribution_pivot.columns]
]

# -----
# 5) Mostrar resultados
# -----

percent_cols_main = [
    "nan_ratio",
    "no_trade_ratio",
    "long_tp_ratio",
    "long_sl_ratio",
    "short_tp_ratio",
    "short_sl_ratio",
    "resolved_trade_ratio",
    "winner_ratio",
    "loser_ratio",
    "winner_share_over_resolved",
    "long_share_over_resolved",
    "short_share_over_resolved",
    "majority_class_ratio",
]

formatters_main = {
    col: "{:.2%}".format
    for col in percent_cols_main
}

print("\nDistribución global por target OPC:")
print(
    df_opc_global_distribution[
        [
            "opc_col",
            "target_role",
            "valid_rows",
            "nan_rows",
            "nan_ratio",
            "no_trade_ratio",
            "long_tp_ratio",
            "long_sl_ratio",
            "short_tp_ratio",
        ]
    ]

```

```

        "short_sl_ratio",
        "resolved_trade_ratio",
        "winner_share_over_resolved",
        "long_share_over_resolved",
        "short_share_over_resolved",
        "majority_class",
        "majority_class_ratio",
    ]
    ].to_string(
        index=False,
        formatters=formatters_main
    )
)

print("\nDistribución por clase en formato largo:")
print(
    df_opc_class_distribution_long.to_string(
        index=False,
        formatters={
            "ratio_over_valid": "{:.2%}".format,
            "ratio_over_total": "{:.2%}".format,
        }
    )
)

print("\nResumen agregado por tipo de target:")
print(
    df_opc_distribution_by_role.to_string(
        index=False,
        formatters={
            "nan_ratio_mean": "{:.2%}".format,
            "no_trade_ratio_mean": "{:.2%}".format,
            "resolved_trade_ratio_mean": "{:.2%}".format,
            "long_tp_ratio_mean": "{:.2%}".format,
            "long_sl_ratio_mean": "{:.2%}".format,
            "short_tp_ratio_mean": "{:.2%}".format,
            "short_sl_ratio_mean": "{:.2%}".format,
            "winner_ratio_mean": "{:.2%}".format,
            "loser_ratio_mean": "{:.2%}".format,
            "winner_share_over_resolved_mean": "{:.2%}".format,
            "long_share_over_resolved_mean": "{:.2%}".format,
            "short_share_over_resolved_mean": "{:.2%}".format,
            "majority_class_ratio_mean": "{:.2%}".format,
        }
    )
)

print("\nRanking de targets más operativos:")
print(
    df_opc_most_active_ranking[
        [
            "opc_col",
            "target_role",
            "resolved_trade_ratio",
            "no_trade_ratio",
            "winner_share_over_resolved",
        ]
    ].to_string(
        index=False,
        formatters={

```

```

        "resolved_trade_ratio": "{:.2%}".format,
        "no_trade_ratio": "{:.2%}".format,
        "winner_share_over_resolved": "{:.2%}".format,
    }
)
)

print("\nRanking de targets más selectivos:")
print(
    df_opc_most_selective_ranking[
        [
            "opc_col",
            "target_role",
            "no_trade_ratio",
            "resolved_trade_ratio",
            "winner_share_over_resolved",
        ]
    ].to_string(
        index=False,
        formatters={
            "no_trade_ratio": "{:.2%}".format,
            "resolved_trade_ratio": "{:.2%}".format,
            "winner_share_over_resolved": "{:.2%}".format,
        }
    )
)

print("\nRanking por proporción ganadora entre trades resueltos:")
print(
    df_opc_winner_share_ranking[
        [
            "opc_col",
            "target_role",
            "winner_share_over_resolved",
            "resolved_trade_ratio",
            "no_trade_ratio",
        ]
    ].to_string(
        index=False,
        formatters={
            "winner_share_over_resolved": "{:.2%}".format,
            "resolved_trade_ratio": "{:.2%}".format,
            "no_trade_ratio": "{:.2%}".format,
        }
    )
)

print("\nDistribución porcentual por clase:")
print(
    df_opc_class_distribution_pivot.to_string(
        index=False,
        formatters={
            col: "{:.2%}".format
            for col in OPC_CLASS_ORDER
            if col in df_opc_class_distribution_pivot.columns
        }
    )
)

# -----

```



```
# 6) Guardado de resultados
# -----

if "MNQ_TARGETS_PATH" in globals():

    MNQ_TARGETS_OPC_GLOBAL_DISTRIBUTION_CSV = (
        MNQ_TARGETS_PATH / "mnq_targets_opc_global_distribution.csv"
    )

    MNQ_TARGETS_OPC_CLASS_DISTRIBUTION_LONG_CSV = (
        MNQ_TARGETS_PATH / "mnq_targets_opc_class_distribution_long.csv"
    )

    MNQ_TARGETS_OPC_DISTRIBUTION_BY_ROLE_CSV = (
        MNQ_TARGETS_PATH / "mnq_targets_opc_distribution_by_role.csv"
    )

    MNQ_TARGETS_OPC_DISTRIBUTION_BY_FAMILY_CSV = (
        MNQ_TARGETS_PATH / "mnq_targets_opc_distribution_by_family.csv"
    )

    MNQ_TARGETS_OPC_MOST_ACTIVE_RANKING_CSV = (
        MNQ_TARGETS_PATH / "mnq_targets_opc_most_active_ranking.csv"
    )

    MNQ_TARGETS_OPC_MOST_SELECTIVE_RANKING_CSV = (
        MNQ_TARGETS_PATH / "mnq_targets_opc_most_selective_ranking.csv"
    )

    MNQ_TARGETS_OPC_WINNER_SHARE_RANKING_CSV = (
        MNQ_TARGETS_PATH / "mnq_targets_opc_winner_share_ranking.csv"
    )

    MNQ_TARGETS_OPC_CLASS_DISTRIBUTION_PIVOT_CSV = (
        MNQ_TARGETS_PATH / "mnq_targets_opc_class_distribution_pivot.csv"
    )

    df_opc_global_distribution.to_csv(
        MNQ_TARGETS_OPC_GLOBAL_DISTRIBUTION_CSV,
        index=False
    )

    df_opc_class_distribution_long.to_csv(
        MNQ_TARGETS_OPC_CLASS_DISTRIBUTION_LONG_CSV,
        index=False
    )

    df_opc_distribution_by_role.to_csv(
        MNQ_TARGETS_OPC_DISTRIBUTION_BY_ROLE_CSV,
        index=False
    )

    df_opc_distribution_by_family.to_csv(
        MNQ_TARGETS_OPC_DISTRIBUTION_BY_FAMILY_CSV,
        index=False
    )

    df_opc_most_active_ranking.to_csv(
        MNQ_TARGETS_OPC_MOST_ACTIVE_RANKING_CSV,
        index=False
    )
```

```
)

df_opc_most_selective_ranking.to_csv(
    MNQ_TARGETS_OPC_MOST_SELECTIVE_RANKING_CSV,
    index=False
)

df_opc_winner_share_ranking.to_csv(
    MNQ_TARGETS_OPC_WINNER_SHARE_RANKING_CSV,
    index=False
)

df_opc_class_distribution_pivot.to_csv(
    MNQ_TARGETS_OPC_CLASS_DISTRIBUTION_PIVOT_CSV,
    index=False
)

print("\nArchivos guardados:")
print(f"Distribución global OPC:           {MNQ_TARGETS_OPC_GLOBAL_DISTRIBUTI")
print(f"Distribución larga por clase:       {MNQ_TARGETS_OPC_CLASS_DISTRIBUTIO")
print(f"Distribución por rol:                 {MNQ_TARGETS_OPC_DISTRIBUTION_BY_R")
print(f"Distribución por familia:            {MNQ_TARGETS_OPC_DISTRIBUTION_BY_F")
print(f"Ranking más operativos:              {MNQ_TARGETS_OPC_MOST_ACTIVE_RANKI")
print(f"Ranking más selectivos:              {MNQ_TARGETS_OPC_MOST_SELECTIVE_RA")
print(f"Ranking winner share:                {MNQ_TARGETS_OPC_WINNER_SHARE_RANK")
print(f"Pivot distribución por clase:        {MNQ_TARGETS_OPC_CLASS_DISTRIBUTIO")

else:
    print("\nMNQ_TARGETS_PATH no está definido. No se guardaron archivos.")

print("\nDistribución global de clases OPC finalizada correctamente.")
```

## Distribución global de clases OPC

=====

Objetos base encontrados correctamente.

## Clases OPC analizadas:

0: NO\_TRADE  
 1: LONG\_TP  
 2: LONG\_SL  
 3: SHORT\_TP  
 4: SHORT\_SL

## Distribución global por target OPC:

| opc_col               | target_role           | valid_rows | nan_rows | nan_ratio | no_trade_ratio | long_tp_ratio | long_sl_ratio | short_tp_ratio | short_sl_ratio | resolved_trade_ratio | winner_share_over_resolved | long_share_over_resolved | short_share_over_resolved | majority_class | majority_class_ratio |
|-----------------------|-----------------------|------------|----------|-----------|----------------|---------------|---------------|----------------|----------------|----------------------|----------------------------|--------------------------|---------------------------|----------------|----------------------|
| opc_p50_h30_tp15_sl10 | principal             | 979602     | 44460    | 4.34%     |                |               |               |                |                |                      |                            |                          |                           |                |                      |
| 67.92%                |                       |            |          |           |                | 13.54%        | 1.54%         | 15.46%         | 1.55%          |                      |                            |                          |                           | 3              |                      |
| 2.08%                 |                       |            |          |           |                |               | 90.38%        |                | 47.00%         |                      |                            |                          |                           | 53.0           |                      |
| 0%                    | NO_TRADE              |            |          |           | 67.92%         |               |               |                |                |                      |                            |                          |                           |                |                      |
| opc_p50_h60_tp15_sl10 | principal             | 935142     | 88920    | 8.68%     |                |               |               |                |                |                      |                            |                          |                           |                |                      |
| 68.80%                |                       |            |          |           |                | 12.87%        | 1.52%         | 15.34%         | 1.48%          |                      |                            |                          |                           | 3              |                      |
| 1.20%                 |                       |            |          |           |                |               | 90.41%        |                | 46.12%         |                      |                            |                          |                           | 53.8           |                      |
| 8%                    | NO_TRADE              |            |          |           | 68.80%         |               |               |                |                |                      |                            |                          |                           |                |                      |
| opc_p50_h90_tp15_sl10 | principal             | 890682     | 133380   | 13.02%    |                |               |               |                |                |                      |                            |                          |                           |                |                      |
| 68.40%                |                       |            |          |           |                | 12.84%        | 1.63%         | 15.55%         | 1.58%          |                      |                            |                          |                           | 3              |                      |
| 1.60%                 |                       |            |          |           |                |               | 89.83%        |                | 45.79%         |                      |                            |                          |                           | 54.2           |                      |
| 1%                    | NO_TRADE              |            |          |           | 68.40%         |               |               |                |                |                      |                            |                          |                           |                |                      |
| opc_p40_h60_tp15_sl10 | comparativo_activo    | 935142     | 88920    | 8.68%     |                |               |               |                |                |                      |                            |                          |                           |                |                      |
| 59.39%                |                       |            |          |           |                | 16.85%        | 2.34%         | 19.09%         | 2.33%          |                      |                            |                          |                           | 4              |                      |
| 0.61%                 |                       |            |          |           |                |               | 88.51%        |                | 47.26%         |                      |                            |                          |                           | 52.7           |                      |
| 4%                    | NO_TRADE              |            |          |           | 59.39%         |               |               |                |                |                      |                            |                          |                           |                |                      |
| opc_p60_h60_tp15_sl10 | comparativo_selectivo | 935142     | 88920    | 8.68%     |                |               |               |                |                |                      |                            |                          |                           |                |                      |
| 76.81%                |                       |            |          |           |                | 9.47%         | 0.94%         | 11.89%         | 0.89%          |                      |                            |                          |                           | 2              |                      |
| 3.19%                 |                       |            |          |           |                |               | 92.10%        |                | 44.87%         |                      |                            |                          |                           | 55.1           |                      |
| 3%                    | NO_TRADE              |            |          |           | 76.81%         |               |               |                |                |                      |                            |                          |                           |                |                      |

## Distribución por clase en formato largo:

| opc_col               | family_name | target_role | percentile | horizon | tp_mult | sl_mult | opc_label | opc_class | count  | ratio_over_valid | ratio_over_total |
|-----------------------|-------------|-------------|------------|---------|---------|---------|-----------|-----------|--------|------------------|------------------|
| opc_p50_h30_tp15_sl10 | p50_h30     | principal   | 50         | 30      |         |         |           |           |        |                  |                  |
| 1.5                   | 1.0         | NO_TRADE    | 0.0        | 665304  |         |         |           |           | 67.92% | 64.97%           |                  |
| opc_p50_h30_tp15_sl10 | p50_h30     | principal   | 50         | 30      |         |         |           |           |        |                  |                  |
| 1.5                   | 1.0         | LONG_TP     | 1.0        | 132636  |         |         |           |           | 13.54% | 12.95%           |                  |
| opc_p50_h30_tp15_sl10 | p50_h30     | principal   | 50         | 30      |         |         |           |           |        |                  |                  |
| 1.5                   | 1.0         | LONG_SL     | 2.0        | 15085   |         |         |           |           | 1.54%  | 1.47%            |                  |
| opc_p50_h30_tp15_sl10 | p50_h30     | principal   | 50         | 30      |         |         |           |           |        |                  |                  |
| 1.5                   | 1.0         | SHORT_TP    | 3.0        | 151418  |         |         |           |           | 15.46% | 14.79%           |                  |
| opc_p50_h30_tp15_sl10 | p50_h30     | principal   | 50         | 30      |         |         |           |           |        |                  |                  |
| 1.5                   | 1.0         | SHORT_SL    | 4.0        | 15159   |         |         |           |           | 1.55%  | 1.48%            |                  |
| opc_p50_h30_tp15_sl10 | p50_h30     | principal   | 50         | 30      |         |         |           |           |        |                  |                  |
| 1.5                   | 1.0         | NaN         | NaN        | 44460   |         |         |           |           | NaN    | 4.34%            |                  |
| opc_p50_h60_tp15_sl10 | p50_h60     | principal   | 50         | 60      |         |         |           |           |        |                  |                  |
| 1.5                   | 1.0         | NO_TRADE    | 0.0        | 643374  |         |         |           |           | 68.80% | 62.83%           |                  |
| opc_p50_h60_tp15_sl10 | p50_h60     | principal   | 50         | 60      |         |         |           |           |        |                  |                  |
| 1.5                   | 1.0         | LONG_TP     | 1.0        | 120384  |         |         |           |           | 12.87% | 11.76%           |                  |
| opc_p50_h60_tp15_sl10 | p50_h60     | principal   | 50         | 60      |         |         |           |           |        |                  |                  |
| 1.5                   | 1.0         | LONG_SL     | 2.0        | 14172   |         |         |           |           | 1.52%  | 1.38%            |                  |
| opc_p50_h60_tp15_sl10 | p50_h60     | principal   | 50         | 60      |         |         |           |           |        |                  |                  |
| 1.5                   | 1.0         | SHORT_TP    | 3.0        | 143412  |         |         |           |           | 15.34% | 14.00%           |                  |

|                       |            |                       |        |    |
|-----------------------|------------|-----------------------|--------|----|
| opc_p50_h60_tp15_sl10 | p50_h60    | principal             | 50     | 60 |
| 1.5 1.0 SHORT_SL      | 4.0 13800  | 1.48%                 | 1.35%  |    |
| opc_p50_h60_tp15_sl10 | p50_h60    | principal             | 50     | 60 |
| 1.5 1.0 NaN           | NaN 88920  | NaN                   | 8.68%  |    |
| opc_p50_h90_tp15_sl10 | p50_h90    | principal             | 50     | 90 |
| 1.5 1.0 NO_TRADE      | 0.0 609262 | 68.40%                | 59.49% |    |
| opc_p50_h90_tp15_sl10 | p50_h90    | principal             | 50     | 90 |
| 1.5 1.0 LONG_TP       | 1.0 114341 | 12.84%                | 11.17% |    |
| opc_p50_h90_tp15_sl10 | p50_h90    | principal             | 50     | 90 |
| 1.5 1.0 LONG_SL       | 2.0 14518  | 1.63%                 | 1.42%  |    |
| opc_p50_h90_tp15_sl10 | p50_h90    | principal             | 50     | 90 |
| 1.5 1.0 SHORT_TP      | 3.0 138461 | 15.55%                | 13.52% |    |
| opc_p50_h90_tp15_sl10 | p50_h90    | principal             | 50     | 90 |
| 1.5 1.0 SHORT_SL      | 4.0 14100  | 1.58%                 | 1.38%  |    |
| opc_p50_h90_tp15_sl10 | p50_h90    | principal             | 50     | 90 |
| 1.5 1.0 NaN           | NaN 133380 | NaN                   | 13.02% |    |
| opc_p40_h60_tp15_sl10 | p40_h60    | comparativo_activo    | 40     | 60 |
| 1.5 1.0 NO_TRADE      | 0.0 555377 | 59.39%                | 54.23% |    |
| opc_p40_h60_tp15_sl10 | p40_h60    | comparativo_activo    | 40     | 60 |
| 1.5 1.0 LONG_TP       | 1.0 157597 | 16.85%                | 15.39% |    |
| opc_p40_h60_tp15_sl10 | p40_h60    | comparativo_activo    | 40     | 60 |
| 1.5 1.0 LONG_SL       | 2.0 21891  | 2.34%                 | 2.14%  |    |
| opc_p40_h60_tp15_sl10 | p40_h60    | comparativo_activo    | 40     | 60 |
| 1.5 1.0 SHORT_TP      | 3.0 178523 | 19.09%                | 17.43% |    |
| opc_p40_h60_tp15_sl10 | p40_h60    | comparativo_activo    | 40     | 60 |
| 1.5 1.0 SHORT_SL      | 4.0 21754  | 2.33%                 | 2.12%  |    |
| opc_p40_h60_tp15_sl10 | p40_h60    | comparativo_activo    | 40     | 60 |
| 1.5 1.0 NaN           | NaN 88920  | NaN                   | 8.68%  |    |
| opc_p60_h60_tp15_sl10 | p60_h60    | comparativo_selectivo | 60     | 60 |
| 1.5 1.0 NO_TRADE      | 0.0 718245 | 76.81%                | 70.14% |    |
| opc_p60_h60_tp15_sl10 | p60_h60    | comparativo_selectivo | 60     | 60 |
| 1.5 1.0 LONG_TP       | 1.0 88560  | 9.47%                 | 8.65%  |    |
| opc_p60_h60_tp15_sl10 | p60_h60    | comparativo_selectivo | 60     | 60 |
| 1.5 1.0 LONG_SL       | 2.0 8768   | 0.94%                 | 0.86%  |    |
| opc_p60_h60_tp15_sl10 | p60_h60    | comparativo_selectivo | 60     | 60 |
| 1.5 1.0 SHORT_TP      | 3.0 111200 | 11.89%                | 10.86% |    |
| opc_p60_h60_tp15_sl10 | p60_h60    | comparativo_selectivo | 60     | 60 |
| 1.5 1.0 SHORT_SL      | 4.0 8369   | 0.89%                 | 0.82%  |    |
| opc_p60_h60_tp15_sl10 | p60_h60    | comparativo_selectivo | 60     | 60 |
| 1.5 1.0 NaN           | NaN 88920  | NaN                   | 8.68%  |    |

Resumen agregado por tipo de target:

| target_role           | n_targets | valid_rows_mean | nan_ratio_mean | no_trade_ratio_mean | resolved_trade_ratio_mean | long_tp_ratio_mean | long_sl_ratio_mean | short_tp_ratio_mean | short_sl_ratio_mean | winner_ratio_mean | loser_ratio_mean | winner_share_over_resolved_mean | long_share_over_resolved_mean | short_share_over_resolved_mean | majority_class_ratio_mean |
|-----------------------|-----------|-----------------|----------------|---------------------|---------------------------|--------------------|--------------------|---------------------|---------------------|-------------------|------------------|---------------------------------|-------------------------------|--------------------------------|---------------------------|
| comparativo_activo    | 1         | 935142.0        | 8.68%          | 59.39%              | 40.61%                    | 16.85%             | 2.34%              | 19.09%              | 2.33%               | 35.94%            | 4.67%            | 88.51%                          | 47.26%                        | 52.74%                         |                           |
| principal             | 3         | 935142.0        | 8.68%          | 68.37%              | 31.63%                    | 13.08%             | 1.56%              | 15.45%              | 1.54%               | 28.53%            | 3.10%            | 90.21%                          | 46.30%                        | 53.70%                         |                           |
| comparativo_selectivo | 1         | 935142.0        | 8.68%          | 76.81%              | 23.19%                    | 9.47%              | 0.94%              | 11.89%              | 0.89%               | 21.36%            | 1.83%            |                                 |                               |                                |                           |

|        |        |        |
|--------|--------|--------|
| 92.10% | 44.87% | 55.13% |
| 76.81% |        |        |

## Ranking de targets más operativos:

| opc_col                   | target_role           | resolved_trade_ratio | no_trade_ratio | w |
|---------------------------|-----------------------|----------------------|----------------|---|
| inner_share_over_resolved |                       |                      |                |   |
| opc_p40_h60_tp15_sl10     | comparativo_activo    | 40.61%               | 59.39%         |   |
| 88.51%                    |                       |                      |                |   |
| opc_p50_h30_tp15_sl10     | principal             | 32.08%               | 67.92%         |   |
| 90.38%                    |                       |                      |                |   |
| opc_p50_h90_tp15_sl10     | principal             | 31.60%               | 68.40%         |   |
| 89.83%                    |                       |                      |                |   |
| opc_p50_h60_tp15_sl10     | principal             | 31.20%               | 68.80%         |   |
| 90.41%                    |                       |                      |                |   |
| opc_p60_h60_tp15_sl10     | comparativo_selectivo | 23.19%               | 76.81%         |   |
| 92.10%                    |                       |                      |                |   |

## Ranking de targets más selectivos:

| opc_col                   | target_role           | no_trade_ratio | resolved_trade_ratio | w |
|---------------------------|-----------------------|----------------|----------------------|---|
| inner_share_over_resolved |                       |                |                      |   |
| opc_p60_h60_tp15_sl10     | comparativo_selectivo | 76.81%         | 23.19%               |   |
| 92.10%                    |                       |                |                      |   |
| opc_p50_h60_tp15_sl10     | principal             | 68.80%         | 31.20%               |   |
| 90.41%                    |                       |                |                      |   |
| opc_p50_h90_tp15_sl10     | principal             | 68.40%         | 31.60%               |   |
| 89.83%                    |                       |                |                      |   |
| opc_p50_h30_tp15_sl10     | principal             | 67.92%         | 32.08%               |   |
| 90.38%                    |                       |                |                      |   |
| opc_p40_h60_tp15_sl10     | comparativo_activo    | 59.39%         | 40.61%               |   |
| 88.51%                    |                       |                |                      |   |

## Ranking por proporción ganadora entre trades resueltos:

| opc_col               | target_role           | winner_share_over_resolved | resolved_t |
|-----------------------|-----------------------|----------------------------|------------|
| rade_ratio            |                       |                            |            |
| opc_p60_h60_tp15_sl10 | comparativo_selectivo | 92.10%                     |            |
| 23.19%                |                       |                            |            |
| 76.81%                |                       |                            |            |
| opc_p50_h60_tp15_sl10 | principal             | 90.41%                     |            |
| 31.20%                |                       |                            |            |
| 68.80%                |                       |                            |            |
| opc_p50_h30_tp15_sl10 | principal             | 90.38%                     |            |
| 32.08%                |                       |                            |            |
| 67.92%                |                       |                            |            |
| opc_p50_h90_tp15_sl10 | principal             | 89.83%                     |            |
| 31.60%                |                       |                            |            |
| 68.40%                |                       |                            |            |
| opc_p40_h60_tp15_sl10 | comparativo_activo    | 88.51%                     |            |
| 40.61%                |                       |                            |            |
| 59.39%                |                       |                            |            |

## Distribución porcentual por clase:

| opc_col               | NO_TRADE | LONG_TP | LONG_SL | SHORT_TP | SHORT_SL |
|-----------------------|----------|---------|---------|----------|----------|
| opc_p40_h60_tp15_sl10 | 59.39%   | 16.85%  | 2.34%   | 19.09%   | 2.33%    |
| opc_p50_h30_tp15_sl10 | 67.92%   | 13.54%  | 1.54%   | 15.46%   | 1.55%    |
| opc_p50_h60_tp15_sl10 | 68.80%   | 12.87%  | 1.52%   | 15.34%   | 1.48%    |
| opc_p50_h90_tp15_sl10 | 68.40%   | 12.84%  | 1.63%   | 15.55%   | 1.58%    |
| opc_p60_h60_tp15_sl10 | 76.81%   | 9.47%   | 0.94%   | 11.89%   | 0.89%    |

## Archivos guardados:

Distribución global OPC: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_opc\_global\_distribution.csv

Distribución larga por clase: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_opc\_class\_distribution\_long.csv

Distribución por rol: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_opc\_distribution\_by\_role.csv

Distribución por familia: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_opc\_distribution\_by\_family.csv  
 Ranking más operativos: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_opc\_most\_active\_ranking.csv  
 Ranking más selectivos: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_opc\_most\_selective\_ranking.csv  
 Ranking winner share: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_opc\_winner\_share\_ranking.csv  
 Pivot distribución por clase: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_opc\_class\_distribution\_pivot.csv

Distribución global de clases OPC finalizada correctamente.

### Observaciones rápidas:

- **NO\_TRADE** domina en todos los targets, especialmente en **p60\_h60** con **76.81%**.
  - Los targets principales **p50** son bastante consistentes: **NO\_TRADE** ~68% y trades resueltos ~31%–32%.
  - **p40\_h60** es el más activo: **40.61%** de trades resueltos.
  - **p60\_h60** es el más selectivo: solo **23.19%** de trades resueltos.
  - Entre trades resueltos, la proporción ganadora es alta en todos los casos: ~88%–92%.
  - **p60\_h60** tiene mejor calidad relativa: **92.10%** de ganadores entre resueltos.
  - **p40\_h60** tiene más oportunidades, pero menor proporción ganadora relativa.
  - Hay leve sesgo hacia **SHORT\_TP** frente a **LONG\_TP** en todos los targets.
  - Las clases perdedoras ( **LONG\_SL** , **SHORT\_SL** ) son muy bajas, alrededor de 1%–2.3%.
  - Los targets **p50** siguen siendo el mejor equilibrio entre frecuencia, estabilidad y selectividad.
- 
- Mejor candidato preliminar: opc\_p50\_h30\_tp15\_sl10
  - Candidato alternativo equilibrado: opc\_p50\_h60\_tp15\_sl10
  - Más activo: opc\_p40\_h60\_tp15\_sl10
  - Más selectivo: opc\_p60\_h60\_tp15\_sl10

## 6.7. Análisis temporal

```
In [29]: # 6.7. Análisis temporal de targets OPC
# =====

import numpy as np
import pandas as pd

print("Análisis temporal de targets OPC")
print("=" * 80)

# -----
# 1) Validar objetos necesarios
# -----

required_objects = [
    "df_targets_opc",
```

```

        "df_opc_metadata",
    ]

    missing_objects = [
        obj for obj in required_objects
        if obj not in globals()
    ]

    if missing_objects:
        raise NameError(
            "Faltan objetos necesarios para analizar OPC temporalmente: "
            f"{missing_objects}"
        )

    print("Objetos base encontrados correctamente.")

    # -----
    # 2) Definir clases OPC
    # -----

    OPC_CLASS_ORDER = [
        "NO_TRADE",
        "LONG_TP",
        "LONG_SL",
        "SHORT_TP",
        "SHORT_SL",
    ]

    OPC_CLASS_MAP = {
        "NO_TRADE": 0,
        "LONG_TP": 1,
        "LONG_SL": 2,
        "SHORT_TP": 3,
        "SHORT_SL": 4,
    }

    # -----
    # 3) Preparar dataframe temporal base
    # -----

    df_opc_temporal_base = df_targets_opc.copy()

    # Crear columna year si no existe
    if "year" not in df_opc_temporal_base.columns:

        if "date" in df_opc_temporal_base.columns:
            date_series_tmp = pd.to_datetime(df_opc_temporal_base["date"])
        else:
            date_series_tmp = pd.to_datetime(df_opc_temporal_base.index)

        df_opc_temporal_base["year"] = date_series_tmp.dt.year

    else:
        date_series_tmp = pd.to_datetime(df_opc_temporal_base["date"]) if "date" in

    # Crear year_quarter
    df_opc_temporal_base["year_quarter"] = (
        date_series_tmp.dt.year.astype(str)
        + "Q"
        + date_series_tmp.dt.quarter.astype(str)
    )

```

```

)

# Crear dataset_split
df_opc_temporal_base["dataset_split"] = np.where(
    df_opc_temporal_base["year"].between(2020, 2024),
    "development_2020_2024",
    np.where(
        df_opc_temporal_base["year"].between(2025, 2026),
        "final_test_2025_2026",
        "other"
    )
)

print("\nColumnas temporales preparadas:")
print("year")
print("year_quarter")
print("dataset_split")

print("\nValores de dataset_split:")
print(df_opc_temporal_base["dataset_split"].value_counts().to_string())

# -----
# 4) Definir períodos temporales a analizar
# -----

period_definitions = [
    {
        "period_type": "dataset_split",
        "period_col": "dataset_split",
    },
    {
        "period_type": "year",
        "period_col": "year",
    },
    {
        "period_type": "quarter",
        "period_col": "year_quarter",
    },
]

# -----
# 5) Función auxiliar para calcular distribución OPC
# -----

def calculate_opc_distribution(opc_label, mask):
    """
    Calcula distribución de clases OPC para un subconjunto temporal.
    """

    tmp = opc_label.loc[mask]

    total_rows = len(tmp)
    valid_rows = int(tmp.notna().sum())
    nan_rows = int(tmp.isna().sum())

    class_counts = {
        label: int((tmp == label).sum())
        for label in OPC_CLASS_ORDER
    }

```



```

class_ratios = {
    label: class_counts[label] / valid_rows if valid_rows > 0 else np.nan
    for label in OPC_CLASS_ORDER
}

no_trade_count = class_counts["NO_TRADE"]

long_tp_count = class_counts["LONG_TP"]
long_sl_count = class_counts["LONG_SL"]
short_tp_count = class_counts["SHORT_TP"]
short_sl_count = class_counts["SHORT_SL"]

resolved_trade_count = (
    long_tp_count
    + long_sl_count
    + short_tp_count
    + short_sl_count
)

winner_count = long_tp_count + short_tp_count
loser_count = long_sl_count + short_sl_count

long_count = long_tp_count + long_sl_count
short_count = short_tp_count + short_sl_count

no_trade_ratio = no_trade_count / valid_rows if valid_rows > 0 else np.nan
resolved_trade_ratio = resolved_trade_count / valid_rows if valid_rows > 0 else np.nan

winner_ratio = winner_count / valid_rows if valid_rows > 0 else np.nan
loser_ratio = loser_count / valid_rows if valid_rows > 0 else np.nan

long_ratio = long_count / valid_rows if valid_rows > 0 else np.nan
short_ratio = short_count / valid_rows if valid_rows > 0 else np.nan

winner_share_over_resolved = (
    winner_count / resolved_trade_count
    if resolved_trade_count > 0
    else np.nan
)

loser_share_over_resolved = (
    loser_count / resolved_trade_count
    if resolved_trade_count > 0
    else np.nan
)

long_share_over_resolved = (
    long_count / resolved_trade_count
    if resolved_trade_count > 0
    else np.nan
)

short_share_over_resolved = (
    short_count / resolved_trade_count
    if resolved_trade_count > 0
    else np.nan
)

majority_class = None
majority_class_count = np.nan

```

```

majority_class_ratio = np.nan

if valid_rows > 0:
    majority_class = max(class_counts, key=class_counts.get)
    majority_class_count = class_counts[majority_class]
    majority_class_ratio = majority_class_count / valid_rows

return {
    "total_rows": total_rows,
    "valid_rows": valid_rows,
    "nan_rows": nan_rows,
    "nan_ratio": nan_rows / total_rows if total_rows > 0 else np.nan,

    "no_trade_count": no_trade_count,
    "long_tp_count": long_tp_count,
    "long_sl_count": long_sl_count,
    "short_tp_count": short_tp_count,
    "short_sl_count": short_sl_count,

    "no_trade_ratio": no_trade_ratio,
    "long_tp_ratio": class_ratios["LONG_TP"],
    "long_sl_ratio": class_ratios["LONG_SL"],
    "short_tp_ratio": class_ratios["SHORT_TP"],
    "short_sl_ratio": class_ratios["SHORT_SL"],

    "resolved_trade_count": resolved_trade_count,
    "resolved_trade_ratio": resolved_trade_ratio,

    "winner_count": winner_count,
    "loser_count": loser_count,
    "winner_ratio": winner_ratio,
    "loser_ratio": loser_ratio,

    "winner_share_over_resolved": winner_share_over_resolved,
    "loser_share_over_resolved": loser_share_over_resolved,

    "long_count": long_count,
    "short_count": short_count,
    "long_ratio": long_ratio,
    "short_ratio": short_ratio,

    "long_share_over_resolved": long_share_over_resolved,
    "short_share_over_resolved": short_share_over_resolved,

    "majority_class": majority_class,
    "majority_class_count": majority_class_count,
    "majority_class_ratio": majority_class_ratio,
}

# -----
# 6) Calcular distribución temporal por target y período
# -----

temporal_records = []

for _, meta_row in df_opc_metadata.iterrows():

    opc_col = meta_row["opc_col"]
    opc_label_col = meta_row["opc_label_col"]

```

```

family_name = meta_row["family_name"]
target_role = meta_row["target_role"]

percentile = meta_row["percentile"]
horizon = meta_row["horizon"]
tp_mult = meta_row["tp_mult"]
sl_mult = meta_row["sl_mult"]

if opc_label_col not in df_opc_temporal_base.columns:
    raise KeyError(f"No existe la columna {opc_label_col}.")

opc_label = df_opc_temporal_base[opc_label_col]

for period_def in period_definitions:

    period_type = period_def["period_type"]
    period_col = period_def["period_col"]

    period_values = sorted(
        df_opc_temporal_base[period_col].dropna().unique().tolist()
    )

    for period_value in period_values:

        mask = df_opc_temporal_base[period_col].eq(period_value)

        metrics = calculate_opc_distribution(
            opc_label=opc_label,
            mask=mask
        )

        temporal_records.append(
            {
                "opc_col": opc_col,
                "opc_label_col": opc_label_col,
                "family_name": family_name,
                "target_role": target_role,
                "percentile": percentile,
                "horizon": horizon,
                "tp_mult": tp_mult,
                "sl_mult": sl_mult,

                "period_type": period_type,
                "period_col": period_col,
                "period_value": period_value,

                **metrics,
            }
        )

df_opc_temporal_distribution = pd.DataFrame(temporal_records)

print("\nFilas generadas en df_opc_temporal_distribution:")
print(len(df_opc_temporal_distribution))

print("\nPeríodos analizados:")
print(
    df_opc_temporal_distribution
    .groupby("period_type")["period_value"]
    .nunique()

```

```

        .to_string()
    )

# -----
# 7) Resumen por split temporal
# -----

df_opc_temporal_by_split = (
    df_opc_temporal_distribution[
        df_opc_temporal_distribution["period_type"] == "dataset_split"
    ]
    .groupby("period_value", as_index=False)
    .agg(
        n_targets=("opc_col", "count"),
        no_trade_ratio_mean=("no_trade_ratio", "mean"),
        resolved_trade_ratio_mean=("resolved_trade_ratio", "mean"),
        long_tp_ratio_mean=("long_tp_ratio", "mean"),
        long_sl_ratio_mean=("long_sl_ratio", "mean"),
        short_tp_ratio_mean=("short_tp_ratio", "mean"),
        short_sl_ratio_mean=("short_sl_ratio", "mean"),
        winner_ratio_mean=("winner_ratio", "mean"),
        loser_ratio_mean=("loser_ratio", "mean"),
        winner_share_over_resolved_mean=("winner_share_over_resolved", "mean"),
        long_share_over_resolved_mean=("long_share_over_resolved", "mean"),
        short_share_over_resolved_mean=("short_share_over_resolved", "mean"),
    )
    .sort_values("period_value")
    .reset_index(drop=True)
)

# -----
# 8) Comparación development vs final_test por target
# -----

split_tmp = df_opc_temporal_distribution[
    df_opc_temporal_distribution["period_type"] == "dataset_split"
].copy()

dev_split_name = "development_2020_2024"
final_split_name = "final_test_2025_2026"

metrics_to_compare = [
    "no_trade_ratio",
    "long_tp_ratio",
    "long_sl_ratio",
    "short_tp_ratio",
    "short_sl_ratio",
    "resolved_trade_ratio",
    "winner_ratio",
    "loser_ratio",
    "winner_share_over_resolved",
    "long_share_over_resolved",
    "short_share_over_resolved",
]

id_cols = [
    "opc_col",
    "family_name",
    "target_role",
    "percentile",

```

```

        "horizon",
        "tp_mult",
        "sl_mult",
    ]

    dev_tmp = (
        split_tmp[split_tmp["period_value"] == dev_split_name]
        [id_cols + metrics_to_compare]
        .copy()
    )

    final_tmp = (
        split_tmp[split_tmp["period_value"] == final_split_name]
        [id_cols + metrics_to_compare]
        .copy()
    )

    dev_tmp = dev_tmp.rename(
        columns={metric: f"{metric}_development" for metric in metrics_to_compare}
    )

    final_tmp = final_tmp.rename(
        columns={metric: f"{metric}_final_test" for metric in metrics_to_compare}
    )

    df_opc_split_comparison = dev_tmp.merge(
        final_tmp,
        on=id_cols,
        how="inner"
    )

    for metric in metrics_to_compare:

        df_opc_split_comparison[f"{metric}_diff"] = (
            df_opc_split_comparison[f"{metric}_final_test"]
            - df_opc_split_comparison[f"{metric}_development"]
        )

        df_opc_split_comparison[f"{metric}_diff_abs"] = (
            df_opc_split_comparison[f"{metric}_diff"].abs()
        )

    # Score simple de cambio en distribución de clases OPC
    class_ratio_cols = [
        "no_trade_ratio",
        "long_tp_ratio",
        "long_sl_ratio",
        "short_tp_ratio",
        "short_sl_ratio",
    ]

    df_opc_split_comparison["split_class_shift_score"] = (
        df_opc_split_comparison[
            [f"{metric}_diff_abs" for metric in class_ratio_cols]
        ]
        .sum(axis=1)
    )

    # -----
    # 9) Estabilidad anual por target

```

```

# -----
year_tmp = df_opc_temporal_distribution[
    df_opc_temporal_distribution["period_type"] == "year"
].copy()

range_metrics = [
    "no_trade_ratio",
    "long_tp_ratio",
    "long_sl_ratio",
    "short_tp_ratio",
    "short_sl_ratio",
    "resolved_trade_ratio",
    "winner_share_over_resolved",
    "long_share_over_resolved",
    "short_share_over_resolved",
]

yearly_stability_records = []

for opc_col, group in year_tmp.groupby("opc_col"):

    first_row = group.iloc[0]

    record = {
        "opc_col": opc_col,
        "family_name": first_row["family_name"],
        "target_role": first_row["target_role"],
        "percentile": first_row["percentile"],
        "horizon": first_row["horizon"],
        "tp_mult": first_row["tp_mult"],
        "sl_mult": first_row["sl_mult"],
        "n_years": group["period_value"].nunique(),
    }

    for metric in range_metrics:
        record[f"{metric}_range"] = group[metric].max() - group[metric].min()
        record[f"{metric}_std"] = group[metric].std()

    record["yearly_class_instability_score"] = sum(
        record[f"{metric}_range"]
        for metric in class_ratio_cols
    )

    yearly_stability_records.append(record)

df_opc_yearly_stability_summary = pd.DataFrame(yearly_stability_records)

df_opc_yearly_stability_summary = (
    df_opc_yearly_stability_summary
    .sort_values("yearly_class_instability_score")
    .reset_index(drop=True)
)

# -----
# 10) Estabilidad trimestral por target
# -----

quarter_tmp = df_opc_temporal_distribution[
    df_opc_temporal_distribution["period_type"] == "quarter"

```

```

].copy()

quarterly_stability_records = []

for opc_col, group in quarter_tmp.groupby("opc_col"):

    first_row = group.iloc[0]

    record = {
        "opc_col": opc_col,
        "family_name": first_row["family_name"],
        "target_role": first_row["target_role"],
        "percentile": first_row["percentile"],
        "horizon": first_row["horizon"],
        "tp_mult": first_row["tp_mult"],
        "sl_mult": first_row["sl_mult"],
        "n_quarters": group["period_value"].nunique(),
    }

    for metric in range_metrics:
        record[f"{metric}_range"] = group[metric].max() - group[metric].min()
        record[f"{metric}_std"] = group[metric].std()

    record["quarterly_class_instability_score"] = sum(
        record[f"{metric}_range"]
        for metric in class_ratio_cols
    )

    quarterly_stability_records.append(record)

df_opc_quarterly_stability_summary = pd.DataFrame(quarterly_stability_records)

df_opc_quarterly_stability_summary = (
    df_opc_quarterly_stability_summary
    .sort_values("quarterly_class_instability_score")
    .reset_index(drop=True)
)

# -----
# 11) Resumen por año, agregado entre targets
# -----

df_opc_temporal_by_year = (
    year_tmp
    .groupby("period_value", as_index=False)
    .agg(
        n_targets=("opc_col", "count"),
        no_trade_ratio_mean=("no_trade_ratio", "mean"),
        resolved_trade_ratio_mean=("resolved_trade_ratio", "mean"),
        long_tp_ratio_mean=("long_tp_ratio", "mean"),
        long_sl_ratio_mean=("long_sl_ratio", "mean"),
        short_tp_ratio_mean=("short_tp_ratio", "mean"),
        short_sl_ratio_mean=("short_sl_ratio", "mean"),
        winner_ratio_mean=("winner_ratio", "mean"),
        loser_ratio_mean=("loser_ratio", "mean"),
        winner_share_over_resolved_mean=("winner_share_over_resolved", "mean"),
    )
    .sort_values("period_value")
    .reset_index(drop=True)
)

```

```

# -----
# 12) Tabla final de estabilidad temporal
# -----

df_opc_temporal_final_table = df_opc_split_comparison.copy()

df_opc_temporal_final_table = df_opc_temporal_final_table.merge(
    df_opc_yearly_stability_summary[
        [
            "opc_col",
            "yearly_class_instability_score",
            "no_trade_ratio_range",
            "resolved_trade_ratio_range",
            "winner_share_over_resolved_range",
        ]
    ],
    on="opc_col",
    how="left"
)

df_opc_temporal_final_table = df_opc_temporal_final_table.merge(
    df_opc_quarterly_stability_summary[
        [
            "opc_col",
            "quarterly_class_instability_score",
        ]
    ],
    on="opc_col",
    how="left"
)

df_opc_temporal_final_table["temporal_instability_score"] = (
    df_opc_temporal_final_table["split_class_shift_score"]
    + df_opc_temporal_final_table["yearly_class_instability_score"]
    + df_opc_temporal_final_table["quarterly_class_instability_score"]
)

# Clasificación relativa de estabilidad temporal
q33 = df_opc_temporal_final_table["temporal_instability_score"].quantile(0.33)
q66 = df_opc_temporal_final_table["temporal_instability_score"].quantile(0.66)

df_opc_temporal_final_table["temporal_stability_profile"] = np.where(
    df_opc_temporal_final_table["temporal_instability_score"] <= q33,
    "más_estable",
    np.where(
        df_opc_temporal_final_table["temporal_instability_score"] <= q66,
        "estabilidad_intermedia",
        "más_inestable"
    )
)

df_opc_temporal_final_table = (
    df_opc_temporal_final_table
    .sort_values("temporal_instability_score")
    .reset_index(drop=True)
)

# -----
# 13) Mostrar resultados principales

```



```

# -----

percent_formatters = {
    col: "{:.2%}".format
    for col in df_opc_temporal_by_split.columns
    if col.endswith("_mean")
}

print("\nResumen por dataset_split:")
print(
    df_opc_temporal_by_split.to_string(
        index=False,
        formatters=percent_formatters
    )
)

print("\nComparación development vs final_test por target:")
print(
    df_opc_split_comparison[
        [
            "opc_col",
            "target_role",
            "no_trade_ratio_development",
            "no_trade_ratio_final_test",
            "no_trade_ratio_diff",
            "resolved_trade_ratio_development",
            "resolved_trade_ratio_final_test",
            "resolved_trade_ratio_diff",
            "winner_share_over_resolved_development",
            "winner_share_over_resolved_final_test",
            "winner_share_over_resolved_diff",
            "split_class_shift_score",
        ]
    ].to_string(
        index=False,
        formatters={
            "no_trade_ratio_development": "{:.2%}".format,
            "no_trade_ratio_final_test": "{:.2%}".format,
            "no_trade_ratio_diff": "{:+.2%}".format,
            "resolved_trade_ratio_development": "{:.2%}".format,
            "resolved_trade_ratio_final_test": "{:.2%}".format,
            "resolved_trade_ratio_diff": "{:+.2%}".format,
            "winner_share_over_resolved_development": "{:.2%}".format,
            "winner_share_over_resolved_final_test": "{:.2%}".format,
            "winner_share_over_resolved_diff": "{:+.2%}".format,
            "split_class_shift_score": "{:.3f}".format,
        }
    )
)

print("\nResumen agregado por año:")
print(
    df_opc_temporal_by_year.to_string(
        index=False,
        formatters={
            "no_trade_ratio_mean": "{:.2%}".format,
            "resolved_trade_ratio_mean": "{:.2%}".format,
            "long_tp_ratio_mean": "{:.2%}".format,
            "long_sl_ratio_mean": "{:.2%}".format,
            "short_tp_ratio_mean": "{:.2%}".format,
        }
    )
)

```

```

        "short_sl_ratio_mean": "{:.2%}".format,
        "winner_ratio_mean": "{:.2%}".format,
        "loser_ratio_mean": "{:.2%}".format,
        "winner_share_over_resolved_mean": "{:.2%}".format,
    }
    )
)

print("\nEstabilidad anual por target:")
print(
    df_opc_yearly_stability_summary[
        [
            "opc_col",
            "target_role",
            "n_years",
            "yearly_class_instability_score",
            "no_trade_ratio_range",
            "resolved_trade_ratio_range",
            "winner_share_over_resolved_range",
        ]
    ].to_string(
        index=False,
        formatters={
            "yearly_class_instability_score": "{:.3f}".format,
            "no_trade_ratio_range": "{:.2%}".format,
            "resolved_trade_ratio_range": "{:.2%}".format,
            "winner_share_over_resolved_range": "{:.2%}".format,
        }
    )
)

print("\nEstabilidad trimestral por target:")
print(
    df_opc_quarterly_stability_summary[
        [
            "opc_col",
            "target_role",
            "n_quarters",
            "quarterly_class_instability_score",
            "no_trade_ratio_range",
            "resolved_trade_ratio_range",
            "winner_share_over_resolved_range",
        ]
    ].to_string(
        index=False,
        formatters={
            "quarterly_class_instability_score": "{:.3f}".format,
            "no_trade_ratio_range": "{:.2%}".format,
            "resolved_trade_ratio_range": "{:.2%}".format,
            "winner_share_over_resolved_range": "{:.2%}".format,
        }
    )
)

print("\nTabla final de estabilidad temporal OPC:")
print(
    df_opc_temporal_final_table[
        [
            "opc_col",
            "target_role",

```

```

        "split_class_shift_score",
        "yearly_class_instability_score",
        "quarterly_class_instability_score",
        "temporal_instability_score",
        "temporal_stability_profile",
    ]
].to_string(
    index=False,
    formatters={
        "split_class_shift_score": "{:.3f}".format,
        "yearly_class_instability_score": "{:.3f}".format,
        "quarterly_class_instability_score": "{:.3f}".format,
        "temporal_instability_score": "{:.3f}".format,
    }
)
)

# -----
# 14) Guardado de resultados
# -----

if "MNQ_TARGETS_PATH" in globals():

    MNQ_TARGETS_OPC_TEMPORAL_DISTRIBUTION_CSV = (
        MNQ_TARGETS_PATH / "mnq_targets_opc_temporal_distribution.csv"
    )

    MNQ_TARGETS_OPC_TEMPORAL_BY_SPLIT_CSV = (
        MNQ_TARGETS_PATH / "mnq_targets_opc_temporal_by_split.csv"
    )

    MNQ_TARGETS_OPC_SPLIT_COMPARISON_CSV = (
        MNQ_TARGETS_PATH / "mnq_targets_opc_split_comparison.csv"
    )

    MNQ_TARGETS_OPC_TEMPORAL_BY_YEAR_CSV = (
        MNQ_TARGETS_PATH / "mnq_targets_opc_temporal_by_year.csv"
    )

    MNQ_TARGETS_OPC_YEARLY_STABILITY_CSV = (
        MNQ_TARGETS_PATH / "mnq_targets_opc_yearly_stability_summary.csv"
    )

    MNQ_TARGETS_OPC_QUARTERLY_STABILITY_CSV = (
        MNQ_TARGETS_PATH / "mnq_targets_opc_quarterly_stability_summary.csv"
    )

    MNQ_TARGETS_OPC_TEMPORAL_FINAL_TABLE_CSV = (
        MNQ_TARGETS_PATH / "mnq_targets_opc_temporal_final_table.csv"
    )

    df_opc_temporal_distribution.to_csv(
        MNQ_TARGETS_OPC_TEMPORAL_DISTRIBUTION_CSV,
        index=False
    )

    df_opc_temporal_by_split.to_csv(
        MNQ_TARGETS_OPC_TEMPORAL_BY_SPLIT_CSV,
        index=False
    )

```

```
df_opc_split_comparison.to_csv(
    MNQ_TARGETS_OPC_SPLIT_COMPARISON_CSV,
    index=False
)

df_opc_temporal_by_year.to_csv(
    MNQ_TARGETS_OPC_TEMPORAL_BY_YEAR_CSV,
    index=False
)

df_opc_yearly_stability_summary.to_csv(
    MNQ_TARGETS_OPC_YEARLY_STABILITY_CSV,
    index=False
)

df_opc_quarterly_stability_summary.to_csv(
    MNQ_TARGETS_OPC_QUARTERLY_STABILITY_CSV,
    index=False
)

df_opc_temporal_final_table.to_csv(
    MNQ_TARGETS_OPC_TEMPORAL_FINAL_TABLE_CSV,
    index=False
)

print("\nArchivos guardados:")
print(f"Distribución temporal OPC:      {MNQ_TARGETS_OPC_TEMPORAL_DISTRIBUTIO")
print(f"Resumen por split:                  {MNQ_TARGETS_OPC_TEMPORAL_BY_SPLIT_CS")
print(f"Comparación split:                   {MNQ_TARGETS_OPC_SPLIT_COMPARISON_CSV")
print(f"Resumen por año:                     {MNQ_TARGETS_OPC_TEMPORAL_BY_YEAR_CSV")
print(f"Estabilidad anual:                   {MNQ_TARGETS_OPC_YEARLY_STABILITY_CSV")
print(f"Estabilidad trimestral:              {MNQ_TARGETS_OPC_QUARTERLY_STABILITY_")
print(f"Tabla final temporal OPC:            {MNQ_TARGETS_OPC_TEMPORAL_FINAL_TABLE")

else:
    print("\nMNQ_TARGETS_PATH no está definido. No se guardaron archivos.")

print("\nAnálisis temporal de targets OPC finalizado correctamente.")
```

## Análisis temporal de targets OPC

=====

Objetos base encontrados correctamente.

## Columnas temporales preparadas:

year  
year\_quarter  
dataset\_split

## Valores de dataset\_split:

dataset\_split  
development\_2020\_2024 818835  
final\_test\_2025\_2026 205227

## Filas generadas en df\_opc\_temporal\_distribution:

175

## Períodos analizados:

period\_type  
dataset\_split 2  
quarter 26  
year 7

## Resumen por dataset\_split:

| period_value          | n_targets | no_trade_ratio_mean | resolved_trade_ratio_mean | long_tp_ratio_mean | long_sl_ratio_mean | short_tp_ratio_mean | short_sl_ratio_mean | winner_ratio_mean | loser_ratio_mean | winner_share_over_resolved_mean | long_share_over_resolved_mean | short_share_over_resolved_mean |
|-----------------------|-----------|---------------------|---------------------------|--------------------|--------------------|---------------------|---------------------|-------------------|------------------|---------------------------------|-------------------------------|--------------------------------|
| development_2020_2024 | 5         | 72.47%              | 27.53%                    | 11.57%             | 1.10%              | 13.71%              | 1.14%               | 25.28%            | 8%               | 2.25%                           | 92.04%                        | 45.87%                         |
| final_test_2025_2026  | 5         | 51.47%              | 48.53%                    | 19.28%             | 3.54%              | 22.47%              | 3.25%               | 41.7%             | 4%               | 6.79%                           | 86.21%                        | 46.94%                         |

## Comparación development vs final\_test por target:

| opc_col               | target_role           | no_trade_ratio_development | no_trade_ratio_final_test | no_trade_ratio_diff | resolved_trade_ratio_development | resolved_trade_ratio_final_test | resolved_trade_ratio_diff | winner_share_over_resolved_development | winner_share_over_resolved_final_test | winner_share_over_resolved_diff | split_class_shift_score |
|-----------------------|-----------------------|----------------------------|---------------------------|---------------------|----------------------------------|---------------------------------|---------------------------|----------------------------------------|---------------------------------------|---------------------------------|-------------------------|
| opc_p50_h30_tp15_sl10 | principal             | 72.05%                     | 51.41%                    | -20.64%             | 27.95%                           | 19.28%                          | -8.67%                    | 92.01%                                 | 86.63%                                | -5.39%                          | 0.413                   |
| opc_p50_h60_tp15_sl10 | principal             | 73.08%                     | 51.73%                    | -21.34%             | 26.92%                           | 19.28%                          | -7.64%                    | 92.34%                                 | 86.12%                                | -6.23%                          | 0.427                   |
| opc_p50_h90_tp15_sl10 | principal             | 72.75%                     | 51.07%                    | -21.67%             | 27.25%                           | 19.28%                          | -7.97%                    | 91.65%                                 | 85.80%                                | -5.85%                          | 0.433                   |
| opc_p40_h60_tp15_sl10 | comparativo_activo    | 63.73%                     | 42.06%                    | -21.67%             | 36.27%                           | 57.94%                          | +21.67%                   | 90.40%                                 | 83.78%                                | -6.62%                          | 0.433                   |
| opc_p60_h60_tp15_sl10 | comparativo_selectivo | 80.75%                     |                           |                     |                                  |                                 |                           |                                        |                                       |                                 |                         |

|        |         |        |        |
|--------|---------|--------|--------|
| 61.06% | -19.69% | 19.25% |        |
| 38.94% | +19.69% |        | 93.80% |
| 88.74% | -5.07%  | 0.394  |        |

## Resumen agregado por año:

| period_value | n_targets | no_trade_ratio_mean | resolved_trade_ratio_mean | long_tp_ratio_mean | long_sl_ratio_mean | short_tp_ratio_mean | short_sl_ratio_mean | winner_ratio_mean | loser_ratio_mean | winner_share_over_resolved_mean |
|--------------|-----------|---------------------|---------------------------|--------------------|--------------------|---------------------|---------------------|-------------------|------------------|---------------------------------|
| 2020         | 5         |                     | 78.00%                    |                    |                    |                     |                     | 22.00%            |                  |                                 |
| 8.91%        |           | 0.96%               | 11.20%                    |                    |                    |                     | 0.93%               |                   | 20.1             |                                 |
| 2%           |           | 1.88%               |                           | 91.62%             |                    |                     |                     |                   |                  |                                 |
| 2021         | 5         |                     | 80.30%                    |                    |                    |                     |                     | 19.70%            |                  |                                 |
| 7.85%        |           | 0.77%               | 10.42%                    |                    |                    |                     | 0.67%               |                   | 18.2             |                                 |
| 6%           |           | 1.44%               |                           | 92.92%             |                    |                     |                     |                   |                  |                                 |
| 2022         | 5         |                     | 53.65%                    |                    |                    |                     |                     | 46.35%            |                  |                                 |
| 19.62%       |           | 2.35%               | 21.82%                    |                    |                    |                     | 2.56%               |                   | 41.4             |                                 |
| 5%           |           | 4.91%               |                           | 89.65%             |                    |                     |                     |                   |                  |                                 |
| 2023         | 5         |                     | 80.03%                    |                    |                    |                     |                     | 19.97%            |                  |                                 |
| 9.11%        |           | 0.51%               | 9.83%                     |                    |                    |                     | 0.53%               |                   | 18.9             |                                 |
| 4%           |           | 1.04%               |                           | 94.96%             |                    |                     |                     |                   |                  |                                 |
| 2024         | 5         |                     | 70.69%                    |                    |                    |                     |                     | 29.31%            |                  |                                 |
| 12.22%       |           | 0.93%               | 15.13%                    |                    |                    |                     | 1.02%               |                   | 27.3             |                                 |
| 5%           |           | 1.96%               |                           | 93.57%             |                    |                     |                     |                   |                  |                                 |
| 2025         | 5         |                     | 55.63%                    |                    |                    |                     |                     | 44.37%            |                  |                                 |
| 17.48%       |           | 3.14%               | 20.92%                    |                    |                    |                     | 2.83%               |                   | 38.3             |                                 |
| 9%           |           | 5.98%               |                           | 86.72%             |                    |                     |                     |                   |                  |                                 |
| 2026         | 5         |                     | 37.96%                    |                    |                    |                     |                     | 62.04%            |                  |                                 |
| 25.12%       |           | 4.84%               | 27.50%                    |                    |                    |                     | 4.58%               |                   | 52.6             |                                 |
| 2%           |           | 9.42%               |                           | 85.02%             |                    |                     |                     |                   |                  |                                 |

## Estabilidad anual por target:

| opc_col               | target_role           | n_years | yearly_class_instability_score |
|-----------------------|-----------------------|---------|--------------------------------|
| opc_p60_h60_tp15_sl10 | comparativo_selectivo | 7       | 0.7                            |
| 92                    |                       | 39.35%  | 7.8                            |
| 8%                    |                       |         |                                |
| opc_p50_h30_tp15_sl10 | principal             | 7       | 0.8                            |
| 30                    |                       | 41.07%  | 9.6                            |
| 2%                    |                       |         |                                |
| opc_p50_h90_tp15_sl10 | principal             | 7       | 0.8                            |
| 78                    |                       | 43.29%  | 9.6                            |
| 7%                    |                       |         |                                |
| opc_p50_h60_tp15_sl10 | principal             | 7       | 0.8                            |
| 79                    |                       | 43.48%  | 10.3                           |
| 7%                    |                       |         |                                |
| opc_p40_h60_tp15_sl10 | comparativo_activo    | 7       | 0.9                            |
| 10                    |                       | 45.17%  | 12.1                           |
| 4%                    |                       |         |                                |

## Estabilidad trimestral por target:

| opc_col               | target_role           | n_quarters | quarterly_class_instability_score |
|-----------------------|-----------------------|------------|-----------------------------------|
| opc_p60_h60_tp15_sl10 | comparativo_selectivo | 26         |                                   |
| 0.939                 |                       | 46.34%     | 1                                 |
| 1.09%                 |                       |            |                                   |
| opc_p50_h30_tp15_sl10 | principal             | 26         |                                   |
| 1.028                 |                       | 49.72%     | 1                                 |
| 2.44%                 |                       |            |                                   |

|                       |                    |        |   |
|-----------------------|--------------------|--------|---|
| opc_p50_h90_tp15_sl10 | principal          | 26     |   |
| 1.079                 | 51.33%             | 51.33% | 1 |
| 2.43%                 |                    |        |   |
| opc_p50_h60_tp15_sl10 | principal          | 26     |   |
| 1.080                 | 51.60%             | 51.60% | 1 |
| 3.06%                 |                    |        |   |
| opc_p40_h60_tp15_sl10 | comparativo_activo | 26     |   |
| 1.170                 | 54.28%             | 54.28% | 1 |
| 5.66%                 |                    |        |   |

Tabla final de estabilidad temporal OPC:

| opc_col               | target_role           | split_class_shift_score | yearly_class_instability_score | quarterly_class_instability_score | temporal_instability_score | temporal_stability_profile |
|-----------------------|-----------------------|-------------------------|--------------------------------|-----------------------------------|----------------------------|----------------------------|
| opc_p60_h60_tp15_sl10 | comparativo_selectivo |                         | 0.394                          |                                   |                            |                            |
| 0.792                 | 0.939                 |                         | 2.126                          |                                   |                            |                            |
| más_estable           |                       |                         |                                |                                   |                            |                            |
| opc_p50_h30_tp15_sl10 | principal             |                         | 0.413                          |                                   |                            |                            |
| 0.830                 | 1.028                 |                         | 2.271                          |                                   |                            |                            |
| más_estable           |                       |                         |                                |                                   |                            |                            |
| opc_p50_h60_tp15_sl10 | principal             |                         | 0.427                          |                                   |                            |                            |
| 0.879                 | 1.080                 |                         | 2.386                          |                                   |                            | estabilida                 |
| d_intermedia          |                       |                         |                                |                                   |                            |                            |
| opc_p50_h90_tp15_sl10 | principal             |                         | 0.433                          |                                   |                            |                            |
| 0.878                 | 1.079                 |                         | 2.391                          |                                   |                            | m                          |
| ás_inestable          |                       |                         |                                |                                   |                            |                            |
| opc_p40_h60_tp15_sl10 | comparativo_activo    |                         | 0.433                          |                                   |                            |                            |
| 0.910                 | 1.170                 |                         | 2.514                          |                                   |                            | m                          |
| ás_inestable          |                       |                         |                                |                                   |                            |                            |

Archivos guardados:

Distribución temporal OPC: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_opc\_temporal\_distribution.csv  
 Resumen por split: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_opc\_temporal\_by\_split.csv  
 Comparación split: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_opc\_split\_comparison.csv  
 Resumen por año: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_opc\_temporal\_by\_year.csv  
 Estabilidad anual: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_opc\_yearly\_stability\_summary.csv  
 Estabilidad trimestral: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_opc\_quarterly\_stability\_summary.csv  
 Tabla final temporal OPC: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_opc\_temporal\_final\_table.csv

Análisis temporal de targets OPC finalizado correctamente.

Observaciones rápidas:

- El análisis temporal confirma que **OPC no es estacionario**.
- En **final\_test\_2025\_2026** baja mucho **NO\_TRADE** : de **72.47%** a **51.47%**.
- Los trades resueltos suben fuerte: de **27.53%** a **48.53%**.
- Pero también aumenta el riesgo: **loser\_ratio** sube de **2.25%** a **6.79%**.
- La proporción ganadora entre resueltos cae: de **92.04%** a **86.21%**.
- 2026 es el año más activo: solo **37.96% NO\_TRADE** y **62.04% trades resueltos**.
- 2021 y 2023 son los años más lentos/selectivos: **NO\_TRADE** cerca de **80%**.

- `opc_p60_h60_tp15_sl10` es el más estable temporalmente, pero también el más selectivo.
- Entre los targets principales, `opc_p50_h30_tp15_sl10` sigue siendo el mejor candidato.
- `opc_p40_h60_tp15_sl10` es el más inestable y probablemente el más ruidoso.
- `opc_p50_h60` y `opc_p50_h90` quedan muy cerca, pero no mejoran claramente a `p50_h30`.

Conclusión rápida:

Mejor target principal hasta ahora:  
`opc_p50_h30_tp15_sl10`

Más estable, pero selectivo:  
`opc_p60_h60_tp15_sl10`

Más activo, pero más inestable:  
`opc_p40_h60_tp15_sl10`

Todavía falta revisar régimen y contexto antes de cerrar una conclusión fuerte.

## 6.8. Análisis por régimen y contexto

Vamos a evaluar si la distribución de clases `OPC` cambia según:

```
régimen intradiario
contrato
familia de contrato
dataset split
régimen + split
contrato + régimen
año + régimen
familia de contrato + régimen
```

Esta celda deja creados estos objetos principales:

```
df_opc_context_analysis
df_opc_regime_analysis
df_opc_by_regime
df_opc_regime_main_candidate
df_opc_by_context_type
df_opc_context_sensitivity_by_target
df_opc_context_sensitivity_by_context_type
df_opc_by_contract_code
df_opc_context_final_table
```

Con esto podremos ver si `opc_p50_h30_tp15_sl10` sigue siendo el mejor candidato cuando se analiza por régimen, contrato y contexto combinado.

```
In [30]: # 6.8. Análisis por régimen y contexto de targets OPC
# =====

import numpy as np
import pandas as pd

print("Análisis por régimen y contexto de targets OPC")
```



```

print("=" * 80)

# -----
# 1) Validar objetos necesarios
# -----

required_objects = [
    "df_targets_opc",
    "df_opc_metadata",
]

missing_objects = [
    obj for obj in required_objects
    if obj not in globals()
]

if missing_objects:
    raise NameError(
        "Faltan objetos necesarios para analizar OPC por contexto: "
        f"{missing_objects}"
    )

print("Objetos base encontrados correctamente.")

# -----
# 2) Definir clases OPC
# -----

OPC_CLASS_ORDER = [
    "NO_TRADE",
    "LONG_TP",
    "LONG_SL",
    "SHORT_TP",
    "SHORT_SL",
]

OPC_CLASS_MAP = {
    "NO_TRADE": 0,
    "LONG_TP": 1,
    "LONG_SL": 2,
    "SHORT_TP": 3,
    "SHORT_SL": 4,
}

REGIME_NAME_MAP = {
    0: "Overnight",
    1: "Pre-market",
    2: "Opening",
    3: "Regular",
    4: "Closing",
}

# -----
# 3) Preparar dataframe base de contexto
# -----

df_opc_context_base = df_targets_opc.copy()

# Crear year si no existe
if "year" not in df_opc_context_base.columns:

```

```

if "date" in df_opc_context_base.columns:
    date_series_tmp = pd.to_datetime(df_opc_context_base["date"])
else:
    date_series_tmp = pd.to_datetime(df_opc_context_base.index)

df_opc_context_base["year"] = date_series_tmp.dt.year

else:
    if "date" in df_opc_context_base.columns:
        date_series_tmp = pd.to_datetime(df_opc_context_base["date"])
    else:
        date_series_tmp = pd.to_datetime(df_opc_context_base.index)

# Crear regime_name si no existe
if "regime_name" not in df_opc_context_base.columns:

    if "regime_id" not in df_opc_context_base.columns:
        raise KeyError("No existe regime_id ni regime_name en df_targets_opc.")

    df_opc_context_base["regime_name"] = (
        df_opc_context_base["regime_id"]
        .map(REGIME_NAME_MAP)
        .fillna("Unknown")
    )

# Crear contract_code si no existe
if "contract_code" not in df_opc_context_base.columns:

    if "contract" not in df_opc_context_base.columns:
        raise KeyError("No existe contract ni contract_code en df_targets_opc.")

    df_opc_context_base["contract_code"] = (
        df_opc_context_base["contract"]
        .astype(str)
        .str[0]
    )

# Crear dataset_split si no existe
df_opc_context_base["dataset_split"] = np.where(
    df_opc_context_base["year"].between(2020, 2024),
    "development_2020_2024",
    np.where(
        df_opc_context_base["year"].between(2025, 2026),
        "final_test_2025_2026",
        "other"
    )
)

# Variables combinadas
df_opc_context_base["year_contract"] = (
    df_opc_context_base["year"].astype(str)
    + "_"
    + df_opc_context_base["contract"].astype(str)
)

df_opc_context_base["regime_dataset_split"] = (
    df_opc_context_base["regime_name"].astype(str)
    + "_"
    + df_opc_context_base["dataset_split"].astype(str)
)

```

```

)

df_opc_context_base["contract_regime"] = (
    df_opc_context_base["contract"].astype(str)
    + "_"
    + df_opc_context_base["regime_name"].astype(str)
)

df_opc_context_base["year_regime"] = (
    df_opc_context_base["year"].astype(str)
    + "_"
    + df_opc_context_base["regime_name"].astype(str)
)

df_opc_context_base["contract_code_regime"] = (
    df_opc_context_base["contract_code"].astype(str)
    + "_"
    + df_opc_context_base["regime_name"].astype(str)
)

print("\nColumnas de contexto preparadas:")
print("regime_name")
print("contract")
print("contract_code")
print("dataset_split")
print("year_contract")
print("regime_dataset_split")
print("contract_regime")
print("year_regime")
print("contract_code_regime")

# -----
# 4) Definir contextos a analizar
# -----

context_definitions = [
    {
        "context_type": "regime",
        "context_col": "regime_name",
        "description": "Régimen intradiario",
    },
    {
        "context_type": "contract",
        "context_col": "contract",
        "description": "Contrato individual",
    },
    {
        "context_type": "contract_code",
        "context_col": "contract_code",
        "description": "Familia de contrato H/M/U/Z",
    },
    {
        "context_type": "dataset_split",
        "context_col": "dataset_split",
        "description": "Development vs final test",
    },
    {
        "context_type": "year_contract",
        "context_col": "year_contract",
        "description": "Año + contrato",
    },

```

```

    },
    {
        "context_type": "regime_dataset_split",
        "context_col": "regime_dataset_split",
        "description": "Régimen + dataset split",
    },
    {
        "context_type": "contract_regime",
        "context_col": "contract_regime",
        "description": "Contrato + régimen intradiario",
    },
    {
        "context_type": "year_regime",
        "context_col": "year_regime",
        "description": "Año + régimen intradiario",
    },
    {
        "context_type": "contract_code_regime",
        "context_col": "contract_code_regime",
        "description": "Familia de contrato + régimen intradiario",
    },
],

df_opc_context_definitions = pd.DataFrame(context_definitions)

print("\nContextos definidos:")
print(df_opc_context_definitions.to_string(index=False))

# -----
# 5) Función auxiliar para calcular distribución OPC
# -----

def calculate_opc_context_distribution(opc_label, mask):
    """
    Calcula distribución de clases OPC para un contexto específico.
    """

    tmp = opc_label.loc[mask]

    total_rows = len(tmp)
    valid_rows = int(tmp.notna().sum())
    nan_rows = int(tmp.isna().sum())

    class_counts = {
        label: int((tmp == label).sum())
        for label in OPC_CLASS_ORDER
    }

    class_ratios = {
        label: class_counts[label] / valid_rows if valid_rows > 0 else np.nan
        for label in OPC_CLASS_ORDER
    }

    no_trade_count = class_counts["NO_TRADE"]

    long_tp_count = class_counts["LONG_TP"]
    long_sl_count = class_counts["LONG_SL"]
    short_tp_count = class_counts["SHORT_TP"]
    short_sl_count = class_counts["SHORT_SL"]

```

```

resolved_trade_count = (
    long_tp_count
    + long_sl_count
    + short_tp_count
    + short_sl_count
)

winner_count = long_tp_count + short_tp_count
loser_count = long_sl_count + short_sl_count

long_count = long_tp_count + long_sl_count
short_count = short_tp_count + short_sl_count

no_trade_ratio = no_trade_count / valid_rows if valid_rows > 0 else np.nan
resolved_trade_ratio = resolved_trade_count / valid_rows if valid_rows > 0 else np.nan

winner_ratio = winner_count / valid_rows if valid_rows > 0 else np.nan
loser_ratio = loser_count / valid_rows if valid_rows > 0 else np.nan

long_ratio = long_count / valid_rows if valid_rows > 0 else np.nan
short_ratio = short_count / valid_rows if valid_rows > 0 else np.nan

winner_share_over_resolved = (
    winner_count / resolved_trade_count
    if resolved_trade_count > 0
    else np.nan
)

loser_share_over_resolved = (
    loser_count / resolved_trade_count
    if resolved_trade_count > 0
    else np.nan
)

long_share_over_resolved = (
    long_count / resolved_trade_count
    if resolved_trade_count > 0
    else np.nan
)

short_share_over_resolved = (
    short_count / resolved_trade_count
    if resolved_trade_count > 0
    else np.nan
)

majority_class = None
majority_class_count = np.nan
majority_class_ratio = np.nan

if valid_rows > 0:
    majority_class = max(class_counts, key=class_counts.get)
    majority_class_count = class_counts[majority_class]
    majority_class_ratio = majority_class_count / valid_rows

return {
    "total_rows": total_rows,
    "valid_rows": valid_rows,
    "nan_rows": nan_rows,
    "nan_ratio": nan_rows / total_rows if total_rows > 0 else np.nan,

```

```

        "no_trade_count": no_trade_count,
        "long_tp_count": long_tp_count,
        "long_sl_count": long_sl_count,
        "short_tp_count": short_tp_count,
        "short_sl_count": short_sl_count,

        "no_trade_ratio": no_trade_ratio,
        "long_tp_ratio": class_ratios["LONG_TP"],
        "long_sl_ratio": class_ratios["LONG_SL"],
        "short_tp_ratio": class_ratios["SHORT_TP"],
        "short_sl_ratio": class_ratios["SHORT_SL"],

        "resolved_trade_count": resolved_trade_count,
        "resolved_trade_ratio": resolved_trade_ratio,

        "winner_count": winner_count,
        "loser_count": loser_count,
        "winner_ratio": winner_ratio,
        "loser_ratio": loser_ratio,

        "winner_share_over_resolved": winner_share_over_resolved,
        "loser_share_over_resolved": loser_share_over_resolved,

        "long_count": long_count,
        "short_count": short_count,
        "long_ratio": long_ratio,
        "short_ratio": short_ratio,

        "long_share_over_resolved": long_share_over_resolved,
        "short_share_over_resolved": short_share_over_resolved,

        "majority_class": majority_class,
        "majority_class_count": majority_class_count,
        "majority_class_ratio": majority_class_ratio,
    }

# -----
# 6) Calcular distribución por contexto
# -----

context_records = []

for _, meta_row in df_opc_metadata.iterrows():

    opc_col = meta_row["opc_col"]
    opc_label_col = meta_row["opc_label_col"]

    family_name = meta_row["family_name"]
    target_role = meta_row["target_role"]

    percentile = meta_row["percentile"]
    horizon = meta_row["horizon"]
    tp_mult = meta_row["tp_mult"]
    sl_mult = meta_row["sl_mult"]

    if opc_label_col not in df_opc_context_base.columns:
        raise KeyError(f"No existe la columna {opc_label_col}.")

    opc_label = df_opc_context_base[opc_label_col]

```

```

for context_def in context_definitions:

    context_type = context_def["context_type"]
    context_col = context_def["context_col"]
    context_description = context_def["description"]

    context_values = (
        df_opc_context_base[context_col]
        .dropna()
        .unique()
        .tolist()
    )

    context_values = sorted(context_values, key=lambda x: str(x))

    for context_value in context_values:

        mask = df_opc_context_base[context_col].eq(context_value)

        metrics = calculate_opc_context_distribution(
            opc_label=opc_label,
            mask=mask
        )

        context_records.append(
            {
                "opc_col": opc_col,
                "opc_label_col": opc_label_col,
                "family_name": family_name,
                "target_role": target_role,
                "percentile": percentile,
                "horizon": horizon,
                "tp_mult": tp_mult,
                "sl_mult": sl_mult,

                "context_type": context_type,
                "context_col": context_col,
                "context_description": context_description,
                "context_value": context_value,

                **metrics,
            }
        )

df_opc_context_analysis = pd.DataFrame(context_records)

print("\nFilas generadas en df_opc_context_analysis:")
print(len(df_opc_context_analysis))

print("\nValores por tipo de contexto:")
print(
    df_opc_context_analysis
    .groupby("context_type")["context_value"]
    .nunique()
    .sort_values(ascending=False)
    .to_string()
)

# -----

```

```

# 7) Tablas principales por régimen
# -----

df_opc_regime_analysis = (
    df_opc_context_analysis[
        df_opc_context_analysis["context_type"] == "regime"
    ]
    .copy()
)

df_opc_by_regime = (
    df_opc_regime_analysis
    .groupby("context_value", as_index=False)
    .agg(
        n_targets=("opc_col", "count"),
        total_rows_mean=("total_rows", "mean"),
        valid_rows_mean=("valid_rows", "mean"),
        nan_ratio_mean=("nan_ratio", "mean"),

        no_trade_ratio_mean=("no_trade_ratio", "mean"),
        resolved_trade_ratio_mean=("resolved_trade_ratio", "mean"),

        long_tp_ratio_mean=("long_tp_ratio", "mean"),
        long_sl_ratio_mean=("long_sl_ratio", "mean"),
        short_tp_ratio_mean=("short_tp_ratio", "mean"),
        short_sl_ratio_mean=("short_sl_ratio", "mean"),

        winner_ratio_mean=("winner_ratio", "mean"),
        loser_ratio_mean=("loser_ratio", "mean"),
        winner_share_over_resolved_mean=("winner_share_over_resolved", "mean"),

        long_share_over_resolved_mean=("long_share_over_resolved", "mean"),
        short_share_over_resolved_mean=("short_share_over_resolved", "mean"),

        majority_class_ratio_mean=("majority_class_ratio", "mean"),
    )
    .rename(columns={"context_value": "regime_name"})
    .reset_index(drop=True)
)

regime_order = [
    "Overnight",
    "Pre-market",
    "Opening",
    "Regular",
    "Closing",
]

df_opc_by_regime["regime_order"] = df_opc_by_regime["regime_name"].map(
    {name: i for i, name in enumerate(regime_order)}
)

df_opc_by_regime = (
    df_opc_by_regime
    .sort_values("regime_order")
    .drop(columns=["regime_order"])
    .reset_index(drop=True)
)

# -----

```



```

# 8) Régimen para target principal p50_h30
# -----

main_opc_candidate = "opc_p50_h30_tp15_sl10"

df_opc_regime_main_candidate = (
    df_opc_regime_analysis[
        df_opc_regime_analysis["opc_col"] == main_opc_candidate
    ]
    .copy()
)

df_opc_regime_main_candidate["regime_order"] = (
    df_opc_regime_main_candidate["context_value"]
    .map({name: i for i, name in enumerate(regime_order)})
)

df_opc_regime_main_candidate = (
    df_opc_regime_main_candidate
    .sort_values("regime_order")
    .drop(columns=["regime_order"])
    .reset_index(drop=True)
)

# -----
# 9) Resumen por context_type
# -----

df_opc_by_context_type = (
    df_opc_context_analysis
    .groupby("context_type", as_index=False)
    .agg(
        n_rows=("opc_col", "count"),
        n_context_values=("context_value", "nunique"),
        n_targets=("opc_col", "nunique"),

        valid_rows_mean=("valid_rows", "mean"),
        nan_ratio_mean=("nan_ratio", "mean"),

        no_trade_ratio_mean=("no_trade_ratio", "mean"),
        resolved_trade_ratio_mean=("resolved_trade_ratio", "mean"),

        long_tp_ratio_mean=("long_tp_ratio", "mean"),
        long_sl_ratio_mean=("long_sl_ratio", "mean"),
        short_tp_ratio_mean=("short_tp_ratio", "mean"),
        short_sl_ratio_mean=("short_sl_ratio", "mean"),

        winner_ratio_mean=("winner_ratio", "mean"),
        loser_ratio_mean=("loser_ratio", "mean"),
        winner_share_over_resolved_mean=("winner_share_over_resolved", "mean"),

        majority_class_ratio_mean=("majority_class_ratio", "mean"),
    )
    .sort_values("resolved_trade_ratio_mean", ascending=False)
    .reset_index(drop=True)
)

# -----
# 10) Sensibilidad por contexto
# -----

```

```

MIN_VALID_ROWS_FOR_CONTEXT_SENSITIVITY = 1000

def value_range(series):
    series = series.dropna()
    if series.empty:
        return np.nan
    return series.max() - series.min()

sensitivity_metrics = [
    "no_trade_ratio",
    "long_tp_ratio",
    "long_sl_ratio",
    "short_tp_ratio",
    "short_sl_ratio",
    "resolved_trade_ratio",
    "winner_share_over_resolved",
    "long_share_over_resolved",
    "short_share_over_resolved",
]

df_context_for_sensitivity = df_opc_context_analysis[
    df_opc_context_analysis["valid_rows"] >= MIN_VALID_ROWS_FOR_CONTEXT_SENSITIVITY
].copy()

sensitivity_records = []

group_cols = [
    "opc_col",
    "family_name",
    "target_role",
    "percentile",
    "horizon",
    "tp_mult",
    "sl_mult",
    "context_type",
]

for group_key, group in df_context_for_sensitivity.groupby(group_cols):

    record = dict(zip(group_cols, group_key))

    record["n_context_values"] = group["context_value"].nunique()
    record["valid_rows_min"] = group["valid_rows"].min()
    record["valid_rows_max"] = group["valid_rows"].max()

    for metric in sensitivity_metrics:
        record[f"{metric}_range"] = value_range(group[metric])
        record[f"{metric}_std"] = group[metric].std()

    # Score simple: suma de rangos de las 5 clases OPC
    record["context_class_sensitivity_score"] = sum(
        record[f"{metric}_range"]
        for metric in [
            "no_trade_ratio",
            "long_tp_ratio",
            "long_sl_ratio",
            "short_tp_ratio",
            "short_sl_ratio",
        ]
    )

```

```

        if pd.isna(record[f"{metric}_range"])
    )

    sensitivity_records.append(record)

df_opc_context_sensitivity_by_target = pd.DataFrame(sensitivity_records)

df_opc_context_sensitivity_by_target = (
    df_opc_context_sensitivity_by_target
    .sort_values(
        [
            "context_type",
            "context_class_sensitivity_score",
        ]
    )
    .reset_index(drop=True)
)

df_opc_context_sensitivity_by_context_type = (
    df_opc_context_sensitivity_by_target
    .groupby("context_type", as_index=False)
    .agg(
        n_targets=("opc_col", "count"),
        n_context_values_mean=("n_context_values", "mean"),

        context_class_sensitivity_score_mean=("context_class_sensitivity_score",
        context_class_sensitivity_score_max=("context_class_sensitivity_score",

        no_trade_ratio_range_mean=("no_trade_ratio_range", "mean"),
        resolved_trade_ratio_range_mean=("resolved_trade_ratio_range", "mean"),
        winner_share_over_resolved_range_mean=("winner_share_over_resolved_range",
        long_share_over_resolved_range_mean=("long_share_over_resolved_range", "
        short_share_over_resolved_range_mean=("short_share_over_resolved_range",

    )
    .sort_values("context_class_sensitivity_score_mean", ascending=False)
    .reset_index(drop=True)
)

df_opc_context_sensitivity_by_target_final = (
    df_opc_context_sensitivity_by_target
    .sort_values("context_class_sensitivity_score")
    .reset_index(drop=True)
)

# -----
# 11) Tablas específicas de contexto
# -----

df_opc_by_contract_code = (
    df_opc_context_analysis[
        df_opc_context_analysis["context_type"] == "contract_code"
    ]
    .groupby("context_value", as_index=False)
    .agg(
        n_targets=("opc_col", "count"),
        no_trade_ratio_mean=("no_trade_ratio", "mean"),
        resolved_trade_ratio_mean=("resolved_trade_ratio", "mean"),
        winner_share_over_resolved_mean=("winner_share_over_resolved", "mean"),
        majority_class_ratio_mean=("majority_class_ratio", "mean"),
    )
)

```

```

        .rename(columns={"context_value": "contract_code"})
        .sort_values("contract_code")
        .reset_index(drop=True)
    )

df_opc_by_dataset_split_regime = (
    df_opc_context_analysis[
        df_opc_context_analysis["context_type"] == "regime_dataset_split"
    ]
    .copy()
)

# -----
# 12) Tabla final de contexto para interpretación
# -----

df_opc_context_final_table = df_opc_context_sensitivity_by_target.copy()

q33 = df_opc_context_final_table["context_class_sensitivity_score"].quantile(0.33)
q66 = df_opc_context_final_table["context_class_sensitivity_score"].quantile(0.66)

df_opc_context_final_table["context_sensitivity_profile"] = np.where(
    df_opc_context_final_table["context_class_sensitivity_score"] <= q33,
    "menor_sensibilidad",
    np.where(
        df_opc_context_final_table["context_class_sensitivity_score"] <= q66,
        "sensibilidad_intermedia",
        "mayor_sensibilidad"
    )
)

df_opc_context_final_table = (
    df_opc_context_final_table
    .sort_values(
        [
            "context_type",
            "context_class_sensitivity_score",
        ]
    )
    .reset_index(drop=True)
)

# -----
# 13) Mostrar resultados principales
# -----

pct_formatter = {
    "nan_ratio_mean": "{:.2%}".format,
    "no_trade_ratio_mean": "{:.2%}".format,
    "resolved_trade_ratio_mean": "{:.2%}".format,
    "long_tp_ratio_mean": "{:.2%}".format,
    "long_sl_ratio_mean": "{:.2%}".format,
    "short_tp_ratio_mean": "{:.2%}".format,
    "short_sl_ratio_mean": "{:.2%}".format,
    "winner_ratio_mean": "{:.2%}".format,
    "loser_ratio_mean": "{:.2%}".format,
    "winner_share_over_resolved_mean": "{:.2%}".format,
    "long_share_over_resolved_mean": "{:.2%}".format,
    "short_share_over_resolved_mean": "{:.2%}".format,
    "majority_class_ratio_mean": "{:.2%}".format,

```

```

}

print("\nResumen OPC por régimen intradiario:")
print(
    df_opc_by_regime.to_string(
        index=False,
        formatters=pct_formatter
    )
)

print(f"\nDetalle por régimen para target principal: {main_opc_candidate}")
print(
    df_opc_regime_main_candidate[
        [
            "context_value",
            "valid_rows",
            "nan_rows",
            "nan_ratio",
            "no_trade_ratio",
            "resolved_trade_ratio",
            "long_tp_ratio",
            "long_sl_ratio",
            "short_tp_ratio",
            "short_sl_ratio",
            "winner_share_over_resolved",
            "long_share_over_resolved",
            "short_share_over_resolved",
        ]
    ].to_string(
        index=False,
        formatters={
            "nan_ratio": "{:.2%}".format,
            "no_trade_ratio": "{:.2%}".format,
            "resolved_trade_ratio": "{:.2%}".format,
            "long_tp_ratio": "{:.2%}".format,
            "long_sl_ratio": "{:.2%}".format,
            "short_tp_ratio": "{:.2%}".format,
            "short_sl_ratio": "{:.2%}".format,
            "winner_share_over_resolved": "{:.2%}".format,
            "long_share_over_resolved": "{:.2%}".format,
            "short_share_over_resolved": "{:.2%}".format,
        }
    )
)

print("\nResumen agregado por tipo de contexto:")
print(
    df_opc_by_context_type.to_string(
        index=False,
        formatters=pct_formatter
    )
)

print("\nSensibilidad promedio por tipo de contexto:")
print(
    df_opc_context_sensitivity_by_context_type.to_string(
        index=False,
        formatters={
            "context_class_sensitivity_score_mean": "{:.3f}".format,
            "context_class_sensitivity_score_max": "{:.3f}".format,
        }
    )
)

```

```

        "no_trade_ratio_range_mean": "{:.2%}".format,
        "resolved_trade_ratio_range_mean": "{:.2%}".format,
        "winner_share_over_resolved_range_mean": "{:.2%}".format,
        "long_share_over_resolved_range_mean": "{:.2%}".format,
        "short_share_over_resolved_range_mean": "{:.2%}".format,
    }
)
)

print("\nResumen por familia de contrato:")
print(
    df_opc_by_contract_code.to_string(
        index=False,
        formatters={
            "no_trade_ratio_mean": "{:.2%}".format,
            "resolved_trade_ratio_mean": "{:.2%}".format,
            "winner_share_over_resolved_mean": "{:.2%}".format,
            "majority_class_ratio_mean": "{:.2%}".format,
        }
    )
)

print("\nTargets más sensibles al contexto:")
print(
    df_opc_context_sensitivity_by_target_final
    .sort_values("context_class_sensitivity_score", ascending=False)
    [
        [
            "opc_col",
            "target_role",
            "context_type",
            "n_context_values",
            "context_class_sensitivity_score",
            "no_trade_ratio_range",
            "resolved_trade_ratio_range",
            "winner_share_over_resolved_range",
        ]
    ]
    .head(15)
    .to_string(
        index=False,
        formatters={
            "context_class_sensitivity_score": "{:.3f}".format,
            "no_trade_ratio_range": "{:.2%}".format,
            "resolved_trade_ratio_range": "{:.2%}".format,
            "winner_share_over_resolved_range": "{:.2%}".format,
        }
    )
)

print("\nTargets menos sensibles al contexto:")
print(
    df_opc_context_sensitivity_by_target_final
    .sort_values("context_class_sensitivity_score", ascending=True)
    [
        [
            "opc_col",
            "target_role",
            "context_type",
            "n_context_values",

```

```

        "context_class_sensitivity_score",
        "no_trade_ratio_range",
        "resolved_trade_ratio_range",
        "winner_share_over_resolved_range",
    ]
]
.head(15)
.to_string(
    index=False,
    formatters={
        "context_class_sensitivity_score": "{:.3f}".format,
        "no_trade_ratio_range": "{:.2%}".format,
        "resolved_trade_ratio_range": "{:.2%}".format,
        "winner_share_over_resolved_range": "{:.2%}".format,
    }
)
)

print("\nTabla final de sensibilidad por contexto:")
print(
    df_opc_context_final_table[
        [
            "opc_col",
            "target_role",
            "context_type",
            "n_context_values",
            "context_class_sensitivity_score",
            "context_sensitivity_profile",
        ]
    ].to_string(
        index=False,
        formatters={
            "context_class_sensitivity_score": "{:.3f}".format,
        }
    )
)

# -----
# 14) Guardado de resultados
# -----

if "MNQ_TARGETS_PATH" in globals():

    MNQ_TARGETS_OPC_CONTEXT_ANALYSIS_CSV = (
        MNQ_TARGETS_PATH / "mnq_targets_opc_context_analysis.csv"
    )

    MNQ_TARGETS_OPC_BY_REGIME_CSV = (
        MNQ_TARGETS_PATH / "mnq_targets_opc_by_regime.csv"
    )

    MNQ_TARGETS_OPC_REGIME_MAIN_CANDIDATE_CSV = (
        MNQ_TARGETS_PATH / "mnq_targets_opc_regime_main_candidate.csv"
    )

    MNQ_TARGETS_OPC_BY_CONTEXT_TYPE_CSV = (
        MNQ_TARGETS_PATH / "mnq_targets_opc_by_context_type.csv"
    )

    MNQ_TARGETS_OPC_CONTEXT_SENSITIVITY_BY_TARGET_CSV = (

```

```

MNQ_TARGETS_PATH / "mnq_targets_opc_context_sensitivity_by_target.csv"
)

MNQ_TARGETS_OPC_CONTEXT_SENSITIVITY_BY_CONTEXT_TYPE_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_opc_context_sensitivity_by_context_type.csv"
)

MNQ_TARGETS_OPC_BY_CONTRACT_CODE_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_opc_by_contract_code.csv"
)

MNQ_TARGETS_OPC_CONTEXT_FINAL_TABLE_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_opc_context_final_table.csv"
)

df_opc_context_analysis.to_csv(
    MNQ_TARGETS_OPC_CONTEXT_ANALYSIS_CSV,
    index=False
)

df_opc_by_regime.to_csv(
    MNQ_TARGETS_OPC_BY_REGIME_CSV,
    index=False
)

df_opc_regime_main_candidate.to_csv(
    MNQ_TARGETS_OPC_REGIME_MAIN_CANDIDATE_CSV,
    index=False
)

df_opc_by_context_type.to_csv(
    MNQ_TARGETS_OPC_BY_CONTEXT_TYPE_CSV,
    index=False
)

df_opc_context_sensitivity_by_target.to_csv(
    MNQ_TARGETS_OPC_CONTEXT_SENSITIVITY_BY_TARGET_CSV,
    index=False
)

df_opc_context_sensitivity_by_context_type.to_csv(
    MNQ_TARGETS_OPC_CONTEXT_SENSITIVITY_BY_CONTEXT_TYPE_CSV,
    index=False
)

df_opc_by_contract_code.to_csv(
    MNQ_TARGETS_OPC_BY_CONTRACT_CODE_CSV,
    index=False
)

df_opc_context_final_table.to_csv(
    MNQ_TARGETS_OPC_CONTEXT_FINAL_TABLE_CSV,
    index=False
)

print("\nArchivos guardados:")
print(f"Análisis contexto OPC: {MNQ_TARGETS_OPC_CONTEXT_ANALYSIS_CSV}")
print(f"Resumen por régimen: {MNQ_TARGETS_OPC_BY_REGIME_CSV}")
print(f"Régimen target principal: {MNQ_TARGETS_OPC_REGIME_MAIN_CANDIDATE_CSV}")
print(f"Resumen por tipo contexto: {MNQ_TARGETS_OPC_BY_CONTEXT_TYPE_CSV}")

```



```
print(f"Sensibilidad por target:      {MNQ_TARGETS_OPC_CONTEXT_SENSITIVITY}")
print(f"Sensibilidad por contexto:    {MNQ_TARGETS_OPC_CONTEXT_SENSITIVITY}")
print(f"Resumen por contract_code:    {MNQ_TARGETS_OPC_BY_CONTRACT_CODE_CS}")
print(f"Tabla final contexto OPC:      {MNQ_TARGETS_OPC_CONTEXT_FINAL_TABLE}")

else:
    print("\nMNQ_TARGETS_PATH no está definido. No se guardaron archivos.")

print("\nAnálisis por régimen y contexto de OPC finalizado correctamente.")
```

## Análisis por régimen y contexto de targets OPC

=====

Objetos base encontrados correctamente.

Columnas de contexto preparadas:

regime\_name  
 contract  
 contract\_code  
 dataset\_split  
 year\_contract  
 regime\_dataset\_split  
 contract\_regime  
 year\_regime  
 contract\_code\_regime

Contextos definidos:

|     | context_type         | context_col          | descripti                               |
|-----|----------------------|----------------------|-----------------------------------------|
| on  | regime               | regime_name          | Régimen intradiar                       |
| io  | contract             | contract             | Contrato individu                       |
| al  | contract_code        | contract_code        | Familia de contrato H/M/                |
| U/Z | dataset_split        | dataset_split        | Development vs final te                 |
| st  | year_contract        | year_contract        | Año + contra                            |
| to  | regime_dataset_split | regime_dataset_split | Régimen + dataset spl                   |
| it  | contract_regime      | contract_regime      | Contrato + régimen intradiar            |
| io  | year_regime          | year_regime          | Año + régimen intradiar                 |
| io  | contract_code_regime | contract_code_regime | Familia de contrato + régimen intradiar |
| io  |                      |                      |                                         |

Filas generadas en df\_opc\_context\_analysis:

1320

Valores por tipo de contexto:

| context_type         |     |
|----------------------|-----|
| contract_regime      | 130 |
| year_regime          | 35  |
| year_contract        | 32  |
| contract             | 26  |
| contract_code_regime | 20  |
| regime_dataset_split | 10  |
| regime               | 5   |
| contract_code        | 4   |
| dataset_split        | 2   |

Resumen OPC por régimen intradiario:

| regime_name | n_targets | total_rows_mean | valid_rows_mean | nan_ratio_mean | no_trade_ratio_mean | resolved_trade_ratio_mean | long_tp_ratio_mean | long_sl_ratio_mean | short_tp_ratio_mean | short_sl_ratio_mean | winner_ratio_mean | loser_ratio_mean | winner_share_over_resolved_mean | long_share_over_resolved_mean | short_share_over_resolved_mean | majority_class_ratio_mean |
|-------------|-----------|-----------------|-----------------|----------------|---------------------|---------------------------|--------------------|--------------------|---------------------|---------------------|-------------------|------------------|---------------------------------|-------------------------------|--------------------------------|---------------------------|
| Overnight   | 5         | 357162.0        | 355680.0        | 0.41%          | 67.62%              | 32.38%                    | 13.61%             | 1.74%              |                     |                     |                   |                  |                                 |                               |                                |                           |

|            |       |          |          |
|------------|-------|----------|----------|
| 15.30%     | 1.73% | 28.91%   | 3.47%    |
| 89.47%     |       | 47.35%   | 52.65%   |
| 67.62%     |       |          |          |
| Pre-market | 5     | 88920.0  | 88920.0  |
| 67.89%     |       | 32.11%   | 13.71%   |
| 15.27%     | 1.53% | 28.98%   | 3.13%    |
| 90.53%     |       | 47.50%   | 52.50%   |
| 67.89%     |       |          |          |
| Opening    | 5     | 88920.0  | 88920.0  |
| 69.28%     |       | 30.72%   | 12.29%   |
| 15.97%     | 1.19% | 28.27%   | 2.46%    |
| 92.21%     |       | 43.89%   | 56.11%   |
| 69.28%     |       |          |          |
| Regular    | 5     | 444600.0 | 401325.6 |
| 68.70%     |       | 31.30%   | 12.72%   |
| 15.55%     | 1.51% | 28.26%   | 3.04%    |
| 90.50%     |       | 45.35%   | 54.65%   |
| 68.70%     |       |          |          |
| Closing    | 5     | 44460.0  | 296.4    |
| 68.96%     |       | 31.04%   | 13.63%   |
| 14.91%     | 1.21% | 28.54%   | 2.50%    |
| 91.96%     |       | 48.04%   | 51.96%   |
| 68.96%     |       |          |          |

Detalle por régimen para target principal: opc\_p50\_h30\_tp15\_sl10

| context_value             | valid_rows    | nan_rows       | nan_ratio      | no_trade_ratio             | resolved_trade_ratio     |
|---------------------------|---------------|----------------|----------------|----------------------------|--------------------------|
| long_tp_ratio             | long_sl_ratio | short_tp_ratio | short_sl_ratio | winner_share_over_resolved | long_share_over_resolved |
| short_share_over_resolved |               |                |                |                            |                          |
| Overnight                 | 355680        | 1482           | 0.41%          | 68.04%                     | 31.96%                   |
| 13.69%                    | 1.54%         | 15.12%         | 1.61%          |                            | 90.15%                   |
| 47.64%                    |               | 52.36%         |                |                            |                          |
| Pre-market                | 88920         | 0              | 0.00%          | 64.38%                     | 35.62%                   |
| 15.24%                    | 2.08%         | 16.12%         | 2.19%          |                            | 88.03%                   |
| 48.60%                    |               | 51.40%         |                |                            |                          |
| Opening                   | 88920         | 0              | 0.00%          | 69.33%                     | 30.67%                   |
| 12.67%                    | 1.30%         | 15.63%         | 1.07%          |                            | 92.25%                   |
| 45.54%                    |               | 54.46%         |                |                            |                          |
| Regular                   | 444600        | 0              | 0.00%          | 68.24%                     | 31.76%                   |
| 13.26%                    | 1.48%         | 15.56%         | 1.46%          |                            | 90.72%                   |
| 46.41%                    |               | 53.59%         |                |                            |                          |
| Closing                   | 1482          | 42978          | 96.67%         | 68.96%                     | 31.04%                   |
| 13.63%                    | 1.28%         | 14.91%         | 1.21%          |                            | 91.96%                   |
| 48.04%                    |               | 51.96%         |                |                            |                          |

Resumen agregado por tipo de contexto:

| context_type         | n_rows                    | n_context_values                | n_targets                 | valid_rows_mean     | nan_ratio_mean      |
|----------------------|---------------------------|---------------------------------|---------------------------|---------------------|---------------------|
| no_trade_ratio_mean  | resolved_trade_ratio_mean | long_tp_ratio_mean              | long_sl_ratio_mean        | short_tp_ratio_mean | short_sl_ratio_mean |
| winner_ratio_mean    | loser_ratio_mean          | winner_share_over_resolved_mean | majority_class_ratio_mean |                     |                     |
| regime_dataset_split | 50                        | 10                              | 5                         | 93514.200000        |                     |
| 21.90%               | 61.84%                    | 38.16%                          | 15.70%                    |                     |                     |
| 2.22%                | 18.17%                    | 2.06%                           | 33.87%                    | 4.29%               |                     |
| 89.74%               | 61.84%                    |                                 |                           |                     |                     |
| dataset_split        | 10                        | 2                               | 5                         | 467571.000000       |                     |
| 8.68%                | 61.97%                    | 38.03%                          | 15.42%                    |                     |                     |
| 2.32%                | 18.09%                    | 2.19%                           | 33.51%                    | 4.52%               |                     |
| 89.13%               | 61.97%                    |                                 |                           |                     |                     |
| year_regime          | 175                       | 35                              | 5                         | 26718.342857        |                     |
| 21.90%               | 65.19%                    | 34.81%                          | 14.55%                    |                     |                     |
| 1.86%                | 16.64%                    | 1.76%                           | 31.19%                    | 3.62%               |                     |

|        |                      |        |        |        |               |
|--------|----------------------|--------|--------|--------|---------------|
| 91.15% |                      | 65.44% |        |        |               |
|        | contract             | 130    | 26     | 5      | 35967.000000  |
| 8.68%  |                      | 67.32% | 32.68% |        | 13.52%        |
| 1.67%  |                      | 15.83% | 1.66%  | 29.35% | 3.33%         |
| 91.50% |                      | 67.37% |        |        |               |
|        | contract_regime      | 650    | 130    | 5      | 7193.400000   |
| 21.90% |                      | 67.54% | 32.46% |        | 13.56%        |
| 1.59%  |                      | 15.76% | 1.56%  | 29.32% | 3.14%         |
| 92.11% |                      | 67.68% |        |        |               |
|        | regime               | 25     | 5      | 5      | 187028.400000 |
| 21.90% |                      | 68.40% | 31.60% |        | 13.11%        |
| 1.52%  |                      | 15.49% | 1.48%  | 28.60% | 3.00%         |
| 90.74% |                      | 68.40% |        |        |               |
|        | contract_code        | 20     | 4      | 5      | 233785.500000 |
| 8.68%  |                      | 68.48% | 31.52% |        | 13.04%        |
| 1.56%  |                      | 15.38% | 1.54%  | 28.42% | 3.11%         |
| 90.64% |                      | 68.48% |        |        |               |
|        | contract_code_regime | 100    | 20     | 5      | 46757.100000  |
| 21.90% |                      | 68.61% | 31.39% |        | 13.04%        |
| 1.49%  |                      | 15.41% | 1.45%  | 28.45% | 2.95%         |
| 91.12% |                      | 68.61% |        |        |               |
|        | year_contract        | 160    | 32     | 5      | 29223.187500  |
| 8.68%  |                      | 69.27% | 30.73% |        | 12.61%        |
| 1.49%  |                      | 15.14% | 1.49%  | 27.75% | 2.98%         |
| 92.31% |                      | 69.36% |        |        |               |

Sensibilidad promedio por tipo de contexto:

|        | context_type         | n_targets | n_context_values_mean | context_class_sensitivity_score_mean | context_class_sensitivity_score_max | no_trade_ratio_range_mean | resolved_trade_ratio_range_mean | winner_share_over_resolved_range_mean | long_share_over_resolved_range_mean | short_share_over_resolved_range_mean |
|--------|----------------------|-----------|-----------------------|--------------------------------------|-------------------------------------|---------------------------|---------------------------------|---------------------------------------|-------------------------------------|--------------------------------------|
|        | contract_regime      | 5         |                       | 104.0                                |                                     |                           |                                 |                                       |                                     |                                      |
| 1.352  |                      |           | 1.484                 |                                      | 62.29%                              |                           |                                 |                                       |                                     |                                      |
| 62.29% |                      |           | 21.14%                |                                      |                                     |                           |                                 |                                       | 39.47%                              |                                      |
| 39.47% |                      |           |                       |                                      |                                     |                           |                                 |                                       |                                     |                                      |
|        | year_contract        | 5         |                       | 32.0                                 |                                     |                           |                                 |                                       |                                     |                                      |
| 1.183  |                      |           | 1.364                 |                                      | 58.13%                              |                           |                                 |                                       |                                     |                                      |
| 58.13% |                      |           | 14.94%                |                                      |                                     |                           |                                 |                                       | 20.85%                              |                                      |
| 20.85% |                      |           |                       |                                      |                                     |                           |                                 |                                       |                                     |                                      |
|        | year_regime          | 5         |                       | 28.0                                 |                                     |                           |                                 |                                       |                                     |                                      |
| 1.113  |                      |           | 1.191                 |                                      | 53.05%                              |                           |                                 |                                       |                                     |                                      |
| 53.05% |                      |           | 14.58%                |                                      |                                     |                           |                                 |                                       | 20.33%                              |                                      |
| 20.33% |                      |           |                       |                                      |                                     |                           |                                 |                                       |                                     |                                      |
|        | contract             | 5         |                       | 26.0                                 |                                     |                           |                                 |                                       |                                     |                                      |
| 1.107  |                      |           | 1.222                 |                                      | 54.89%                              |                           |                                 |                                       |                                     |                                      |
| 54.89% |                      |           | 12.38%                |                                      |                                     |                           |                                 |                                       | 13.71%                              |                                      |
| 13.71% |                      |           |                       |                                      |                                     |                           |                                 |                                       |                                     |                                      |
|        | regime_dataset_split | 5         |                       | 8.2                                  |                                     |                           |                                 |                                       |                                     |                                      |
| 0.551  |                      |           | 0.598                 |                                      | 26.41%                              |                           |                                 |                                       |                                     |                                      |
| 26.41% |                      |           | 9.18%                 |                                      |                                     |                           |                                 |                                       | 6.45%                               |                                      |
| 6.45%  |                      |           |                       |                                      |                                     |                           |                                 |                                       |                                     |                                      |
|        | dataset_split        | 5         |                       | 2.0                                  |                                     |                           |                                 |                                       |                                     |                                      |
| 0.420  |                      |           | 0.433                 |                                      | 21.00%                              |                           |                                 |                                       |                                     |                                      |
| 21.00% |                      |           | 5.83%                 |                                      |                                     |                           |                                 |                                       | 1.07%                               |                                      |
| 1.07%  |                      |           |                       |                                      |                                     |                           |                                 |                                       |                                     |                                      |
|        | contract_code_regime | 5         |                       | 16.0                                 |                                     |                           |                                 |                                       |                                     |                                      |
| 0.331  |                      |           | 0.360                 |                                      | 15.93%                              |                           |                                 |                                       |                                     |                                      |
| 15.93% |                      |           | 8.26%                 |                                      |                                     |                           |                                 |                                       | 9.42%                               |                                      |
| 9.42%  |                      |           |                       |                                      |                                     |                           |                                 |                                       |                                     |                                      |
|        | contract_code        | 5         |                       | 4.0                                  |                                     |                           |                                 |                                       |                                     |                                      |

|        |        |       |        |       |
|--------|--------|-------|--------|-------|
| 0.258  |        | 0.266 | 12.88% |       |
| 12.88% |        | 5.44% |        | 3.61% |
| 3.61%  |        |       |        |       |
|        | regime | 5     | 4.2    |       |
| 0.065  |        | 0.106 | 2.45%  |       |
| 2.45%  |        | 3.16% |        | 3.96% |
| 3.96%  |        |       |        |       |

Resumen por familia de contrato:

| contract_code | n_targets | no_trade_ratio_mean | resolved_trade_ratio_mean | winner_share_over_resolved_mean | majority_class_ratio_mean |
|---------------|-----------|---------------------|---------------------------|---------------------------------|---------------------------|
| H             | 5         | 62.31%              |                           | 37.69%                          |                           |
| 87.79%        |           | 62.31%              |                           |                                 |                           |
| M             | 5         | 67.20%              |                           | 32.80%                          |                           |
| 89.80%        |           | 67.20%              |                           |                                 |                           |
| U             | 5         | 75.18%              |                           | 24.82%                          |                           |
| 93.22%        |           | 75.18%              |                           |                                 |                           |
| Z             | 5         | 69.22%              |                           | 30.78%                          |                           |
| 91.77%        |           | 69.22%              |                           |                                 |                           |

Targets más sensibles al contexto:

| opc_col               | target_role           | context_type    | n_context_values | text_class_sensitivity_score | no_trade_ratio_range | resolved_trade_ratio_range | winner_share_over_resolved_range |
|-----------------------|-----------------------|-----------------|------------------|------------------------------|----------------------|----------------------------|----------------------------------|
| opc_p40_h60_tp15_sl10 | comparativo_activo    | contract_regime | 104              | 1.484                        | 66.36%               | 66.36%                     | 2                                |
| 0.43%                 |                       |                 |                  |                              |                      |                            |                                  |
| opc_p50_h90_tp15_sl10 | principal             | contract_regime | 104              | 1.434                        | 63.53%               | 63.53%                     | 2                                |
| 9.71%                 |                       |                 |                  |                              |                      |                            |                                  |
| opc_p40_h60_tp15_sl10 | comparativo_activo    | year_contract   | 32               | 1.364                        | 66.76%               | 66.76%                     | 1                                |
| 5.56%                 |                       |                 |                  |                              |                      |                            |                                  |
| opc_p50_h60_tp15_sl10 | principal             | contract_regime | 104              | 1.361                        | 63.28%               | 63.28%                     | 1                                |
| 8.56%                 |                       |                 |                  |                              |                      |                            |                                  |
| opc_p50_h30_tp15_sl10 | principal             | contract_regime | 104              | 1.277                        | 61.08%               | 61.08%                     | 1                                |
| 9.95%                 |                       |                 |                  |                              |                      |                            |                                  |
| opc_p40_h60_tp15_sl10 | comparativo_activo    | contract        | 26               | 1.222                        | 60.61%               | 60.61%                     | 1                                |
| 4.11%                 |                       |                 |                  |                              |                      |                            |                                  |
| opc_p50_h90_tp15_sl10 | principal             | year_contract   | 32               | 1.211                        | 59.31%               | 59.31%                     | 1                                |
| 6.84%                 |                       |                 |                  |                              |                      |                            |                                  |
| opc_p60_h60_tp15_sl10 | comparativo_selectivo | contract_regime | 104              | 1.204                        | 57.22%               | 57.22%                     | 1                                |
| 7.05%                 |                       |                 |                  |                              |                      |                            |                                  |
| opc_p50_h60_tp15_sl10 | principal             | year_contract   | 32               | 1.192                        | 58.80%               | 58.80%                     | 1                                |
| 4.92%                 |                       |                 |                  |                              |                      |                            |                                  |
| opc_p50_h90_tp15_sl10 | principal             | year_regime     | 28               | 1.191                        | 56.89%               | 56.89%                     | 1                                |
| 6.21%                 |                       |                 |                  |                              |                      |                            |                                  |
| opc_p40_h60_tp15_sl10 | comparativo_activo    | year_regime     | 28               | 1.178                        | 55.60%               | 55.60%                     | 1                                |
| 5.51%                 |                       |                 |                  |                              |                      |                            |                                  |
| opc_p50_h30_tp15_sl10 | principal             | year_contract   | 32               | 1.168                        | 57.41%               | 57.41%                     | 1                                |
| 4.67%                 |                       |                 |                  |                              |                      |                            |                                  |

|                       |           |             |    |   |
|-----------------------|-----------|-------------|----|---|
| opc_p50_h90_tp15_sl10 | principal | contract    | 26 |   |
| 1.166                 | 57.54%    | 57.54%      |    | 1 |
| 3.34%                 |           |             |    |   |
| opc_p50_h60_tp15_sl10 | principal | year_regime | 28 |   |
| 1.134                 | 54.06%    | 54.06%      |    | 1 |
| 3.99%                 |           |             |    |   |
| opc_p50_h60_tp15_sl10 | principal | contract    | 26 |   |
| 1.133                 | 56.25%    | 56.25%      |    | 1 |
| 2.18%                 |           |             |    |   |

Targets menos sensibles al contexto:

| opc_col                           | target_role           | context_type               | n_context_value |        |
|-----------------------------------|-----------------------|----------------------------|-----------------|--------|
| s context_class_sensitivity_score | no_trade_ratio_range  | resolved_trade_ratio_range |                 |        |
| winner_share_over_resolved_range  |                       |                            |                 |        |
| opc_p60_h60_tp15_sl10             | comparativo_selectivo | regime                     |                 |        |
| 4                                 | 0.046                 | 1.74%                      |                 | 1.74%  |
| 2.56%                             |                       |                            |                 |        |
| opc_p50_h60_tp15_sl10             | principal             | regime                     |                 |        |
| 4                                 | 0.047                 | 1.44%                      |                 | 1.44%  |
| 2.51%                             |                       |                            |                 |        |
| opc_p40_h60_tp15_sl10             | comparativo_activo    | regime                     |                 |        |
| 4                                 | 0.051                 | 1.28%                      |                 | 1.28%  |
| 2.45%                             |                       |                            |                 |        |
| opc_p50_h90_tp15_sl10             | principal             | regime                     |                 |        |
| 4                                 | 0.074                 | 2.82%                      |                 | 2.82%  |
| 4.07%                             |                       |                            |                 |        |
| opc_p50_h30_tp15_sl10             | principal             | regime                     |                 |        |
| 5                                 | 0.106                 | 4.95%                      |                 | 4.95%  |
| 4.22%                             |                       |                            |                 |        |
| opc_p60_h60_tp15_sl10             | comparativo_selectivo | contract_code              |                 |        |
| 4                                 | 0.248                 | 12.40%                     |                 | 12.40% |
| 4.65%                             |                       |                            |                 |        |
| opc_p40_h60_tp15_sl10             | comparativo_activo    | contract_code              |                 |        |
| 4                                 | 0.255                 | 12.75%                     |                 | 12.75% |
| 5.95%                             |                       |                            |                 |        |
| opc_p50_h30_tp15_sl10             | principal             | contract_code              |                 |        |
| 4                                 | 0.256                 | 12.78%                     |                 | 12.78% |
| 5.09%                             |                       |                            |                 |        |
| opc_p50_h60_tp15_sl10             | principal             | contract_code              |                 |        |
| 4                                 | 0.263                 | 13.15%                     |                 | 13.15% |
| 5.56%                             |                       |                            |                 |        |
| opc_p50_h90_tp15_sl10             | principal             | contract_code              |                 |        |
| 4                                 | 0.266                 | 13.31%                     |                 | 13.31% |
| 5.93%                             |                       |                            |                 |        |
| opc_p60_h60_tp15_sl10             | comparativo_selectivo | contract_code_regime       |                 |        |
| 6                                 | 0.299                 | 14.67%                     |                 | 1      |
| 6.50%                             |                       |                            |                 | 14.67% |
| opc_p40_h60_tp15_sl10             | comparativo_activo    | contract_code_regime       |                 |        |
| 6                                 | 0.325                 | 15.88%                     |                 | 1      |
| 7.59%                             |                       |                            |                 | 15.88% |
| opc_p50_h90_tp15_sl10             | principal             | contract_code_regime       |                 |        |
| 6                                 | 0.334                 | 15.96%                     |                 | 1      |
| 9.86%                             |                       |                            |                 | 15.96% |
| opc_p50_h60_tp15_sl10             | principal             | contract_code_regime       |                 |        |
| 6                                 | 0.335                 | 16.27%                     |                 | 1      |
| 7.45%                             |                       |                            |                 | 16.27% |
| opc_p50_h30_tp15_sl10             | principal             | contract_code_regime       |                 |        |
| 6                                 | 0.360                 | 16.84%                     |                 | 1      |
| 9.90%                             |                       |                            |                 | 16.84% |

Tabla final de sensibilidad por contexto:

| opc_col                           | target_role                 | context_type            | n_context_value |
|-----------------------------------|-----------------------------|-------------------------|-----------------|
| s context_class_sensitivity_score | context_sensitivity_profile |                         |                 |
| opc_p60_h60_tp15_sl10             | comparativo_selectivo       | contract                | 2               |
| 6                                 | 0.972                       | sensibilidad_intermedia |                 |
| opc_p50_h30_tp15_sl10             | principal                   | contract                | 2               |
| 6                                 | 1.044                       | sensibilidad_intermedia |                 |
| opc_p50_h60_tp15_sl10             | principal                   | contract                | 2               |
| 6                                 | 1.133                       | mayor_sensibilidad      |                 |
| opc_p50_h90_tp15_sl10             | principal                   | contract                | 2               |
| 6                                 | 1.166                       | mayor_sensibilidad      |                 |
| opc_p40_h60_tp15_sl10             | comparativo_activo          | contract                | 2               |
| 6                                 | 1.222                       | mayor_sensibilidad      |                 |
| opc_p60_h60_tp15_sl10             | comparativo_selectivo       | contract_code           |                 |
| 4                                 | 0.248                       | menor_sensibilidad      |                 |
| opc_p40_h60_tp15_sl10             | comparativo_activo          | contract_code           |                 |
| 4                                 | 0.255                       | menor_sensibilidad      |                 |
| opc_p50_h30_tp15_sl10             | principal                   | contract_code           |                 |
| 4                                 | 0.256                       | menor_sensibilidad      |                 |
| opc_p50_h60_tp15_sl10             | principal                   | contract_code           |                 |
| 4                                 | 0.263                       | menor_sensibilidad      |                 |
| opc_p50_h90_tp15_sl10             | principal                   | contract_code           |                 |
| 4                                 | 0.266                       | menor_sensibilidad      |                 |
| opc_p60_h60_tp15_sl10             | comparativo_selectivo       | contract_code_regime    | 1               |
| 6                                 | 0.299                       | menor_sensibilidad      |                 |
| opc_p40_h60_tp15_sl10             | comparativo_activo          | contract_code_regime    | 1               |
| 6                                 | 0.325                       | menor_sensibilidad      |                 |
| opc_p50_h90_tp15_sl10             | principal                   | contract_code_regime    | 1               |
| 6                                 | 0.334                       | menor_sensibilidad      |                 |
| opc_p50_h60_tp15_sl10             | principal                   | contract_code_regime    | 1               |
| 6                                 | 0.335                       | menor_sensibilidad      |                 |
| opc_p50_h30_tp15_sl10             | principal                   | contract_code_regime    | 1               |
| 6                                 | 0.360                       | menor_sensibilidad      |                 |
| opc_p60_h60_tp15_sl10             | comparativo_selectivo       | contract_regime         | 10              |
| 4                                 | 1.204                       | mayor_sensibilidad      |                 |
| opc_p50_h30_tp15_sl10             | principal                   | contract_regime         | 10              |
| 4                                 | 1.277                       | mayor_sensibilidad      |                 |
| opc_p50_h60_tp15_sl10             | principal                   | contract_regime         | 10              |
| 4                                 | 1.361                       | mayor_sensibilidad      |                 |
| opc_p50_h90_tp15_sl10             | principal                   | contract_regime         | 10              |
| 4                                 | 1.434                       | mayor_sensibilidad      |                 |
| opc_p40_h60_tp15_sl10             | comparativo_activo          | contract_regime         | 10              |
| 4                                 | 1.484                       | mayor_sensibilidad      |                 |
| opc_p60_h60_tp15_sl10             | comparativo_selectivo       | dataset_split           |                 |
| 2                                 | 0.394                       | sensibilidad_intermedia |                 |
| opc_p50_h30_tp15_sl10             | principal                   | dataset_split           |                 |
| 2                                 | 0.413                       | sensibilidad_intermedia |                 |
| opc_p50_h60_tp15_sl10             | principal                   | dataset_split           |                 |
| 2                                 | 0.427                       | sensibilidad_intermedia |                 |
| opc_p40_h60_tp15_sl10             | comparativo_activo          | dataset_split           |                 |
| 2                                 | 0.433                       | sensibilidad_intermedia |                 |
| opc_p50_h90_tp15_sl10             | principal                   | dataset_split           |                 |
| 2                                 | 0.433                       | sensibilidad_intermedia |                 |
| opc_p60_h60_tp15_sl10             | comparativo_selectivo       | regime                  |                 |
| 4                                 | 0.046                       | menor_sensibilidad      |                 |
| opc_p50_h60_tp15_sl10             | principal                   | regime                  |                 |
| 4                                 | 0.047                       | menor_sensibilidad      |                 |
| opc_p40_h60_tp15_sl10             | comparativo_activo          | regime                  |                 |
| 4                                 | 0.051                       | menor_sensibilidad      |                 |
| opc_p50_h90_tp15_sl10             | principal                   | regime                  |                 |

|                       |                       |                         |   |
|-----------------------|-----------------------|-------------------------|---|
| 4                     | 0.074                 | menor_sensibilidad      |   |
| opc_p50_h30_tp15_sl10 | principal             | regime                  |   |
| 5                     | 0.106                 | menor_sensibilidad      |   |
| opc_p60_h60_tp15_sl10 | comparativo_selectivo | regime_dataset_split    |   |
| 8                     | 0.516                 | sensibilidad_intermedia |   |
| opc_p50_h30_tp15_sl10 | principal             | regime_dataset_split    |   |
| 9                     | 0.517                 | sensibilidad_intermedia |   |
| opc_p50_h60_tp15_sl10 | principal             | regime_dataset_split    |   |
| 8                     | 0.561                 | sensibilidad_intermedia |   |
| opc_p40_h60_tp15_sl10 | comparativo_activo    | regime_dataset_split    |   |
| 8                     | 0.565                 | sensibilidad_intermedia |   |
| opc_p50_h90_tp15_sl10 | principal             | regime_dataset_split    |   |
| 8                     | 0.598                 | sensibilidad_intermedia |   |
| opc_p60_h60_tp15_sl10 | comparativo_selectivo | year_contract           | 3 |
| 2                     | 0.981                 | sensibilidad_intermedia |   |
| opc_p50_h30_tp15_sl10 | principal             | year_contract           | 3 |
| 2                     | 1.168                 | mayor_sensibilidad      |   |
| opc_p50_h60_tp15_sl10 | principal             | year_contract           | 3 |
| 2                     | 1.192                 | mayor_sensibilidad      |   |
| opc_p50_h90_tp15_sl10 | principal             | year_contract           | 3 |
| 2                     | 1.211                 | mayor_sensibilidad      |   |
| opc_p40_h60_tp15_sl10 | comparativo_activo    | year_contract           | 3 |
| 2                     | 1.364                 | mayor_sensibilidad      |   |
| opc_p60_h60_tp15_sl10 | comparativo_selectivo | year_regime             | 2 |
| 8                     | 1.022                 | sensibilidad_intermedia |   |
| opc_p50_h30_tp15_sl10 | principal             | year_regime             | 2 |
| 8                     | 1.041                 | sensibilidad_intermedia |   |
| opc_p50_h60_tp15_sl10 | principal             | year_regime             | 2 |
| 8                     | 1.134                 | mayor_sensibilidad      |   |
| opc_p40_h60_tp15_sl10 | comparativo_activo    | year_regime             | 2 |
| 8                     | 1.178                 | mayor_sensibilidad      |   |
| opc_p50_h90_tp15_sl10 | principal             | year_regime             | 2 |
| 8                     | 1.191                 | mayor_sensibilidad      |   |

Archivos guardados:

Análisis contexto OPC: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_opc\_context\_analysis.csv  
 Resumen por régimen: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_opc\_by\_regime.csv  
 Régimen target principal: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_opc\_regime\_main\_candidate.csv  
 Resumen por tipo contexto: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_opc\_by\_context\_type.csv  
 Sensibilidad por target: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_opc\_context\_sensitivity\_by\_target.csv  
 Sensibilidad por contexto: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_opc\_context\_sensitivity\_by\_context\_type.csv  
 Resumen por contract\_code: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_opc\_by\_contract\_code.csv  
 Tabla final contexto OPC: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_opc\_context\_final\_table.csv

Análisis por régimen y contexto de OPC finalizado correctamente.

## Observaciones principales:

- El régimen intradiario, por sí solo, **no cambia demasiado** la distribución de OPC ; la sensibilidad por regime es la más baja.



- **Opening** muestra mejor calidad relativa: mayor **winner\_share\_over\_resolved** con **92.21%**.
- **Pre-market** es el régimen más operativo para **opc\_p50\_h30**, con **35.62%** de trades resueltos.
- **Closing** debe tratarse con cautela: tiene **99.33% de NaN promedio** por falta de ventana futura válida.
- La mayor sensibilidad aparece en contextos combinados, especialmente **contract\_regime**, **year\_contract** y **year\_regime**.
- La familia de contrato **H/M/U/Z** sola no explica tanto; **contract\_code** tiene baja sensibilidad.
- La familia **H** es la más activa: **37.69%** de trades resueltos.
- La familia **U** es la más selectiva: **75.18% NO\_TRADE**, pero mejor **winner\_share** con **93.22%**.
- **opc\_p40\_h60** vuelve a aparecer como el más sensible/inestable al contexto.
- **opc\_p60\_h60** suele ser el más estable, pero también el más selectivo.
- Entre los targets principales, **opc\_p50\_h30\_tp15\_sl10** sigue siendo el candidato más razonable: buen equilibrio y menor sensibilidad que **p50\_h60** y **p50\_h90** en varios contextos.
- En resumen: el contexto importante no es solo el régimen, sino **régimen combinado con contrato o año**.

## 6.9. Observaciones e interpretación

El análisis del target **OPC** permitió integrar en una sola variable la dirección operativa del movimiento y el resultado asociado a esa dirección. A diferencia de **DIR**, que solo identifica si existe una dirección significativa, y de **BAR**, que evalúa el resultado operativo condicionado a una dirección, **OPC** resume ambos elementos en una única categoría operativa.

La lógica de **OPC** permite clasificar cada observación en cinco clases principales:

```
NO_TRADE
LONG_TP
LONG_SL
SHORT_TP
SHORT_SL
```

Esta estructura facilita una lectura operativa más directa, ya que permite distinguir entre operaciones **LONG** ganadoras, operaciones **LONG** perdedoras, operaciones **SHORT** ganadoras, operaciones **SHORT** perdedoras y observaciones sin operación válida o sin resolución operativa.

La construcción de **OPC** se realizó únicamente sobre el universo reducido de targets candidatos definido previamente. Esto evitó volver a expandir el análisis sobre todas las combinaciones originales de **DIR** y **BAR**, manteniendo el estudio enfocado en las familias operativas más relevantes.

Las familias evaluadas fueron:

```
dir_p50_h30 → opc_p50_h30_tp15_sl10  
dir_p50_h60 → opc_p50_h60_tp15_sl10  
dir_p50_h90 → opc_p50_h90_tp15_sl10  
dir_p40_h60 → opc_p40_h60_tp15_sl10  
dir_p60_h60 → opc_p60_h60_tp15_sl10
```

La validación de consistencia confirmó que los targets `OPC` fueron construidos correctamente. Todas las columnas necesarias existen, las etiquetas y clases numéricas son válidas, no se detectaron inconsistencias entre etiqueta y clase, y la relación lógica `DIR + BAR → OPC` quedó correctamente mapeada.

También se verificó que los casos `DIR = 0` se transforman correctamente en `NO_TRADE`, que los casos `BAR = 0` también se agrupan como `NO_TRADE`, y que no existen casos inesperados de `DIR` operativo con `BAR` faltante. Esto confirma que la construcción de `OPC` es metodológicamente consistente.

En la distribución global de clases, la clase dominante en todos los targets es `NO_TRADE`. Esto era esperable, ya que `NO_TRADE` agrupa tanto las observaciones sin dirección operativa clara como aquellas que, aun teniendo dirección, no alcanzan ni `TP` ni `SL` dentro del horizonte evaluado.

Los targets principales basados en `p50` muestran una estructura bastante consistente. En ellos, `NO_TRADE` se ubica alrededor de dos tercios de las observaciones válidas, mientras que los trades resueltos representan aproximadamente un tercio del total. Esta proporción resulta razonable para un target operativo intradiario, ya que evita tanto un exceso de señales como una selectividad extrema.

El target `opc_p40_h60_tp15_sl10` es el más activo. Presenta una mayor proporción de trades resueltos y una menor proporción de `NO_TRADE`. Sin embargo, esta mayor actividad también puede implicar más ruido y mayor sensibilidad al contexto. Por este motivo, se mantiene como variante comparativa activa, pero no como candidato principal.

El target `opc_p60_h60_tp15_sl10` es el más selectivo. Presenta la mayor proporción de `NO_TRADE` y la menor frecuencia de trades resueltos. A cambio, muestra una mayor proporción de operaciones ganadoras entre los trades resueltos. Esto sugiere que puede ser una variante más limpia, pero con menos muestras útiles y menor frecuencia operativa.

Dentro de los targets principales, `opc_p50_h30_tp15_sl10` aparece como el candidato más equilibrado. Tiene menor pérdida de datos por ventanas futuras incompletas, mantiene una frecuencia operativa razonable y conserva una alta proporción de trades ganadores entre los casos resueltos. Además, al trabajar con un horizonte de 30 minutos, reduce la exposición temporal frente a los horizontes de 60 y 90 minutos.

El análisis temporal mostró que `OPC` no es estacionario. En el período `final_test_2025_2026` disminuye de forma importante la proporción de `NO_TRADE` y aumenta la proporción de trades resueltos. Sin embargo, este aumento de actividad viene acompañado por un aumento en las clases perdedoras. Por lo tanto, el período

final no debe interpretarse simplemente como más favorable, sino como más activo y también más riesgoso.

Por año, se observa que 2022, 2025 y especialmente 2026 presentan mayor actividad operativa, mientras que 2021 y 2023 presentan una proporción más alta de `NO_TRADE`. Esto confirma que el target `OPC` está afectado por el contexto histórico del mercado y que su distribución cambia de forma relevante entre períodos.

El análisis de estabilidad temporal mostró que `opc_p60_h60_tp15_s110` es el target más estable, pero también el más selectivo. Entre los targets principales, `opc_p50_h30_tp15_s110` mostró el mejor equilibrio entre estabilidad relativa, frecuencia operativa y utilidad práctica. Por el contrario, `opc_p40_h60_tp15_s110` fue el más inestable, lo que refuerza su interpretación como variante activa pero más sensible al ruido.

El análisis por régimen intradiario mostró que el régimen, por sí solo, no explica grandes diferencias en la distribución de `OPC`. Las métricas entre `Overnight`, `Pre-market`, `Opening` y `Regular` son relativamente consistentes. Sin embargo, el régimen `Closing` debe interpretarse con cautela, ya que tiene muy pocas observaciones válidas debido a la falta de ventana futura suficiente, especialmente en horizontes más largos.

Desde una perspectiva operativa, `Pre-market` aparece como un régimen relativamente más activo para el target principal `opc_p50_h30_tp15_s110`, mientras que `Opening` muestra una buena proporción de trades ganadores entre eventos resueltos. Aun así, estas diferencias no son tan fuertes como para seleccionar un target únicamente en función del régimen.

El análisis por contexto mostró que la mayor sensibilidad aparece en combinaciones como `contract_regime`, `year_contract` y `year_regime`. Esto indica que el comportamiento de `OPC` no depende solamente del régimen intradiario o del contrato de forma aislada, sino de la interacción entre contexto temporal, contrato y régimen de mercado.

La familia de contrato `H/M/U/Z` por sí sola no parece explicar demasiado. Sin embargo, al combinar contrato con régimen o año, la sensibilidad aumenta de forma significativa. Esto sugiere que, en etapas posteriores de modelado, será importante incluir variables de contexto que representen el entorno temporal y operativo de cada observación.

En términos operativos, el target `opc_p50_h30_tp15_s110` se mantiene como el mejor candidato principal para continuar. Su ventaja no proviene de ser el más activo ni el más selectivo, sino de ofrecer el mejor equilibrio general entre frecuencia operativa, baja pérdida de datos, horizonte corto, interpretación clara y estabilidad relativa.

La lectura final de los targets `OPC` puede resumirse de la siguiente manera:

- `opc_p50_h30_tp15_s110`  
→ candidato principal más equilibrado.

- `opc_p50_h60_tp15_s110`  
→ alternativa intermedia, pero sin ventaja clara frente a H30.

opc\_p50\_h90\_tp15\_sl10

→ alternativa de horizonte más largo, con mayor exposición temporal.

opc\_p40\_h60\_tp15\_sl10

→ variante más activa, pero más sensible e inestable.

opc\_p60\_h60\_tp15\_sl10

→ variante más selectiva y estable, pero con menos oportunidades.

A pesar de estos resultados, no debe declararse todavía un target definitivo. El análisis realizado hasta este punto es descriptivo, estadístico y operativo. La selección final deberá depender de una etapa posterior de evaluación predictiva, utilizando únicamente información disponible en el instante `t`, sin usar `DIR`, `BAR` u otros targets como features, y respetando una separación temporal estricta entre entrenamiento, validación y prueba.

En síntesis, `OPC` aporta una representación operativa más completa que `DIR` y `BAR` por separado, ya que combina dirección y resultado en una sola clase nominal. Sin embargo, también introduce mayor complejidad, mayor desbalance de clases y mayor sensibilidad al contexto. Por este motivo, debe considerarse un target valioso para investigación posterior, pero no una solución definitiva por sí solo.

Bajo los análisis realizados, el target que queda mejor posicionado para avanzar hacia etapas posteriores es:

opc\_p50\_h30\_tp15\_sl10

Este target deberá ser evaluado posteriormente en términos de señal predictiva, robustez fuera de muestra y utilidad operativa real.

## 7. Conclusiones finales del stage de targets

En este stage se construyeron, validaron y analizaron tres familias de targets para el estudio intradiario del MNQ:

`DIR` → dirección significativa del movimiento futuro

`BAR` → resultado operativo por barreras TP/SL

`OPC` → combinación entre dirección y resultado operativo

El objetivo principal no fue entrenar modelos ni seleccionar todavía un target definitivo, sino definir un universo reducido de targets candidatos, construido de forma consistente y con interpretación operativa clara.

A lo largo del análisis se mantuvieron criterios metodológicos centrales:

No cruzar días.

Usar ventanas futuras bien definidas.

Calibrar thresholds con 2020–2024.

Reservar 2025–2026 como período final fuera de muestra.

No usar targets como features.

No encadenar modelos en esta etapa.

Evaluar primero la estructura estadística antes de entrenar modelos.

Esto permitió separar correctamente dos problemas distintos: la construcción de targets y la evaluación predictiva posterior.

## 7.1. Resumen del proceso realizado

El análisis comenzó con la familia `DIR`, cuyo objetivo fue identificar si dentro de una ventana futura existía una dirección significativa del movimiento. Este target permitió distinguir entre señales `LONG`, señales `SHORT` y casos sin dirección operativa clara.

Luego se construyó la familia `BAR`, que evaluó el resultado operativo de las señales generadas por `DIR`. Para cada dirección válida, `BAR` permitió identificar si se alcanzaba primero el `Take Profit`, el `Stop Loss` o si no había resolución dentro del horizonte evaluado.

Finalmente se construyó la familia `OPC`, que combinó la dirección operativa y el resultado de la operación en una sola clase nominal. De esta forma, cada observación pudo clasificarse como `NO_TRADE`, `LONG_TP`, `LONG_SL`, `SHORT_TP` o `SHORT_SL`.

La construcción de `OPC` confirmó que la clase dominante es `NO_TRADE`, mientras que los targets principales `p50` mantienen una proporción de trades resueltos cercana al 31%–32%. En particular, `opc_p50_h30_tp15_sl10` mostró 32.08% de trades resueltos, 67.92% de `NO_TRADE` y 90.38% de ganadores entre trades resueltos.

## 7.2. Targets DIR que pasan a la siguiente etapa

Los targets `DIR` principales que pasan a la siguiente etapa son:

```
dir_p50_h30
dir_p50_h60
dir_p50_h90
```

Estas variantes representan el núcleo principal del análisis direccional. Se basan en `p50`, que ofrece un equilibrio razonable entre frecuencia operativa y selectividad.

Además, se conservan dos variantes comparativas:

```
dir_p40_h60
dir_p60_h60
```

La variante `dir_p40_h60` representa un escenario más activo, mientras que `dir_p60_h60` representa un escenario más selectivo. Ambas permiten evaluar cómo cambia el comportamiento del target cuando se reduce o aumenta la exigencia del threshold.

Por lo tanto, los targets `DIR` que pasan a la siguiente etapa son:

```
Targets DIR principales:
dir_p50_h30
dir_p50_h60
```

```
dir_p50_h90
```

Targets DIR comparativos:

```
dir_p40_h60
```

```
dir_p60_h60
```

## 7.3. Targets BAR que pasan a la siguiente etapa

Para los targets **BAR**, la configuración seleccionada como principal fue:

```
TP 1.5 / SL 1.0
```

Esta configuración representa un punto intermedio entre exigencia operativa y frecuencia de resolución. No es tan simple como **TP 1.0 / SL 1.0**, que queda interpretada como benchmark metodológico, ni tan restrictiva como **TP 2.0 / SL 1.0** o **TP 2.5 / SL 1.0**.

Los targets **BAR** principales que pasan a la siguiente etapa son:

```
bar_p50_h30_tp15_sl10
```

```
bar_p50_h60_tp15_sl10
```

```
bar_p50_h90_tp15_sl10
```

Además, se conservan dos variantes comparativas:

```
bar_p40_h60_tp15_sl10
```

```
bar_p60_h60_tp15_sl10
```

La primera representa una versión más activa, mientras que la segunda representa una versión más selectiva.

Por lo tanto, los targets **BAR** que pasan a la siguiente etapa son:

Targets BAR principales:

```
bar_p50_h30_tp15_sl10
```

```
bar_p50_h60_tp15_sl10
```

```
bar_p50_h90_tp15_sl10
```

Targets BAR comparativos:

```
bar_p40_h60_tp15_sl10
```

```
bar_p60_h60_tp15_sl10
```

## 7.4. Targets OPC que pasan a la siguiente etapa

El análisis de **OPC** permitió evaluar la versión más completa de los targets, ya que combina dirección y resultado operativo en una sola clase.

Los targets **OPC** evaluados fueron:

```
opc_p50_h30_tp15_sl10
```

```
opc_p50_h60_tp15_sl10
```

```
opc_p50_h90_tp15_sl10
```

```
opc_p40_h60_tp15_sl10
```

```
opc_p60_h60_tp15_sl10
```

Dentro de este grupo, `opc_p50_h30_tp15_sl10` queda como el candidato principal más equilibrado. Presenta menor pérdida de datos por ventanas futuras incompletas, una frecuencia operativa razonable, alta proporción de trades ganadores entre resueltos y menor exposición temporal que los horizontes de 60 y 90 minutos.

La variante `opc_p40_h60_tp15_sl10` es la más activa, con 40.61% de trades resueltos, pero también es más sensible al contexto. La variante `opc_p60_h60_tp15_sl10` es la más selectiva, con 23.19% de trades resueltos y 92.10% de ganadores entre resueltos, pero reduce bastante la cantidad de oportunidades.

Por lo tanto, los targets `OPC` que pasan a la siguiente etapa son:

Target OPC principal:

`opc_p50_h30_tp15_sl10`

Targets OPC alternativos:

`opc_p50_h60_tp15_sl10`

`opc_p50_h90_tp15_sl10`

Targets OPC comparativos:

`opc_p40_h60_tp15_sl10`

`opc_p60_h60_tp15_sl10`

## 7.5. Universo final de targets candidatos

El universo final de targets candidatos queda organizado por familias operativas:

Familia principal p50\_h30:

`dir_p50_h30`

`bar_p50_h30_tp15_sl10`

`opc_p50_h30_tp15_sl10`

Familia principal p50\_h60:

`dir_p50_h60`

`bar_p50_h60_tp15_sl10`

`opc_p50_h60_tp15_sl10`

Familia principal p50\_h90:

`dir_p50_h90`

`bar_p50_h90_tp15_sl10`

`opc_p50_h90_tp15_sl10`

Familia comparativa activa p40\_h60:

`dir_p40_h60`

`bar_p40_h60_tp15_sl10`

`opc_p40_h60_tp15_sl10`

Familia comparativa selectiva p60\_h60:

`dir_p60_h60`

`bar_p60_h60_tp15_sl10`

`opc_p60_h60_tp15_sl10`

En total, pasan a la siguiente etapa:

5 targets DIR

5 targets BAR

5 targets OPC

Total: 15 targets candidatos

Esta estructura permite comparar tres enfoques diferentes:

DIR → predicción direccional

BAR → predicción de resultado operativo condicionado por dirección

OPC → predicción operativa completa en una sola clase

## 7.6. Target operativo principal del stage

Bajo el análisis realizado hasta este punto, el target mejor posicionado como candidato operativo principal es:

opc\_p50\_h30\_tp15\_sl10

Este target no se selecciona porque sea el más activo ni porque sea el más selectivo. Se selecciona preliminarmente porque ofrece el mejor equilibrio general entre:

frecuencia operativa

baja pérdida de datos

horizonte corto

menor exposición temporal

interpretación clara

proporción razonable de trades resueltos

alta proporción de ganadores entre resueltos

sensibilidad contextual aceptable frente a H60 y H90

Además, el análisis por régimen y contexto mostró que el régimen intradiario por sí solo no genera grandes diferencias en la distribución de `OPC`, mientras que las mayores sensibilidades aparecen en combinaciones como `contract_regime`, `year_contract` y `year_regime`. Esto sugiere que, para etapas posteriores, será importante incorporar variables de contexto temporal y operativo dentro de las features.

La lectura operativa final queda así:

Target principal:

opc\_p50\_h30\_tp15\_sl10

Alternativa intermedia:

opc\_p50\_h60\_tp15\_sl10

Alternativa de horizonte largo:

opc\_p50\_h90\_tp15\_sl10

Escenario activo:

opc\_p40\_h60\_tp15\_sl10

Escenario selectivo:

opc\_p60\_h60\_tp15\_sl10



## 7.7. Consideraciones para la siguiente etapa

Aunque `opc_p50_h30_tp15_sl10` queda como el mejor candidato preliminar, no debe interpretarse todavía como target definitivo.

Hasta este punto, el análisis fue descriptivo, estadístico y operativo. La siguiente etapa deberá responder una pregunta diferente:

¿Existe información disponible en  $t$  que permita predecir estos targets de manera útil?

Para responder esta pregunta será necesario construir features causales y evaluar modelos predictivos respetando una separación temporal estricta.

La siguiente etapa deberá cumplir estas condiciones:

- Usar solo información disponible hasta  $t$ .
- No usar DIR, BAR ni OPC como features.
- No usar predicciones de un modelo como entrada de otro modelo.
- No usar splits aleatorios.
- No ajustar decisiones metodológicas con 2025–2026.
- Evaluar estabilidad fuera de muestra.
- Comparar resultados por año, régimen, contrato y contexto.

En particular, deberán evaluarse por separado tres problemas predictivos:

Modelo DIR:

$X_t \rightarrow \text{DIR}$

Modelo BAR:

$X_t \rightarrow \text{BAR}$

Modelo OPC:

$X_t \rightarrow \text{OPC}$

Esto permitirá determinar si conviene avanzar con un enfoque direccional, operativo por barreras o directamente con un target combinado.

## 7.8. Cierre del stage

Este stage permitió transformar un universo amplio de targets en un conjunto reducido, coherente y metodológicamente justificable.

Se construyeron y analizaron targets `DIR`, `BAR` y `OPC`, se validó su consistencia, se estudió su distribución, su estabilidad temporal y su comportamiento por régimen y contexto.

La principal conclusión es que los targets basados en `p50` y `TP 1.5 / SL 1.0` ofrecen el mejor equilibrio general para continuar. Dentro de ellos, `opc_p50_h30_tp15_sl10` queda como el candidato operativo principal del stage.

Sin embargo, la selección final dependerá de la etapa posterior de modelado. Un target puede estar bien construido y ser operativamente razonable, pero aun así no ser

fácilmente predecible.

Por este motivo, el cierre de este stage no define un target definitivo. Define un universo reducido de candidatos y una hipótesis principal para investigar:

Hipótesis principal:

opc\_p50\_h30\_tp15\_sl10 podría ser el target operativo más razonable para modelado predictivo inicial.

Hipótesis alternativas:

dir\_p50\_h30 y bar\_p50\_h30\_tp15\_sl10 pueden servir para comparar si conviene separar dirección y resultado operativo.

Hipótesis comparativas:

p40\_h60 permite evaluar un escenario más activo.

p60\_h60 permite evaluar un escenario más selectivo.

Con esto, el stage de targets queda cerrado y la siguiente etapa deberá enfocarse en feature engineering, evaluación de señal predictiva y validación fuera de muestra.

## 7.9. Guardado final de targets preseleccionados para siguiente stage

```
In [39]: # 7.9. Guardado final de targets preseleccionados para siguiente stage
# =====

import json
import pandas as pd
from pathlib import Path

print("Guardado final de targets preseleccionados")
print("=" * 80)

# -----
# 1) Validar objeto base
# -----

required_objects = [
    "df_targets_opc",
]

missing_objects = [
    obj for obj in required_objects
    if obj not in globals()
]

if missing_objects:
    raise NameError(
        "Faltan objetos necesarios para guardar targets seleccionados: "
        f"{missing_objects}"
    )

# -----
# 2) Definir targets preseleccionados
# -----

selected_dir_cols = [
```

```

        "dir_p50_h30",
        "dir_p50_h60",
        "dir_p50_h90",
        "dir_p40_h60",
        "dir_p60_h60",
    ]

    selected_bar_cols = [
        "bar_p50_h30_tp15_sl10",
        "bar_p50_h60_tp15_sl10",
        "bar_p50_h90_tp15_sl10",
        "bar_p40_h60_tp15_sl10",
        "bar_p60_h60_tp15_sl10",
    ]

    selected_opc_cols = [
        "opc_p50_h30_tp15_sl10",
        "opc_p50_h60_tp15_sl10",
        "opc_p50_h90_tp15_sl10",
        "opc_p40_h60_tp15_sl10",
        "opc_p60_h60_tp15_sl10",
    ]

    selected_opc_label_cols = [
        f"{col}_label"
        for col in selected_opc_cols
    ]

    # -----
    # 3) Columnas de contexto mínimas
    # -----

    context_cols = [
        "date",
        "minute_of_day",
        "regime_id",
        "regime_name",
        "contract",
        "contract_code",
        "year",
    ]

    context_cols = [
        col for col in context_cols
        if col in df_targets_opc.columns
    ]

    # -----
    # 4) Validar columnas
    # -----

    all_selected_cols = (
        context_cols
        + selected_dir_cols
        + selected_bar_cols
        + selected_opc_label_cols
        + selected_opc_cols
    )

    missing_cols = [

```

```

col for col in all_selected_cols
if col not in df_targets_opc.columns
]

if missing_cols:
    raise KeyError(
        "Faltan columnas en df_targets_opc:\n"
        f"{missing_cols}"
    )

# -----
# 5) Crear dataframe final de targets
# -----

df_targets_selected_final = df_targets_opc[all_selected_cols].copy()

# Crear dataset_split si no existe
if "dataset_split" not in df_targets_selected_final.columns:
    df_targets_selected_final["dataset_split"] = pd.Series(
        index=df_targets_selected_final.index,
        data=pd.NA
    )

    df_targets_selected_final.loc[
        df_targets_selected_final["year"].between(2020, 2024),
        "dataset_split"
    ] = "development_2020_2024"

    df_targets_selected_final.loc[
        df_targets_selected_final["year"].between(2025, 2026),
        "dataset_split"
    ] = "final_test_2025_2026"

# Reordenar para dejar dataset_split junto al contexto
ordered_cols = []

for col in [
    "date",
    "year",
    "dataset_split",
    "minute_of_day",
    "regime_id",
    "regime_name",
    "contract",
    "contract_code",
]:
    if col in df_targets_selected_final.columns:
        ordered_cols.append(col)

ordered_cols += selected_dir_cols
ordered_cols += selected_bar_cols
ordered_cols += selected_opc_label_cols
ordered_cols += selected_opc_cols

df_targets_selected_final = df_targets_selected_final[ordered_cols].copy()

print("\nShape df_targets_selected_final:")
print(df_targets_selected_final.shape)

print("\nColumnas finales:")

```

```

print(df_targets_selected_final.columns.tolist())

print("\nVista previa:")
print(df_targets_selected_final.head().to_string())

# -----
# 6) Crear metadata final
# -----

metadata_records = []

families = [
    {
        "family_name": "p50_h30",
        "target_role": "principal",
        "dir_col": "dir_p50_h30",
        "bar_col": "bar_p50_h30_tp15_sl10",
        "opc_col": "opc_p50_h30_tp15_sl10",
        "opc_label_col": "opc_p50_h30_tp15_sl10_label",
        "percentile": 50,
        "horizon": 30,
        "tp_mult": 1.5,
        "sl_mult": 1.0,
    },
    {
        "family_name": "p50_h60",
        "target_role": "principal",
        "dir_col": "dir_p50_h60",
        "bar_col": "bar_p50_h60_tp15_sl10",
        "opc_col": "opc_p50_h60_tp15_sl10",
        "opc_label_col": "opc_p50_h60_tp15_sl10_label",
        "percentile": 50,
        "horizon": 60,
        "tp_mult": 1.5,
        "sl_mult": 1.0,
    },
    {
        "family_name": "p50_h90",
        "target_role": "principal",
        "dir_col": "dir_p50_h90",
        "bar_col": "bar_p50_h90_tp15_sl10",
        "opc_col": "opc_p50_h90_tp15_sl10",
        "opc_label_col": "opc_p50_h90_tp15_sl10_label",
        "percentile": 50,
        "horizon": 90,
        "tp_mult": 1.5,
        "sl_mult": 1.0,
    },
    {
        "family_name": "p40_h60",
        "target_role": "comparativo_activo",
        "dir_col": "dir_p40_h60",
        "bar_col": "bar_p40_h60_tp15_sl10",
        "opc_col": "opc_p40_h60_tp15_sl10",
        "opc_label_col": "opc_p40_h60_tp15_sl10_label",
        "percentile": 40,
        "horizon": 60,
        "tp_mult": 1.5,
        "sl_mult": 1.0,
    },
]

```

```

    {
        "family_name": "p60_h60",
        "target_role": "comparativo_selectivo",
        "dir_col": "dir_p60_h60",
        "bar_col": "bar_p60_h60_tp15_sl10",
        "opc_col": "opc_p60_h60_tp15_sl10",
        "opc_label_col": "opc_p60_h60_tp15_sl10_label",
        "percentile": 60,
        "horizon": 60,
        "tp_mult": 1.5,
        "sl_mult": 1.0,
    },
]

for family in families:

    metadata_records.append({
        **family,
        "target_type": "DIR",
        "target_col": family["dir_col"],
        "target_description": "Dirección significativa del movimiento futuro.",
        "class_definition": "-1=SHORT, 0=NO_SIGNAL, 1=LONG",
    })

    metadata_records.append({
        **family,
        "target_type": "BAR",
        "target_col": family["bar_col"],
        "target_description": "Resultado operativo por barreras TP/SL.",
        "class_definition": "-1=SL, 0=NO_EVENT, 1=TP",
    })

    metadata_records.append({
        **family,
        "target_type": "OPC",
        "target_col": family["opc_col"],
        "target_description": "Target operativo combinado DIR + BAR.",
        "class_definition": "0=NO_TRADE, 1=LONG_TP, 2=LONG_SL, 3=SHORT_TP, 4=SHO

df_targets_selected_metadata = pd.DataFrame(metadata_records)

# -----
# 7) Resumen de decisión del stage
# -----

stage_targets_decision = {
    "stage": "04_targets",
    "description": "Targets preseleccionados para evaluación predictiva posterior",
    "main_operational_candidate": "opc_p50_h30_tp15_sl10",
    "main_dir_candidate": "dir_p50_h30",
    "main_bar_candidate": "bar_p50_h30_tp15_sl10",
    "selected_dir_targets": selected_dir_cols,
    "selected_bar_targets": selected_bar_cols,
    "selected_opc_targets": selected_opc_cols,
    "selected_opc_label_targets": selected_opc_label_cols,
    "families": families,
    "methodological_rules": [
        "Targets contruidos sin cruzar días.",
        "Thresholds calibrados con 2020-2024.",
    ],
}

```

```

        "2025-2026 reservado como final_test.",
        "No usar targets DIR, BAR u OPC como features.",
        "No usar predicciones de otros modelos como features en esta etapa.",
        "Cada target debe evaluarse como problema predictivo independiente.",
    ],
}

# -----
# 8) Guardado
# -----

if "MNQ_TARGETS_PATH" not in globals():
    raise NameError("MNQ_TARGETS_PATH no está definido.")

MNQ_TARGETS_PATH.mkdir(parents=True, exist_ok=True)

MNQ_TARGETS_SELECTED_FINAL_PARQUET = (
    MNQ_TARGETS_PATH / "mnq_targets_selected_final.parquet"
)

MNQ_TARGETS_SELECTED_FINAL_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_selected_final.csv"
)

MNQ_TARGETS_SELECTED_METADATA_CSV = (
    MNQ_TARGETS_PATH / "mnq_targets_selected_metadata.csv"
)

MNQ_TARGETS_SELECTED_METADATA_PARQUET = (
    MNQ_TARGETS_PATH / "mnq_targets_selected_metadata.parquet"
)

MNQ_TARGETS_STAGE_DECISION_JSON = (
    MNQ_TARGETS_PATH / "mnq_targets_stage_decision.json"
)

df_targets_selected_final.to_parquet(MNQ_TARGETS_SELECTED_FINAL_PARQUET)
df_targets_selected_final.to_csv(MNQ_TARGETS_SELECTED_FINAL_CSV)

df_targets_selected_metadata.to_parquet(MNQ_TARGETS_SELECTED_METADATA_PARQUET)
df_targets_selected_metadata.to_csv(MNQ_TARGETS_SELECTED_METADATA_CSV, index=False)

with MNQ_TARGETS_STAGE_DECISION_JSON.open("w", encoding="utf-8") as f:
    json.dump(
        stage_targets_decision,
        f,
        indent=2,
        ensure_ascii=False
    )

print("\nArchivos guardados:")
print(f"Targets seleccionados parquet: {MNQ_TARGETS_SELECTED_FINAL_PARQUET}")
print(f"Targets seleccionados csv: {MNQ_TARGETS_SELECTED_FINAL_CSV}")
print(f"Metadata seleccionados parquet: {MNQ_TARGETS_SELECTED_METADATA_PARQUET}")
print(f"Metadata seleccionados csv: {MNQ_TARGETS_SELECTED_METADATA_CSV}")
print(f"Decisión del stage json: {MNQ_TARGETS_STAGE_DECISION_JSON}")

print("\nGuardado final de targets preseleccionados finalizado correctamente.")

```

## Guardado final de targets preseleccionados

=====

Shape df\_targets\_selected\_final:  
(1024062, 26)

Columnas finales:

```
['date', 'year', 'dataset_split', 'minute_of_day', 'regime_id', 'contract', 'dir_p50_h30', 'dir_p50_h60', 'dir_p50_h90', 'dir_p40_h60', 'dir_p60_h60', 'bar_p50_h30_tp15_sl10', 'bar_p50_h60_tp15_sl10', 'bar_p50_h90_tp15_sl10', 'bar_p40_h60_tp15_sl10', 'bar_p60_h60_tp15_sl10', 'opc_p50_h30_tp15_sl10_label', 'opc_p50_h60_tp15_sl10_label', 'opc_p50_h90_tp15_sl10_label', 'opc_p40_h60_tp15_sl10_label', 'opc_p60_h60_tp15_sl10_label', 'opc_p50_h30_tp15_sl10', 'opc_p50_h60_tp15_sl10', 'opc_p50_h90_tp15_sl10', 'opc_p40_h60_tp15_sl10', 'opc_p60_h60_tp15_sl10']
```

Vista previa:

|           | date       | year           | dataset_split | minute_of_day |                       |             |               |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  |   |
|-----------|------------|----------------|---------------|---------------|-----------------------|-------------|---------------|---------------|---------------|-------------------|-------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|--|---|
| regime_id | contract   | dir_p50_h30    | dir_p50_h60   | dir_p50_h90   | dir_p40_h60           | dir_p60_h60 | h30_tp15_sl10 | h60_tp15_sl10 | h90_tp15_sl10 | p40_h60_tp15_sl10 | p60_h60_tp15_sl10 | opc_p50_h30_tp15_sl10_label | opc_p50_h60_tp15_sl10_label | opc_p50_h90_tp15_sl10_label | opc_p40_h60_tp15_sl10_label | opc_p60_h60_tp15_sl10_label | opc_p50_h30_tp15_sl10 | opc_p50_h60_tp15_sl10 | opc_p50_h90_tp15_sl10 | opc_p40_h60_tp15_sl10 | opc_p60_h60_tp15_sl10 |  |   |
| datetime  | 2020-01-02 | 04:30:00-05:00 | 2020-01-02    | 2020          | development_2020_2024 |             | 270           |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  |   |
|           | 0          | H20            | 0             | 0             | 0                     | 0           | 0             |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  |   |
|           | <NA>       |                | <NA>          |               | <NA>                  |             | <NA>          |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  |   |
|           | <NA>       |                | NO_TRADE      |               |                       | NO_TRADE    |               |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  | N |
|           | O_TRADE    |                | NO_TRADE      |               |                       | NO_TRADE    |               |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  |   |
|           | 0.0        |                | 0.0           |               |                       | 0.0         |               |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  |   |
|           | 0.0        |                |               |               |                       |             |               |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  |   |
|           | 2020-01-02 | 04:31:00-05:00 | 2020-01-02    | 2020          | development_2020_2024 |             | 271           |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  |   |
|           | 0          | H20            | 0             | 0             | 0                     | 0           | 0             |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  |   |
|           | <NA>       |                | <NA>          |               | <NA>                  |             | <NA>          |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  |   |
|           | <NA>       |                | NO_TRADE      |               |                       | NO_TRADE    |               |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  | N |
|           | O_TRADE    |                | NO_TRADE      |               |                       | NO_TRADE    |               |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  |   |
|           | 0.0        |                | 0.0           |               |                       | 0.0         |               |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  |   |
|           | 0.0        |                |               |               |                       |             |               |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  |   |
|           | 2020-01-02 | 04:32:00-05:00 | 2020-01-02    | 2020          | development_2020_2024 |             | 272           |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  |   |
|           | 0          | H20            | 0             | 0             | 0                     | 0           | 0             |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  |   |
|           | <NA>       |                | <NA>          |               | <NA>                  |             | <NA>          |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  |   |
|           | <NA>       |                | NO_TRADE      |               |                       | NO_TRADE    |               |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  | N |
|           | O_TRADE    |                | NO_TRADE      |               |                       | NO_TRADE    |               |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  |   |
|           | 0.0        |                | 0.0           |               |                       | 0.0         |               |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  |   |
|           | 0.0        |                |               |               |                       |             |               |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  |   |
|           | 2020-01-02 | 04:33:00-05:00 | 2020-01-02    | 2020          | development_2020_2024 |             | 273           |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  |   |
|           | 0          | H20            | 0             | 0             | 0                     | 0           | 0             |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  |   |
|           | <NA>       |                | <NA>          |               | <NA>                  |             | <NA>          |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  |   |
|           | <NA>       |                | NO_TRADE      |               |                       | NO_TRADE    |               |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  | N |
|           | O_TRADE    |                | NO_TRADE      |               |                       | NO_TRADE    |               |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  |   |
|           | 0.0        |                | 0.0           |               |                       | 0.0         |               |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  |   |
|           | 0.0        |                |               |               |                       |             |               |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  |   |
|           | 2020-01-02 | 04:34:00-05:00 | 2020-01-02    | 2020          | development_2020_2024 |             | 274           |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  |   |
|           | 0          | H20            | 0             | 0             | 0                     | 0           | 0             |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  |   |
|           | <NA>       |                | <NA>          |               | <NA>                  |             | <NA>          |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  |   |
|           | <NA>       |                | NO_TRADE      |               |                       | NO_TRADE    |               |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  | N |
|           | O_TRADE    |                | NO_TRADE      |               |                       | NO_TRADE    |               |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  |   |
|           | 0.0        |                | 0.0           |               |                       | 0.0         |               |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  |   |
|           | 0.0        |                |               |               |                       |             |               |               |               |                   |                   |                             |                             |                             |                             |                             |                       |                       |                       |                       |                       |  |   |



Archivos guardados:

Targets seleccionados parquet: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_selected\_final.parquet

Targets seleccionados csv: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_selected\_final.csv

Metadata seleccionados parquet: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_selected\_metadata.parquet

Metadata seleccionados csv: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_selected\_metadata.csv

Decisión del stage json: c:\Users\heguu\OneDrive\Escritorio\neural\_profit\_local\data\04\_mnq\_targets\mnq\_targets\_stage\_decision.json

Guardado final de targets preseleccionados finalizado correctamente.